VROON

FINAL

STABILITY BOOKLET

APPROVED
on behalf of the
government
of the vessel's registry
subject to conditions
of ABS letter

ABS
FOUNDED 1862

This document satisfies
the stability requirements of
REG. 10 (2), ANNEX I
of the International
Convention on
Load Lines, 1966
as amended by the
1988 Load Line Protocol

AS AMENDED
on pages 14, 21, 40, 41
74-87, 118-204, 261

YARD NO. N571

"VOS PASSION"

ULSTEIN®

See ABS Singapore Letter Ref 1149174 Dated 04-APR-2016

Ulstein Design & Solutions AS
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N-6067 ULSTEINVIK
NORWAY

RESTRICTED

ULSTEIN®

FINAL STABILITY BOOKLET

for

ULSTEIN PX121

YARD NO. N571

"VOS PASSION"

Calculation :	**Approved :**
Adrian Fotache _	 _

Ulsteinvik, 10.3.2016 *Dwg. no. U10747.4_152-200-01_A*

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Ulsteinvik 26.11.2015

Ulstein Design & Solutions AS kindly advice you to familiarize yourself with this stability manual.

Especially Chapter 6 should be read with high attention.

All loading conditions in this manual are examples only and should be used as such.

When comparing the conditions and calculations in this manual with your own, please check carefully the centre of gravity of all loads and the amount of liquids and corresponding free surface moments.

Compliance with the stability criteria as shown does not absolve the Master from his responsibilities. Master should therefore exercise prudence and good seamanship, having regard to the season of year, weather forecasts and navigational zone, and should take appropriate action as to speed, course and loads warranted by the prevailing circumstances and the ships capacities.

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See ABS Singapore Letter Ref 1499171 Dated 04-APR-2016

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1 NOMENCLATURE

SYMBOL	DESCRIPTION	TYPICAL UNIT
A	Area	m2
AP	Aft perpendicular	-
ATTV	Attained value	-
B	Moulded breadth	meters
BD	Buoyancy distribution	tonnes
BDWL	Breadth at design waterline	meters
BEND	Bending moment	tonn*m
BMT	Trasverse metacentric radius	m
CASE	Initial condition / damage case	-
CB	Block coefficient	-
CG	Centre of gravity	m
CGX	CGX of volume	m
CGY	CGY of volume	m
CGZ	CGZ of volume	m
CI	Common angle in interval	degrees
CM	Midship section coefficient	-
CP	Prismatic coefficient	-
CROSSTM	Cross-flooding time (if applicable)	sec
CW	Waterplane coefficient	-
D	Depth to freeboard deck	meters
d	Internal diameter	meters
D	Outer diameter	meters
DAM	Damage case	-
DCRI	Determining criterion	-
Delta	Angle of heel	degrees
Delta-f	Flooding angle	degrees
Delta-max	Angle where GZ occurs at maximum level	degrees
DES	Descriptive text	-
dGM	Reduction in GM0	meters
DGM	Reduction in GM0	meters
DGZ	Reduction in GZ	meters
DISP	Displacement	tonnes
DSP0	Intact displacement	tonnes
DW	Dead-weight	tonnes
DWL	Design water line	-
EPHI	E(phi), area below GZ-curve	m*radians
FA	Flooding angle	degrees
FAWE	Flooding angle of weathertight opening	degrees
FAUN	Flooding angle of unprotected opening	degrees
FP	Fore perpendicular	-
FR	Building frame number	#
FRA	Total frame area	m2
FRMAX	Maximum frame no.	#
FRMIN	Minimum frame no.	#
FRSM	Free surface moment	m4, tonn*m
FSM	Free surface moment	m4, tonn*m
FSMOM	Free surface moment	m4, tonn*m
GM	Metacentric height	m
GM0	Uncorrected GM	m
GSTYPE	Geometric subtype	-
GT	Gross tonnage	GT
GTYPE	Geometric type	-
GZ	Righting lever	meters
HEEL	Heeling angle	degrees
HEEL0	Initial heeling angle	degrees
HPHI	Righting lever curve	meters
IMAX	Maximum moment of inertia of surface	m4
IMMA	Immersion angle	degrees
IMMR	Reserve to immersion	meters
IMRES	Reserve to immersion of openings	m
INIT	Initial condition	-
IYMAX	Maximum Iy of surface	m4
KG	Distance from keel to centre of gravity	meters
KM	Transverse metacentric height = KMT	meters
KMT	Transverse metacentric height	m
KN	Distance from keel to metacentric axis	meters
L	Length acc. International Load Line Convention	meters
LCB	Longitudinal centre of buoyancy	meters
LCB	Longitudinal centre of buoyancy	m
LCF	Longitudinal centre of flotation	meters
LCG	Longitudinal centre of gravity	meters
LPP	Length between perpendiculars	meters
LREF	Distance from AP to FP = LPP	meters
LWL	Length of waterline	meters

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MAXKG	Maximum KG	meters
MCT	Moment to change trim one centimeters	tonn*m/cm
MINGM	Minimum metacentric height	meters
ML	Longitudinal moment of weights	tonn*m
MS	Residual stability	meters
MT	Transversal moment of weights	meters
MV	Vertical moment of weights	tonn*m
NAME	NAPA definition of name	-
NT	Net tonnage	NT
OPEN	Critical opening	-
OPNAME	Name of opening	-
ORNT	Orientation	-
OTYPE	Type of opening	-
PERM	Permeability	-
PHASE	Flooding phase	-
PHI	Angle of heel	rad, degrees
REDPD	Reduction per 1 degree	meters
REFQ	Reference coordinate	-
REQ	Required value	meters, degrees
RESFLD	Reserve to downflooding	meters
RESMRG	Reserve to immersion of margin line	meters
RHO	Density	tonnes/m3
SC	Side condition	-
SHEAR	Shear force	tonnes
SIDE	Side of ship SB/PS	-
SM	Simpsons multiplicator	-
SS	Status of calculation sections	-
STAGE	Flooding stage	-
SWL	Draught at summer water line	meters
T	Draught, moulded	meters
T0	Initial draught	meters
TA	Draught at AP	meters
TCB	Transversal centre of buoyancy	meters
TCG	Transversal centre of gravity	meters
TCP	Change of displacement/change of draught	tonnes/cm
TDWL	Draught at design waterline	meters
TF	Draught at FP	meters
TK	Draught below keel	meters
TR	Trim	meters
TR0	Initial trim	meters
TRF	Trim factor = DISP/(100*MCT)	-
TRFAP	Trim factor aft = DISP*LCF/(100*MCT*L)	-
TRFFP	Trim factor fwd = DISP*(L-LCF)/(100*MCT*L)	-
V	Volume	m3
VCB	Vertical centre of buoyancy	meters
VCG	Vertical centre of gravity above baseline	meters
VNET	Net volume	m3
VOL	Volume	m3
VOLT	Total volume, Displacement in fresh water	m3, tonnes
W	Weight	tonnes
WD	Weight distribution	tonnes
WLA	Waterline area	m2
WSA	Wetted surface area	m2
X	Distance in longitudinal direction	meters
X	Longitudinal coordinate	meters
XCG	Longitudinal centre of gravity	meters
XM	Longitudinal centre of gravity of load	meters
XMIN	Maximum X	meters
XMIN	Minimum X	meters
XREF	Distance from AP to midship	meters
Y	Distance in transverse direction	meters
Y	Transversal coordinate	meters
YCG	Transversal centre of gravity	meters
YM	Transversal centre of gravity of load	meters
Z	Distance in vertical direction	meters
Z	Vertical coordinate	meters
ZCG	Vertical centre of gravity	meters
ZM	Vertical centre of gravity of load	meters
ZMAX	Maximum Z	meters
ZMIN	Minimum Z	meters

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2 GENERAL PARTICULARS

2.1 GENERAL PARTICULARS

SHIP IDENTIFICATION

Ship name	VOS PASSION
Ship type	ULSTEIN PX121
Ship purpose	OFFSHORE SERVICE VESSEL
Signal letters	ZDPX7
Official number (IMO number)	9709130
Port of registry	GIBRALTAR
Flag	GIBRALTAR
Classification society	ABS
Classification society identification number	3201405

Classification, Class notations :
ABS + 1A1,OFFSHORE SUPORT VESSEL (SUPPLY - HNLS,FFV1), +AMS, +DPS-2, +ACCU, ENVIRO, GP, HAB(WB), UWILD ,CRC

OWNERS IDENTIFICATION

Name of the owner	VROON B.V.
Postal address	
Country	GIBRALTAR

BUILDERS IDENTIFICATION

Name of the builder	COSCO (GUANGDONG) SHIPYARD CO., LTD
Postal address	DASHENG INDUSTRIAL ZONE, MACHONG TOWN
Country	DONGGUAN, GUANGDONG PROVINCE, CHINA

Yard number	N571
Date keel laid	10.11.2014
Name of the sister vessel (if any)	VOS PACE

MAIN PARTICULARS, MOULDED VALUES

Length over all	83.314 m
Length between perpendiculars	79.612 m
Length of the vessel acc. ICLL 1966	79.614 m
Length of the vessel according to class	79.554 m
Breadth moulded	18 m
Maximum breadth	18 m
Moulded depth to maindeck	8 m
Summer load draught	6.7 m
Keel thickness	12 mm
Mean thickness of the shell	12 mm

DEADWEIGHT AT SUMMER LOAD DRAUGHT

Lightweight	2704.9 t
Deadweight	4138.4 t
Block coefficient	0.693
Displacement, moulded	6817.4 t
Displacement, total	6843.3 t
Volume, moulded	6651.1 cbm
Volume, total	6676.4 cbm
Assumed water density	1.025 t/cbm

VESSEL LIMITATIONS

Trim limit	-1.5 m aft, 1.5 m forward
Lowest service draught	3.2 m
Lowest draught forward due to slamming	3.5 m
Lowest draught aft due to propeller immersion	3.6 m

TONNAGE CAPACITIES

Gross tonnage	3638
Net tonnage	1295

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2.2 GENERAL REMARKS

This stability booklet is created with use of the computer program NAPA (Release 2015.22).
This program has been checked and approved by the following authorities:

- DNVGL
- Lloyd's Register
- Maritime and Coastguard Agency
- Nippon Kaiji Kyokai
- Norwegian Maritime Administration
- Russian Maritime Register of Shipping

The vessel complies with necessary stability requirements developed by Authorities. For more details reg. stability requirements see Chapter 6) INSTRUCTIONS FOR MASTER.
One copy of this booklet should be kept on board together with the approval letters from the classification society and national authorities.

THE MASTER SHOULD READ THE BOOKLET CAREFULLY!

NOTE:
The calculation of the stability curve and the metacentric height, GM, take the actual floating position into account. Therefore small deviation to manual calculations can NOT be prevented:

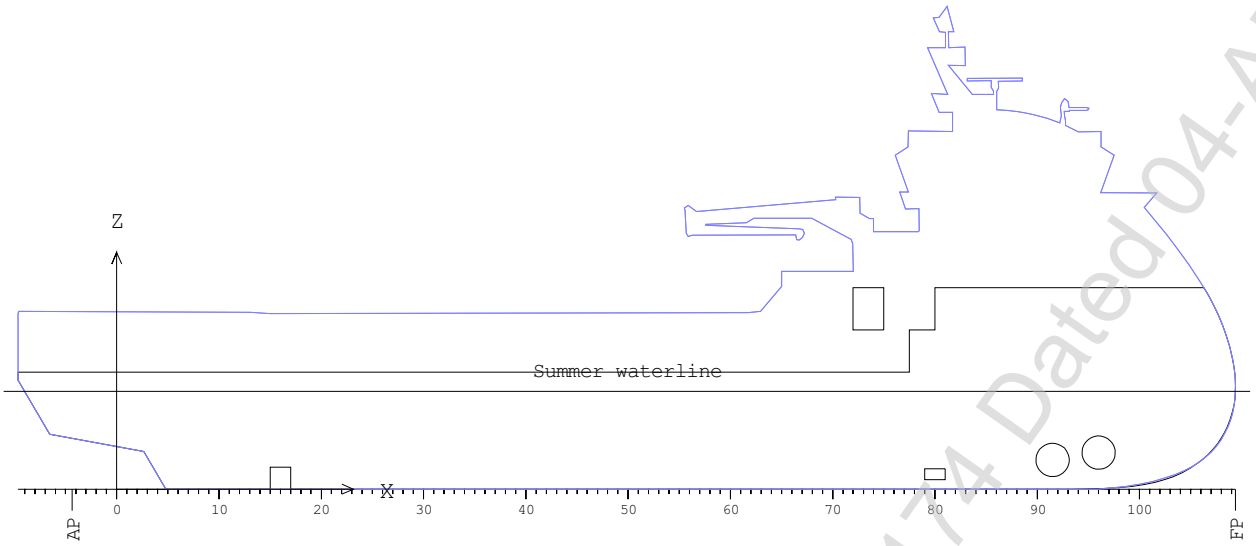
The real free surface moment is calculated for the tanks at their actual filling. The correction for different heeling angles is got by calculating the real new center of gravity for the load in the heeled tank. The correction for the initial GM is got by calculating the real moment of inertia for the free surface in the tank.

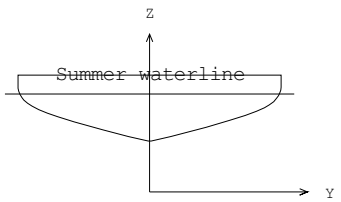
In addition, according to regulations for stability calculations, some free surface moments may be assumed more conservative, and therefore final GM may differ from the one calculated manually.

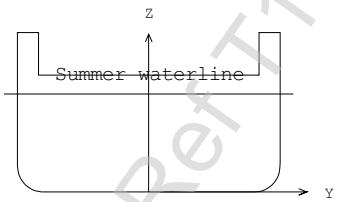
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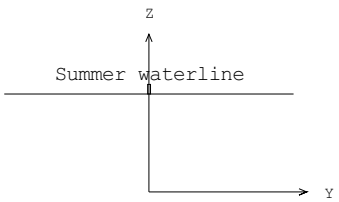
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2.3 REFERENCE SYSTEM









Main characteristics of the vessel

Length between perpendiculars	79.61	m
Length over all	83.31	m
Forward perpendicular at	76.56	m
Aft perpendicular at	-3.05	m
x coordinate of frame 0	0.00	m
Breadth moulded	18.00	m
Maximum breadth	18.00	m
Depth moulded	8.00	m
Summer draught	6.70	m
Design draught	6.00	m
Mean thickness of shell plating	12.0	mm
Thickness of keel plate	12.0	mm
Assumed density of water	1.025	ton/cbm

Sign convention

Zero references are as follows:
In longitudinal direction frame 0 (#0), positive direction is forward, negative aft
In transverse direction center line (CL), positive is SB, negative is PS
In vertical direction base line (BL), positive upwards, negative downwards

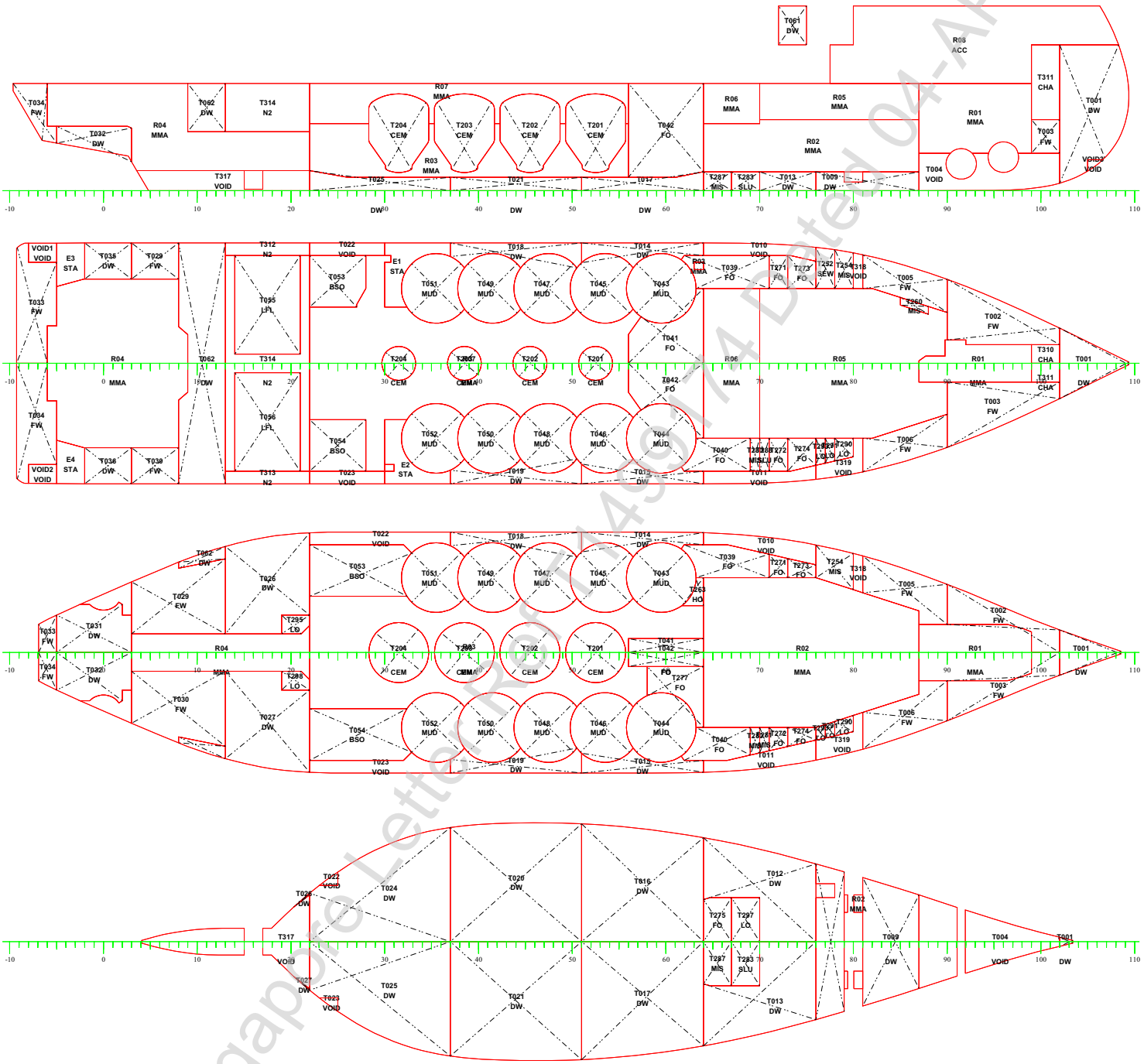
Trim by bow is assumed as positive (+)
Trim by stern is assumed as negative (-)
Heel towards starboard side (SB) is assumed as positive (+)
Heel towards port side (PS) is assumed as negative (-)

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3 ARRANGEMENT

3.1 COMPARTMENT ARRANGEMENT



See ABS Singapore Letter of 14961 74 Dated 04-APR-2016

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3.2 COMPARTMENT TABLE

NAME	DES	PURP	VOLM m3	VNET m3	PERM	RHO t/m3	FRMAX #	FRMIN #	CGX m	CGY m	CGZ m	FSMAX tm
CAPACITY OF Accomodation (RHO=1)												
R08	Acc. superstructure	ACC	1470.7	1426.6	0.95	1.000	108.3	74.0	61.56	-0.06	11.05	6131
CAPACITY OF Base oil (RHO=0.83)												
T053	STORAGE TANK PS	BSO	131.2	128.6	0.95	0.830	32.9	22.0	18.71	-6.02	4.82	28
T054	STORAGE TANK SB	BSO	131.0	128.3	0.95	0.830	32.9	22.0	18.69	6.02	4.84	28
SUBTOTAL		BSO	262.2	256.9					18.70	-0.01	4.83	
CAPACITY OF Dry bulk (RHO=2.4)												
T201	DRY BULK TANK	CEM	66.3	65.0	0.95	2.400	55.7	49.3	36.75	0.00	4.67	47
T202	DRY BULK TANK	CEM	66.3	65.0	0.95	2.400	48.7	42.3	31.85	0.00	4.67	47
T203	DRY BULK TANK	CEM	66.3	65.0	0.95	2.400	41.7	35.3	26.95	0.00	4.67	47
T204	DRY BULK TANK	CEM	66.3	65.0	0.95	2.400	34.7	28.3	22.05	0.00	4.67	47
SUBTOTAL		CEM	265.2	259.9					29.40	0.00	4.67	
CAPACITY OF Chain Locker (RHO=1.025)												
T310	CHAIN LOCKER PS	CHA	16.5	16.0	0.60	1.025	102.0	99.0	70.35	-0.70	8.10	0
T311	CHAIN LOCKER SB	CHA	16.5	16.0	0.60	1.025	102.0	99.0	70.35	0.70	8.10	0
SUBTOTAL		CHA	32.9	31.9					70.35	0.00	8.10	
CAPACITY OF Drill water (RHO=1)												
T001	FOREPEAK TANK	DW	105.6	102.4	0.95	1.000	109.4	102.0	73.11	0.00	7.21	33
T009	DB TANK CE	DW	77.8	75.4	0.95	1.000	87.0	76.0	56.90	0.09	0.73	596
T012	DB TANK PS	DW	56.5	54.8	0.95	1.000	76.0	64.0	49.41	-3.85	0.76	200
T013	DB TANK SB	DW	56.5	54.8	0.95	1.000	76.0	64.0	49.41	3.85	0.76	200
T014	WING TANK PS	DW	61.9	60.1	0.95	1.000	64.0	51.0	40.15	-8.42	4.73	1
T015	WING TANK SB	DW	61.8	60.0	0.95	1.000	64.0	51.0	40.15	8.42	4.73	1
T016	DB TANK PS	DW	86.1	83.5	0.95	1.000	64.0	51.0	40.20	-4.33	0.62	462
T017	DB TANK SB	DW	86.1	83.5	0.95	1.000	64.0	51.0	40.20	4.33	0.62	462
T018	WING TANK PS	DW	74.7	72.5	0.95	1.000	51.0	37.0	30.79	-8.37	4.73	2
T019	WING TANK SB	DW	74.7	72.5	0.95	1.000	51.0	37.0	30.79	8.37	4.73	2
T020	DB TANK PS	DW	97.3	94.4	0.95	1.000	51.0	37.0	30.84	-4.63	0.61	568
T021	DB TANK SB	DW	97.3	94.4	0.95	1.000	51.0	37.0	30.84	4.63	0.61	568
T024	DB TANK PS	DW	76.3	74.0	0.95	1.000	37.0	22.0	21.46	-3.85	0.81	344
T025	DB TANK SB	DW	76.3	74.0	0.95	1.000	37.0	22.0	21.46	3.85	0.81	344
T026	WING TANK PS	DW	92.9	90.1	0.95	1.000	22.0	13.0	12.42	-4.45	3.13	178
T027	WING TANK SB	DW	92.9	90.1	0.95	1.000	22.0	13.0	12.42	4.45	3.13	178
T031	WING TANK PS	DW	27.3	26.5	0.95	1.000	3.0	-5.0	-0.65	-1.76	4.18	34
T032	WING TANK SB	DW	27.3	26.5	0.95	1.000	3.0	-5.0	-0.65	1.76	4.18	34
T035	WING TANK PS	DW	23.8	23.1	0.95	1.000	3.0	-2.0	0.39	-7.51	6.69	6
T036	WING TANK SB	DW	23.8	23.1	0.95	1.000	3.0	-2.0	0.39	7.51	6.69	6
T061	ROLL RED. TANK	DW	129.1	125.2	0.95	1.000	76.0	72.0	51.66	0.00	12.35	1325
T062	ROLL RED. TANK	DW	211.8	205.4	0.95	1.000	13.0	8.0	7.44	0.00	6.21	1694
SUBTOTAL		DW	1718.0	1666.5					31.45	0.00	3.83	

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NAME	DES	PURP	VOLM m3	VNET m3	PERM	RHO t/m3	FRMAX #	FRMIN #	CGX m	CGY m	CGZ m	FSMAX tm
CAPACITY OF Diesel Oil (RHO=0.86)												
T039	STORAGE TANK PS	FO	81.9	79.4	0.95	0.860	71.0	61.6	46.68	-6.76	4.97	6
T040	STORAGE TANK SB	FO	65.9	63.9	0.95	0.860	69.0	61.6	46.00	6.80	4.93	5
T041	STORAGE TANK PS	FO	83.8	81.3	0.95	0.860	64.0	56.0	42.17	-1.52	5.53	27
T042	STORAGE TANK SB	FO	83.8	81.3	0.95	0.860	64.0	56.0	42.17	1.52	5.53	27
T271	FO SETTLING TANK PS	FO	14.6	14.2	0.95	0.860	73.0	71.0	50.39	-6.62	5.66	1
T272	FO SETTLING TANK SB	FO	14.6	14.2	0.95	0.860	73.0	71.0	50.39	6.62	5.66	1
T273	FO SERVICE TANK PS	FO	18.2	17.6	0.95	0.860	76.0	73.0	52.10	-6.50	5.84	2
T274	FO SERVICE TANK SB	FO	18.2	17.6	0.95	0.860	76.0	73.0	52.10	6.50	5.84	2
T275	FO DRAIN TANK	FO	9.6	9.3	0.95	0.860	67.0	64.0	45.85	-1.64	0.70	5
T277	FO OVERFLOW TANK	FO	42.7	41.4	0.95	0.860	64.0	58.0	42.99	2.51	3.06	10
T279	EMERG. GEN. FO TANK	FO	2.3	2.2	0.95	0.860	89.0	87.0	61.59	-1.38	24.60	0
SUBTOTAL		FO	435.5	422.5					45.24	-0.04	5.12	
CAPACITY OF Fresh Water (RHO=1)												
T002	WING TANK PS	FW	117.2	113.7	0.95	1.000	102.0	90.0	67.06	-2.88	6.32	45
T003	WING TANK SB	FW	118.5	115.0	0.95	1.000	102.0	90.0	67.02	2.86	6.32	46
T005	WING TANK PS	FW	101.2	98.2	0.95	1.000	91.0	81.0	59.97	-5.07	4.86	34
T006	WING TANK SB	FW	101.2	98.2	0.95	1.000	91.0	81.0	59.97	5.07	4.86	34
T029	WING TANK PS	FW	78.5	76.2	0.95	1.000	13.0	3.0	5.26	-4.93	4.67	83
T030	WING TANK SB	FW	78.5	76.2	0.95	1.000	13.0	3.0	5.26	4.93	4.67	83
T033	WING TANK PS	FW	60.9	59.1	0.95	1.000	-5.0	-9.6	-4.96	-3.92	6.52	126
T034	WING TANK SB	FW	60.9	59.1	0.95	1.000	-5.0	-9.6	-4.96	3.92	6.52	126
SUBTOTAL		FW	716.9	695.4					39.28	0.00	5.58	
CAPACITY OF Hydraulic oil (RHO=0.924)												
T263	HYDRAULIC OIL STORE TANK	HO	5.3	5.1	0.95	0.924	64.0	61.7	44.30	-4.34	2.76	1
CAPACITY OF Low flashpoint liquid (RHO=1.2)												
T055	LFL TANK PS	LFL	79.2	77.6	0.95	1.200	21.0	14.0	12.25	-4.38	6.10	195
T056	LFL TANK SB	LFL	79.2	77.6	0.95	1.200	21.0	14.0	12.25	4.38	6.10	195
SUBTOTAL		LFL	158.5	155.3					12.25	0.00	6.10	
CAPACITY OF Lubricating oil (RHO=0.924)												
T290	LO STORE TANK MAIN ENG.	LO	9.4	9.1	0.95	0.924	80.0	78.0	55.29	5.97	5.66	1
T291	LO STORE TANK MAIN ENG.	LO	4.9	4.8	0.95	0.924	78.0	77.0	54.25	6.20	5.82	1
T293	LO STORE TANK THRUSTER	LO	5.1	5.0	0.95	0.924	77.0	76.0	53.55	6.34	5.92	1
T295	LO STORE TANK AZIMUTH TH.	LO	6.3	6.1	0.95	0.924	22.0	19.0	14.33	-2.11	3.28	0
T297	LO DRAIN TANK	LO	9.7	9.4	0.95	0.924	70.0	67.0	47.95	-1.64	0.70	6
T298	DIRTY LO TANK	LO	6.3	6.1	0.95	0.924	22.0	19.0	14.33	2.11	3.28	0
SUBTOTAL		LO	41.8	40.5					40.85	2.47	3.84	
CAPACITY OF Miscellaneous (RHO=1)												
T254	GREY WATER TANK	MIS	33.3	32.3	0.95	1.000	80.0	76.0	54.83	-6.49	4.59	6
T280	BILGE WATER HOLDING TANK	MIS	9.1	8.9	0.95	1.000	70.0	69.0	48.65	6.66	5.17	1
T281	BILGE WATER SETTLING TANK	MIS	4.1	4.0	0.95	1.000	71.0	70.0	49.34	6.41	3.68	1
T287	BILGE WATER DRAIN TANK	MIS	9.6	9.3	0.95	1.000	67.0	64.0	45.85	1.64	0.70	6
T260	WATER MIST TANK	MIS	2.5	2.4	0.95	1.000	88.0	85.0	60.55	-4.30	6.30	0
SUBTOTAL		MIS	58.5	56.8					52.25	-2.11	4.05	

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NAME	DES	PURP	VOLM m3	VNET m3	PERM	RHO t/m3	FRMAX #	FRMIN #	CGX m	CGY m	CGZ m	FSMAX tm
CAPACITY OF Machinery space (RHO=1)												
R01	Bowthruster room	MMA	140.4	136.2	0.85	1.000	99.0	87.0	64.96	-0.01	5.25	49
R02	Engine room lower	MMA	618.1	599.6	0.85	1.000	87.0	63.5	52.34	0.00	3.30	1446
R03	Cargo pump room lower	MMA	594.8	577.0	0.85	1.000	64.0	22.0	27.83	-0.13	2.90	978
R04	Propulsion room	MMA	505.8	490.6	0.85	1.000	22.0	-6.0	2.77	0.00	5.65	1493
R05	Engine room upper	MMA	379.6	368.3	0.85	1.000	91.2	70.0	55.38	0.02	6.66	1385
R06	Switchboard room	MMA	141.1	136.9	0.85	1.000	70.0	64.0	46.90	0.00	6.50	492
R07	Cargo pump room upper	MMA	416.1	403.7	0.85	1.000	57.3	22.0	25.88	0.01	6.71	1340
SUBTOTAL		MMA	2796.0	2712.1					34.99	-0.02	4.86	
CAPACITY OF Mud (RHO=2.8)												
T043	LIQUID CARGO TANK PS	MUD	142.9	140.1	0.95	2.800	63.2	55.8	41.65	-5.57	4.63	100
T044	LIQUID CARGO TANK SB	MUD	142.9	140.1	0.95	2.800	63.2	55.8	41.65	5.57	4.63	100
T045	LIQUID CARGO TANK PS	MUD	129.7	127.2	0.95	2.800	56.5	49.8	37.22	-5.58	4.61	98
T046	LIQUID CARGO TANK SB	MUD	126.8	124.3	0.95	2.800	56.5	49.8	37.18	5.58	4.54	98
T047	LIQUID CARGO TANK PS	MUD	129.7	127.2	0.95	2.800	50.5	43.8	33.02	-5.58	4.61	98
T048	LIQUID CARGO TANK SB	MUD	129.7	127.2	0.95	2.800	50.5	43.8	33.02	5.58	4.61	98
T049	LIQUID CARGO TANK PS	MUD	129.7	127.2	0.95	2.800	44.5	37.8	28.82	-5.58	4.61	98
T050	LIQUID CARGO TANK SB	MUD	129.7	127.2	0.95	2.800	44.5	37.8	28.82	5.58	4.61	98
T051	LIQUID CARGO TANK PS	MUD	128.7	126.2	0.95	2.800	38.5	31.8	24.63	-5.56	4.63	98
T052	LIQUID CARGO TANK SB	MUD	128.7	126.2	0.95	2.800	38.5	31.8	24.63	5.56	4.63	98
SUBTOTAL		MUD	1319.0	1292.6					33.24	-0.01	4.61	
CAPACITY OF Nitrogen (RHO=0.0013)												
T312	CD VOID OUTSIDE LFL TANK.N2		20.9	20.2	0.95	0.001	22.0	13.0	12.30	-8.51	6.25	0
T313	CD VOID OUTSIDE LFL TANK.N2		20.9	20.2	0.95	0.001	22.0	13.0	12.30	8.51	6.25	0
T314	CD VOID AROUND LFL TANKS.N2		206.7	200.5	0.95	0.001	22.0	13.0	12.25	0.00	6.28	3
SUBTOTAL		N2	248.4	241.0					12.26	0.00	6.27	
CAPACITY OF Pass. Corridor (RHO=1)												
RAIL1	Closed cargo rail PS	PCO	58.9	57.1	0.95	1.000	57.0	37.0	32.90	-8.27	9.45	4
RAIL2	Closed cargo rail SB	PCO	58.9	57.1	0.95	1.000	57.0	37.0	32.90	8.27	9.45	4
SUBTOTAL		PCO	117.7	114.2					32.90	0.00	9.45	
CAPACITY OF Sewage (RHO=1)												
T252	BLACK WATER TANK	SEW	11.0	10.7	0.95	1.000	78.0	76.0	53.89	-7.06	6.69	4
CAPACITY OF Sludge (RHO=1)												
T283	SLUDGE DRAIN TANK	SLU	9.7	9.4	0.95	1.000	70.0	67.0	47.95	1.64	0.70	6
T288	SLUDGE SETTLING TANK	SLU	4.6	4.5	0.95	1.000	71.0	70.0	49.35	6.81	6.66	1
SUBTOTAL		SLU	14.3	13.8					48.40	3.31	2.62	
CAPACITY OF Stairs (RHO=1)												
E1	Emrg. exit 1 PS	STA	31.5	30.6	0.95	1.000	37.0	30.0	22.52	-7.26	6.49	21
E2	Emrg. exit 1 SB	STA	30.8	29.9	0.95	1.000	37.0	30.0	22.54	7.25	6.50	21
E3	Emrg. exit 2 PS	STA	14.6	14.2	0.95	1.000	-2.0	-5.0	-2.48	-7.34	6.78	5
E4	Emrg. exit 2 SB	STA	14.6	14.2	0.95	1.000	-2.0	-5.0	-2.48	7.34	6.78	5
SUBTOTAL		STA	91.5	88.8					14.56	-0.06	6.59	

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NAME	DES	PURP	VOLM m3	VNET m3	PERM	RHO t/m3	FRMAX #	FRMIN #	CGX m	CGY m	CGZ m	FSMAX tm
CAPACITY OF Void (RHO=1.025)												
VOID1	Capstan void PS	VOID	5.6	5.5	0.95	1.025	-5.0	-8.0	-4.54	-8.20	7.04	1
VOID2	Capstan void SB	VOID	5.6	5.5	0.95	1.025	-5.0	-8.0	-4.54	8.20	7.04	1
VOID3	Forepeak void CE	VOID	0.2	0.2	0.95	1.025	106.6	105.0	73.83	0.00	2.04	0
T004	DB TANK CE	VOID	75.8	73.5	0.95	1.025	102.0	87.0	65.09	0.00	1.38	95
T010	WING TANK PS	VOID	61.0	59.2	0.95	1.025	76.0	64.0	49.30	-7.86	4.45	8
T011	WING TANK SB	VOID	61.0	59.2	0.95	1.025	76.0	64.0	49.30	7.86	4.45	8
T022	WING TANK PS	VOID	61.8	60.0	0.95	1.025	37.0	22.0	19.27	-7.89	3.91	21
T023	WING TANK SB	VOID	61.8	60.0	0.95	1.025	37.0	22.0	19.27	7.89	3.91	21
T317	CD VOID CE	VOID	34.1	33.1	0.95	1.025	22.0	3.6	10.60	0.00	0.83	37
T318	CD VOID PS	VOID	10.8	10.4	0.95	1.025	81.0	80.0	56.35	-6.25	4.99	2
T319	CD VOID SB	VOID	33.5	32.5	0.95	1.025	81.0	76.0	55.09	6.88	4.89	5
SUBTOTAL		VOID	411.3	398.9					39.17	0.40	3.54	
TOTAL			10174.8	9889.6					37.51	-0.01	5.67	

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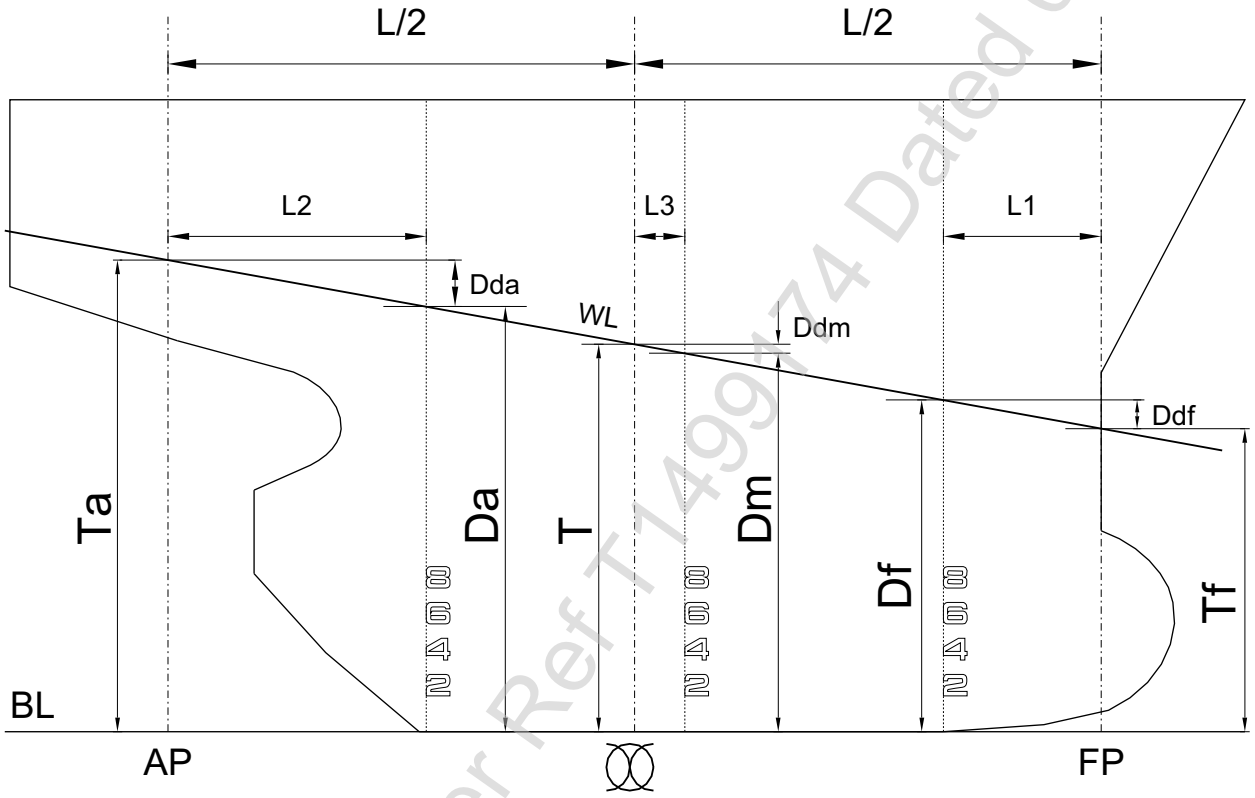
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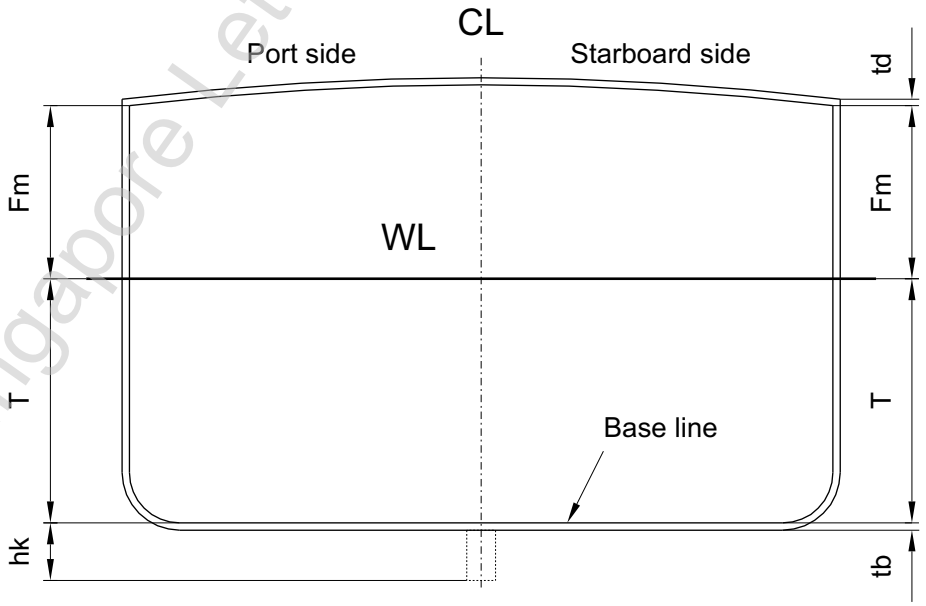
4 DRAUGHT MARKS

4.1 DRAUGHT MARKS IN GENERAL

Relevant documentation showing the location of the draft marks should be onboard. Draught marks drawing is attached within this stability booklet, see attachments. In addition, there is a sketch of the draught marks within this chapter. The attained draft, which should be used in the hydrostatic curves and cross curves can then be calculated according to the following formulas:

EXPLANATORY SKETCHES:





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WHERE:
L = Length between perpendiculars = 79.612 m
L1 = Distance from FP to fore draft mark = -4.461 m
L2 = Distance from AP to aft draft mark = 7.251 m
L3 = Distance from L/2 to midship draught marks = 1.745 m
tb = Line of bottom of keel plate below baseline = 0.012 m

Da = Draught reading aft above base line
Df = Draught reading forward above base line
Dm = Draught reading amidship above base line
tra = Apparent trim (positive by bow)
Dda = Longitudinal correction for draught marks aft
Ddf = Longitudinal correction for draught marks forward
Ddm = Longitudinal correction for draught marks amidship
Ta = Moulded draught at aft perpendicular
Tf = Moulded draught at forward perpendicular
T = Mean moulded draught amidship
tr = Trim (positive by bow)
fm = moulded freeboard
hk = height of keel/nozzle/skeg/propeller/... below base line
td = deck thickness amidship

Da = aft reading - tb
Df = forward reading - tb
Dm = midship reading - tb
tra = Df - Da
dDf = L1/(L-(L1+L2))* tra = 0.094387 * tra
dDa = L2/(L-(L1+L2))* tra = -0.058069 * tra
dDm = L3/(L-(L1+L2))* tra = 0.022715 * tra

ATTAINED DRAFT AND TRIM TO BE USED IN THE HYDROSTATIC CURVES:
Draft at AP (m) : Ta = Da - dDa
Draft at FP (m) : Tf = Df - dDf
Draft midships (m) : T = (Ta+Tf)/2
Trim : tr = Tf - Ta

EXAMPLE OF USE:

Da = Draft at Aft draft mark from BL = 5.7 - 0.012 = 5.688 m

Df = Draft at Fore draft mark from BL = 6.1 - 0.012 = 6.088 m

tra = Apparent Trim = 6.088 - 5.688 = 0.4 m

ATTAINED DRAFT AND TRIM TO BE USED IN THE HYDROSTATIC CURVES:

Draft at AP (Ta) : Da - dDa = 5.688 - (0.094387 * 0.4) = 5.65 m

Draft at FP (Tf) : Df - dDf = 6.088 - (-0.058069 * 0.4) = 6.111 m

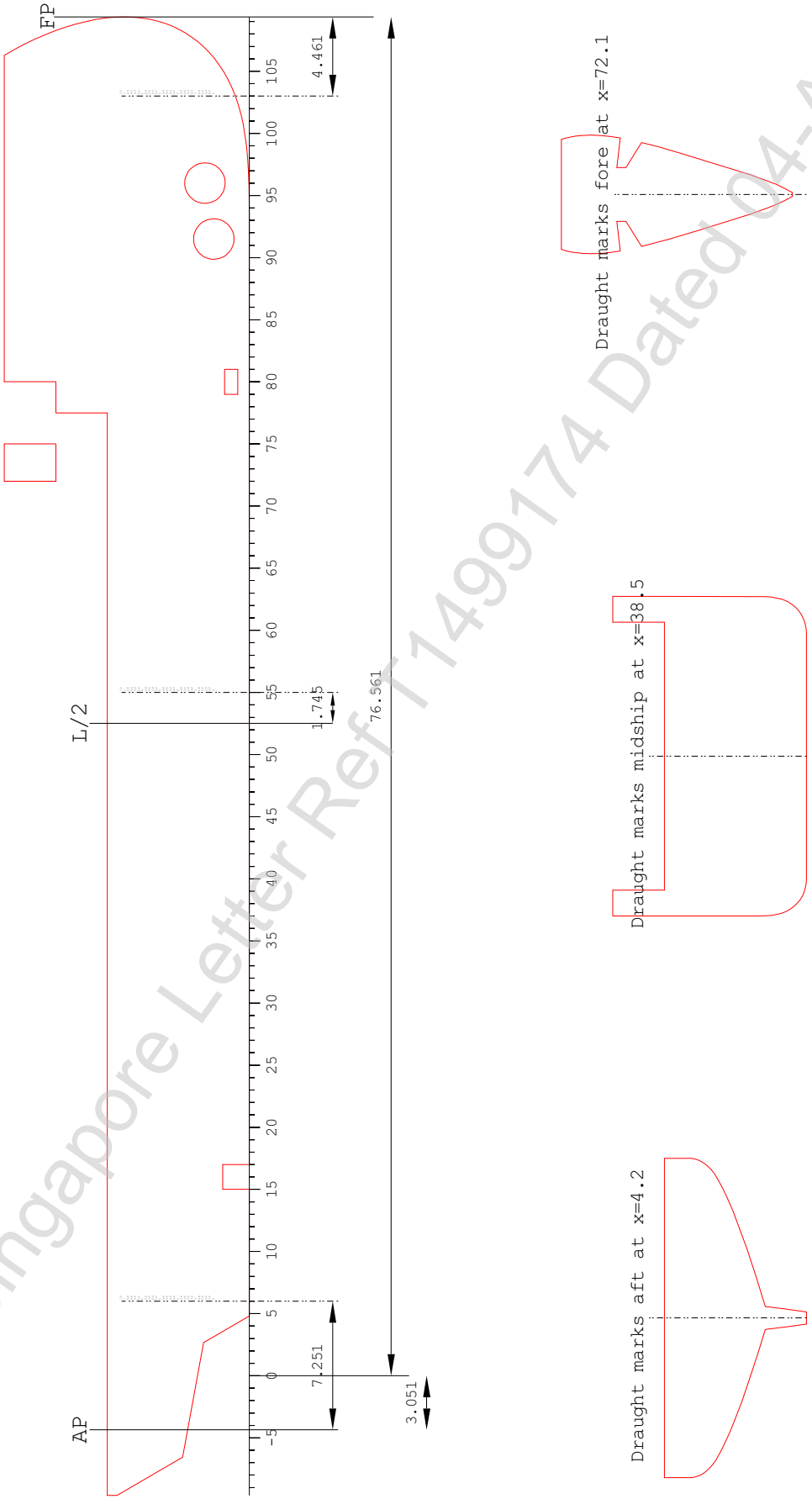
Draft midships : T = (Tf + Ta) / 2 = (6.111 + 5.65) / 2 = 5.881 m

Trim : Tf - Ta = 0.461 m

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4.2 DRAUGHT MARKS LOCATION

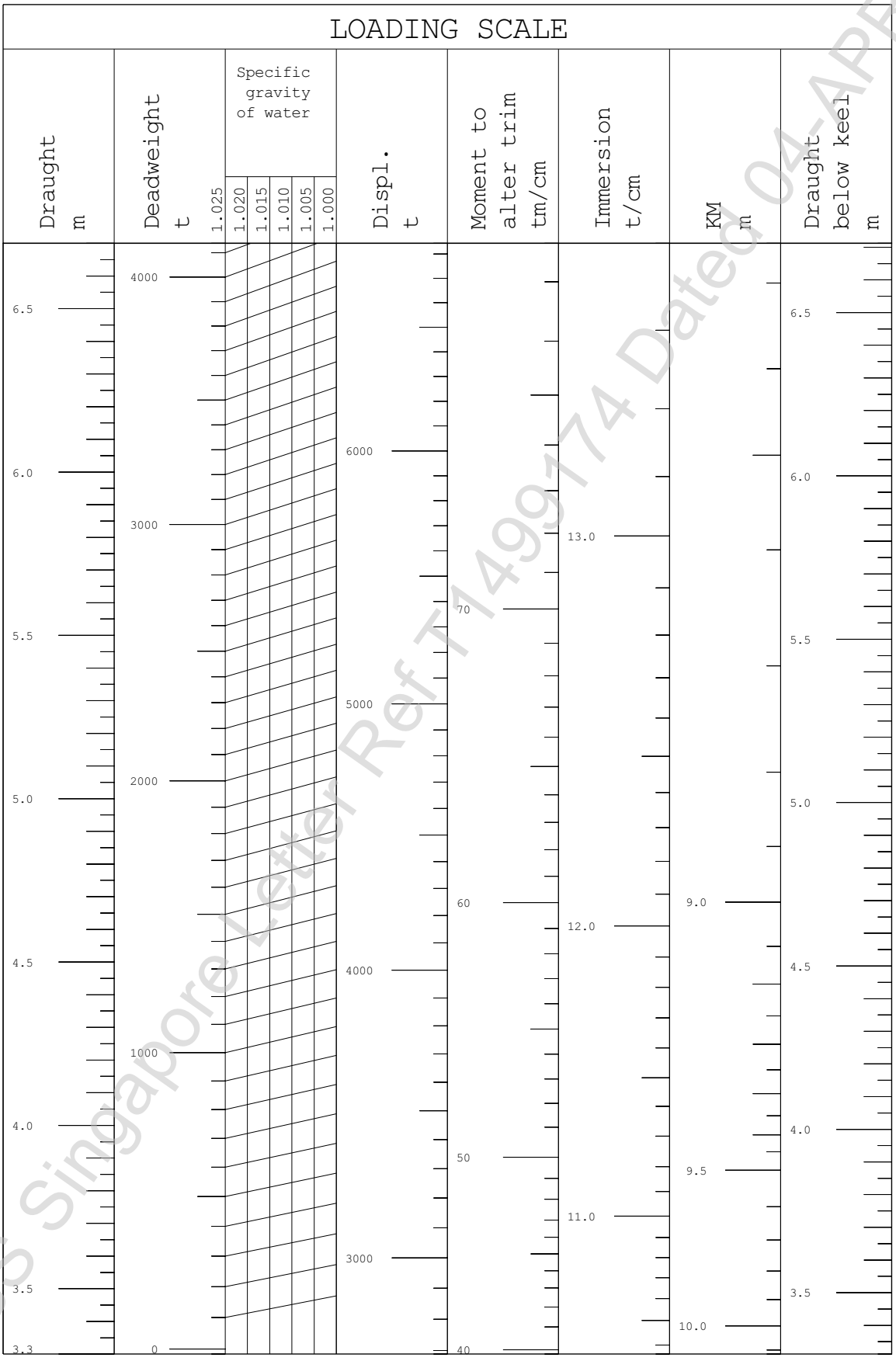


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5 DEADWEIGHT SCALE

Not part of ABS review



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6 INSTRUCTIONS FOR MASTER

6.1 GENERAL

COMPLIANCE WITH THE STABILITY CRITERIA DOES NOT ENSURE IMMUNITY AGAINST CAPSIZING, REGARDLESS OF THE CIRCUMSTANCES, OR ABSOLVE THE MASTER FROM HIS RESPONSIBILITIES.
Master should therefore exercise prudence and good seamanship, having regard to the season of year, weather forecasts and the navigational zone, and should take the appropriate action as to speed and course warranted by the prevailing circumstances.

Care should be taken to ensure that the cargo allocated to the ship is capable of being stowed, so that compliance with the criteria can be achieved. If necessary, the amount should be limited to the extent that ballast weight may be required.

Before a voyage commences, care should be taken to ensure that the cargo and sizeable pieces of equipment have been properly stowed or lashed, so as to minimize the possibility of both longitudinal and lateral shifting while at sea, under the effect of acceleration caused by rolling and pitching.

It is important to ensure that all ballast tanks in any condition, which are to be full, are in fact pressed full. Those tanks which are dry should be completely dry.

When a tank is to be ballasted at sea to counteract the loss of consumables, it should be noted that:

- The tank will have a complete free surface as ballasting commences.

- The free surface will adversely effect the stability of the vessel before any benefit can be assumed by the weight of the ballast water taken on board.

6.2 PRINCIPLES FOR STABILITY

The loading conditions in this booklet are representative for operating conditions, and it is for the Master to either compare his actual condition with a similar in the Stability Booklet or use a Simplified Stability Calculation to estimate the stability of an actual condition of the vessel.

When simplified stability calculations are performed, the master should be aware of the influence of free surface moments. Since the loading conditions in this booklet are calculated with actual free surface moments (at each heeling angle), small deviation from the simplified calculations can NOT be prevented.

Any loading condition calculated by the master must comply with the limiting GM-curves.

NOTE :
BEFORE E.G. CARRYING HEAVY DECK LOADS AND PERFORMING HEAVY ANCHOR HANDLING, CRANE OPERATION OR ANY OTHER HEAVY OPERATION, IF APPLICABLE, WHICH EXPECTED TO HAVE A SEVERE EFFECT ON THE VESSEL STABILITY, APPROPRIATE CALCULATIONS HAVE TO BE DONE TO ENSURE THAT THE OPERATIONAL CONDITIONS FULFILS THE STABILITY CRITERIA.

6.3 FREE SURFACE HANDLING

Provided a tank is completely filled with liquid, no movement of the liquid is possible and the effect on the ships stability is precisely the same as if the tank contained solid material.

Immediately a quantity of the liquid is withdrawn from the tank the situation changes completely and the stability of the ship is adversely affected by what is know as the FREE SURFACE EFFECT. This adverse effects on the stability is referred to as a loss in GM or as a virtual rise in VCG and is calculated as follows:

$$\text{GM Reduction [m]} = \frac{\text{Sum of free surface moments [tm]}}{\text{Displacement [t]}}$$

Tanks with filling levels fixed (e.g. liquid cargo, water ballast). The free surface correction should be determined for the actual/real filling level to be used in each tank.

Tanks with filling levels variable (e.g. consumable liquids such as fuel oil, diesel oil, and fresh water, and also liquid cargo and water ballast during liquid transfer operations). The free surface correction should be the maximum value attainable between the filling limits envisaged for each tank, consistent with any operating instructions.

In calculating the free surface effects in tanks containing consumable liquids, it should be assumed that for

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each type of liquid at least one transverse pair or a single centreline tank has a free surface and the tank or combination of tanks taken into account should be those where the effect of free surfaces is the greatest.

Where water ballast tanks, including anti-rolling tanks and anti-heeling tanks, are to be filled or discharged during the course of a voyage, the free surface effects should be calculated to take account of the most onerous transitory stage relating to such operations.

For vessels engaged in liquid transfer operations, the free surface corrections at any stage of the liquid transfer operations may be determined in accordance with the filling level in each tank at that stage of the transfer operation.

NOTE: The free surface effects of all slack tanks should be taken into account in both the arrival and the departure conditions.

NOTE: When it is intended to introduce or discharge ballast water during a voyage the Master should, prior to departure, ensure that the required stability criteria will be maintained at all times.

Due to the loss of stability by partially filled tanks it is strictly recommended that free surfaces should be kept to the minimum. If possible, only one center tank/tank pair of each kind of liquid should have a free surface. As much as practical the tanks with altering content between departure and arrival with largest free surface effect should be emptied first. This is in order to avoid large free surfaces as soon as possible during the voyage. Loading conditions within this document are following the same principle as far as practical.

Free surface values for all filling levels can be found in the TANK CAPACITY TABLES.

The FREE SURFACE MOMENTS used in simplified stability calculations (Form1, Form2 and Form3) is the moment of inertia of the actual tank (m4) multiplied with the specific gravity of the liquid (t/m3). Maximum values for moment of inertia, given on tank plan drawing could be used as conservative way of maintaining the vessels stability.

6.4 DECK CARGO

When loading deck cargo, attention should be paid to the need for maintaining freeboard at the stern with regard to the safety of the crew working on deck, and also to the adverse effect of trim by stern has on the stability of the vessel. the vessel should not be operated at a trim exceeding the values of the min. GM curves. In addition it should be ensured that the stern trim limitation according to the rules is always enforced.

All doors, hatches and other openings onto exposed cargo deck should be kept closed and secured at sea (except for access) and it should be ensured that the disposition of the deck cargo does not impede the operation of these fittings.

When loading deck cargoes, care should be taken to avoid any obstruction of the freeing ports or areas necessary for the drainage of pipe stowage positions.

When pipe deck cargoes (or other cargoes liable to provide a temporary trap for water) are carried, adequate areas should be provided between adjacent stowage racks and also between the vessels structure and stowage racks to ensure free drainage of the pipe stowage positions. Such areas should be in excess of the freeing port area provided.

When pipes are carried on deck, a volume of entrapped water should be assumed. The centre of gravity of this entrapped water should be assumed to be at a height above the cargo deck equal to one half the height to the top of the (pipe) deck cargo. A suitable allowance should also be made where other deck cargoes which may trap water are carried.

The effect of the weight of this trapped water on the vertical centre of gravity and trim of vessel must be taken into account.

The weight of this water is included in the cargo dead-weight permitted by the position of the Load Line Mark.

The minimum GM limitations should not be exceeded and should include an allowance for the effect of entrapped water in deck cargoes in the total VCG for the relevant conditions.

When cargo is discharged at sea, great care must be taken to maintain the vessel in a stable condition during the discharging operation. Attention should be paid to the trimming of the vessel during unloading.

- Cargo should be unloaded from the top, i.e. deck cargo should be discharged (heaviest items first) before any underdeck cargo.

- It should be noted that when liquid cargo is to be discharged, as soon as pumping is commenced a full free surface will exist in those tanks being pumped and the effect this has on the stability of the vessel taken into account.

- The loading pattern of cargo should be such that the freeboard at plimsoll marks is at least 1.3 metres.

- Tanks containing slack liquids should be kept to a minimum.

- Necessary ballasting should be carried out before taking on cargo at sea.

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6.5 ICING

If the vessel is engaged in waters, where the possibility of top-side icing may occur, then the Master should take such action as deemed necessary either to remove this ice, or at least, keep it to a minimum, bearing in mind that this top-side icing will adversely affect the stability of the vessel.

Ice formation may also occur in conditions of snowfall, sea fog (including arctic sea smoke), a drastic fall in ambient temperature, as well as from the freezing of drops of rain on impact with the vessel's structure. Ice formation may sometimes be caused or accentuated by water shipped on board and retained on deck.

Intensive ice formation generally occurs on stem, bulwark and bulwark rail, front walls of superstructures and deck-houses, hawse holes, anchors, deck gear, forecastle deck and upper deck, freeing ports, aerials, stays, shrouds, masts and spars.

The most intensive ice formation takes place when wind and sea come from ahead. In beam and quartering winds, ice accumulates quicker on the windward side of the vessel, thus leading to a constant list which is extremely dangerous.

Consequences of the ice accretion negatively influencing vessel seaworthiness:
1. an increase in the weight of the vessel due to accumulation of ice on the vessel's surfaces which causes the reduction of freeboard and buoyancy

2. a rise of the vessel's centre of gravity due to the high location of ice on the vessel's structures with corresponding reduction in the level of stability;
3. an increase of windage area due to ice formation on the upper parts of the vessel and hence an increase in the heeling moment due to the action of the wind

4. a change of trim due to uneven distribution of ice along the vessel's length;
5. the development of a constant list due to uneven distribution of ice across the breadth of the vessel and
6. impairment of the maneuverability and reduction of the speed of the vessel.

6.6 DOCKING

When docking the vessel, as much water ballast as possible should be removed. Fuel oil and fresh water should be kept to a minimum. Deck cargo or any other cargo should be removed or to kept at bare minimum. This to avoid damages to hull in bottom of the vessel.

Regarding location of supports and loading condition details for docking see DOCKING PLAN drawing.

6.7 METRIC CONVERSIONS

MULTIPLY BY	TO CONVERT FROM	TO OBTAIN	
0.03937	Millimeters	Inches	25.4
0.3937	Centimeters	Inches	2.54
3.2808	Meters	Feet	0.3048
2.2046	Kilograms	Pounds	0.45359
0.0009842	Kilograms	Tons (2240 lb.)	1016.047
0.9842	Metric Tonnes (1000 kg)	Tons (2240 lb.)	1.016
2.4998	Metric Tonnes per Centimeter of immersion	Tons per inch	0.400
8.2014	Moment to change trim one centimeter	Moment per Inch	0.122
187.967	Meter Radians	Feet Degrees	0.0053
0.01745	Meter Degrees	Meter Radians	57.3
	TO OBTAIN	TO CONVERT FROM	MULTIPLY BY ABOVE

6.8 RELATIONS BETWEEN WEIGHT AND VOLUME

1000	Cubic millimeters		=	1	Cubic centimeter
1	Cubic centimeter of freshwater	(SG = 1.000)	=	1	gram
1000	Cubic centimeter of freshwater	(SG = 1.000)	=	1	Kilogram
1	Cubic meter of freshwater	(SG = 1.000)	=	1	Tonne
1	Cubic meter of saltwater	(SG = 1.025)	=	1.025	Tonnes
1	Cubic meter of saltwater	(SG = 1.025)	=	1.025	Tonnes
1	Tonne of saltwater	(SG = 1.025)	=	0.975	Cubic meters
1	Cubic meter		=	35.316	Cubic feet
1	Cubic foot		=	0.0283	Cubic meters

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6.9 WATERTIGHT INTEGRITY

The stability calculations are based upon the geometric definition of the hull with relevant appendages. The following definitions are used through the calculations:
STABHULL Geometric definition for the intact stability calculations.
DAMHULL Geometric definition for the damage stability calculations.
STABHULL Geometric definition for the hydrostatic tables.

The main difference of the above definitions is the above waterline volumes. Wheather or not such a volume should be included in the buoyancy/stability calculations depends on its watertight integrity.
The requirements due to immersion angles of relevant openings depends on the opening type. The NAPA program has four different opening types:
- UNPROTECTED
- WEATHERTIGHT
- WATERTIGHT

The different opening types are handled in the following way:

UNPROTECTED:
Unprotected openings are taken into account in all calculations, lists and plots, wherever openings are referenced.

WEATHERTIGHT:
Weathertight openings do not restrict the calculation of the GZ curve and area calculations, but may not be submerged in the floating position.

WATERTIGHT:
Watertight openings are ignored in all calculations. These openings appear only in some lists and as additional information.

The program follows the underlying logic in taking relevant openings into account.
The opening is taken into account if:
- It leads from the sea to an undamaged compartment.
- It leads from a damaged compartment to an undamaged compartment.
- Data about connection is missing (no record CONNECT)

The opening is ignored if:
- It connects damaged compartments.
- It connects undamaged compartments.
- It leads from the sea to a damaged compartment.

You can find list of openings used in the calculations in accordance with the relevant criterias in this chapter.

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6.10 PIPECARGOS ON DECK - EFFECT OF ENTRAPPED WATER

When pipes are carried on deck, a quantity of water has to be assumed to be trapped in and around the pipes, equal to a certain percentage of the net volume of the pipe deck cargo.

The net volume should be taken as the internal volume of the pipes plus the volume between the pipes. The percentage of the net volume used to obtain the quantity of trapped water is based on the freeboard amidships, as follows,

- 30 % if the summer freeboard amidships to the main deck is equal to or less than 0.015*L.

- 10 % if the summer freeboard amidships to the main deck is equal to or greater than 0.03*L.

- For intermediate values of the freeboard amidships, the percentage may be obtained by linear interpolation. For this vessel the value is 28.2 %.

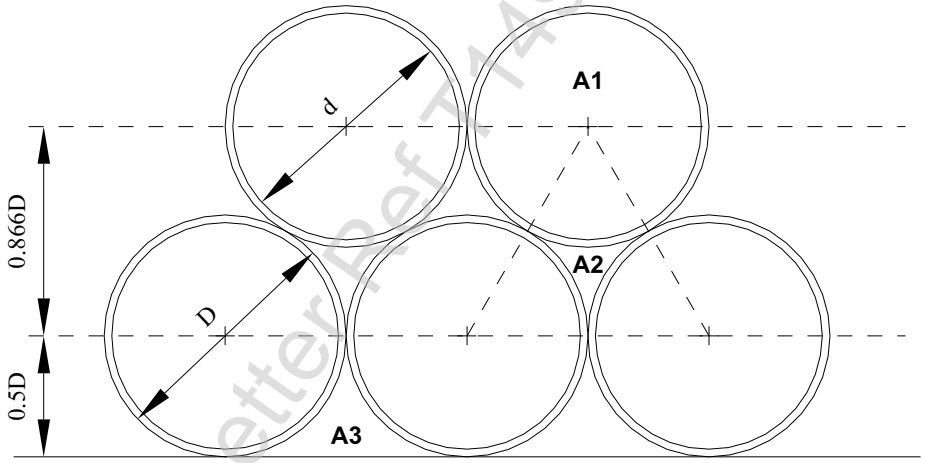
Depth..... 8.000 [m]
Draught..... 6.700 [m]
Length..... 79.614 [m]
Freeboard at SWL. 1.300 [m]
Freeboard/L..... 0.0163 [m]

CALCULATION FORM OF ENTRAPPED WATER :

OUTSIDE DIAMETER OF PIPES IN METERS ..... D = _____ [m]

INSIDE DIAMETER OF PIPES IN METERS ..... d = _____ [m]

DERIVATION OF AREAS (SEE SKETCH) :



INTERNAL SECTIONAL AREA OF ONE PIPE :

$A1 = (\Pi/4) \times d^2 = (3.14159/4) \times d^2 =$ $= 0.785398 \times d^2 =$ _____ [m²]

SECTIONAL AREA OF ONE SPACE BETWEEN PIPES :

$A2 = \frac{1}{2}D \times (\text{SIN } 60^\circ \times D) - \frac{1}{2} \times (\Pi/4) \times D^2 =$
$= \frac{1}{2} \times 0.866025 \times D^2 - \frac{1}{2} \times 0.785398 \times D^2 =$ $= 0.040314 \times D^2 =$ _____ [m²]

SECTIONAL AREA OF ONE SPACE BETWEEN PIPES AND DECK :

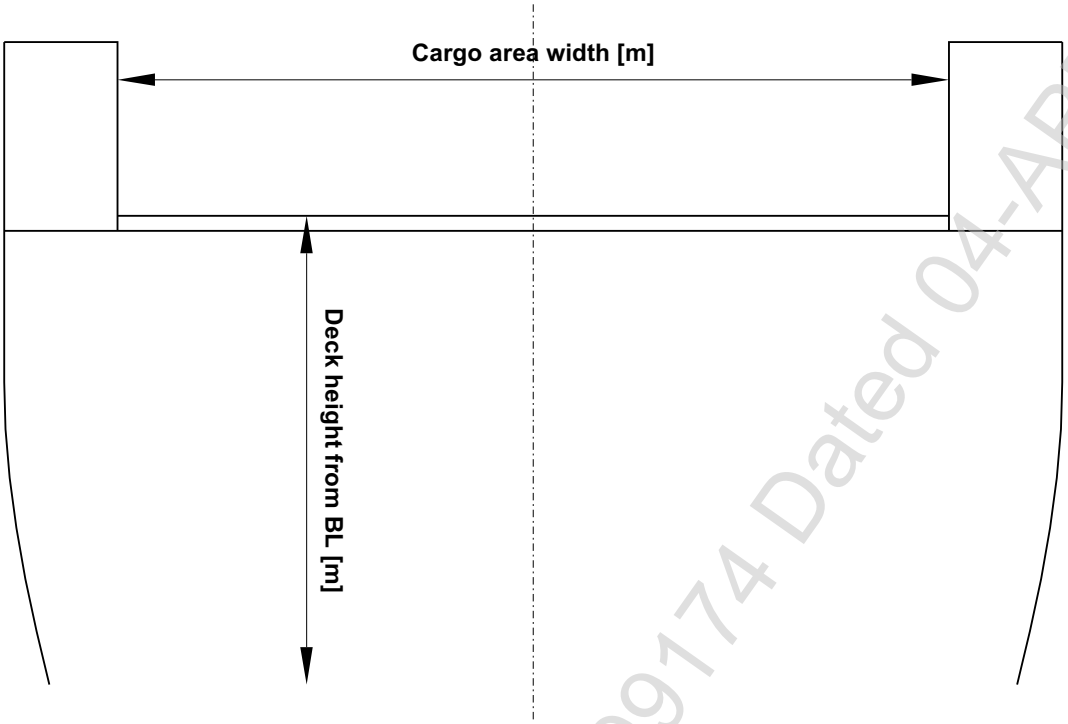
$A3 = \frac{1}{2}D \times D - \frac{1}{2} \times (\Pi/4) \times D^2 =$
$= \frac{1}{2} \times D^2 - \frac{1}{2} \times 0.785398 \times D^2 =$ $= 0.107301 \times D^2 =$ _____ [m²]

DERIVATION OF VOLUME : The volume of entrapped water is found by multiplying A1, A2 and A3 respectively by length of pipes in meters and actual number of pipes or spaces. (See condition sheet on next page giving details of pipe deck cargo.)

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PIPE DECK CARGO FOR LOADING CONDITION No. :



OUTSIDE DIAMETER OF PIPES IN METERS ..... D = _____ [inch] = _____ [m]

PIPE WALL THICKNESS ..... = _____ [inch] = _____ [m]

INSIDE DIAMETER OF PIPES IN METERS ..... d = _____ [inch] = _____ [m]

LENGTH OF THE PIPE ..... L = _____ [feet] = _____ [m]

WIEGHT OF PIPE ..... = _____ [t]

POSITION		Pcs	Weight[t]	LCG [m]	VCG [m]
BAY No.1	Layer No.1				
	Layer No.2				
	Layer No. ...				
	TOTAL				
BAY No.2	Layer No.1				
	Layer No.2				
	Layer No. ...				
	TOTAL				
GRAND TOTAL					

ENTRAPED WATER :

INSIDE PIPES V1 = A1 × L × _____ = _____ [m³]

BETWEEN PIPES V2 = A2 × L × _____ = _____ [m³]

BETWEEN PIPES AND DECK V3 = A3 × L × _____ = _____ [m³]

TOTAL VOLUME V = _____ [m³]

ASSUMED WEIGHT = V × (_____ % / 100) × 1.025 = _____ [t]

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6.11 SPECIAL CONSIDERATIONS USING PASSIVE ROLL REDUCTION TANKS

ROLL REDUCTION TANK(S) MUST NOT BE USED IN ANY HEAVY OPERATIONS OR HEAVY WEATHER CONDITIONS!!!

Not part of ABS review (page 21-24)

This vessel is equipped with an Ulstein Passive Roll Reduction System designed and tested by Marintek, and a copy of the results in booklet form has been supplied to the vessel. The master should refer to this booklet for operation information. Please read this chapter carefully. In all loading conditions where the roll reduction tank(s) are included, the tank(s) should be filled to their working level and the full free surface loss taken into account when assessing the vessels stability.

The working level of the tanks varies with the displacement of the vessel and is found within this chapter. Filling and/or discharging of the tank(s) is through the ballast/fire pumps situated in engine room. To eliminate the free surface loss when stability is critical the time to fill or empty the tanks from working level should be as short as possible.

The limitations on use of roll-reduction tanks should be observed. When the tanks are in use, the FREE SURFACE EFFECT should be taken into account in the calculations.

Any loading condition calculated by the master must comply with the limiting GM-curves, which can be found in Chapter 8 of this stability manual. If the actual GM (corrected for FREE SURFACE) is on or above the minimum GM value the stability of the vessel is satisfactory.

Free surface values for all filling levels can be found in the Chapter 11 TANK CAPACITY TABLES.

Due to the shape of a passive roll reduction tank, its use reduces the GM significantly and should be considered specially regarding the vessels stability.

On general basis it is the designers advice that due to possible sudden changes in weather during operation, possible obstructions on crane loads or anchor loads, and possible heavier loads than anticipated, roll reduction tanks should not be used during operation. This to preserve stability reserves.

Roll reduction tanks must neither be used in any conditions with an initial GM0 (prior to free surface correction) close to minimum allowed GM.

NOTE: BEFORE CARRYING HEAVY DECK LOADS AND PERFORMING HEAVY ANCHOR HANDLING, CRANE OPERATION AND ANY OTHER HEAVY OPERATION EXPECTED TO HAVE A SEVERE EFFECT ON THE VESSEL, SPECIAL STABILITY CALCULATIONS HAS TO BE DONE TO ENSURE THAT THE CONDITION FULFILS THE STABILITY CRITERION.

In case of grounding, collision or other outside influence on the vessel, the tank must be emptied immediately.

When it is intended to introduce or discharge ballast water during a voyage the Master should, prior to departure, ensure that the required stability criteria will be maintained at all times.

6.11.1 INSTRUCTIONS IN USE OF ULSTEIN ROLL REDUCTION SYSTEM

Find the period of rolling Tr by:

- Take a reading from the roll indicator.

- Measure the time of one rolling movement from one extreme to the other and back.

In the diagram which is delivered with the tank (on the next pages), you will find the optimal/best possible level of liquid for any period of rolling.

Measure the level of liquid in the tank, and adjust it if necessary.

This procedure should be repeated whenever the stability of the ship is altered by loading, unloading or ballasting.

In all loading conditions where the stabilizing tanks are included, the tanks should be filled to their working level and the full free surface loss taken into account when assessing the vessels stability.

Filling and/or discharging of the ballast tanks is through the vessels fire/ballast pumps. To eliminate the free surface loss when stability is critical the time to fill or empty the tanks from working level should be as short as possible.

The tank will have a complete free surface as ballasting commences, or as soon pumping starts from a full tank.

It is important to ensure that all ballast tanks in any condition, which are to be full, are in fact pressed full. Those tanks which are supposed to be dry should be completely dry.

The free surface will adversely effect the stability of the vessel before any benefit can be assumed by the weight of the ballast water taken onboard. Provided a tank is completely filled with liquid, no movement of the liquid is possible and the effect on the ships stability is precisely the same as if the tank contained solid material.

The FREE SURFACE MOMENTS used in simplified stability calculations in Chapter 7 (Form1, Form2 and Form3) is the

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moment of inertia of the actual tank (m4) multiplied with the specific gravity of the liquid (t/m3), and can be found in the Chapter 11 TANK CAPACITY TABLES.

The onboard loading computer may be the fastest and most correct tool to check the vessels stability for any condition, and must be checked continuously and in advance of every operation.

6.11.2 CONDITIONS WHICH INFLUENCE THE FUNCTIONALITY OF THE SYSTEM

Too much or too little liquid reduces effect.

When the rolling periods are very short, it may happen that the liquid just splashes around without any effect. If the tank is placed fairly high (above the centre of gravity), one solution may be to fill it up. This will give a longer, and more agreeable period of rolling.

This effect is not caused by the tank as such, but the fact that placing the weight higher, reduces an extremely good stability. Changing ballast from tanks placed low in the vessel, to tanks higher up will enhance this effect.

Increased KG, decreased GM implies longer rolling period.

Extremely long periods of rolling indicates that the stability is unsatisfactory (low GM). A stability check can be done by comparing the actual metacentric height GM with the minimum GM. Minimum GM is the least metacentric height allowed by the authorities on a given displacement.

The consideration of comfortable motion and better working conditions for the crew is important, but MUST NOT under any circumstances put the ships stability aside.

Note how the use of these tanks dramatically decreases the loads one can put onto the vessel, i.e. crane loads, anchorhandling loads, deck loads etc.

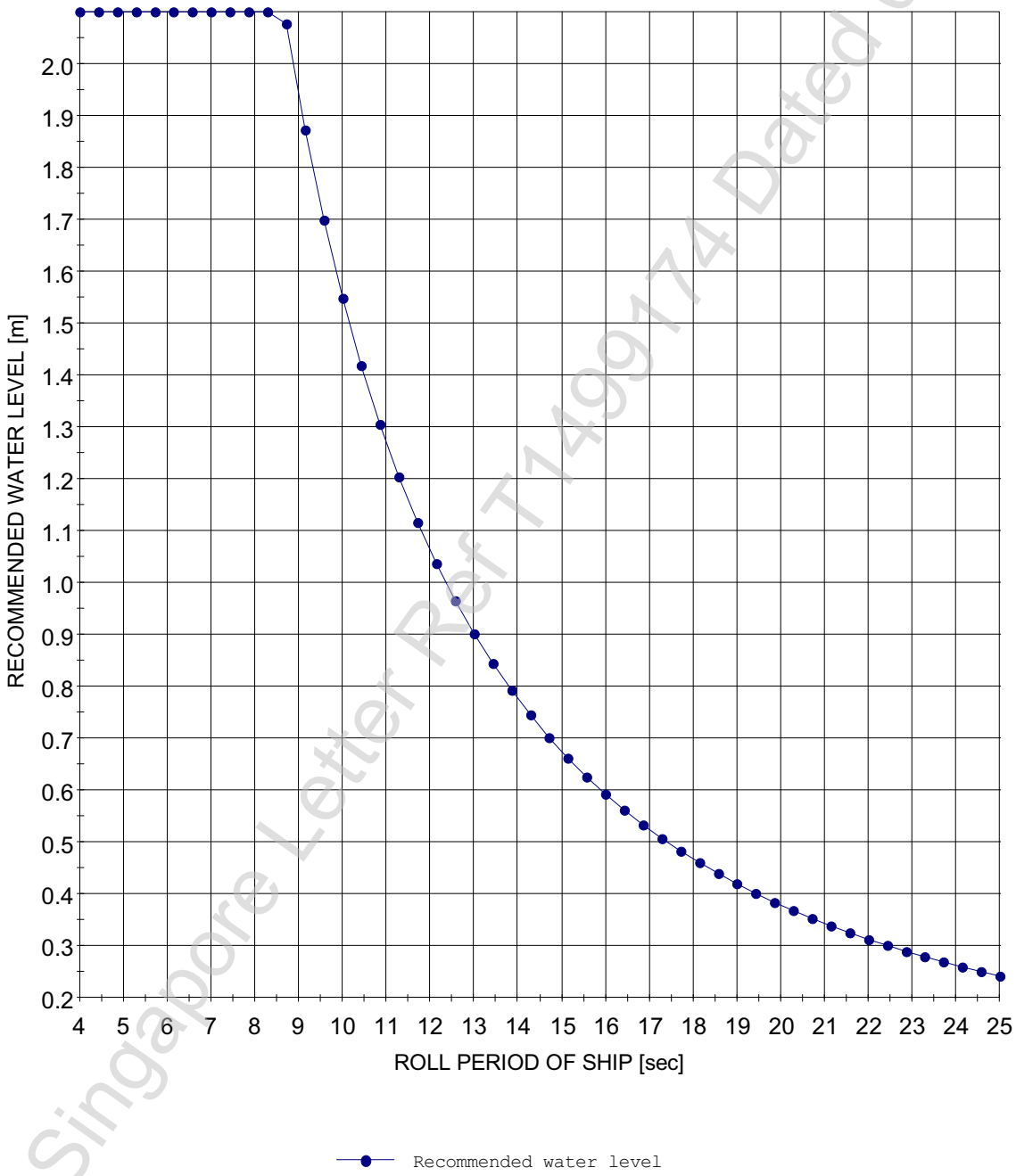
Optimal liquid levels is shown on next page.

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OPTIMUM WATER LEVEL	ENCL.	A.1
	REPORT	PX121
	DATE	07.10.14
	REF.	U10747

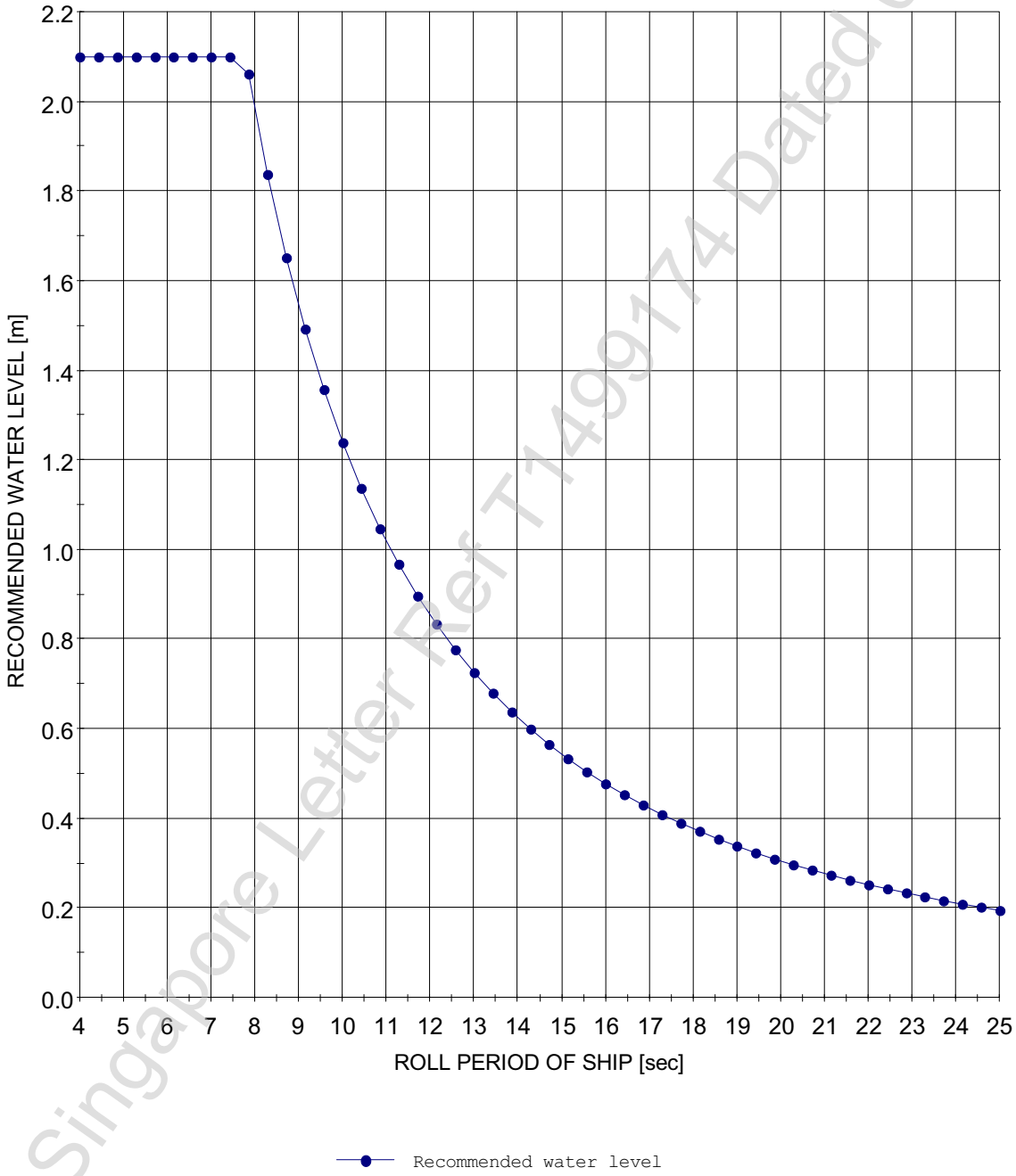
T062



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OPTIMUM WATER LEVEL	ENCL.	A.2
	REPORT	PX121
	DATE	07.10.14
	REF.	U10747

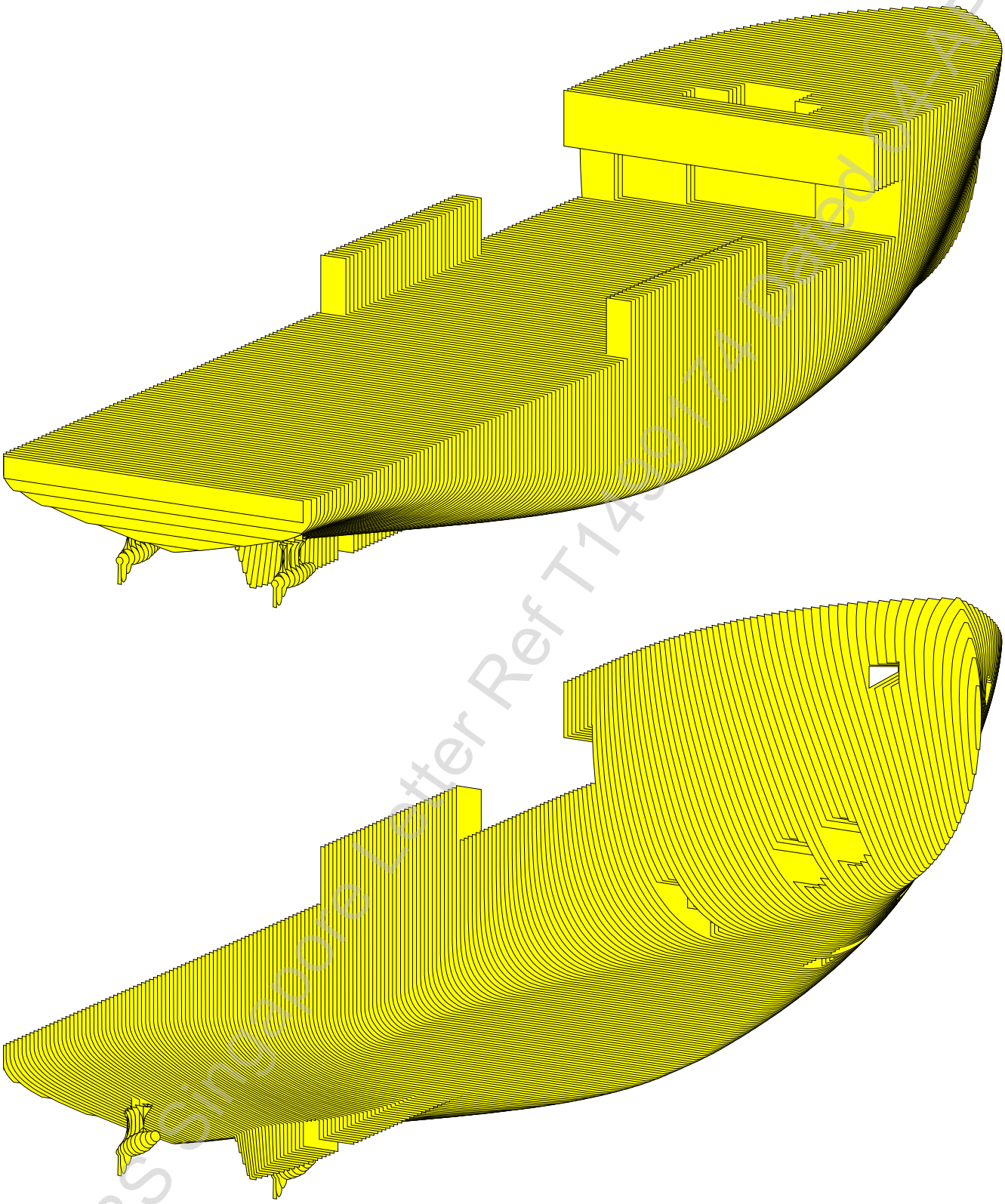
T061



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6.12 DESIGN BASIS

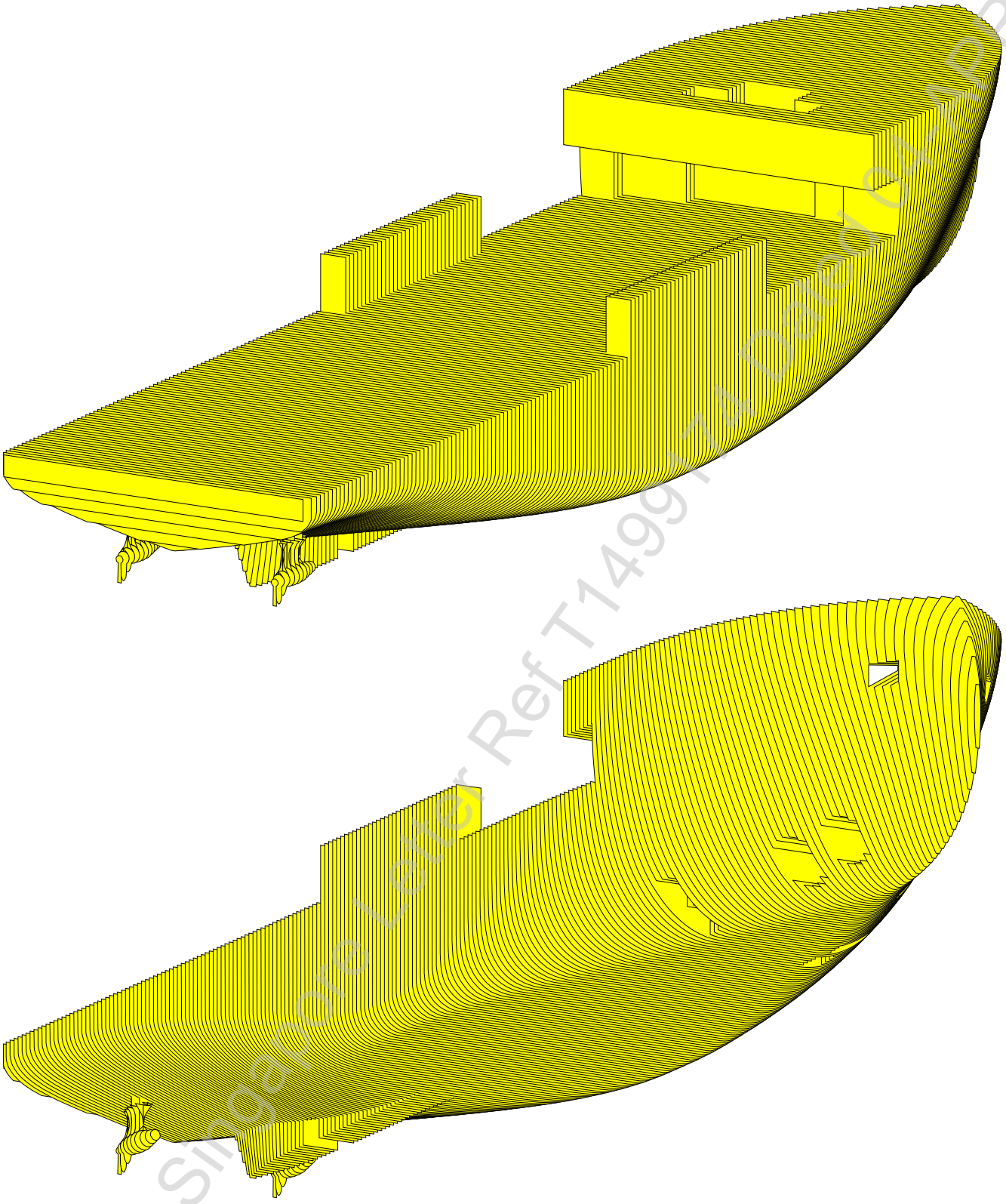
6.12.1 BUOYANT HULL DEFINITION - INTACT STABILITY



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6.12.2 BUOYANT HULL DEFINITION - DAMAGE STABILITY



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6.12.3 USED STABILITY CRITERIA

Intact stability criteria description :

* 2008IS-B2.4.5
It is important that the stability requirements for this vessel complies with the criteria listed below, which are in accordance with INTACT STABILITY CODE 2008, Part B, Reg. 2.4 Recommended design criteria for Offshore supply vessels.
****

** V.AREA15-30
The area under the curve of righting levers (GZ curve) should not be less than 0.070 metre-radians up to an angle of 15 deg when the maximum righting lever (GZ) occurs at 15 deg and 0.055 metre-radians up to an angle of 30 deg when the maximum righting lever (GZ) occurs at 30 deg or above. Where the maximum righting ever (GZ) occurs at angles of between 15 deg and 30 deg, the corresponding area under the righting lever curve should be:

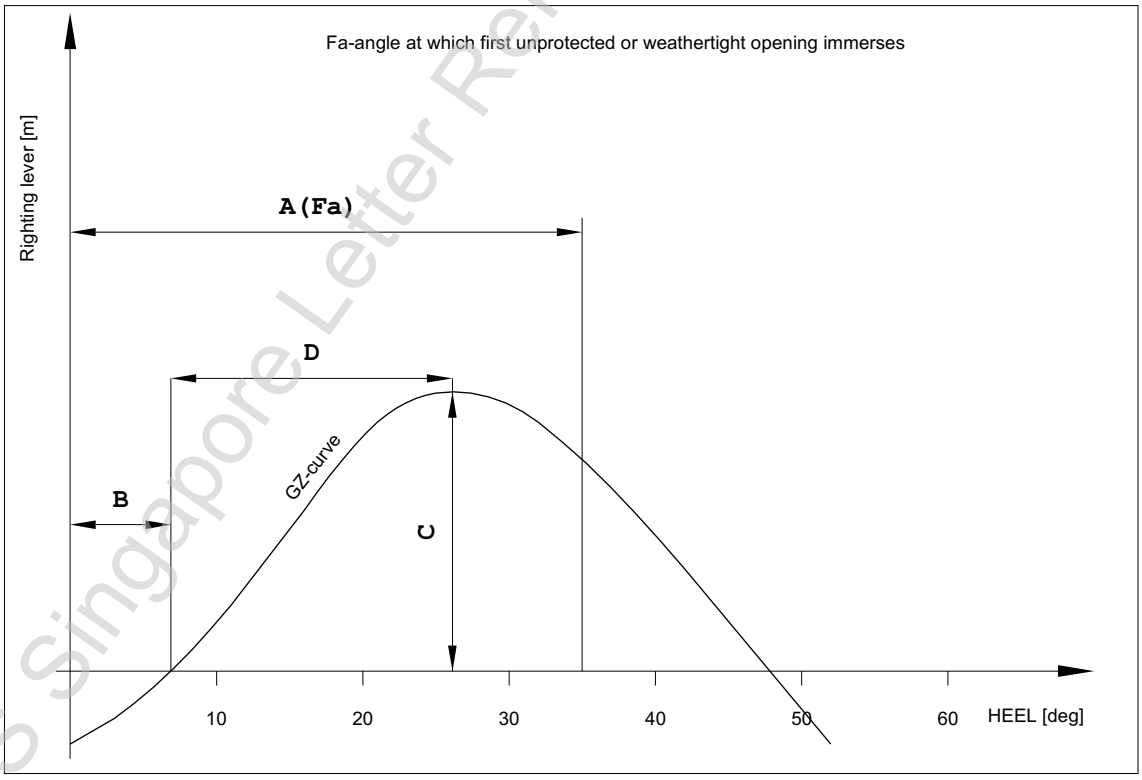
0.055 + 0.001 (30 deg - PHImax) meter-radians
****

** V.AREA3040
The area under the righting lever curve (GZ curve) between the angles of heel of 30 deg and 40 deg, or between 30 deg and PHIf (angle of flooding) if this angle is less than 40 deg, should be not less than 0.03 meter-radians.
****

** V.GZ0.2
The righting lever (GZ) should be at least 0.20 m at an angle of heel equal to or greater than 30 deg.
****

** V.POSMAX15
The maximum righting lever (GZ) should occur at an angle of heel not less than 15 deg.
****

** V.GM0.15
The initial transverse metacentric height (GMo) should not be less than 0.15 m.



****

** IMOWEATHER
Details of the weather criteria and ships windage area are included in following chapter.
****

** 2008IS-A2.3.1.2
Details of the weather criteria and ships windage area are included in following chapter.

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****

** ABSFIFI1
Each vessel is to have adequate stability for all loading conditions, with all firefighting monitors operating at maximum output multiplied by a factor of 1.1 in the direction most unfavorable to the stability of the vessel. The thruster(s) are to be considered operating at the power needed to counter-act that force. For the calculation purposes, the total thruster force should be vertically located at the location of the lowest available thruster.

The first intercept (steady heeling anlge due to heeling moment of Fi-Fi monitors) must occure before half of the freeboard is submerged.

****

** ABSFIFI2
As a consquence of heeling moment of Fi-Fi monitors, all loading conditions area of the residual stability (area between righting arm and heeling arm curves beyond the angle of the first intercept) up to an angle of heel 40 deg beyond the angle of the first intercept; or the angle of downflooding if this angle is less than 40 deg beyond the angle of the first intercept, should not be less than 0.09 meter-radians.

HEELING MOMENTS / FIRE FIGHTING OPERATIONS

The heeling force from the Fi-fi monitors is 15 kN for each monitor full capacity, that for two monitors multiplied by a factor of 1.1 gives 33 kN.
The Fi-fi monitors total force is vertically located at 27.196 m from BL.
The total thruster force is vertically located at the location of the lowest thruster, i.e. 2.00 m ab. BL.
The total heeling moment due to operation of all fire fighting monitors and thrusters is 831.468 kNm.
****

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Deterministic damage stability criteria description :

*** MSC.335[90]
This vessel should comply with damage stability criteria as stated in IMO MSC.335(90) AMMENDMENTS TO THE GUIDELINES FOR THE DESIGN AND CONSTRUCTION OF OFFSHORE SUPPLY VESSELS, 2006. (RESOLUTION MSC.235(82))

Damage should be assumed to occur anywhere in the vessel's length between transverse watertight bulkheads. The longitudinal extent of damage should be assumed for vessels with the length (L) greater than 43 m, and less than 80 m:3 m plus 3% of the vessel's length. For those with length (L) not greater than 43 m, 10% of the vessel's length and for those with length from 80 to 100 m: 1/3 L**2/3.
For this vessel the longitudinal extent of damage of 5.388 m
was calculated.

The vertical extent of damage should be assumed from the underside of the cargo deck, or the continuation thereof, for the full depth of the vessel.

The transverse extent of damage should be assumed as 760 mm for vessels with length less than 80 m and B/20, but not less than 760 m for vessels with length from 80 to 100 m.
The transverse extent should be measured inboard from the side of the vessel perpendicularly to the centerline at the level of the summer load waterline.

If pipes, ducts or tunnels are situated within the assumed extent of damage, arrangements should be made to ensure that progressive flooding cannot thereby extend to compartments other than those assumed to be floodable for each case of damage.

If a damage of a lesser extent than that specified above results in a more severe condition, such lesser extent should be assumed. Where a transverse watertight bulkhead is located within the transverse extent of assumed damage and is stepped in way of a double bottom or side tank by more than 3.05 m, the double bottom or side tanks adjacent to the stepped portion of the transverse watertight bulkhead should be considered as flooded simultaneously.
****

**** V.PROGR.A469
The final waterline, taking into account sinkage, heel and trim, should be below lower edge of any opening through which progressive flooding may take place. Such openings should include airpipes and those which are capable of being closed by means of watertight doors or hatch covers and may exclude those openings closed by means of watertight manhole covers and flush scuttles, small watertight cargo tank hatch covers which maintain the high integrity of the deck, remotely operated sliding doors and side scuttles of the nonopening type.
****

**** V.MAXHEEL.A469
In the final stage of flooding, the angle of heel due to unsymmetrical flooding should not exceed 15 degrees. This angle may be increased up to 17 degrees if no deck immersion occurs.
****

**** V.RANGE.A469
The stability in the final stage of flooding should be investigated and may be regarded as sufficient if the righting lever curve has at least a range of 20 degrees beyond the position of equilibrium. Unprotected openings should not become immersed at an angle of heel within the prescribed minimum range of residual stability unless the space in question has been included as a floodable space in calculations for damage stability.
****

**** V.MINGZ.A469
The stability in the final stage of flooding should be investigated and may be regarded as sufficient if the righting lever curve has at least a range of 20 degrees beyond the position of equilibrium in association with a maximum residual righting lever of at least 100 mm within this range. Unprotected openings should not become immersed at an angle of heel within the prescribed minimum range of residual stability unless the space in question has been included as a floodable space in calculations for damage stability.
****

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6.12.4 WEATHER CRITERION DETAILS

GENERAL

The ability of a ship to withstand the combined effects of beam wind and rolling should be demonstrated for each standard condition of loading, with reference to the figure as follows:

The ship is subjected to a steady wind pressure acting perpendicular to the ships centreline which result in a steady wind heeling lever (Lw1).

From the resultant angle of equilibrium (Theta0) the ship is assumed to roll owing to wave action to an angle of roll (Theta1) to windward. Attention should be paid to the effect of steady wind so that excessive resultant angles of heel are avoided:

The angle of heel under action of steady wind (Theta0) should be limited to a certain angle to the satisfaction of the Administration. As a guide, 16 degrees or 80% of the angle of deck edge immersion, whichever is less, is suggested.

The ship is then subjected to a gust wind pressure which results in a gust wind heeling lever (Lw2). Under these circumstances, area B should be equal to or greater than area A.
Proper free surface effects should be accounted for in the standard conditions of loading.

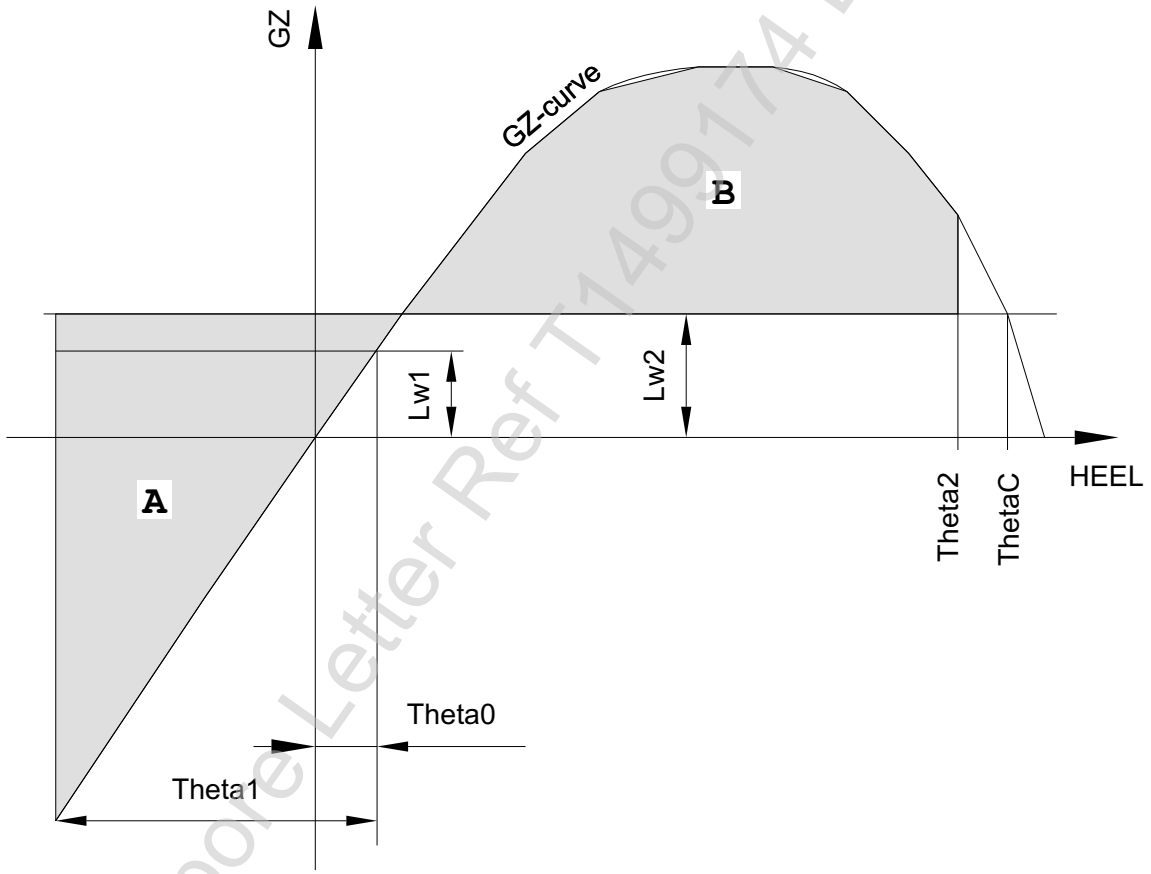


Figure - Severe wind and rolling

The angles in the above figure are defined as follows:
ThetaC = Angle of second intercept between wind heeling lever Lw2 and GZ curves.
Theta0 = angle of heel under action of steady wind
Theta1 = angle of roll to windward due to wave action
for this vessel total bilge keel(s) area of 26.34 m2 is considered in Theta1 angle calculation
Theta2 = angle of downflooding (ThetaF) or 50 degrees or ThetaC whichever is less, where:
ThetaF = Angle of heel at which openings in the hull, superstructure or deckhouses which cannot be closed weathertight immerse.

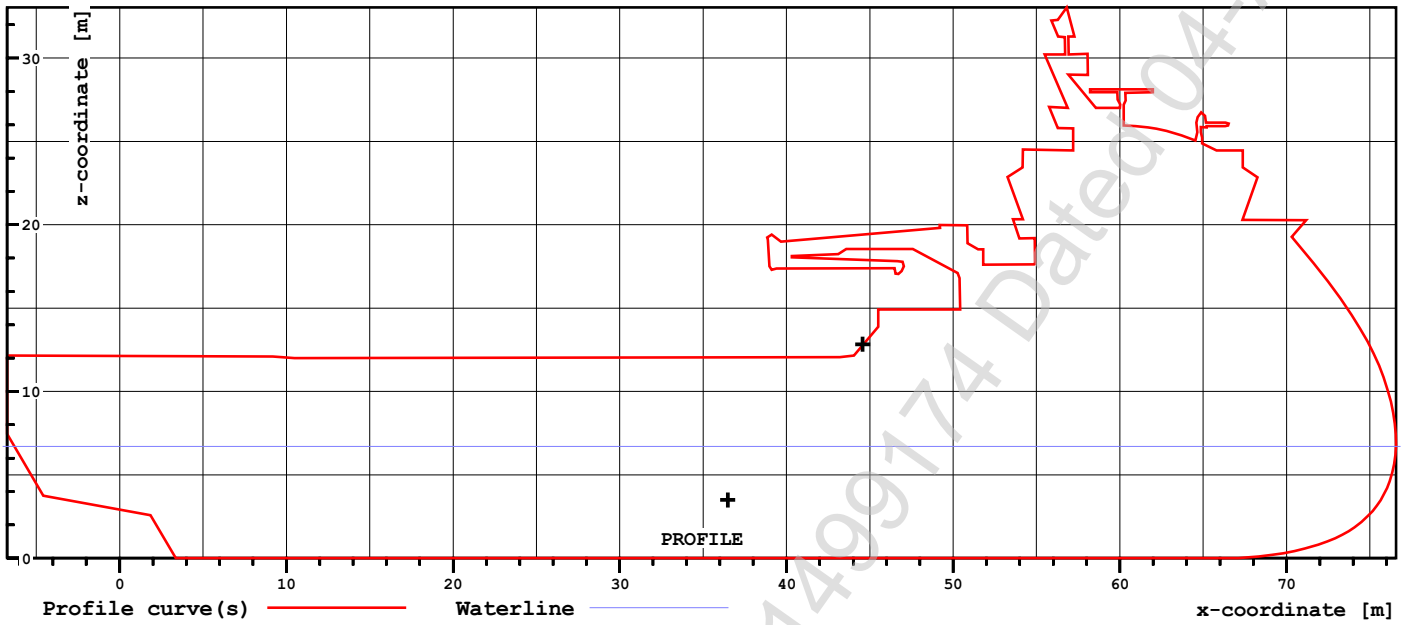
In applying this criterion, small openings through which progressive flooding cannot take place need not be considered as open.
lw1 = P * A * Z / 100 * g * disp lw2 = 1,5 * lw1
The wind heeling levers lw1 and lw2 are constant values at all angles of inclination and should be calculated as follows:
P = 504 N/m2 (wind speed = 29 m/s). The value of P, used for ships in restricted service and/or for ships with very large windage areas (due to coherence length for wind speed), may be reduced subject to the approval of the Society

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A = projected lateral area of the portion of the ship and deck cargo above the waterline (m2)
Z = vertical distance from the centre of A to the centre of the underwater lateral area or approximately to a point at one half the draught (m)
disp = displacement (t)
g = 9.81 m/s2
Ak = 26.34 m2

DETAILS OF WIND AREA(S) - INTACT STABILITY

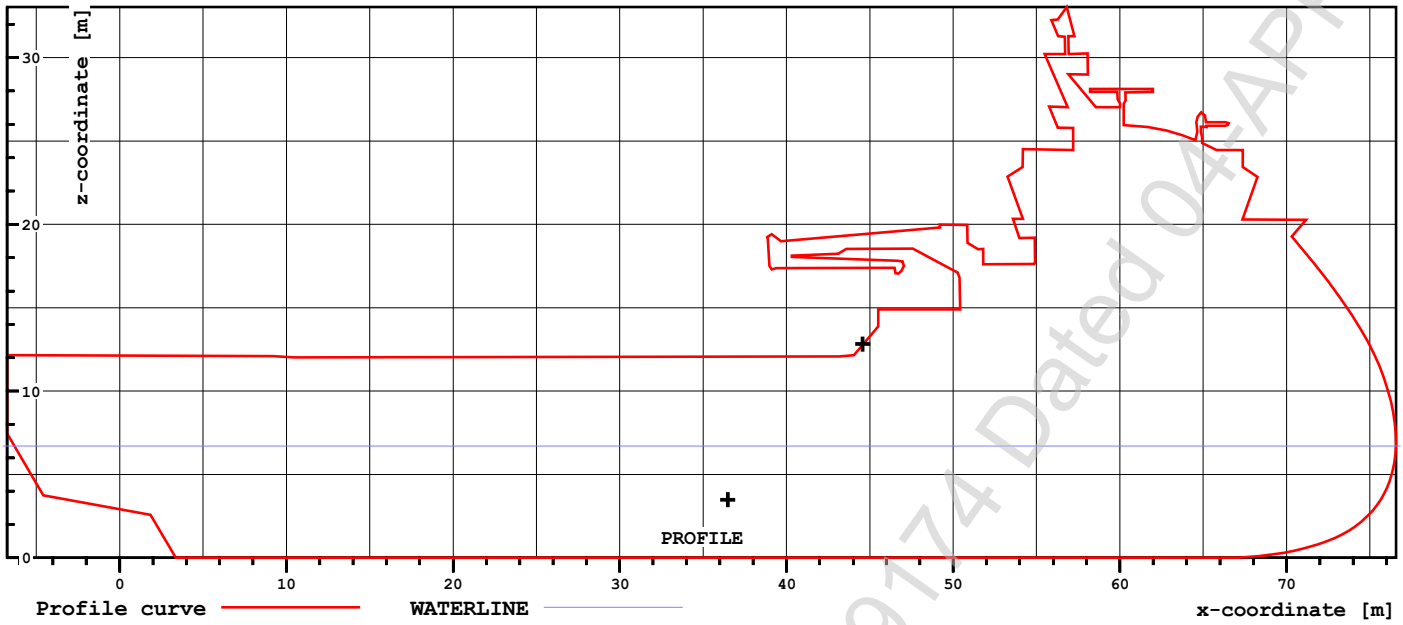


NAME	LCOND	COEF	AREA m2	ZCG m	LEVER m	MOM m3
PROFILE	T=3.2, TR=0	1.000	1020.3	10.635	8.997	9180.1
PROFILE	T=3.3, TR=0	1.000	1012.6	10.691	9.000	9113.9
PROFILE	T=3.4, TR=0	1.000	1004.8	10.748	9.004	9047.4
PROFILE	T=3.5, TR=0	1.000	997.0	10.806	9.008	8980.7
PROFILE	T=3.6, TR=0	1.000	989.0	10.864	9.012	8913.7
PROFILE	T=3.7, TR=0	1.000	981.1	10.922	9.017	8846.5
PROFILE	T=3.8, TR=0	1.000	973.0	10.982	9.022	8779.1
PROFILE	T=3.9, TR=0	1.000	965.0	11.041	9.028	8711.9
PROFILE	T=4, TR=0	1.000	956.9	11.101	9.034	8644.8
PROFILE	T=4.1, TR=0	1.000	948.9	11.161	9.040	8578.0
PROFILE	T=4.2, TR=0	1.000	940.8	11.221	9.047	8511.2
PROFILE	T=4.3, TR=0	1.000	932.7	11.282	9.054	8444.7
PROFILE	T=4.4, TR=0	1.000	924.6	11.342	9.062	8378.2
PROFILE	T=4.5, TR=0	1.000	916.5	11.403	9.069	8311.9
PROFILE	T=4.6, TR=0	1.000	908.3	11.465	9.078	8245.7
PROFILE	T=4.7, TR=0	1.000	900.2	11.526	9.086	8179.6
PROFILE	T=4.8, TR=0	1.000	892.1	11.588	9.095	8113.5
PROFILE	T=4.9, TR=0	1.000	883.9	11.650	9.104	8047.6
PROFILE	T=5, TR=0	1.000	875.8	11.713	9.114	7981.8
PROFILE	T=5.1, TR=0	1.000	867.6	11.775	9.124	7916.0
PROFILE	T=5.2, TR=0	1.000	859.4	11.838	9.134	7850.4
PROFILE	T=5.3, TR=0	1.000	851.2	11.902	9.145	7784.7
PROFILE	T=5.4, TR=0	1.000	843.1	11.965	9.156	7719.2
PROFILE	T=5.5, TR=0	1.000	834.9	12.029	9.168	7653.7
PROFILE	T=5.6, TR=0	1.000	826.6	12.094	9.180	7588.3
PROFILE	T=5.7, TR=0	1.000	818.4	12.158	9.192	7523.0
PROFILE	T=5.8, TR=0	1.000	810.2	12.223	9.205	7457.6
PROFILE	T=5.9, TR=0	1.000	802.0	12.289	9.218	7392.4
PROFILE	T=6, TR=0	1.000	793.7	12.354	9.231	7327.2
PROFILE	T=6.1, TR=0	1.000	785.5	12.421	9.245	7262.0
PROFILE	T=6.2, TR=0	1.000	777.2	12.487	9.259	7196.9
PROFILE	T=6.3, TR=0	1.000	769.0	12.554	9.274	7131.8
PROFILE	T=6.4, TR=0	1.000	760.7	12.622	9.290	7066.8
PROFILE	T=6.5, TR=0	1.000	752.5	12.689	9.305	7001.8
PROFILE	T=6.6, TR=0	1.000	744.2	12.758	9.321	6936.9
PROFILE	T=6.7, TR=0	1.000	735.9	12.826	9.338	6871.9
NAME	LCOND	COEF	AREA m2	ZCG m	LEVER m	MOM m3

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DETAILS OF WIND AREA(S) - DAMAGE STABILITY



NAME	LCOND	COEF	AREA m2	ZCG m	LEVER m	MOM m3
PROFILE	T=3.2, TR=0	1.000	1020.3	10.635	8.997	9180.1
PROFILE	T=3.3, TR=0	1.000	1012.6	10.691	9.000	9113.9
PROFILE	T=3.4, TR=0	1.000	1004.8	10.748	9.004	9047.4
PROFILE	T=3.5, TR=0	1.000	997.0	10.806	9.008	8980.7
PROFILE	T=3.6, TR=0	1.000	989.0	10.864	9.012	8913.7
PROFILE	T=3.7, TR=0	1.000	981.1	10.922	9.017	8846.5
PROFILE	T=3.8, TR=0	1.000	973.0	10.982	9.022	8779.1
PROFILE	T=3.9, TR=0	1.000	965.0	11.041	9.028	8711.9
PROFILE	T=4, TR=0	1.000	956.9	11.101	9.034	8644.8
PROFILE	T=4.1, TR=0	1.000	948.9	11.161	9.040	8578.0
PROFILE	T=4.2, TR=0	1.000	940.8	11.221	9.047	8511.2
PROFILE	T=4.3, TR=0	1.000	932.7	11.282	9.054	8444.7
PROFILE	T=4.4, TR=0	1.000	924.6	11.342	9.062	8378.2
PROFILE	T=4.5, TR=0	1.000	916.5	11.403	9.069	8311.9
PROFILE	T=4.6, TR=0	1.000	908.3	11.465	9.078	8245.7
PROFILE	T=4.7, TR=0	1.000	900.2	11.526	9.086	8179.6
PROFILE	T=4.8, TR=0	1.000	892.1	11.588	9.095	8113.5
PROFILE	T=4.9, TR=0	1.000	883.9	11.650	9.104	8047.6
PROFILE	T=5, TR=0	1.000	875.8	11.713	9.114	7981.8
PROFILE	T=5.1, TR=0	1.000	867.6	11.775	9.124	7916.0
PROFILE	T=5.2, TR=0	1.000	859.4	11.838	9.134	7850.4
PROFILE	T=5.3, TR=0	1.000	851.2	11.902	9.145	7784.7
PROFILE	T=5.4, TR=0	1.000	843.1	11.965	9.156	7719.2
PROFILE	T=5.5, TR=0	1.000	834.9	12.029	9.168	7653.7
PROFILE	T=5.6, TR=0	1.000	826.6	12.094	9.180	7588.3
PROFILE	T=5.7, TR=0	1.000	818.4	12.158	9.192	7523.0
PROFILE	T=5.8, TR=0	1.000	810.2	12.223	9.205	7457.6
PROFILE	T=5.9, TR=0	1.000	802.0	12.289	9.218	7392.4
PROFILE	T=6, TR=0	1.000	793.7	12.354	9.231	7327.2
PROFILE	T=6.1, TR=0	1.000	785.5	12.421	9.245	7262.0
PROFILE	T=6.2, TR=0	1.000	777.2	12.487	9.259	7196.9
PROFILE	T=6.3, TR=0	1.000	769.0	12.554	9.274	7131.8
PROFILE	T=6.4, TR=0	1.000	760.7	12.622	9.290	7066.8
PROFILE	T=6.5, TR=0	1.000	752.5	12.689	9.305	7001.8
PROFILE	T=6.6, TR=0	1.000	744.2	12.758	9.321	6936.9
PROFILE	T=6.7, TR=0	1.000	735.9	12.826	9.338	6871.9

NAME	LCOND	COEF	AREA m2	ZCG m	LEVER m	MOM m3

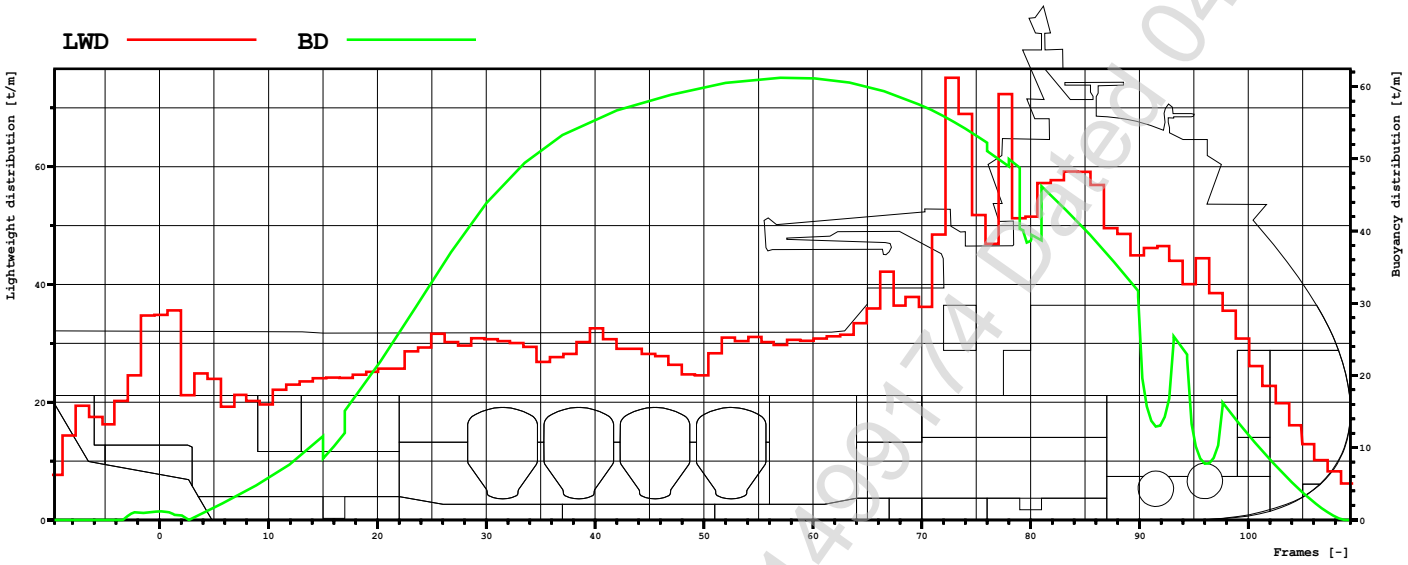
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6.12.5 LIGHTWEIGHT PARTICULARS

lightweight	t	2704.9
cgx of lightweight	m	39.15
cgy of lightweight	m	-0.01
cgz of lightweight	m	7.08

LIGHTWEIGHT DISTRIBUTION DIAGRAM



LIGHTWEIGHT DISTRIBUTION TABLE

X	x-coordinate	m
FR	frame number	#
BD	buoyancy distribution	t/m
LWD	lightweight distribution	t/m

X m	FR #	BD t/m	LWD t/m
-5.600	-8.00	0.00	14.35
-4.200	-6.00	0.00	17.50
-2.800	-4.00	0.00	20.21
-1.400	-2.00	1.02	24.56
0.000	0.00	1.15	34.85
1.400	2.00	0.61	21.18
2.800	4.00	0.97	24.86
4.200	6.00	2.47	19.23
5.600	8.00	4.10	20.21
7.000	10.00	5.86	19.61
8.400	12.00	7.74	22.99
9.800	14.00	10.32	23.51
11.200	16.00	10.18	24.19
12.600	18.00	17.20	24.64
14.000	20.00	21.37	25.16
15.400	22.00	25.92	25.69
16.800	24.00	30.55	29.30
18.200	26.00	35.26	31.62
19.600	28.00	39.64	29.60
21.000	30.00	43.80	30.72
22.400	32.00	47.02	30.35
23.800	34.00	49.98	29.37
25.200	36.00	52.17	27.64
26.600	38.00	53.95	28.17
28.000	40.00	55.32	32.53
29.400	42.00	56.69	29.07
30.800	44.00	57.56	29.07
32.200	46.00	58.43	27.79
33.600	48.00	59.20	24.71

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X m	FR #	BD t/m	LWD t/m
35.000	50.00	59.86	24.56
36.400	52.00	60.52	30.95
37.800	54.00	60.80	30.35
39.200	56.00	61.08	30.20
40.600	58.00	61.20	30.57
42.000	60.00	61.15	30.42
43.400	62.00	60.81	31.17
44.800	64.00	60.31	33.43
46.200	66.00	59.56	35.91
47.600	68.00	58.53	36.43
49.000	70.00	57.36	36.21
50.400	72.00	55.90	48.45
51.800	74.00	54.17	68.96
53.200	76.00	52.20	46.87
54.600	78.00	49.98	72.34
56.000	80.00	38.62	51.53
57.400	82.00	44.73	57.69
58.800	84.00	41.75	59.19
60.200	86.00	38.54	56.86
61.600	88.00	35.10	48.60
63.000	90.00	27.30	44.92
64.400	92.00	13.34	46.50
65.800	94.00	23.65	40.04
67.200	96.00	7.80	44.47
68.600	98.00	15.53	35.53
70.000	100.00	11.79	30.80
71.400	102.00	8.32	22.76
72.800	104.00	5.20	16.07
74.200	106.00	2.48	12.84
75.600	108.00	0.46	8.26

NOTE: Report from measurement of lightship particulars is included in appendix of this stability booklet.

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6.12.6 PERMISSIBLE STRENGTH LIMITS

DIAGRAM OF PERMISSIBLE STRENGTH LIMITS

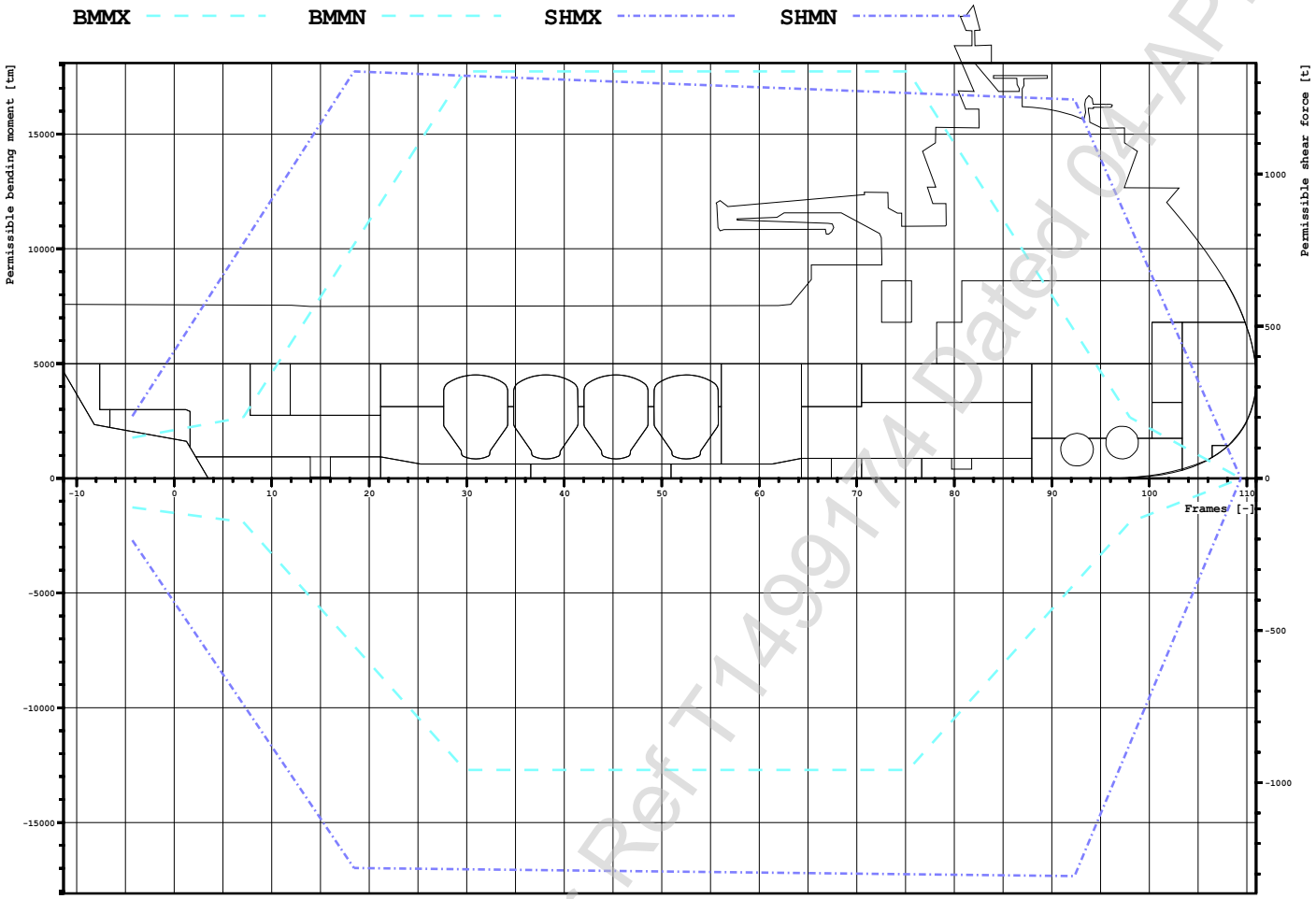


TABLE OF PERMISSIBLE STRENGTH LIMITS

X	x-coordinate	m
FR	frame number	#
BMMX	max. permissible bending moment	tm
BMMN	min. permissible bending moment	tm
SHMX	max. permissible shear force	t
SHMN	min. permissible shear force	t

X	FR	BMMX	BMMN	SHMX	SHMN
m	#	tm	tm	t	t
-3.000	-4.29	1774	-1271	204	-204
-2.800	-4.00	1796	-1287	218	-217
-1.400	-2.00	1952	-1399	318	-312
0.000	0.00	2108	-1510	418	-407
1.400	2.00	2264	-1622	517	-502
2.800	4.00	2420	-1734	617	-596
4.200	6.00	2576	-1846	717	-691
4.960	7.09	2660	-1906	771	-742
5.600	8.00	3267	-2341	817	-786
7.000	10.00	4594	-3292	916	-881
8.400	12.00	5921	-4242	1016	-975
9.800	14.00	7247	-5193	1116	-1070
11.200	16.00	8574	-6144	1215	-1165
12.600	18.00	9901	-7094	1315	-1259
12.910	18.44	10195	-7305	1337	-1280
14.000	20.00	11228	-8045	1335	-1281
15.400	22.00	12554	-8996	1333	-1282
16.800	24.00	13881	-9946	1330	-1282
18.200	26.00	15208	-10897	1328	-1283
19.600	28.00	16535	-11848	1325	-1284

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20.868	29.81	17736	-12709	1323	-1284
21.000	30.00	17736	-12709	1323	-1284
22.400	32.00	17736	-12709	1320	-1285
23.800	34.00	17736	-12709	1318	-1286
25.200	36.00	17736	-12709	1315	-1287
26.600	38.00	17736	-12709	1313	-1287
28.000	40.00	17736	-12709	1310	-1288
29.400	42.00	17736	-12709	1308	-1289
30.800	44.00	17736	-12709	1305	-1289
32.200	46.00	17736	-12709	1303	-1290
33.600	48.00	17736	-12709	1300	-1291
35.000	50.00	17736	-12709	1298	-1292
36.400	52.00	17736	-12709	1295	-1292
37.800	54.00	17736	-12709	1293	-1293
39.200	56.00	17736	-12709	1290	-1294
40.600	58.00	17736	-12709	1288	-1294
42.000	60.00	17736	-12709	1285	-1295
43.400	62.00	17736	-12709	1283	-1296
44.800	64.00	17736	-12709	1280	-1296
46.200	66.00	17736	-12709	1278	-1297
47.600	68.00	17736	-12709	1275	-1298
49.000	70.00	17736	-12709	1273	-1299
50.400	72.00	17736	-12709	1270	-1299
51.800	74.00	17736	-12709	1268	-1300
52.692	75.27	17736	-12709	1266	-1300
53.200	76.00	17255	-12364	1265	-1301
54.600	78.00	15928	-11413	1263	-1301
56.000	80.00	14601	-10463	1260	-1302
57.400	82.00	13275	-9512	1258	-1303
58.800	84.00	11948	-8561	1255	-1304
60.200	86.00	10621	-7610	1253	-1304
61.600	88.00	9294	-6660	1250	-1305
63.000	90.00	7968	-5709	1248	-1306
64.400	92.00	6641	-4758	1245	-1306
64.630	92.33	6423	-4602	1245	-1306
65.800	94.00	5314	-3808	1123	-1178
67.200	96.00	3987	-2857	977	-1025
68.600	98.00	2660	-1906	831	-872
70.000	100.00	2193	-1571	685	-718
71.400	102.00	1725	-1236	538	-565
72.800	104.00	1257	-900	392	-412
74.200	106.00	789	-565	246	-258
75.600	108.00	321	-230	100	-105
76.560	109.37	0	0	0	0

X	FR	BMMX	BMMN	SHMX	SHMN
m	#	tm	tm	t	t

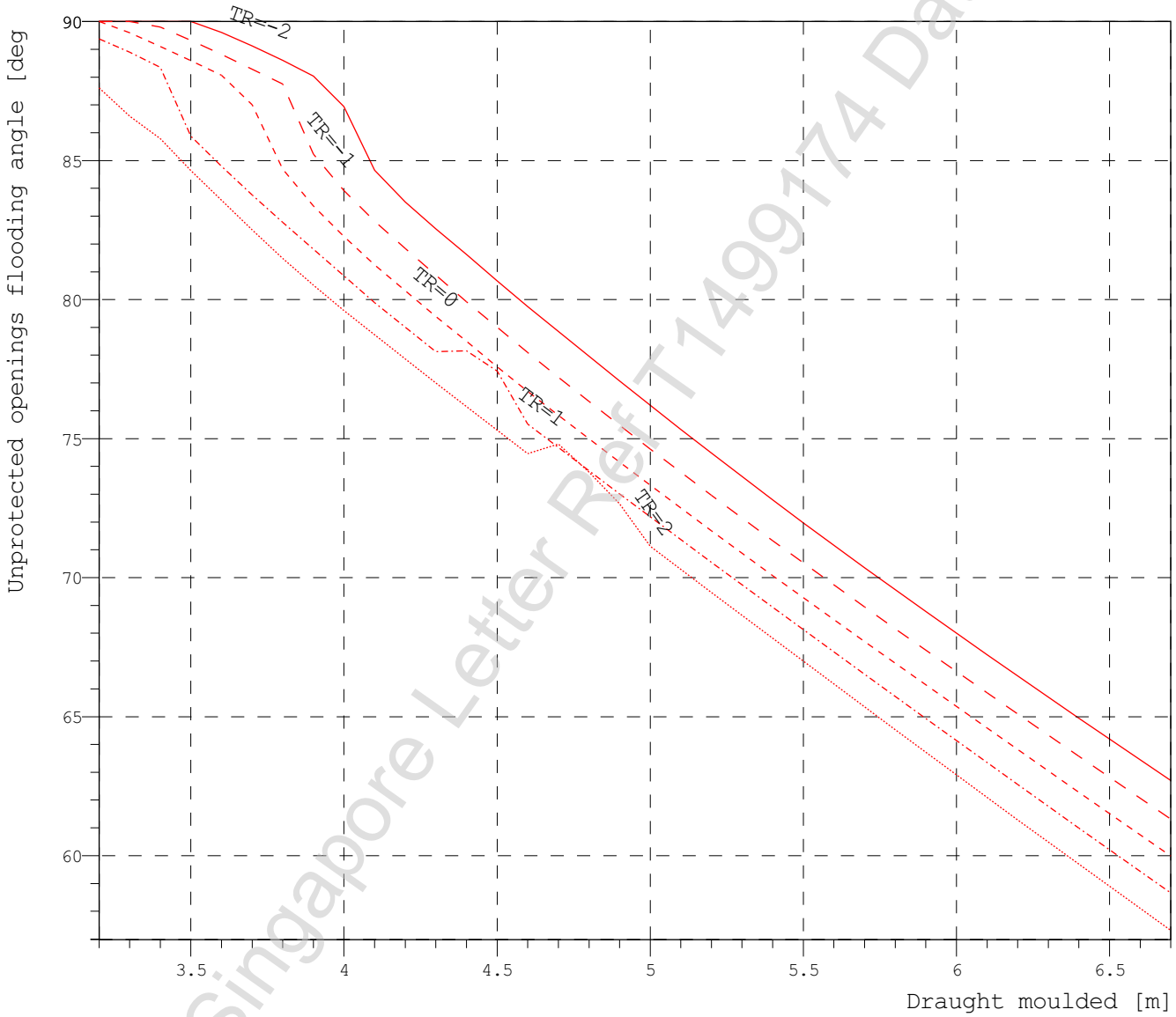
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6.12.7 RELEVANT OPENINGS LIST - INTACT STABILITY

NAME	DESCRIPTION	TYPE	FRAME #	X m	Y m	Z m	CONN
V-421G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	79.84	55.890	2.093	14.288	
V-424G	SUPPLY AIR ENGINE ROOM PS	UNPROTECTED	77.49	54.243	-3.413	14.288	
V-425G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	77.49	54.243	3.767	14.288	
V-421E	EXHAUST AIR ENGINE ROOM PS	UNPROTECTED	79.99	55.993	-2.538	17.088	
V-423E	EXHAUST AIR ENGINE ROOM SB	UNPROTECTED	79.99	55.993	3.423	17.088	
D-651	ACCOMMODATION/C-DECK	UNPROTECTED	80.00	56.000	3.771	16.906	

6.12.8 DOWNFLOODING ANGLES DIAGRAM



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6.12.9 RELEVANT OPENINGS LIST - DAMAGE STABILITY

NAME	DESCRIPTION	TYPE	FRAME #	X m	Y m	Z m	CONN
V-100A	CENTR.UNIT AIR COND.PLANT FO.	WEATHERTIGHT	75.99	53.193	8.186	14.782	
V-100E	CENTR.UNIT AIR COND.PLANT FO.	WEATHERTIGHT	73.25	51.275	3.405	10.514	
V-101K	OVERPRESSURE IN ACC.	WEATHERTIGHT	77.31	54.115	1.699	10.566	
V-101L	OVERPRESSURE IN ACC.	WEATHERTIGHT	79.68	55.773	1.532	13.416	
V-101M	OVERPRESSURE IN ACC.	WEATHERTIGHT	79.75	55.825	0.433	16.266	
V-101N	OVERPRESSURE IN ACC.	WEATHERTIGHT	79.75	55.825	3.100	18.866	
V-114A	EXHAUST FAN HOSPITAL. MAIN D.	WEATHERTIGHT	72.64	50.850	6.753	10.616	
V-119A	EXHAUST FAN SMOKERS LOUNGE	WEATHERTIGHT	75.53	52.874	3.674	12.849	
V-140M	SUP. TUMBLE DRIERS	WEATHERTIGHT	76.80	53.758	5.250	10.616	
V-140K	EXH.TUMBLE DRIERS	WEATHERTIGHT	76.80	53.758	4.700	10.616	
V-140L	EXH.TUMBLE DRIERS	WEATHERTIGHT	76.80	53.758	4.380	10.616	
V-213A	SUPPLY FAN GALLEY	WEATHERTIGHT	74.55	52.186	-6.893	14.789	
V-218A1	SUPPLY AIR FOREPEAK STORE	WEATHERTIGHT	100.50	70.348	-3.501	14.757	
V-218S	EXHAUST FAN FOREPEAK STORE	WEATHERTIGHT	100.50	70.349	3.421	14.575	
V-231A	EXHAUST FAN FIRE STORE MAIN .	WEATHERTIGHT	77.31	54.118	-1.562	10.431	
V-231S	SUP.STORES MAIN DECK	WEATHERTIGHT	79.66	55.760	-0.929	11.839	
V-237A	EXHAUST FAN CO2 ROOM	WEATHERTIGHT	72.82	50.974	-7.861	10.630	
V-237S	SUPPLY AIR CO2 ROOM	WEATHERTIGHT	74.26	51.984	-6.573	10.596	
V-351A	SUPPLY FAN ENGINE CONTROL RO.	WEATHERTIGHT	75.01	52.506	-6.464	10.616	
V-351E	EXHAUST ENGINE CONTROL ROOM	WEATHERTIGHT	72.83	50.984	-7.660	10.616	
V-353A	EXHAUST FAN SWITCHBOARD ROOM	WEATHERTIGHT	67.35	47.146	7.297	8.959	
V-353S	SUPPLY SWITCHBOARD ROOM	WEATHERTIGHT	70.53	49.374	-7.229	11.900	
V-420G	SUPPLY AIR ENGINE ROOM PS	WEATHERTIGHT	79.84	55.890	-1.743	14.288	
V-421E	EXHAUST AIR ENGINE ROOM PS	UNPROTECTED	79.99	55.993	-2.538	17.088	
V-421G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	79.84	55.890	2.093	14.288	
V-423E	EXHAUST AIR ENGINE ROOM SB	UNPROTECTED	79.99	55.993	3.423	17.088	
V-424G	SUPPLY AIR ENGINE ROOM PS	UNPROTECTED	77.49	54.243	-3.413	14.288	
V-425G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	77.49	54.243	3.767	14.288	
V-428A	SUPPLY FAN BOWTH.ROOM	WEATHERTIGHT	76.73	53.709	3.995	9.441	
V-428E	EXHAUST BOWTHRUSTER ROOM	WEATHERTIGHT	79.84	55.885	-6.807	18.827	
V-433A	SUPPLY AIR GARBAGE ROOM	WEATHERTIGHT	75.99	53.192	-6.882	10.587	
V-444A	EXHAUST FAN ENGINE WORKSHOP	WEATHERTIGHT	79.74	55.818	-6.541	18.777	
V-446A	EXHAUST FAN PAINT STORE	WEATHERTIGHT	48.54	33.980	8.680	11.909	
V-446S	SUPPLY AIR PAINT STORE	WEATHERTIGHT	74.68	52.275	5.970	10.630	
V-448A	EXHAUST FAN GARBAGE ROOM	WEATHERTIGHT	73.29	51.301	-6.127	10.617	
V-449A	EXHAUST FAN WORKSHOP/DECK ST.	WEATHERTIGHT	76.82	53.775	-3.190	10.566	
V-449S	SUPPLY AIR CO2 ROOM	WEATHERTIGHT	75.12	52.582	-5.438	10.616	
V-450A	EXHAUST FAN CHEMICAL STORE	WEATHERTIGHT	46.47	32.528	8.700	11.909	
V-450S	SUPPLY AIR CHEMICAL STORE	WEATHERTIGHT	75.84	53.090	8.690	10.630	
V-480\B2	VENT.ACETYLEN ROOM	WEATHERTIGHT	72.68	50.875	-3.995	10.580	
V-480\B1	VENT.ACETYLEN ROOM	WEATHERTIGHT	72.68	50.875	-3.710	10.580	
V-481\B2	VENT.OXYGEN ROOM	WEATHERTIGHT	72.68	50.875	-5.055	10.630	
V-481\B1	VENT.OXYGEN ROOM	WEATHERTIGHT	72.68	50.875	-4.580	10.580	
V-512A1	EXHAUST FAN CARGO PUMP ROOM .	WEATHERTIGHT	33.91	23.738	7.880	11.749	
V-512A2	EXHAUST FAN CARGO PUMP ROOM .	WEATHERTIGHT	31.90	22.328	7.550	11.749	
V-512S	SUPPLY AIR CARGO PUMP ROOM	WEATHERTIGHT	63.88	44.713	-7.880	12.912	
V-513A1	EXHAUST FAN CARGO PUMP ROOM .	WEATHERTIGHT	30.91	21.638	-7.885	11.749	
V-513A2	EXHAUST FAN CARGO PUMP ROOM .	WEATHERTIGHT	32.90	23.028	-7.888	11.749	
V-513S1	SUPPLY AIR CARGO PUMP ROOM U.	WEATHERTIGHT	57.92	40.541	8.026	10.747	
V-513S2	SUPPLY AIR CARGO PUMP ROOM U.	WEATHERTIGHT	58.92	41.243	8.026	10.747	
V-514\B1	VENT.ENTRANCE PS	WEATHERTIGHT	34.36	24.053	-8.720	10.273	
V-514\B2	VENT.ENTRANCE PS	WEATHERTIGHT	30.79	21.550	-8.711	10.498	
V-515\B1	VENT.ENTRANCE SB	WEATHERTIGHT	29.22	20.455	8.675	10.420	
V-515\B2	VENT.ENTRANCE SB	WEATHERTIGHT	33.58	23.505	8.721	10.338	
V-612A	SUPPLY FAN PROPULSION ROOM	WEATHERTIGHT	-5.68	-3.974	-8.399	10.333	
V-612E	EXHAUST PROPULSION ROOM	WEATHERTIGHT	9.43	6.600	-7.796	8.936	
V-617\B1	VENT.PROP.ROOM ENTRANCEPS	WEATHERTIGHT	-4.57	-3.200	-7.677	10.443	
V-617\B2	VENT.PROP.ROOM ENTRANCEPS	WEATHERTIGHT	-2.53	-1.770	-8.748	10.443	
V-618\B2	VENT.PROP.ROOM ENTRANCE SB	WEATHERTIGHT	-7.47	-5.231	8.700	11.993	
V-618\B1	VENT.PROP.ROOM ENTRANCE SB	WEATHERTIGHT	-7.86	-5.500	7.977	11.993	
AP-001	FOREPEAK TANK	WEATHERTIGHT	102.74	71.919	1.728	14.899	SEA, T001
AP-002	WING TANK PS	WEATHERTIGHT	100.05	70.038	-3.988	14.887	SEA, T002
AP-003	WING TANK SB	WEATHERTIGHT	100.05	70.038	3.988	14.897	SEA, T003
AP-004	DB TANK CE VOID	WEATHERTIGHT	95.27	66.689	-5.115	14.375	SEA, T004
AP-005	WING TANK PS	WEATHERTIGHT	95.57	66.902	-6.002	14.520	SEA, T005
AP-006	WING TANK SB	WEATHERTIGHT	95.64	66.947	5.971	14.520	SEA, T006
AP-009	DB TANK CE	WEATHERTIGHT	74.45	52.115	-7.917	14.525	SEA, T009
AP-010	WING TANK PS	WEATHERTIGHT	67.44	47.211	-8.731	8.953	SEA, T010
AP-011	WING TANK SB	WEATHERTIGHT	67.44	47.211	8.754	8.953	SEA, T011
AP-012	DB TANK PS	WEATHERTIGHT	71.43	50.000	-7.250	9.047	SEA, T012
NAME	DESCRIPTION	TYPE	FRAME #	X m	Y m	Z m	CONN

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			#	m	m	m	
AP-013	DB TANK SB	WEATHERTIGHT	71.45	50.015	7.238	9.030	SEA, T013
AP-014	WING TANK PS	WEATHERTIGHT	61.54	43.078	-7.815	9.372	SEA, T014
AP-015	WING TANK SB	WEATHERTIGHT	61.54	43.078	7.815	9.372	SEA, T015
AP-016	DB TANK PS	WEATHERTIGHT	62.72	43.905	-7.745	9.372	SEA, T016
AP-017	DB TANK SB	WEATHERTIGHT	62.58	43.809	7.815	9.372	SEA, T017
AP-018	WING TANK PS	WEATHERTIGHT	44.66	31.265	-7.803	11.668	SEA, T018
AP-019	WING TANK SB	WEATHERTIGHT	44.66	31.265	7.803	11.668	SEA, T019
AP-020	DB TANK PS	WEATHERTIGHT	38.39	26.870	-7.800	11.690	SEA, T020
AP-021	DB TANK SB	WEATHERTIGHT	38.39	26.870	7.800	11.690	SEA, T021
AP-022	WING TANK PS	WEATHERTIGHT	27.43	19.200	-8.702	8.936	SEA, T022
AP-023	WING TANK SB	WEATHERTIGHT	27.43	19.200	8.702	8.936	SEA, T023
AP-024	DB TANK PS	WEATHERTIGHT	26.43	18.498	-8.000	11.882	SEA, T024
AP-025	DB TANK SB	WEATHERTIGHT	25.46	17.820	7.800	9.368	SEA, T025
AP-026	WING TANK PS	WEATHERTIGHT	21.41	14.986	-7.800	9.372	SEA, T026
AP-027	WING TANK SB	WEATHERTIGHT	21.41	14.986	7.800	9.372	SEA, T027
AP-029	WING TANK PS	WEATHERTIGHT	7.50	5.250	-7.800	9.372	SEA, T029
AP-030	WING TANK SB	WEATHERTIGHT	7.50	5.250	7.801	9.168	SEA, T030
AP-031	WING TANK PS	WEATHERTIGHT	-1.33	-0.930	-7.800	9.372	SEA, T031
AP-032	WING TANK SB	WEATHERTIGHT	-6.96	-4.871	8.032	12.072	SEA, T032
AP-033	WING TANK PS	WEATHERTIGHT	-8.39	-5.873	-7.873	9.372	SEA, T033
AP-034	WING TANK SB	WEATHERTIGHT	-8.39	-5.873	7.880	9.372	SEA, T034
AP-035	WING TANK PS	WEATHERTIGHT	1.37	0.960	-7.691	9.372	SEA, T035
AP-036	WING TANK SB	WEATHERTIGHT	1.66	1.163	7.810	9.372	SEA, T036
AP-043	LIQUID CARGO TANK PS	WEATHERTIGHT	58.61	41.028	-7.803	11.768	SEA, T043
AP-044	LIQUID CARGO TANK SB	WEATHERTIGHT	59.40	41.578	7.840	11.768	SEA, T044
AP-045	LIQUID CARGO TANK PS	WEATHERTIGHT	52.47	36.728	-7.803	11.768	SEA, T045
AP-046	LIQUID CARGO TANK SB	WEATHERTIGHT	52.47	36.728	7.803	11.768	SEA, T046
AP-047	LIQUID CARGO TANK PS	WEATHERTIGHT	47.54	33.278	-7.803	11.768	SEA, T047
AP-048	LIQUID CARGO TANK SB	WEATHERTIGHT	47.54	33.278	7.803	11.768	SEA, T048
AP-049	LIQUID CARGO TANK PS	WEATHERTIGHT	41.54	29.078	-7.803	11.768	SEA, T049
AP-050	LIQUID CARGO TANK SB	WEATHERTIGHT	41.54	29.078	7.803	11.768	SEA, T050
AP-051	LIQUID CARGO TANK PS	WEATHERTIGHT	34.34	24.040	-7.800	11.770	SEA, T051
AP-052	LIQUID CARGO TANK SB	WEATHERTIGHT	34.36	24.053	7.803	11.768	SEA, T052
AP-053	STORAGE TANK PS	WEATHERTIGHT	25.44	17.806	-8.000	11.950	SEA, T053
AP-054	STORAGE TANK SB	WEATHERTIGHT	24.79	17.350	7.800	11.790	SEA, T054
AP-055	LFL TANK PS	WEATHERTIGHT	8.37	5.862	8.701	11.851	SEA, T055
AP-056	LFL TANK SB	WEATHERTIGHT	8.95	6.262	8.701	11.851	SEA, T056
AP-061A	T061 DW/WB/FW	WEATHERTIGHT	74.17	51.918	-8.417	16.447	SEA, T061
AP-061B	T061 DW/WB/FW	WEATHERTIGHT	75.50	52.852	8.667	16.447	SEA, T061
AP-062A	ROLL RED. TANK	WEATHERTIGHT	8.64	6.050	-7.800	11.692	SEA, T062
AP-062B	ROLL RED. TANK	WEATHERTIGHT	9.50	6.650	7.800	11.692	SEA, T062
AP-252	BLACK WATER TANK	WEATHERTIGHT	85.83	60.080	-3.500	26.060	SEA, T252
AP-254	GREY WATER TANK	WEATHERTIGHT	86.24	60.370	-3.500	26.060	SEA, T254
AP-263	HYDRAULIC OIL STORE TANK	WEATHERTIGHT	68.13	47.690	-7.260	8.912	SEA, T263
AP-275	FO DRAIN TANK	WEATHERTIGHT	77.26	54.080	2.600	8.986	SEA, T275
AP-277	FO OVERFLOW TANK	WEATHERTIGHT	62.47	43.732	7.307	9.447	SEA, T277
AP-280	BILGE WATER HOLDING TANK	WEATHERTIGHT	68.20	47.738	7.275	8.951	SEA, T280
AP-283	SLUDGE DRAIN TANK	WEATHERTIGHT	85.52	59.865	-3.500	26.060	SEA, T283
AP-287	BILGE WATER DRAIN TANK	WEATHERTIGHT	72.27	50.590	3.084	8.916	SEA, T287
AP-290	LO STORE TANK MAIN ENG	WEATHERTIGHT	75.69	52.985	7.410	8.927	SEA, T290
AP-291	LO STORE TANK MAIN ENG	WEATHERTIGHT	75.69	52.985	7.170	8.927	SEA, T291
AP-293	LO STORE TANK THRUSTER	WEATHERTIGHT	75.69	52.985	6.878	8.927	SEA, T293
AP-295	LO STORE TANK AZIMUTH THR.	WEATHERTIGHT	25.91	18.140	-8.085	11.929	SEA, T295
AP-297	LO DRAIN TANK	WEATHERTIGHT	77.26	54.080	-2.944	9.060	SEA, T297
AP-298	DIRTY LO TANK	WEATHERTIGHT	24.26	16.983	7.735	11.878	SEA, T298
AP-312	CD VOID OUTSIDE LFL TANKS PS	WEATHERTIGHT	14.54	10.175	-8.760	11.690	SEA, T312
AP-313	CD VOID OUTSIDE LFL TANKS SB	WEATHERTIGHT	18.69	13.080	8.743	11.688	SEA, T313
AP-314	CD VOID AROUND LFL TANKS CE	WEATHERTIGHT	16.78	11.743	7.827	11.688	SEA, T314
AP-317	CD VOID CE	WEATHERTIGHT	23.94	16.758	7.221	11.942	SEA, T317
AP-318	CD VOID PS	WEATHERTIGHT	71.44	50.007	-7.137	11.484	SEA, T318
AP-319	CD VOID SB	WEATHERTIGHT	72.26	50.583	3.978	8.896	SEA, T319
AP-320	VOID PS	WEATHERTIGHT	1.81	1.266	-7.721	9.234	SEA, T320
AP-321	VOID SB	WEATHERTIGHT	-7.05	-4.934	8.727	11.919	SEA, T321
AP-411A	CLOSED CARGO RAIL PS	WEATHERTIGHT	56.69	39.683	-7.880	11.666	SEA, RAIL1
AP-411B	CLOSED CARGO RAIL PS	WEATHERTIGHT	37.29	26.100	-7.880	11.666	SEA, RAIL1
AP-412A	CLOSED CARGO RAIL SB	WEATHERTIGHT	53.40	37.383	7.780	11.666	SEA, RAIL2
AP-412B	CLOSED CARGO RAIL SB	WEATHERTIGHT	37.30	26.110	7.880	11.666	SEA, RAIL2
D-651	ACCOMMODATION/C-DECK	UNPROTECTED	80.00	56.000	3.771	16.906	
D-351	CO2 ROOM/M-DECK	WEATHERTIGHT	73.99	51.792	-8.360	8.600	
D-352	CHEMICAL/M-DECK	WEATHERTIGHT	75.99	53.192	8.375	8.600	

NAME	DESCRIPTION	TYPE	FRAME #	X m	Y m	Z m	CONN
D-353	DECK WORKSHOP/M-DECK	WEATHERTIGHT	75.99	53.192	-6.550	8.600	
D-354	PAINT/M-DECK	WEATHERTIGHT	75.99	53.192	6.600	8.600	
D-355	AC/M-DECK	WEATHERTIGHT	75.99	53.192	-5.130	8.600	

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D-356	WARDROBE/M-DECK	WEATHERTIGHT	76.99	53.893	5.270	8.600
D-357	OX/M-DECK	WEATHERTIGHT	75.99	53.192	-4.260	8.600
D-358	ACCOMODATION/M-DECK	WEATHERTIGHT	77.49	54.243	1.800	8.600
D-359	FIRE STORE/M-DECK	WEATHERTIGHT	77.49	54.243	-1.994	8.600
D-360	SB CARGORAIL #57/M-DECK	WEATHERTIGHT	56.98	39.885	8.560	8.600
D-361	PS CARGORAIL #57/M-DECK	WEATHERTIGHT	56.98	39.885	-8.560	8.600
D-362	SB CARGORAIL #37/M-DECK	WEATHERTIGHT	37.00	25.900	8.550	8.600
D-363	PS CARGORAIL #37/M-DECK	WEATHERTIGHT	37.00	25.900	-8.550	8.600
D-451	DRY PROV./A-DECK	WEATHERTIGHT	77.23	54.059	-3.500	11.300
D-452	ACCOMMODATION/A-DECK	WEATHERTIGHT	79.99	55.993	1.930	11.300
D-454	AC ROOM/A-DECK	WEATHERTIGHT	77.25	54.077	3.843	11.300
D-551	ACCOMMODATION/B-DECK	WEATHERTIGHT	95.00	66.500	-2.606	14.359
D-552	ACCOMMODATION/B-DECK	WEATHERTIGHT	79.99	55.993	1.177	14.140
D-553	ACCOMMODATION/B-DECK	WEATHERTIGHT	74.99	52.493	-4.659	14.165

Longitudinal extent 5.387m; transverse extent 760mm; vertical extent 8.00m

ZONE1
001FPTK.C
002FW.P

ZONE2
002FW.P

ZONE3
005FW.P

ZONE4
010VOID.P
252SEWAGE.P
254GREY.P
318CDVOID.P

AMENDED BY ABS

ZONE5
005FW.P
252SEWAGE.P
254GREY.P
318CDVOID.P

ZONE6
010VOID.P

ZONE7
016DWWB.P
014DWWB.P

ZONE8
020DWWB.P
024DWWB.P
018DWWB.P
EEMID.P
022VOID.P

ZONE9
024DWWB.P
EEMID.P
022VOID.P

ZONE10
026DWWB.P
312CDVOID.P

ZONE11
029FW.P
062ROLLRED.C

ZONE12
035DWWB.P
033FW.P
EEAFT.P
RECESS.P

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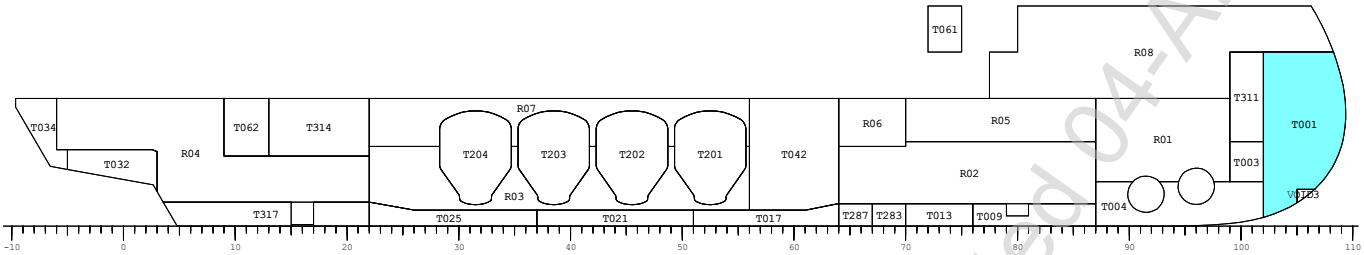
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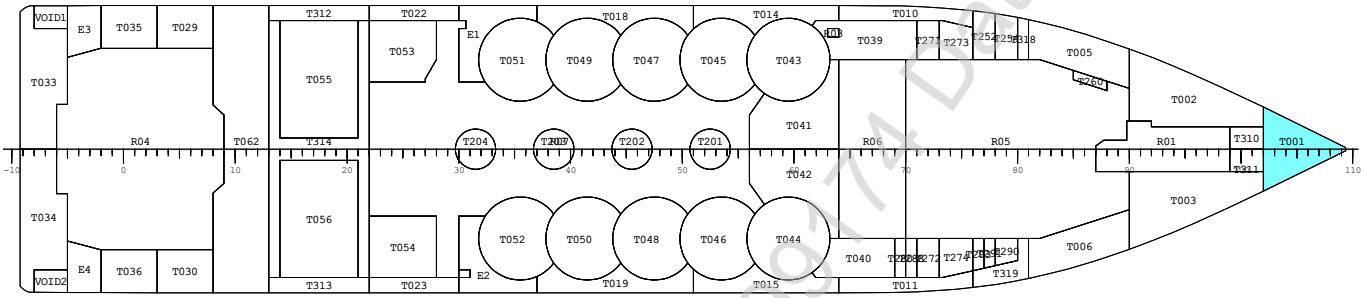
6.12.10 DEFINITIONS OF DETERMINISTIC DAMAGE CASES

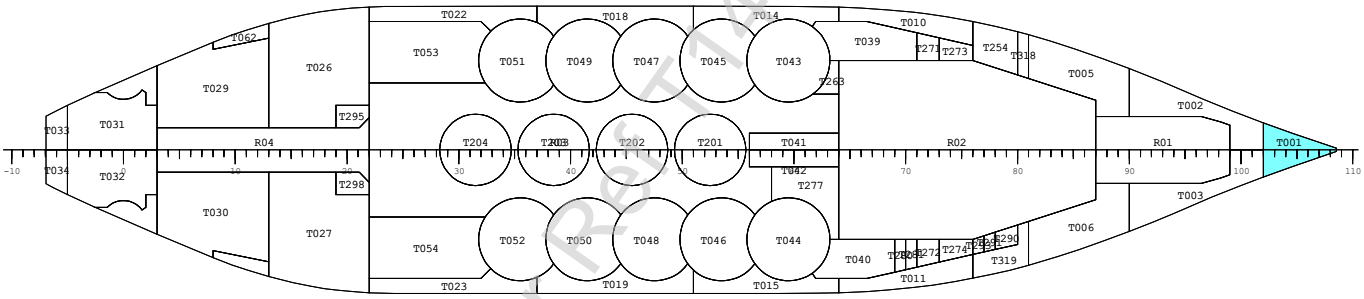
Refer to page 40 for damage cases

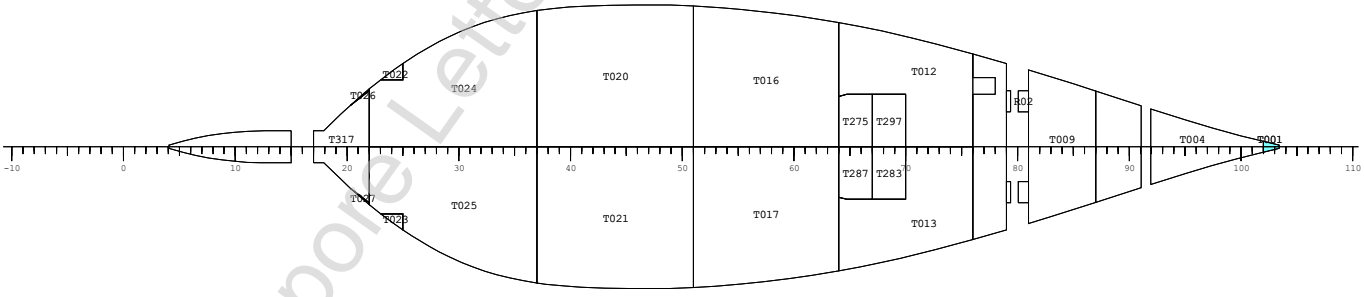
On the following pages the definition of the damage group and damage cases is given. The damage cases is also illustrated graphically with additional information of the flooded compartments.

DAMAGE CASE D01







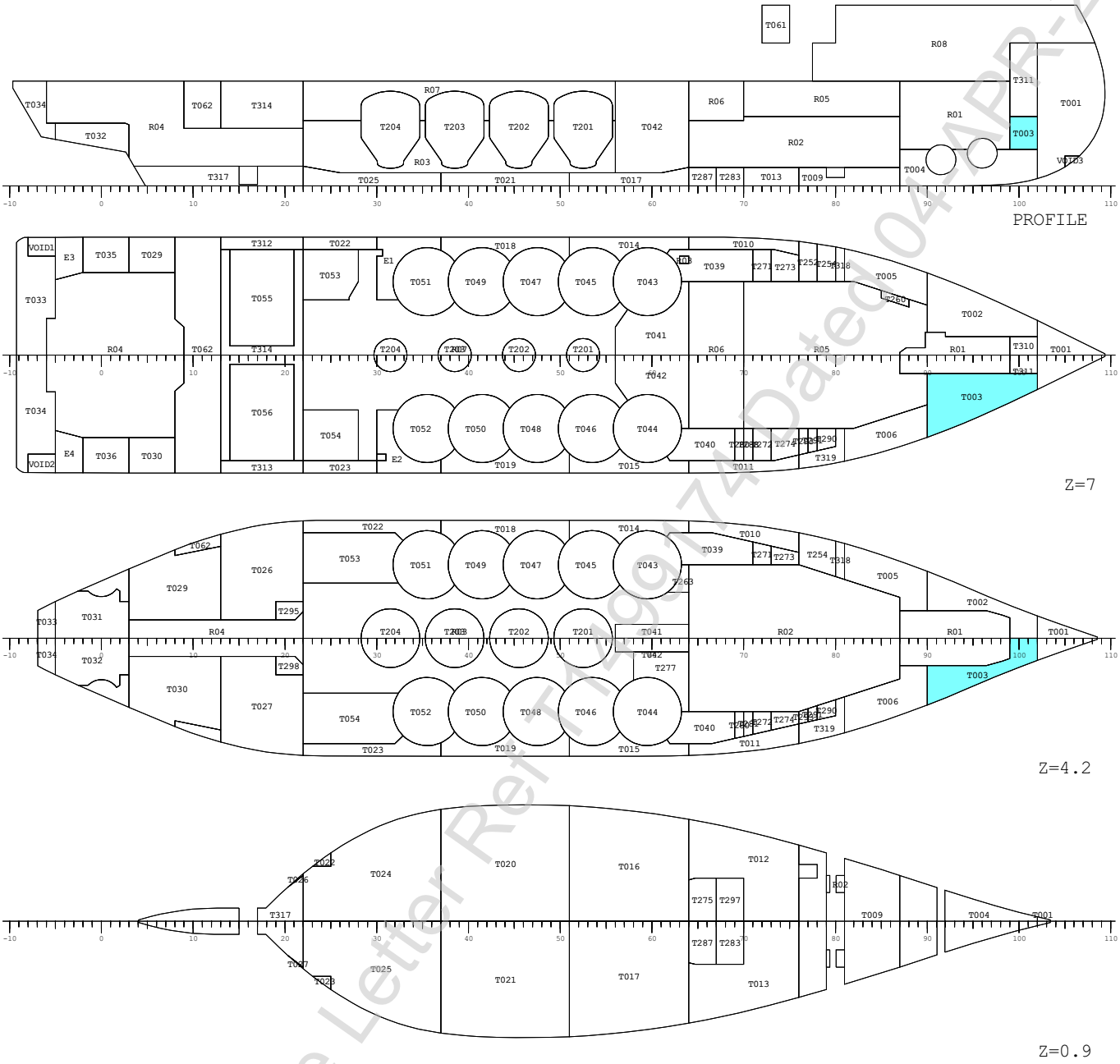


DAM	COMP	DES	PERM	VOL m3	XCG m	YCG m	ZCG m	STAGE
D01	T001	FOREPEAK TANK	0.95	105.6	73.11	0.00	7.21	1

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DAMAGE CASE D02

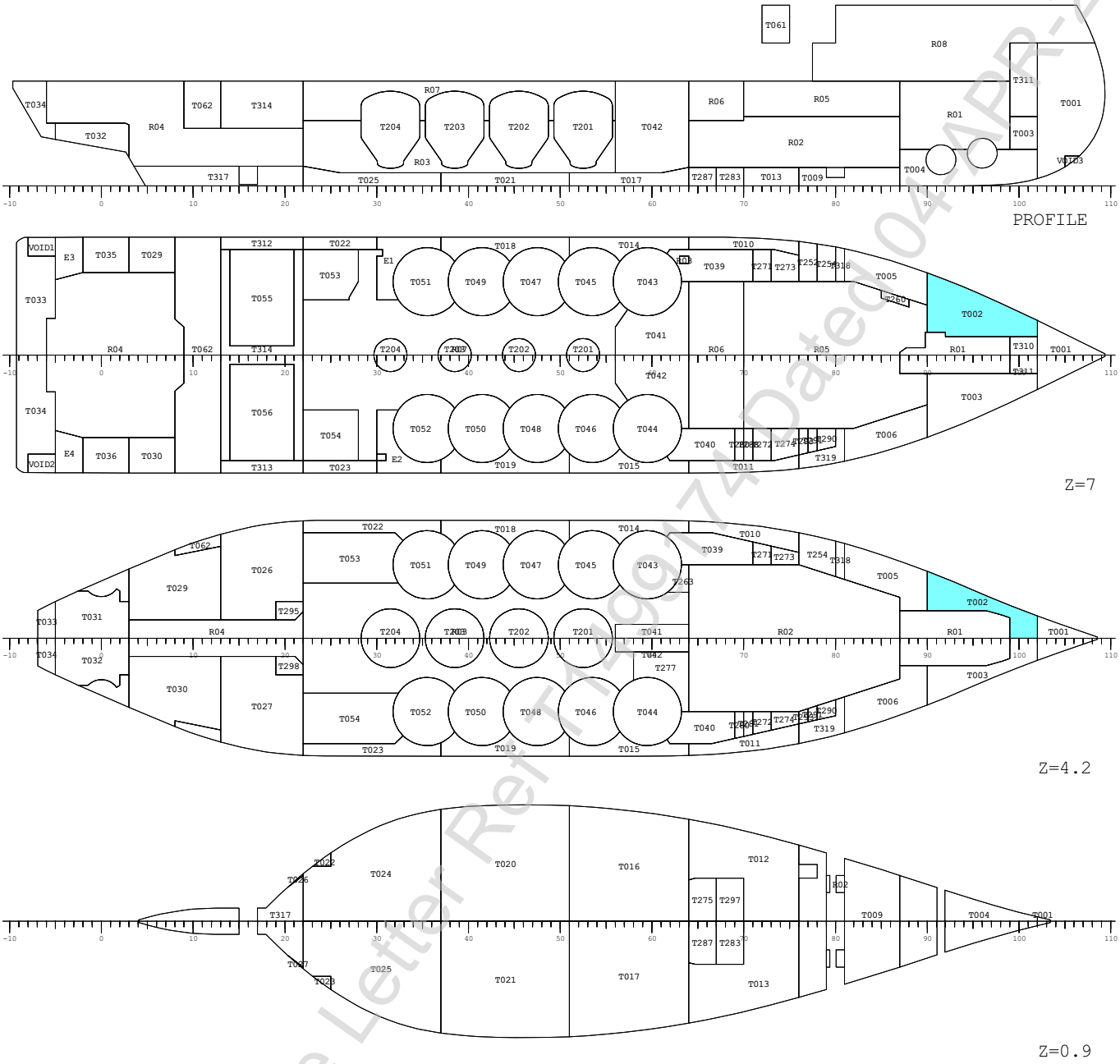


DAM	COMP	DES	PERM	VOL m3	XCG m	YCG m	ZCG m	STAGE
D02	T003	WING TANK SB	0.95	118.5	67.02	2.86	6.32	1

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DAMAGE CASE D03

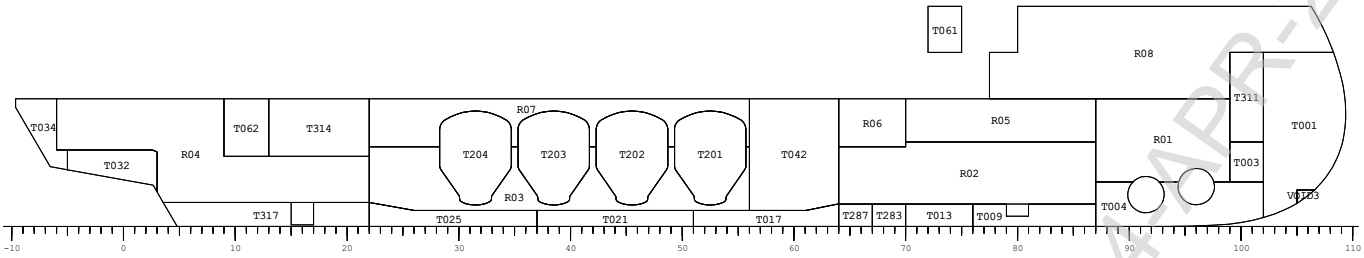


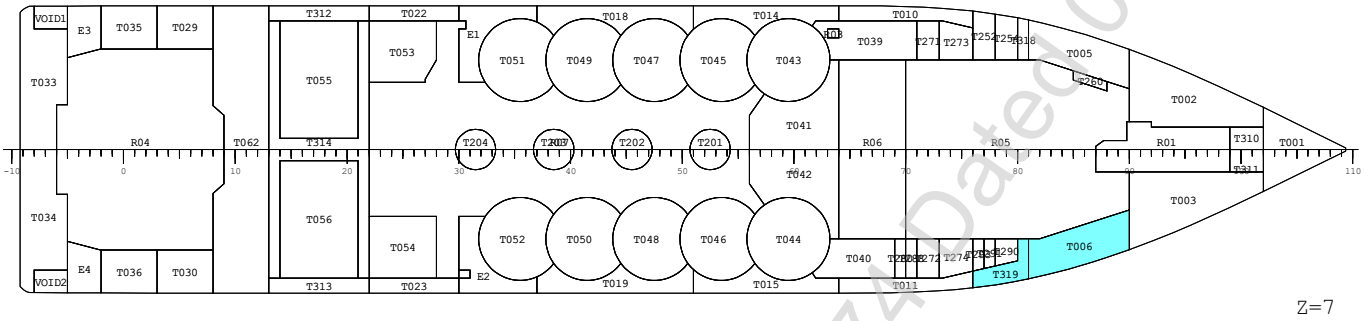
DAM	COMP	DES	PERM	VOL m3	XCG m	YCG m	ZCG m	STAGE
D03	T002	WING TANK PS	0.95	117.2	67.06	-2.88	6.32	1

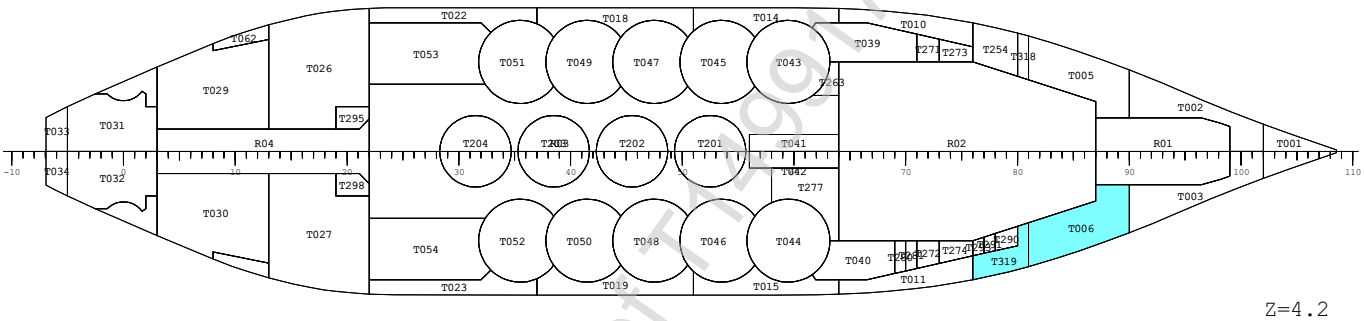
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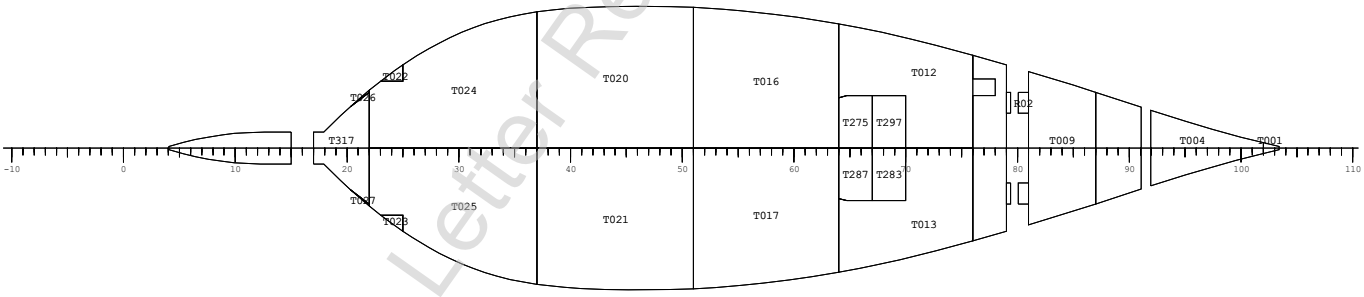
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DAMAGE CASE D04







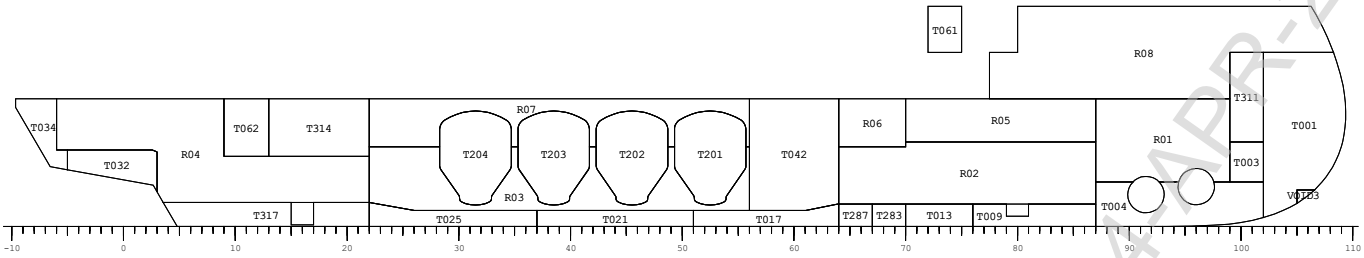


DAM	COMP	DES	PERM	VOL m3	XCG m	YCG m	ZCG m	STAGE
D04	T006	WING TANK SB	0.95	101.2	59.97	5.07	4.86	1
D04	T319	CD VOID SB	0.95	33.5	55.09	6.88	4.89	1

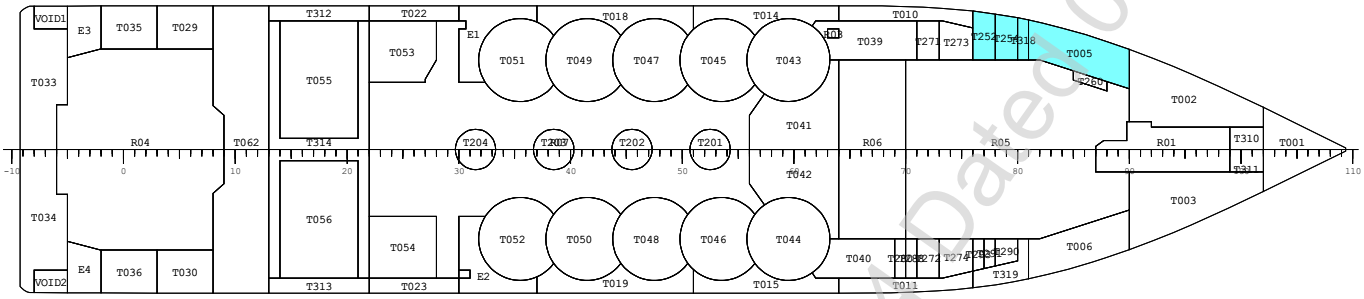
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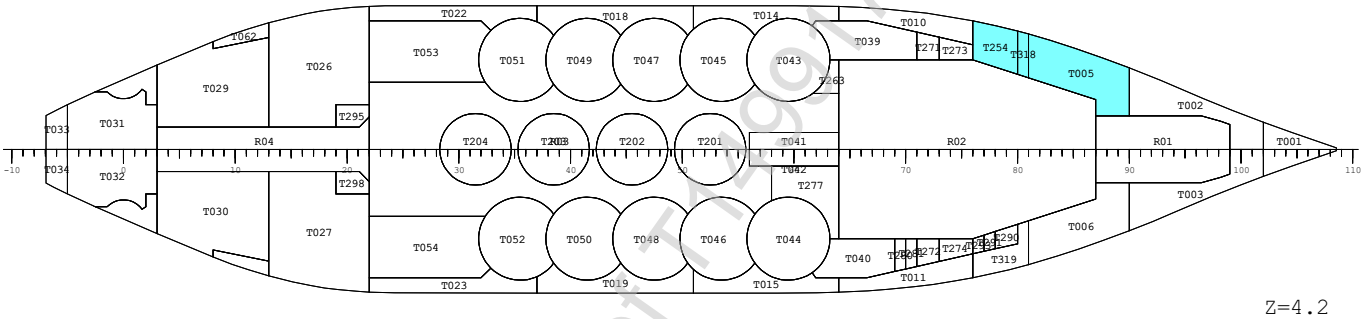
DAMAGE CASE D05



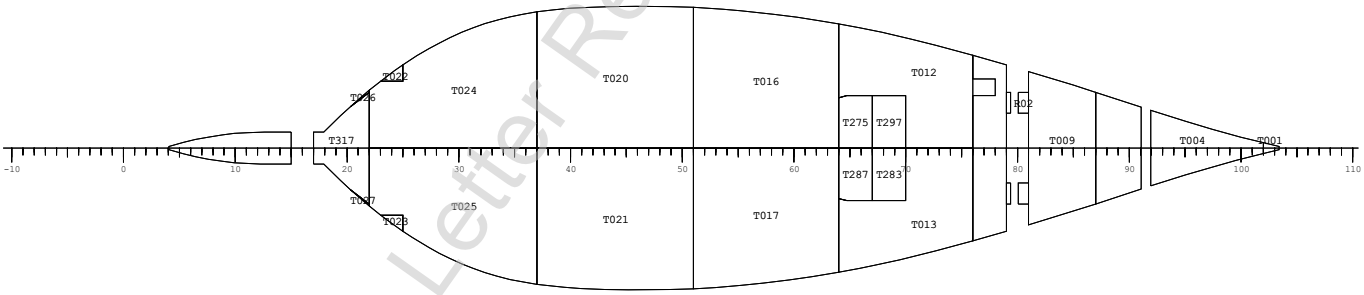
PROFILE



Z=7



Z=4.2



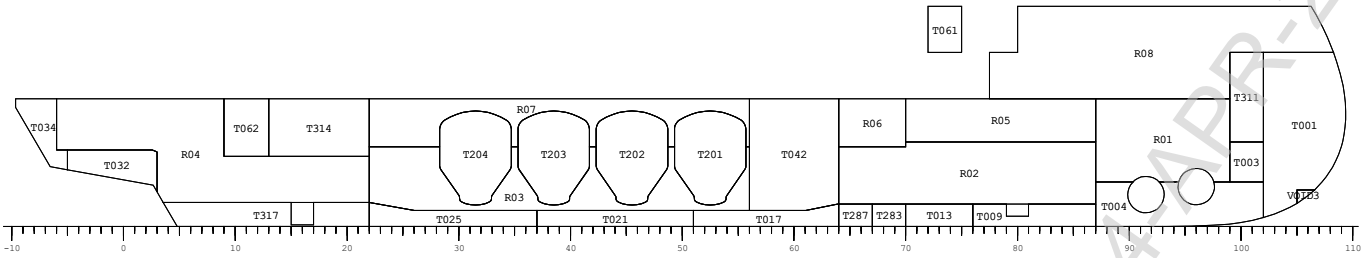
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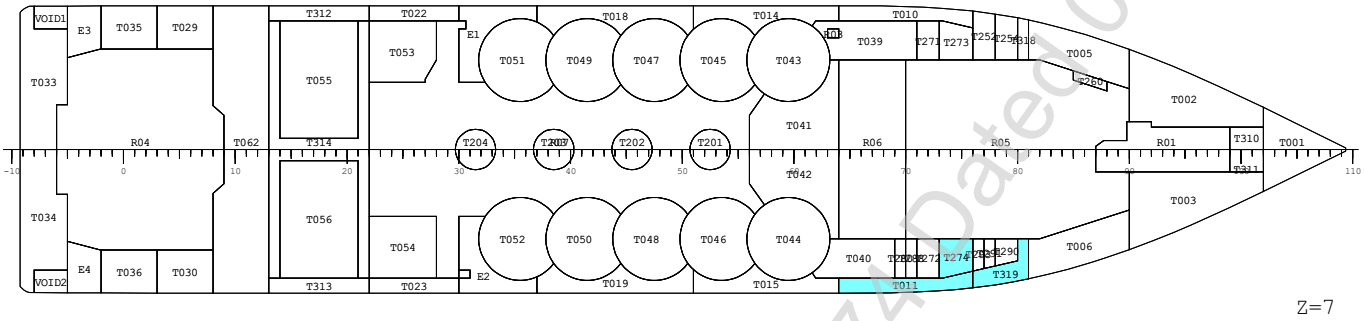
DAM	COMP	DES	PERM	VOL m3	XCG m	YCG m	ZCG m	STAGE
D05	T005	WING TANK PS	0.95	101.2	59.97	-5.07	4.86	1
D05	T318	CD VOID PS	0.95	10.8	56.35	-6.25	4.99	1
D05	T254	GREY WATER TANK	0.95	33.3	54.83	-6.49	4.59	1
D05	T252	BLACK WATER TANK	0.95	11.0	53.89	-7.06	6.69	1

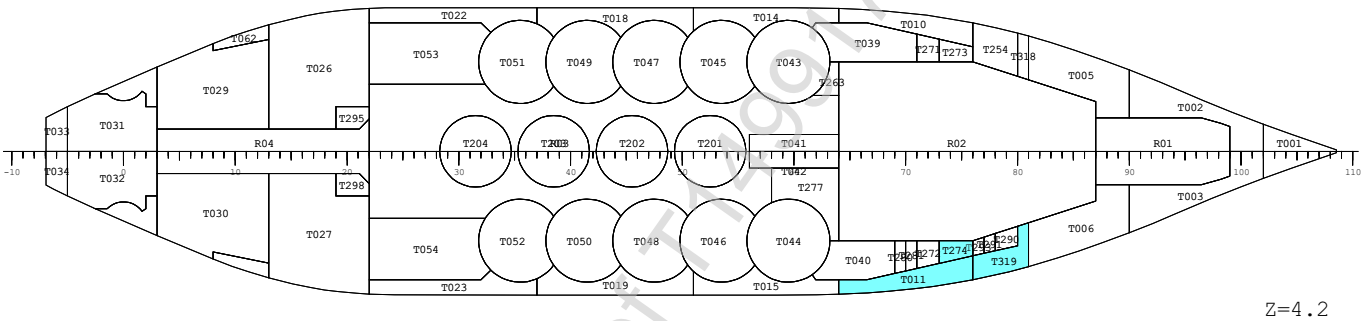
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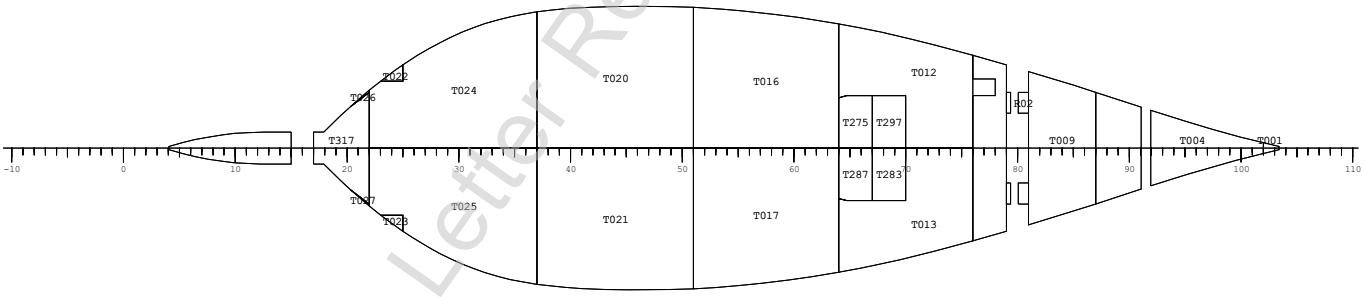
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DAMAGE CASE D06







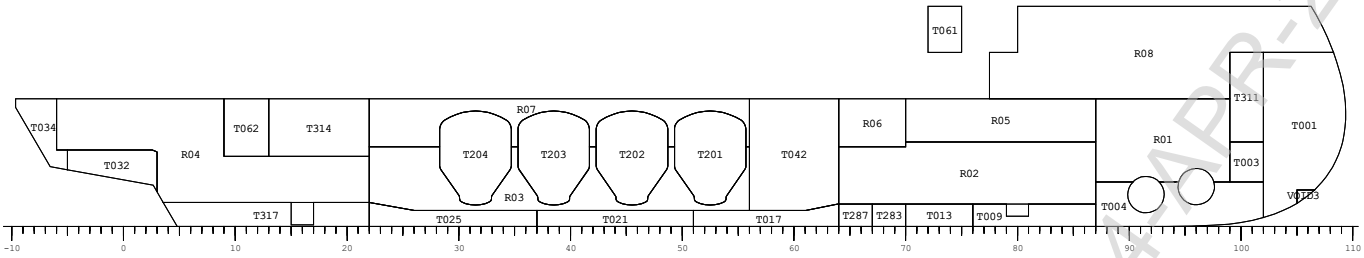


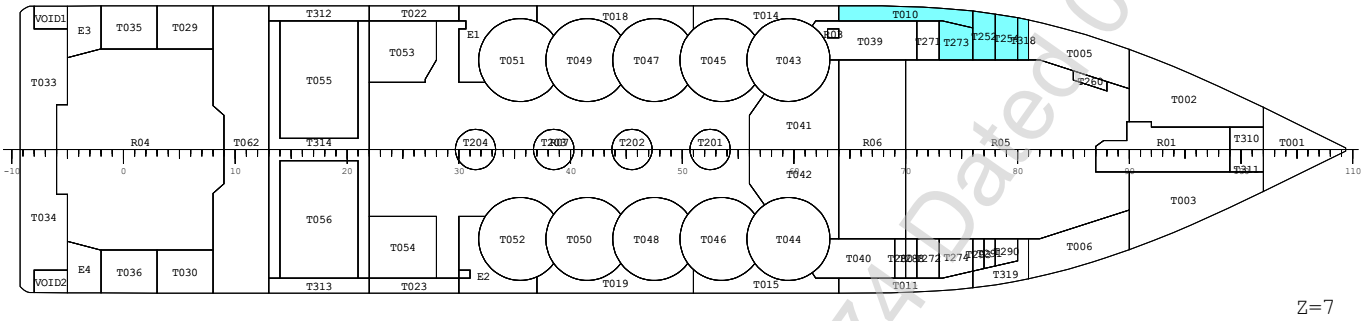
DAM	COMP	DES	PERM	VOL m3	XCG m	YCG m	ZCG m	STAGE
D06	T011	WING TANK SB	0.95	61.0	49.30	7.86	4.45	1
D06	T274	FO SERVICE TANK SB	0.95	18.2	52.10	6.50	5.84	1
D06	T319	CD VOID SB	0.95	33.5	55.09	6.88	4.89	1

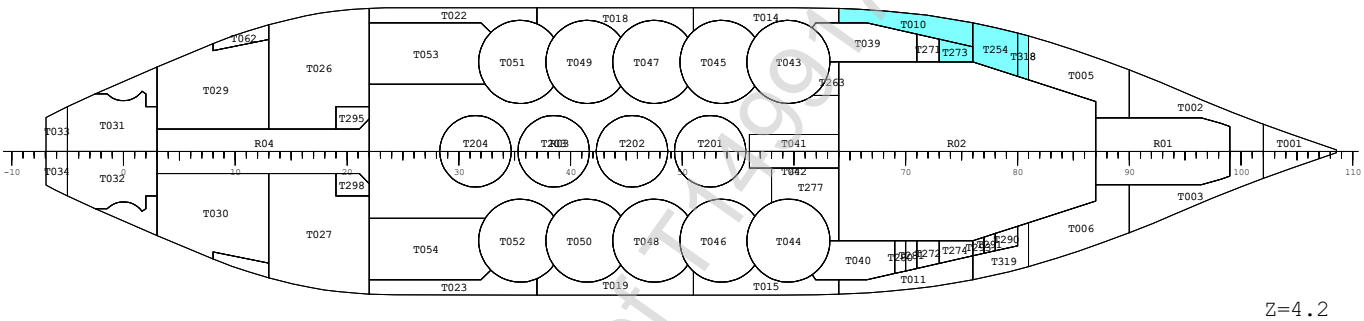
See ABS Singapore Letter Ref: 174Dated 04-APR-2016

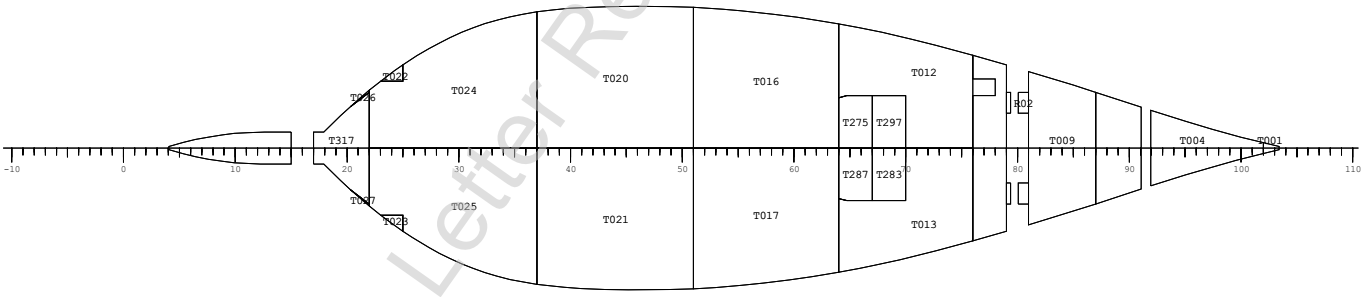
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DAMAGE CASE D07







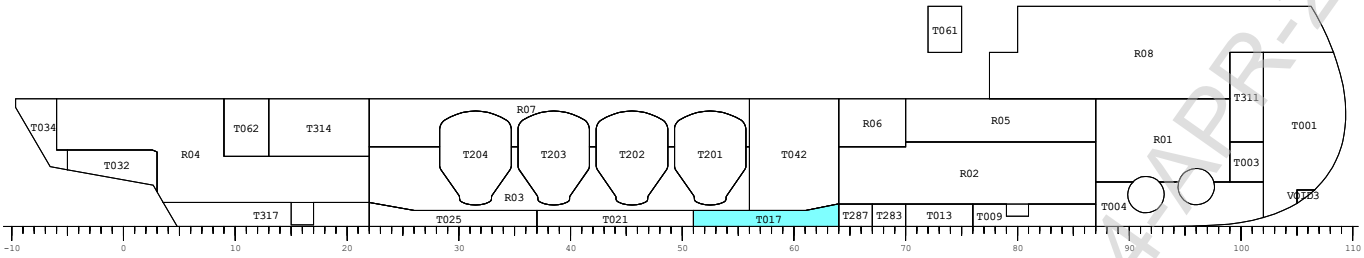


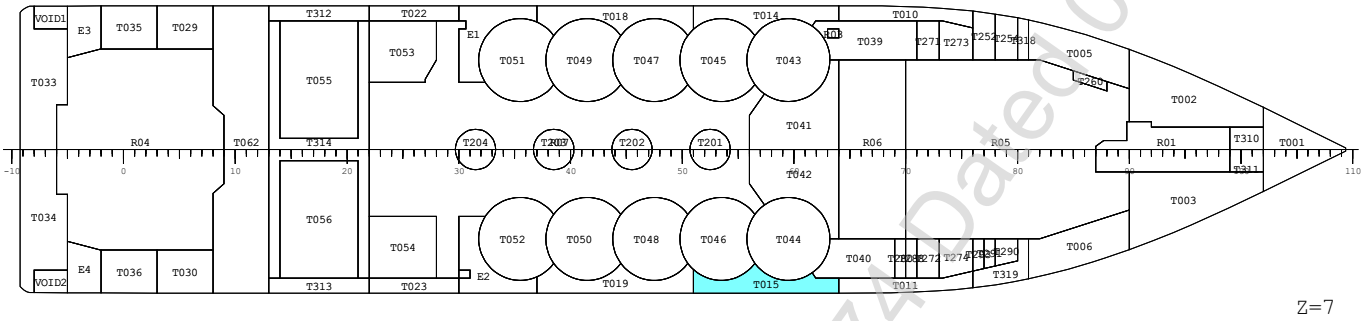
DAM	COMP	DES	PERM	VOL m3	XCG m	YCG m	ZCG m	STAGE
D07	T010	WING TANK PS	0.95	61.0	49.30	-7.86	4.45	1
D07	T273	FO SERVICE TANK PS	0.95	18.2	52.10	-6.50	5.84	1
D07	T252	BLACK WATER TANK	0.95	11.0	53.89	-7.06	6.69	1
D07	T254	GREY WATER TANK	0.95	33.3	54.83	-6.49	4.59	1
D07	T318	CD VOID PS	0.95	10.8	56.35	-6.25	4.99	1

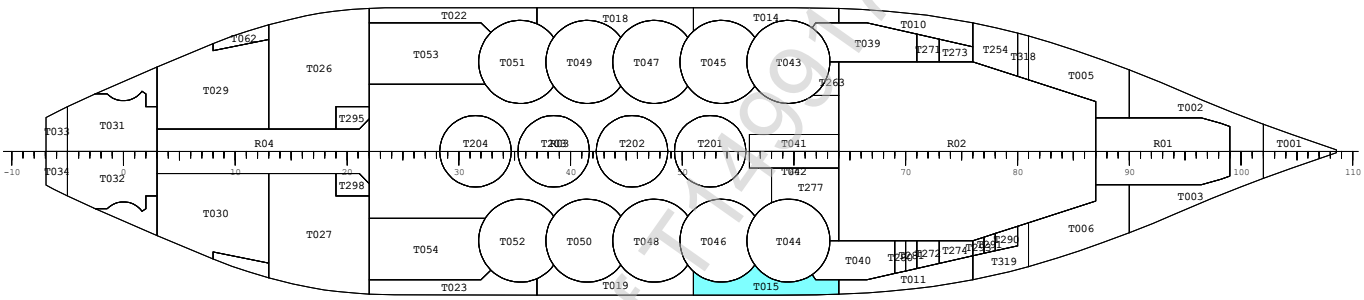
See ABS Singapore Letter Ref: EXP-174 Dated 04-APR-2016

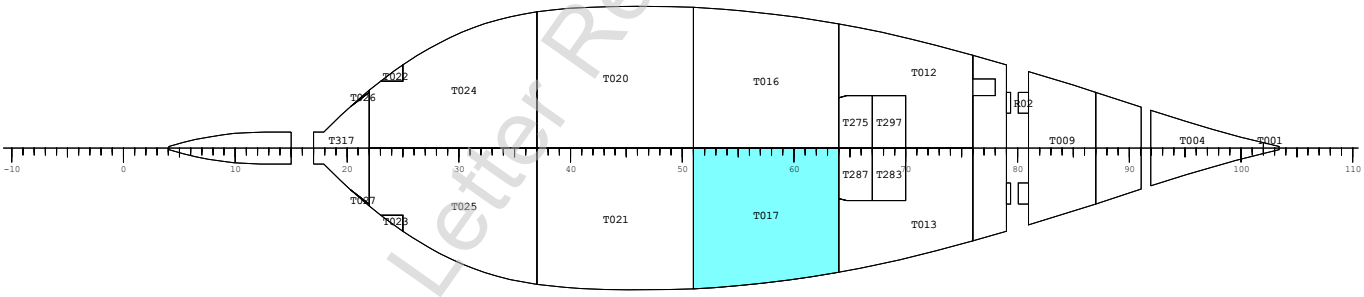
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DAMAGE CASE D08







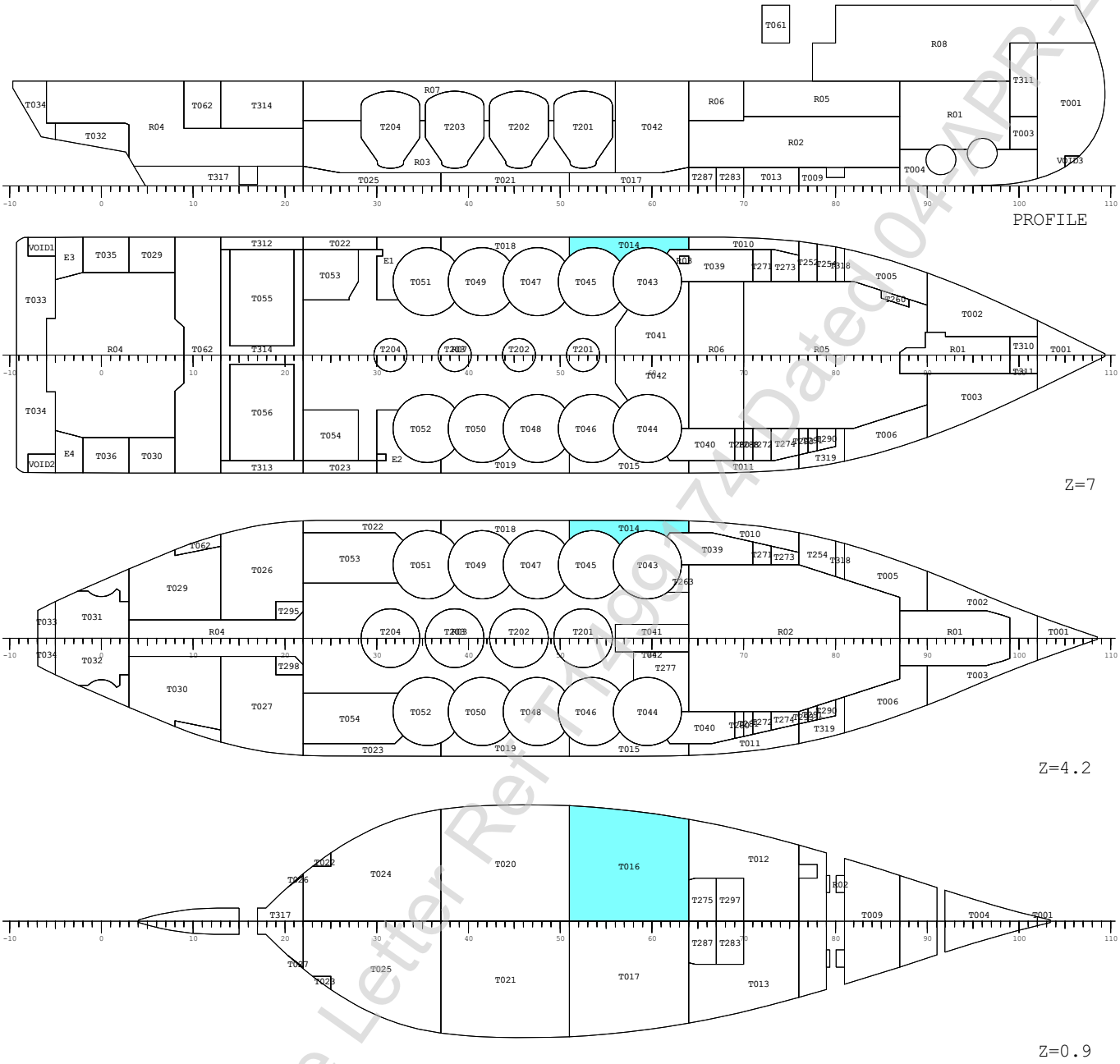


DAM	COMP	DES	PERM	VOL m3	XCG m	YCG m	ZCG m	STAGE
D08	T017	DB TANK SB	0.95	86.1	40.20	4.33	0.62	1
D08	T015	WING TANK SB	0.95	61.8	40.15	8.42	4.73	1

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DAMAGE CASE D09

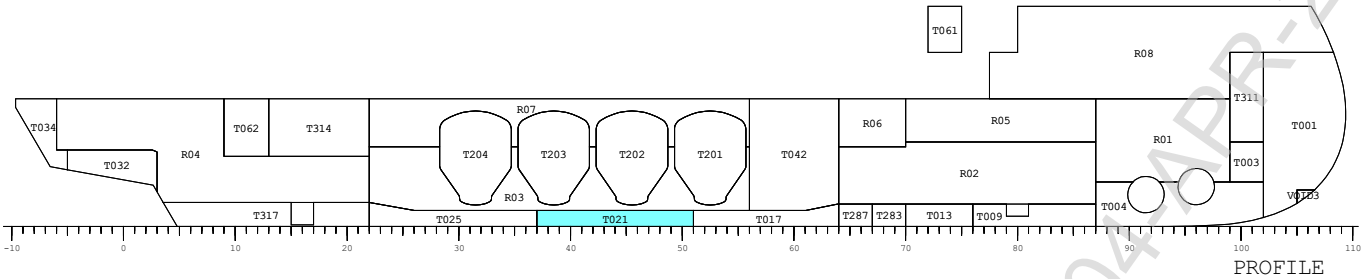


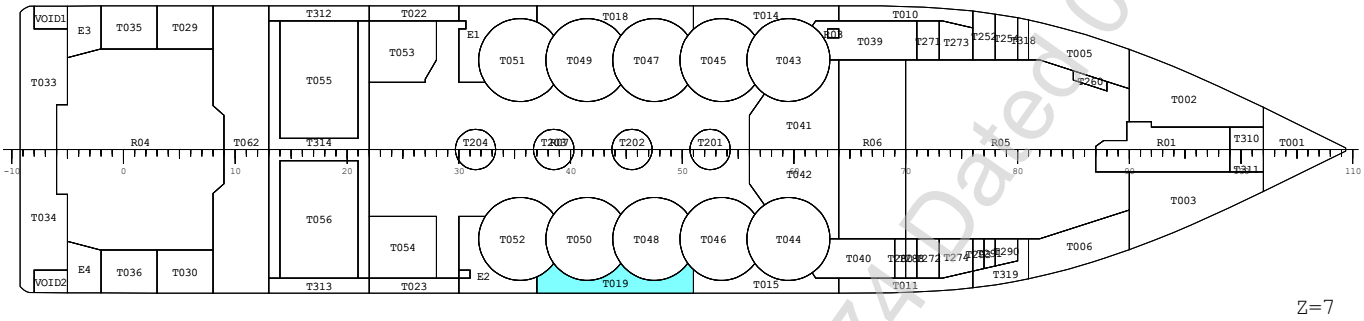
DAM	COMP	DES	PERM	VOL m3	XCG m	YCG m	ZCG m	STAGE
D09	T016	DB TANK PS	0.95	86.1	40.20	-4.33	0.62	1
D09	T014	WING TANK PS	0.95	61.9	40.15	-8.42	4.73	1

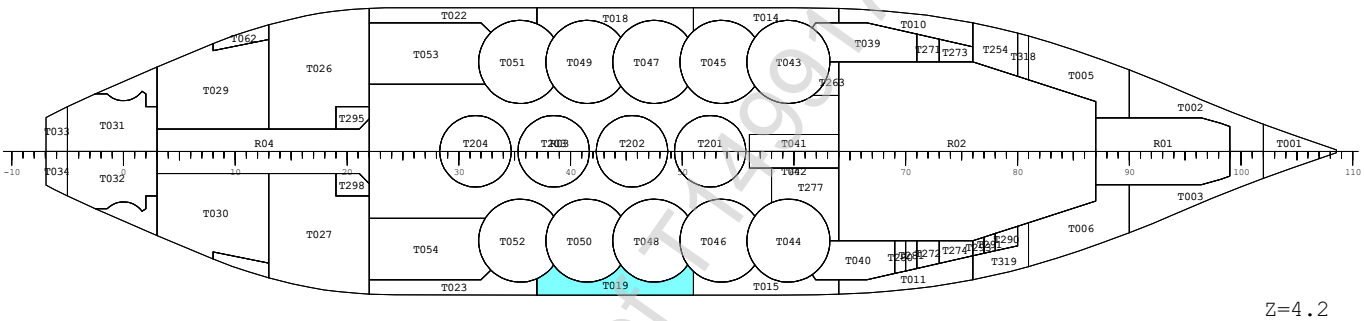
See ABS Singapore Letter Ref: Engg/74 Dated 04-APR-2016

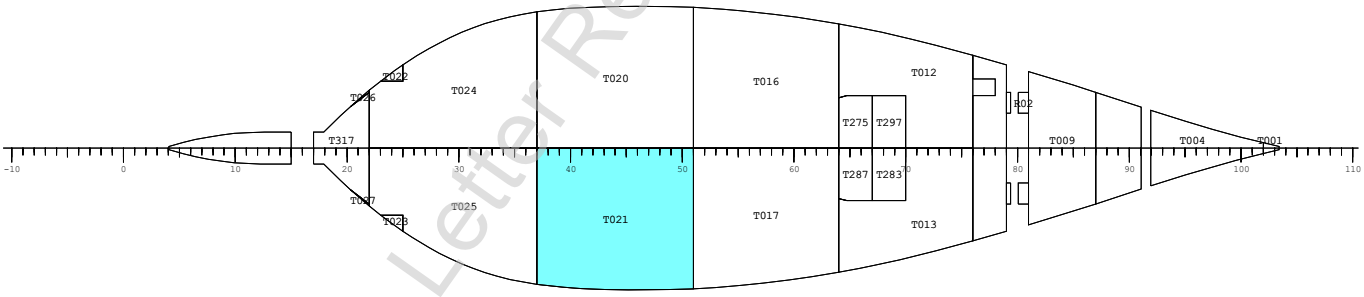
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DAMAGE CASE D10







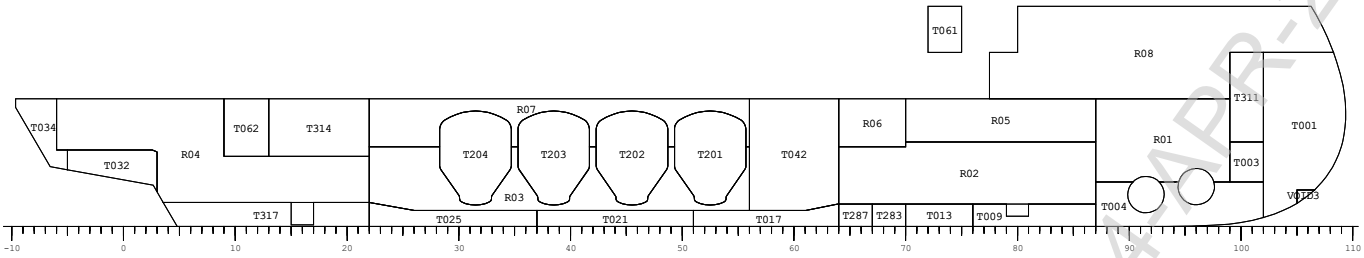


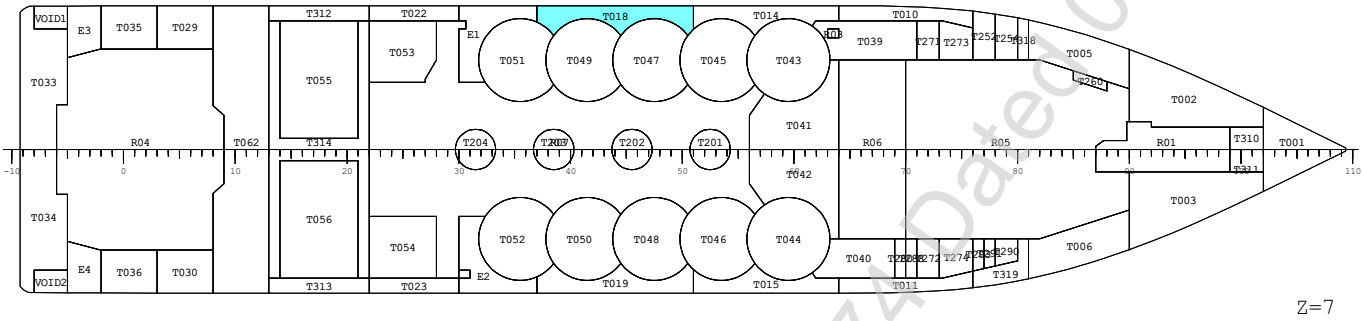
DAM	COMP	DES	PERM	VOL m3	XCG m	YCG m	ZCG m	STAGE
D10	T019	WING TANK SB	0.95	74.7	30.79	8.37	4.73	1
D10	T021	DB TANK SB	0.95	97.3	30.84	4.63	0.61	1

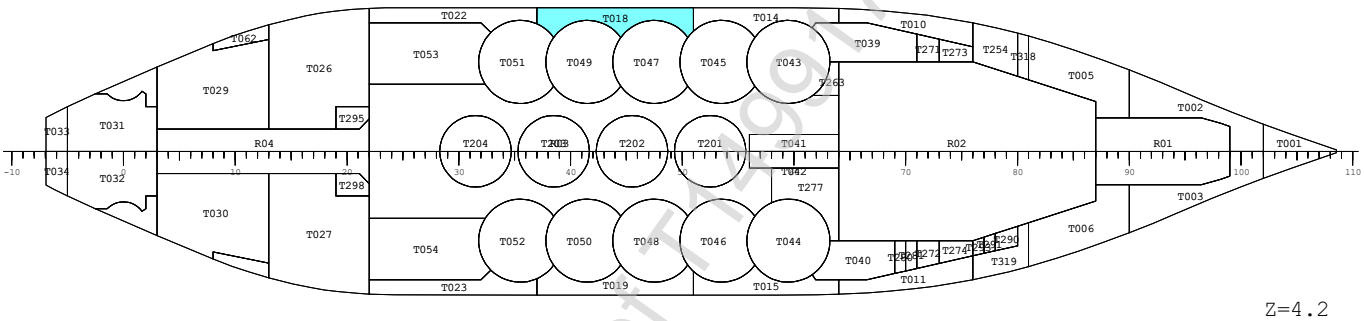
See ABS Singapore Letter Ref: KL99174 Dated 04-APR-2016

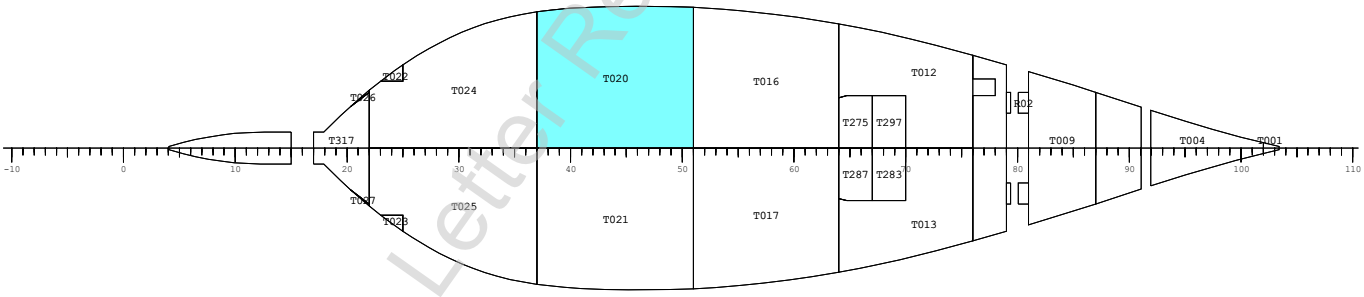
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DAMAGE CASE D11







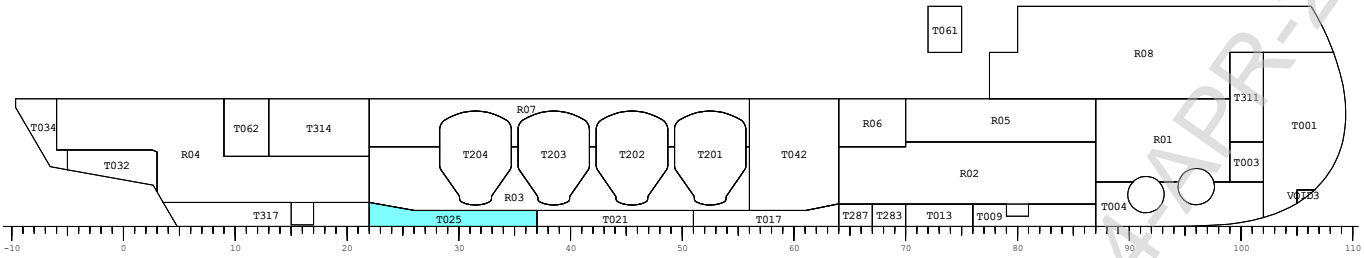


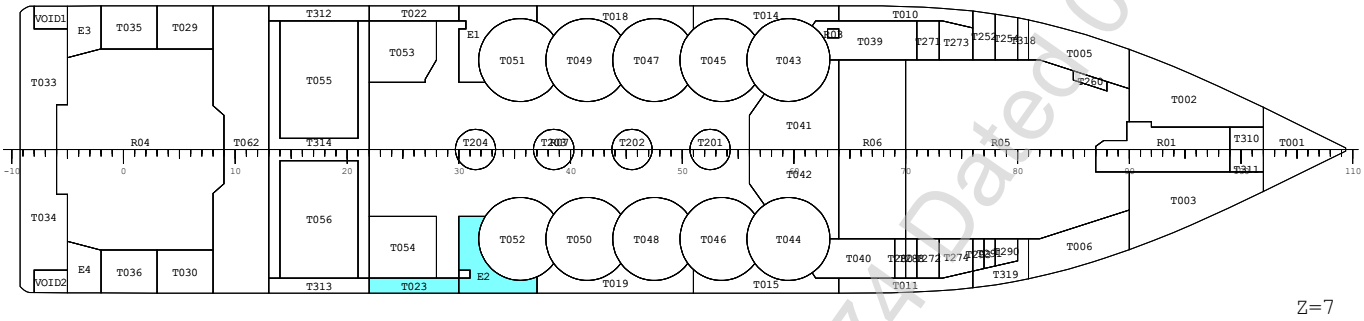
DAM	COMP	DES	PERM	VOL m3	XCG m	YCG m	ZCG m	STAGE
D11	T018	WING TANK PS	0.95	74.7	30.79	-8.37	4.73	1
D11	T020	DB TANK PS	0.95	97.3	30.84	-4.63	0.61	1

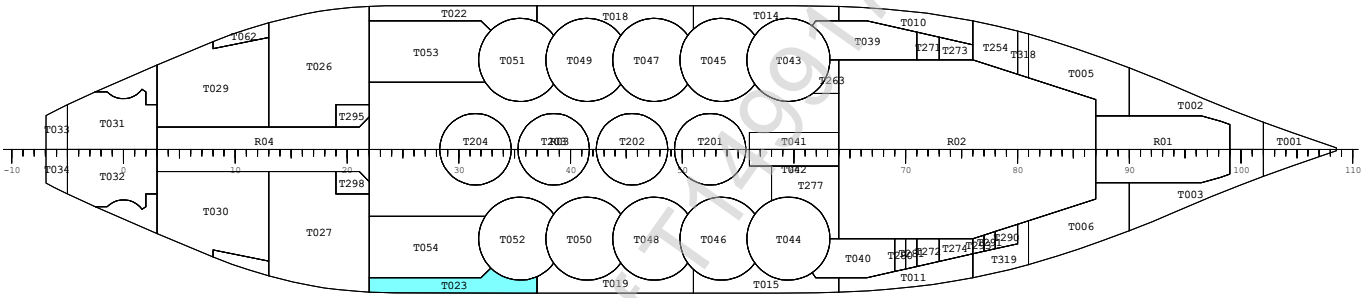
See ABS Singapore Letter Ref: 174Dated 04-APR-2016

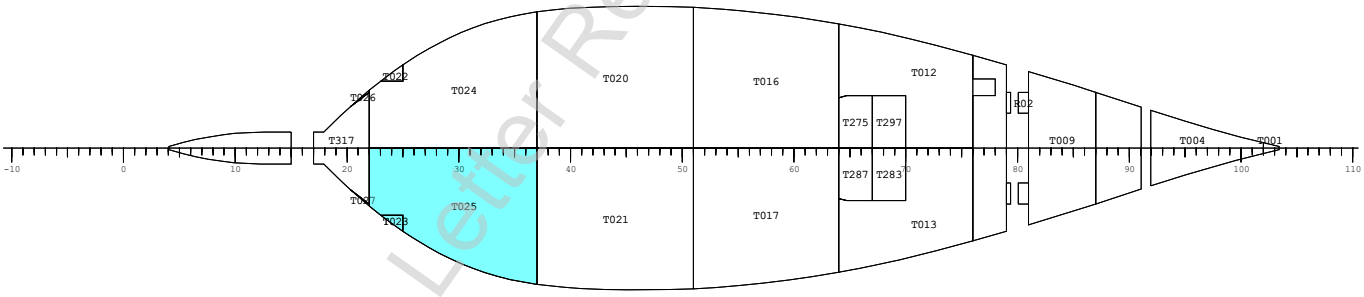
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DAMAGE CASE D12







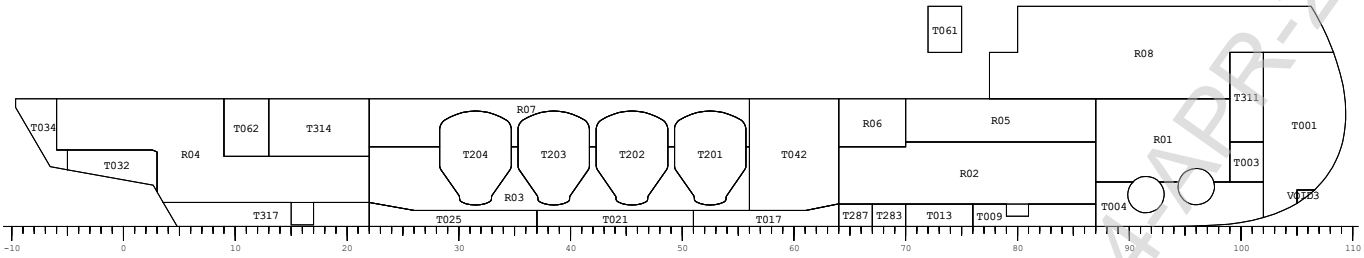


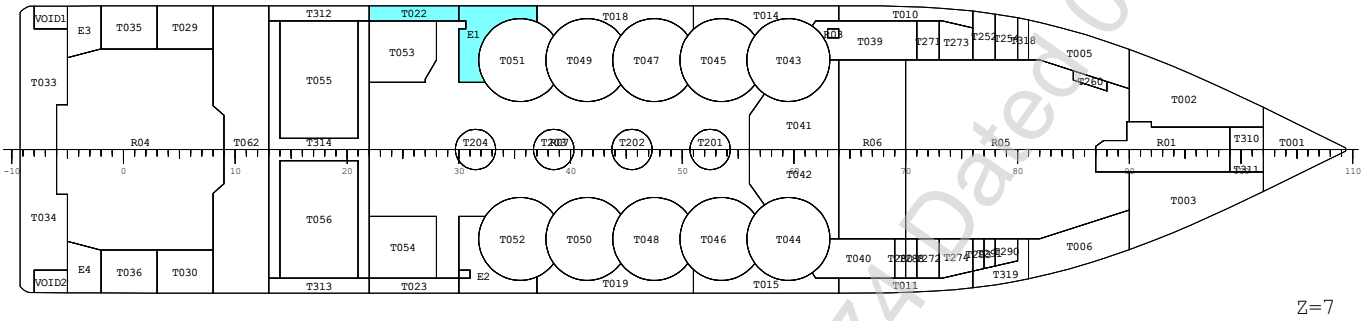
DAM	COMP	DES	PERM	VOL m3	XCG m	YCG m	ZCG m	STAGE
D12	T023	WING TANK SB	0.95	61.8	19.27	7.89	3.91	1
D12	T025	DB TANK SB	0.95	76.3	21.46	3.85	0.81	1
D12	E2	Emrg. exit 1 SB	0.95	30.8	22.54	7.25	6.50	1

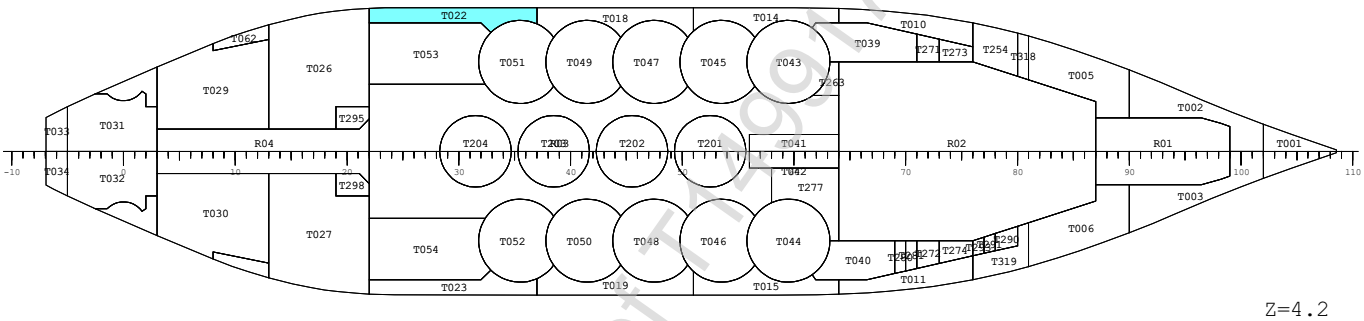
See ABS Singapore Letter Ref: 174Dated 04-APR-2016

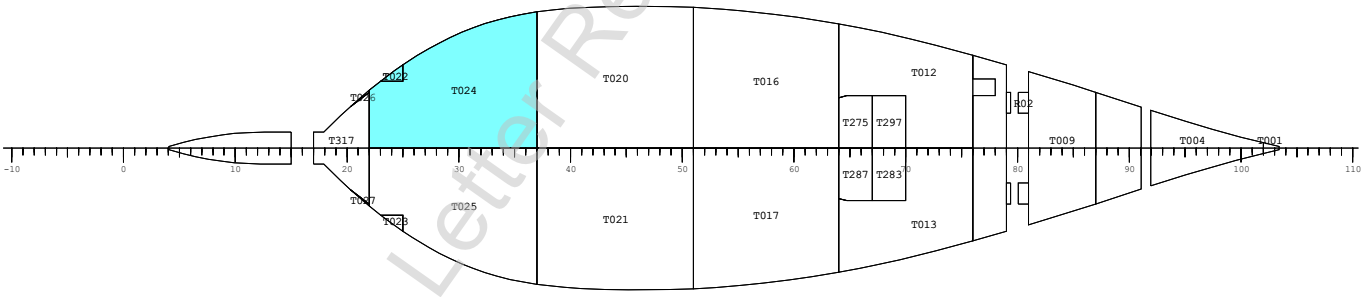
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DAMAGE CASE D13







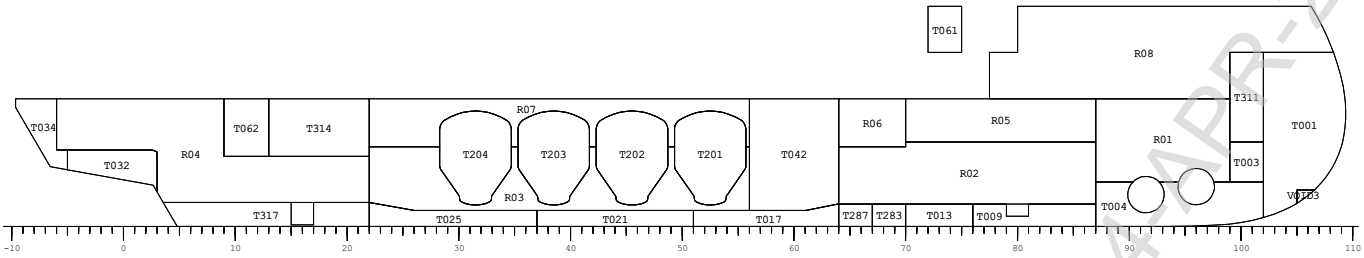


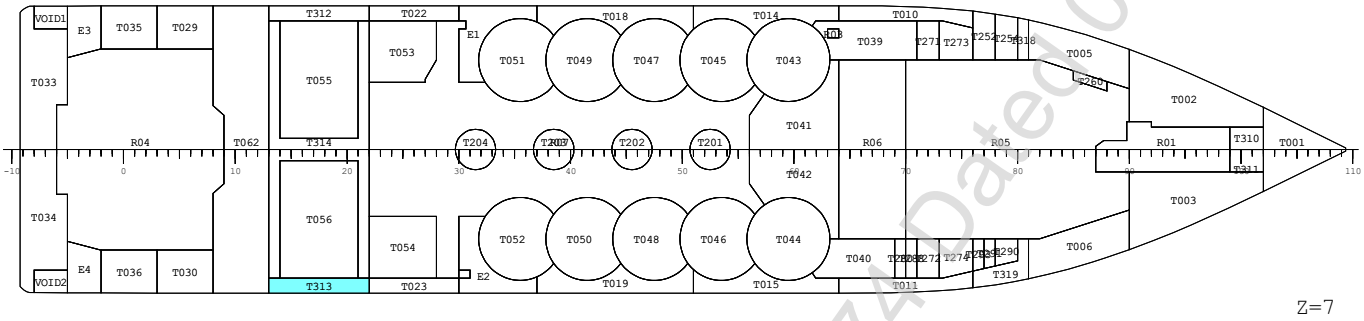
DAM	COMP	DES	PERM	VOL m3	XCG m	YCG m	ZCG m	STAGE
D13	T022	WING TANK PS	0.95	61.8	19.27	-7.89	3.91	1
D13	T024	DB TANK PS	0.95	76.3	21.46	-3.85	0.81	1
D13	E1	Emrg. exit 1 PS	0.95	31.5	22.52	-7.26	6.49	1

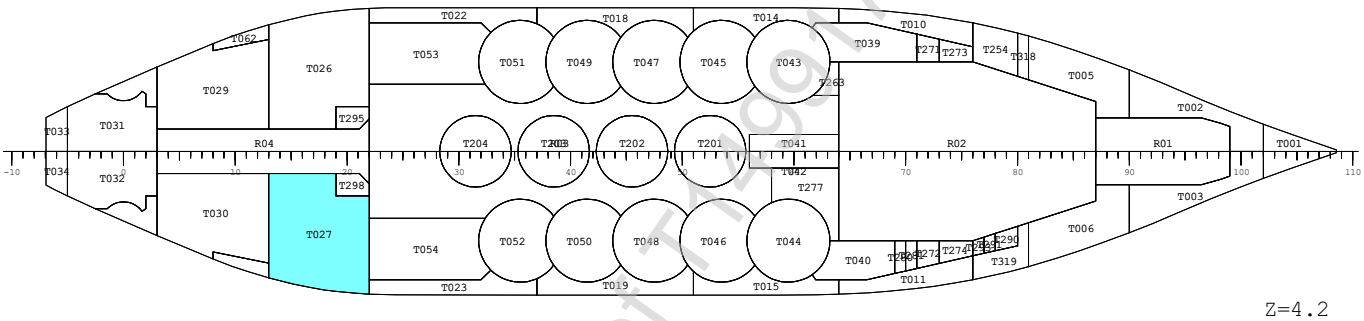
See ABS Singapore Letter Ref: Engg/174 Dated 04-APR-2016

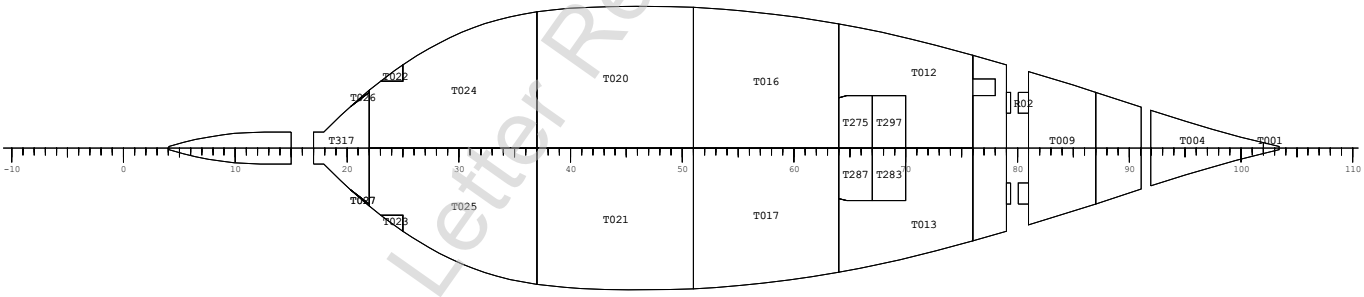
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DAMAGE CASE D14







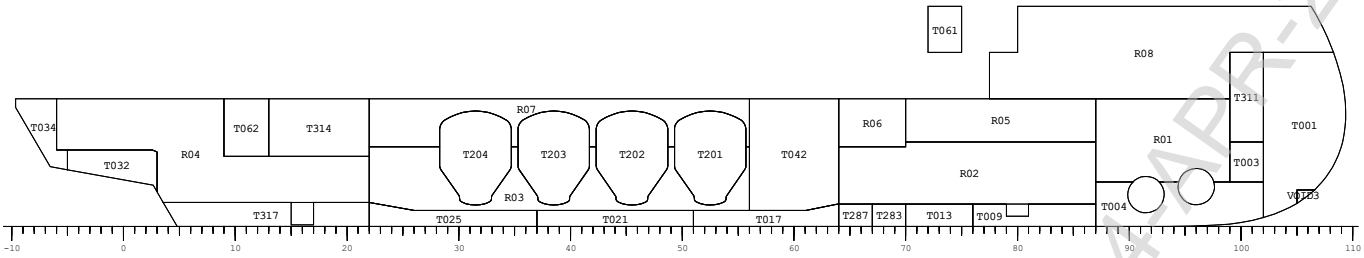


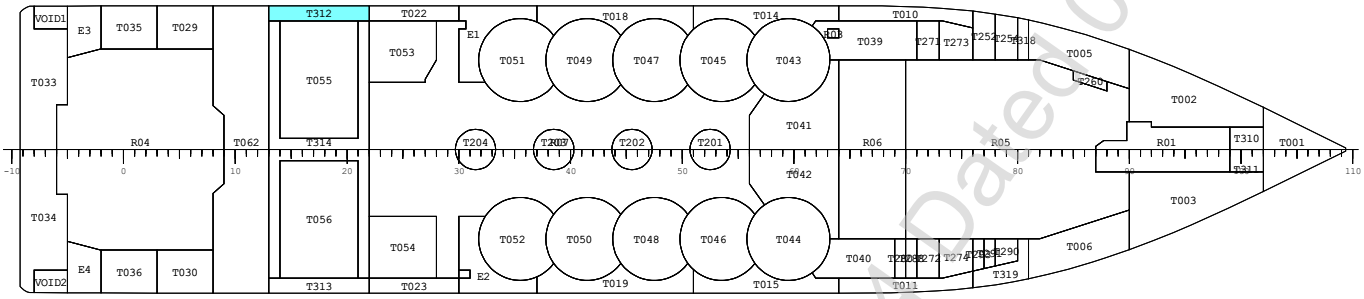
DAM	COMP	DES	PERM	VOL m3	XCG m	YCG m	ZCG m	STAGE
D14	T027	WING TANK SB	0.95	92.9	12.42	4.45	3.13	1
D14	T313	CD VOID OUTSIDE LFL TANK.	0.95	20.9	12.30	8.51	6.25	1

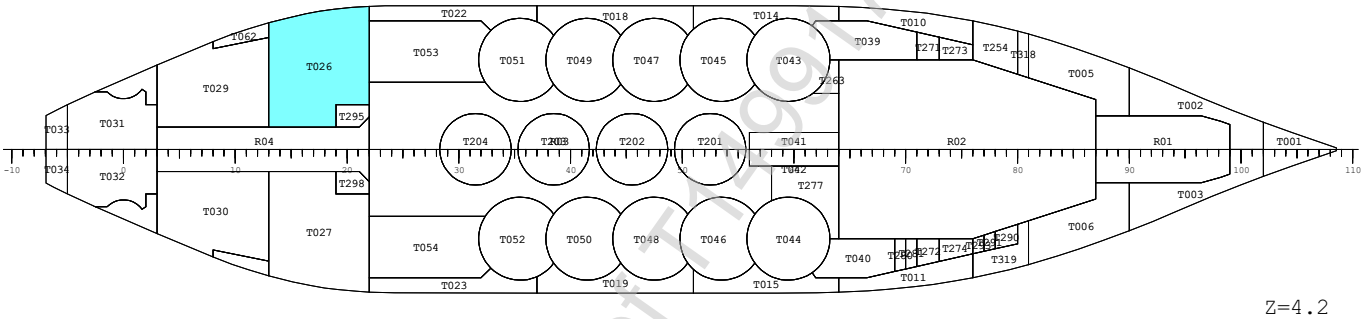
See ABS Singapore Letter Ref: KNgg174Dated 04-APR-2016

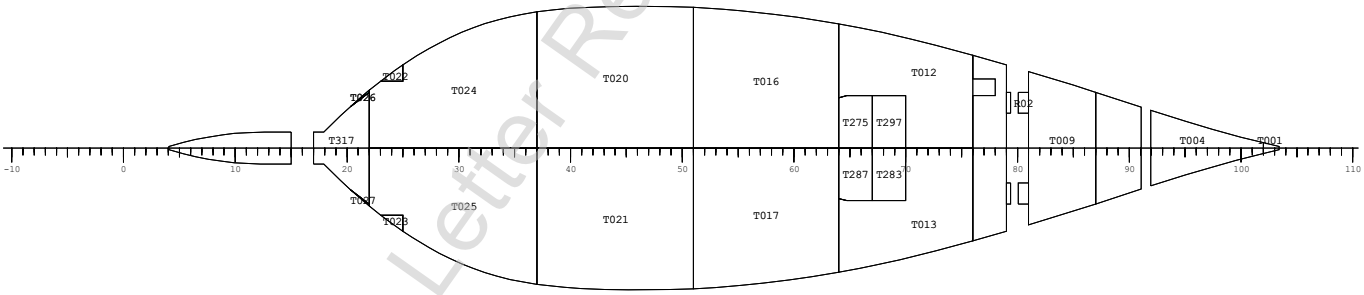
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DAMAGE CASE D15







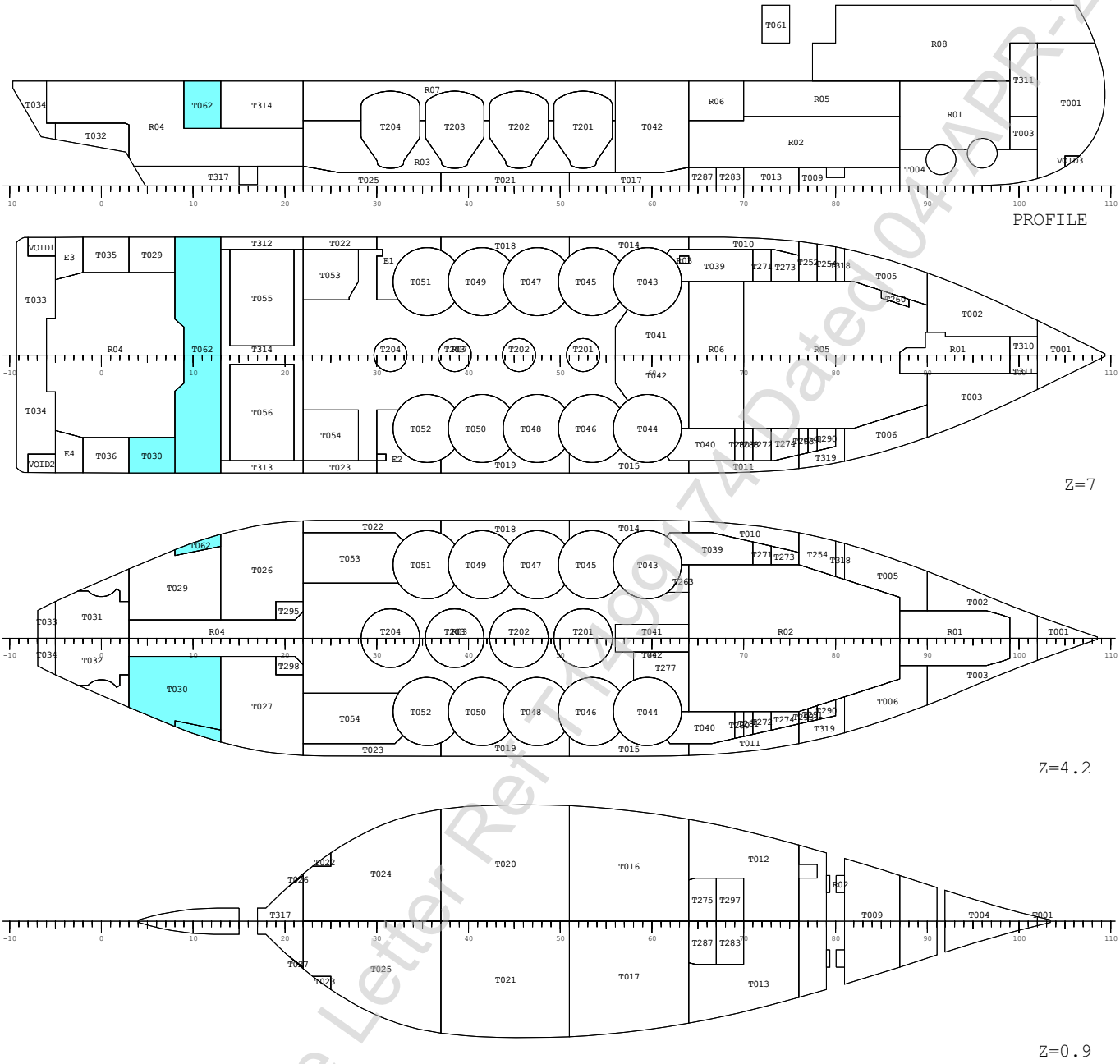


DAM	COMP	DES	PERM	VOL m3	XCG m	YCG m	ZCG m	STAGE
D15	T026	WING TANK PS	0.95	92.9	12.42	-4.45	3.13	1
D15	T312	CD VOID OUTSIDE LFL TANK.	0.95	20.9	12.30	-8.51	6.25	1

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DAMAGE CASE D16

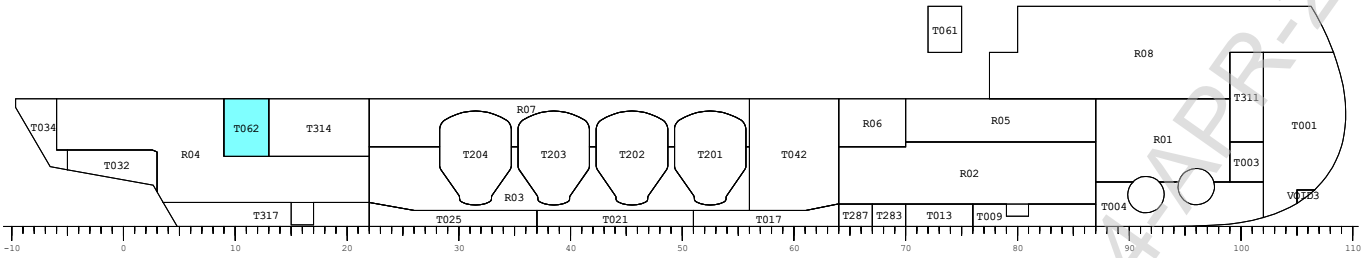


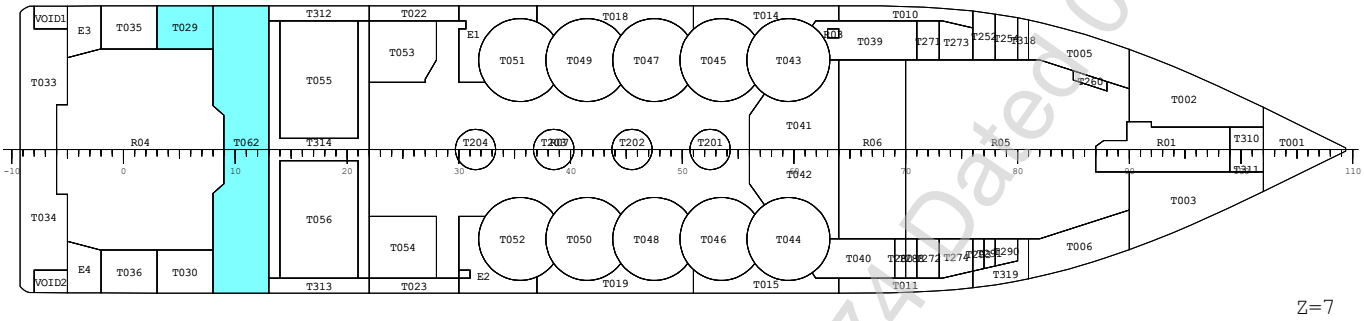
DAM	COMP	DES	PERM	VOL m3	XCG m	YCG m	ZCG m	STAGE
D16	T062	ROLL RED. TANK	0.95	211.8	7.44	0.00	6.21	1
D16	T030	WING TANK SB	0.95	78.5	5.26	4.93	4.67	1

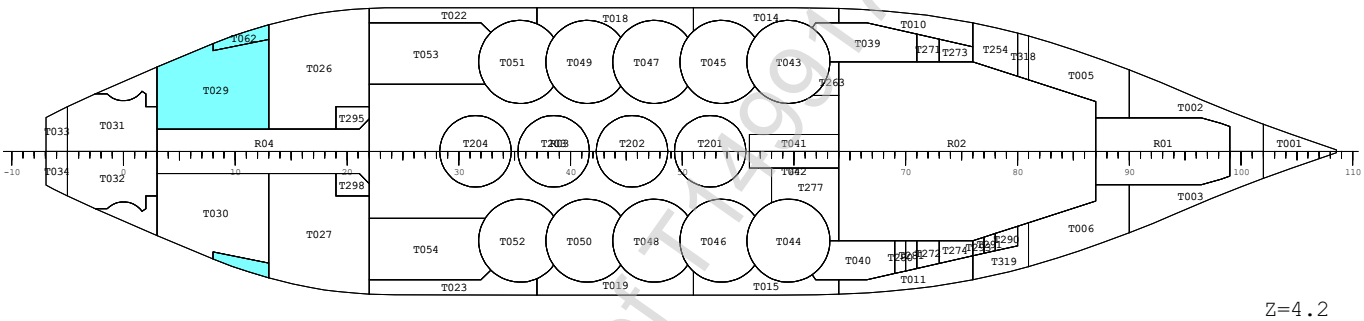
See ABS Singapore Letter Ref: Page 174 Dated 04-APR-2016

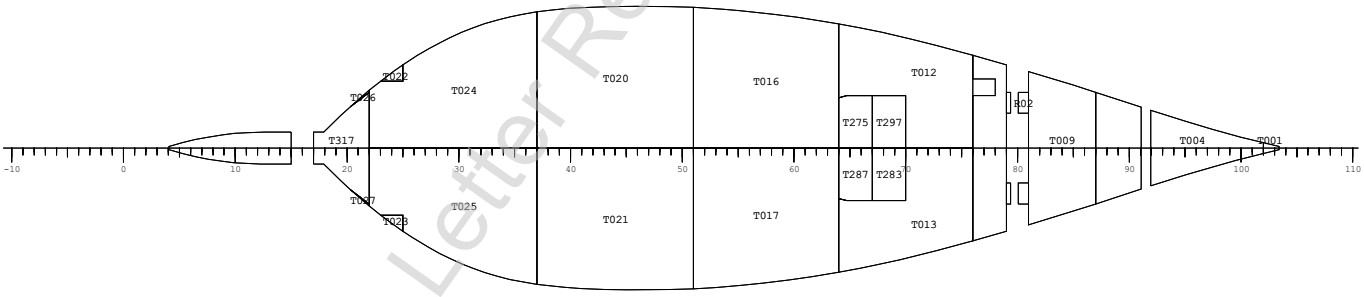
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DAMAGE CASE D17







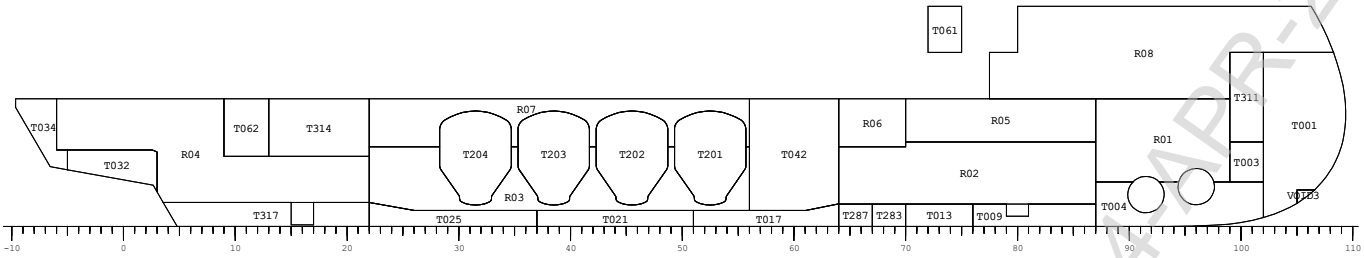


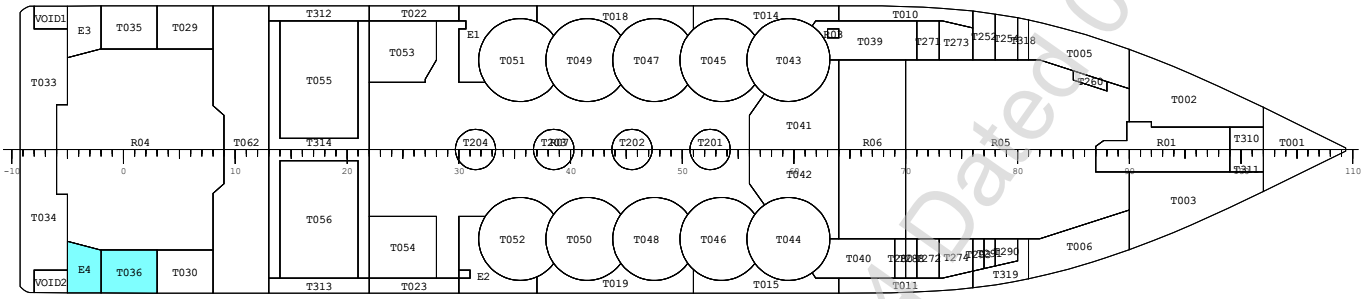
DAM	COMP	DES	PERM	VOL m3	XCG m	YCG m	ZCG m	STAGE
D17	T062	ROLL RED. TANK	0.95	211.8	7.44	0.00	6.21	1
D17	T029	WING TANK PS	0.95	78.5	5.26	-4.93	4.67	1

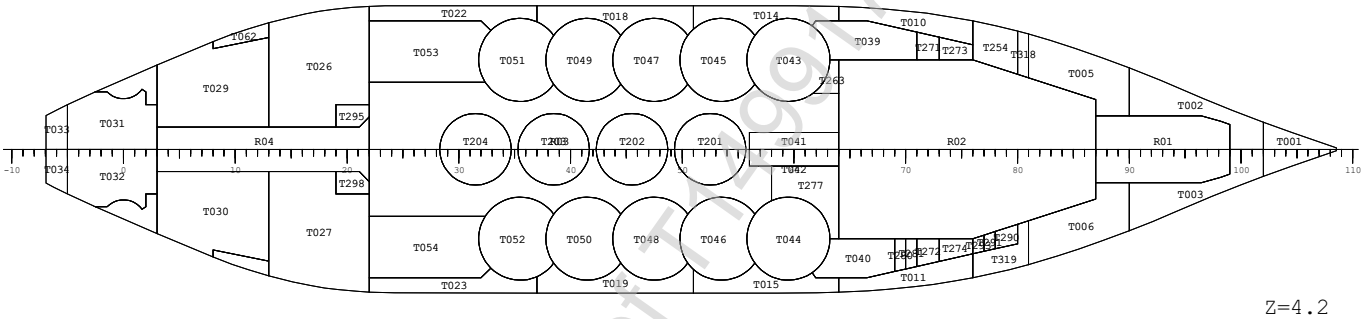
See ABS Singapore Letter Ref: Page 174 Dated 04-APR-2016

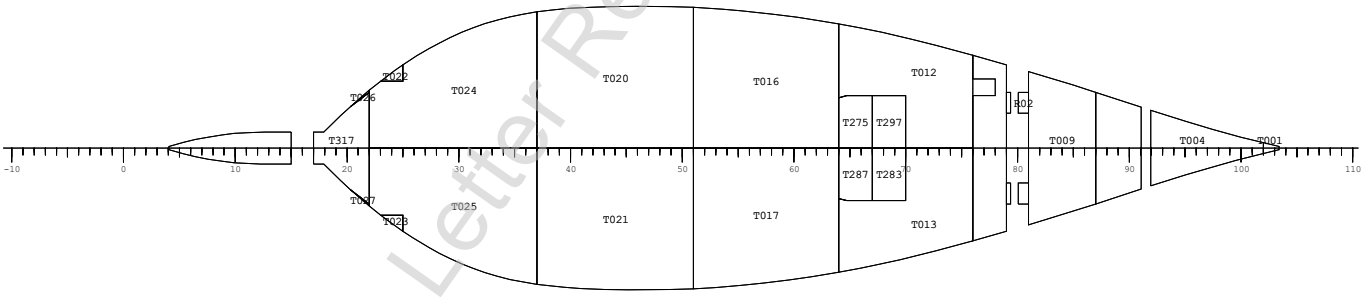
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DAMAGE CASE D18







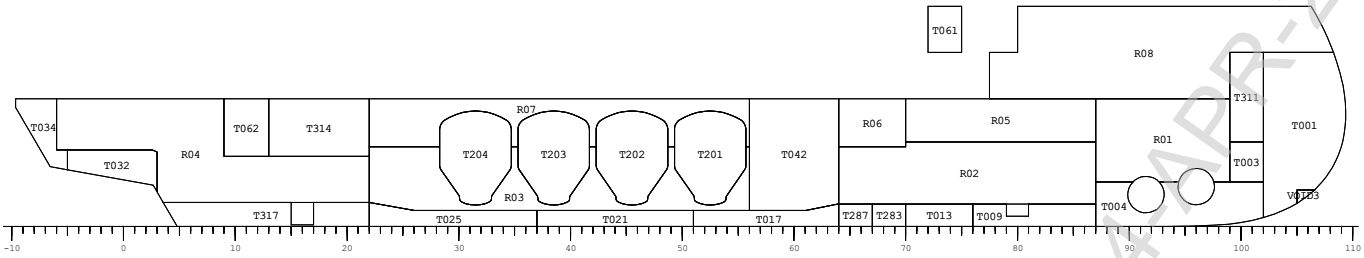


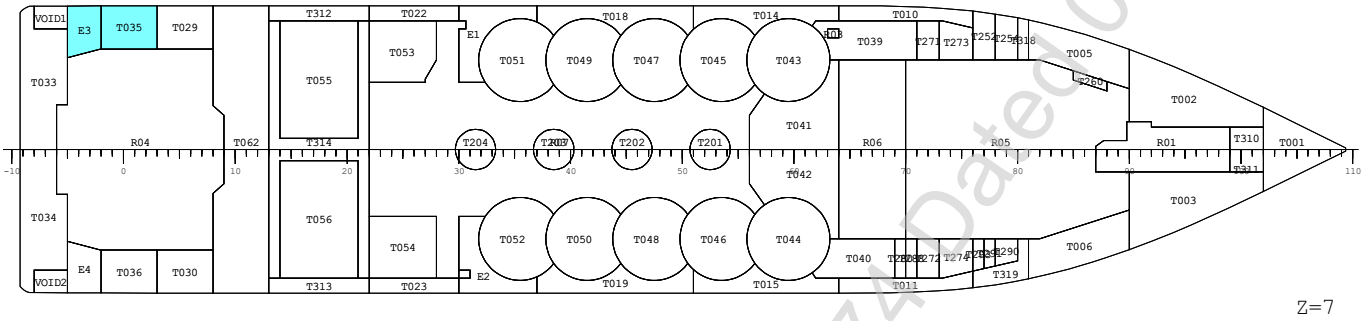
DAM	COMP	DES	PERM	VOL m3	XCG m	YCG m	ZCG m	STAGE
D18	T036	WING TANK SB	0.95	23.8	0.39	7.51	6.69	1
D18	E4	Emrg. exit 2 SB	0.95	14.6	-2.48	7.34	6.78	1

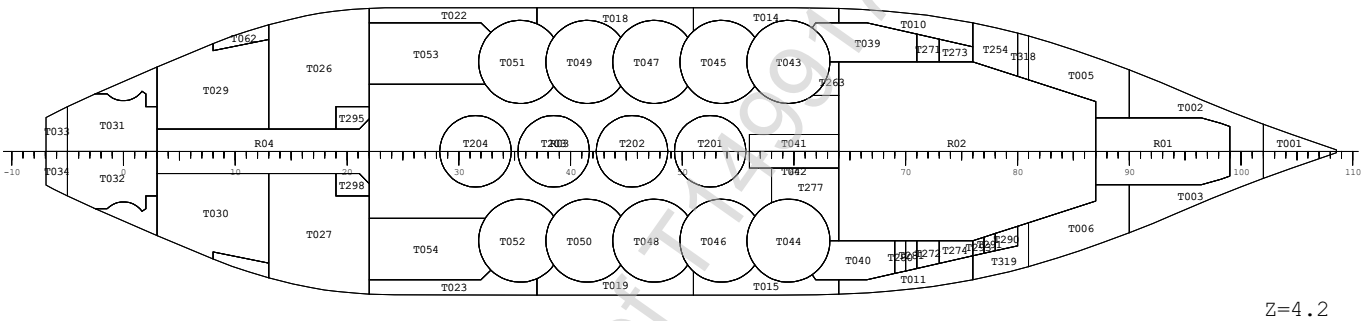
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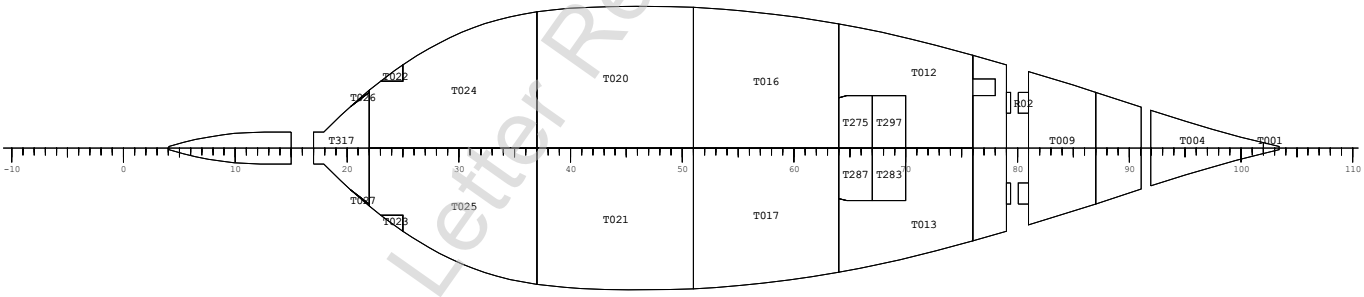
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DAMAGE CASE D19







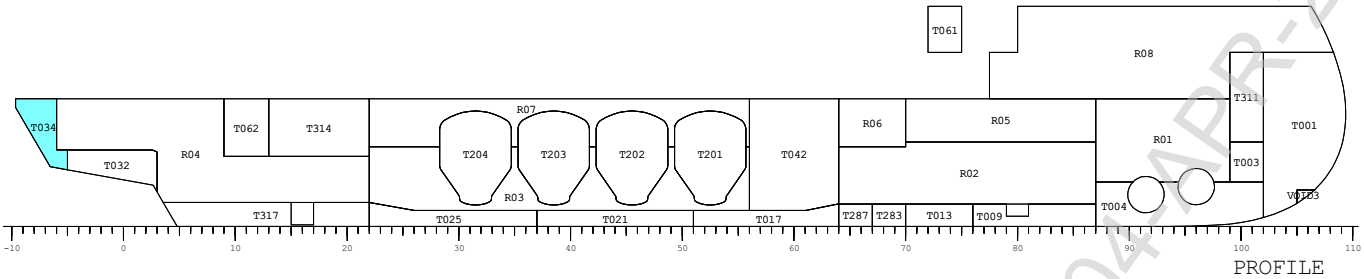


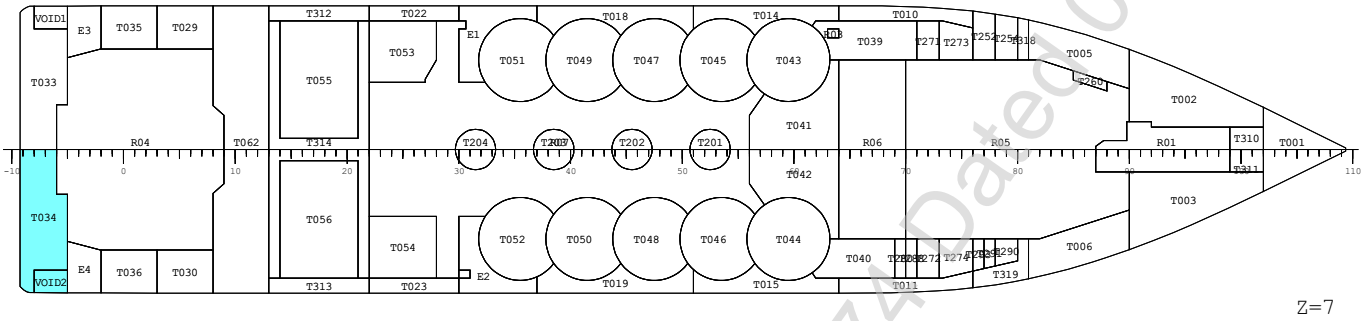
DAM	COMP	DES	PERM	VOL m3	XCG m	YCG m	ZCG m	STAGE
D19	T035	WING TANK PS	0.95	23.8	0.39	-7.51	6.69	1
D19	E3	Emrg. exit 2 PS	0.95	14.6	-2.48	-7.34	6.78	1

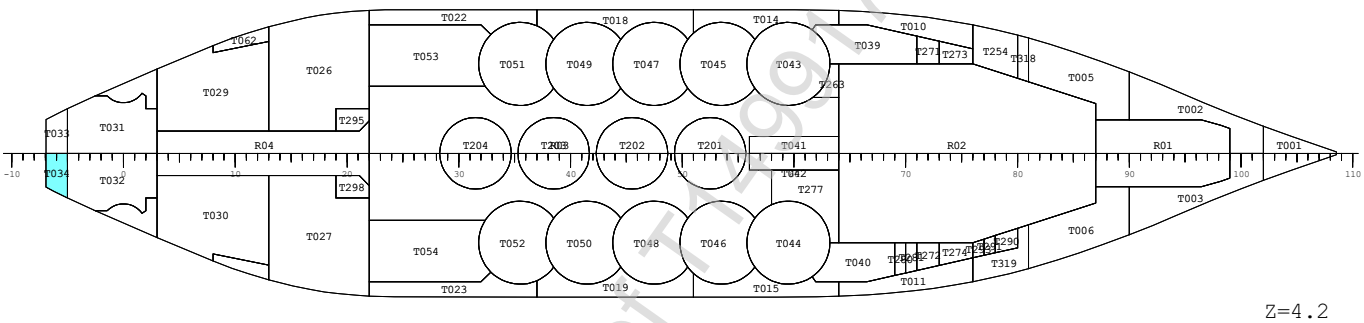
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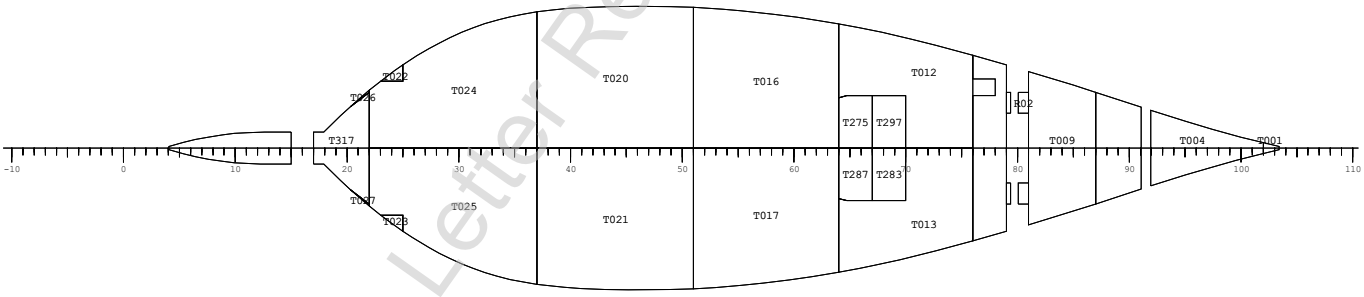
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DAMAGE CASE D20







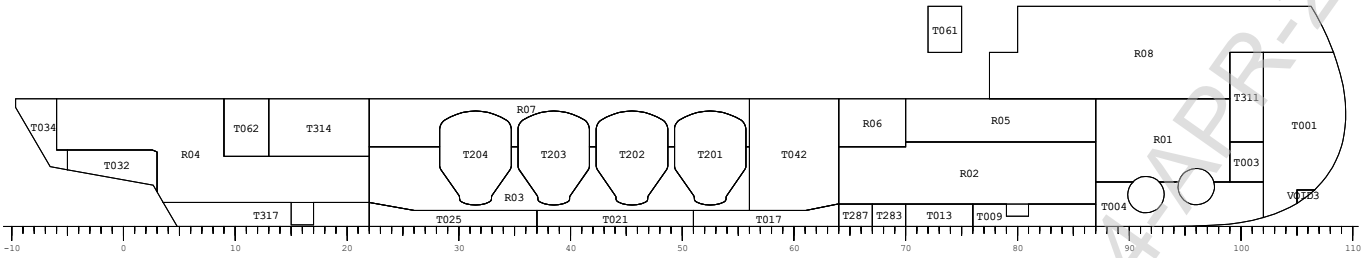


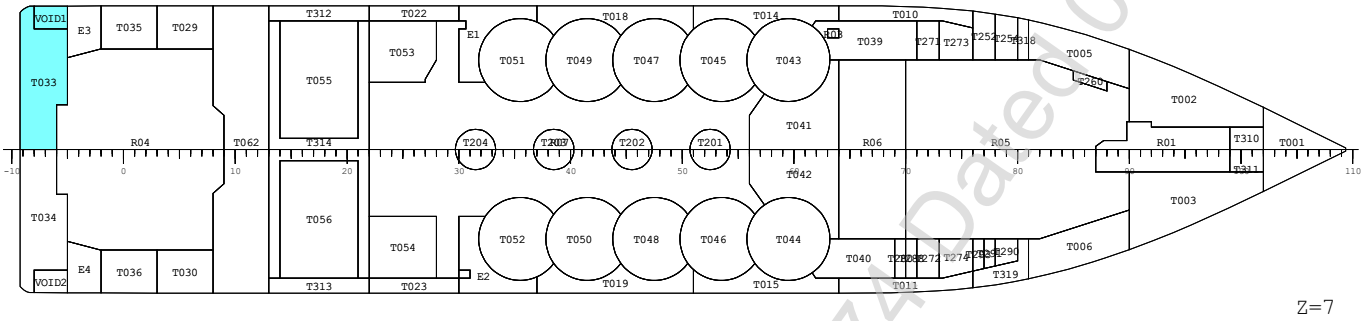
DAM	COMP	DES	PERM	VOL m3	XCG m	YCG m	ZCG m	STAGE
D20	VOID2	Capstan void SB	0.95	5.6	-4.54	8.20	7.04	1
D20	T034	WING TANK SB	0.95	60.9	-4.96	3.92	6.52	1

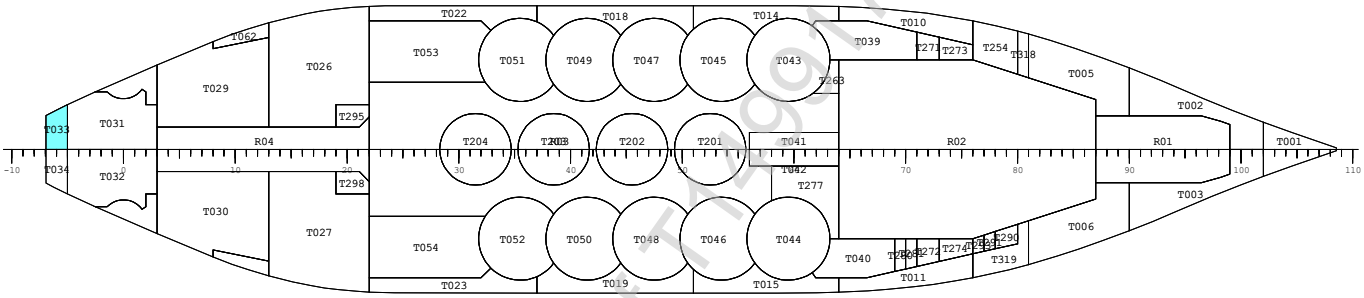
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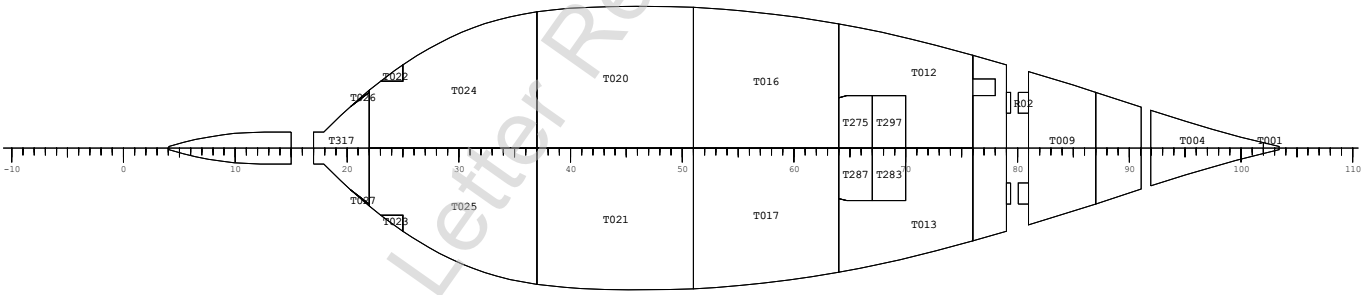
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DAMAGE CASE D21









DAM	COMP	DES	PERM	VOL m3	XCG m	YCG m	ZCG m	STAGE
D21	VOID1	Capstan void PS	0.95	5.6	-4.54	-8.20	7.04	1
D21	T033	WING TANK PS	0.95	60.9	-4.96	-3.92	6.52	1

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7 MANUAL DETERMINATION OF STABILITY ON BOARD

7.1 GENERAL

Using the stability forms (form1, form2 and form3) together with the tank plan/tank capacities hydrostatic tables and cross-curves it is possible to carry out stability calculations for different loading conditions compared with the loading conditions calculated in advance.

The documents are composed in such a way that the stability calculations may be carried out schematically.

The tables contain all necessary basic data for the calculations which have to be logged in form1, form2 and form3.

THE FOLLOWING TABLES AND FORMS ARE USED IN THE CALCULATIONS

- Drawing tank plan (chap. 13)
- Tank capacities (chap. 11)
- Tables of hydrostatics (chap. 10)
- Cross curves of stability (chap. 9)
- Form1 : Blank form calculation of loaded ship (chap. 7)
- Form2 : Blank form hydrostatic values (chap. 7)
- Form3 : Blank form GZ-curve (chap. 7)

INSTRUCTIONS FOR STABILITY CALCULATIONS

To get correct result of these calculations it is recommended to read the instructions very carefully and follow them step by step.

7.2 CALCULATION OF LOADED SHIP - FORM1

- Enter the individual weights (light ship, crew and provisions, tank capacities, cargo and miscellaneous weights) in the column WEIGHT.

- Ascertain the vertical centre of gravity above baseline of the individual weights and enter same in the column VCG.

- Multiply the individual weight by the corresponding vertical centre of gravity. Enter in column MV.

- Add the individual weights. Enter in column TOTAL SUM (1).

- Add the vertical moments. Enter in column TOTAL SUM (3).

- Divide (3) / (1) = (6) = VCG. Enter into position VCG (6).

VCG = Vertical centre of gravity above baseline of the loaded vessel.

NOTE:

The next four items are required only to be considered if a trim calculation is to be carried out.

- Ascertain the longitudinal centers of gravity of the individual weights and enter same in column LCG.

- Multiply the individual weight by the corresponding longitudinal centre of gravity. Enter in column ML.

- Add the longitudinal moments. Enter in column TOTAL SUM (2).

- Divide (2) / (1) = (5) = LCG. Enter into position LCG (5).

LCG = longitudinal centre of gravity of the loaded vessel measured from AP.

CENTRES OF GRAVITY FOR CONSUMABLES

Completely filled consumable tanks:

Weight, vertical and longitudinal centre of gravity are to be taken directly from drawing tank plan.

Partially filled consumable tanks:

Centres of gravity to be taken from the capacity tables of the individual consumable tanks.

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INFLUENCE OF FREE SURFACES

For partially filled consumable tanks the value FSM (Moment of free surface) of drawing TANK PLAN to be applied to FORM1 column FSM or to be taken form the capacity tables.

- Enter the density for all partially filled tanks into FORM1 column RHO.

- Get the moment of inertia IT for the actual tank from drawing TANK PLAN or from the capacity tables.

- Multiply the value IT by the corresponding density of the actual tank. Enter into FORM1 column FSM.

- Add the values of FSM. Enter in column TOTAL SUM (4).

- Divide (4) / (1) = (7) = dGM.

dGM = loss of stability due to free surface effect.

7.3 HYDROSTATIC VALUES - FORM2

Transfer the values of displacement, LCG, VCG, and dGM calculated in FORM1 to the positions (1) (5) (6) and (7) of FORM2 respectively.

- Depending on the specific gravity of water, locate the displacement (1) in the table of hydrostatics for trim = 0 (app. 1) due to following:

S.G. = 1.000 : Freshwater : Use column VOLT
S.G. = 1.025 : Seawater : Use column DISP

- From the located displacement read off the corresponding values and enter into FORM2 as follows:

- Draught T to position (8)

If a trim calculation is NOT to be carried out : - Transverse metacentric height KMT to column (21) first row.

If a trim calculation is to be carried out: - Longitudinal centre of buoyancy LCB to position (9)

- Trim factor aft TRFAP to position (10)

- Trim factor forward TRFFP to position (11)

TRIM CALCULATION

- Subtract (9) - (5) = (12) = Longitudinal moment arm. Enter with sign into position (12).

- Multiply and add (8) + (10)*(12) = (13) = Draught AP. Enter into position (13).

- Multiply and subtract (8) - (11)*(12) = (14) = Draught FP. Enter into position (14).

- Subtract (14) - (13) = (15) = Trim. Enter with sign into position (15).

NOTE: For trim calculation the rule for sign is to be observed: Trim by head is positive.

CROSS CURVES (KN)

The cross curves data available for the mean draught in app. 2 (KN-values) is to be applied to FORM2 column (19) (20) and (21), depending of the trim and specific gravity of water, as follows:

SG = 1.000 : Freshwater : Use column two. SG = 1.025 : Seawater : Use column one.

Trim calculation not considered :

- Use the table for TRIM=0

- Locate the correct row corresponding to actual displacement (1) and specific gravity of water.

- Read off the KN-values for each heeling angle and enter into FORM2 column (21).

Trim calculation considered :

It is necessary to interpolate between given trim values listed in the cross curve tables:

- locate the table for the nearest trim BELOW actual trim (15), and enter same into FORM2 position (16).

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- locate the table for the nearest trim ABOVE actual trim (15), and enter same into FORM2 position (17).

- Calculate interpolating constant (18) = ((15)-(16)) / ((17)-(16)), and enter into FORM2 position (18).

- Lookup the cross curve table for low trim value(16).

- Locate the correct row corresponding to actual displacement (1) and specific gravity of water.

- Read off the KN-values for each heeling angle and enter into FORM2 column (19).

- Lookup the cross curve table for high trim value(17).

- Locate the correct row corresponding to actual displacement (1) and specific gravity of water.

- Read off the KN-values for each heeling angle and enter into FORM2 column (20).

- Lookup the HYDROSTATIC TABLE in app. 2 for low trim value(16).

- Locate the correct row corresponding to actual displacement (1) and specific gravity of water.

- Read off the value of KMT and enter same into FORM2 column (19), first row.

- Lookup the HYDROSTATIC TABLE in app. 2 for high trim value(17).

- Locate the correct row corresponding to actual displacement (1) and specific gravity of water.

- Read off the value of KMT and enter same into FORM2 column (20), first row.

- For each row in the interpolation table in FORM2 calculate:
(21) = (19) + ((20)-(19)) * (18), and enter into column (21).

7.4 GZ-CURVE - FORM3

CALCULATION OF GM

The value of KMT and cross curve data calculated in FORM2 are to be applied to calculation of lever arm curve (GZ-curve) in FORM3:

- Transfer the values of displacement, LCG, VCG and dGM calculated in FORM1 to the position (1) (5) (6) and (7) of FORM2 respectively.

- Transfer the value of KMT from FORM2 column(21) first row to FORM3 position (22).

- Transfer the KN-values from FORM2 column(21) to the corresponding heeling angles in FORM3 row(27).

- Subtract (22) - (6) = (23) = GM0. Enter into FORM3 position (23).

- Subtract (23) - (7) = (24) = GM. Enter into FORM3 position (24).

- Add (6) + (7) = (25). Enter into FORM3 position (25).

GM0 = Initial metacentric height
GM = Metacentric height corrected for free surface effect.
KG = Centre of gravity for loaded ship corrected for free surface effect.

KG is measured from the baseline.
GM and GM0 is measured from the centre of gravity for loaded ship.

DRAWING OF LEVER ARM CURVE (GZ)

Calculation of table for the righting lever arm (GZ) :

- Multiply each value in row(26) with the value of KG(25), and enter results in row (28).

- For each heeling angle subtract (27) - (28) = (29) = GZ, and enter results in row (29).

Drawing the lever arm curve:

- Offset a point on the dashed line marked GM in such way that the point is corresponding with the value of GM(24) on the righting lever axis.

- Draw a straight line from this point to the origin of the chart diagram.

- Tag the points listed in row(29) into the chart diagram.

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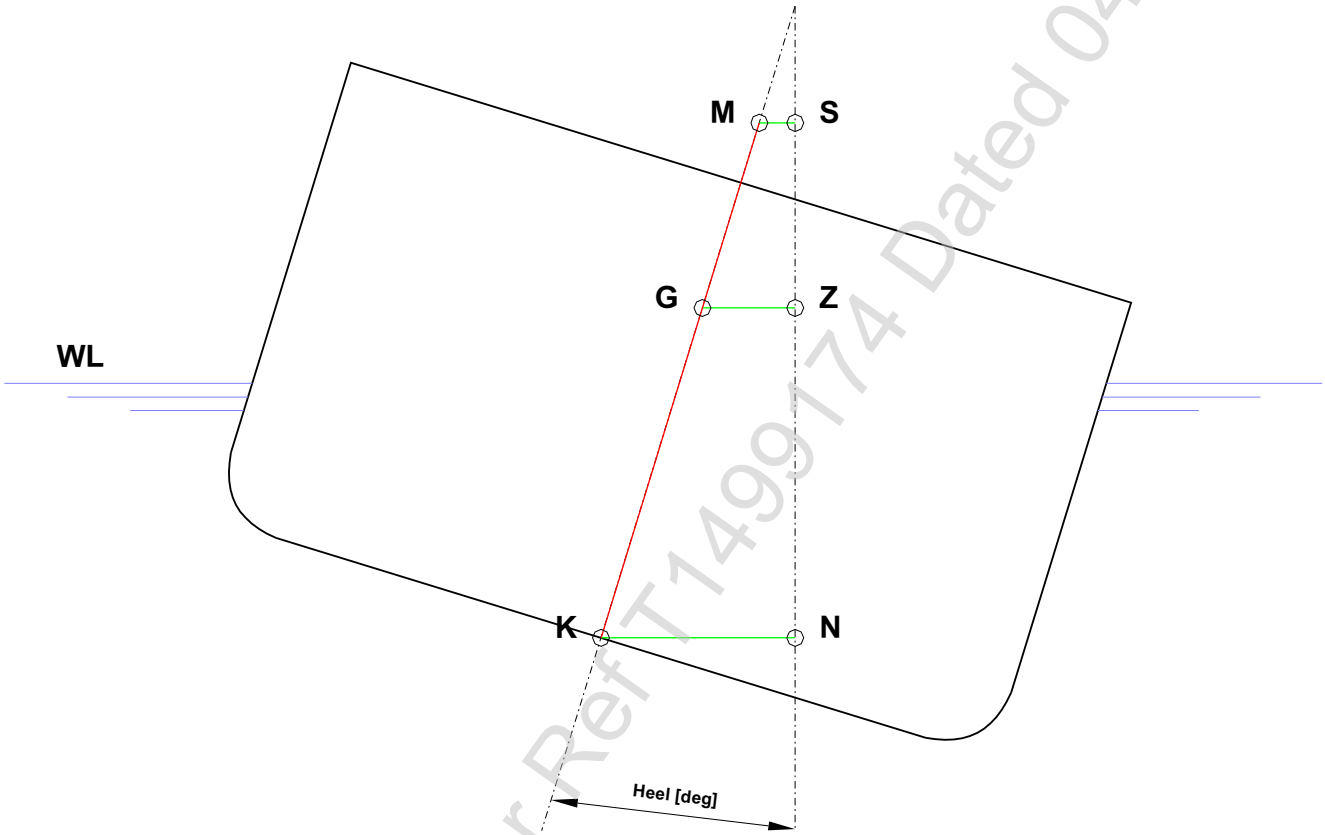
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- Draw a smooth curve through the points with the straight GM-line forming a tangent to the curve.

The straight GM-line states the actual initial stability.

STABILITY COMPONENTS

The following diagram shows the different components for the stability.



G - vessels center of gravity
M - metacenter
K - origin
GM - metacentric height
KG - vessels vertical center of gravity

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FORM 1 CALCULATION OF LOADED SHIP

ITEM	VOLUME [m³]	RHO [t/m³]	WEIGHT [t]	LCG [m]	ML [tm]	VCG [m]	MV [tm]	FSM [tm]
LIGHT SHIP								
CREW + EFFECTS								
TOTAL SUM								
			(1)		(2)		(3)	(4)

DISPLACEMENT (1) => (1) _____ [t]
LCG (5) => (2)/(1) _____ [m]
VCG (6) => (3)/(1) _____ [m]
dGM (7) => (4)/(1) _____ [m]

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FORM 2 HYDROSTATIC VALUES

VALUES CALCUALTED IN FORM 1		VALUES ON EVEN KEEL	
DISPLACEMENT	(1) _____ [t]	DRAUGHT	(8) _____ [m]
LCG	(5) _____ [m]	LCB	(9) _____ [m]
VCG	(6) _____ [m]	TRFAP	(10) _____ [-]
dGM	(7) _____ [m]	TRFFP	(11) _____ [-]

LONGITUDINAL MOMENT ARM (12) => (9) - (5) _____ [m]

DRAUGHT AP (13) => (8) + (10)*(12) _____ [m]

DRAUGHT FP (14) => (8) - (11)*(12) _____ [m]

TRIM (15) => (14) - (13) _____ [m]

NEAREST TRIM VALUES LISTED IN HYDROSTATIC TABLES: **INTERPOLATION CONSTANT:**

LOW TRIM VALUE (16) _____ [m]

HIGH TRIM VALUE (17) _____ [m]

$$\text{CONST (18)} => \frac{(15) - (16)}{(17) - (16)} = _____$$

INTERPOLATION BETWEEN GIVEN TRIM VALUES

ITEM	LOW TRIM VALUE	HIGH TRIM VALUE	ACTUAL TRIM
KMT			
KN at 10 degrees			
KN at 20 degrees			
KN at 30 degrees			
KN at 40 degrees			
KN at 50 degrees			
KN at 60 degrees			
KN at 70 degrees			
	(19)	(20)	(21)

(21) => (19) + { (20) - (19) }*(18)

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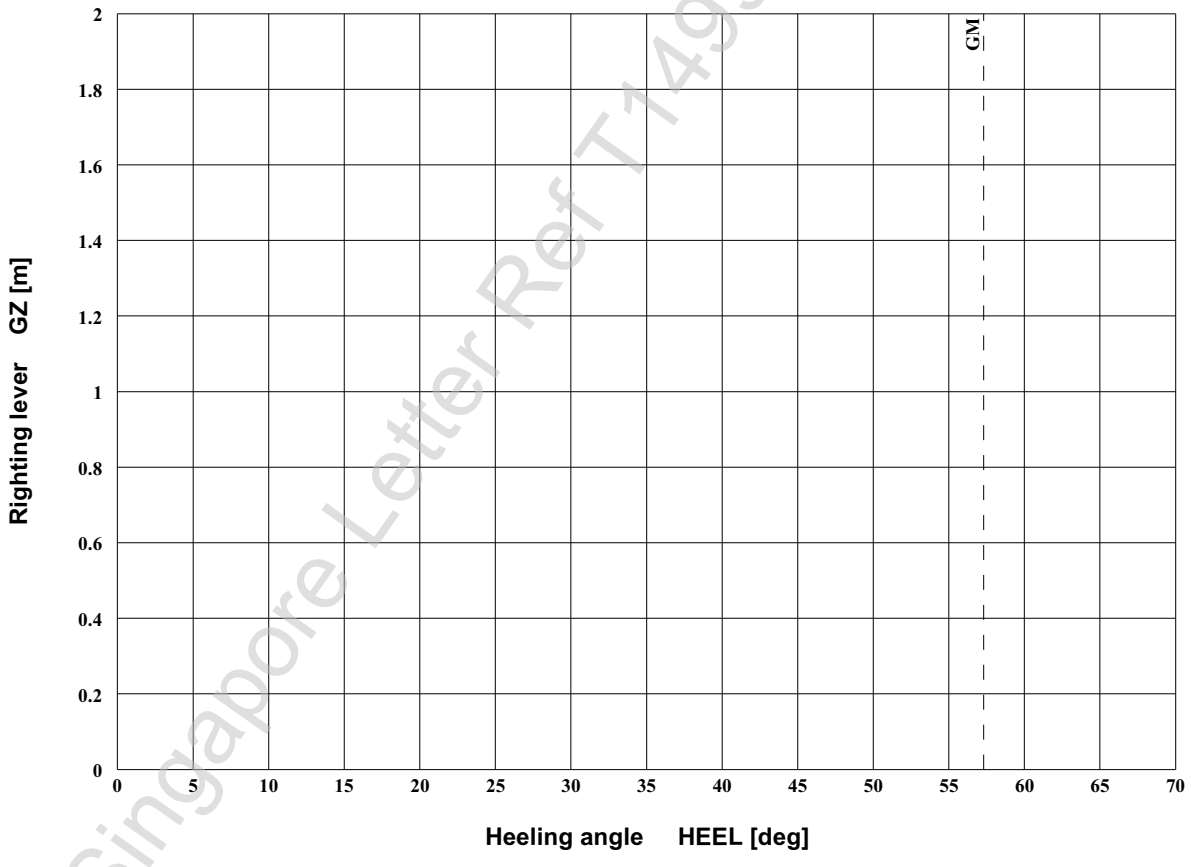
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FORM 3 GZ-CURVE

VALUES CALCUALTED IN FORM 1

DISPLACEMENT	**(1) _____ [t]**	**KMT FROM FORM 2**	**(22) _____ [m]**
LCG	**(5) _____ [m]**	**GM0**	**(23) => (22) - (6) _____ [m]**
VCG	**(6) _____ [m]**	**GM**	**(24) => (23) - (7) _____ [m]**
dGM	**(7) _____ [m]**	**KG**	**(25) => (6) - (7) _____ [m]**

		10 deg	20 deg	30 deg	40 deg	50 deg	60 deg	70 deg
SIN (HEEL)	**(26)**	**0.173648**	**0.34202**	**0.5**	**0.642788**	**0.766044**	**0.866025**	**0.939693**
KN FROM FORM 2	**(27)**							
KG * SIN (HEEL)	**(28) => (25) * (26)**							
GZ	**(29) => (27) - (28)**							



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7.5 COMPLIANCE WITH STABILITY CRITERIA

7.5.1 GENERAL

In order to carry out a stability check for the actual loading condition the Master should follow one of the following instructions :

- Check against MINGM-curve
- Check against the loading conditions calculated in advance
- Check against relevant criteria (Both intact and damage criteria !)

Instructions for calculation of GM and drawing of GZ-curve are given in chapter 7. It is of great importance for the stability that the ship is balanced transversely, giving zero heeling moment in all conditions.

To calculate the heeling moment due to loaded ship :

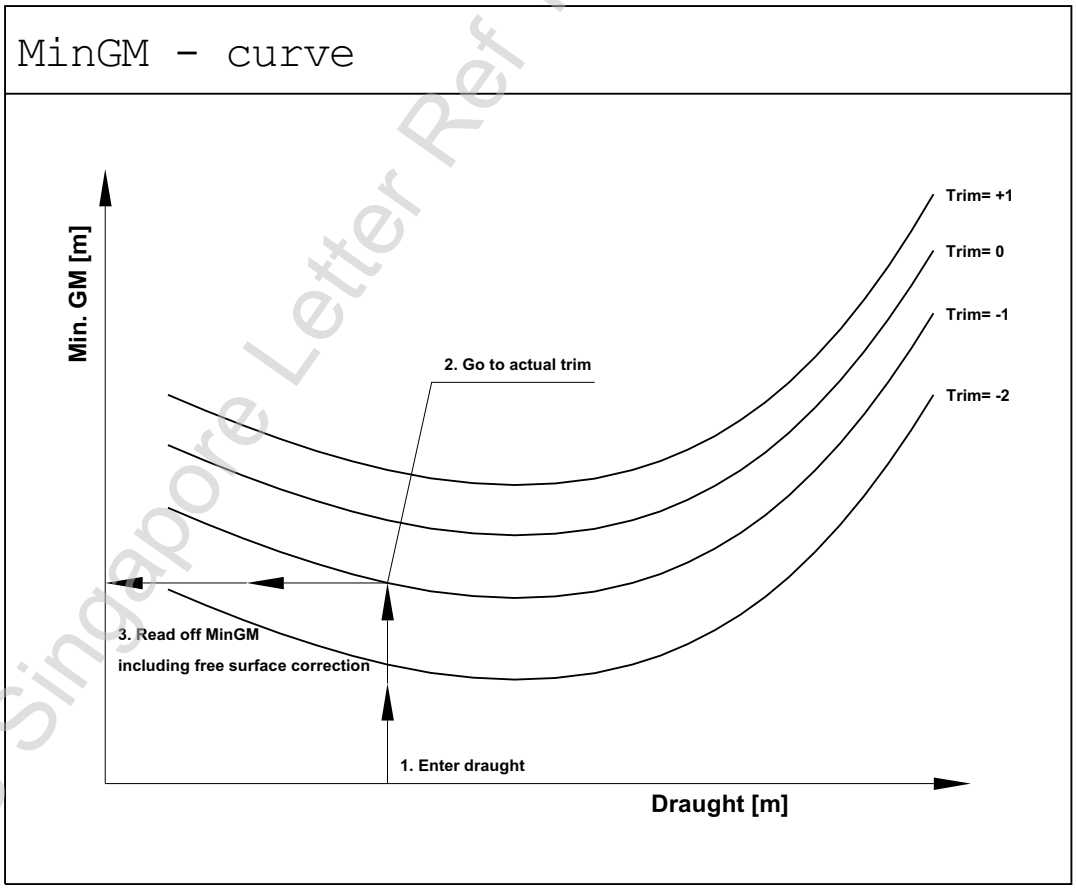
- For every tank and mass load :
Heeling moment = Weight of tank or mass multiplied with the transverse centre of gravity TCG with sign (Weight and TCG given in ch. 2 or app. 3).

- For loaded ship: Heeling moment = sum of heeling moments calculated above.

If the GZ-curve is calculated (ch. 7) it is possible to check the stability against relevant stability criteria.
NOTE: When carrying heavy deck loads, special stability calculations has to be done to ensure that the condition fulfils the stability criterion.

7.5.2 CHECK AGAINST MINIMUM GM CURVE

Calculation of GM using standard forms is described in chapter 7. The calculated GM is compared to the MINGM-curve as shown below. If the actual GM (corrected for free surface) is on or above the minimum GM value the stability of the vessel is satisfactory.



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7.5.3 CALCULATION OF AREA BELOW GZ-CURVE

In order to compare the actual GZ-curve against stability criteria it is necessary to calculate the area below the GZ-curve.

On the following page an example showing how to calculate the area below 40 degrees is described.

To calculate the area up to DELTA degrees do following :

- Calculate CI = Common angle in interval = DELTA / 8
- Replace column I with angles in interval CI, i.e. 0*CI , 1*CI, 2*CI ... 8*CI.
- Read off the GZ-values at given angles of heel, and enter same into column III.
- For each angle of heel multiply column II and column III, and enter the result into column IV.
- Calculate the sum of column IV and enter into position SUM IV =....
- Calculate the area by use of the formula:

AREA = (CI/3) * (SUM IV) * 0.01745 = ... meter radians

If you want the area between 30 and 40 degrees, do following :

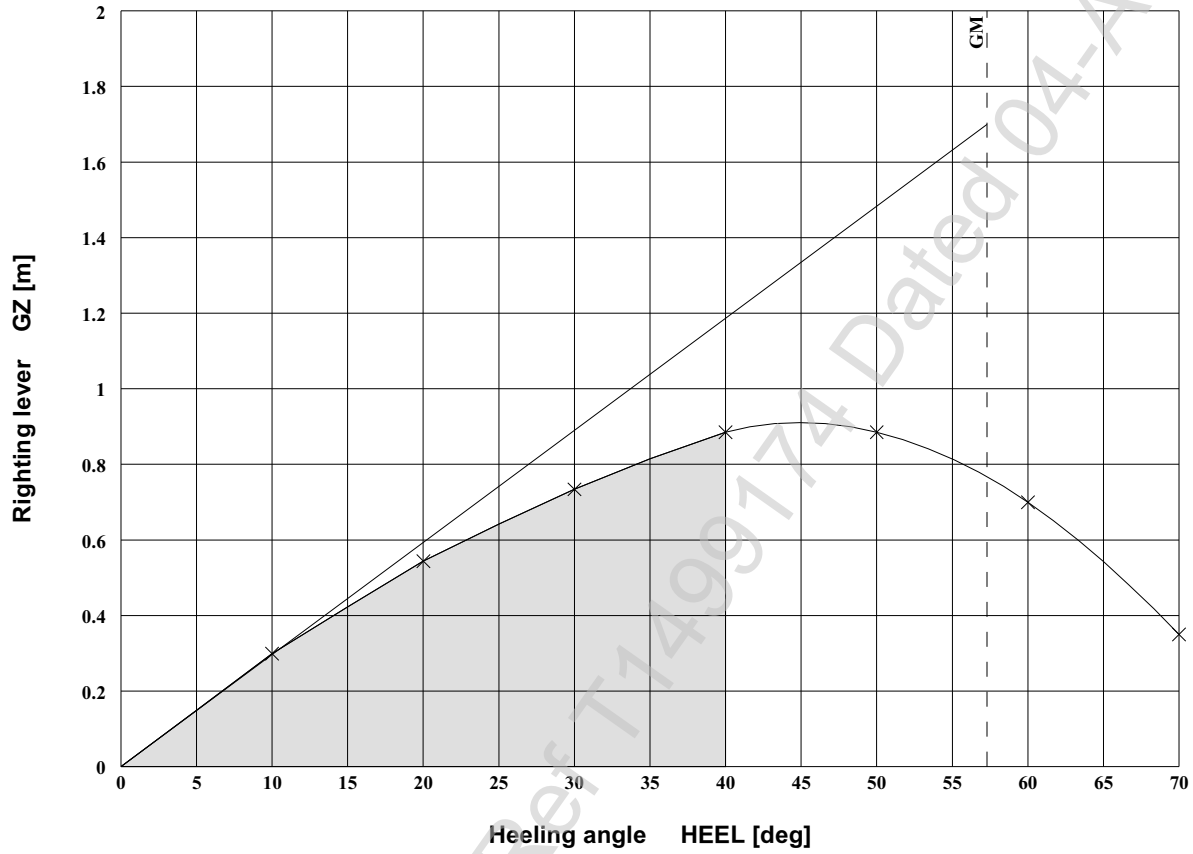
- Use the instructions above to calculate the area for DELTA = 30 deg
- Use the same procedure for DELTA = 40 degrees.
- Area between 30 and 40 degrees is given by: AREA40-30 = AREA40 - AREA30 = ...

The calculations should be logged into the given form at the end of this chapter.

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GZ - curve



SM = SIMPSONS MULTIPLICATOR, WHICH ARE STANDARD
FACTORS FOR THE PARTICULAR SIMPSONS RULE
BEEING APPLIED.

CI = COMMON ANGLE IN INTERVAL
(=5 DEGREES IN THIS EXAMPLE)

TO CONVERT METER DEGREES TO METER RADIANS :
MULTIPLY WITH 0.01745329

CALCULATED AREA IS SHADED ON GZ-CURVE.

EXAMPLE SHOWING AREA BELOW GZ-CURVE FROM 0 deg UP TO 40 deg

I	II	III	IV
HEEL [deg]	SM [-]	GZ [m]	PRODUCT II×III [mdeg]
0	1	0	0
5	4	0.149	0.596
10	2	0.299	0.598
15	4	0.423	1.692
20	2	0.544	1.088
25	4	0.642	2.568
30	2	0.734	1.468
35	4	0.815	3.26
40	1	0.885	0.885
TOTAL SUM			12.155

AREA = (CI/3) * (SUM IV) = (5/3) * (12.155) = 20.258 METER DEGREES = 0.3535 METER RADIANS

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FORM 4	AREA BELOW GZ - CURVE		
VALUES IN COLUMN I AND III READED FROM GZ-CURVE AT FORM 3			
I	**II**	**III**	**IV**
HEEL [deg]	**SM [-]**	**GZ [m]**	**PRODUCT II×III [mdeg]**
0	**1**	**0.000**	**0.000**
	4		
	2		
	4		
	2		
	4		
	2		
	4		
	1		
TOTAL SUM			

UPPER ANGLE OF CALCULATED AREA **DELTA _____ [deg]**

COMMON ITERVAL ANGLE **CI => DELTA/8 _____ [deg]**

AREA BELOW GZ-CURVE UP TO DELTA **(CI/3) × (SUM IV) × 0.01745 _____ [mrad]**

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8 GM LIMIT CURVES

8.1 INTERPOLATION BETWEEN CURVES

Please follow the checklist below being sure of using the curves correctly:

1. Calculate the GM using standard forms described in chapter 8.

2. Make sure the calculated GM is corrected for free surfaces.

3. Make sure the ship is transversely balanced giving zero heeling moment.

4. If the actual trim is between the values of trim listed, the maximum GM-value for the nearest trim values may be used.

OR:

Use the interpolation formula below to calculate the minimum GM for the actual trim.

5. If the minimum GM-value is LESS THAN the actual GM (corrected for free surfaces) the stability of the vessel is satisfactory.

$$\text{MinGM [m]} = GM1 + \frac{(GM2 - GM1) * (TR - TR1)}{(TR2 - TR1)}$$

TR1, TR2 and TR are trim values LOW, HIGH and ACTUAL respectively.
GM1 and GM2 are minGM corresponding to trim values TR1 and TR2.
TR1 and TR2 have to be chosen in such way giving TR2>TR>TR1.

Explanation of symbols used in GM LIMIT TABLES :

T - draught
IGMLIM - GM limit values for intact stability
ICRLIM - intact criteria limiting the GM for corresponding GM limit value
ITEXT - text describing the corresponding intact criteria
GMLIM - overall resulting GM limit value for corresponding draught

DGMLIM - GM limit values for deterministic damage stability
DCRLIM - deterministic damage criteria limiting the GM for corresponding GM limit value
DTEXT - text describing the corresponding deterministic damage criteria

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8.2 GM LIMITS - TRIM = -1.5 m

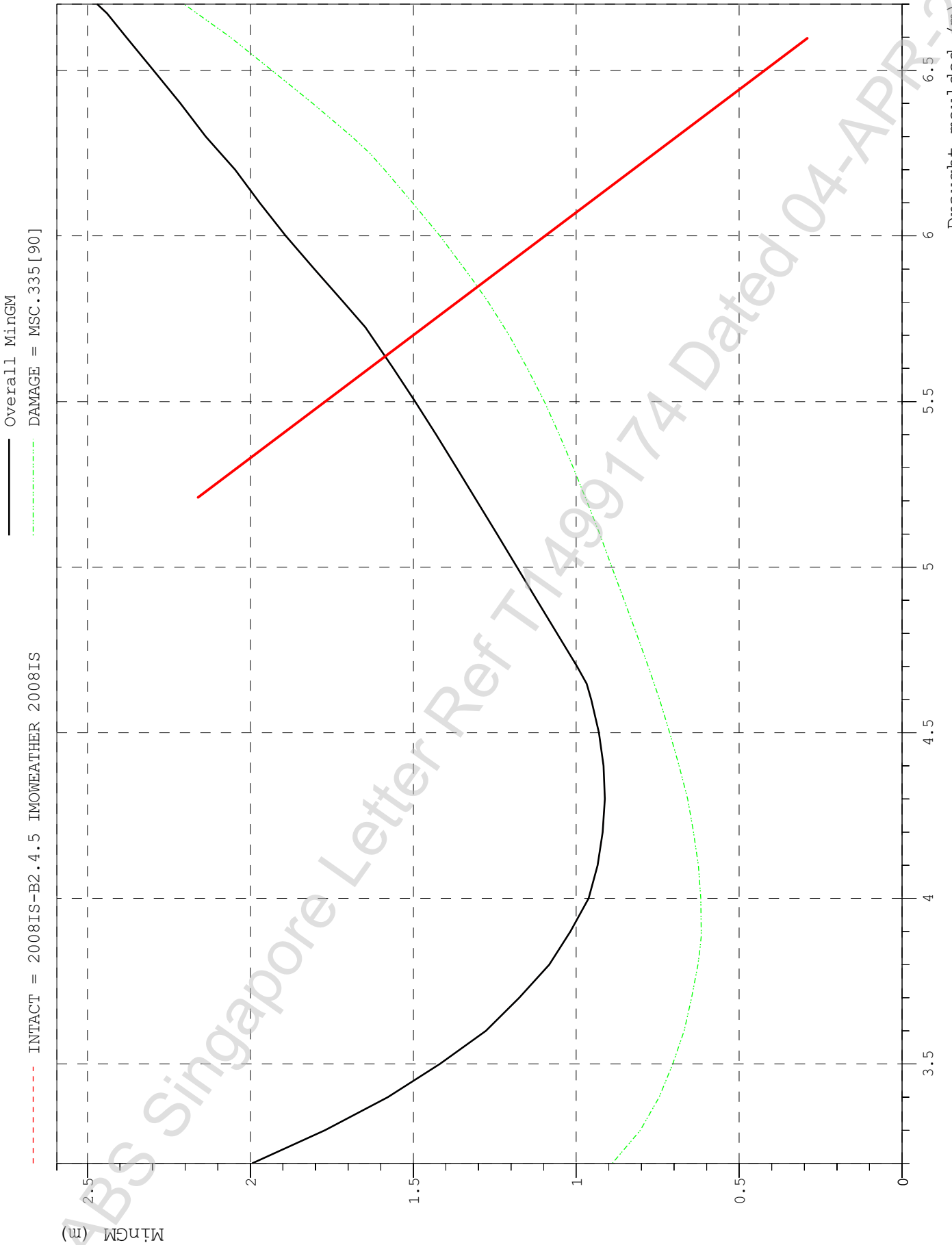
Refer to Intact Stability Minimum GM tables enclosed in ABS review letter ZL/T1499174 dated 04 April 2016

Damage GM table is not part of ABS review (TYP.)

T m	IGMLIM m	ICRLIM	DGMLIM m	DDAM	DCRLIM	GMLIM m
3.200	1.995	IMOWEATHER	0.889	D11	V.RANGE.A469	1.995
3.300	1.771	IMOWEATHER	0.804	D11	V.RANGE.A469	1.771
3.400	1.579	IMOWEATHER	0.746	D11	V.RANGE.A469	1.579
3.500	1.418	IMOWEATHER	0.704	D11	V.RANGE.A469	1.418
3.600	1.277	IMOWEATHER	0.669	D11	V.RANGE.A469	1.277
3.700	1.175	IMOWEATHER	0.646	D11	V.RANGE.A469	1.175
3.800	1.083	IMOWEATHER	0.627	D11	V.RANGE.A469	1.083
3.846	1.053	IMOWEATHER	0.621			1.053
3.885	1.027	IMOWEATHER	0.617			1.027
3.900	1.018	IMOWEATHER	0.616	D07	V.MINGZ.A469	1.018
4.000	0.962	IMOWEATHER	0.618	D07	V.MINGZ.A469	0.962
4.100	0.935	IMOWEATHER	0.626	D07	V.MINGZ.A469	0.935
4.200	0.919	IMOWEATHER	0.640	D07	V.MINGZ.A469	0.919
4.207	0.918	IMOWEATHER	0.641			0.918
4.300	0.912	IMOWEATHER	0.659	D10	V.RANGE.A469	0.912
4.400	0.916	IMOWEATHER	0.684	D10	V.RANGE.A469	0.916
4.500	0.931	IMOWEATHER	0.713	D10	V.RANGE.A469	0.931
4.600	0.954	IMOWEATHER	0.744	D10	V.RANGE.A469	0.954
4.648	0.968		0.761	D10	V.RANGE.A469	0.968
4.700	0.997	V.AREA3040	0.779	D10	V.RANGE.A469	0.997
4.800	1.059	V.AREA3040	0.816	D10	V.RANGE.A469	1.059
4.900	1.121	V.AREA3040	0.853	D10	V.RANGE.A469	1.121
5.000	1.182	V.AREA3040	0.891	D10	V.RANGE.A469	1.182
5.100	1.243	V.AREA3040	0.928	D10	V.RANGE.A469	1.243
5.200	1.305	V.AREA3040	0.968	D10	V.RANGE.A469	1.305
5.300	1.367	V.AREA3040	1.009	D10	V.RANGE.A469	1.367
5.400	1.430	V.AREA3040	1.052	D10	V.RANGE.A469	1.430
5.500	1.494	V.AREA3040	1.098	D10	V.RANGE.A469	1.494
5.600	1.562	V.AREA3040	1.150	D10	V.RANGE.A469	1.562
5.700	1.630	V.AREA3040	1.206	D10	V.RANGE.A469	1.630
5.724	1.647		1.221	D10	V.RANGE.A469	1.647
5.800	1.714	IMOWEATHER	1.270	D10	V.RANGE.A469	1.714
5.871	1.777	IMOWEATHER	1.320			1.777
5.900	1.803	IMOWEATHER	1.342	D12	V.RANGE.A469	1.803
6.000	1.891	IMOWEATHER	1.419	D12	V.RANGE.A469	1.891
6.100	1.972	IMOWEATHER	1.502	D12	V.RANGE.A469	1.972
6.200	2.047	IMOWEATHER	1.588	D12	V.RANGE.A469	2.047
6.251	2.093	IMOWEATHER	1.635			2.093
6.300	2.137	IMOWEATHER	1.689	D16	V.RANGE.A469	2.137
6.400	2.215	IMOWEATHER	1.809	D16	V.RANGE.A469	2.215
6.500	2.297	IMOWEATHER	1.934	D16	V.RANGE.A469	2.297
6.600	2.381	IMOWEATHER	2.062	D16	V.RANGE.A469	2.381
6.672	2.440		2.164	D16	V.RANGE.A469	2.440
6.700	2.470	V.AREA15-30	2.203	D16	V.RANGE.A469	2.470

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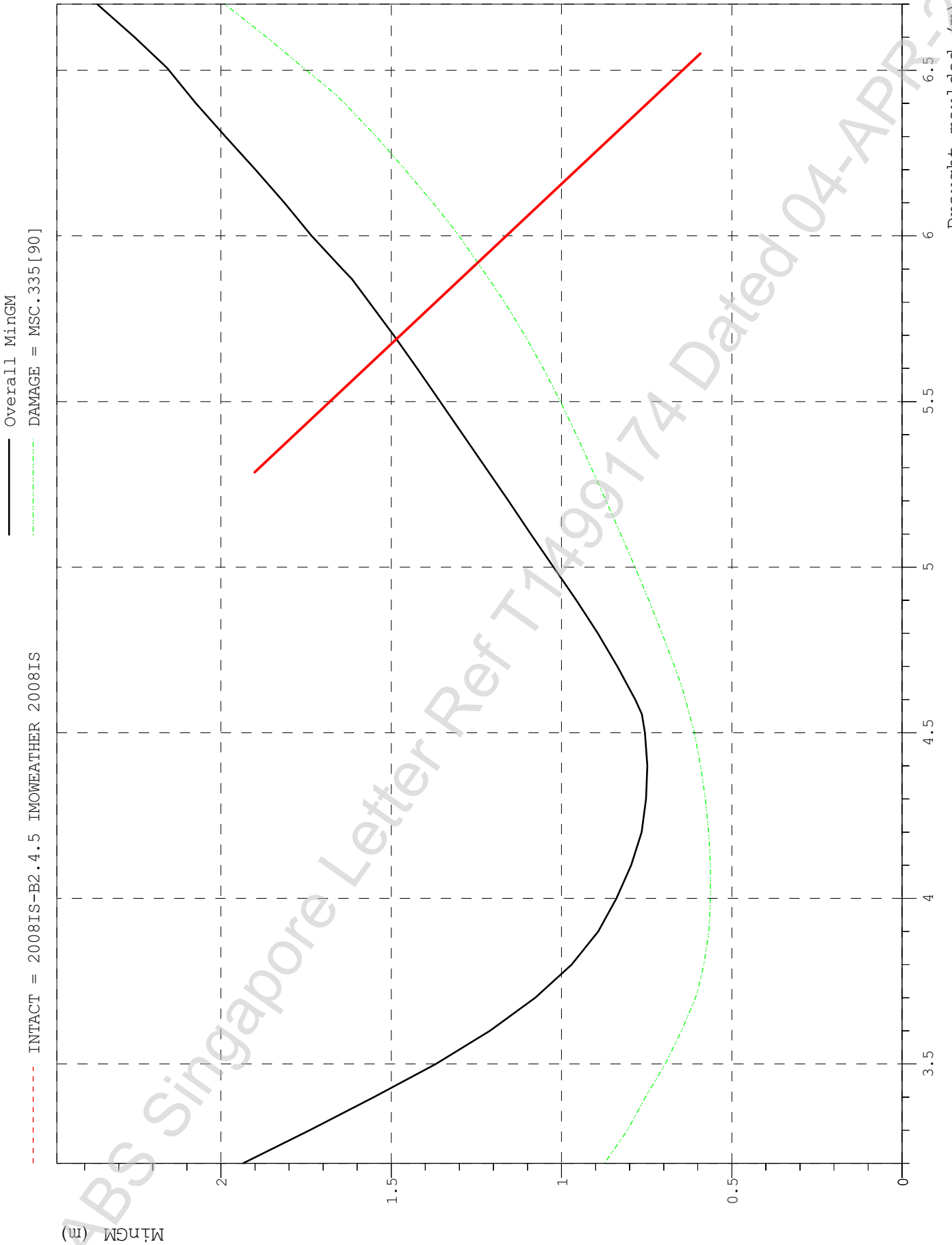
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8.3 GM LIMITS - TRIM = -1 m

T m	IGMLIM m	ICRLIM	DGMLIM m	DDAM	DCRLIM	GMLIM m
3.200	1.935	IMOWEATHER	0.874	D11	V.RANGE.A469	1.935
3.300	1.740	IMOWEATHER	0.807	D11	V.RANGE.A469	1.740
3.400	1.551	IMOWEATHER	0.754	D11	V.RANGE.A469	1.551
3.500	1.368	IMOWEATHER	0.695	D11	V.RANGE.A469	1.368
3.600	1.211	IMOWEATHER	0.648	D11	V.RANGE.A469	1.211
3.700	1.077	IMOWEATHER	0.607	D11	V.RANGE.A469	1.077
3.725	1.050	IMOWEATHER	0.599			1.050
3.800	0.971	IMOWEATHER	0.582	D07	V.MINGZ.A469	0.971
3.900	0.892	IMOWEATHER	0.567	D07	V.MINGZ.A469	0.892
4.000	0.839	IMOWEATHER	0.563	D07	V.MINGZ.A469	0.839
4.100	0.796	IMOWEATHER	0.563	D07	V.MINGZ.A469	0.796
4.200	0.765	IMOWEATHER	0.568	D07	V.MINGZ.A469	0.765
4.300	0.752	IMOWEATHER	0.578	D07	V.MINGZ.A469	0.752
4.400	0.748	IMOWEATHER	0.592	D07	V.MINGZ.A469	0.748
4.500	0.755	IMOWEATHER	0.610	D07	V.MINGZ.A469	0.755
4.556	0.764		0.625	D07	V.MINGZ.A469	0.764
4.600	0.784	V.AREA3040	0.637	D07	V.MINGZ.A469	0.784
4.633	0.801	V.AREA3040	0.645			0.801
4.700	0.836	V.AREA3040	0.668	D10	V.RANGE.A469	0.836
4.800	0.894	V.AREA3040	0.705	D10	V.RANGE.A469	0.894
4.900	0.956	V.AREA3040	0.744	D10	V.RANGE.A469	0.956
5.000	1.023	V.AREA3040	0.784	D10	V.RANGE.A469	1.023
5.100	1.090	V.AREA3040	0.827	D10	V.RANGE.A469	1.090
5.200	1.156	V.AREA3040	0.869	D10	V.RANGE.A469	1.156
5.300	1.222	V.AREA3040	0.913	D10	V.RANGE.A469	1.222
5.400	1.290	V.AREA3040	0.957	D10	V.RANGE.A469	1.290
5.500	1.356	V.AREA3040	1.003	D10	V.RANGE.A469	1.356
5.600	1.424	V.AREA3040	1.053	D10	V.RANGE.A469	1.424
5.700	1.493	V.AREA3040	1.108	D10	V.RANGE.A469	1.493
5.800	1.565	V.AREA3040	1.167	D10	V.RANGE.A469	1.565
5.869	1.615		1.213	D10	V.RANGE.A469	1.615
5.900	1.643	IMOWEATHER	1.233	D10	V.RANGE.A469	1.643
6.000	1.734	IMOWEATHER	1.301	D10	V.RANGE.A469	1.734
6.022	1.752	IMOWEATHER	1.317			1.752
6.100	1.814	IMOWEATHER	1.377	D12	V.RANGE.A469	1.814
6.200	1.899	IMOWEATHER	1.459	D12	V.RANGE.A469	1.899
6.300	1.987	IMOWEATHER	1.543	D12	V.RANGE.A469	1.987
6.400	2.073	IMOWEATHER	1.637	D12	V.RANGE.A469	2.073
6.428	2.095	IMOWEATHER	1.664			2.095
6.500	2.152	IMOWEATHER	1.749	D16	V.RANGE.A469	2.152
6.507	2.158		1.758	D16	V.RANGE.A469	2.158
6.600	2.253	V.AREA15-30	1.866	D16	V.RANGE.A469	2.253
6.700	2.363	V.AREA15-30	1.988	D16	V.RANGE.A469	2.363

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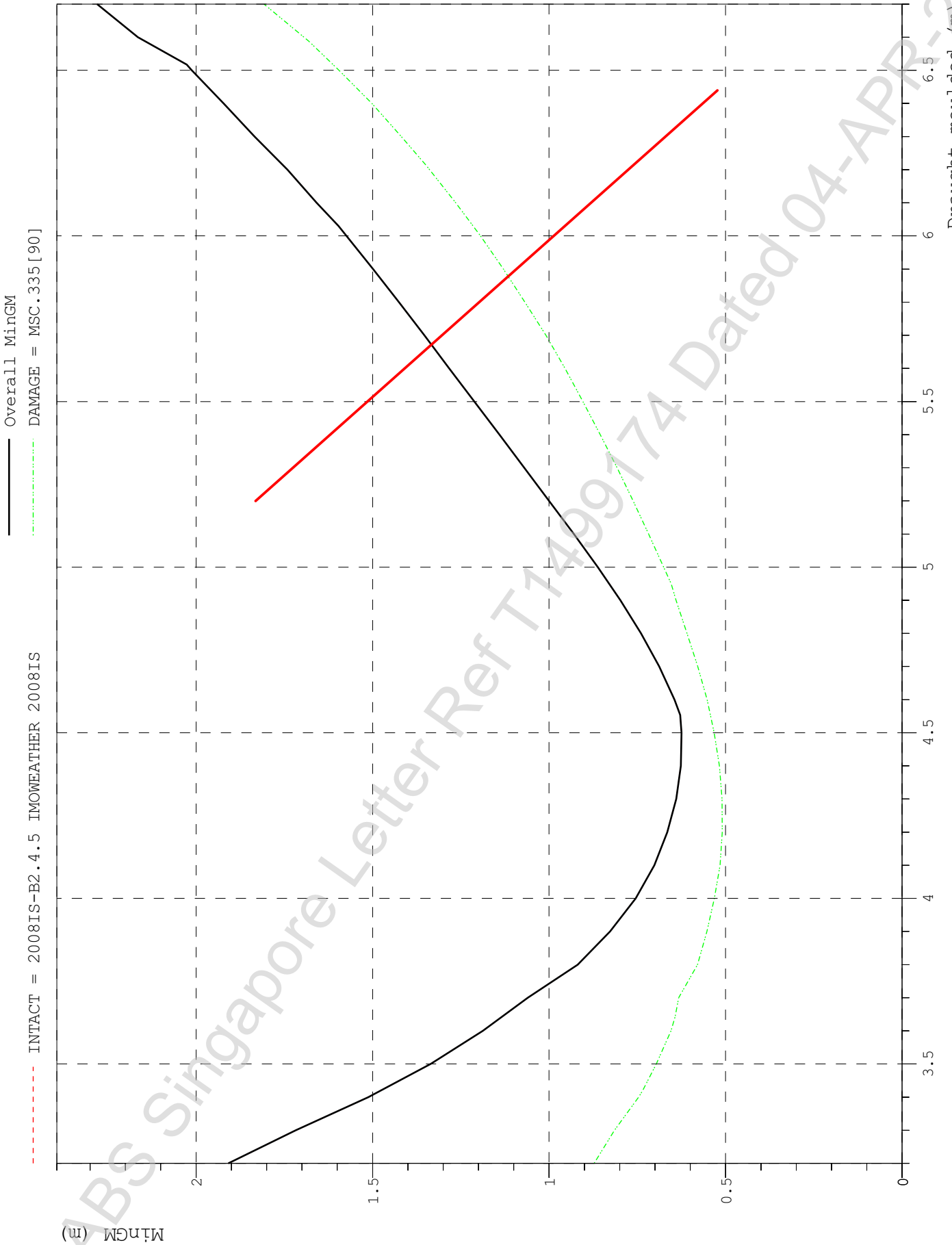
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8.4 GM LIMITS - TRIM = -0.5 m

T m	IGMLIM m	ICRLIM	DGMLIM m	DDAM	DCRLIM	GMLIM m
3.200	1.908	IMOWEATHER	0.872	D11	V.RANGE.A469	1.908
3.300	1.717	IMOWEATHER	0.815	D11	V.RANGE.A469	1.717
3.400	1.512	IMOWEATHER	0.747	D11	V.RANGE.A469	1.512
3.419	1.479	IMOWEATHER	0.736			1.479
3.500	1.336	IMOWEATHER	0.696	D07	V.MINGZ.A469	1.336
3.600	1.187	IMOWEATHER	0.655	D07	V.MINGZ.A469	1.187
3.643	1.133	IMOWEATHER	0.643			1.133
3.700	1.060	IMOWEATHER	0.633	D07	V.RANGE.A469	1.060
3.800	0.919	IMOWEATHER	0.580	D07	V.RANGE.A469	0.919
3.900	0.827	IMOWEATHER	0.553	D07	V.RANGE.A469	0.827
4.000	0.755	IMOWEATHER	0.532	D07	V.RANGE.A469	0.755
4.100	0.702	IMOWEATHER	0.516	D07	V.RANGE.A469	0.702
4.200	0.665	IMOWEATHER	0.509	D07	V.RANGE.A469	0.665
4.300	0.640	IMOWEATHER	0.510	D07	V.RANGE.A469	0.640
4.400	0.627	IMOWEATHER	0.518			0.627
4.400	0.627	IMOWEATHER	0.518	D07	V.MINGZ.A469	0.627
4.500	0.625	IMOWEATHER	0.533	D07	V.MINGZ.A469	0.625
4.553	0.629		0.543	D07	V.MINGZ.A469	0.629
4.600	0.645	V.AREA3040	0.552	D07	V.MINGZ.A469	0.645
4.700	0.689	V.AREA3040	0.579	D07	V.MINGZ.A469	0.689
4.800	0.740	V.AREA3040	0.609	D07	V.MINGZ.A469	0.740
4.900	0.798	V.AREA3040	0.641	D07	V.MINGZ.A469	0.798
4.950	0.830	V.AREA3040	0.654			0.830
5.000	0.862	V.AREA3040	0.676	D10	V.RANGE.A469	0.862
5.100	0.930	V.AREA3040	0.718	D10	V.RANGE.A469	0.930
5.200	1.000	V.AREA3040	0.762	D10	V.RANGE.A469	1.000
5.300	1.070	V.AREA3040	0.807	D10	V.RANGE.A469	1.070
5.400	1.140	V.AREA3040	0.855	D10	V.RANGE.A469	1.140
5.500	1.212	V.AREA3040	0.904	D10	V.RANGE.A469	1.212
5.600	1.283	V.AREA3040	0.955	D10	V.RANGE.A469	1.283
5.700	1.354	V.AREA3040	1.009	D10	V.RANGE.A469	1.354
5.800	1.425	V.AREA3040	1.069	D10	V.RANGE.A469	1.425
5.900	1.498	V.AREA3040	1.129	D10	V.RANGE.A469	1.498
6.000	1.574	V.AREA3040	1.195	D10	V.RANGE.A469	1.574
6.030	1.596		1.215	D10	V.RANGE.A469	1.596
6.100	1.659	IMOWEATHER	1.265	D10	V.RANGE.A469	1.659
6.200	1.741	IMOWEATHER	1.338	D10	V.RANGE.A469	1.741
6.213	1.753	IMOWEATHER	1.348			1.753
6.300	1.834	IMOWEATHER	1.418	D12	V.RANGE.A469	1.834
6.400	1.922	IMOWEATHER	1.502	D12	V.RANGE.A469	1.922
6.500	2.012	IMOWEATHER	1.595	D12	V.RANGE.A469	2.012
6.517	2.026		1.612	D12	V.RANGE.A469	2.026
6.588	2.145		1.681			2.145
6.600	2.165	V.AREA15-30	1.694	D16	V.RANGE.A469	2.165
6.700	2.280	V.AREA15-30	1.808	D16	V.RANGE.A469	2.280

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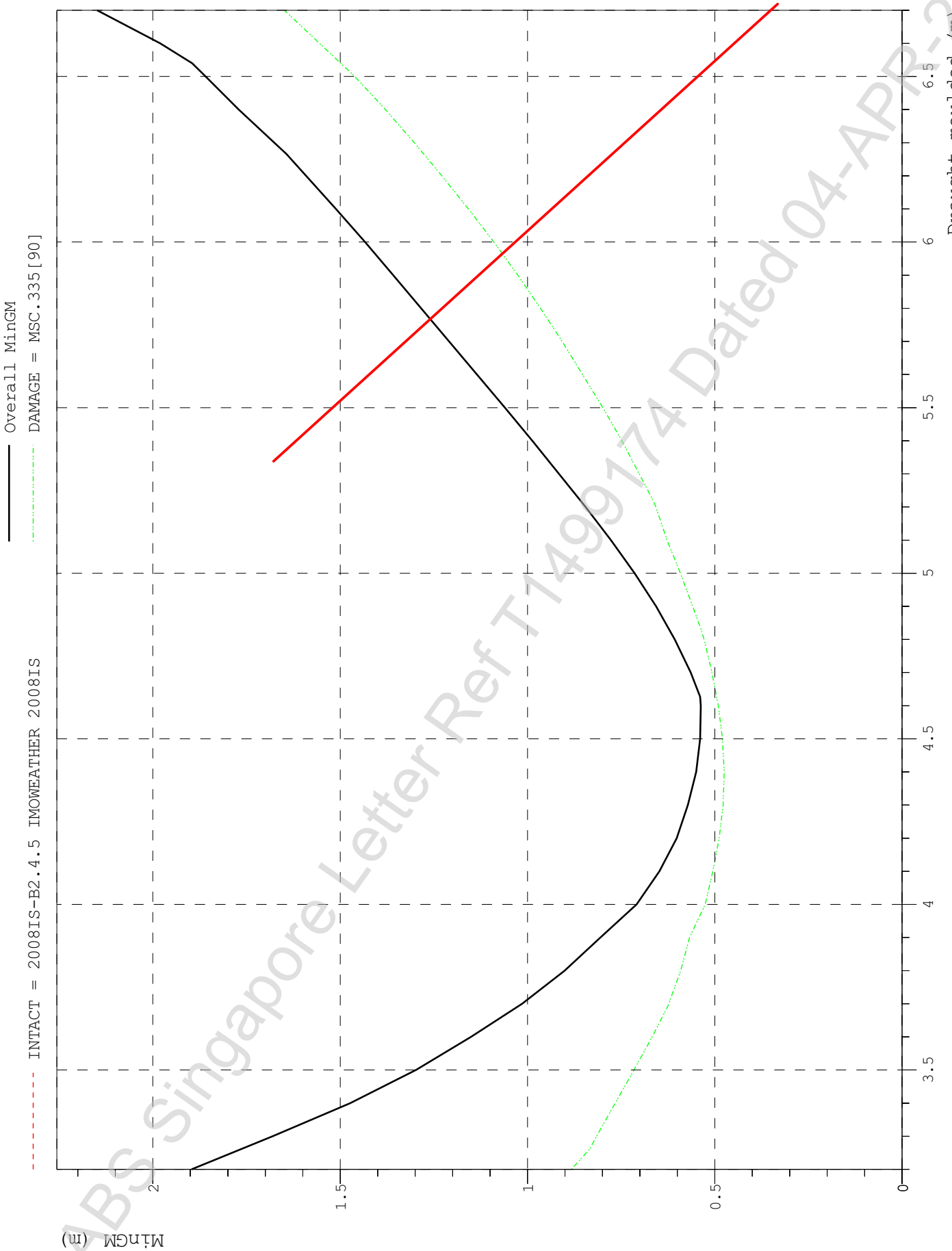
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8.5 GM LIMITS - TRIM = 0 m

T m	IGMLIM m	ICRLIM	DGMLIM m	DDAM	DCRLIM	GMLIM m
3.200	1.897	IMOWEATHER	0.883	D11	V.RANGE.A469	1.897
3.262	1.762	IMOWEATHER	0.834			1.762
3.300	1.681	IMOWEATHER	0.816	D07	V.MINGZ.A469	1.681
3.400	1.473	IMOWEATHER	0.766	D07	V.MINGZ.A469	1.473
3.500	1.297	IMOWEATHER	0.716	D07	V.MINGZ.A469	1.297
3.600	1.151	IMOWEATHER	0.667	D07	V.MINGZ.A469	1.151
3.693	1.024	IMOWEATHER	0.625			1.024
3.700	1.015	IMOWEATHER	0.623	D07	V.RANGE.A469	1.015
3.800	0.901	IMOWEATHER	0.591	D07	V.RANGE.A469	0.901
3.900	0.805	IMOWEATHER	0.567	D07	V.RANGE.A469	0.805
4.000	0.709	IMOWEATHER	0.526	D07	V.RANGE.A469	0.709
4.100	0.648	IMOWEATHER	0.505	D07	V.RANGE.A469	0.648
4.200	0.602	IMOWEATHER	0.488	D07	V.RANGE.A469	0.602
4.300	0.572	IMOWEATHER	0.478	D07	V.RANGE.A469	0.572
4.400	0.550	IMOWEATHER	0.475	D07	V.RANGE.A469	0.550
4.500	0.539	IMOWEATHER	0.480	D07	V.RANGE.A469	0.539
4.600	0.537	IMOWEATHER	0.490	D07	V.RANGE.A469	0.537
4.627	0.539		0.495	D07	V.RANGE.A469	0.539
4.700	0.564	V.AREA3040	0.508	D07	V.RANGE.A469	0.564
4.800	0.607	V.AREA3040	0.529	D07	V.RANGE.A469	0.607
4.825	0.619	V.AREA3040	0.535			0.619
4.900	0.657	V.AREA3040	0.559	D07	V.MINGZ.A469	0.657
5.000	0.714	V.AREA3040	0.592	D07	V.MINGZ.A469	0.714
5.100	0.777	V.AREA3040	0.628	D07	V.MINGZ.A469	0.777
5.200	0.845	V.AREA3040	0.657	D07	V.MINGZ.A469	0.845
5.204	0.847	V.AREA3040	0.658			0.847
5.222	0.861	V.AREA3040	0.666			0.861
5.300	0.916	V.AREA3040	0.701	D10	V.RANGE.A469	0.916
5.400	0.987	V.AREA3040	0.748	D10	V.RANGE.A469	0.987
5.500	1.060	V.AREA3040	0.799	D10	V.RANGE.A469	1.060
5.600	1.135	V.AREA3040	0.853	D10	V.RANGE.A469	1.135
5.700	1.209	V.AREA3040	0.907	D10	V.RANGE.A469	1.209
5.800	1.283	V.AREA3040	0.965	D10	V.RANGE.A469	1.283
5.900	1.358	V.AREA3040	1.026	D10	V.RANGE.A469	1.358
6.000	1.434	V.AREA3040	1.090	D10	V.RANGE.A469	1.434
6.100	1.511	V.AREA3040	1.158	D10	V.RANGE.A469	1.511
6.200	1.591	V.AREA3040	1.229	D10	V.RANGE.A469	1.591
6.266	1.644		1.278	D10	V.RANGE.A469	1.644
6.300	1.676	IMOWEATHER	1.303	D10	V.RANGE.A469	1.676
6.361	1.734	IMOWEATHER	1.349			1.734
6.400	1.771	IMOWEATHER	1.380	D12	V.RANGE.A469	1.771
6.500	1.859	IMOWEATHER	1.461	D12	V.RANGE.A469	1.859
6.540	1.895		1.499	D12	V.RANGE.A469	1.895
6.600	1.980	V.AREA15-30	1.555	D12	V.RANGE.A469	1.980
6.700	2.149	V.AREA15-30	1.649	D12	V.RANGE.A469	2.149

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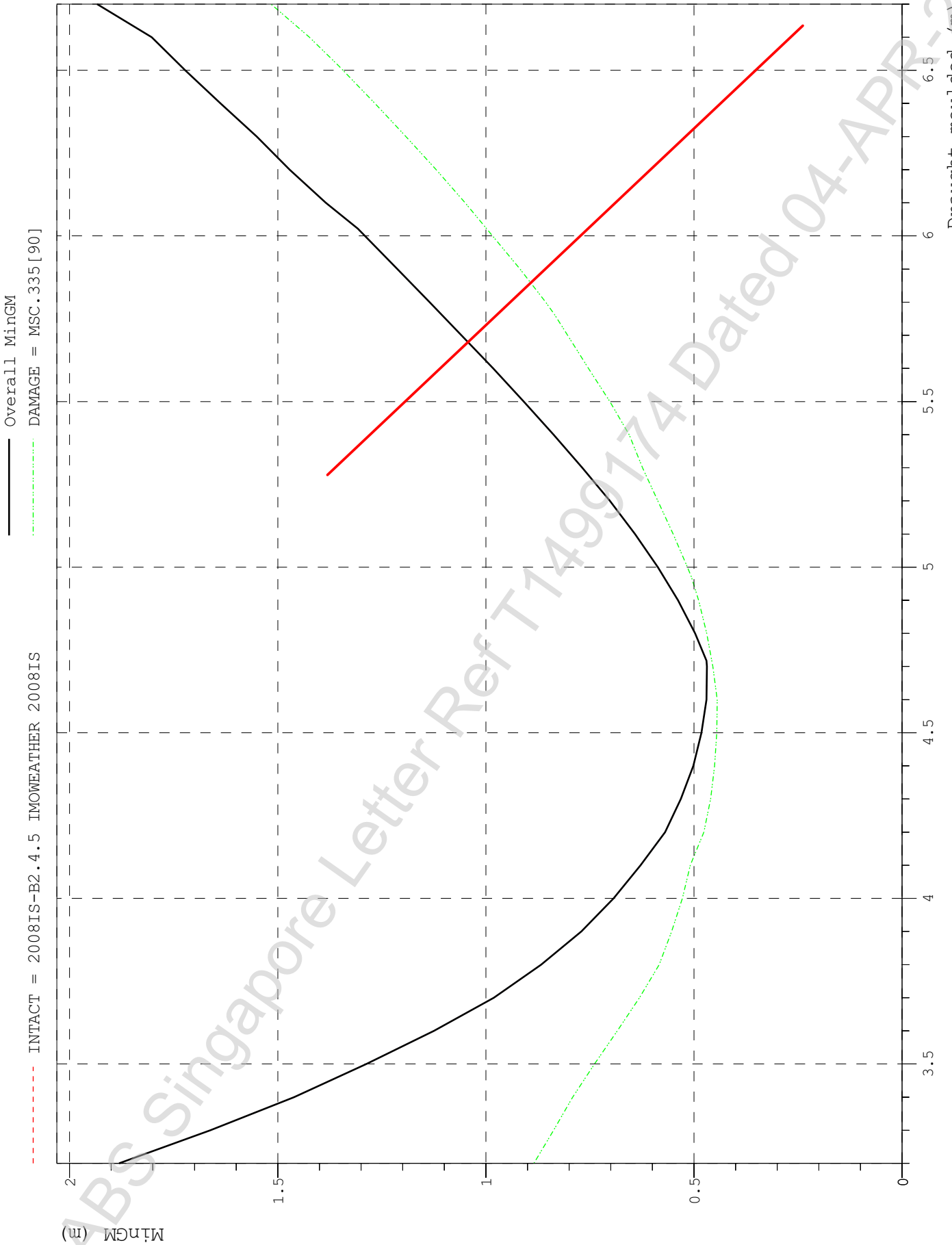
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8.6 GM LIMITS - TRIM = 0.5 m

T m	IGMLIM m	ICRLIM	DGMLIM m	DDAM	DCRLIM	GMLIM m
3.200	1.881	IMOWEATHER	0.884	D07	V.MINGZ.A469	1.881
3.300	1.662	IMOWEATHER	0.837	D07	V.MINGZ.A469	1.662
3.400	1.461	IMOWEATHER	0.791	D07	V.MINGZ.A469	1.461
3.500	1.287	IMOWEATHER	0.739	D07	V.MINGZ.A469	1.287
3.600	1.124	IMOWEATHER	0.684	D07	V.MINGZ.A469	1.124
3.700	0.981	IMOWEATHER	0.631	D07	V.MINGZ.A469	0.981
3.798	0.869	IMOWEATHER	0.585			0.869
3.800	0.867	IMOWEATHER	0.584	D07	V.RANGE.A469	0.867
3.900	0.771	IMOWEATHER	0.554	D07	V.RANGE.A469	0.771
4.000	0.693	IMOWEATHER	0.528	D07	V.RANGE.A469	0.693
4.100	0.629	IMOWEATHER	0.509	D07	V.RANGE.A469	0.629
4.200	0.569	IMOWEATHER	0.477	D07	V.RANGE.A469	0.569
4.300	0.532	IMOWEATHER	0.460	D07	V.RANGE.A469	0.532
4.400	0.502	IMOWEATHER	0.451	D07	V.RANGE.A469	0.502
4.500	0.482	IMOWEATHER	0.445	D07	V.RANGE.A469	0.482
4.600	0.470	IMOWEATHER	0.444	D07	V.RANGE.A469	0.470
4.700	0.469	IMOWEATHER	0.455	D07	V.RANGE.A469	0.469
4.718	0.470		0.457	D07	V.RANGE.A469	0.470
4.800	0.497	V.AREA3040	0.470	D07	V.RANGE.A469	0.497
4.900	0.539	V.AREA3040	0.489	D07	V.RANGE.A469	0.539
4.971	0.573	V.AREA3040	0.506			0.573
5.000	0.587	V.AREA3040	0.516	D07	V.MINGZ.A469	0.587
5.100	0.642	V.AREA3040	0.550	D07	V.MINGZ.A469	0.642
5.200	0.702	V.AREA3040	0.587	D07	V.MINGZ.A469	0.702
5.300	0.768	V.AREA3040	0.624	D07	V.MINGZ.A469	0.768
5.400	0.838	V.AREA3040	0.656	D07	V.MINGZ.A469	0.838
5.500	0.909	V.AREA3040	0.702	D07	V.MINGZ.A469	0.909
5.600	0.983	V.AREA3040	0.754	D07	V.MINGZ.A469	0.983
5.700	1.059	V.AREA3040	0.804	D07	V.MINGZ.A469	1.059
5.763	1.107	V.AREA3040	0.836			1.107
5.800	1.135	V.AREA3040	0.856	D10	V.RANGE.A469	1.135
5.900	1.212	V.AREA3040	0.918	D10	V.RANGE.A469	1.212
6.000	1.290	V.AREA3040	0.984	D10	V.RANGE.A469	1.290
6.022	1.308		0.998	D10	V.RANGE.A469	1.308
6.100	1.384	V.AREA15-30	1.050	D10	V.RANGE.A469	1.384
6.200	1.471	V.AREA15-30	1.119	D10	V.RANGE.A469	1.471
6.300	1.550	V.AREA15-30	1.192	D10	V.RANGE.A469	1.550
6.400	1.638	V.AREA15-30	1.267	D10	V.RANGE.A469	1.638
6.500	1.722	V.AREA15-30	1.343	D10	V.RANGE.A469	1.722
6.513	1.733	V.AREA15-30	1.353			1.733
6.600	1.803	V.AREA15-30	1.424	D12	V.RANGE.A469	1.803
6.700	1.934	V.AREA15-30	1.516	D12	V.RANGE.A469	1.934

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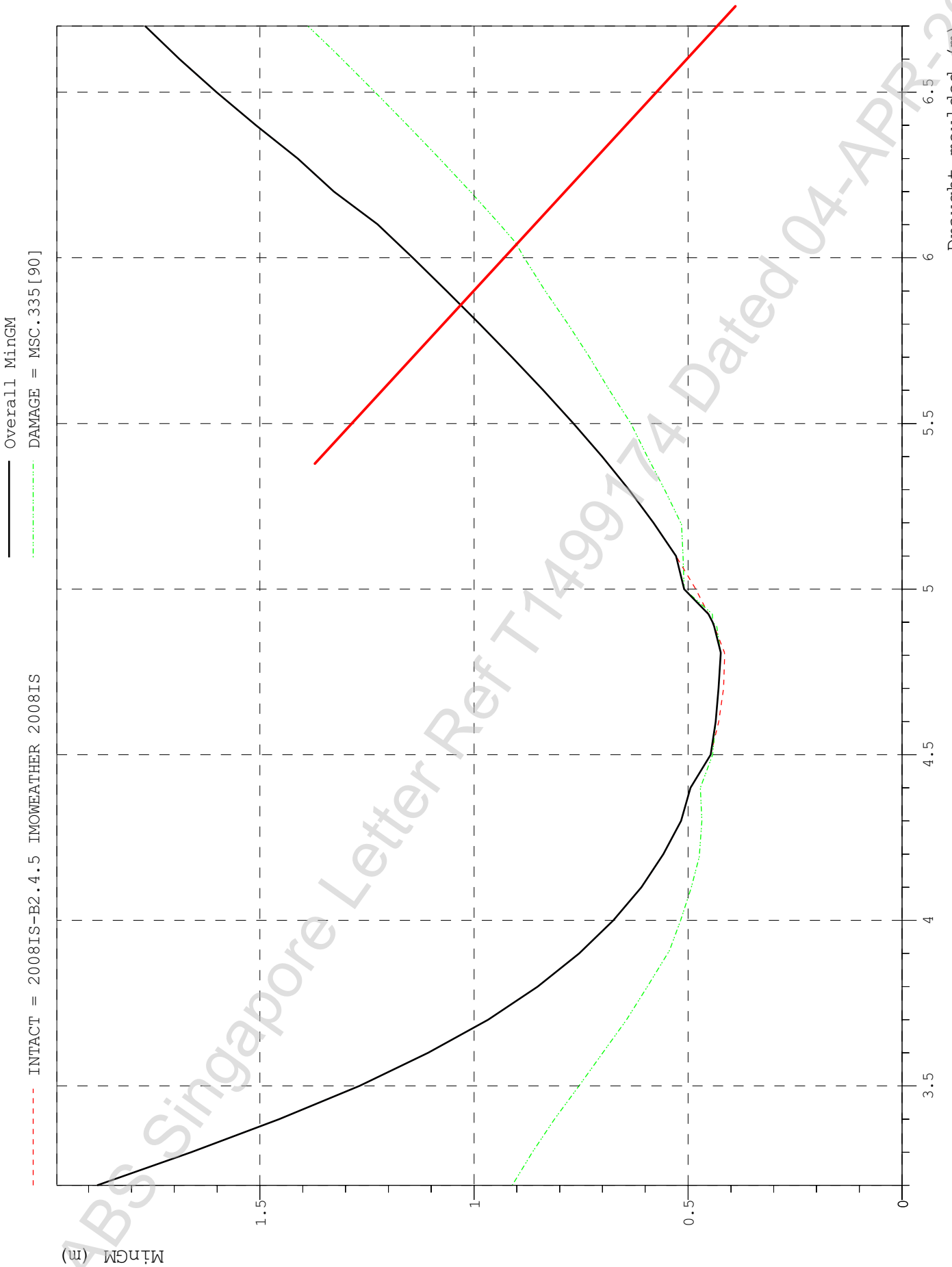
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8.7 GM LIMITS - TRIM = 1 m

T m	IGMLIM m	ICRLIM	DGMLIM m	DDAM	DCRLIM	GMLIM m
3.200	1.880	IMOWEATHER	0.911	D07	V.MINGZ.A469	1.880
3.300	1.661	IMOWEATHER	0.863	D07	V.MINGZ.A469	1.661
3.400	1.455	IMOWEATHER	0.812	D07	V.MINGZ.A469	1.455
3.500	1.267	IMOWEATHER	0.755	D07	V.MINGZ.A469	1.267
3.600	1.107	IMOWEATHER	0.699	D07	V.MINGZ.A469	1.107
3.700	0.967	IMOWEATHER	0.643	D07	V.MINGZ.A469	0.967
3.800	0.852	IMOWEATHER	0.595	D07	V.MINGZ.A469	0.852
3.900	0.754	IMOWEATHER	0.548	D07	V.MINGZ.A469	0.754
3.909	0.747	IMOWEATHER	0.544			0.747
4.000	0.674	IMOWEATHER	0.517	D07	V.RANGE.A469	0.674
4.100	0.609	IMOWEATHER	0.492	D07	V.RANGE.A469	0.609
4.192	0.562	IMOWEATHER	0.474			0.562
4.200	0.558	IMOWEATHER	0.474	D07	V.MAXHEEL.A4.	0.558
4.300	0.516	IMOWEATHER	0.468	D07	V.MAXHEEL.A4.	0.516
4.400	0.494	IMOWEATHER	0.471	D07	V.MAXHEEL.A4.	0.494
4.500	0.447	IMOWEATHER	0.443	D07	V.MAXHEEL.A4.	0.447
4.600	0.428	IMOWEATHER	0.435	D07	V.MAXHEEL.A4.	0.435
4.700	0.417	IMOWEATHER	0.429	D07	V.MAXHEEL.A4.	0.429
4.800	0.414	IMOWEATHER	0.424	D07	V.MAXHEEL.A4.	0.424
4.805	0.415	IMOWEATHER	0.424			0.424
4.809	0.415		0.424			0.424
4.888	0.439		0.433			0.439
4.900	0.443	V.AREA3040	0.443	D05	V.MAXHEEL.A4.	0.443
4.925	0.452	V.AREA3040	0.443			0.452
5.000	0.482	V.AREA3040	0.509	D07	V.MAXHEEL.A4.	0.509
5.100	0.528	V.AREA3040	0.512	D07	V.MAXHEEL.A4.	0.528
5.196	0.578	V.AREA3040	0.515			0.578
5.200	0.581	V.AREA3040	0.517	D07	V.MINGZ.A469	0.581
5.300	0.638	V.AREA3040	0.555	D07	V.MINGZ.A469	0.638
5.400	0.700	V.AREA3040	0.595	D07	V.MINGZ.A469	0.700
5.500	0.767	V.AREA3040	0.634	D07	V.MINGZ.A469	0.767
5.600	0.838	V.AREA3040	0.684	D07	V.MINGZ.A469	0.838
5.700	0.911	V.AREA3040	0.730	D07	V.MINGZ.A469	0.911
5.800	0.987	V.AREA3040	0.780	D07	V.MINGZ.A469	0.987
5.900	1.064	V.AREA3040	0.834	D07	V.MINGZ.A469	1.064
6.000	1.143	V.AREA3040	0.883	D07	V.MINGZ.A469	1.143
6.047	1.181	V.AREA3040	0.904			1.181
6.100	1.224	V.AREA3040	0.939	D10	V.RANGE.A469	1.224
6.102	1.225		0.940	D10	V.RANGE.A469	1.225
6.200	1.327	V.AREA15-30	1.008	D10	V.RANGE.A469	1.327
6.300	1.412	V.AREA15-30	1.081	D10	V.RANGE.A469	1.412
6.400	1.509	V.AREA15-30	1.155	D10	V.RANGE.A469	1.509
6.500	1.601	V.AREA15-30	1.231	D10	V.RANGE.A469	1.601
6.600	1.688	V.AREA15-30	1.308	D10	V.RANGE.A469	1.688
6.622	1.705	V.AREA15-30	1.325			1.705
6.700	1.768	V.AREA15-30	1.388	D12	V.RANGE.A469	1.768

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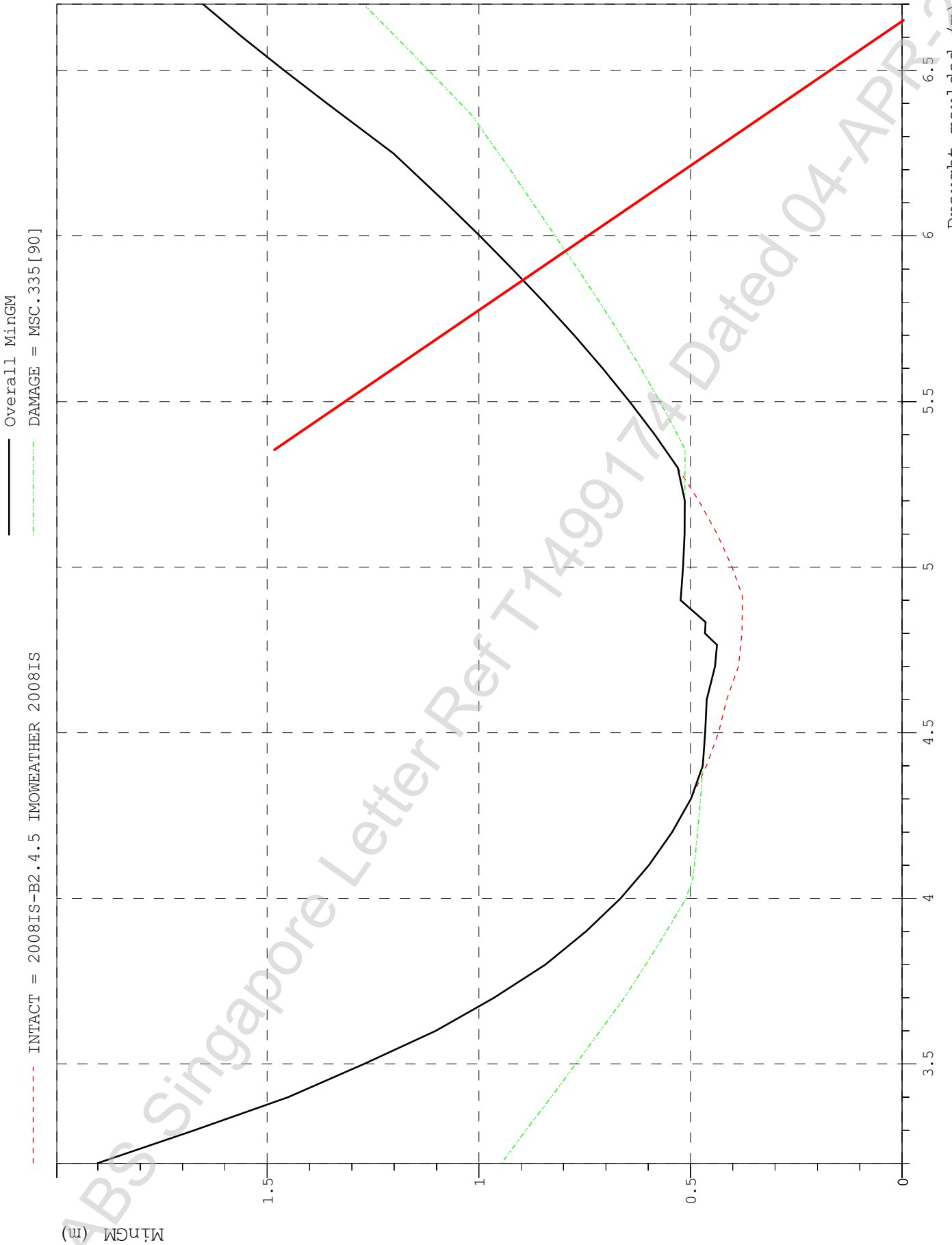
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8.8 GM LIMITS - TRIM = 1.5 m

T m	IGMLIM m	ICRLIM	DGMLIM m	DDAM	DCRLIM	GMLIM m
3.200	1.902	IMOWEATHER	0.947	D07	V.MINGZ.A469	1.902
3.300	1.672	IMOWEATHER	0.889	D07	V.MINGZ.A469	1.672
3.400	1.450	IMOWEATHER	0.829	D07	V.MINGZ.A469	1.450
3.500	1.271	IMOWEATHER	0.771	D07	V.MINGZ.A469	1.271
3.600	1.102	IMOWEATHER	0.713	D07	V.MINGZ.A469	1.102
3.700	0.964	IMOWEATHER	0.656	D07	V.MINGZ.A469	0.964
3.800	0.843	IMOWEATHER	0.605	D07	V.MINGZ.A469	0.843
3.900	0.747	IMOWEATHER	0.557	D07	V.MINGZ.A469	0.747
3.995	0.669	IMOWEATHER	0.513			0.669
4.000	0.665	IMOWEATHER	0.511	D07	V.RANGE.A469	0.665
4.058	0.626	IMOWEATHER	0.495			0.626
4.100	0.598	IMOWEATHER	0.491	D07	V.MAXHEEL.A4.	0.598
4.200	0.544	IMOWEATHER	0.483	D07	V.MAXHEEL.A4.	0.544
4.300	0.499	IMOWEATHER	0.476	D07	V.MAXHEEL.A4.	0.499
4.400	0.462	IMOWEATHER	0.471	D07	V.MAXHEEL.A4.	0.471
4.500	0.434	IMOWEATHER	0.466	D07	V.MAXHEEL.A4.	0.466
4.600	0.415	IMOWEATHER	0.462	D07	V.MAXHEEL.A4.	0.462
4.700	0.386	IMOWEATHER	0.442	D07	V.MAXHEEL.A4.	0.442
4.765	0.381	IMOWEATHER	0.437			0.437
4.800	0.378	IMOWEATHER	0.466	D05	V.MAXHEEL.A4.	0.466
4.834	0.378	IMOWEATHER	0.465			0.465
4.900	0.377	IMOWEATHER	0.523	D07	V.MAXHEEL.A4.	0.523
4.915	0.378		0.523	D07	V.MAXHEEL.A4.	0.523
5.000	0.401	V.AREA3040	0.518	D07	V.MAXHEEL.A4.	0.518
5.100	0.437	V.AREA3040	0.514	D07	V.MAXHEEL.A4.	0.514
5.200	0.480	V.AREA3040	0.513	D07	V.MAXHEEL.A4.	0.513
5.300	0.530	V.AREA3040	0.513	D07	V.MAXHEEL.A4.	0.530
5.356	0.560	V.AREA3040	0.513			0.560
5.400	0.584	V.AREA3040	0.531	D07	V.MINGZ.A469	0.584
5.500	0.644	V.AREA3040	0.573	D07	V.MINGZ.A469	0.644
5.600	0.707	V.AREA3040	0.617	D07	V.MINGZ.A469	0.707
5.700	0.775	V.AREA3040	0.664	D07	V.MINGZ.A469	0.775
5.800	0.846	V.AREA3040	0.715	D07	V.MINGZ.A469	0.846
5.900	0.920	V.AREA3040	0.766	D07	V.MINGZ.A469	0.920
6.000	0.998	V.AREA3040	0.820	D07	V.MINGZ.A469	0.998
6.100	1.078	V.AREA3040	0.873	D07	V.MINGZ.A469	1.078
6.200	1.161	V.AREA3040	0.926	D07	V.MINGZ.A469	1.161
6.249	1.202		0.953	D07	V.MINGZ.A469	1.202
6.300	1.255	V.AREA15-30	0.981	D07	V.MINGZ.A469	1.255
6.360	1.317	V.AREA15-30	1.013			1.317
6.400	1.359	V.AREA15-30	1.043	D10	V.RANGE.A469	1.359
6.500	1.460	V.AREA15-30	1.119	D10	V.RANGE.A469	1.460
6.600	1.559	V.AREA15-30	1.195	D10	V.RANGE.A469	1.559
6.700	1.652	V.AREA15-30	1.273	D10	V.RANGE.A469	1.652

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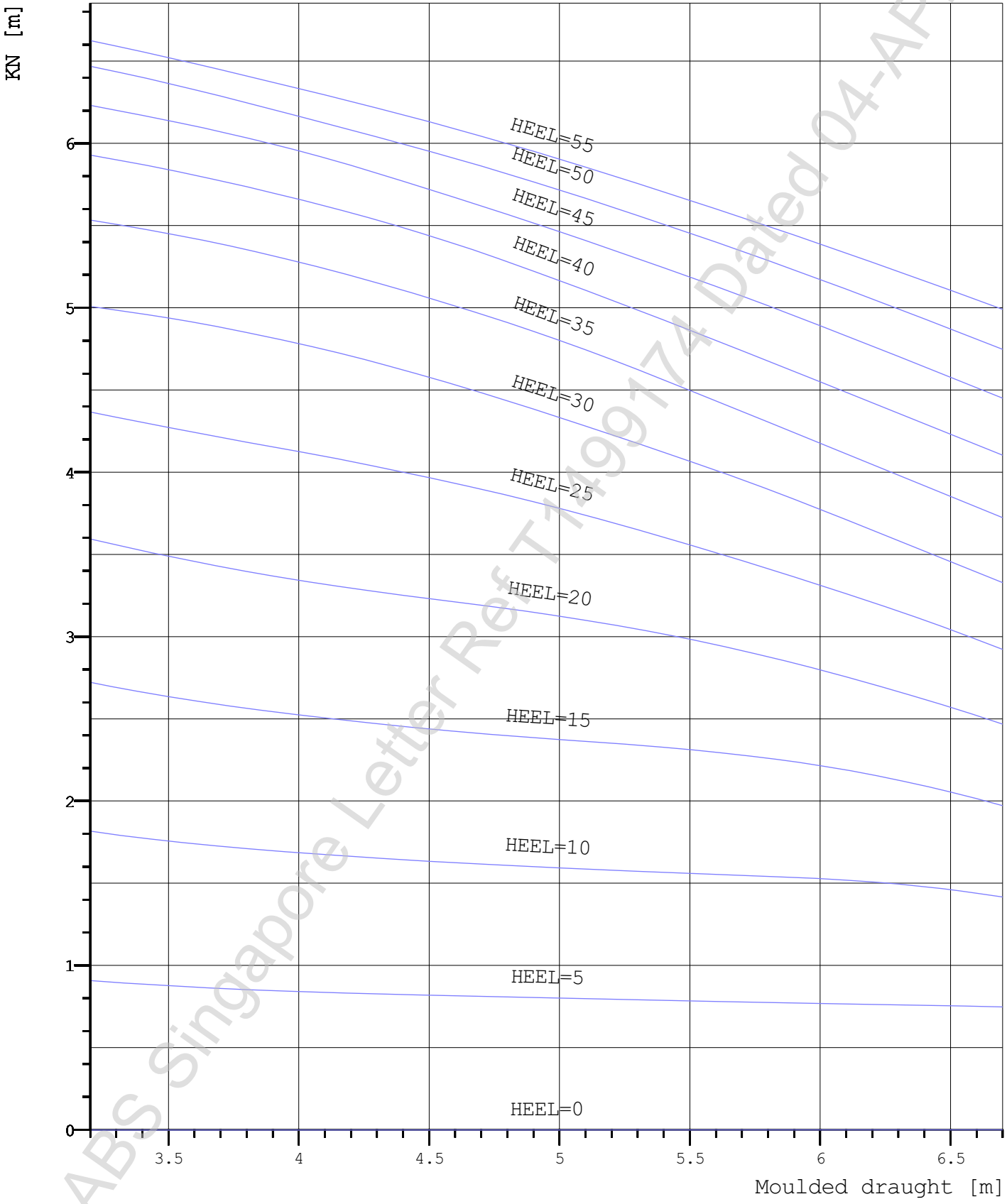


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9 CROSS CURVES

9.1 CROSS CURVES, TRIM=-1.5 m



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CROSS CURVES TABLE, TRIM=-1.5 m

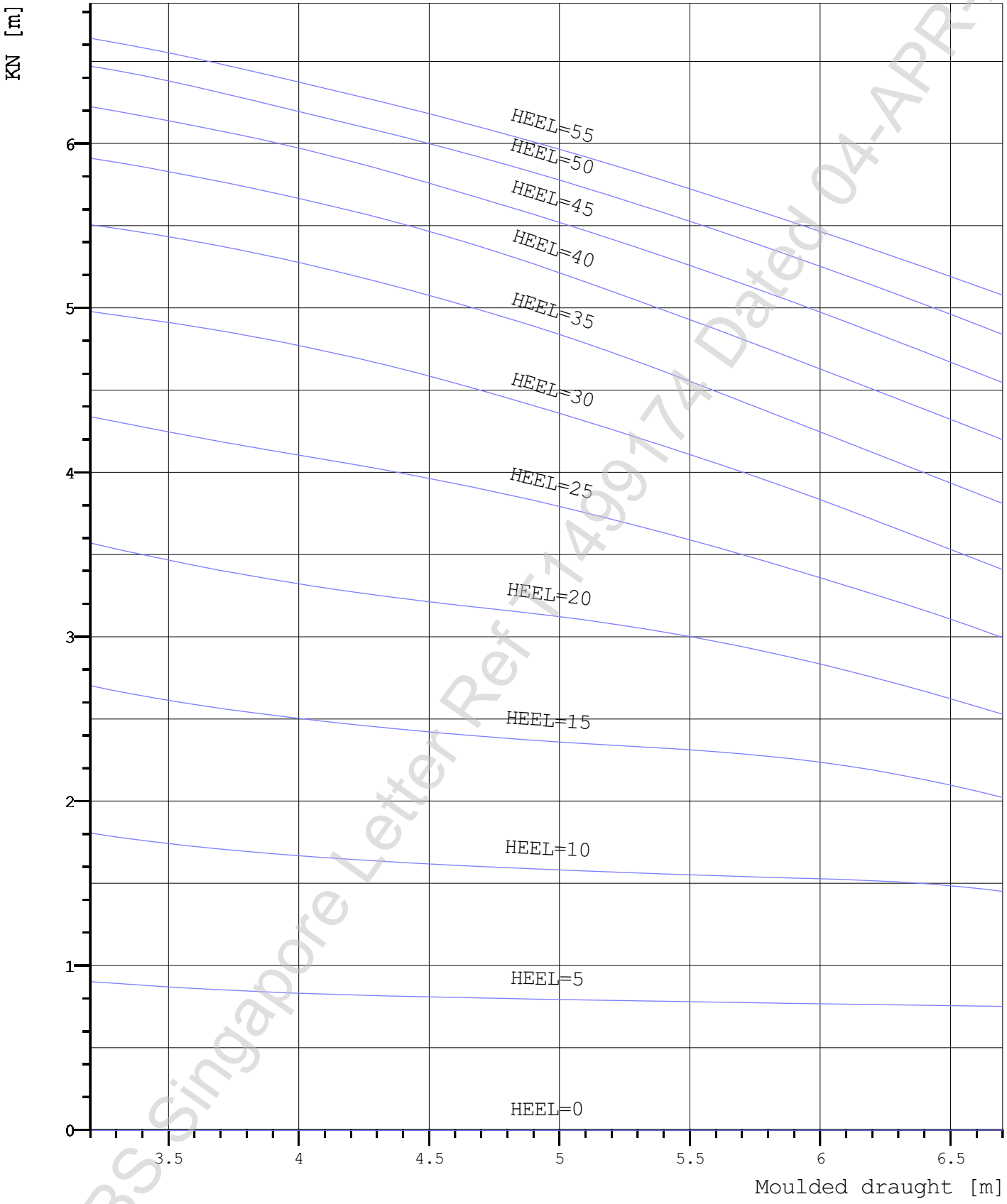
T m	HEEL=0 m	HEEL=5 m	HEEL=10 m	HEEL=15 m	HEEL=20 m	HEEL=25 m	HEEL=30 m	HEEL=35 m	HEEL=40 m	HEEL=45 m	HEEL=50 m	HEEL=55 m
3.200	-0.002	0.907	1.816	2.721	3.594	4.366	5.008	5.533	5.929	6.232	6.468	6.624
3.300	-0.002	0.896	1.794	2.690	3.557	4.334	4.985	5.507	5.901	6.201	6.436	6.592
3.400	-0.002	0.886	1.774	2.662	3.522	4.303	4.962	5.480	5.871	6.170	6.402	6.558
3.500	-0.002	0.876	1.756	2.635	3.489	4.272	4.938	5.451	5.840	6.138	6.365	6.522
3.600	-0.002	0.868	1.740	2.610	3.457	4.242	4.911	5.421	5.807	6.105	6.327	6.485
3.700	-0.002	0.860	1.725	2.586	3.426	4.212	4.881	5.388	5.772	6.071	6.287	6.448
3.800	-0.002	0.853	1.711	2.564	3.396	4.182	4.850	5.353	5.736	6.034	6.247	6.410
3.900	-0.002	0.847	1.698	2.543	3.369	4.154	4.817	5.317	5.699	5.995	6.206	6.371
4.000	-0.002	0.841	1.686	2.523	3.343	4.125	4.782	5.278	5.660	5.954	6.165	6.333
4.100	-0.002	0.836	1.674	2.505	3.318	4.096	4.745	5.238	5.620	5.911	6.123	6.294
4.200	-0.002	0.831	1.663	2.487	3.295	4.065	4.707	5.196	5.578	5.866	6.082	6.255
4.300	-0.002	0.827	1.653	2.470	3.272	4.034	4.666	5.152	5.534	5.819	6.039	6.215
4.400	-0.001	0.823	1.643	2.454	3.251	4.001	4.622	5.106	5.487	5.770	5.996	6.174
4.500	-0.001	0.819	1.634	2.439	3.231	3.967	4.577	5.060	5.439	5.721	5.953	6.131
4.600	-0.001	0.815	1.625	2.424	3.211	3.932	4.531	5.011	5.388	5.670	5.908	6.088
4.700	-0.001	0.811	1.616	2.410	3.191	3.896	4.483	4.961	5.335	5.620	5.862	6.043
4.800	-0.001	0.808	1.608	2.397	3.170	3.859	4.434	4.910	5.280	5.568	5.815	5.998
4.900	-0.001	0.804	1.600	2.385	3.148	3.820	4.384	4.857	5.223	5.516	5.766	5.951
5.000	-0.001	0.801	1.593	2.374	3.124	3.780	4.333	4.801	5.165	5.463	5.717	5.904
5.100	-0.001	0.797	1.585	2.362	3.099	3.738	4.281	4.744	5.106	5.409	5.666	5.856
5.200	-0.001	0.794	1.578	2.351	3.073	3.695	4.228	4.685	5.046	5.355	5.614	5.807
5.300	-0.001	0.790	1.572	2.340	3.045	3.650	4.175	4.624	4.986	5.299	5.562	5.756
5.400	-0.001	0.787	1.565	2.327	3.016	3.605	4.121	4.562	4.925	5.244	5.508	5.705
5.500	-0.001	0.784	1.559	2.313	2.984	3.558	4.067	4.498	4.864	5.187	5.454	5.653
5.600	-0.001	0.780	1.553	2.297	2.951	3.511	4.011	4.434	4.802	5.130	5.399	5.601
5.700	-0.001	0.777	1.547	2.279	2.915	3.462	3.954	4.369	4.739	5.072	5.343	5.549
5.800	-0.001	0.774	1.541	2.260	2.878	3.413	3.896	4.305	4.677	5.012	5.287	5.495
5.900	-0.001	0.771	1.535	2.239	2.838	3.363	3.835	4.240	4.614	4.952	5.230	5.442
6.000	-0.001	0.768	1.528	2.215	2.798	3.312	3.774	4.175	4.550	4.892	5.172	5.387
6.100	-0.001	0.765	1.519	2.189	2.755	3.260	3.711	4.110	4.487	4.830	5.113	5.332
6.200	-0.001	0.763	1.508	2.159	2.711	3.208	3.648	4.046	4.423	4.768	5.054	5.277
6.300	-0.001	0.760	1.494	2.127	2.665	3.155	3.584	3.981	4.359	4.705	4.994	5.221
6.400	-0.001	0.757	1.479	2.092	2.618	3.099	3.520	3.916	4.296	4.642	4.933	5.164
6.500	-0.001	0.755	1.461	2.054	2.570	3.042	3.456	3.851	4.232	4.578	4.871	5.107

T m	HEEL=0 m	HEEL=5 m	HEEL=10 m	HEEL=15 m	HEEL=20 m	HEEL=25 m	HEEL=30 m	HEEL=35 m	HEEL=40 m	HEEL=45 m	HEEL=50 m	HEEL=55 m
6.600	-0.001	0.752	1.440	2.014	2.519	2.983	3.392	3.787	4.168	4.515	4.810	5.049
6.700	-0.001	0.747	1.415	1.971	2.468	2.921	3.328	3.723	4.104	4.451	4.747	4.991

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9.2 CROSS CURVES, TRIM=-1 m



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CROSS CURVES TABLE, TRIM=-1 m

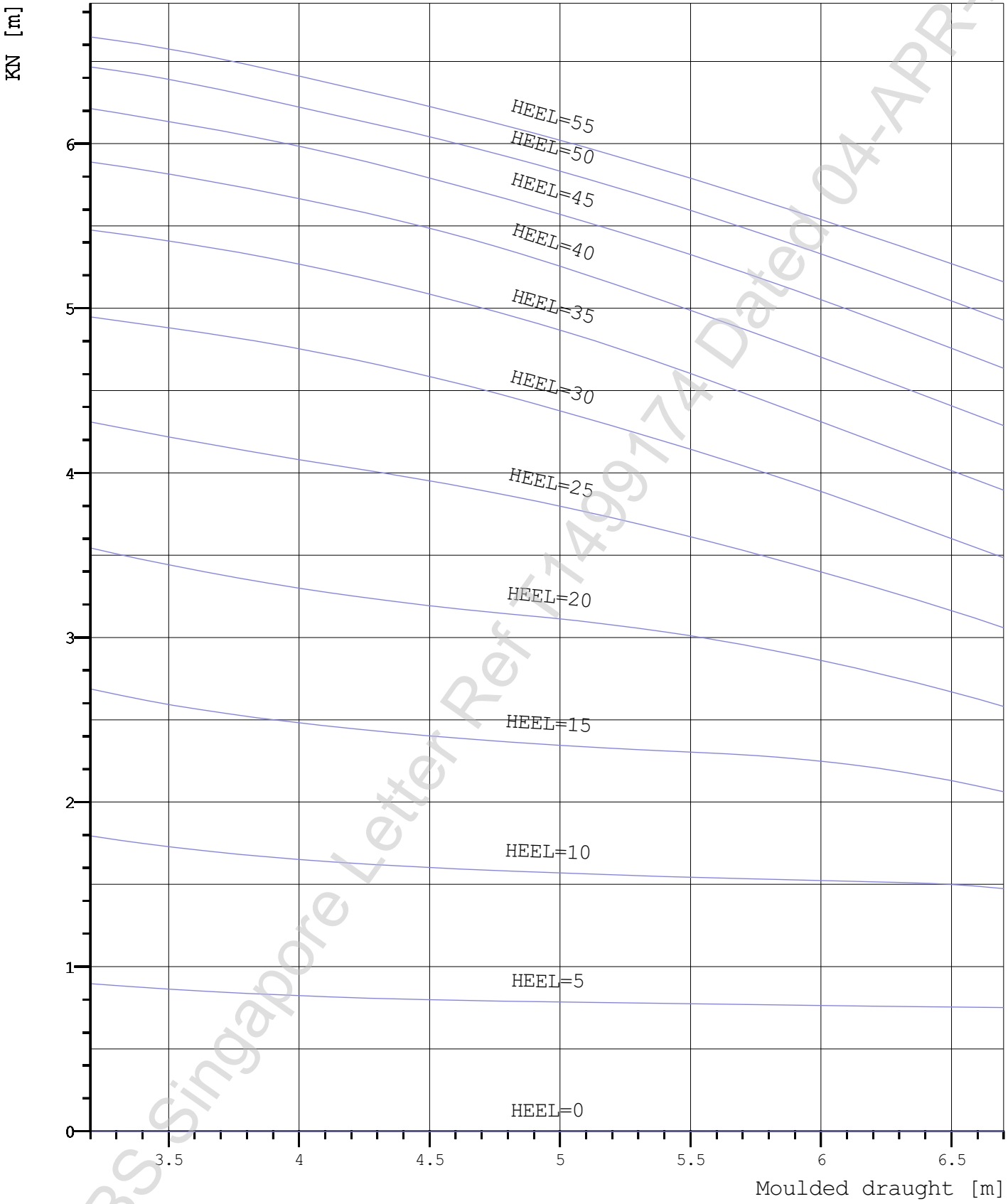
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3.200	-0.002	0.902	1.805	2.702	3.570	4.338	4.978	5.506	5.911	6.225	6.470	6.640
3.300	-0.002	0.891	1.782	2.670	3.534	4.307	4.956	5.483	5.886	6.197	6.443	6.613
3.400	-0.002	0.880	1.761	2.641	3.499	4.277	4.934	5.459	5.858	6.168	6.413	6.583
3.500	-0.002	0.870	1.741	2.613	3.466	4.246	4.911	5.433	5.830	6.138	6.381	6.552
3.600	-0.002	0.861	1.724	2.588	3.434	4.216	4.887	5.405	5.799	6.108	6.346	6.519
3.700	-0.002	0.852	1.708	2.564	3.404	4.187	4.861	5.376	5.768	6.077	6.309	6.484
3.800	-0.002	0.845	1.694	2.542	3.375	4.159	4.833	5.345	5.735	6.044	6.272	6.448
3.900	-0.002	0.838	1.680	2.522	3.348	4.131	4.803	5.312	5.701	6.009	6.234	6.412
4.000	-0.002	0.832	1.668	2.503	3.322	4.104	4.771	5.277	5.666	5.972	6.195	6.374
4.100	-0.002	0.826	1.657	2.484	3.297	4.078	4.738	5.240	5.629	5.933	6.157	6.337
4.200	-0.002	0.821	1.646	2.467	3.274	4.051	4.702	5.201	5.591	5.892	6.119	6.299
4.300	-0.002	0.817	1.636	2.451	3.252	4.023	4.665	5.161	5.552	5.849	6.080	6.261
4.400	-0.001	0.813	1.627	2.435	3.232	3.993	4.626	5.119	5.510	5.804	6.040	6.221
4.500	-0.001	0.809	1.618	2.421	3.212	3.963	4.586	5.076	5.466	5.759	6.000	6.181
4.600	-0.001	0.805	1.610	2.407	3.194	3.931	4.543	5.031	5.420	5.712	5.958	6.140
4.700	-0.001	0.802	1.602	2.394	3.177	3.898	4.499	4.985	5.371	5.665	5.915	6.098
4.800	-0.001	0.799	1.595	2.382	3.160	3.865	4.453	4.938	5.321	5.618	5.870	6.055
4.900	-0.001	0.796	1.588	2.371	3.142	3.829	4.406	4.889	5.268	5.569	5.825	6.010
5.000	-0.001	0.794	1.581	2.360	3.122	3.793	4.358	4.838	5.214	5.520	5.778	5.965
5.100	-0.001	0.791	1.575	2.349	3.102	3.755	4.310	4.785	5.158	5.470	5.730	5.919
5.200	-0.001	0.788	1.569	2.340	3.079	3.715	4.260	4.730	5.102	5.419	5.681	5.871
5.300	-0.001	0.785	1.563	2.331	3.055	3.674	4.210	4.673	5.045	5.366	5.631	5.823
5.400	-0.001	0.783	1.557	2.322	3.029	3.632	4.160	4.614	4.987	5.313	5.579	5.774
5.500	-0.001	0.780	1.552	2.312	3.001	3.589	4.108	4.554	4.929	5.259	5.527	5.724
5.600	-0.001	0.777	1.547	2.301	2.971	3.545	4.056	4.493	4.870	5.204	5.474	5.673
5.700	-0.001	0.775	1.541	2.288	2.940	3.500	4.003	4.431	4.810	5.148	5.420	5.622
5.800	-0.001	0.772	1.537	2.273	2.906	3.454	3.948	4.370	4.750	5.091	5.365	5.570
5.900	-0.001	0.769	1.532	2.256	2.871	3.407	3.892	4.308	4.690	5.033	5.310	5.518
6.000	-0.001	0.767	1.527	2.236	2.833	3.359	3.834	4.246	4.629	4.975	5.254	5.465
6.100	-0.001	0.764	1.522	2.214	2.794	3.310	3.774	4.184	4.568	4.915	5.197	5.412
6.200	-0.001	0.762	1.516	2.189	2.754	3.261	3.714	4.122	4.507	4.855	5.139	5.357
6.300	-0.001	0.760	1.508	2.161	2.712	3.211	3.653	4.059	4.446	4.794	5.080	5.303
6.400	-0.001	0.758	1.498	2.130	2.668	3.160	3.592	3.997	4.384	4.732	5.021	5.247

T m	HEEL=0 m	HEEL=5 m	HEEL=10 m	HEEL=15 m	HEEL=20 m	HEEL=25 m	HEEL=30 m	HEEL=35 m	HEEL=40 m	HEEL=45 m	HEEL=50 m	HEEL=55 m
6.500	-0.001	0.755	1.485	2.097	2.623	3.106	3.532	3.935	4.322	4.670	4.961	5.191
6.600	-0.001	0.753	1.469	2.061	2.576	3.051	3.470	3.873	4.260	4.608	4.900	5.135
6.700	-0.001	0.751	1.450	2.022	2.528	2.994	3.409	3.811	4.198	4.545	4.839	5.077

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9.3 CROSS CURVES, TRIM=-0.5 m



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CROSS CURVES TABLE, TRIM=-0.5 m

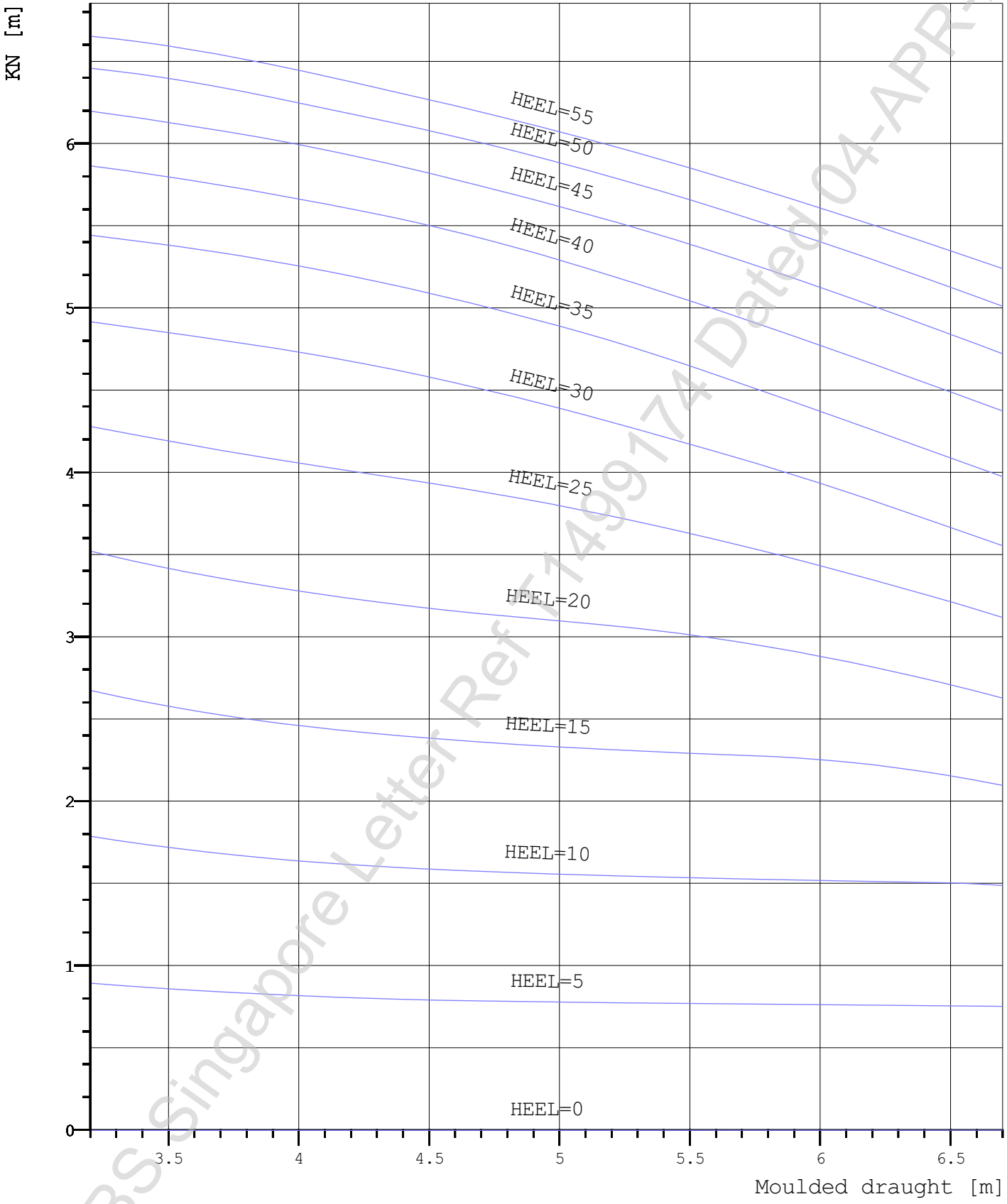
T m	HEEL=0 m	HEEL=5 m	HEEL=10 m	HEEL=15 m	HEEL=20 m	HEEL=25 m	HEEL=30 m	HEEL=35 m	HEEL=40 m	HEEL=45 m	HEEL=50 m	HEEL=55 m
3.200	-0.002	0.897	1.795	2.687	3.544	4.310	4.947	5.476	5.889	6.213	6.467	6.649
3.300	-0.002	0.885	1.771	2.653	3.508	4.279	4.925	5.455	5.866	6.188	6.444	6.627
3.400	-0.002	0.874	1.749	2.622	3.474	4.248	4.903	5.432	5.841	6.161	6.419	6.602
3.500	-0.002	0.864	1.729	2.593	3.441	4.218	4.881	5.409	5.815	6.134	6.391	6.575
3.600	-0.002	0.855	1.711	2.567	3.410	4.190	4.858	5.384	5.788	6.107	6.360	6.546
3.700	-0.002	0.846	1.694	2.543	3.381	4.161	4.835	5.358	5.759	6.079	6.328	6.515
3.800	-0.002	0.838	1.678	2.521	3.352	4.134	4.810	5.330	5.729	6.049	6.293	6.482
3.900	-0.002	0.830	1.664	2.500	3.325	4.107	4.782	5.300	5.698	6.018	6.258	6.447
4.000	-0.002	0.824	1.651	2.481	3.300	4.080	4.754	5.269	5.666	5.984	6.222	6.412
4.100	-0.002	0.817	1.640	2.463	3.276	4.055	4.723	5.235	5.633	5.949	6.187	6.375
4.200	-0.002	0.812	1.629	2.446	3.253	4.030	4.692	5.200	5.599	5.912	6.151	6.339
4.300	-0.002	0.807	1.619	2.431	3.232	4.005	4.658	5.164	5.564	5.874	6.116	6.302
4.400	-0.001	0.803	1.611	2.416	3.211	3.979	4.623	5.125	5.526	5.833	6.079	6.264
4.500	-0.001	0.799	1.602	2.402	3.193	3.952	4.586	5.086	5.487	5.792	6.041	6.226
4.600	-0.001	0.796	1.595	2.389	3.175	3.923	4.547	5.045	5.445	5.749	6.002	6.187
4.700	-0.001	0.793	1.587	2.377	3.158	3.894	4.507	5.002	5.400	5.706	5.962	6.147
4.800	-0.001	0.790	1.581	2.366	3.143	3.863	4.465	4.959	5.354	5.661	5.920	6.106
4.900	-0.001	0.788	1.575	2.355	3.128	3.831	4.422	4.914	5.306	5.617	5.877	6.064
5.000	-0.001	0.785	1.569	2.345	3.113	3.798	4.377	4.867	5.255	5.571	5.833	6.021
5.100	-0.001	0.783	1.563	2.336	3.095	3.763	4.332	4.819	5.204	5.524	5.788	5.976
5.200	-0.001	0.781	1.558	2.327	3.077	3.728	4.286	4.768	5.151	5.477	5.741	5.931
5.300	-0.001	0.779	1.553	2.318	3.056	3.690	4.239	4.715	5.098	5.428	5.694	5.885
5.400	-0.001	0.777	1.548	2.310	3.034	3.652	4.191	4.660	5.044	5.378	5.645	5.838
5.500	-0.001	0.775	1.543	2.303	3.010	3.612	4.143	4.603	4.988	5.326	5.595	5.790
5.600	-0.001	0.773	1.539	2.296	2.984	3.572	4.094	4.546	4.932	5.273	5.544	5.741
5.700	-0.001	0.771	1.534	2.288	2.956	3.530	4.045	4.488	4.876	5.219	5.492	5.691
5.800	-0.001	0.769	1.530	2.277	2.926	3.487	3.994	4.429	4.819	5.165	5.439	5.641
5.900	-0.001	0.767	1.526	2.264	2.895	3.443	3.941	4.370	4.761	5.109	5.385	5.590
6.000	-0.001	0.765	1.523	2.249	2.861	3.399	3.887	4.311	4.703	5.053	5.331	5.539
6.100	-0.001	0.763	1.519	2.231	2.826	3.353	3.831	4.252	4.645	4.995	5.275	5.487
6.200	-0.001	0.761	1.515	2.210	2.789	3.307	3.774	4.192	4.587	4.937	5.219	5.434
6.300	-0.001	0.759	1.512	2.186	2.750	3.260	3.717	4.133	4.527	4.878	5.162	5.381

T m	HEEL=0 m	HEEL=5 m	HEEL=10 m	HEEL=15 m	HEEL=20 m	HEEL=25 m	HEEL=30 m	HEEL=35 m	HEEL=40 m	HEEL=45 m	HEEL=50 m	HEEL=55 m
6.400	-0.001	0.757	1.507	2.159	2.710	3.212	3.659	4.073	4.468	4.818	5.104	5.326
6.500	-0.001	0.755	1.499	2.130	2.669	3.163	3.601	4.014	4.408	4.758	5.046	5.272
6.600	-0.001	0.753	1.488	2.098	2.626	3.112	3.543	3.954	4.348	4.697	4.987	5.216
6.700	-0.001	0.752	1.474	2.063	2.581	3.059	3.484	3.894	4.288	4.636	4.927	5.160

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9.4 CROSS CURVES, TRIM=0 m



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CROSS CURVES TABLE, TRIM=0 m

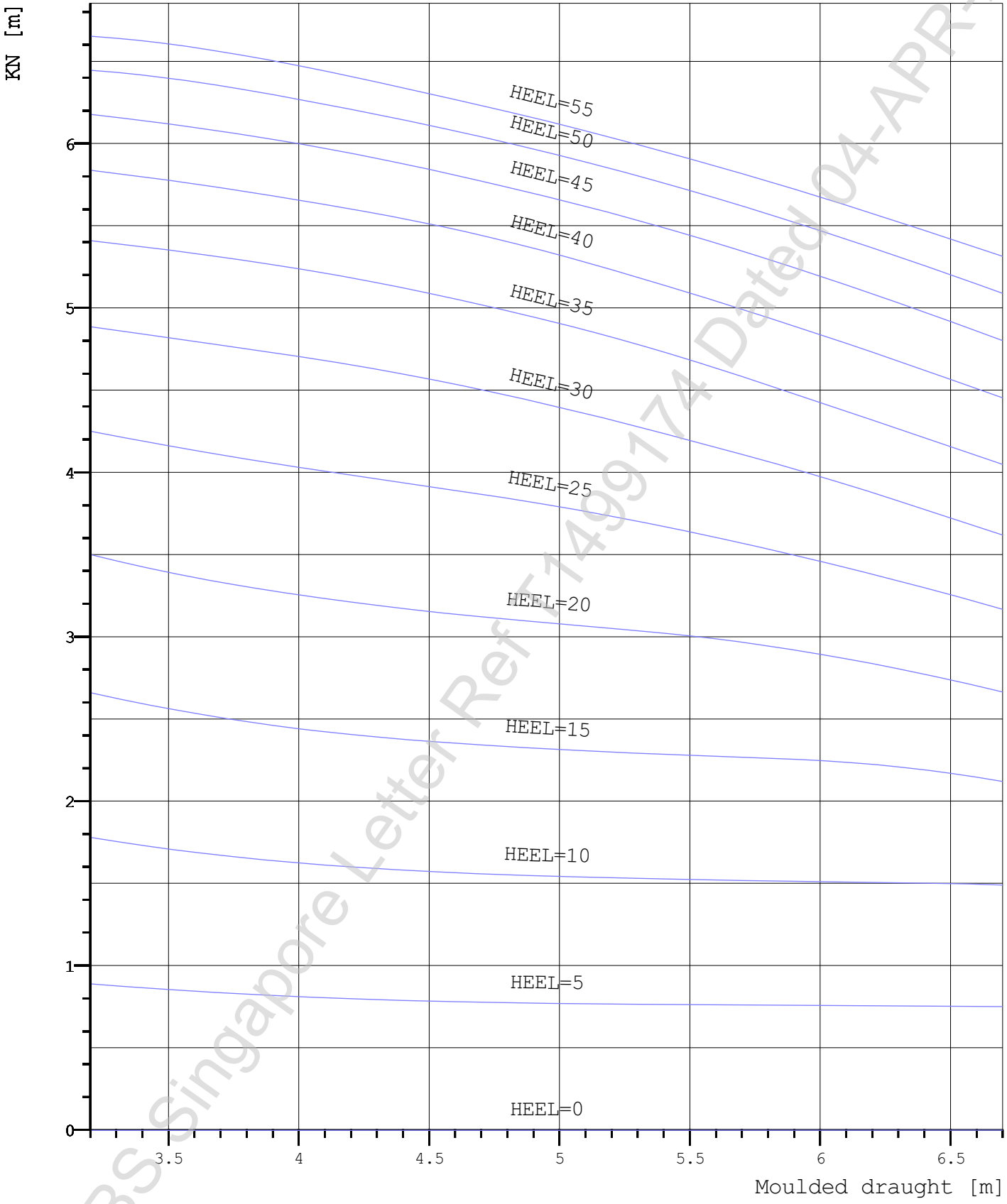
T m	HEEL=0 m	HEEL=5 m	HEEL=10 m	HEEL=15 m	HEEL=20 m	HEEL=25 m	HEEL=30 m	HEEL=35 m	HEEL=40 m	HEEL=45 m	HEEL=50 m	HEEL=55 m
3.200	-0.002	0.893	1.787	2.673	3.521	4.279	4.916	5.443	5.864	6.197	6.458	6.653
3.300	-0.002	0.880	1.762	2.639	3.483	4.249	4.894	5.423	5.843	6.175	6.440	6.636
3.400	-0.002	0.869	1.739	2.607	3.448	4.219	4.872	5.403	5.821	6.152	6.419	6.616
3.500	-0.002	0.859	1.719	2.577	3.416	4.190	4.850	5.382	5.797	6.128	6.396	6.593
3.600	-0.002	0.849	1.699	2.549	3.385	4.162	4.828	5.359	5.773	6.103	6.370	6.567
3.700	-0.002	0.840	1.681	2.524	3.356	4.134	4.805	5.336	5.747	6.078	6.342	6.540
3.800	-0.002	0.832	1.665	2.500	3.328	4.107	4.781	5.311	5.720	6.051	6.311	6.510
3.900	-0.002	0.824	1.650	2.479	3.302	4.081	4.756	5.284	5.691	6.023	6.279	6.478
4.000	-0.002	0.817	1.636	2.460	3.278	4.056	4.731	5.255	5.662	5.993	6.246	6.445
4.100	-0.002	0.810	1.624	2.442	3.254	4.031	4.704	5.225	5.633	5.962	6.214	6.410
4.200	-0.002	0.805	1.613	2.425	3.232	4.007	4.675	5.194	5.602	5.928	6.181	6.375
4.300	-0.002	0.799	1.603	2.410	3.211	3.983	4.645	5.161	5.570	5.894	6.148	6.339
4.400	-0.002	0.795	1.595	2.396	3.191	3.959	4.613	5.126	5.537	5.857	6.113	6.303
4.500	-0.001	0.791	1.587	2.383	3.173	3.935	4.580	5.090	5.501	5.819	6.078	6.266
4.600	-0.001	0.787	1.579	2.371	3.156	3.909	4.544	5.052	5.464	5.780	6.042	6.229
4.700	-0.001	0.784	1.573	2.359	3.140	3.883	4.508	5.013	5.424	5.741	6.004	6.191
4.800	-0.001	0.782	1.566	2.349	3.125	3.856	4.470	4.973	5.382	5.700	5.965	6.152
4.900	-0.001	0.779	1.561	2.339	3.110	3.827	4.430	4.932	5.337	5.659	5.925	6.112
5.000	-0.001	0.777	1.555	2.330	3.096	3.797	4.390	4.889	5.291	5.617	5.883	6.071
5.100	-0.001	0.775	1.551	2.321	3.082	3.766	4.348	4.845	5.243	5.573	5.840	6.029
5.200	-0.001	0.774	1.546	2.313	3.067	3.733	4.305	4.799	5.195	5.529	5.796	5.986
5.300	-0.001	0.772	1.542	2.305	3.050	3.699	4.261	4.750	5.145	5.483	5.751	5.942
5.400	-0.001	0.770	1.538	2.298	3.032	3.664	4.217	4.699	5.094	5.436	5.705	5.897
5.500	-0.001	0.769	1.534	2.291	3.011	3.628	4.172	4.647	5.042	5.387	5.657	5.851
5.600	-0.001	0.768	1.530	2.285	2.989	3.591	4.126	4.593	4.990	5.337	5.609	5.804
5.700	-0.001	0.766	1.526	2.279	2.965	3.553	4.080	4.538	4.936	5.286	5.559	5.757
5.800	-0.001	0.765	1.523	2.272	2.939	3.514	4.033	4.482	4.882	5.233	5.508	5.708
5.900	-0.001	0.763	1.520	2.263	2.911	3.473	3.984	4.427	4.828	5.180	5.456	5.658
6.000	-0.001	0.762	1.517	2.252	2.881	3.432	3.934	4.371	4.773	5.125	5.403	5.608
6.100	-0.001	0.760	1.514	2.238	2.849	3.390	3.882	4.314	4.717	5.070	5.349	5.558
6.200	-0.001	0.758	1.511	2.221	2.816	3.347	3.829	4.258	4.661	5.014	5.295	5.506

T m	HEEL=0 m	HEEL=5 m	HEEL=10 m	HEEL=15 m	HEEL=20 m	HEEL=25 m	HEEL=30 m	HEEL=35 m	HEEL=40 m	HEEL=45 m	HEEL=50 m	HEEL=55 m
6.300	-0.001	0.757	1.508	2.201	2.781	3.303	3.774	4.201	4.604	4.956	5.239	5.454
6.400	-0.001	0.755	1.505	2.179	2.745	3.258	3.720	4.144	4.547	4.899	5.183	5.401
6.500	-0.001	0.754	1.502	2.154	2.707	3.213	3.665	4.087	4.489	4.840	5.126	5.348
6.600	-0.001	0.752	1.496	2.126	2.667	3.166	3.609	4.030	4.431	4.781	5.069	5.294
6.700	-0.001	0.751	1.487	2.096	2.626	3.116	3.554	3.973	4.372	4.721	5.010	5.239

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9.5 CROSS CURVES, TRIM=0.5 m



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CROSS CURVES TABLE, TRIM=0.5 m

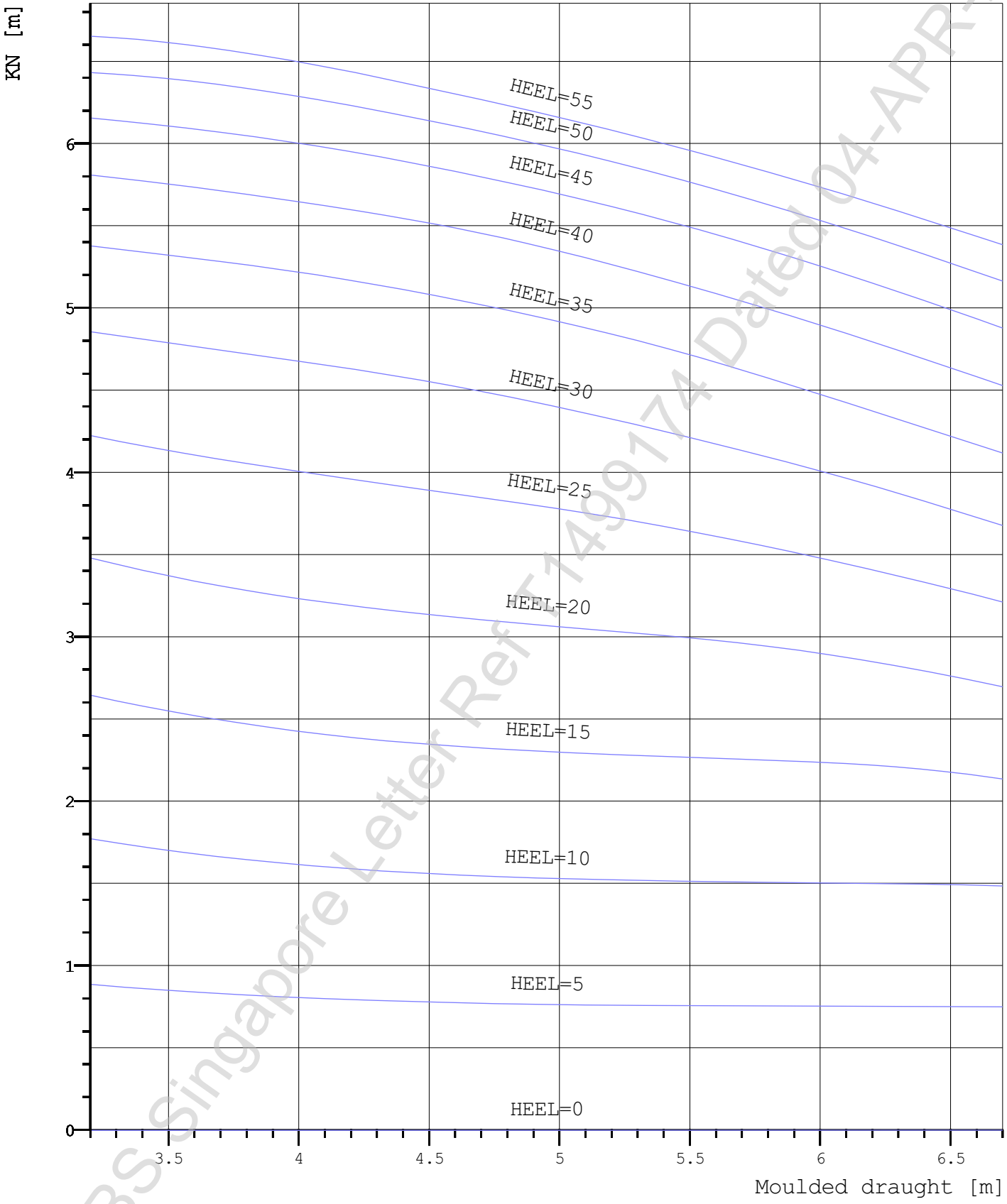
T m	HEEL=0 m	HEEL=5 m	HEEL=10 m	HEEL=15 m	HEEL=20 m	HEEL=25 m	HEEL=30 m	HEEL=35 m	HEEL=40 m	HEEL=45 m	HEEL=50 m	HEEL=55 m
3.200	-0.002	0.889	1.779	2.659	3.500	4.250	4.885	5.410	5.837	6.177	6.446	6.653
3.300	-0.002	0.876	1.754	2.625	3.462	4.219	4.863	5.391	5.818	6.159	6.432	6.641
3.400	-0.002	0.864	1.731	2.593	3.426	4.190	4.841	5.372	5.798	6.140	6.416	6.625
3.500	-0.002	0.853	1.709	2.562	3.392	4.161	4.819	5.352	5.776	6.119	6.397	6.606
3.600	-0.002	0.843	1.689	2.534	3.360	4.133	4.797	5.331	5.754	6.097	6.375	6.583
3.700	-0.002	0.834	1.671	2.508	3.331	4.107	4.774	5.310	5.731	6.074	6.352	6.559
3.800	-0.002	0.826	1.654	2.483	3.304	4.080	4.751	5.287	5.707	6.050	6.326	6.532
3.900	-0.002	0.818	1.638	2.461	3.279	4.055	4.728	5.263	5.681	6.025	6.298	6.504
4.000	-0.002	0.811	1.624	2.440	3.255	4.031	4.704	5.238	5.656	5.999	6.268	6.473
4.100	-0.002	0.804	1.611	2.422	3.232	4.007	4.679	5.211	5.629	5.971	6.238	6.441
4.200	-0.002	0.798	1.600	2.405	3.211	3.983	4.654	5.183	5.601	5.941	6.207	6.408
4.300	-0.002	0.793	1.589	2.390	3.191	3.959	4.627	5.153	5.573	5.910	6.176	6.373
4.400	-0.002	0.788	1.580	2.376	3.172	3.936	4.598	5.121	5.543	5.877	6.144	6.338
4.500	-0.001	0.784	1.572	2.364	3.154	3.913	4.568	5.088	5.511	5.842	6.111	6.303
4.600	-0.001	0.780	1.564	2.352	3.137	3.890	4.536	5.054	5.477	5.807	6.076	6.267
4.700	-0.001	0.776	1.558	2.341	3.121	3.866	4.503	5.019	5.441	5.771	6.041	6.231
4.800	-0.001	0.773	1.552	2.332	3.107	3.842	4.468	4.982	5.403	5.734	6.005	6.194
4.900	-0.001	0.771	1.547	2.322	3.092	3.817	4.432	4.944	5.363	5.696	5.967	6.156
5.000	-0.001	0.769	1.542	2.314	3.079	3.790	4.395	4.905	5.321	5.657	5.928	6.117
5.100	-0.001	0.767	1.537	2.306	3.065	3.762	4.357	4.865	5.277	5.617	5.887	6.077
5.200	-0.001	0.766	1.533	2.298	3.051	3.733	4.318	4.823	5.232	5.576	5.846	6.036
5.300	-0.001	0.765	1.530	2.291	3.037	3.702	4.277	4.778	5.186	5.533	5.803	5.994
5.400	-0.001	0.763	1.526	2.285	3.022	3.671	4.236	4.732	5.139	5.488	5.759	5.951
5.500	-0.001	0.762	1.523	2.279	3.005	3.638	4.194	4.684	5.091	5.442	5.714	5.907
5.600	-0.001	0.761	1.520	2.273	2.987	3.604	4.152	4.634	5.042	5.395	5.668	5.862
5.700	-0.001	0.760	1.517	2.268	2.966	3.569	4.109	4.583	4.992	5.346	5.620	5.816
5.800	-0.001	0.759	1.514	2.261	2.944	3.533	4.065	4.530	4.941	5.296	5.571	5.770
5.900	-0.001	0.758	1.512	2.255	2.920	3.496	4.020	4.478	4.889	5.245	5.521	5.722
6.000	-0.001	0.758	1.509	2.247	2.893	3.459	3.974	4.425	4.837	5.193	5.470	5.674
6.100	-0.001	0.757	1.507	2.237	2.866	3.420	3.926	4.372	4.784	5.139	5.418	5.624

T m	HEEL=0 m	HEEL=5 m	HEEL=10 m	HEEL=15 m	HEEL=20 m	HEEL=25 m	HEEL=30 m	HEEL=35 m	HEEL=40 m	HEEL=45 m	HEEL=50 m	HEEL=55 m
6.200	-0.001	0.755	1.505	2.224	2.836	3.380	3.877	4.318	4.730	5.085	5.365	5.574
6.300	-0.001	0.754	1.503	2.208	2.805	3.339	3.826	4.264	4.676	5.030	5.311	5.523
6.400	-0.001	0.753	1.501	2.190	2.772	3.298	3.775	4.210	4.621	4.974	5.257	5.472
6.500	-0.001	0.752	1.498	2.169	2.738	3.256	3.723	4.156	4.565	4.917	5.202	5.420
6.600	-0.001	0.751	1.494	2.145	2.702	3.212	3.670	4.102	4.509	4.860	5.146	5.367
6.700	-0.001	0.750	1.489	2.119	2.664	3.167	3.618	4.048	4.452	4.802	5.089	5.314

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9.6 CROSS CURVES, TRIM=1 m



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CROSS CURVES TABLE, TRIM=1 m

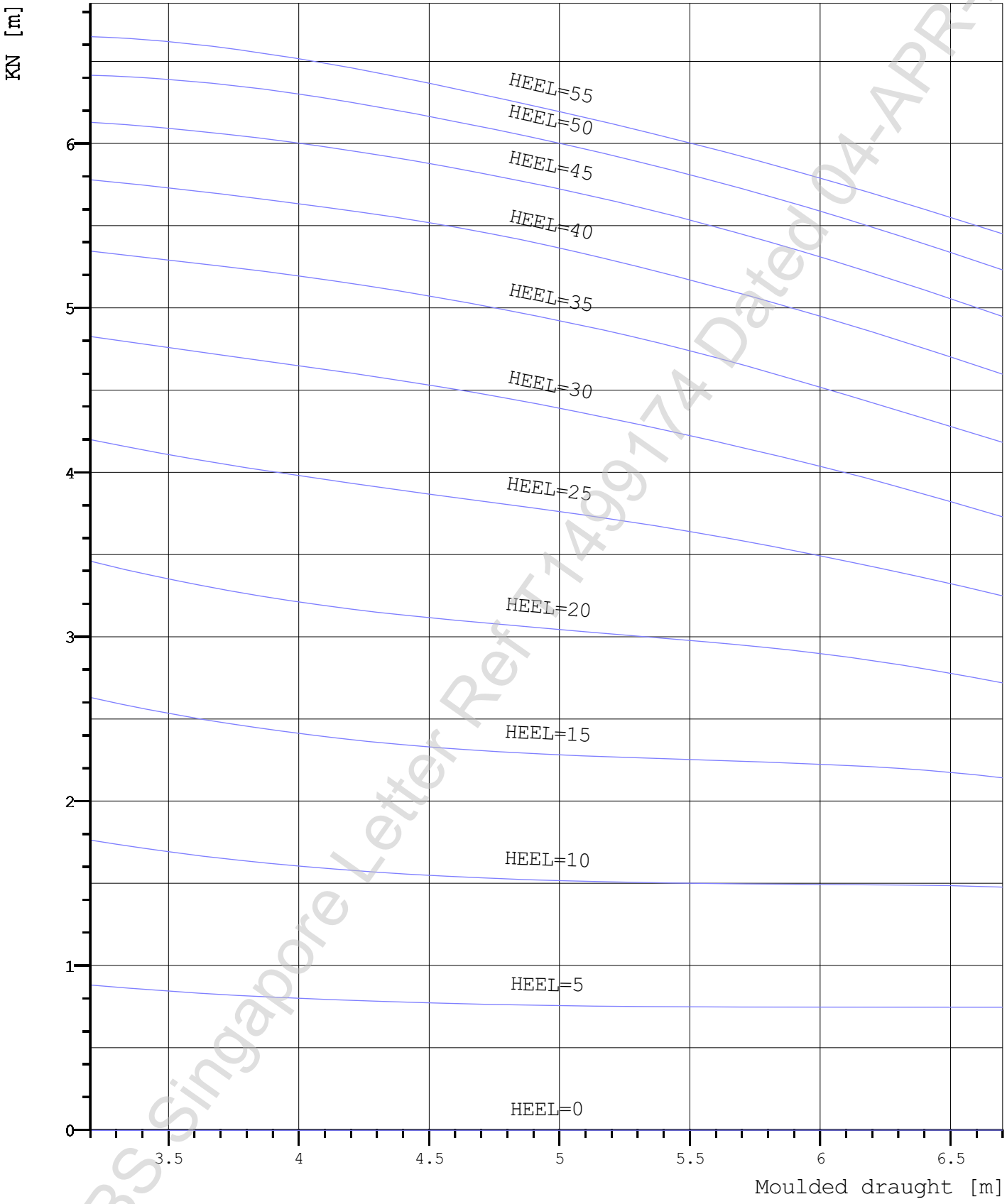
T m	HEEL=0 m	HEEL=5 m	HEEL=10 m	HEEL=15 m	HEEL=20 m	HEEL=25 m	HEEL=30 m	HEEL=35 m	HEEL=40 m	HEEL=45 m	HEEL=50 m	HEEL=55 m
3.200	-0.002	0.885	1.772	2.645	3.480	4.224	4.855	5.377	5.809	6.155	6.432	6.652
3.300	-0.002	0.872	1.746	2.611	3.441	4.192	4.833	5.359	5.791	6.140	6.422	6.643
3.400	-0.002	0.860	1.722	2.578	3.405	4.162	4.810	5.340	5.773	6.124	6.409	6.630
3.500	-0.002	0.849	1.700	2.548	3.371	4.133	4.788	5.321	5.753	6.107	6.395	6.614
3.600	-0.002	0.838	1.680	2.520	3.339	4.105	4.766	5.302	5.733	6.088	6.377	6.595
3.700	-0.002	0.829	1.661	2.493	3.309	4.079	4.744	5.283	5.713	6.068	6.358	6.574
3.800	-0.002	0.820	1.644	2.469	3.281	4.054	4.721	5.262	5.691	6.047	6.336	6.550
3.900	-0.002	0.813	1.628	2.446	3.255	4.029	4.699	5.240	5.669	6.025	6.312	6.524
4.000	-0.002	0.805	1.613	2.425	3.232	4.005	4.676	5.217	5.646	6.002	6.286	6.497
4.100	-0.002	0.799	1.600	2.405	3.210	3.982	4.653	5.193	5.622	5.977	6.259	6.467
4.200	-0.002	0.793	1.588	2.388	3.189	3.958	4.629	5.167	5.597	5.950	6.231	6.436
4.300	-0.002	0.787	1.577	2.372	3.170	3.935	4.604	5.140	5.572	5.922	6.201	6.404
4.400	-0.002	0.782	1.568	2.358	3.152	3.913	4.579	5.112	5.545	5.893	6.171	6.370
4.500	-0.001	0.778	1.559	2.345	3.135	3.890	4.551	5.082	5.517	5.862	6.139	6.336
4.600	-0.001	0.774	1.551	2.334	3.119	3.868	4.523	5.051	5.486	5.830	6.107	6.302
4.700	-0.001	0.770	1.544	2.324	3.103	3.846	4.493	5.019	5.454	5.797	6.073	6.267
4.800	-0.001	0.767	1.539	2.314	3.089	3.824	4.461	4.986	5.420	5.763	6.039	6.231
4.900	-0.001	0.764	1.533	2.306	3.075	3.801	4.429	4.951	5.383	5.729	6.004	6.195
5.000	-0.001	0.762	1.529	2.298	3.061	3.778	4.395	4.916	5.345	5.693	5.967	6.158
5.100	-0.001	0.760	1.525	2.290	3.047	3.753	4.360	4.879	5.305	5.656	5.929	6.120
5.200	-0.001	0.758	1.521	2.284	3.034	3.727	4.324	4.841	5.264	5.617	5.890	6.081
5.300	-0.001	0.757	1.518	2.277	3.021	3.699	4.288	4.801	5.221	5.576	5.850	6.041
5.400	-0.001	0.756	1.515	2.272	3.007	3.671	4.250	4.759	5.177	5.535	5.808	6.000
5.500	-0.001	0.755	1.512	2.266	2.993	3.641	4.212	4.715	5.133	5.491	5.765	5.958
5.600	-0.001	0.755	1.510	2.261	2.978	3.611	4.172	4.669	5.087	5.447	5.721	5.915
5.700	-0.001	0.754	1.507	2.255	2.961	3.579	4.132	4.622	5.041	5.401	5.675	5.871
5.800	-0.001	0.753	1.505	2.249	2.942	3.547	4.092	4.573	4.994	5.353	5.629	5.826
5.900	-0.001	0.753	1.503	2.243	2.921	3.513	4.051	4.523	4.945	5.305	5.581	5.781
6.000	-0.001	0.752	1.502	2.236	2.899	3.479	4.008	4.474	4.896	5.255	5.532	5.734

T m	HEEL=0 m	HEEL=5 m	HEEL=10 m	HEEL=15 m	HEEL=20 m	HEEL=25 m	HEEL=30 m	HEEL=35 m	HEEL=40 m	HEEL=45 m	HEEL=50 m	HEEL=55 m
6.100	-0.001	0.752	1.500	2.228	2.875	3.443	3.964	4.424	4.846	5.203	5.482	5.687
6.200	-0.001	0.752	1.498	2.219	2.849	3.407	3.919	4.373	4.794	5.151	5.431	5.638
6.300	-0.001	0.751	1.497	2.207	2.821	3.369	3.872	4.322	4.742	5.098	5.379	5.588
6.400	-0.001	0.750	1.495	2.193	2.792	3.331	3.824	4.271	4.690	5.044	5.326	5.538
6.500	-0.001	0.750	1.492	2.176	2.761	3.292	3.775	4.220	4.636	4.989	5.272	5.488
6.600	-0.001	0.749	1.489	2.156	2.729	3.252	3.726	4.169	4.582	4.934	5.218	5.436
6.700	-0.001	0.748	1.484	2.134	2.695	3.211	3.676	4.117	4.527	4.877	5.163	5.384

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9.7 CROSS CURVES, TRIM=1.5 m



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CROSS CURVES TABLE, TRIM=1.5 m

T m	HEEL=0 m	HEEL=5 m	HEEL=10 m	HEEL=15 m	HEEL=20 m	HEEL=25 m	HEEL=30 m	HEEL=35 m	HEEL=40 m	HEEL=45 m	HEEL=50 m	HEEL=55 m
3.200	-0.002	0.882	1.763	2.631	3.460	4.199	4.826	5.345	5.780	6.130	6.416	6.650
3.300	-0.002	0.868	1.738	2.596	3.421	4.167	4.803	5.327	5.764	6.118	6.410	6.643
3.400	-0.002	0.856	1.714	2.564	3.385	4.136	4.780	5.309	5.747	6.106	6.401	6.633
3.500	-0.002	0.844	1.692	2.534	3.351	4.107	4.758	5.291	5.730	6.092	6.390	6.620
3.600	-0.002	0.834	1.672	2.506	3.319	4.080	4.736	5.273	5.712	6.077	6.377	6.604
3.700	-0.002	0.824	1.653	2.480	3.290	4.053	4.714	5.255	5.693	6.061	6.361	6.585
3.800	-0.002	0.816	1.635	2.456	3.262	4.028	4.692	5.236	5.674	6.043	6.343	6.563
3.900	-0.002	0.808	1.619	2.433	3.235	4.004	4.670	5.216	5.654	6.023	6.323	6.540
4.000	-0.002	0.801	1.604	2.412	3.211	3.980	4.648	5.195	5.634	6.002	6.301	6.515
4.100	-0.002	0.794	1.591	2.392	3.189	3.957	4.626	5.173	5.613	5.980	6.277	6.489
4.200	-0.002	0.788	1.579	2.374	3.168	3.934	4.603	5.149	5.591	5.957	6.251	6.460
4.300	-0.002	0.782	1.568	2.357	3.149	3.911	4.580	5.125	5.568	5.932	6.223	6.430
4.400	-0.002	0.777	1.558	2.343	3.132	3.889	4.556	5.099	5.544	5.906	6.194	6.399
4.500	-0.001	0.773	1.549	2.329	3.116	3.867	4.531	5.073	5.518	5.878	6.164	6.366
4.600	-0.001	0.769	1.540	2.317	3.100	3.846	4.505	5.045	5.491	5.849	6.134	6.333
4.700	-0.001	0.765	1.533	2.307	3.086	3.825	4.478	5.015	5.462	5.819	6.102	6.299
4.800	-0.001	0.761	1.527	2.298	3.071	3.804	4.450	4.985	5.432	5.789	6.069	6.265
4.900	-0.001	0.758	1.521	2.289	3.057	3.783	4.420	4.954	5.399	5.757	6.036	6.230
5.000	-0.001	0.756	1.516	2.282	3.043	3.761	4.390	4.922	5.364	5.723	6.001	6.194
5.100	-0.001	0.754	1.512	2.275	3.030	3.739	4.358	4.889	5.328	5.689	5.966	6.158
5.200	-0.001	0.752	1.509	2.269	3.016	3.716	4.326	4.854	5.290	5.652	5.929	6.121
5.300	-0.001	0.750	1.506	2.263	3.004	3.691	4.292	4.817	5.251	5.615	5.891	6.083
5.400	-0.001	0.749	1.503	2.258	2.991	3.666	4.258	4.779	5.211	5.575	5.851	6.044
5.500	-0.001	0.749	1.501	2.253	2.978	3.640	4.223	4.739	5.170	5.535	5.811	6.004
5.600	-0.001	0.748	1.499	2.248	2.964	3.612	4.187	4.698	5.128	5.493	5.769	5.963
5.700	-0.001	0.748	1.497	2.242	2.950	3.584	4.151	4.655	5.085	5.449	5.725	5.921
5.800	-0.001	0.747	1.496	2.236	2.934	3.554	4.114	4.610	5.041	5.404	5.681	5.878
5.900	-0.001	0.747	1.494	2.230	2.916	3.524	4.076	4.564	4.996	5.358	5.635	5.834

T m	HEEL=0 m	HEEL=5 m	HEEL=10 m	HEEL=15 m	HEEL=20 m	HEEL=25 m	HEEL=30 m	HEEL=35 m	HEEL=40 m	HEEL=45 m	HEEL=50 m	HEEL=55 m
6.000	-0.001	0.747	1.493	2.224	2.897	3.492	4.037	4.517	4.950	5.310	5.588	5.789
6.100	-0.001	0.747	1.492	2.217	2.877	3.460	3.996	4.470	4.902	5.262	5.540	5.744
6.200	-0.001	0.747	1.491	2.209	2.854	3.427	3.955	4.423	4.853	5.212	5.491	5.697
6.300	-0.001	0.747	1.490	2.199	2.830	3.393	3.912	4.375	4.803	5.161	5.441	5.649
6.400	-0.001	0.747	1.488	2.188	2.805	3.358	3.867	4.327	4.753	5.109	5.390	5.600
6.500	-0.001	0.747	1.485	2.175	2.777	3.323	3.822	4.279	4.702	5.056	5.338	5.551
6.600	-0.001	0.746	1.482	2.159	2.749	3.286	3.776	4.230	4.650	5.002	5.285	5.501
6.700	-0.001	0.746	1.477	2.141	2.718	3.249	3.729	4.181	4.597	4.948	5.232	5.450

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10 HYDROSTATICS

MAIN CHARACTERISTICS OF THE VESSEL:

Length between perpendiculars	79.61	m
Breadth (moulded)	18.00	m
Design draught (moulded)	6.00	m
X-coordinate of aft perpendicular	-3.05	m
X-coordinate of reference point (XREF)	36.75	m
X-coordinate of midship section (XMID)	36.75	m
Thickness of keelplate	0.012	m
Mean thickness of shell plating	0.012	m
Seawater density	1.025	ton/m3

CALCULATION OBJECT:

Calculations are based on STABHULL date 2015-11-19 time 14:45

Shell thickness used in the calculation	12.0	mm
X-coord. of aft end of DWL	-5.90	m
X-coord. of fore end of DWL	76.50	m
Calc. sections	143	
Plate thickness	12.0	mm

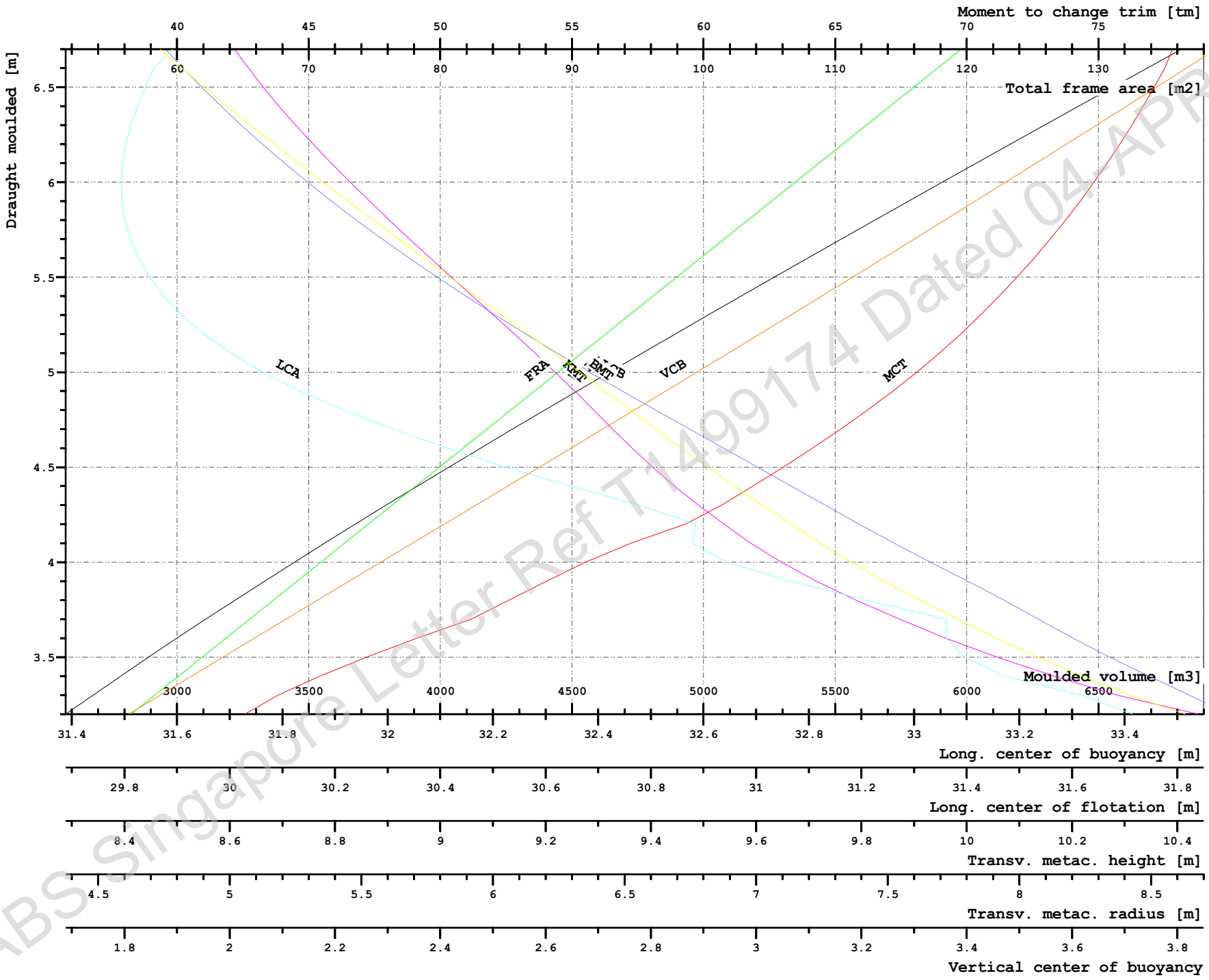
EXPLANATION OF SYMBOLS:

T	draught, moulded	m
TK	draught below keel	m
DISP	total displacement	t
VOLT	total volume	m3
DW	deadweight	t
LCF	long. centre of flotation	m
LCB	longitudinal centre of buoyancy	m
VCB	vertical center of buoyancy	m
KMT	transv. metac. height	m
IMAX	max. mom. of inertia of surface	m4
BMT	transv. metac. radius	m
LWL	length of waterline	m
WLA	waterline area	m2
WSA	wetted surface area	m2
FRA	total frame area	m2
MCT	moment to change trim	tm/cm
TCP	change of displacement/change of draught	t/cm
TRFAP	trim factor aft	
TRFFP	trim factor fore	
CB	block coefficient	
CP	prismatic coefficient	
CW	waterplane coefficient	

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10.1 HYDROSTATICS, TRIM = -1.5 m



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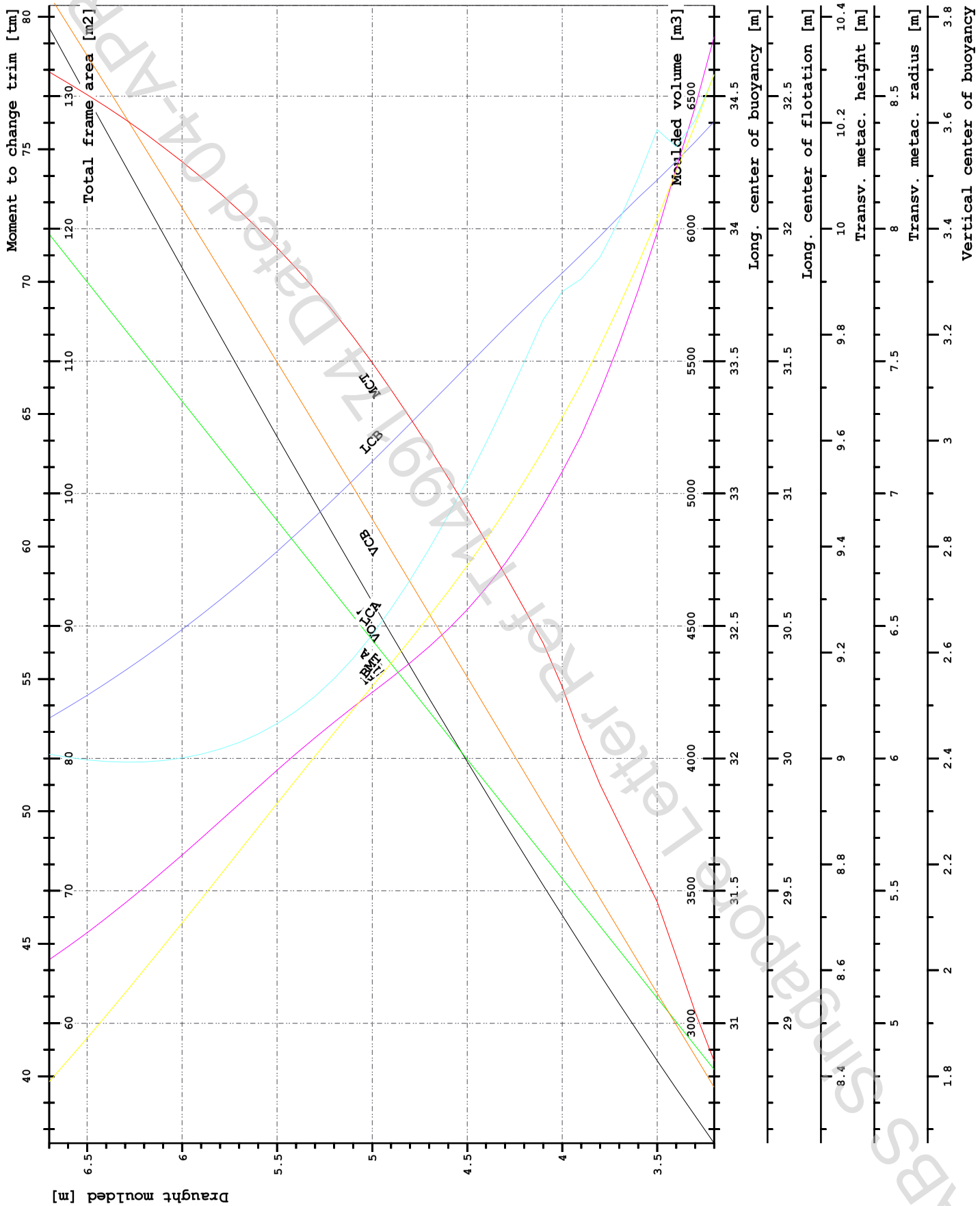
HYDROSTATICS TABLE, TRIM = -1.5 m

T m	TK m	DISP t	VOLT m3	DW t	LCF m	LCB m	VCB m	KMT m	IMAX m4	BMT m	LWL m	WLA m2	WSA m2	FRA m2	MCT tm/cm	TCP t/cm	TRFAP m	TRFFP m	CB	CP	CW
3.200	3.212	2658	2593	-47	31.715	33.598	1.809	10.439	330683	8.631	79.515	1034	1387	56.5	42.58	10.60	0.272640	0.351692	0.562	0.576	0.719
3.300	3.312	2765	2697	60	31.619	33.523	1.868	10.287	340418	8.419	79.668	1046	1419	58.3	43.83	10.72	0.274710	0.356098	0.567	0.580	0.728
3.400	3.412	2872	2802	167	31.472	33.447	1.927	10.166	352710	8.239	79.817	1060	1443	60.1	45.41	10.86	0.274289	0.358233	0.572	0.585	0.737
3.500	3.512	2982	2909	277	31.396	33.372	1.987	10.058	366548	8.071	79.962	1075	1467	61.9	47.19	11.02	0.273364	0.358414	0.576	0.589	0.748
3.600	3.612	3093	3017	388	31.361	33.302	2.046	9.959	381388	7.913	80.104	1091	1492	63.7	49.10	11.18	0.272235	0.357573	0.581	0.594	0.759
3.700	3.712	3205	3127	500	31.361	33.236	2.106	9.873	397489	7.766	80.243	1108	1520	65.5	51.18	11.35	0.270724	0.355598	0.586	0.599	0.771
3.800	3.812	3319	3238	615	31.208	33.171	2.166	9.789	408557	7.623	80.378	1120	1543	67.3	52.60	11.48	0.271557	0.359502	0.591	0.603	0.779
3.900	3.912	3435	3351	730	31.066	33.102	2.226	9.714	419660	7.488	80.510	1132	1568	69.1	54.03	11.60	0.272422	0.363267	0.596	0.608	0.788
4.000	4.012	3551	3464	846	30.944	33.031	2.287	9.648	431127	7.362	80.640	1144	1591	70.9	55.51	11.72	0.273178	0.366572	0.601	0.613	0.796
4.100	4.112	3669	3580	964	30.879	32.962	2.347	9.590	444550	7.243	80.766	1157	1615	72.7	57.24	11.86	0.273206	0.367833	0.606	0.617	0.805
4.200	4.212	3788	3696	1083	30.885	32.895	2.407	9.539	460608	7.132	80.889	1172	1642	74.5	59.30	12.01	0.272309	0.366510	0.611	0.622	0.815
4.300	4.312	3909	3814	1204	30.777	32.831	2.468	9.490	471336	7.022	81.010	1183	1664	76.3	60.68	12.13	0.273723	0.370471	0.616	0.627	0.824
4.400	4.412	4031	3933	1326	30.647	32.766	2.528	9.445	480569	6.917	81.128	1193	1685	78.1	61.87	12.23	0.275767	0.375729	0.621	0.631	0.831
4.500	4.512	4154	4052	1449	30.526	32.702	2.588	9.403	489523	6.815	81.244	1202	1706	79.9	63.03	12.32	0.277959	0.381092	0.625	0.636	0.837
4.600	4.612	4277	4173	1572	30.414	32.637	2.648	9.364	498131	6.716	81.357	1211	1726	81.7	64.13	12.41	0.280348	0.386591	0.630	0.641	0.843
4.700	4.712	4402	4295	1697	30.310	32.573	2.708	9.327	506453	6.619	81.468	1219	1747	83.5	65.21	12.50	0.282892	0.392186	0.635	0.645	0.849
4.800	4.812	4527	4417	1822	30.217	32.509	2.768	9.291	514381	6.523	81.576	1227	1769	85.3	66.23	12.58	0.285660	0.397941	0.639	0.649	0.855
4.900	4.912	4653	4540	1948	30.134	32.446	2.828	9.255	521878	6.427	81.683	1235	1789	87.1	67.19	12.65	0.288684	0.403871	0.643	0.654	0.860
5.000	5.012	4780	4664	2075	30.063	32.383	2.888	9.218	528928	6.330	81.787	1241	1808	88.9	68.10	12.72	0.291975	0.409979	0.648	0.658	0.865
5.100	5.112	4908	4788	2203	30.002	32.322	2.947	9.180	535576	6.233	81.889	1248	1827	90.7	68.96	12.79	0.295500	0.416238	0.652	0.662	0.869
5.200	5.212	5036	4913	2331	29.951	32.263	3.006	9.141	541825	6.135	81.989	1254	1846	92.5	69.76	12.85	0.299258	0.422644	0.656	0.666	0.873
5.300	5.312	5165	5039	2460	29.909	32.205	3.066	9.101	547702	6.036	82.087	1259	1864	94.3	70.52	12.91	0.303230	0.429185	0.660	0.670	0.877
5.400	5.412	5294	5165	2589	29.875	32.148	3.124	9.061	553256	5.937	82.184	1264	1882	96.2	71.23	12.96	0.307378	0.435834	0.665	0.674	0.881
5.500	5.512	5424	5292	2719	29.847	32.094	3.183	9.021	558529	5.838	82.278	1269	1900	98.0	71.91	13.01	0.311680	0.442566	0.668	0.678	0.884
5.600	5.612	5554	5419	2849	29.826	32.041	3.242	8.981	563527	5.739	82.371	1274	1918	99.8	72.55	13.05	0.316134	0.449376	0.672	0.682	0.887
5.700	5.712	5685	5546	2980	29.812	31.990	3.300	8.942	568256	5.641	82.461	1278	1935	101.6	73.16	13.10	0.320735	0.456269	0.676	0.685	0.890
5.800	5.812	5816	5674	3111	29.801	31.941	3.358	8.903	572806	5.545	82.551	1282	1958	103.4	73.75	13.14	0.325419	0.463197	0.680	0.689	0.893
5.900	5.912	5947	5802	3243	29.794	31.893	3.416	8.866	577136	5.449	82.638	1285	1976	105.2	74.31	13.18	0.330223	0.470183	0.683	0.692	0.896
6.000	6.012	6079	5931	3375	29.794	31.848	3.474	8.829	581225	5.354	82.724	1289	1994	107.0	74.83	13.21	0.335164	0.477240	0.687	0.696	0.898
6.100	6.112	6212	6060	3507	29.796	31.804	3.532	8.793	585124	5.261	82.808	1292	2012	108.8	75.33	13.24	0.340204	0.484348	0.691	0.699	0.900
6.200	6.212	6344	6190	3639	29.802	31.762	3.589	8.758	588845	5.169	82.891	1295	2030	110.6	75.81	13.28	0.345335	0.491497	0.694	0.703	0.902
6.300	6.312	6477	6319	3772	29.812	31.722	3.647	8.725	592384	5.078	82.972	1298	2047	112.4	76.27	13.31	0.350567	0.498689	0.697	0.706	0.904
6.400	6.412	6610	6449	3905	29.825	31.683	3.704	8.693	595762	4.989	83.051	1301	2064	114.2	76.70	13.33	0.355886	0.505919	0.700	0.709	0.906
6.500	6.512	6744	6579	4039	29.840	31.646	3.761	8.663	599000	4.902	83.129	1304	2082	116.0	77.12	13.36	0.361274	0.513179	0.704	0.712	0.908
6.600	6.612	6878	6710	4173	29.856	31.611	3.817	8.636	602174	4.818	83.205	1306	2099	117.8	77.53	13.39	0.366672	0.520424	0.707	0.715	0.910
6.700	6.712	7012	6841	4307	29.892	31.578	3.874	8.609	604346	4.735	83.242	1308	2119	119.6	77.81	13.41	0.372878	0.528246	0.710	0.718	0.911

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See ABS Singapore Letter Ref: 19169174 Dated MAR-2016

10.2 HYDROSTATICS, TRIM = -1 m



See ABS Singapore Letter Ref: TA99124 Dated 04-APR-2016

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HYDROSTATICS TABLE, TRIM = -1 m

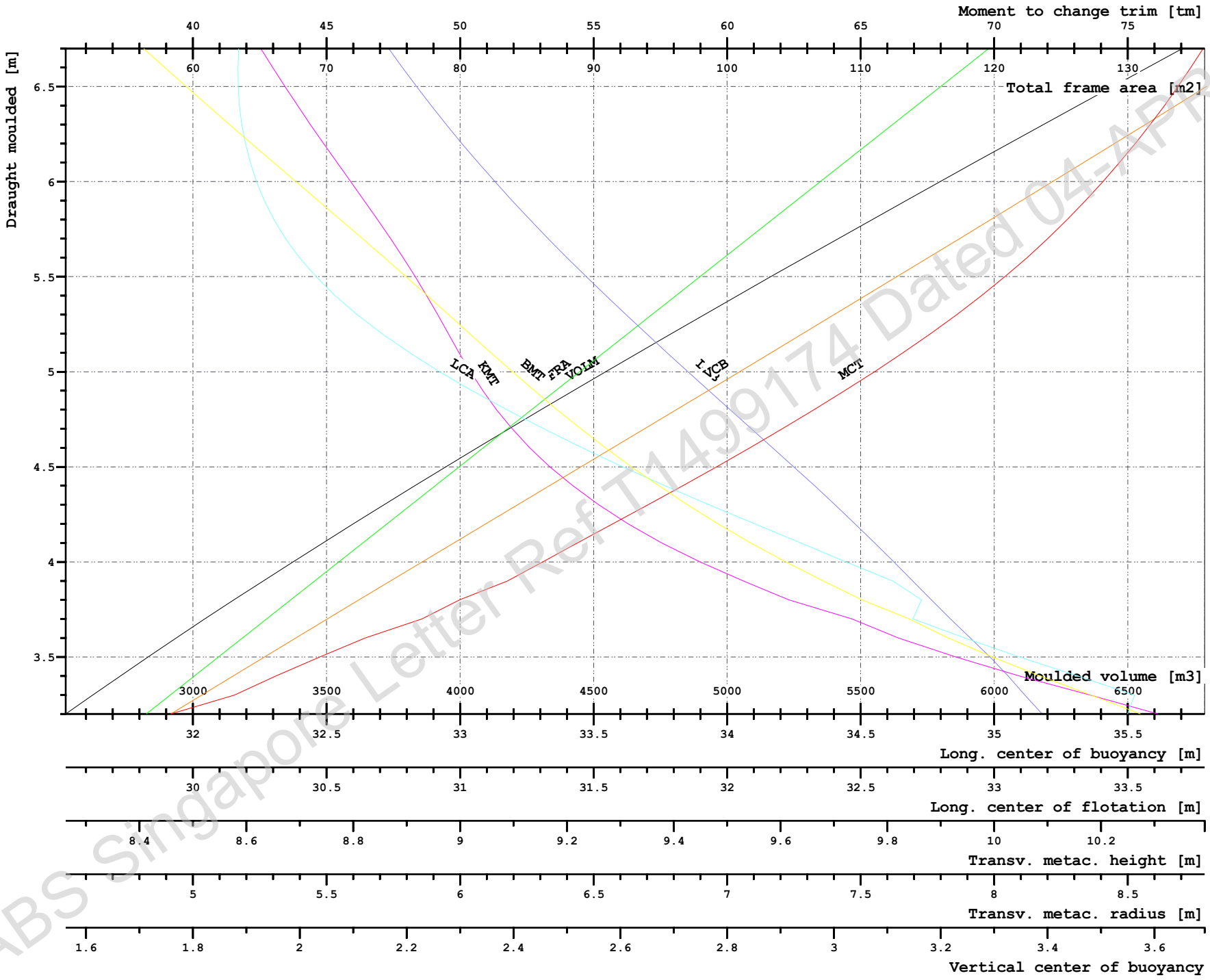
T m	TK m	DISP t	VOLT m3	DW t	LCF m	LCB m	VCB m	KMT m	IMAX m4	BMT m	LWL m	WLA m2	WSA m2	FRA m2	MCT tm/cm	TCP t/cm	TRFAP m	TRFFP m	CB	CP	CW
3.200	3.212	2628	2563	-77	32.581	34.404	1.779	10.363	315110	8.583	79.416	1017	1369	56.5	40.57	10.42	0.289871	0.357791	0.555	0.569	0.707
3.300	3.312	2732	2666	27	32.410	34.328	1.838	10.231	329356	8.393	79.725	1033	1395	58.3	42.40	10.58	0.287010	0.357348	0.560	0.573	0.718
3.400	3.412	2839	2770	134	32.310	34.255	1.897	10.114	345782	8.217	79.866	1051	1423	60.1	44.52	10.77	0.283269	0.354484	0.565	0.578	0.731
3.500	3.512	2948	2876	243	32.374	34.184	1.956	9.993	361761	8.037	80.003	1068	1461	61.9	46.58	10.95	0.281638	0.351305	0.570	0.583	0.743
3.600	3.612	3058	2984	353	32.192	34.117	2.016	9.884	373359	7.868	80.137	1081	1484	63.7	48.07	11.08	0.281644	0.354569	0.575	0.587	0.752
3.700	3.712	3170	3092	465	32.031	34.046	2.075	9.783	384709	7.708	80.268	1093	1509	65.5	49.53	11.21	0.281993	0.357932	0.580	0.592	0.761
3.800	3.812	3282	3202	577	31.893	33.973	2.135	9.692	396158	7.557	80.396	1106	1531	67.3	51.01	11.33	0.282440	0.361035	0.585	0.597	0.770
3.900	3.912	3396	3313	691	31.810	33.902	2.194	9.610	409505	7.416	80.521	1119	1555	69.1	52.72	11.47	0.282052	0.362065	0.589	0.601	0.779
4.000	4.012	3511	3426	807	31.763	33.833	2.254	9.541	424916	7.287	80.644	1135	1582	70.9	54.71	11.63	0.280678	0.361175	0.594	0.606	0.789
4.100	4.112	3628	3540	923	31.657	33.766	2.314	9.478	437696	7.164	80.763	1148	1605	72.7	56.35	11.76	0.280702	0.363164	0.599	0.611	0.799
4.200	4.212	3747	3655	1042	31.496	33.697	2.374	9.420	447829	7.046	80.880	1158	1629	74.5	57.66	11.87	0.281969	0.367823	0.604	0.615	0.807
4.300	4.312	3866	3772	1161	31.341	33.626	2.434	9.368	457745	6.934	80.995	1169	1651	76.3	58.93	11.98	0.283368	0.372592	0.609	0.620	0.814
4.400	4.412	3986	3889	1281	31.192	33.554	2.494	9.322	467450	6.828	81.107	1179	1673	78.1	60.18	12.08	0.284891	0.377452	0.614	0.624	0.821
4.500	4.512	4108	4007	1403	31.051	33.482	2.554	9.281	476924	6.727	81.216	1189	1695	79.9	61.40	12.19	0.286555	0.382407	0.618	0.629	0.828
4.600	4.612	4230	4127	1525	30.917	33.408	2.614	9.244	486140	6.630	81.324	1198	1716	81.7	62.59	12.28	0.288364	0.387477	0.623	0.633	0.834
4.700	4.712	4353	4247	1648	30.791	33.336	2.673	9.211	495083	6.538	81.429	1207	1737	83.5	63.74	12.38	0.290324	0.392649	0.628	0.638	0.841
4.800	4.812	4478	4368	1773	30.674	33.264	2.733	9.181	503709	6.448	81.532	1216	1758	85.3	64.85	12.46	0.292477	0.397951	0.632	0.642	0.847
4.900	4.912	4603	4490	1898	30.566	33.192	2.793	9.152	511997	6.359	81.633	1224	1778	87.1	65.92	12.55	0.294830	0.403394	0.636	0.647	0.853
5.000	5.012	4729	4613	2024	30.467	33.121	2.852	9.124	519964	6.272	81.732	1232	1798	88.9	66.95	12.63	0.297374	0.408957	0.641	0.651	0.858
5.100	5.112	4855	4737	2150	30.378	33.050	2.912	9.097	527528	6.185	81.830	1240	1820	90.7	67.92	12.71	0.300168	0.414685	0.645	0.655	0.863
5.200	5.212	4983	4861	2278	30.301	32.981	2.971	9.068	534633	6.097	81.925	1246	1839	92.5	68.83	12.78	0.303251	0.420611	0.649	0.659	0.868
5.300	5.312	5111	4986	2406	30.235	32.913	3.030	9.039	541341	6.009	82.018	1253	1858	94.3	69.70	12.84	0.306579	0.426688	0.654	0.663	0.873
5.400	5.412	5239	5112	2534	30.178	32.846	3.089	9.009	547675	5.920	82.110	1259	1877	96.2	70.51	12.90	0.310136	0.432903	0.658	0.667	0.877
5.500	5.512	5369	5238	2664	30.131	32.782	3.148	8.978	553623	5.830	82.200	1264	1896	98.0	71.28	12.96	0.313930	0.439265	0.662	0.671	0.881
5.600	5.612	5498	5364	2794	30.092	32.719	3.207	8.946	559221	5.739	82.288	1269	1914	99.8	72.00	13.01	0.317931	0.445754	0.666	0.675	0.884
5.700	5.712	5629	5492	2924	30.061	32.658	3.265	8.913	564540	5.648	82.374	1274	1932	101.6	72.68	13.06	0.322091	0.452330	0.669	0.679	0.888
5.800	5.812	5760	5619	3055	30.035	32.599	3.323	8.881	569582	5.558	82.459	1279	1950	103.4	73.33	13.11	0.326406	0.458995	0.673	0.682	0.891
5.900	5.912	5891	5747	3186	30.016	32.541	3.382	8.849	574355	5.468	82.542	1283	1967	105.2	73.95	13.15	0.330875	0.465747	0.677	0.686	0.894
6.000	6.012	6023	5876	3318	30.002	32.486	3.440	8.818	578891	5.378	82.624	1287	1990	107.0	74.53	13.19	0.335476	0.472568	0.681	0.689	0.896
6.100	6.112	6155	6004	3450	29.992	32.433	3.497	8.787	583231	5.289	82.704	1290	2008	108.8	75.09	13.23	0.340182	0.479440	0.684	0.693	0.899
6.200	6.212	6287	6134	3582	29.986	32.381	3.555	8.756	587352	5.201	82.782	1294	2026	110.6	75.62	13.26	0.345010	0.486377	0.688	0.696	0.901
6.300	6.312	6420	6263	3715	29.986	32.331	3.613	8.727	591263	5.114	82.859	1297	2044	112.4	76.12	13.30	0.349959	0.493375	0.691	0.700	0.904
6.400	6.412	6553	6393	3848	29.988	32.284	3.670	8.698	594999	5.028	82.934	1300	2062	114.2	76.61	13.33	0.355002	0.500419	0.694	0.703	0.906
6.500	6.512	6686	6523	3982	29.994	32.238	3.727	8.671	598555	4.944	83.008	1303	2079	116.0	77.06	13.36	0.360145	0.507508	0.698	0.706	0.908
6.600	6.612	6820	6654	4115	30.004	32.194	3.784	8.645	601944	4.860	83.080	1306	2096	117.8	77.50	13.39	0.365383	0.514639	0.701	0.709	0.910
6.700	6.712	6954	6785	4249	30.016	32.152	3.841	8.620	605180	4.779	83.150	1309	2113	119.6	77.92	13.41	0.370707	0.521808	0.704	0.712	0.912

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10.3 HYDROSTATICS, TRIM = -0.5 m



See ABS Singapore Letter Ref T1499174 Dated 04-APR-2016

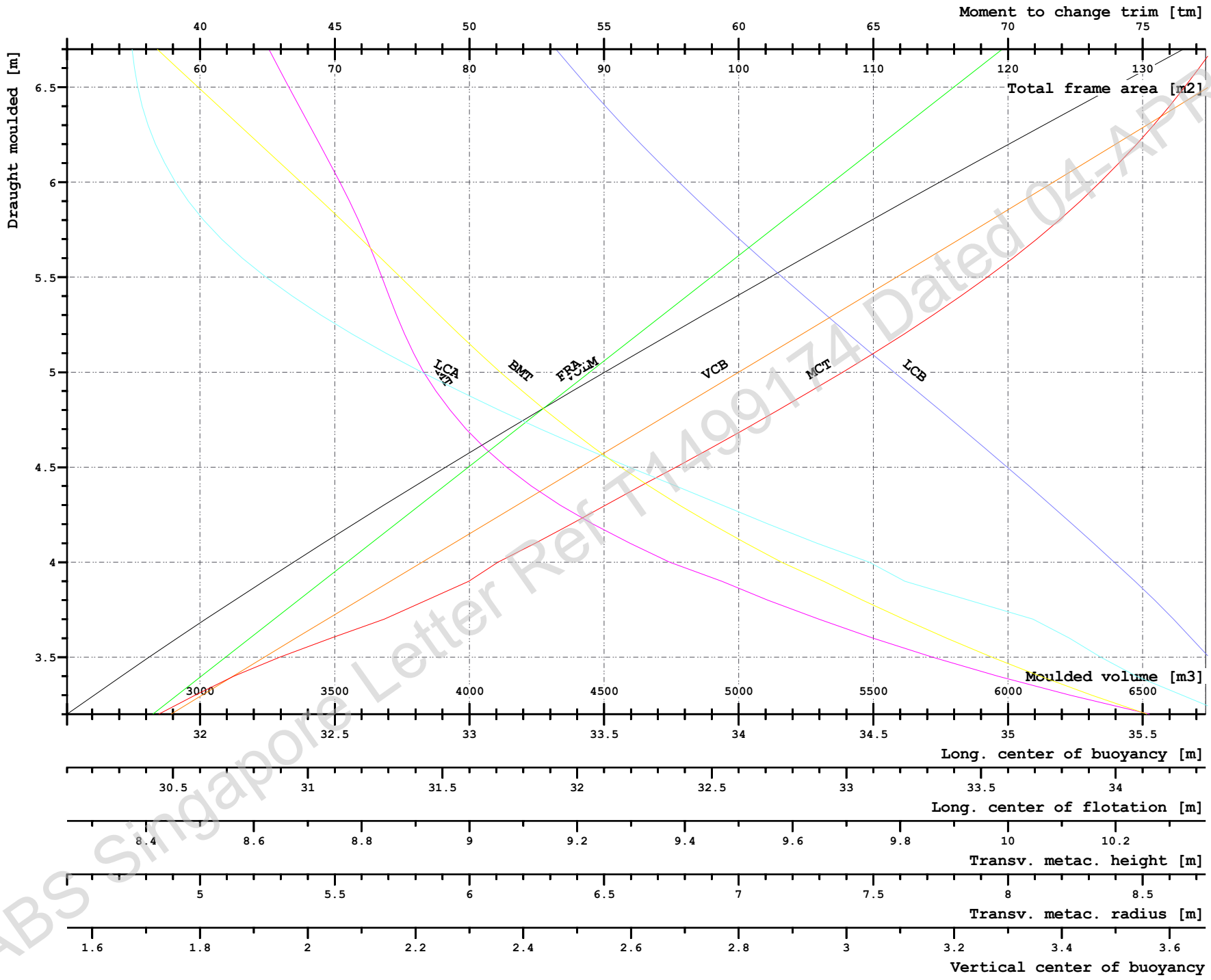
HYDROSTATICS TABLE, TRIM = -0.5 m

T m	TK m	DISP t	VOLT m3	DW t	LCF m	LCB m	VCB m	KMT m	IMAX m4	BMT m	LWL m	WLA m2	WSA m2	FRA m2	MCT tm/cm	TCP t/cm	TRFAP m	TRFFP m	CB	CP	CW
3.200	3.212	2603	2540	-102	33.500	35.179	1.758	10.306	304242	8.548	78.138	1005	1354	56.5	39.17	10.30	0.305113	0.359464	0.550	0.564	0.699
3.300	3.312	2707	2641	2	33.525	35.115	1.817	10.180	322670	8.364	78.788	1025	1385	58.3	41.54	10.51	0.299394	0.352266	0.555	0.568	0.712
3.400	3.412	2813	2745	108	33.313	35.052	1.875	10.047	334736	8.172	79.434	1039	1410	60.1	43.10	10.65	0.298152	0.354587	0.560	0.573	0.723
3.500	3.512	2920	2849	215	33.095	34.984	1.934	9.928	347558	7.994	80.028	1053	1437	61.9	44.75	10.79	0.296306	0.356308	0.565	0.577	0.733
3.600	3.612	3029	2955	324	32.890	34.910	1.992	9.820	360759	7.828	80.154	1067	1463	63.7	46.45	10.94	0.294383	0.357697	0.569	0.582	0.743
3.700	3.712	3139	3062	434	32.696	34.836	2.052	9.733	377262	7.682	80.278	1084	1492	65.5	48.57	11.11	0.290169	0.356065	0.574	0.586	0.753
3.800	3.812	3251	3171	546	32.728	34.765	2.111	9.615	388263	7.505	80.399	1096	1524	67.3	49.99	11.23	0.292233	0.358019	0.579	0.591	0.762
3.900	3.912	3364	3282	659	32.620	34.695	2.170	9.531	402116	7.360	80.518	1110	1548	69.1	51.77	11.38	0.291105	0.358595	0.584	0.596	0.773
4.000	4.012	3478	3393	773	32.445	34.624	2.229	9.450	412392	7.220	80.634	1121	1570	70.9	53.10	11.49	0.292059	0.362987	0.589	0.600	0.781
4.100	4.112	3593	3506	889	32.275	34.552	2.289	9.377	422511	7.088	80.747	1132	1592	72.7	54.40	11.60	0.293121	0.367460	0.593	0.605	0.788
4.200	4.212	3710	3619	1005	32.103	34.478	2.348	9.315	432778	6.966	80.858	1143	1614	74.5	55.72	11.72	0.294008	0.371816	0.598	0.609	0.796
4.300	4.312	3828	3734	1123	31.933	34.404	2.408	9.260	443021	6.852	80.967	1154	1636	76.3	57.04	11.83	0.294878	0.376175	0.603	0.614	0.803
4.400	4.412	3946	3850	1241	31.768	34.327	2.467	9.211	453102	6.744	81.073	1164	1660	78.1	58.34	11.93	0.295861	0.380613	0.607	0.618	0.811
4.500	4.512	4066	3967	1361	31.609	34.249	2.526	9.168	462969	6.642	81.177	1174	1682	79.9	59.61	12.04	0.296993	0.385177	0.612	0.623	0.818
4.600	4.612	4187	4085	1482	31.458	34.170	2.586	9.131	472623	6.545	81.279	1184	1704	81.7	60.85	12.14	0.298268	0.389842	0.617	0.627	0.825
4.700	4.712	4309	4204	1604	31.313	34.091	2.645	9.098	482054	6.452	81.379	1194	1726	83.5	62.06	12.24	0.299690	0.394609	0.621	0.631	0.831
4.800	4.812	4432	4324	1727	31.175	34.011	2.704	9.068	491252	6.364	81.477	1203	1747	85.3	63.25	12.33	0.301258	0.399484	0.626	0.636	0.838
4.900	4.912	4556	4445	1851	31.045	33.932	2.764	9.043	500200	6.279	81.573	1212	1768	87.1	64.40	12.43	0.302980	0.404459	0.630	0.640	0.844
5.000	5.012	4681	4566	1976	30.924	33.853	2.823	9.020	508840	6.197	81.668	1221	1789	88.9	65.51	12.51	0.304898	0.409564	0.634	0.644	0.850
5.100	5.112	4806	4689	2101	30.811	33.775	2.882	8.998	517143	6.116	81.760	1229	1809	90.7	66.58	12.60	0.307034	0.414816	0.639	0.648	0.856
5.200	5.212	4933	4812	2228	30.708	33.698	2.941	8.978	525153	6.037	81.851	1237	1829	92.5	67.61	12.68	0.309354	0.420177	0.643	0.652	0.861
5.300	5.312	5060	4936	2355	30.615	33.621	3.000	8.958	532803	5.958	81.940	1244	1849	94.3	68.60	12.75	0.311907	0.425688	0.647	0.656	0.867
5.400	5.412	5188	5061	2483	30.532	33.546	3.059	8.937	540015	5.878	82.027	1251	1871	96.2	69.53	12.82	0.314749	0.431389	0.651	0.660	0.871
5.500	5.512	5316	5187	2611	30.461	33.473	3.118	8.916	546807	5.798	82.112	1258	1890	98.0	70.40	12.89	0.317865	0.437267	0.655	0.664	0.876
5.600	5.612	5445	5313	2741	30.399	33.400	3.176	8.894	553220	5.717	82.196	1264	1909	99.8	71.23	12.95	0.321222	0.443296	0.659	0.668	0.880
5.700	5.712	5575	5439	2870	30.347	33.330	3.235	8.870	559241	5.635	82.278	1269	1927	101.6	72.00	13.01	0.324831	0.449482	0.663	0.672	0.884
5.800	5.812	5706	5566	3001	30.304	33.261	3.293	8.845	564900	5.552	82.359	1274	1945	103.4	72.73	13.06	0.328667	0.455809	0.667	0.676	0.888
5.900	5.912	5836	5694	3131	30.267	33.195	3.352	8.820	570257	5.469	82.438	1279	1963	105.2	73.42	13.11	0.332685	0.462241	0.671	0.679	0.891
6.000	6.012	5968	5822	3263	30.238	33.130	3.410	8.795	575339	5.386	82.515	1283	1981	107.0	74.07	13.16	0.336863	0.468766	0.674	0.683	0.894
6.100	6.112	6099	5951	3395	30.214	33.068	3.468	8.770	580149	5.302	82.591	1288	1999	108.8	74.69	13.20	0.341204	0.475383	0.678	0.687	0.897
6.200	6.212	6232	6080	3527	30.196	33.007	3.525	8.745	584706	5.219	82.665	1292	2017	110.6	75.28	13.24	0.345694	0.482085	0.682	0.690	0.900
6.300	6.312	6364	6209	3659	30.183	32.948	3.583	8.720	589058	5.137	82.738	1295	2039	112.4	75.84	13.28	0.350299	0.488845	0.685	0.693	0.902
6.400	6.412	6497	6339	3792	30.174	32.892	3.640	8.696	593226	5.055	82.809	1299	2058	114.2	76.38	13.31	0.355004	0.495651	0.688	0.697	0.905
6.500	6.512	6630	6469	3926	30.169	32.837	3.698	8.672	597150	4.974	82.878	1302	2075	116.0	76.88	13.35	0.359864	0.502542	0.692	0.700	0.907
6.600	6.612	6764	6599	4059	30.169	32.784	3.755	8.649	600902	4.894	82.947	1305	2093	117.8	77.37	13.38	0.364819	0.509478	0.695	0.703	0.909
6.700	6.712	6898	6730	4193	30.171	32.733	3.812	8.627	604486	4.815	83.013	1308	2111	119.6	77.83	13.41	0.369866	0.516460	0.698	0.706	0.911

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See ABS Singapore Letter Rpt 11-0174 Dated 04-APR-2016

10.4 HYDROSTATICS, TRIM = 0 m



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See ABS Singapore Letter Ref T1499174 Dated 04-APR-2016

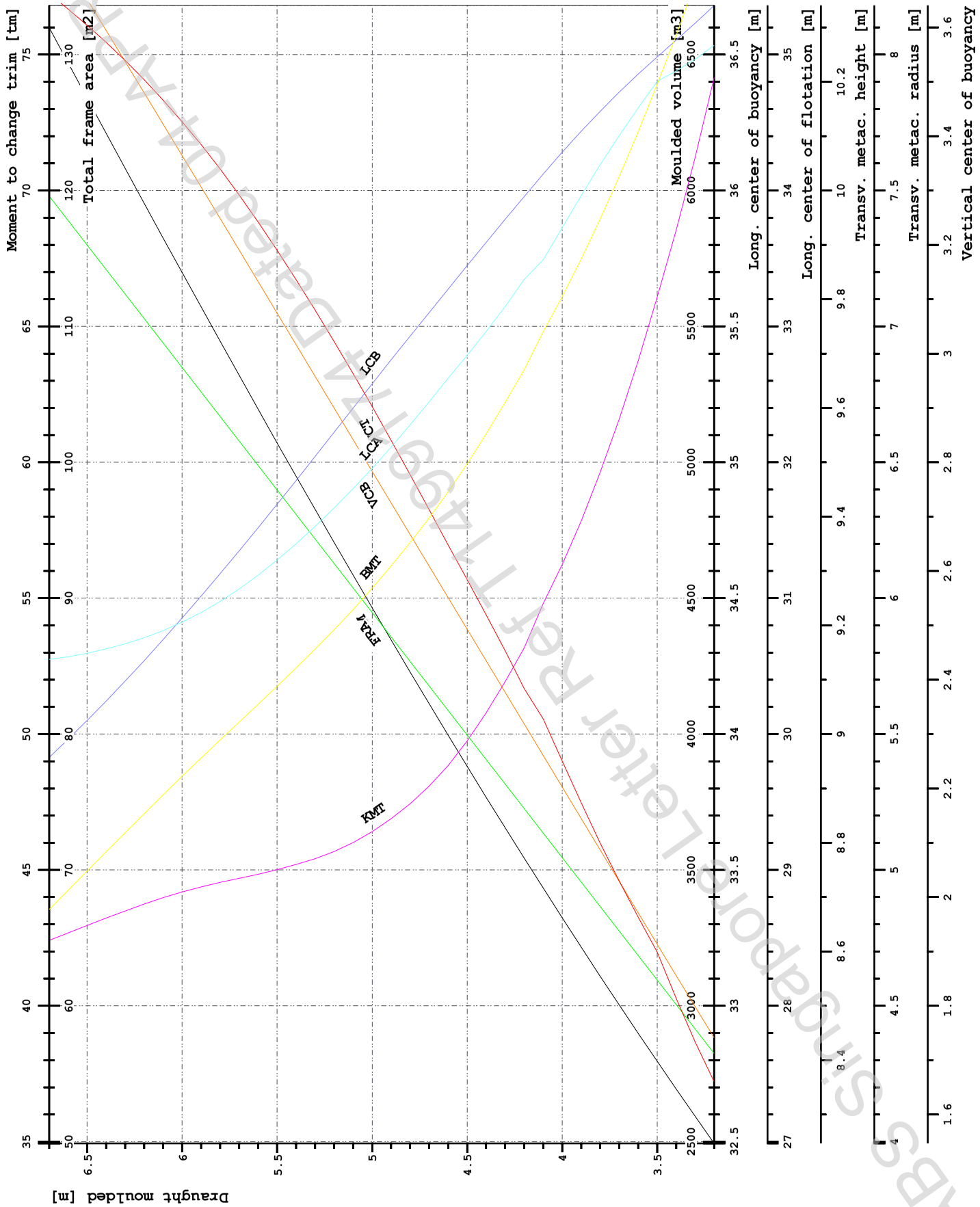
HYDROSTATICS TABLE, TRIM = 0 m

T m	TK m	DISP t	VOLT m3	DW t	LCF m	LCB m	VCB m	KMT m	IMAX m4	BMT m	LWL m	WLA m2	WSA m2	FRA m2	MCT tm/cm	TCP t/cm	TRFAP m	TRFFP m	CB	CP	CW
3.200	3.212	2586	2522	-119	34.419	35.942	1.746	10.263	298787	8.517	76.946	998	1346	56.5	38.47	10.23	0.316336	0.355782	0.547	0.560	0.695
3.300	3.312	2689	2623	-16	34.247	35.881	1.804	10.116	309226	8.312	77.568	1011	1371	58.3	39.81	10.36	0.316372	0.358926	0.551	0.564	0.704
3.400	3.412	2793	2725	88	34.077	35.814	1.861	9.982	320347	8.120	78.187	1024	1395	60.1	41.24	10.50	0.315769	0.361320	0.556	0.569	0.712
3.500	3.512	2898	2828	193	33.945	35.747	1.919	9.860	333623	7.941	78.803	1039	1421	61.9	42.95	10.65	0.313549	0.361187	0.560	0.573	0.723
3.600	3.612	3006	2932	301	33.829	35.681	1.978	9.750	348438	7.772	79.416	1054	1450	63.7	44.86	10.81	0.310355	0.359607	0.565	0.577	0.733
3.700	3.712	3114	3038	410	33.693	35.615	2.036	9.649	363806	7.613	80.028	1070	1476	65.5	46.84	10.97	0.306881	0.358034	0.570	0.582	0.745
3.800	3.812	3225	3146	520	33.455	35.545	2.095	9.554	375951	7.459	80.389	1083	1501	67.3	48.40	11.10	0.305499	0.360739	0.574	0.586	0.754
3.900	3.912	3337	3255	632	33.217	35.471	2.154	9.469	388166	7.315	80.501	1096	1526	69.1	49.98	11.24	0.304144	0.363478	0.579	0.591	0.763
4.000	4.012	3449	3365	744	33.085	35.395	2.212	9.372	396578	7.160	80.611	1106	1556	70.9	51.06	11.33	0.306645	0.368923	0.584	0.595	0.770
4.100	4.112	3563	3476	858	32.890	35.318	2.271	9.298	407363	7.027	80.719	1117	1579	72.7	52.45	11.45	0.306717	0.372679	0.588	0.600	0.778
4.200	4.212	3678	3589	973	32.711	35.239	2.330	9.230	417635	6.900	80.824	1128	1601	74.5	53.77	11.56	0.307288	0.376791	0.593	0.604	0.785
4.300	4.312	3794	3702	1090	32.539	35.159	2.389	9.169	427676	6.780	80.927	1138	1623	76.3	55.06	11.67	0.308056	0.381048	0.598	0.608	0.793
4.400	4.412	3912	3816	1207	32.368	35.079	2.447	9.115	437720	6.668	81.028	1149	1645	78.1	56.36	11.78	0.308792	0.385292	0.602	0.613	0.800
4.500	4.512	4030	3932	1325	32.194	34.997	2.506	9.069	447917	6.563	81.127	1159	1667	79.9	57.67	11.88	0.309361	0.389423	0.607	0.617	0.807
4.600	4.612	4149	4048	1444	32.027	34.915	2.565	9.029	457938	6.464	81.224	1170	1689	81.7	58.96	11.99	0.310071	0.393655	0.611	0.621	0.814
4.700	4.712	4269	4165	1565	31.866	34.832	2.624	8.994	467776	6.371	81.319	1180	1713	83.5	60.23	12.09	0.310918	0.397994	0.615	0.626	0.821
4.800	4.812	4391	4284	1686	31.711	34.747	2.682	8.964	477416	6.282	81.412	1189	1735	85.3	61.47	12.19	0.311919	0.402437	0.620	0.630	0.828
4.900	4.912	4513	4403	1808	31.564	34.662	2.741	8.938	486821	6.197	81.504	1199	1756	87.1	62.68	12.29	0.313093	0.407000	0.624	0.634	0.835
5.000	5.012	4637	4524	1932	31.424	34.577	2.800	8.915	496013	6.115	81.593	1208	1778	88.9	63.86	12.38	0.314416	0.411663	0.628	0.638	0.841
5.100	5.112	4761	4645	2056	31.290	34.492	2.859	8.896	504992	6.038	81.681	1217	1799	90.7	65.02	12.47	0.315885	0.416419	0.633	0.642	0.847
5.200	5.212	4886	4767	2182	31.165	34.409	2.918	8.880	513676	5.962	81.768	1225	1820	92.5	66.14	12.56	0.317550	0.421302	0.637	0.646	0.853
5.300	5.312	5012	4890	2308	31.049	34.325	2.976	8.865	522025	5.889	81.852	1234	1840	94.3	67.21	12.64	0.319446	0.426342	0.641	0.650	0.859
5.400	5.412	5139	5014	2434	30.943	34.243	3.035	8.851	530059	5.816	81.935	1241	1860	96.2	68.24	12.72	0.321554	0.431517	0.645	0.654	0.865
5.500	5.512	5267	5138	2562	30.845	34.162	3.093	8.838	537774	5.744	82.016	1249	1880	98.0	69.24	12.80	0.323875	0.436824	0.649	0.658	0.870
5.600	5.612	5395	5264	2690	30.757	34.082	3.152	8.824	545084	5.672	82.096	1256	1902	99.8	70.18	12.87	0.326474	0.442314	0.653	0.662	0.875
5.700	5.712	5524	5390	2819	30.680	34.003	3.210	8.810	551964	5.599	82.174	1262	1921	101.6	71.07	12.94	0.329365	0.447996	0.657	0.666	0.879
5.800	5.812	5654	5516	2949	30.614	33.927	3.268	8.794	558449	5.526	82.250	1268	1940	103.4	71.90	13.00	0.332525	0.453844	0.661	0.670	0.883
5.900	5.912	5784	5643	3079	30.557	33.851	3.326	8.778	564557	5.451	82.325	1274	1959	105.2	72.69	13.05	0.335935	0.459846	0.665	0.673	0.887
6.000	6.012	5915	5771	3210	30.509	33.778	3.384	8.760	570288	5.375	82.398	1279	1977	107.0	73.42	13.11	0.339593	0.466006	0.668	0.677	0.891
6.100	6.112	6046	5899	3341	30.468	33.707	3.442	8.741	575685	5.299	82.470	1284	1995	108.8	74.12	13.16	0.343464	0.472300	0.672	0.681	0.894
6.200	6.212	6178	6027	3473	30.435	33.637	3.500	8.722	580810	5.222	82.540	1288	2013	110.6	74.78	13.20	0.347502	0.478685	0.676	0.684	0.897
6.300	6.312	6310	6156	3605	30.407	33.570	3.558	8.703	585673	5.145	82.608	1292	2031	112.4	75.41	13.25	0.351703	0.485160	0.679	0.688	0.900
6.400	6.412	6443	6286	3738	30.385	33.505	3.615	8.684	590275	5.069	82.676	1296	2049	114.2	76.00	13.29	0.356065	0.491727	0.683	0.691	0.903
6.500	6.512	6576	6416	3871	30.369	33.442	3.672	8.665	594643	4.992	82.742	1300	2066	116.0	76.56	13.33	0.360569	0.498372	0.686	0.694	0.906
6.600	6.612	6709	6546	4005	30.356	33.380	3.730	8.646	598815	4.916	82.806	1304	2089	117.8	77.10	13.36	0.365187	0.505077	0.689	0.698	0.908
6.700	6.712	6843	6676	4138	30.348	33.321	3.787	8.628	602794	4.841	82.869	1307	2107	119.6	77.61	13.39	0.369917	0.511841	0.693	0.701	0.910

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10.5 HYDROSTATICS, TRIM = 0.5 m



See ABS Singapore Letter Ref T14991A Dated 04 APR 2016

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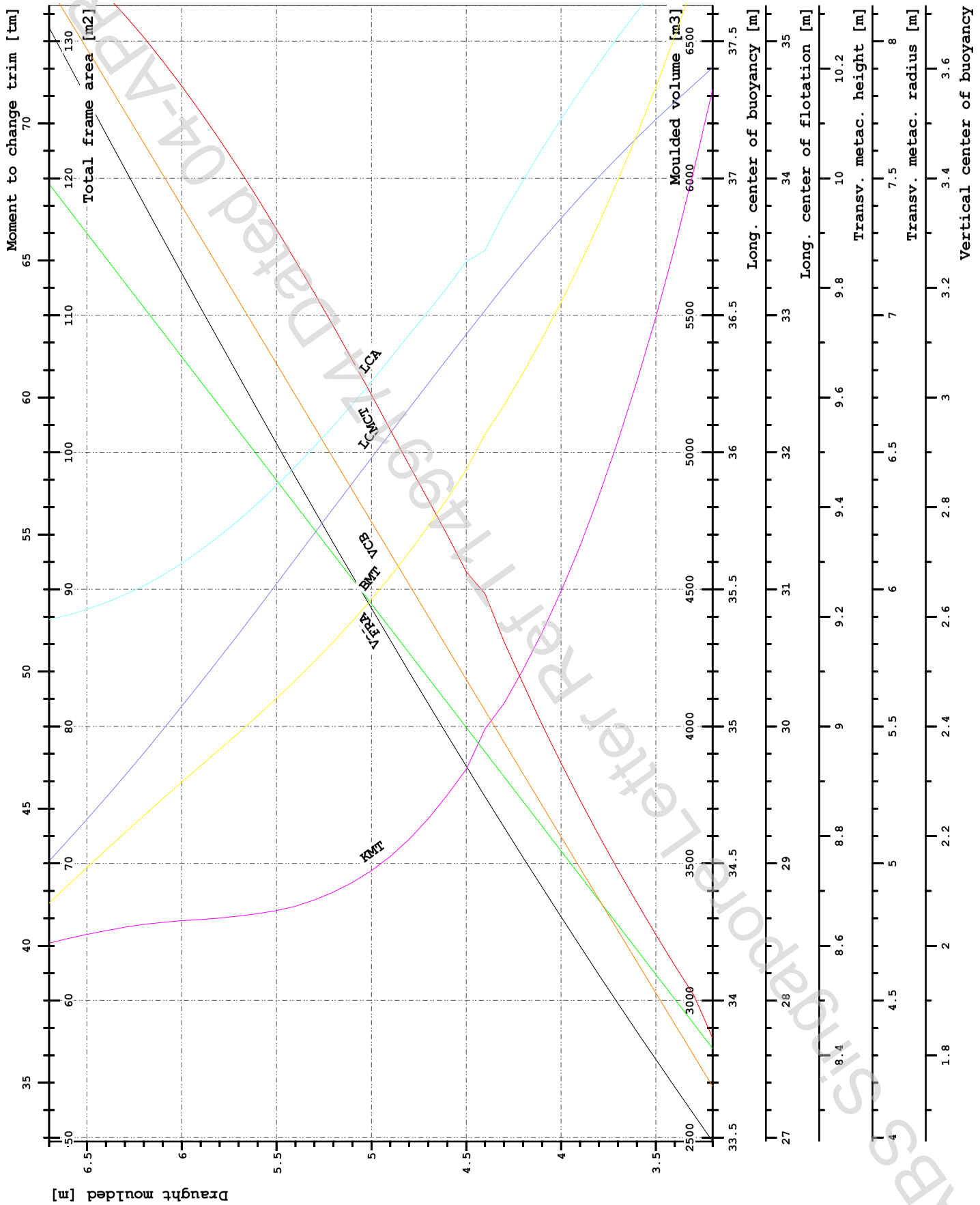
HYDROSTATICS TABLE, TRIM = 0.5 m

T	TK	DISP	VOLT	DW	LCF	LCB	VCB	KMT	IMAX	BMT	LWL	WLA	WSA	FRA	MCT	TCP	TRFAP	TRFFP	CB	CP	CW
m	m	t	m3	t	m	m	m	m	m4	m	m	m2	m2	m2	tm/cm	t/cm	m	m			
3.200	3.212	2573	2510	-132	35.068	36.681	1.741	10.208	289013	8.467	76.914	987	1334	56.5	37.21	10.12	0.331025	0.360325	0.544	0.557	0.687
3.300	3.312	2674	2609	-31	34.968	36.616	1.798	10.062	300243	8.264	76.952	1001	1359	58.3	38.66	10.26	0.330374	0.361440	0.548	0.561	0.696
3.400	3.412	2778	2710	73	34.879	36.554	1.855	9.928	312770	8.072	77.020	1015	1385	60.1	40.27	10.41	0.328634	0.361135	0.553	0.565	0.706
3.500	3.512	2882	2812	178	34.804	36.492	1.913	9.804	326043	7.891	77.611	1030	1411	61.9	41.98	10.56	0.326509	0.360158	0.557	0.570	0.717
3.600	3.612	2989	2916	284	34.608	36.428	1.971	9.687	336014	7.716	78.199	1042	1435	63.7	43.26	10.68	0.326782	0.364049	0.562	0.574	0.725
3.700	3.712	3096	3020	391	34.403	36.362	2.029	9.579	346405	7.551	78.785	1053	1458	65.5	44.60	10.80	0.326586	0.367595	0.566	0.578	0.733
3.800	3.812	3205	3126	500	34.189	36.291	2.086	9.481	357253	7.395	79.369	1065	1483	67.3	46.00	10.92	0.325904	0.370808	0.571	0.583	0.741
3.900	3.912	3314	3234	609	33.963	36.218	2.144	9.392	368666	7.248	79.951	1077	1507	69.1	47.47	11.04	0.324650	0.373620	0.575	0.587	0.750
4.000	4.012	3425	3342	721	33.727	36.141	2.202	9.312	380610	7.110	80.531	1090	1532	70.9	49.00	11.17	0.322927	0.376098	0.580	0.591	0.758
4.100	4.112	3538	3452	833	33.489	36.061	2.260	9.240	392712	6.980	80.679	1102	1557	72.7	50.56	11.30	0.321153	0.378554	0.584	0.595	0.767
4.200	4.212	3651	3562	947	33.346	35.977	2.318	9.158	401367	6.840	80.778	1112	1587	74.5	51.68	11.40	0.323043	0.383561	0.589	0.600	0.774
4.300	4.312	3766	3674	1061	33.146	35.894	2.376	9.096	412211	6.720	80.876	1123	1610	76.3	53.07	11.51	0.322628	0.386970	0.593	0.604	0.782
4.400	4.412	3882	3787	1177	32.963	35.810	2.435	9.039	422476	6.604	80.972	1134	1633	78.1	54.39	11.62	0.322818	0.390803	0.598	0.608	0.789
4.500	4.512	3998	3901	1293	32.788	35.724	2.493	8.988	432465	6.495	81.066	1144	1655	79.9	55.68	11.73	0.323271	0.394829	0.602	0.612	0.797
4.600	4.612	4116	4016	1411	32.619	35.638	2.551	8.942	442327	6.391	81.158	1154	1676	81.7	56.95	11.83	0.323837	0.398930	0.606	0.616	0.804
4.700	4.712	4235	4132	1530	32.445	35.552	2.609	8.904	452471	6.295	81.249	1165	1698	83.5	58.26	11.94	0.324121	0.402836	0.610	0.621	0.811
4.800	4.812	4355	4249	1650	32.276	35.464	2.667	8.872	462466	6.205	81.338	1175	1720	85.3	59.54	12.04	0.324533	0.406836	0.615	0.625	0.818
4.900	4.912	4476	4366	1771	32.112	35.377	2.725	8.844	472301	6.119	81.425	1185	1742	87.1	60.81	12.14	0.325084	0.410926	0.619	0.629	0.825
5.000	5.012	4597	4485	1893	31.956	35.288	2.783	8.821	481923	6.037	81.510	1194	1766	88.9	62.05	12.24	0.325821	0.415145	0.623	0.633	0.832
5.100	5.112	4720	4605	2016	31.807	35.199	2.842	8.800	491332	5.959	81.593	1204	1787	90.7	63.26	12.34	0.326724	0.419484	0.627	0.637	0.838
5.200	5.212	4844	4726	2139	31.664	35.110	2.900	8.784	500520	5.884	81.675	1213	1809	92.5	64.44	12.43	0.327796	0.423939	0.631	0.641	0.844
5.300	5.312	4969	4848	2264	31.528	35.021	2.958	8.771	509507	5.813	81.755	1221	1830	94.3	65.60	12.52	0.329017	0.428489	0.635	0.645	0.851
5.400	5.412	5095	4971	2390	31.400	34.933	3.016	8.760	518224	5.744	81.834	1230	1851	96.2	66.72	12.61	0.330436	0.433171	0.639	0.649	0.856
5.500	5.512	5221	5094	2516	31.280	34.846	3.075	8.751	526606	5.676	81.911	1238	1871	98.0	67.80	12.69	0.332095	0.438012	0.643	0.653	0.862
5.600	5.612	5349	5218	2644	31.170	34.759	3.133	8.742	534674	5.610	81.986	1246	1891	99.8	68.84	12.77	0.333980	0.442995	0.647	0.656	0.867
5.700	5.712	5477	5343	2772	31.068	34.674	3.191	8.735	542431	5.544	82.060	1253	1911	101.6	69.84	12.84	0.336085	0.448115	0.651	0.660	0.873
5.800	5.812	5605	5469	2901	30.976	34.590	3.249	8.727	549834	5.479	82.132	1260	1931	103.4	70.79	12.91	0.338440	0.453394	0.655	0.664	0.877
5.900	5.912	5735	5595	3030	30.894	34.508	3.306	8.719	556813	5.412	82.203	1266	1953	105.2	71.69	12.98	0.341095	0.458874	0.659	0.668	0.882
6.000	6.012	5865	5722	3160	30.823	34.427	3.364	8.709	563379	5.345	82.272	1272	1972	107.0	72.53	13.04	0.344042	0.464541	0.663	0.671	0.886
6.100	6.112	5996	5849	3291	30.761	34.348	3.422	8.699	569570	5.277	82.340	1278	1990	108.8	73.33	13.10	0.347252	0.470365	0.666	0.675	0.890
6.200	6.212	6127	5978	3422	30.709	34.270	3.480	8.687	575381	5.207	82.406	1283	2009	110.6	74.08	13.15	0.350724	0.476352	0.670	0.678	0.894
6.300	6.312	6259	6106	3554	30.664	34.195	3.537	8.675	580857	5.137	82.471	1288	2027	112.4	74.79	13.20	0.354420	0.482478	0.674	0.682	0.897
6.400	6.412	6391	6235	3686	30.626	34.121	3.595	8.662	586032	5.067	82.535	1293	2045	114.2	75.45	13.25	0.358312	0.488723	0.677	0.685	0.900
6.500	6.512	6524	6365	3819	30.595	34.050	3.652	8.648	590932	4.996	82.597	1297	2063	116.0	76.08	13.29	0.362380	0.495070	0.681	0.689	0.903
6.600	6.612	6657	6494	3952	30.569	33.981	3.709	8.634	595583	4.925	82.658	1301	2080	117.8	76.68	13.33	0.366606	0.501506	0.684	0.692	0.906
6.700	6.712	6790	6625	4085	30.549	33.914	3.766	8.621	599985	4.855	82.718	1305	2098	119.6	77.25	13.37	0.370990	0.508034	0.687	0.695	0.909

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See ABS Singapore Letter Ref. 14699174 Rapid QUAR-2016

10.6 HYDROSTATICS, TRIM = 1 m



See ABS Shipboard Letter Perf 17499 Dated 04-APR-2016

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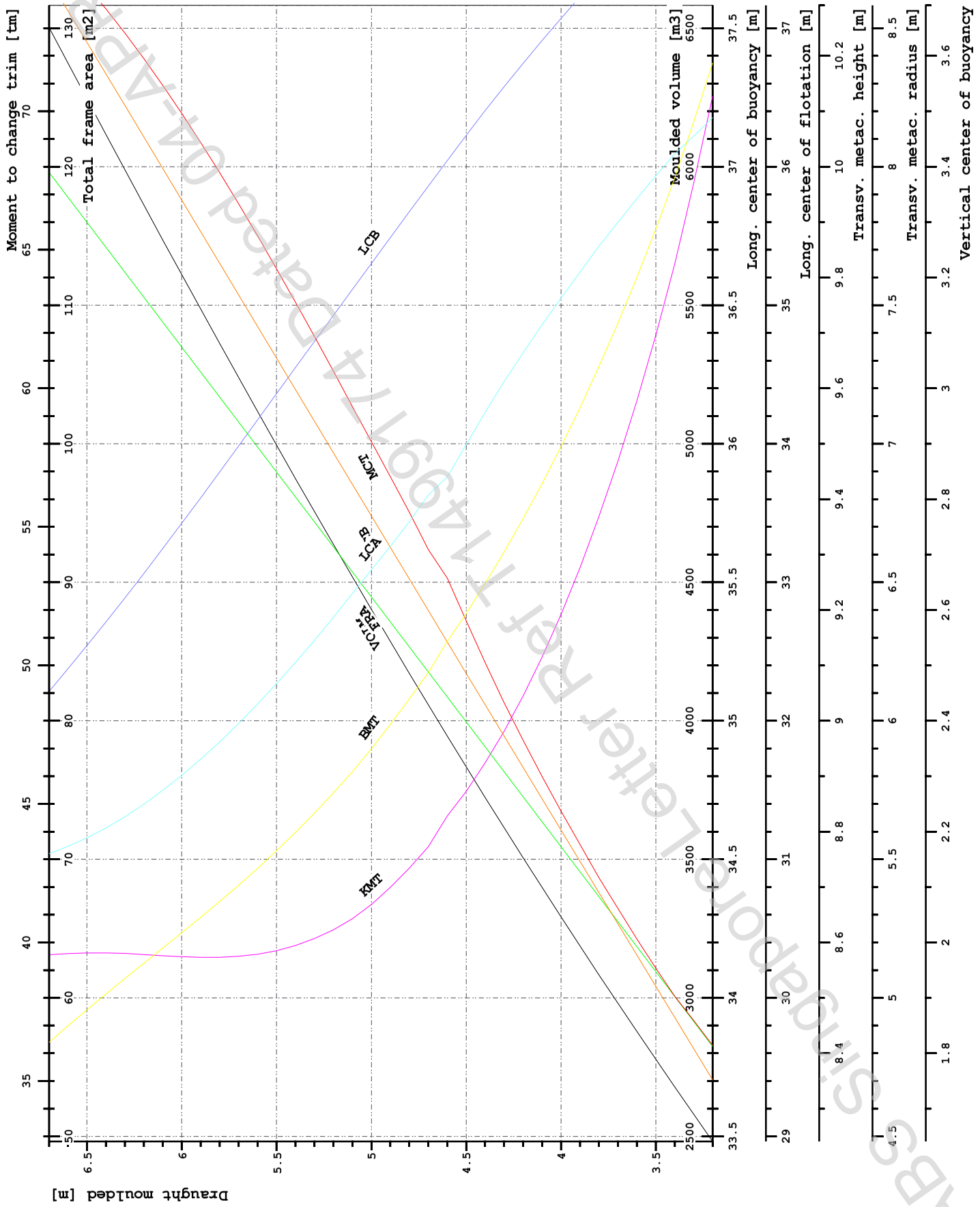
HYDROSTATICS TABLE, TRIM = 1 m

T m	TK m	DISP t	VOLT m3	DW t	LCF m	LCB m	VCB m	KMT m	IMAX m4	BMT m	LWL m	WLA m2	WSA m2	FRA m2	MCT tm/cm	TCP t/cm	TRFAP m	TRFFP m	CB	CP	CW
3.200	3.212	2564	2501	-141	35.774	37.403	1.743	10.162	284534	8.419	77.146	981	1326	56.5	36.63	10.06	0.341329	0.358570	0.542	0.555	0.682
3.300	3.312	2665	2600	-40	35.738	37.341	1.800	10.015	296595	8.215	77.157	996	1352	58.3	38.19	10.21	0.340070	0.357905	0.546	0.559	0.693
3.400	3.412	2768	2700	63	35.571	37.279	1.857	9.876	305046	8.019	77.185	1006	1375	60.1	39.27	10.31	0.341894	0.362865	0.551	0.564	0.700
3.500	3.512	2872	2802	167	35.396	37.214	1.914	9.747	313834	7.833	77.212	1017	1398	61.9	40.41	10.43	0.343209	0.367479	0.555	0.568	0.708
3.600	3.612	2976	2904	271	35.215	37.147	1.971	9.629	322924	7.658	77.234	1028	1420	63.7	41.58	10.54	0.344093	0.371795	0.560	0.572	0.715
3.700	3.712	3082	3007	377	35.031	37.077	2.028	9.520	332196	7.493	77.620	1039	1443	65.5	42.77	10.65	0.344727	0.375946	0.564	0.576	0.723
3.800	3.812	3189	3112	484	34.838	37.005	2.085	9.420	341889	7.336	78.181	1050	1467	67.3	44.02	10.76	0.344828	0.379728	0.568	0.580	0.731
3.900	3.912	3298	3217	593	34.640	36.931	2.142	9.329	351878	7.187	78.739	1061	1490	69.1	45.30	10.88	0.344599	0.383268	0.572	0.584	0.738
4.000	4.012	3407	3324	702	34.431	36.854	2.199	9.247	362421	7.048	79.296	1073	1514	70.9	46.66	10.99	0.343748	0.386383	0.577	0.588	0.746
4.100	4.112	3517	3432	813	34.215	36.774	2.256	9.171	373312	6.915	79.851	1084	1538	72.7	48.06	11.11	0.342565	0.389261	0.581	0.592	0.755
4.200	4.212	3629	3541	924	33.983	36.692	2.314	9.104	384963	6.790	80.405	1096	1563	74.5	49.56	11.24	0.340615	0.391599	0.585	0.596	0.763
4.300	4.312	3742	3651	1037	33.749	36.607	2.371	9.042	396843	6.671	80.815	1108	1588	76.3	51.09	11.36	0.338550	0.393858	0.589	0.600	0.771
4.400	4.412	3856	3762	1151	33.473	36.518	2.428	8.995	410364	6.567	80.906	1122	1614	78.1	52.83	11.50	0.334864	0.395036	0.594	0.604	0.779
4.500	4.512	3972	3875	1267	33.392	36.430	2.486	8.922	416634	6.436	80.996	1129	1642	79.9	53.64	11.57	0.338920	0.401469	0.598	0.608	0.786
4.600	4.612	4088	3988	1383	33.204	36.341	2.543	8.875	427024	6.331	81.083	1140	1664	81.7	54.98	11.68	0.338593	0.404926	0.602	0.612	0.793
4.700	4.712	4205	4103	1500	33.028	36.251	2.601	8.832	436996	6.231	81.169	1150	1686	83.5	56.26	11.78	0.338712	0.408688	0.606	0.616	0.800
4.800	4.812	4323	4218	1619	32.859	36.161	2.658	8.794	446798	6.136	81.254	1160	1708	85.3	57.52	11.89	0.339005	0.412567	0.610	0.620	0.807
4.900	4.912	4443	4334	1738	32.687	36.070	2.716	8.763	456777	6.047	81.336	1170	1729	87.1	58.81	11.99	0.339117	0.416324	0.614	0.624	0.814
5.000	5.012	4563	4452	1858	32.516	35.979	2.773	8.737	466757	5.964	81.417	1180	1751	88.9	60.09	12.09	0.339224	0.420088	0.618	0.628	0.821
5.100	5.112	4684	4570	1979	32.351	35.888	2.831	8.716	476561	5.885	81.496	1189	1773	90.7	61.36	12.19	0.339499	0.423967	0.622	0.632	0.828
5.200	5.212	4807	4689	2102	32.193	35.797	2.889	8.698	486174	5.809	81.574	1199	1796	92.5	62.59	12.29	0.339951	0.427961	0.626	0.636	0.835
5.300	5.312	4930	4810	2225	32.041	35.704	2.946	8.683	495573	5.737	81.650	1208	1818	94.3	63.80	12.38	0.340592	0.432089	0.630	0.640	0.841
5.400	5.412	5054	4931	2349	31.897	35.612	3.004	8.672	504750	5.668	81.724	1217	1840	96.2	64.99	12.47	0.341419	0.436344	0.634	0.643	0.847
5.500	5.512	5180	5053	2475	31.758	35.520	3.061	8.664	513736	5.603	81.797	1226	1861	98.0	66.14	12.56	0.342398	0.440697	0.638	0.647	0.853
5.600	5.612	5306	5176	2601	31.627	35.428	3.119	8.658	522471	5.539	81.868	1234	1882	99.8	67.27	12.65	0.343575	0.445180	0.642	0.651	0.859
5.700	5.712	5433	5300	2728	31.505	35.338	3.177	8.654	530906	5.477	81.938	1242	1902	101.6	68.35	12.73	0.344982	0.449812	0.646	0.655	0.865
5.800	5.812	5560	5425	2856	31.391	35.248	3.234	8.651	539023	5.416	82.006	1250	1923	103.4	69.40	12.81	0.346627	0.454597	0.650	0.659	0.870
5.900	5.912	5689	5550	2984	31.286	35.160	3.292	8.648	546818	5.356	82.073	1257	1942	105.2	70.40	12.88	0.348513	0.459536	0.654	0.662	0.875
6.000	6.012	5818	5676	3113	31.189	35.073	3.349	8.646	554292	5.296	82.139	1264	1962	107.0	71.36	12.95	0.350634	0.464622	0.657	0.666	0.880
6.100	6.112	5948	5803	3243	31.103	34.987	3.407	8.643	561374	5.236	82.203	1270	1981	108.8	72.28	13.02	0.353047	0.469893	0.661	0.670	0.885
6.200	6.212	6078	5930	3374	31.027	34.903	3.464	8.639	568039	5.175	82.265	1276	2003	110.6	73.13	13.08	0.355767	0.475364	0.665	0.673	0.889
6.300	6.312	6210	6058	3505	30.961	34.820	3.521	8.634	574308	5.112	82.327	1282	2022	112.4	73.94	13.14	0.358776	0.481016	0.668	0.677	0.893
6.400	6.412	6341	6187	3636	30.904	34.740	3.579	8.628	580203	5.049	82.387	1287	2040	114.2	74.70	13.20	0.362050	0.486835	0.672	0.680	0.897
6.500	6.512	6473	6316	3769	30.854	34.661	3.636	8.621	585758	4.985	82.446	1292	2059	116.0	75.42	13.25	0.365562	0.492801	0.675	0.683	0.900
6.600	6.612	6606	6445	3901	30.813	34.584	3.693	8.613	590986	4.920	82.503	1297	2077	117.8	76.09	13.29	0.369300	0.498905	0.679	0.687	0.903
6.700	6.712	6739	6575	4034	30.778	34.509	3.750	8.604	595925	4.855	82.560	1301	2095	119.6	76.73	13.34	0.373231	0.505128	0.682	0.690	0.906

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10.7 HYDROSTATICS, TRIM = 1.5 m



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HYDROSTATICS TABLE, TRIM = 1.5 m

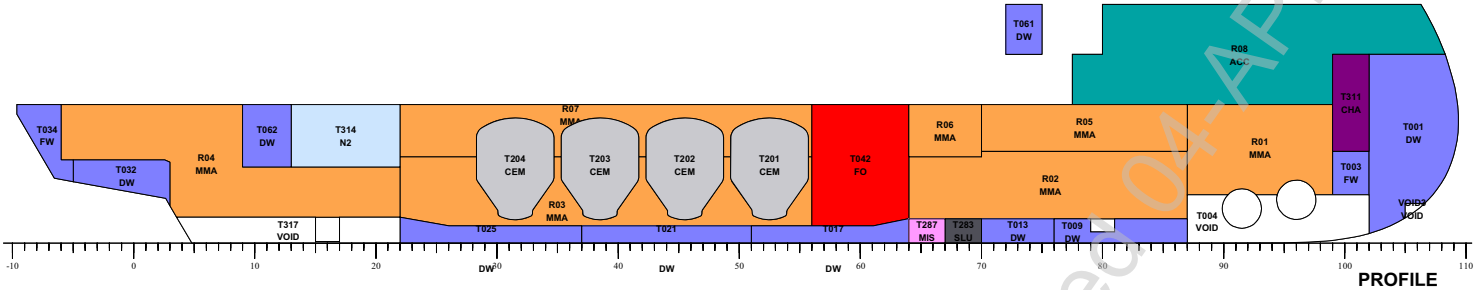
T m	TK m	DISP t	VOLT m3	DW t	LCF m	LCB m	VCB m	KMT m	IMAX m4	BMT m	LWL m	WLA m2	WSA m2	FRA m2	MCT tm/cm	TCP t/cm	TRFAP m	TRFFP m	CB	CP	CW
3.200	3.212	2560	2497	-145	36.355	38.120	1.752	10.127	281828	8.375	77.516	977	1314	56.5	36.29	10.01	0.349196	0.356294	0.541	0.554	0.679
3.300	3.312	2660	2596	-44	36.222	38.050	1.808	9.972	288740	8.163	77.444	986	1341	58.3	37.18	10.11	0.353034	0.362621	0.545	0.558	0.686
3.400	3.412	2762	2695	57	36.093	37.981	1.865	9.825	295554	7.961	77.407	996	1363	60.1	38.05	10.20	0.356881	0.368958	0.550	0.562	0.693
3.500	3.512	2865	2795	160	35.935	37.911	1.921	9.696	303432	7.775	77.397	1006	1386	61.9	39.07	10.31	0.359074	0.374169	0.554	0.566	0.700
3.600	3.612	2968	2896	263	35.769	37.839	1.977	9.577	311681	7.600	77.408	1016	1408	63.7	40.13	10.41	0.360668	0.378981	0.558	0.570	0.707
3.700	3.712	3073	2998	368	35.597	37.765	2.033	9.468	320242	7.435	77.427	1026	1430	65.5	41.23	10.52	0.361788	0.383470	0.562	0.574	0.714
3.800	3.812	3178	3101	474	35.426	37.690	2.090	9.368	328850	7.278	77.440	1037	1453	67.3	42.34	10.62	0.362824	0.387891	0.566	0.578	0.721
3.900	3.912	3285	3205	580	35.239	37.614	2.146	9.276	338105	7.130	77.599	1047	1476	69.1	43.53	10.73	0.362980	0.391715	0.570	0.582	0.729
4.000	4.012	3393	3310	688	35.052	37.535	2.202	9.192	347475	6.990	78.135	1058	1499	70.9	44.74	10.84	0.363007	0.395447	0.574	0.586	0.736
4.100	4.112	3502	3417	797	34.859	37.455	2.259	9.115	357209	6.856	78.669	1069	1522	72.7	45.99	10.95	0.362605	0.398872	0.578	0.589	0.744
4.200	4.212	3612	3524	907	34.656	37.373	2.315	9.045	367409	6.729	79.201	1080	1546	74.5	47.30	11.07	0.361670	0.401936	0.582	0.593	0.751
4.300	4.312	3723	3633	1018	34.448	37.288	2.372	8.981	377938	6.609	79.732	1091	1569	76.3	48.66	11.18	0.360425	0.404770	0.586	0.597	0.759
4.400	4.412	3836	3742	1131	34.225	37.202	2.429	8.924	389199	6.495	80.261	1102	1594	78.1	50.11	11.30	0.358407	0.407064	0.590	0.601	0.767
4.500	4.512	3949	3853	1244	33.995	37.113	2.485	8.873	400828	6.387	80.788	1114	1618	79.9	51.61	11.42	0.356106	0.409161	0.594	0.605	0.775
4.600	4.612	4064	3965	1359	33.760	37.022	2.542	8.828	412794	6.286	80.999	1126	1643	81.7	53.15	11.54	0.353574	0.411106	0.598	0.609	0.784
4.700	4.712	4180	4078	1475	33.629	36.929	2.599	8.773	420830	6.174	81.081	1134	1673	83.5	54.18	11.63	0.355444	0.416018	0.602	0.612	0.790
4.800	4.812	4297	4192	1592	33.437	36.837	2.656	8.734	431301	6.078	81.161	1145	1695	85.3	55.53	11.74	0.354637	0.419128	0.606	0.616	0.797
4.900	4.912	4415	4307	1710	33.259	36.744	2.713	8.699	441298	5.987	81.239	1155	1717	87.1	56.82	11.84	0.354367	0.422608	0.610	0.620	0.804
5.000	5.012	4533	4423	1828	33.089	36.651	2.769	8.669	451051	5.899	81.315	1165	1739	88.9	58.07	11.94	0.354370	0.426270	0.614	0.624	0.811
5.100	5.112	4653	4540	1948	32.920	36.557	2.826	8.643	460838	5.817	81.390	1174	1761	90.7	59.33	12.04	0.354345	0.429911	0.618	0.628	0.818
5.200	5.212	4774	4658	2069	32.748	36.463	2.883	8.623	470786	5.740	81.464	1184	1782	92.5	60.61	12.14	0.354161	0.433453	0.622	0.631	0.825
5.300	5.312	4896	4776	2191	32.581	36.369	2.940	8.607	480574	5.667	81.536	1194	1804	94.3	61.87	12.24	0.354148	0.437109	0.626	0.635	0.831
5.400	5.412	5019	4896	2314	32.422	36.275	2.997	8.595	490175	5.597	81.606	1203	1825	96.2	63.11	12.33	0.354318	0.440887	0.630	0.639	0.838
5.500	5.512	5142	5017	2437	32.269	36.181	3.054	8.585	499554	5.531	81.675	1212	1849	98.0	64.32	12.42	0.354705	0.444812	0.634	0.643	0.844
5.600	5.612	5267	5139	2562	32.122	36.086	3.111	8.579	508747	5.467	81.742	1221	1870	99.8	65.50	12.52	0.355260	0.448848	0.638	0.646	0.850
5.700	5.712	5393	5261	2688	31.982	35.992	3.168	8.575	517723	5.407	81.808	1230	1892	101.6	66.66	12.60	0.356002	0.453014	0.641	0.650	0.856
5.800	5.812	5519	5385	2814	31.848	35.898	3.225	8.573	526477	5.348	81.873	1238	1913	103.4	67.78	12.69	0.356930	0.457299	0.645	0.654	0.862
5.900	5.912	5647	5509	2942	31.723	35.804	3.283	8.573	534952	5.291	81.936	1246	1934	105.2	68.87	12.77	0.358092	0.461735	0.649	0.657	0.868
6.000	6.012	5775	5634	3070	31.606	35.712	3.340	8.574	543110	5.235	81.998	1254	1954	107.0	69.93	12.85	0.359502	0.466326	0.653	0.661	0.873
6.100	6.112	5903	5759	3199	31.498	35.621	3.397	8.576	550953	5.179	82.058	1261	1974	108.8	70.93	12.92	0.361159	0.471077	0.656	0.664	0.878
6.200	6.212	6033	5886	3328	31.398	35.532	3.454	8.578	558465	5.125	82.117	1268	1993	110.6	71.90	12.99	0.363067	0.475995	0.660	0.668	0.883
6.300	6.312	6163	6013	3458	31.307	35.443	3.511	8.580	565629	5.070	82.175	1274	2013	112.4	72.82	13.06	0.365244	0.481077	0.663	0.672	0.888
6.400	6.412	6294	6141	3589	31.226	35.357	3.567	8.581	572401	5.014	82.232	1280	2034	114.2	73.70	13.12	0.367720	0.486350	0.667	0.675	0.892
6.500	6.512	6426	6269	3721	31.155	35.271	3.624	8.581	578754	4.957	82.288	1286	2053	116.0	74.51	13.18	0.370515	0.491830	0.670	0.678	0.896
6.600	6.612	6558	6398	3853	31.093	35.188	3.681	8.580	584725	4.899	82.343	1291	2072	117.8	75.28	13.24	0.373591	0.497490	0.674	0.682	0.900
6.700	6.712	6690	6527	3985	31.040	35.106	3.738	8.578	590346	4.840	82.397	1296	2090	119.6	76.01	13.29	0.376923	0.503308	0.677	0.685	0.903

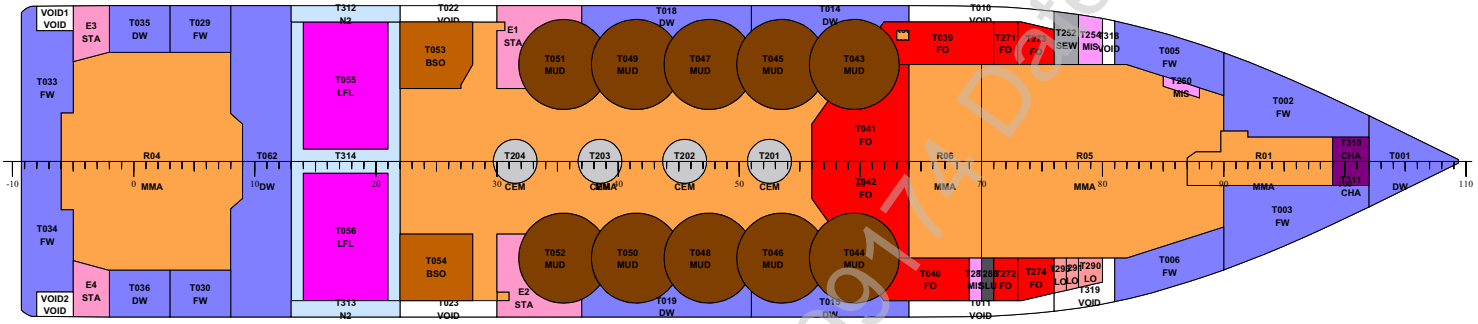
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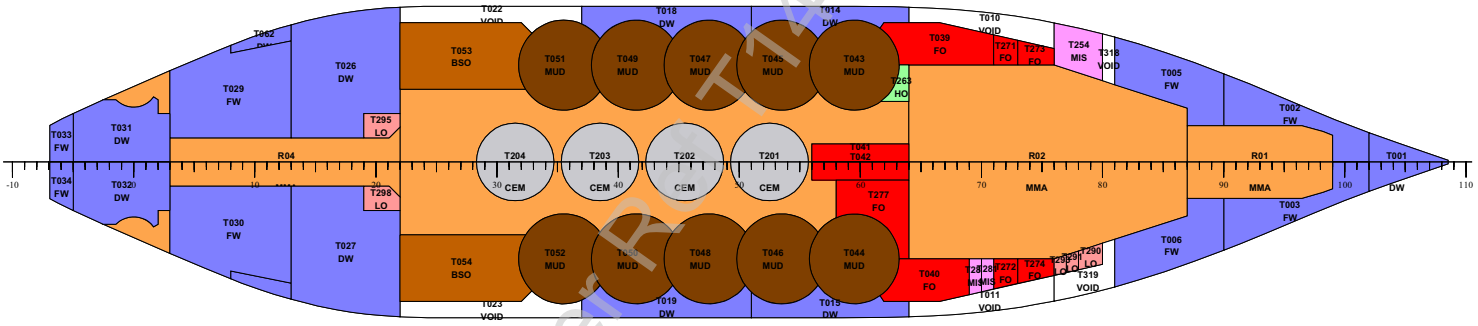
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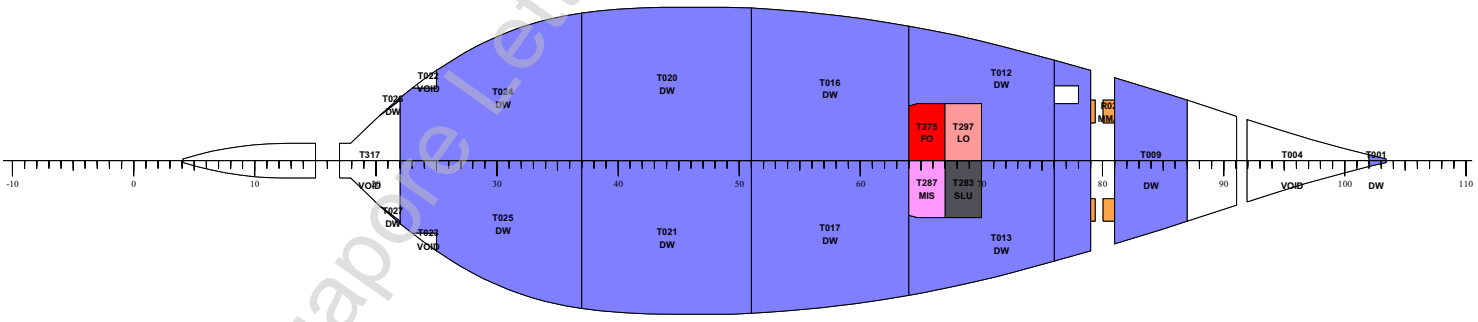
11 TANK TABLES

11.1 TANK ARRANGEMENT









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11.2 TANK TABLES

Not part of ABS review (Page 118-204)

TANK NO. 001 : FOREPEAK TANK : NET VOLUME= 102.42 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	0.555	0.000	0.0	0.000	0.000	0.000	0.0
1.7	2.017	1.461	2.5	72.168	0.000	1.562	0.4
3.4	2.552	1.997	4.2	72.319	0.000	1.936	0.9
5.1	2.931	2.375	5.1	72.449	0.000	2.206	1.3
6.8	3.252	2.697	5.9	72.535	0.000	2.429	1.8
8.5	3.535	2.980	6.5	72.601	0.000	2.622	2.3
10.2	3.791	3.236	7.1	72.654	0.000	2.796	2.8
11.9	4.027	3.472	7.7	72.697	0.000	2.955	3.4
13.6	4.248	3.693	8.2	72.734	0.000	3.103	4.0
15.3	4.455	3.900	8.7	72.767	0.000	3.242	4.5
17.0	4.652	4.096	9.1	72.795	0.000	3.373	5.1
18.7	4.839	4.284	9.6	72.820	0.000	3.498	5.8
20.4	5.018	4.463	10.0	72.843	0.000	3.617	6.4
22.1	5.190	4.635	10.4	72.863	0.000	3.732	7.1
23.8	5.356	4.801	10.8	72.882	0.000	3.842	7.7
25.5	5.516	4.961	11.1	72.898	0.000	3.948	8.4
27.2	5.671	5.116	11.5	72.914	0.000	4.051	9.1
28.9	5.822	5.267	11.8	72.928	0.000	4.151	9.8
30.6	5.969	5.414	12.1	72.941	0.000	4.248	10.6
32.3	6.112	5.557	12.4	72.953	0.000	4.342	11.3
34.0	6.251	5.696	12.7	72.964	0.000	4.434	12.0
35.7	6.388	5.833	13.0	72.974	0.000	4.524	12.8
37.4	6.521	5.966	13.3	72.983	0.000	4.612	13.6
39.1	6.652	6.097	13.5	72.992	0.000	4.697	14.3
40.8	6.781	6.226	13.8	73.000	0.000	4.782	15.1
42.5	6.907	6.352	14.0	73.007	0.000	4.864	15.9
44.2	7.031	6.476	14.2	73.014	0.000	4.945	16.7
45.9	7.153	6.598	14.5	73.021	0.000	5.025	17.5
47.6	7.274	6.718	14.7	73.026	0.000	5.103	18.3
49.3	7.392	6.837	14.9	73.032	0.000	5.180	19.2
51.0	7.509	6.954	15.1	73.037	0.000	5.255	20.0
52.7	7.624	7.069	15.3	73.041	0.000	5.330	20.8
54.4	7.738	7.183	15.5	73.046	0.000	5.403	21.6
56.1	7.851	7.296	15.6	73.050	0.000	5.476	22.5
57.8	7.963	7.407	15.8	73.053	0.000	5.547	23.3
59.5	8.073	7.518	16.0	73.056	0.000	5.618	24.1
61.2	8.182	7.627	16.1	73.059	0.000	5.688	25.0
62.9	8.290	7.735	16.3	73.062	0.000	5.757	25.8
64.6	8.397	7.842	16.4	73.065	0.000	5.825	26.6
66.3	8.503	7.948	16.6	73.067	0.000	5.892	27.5
68.0	8.609	8.054	16.7	73.069	0.000	5.959	28.3
69.7	8.714	8.158	16.8	73.070	0.000	6.025	29.1
71.4	8.817	8.262	16.9	73.072	0.000	6.090	29.9
73.1	8.921	8.366	17.0	73.073	0.000	6.154	30.7
74.8	9.023	8.468	17.1	73.074	0.000	6.218	31.5
76.5	9.125	8.570	17.2	73.075	0.000	6.282	32.3
78.2	9.227	8.672	17.3	73.076	0.000	6.345	33.0
79.9	9.328	8.773	17.2	73.076	0.000	6.407	31.8
81.6	9.432	8.877	16.7	73.077	0.000	6.469	28.7
83.3	9.538	8.983	16.3	73.078	0.000	6.531	25.8
85.0	9.646	9.091	15.9	73.080	0.000	6.592	23.6
86.7	9.758	9.203	15.6	73.082	0.000	6.653	21.7
88.4	9.872	9.316	15.2	73.084	0.000	6.714	20.1
90.1	9.988	9.433	14.8	73.087	0.000	6.774	18.8
91.8	10.108	9.553	14.5	73.090	0.000	6.835	17.8
93.5	10.230	9.675	14.2	73.093	0.000	6.896	17.1
95.2	10.354	9.799	13.9	73.097	0.000	6.956	16.6
96.9	10.481	9.926	13.7	73.101	0.000	7.017	16.3
98.6	10.609	10.054	13.6	73.104	0.000	7.078	16.4
100.3	10.738	10.183	13.6	73.107	0.000	7.139	16.5
102.0	10.867	10.312	13.5	73.110	0.000	7.200	16.5
102.4	10.900	10.345	13.4	73.111	0.000	7.215	16.5

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TANK NO. 002 : WING TANK PS : NET VOLUME= 113.67 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	2.800	0.000	5.5	68.949	-1.359	2.801	4.4
1.9	3.097	0.297	7.7	68.442	-1.593	2.954	8.4
3.8	3.308	0.508	11.4	67.974	-1.791	3.077	15.0
5.7	3.474	0.674	12.3	67.599	-1.946	3.182	16.0
7.6	3.628	0.828	13.6	67.409	-2.040	3.277	17.1
9.5	3.766	0.966	14.4	67.298	-2.106	3.361	17.9
11.4	3.901	1.101	14.7	67.223	-2.157	3.440	18.8
13.3	4.032	1.232	15.1	67.168	-2.199	3.515	19.6
15.2	4.160	1.360	15.5	67.126	-2.235	3.588	20.4
17.1	4.284	1.484	15.9	67.093	-2.266	3.658	21.2
19.0	4.406	1.606	16.4	67.067	-2.294	3.727	22.1
20.9	4.524	1.724	16.8	67.047	-2.319	3.794	22.9
22.8	4.639	1.839	17.2	67.030	-2.341	3.860	23.8
24.7	4.751	1.951	17.6	67.016	-2.363	3.924	24.6
26.6	4.861	2.061	18.0	67.004	-2.383	3.987	25.5
28.5	4.969	2.169	18.4	66.994	-2.402	4.049	26.3
30.4	5.074	2.274	18.8	66.986	-2.419	4.110	27.1
32.3	5.178	2.378	19.1	66.979	-2.437	4.169	28.0
34.2	5.279	2.479	19.5	66.973	-2.453	4.228	28.8
36.1	5.376	2.576	20.4	66.941	-2.474	4.286	20.0
38.0	5.472	2.672	20.7	66.907	-2.495	4.343	20.7
39.9	5.566	2.766	21.0	66.877	-2.515	4.399	21.5
41.8	5.658	2.858	21.4	66.850	-2.534	4.454	22.2
43.7	5.749	2.949	21.7	66.826	-2.552	4.509	22.9
45.6	5.839	3.039	22.0	66.804	-2.569	4.562	23.6
47.5	5.927	3.127	22.3	66.785	-2.585	4.615	24.3
49.4	6.015	3.215	22.6	66.767	-2.601	4.667	25.1
51.3	6.101	3.301	22.9	66.751	-2.616	4.719	25.8
53.2	6.186	3.386	23.2	66.737	-2.631	4.770	26.5
55.1	6.270	3.470	23.5	66.724	-2.645	4.820	27.3
57.0	6.353	3.553	23.7	66.712	-2.659	4.870	28.0
58.9	6.435	3.635	24.0	66.701	-2.672	4.919	28.7
60.8	6.516	3.716	24.3	66.691	-2.685	4.967	29.5
62.7	6.596	3.796	24.6	66.682	-2.698	5.016	30.2
64.6	6.676	3.876	24.8	66.674	-2.710	5.063	31.0
66.5	6.754	3.954	25.1	66.666	-2.722	5.110	31.7
68.4	6.832	4.032	25.3	66.659	-2.734	5.157	32.5
70.3	6.909	4.109	25.6	66.653	-2.745	5.203	33.2
72.2	6.985	4.185	25.9	66.647	-2.756	5.249	34.0
74.1	7.060	4.260	26.1	66.642	-2.767	5.295	34.7
76.0	7.135	4.335	26.4	66.637	-2.778	5.340	35.5
77.9	7.209	4.409	26.6	66.632	-2.788	5.385	36.2
79.8	7.282	4.482	26.8	66.628	-2.799	5.429	37.0
81.7	7.355	4.555	27.1	66.624	-2.809	5.473	37.7
83.6	7.427	4.627	27.3	66.621	-2.819	5.516	38.5
85.5	7.498	4.698	27.6	66.617	-2.829	5.560	39.3
87.4	7.569	4.769	27.8	66.615	-2.838	5.603	40.0
89.3	7.639	4.839	28.0	66.612	-2.848	5.645	40.8
91.2	7.709	4.909	28.2	66.609	-2.857	5.687	41.6
93.1	7.778	4.978	28.5	66.607	-2.867	5.729	42.3
95.0	7.847	5.047	28.7	66.605	-2.876	5.771	43.1
96.9	7.915	5.115	28.9	66.603	-2.885	5.812	43.9
98.8	7.982	5.182	29.1	66.601	-2.894	5.853	44.6
100.7	8.316	5.516	4.7	66.652	-2.890	5.896	2.1
102.6	8.724	5.924	5.0	66.719	-2.885	5.945	2.6
104.5	9.108	6.308	5.2	66.783	-2.880	5.999	3.0
106.4	9.476	6.676	5.4	66.845	-2.877	6.058	3.4
108.3	9.836	7.036	5.4	66.905	-2.874	6.121	3.8
110.2	10.203	7.403	5.3	66.962	-2.873	6.188	4.2
112.1	10.586	7.786	5.1	67.015	-2.874	6.259	4.3
113.7	10.900	8.100	5.2	67.058	-2.876	6.321	4.6

See ABS Singapore letter Ref 1469174 Dated 04-APR-2016

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TANK NO. 003 : WING TANK SB : NET VOLUME= 114.96 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	2.800	0.000	5.5	68.949	1.359	2.801	4.4
1.9	3.097	0.297	7.7	68.438	1.594	2.954	8.5
3.8	3.308	0.508	11.4	67.973	1.792	3.077	15.0
5.7	3.474	0.674	12.3	67.598	1.946	3.182	16.0
7.6	3.628	0.828	13.6	67.408	2.040	3.277	17.1
9.5	3.766	0.966	14.4	67.298	2.106	3.361	17.9
11.4	3.901	1.101	14.7	67.223	2.157	3.440	18.8
13.3	4.032	1.232	15.1	67.168	2.199	3.515	19.6
15.2	4.160	1.360	15.5	67.126	2.235	3.588	20.4
17.1	4.284	1.484	15.9	67.093	2.266	3.658	21.2
19.0	4.406	1.606	16.4	67.067	2.294	3.727	22.1
20.9	4.524	1.724	16.8	67.046	2.319	3.794	22.9
22.8	4.639	1.839	17.2	67.029	2.341	3.860	23.8
24.7	4.751	1.951	17.6	67.015	2.363	3.924	24.6
26.6	4.861	2.061	18.0	67.004	2.383	3.987	25.5
28.5	4.969	2.169	18.4	66.994	2.402	4.049	26.3
30.4	5.074	2.274	18.8	66.985	2.419	4.110	27.1
32.3	5.177	2.377	19.1	66.978	2.437	4.169	28.0
34.2	5.279	2.479	19.5	66.972	2.453	4.228	28.8
36.1	5.375	2.575	20.9	66.938	2.473	4.286	20.8
38.0	5.468	2.668	21.2	66.901	2.492	4.343	21.6
39.9	5.560	2.760	21.5	66.868	2.511	4.399	22.3
41.8	5.650	2.850	21.8	66.839	2.528	4.453	23.0
43.7	5.739	2.939	22.1	66.813	2.545	4.507	23.7
45.6	5.827	3.027	22.4	66.789	2.561	4.561	24.5
47.5	5.914	3.114	22.7	66.768	2.577	4.613	25.2
49.4	6.000	3.200	23.0	66.749	2.592	4.665	25.9
51.3	6.084	3.284	23.3	66.731	2.606	4.716	26.7
53.2	6.168	3.368	23.6	66.716	2.620	4.766	27.4
55.1	6.250	3.450	23.9	66.701	2.633	4.816	28.1
57.0	6.332	3.532	24.2	66.688	2.646	4.865	28.9
58.9	6.412	3.612	24.4	66.676	2.659	4.914	29.6
60.8	6.492	3.692	24.7	66.666	2.671	4.962	30.4
62.7	6.571	3.771	25.0	66.656	2.683	5.009	31.1
64.6	6.649	3.849	25.2	66.646	2.695	5.056	31.9
66.5	6.726	3.926	25.5	66.638	2.706	5.103	32.6
68.4	6.803	4.003	25.7	66.630	2.717	5.149	33.4
70.3	6.879	4.079	26.0	66.623	2.728	5.195	34.1
72.2	6.954	4.154	26.2	66.617	2.739	5.240	34.9
74.1	7.028	4.228	26.5	66.611	2.749	5.285	35.6
76.0	7.101	4.301	26.7	66.605	2.759	5.329	36.4
77.9	7.174	4.374	27.0	66.600	2.770	5.374	37.1
79.8	7.247	4.447	27.2	66.596	2.780	5.417	37.9
81.7	7.318	4.518	27.5	66.591	2.789	5.461	38.7
83.6	7.389	4.589	27.7	66.587	2.799	5.504	39.5
85.5	7.460	4.660	27.9	66.584	2.808	5.546	40.2
87.4	7.530	4.730	28.2	66.580	2.818	5.589	41.0
89.3	7.599	4.799	28.4	66.577	2.827	5.631	41.8
91.2	7.668	4.868	28.6	66.574	2.836	5.673	42.6
93.1	7.736	4.936	28.8	66.572	2.845	5.714	43.3
95.0	7.804	5.004	29.0	66.570	2.854	5.755	44.1
96.9	7.871	5.071	29.3	66.567	2.862	5.796	44.9
98.8	7.937	5.137	29.5	66.565	2.871	5.836	45.7
100.7	8.025	5.225	4.4	66.568	2.879	5.877	1.9
102.6	8.451	5.651	4.8	66.636	2.872	5.920	2.3
104.5	8.851	6.051	5.0	66.702	2.867	5.970	2.7
106.4	9.229	6.429	5.3	66.766	2.863	6.025	3.1
108.3	9.593	6.793	5.4	66.827	2.860	6.084	3.5
110.2	9.954	7.154	5.4	66.886	2.859	6.148	4.0
112.1	10.325	7.525	5.1	66.941	2.858	6.216	4.2
114.0	10.710	7.910	5.1	66.994	2.860	6.287	4.4
115.0	10.900	8.100	5.2	67.020	2.861	6.325	4.6

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TANK NO. 004 : DB TANK CE : NET VOLUME= 73.54 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	0.000	0.000	6.8	62.616	0.000	0.001	1.8
1.2	0.102	0.102	15.7	63.278	0.000	0.057	10.3
2.4	0.173	0.173	19.0	63.476	0.000	0.098	15.6
3.6	0.234	0.234	21.5	63.603	0.000	0.134	20.3
4.8	0.289	0.289	23.4	63.698	0.000	0.166	24.6
6.0	0.340	0.340	25.1	63.775	0.000	0.196	28.6
7.2	0.388	0.388	26.6	63.839	0.000	0.224	32.5
8.4	0.434	0.434	27.9	63.893	0.000	0.251	36.2
9.6	0.477	0.477	29.1	63.940	0.000	0.276	39.8
10.8	0.519	0.519	30.2	63.984	0.000	0.301	43.3
12.0	0.559	0.559	31.2	64.024	0.000	0.325	46.7
13.2	0.598	0.598	32.2	64.059	0.000	0.348	50.0
14.4	0.636	0.636	33.1	64.092	0.000	0.370	53.3
15.6	0.673	0.673	33.9	64.122	0.000	0.392	56.5
16.8	0.709	0.709	34.7	64.149	0.000	0.413	59.7
18.0	0.744	0.744	35.5	64.174	0.000	0.434	62.8
19.2	0.779	0.779	36.2	64.198	0.000	0.455	65.9
20.4	0.813	0.813	36.9	64.220	0.000	0.475	69.0
21.6	0.846	0.846	37.5	64.241	0.000	0.495	72.1
22.8	0.879	0.879	37.5	64.260	0.000	0.514	73.8
24.0	0.913	0.913	37.2	64.280	0.000	0.533	74.9
25.2	0.949	0.949	35.1	64.301	0.000	0.552	72.3
26.4	0.984	0.984	34.9	64.321	0.000	0.571	73.6
27.6	1.020	1.020	34.8	64.341	0.000	0.590	75.2
28.8	1.055	1.055	34.8	64.360	0.000	0.608	76.8
30.0	1.091	1.091	34.9	64.379	0.000	0.627	78.5
31.2	1.126	1.126	34.9	64.398	0.000	0.645	80.2
32.4	1.162	1.162	35.0	64.416	0.000	0.664	82.0
33.6	1.197	1.197	35.1	64.434	0.000	0.682	83.9
34.8	1.232	1.232	35.2	64.451	0.000	0.701	85.8
36.0	1.267	1.267	35.3	64.468	0.000	0.719	87.7
37.2	1.302	1.302	35.5	64.485	0.000	0.737	89.6
38.4	1.337	1.337	35.6	64.501	0.000	0.755	91.6
39.6	1.372	1.372	34.5	64.517	0.000	0.774	92.2
40.8	1.410	1.410	26.7	64.534	0.000	0.792	33.9
42.0	1.458	1.458	25.8	64.566	0.000	0.810	33.2
43.2	1.507	1.507	25.1	64.596	0.000	0.829	32.8
44.4	1.556	1.556	24.6	64.625	0.000	0.848	32.5
45.6	1.607	1.607	24.1	64.652	0.000	0.867	32.4
46.8	1.659	1.659	23.8	64.678	0.000	0.887	32.3
48.0	1.711	1.711	23.4	64.704	0.000	0.907	32.3
49.2	1.764	1.764	23.2	64.728	0.000	0.927	32.3
50.4	1.818	1.818	23.0	64.752	0.000	0.948	32.5
51.6	1.872	1.872	22.9	64.775	0.000	0.968	32.7
52.8	1.926	1.926	22.7	64.797	0.000	0.990	32.9
54.0	1.981	1.981	22.7	64.819	0.000	1.011	33.4
55.2	2.035	2.035	22.7	64.840	0.000	1.033	33.8
56.4	2.089	2.089	22.8	64.860	0.000	1.055	34.4
57.6	2.144	2.144	22.9	64.880	0.000	1.077	35.2
58.8	2.197	2.197	23.0	64.899	0.000	1.099	36.0
60.0	2.251	2.251	23.2	64.918	0.000	1.121	36.9
61.2	2.304	2.304	23.5	64.937	0.000	1.144	38.0
62.4	2.356	2.356	23.8	64.954	0.000	1.167	39.2
63.6	2.408	2.408	24.1	64.972	0.000	1.190	40.5
64.8	2.459	2.459	24.5	64.988	0.000	1.213	42.1
66.0	2.509	2.509	25.0	65.005	0.000	1.236	43.7
67.2	2.558	2.558	25.4	65.020	0.000	1.259	45.5
68.4	2.606	2.606	26.0	65.035	0.000	1.282	47.6
69.6	2.653	2.653	26.6	65.050	0.000	1.306	49.8
70.8	2.699	2.699	27.2	65.064	0.000	1.329	52.2
72.0	2.744	2.744	27.9	65.077	0.000	1.352	54.8
73.2	2.788	2.788	28.9	65.091	0.000	1.375	57.9
73.5	2.800	2.800	29.0	65.094	0.000	1.382	58.7

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TANK NO. 005 : WING TANK PS : NET VOLUME= 98.16 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.400	0.000	7.2	60.213	-3.635	1.401	4.9
1.6	1.611	0.211	8.4	60.181	-3.700	1.508	6.1
3.2	1.796	0.396	9.4	60.158	-3.754	1.607	7.2
4.8	1.963	0.563	10.3	60.140	-3.801	1.698	8.2
6.4	2.118	0.718	11.0	60.125	-3.843	1.784	9.2
8.0	2.263	0.863	11.7	60.113	-3.881	1.865	10.2
9.6	2.400	1.000	12.3	60.103	-3.916	1.943	11.1
11.2	2.531	1.131	12.9	60.093	-3.948	2.018	12.0
12.8	2.656	1.256	13.4	60.085	-3.979	2.090	12.9
14.4	2.777	1.377	13.9	60.078	-4.007	2.159	13.8
16.0	2.890	1.490	15.1	60.083	-4.030	2.227	15.8
17.6	2.996	1.596	15.9	60.094	-4.050	2.292	17.1
19.2	3.097	1.697	16.8	60.109	-4.067	2.355	18.7
20.8	3.194	1.794	17.2	60.126	-4.082	2.416	19.5
22.4	3.289	1.889	17.5	60.139	-4.098	2.475	20.4
24.0	3.391	1.991	16.2	60.132	-4.119	2.532	18.4
25.6	3.491	2.091	16.6	60.123	-4.141	2.589	19.2
27.2	3.590	2.190	16.9	60.116	-4.162	2.645	20.0
28.8	3.686	2.286	17.2	60.108	-4.182	2.700	20.7
30.4	3.781	2.381	17.5	60.102	-4.201	2.755	21.5
32.0	3.875	2.475	17.8	60.096	-4.219	2.808	22.2
33.6	3.966	2.566	18.1	60.091	-4.237	2.861	23.0
35.2	4.057	2.657	18.4	60.086	-4.255	2.914	23.7
36.8	4.146	2.746	18.7	60.081	-4.271	2.965	24.4
38.4	4.233	2.833	18.9	60.077	-4.288	3.016	25.1
40.0	4.321	2.921	19.1	60.073	-4.304	3.067	25.2
41.6	4.406	3.006	19.3	60.069	-4.320	3.117	25.9
43.2	4.491	3.091	19.6	60.065	-4.335	3.166	26.5
44.8	4.575	3.175	19.8	60.062	-4.350	3.215	27.2
46.4	4.658	3.258	20.0	60.059	-4.365	3.263	27.9
48.0	4.740	3.340	20.2	60.056	-4.379	3.311	28.5
49.6	4.821	3.421	20.4	60.053	-4.393	3.358	29.2
51.2	4.902	3.502	20.6	60.050	-4.406	3.405	29.8
52.8	4.981	3.581	21.0	60.048	-4.419	3.452	31.1
54.4	5.059	3.659	21.2	60.046	-4.432	3.498	31.7
56.0	5.137	3.737	21.3	60.044	-4.444	3.544	32.4
57.6	5.214	3.814	21.5	60.042	-4.456	3.589	33.0
59.2	5.290	3.890	21.7	60.040	-4.468	3.634	33.6
60.8	5.420	4.020	12.1	60.035	-4.499	3.679	7.8
62.4	5.554	4.154	12.5	60.030	-4.532	3.726	8.1
64.0	5.684	4.284	12.8	60.025	-4.563	3.773	8.5
65.6	5.812	4.412	13.0	60.021	-4.594	3.821	8.9
67.2	5.937	4.537	13.3	60.016	-4.624	3.870	9.2
68.8	6.060	4.660	13.6	60.013	-4.653	3.920	9.6
70.4	6.180	4.780	13.9	60.009	-4.681	3.970	10.0
72.0	6.298	4.898	14.1	60.006	-4.708	4.020	10.3
73.6	6.414	5.014	14.4	60.002	-4.735	4.071	10.7
75.2	6.528	5.128	14.6	59.999	-4.761	4.122	11.1
76.8	6.640	5.240	14.8	59.997	-4.786	4.173	11.4
78.4	6.750	5.350	15.1	59.994	-4.810	4.225	11.8
80.0	6.859	5.459	15.3	59.991	-4.834	4.276	12.1
81.6	6.966	5.566	15.5	59.989	-4.857	4.328	12.5
83.2	7.072	5.672	15.7	59.987	-4.880	4.380	12.9
84.8	7.176	5.776	15.9	59.985	-4.902	4.431	13.2
86.4	7.279	5.879	16.1	59.983	-4.923	4.483	13.6
88.0	7.381	5.981	16.3	59.981	-4.944	4.535	13.9
89.6	7.481	6.081	16.5	59.979	-4.965	4.587	14.3
91.2	7.580	6.180	16.7	59.978	-4.985	4.638	14.6
92.8	7.679	6.279	16.9	59.976	-5.004	4.690	15.0
94.4	7.776	6.376	17.1	59.975	-5.023	4.741	15.3
96.0	7.872	6.472	17.3	59.973	-5.042	4.793	15.7
97.6	7.967	6.567	17.4	59.972	-5.061	4.844	16.0
98.2	8.000	6.600	17.5	59.972	-5.067	4.862	16.2

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TANK NO. 006 : WING TANK SB : NET VOLUME= 98.16 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.400	0.000	7.2	60.213	3.635	1.401	4.9
1.6	1.611	0.211	8.4	60.181	3.700	1.508	6.1
3.2	1.796	0.396	9.4	60.158	3.754	1.607	7.2
4.8	1.963	0.563	10.3	60.140	3.801	1.698	8.2
6.4	2.118	0.718	11.0	60.125	3.843	1.784	9.2
8.0	2.263	0.863	11.7	60.113	3.881	1.865	10.2
9.6	2.400	1.000	12.3	60.103	3.916	1.943	11.1
11.2	2.531	1.131	12.9	60.093	3.948	2.018	12.0
12.8	2.656	1.256	13.4	60.085	3.979	2.090	12.9
14.4	2.777	1.377	13.9	60.078	4.007	2.159	13.8
16.0	2.890	1.490	15.1	60.083	4.030	2.227	15.8
17.6	2.996	1.596	15.9	60.094	4.050	2.292	17.1
19.2	3.097	1.697	16.8	60.109	4.067	2.355	18.7
20.8	3.194	1.794	17.2	60.126	4.082	2.416	19.5
22.4	3.289	1.889	17.5	60.139	4.098	2.475	20.4
24.0	3.391	1.991	16.2	60.132	4.119	2.532	18.4
25.6	3.491	2.091	16.6	60.123	4.141	2.589	19.2
27.2	3.590	2.190	16.9	60.116	4.162	2.645	20.0
28.8	3.686	2.286	17.2	60.108	4.182	2.700	20.7
30.4	3.781	2.381	17.5	60.102	4.201	2.755	21.5
32.0	3.875	2.475	17.8	60.096	4.219	2.808	22.2
33.6	3.966	2.566	18.1	60.091	4.237	2.861	23.0
35.2	4.057	2.657	18.4	60.086	4.255	2.914	23.7
36.8	4.146	2.746	18.7	60.081	4.271	2.965	24.4
38.4	4.233	2.833	18.9	60.077	4.288	3.016	25.1
40.0	4.321	2.921	19.1	60.073	4.304	3.067	25.2
41.6	4.406	3.006	19.3	60.069	4.320	3.117	25.9
43.2	4.491	3.091	19.6	60.065	4.335	3.166	26.5
44.8	4.575	3.175	19.8	60.062	4.350	3.215	27.2
46.4	4.658	3.258	20.0	60.059	4.365	3.263	27.9
48.0	4.740	3.340	20.2	60.056	4.379	3.311	28.5
49.6	4.821	3.421	20.4	60.053	4.393	3.358	29.2
51.2	4.902	3.502	20.6	60.050	4.406	3.405	29.8
52.8	4.981	3.581	21.0	60.048	4.419	3.452	31.1
54.4	5.059	3.659	21.2	60.046	4.432	3.498	31.7
56.0	5.137	3.737	21.3	60.044	4.444	3.544	32.4
57.6	5.214	3.814	21.5	60.042	4.456	3.589	33.0
59.2	5.290	3.890	21.7	60.040	4.468	3.634	33.6
60.8	5.420	4.020	12.1	60.035	4.499	3.679	7.8
62.4	5.554	4.154	12.5	60.030	4.532	3.726	8.1
64.0	5.684	4.284	12.8	60.025	4.563	3.773	8.5
65.6	5.812	4.412	13.0	60.021	4.594	3.821	8.9
67.2	5.937	4.537	13.3	60.016	4.624	3.870	9.2
68.8	6.060	4.660	13.6	60.013	4.653	3.920	9.6
70.4	6.180	4.780	13.9	60.009	4.681	3.970	10.0
72.0	6.298	4.898	14.1	60.006	4.708	4.020	10.3
73.6	6.414	5.014	14.4	60.002	4.735	4.071	10.7
75.2	6.528	5.128	14.6	59.999	4.761	4.122	11.1
76.8	6.640	5.240	14.8	59.997	4.786	4.173	11.4
78.4	6.750	5.350	15.1	59.994	4.810	4.225	11.8
80.0	6.859	5.459	15.3	59.991	4.834	4.276	12.1
81.6	6.966	5.566	15.5	59.989	4.857	4.328	12.5
83.2	7.072	5.672	15.7	59.987	4.880	4.380	12.9
84.8	7.176	5.776	15.9	59.985	4.902	4.431	13.2
86.4	7.279	5.879	16.1	59.983	4.923	4.483	13.6
88.0	7.381	5.981	16.3	59.981	4.944	4.535	13.9
89.6	7.481	6.081	16.5	59.979	4.965	4.587	14.3
91.2	7.580	6.180	16.7	59.978	4.985	4.638	14.6
92.8	7.679	6.279	16.9	59.976	5.004	4.690	15.0
94.4	7.776	6.376	17.1	59.975	5.023	4.741	15.3
96.0	7.872	6.472	17.3	59.973	5.042	4.793	15.7
97.6	7.967	6.567	17.4	59.972	5.061	4.844	16.0
98.2	8.000	6.600	17.5	59.972	5.067	4.862	16.2

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TANK NO. 009 : DB TANK CE : NET VOLUME= 75.42 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	0.000	0.000	34.5	56.474	0.000	0.001	69.3
1.3	0.035	0.035	40.7	56.544	0.009	0.018	107.5
2.6	0.067	0.067	43.6	56.577	0.022	0.035	129.1
3.9	0.097	0.097	45.6	56.600	0.033	0.050	146.6
5.2	0.126	0.126	47.2	56.618	0.042	0.066	161.2
6.5	0.154	0.154	48.6	56.633	0.049	0.081	174.3
7.8	0.181	0.181	49.6	56.647	0.056	0.095	185.2
9.1	0.208	0.208	50.6	56.658	0.062	0.109	195.9
10.4	0.234	0.234	51.5	56.669	0.067	0.123	205.9
11.7	0.260	0.260	52.3	56.678	0.071	0.137	216.6
13.0	0.285	0.285	53.1	56.686	0.075	0.151	227.0
14.3	0.310	0.310	53.9	56.693	0.077	0.164	237.2
15.6	0.335	0.335	54.6	56.699	0.080	0.177	247.3
16.9	0.359	0.359	55.3	56.705	0.081	0.190	257.2
18.2	0.384	0.384	55.9	56.711	0.083	0.203	266.7
19.5	0.407	0.407	56.6	56.716	0.084	0.216	276.1
20.8	0.431	0.431	57.2	56.720	0.085	0.229	285.4
22.1	0.454	0.454	57.7	56.725	0.085	0.241	294.5
23.4	0.477	0.477	58.3	56.729	0.086	0.254	303.3
24.7	0.500	0.500	55.1	56.732	0.086	0.266	281.8
26.0	0.525	0.525	55.7	56.739	0.087	0.278	291.0
27.3	0.548	0.548	56.2	56.744	0.088	0.291	300.1
28.6	0.572	0.572	56.7	56.750	0.088	0.303	309.0
29.9	0.596	0.596	57.2	56.755	0.089	0.315	317.9
31.2	0.619	0.619	57.7	56.759	0.089	0.327	326.6
32.5	0.642	0.642	58.1	56.764	0.089	0.340	335.3
33.8	0.667	0.667	52.5	56.771	0.090	0.352	334.6
35.1	0.692	0.692	53.0	56.778	0.090	0.364	344.0
36.4	0.718	0.718	53.4	56.785	0.091	0.376	353.2
37.7	0.743	0.743	53.9	56.792	0.091	0.388	362.4
39.0	0.767	0.767	54.3	56.798	0.092	0.400	371.5
40.3	0.792	0.792	54.8	56.804	0.092	0.413	380.5
41.6	0.816	0.816	55.2	56.810	0.092	0.425	389.4
42.9	0.840	0.840	55.6	56.815	0.093	0.437	398.2
44.2	0.864	0.864	56.0	56.820	0.093	0.449	406.9
45.5	0.888	0.888	56.4	56.825	0.093	0.462	415.5
46.8	0.912	0.912	56.8	56.830	0.093	0.474	424.1
48.1	0.935	0.935	57.2	56.834	0.093	0.486	432.6
49.4	0.959	0.959	57.6	56.838	0.094	0.498	441.0
50.7	0.982	0.982	57.9	56.842	0.094	0.510	449.3
52.0	1.005	1.005	58.3	56.846	0.094	0.522	457.5
53.3	1.028	1.028	58.7	56.850	0.094	0.534	465.7
54.6	1.051	1.051	59.0	56.854	0.094	0.546	473.8
55.9	1.073	1.073	59.3	56.857	0.094	0.558	481.9
57.2	1.096	1.096	59.7	56.860	0.094	0.570	489.9
58.5	1.118	1.118	60.0	56.863	0.094	0.582	497.7
59.8	1.141	1.141	60.3	56.867	0.094	0.594	505.6
61.1	1.163	1.163	60.6	56.869	0.094	0.606	513.4
62.4	1.185	1.185	60.9	56.872	0.094	0.618	521.2
63.7	1.207	1.207	61.2	56.875	0.094	0.630	528.9
65.0	1.229	1.229	61.5	56.878	0.094	0.641	536.5
66.3	1.250	1.250	61.8	56.880	0.093	0.653	544.0
67.6	1.272	1.272	62.1	56.883	0.093	0.665	551.5
68.9	1.293	1.293	62.4	56.885	0.093	0.676	559.0
70.2	1.315	1.315	62.7	56.888	0.093	0.688	566.4
71.5	1.336	1.336	63.0	56.890	0.093	0.700	573.9
72.8	1.357	1.357	63.2	56.892	0.093	0.711	581.2
74.1	1.379	1.379	63.5	56.895	0.093	0.723	588.5
75.4	1.400	1.400	63.7	56.897	0.093	0.734	595.1

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TANK NO. 010 : WING TANK PS : NET VOLUME= 59.16 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.400	0.000	9.7	49.061	-6.767	1.401	3.8
1.0	1.505	0.105	9.9	49.114	-6.789	1.453	4.1
2.0	1.609	0.209	10.1	49.166	-6.810	1.505	4.5
3.0	1.710	0.310	10.3	49.217	-6.830	1.557	4.8
4.0	1.809	0.409	10.6	49.260	-6.848	1.607	5.2
5.0	1.905	0.505	10.9	49.294	-6.866	1.657	5.5
6.0	1.998	0.598	11.3	49.321	-6.883	1.706	5.9
7.0	2.088	0.688	11.5	49.345	-6.899	1.754	6.2
8.0	2.176	0.776	11.8	49.367	-6.915	1.802	6.5
9.0	2.263	0.863	12.0	49.387	-6.930	1.848	6.8
10.0	2.348	0.948	12.2	49.405	-6.944	1.894	7.1
11.0	2.432	1.032	12.4	49.423	-6.958	1.939	7.4
12.0	2.517	1.117	10.2	49.434	-6.975	1.983	3.5
13.0	2.618	1.218	10.2	49.423	-7.010	2.028	3.5
14.0	2.719	1.319	10.3	49.414	-7.043	2.074	3.5
15.0	2.819	1.419	10.3	49.408	-7.074	2.120	3.5
16.0	2.919	1.519	10.3	49.403	-7.103	2.167	3.5
17.0	3.018	1.618	10.3	49.400	-7.131	2.214	3.4
18.0	3.118	1.718	10.3	49.399	-7.158	2.262	3.4
19.0	3.218	1.818	10.3	49.398	-7.183	2.310	3.4
20.0	3.319	1.919	10.3	49.399	-7.208	2.357	3.3
21.0	3.419	2.019	10.2	49.400	-7.232	2.406	3.2
22.0	3.521	2.121	10.2	49.402	-7.256	2.454	3.2
23.0	3.622	2.222	10.1	49.405	-7.278	2.503	3.1
24.0	3.725	2.325	10.0	49.408	-7.300	2.551	3.0
25.0	3.828	2.428	10.0	49.410	-7.322	2.600	2.9
26.0	3.932	2.532	9.9	49.412	-7.343	2.650	2.8
27.0	4.036	2.636	9.8	49.415	-7.364	2.699	2.7
28.0	4.142	2.742	9.7	49.417	-7.384	2.749	2.6
29.0	4.248	2.848	9.7	49.419	-7.404	2.799	2.5
30.0	4.355	2.955	9.6	49.421	-7.423	2.849	2.4
31.0	4.463	3.063	9.5	49.422	-7.442	2.899	2.3
32.0	4.573	3.173	9.4	49.424	-7.461	2.950	2.2
33.0	4.683	3.283	9.3	49.425	-7.480	3.000	2.1
34.0	4.795	3.395	9.2	49.425	-7.498	3.052	2.0
35.0	4.908	3.508	9.1	49.425	-7.516	3.103	1.9
36.0	5.022	3.622	9.0	49.425	-7.533	3.155	1.8
37.0	5.137	3.737	8.9	49.424	-7.550	3.207	1.7
38.0	5.254	3.854	8.8	49.423	-7.567	3.259	1.6
39.0	5.372	3.972	8.7	49.421	-7.584	3.312	1.5
40.0	5.492	4.092	8.6	49.419	-7.601	3.365	1.4
41.0	5.613	4.213	8.5	49.416	-7.617	3.418	1.3
42.0	5.736	4.336	8.4	49.413	-7.633	3.472	1.2
43.0	5.859	4.459	8.3	49.409	-7.649	3.526	1.1
44.0	5.984	4.584	8.2	49.405	-7.664	3.580	1.0
45.0	6.111	4.711	8.1	49.400	-7.679	3.635	1.0
46.0	6.239	4.839	8.0	49.395	-7.694	3.690	0.9
47.0	6.368	4.968	8.0	49.389	-7.709	3.746	0.8
48.0	6.498	5.098	7.9	49.383	-7.723	3.802	0.8
49.0	6.630	5.230	7.8	49.376	-7.737	3.858	0.7
50.0	6.762	5.362	7.8	49.369	-7.751	3.915	0.7
51.0	6.895	5.495	7.7	49.362	-7.765	3.972	0.6
52.0	7.029	5.629	7.7	49.354	-7.778	4.030	0.6
53.0	7.164	5.764	7.6	49.347	-7.791	4.088	0.6
54.0	7.299	5.899	7.6	49.339	-7.804	4.146	0.6
55.0	7.434	6.034	7.6	49.331	-7.816	4.204	0.5
56.0	7.570	6.170	7.6	49.323	-7.828	4.263	0.5
57.0	7.707	6.307	7.6	49.315	-7.840	4.322	0.5
58.0	7.842	6.442	7.6	49.307	-7.851	4.382	0.5
59.0	7.978	6.578	7.6	49.300	-7.862	4.442	0.5
59.2	8.000	6.600	7.6	49.299	-7.864	4.452	0.5

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TANK NO. 011 : WING TANK SB : NET VOLUME= 59.16 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.400	0.000	9.7	49.061	6.767	1.401	3.8
1.0	1.505	0.105	9.9	49.114	6.789	1.453	4.1
2.0	1.609	0.209	10.1	49.166	6.810	1.505	4.5
3.0	1.710	0.310	10.3	49.217	6.830	1.557	4.8
4.0	1.809	0.409	10.6	49.260	6.848	1.607	5.2
5.0	1.905	0.505	10.9	49.294	6.866	1.657	5.5
6.0	1.998	0.598	11.3	49.321	6.883	1.706	5.9
7.0	2.088	0.688	11.5	49.345	6.899	1.754	6.2
8.0	2.176	0.776	11.8	49.367	6.915	1.802	6.5
9.0	2.263	0.863	12.0	49.387	6.930	1.848	6.8
10.0	2.348	0.948	12.2	49.405	6.944	1.894	7.1
11.0	2.432	1.032	12.4	49.423	6.958	1.939	7.4
12.0	2.517	1.117	10.2	49.434	6.975	1.983	3.5
13.0	2.618	1.218	10.2	49.423	7.010	2.028	3.5
14.0	2.719	1.319	10.3	49.414	7.043	2.074	3.5
15.0	2.819	1.419	10.3	49.408	7.074	2.120	3.5
16.0	2.919	1.519	10.3	49.403	7.103	2.167	3.5
17.0	3.018	1.618	10.3	49.400	7.131	2.214	3.4
18.0	3.118	1.718	10.3	49.399	7.158	2.262	3.4
19.0	3.218	1.818	10.3	49.398	7.183	2.310	3.4
20.0	3.319	1.919	10.3	49.399	7.208	2.357	3.3
21.0	3.419	2.019	10.2	49.400	7.232	2.406	3.2
22.0	3.521	2.121	10.2	49.402	7.256	2.454	3.2
23.0	3.622	2.222	10.1	49.405	7.278	2.503	3.1
24.0	3.725	2.325	10.0	49.408	7.300	2.551	3.0
25.0	3.828	2.428	10.0	49.410	7.322	2.600	2.9
26.0	3.932	2.532	9.9	49.412	7.343	2.650	2.8
27.0	4.036	2.636	9.8	49.415	7.364	2.699	2.7
28.0	4.142	2.742	9.7	49.417	7.384	2.749	2.6
29.0	4.248	2.848	9.7	49.419	7.404	2.799	2.5
30.0	4.355	2.955	9.6	49.421	7.423	2.849	2.4
31.0	4.463	3.063	9.5	49.422	7.442	2.899	2.3
32.0	4.573	3.173	9.4	49.424	7.461	2.950	2.2
33.0	4.683	3.283	9.3	49.425	7.480	3.000	2.1
34.0	4.795	3.395	9.2	49.425	7.498	3.052	2.0
35.0	4.908	3.508	9.1	49.425	7.516	3.103	1.9
36.0	5.022	3.622	9.0	49.425	7.533	3.155	1.8
37.0	5.137	3.737	8.9	49.424	7.550	3.207	1.7
38.0	5.254	3.854	8.8	49.423	7.567	3.259	1.6
39.0	5.372	3.972	8.7	49.421	7.584	3.312	1.5
40.0	5.492	4.092	8.6	49.419	7.601	3.365	1.4
41.0	5.613	4.213	8.5	49.416	7.617	3.418	1.3
42.0	5.736	4.336	8.4	49.413	7.633	3.472	1.2
43.0	5.859	4.459	8.3	49.409	7.649	3.526	1.1
44.0	5.984	4.584	8.2	49.405	7.664	3.580	1.0
45.0	6.111	4.711	8.1	49.400	7.679	3.635	1.0
46.0	6.239	4.839	8.0	49.395	7.694	3.690	0.9
47.0	6.368	4.968	8.0	49.389	7.709	3.746	0.8
48.0	6.498	5.098	7.9	49.383	7.723	3.802	0.8
49.0	6.630	5.230	7.8	49.376	7.737	3.858	0.7
50.0	6.762	5.362	7.8	49.369	7.751	3.915	0.7
51.0	6.895	5.495	7.7	49.362	7.765	3.972	0.6
52.0	7.029	5.629	7.7	49.354	7.778	4.030	0.6
53.0	7.164	5.764	7.6	49.347	7.791	4.088	0.6
54.0	7.299	5.899	7.6	49.339	7.804	4.146	0.6
55.0	7.434	6.034	7.6	49.331	7.816	4.204	0.5
56.0	7.570	6.170	7.6	49.323	7.828	4.263	0.5
57.0	7.707	6.307	7.6	49.315	7.840	4.322	0.5
58.0	7.842	6.442	7.6	49.307	7.851	4.382	0.5
59.0	7.978	6.578	7.6	49.300	7.862	4.442	0.5
59.2	8.000	6.600	7.6	49.299	7.864	4.452	0.5

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TANK NO. 012 : DB TANK PS : NET VOLUME= 54.84 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	0.000	0.000	23.0	49.687	-2.552	0.001	46.7
0.9	0.037	0.037	26.9	49.603	-2.771	0.019	61.8
1.8	0.070	0.070	28.7	49.575	-2.865	0.036	69.9
2.7	0.102	0.102	30.0	49.556	-2.933	0.053	76.1
3.6	0.132	0.132	31.1	49.543	-2.988	0.069	81.4
4.5	0.161	0.161	32.1	49.532	-3.035	0.085	86.2
5.4	0.190	0.190	32.9	49.523	-3.076	0.100	90.5
6.3	0.218	0.218	33.6	49.515	-3.113	0.115	94.5
7.2	0.245	0.245	34.3	49.508	-3.146	0.129	98.3
8.1	0.272	0.272	34.9	49.502	-3.177	0.144	101.8
9.0	0.298	0.298	35.5	49.497	-3.206	0.158	105.2
9.9	0.324	0.324	36.0	49.492	-3.233	0.172	108.4
10.8	0.350	0.350	36.5	49.487	-3.258	0.186	111.5
11.7	0.375	0.375	37.0	49.483	-3.282	0.199	114.4
12.6	0.400	0.400	37.4	49.479	-3.304	0.213	117.3
13.5	0.425	0.425	37.9	49.476	-3.326	0.226	120.0
14.4	0.449	0.449	38.3	49.472	-3.346	0.239	122.7
15.3	0.473	0.473	38.6	49.469	-3.366	0.252	125.2
16.2	0.497	0.497	39.0	49.466	-3.384	0.265	127.7
17.1	0.521	0.521	39.4	49.464	-3.402	0.278	130.1
18.0	0.544	0.544	39.7	49.461	-3.420	0.291	132.5
18.9	0.568	0.568	40.0	49.459	-3.436	0.303	134.8
19.8	0.591	0.591	40.4	49.457	-3.452	0.316	137.0
20.7	0.614	0.614	40.7	49.455	-3.468	0.328	139.2
21.6	0.636	0.636	41.0	49.452	-3.483	0.341	141.4
22.5	0.659	0.659	41.2	49.451	-3.498	0.353	143.5
23.4	0.681	0.681	41.5	49.449	-3.512	0.365	145.5
24.3	0.704	0.704	41.8	49.447	-3.526	0.377	147.5
25.2	0.726	0.726	42.1	49.445	-3.539	0.389	149.5
26.1	0.748	0.748	42.3	49.444	-3.552	0.401	151.4
27.0	0.770	0.770	42.6	49.442	-3.565	0.413	153.3
27.9	0.791	0.791	42.8	49.441	-3.578	0.425	155.2
28.8	0.813	0.813	43.0	49.439	-3.590	0.437	157.0
29.7	0.834	0.834	43.3	49.438	-3.602	0.449	158.8
30.6	0.856	0.856	43.5	49.437	-3.613	0.460	160.6
31.5	0.877	0.877	43.7	49.436	-3.624	0.472	162.3
32.4	0.898	0.898	43.9	49.434	-3.635	0.483	164.0
33.3	0.919	0.919	44.1	49.433	-3.646	0.495	165.7
34.2	0.940	0.940	44.3	49.432	-3.657	0.506	167.4
35.1	0.961	0.961	44.5	49.431	-3.667	0.518	169.1
36.0	0.982	0.982	44.7	49.430	-3.677	0.529	170.7
36.9	1.003	1.003	44.9	49.429	-3.687	0.540	172.3
37.8	1.023	1.023	45.1	49.428	-3.697	0.552	173.8
38.7	1.044	1.044	45.3	49.427	-3.707	0.563	175.4
39.6	1.064	1.064	45.5	49.426	-3.716	0.574	176.9
40.5	1.085	1.085	45.6	49.425	-3.725	0.585	178.4
41.4	1.105	1.105	45.8	49.425	-3.734	0.596	179.9
42.3	1.125	1.125	46.0	49.424	-3.743	0.607	181.4
43.2	1.145	1.145	46.2	49.423	-3.752	0.618	182.8
44.1	1.165	1.165	46.3	49.422	-3.761	0.629	184.3
45.0	1.185	1.185	46.5	49.421	-3.769	0.640	185.7
45.9	1.205	1.205	46.6	49.421	-3.778	0.651	187.1
46.8	1.225	1.225	46.8	49.420	-3.786	0.662	188.5
47.7	1.245	1.245	46.9	49.419	-3.794	0.673	189.8
48.6	1.265	1.265	47.1	49.419	-3.802	0.684	191.2
49.5	1.284	1.284	47.2	49.418	-3.810	0.694	192.5
50.4	1.304	1.304	47.4	49.417	-3.818	0.705	193.8
51.3	1.324	1.324	47.5	49.417	-3.825	0.716	195.2
52.2	1.343	1.343	47.7	49.416	-3.833	0.726	196.5
53.1	1.363	1.363	47.8	49.416	-3.840	0.737	197.7
54.0	1.382	1.382	48.0	49.415	-3.847	0.748	199.0
54.8	1.400	1.400	48.1	49.415	-3.854	0.757	200.1

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TANK NO. 013 : DB TANK SB : NET VOLUME= 54.84 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	0.000	0.000	23.0	49.687	2.552	0.001	46.7
0.9	0.037	0.037	26.9	49.603	2.771	0.019	61.8
1.8	0.070	0.070	28.7	49.575	2.865	0.036	69.9
2.7	0.102	0.102	30.0	49.556	2.933	0.053	76.1
3.6	0.132	0.132	31.1	49.543	2.988	0.069	81.4
4.5	0.161	0.161	32.1	49.532	3.035	0.085	86.2
5.4	0.190	0.190	32.9	49.523	3.076	0.100	90.5
6.3	0.218	0.218	33.6	49.515	3.113	0.115	94.5
7.2	0.245	0.245	34.3	49.508	3.146	0.129	98.3
8.1	0.272	0.272	34.9	49.502	3.177	0.144	101.8
9.0	0.298	0.298	35.5	49.497	3.206	0.158	105.2
9.9	0.324	0.324	36.0	49.492	3.233	0.172	108.4
10.8	0.350	0.350	36.5	49.487	3.258	0.186	111.5
11.7	0.375	0.375	37.0	49.483	3.282	0.199	114.4
12.6	0.400	0.400	37.4	49.479	3.304	0.213	117.3
13.5	0.425	0.425	37.9	49.476	3.326	0.226	120.0
14.4	0.449	0.449	38.3	49.472	3.346	0.239	122.7
15.3	0.473	0.473	38.6	49.469	3.366	0.252	125.2
16.2	0.497	0.497	39.0	49.466	3.384	0.265	127.7
17.1	0.521	0.521	39.4	49.464	3.402	0.278	130.1
18.0	0.544	0.544	39.7	49.461	3.420	0.291	132.5
18.9	0.568	0.568	40.0	49.459	3.436	0.303	134.8
19.8	0.591	0.591	40.4	49.457	3.452	0.316	137.0
20.7	0.614	0.614	40.7	49.455	3.468	0.328	139.2
21.6	0.636	0.636	41.0	49.452	3.483	0.341	141.4
22.5	0.659	0.659	41.2	49.451	3.498	0.353	143.5
23.4	0.681	0.681	41.5	49.449	3.512	0.365	145.5
24.3	0.704	0.704	41.8	49.447	3.526	0.377	147.5
25.2	0.726	0.726	42.1	49.445	3.539	0.389	149.5
26.1	0.748	0.748	42.3	49.444	3.552	0.401	151.4
27.0	0.770	0.770	42.6	49.442	3.565	0.413	153.3
27.9	0.791	0.791	42.8	49.441	3.578	0.425	155.2
28.8	0.813	0.813	43.0	49.439	3.590	0.437	157.0
29.7	0.834	0.834	43.3	49.438	3.602	0.449	158.8
30.6	0.856	0.856	43.5	49.437	3.613	0.460	160.6
31.5	0.877	0.877	43.7	49.436	3.624	0.472	162.3
32.4	0.898	0.898	43.9	49.434	3.635	0.483	164.0
33.3	0.919	0.919	44.1	49.433	3.646	0.495	165.7
34.2	0.940	0.940	44.3	49.432	3.657	0.506	167.4
35.1	0.961	0.961	44.5	49.431	3.667	0.518	169.1
36.0	0.982	0.982	44.7	49.430	3.677	0.529	170.7
36.9	1.003	1.003	44.9	49.429	3.687	0.540	172.3
37.8	1.023	1.023	45.1	49.428	3.697	0.552	173.8
38.7	1.044	1.044	45.3	49.427	3.707	0.563	175.4
39.6	1.064	1.064	45.5	49.426	3.716	0.574	176.9
40.5	1.085	1.085	45.6	49.425	3.725	0.585	178.4
41.4	1.105	1.105	45.8	49.425	3.734	0.596	179.9
42.3	1.125	1.125	46.0	49.424	3.743	0.607	181.4
43.2	1.145	1.145	46.2	49.423	3.752	0.618	182.8
44.1	1.165	1.165	46.3	49.422	3.761	0.629	184.3
45.0	1.185	1.185	46.5	49.421	3.769	0.640	185.7
45.9	1.205	1.205	46.6	49.421	3.778	0.651	187.1
46.8	1.225	1.225	46.8	49.420	3.786	0.662	188.5
47.7	1.245	1.245	46.9	49.419	3.794	0.673	189.8
48.6	1.265	1.265	47.1	49.419	3.802	0.684	191.2
49.5	1.284	1.284	47.2	49.418	3.810	0.694	192.5
50.4	1.304	1.304	47.4	49.417	3.818	0.705	193.8
51.3	1.324	1.324	47.5	49.417	3.825	0.716	195.2
52.2	1.343	1.343	47.7	49.416	3.833	0.726	196.5
53.1	1.363	1.363	47.8	49.416	3.840	0.737	197.7
54.0	1.382	1.382	48.0	49.415	3.847	0.748	199.0
54.8	1.400	1.400	48.1	49.415	3.854	0.757	200.1

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TANK NO. 014 : WING TANK PS : NET VOLUME= 60.06 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.347	0.000	0.0	0.000	0.000	0.000	0.0
1.0	1.549	0.201	9.0	40.600	-8.095	1.487	1.2
2.0	1.662	0.315	9.2	40.394	-8.163	1.546	1.2
3.0	1.774	0.426	9.3	40.325	-8.194	1.603	1.2
4.0	1.884	0.537	9.4	40.289	-8.215	1.660	1.2
5.0	1.994	0.646	9.4	40.268	-8.232	1.716	1.2
6.0	2.103	0.755	9.5	40.252	-8.246	1.771	1.2
7.0	2.212	0.864	9.5	40.241	-8.259	1.826	1.2
8.0	2.321	0.973	9.5	40.231	-8.271	1.881	1.2
9.0	2.430	1.082	9.4	40.222	-8.281	1.936	1.2
10.0	2.539	1.191	9.4	40.214	-8.291	1.991	1.1
11.0	2.649	1.301	9.4	40.205	-8.300	2.046	1.1
12.0	2.759	1.411	9.3	40.198	-8.308	2.101	1.1
13.0	2.869	1.522	9.3	40.190	-8.316	2.155	1.1
14.0	2.980	1.633	9.3	40.182	-8.323	2.210	1.1
15.0	3.092	1.744	9.2	40.174	-8.330	2.265	1.1
16.0	3.203	1.856	9.2	40.167	-8.337	2.321	1.1
17.0	3.315	1.967	9.3	40.161	-8.342	2.376	1.1
18.0	3.426	2.078	9.3	40.157	-8.348	2.431	1.1
19.0	3.536	2.189	9.3	40.153	-8.352	2.486	1.2
20.0	3.647	2.299	9.4	40.150	-8.357	2.542	1.2
21.0	3.757	2.409	9.4	40.148	-8.361	2.597	1.2
22.0	3.867	2.519	9.4	40.147	-8.365	2.652	1.2
23.0	3.976	2.629	9.4	40.146	-8.368	2.707	1.2
24.0	4.085	2.738	9.4	40.145	-8.371	2.762	1.2
25.0	4.195	2.847	9.4	40.144	-8.374	2.817	1.2
26.0	4.304	2.956	9.5	40.144	-8.377	2.873	1.2
27.0	4.413	3.065	9.5	40.144	-8.379	2.928	1.2
28.0	4.522	3.174	9.5	40.144	-8.382	2.983	1.2
29.0	4.630	3.283	9.5	40.144	-8.384	3.038	1.2
30.0	4.739	3.392	9.5	40.144	-8.386	3.092	1.2
31.0	4.848	3.500	9.5	40.144	-8.388	3.147	1.2
32.0	4.957	3.609	9.5	40.144	-8.390	3.202	1.2
33.0	5.065	3.718	9.5	40.144	-8.392	3.257	1.2
34.0	5.174	3.826	9.5	40.144	-8.394	3.312	1.2
35.0	5.282	3.935	9.5	40.145	-8.395	3.366	1.2
36.0	5.391	4.043	9.5	40.145	-8.397	3.421	1.2
37.0	5.499	4.152	9.5	40.145	-8.398	3.476	1.2
38.0	5.608	4.260	9.5	40.145	-8.399	3.531	1.2
39.0	5.716	4.369	9.5	40.146	-8.401	3.585	1.2
40.0	5.825	4.477	9.5	40.146	-8.402	3.640	1.2
41.0	5.933	4.586	9.5	40.146	-8.403	3.694	1.2
42.0	6.042	4.694	9.5	40.146	-8.404	3.749	1.2
43.0	6.150	4.803	9.5	40.147	-8.405	3.804	1.2
44.0	6.259	4.911	9.5	40.147	-8.406	3.858	1.2
45.0	6.367	5.020	9.5	40.147	-8.407	3.913	1.2
46.0	6.476	5.128	9.5	40.147	-8.408	3.967	1.2
47.0	6.584	5.236	9.5	40.147	-8.409	4.022	1.2
48.0	6.692	5.345	9.5	40.148	-8.410	4.076	1.2
49.0	6.801	5.453	9.5	40.148	-8.410	4.131	1.2
50.0	6.909	5.562	9.5	40.148	-8.411	4.185	1.2
51.0	7.018	5.670	9.5	40.148	-8.412	4.240	1.2
52.0	7.126	5.779	9.5	40.148	-8.413	4.294	1.2
53.0	7.235	5.887	9.5	40.149	-8.413	4.349	1.2
54.0	7.343	5.996	9.5	40.149	-8.414	4.403	1.2
55.0	7.451	6.104	9.5	40.149	-8.415	4.458	1.2
56.0	7.560	6.212	9.5	40.149	-8.415	4.512	1.2
57.0	7.668	6.321	9.5	40.149	-8.416	4.566	1.2
58.0	7.777	6.429	9.5	40.149	-8.416	4.621	1.2
59.0	7.885	6.538	9.5	40.149	-8.417	4.675	1.2
60.1	8.000	6.653	9.5	40.150	-8.418	4.733	1.2

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TANK NO. 015 : WING TANK SB : NET VOLUME= 59.97 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.347	0.000	0.0	0.000	0.000	0.000	0.0
1.0	1.548	0.201	9.0	40.599	8.094	1.487	1.2
2.0	1.662	0.314	9.2	40.394	8.162	1.546	1.2
3.0	1.774	0.426	9.3	40.325	8.194	1.603	1.2
4.0	1.884	0.537	9.4	40.289	8.215	1.660	1.2
5.0	1.994	0.646	9.4	40.267	8.232	1.716	1.2
6.0	2.103	0.756	9.5	40.252	8.246	1.771	1.2
7.0	2.212	0.864	9.5	40.240	8.259	1.826	1.2
8.0	2.321	0.973	9.5	40.231	8.271	1.881	1.2
9.0	2.430	1.082	9.5	40.222	8.281	1.936	1.2
10.0	2.539	1.191	9.4	40.214	8.291	1.991	1.1
11.0	2.649	1.301	9.4	40.206	8.300	2.046	1.1
12.0	2.759	1.411	9.3	40.198	8.308	2.101	1.1
13.0	2.869	1.522	9.3	40.190	8.316	2.155	1.1
14.0	2.980	1.633	9.3	40.182	8.323	2.210	1.1
15.0	3.092	1.744	9.2	40.175	8.330	2.265	1.1
16.0	3.203	1.856	9.2	40.167	8.336	2.321	1.1
17.0	3.315	1.967	9.3	40.161	8.342	2.376	1.1
18.0	3.426	2.078	9.3	40.157	8.348	2.431	1.1
19.0	3.536	2.189	9.3	40.153	8.352	2.486	1.2
20.0	3.647	2.299	9.4	40.150	8.357	2.542	1.2
21.0	3.757	2.409	9.4	40.148	8.361	2.597	1.2
22.0	3.866	2.519	9.4	40.147	8.364	2.652	1.2
23.0	3.976	2.629	9.4	40.146	8.368	2.707	1.2
24.0	4.085	2.738	9.4	40.145	8.371	2.762	1.2
25.0	4.195	2.847	9.4	40.144	8.374	2.817	1.2
26.0	4.304	2.956	9.5	40.144	8.377	2.873	1.2
27.0	4.413	3.065	9.5	40.144	8.379	2.928	1.2
28.0	4.522	3.174	9.5	40.144	8.382	2.983	1.2
29.0	4.630	3.283	9.5	40.144	8.384	3.037	1.2
30.0	4.739	3.392	9.5	40.144	8.386	3.092	1.2
31.0	4.848	3.500	9.5	40.144	8.388	3.147	1.2
32.0	4.957	3.609	9.5	40.144	8.390	3.202	1.2
33.0	5.065	3.718	9.5	40.144	8.392	3.257	1.2
34.0	5.174	3.826	9.5	40.144	8.393	3.312	1.2
35.0	5.282	3.935	9.5	40.145	8.395	3.366	1.2
36.0	5.391	4.043	9.5	40.145	8.397	3.421	1.2
37.0	5.499	4.152	9.5	40.145	8.398	3.476	1.2
38.0	5.608	4.260	9.5	40.145	8.399	3.531	1.2
39.0	5.716	4.369	9.5	40.146	8.401	3.585	1.2
40.0	5.825	4.477	9.5	40.146	8.402	3.640	1.2
41.0	5.933	4.586	9.5	40.146	8.403	3.694	1.2
42.0	6.042	4.694	9.5	40.146	8.404	3.749	1.2
43.0	6.150	4.803	9.5	40.147	8.405	3.804	1.2
44.0	6.259	4.911	9.5	40.147	8.406	3.858	1.2
45.0	6.367	5.020	9.5	40.147	8.407	3.913	1.2
46.0	6.475	5.128	9.5	40.147	8.408	3.967	1.2
47.0	6.584	5.236	9.5	40.147	8.409	4.022	1.2
48.0	6.692	5.345	9.5	40.148	8.410	4.076	1.2
49.0	6.801	5.453	9.5	40.148	8.410	4.131	1.2
50.0	6.909	5.562	9.5	40.148	8.411	4.185	1.2
51.0	7.018	5.670	9.5	40.148	8.412	4.240	1.2
52.0	7.126	5.779	9.4	40.148	8.413	4.294	1.2
53.0	7.236	5.889	9.4	40.149	8.414	4.349	1.2
54.0	7.346	5.998	9.4	40.149	8.414	4.403	1.2
55.0	7.455	6.108	9.4	40.150	8.415	4.458	1.2
56.0	7.565	6.217	9.4	40.150	8.416	4.512	1.2
57.0	7.674	6.327	9.4	40.150	8.417	4.567	1.2
58.0	7.784	6.436	9.4	40.151	8.417	4.621	1.2
59.0	7.894	6.546	9.4	40.151	8.418	4.676	1.2
60.0	8.000	6.653	9.4	40.151	8.418	4.729	1.2

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TANK NO. 016 : DB TANK PS : NET VOLUME= 83.50 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	0.000	0.000	60.1	40.016	-3.304	0.001	228.5
1.4	0.023	0.023	62.7	40.040	-3.402	0.012	256.3
2.8	0.046	0.046	64.0	40.049	-3.452	0.023	271.3
4.2	0.069	0.069	64.9	40.056	-3.488	0.035	283.0
5.6	0.091	0.091	65.7	40.061	-3.517	0.046	292.7
7.0	0.113	0.113	66.4	40.065	-3.542	0.057	301.3
8.4	0.134	0.134	67.0	40.069	-3.565	0.068	309.0
9.8	0.156	0.156	67.5	40.072	-3.586	0.079	316.2
11.2	0.177	0.177	68.0	40.075	-3.604	0.090	322.8
12.6	0.198	0.198	68.5	40.078	-3.622	0.101	329.0
14.0	0.219	0.219	68.9	40.080	-3.638	0.112	334.9
15.4	0.240	0.240	69.3	40.083	-3.653	0.122	340.5
16.8	0.261	0.261	69.6	40.085	-3.668	0.133	345.8
18.2	0.281	0.281	70.0	40.087	-3.682	0.144	350.8
19.6	0.302	0.302	70.3	40.089	-3.695	0.154	355.6
21.0	0.323	0.323	70.6	40.090	-3.707	0.165	360.3
22.4	0.343	0.343	70.9	40.092	-3.719	0.175	364.7
23.8	0.363	0.363	71.2	40.094	-3.731	0.186	369.0
25.2	0.383	0.383	71.5	40.095	-3.742	0.196	373.2
26.6	0.404	0.404	71.8	40.097	-3.753	0.207	377.2
28.0	0.424	0.424	72.0	40.098	-3.763	0.217	381.1
29.4	0.444	0.444	72.3	40.100	-3.773	0.227	384.8
30.8	0.464	0.464	72.5	40.101	-3.783	0.238	388.5
32.2	0.483	0.483	72.7	40.102	-3.792	0.248	392.1
33.6	0.503	0.503	73.0	40.103	-3.801	0.258	395.6
35.0	0.523	0.523	73.2	40.105	-3.810	0.268	399.0
36.4	0.543	0.543	73.4	40.106	-3.819	0.278	402.3
37.8	0.562	0.562	73.6	40.107	-3.827	0.289	405.5
39.2	0.582	0.582	73.8	40.108	-3.835	0.299	408.7
40.6	0.602	0.602	74.0	40.109	-3.843	0.309	411.7
42.0	0.621	0.621	74.2	40.110	-3.851	0.319	414.8
43.4	0.640	0.640	74.3	40.111	-3.859	0.329	417.6
44.8	0.660	0.660	74.5	40.112	-3.866	0.339	420.5
46.2	0.679	0.679	74.7	40.113	-3.873	0.349	423.3
47.6	0.698	0.698	74.8	40.114	-3.881	0.359	426.0
49.0	0.718	0.718	75.0	40.115	-3.887	0.369	428.7
50.4	0.737	0.737	75.2	40.116	-3.894	0.379	431.3
51.8	0.756	0.756	75.3	40.116	-3.901	0.389	433.8
53.2	0.775	0.775	75.5	40.117	-3.907	0.399	436.3
54.6	0.794	0.794	75.6	40.118	-3.914	0.409	438.8
56.0	0.813	0.813	75.8	40.119	-3.920	0.419	441.2
57.4	0.833	0.833	75.9	40.120	-3.926	0.428	443.5
58.8	0.852	0.852	76.0	40.121	-3.932	0.438	445.8
60.2	0.870	0.870	76.2	40.121	-3.938	0.448	448.1
61.6	0.889	0.889	76.3	40.122	-3.944	0.458	450.3
63.0	0.908	0.908	76.4	40.123	-3.950	0.468	452.5
64.4	0.927	0.927	76.5	40.124	-3.955	0.477	454.6
65.8	0.946	0.946	76.7	40.124	-3.961	0.487	456.7
67.2	0.965	0.965	76.8	40.125	-3.966	0.497	458.7
68.6	0.984	0.984	76.9	40.126	-3.972	0.507	460.7
70.0	1.003	1.003	56.2	40.127	-3.979	0.516	225.9
71.4	1.029	1.029	54.0	40.135	-4.006	0.526	215.4
72.8	1.057	1.057	51.8	40.143	-4.033	0.536	206.7
74.2	1.085	1.085	49.4	40.152	-4.061	0.546	199.8
75.6	1.115	1.115	46.9	40.162	-4.090	0.557	194.7
77.0	1.147	1.147	44.2	40.174	-4.120	0.567	191.5
78.4	1.181	1.181	41.4	40.186	-4.150	0.578	190.1
79.8	1.226	1.226	29.1	40.190	-4.194	0.589	62.8
81.2	1.281	1.281	23.4	40.194	-4.241	0.600	58.0
82.6	1.355	1.355	15.8	40.200	-4.290	0.612	57.2
83.5	1.499	1.499	0.0	40.199	-4.326	0.621	0.0

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TANK NO. 017 : DB TANK SB : NET VOLUME= 83.50 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	0.000	0.000	60.1	40.016	3.304	0.001	228.5
1.4	0.023	0.023	62.7	40.040	3.402	0.012	256.3
2.8	0.046	0.046	64.0	40.049	3.452	0.023	271.3
4.2	0.069	0.069	64.9	40.056	3.488	0.035	283.0
5.6	0.091	0.091	65.7	40.061	3.517	0.046	292.7
7.0	0.113	0.113	66.4	40.065	3.542	0.057	301.3
8.4	0.134	0.134	67.0	40.069	3.565	0.068	309.0
9.8	0.156	0.156	67.5	40.072	3.586	0.079	316.2
11.2	0.177	0.177	68.0	40.075	3.604	0.090	322.8
12.6	0.198	0.198	68.5	40.078	3.622	0.101	329.0
14.0	0.219	0.219	68.9	40.080	3.638	0.112	334.9
15.4	0.240	0.240	69.3	40.083	3.653	0.122	340.5
16.8	0.261	0.261	69.6	40.085	3.668	0.133	345.8
18.2	0.281	0.281	70.0	40.087	3.682	0.144	350.8
19.6	0.302	0.302	70.3	40.089	3.695	0.154	355.6
21.0	0.323	0.323	70.6	40.090	3.707	0.165	360.3
22.4	0.343	0.343	70.9	40.092	3.719	0.175	364.7
23.8	0.363	0.363	71.2	40.094	3.731	0.186	369.0
25.2	0.383	0.383	71.5	40.095	3.742	0.196	373.2
26.6	0.404	0.404	71.8	40.097	3.753	0.207	377.2
28.0	0.424	0.424	72.0	40.098	3.763	0.217	381.1
29.4	0.444	0.444	72.3	40.100	3.773	0.227	384.8
30.8	0.464	0.464	72.5	40.101	3.783	0.238	388.5
32.2	0.483	0.483	72.7	40.102	3.792	0.248	392.1
33.6	0.503	0.503	73.0	40.103	3.801	0.258	395.6
35.0	0.523	0.523	73.2	40.105	3.810	0.268	399.0
36.4	0.543	0.543	73.4	40.106	3.819	0.278	402.3
37.8	0.562	0.562	73.6	40.107	3.827	0.289	405.5
39.2	0.582	0.582	73.8	40.108	3.835	0.299	408.7
40.6	0.602	0.602	74.0	40.109	3.843	0.309	411.7
42.0	0.621	0.621	74.2	40.110	3.851	0.319	414.8
43.4	0.640	0.640	74.3	40.111	3.859	0.329	417.6
44.8	0.660	0.660	74.5	40.112	3.866	0.339	420.5
46.2	0.679	0.679	74.7	40.113	3.873	0.349	423.3
47.6	0.698	0.698	74.8	40.114	3.881	0.359	426.0
49.0	0.718	0.718	75.0	40.115	3.887	0.369	428.7
50.4	0.737	0.737	75.2	40.116	3.894	0.379	431.3
51.8	0.756	0.756	75.3	40.116	3.901	0.389	433.8
53.2	0.775	0.775	75.5	40.117	3.907	0.399	436.3
54.6	0.794	0.794	75.6	40.118	3.914	0.409	438.8
56.0	0.813	0.813	75.8	40.119	3.920	0.419	441.2
57.4	0.833	0.833	75.9	40.120	3.926	0.428	443.5
58.8	0.852	0.852	76.0	40.121	3.932	0.438	445.8
60.2	0.870	0.870	76.2	40.121	3.938	0.448	448.1
61.6	0.889	0.889	76.3	40.122	3.944	0.458	450.3
63.0	0.908	0.908	76.4	40.123	3.950	0.468	452.5
64.4	0.927	0.927	76.5	40.124	3.955	0.477	454.6
65.8	0.946	0.946	76.7	40.124	3.961	0.487	456.7
67.2	0.965	0.965	76.8	40.125	3.966	0.497	458.7
68.6	0.984	0.984	76.9	40.126	3.972	0.507	460.7
70.0	1.003	1.003	56.2	40.127	3.979	0.516	225.9
71.4	1.029	1.029	54.0	40.135	4.006	0.526	215.4
72.8	1.057	1.057	51.8	40.143	4.033	0.536	206.7
74.2	1.085	1.085	49.4	40.152	4.061	0.546	199.8
75.6	1.115	1.115	46.9	40.162	4.090	0.557	194.7
77.0	1.147	1.147	44.2	40.174	4.120	0.567	191.5
78.4	1.181	1.181	41.4	40.186	4.150	0.578	190.1
79.8	1.226	1.226	29.1	40.190	4.194	0.589	62.8
81.2	1.281	1.281	23.4	40.194	4.241	0.600	58.0
82.6	1.355	1.355	15.8	40.200	4.290	0.612	57.2
83.5	1.499	1.499	0.0	40.199	4.326	0.621	0.0

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TANK NO. 018 : WING TANK PS : NET VOLUME= 72.47 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.347	0.000	0.0	0.000	0.000	0.000	0.0
1.2	1.564	0.217	10.8	31.058	-8.231	1.501	1.8
2.4	1.677	0.330	11.1	31.018	-8.293	1.561	1.8
3.6	1.788	0.440	11.3	30.969	-8.317	1.618	1.9
4.8	1.896	0.549	11.4	30.929	-8.330	1.674	1.9
6.0	2.005	0.657	11.4	30.902	-8.339	1.729	1.9
7.2	2.113	0.766	11.4	30.883	-8.344	1.784	1.9
8.4	2.221	0.874	11.4	30.869	-8.349	1.839	1.9
9.6	2.330	0.982	11.4	30.858	-8.352	1.894	1.9
10.8	2.438	1.090	11.4	30.850	-8.354	1.948	1.9
12.0	2.546	1.199	11.4	30.844	-8.356	2.002	1.9
13.2	2.654	1.307	11.4	30.838	-8.358	2.057	1.9
14.4	2.762	1.415	11.4	30.834	-8.359	2.111	1.9
15.6	2.871	1.523	11.4	30.830	-8.360	2.165	1.9
16.8	2.979	1.631	11.4	30.827	-8.361	2.220	1.9
18.0	3.087	1.740	11.4	30.824	-8.362	2.274	1.9
19.2	3.195	1.848	11.4	30.822	-8.363	2.328	1.9
20.4	3.304	1.956	11.4	30.819	-8.363	2.382	1.9
21.6	3.412	2.064	11.4	30.817	-8.364	2.436	1.9
22.8	3.520	2.173	11.4	30.816	-8.364	2.491	1.9
24.0	3.628	2.281	11.4	30.814	-8.365	2.545	1.9
25.2	3.737	2.389	11.4	30.813	-8.365	2.599	1.9
26.4	3.845	2.497	11.4	30.812	-8.366	2.653	1.9
27.6	3.953	2.606	11.4	30.810	-8.366	2.707	1.9
28.8	4.061	2.714	11.4	30.809	-8.366	2.761	1.9
30.0	4.169	2.822	11.4	30.808	-8.366	2.816	1.9
31.2	4.278	2.930	11.4	30.807	-8.367	2.870	1.9
32.4	4.386	3.038	11.4	30.807	-8.367	2.924	1.9
33.6	4.494	3.147	11.4	30.806	-8.367	2.978	1.9
34.8	4.602	3.255	11.4	30.805	-8.367	3.032	1.9
36.0	4.711	3.363	11.4	30.804	-8.368	3.086	1.9
37.2	4.819	3.471	11.4	30.804	-8.368	3.140	1.9
38.4	4.927	3.580	11.4	30.803	-8.368	3.195	1.9
39.6	5.035	3.688	11.4	30.803	-8.368	3.249	1.9
40.8	5.144	3.796	11.4	30.802	-8.368	3.303	1.9
42.0	5.252	3.904	11.4	30.802	-8.368	3.357	1.9
43.2	5.360	4.013	11.4	30.801	-8.369	3.411	1.9
44.4	5.468	4.121	11.4	30.801	-8.369	3.465	1.9
45.6	5.576	4.229	11.4	30.800	-8.369	3.519	1.9
46.8	5.685	4.337	11.4	30.800	-8.369	3.574	1.9
48.0	5.793	4.445	11.4	30.800	-8.369	3.628	1.9
49.2	5.901	4.554	11.4	30.799	-8.369	3.682	1.9
50.4	6.009	4.662	11.4	30.799	-8.369	3.736	1.9
51.6	6.118	4.770	11.4	30.798	-8.369	3.790	1.9
52.8	6.226	4.878	11.4	30.798	-8.369	3.844	1.9
54.0	6.334	4.987	11.4	30.798	-8.370	3.898	1.9
55.2	6.442	5.095	11.4	30.798	-8.370	3.952	1.9
56.4	6.551	5.203	11.4	30.797	-8.370	4.007	1.9
57.6	6.659	5.311	11.4	30.797	-8.370	4.061	1.9
58.8	6.767	5.419	11.4	30.797	-8.370	4.115	1.9
60.0	6.875	5.528	11.4	30.797	-8.370	4.169	1.9
61.2	6.983	5.636	11.4	30.796	-8.370	4.223	1.9
62.4	7.092	5.744	11.4	30.796	-8.370	4.277	1.9
63.6	7.200	5.852	11.4	30.796	-8.370	4.331	1.9
64.8	7.308	5.961	11.4	30.796	-8.370	4.385	1.9
66.0	7.416	6.069	11.4	30.795	-8.370	4.440	1.9
67.2	7.525	6.177	11.4	30.795	-8.370	4.494	1.9
68.4	7.633	6.285	11.4	30.795	-8.370	4.548	1.9
69.6	7.741	6.394	11.4	30.795	-8.370	4.602	1.9
70.8	7.849	6.502	11.4	30.795	-8.370	4.656	1.9
72.0	7.958	6.610	11.4	30.795	-8.371	4.710	1.9
72.5	8.000	6.653	11.4	30.795	-8.371	4.731	1.9

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TANK NO. 019 : WING TANK SB : NET VOLUME= 72.47 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.347	0.000	0.0	0.000	0.000	0.000	0.0
1.2	1.564	0.217	10.8	31.058	8.231	1.501	1.8
2.4	1.677	0.330	11.1	31.018	8.293	1.561	1.8
3.6	1.788	0.440	11.3	30.969	8.317	1.618	1.9
4.8	1.896	0.549	11.4	30.929	8.330	1.674	1.9
6.0	2.005	0.657	11.4	30.902	8.339	1.729	1.9
7.2	2.113	0.766	11.4	30.883	8.344	1.784	1.9
8.4	2.221	0.874	11.4	30.869	8.349	1.839	1.9
9.6	2.330	0.982	11.4	30.858	8.352	1.894	1.9
10.8	2.438	1.090	11.4	30.850	8.354	1.948	1.9
12.0	2.546	1.199	11.4	30.844	8.356	2.002	1.9
13.2	2.654	1.307	11.4	30.838	8.358	2.057	1.9
14.4	2.762	1.415	11.4	30.834	8.359	2.111	1.9
15.6	2.871	1.523	11.4	30.830	8.360	2.165	1.9
16.8	2.979	1.631	11.4	30.827	8.361	2.220	1.9
18.0	3.087	1.740	11.4	30.824	8.362	2.274	1.9
19.2	3.195	1.848	11.4	30.822	8.363	2.328	1.9
20.4	3.304	1.956	11.4	30.819	8.363	2.382	1.9
21.6	3.412	2.064	11.4	30.817	8.364	2.436	1.9
22.8	3.520	2.173	11.4	30.816	8.364	2.491	1.9
24.0	3.628	2.281	11.4	30.814	8.365	2.545	1.9
25.2	3.737	2.389	11.4	30.813	8.365	2.599	1.9
26.4	3.845	2.497	11.4	30.812	8.366	2.653	1.9
27.6	3.953	2.606	11.4	30.810	8.366	2.707	1.9
28.8	4.061	2.714	11.4	30.809	8.366	2.761	1.9
30.0	4.169	2.822	11.4	30.808	8.366	2.816	1.9
31.2	4.278	2.930	11.4	30.807	8.367	2.870	1.9
32.4	4.386	3.038	11.4	30.807	8.367	2.924	1.9
33.6	4.494	3.147	11.4	30.806	8.367	2.978	1.9
34.8	4.602	3.255	11.4	30.805	8.367	3.032	1.9
36.0	4.711	3.363	11.4	30.804	8.368	3.086	1.9
37.2	4.819	3.471	11.4	30.804	8.368	3.140	1.9
38.4	4.927	3.580	11.4	30.803	8.368	3.195	1.9
39.6	5.035	3.688	11.4	30.803	8.368	3.249	1.9
40.8	5.144	3.796	11.4	30.802	8.368	3.303	1.9
42.0	5.252	3.904	11.4	30.802	8.368	3.357	1.9
43.2	5.360	4.013	11.4	30.801	8.369	3.411	1.9
44.4	5.468	4.121	11.4	30.801	8.369	3.465	1.9
45.6	5.576	4.229	11.4	30.800	8.369	3.519	1.9
46.8	5.685	4.337	11.4	30.800	8.369	3.574	1.9
48.0	5.793	4.445	11.4	30.800	8.369	3.628	1.9
49.2	5.901	4.554	11.4	30.799	8.369	3.682	1.9
50.4	6.009	4.662	11.4	30.799	8.369	3.736	1.9
51.6	6.118	4.770	11.4	30.798	8.369	3.790	1.9
52.8	6.226	4.878	11.4	30.798	8.369	3.844	1.9
54.0	6.334	4.987	11.4	30.798	8.370	3.898	1.9
55.2	6.442	5.095	11.4	30.798	8.370	3.952	1.9
56.4	6.551	5.203	11.4	30.797	8.370	4.007	1.9
57.6	6.659	5.311	11.4	30.797	8.370	4.061	1.9
58.8	6.767	5.419	11.4	30.797	8.370	4.115	1.9
60.0	6.875	5.528	11.4	30.797	8.370	4.169	1.9
61.2	6.983	5.636	11.4	30.796	8.370	4.223	1.9
62.4	7.092	5.744	11.4	30.796	8.370	4.277	1.9
63.6	7.200	5.852	11.4	30.796	8.370	4.331	1.9
64.8	7.308	5.961	11.4	30.796	8.370	4.385	1.9
66.0	7.416	6.069	11.4	30.795	8.370	4.440	1.9
67.2	7.525	6.177	11.4	30.795	8.370	4.494	1.9
68.4	7.633	6.285	11.4	30.795	8.370	4.548	1.9
69.6	7.741	6.394	11.4	30.795	8.370	4.602	1.9
70.8	7.849	6.502	11.4	30.795	8.370	4.656	1.9
72.0	7.958	6.610	11.4	30.795	8.371	4.710	1.9
72.5	8.000	6.653	11.4	30.795	8.371	4.731	1.9

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TANK NO. 020 : DB TANK PS : NET VOLUME= 94.37 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	0.000	0.000	70.4	31.000	-3.587	0.001	313.4
1.6	0.023	0.023	73.3	30.951	-3.686	0.012	346.8
3.2	0.045	0.045	74.7	30.933	-3.736	0.023	365.7
4.8	0.067	0.067	75.7	30.921	-3.772	0.034	379.8
6.4	0.089	0.089	76.5	30.913	-3.801	0.045	391.3
8.0	0.110	0.110	77.1	30.907	-3.826	0.056	401.2
9.6	0.132	0.132	77.7	30.902	-3.848	0.067	410.3
11.2	0.153	0.153	78.3	30.898	-3.867	0.078	418.3
12.8	0.174	0.174	78.8	30.894	-3.885	0.088	425.9
14.4	0.195	0.195	79.2	30.891	-3.901	0.099	432.7
16.0	0.215	0.215	79.6	30.888	-3.916	0.110	439.2
17.6	0.236	0.236	80.0	30.885	-3.931	0.120	445.3
19.2	0.257	0.257	80.3	30.883	-3.944	0.131	451.3
20.8	0.277	0.277	80.7	30.881	-3.957	0.141	456.7
22.4	0.298	0.298	81.0	30.879	-3.969	0.152	462.0
24.0	0.318	0.318	81.3	30.877	-3.980	0.162	467.1
25.6	0.338	0.338	81.6	30.875	-3.991	0.172	471.9
27.2	0.358	0.358	81.8	30.873	-4.002	0.183	476.5
28.8	0.378	0.378	82.1	30.872	-4.012	0.193	481.0
30.4	0.399	0.399	82.3	30.870	-4.021	0.203	485.3
32.0	0.419	0.419	82.6	30.869	-4.031	0.214	489.5
33.6	0.439	0.439	82.8	30.868	-4.040	0.224	493.5
35.2	0.458	0.458	83.0	30.867	-4.049	0.234	497.4
36.8	0.478	0.478	83.2	30.865	-4.057	0.244	501.2
38.4	0.498	0.498	83.4	30.864	-4.065	0.254	504.8
40.0	0.518	0.518	83.6	30.863	-4.073	0.265	508.2
41.6	0.537	0.537	83.8	30.862	-4.081	0.275	511.5
43.2	0.557	0.557	84.0	30.861	-4.088	0.285	514.8
44.8	0.577	0.577	84.2	30.860	-4.095	0.295	518.0
46.4	0.596	0.596	84.3	30.859	-4.102	0.305	521.2
48.0	0.616	0.616	84.5	30.858	-4.109	0.315	524.1
49.6	0.635	0.635	84.7	30.857	-4.116	0.325	527.0
51.2	0.655	0.655	84.8	30.857	-4.122	0.335	529.9
52.8	0.674	0.674	85.0	30.856	-4.129	0.345	532.7
54.4	0.694	0.694	85.1	30.855	-4.135	0.355	535.2
56.0	0.713	0.713	85.2	30.854	-4.141	0.365	537.7
57.6	0.732	0.732	85.4	30.853	-4.147	0.375	540.2
59.2	0.752	0.752	85.5	30.853	-4.153	0.385	542.6
60.8	0.771	0.771	85.6	30.852	-4.158	0.395	545.0
62.4	0.790	0.790	85.7	30.851	-4.164	0.405	547.2
64.0	0.809	0.809	85.8	30.851	-4.169	0.414	549.3
65.6	0.829	0.829	86.0	30.850	-4.174	0.424	551.4
67.2	0.848	0.848	86.1	30.849	-4.179	0.434	553.5
68.8	0.867	0.867	86.2	30.849	-4.184	0.444	555.6
70.4	0.886	0.886	86.3	30.848	-4.189	0.454	557.5
72.0	0.905	0.905	86.4	30.848	-4.194	0.464	559.3
73.6	0.924	0.924	86.5	30.847	-4.199	0.473	561.1
75.2	0.943	0.943	86.5	30.847	-4.203	0.483	562.8
76.8	0.962	0.962	86.6	30.846	-4.208	0.493	564.5
78.4	0.981	0.981	86.7	30.846	-4.212	0.503	566.2
80.0	1.001	1.001	58.8	30.845	-4.217	0.513	176.4
81.6	1.030	1.030	55.4	30.845	-4.251	0.522	147.8
83.2	1.060	1.060	51.8	30.844	-4.287	0.532	120.9
84.8	1.093	1.093	48.0	30.844	-4.326	0.543	95.8
86.4	1.129	1.129	43.7	30.844	-4.368	0.553	72.7
88.0	1.169	1.169	39.1	30.844	-4.412	0.564	51.8
89.6	1.215	1.215	33.7	30.844	-4.459	0.575	33.3
91.2	1.269	1.269	27.3	30.844	-4.510	0.587	17.8
92.8	1.340	1.340	18.9	30.844	-4.566	0.599	5.9
94.4	1.917	1.917	0.0	30.838	-4.629	0.613	0.0

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TANK NO. 021 : DB TANK SB : NET VOLUME= 94.37 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	0.000	0.000	70.4	31.000	3.587	0.001	313.4
1.6	0.023	0.023	73.3	30.951	3.686	0.012	346.8
3.2	0.045	0.045	74.7	30.933	3.736	0.023	365.7
4.8	0.067	0.067	75.7	30.921	3.772	0.034	379.8
6.4	0.089	0.089	76.5	30.913	3.801	0.045	391.3
8.0	0.110	0.110	77.1	30.907	3.826	0.056	401.2
9.6	0.132	0.132	77.7	30.902	3.848	0.067	410.3
11.2	0.153	0.153	78.3	30.898	3.867	0.078	418.3
12.8	0.174	0.174	78.8	30.894	3.885	0.088	425.9
14.4	0.195	0.195	79.2	30.891	3.901	0.099	432.7
16.0	0.215	0.215	79.6	30.888	3.916	0.110	439.2
17.6	0.236	0.236	80.0	30.885	3.931	0.120	445.3
19.2	0.257	0.257	80.3	30.883	3.944	0.131	451.3
20.8	0.277	0.277	80.7	30.881	3.957	0.141	456.7
22.4	0.298	0.298	81.0	30.879	3.969	0.152	462.0
24.0	0.318	0.318	81.3	30.877	3.980	0.162	467.1
25.6	0.338	0.338	81.6	30.875	3.991	0.172	471.9
27.2	0.358	0.358	81.8	30.873	4.002	0.183	476.5
28.8	0.378	0.378	82.1	30.872	4.012	0.193	481.0
30.4	0.399	0.399	82.3	30.870	4.021	0.203	485.3
32.0	0.419	0.419	82.6	30.869	4.031	0.214	489.5
33.6	0.439	0.439	82.8	30.868	4.040	0.224	493.5
35.2	0.458	0.458	83.0	30.867	4.049	0.234	497.4
36.8	0.478	0.478	83.2	30.865	4.057	0.244	501.2
38.4	0.498	0.498	83.4	30.864	4.065	0.254	504.8
40.0	0.518	0.518	83.6	30.863	4.073	0.265	508.2
41.6	0.537	0.537	83.8	30.862	4.081	0.275	511.5
43.2	0.557	0.557	84.0	30.861	4.088	0.285	514.8
44.8	0.577	0.577	84.2	30.860	4.095	0.295	518.0
46.4	0.596	0.596	84.3	30.859	4.102	0.305	521.2
48.0	0.616	0.616	84.5	30.858	4.109	0.315	524.1
49.6	0.635	0.635	84.7	30.857	4.116	0.325	527.0
51.2	0.655	0.655	84.8	30.857	4.122	0.335	529.9
52.8	0.674	0.674	85.0	30.856	4.129	0.345	532.7
54.4	0.694	0.694	85.1	30.855	4.135	0.355	535.2
56.0	0.713	0.713	85.2	30.854	4.141	0.365	537.7
57.6	0.732	0.732	85.4	30.853	4.147	0.375	540.2
59.2	0.752	0.752	85.5	30.853	4.153	0.385	542.6
60.8	0.771	0.771	85.6	30.852	4.158	0.395	545.0
62.4	0.790	0.790	85.7	30.851	4.164	0.405	547.2
64.0	0.809	0.809	85.8	30.851	4.169	0.414	549.3
65.6	0.829	0.829	86.0	30.850	4.174	0.424	551.4
67.2	0.848	0.848	86.1	30.849	4.179	0.434	553.5
68.8	0.867	0.867	86.2	30.849	4.184	0.444	555.6
70.4	0.886	0.886	86.3	30.848	4.189	0.454	557.5
72.0	0.905	0.905	86.4	30.848	4.194	0.464	559.3
73.6	0.924	0.924	86.5	30.847	4.199	0.473	561.1
75.2	0.943	0.943	86.5	30.847	4.203	0.483	562.8
76.8	0.962	0.962	86.6	30.846	4.208	0.493	564.5
78.4	0.981	0.981	86.7	30.846	4.212	0.503	566.2
80.0	1.001	1.001	58.8	30.845	4.217	0.513	176.4
81.6	1.030	1.030	55.4	30.845	4.251	0.522	147.8
83.2	1.060	1.060	51.8	30.844	4.287	0.532	120.9
84.8	1.093	1.093	48.0	30.844	4.326	0.543	95.8
86.4	1.129	1.129	43.7	30.844	4.368	0.553	72.7
88.0	1.169	1.169	39.1	30.844	4.412	0.564	51.8
89.6	1.215	1.215	33.7	30.844	4.459	0.575	33.3
91.2	1.269	1.269	27.3	30.844	4.510	0.587	17.8
92.8	1.340	1.340	18.9	30.844	4.566	0.599	5.9
94.4	1.917	1.917	0.0	30.838	4.629	0.613	0.0

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TANK NO. 022 : WING TANK PS : NET VOLUME= 59.97 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	0.647	0.000	0.0	0.000	0.000	0.000	0.0
1.0	1.233	0.585	5.5	17.688	-5.424	1.097	6.3
2.0	1.390	0.742	7.6	17.850	-5.687	1.206	9.3
3.0	1.513	0.865	9.1	17.927	-5.836	1.289	11.4
4.0	1.618	0.971	10.4	17.979	-5.944	1.358	13.2
5.0	1.713	1.066	11.5	18.025	-6.031	1.420	15.0
6.0	1.799	1.152	12.5	18.071	-6.105	1.476	16.8
7.0	1.879	1.232	13.4	18.119	-6.171	1.528	18.4
8.0	1.954	1.306	14.1	18.169	-6.233	1.576	19.4
9.0	2.025	1.377	14.9	18.222	-6.293	1.622	20.3
10.0	2.092	1.445	15.6	18.278	-6.350	1.666	20.7
11.0	2.157	1.510	16.2	18.334	-6.405	1.708	20.8
12.0	2.220	1.572	16.6	18.389	-6.459	1.748	20.3
13.0	2.281	1.634	17.0	18.442	-6.510	1.786	19.6
14.0	2.341	1.694	17.3	18.495	-6.560	1.824	18.7
15.0	2.401	1.753	17.2	18.545	-6.609	1.860	17.5
16.0	2.461	1.814	16.8	18.593	-6.658	1.896	16.1
17.0	2.523	1.876	16.4	18.638	-6.706	1.931	14.8
18.0	2.587	1.940	16.0	18.681	-6.753	1.966	13.4
19.0	2.652	2.005	15.5	18.721	-6.800	2.000	12.0
20.0	2.720	2.072	15.1	18.760	-6.846	2.034	10.6
21.0	2.789	2.142	14.6	18.798	-6.891	2.069	9.3
22.0	2.861	2.214	14.0	18.834	-6.935	2.103	8.0
23.0	2.936	2.289	13.5	18.871	-6.979	2.138	6.8
24.0	3.014	2.366	13.1	18.907	-7.023	2.173	5.6
25.0	3.094	2.447	12.6	18.944	-7.065	2.208	4.6
26.0	3.177	2.530	12.2	18.982	-7.107	2.244	3.7
27.0	3.264	2.616	11.7	19.022	-7.149	2.280	2.9
28.0	3.354	2.706	11.3	19.063	-7.190	2.316	2.3
29.0	3.446	2.798	11.0	19.104	-7.229	2.354	1.8
30.0	3.541	2.894	10.7	19.148	-7.268	2.392	1.4
31.0	3.639	2.991	10.4	19.192	-7.305	2.430	1.1
32.0	3.738	3.091	10.3	19.236	-7.341	2.470	1.0
33.0	3.839	3.192	10.1	19.280	-7.376	2.510	0.9
34.0	3.941	3.294	10.1	19.323	-7.409	2.550	0.9
35.0	4.043	3.396	10.1	19.363	-7.440	2.591	0.9
36.0	4.145	3.497	10.2	19.401	-7.469	2.633	0.9
37.0	4.246	3.598	10.2	19.436	-7.497	2.675	0.9
38.0	4.346	3.699	10.2	19.469	-7.524	2.718	0.9
39.0	4.447	3.800	10.2	19.501	-7.549	2.761	0.9
40.0	4.548	3.901	10.2	19.530	-7.573	2.805	0.9
41.0	4.649	4.001	10.2	19.558	-7.595	2.848	0.9
42.0	4.749	4.102	10.2	19.585	-7.617	2.892	0.9
43.0	4.850	4.203	10.2	19.611	-7.638	2.937	0.9
44.0	4.951	4.303	10.2	19.635	-7.657	2.981	0.9
45.0	5.099	4.451	5.3	19.630	-7.676	3.026	0.4
46.0	5.292	4.645	5.3	19.599	-7.695	3.074	0.4
47.0	5.486	4.839	5.3	19.570	-7.712	3.123	0.4
48.0	5.680	5.033	5.3	19.541	-7.729	3.174	0.4
49.0	5.874	5.226	5.3	19.514	-7.746	3.227	0.4
50.0	6.068	5.420	5.3	19.487	-7.761	3.282	0.4
51.0	6.261	5.614	5.3	19.462	-7.776	3.339	0.4
52.0	6.455	5.808	5.3	19.438	-7.791	3.397	0.4
53.0	6.649	6.002	5.3	19.414	-7.804	3.456	0.4
54.0	6.843	6.195	5.3	19.392	-7.818	3.517	0.4
55.0	7.036	6.389	5.3	19.370	-7.831	3.579	0.4
56.0	7.230	6.583	5.3	19.349	-7.843	3.643	0.4
57.0	7.424	6.777	5.3	19.329	-7.855	3.708	0.4
58.0	7.618	6.970	5.3	19.310	-7.866	3.773	0.4
59.0	7.812	7.164	5.3	19.291	-7.878	3.840	0.4
60.0	8.000	7.353	5.3	19.273	-7.888	3.906	0.4

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TANK NO. 023 : WING TANK SB : NET VOLUME= 59.97 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	0.647	0.000	0.0	0.000	0.000	0.000	0.0
1.0	1.233	0.585	5.5	17.688	5.424	1.097	6.3
2.0	1.390	0.742	7.6	17.850	5.687	1.206	9.3
3.0	1.513	0.865	9.1	17.927	5.836	1.289	11.4
4.0	1.618	0.971	10.4	17.979	5.944	1.358	13.2
5.0	1.713	1.066	11.5	18.025	6.031	1.420	15.0
6.0	1.799	1.152	12.5	18.071	6.105	1.476	16.8
7.0	1.879	1.232	13.4	18.119	6.171	1.528	18.4
8.0	1.954	1.306	14.1	18.169	6.233	1.576	19.4
9.0	2.025	1.377	14.9	18.222	6.293	1.622	20.3
10.0	2.092	1.445	15.6	18.278	6.350	1.666	20.7
11.0	2.157	1.510	16.2	18.334	6.405	1.708	20.8
12.0	2.220	1.572	16.6	18.389	6.459	1.748	20.3
13.0	2.281	1.634	17.0	18.442	6.510	1.786	19.6
14.0	2.341	1.694	17.3	18.495	6.560	1.824	18.7
15.0	2.401	1.753	17.2	18.545	6.609	1.860	17.5
16.0	2.461	1.814	16.8	18.593	6.658	1.896	16.1
17.0	2.523	1.876	16.4	18.638	6.706	1.931	14.8
18.0	2.587	1.940	16.0	18.681	6.753	1.966	13.4
19.0	2.652	2.005	15.5	18.721	6.800	2.000	12.0
20.0	2.720	2.072	15.1	18.760	6.846	2.034	10.6
21.0	2.789	2.142	14.6	18.798	6.891	2.069	9.3
22.0	2.861	2.214	14.0	18.834	6.935	2.103	8.0
23.0	2.936	2.289	13.5	18.871	6.979	2.138	6.8
24.0	3.014	2.366	13.1	18.907	7.023	2.173	5.6
25.0	3.094	2.447	12.6	18.944	7.065	2.208	4.6
26.0	3.177	2.530	12.2	18.982	7.107	2.244	3.7
27.0	3.264	2.616	11.7	19.022	7.149	2.280	2.9
28.0	3.354	2.706	11.3	19.063	7.190	2.316	2.3
29.0	3.446	2.798	11.0	19.104	7.229	2.354	1.8
30.0	3.541	2.894	10.7	19.148	7.268	2.392	1.4
31.0	3.639	2.991	10.4	19.192	7.305	2.430	1.1
32.0	3.738	3.091	10.3	19.236	7.341	2.470	1.0
33.0	3.839	3.192	10.1	19.280	7.376	2.510	0.9
34.0	3.941	3.294	10.1	19.323	7.409	2.550	0.9
35.0	4.043	3.396	10.1	19.363	7.440	2.591	0.9
36.0	4.145	3.497	10.2	19.401	7.469	2.633	0.9
37.0	4.246	3.598	10.2	19.436	7.497	2.675	0.9
38.0	4.346	3.699	10.2	19.469	7.524	2.718	0.9
39.0	4.447	3.800	10.2	19.501	7.549	2.761	0.9
40.0	4.548	3.901	10.2	19.530	7.573	2.805	0.9
41.0	4.649	4.001	10.2	19.558	7.595	2.848	0.9
42.0	4.749	4.102	10.2	19.585	7.617	2.892	0.9
43.0	4.850	4.203	10.2	19.611	7.638	2.937	0.9
44.0	4.951	4.303	10.2	19.635	7.657	2.981	0.9
45.0	5.099	4.451	5.3	19.630	7.676	3.026	0.4
46.0	5.292	4.645	5.3	19.599	7.695	3.074	0.4
47.0	5.486	4.839	5.3	19.570	7.712	3.123	0.4
48.0	5.680	5.033	5.3	19.541	7.729	3.174	0.4
49.0	5.874	5.226	5.3	19.514	7.746	3.227	0.4
50.0	6.068	5.420	5.3	19.487	7.761	3.282	0.4
51.0	6.261	5.614	5.3	19.462	7.776	3.339	0.4
52.0	6.455	5.808	5.3	19.438	7.791	3.397	0.4
53.0	6.649	6.002	5.3	19.414	7.804	3.456	0.4
54.0	6.843	6.195	5.3	19.392	7.818	3.517	0.4
55.0	7.036	6.389	5.3	19.370	7.831	3.579	0.4
56.0	7.230	6.583	5.3	19.349	7.843	3.643	0.4
57.0	7.424	6.777	5.3	19.329	7.855	3.708	0.4
58.0	7.618	6.970	5.3	19.310	7.866	3.773	0.4
59.0	7.812	7.164	5.3	19.291	7.878	3.840	0.4
60.0	8.000	7.353	5.3	19.273	7.888	3.906	0.4

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TANK NO. 024 : DB TANK PS : NET VOLUME= 74.02 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	0.000	0.000	17.6	22.250	-1.263	0.001	27.9
1.2	0.052	0.052	28.5	22.696	-1.871	0.028	80.3
2.4	0.092	0.092	32.9	22.701	-2.056	0.050	101.4
3.6	0.128	0.128	36.3	22.680	-2.179	0.070	118.7
4.8	0.161	0.161	39.0	22.647	-2.269	0.089	132.6
6.0	0.191	0.191	41.1	22.612	-2.341	0.106	144.2
7.2	0.221	0.221	43.0	22.577	-2.400	0.123	154.3
8.4	0.249	0.249	44.7	22.542	-2.451	0.139	163.6
9.6	0.276	0.276	46.2	22.508	-2.494	0.154	172.1
10.8	0.303	0.303	47.6	22.475	-2.533	0.169	179.9
12.0	0.328	0.328	49.0	22.443	-2.568	0.184	187.6
13.2	0.353	0.353	50.3	22.411	-2.600	0.198	194.8
14.4	0.377	0.377	51.5	22.380	-2.628	0.212	201.7
15.6	0.401	0.401	52.6	22.349	-2.655	0.226	208.1
16.8	0.425	0.425	53.7	22.320	-2.680	0.239	214.3
18.0	0.447	0.447	54.8	22.291	-2.704	0.252	220.7
19.2	0.470	0.470	55.8	22.263	-2.725	0.265	226.7
20.4	0.492	0.492	56.8	22.235	-2.746	0.278	232.5
21.6	0.513	0.513	57.8	22.208	-2.766	0.290	238.2
22.8	0.534	0.534	58.7	22.181	-2.784	0.303	243.7
24.0	0.555	0.555	59.6	22.155	-2.802	0.315	248.9
25.2	0.576	0.576	60.4	22.130	-2.820	0.327	254.0
26.4	0.596	0.596	61.1	22.106	-2.836	0.338	259.1
27.6	0.616	0.616	61.9	22.083	-2.853	0.350	263.9
28.8	0.636	0.636	62.6	22.060	-2.868	0.362	268.7
30.0	0.656	0.656	63.3	22.039	-2.884	0.373	273.3
31.2	0.675	0.675	63.9	22.018	-2.899	0.384	277.9
32.4	0.695	0.695	64.5	21.998	-2.913	0.395	282.4
33.6	0.714	0.714	65.1	21.978	-2.927	0.406	286.8
34.8	0.733	0.733	65.7	21.960	-2.941	0.417	291.1
36.0	0.752	0.752	66.2	21.942	-2.955	0.428	295.4
37.2	0.770	0.770	66.7	21.924	-2.968	0.439	299.6
38.4	0.789	0.789	67.1	21.908	-2.981	0.450	303.6
39.6	0.807	0.807	67.6	21.892	-2.994	0.460	307.6
40.8	0.825	0.825	68.0	21.877	-3.006	0.471	311.6
42.0	0.843	0.843	68.4	21.862	-3.019	0.481	315.6
43.2	0.861	0.861	68.8	21.848	-3.031	0.491	319.4
44.4	0.879	0.879	69.2	21.834	-3.043	0.502	323.1
45.6	0.897	0.897	69.5	21.821	-3.054	0.512	326.9
46.8	0.915	0.915	69.9	21.808	-3.065	0.522	330.6
48.0	0.933	0.933	70.3	21.796	-3.077	0.532	334.3
49.2	0.950	0.950	70.5	21.784	-3.087	0.542	337.0
50.4	0.968	0.968	70.6	21.773	-3.098	0.552	339.6
51.6	0.985	0.985	70.8	21.763	-3.108	0.562	342.1
52.8	1.004	1.004	48.6	21.751	-3.122	0.572	185.3
54.0	1.030	1.030	45.8	21.734	-3.154	0.581	175.3
55.2	1.058	1.058	43.5	21.717	-3.187	0.591	172.5
56.4	1.087	1.087	40.1	21.699	-3.222	0.602	157.1
57.6	1.119	1.119	37.8	21.681	-3.257	0.612	156.6
58.8	1.153	1.153	34.6	21.662	-3.293	0.623	146.2
60.0	1.190	1.190	32.6	21.642	-3.329	0.634	146.9
61.2	1.230	1.230	28.5	21.620	-3.366	0.645	117.4
62.4	1.275	1.275	26.3	21.598	-3.404	0.657	116.6
63.6	1.324	1.324	23.6	21.575	-3.442	0.669	98.5
64.8	1.379	1.379	22.0	21.550	-3.481	0.682	95.7
66.0	1.440	1.440	18.9	21.529	-3.522	0.695	56.1
67.2	1.508	1.508	17.0	21.513	-3.566	0.709	41.5
68.4	1.585	1.585	15.3	21.500	-3.613	0.723	36.5
69.6	1.671	1.671	13.5	21.487	-3.661	0.739	30.9
70.8	1.771	1.771	11.3	21.477	-3.709	0.756	24.3
72.0	1.895	1.895	8.6	21.468	-3.759	0.773	16.6
73.2	2.078	2.078	5.0	21.461	-3.812	0.793	7.5
74.0	2.751	2.751	0.0	21.463	-3.854	0.809	0.0

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TANK NO. 025 : DB TANK SB : NET VOLUME= 74.02 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	0.000	0.000	17.6	22.250	1.263	0.001	27.9
1.2	0.052	0.052	28.5	22.696	1.871	0.028	80.3
2.4	0.092	0.092	32.9	22.701	2.056	0.050	101.4
3.6	0.128	0.128	36.3	22.680	2.179	0.070	118.7
4.8	0.161	0.161	39.0	22.647	2.269	0.089	132.6
6.0	0.191	0.191	41.1	22.612	2.341	0.106	144.2
7.2	0.221	0.221	43.0	22.577	2.400	0.123	154.3
8.4	0.249	0.249	44.7	22.542	2.451	0.139	163.6
9.6	0.276	0.276	46.2	22.508	2.494	0.154	172.1
10.8	0.303	0.303	47.6	22.475	2.533	0.169	179.9
12.0	0.328	0.328	49.0	22.443	2.568	0.184	187.6
13.2	0.353	0.353	50.3	22.411	2.600	0.198	194.8
14.4	0.377	0.377	51.5	22.380	2.628	0.212	201.7
15.6	0.401	0.401	52.6	22.349	2.655	0.226	208.1
16.8	0.425	0.425	53.7	22.320	2.680	0.239	214.3
18.0	0.447	0.447	54.8	22.291	2.704	0.252	220.7
19.2	0.470	0.470	55.8	22.263	2.725	0.265	226.7
20.4	0.492	0.492	56.8	22.235	2.746	0.278	232.5
21.6	0.513	0.513	57.8	22.208	2.766	0.290	238.2
22.8	0.534	0.534	58.7	22.181	2.784	0.303	243.7
24.0	0.555	0.555	59.6	22.155	2.802	0.315	248.9
25.2	0.576	0.576	60.4	22.130	2.820	0.327	254.0
26.4	0.596	0.596	61.1	22.106	2.836	0.338	259.1
27.6	0.616	0.616	61.9	22.083	2.853	0.350	263.9
28.8	0.636	0.636	62.6	22.060	2.868	0.362	268.7
30.0	0.656	0.656	63.3	22.039	2.884	0.373	273.3
31.2	0.675	0.675	63.9	22.018	2.899	0.384	277.9
32.4	0.695	0.695	64.5	21.998	2.913	0.395	282.4
33.6	0.714	0.714	65.1	21.978	2.927	0.406	286.8
34.8	0.733	0.733	65.7	21.960	2.941	0.417	291.1
36.0	0.752	0.752	66.2	21.942	2.955	0.428	295.4
37.2	0.770	0.770	66.7	21.924	2.968	0.439	299.6
38.4	0.789	0.789	67.1	21.908	2.981	0.450	303.6
39.6	0.807	0.807	67.6	21.892	2.994	0.460	307.6
40.8	0.825	0.825	68.0	21.877	3.006	0.471	311.6
42.0	0.843	0.843	68.4	21.862	3.019	0.481	315.6
43.2	0.861	0.861	68.8	21.848	3.031	0.491	319.4
44.4	0.879	0.879	69.2	21.834	3.043	0.502	323.1
45.6	0.897	0.897	69.5	21.821	3.054	0.512	326.9
46.8	0.915	0.915	69.9	21.808	3.065	0.522	330.6
48.0	0.933	0.933	70.3	21.796	3.077	0.532	334.3
49.2	0.950	0.950	70.5	21.784	3.087	0.542	337.0
50.4	0.968	0.968	70.6	21.773	3.098	0.552	339.6
51.6	0.985	0.985	70.8	21.763	3.108	0.562	342.1
52.8	1.004	1.004	48.6	21.751	3.122	0.572	185.3
54.0	1.030	1.030	45.8	21.734	3.154	0.581	175.3
55.2	1.058	1.058	43.5	21.717	3.187	0.591	172.5
56.4	1.087	1.087	40.1	21.699	3.222	0.602	157.1
57.6	1.119	1.119	37.8	21.681	3.257	0.612	156.6
58.8	1.153	1.153	34.6	21.662	3.293	0.623	146.2
60.0	1.190	1.190	32.6	21.642	3.329	0.634	146.9
61.2	1.230	1.230	28.5	21.620	3.366	0.645	117.4
62.4	1.275	1.275	26.3	21.598	3.404	0.657	116.6
63.6	1.324	1.324	23.6	21.575	3.442	0.669	98.5
64.8	1.379	1.379	22.0	21.550	3.481	0.682	95.7
66.0	1.440	1.440	18.9	21.529	3.522	0.695	56.1
67.2	1.508	1.508	17.0	21.513	3.566	0.709	41.5
68.4	1.585	1.585	15.3	21.500	3.613	0.723	36.5
69.6	1.671	1.671	13.5	21.487	3.661	0.739	30.9
70.8	1.771	1.771	11.3	21.477	3.709	0.756	24.3
72.0	1.895	1.895	8.6	21.468	3.759	0.773	16.6
73.2	2.078	2.078	5.0	21.461	3.812	0.793	7.5
74.0	2.751	2.751	0.0	21.463	3.854	0.809	0.0

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TANK NO. 026 : WING TANK PS : NET VOLUME= 90.14 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	0.872	0.000	0.0	0.000	0.000	0.000	0.0
1.5	1.360	0.488	7.2	13.655	-2.862	1.213	6.3
3.0	1.527	0.655	13.3	13.478	-2.926	1.333	14.2
4.5	1.635	0.763	15.4	13.421	-2.926	1.416	18.6
6.0	1.730	0.858	17.3	13.353	-2.949	1.483	22.7
7.5	1.815	0.943	18.9	13.290	-2.981	1.541	26.8
9.0	1.894	1.022	20.3	13.235	-3.016	1.593	30.6
10.5	1.968	1.096	21.6	13.187	-3.053	1.642	34.4
12.0	2.038	1.166	22.7	13.144	-3.089	1.687	38.2
13.5	2.104	1.232	23.8	13.106	-3.126	1.729	42.0
15.0	2.168	1.296	24.8	13.073	-3.162	1.770	45.8
16.5	2.233	1.361	22.9	13.033	-3.206	1.809	41.9
18.0	2.299	1.427	23.9	12.988	-3.256	1.847	45.5
19.5	2.363	1.491	24.8	12.949	-3.304	1.884	49.1
21.0	2.424	1.552	25.7	12.914	-3.349	1.921	52.6
22.5	2.483	1.611	26.5	12.883	-3.393	1.956	56.2
24.0	2.541	1.669	27.3	12.854	-3.434	1.991	59.7
25.5	2.597	1.725	28.0	12.828	-3.474	2.025	63.2
27.0	2.651	1.779	28.7	12.804	-3.512	2.058	66.7
28.5	2.704	1.832	29.4	12.782	-3.549	2.091	70.2
30.0	2.756	1.884	30.1	12.761	-3.584	2.123	73.6
31.5	2.807	1.935	30.7	12.742	-3.619	2.154	77.1
33.0	2.857	1.985	31.3	12.725	-3.652	2.185	80.5
34.5	2.906	2.034	31.8	12.708	-3.684	2.215	83.9
36.0	2.954	2.082	32.4	12.692	-3.715	2.245	87.2
37.5	3.002	2.130	32.9	12.678	-3.745	2.274	90.5
39.0	3.048	2.176	33.4	12.664	-3.775	2.303	93.8
40.5	3.094	2.222	33.9	12.650	-3.803	2.332	97.1
42.0	3.139	2.267	34.4	12.638	-3.831	2.360	100.3
43.5	3.184	2.312	34.8	12.626	-3.858	2.387	103.5
45.0	3.228	2.356	35.3	12.615	-3.884	2.415	106.6
46.5	3.272	2.400	35.7	12.604	-3.910	2.442	109.7
48.0	3.315	2.443	36.1	12.594	-3.935	2.468	112.7
49.5	3.358	2.486	36.5	12.584	-3.959	2.495	115.7
51.0	3.400	2.528	36.8	12.575	-3.983	2.521	118.6
52.5	3.442	2.570	37.2	12.566	-4.006	2.546	121.5
54.0	3.483	2.611	37.5	12.557	-4.029	2.572	124.4
55.5	3.524	2.652	37.9	12.549	-4.051	2.597	127.2
57.0	3.565	2.693	38.2	12.541	-4.073	2.622	130.0
58.5	3.605	2.733	38.5	12.533	-4.094	2.647	132.7
60.0	3.645	2.773	38.8	12.526	-4.115	2.671	135.4
61.5	3.685	2.813	39.1	12.519	-4.135	2.695	138.0
63.0	3.724	2.852	39.4	12.512	-4.154	2.719	140.5
64.5	3.763	2.891	39.6	12.505	-4.174	2.743	143.1
66.0	3.802	2.930	39.9	12.499	-4.193	2.767	145.5
67.5	3.841	2.969	40.2	12.493	-4.211	2.790	147.9
69.0	3.879	3.007	40.4	12.487	-4.229	2.813	150.3
70.5	3.917	3.045	40.6	12.481	-4.247	2.837	152.6
72.0	3.955	3.083	40.9	12.475	-4.264	2.859	154.8
73.5	3.993	3.121	41.1	12.470	-4.281	2.882	157.1
75.0	4.030	3.159	41.3	12.464	-4.298	2.905	159.2
76.5	4.068	3.196	41.5	12.459	-4.314	2.927	161.3
78.0	4.105	3.233	41.7	12.454	-4.330	2.950	163.3
79.5	4.142	3.270	41.9	12.449	-4.346	2.972	165.3
81.0	4.179	3.307	42.1	12.444	-4.361	2.994	167.3
82.5	4.215	3.344	42.3	12.439	-4.376	3.016	169.1
84.0	4.252	3.380	42.4	12.435	-4.391	3.037	170.9
85.5	4.288	3.416	42.6	12.430	-4.406	3.059	172.7
87.0	4.325	3.453	42.8	12.426	-4.420	3.080	174.4
88.5	4.361	3.489	42.9	12.422	-4.434	3.102	176.0
90.1	4.400	3.528	43.1	12.417	-4.448	3.125	177.6

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TANK NO. 027 : WING TANK SB : NET VOLUME= 90.14 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	0.872	0.000	0.0	0.000	0.000	0.000	0.0
1.5	1.360	0.488	7.2	13.655	2.862	1.213	6.3
3.0	1.527	0.655	13.3	13.478	2.926	1.333	14.2
4.5	1.635	0.763	15.4	13.421	2.926	1.416	18.6
6.0	1.730	0.858	17.3	13.353	2.949	1.483	22.7
7.5	1.815	0.943	18.9	13.290	2.981	1.541	26.8
9.0	1.894	1.022	20.3	13.235	3.016	1.593	30.6
10.5	1.968	1.096	21.6	13.187	3.053	1.642	34.4
12.0	2.038	1.166	22.7	13.144	3.089	1.687	38.2
13.5	2.104	1.232	23.8	13.106	3.126	1.729	42.0
15.0	2.168	1.296	24.8	13.073	3.162	1.770	45.8
16.5	2.233	1.361	22.9	13.033	3.206	1.809	41.9
18.0	2.299	1.427	23.9	12.988	3.256	1.847	45.5
19.5	2.363	1.491	24.8	12.949	3.304	1.884	49.1
21.0	2.424	1.552	25.7	12.914	3.349	1.921	52.6
22.5	2.483	1.611	26.5	12.883	3.393	1.956	56.2
24.0	2.541	1.669	27.3	12.854	3.434	1.991	59.7
25.5	2.597	1.725	28.0	12.828	3.474	2.025	63.2
27.0	2.651	1.779	28.7	12.804	3.512	2.058	66.7
28.5	2.704	1.832	29.4	12.782	3.549	2.091	70.2
30.0	2.756	1.884	30.1	12.761	3.584	2.123	73.6
31.5	2.807	1.935	30.7	12.742	3.619	2.154	77.1
33.0	2.857	1.985	31.3	12.725	3.652	2.185	80.5
34.5	2.906	2.034	31.8	12.708	3.684	2.215	83.9
36.0	2.954	2.082	32.4	12.692	3.715	2.245	87.2
37.5	3.002	2.130	32.9	12.678	3.745	2.274	90.5
39.0	3.048	2.176	33.4	12.664	3.775	2.303	93.8
40.5	3.094	2.222	33.9	12.650	3.803	2.332	97.1
42.0	3.139	2.267	34.4	12.638	3.831	2.360	100.3
43.5	3.184	2.312	34.8	12.626	3.858	2.387	103.5
45.0	3.228	2.356	35.3	12.615	3.884	2.415	106.6
46.5	3.272	2.400	35.7	12.604	3.910	2.442	109.7
48.0	3.315	2.443	36.1	12.594	3.935	2.468	112.7
49.5	3.358	2.486	36.5	12.584	3.959	2.495	115.7
51.0	3.400	2.528	36.8	12.575	3.983	2.521	118.6
52.5	3.442	2.570	37.2	12.566	4.006	2.546	121.5
54.0	3.483	2.611	37.5	12.557	4.029	2.572	124.4
55.5	3.524	2.652	37.9	12.549	4.051	2.597	127.2
57.0	3.565	2.693	38.2	12.541	4.073	2.622	130.0
58.5	3.605	2.733	38.5	12.533	4.094	2.647	132.7
60.0	3.645	2.773	38.8	12.526	4.115	2.671	135.4
61.5	3.685	2.813	39.1	12.519	4.135	2.695	138.0
63.0	3.724	2.852	39.4	12.512	4.154	2.719	140.5
64.5	3.763	2.891	39.6	12.505	4.174	2.743	143.1
66.0	3.802	2.930	39.9	12.499	4.193	2.767	145.5
67.5	3.841	2.969	40.2	12.493	4.211	2.790	147.9
69.0	3.879	3.007	40.4	12.487	4.229	2.813	150.3
70.5	3.917	3.045	40.6	12.481	4.247	2.837	152.6
72.0	3.955	3.083	40.9	12.475	4.264	2.859	154.8
73.5	3.993	3.121	41.1	12.470	4.281	2.882	157.1
75.0	4.030	3.159	41.3	12.464	4.298	2.905	159.2
76.5	4.068	3.196	41.5	12.459	4.314	2.927	161.3
78.0	4.105	3.233	41.7	12.454	4.330	2.950	163.3
79.5	4.142	3.270	41.9	12.449	4.346	2.972	165.3
81.0	4.179	3.307	42.1	12.444	4.361	2.994	167.3
82.5	4.215	3.344	42.3	12.439	4.376	3.016	169.1
84.0	4.252	3.380	42.4	12.435	4.391	3.037	170.9
85.5	4.288	3.416	42.6	12.430	4.406	3.059	172.7
87.0	4.325	3.453	42.8	12.426	4.420	3.080	174.4
88.5	4.361	3.489	42.9	12.422	4.434	3.102	176.0
90.1	4.400	3.528	43.1	12.417	4.448	3.125	177.6

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TANK NO. 029 : WING TANK PS : NET VOLUME= 76.16 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.601	0.000	0.0	0.000	0.000	0.000	0.0
1.3	2.371	0.771	5.1	8.114	-2.047	2.175	1.8
2.6	2.580	0.979	7.9	7.847	-2.201	2.329	4.3
3.9	2.727	1.126	10.4	7.655	-2.305	2.438	7.2
5.2	2.844	1.243	12.5	7.499	-2.386	2.525	10.2
6.5	2.943	1.342	14.5	7.366	-2.454	2.599	13.5
7.8	3.030	1.429	16.3	7.249	-2.512	2.664	16.8
9.1	3.109	1.508	17.9	7.148	-2.566	2.722	20.0
10.4	3.181	1.580	19.3	7.062	-2.616	2.775	23.2
11.7	3.248	1.647	20.6	6.987	-2.664	2.824	26.5
13.0	3.311	1.710	21.9	6.921	-2.709	2.869	29.8
14.3	3.371	1.770	23.0	6.863	-2.752	2.912	33.1
15.6	3.428	1.827	24.1	6.810	-2.793	2.953	36.4
16.9	3.482	1.881	25.1	6.764	-2.833	2.991	39.8
18.2	3.535	1.934	26.1	6.721	-2.872	3.028	43.2
19.5	3.585	1.984	27.0	6.683	-2.909	3.064	46.6
20.8	3.634	2.033	27.9	6.647	-2.945	3.098	49.8
22.1	3.681	2.081	28.6	6.614	-2.980	3.131	52.8
23.4	3.728	2.127	29.4	6.584	-3.014	3.163	55.7
24.7	3.773	2.172	30.1	6.555	-3.046	3.194	58.7
26.0	3.817	2.216	30.7	6.528	-3.077	3.224	60.2
27.3	3.860	2.259	31.2	6.502	-3.106	3.253	61.8
28.6	3.902	2.302	31.7	6.477	-3.134	3.282	63.5
29.9	3.944	2.344	32.2	6.454	-3.161	3.309	65.1
31.2	3.986	2.385	32.6	6.431	-3.186	3.337	66.4
32.5	4.027	2.426	33.0	6.409	-3.210	3.364	67.5
33.8	4.067	2.466	33.4	6.388	-3.234	3.390	68.8
35.1	4.107	2.506	33.7	6.367	-3.256	3.416	70.1
36.4	4.146	2.546	34.1	6.347	-3.277	3.441	71.5
37.7	4.186	2.585	34.4	6.328	-3.297	3.466	73.0
39.0	4.224	2.624	34.7	6.310	-3.317	3.491	74.5
40.3	4.263	2.662	35.1	6.292	-3.336	3.515	76.1
41.6	4.301	2.700	35.4	6.275	-3.355	3.539	77.8
42.9	4.338	2.738	35.7	6.258	-3.373	3.563	79.5
44.2	4.376	2.775	36.0	6.241	-3.391	3.586	81.3
45.5	4.427	2.826	17.2	6.208	-3.408	3.609	36.6
46.8	4.509	2.908	15.5	6.142	-3.433	3.633	29.1
48.1	4.604	3.003	12.6	6.078	-3.469	3.658	17.8
49.4	4.733	3.132	8.2	6.016	-3.523	3.684	6.1
50.7	4.970	3.369	4.9	5.960	-3.601	3.714	1.0
52.0	5.211	3.610	6.2	5.911	-3.689	3.748	1.8
53.3	5.410	3.809	7.2	5.863	-3.776	3.787	2.6
54.6	5.588	3.987	7.8	5.817	-3.862	3.827	3.4
55.9	5.754	4.153	8.3	5.772	-3.946	3.870	4.0
57.2	5.911	4.310	8.7	5.729	-4.027	3.915	4.5
58.5	6.062	4.462	9.0	5.688	-4.106	3.961	4.9
59.8	6.210	4.609	9.2	5.649	-4.182	4.008	5.3
61.1	6.355	4.754	9.3	5.611	-4.255	4.057	5.5
62.4	6.498	4.897	9.4	5.574	-4.325	4.106	5.6
63.7	6.640	5.039	9.4	5.539	-4.393	4.156	5.7
65.0	6.782	5.181	9.4	5.505	-4.458	4.207	5.7
66.3	6.924	5.323	9.4	5.473	-4.521	4.259	5.7
67.6	7.066	5.465	9.4	5.442	-4.581	4.312	5.7
68.9	7.208	5.607	9.4	5.412	-4.639	4.365	5.7
70.2	7.349	5.749	9.4	5.383	-4.695	4.419	5.7
71.5	7.491	5.890	9.4	5.355	-4.748	4.474	5.7
72.8	7.633	6.032	9.4	5.328	-4.800	4.529	5.7
74.1	7.775	6.174	9.4	5.302	-4.850	4.585	5.7
75.4	7.917	6.316	9.4	5.277	-4.898	4.641	5.7
76.2	8.000	6.399	9.4	5.263	-4.926	4.674	5.7

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TANK NO. 030 : WING TANK SB : NET VOLUME= 76.16 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.601	0.000	0.0	0.000	0.000	0.000	0.0
1.3	2.371	0.771	5.1	8.114	2.047	2.175	1.8
2.6	2.580	0.979	7.9	7.847	2.201	2.329	4.3
3.9	2.727	1.126	10.4	7.655	2.305	2.438	7.2
5.2	2.844	1.243	12.5	7.499	2.386	2.525	10.2
6.5	2.943	1.342	14.5	7.366	2.454	2.599	13.5
7.8	3.030	1.429	16.3	7.249	2.512	2.664	16.8
9.1	3.109	1.508	17.9	7.148	2.566	2.722	20.0
10.4	3.181	1.580	19.3	7.062	2.616	2.775	23.2
11.7	3.248	1.647	20.6	6.987	2.664	2.824	26.5
13.0	3.311	1.710	21.9	6.921	2.709	2.869	29.8
14.3	3.371	1.770	23.0	6.863	2.752	2.912	33.1
15.6	3.428	1.827	24.1	6.810	2.793	2.953	36.4
16.9	3.482	1.881	25.1	6.764	2.833	2.991	39.8
18.2	3.535	1.934	26.1	6.721	2.872	3.028	43.2
19.5	3.585	1.984	27.0	6.683	2.909	3.064	46.6
20.8	3.634	2.033	27.9	6.647	2.945	3.098	49.8
22.1	3.681	2.081	28.6	6.614	2.980	3.131	52.8
23.4	3.728	2.127	29.4	6.584	3.014	3.163	55.7
24.7	3.773	2.172	30.1	6.555	3.046	3.194	58.7
26.0	3.817	2.216	30.7	6.528	3.077	3.224	60.2
27.3	3.860	2.259	31.2	6.502	3.106	3.253	61.8
28.6	3.902	2.302	31.7	6.477	3.134	3.282	63.5
29.9	3.944	2.344	32.2	6.454	3.161	3.309	65.1
31.2	3.986	2.385	32.6	6.431	3.186	3.337	66.4
32.5	4.027	2.426	33.0	6.409	3.210	3.364	67.5
33.8	4.067	2.466	33.4	6.388	3.234	3.390	68.8
35.1	4.107	2.506	33.7	6.367	3.256	3.416	70.1
36.4	4.146	2.546	34.1	6.347	3.277	3.441	71.5
37.7	4.186	2.585	34.4	6.328	3.297	3.466	73.0
39.0	4.224	2.624	34.7	6.310	3.317	3.491	74.5
40.3	4.263	2.662	35.1	6.292	3.336	3.515	76.1
41.6	4.301	2.700	35.4	6.275	3.355	3.539	77.8
42.9	4.338	2.738	35.7	6.258	3.373	3.563	79.5
44.2	4.376	2.775	36.0	6.241	3.391	3.586	81.3
45.5	4.427	2.826	17.2	6.208	3.408	3.609	36.6
46.8	4.509	2.908	15.5	6.142	3.433	3.633	29.1
48.1	4.604	3.003	12.6	6.078	3.469	3.658	17.8
49.4	4.733	3.132	8.2	6.016	3.523	3.684	6.1
50.7	4.970	3.369	4.9	5.960	3.601	3.714	1.0
52.0	5.211	3.610	6.2	5.911	3.689	3.748	1.8
53.3	5.410	3.809	7.2	5.863	3.776	3.787	2.6
54.6	5.588	3.987	7.8	5.817	3.862	3.827	3.4
55.9	5.754	4.153	8.3	5.772	3.946	3.870	4.0
57.2	5.911	4.310	8.7	5.729	4.027	3.915	4.5
58.5	6.062	4.462	9.0	5.688	4.106	3.961	4.9
59.8	6.210	4.609	9.2	5.649	4.182	4.008	5.3
61.1	6.355	4.754	9.3	5.611	4.255	4.057	5.5
62.4	6.498	4.897	9.4	5.574	4.325	4.106	5.6
63.7	6.640	5.039	9.4	5.539	4.393	4.156	5.7
65.0	6.782	5.181	9.4	5.505	4.458	4.207	5.7
66.3	6.924	5.323	9.4	5.473	4.521	4.259	5.7
67.6	7.066	5.465	9.4	5.442	4.581	4.312	5.7
68.9	7.208	5.607	9.4	5.412	4.639	4.365	5.7
70.2	7.349	5.749	9.4	5.383	4.695	4.419	5.7
71.5	7.491	5.890	9.4	5.355	4.748	4.474	5.7
72.8	7.633	6.032	9.4	5.328	4.800	4.529	5.7
74.1	7.775	6.174	9.4	5.302	4.850	4.585	5.7
75.4	7.917	6.316	9.4	5.277	4.898	4.641	5.7
76.2	8.000	6.399	9.4	5.263	4.926	4.674	5.7

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TANK NO. 031 : WING TANK PS : NET VOLUME= 26.52 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	2.170	0.000	0.0	0.000	0.000	0.000	0.0
0.5	3.058	0.888	2.8	1.350	-0.459	2.917	0.5
1.0	3.201	1.031	4.5	1.145	-0.580	3.026	1.3
1.5	3.301	1.130	5.9	1.000	-0.663	3.101	2.2
2.0	3.380	1.209	7.2	0.883	-0.729	3.161	3.2
2.5	3.446	1.276	8.3	0.781	-0.783	3.212	4.1
3.0	3.505	1.335	9.3	0.694	-0.828	3.256	4.9
3.5	3.558	1.388	10.2	0.610	-0.866	3.296	5.8
4.0	3.607	1.437	11.1	0.533	-0.899	3.331	6.6
4.5	3.652	1.482	11.9	0.463	-0.930	3.365	7.5
5.0	3.694	1.524	12.6	0.401	-0.960	3.395	8.4
5.5	3.734	1.563	13.3	0.345	-0.989	3.424	9.4
6.0	3.772	1.601	13.9	0.294	-1.016	3.452	10.3
6.5	3.808	1.638	14.5	0.248	-1.043	3.478	11.0
7.0	3.843	1.673	14.9	0.205	-1.068	3.503	11.5
7.5	3.877	1.707	15.3	0.165	-1.091	3.526	11.9
8.0	3.910	1.740	15.7	0.127	-1.113	3.549	12.3
8.5	3.943	1.773	16.0	0.092	-1.133	3.572	12.7
9.0	3.975	1.804	16.4	0.059	-1.153	3.593	13.2
9.5	4.006	1.836	16.7	0.028	-1.172	3.614	13.7
10.0	4.036	1.866	17.0	-0.002	-1.190	3.634	14.2
10.5	4.067	1.896	17.3	-0.030	-1.207	3.654	14.7
11.0	4.096	1.926	17.5	-0.057	-1.223	3.674	15.2
11.5	4.125	1.955	17.8	-0.083	-1.239	3.693	15.8
12.0	4.154	1.984	18.0	-0.107	-1.255	3.711	16.3
12.5	4.183	2.012	18.2	-0.130	-1.270	3.730	16.8
13.0	4.211	2.041	18.4	-0.153	-1.284	3.748	17.2
13.5	4.239	2.068	18.6	-0.174	-1.298	3.765	17.6
14.0	4.266	2.096	18.7	-0.194	-1.312	3.783	18.0
14.5	4.294	2.124	18.8	-0.214	-1.325	3.800	18.4
15.0	4.321	2.151	19.0	-0.233	-1.337	3.817	18.9
15.5	4.348	2.178	19.1	-0.251	-1.349	3.833	19.4
16.0	4.375	2.205	19.3	-0.268	-1.361	3.850	20.0
16.5	4.402	2.231	19.4	-0.285	-1.373	3.866	20.5
17.0	4.428	2.258	19.5	-0.301	-1.384	3.882	21.2
17.5	4.455	2.284	19.7	-0.316	-1.395	3.898	21.8
18.0	4.481	2.310	19.8	-0.331	-1.406	3.914	22.5
18.5	4.507	2.336	19.9	-0.346	-1.417	3.930	23.3
19.0	4.532	2.362	20.0	-0.360	-1.427	3.945	24.1
19.5	4.558	2.388	20.2	-0.374	-1.438	3.961	24.9
20.0	4.583	2.413	20.3	-0.387	-1.448	3.976	25.8
20.5	4.609	2.439	20.4	-0.400	-1.458	3.991	26.7
21.0	4.634	2.464	20.5	-0.413	-1.468	4.006	27.6
21.5	4.659	2.489	20.7	-0.425	-1.478	4.021	28.6
22.0	4.684	2.514	20.8	-0.438	-1.488	4.036	29.6
22.5	4.709	2.538	20.7	-0.449	-1.498	4.050	30.5
23.0	4.734	2.563	20.7	-0.462	-1.508	4.065	31.4
23.5	4.759	2.588	20.6	-0.474	-1.518	4.079	32.4
24.0	4.784	2.613	20.5	-0.487	-1.528	4.094	33.3
24.5	4.831	2.661	5.8	-0.506	-1.551	4.108	3.3
25.0	4.918	2.747	6.0	-0.534	-1.595	4.124	4.4
25.5	5.007	2.837	5.5	-0.568	-1.642	4.140	4.9
26.0	5.114	2.943	4.1	-0.609	-1.693	4.158	3.0
26.5	5.393	3.222	0.0	-0.653	-1.757	4.178	0.0

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TANK NO. 032 : WING TANK SB : NET VOLUME= 26.52 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	2.170	0.000	0.0	0.000	0.000	0.000	0.0
0.5	3.058	0.888	2.8	1.350	0.459	2.917	0.5
1.0	3.201	1.031	4.5	1.145	0.580	3.026	1.3
1.5	3.301	1.130	5.9	1.000	0.663	3.101	2.2
2.0	3.380	1.209	7.2	0.883	0.729	3.161	3.2
2.5	3.446	1.276	8.3	0.781	0.783	3.212	4.1
3.0	3.505	1.335	9.3	0.694	0.828	3.256	4.9
3.5	3.558	1.388	10.2	0.610	0.866	3.296	5.8
4.0	3.607	1.437	11.1	0.533	0.899	3.331	6.6
4.5	3.652	1.482	11.9	0.463	0.930	3.365	7.5
5.0	3.694	1.524	12.6	0.401	0.960	3.395	8.4
5.5	3.734	1.563	13.3	0.345	0.989	3.424	9.4
6.0	3.772	1.601	13.9	0.294	1.016	3.452	10.3
6.5	3.808	1.638	14.5	0.248	1.043	3.478	11.0
7.0	3.843	1.673	14.9	0.205	1.068	3.503	11.5
7.5	3.877	1.707	15.3	0.165	1.091	3.526	11.9
8.0	3.910	1.740	15.7	0.127	1.113	3.549	12.3
8.5	3.943	1.773	16.0	0.092	1.133	3.572	12.7
9.0	3.975	1.804	16.4	0.059	1.153	3.593	13.2
9.5	4.006	1.836	16.7	0.028	1.172	3.614	13.7
10.0	4.036	1.866	17.0	-0.002	1.190	3.634	14.2
10.5	4.067	1.896	17.3	-0.030	1.207	3.654	14.7
11.0	4.096	1.926	17.5	-0.057	1.223	3.674	15.2
11.5	4.125	1.955	17.8	-0.083	1.239	3.693	15.8
12.0	4.154	1.984	18.0	-0.107	1.255	3.711	16.3
12.5	4.183	2.012	18.2	-0.130	1.270	3.730	16.8
13.0	4.211	2.041	18.4	-0.153	1.284	3.748	17.2
13.5	4.239	2.068	18.6	-0.174	1.298	3.765	17.6
14.0	4.266	2.096	18.7	-0.194	1.312	3.783	18.0
14.5	4.294	2.124	18.8	-0.214	1.325	3.800	18.4
15.0	4.321	2.151	19.0	-0.233	1.337	3.817	18.9
15.5	4.348	2.178	19.1	-0.251	1.349	3.833	19.4
16.0	4.375	2.205	19.3	-0.268	1.361	3.850	20.0
16.5	4.402	2.231	19.4	-0.285	1.373	3.866	20.5
17.0	4.428	2.258	19.5	-0.301	1.384	3.882	21.2
17.5	4.455	2.284	19.7	-0.316	1.395	3.898	21.8
18.0	4.481	2.310	19.8	-0.331	1.406	3.914	22.5
18.5	4.507	2.336	19.9	-0.346	1.417	3.930	23.3
19.0	4.532	2.362	20.0	-0.360	1.427	3.945	24.1
19.5	4.558	2.388	20.2	-0.374	1.438	3.961	24.9
20.0	4.583	2.413	20.3	-0.387	1.448	3.976	25.8
20.5	4.609	2.439	20.4	-0.400	1.458	3.991	26.7
21.0	4.634	2.464	20.5	-0.413	1.468	4.006	27.6
21.5	4.659	2.489	20.7	-0.425	1.478	4.021	28.6
22.0	4.684	2.514	20.8	-0.438	1.488	4.036	29.6
22.5	4.709	2.538	20.7	-0.449	1.498	4.050	30.5
23.0	4.734	2.563	20.7	-0.462	1.508	4.065	31.4
23.5	4.759	2.588	20.6	-0.474	1.518	4.079	32.4
24.0	4.784	2.613	20.5	-0.487	1.528	4.094	33.3
24.5	4.831	2.661	5.8	-0.506	1.551	4.108	3.3
25.0	4.918	2.747	6.0	-0.534	1.595	4.124	4.4
25.5	5.007	2.837	5.5	-0.568	1.642	4.140	4.9
26.0	5.114	2.943	4.1	-0.609	1.693	4.158	3.0
26.5	5.393	3.222	0.0	-0.653	1.757	4.178	0.0

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TANK NO. 033 : WING TANK PS : NET VOLUME= 59.08 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	3.554	0.000	0.0	0.000	0.000	0.000	0.0
1.0	4.237	0.683	3.6	-4.067	-0.907	4.033	2.1
2.0	4.471	0.917	5.3	-4.129	-1.232	4.196	5.6
3.0	4.643	1.089	6.8	-4.171	-1.467	4.316	10.0
4.0	4.785	1.231	8.0	-4.202	-1.656	4.413	15.0
5.0	4.935	1.382	7.4	-4.256	-1.880	4.500	16.9
6.0	5.062	1.508	8.7	-4.302	-2.085	4.585	22.7
7.0	5.172	1.618	9.7	-4.340	-2.261	4.664	28.7
8.0	5.272	1.718	10.7	-4.371	-2.413	4.734	34.9
9.0	5.364	1.810	11.6	-4.397	-2.546	4.798	41.0
10.0	5.450	1.896	12.5	-4.420	-2.665	4.859	47.2
11.0	5.530	1.976	13.3	-4.441	-2.772	4.916	53.3
12.0	5.605	2.051	14.0	-4.460	-2.871	4.970	59.3
13.0	5.677	2.123	14.7	-4.478	-2.961	5.022	64.7
14.0	5.747	2.193	15.0	-4.495	-3.041	5.071	67.1
15.0	5.814	2.261	15.4	-4.511	-3.110	5.118	68.9
16.0	5.881	2.327	15.7	-4.526	-3.171	5.163	70.9
17.0	5.946	2.392	16.0	-4.541	-3.225	5.206	72.9
18.0	6.010	2.456	16.4	-4.556	-3.273	5.249	75.0
19.0	6.072	2.518	16.7	-4.570	-3.317	5.291	77.2
20.0	6.133	2.579	17.0	-4.584	-3.357	5.332	79.4
21.0	6.193	2.639	17.3	-4.598	-3.394	5.372	81.6
22.0	6.252	2.698	17.7	-4.611	-3.428	5.411	83.9
23.0	6.309	2.756	18.0	-4.624	-3.459	5.449	86.3
24.0	6.366	2.812	18.3	-4.637	-3.488	5.486	88.6
25.0	6.422	2.868	18.6	-4.649	-3.515	5.522	90.9
26.0	6.477	2.923	18.9	-4.662	-3.540	5.558	93.2
27.0	6.531	2.977	19.2	-4.674	-3.563	5.593	95.5
28.0	6.584	3.031	19.5	-4.686	-3.586	5.627	97.7
29.0	6.637	3.083	19.8	-4.697	-3.607	5.661	99.9
30.0	6.688	3.135	20.1	-4.709	-3.626	5.695	102.1
31.0	6.739	3.186	20.4	-4.720	-3.645	5.727	104.1
32.0	6.790	3.236	20.6	-4.731	-3.663	5.760	106.0
33.0	6.839	3.286	20.9	-4.742	-3.680	5.792	107.8
34.0	6.888	3.335	21.2	-4.753	-3.696	5.823	109.6
35.0	6.937	3.383	21.4	-4.763	-3.711	5.855	111.3
36.0	6.985	3.431	21.7	-4.774	-3.725	5.885	112.9
37.0	7.032	3.478	21.9	-4.784	-3.739	5.916	114.4
38.0	7.079	3.525	22.1	-4.794	-3.751	5.946	115.8
39.0	7.125	3.571	22.4	-4.804	-3.764	5.975	117.2
40.0	7.171	3.617	22.6	-4.814	-3.775	6.004	118.6
41.0	7.216	3.663	22.8	-4.823	-3.786	6.033	119.9
42.0	7.261	3.708	23.0	-4.833	-3.797	6.062	121.2
43.0	7.306	3.752	23.3	-4.842	-3.807	6.091	122.5
44.0	7.350	3.796	23.5	-4.852	-3.816	6.119	123.7
45.0	7.394	3.840	23.7	-4.861	-3.826	6.147	124.9
46.0	7.437	3.883	23.9	-4.870	-3.834	6.174	126.0
47.0	7.480	3.927	24.0	-4.879	-3.843	6.202	126.4
48.0	7.523	3.970	24.0	-4.887	-3.851	6.229	126.4
49.0	7.566	4.013	24.0	-4.895	-3.858	6.256	126.4
50.0	7.609	4.056	24.0	-4.903	-3.865	6.282	126.4
51.0	7.652	4.099	24.0	-4.911	-3.872	6.309	126.4
52.0	7.695	4.142	24.0	-4.918	-3.879	6.335	126.4
53.0	7.738	4.185	24.0	-4.925	-3.886	6.361	126.4
54.0	7.781	4.228	24.0	-4.931	-3.892	6.387	126.4
55.0	7.824	4.271	24.0	-4.938	-3.898	6.413	126.4
56.0	7.868	4.314	24.0	-4.944	-3.904	6.438	126.4
57.0	7.911	4.357	24.0	-4.950	-3.909	6.464	126.4
58.0	7.954	4.400	24.0	-4.956	-3.915	6.489	126.4
59.1	8.000	4.446	24.0	-4.962	-3.920	6.516	126.4

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TANK NO. 034 : WING TANK SB : NET VOLUME= 59.08 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	3.554	0.000	0.0	0.000	0.000	0.000	0.0
1.0	4.237	0.683	3.6	-4.067	0.907	4.033	2.1
2.0	4.471	0.917	5.3	-4.129	1.232	4.196	5.6
3.0	4.643	1.089	6.8	-4.171	1.467	4.316	10.0
4.0	4.785	1.231	8.0	-4.202	1.656	4.413	15.0
5.0	4.935	1.382	7.4	-4.256	1.880	4.500	16.9
6.0	5.062	1.508	8.7	-4.302	2.085	4.585	22.7
7.0	5.172	1.618	9.7	-4.340	2.261	4.664	28.7
8.0	5.272	1.718	10.7	-4.371	2.413	4.734	34.9
9.0	5.364	1.810	11.6	-4.397	2.546	4.798	41.0
10.0	5.450	1.896	12.5	-4.420	2.665	4.859	47.2
11.0	5.530	1.976	13.3	-4.441	2.772	4.916	53.3
12.0	5.605	2.051	14.0	-4.460	2.871	4.970	59.3
13.0	5.677	2.123	14.7	-4.478	2.961	5.022	64.7
14.0	5.747	2.193	15.0	-4.495	3.041	5.071	67.1
15.0	5.814	2.261	15.4	-4.511	3.110	5.118	68.9
16.0	5.881	2.327	15.7	-4.526	3.171	5.163	70.9
17.0	5.946	2.392	16.0	-4.541	3.225	5.206	72.9
18.0	6.010	2.456	16.4	-4.556	3.273	5.249	75.0
19.0	6.072	2.518	16.7	-4.570	3.317	5.291	77.2
20.0	6.133	2.579	17.0	-4.584	3.357	5.332	79.4
21.0	6.193	2.639	17.3	-4.598	3.394	5.372	81.6
22.0	6.252	2.698	17.7	-4.611	3.428	5.411	83.9
23.0	6.309	2.756	18.0	-4.624	3.459	5.449	86.3
24.0	6.366	2.812	18.3	-4.637	3.488	5.486	88.6
25.0	6.422	2.868	18.6	-4.649	3.515	5.522	90.9
26.0	6.477	2.923	18.9	-4.662	3.540	5.558	93.2
27.0	6.531	2.977	19.2	-4.674	3.563	5.593	95.5
28.0	6.584	3.031	19.5	-4.686	3.586	5.627	97.7
29.0	6.637	3.083	19.8	-4.697	3.607	5.661	99.9
30.0	6.688	3.135	20.1	-4.709	3.626	5.695	102.1
31.0	6.739	3.186	20.4	-4.720	3.645	5.727	104.1
32.0	6.790	3.236	20.6	-4.731	3.663	5.760	106.0
33.0	6.839	3.286	20.9	-4.742	3.680	5.792	107.8
34.0	6.888	3.335	21.2	-4.753	3.696	5.823	109.6
35.0	6.937	3.383	21.4	-4.763	3.711	5.855	111.3
36.0	6.985	3.431	21.7	-4.774	3.725	5.885	112.9
37.0	7.032	3.478	21.9	-4.784	3.739	5.916	114.4
38.0	7.079	3.525	22.1	-4.794	3.751	5.946	115.8
39.0	7.125	3.571	22.4	-4.804	3.764	5.975	117.2
40.0	7.171	3.617	22.6	-4.814	3.775	6.004	118.6
41.0	7.216	3.663	22.8	-4.823	3.786	6.033	119.9
42.0	7.261	3.708	23.0	-4.833	3.797	6.062	121.2
43.0	7.306	3.752	23.3	-4.842	3.807	6.091	122.5
44.0	7.350	3.796	23.5	-4.852	3.816	6.119	123.7
45.0	7.394	3.840	23.7	-4.861	3.826	6.147	124.9
46.0	7.437	3.883	23.9	-4.870	3.834	6.174	126.0
47.0	7.480	3.927	24.0	-4.879	3.843	6.202	126.4
48.0	7.523	3.970	24.0	-4.887	3.851	6.229	126.4
49.0	7.566	4.013	24.0	-4.895	3.858	6.256	126.4
50.0	7.609	4.056	24.0	-4.903	3.865	6.282	126.4
51.0	7.652	4.099	24.0	-4.911	3.872	6.309	126.4
52.0	7.695	4.142	24.0	-4.918	3.879	6.335	126.4
53.0	7.738	4.185	24.0	-4.925	3.886	6.361	126.4
54.0	7.781	4.228	24.0	-4.931	3.892	6.387	126.4
55.0	7.824	4.271	24.0	-4.938	3.898	6.413	126.4
56.0	7.868	4.314	24.0	-4.944	3.904	6.438	126.4
57.0	7.911	4.357	24.0	-4.950	3.909	6.464	126.4
58.0	7.954	4.400	24.0	-4.956	3.915	6.489	126.4
59.1	8.000	4.446	24.0	-4.962	3.920	6.516	126.4

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TANK NO. 035 : WING TANK PS : NET VOLUME= 23.12 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	4.593	0.000	0.0	0.000	0.000	0.000	0.0
0.4	5.102	0.509	2.5	1.079	-6.602	4.976	0.2
0.8	5.238	0.645	3.6	0.862	-6.688	5.076	0.4
1.2	5.342	0.749	4.4	0.759	-6.757	5.148	0.7
1.6	5.429	0.836	5.0	0.696	-6.815	5.207	1.0
2.0	5.507	0.914	5.5	0.652	-6.866	5.260	1.2
2.4	5.579	0.986	6.0	0.619	-6.910	5.307	1.5
2.8	5.646	1.052	6.4	0.594	-6.950	5.351	1.8
3.2	5.709	1.115	6.7	0.573	-6.987	5.392	2.1
3.6	5.769	1.176	7.0	0.556	-7.020	5.430	2.4
4.0	5.827	1.233	7.3	0.541	-7.050	5.467	2.6
4.4	5.882	1.289	7.5	0.529	-7.078	5.502	2.9
4.8	5.937	1.344	7.7	0.518	-7.104	5.536	3.1
5.2	5.990	1.397	7.9	0.508	-7.128	5.569	3.4
5.6	6.041	1.448	8.0	0.499	-7.150	5.601	3.6
6.0	6.092	1.499	8.2	0.492	-7.171	5.632	3.8
6.4	6.142	1.549	8.3	0.485	-7.190	5.662	4.0
6.8	6.191	1.598	8.5	0.478	-7.208	5.692	4.1
7.2	6.240	1.647	8.6	0.473	-7.226	5.721	4.3
7.6	6.287	1.694	8.7	0.467	-7.242	5.750	4.5
8.0	6.335	1.741	8.8	0.463	-7.257	5.778	4.6
8.4	6.381	1.788	8.9	0.458	-7.272	5.805	4.8
8.8	6.428	1.834	9.0	0.454	-7.285	5.833	4.9
9.2	6.473	1.880	9.0	0.450	-7.298	5.859	5.0
9.6	6.519	1.926	9.1	0.447	-7.311	5.886	5.1
10.0	6.564	1.971	9.2	0.443	-7.322	5.912	5.2
10.4	6.609	2.016	9.2	0.440	-7.334	5.938	5.3
10.8	6.654	2.061	9.3	0.437	-7.344	5.964	5.4
11.2	6.698	2.105	9.3	0.434	-7.354	5.989	5.5
11.6	6.742	2.149	9.3	0.432	-7.364	6.014	5.5
12.0	6.787	2.193	9.4	0.429	-7.373	6.039	5.6
12.4	6.831	2.237	9.4	0.427	-7.381	6.064	5.6
12.8	6.874	2.281	9.4	0.424	-7.390	6.089	5.7
13.2	6.918	2.325	9.4	0.422	-7.397	6.113	5.7
13.6	6.962	2.369	9.4	0.420	-7.405	6.138	5.7
14.0	7.006	2.413	9.4	0.418	-7.412	6.162	5.7
14.4	7.049	2.456	9.4	0.416	-7.418	6.186	5.7
14.8	7.093	2.500	9.4	0.415	-7.425	6.210	5.7
15.2	7.137	2.543	9.4	0.413	-7.430	6.234	5.7
15.6	7.180	2.587	9.4	0.411	-7.436	6.257	5.7
16.0	7.224	2.631	9.4	0.410	-7.441	6.281	5.7
16.4	7.267	2.674	9.4	0.408	-7.447	6.304	5.7
16.8	7.311	2.718	9.4	0.407	-7.451	6.328	5.7
17.2	7.355	2.762	9.4	0.406	-7.456	6.351	5.7
17.6	7.398	2.805	9.4	0.404	-7.460	6.375	5.7
18.0	7.442	2.849	9.4	0.403	-7.465	6.398	5.7
18.4	7.486	2.893	9.4	0.402	-7.469	6.421	5.7
18.8	7.529	2.936	9.4	0.401	-7.472	6.444	5.7
19.2	7.573	2.980	9.4	0.400	-7.476	6.467	5.7
19.6	7.617	3.023	9.4	0.399	-7.480	6.490	5.7
20.0	7.660	3.067	9.4	0.398	-7.483	6.513	5.7
20.4	7.704	3.111	9.4	0.397	-7.486	6.536	5.7
20.8	7.747	3.154	9.4	0.396	-7.490	6.559	5.7
21.2	7.791	3.198	9.4	0.395	-7.493	6.582	5.7
21.6	7.835	3.242	9.4	0.394	-7.496	6.605	5.7
22.0	7.878	3.285	9.4	0.393	-7.498	6.627	5.7
22.4	7.922	3.329	9.4	0.393	-7.501	6.650	5.7
22.8	7.966	3.373	9.4	0.392	-7.504	6.673	5.7
23.1	8.000	3.407	9.4	0.391	-7.506	6.691	5.7

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TANK NO. 036 : WING TANK SB : NET VOLUME= 23.12 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	4.593	0.000	0.0	0.000	0.000	0.000	0.0
0.4	5.102	0.509	2.5	1.079	6.602	4.976	0.2
0.8	5.238	0.645	3.6	0.862	6.688	5.076	0.4
1.2	5.342	0.749	4.4	0.759	6.757	5.148	0.7
1.6	5.429	0.836	5.0	0.696	6.815	5.207	1.0
2.0	5.507	0.914	5.5	0.652	6.866	5.260	1.2
2.4	5.579	0.986	6.0	0.619	6.910	5.307	1.5
2.8	5.646	1.052	6.4	0.594	6.950	5.351	1.8
3.2	5.709	1.115	6.7	0.573	6.987	5.392	2.1
3.6	5.769	1.176	7.0	0.556	7.020	5.430	2.4
4.0	5.827	1.233	7.3	0.541	7.050	5.467	2.6
4.4	5.882	1.289	7.5	0.529	7.078	5.502	2.9
4.8	5.937	1.344	7.7	0.518	7.104	5.536	3.1
5.2	5.990	1.397	7.9	0.508	7.128	5.569	3.4
5.6	6.041	1.448	8.0	0.499	7.150	5.601	3.6
6.0	6.092	1.499	8.2	0.492	7.171	5.632	3.8
6.4	6.142	1.549	8.3	0.485	7.190	5.662	4.0
6.8	6.191	1.598	8.5	0.478	7.208	5.692	4.1
7.2	6.240	1.647	8.6	0.473	7.226	5.721	4.3
7.6	6.287	1.694	8.7	0.467	7.242	5.750	4.5
8.0	6.335	1.741	8.8	0.463	7.257	5.778	4.6
8.4	6.381	1.788	8.9	0.458	7.272	5.805	4.8
8.8	6.428	1.834	9.0	0.454	7.285	5.833	4.9
9.2	6.473	1.880	9.0	0.450	7.298	5.859	5.0
9.6	6.519	1.926	9.1	0.447	7.311	5.886	5.1
10.0	6.564	1.971	9.2	0.443	7.322	5.912	5.2
10.4	6.609	2.016	9.2	0.440	7.334	5.938	5.3
10.8	6.654	2.061	9.3	0.437	7.344	5.964	5.4
11.2	6.698	2.105	9.3	0.434	7.354	5.989	5.5
11.6	6.742	2.149	9.3	0.432	7.364	6.014	5.5
12.0	6.787	2.193	9.4	0.429	7.373	6.039	5.6
12.4	6.831	2.237	9.4	0.427	7.381	6.064	5.6
12.8	6.874	2.281	9.4	0.424	7.390	6.089	5.7
13.2	6.918	2.325	9.4	0.422	7.397	6.113	5.7
13.6	6.962	2.369	9.4	0.420	7.405	6.138	5.7
14.0	7.006	2.413	9.4	0.418	7.412	6.162	5.7
14.4	7.049	2.456	9.4	0.416	7.418	6.186	5.7
14.8	7.093	2.500	9.4	0.415	7.425	6.210	5.7
15.2	7.137	2.543	9.4	0.413	7.430	6.234	5.7
15.6	7.180	2.587	9.4	0.411	7.436	6.257	5.7
16.0	7.224	2.631	9.4	0.410	7.441	6.281	5.7
16.4	7.267	2.674	9.4	0.408	7.447	6.304	5.7
16.8	7.311	2.718	9.4	0.407	7.451	6.328	5.7
17.2	7.355	2.762	9.4	0.406	7.456	6.351	5.7
17.6	7.398	2.805	9.4	0.404	7.460	6.375	5.7
18.0	7.442	2.849	9.4	0.403	7.465	6.398	5.7
18.4	7.486	2.893	9.4	0.402	7.469	6.421	5.7
18.8	7.529	2.936	9.4	0.401	7.472	6.444	5.7
19.2	7.573	2.980	9.4	0.400	7.476	6.467	5.7
19.6	7.617	3.023	9.4	0.399	7.480	6.490	5.7
20.0	7.660	3.067	9.4	0.398	7.483	6.513	5.7
20.4	7.704	3.111	9.4	0.397	7.486	6.536	5.7
20.8	7.747	3.154	9.4	0.396	7.490	6.559	5.7
21.2	7.791	3.198	9.4	0.395	7.493	6.582	5.7
21.6	7.835	3.242	9.4	0.394	7.496	6.605	5.7
22.0	7.878	3.285	9.4	0.393	7.498	6.627	5.7
22.4	7.922	3.329	9.4	0.393	7.501	6.650	5.7
22.8	7.966	3.373	9.4	0.392	7.504	6.673	5.7
23.1	8.000	3.407	9.4	0.391	7.506	6.691	5.7

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TANK NO. 039 : STORAGE TANK PS : NET VOLUME= 79.40 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.285	0.000	0.0	0.000	0.000	0.000	0.0
1.3	1.583	0.298	7.5	46.286	-6.302	1.487	1.6
2.6	1.754	0.469	8.2	46.358	-6.333	1.579	2.0
3.9	1.913	0.628	8.6	46.385	-6.358	1.664	2.3
5.2	2.066	0.781	9.0	46.399	-6.378	1.745	2.5
6.5	2.212	0.927	9.3	46.407	-6.397	1.824	2.8
7.8	2.353	1.068	9.7	46.413	-6.415	1.901	3.1
9.1	2.489	1.204	10.0	46.417	-6.431	1.975	3.3
10.4	2.620	1.335	10.4	46.420	-6.447	2.047	3.6
11.7	2.748	1.463	10.7	46.422	-6.462	2.118	3.9
13.0	2.872	1.587	11.0	46.424	-6.477	2.187	4.2
14.3	2.992	1.707	11.3	46.426	-6.491	2.255	4.4
15.6	3.110	1.825	11.5	46.428	-6.505	2.322	4.7
16.9	3.225	1.940	11.8	46.430	-6.518	2.387	4.9
18.2	3.338	2.053	12.0	46.433	-6.531	2.451	5.1
19.5	3.449	2.164	12.2	46.436	-6.542	2.513	5.2
20.8	3.558	2.273	12.4	46.440	-6.554	2.575	5.4
22.1	3.666	2.381	12.5	46.445	-6.564	2.636	5.5
23.4	3.772	2.487	12.7	46.449	-6.574	2.696	5.7
24.7	3.876	2.591	12.9	46.454	-6.584	2.756	5.9
26.0	3.980	2.695	13.0	46.459	-6.593	2.814	6.0
27.3	4.082	2.797	13.1	46.464	-6.602	2.872	6.0
28.6	4.184	2.899	13.2	46.470	-6.610	2.930	6.1
29.9	4.284	2.999	13.4	46.475	-6.618	2.986	6.2
31.2	4.384	3.099	13.5	46.481	-6.626	3.043	6.3
32.5	4.484	3.199	13.6	46.487	-6.633	3.098	6.4
33.8	4.590	3.305	12.4	46.500	-6.640	3.153	6.1
35.1	4.698	3.413	12.5	46.514	-6.648	3.209	6.1
36.4	4.805	3.520	12.6	46.527	-6.655	3.264	6.2
37.7	4.911	3.626	12.7	46.540	-6.662	3.319	6.3
39.0	5.015	3.730	13.6	46.552	-6.668	3.374	6.7
40.3	5.113	3.828	13.7	46.559	-6.674	3.428	6.8
41.6	5.211	3.926	13.8	46.565	-6.679	3.482	6.9
42.9	5.308	4.023	13.9	46.572	-6.684	3.536	7.0
44.2	5.404	4.119	13.9	46.579	-6.689	3.590	7.0
45.5	5.501	4.216	13.9	46.585	-6.693	3.643	7.1
46.8	5.597	4.312	13.9	46.591	-6.698	3.696	7.1
48.1	5.693	4.408	13.9	46.597	-6.702	3.749	7.1
49.4	5.789	4.504	14.0	46.602	-6.706	3.801	7.1
50.7	5.885	4.600	14.0	46.608	-6.710	3.853	7.1
52.0	5.981	4.696	14.0	46.613	-6.713	3.905	7.2
53.3	6.077	4.792	14.0	46.618	-6.717	3.957	7.2
54.6	6.173	4.888	14.0	46.622	-6.720	4.009	7.2
55.9	6.269	4.984	14.0	46.627	-6.723	4.060	7.2
57.2	6.364	5.079	14.0	46.631	-6.726	4.111	7.2
58.5	6.460	5.175	14.0	46.635	-6.729	4.163	7.2
59.8	6.556	5.271	14.0	46.639	-6.732	4.214	7.2
61.1	6.652	5.367	14.0	46.643	-6.735	4.264	7.2
62.4	6.748	5.463	14.0	46.647	-6.737	4.315	7.2
63.7	6.843	5.558	14.0	46.650	-6.740	4.366	7.2
65.0	6.939	5.654	14.0	46.654	-6.742	4.416	7.2
66.3	7.035	5.750	14.0	46.657	-6.744	4.467	7.2
67.6	7.131	5.846	14.0	46.660	-6.746	4.517	7.2
68.9	7.226	5.941	14.0	46.663	-6.748	4.567	7.2
70.2	7.322	6.037	14.0	46.666	-6.750	4.617	7.2
71.5	7.418	6.133	14.0	46.668	-6.752	4.667	7.2
72.8	7.514	6.229	14.0	46.671	-6.754	4.717	7.2
74.1	7.609	6.324	14.0	46.674	-6.756	4.767	7.2
75.4	7.705	6.420	14.0	46.676	-6.758	4.817	7.2
76.7	7.801	6.516	14.0	46.679	-6.759	4.867	7.2
78.0	7.897	6.612	14.0	46.681	-6.761	4.917	7.2
79.4	8.000	6.715	14.0	46.683	-6.763	4.970	7.2

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TANK NO. 040 : STORAGE TANK SB : NET VOLUME= 63.89 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.283	0.000	0.0	0.000	0.000	0.000	0.0
1.1	1.582	0.299	6.3	45.821	6.357	1.486	1.4
2.2	1.754	0.471	6.9	45.876	6.390	1.578	1.8
3.3	1.916	0.633	7.2	45.895	6.417	1.664	2.0
4.4	2.071	0.788	7.5	45.903	6.439	1.746	2.2
5.5	2.220	0.937	7.8	45.907	6.458	1.826	2.4
6.6	2.363	1.080	8.0	45.908	6.477	1.904	2.7
7.7	2.502	1.219	8.3	45.908	6.495	1.980	2.9
8.8	2.636	1.353	8.6	45.908	6.512	2.053	3.2
9.9	2.767	1.483	8.8	45.907	6.528	2.125	3.4
11.0	2.893	1.610	9.1	45.906	6.543	2.196	3.6
12.1	3.017	1.734	9.3	45.905	6.559	2.265	3.9
13.2	3.138	1.854	9.5	45.904	6.573	2.333	4.1
14.3	3.256	1.972	9.7	45.904	6.587	2.399	4.3
15.4	3.372	2.089	9.8	45.904	6.600	2.464	4.4
16.5	3.486	2.203	10.0	45.905	6.612	2.529	4.5
17.6	3.599	2.316	10.1	45.907	6.624	2.592	4.7
18.7	3.711	2.427	10.2	45.909	6.634	2.655	4.8
19.8	3.821	2.538	10.3	45.912	6.645	2.716	4.9
20.9	3.931	2.647	10.4	45.915	6.654	2.777	4.9
22.0	4.039	2.756	10.5	45.918	6.663	2.838	5.0
23.1	4.147	2.863	10.6	45.921	6.672	2.898	5.1
24.2	4.254	2.970	10.6	45.924	6.680	2.957	5.2
25.3	4.360	3.077	10.7	45.928	6.687	3.015	5.2
26.4	4.466	3.182	10.8	45.931	6.694	3.074	5.3
27.5	4.571	3.287	10.8	45.935	6.701	3.131	5.4
28.6	4.675	3.392	10.9	45.939	6.708	3.189	5.4
29.7	4.779	3.496	10.9	45.943	6.714	3.246	5.5
30.8	4.883	3.600	10.9	45.946	6.720	3.302	5.5
31.9	4.987	3.704	10.9	45.949	6.725	3.359	5.5
33.0	5.091	3.808	10.9	45.953	6.730	3.415	5.5
34.1	5.195	3.911	10.9	45.956	6.735	3.471	5.5
35.2	5.298	4.015	10.9	45.959	6.740	3.526	5.5
36.3	5.402	4.119	10.9	45.961	6.744	3.581	5.5
37.4	5.505	4.222	10.9	45.964	6.748	3.636	5.5
38.5	5.609	4.326	10.9	45.967	6.752	3.691	5.5
39.6	5.713	4.429	10.9	45.969	6.756	3.746	5.5
40.7	5.816	4.533	10.9	45.971	6.759	3.801	5.5
41.8	5.920	4.636	10.9	45.973	6.762	3.855	5.5
42.9	6.023	4.740	10.9	45.975	6.765	3.909	5.5
44.0	6.127	4.844	10.9	45.977	6.768	3.963	5.5
45.1	6.231	4.947	10.9	45.979	6.771	4.017	5.5
46.2	6.334	5.051	10.9	45.980	6.774	4.071	5.5
47.3	6.438	5.154	10.9	45.982	6.776	4.125	5.5
48.4	6.541	5.258	10.9	45.984	6.778	4.179	5.5
49.5	6.645	5.361	10.9	45.985	6.781	4.233	5.5
50.6	6.748	5.465	10.9	45.987	6.783	4.286	5.5
51.7	6.852	5.569	10.9	45.988	6.785	4.340	5.5
52.8	6.956	5.672	10.9	45.989	6.787	4.393	5.5
53.9	7.059	5.776	10.9	45.990	6.789	4.446	5.5
55.0	7.163	5.879	10.9	45.992	6.791	4.500	5.5
56.1	7.266	5.983	10.9	45.993	6.793	4.553	5.5
57.2	7.370	6.087	10.9	45.994	6.794	4.606	5.5
58.3	7.473	6.190	10.9	45.995	6.796	4.659	5.5
59.4	7.577	6.294	10.9	45.996	6.798	4.712	5.5
60.5	7.681	6.397	10.9	45.997	6.799	4.765	5.5
61.6	7.784	6.501	10.9	45.998	6.801	4.818	5.5
62.7	7.888	6.604	10.9	45.999	6.802	4.871	5.5
63.9	8.000	6.717	10.9	46.000	6.803	4.929	5.5

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TANK NO. 041 : STORAGE TANK PS : NET VOLUME= 81.28 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.000	0.000	3.7	40.960	-0.525	1.001	0.3
1.4	1.315	0.315	5.6	41.407	-0.525	1.166	0.5
2.8	1.564	0.564	5.9	41.681	-0.525	1.304	0.5
4.2	1.810	0.810	5.9	41.788	-0.525	1.431	0.5
5.6	2.055	1.055	5.9	41.841	-0.525	1.557	0.5
7.0	2.301	1.301	5.9	41.873	-0.525	1.681	0.5
8.4	2.546	1.546	5.9	41.894	-0.525	1.805	0.5
9.8	2.792	1.792	5.9	41.909	-0.525	1.928	0.5
11.2	3.037	2.037	5.9	41.920	-0.525	2.051	0.5
12.6	3.283	2.283	5.9	41.929	-0.525	2.175	0.5
14.0	3.528	2.528	5.9	41.936	-0.525	2.298	0.5
15.4	3.774	2.774	5.9	41.942	-0.525	2.421	0.5
16.8	4.019	3.019	5.9	41.947	-0.525	2.544	0.5
18.2	4.264	3.264	5.9	41.951	-0.525	2.667	0.5
19.6	4.510	3.510	5.9	41.955	-0.525	2.789	0.5
21.0	4.755	3.755	5.9	41.958	-0.525	2.912	0.5
22.4	5.000	4.000	20.7	41.960	-0.525	3.035	30.9
23.8	5.070	4.070	20.7	41.974	-0.608	3.153	30.9
25.2	5.140	4.140	20.7	41.986	-0.681	3.261	30.9
26.6	5.210	4.210	20.7	41.996	-0.747	3.362	30.9
28.0	5.280	4.280	20.7	42.006	-0.806	3.456	30.9
29.4	5.350	4.350	20.7	42.015	-0.859	3.545	30.9
30.8	5.419	4.419	20.7	42.023	-0.908	3.628	30.9
32.2	5.489	4.489	20.7	42.030	-0.952	3.708	30.9
33.6	5.559	4.559	20.7	42.037	-0.993	3.783	30.9
35.0	5.629	4.629	20.7	42.043	-1.030	3.856	30.9
36.4	5.699	4.699	20.7	42.049	-1.065	3.925	30.9
37.8	5.769	4.769	20.7	42.054	-1.097	3.992	30.9
39.2	5.839	4.839	20.7	42.059	-1.126	4.057	30.9
40.6	5.908	4.908	20.7	42.063	-1.154	4.120	30.9
42.0	5.978	4.978	20.7	42.067	-1.180	4.180	30.9
43.4	6.048	5.048	20.7	42.071	-1.204	4.240	30.9
44.8	6.118	5.118	20.7	42.075	-1.227	4.297	30.9
46.2	6.188	5.188	20.7	42.079	-1.248	4.353	30.9
47.6	6.258	5.258	20.7	42.082	-1.268	4.408	30.9
49.0	6.328	5.328	20.7	42.085	-1.287	4.462	30.9
50.4	6.397	5.397	20.7	42.088	-1.304	4.515	30.9
51.8	6.467	5.467	20.7	42.091	-1.321	4.567	30.9
53.2	6.537	5.537	20.7	42.093	-1.337	4.618	30.9
54.6	6.607	5.607	20.7	42.096	-1.352	4.668	30.9
56.0	6.680	5.680	19.7	42.101	-1.365	4.717	29.1
57.4	6.753	5.753	19.7	42.106	-1.378	4.766	29.1
58.8	6.826	5.826	19.7	42.111	-1.389	4.814	29.1
60.2	6.899	5.899	19.7	42.116	-1.400	4.862	29.1
61.6	6.972	5.972	19.7	42.120	-1.411	4.909	29.1
63.0	7.046	6.046	19.7	42.125	-1.421	4.956	29.1
64.4	7.119	6.119	19.7	42.129	-1.431	5.002	29.1
65.8	7.192	6.192	19.7	42.133	-1.440	5.048	29.1
67.2	7.265	6.265	19.7	42.137	-1.449	5.093	29.1
68.6	7.338	6.338	19.7	42.140	-1.457	5.138	29.1
70.0	7.411	6.411	19.7	42.144	-1.466	5.183	29.1
71.4	7.484	6.484	19.7	42.147	-1.474	5.227	29.1
72.8	7.557	6.557	19.7	42.150	-1.481	5.271	29.1
74.2	7.630	6.630	19.7	42.153	-1.488	5.315	29.1
75.6	7.704	6.704	19.7	42.156	-1.495	5.359	29.1
77.0	7.777	6.777	19.7	42.159	-1.502	5.402	29.1
78.4	7.850	6.850	19.7	42.162	-1.509	5.445	29.1
79.8	7.923	6.923	19.7	42.164	-1.515	5.488	29.1
81.3	8.000	7.000	19.7	42.167	-1.521	5.533	29.1

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TANK NO. 042 : STORAGE TANK SB : NET VOLUME= 81.28 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.000	0.000	3.7	40.960	0.525	1.001	0.3
1.4	1.315	0.315	5.6	41.407	0.525	1.166	0.5
2.8	1.564	0.564	5.9	41.681	0.525	1.304	0.5
4.2	1.810	0.810	5.9	41.788	0.525	1.431	0.5
5.6	2.055	1.055	5.9	41.841	0.525	1.557	0.5
7.0	2.301	1.301	5.9	41.873	0.525	1.681	0.5
8.4	2.546	1.546	5.9	41.894	0.525	1.805	0.5
9.8	2.792	1.792	5.9	41.909	0.525	1.928	0.5
11.2	3.037	2.037	5.9	41.920	0.525	2.051	0.5
12.6	3.283	2.283	5.9	41.929	0.525	2.175	0.5
14.0	3.528	2.528	5.9	41.936	0.525	2.298	0.5
15.4	3.774	2.774	5.9	41.942	0.525	2.421	0.5
16.8	4.019	3.019	5.9	41.947	0.525	2.544	0.5
18.2	4.264	3.264	5.9	41.951	0.525	2.667	0.5
19.6	4.510	3.510	5.9	41.955	0.525	2.789	0.5
21.0	4.755	3.755	5.9	41.958	0.525	2.912	0.5
22.4	5.000	4.000	20.7	41.960	0.525	3.035	30.9
23.8	5.070	4.070	20.7	41.974	0.608	3.153	30.9
25.2	5.140	4.140	20.7	41.986	0.681	3.261	30.9
26.6	5.210	4.210	20.7	41.996	0.747	3.362	30.9
28.0	5.280	4.280	20.7	42.006	0.806	3.456	30.9
29.4	5.350	4.350	20.7	42.015	0.859	3.545	30.9
30.8	5.419	4.419	20.7	42.023	0.908	3.628	30.9
32.2	5.489	4.489	20.7	42.030	0.952	3.708	30.9
33.6	5.559	4.559	20.7	42.037	0.993	3.783	30.9
35.0	5.629	4.629	20.7	42.043	1.030	3.856	30.9
36.4	5.699	4.699	20.7	42.049	1.065	3.925	30.9
37.8	5.769	4.769	20.7	42.054	1.097	3.992	30.9
39.2	5.839	4.839	20.7	42.059	1.126	4.057	30.9
40.6	5.908	4.908	20.7	42.063	1.154	4.120	30.9
42.0	5.978	4.978	20.7	42.067	1.180	4.180	30.9
43.4	6.048	5.048	20.7	42.071	1.204	4.240	30.9
44.8	6.118	5.118	20.7	42.075	1.227	4.297	30.9
46.2	6.188	5.188	20.7	42.079	1.248	4.353	30.9
47.6	6.258	5.258	20.7	42.082	1.268	4.408	30.9
49.0	6.328	5.328	20.7	42.085	1.287	4.462	30.9
50.4	6.397	5.397	20.7	42.088	1.304	4.515	30.9
51.8	6.467	5.467	20.7	42.091	1.321	4.567	30.9
53.2	6.537	5.537	20.7	42.093	1.337	4.618	30.9
54.6	6.607	5.607	20.7	42.096	1.352	4.668	30.9
56.0	6.680	5.680	19.7	42.101	1.365	4.717	29.1
57.4	6.753	5.753	19.7	42.106	1.378	4.766	29.1
58.8	6.826	5.826	19.7	42.111	1.389	4.814	29.1
60.2	6.899	5.899	19.7	42.116	1.400	4.862	29.1
61.6	6.972	5.972	19.7	42.120	1.411	4.909	29.1
63.0	7.046	6.046	19.7	42.125	1.421	4.956	29.1
64.4	7.119	6.119	19.7	42.129	1.431	5.002	29.1
65.8	7.192	6.192	19.7	42.133	1.440	5.048	29.1
67.2	7.265	6.265	19.7	42.137	1.449	5.093	29.1
68.6	7.338	6.338	19.7	42.140	1.457	5.138	29.1
70.0	7.411	6.411	19.7	42.144	1.466	5.183	29.1
71.4	7.484	6.484	19.7	42.147	1.474	5.227	29.1
72.8	7.557	6.557	19.7	42.150	1.481	5.271	29.1
74.2	7.630	6.630	19.7	42.153	1.488	5.315	29.1
75.6	7.704	6.704	19.7	42.156	1.495	5.359	29.1
77.0	7.777	6.777	19.7	42.159	1.502	5.402	29.1
78.4	7.850	6.850	19.7	42.162	1.509	5.445	29.1
79.8	7.923	6.923	19.7	42.164	1.515	5.488	29.1
81.3	8.000	7.000	19.7	42.167	1.521	5.533	29.1

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TANK NO. 043 : LIQUID CARGO TANK PS : NET VOLUME= 140.09 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.012	0.000	0.0	0.000	0.000	0.000	0.0
2.3	1.328	0.316	17.0	41.597	-4.540	1.234	17.9
4.6	1.450	0.438	20.1	41.620	-4.965	1.312	29.5
6.9	1.566	0.554	20.2	41.625	-5.135	1.378	30.1
9.2	1.682	0.670	20.3	41.628	-5.223	1.439	30.6
11.5	1.798	0.786	20.4	41.630	-5.278	1.499	31.1
13.8	1.913	0.900	20.5	41.632	-5.316	1.559	31.6
16.1	2.027	1.015	20.6	41.634	-5.345	1.617	32.1
18.4	2.141	1.129	20.7	41.635	-5.368	1.676	32.5
20.7	2.254	1.242	20.8	41.636	-5.387	1.734	33.0
23.0	2.367	1.355	20.8	41.637	-5.403	1.791	33.5
25.3	2.479	1.467	20.9	41.638	-5.418	1.849	34.0
27.6	2.591	1.579	21.0	41.638	-5.430	1.906	34.5
29.9	2.703	1.691	21.1	41.639	-5.441	1.963	34.8
32.2	2.814	1.802	21.1	41.639	-5.452	2.020	35.1
34.5	2.925	1.913	21.1	41.640	-5.461	2.077	35.4
36.8	3.036	2.024	21.2	41.641	-5.469	2.133	35.6
39.1	3.147	2.135	21.2	41.641	-5.476	2.189	35.9
41.4	3.257	2.245	21.2	41.642	-5.483	2.246	35.9
43.7	3.368	2.356	21.2	41.642	-5.489	2.302	35.9
46.0	3.478	2.466	21.2	41.642	-5.495	2.358	35.9
48.3	3.589	2.577	21.2	41.643	-5.500	2.414	35.9
50.6	3.699	2.687	21.2	41.643	-5.505	2.470	35.9
52.9	3.810	2.798	21.2	41.643	-5.509	2.526	35.9
55.2	3.920	2.908	21.2	41.644	-5.512	2.581	35.9
57.5	4.031	3.019	21.2	41.644	-5.516	2.637	35.9
59.8	4.142	3.129	21.2	41.644	-5.519	2.693	35.9
62.1	4.252	3.240	21.2	41.644	-5.522	2.749	35.9
64.4	4.363	3.350	21.2	41.645	-5.525	2.804	35.9
66.7	4.473	3.461	21.2	41.645	-5.528	2.860	35.9
69.0	4.584	3.571	21.2	41.645	-5.530	2.916	35.9
71.3	4.694	3.682	21.2	41.645	-5.532	2.971	35.9
73.6	4.805	3.793	21.2	41.645	-5.534	3.027	35.9
75.9	4.915	3.903	21.2	41.645	-5.536	3.082	35.9
78.2	5.026	4.014	21.2	41.646	-5.538	3.138	35.9
80.5	5.136	4.124	21.2	41.646	-5.540	3.193	35.9
82.8	5.247	4.235	21.2	41.646	-5.542	3.249	35.9
85.1	5.357	4.345	21.2	41.646	-5.543	3.304	35.9
87.4	5.468	4.456	21.2	41.646	-5.545	3.360	35.9
89.7	5.579	4.566	21.2	41.646	-5.546	3.415	35.9
92.0	5.689	4.677	21.2	41.646	-5.547	3.471	35.9
94.3	5.800	4.787	21.2	41.646	-5.549	3.526	35.9
96.6	5.910	4.898	21.2	41.646	-5.550	3.582	35.9
98.9	6.021	5.008	21.2	41.647	-5.551	3.637	35.9
101.2	6.131	5.119	21.2	41.647	-5.552	3.693	35.9
103.5	6.242	5.230	21.2	41.647	-5.553	3.748	35.9
105.8	6.352	5.340	21.2	41.647	-5.554	3.803	35.9
108.1	6.463	5.451	21.2	41.647	-5.555	3.859	35.9
110.4	6.573	5.561	21.2	41.647	-5.556	3.914	35.9
112.7	6.684	5.672	21.2	41.647	-5.557	3.970	35.9
115.0	6.794	5.782	21.2	41.647	-5.558	4.025	35.9
117.3	6.905	5.893	21.2	41.647	-5.559	4.080	35.9
119.6	7.015	6.003	21.2	41.647	-5.560	4.136	35.9
121.9	7.126	6.114	21.2	41.647	-5.560	4.191	35.9
124.2	7.237	6.224	21.2	41.647	-5.561	4.246	35.9
126.5	7.347	6.335	21.2	41.647	-5.562	4.302	35.9
128.8	7.458	6.445	21.2	41.647	-5.562	4.357	35.9
131.1	7.568	6.556	21.2	41.647	-5.563	4.413	35.9
133.4	7.679	6.666	21.2	41.647	-5.564	4.468	35.9
135.7	7.789	6.777	21.2	41.647	-5.564	4.523	35.9
138.0	7.900	6.888	21.2	41.648	-5.565	4.579	35.9
140.1	8.000	6.988	21.2	41.648	-5.566	4.629	35.9

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TANK NO. 044 : LIQUID CARGO TANK SB : NET VOLUME= 140.09 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.012	0.000	0.0	0.000	0.000	0.000	0.0
2.3	1.328	0.316	17.0	41.597	4.540	1.234	17.9
4.6	1.450	0.438	20.1	41.620	4.965	1.312	29.5
6.9	1.566	0.554	20.2	41.625	5.135	1.378	30.1
9.2	1.682	0.670	20.3	41.628	5.223	1.439	30.6
11.5	1.798	0.786	20.4	41.630	5.278	1.499	31.1
13.8	1.913	0.900	20.5	41.632	5.316	1.559	31.6
16.1	2.027	1.015	20.6	41.634	5.345	1.617	32.1
18.4	2.141	1.129	20.7	41.635	5.368	1.676	32.5
20.7	2.254	1.242	20.8	41.636	5.387	1.734	33.0
23.0	2.367	1.355	20.8	41.637	5.403	1.791	33.5
25.3	2.479	1.467	20.9	41.638	5.418	1.849	34.0
27.6	2.591	1.579	21.0	41.638	5.430	1.906	34.5
29.9	2.703	1.691	21.1	41.639	5.441	1.963	34.8
32.2	2.814	1.802	21.1	41.639	5.452	2.020	35.1
34.5	2.925	1.913	21.1	41.640	5.461	2.077	35.4
36.8	3.036	2.024	21.2	41.641	5.469	2.133	35.6
39.1	3.147	2.135	21.2	41.641	5.476	2.189	35.9
41.4	3.257	2.245	21.2	41.642	5.483	2.246	35.9
43.7	3.368	2.356	21.2	41.642	5.489	2.302	35.9
46.0	3.478	2.466	21.2	41.642	5.495	2.358	35.9
48.3	3.589	2.577	21.2	41.643	5.500	2.414	35.9
50.6	3.699	2.687	21.2	41.643	5.505	2.470	35.9
52.9	3.810	2.798	21.2	41.643	5.509	2.526	35.9
55.2	3.920	2.908	21.2	41.644	5.512	2.581	35.9
57.5	4.031	3.019	21.2	41.644	5.516	2.637	35.9
59.8	4.142	3.129	21.2	41.644	5.519	2.693	35.9
62.1	4.252	3.240	21.2	41.644	5.522	2.749	35.9
64.4	4.363	3.350	21.2	41.645	5.525	2.804	35.9
66.7	4.473	3.461	21.2	41.645	5.528	2.860	35.9
69.0	4.584	3.571	21.2	41.645	5.530	2.916	35.9
71.3	4.694	3.682	21.2	41.645	5.532	2.971	35.9
73.6	4.805	3.793	21.2	41.645	5.534	3.027	35.9
75.9	4.915	3.903	21.2	41.645	5.536	3.082	35.9
78.2	5.026	4.014	21.2	41.646	5.538	3.138	35.9
80.5	5.136	4.124	21.2	41.646	5.540	3.193	35.9
82.8	5.247	4.235	21.2	41.646	5.542	3.249	35.9
85.1	5.357	4.345	21.2	41.646	5.543	3.304	35.9
87.4	5.468	4.456	21.2	41.646	5.545	3.360	35.9
89.7	5.579	4.566	21.2	41.646	5.546	3.415	35.9
92.0	5.689	4.677	21.2	41.646	5.547	3.471	35.9
94.3	5.800	4.787	21.2	41.646	5.549	3.526	35.9
96.6	5.910	4.898	21.2	41.646	5.550	3.582	35.9
98.9	6.021	5.008	21.2	41.647	5.551	3.637	35.9
101.2	6.131	5.119	21.2	41.647	5.552	3.693	35.9
103.5	6.242	5.230	21.2	41.647	5.553	3.748	35.9
105.8	6.352	5.340	21.2	41.647	5.554	3.803	35.9
108.1	6.463	5.451	21.2	41.647	5.555	3.859	35.9
110.4	6.573	5.561	21.2	41.647	5.556	3.914	35.9
112.7	6.684	5.672	21.2	41.647	5.557	3.970	35.9
115.0	6.794	5.782	21.2	41.647	5.558	4.025	35.9
117.3	6.905	5.893	21.2	41.647	5.559	4.080	35.9
119.6	7.015	6.003	21.2	41.647	5.560	4.136	35.9
121.9	7.126	6.114	21.2	41.647	5.560	4.191	35.9
124.2	7.237	6.224	21.2	41.647	5.561	4.246	35.9
126.5	7.347	6.335	21.2	41.647	5.562	4.302	35.9
128.8	7.458	6.445	21.2	41.647	5.562	4.357	35.9
131.1	7.568	6.556	21.2	41.647	5.563	4.413	35.9
133.4	7.679	6.666	21.2	41.647	5.564	4.468	35.9
135.7	7.789	6.777	21.2	41.647	5.564	4.523	35.9
138.0	7.900	6.888	21.2	41.648	5.565	4.579	35.9
140.1	8.000	6.988	21.2	41.648	5.566	4.629	35.9

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TANK NO. 045 : LIQUID CARGO TANK PS : NET VOLUME= 127.15 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.012	0.000	0.0	0.000	0.000	0.000	0.0
2.1	1.325	0.313	14.8	37.242	-4.464	1.228	16.0
4.2	1.447	0.435	19.1	37.221	-4.938	1.309	34.9
6.3	1.559	0.547	19.1	37.221	-5.159	1.374	34.9
8.4	1.671	0.659	19.1	37.221	-5.269	1.434	34.9
10.5	1.783	0.771	19.1	37.221	-5.335	1.493	34.9
12.6	1.895	0.883	19.1	37.221	-5.379	1.550	34.9
14.7	2.007	0.995	19.1	37.221	-5.411	1.608	34.9
16.8	2.119	1.107	19.1	37.221	-5.435	1.665	34.9
18.9	2.231	1.219	19.1	37.221	-5.453	1.721	34.9
21.0	2.343	1.330	19.1	37.221	-5.468	1.778	34.9
23.1	2.455	1.442	19.1	37.221	-5.480	1.834	34.9
25.2	2.567	1.554	19.1	37.221	-5.490	1.891	34.9
27.3	2.678	1.666	19.1	37.221	-5.498	1.947	34.9
29.4	2.790	1.778	19.1	37.221	-5.505	2.003	34.9
31.5	2.902	1.890	19.1	37.221	-5.512	2.059	34.9
33.6	3.014	2.002	19.1	37.221	-5.517	2.116	34.9
35.7	3.126	2.114	19.1	37.221	-5.522	2.172	34.9
37.8	3.238	2.226	19.1	37.221	-5.526	2.228	34.9
39.9	3.350	2.338	19.1	37.221	-5.530	2.284	34.9
42.0	3.462	2.450	19.1	37.221	-5.534	2.340	34.9
44.1	3.574	2.562	19.1	37.221	-5.537	2.396	34.9
46.2	3.686	2.673	19.1	37.221	-5.540	2.452	34.9
48.3	3.798	2.785	19.1	37.221	-5.542	2.508	34.9
50.4	3.910	2.897	19.1	37.221	-5.545	2.564	34.9
52.5	4.021	3.009	19.1	37.221	-5.547	2.620	34.9
54.6	4.133	3.121	19.1	37.221	-5.549	2.676	34.9
56.7	4.245	3.233	19.1	37.221	-5.551	2.732	34.9
58.8	4.357	3.345	19.1	37.221	-5.553	2.788	34.9
60.9	4.469	3.457	19.1	37.221	-5.554	2.844	34.9
63.0	4.581	3.569	19.1	37.221	-5.556	2.901	34.9
65.1	4.693	3.681	19.1	37.221	-5.557	2.957	34.9
67.2	4.805	3.793	19.1	37.221	-5.559	3.013	34.9
69.3	4.917	3.905	19.1	37.221	-5.560	3.069	34.9
71.4	5.029	4.016	19.1	37.221	-5.561	3.125	34.9
73.5	5.141	4.128	19.1	37.221	-5.562	3.181	34.9
75.6	5.253	4.240	19.1	37.221	-5.563	3.237	34.9
77.7	5.364	4.352	19.1	37.221	-5.564	3.293	34.9
79.8	5.476	4.464	19.1	37.221	-5.565	3.349	34.9
81.9	5.588	4.576	19.1	37.221	-5.566	3.405	34.9
84.0	5.700	4.688	19.1	37.221	-5.567	3.461	34.9
86.1	5.812	4.800	19.1	37.221	-5.568	3.517	34.9
88.2	5.924	4.912	19.1	37.221	-5.568	3.573	34.9
90.3	6.036	5.024	19.1	37.221	-5.569	3.628	34.9
92.4	6.148	5.136	19.1	37.221	-5.570	3.684	34.9
94.5	6.260	5.248	19.1	37.221	-5.571	3.740	34.9
96.6	6.372	5.359	19.1	37.221	-5.571	3.796	34.9
98.7	6.484	5.471	19.1	37.221	-5.572	3.852	34.9
100.8	6.595	5.583	19.1	37.221	-5.572	3.908	34.9
102.9	6.707	5.695	19.1	37.221	-5.573	3.964	34.9
105.0	6.819	5.807	19.1	37.221	-5.574	4.020	34.9
107.1	6.931	5.919	19.1	37.221	-5.574	4.076	34.9
109.2	7.043	6.031	19.1	37.221	-5.575	4.132	34.9
111.3	7.155	6.143	19.1	37.221	-5.575	4.188	34.9
113.4	7.267	6.255	19.1	37.221	-5.575	4.244	34.9
115.5	7.379	6.367	19.1	37.221	-5.576	4.300	34.9
117.6	7.491	6.479	19.1	37.221	-5.576	4.356	34.9
119.7	7.603	6.590	19.1	37.221	-5.577	4.412	34.9
121.8	7.715	6.702	19.1	37.221	-5.577	4.468	34.9
123.9	7.827	6.814	19.1	37.221	-5.578	4.524	34.9
126.0	7.938	6.926	19.1	37.221	-5.578	4.580	34.9
127.2	8.000	6.988	19.1	37.221	-5.578	4.611	34.9

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TANK NO. 046 : LIQUID CARGO TANK SB : NET VOLUME= 124.31 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.012	0.000	0.0	0.000	0.000	0.000	0.0
2.1	1.325	0.313	14.8	37.242	4.464	1.228	16.0
4.2	1.447	0.435	19.1	37.221	4.938	1.309	34.9
6.3	1.559	0.547	19.1	37.221	5.159	1.374	34.9
8.4	1.671	0.659	19.1	37.221	5.269	1.434	34.9
10.5	1.783	0.771	19.1	37.221	5.335	1.493	34.9
12.6	1.895	0.883	19.1	37.221	5.379	1.550	34.9
14.7	2.007	0.995	19.1	37.221	5.411	1.608	34.9
16.8	2.119	1.107	19.1	37.221	5.435	1.665	34.9
18.9	2.231	1.219	19.1	37.221	5.453	1.721	34.9
21.0	2.343	1.331	19.1	37.221	5.468	1.778	34.9
23.1	2.455	1.442	19.1	37.221	5.480	1.834	34.9
25.2	2.567	1.554	19.1	37.221	5.490	1.891	34.9
27.3	2.678	1.666	19.1	37.221	5.498	1.947	34.9
29.4	2.790	1.778	19.1	37.221	5.505	2.003	34.9
31.5	2.902	1.890	19.1	37.221	5.512	2.059	34.9
33.6	3.014	2.002	19.1	37.221	5.517	2.116	34.9
35.7	3.126	2.114	19.1	37.221	5.522	2.172	34.9
37.8	3.238	2.226	19.1	37.221	5.526	2.228	34.9
39.9	3.350	2.338	19.1	37.221	5.530	2.284	34.9
42.0	3.462	2.450	19.1	37.221	5.534	2.340	34.9
44.1	3.574	2.562	19.1	37.221	5.537	2.396	34.9
46.2	3.686	2.673	19.1	37.221	5.540	2.452	34.9
48.3	3.798	2.785	19.1	37.221	5.542	2.508	34.9
50.4	3.910	2.897	19.1	37.221	5.545	2.564	34.9
52.5	4.021	3.009	19.1	37.221	5.547	2.620	34.9
54.6	4.133	3.121	19.1	37.221	5.549	2.676	34.9
56.7	4.245	3.233	19.1	37.221	5.551	2.732	34.9
58.8	4.357	3.345	19.1	37.221	5.553	2.788	34.9
60.9	4.469	3.457	19.1	37.221	5.554	2.844	34.9
63.0	4.581	3.569	19.1	37.221	5.556	2.900	34.9
65.1	4.693	3.681	19.1	37.221	5.557	2.957	34.9
67.2	4.805	3.793	19.1	37.221	5.559	3.013	34.9
69.3	4.917	3.905	19.1	37.221	5.560	3.069	34.9
71.4	5.029	4.016	19.1	37.221	5.561	3.125	34.9
73.5	5.141	4.128	19.1	37.221	5.562	3.181	34.9
75.6	5.252	4.240	19.1	37.221	5.563	3.237	34.9
77.7	5.364	4.352	19.1	37.221	5.564	3.293	34.9
79.8	5.476	4.464	19.1	37.221	5.565	3.349	34.9
81.9	5.588	4.576	19.1	37.221	5.566	3.405	34.9
84.0	5.700	4.688	19.1	37.221	5.567	3.461	34.9
86.1	5.812	4.800	19.1	37.221	5.568	3.516	34.9
88.2	5.924	4.912	19.1	37.221	5.568	3.572	34.9
90.3	6.036	5.024	19.1	37.221	5.569	3.628	34.9
92.4	6.148	5.136	19.1	37.221	5.570	3.684	34.9
94.5	6.260	5.247	19.1	37.221	5.571	3.740	34.9
96.6	6.372	5.359	19.1	37.221	5.571	3.796	34.9
98.7	6.484	5.471	19.1	37.221	5.572	3.852	34.9
100.8	6.595	5.583	19.1	37.221	5.572	3.908	34.9
102.9	6.707	5.695	19.1	37.221	5.573	3.964	34.9
105.0	6.819	5.807	19.1	37.221	5.574	4.020	34.9
107.1	6.931	5.919	19.1	37.221	5.574	4.076	34.9
109.2	7.043	6.031	19.1	37.221	5.575	4.132	34.9
111.3	7.166	6.154	15.9	37.218	5.575	4.188	29.3
113.4	7.301	6.289	15.9	37.212	5.575	4.245	29.3
115.5	7.435	6.423	15.9	37.206	5.576	4.302	29.3
117.6	7.570	6.558	15.9	37.200	5.576	4.359	29.3
119.7	7.704	6.692	15.9	37.195	5.577	4.416	29.3
121.8	7.839	6.827	15.9	37.190	5.577	4.474	29.3
123.9	7.974	6.961	15.9	37.185	5.578	4.532	29.3
124.3	8.000	6.988	15.9	37.184	5.578	4.544	29.3

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TANK NO. 047 : LIQUID CARGO TANK PS : NET VOLUME= 127.15 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.012	0.000	0.0	0.000	0.000	0.000	0.0
2.1	1.325	0.313	14.8	33.042	-4.464	1.228	16.0
4.2	1.447	0.435	19.1	33.021	-4.938	1.309	34.9
6.3	1.559	0.547	19.1	33.021	-5.159	1.374	34.9
8.4	1.671	0.659	19.1	33.021	-5.269	1.434	34.9
10.5	1.783	0.771	19.1	33.021	-5.335	1.493	34.9
12.6	1.895	0.883	19.1	33.021	-5.379	1.550	34.9
14.7	2.007	0.995	19.1	33.021	-5.411	1.608	34.9
16.8	2.119	1.107	19.1	33.021	-5.435	1.665	34.9
18.9	2.231	1.219	19.1	33.021	-5.453	1.721	34.9
21.0	2.343	1.330	19.1	33.021	-5.468	1.778	34.9
23.1	2.455	1.442	19.1	33.021	-5.480	1.834	34.9
25.2	2.567	1.554	19.1	33.021	-5.490	1.891	34.9
27.3	2.678	1.666	19.1	33.021	-5.498	1.947	34.9
29.4	2.790	1.778	19.1	33.021	-5.505	2.003	34.9
31.5	2.902	1.890	19.1	33.021	-5.512	2.059	34.9
33.6	3.014	2.002	19.1	33.021	-5.517	2.116	34.9
35.7	3.126	2.114	19.1	33.021	-5.522	2.172	34.9
37.8	3.238	2.226	19.1	33.021	-5.526	2.228	34.9
39.9	3.350	2.338	19.1	33.021	-5.530	2.284	34.9
42.0	3.462	2.450	19.1	33.021	-5.534	2.340	34.9
44.1	3.574	2.562	19.1	33.021	-5.537	2.396	34.9
46.2	3.686	2.673	19.1	33.021	-5.540	2.452	34.9
48.3	3.798	2.785	19.1	33.021	-5.542	2.508	34.9
50.4	3.910	2.897	19.1	33.021	-5.545	2.564	34.9
52.5	4.021	3.009	19.1	33.021	-5.547	2.620	34.9
54.6	4.133	3.121	19.1	33.021	-5.549	2.676	34.9
56.7	4.245	3.233	19.1	33.021	-5.551	2.732	34.9
58.8	4.357	3.345	19.1	33.021	-5.553	2.788	34.9
60.9	4.469	3.457	19.1	33.021	-5.554	2.844	34.9
63.0	4.581	3.569	19.1	33.021	-5.556	2.901	34.9
65.1	4.693	3.681	19.1	33.021	-5.557	2.957	34.9
67.2	4.805	3.793	19.1	33.021	-5.559	3.013	34.9
69.3	4.917	3.905	19.1	33.021	-5.560	3.069	34.9
71.4	5.029	4.016	19.1	33.021	-5.561	3.125	34.9
73.5	5.141	4.128	19.1	33.021	-5.562	3.181	34.9
75.6	5.253	4.240	19.1	33.021	-5.563	3.237	34.9
77.7	5.364	4.352	19.1	33.021	-5.564	3.293	34.9
79.8	5.476	4.464	19.1	33.021	-5.565	3.349	34.9
81.9	5.588	4.576	19.1	33.021	-5.566	3.405	34.9
84.0	5.700	4.688	19.1	33.021	-5.567	3.461	34.9
86.1	5.812	4.800	19.1	33.021	-5.568	3.517	34.9
88.2	5.924	4.912	19.1	33.021	-5.568	3.573	34.9
90.3	6.036	5.024	19.1	33.021	-5.569	3.628	34.9
92.4	6.148	5.136	19.1	33.021	-5.570	3.684	34.9
94.5	6.260	5.248	19.1	33.021	-5.571	3.740	34.9
96.6	6.372	5.359	19.1	33.021	-5.571	3.796	34.9
98.7	6.484	5.471	19.1	33.021	-5.572	3.852	34.9
100.8	6.595	5.583	19.1	33.021	-5.572	3.908	34.9
102.9	6.707	5.695	19.1	33.021	-5.573	3.964	34.9
105.0	6.819	5.807	19.1	33.021	-5.574	4.020	34.9
107.1	6.931	5.919	19.1	33.021	-5.574	4.076	34.9
109.2	7.043	6.031	19.1	33.021	-5.575	4.132	34.9
111.3	7.155	6.143	19.1	33.021	-5.575	4.188	34.9
113.4	7.267	6.255	19.1	33.021	-5.575	4.244	34.9
115.5	7.379	6.367	19.1	33.021	-5.576	4.300	34.9
117.6	7.491	6.479	19.1	33.021	-5.576	4.356	34.9
119.7	7.603	6.591	19.1	33.021	-5.577	4.412	34.9
121.8	7.715	6.702	19.1	33.021	-5.577	4.468	34.9
123.9	7.827	6.814	19.1	33.021	-5.578	4.524	34.9
126.0	7.938	6.926	19.1	33.021	-5.578	4.580	34.9
127.2	8.000	6.988	19.1	33.021	-5.578	4.611	34.9

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TANK NO. 048 : LIQUID CARGO TANK SB : NET VOLUME= 127.15 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.012	0.000	0.0	0.000	0.000	0.000	0.0
2.1	1.325	0.313	14.8	33.042	4.464	1.228	16.0
4.2	1.447	0.435	19.1	33.021	4.938	1.309	34.9
6.3	1.559	0.547	19.1	33.021	5.159	1.374	34.9
8.4	1.671	0.659	19.1	33.021	5.269	1.434	34.9
10.5	1.783	0.771	19.1	33.021	5.335	1.493	34.9
12.6	1.895	0.883	19.1	33.021	5.379	1.550	34.9
14.7	2.007	0.995	19.1	33.021	5.411	1.608	34.9
16.8	2.119	1.107	19.1	33.021	5.435	1.665	34.9
18.9	2.231	1.219	19.1	33.021	5.453	1.721	34.9
21.0	2.343	1.330	19.1	33.021	5.468	1.778	34.9
23.1	2.455	1.442	19.1	33.021	5.480	1.834	34.9
25.2	2.567	1.554	19.1	33.021	5.490	1.891	34.9
27.3	2.678	1.666	19.1	33.021	5.498	1.947	34.9
29.4	2.790	1.778	19.1	33.021	5.505	2.003	34.9
31.5	2.902	1.890	19.1	33.021	5.512	2.059	34.9
33.6	3.014	2.002	19.1	33.021	5.517	2.116	34.9
35.7	3.126	2.114	19.1	33.021	5.522	2.172	34.9
37.8	3.238	2.226	19.1	33.021	5.526	2.228	34.9
39.9	3.350	2.338	19.1	33.021	5.530	2.284	34.9
42.0	3.462	2.450	19.1	33.021	5.534	2.340	34.9
44.1	3.574	2.562	19.1	33.021	5.537	2.396	34.9
46.2	3.686	2.673	19.1	33.021	5.540	2.452	34.9
48.3	3.798	2.785	19.1	33.021	5.542	2.508	34.9
50.4	3.910	2.897	19.1	33.021	5.545	2.564	34.9
52.5	4.021	3.009	19.1	33.021	5.547	2.620	34.9
54.6	4.133	3.121	19.1	33.021	5.549	2.676	34.9
56.7	4.245	3.233	19.1	33.021	5.551	2.732	34.9
58.8	4.357	3.345	19.1	33.021	5.553	2.788	34.9
60.9	4.469	3.457	19.1	33.021	5.554	2.844	34.9
63.0	4.581	3.569	19.1	33.021	5.556	2.901	34.9
65.1	4.693	3.681	19.1	33.021	5.557	2.957	34.9
67.2	4.805	3.793	19.1	33.021	5.559	3.013	34.9
69.3	4.917	3.905	19.1	33.021	5.560	3.069	34.9
71.4	5.029	4.016	19.1	33.021	5.561	3.125	34.9
73.5	5.141	4.128	19.1	33.021	5.562	3.181	34.9
75.6	5.253	4.240	19.1	33.021	5.563	3.237	34.9
77.7	5.364	4.352	19.1	33.021	5.564	3.293	34.9
79.8	5.476	4.464	19.1	33.021	5.565	3.349	34.9
81.9	5.588	4.576	19.1	33.021	5.566	3.405	34.9
84.0	5.700	4.688	19.1	33.021	5.567	3.461	34.9
86.1	5.812	4.800	19.1	33.021	5.568	3.517	34.9
88.2	5.924	4.912	19.1	33.021	5.568	3.573	34.9
90.3	6.036	5.024	19.1	33.021	5.569	3.628	34.9
92.4	6.148	5.136	19.1	33.021	5.570	3.684	34.9
94.5	6.260	5.248	19.1	33.021	5.571	3.740	34.9
96.6	6.372	5.359	19.1	33.021	5.571	3.796	34.9
98.7	6.484	5.471	19.1	33.021	5.572	3.852	34.9
100.8	6.595	5.583	19.1	33.021	5.572	3.908	34.9
102.9	6.707	5.695	19.1	33.021	5.573	3.964	34.9
105.0	6.819	5.807	19.1	33.021	5.574	4.020	34.9
107.1	6.931	5.919	19.1	33.021	5.574	4.076	34.9
109.2	7.043	6.031	19.1	33.021	5.575	4.132	34.9
111.3	7.155	6.143	19.1	33.021	5.575	4.188	34.9
113.4	7.267	6.255	19.1	33.021	5.575	4.244	34.9
115.5	7.379	6.367	19.1	33.021	5.576	4.300	34.9
117.6	7.491	6.479	19.1	33.021	5.576	4.356	34.9
119.7	7.603	6.591	19.1	33.021	5.577	4.412	34.9
121.8	7.715	6.702	19.1	33.021	5.577	4.468	34.9
123.9	7.827	6.814	19.1	33.021	5.578	4.524	34.9
126.0	7.938	6.926	19.1	33.021	5.578	4.580	34.9
127.2	8.000	6.988	19.1	33.021	5.578	4.611	34.9

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TANK NO. 049 : LIQUID CARGO TANK PS : NET VOLUME= 127.15 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.012	0.000	0.0	0.000	0.000	0.000	0.0
2.1	1.325	0.313	14.8	28.842	-4.464	1.228	16.0
4.2	1.447	0.435	19.1	28.821	-4.938	1.309	34.9
6.3	1.559	0.547	19.1	28.821	-5.159	1.374	34.9
8.4	1.671	0.659	19.1	28.821	-5.269	1.434	34.9
10.5	1.783	0.771	19.1	28.821	-5.335	1.493	34.9
12.6	1.895	0.883	19.1	28.821	-5.379	1.550	34.9
14.7	2.007	0.995	19.1	28.821	-5.411	1.608	34.9
16.8	2.119	1.107	19.1	28.821	-5.435	1.665	34.9
18.9	2.231	1.219	19.1	28.821	-5.453	1.721	34.9
21.0	2.343	1.330	19.1	28.821	-5.468	1.778	34.9
23.1	2.455	1.442	19.1	28.821	-5.480	1.834	34.9
25.2	2.567	1.554	19.1	28.821	-5.490	1.891	34.9
27.3	2.678	1.666	19.1	28.821	-5.498	1.947	34.9
29.4	2.790	1.778	19.1	28.821	-5.505	2.003	34.9
31.5	2.902	1.890	19.1	28.821	-5.512	2.059	34.9
33.6	3.014	2.002	19.1	28.821	-5.517	2.116	34.9
35.7	3.126	2.114	19.1	28.821	-5.522	2.172	34.9
37.8	3.238	2.226	19.1	28.821	-5.526	2.228	34.9
39.9	3.350	2.338	19.1	28.821	-5.530	2.284	34.9
42.0	3.462	2.450	19.1	28.821	-5.534	2.340	34.9
44.1	3.574	2.562	19.1	28.821	-5.537	2.396	34.9
46.2	3.686	2.673	19.1	28.821	-5.540	2.452	34.9
48.3	3.798	2.785	19.1	28.821	-5.542	2.508	34.9
50.4	3.910	2.897	19.1	28.821	-5.545	2.564	34.9
52.5	4.021	3.009	19.1	28.821	-5.547	2.620	34.9
54.6	4.133	3.121	19.1	28.821	-5.549	2.676	34.9
56.7	4.245	3.233	19.1	28.821	-5.551	2.732	34.9
58.8	4.357	3.345	19.1	28.821	-5.553	2.788	34.9
60.9	4.469	3.457	19.1	28.821	-5.554	2.844	34.9
63.0	4.581	3.569	19.1	28.821	-5.556	2.901	34.9
65.1	4.693	3.681	19.1	28.821	-5.557	2.957	34.9
67.2	4.805	3.793	19.1	28.821	-5.559	3.013	34.9
69.3	4.917	3.905	19.1	28.821	-5.560	3.069	34.9
71.4	5.029	4.016	19.1	28.821	-5.561	3.125	34.9
73.5	5.141	4.128	19.1	28.821	-5.562	3.181	34.9
75.6	5.253	4.240	19.1	28.821	-5.563	3.237	34.9
77.7	5.364	4.352	19.1	28.821	-5.564	3.293	34.9
79.8	5.476	4.464	19.1	28.821	-5.565	3.349	34.9
81.9	5.588	4.576	19.1	28.821	-5.566	3.405	34.9
84.0	5.700	4.688	19.1	28.821	-5.567	3.461	34.9
86.1	5.812	4.800	19.1	28.821	-5.568	3.517	34.9
88.2	5.924	4.912	19.1	28.821	-5.568	3.573	34.9
90.3	6.036	5.024	19.1	28.821	-5.569	3.628	34.9
92.4	6.148	5.136	19.1	28.821	-5.570	3.684	34.9
94.5	6.260	5.248	19.1	28.821	-5.571	3.740	34.9
96.6	6.372	5.359	19.1	28.821	-5.571	3.796	34.9
98.7	6.484	5.471	19.1	28.821	-5.572	3.852	34.9
100.8	6.595	5.583	19.1	28.821	-5.572	3.908	34.9
102.9	6.707	5.695	19.1	28.821	-5.573	3.964	34.9
105.0	6.819	5.807	19.1	28.821	-5.574	4.020	34.9
107.1	6.931	5.919	19.1	28.821	-5.574	4.076	34.9
109.2	7.043	6.031	19.1	28.821	-5.575	4.132	34.9
111.3	7.155	6.143	19.1	28.821	-5.575	4.188	34.9
113.4	7.267	6.255	19.1	28.821	-5.575	4.244	34.9
115.5	7.379	6.367	19.1	28.821	-5.576	4.300	34.9
117.6	7.491	6.479	19.1	28.821	-5.576	4.356	34.9
119.7	7.603	6.590	19.1	28.821	-5.577	4.412	34.9
121.8	7.715	6.702	19.1	28.821	-5.577	4.468	34.9
123.9	7.827	6.814	19.1	28.821	-5.578	4.524	34.9
126.0	7.938	6.926	19.1	28.821	-5.578	4.580	34.9
127.2	8.000	6.988	19.1	28.821	-5.578	4.611	34.9

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TANK NO. 050 : LIQUID CARGO TANK SB : NET VOLUME= 127.15 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.012	0.000	0.0	0.000	0.000	0.000	0.0
2.1	1.325	0.313	14.8	28.842	4.464	1.228	16.0
4.2	1.447	0.435	19.1	28.821	4.938	1.309	34.9
6.3	1.559	0.547	19.1	28.821	5.159	1.374	34.9
8.4	1.671	0.659	19.1	28.821	5.269	1.434	34.9
10.5	1.783	0.771	19.1	28.821	5.335	1.493	34.9
12.6	1.895	0.883	19.1	28.821	5.379	1.550	34.9
14.7	2.007	0.995	19.1	28.821	5.411	1.608	34.9
16.8	2.119	1.107	19.1	28.821	5.435	1.665	34.9
18.9	2.231	1.219	19.1	28.821	5.453	1.721	34.9
21.0	2.343	1.330	19.1	28.821	5.468	1.778	34.9
23.1	2.455	1.442	19.1	28.821	5.480	1.834	34.9
25.2	2.567	1.554	19.1	28.821	5.490	1.891	34.9
27.3	2.678	1.666	19.1	28.821	5.498	1.947	34.9
29.4	2.790	1.778	19.1	28.821	5.505	2.003	34.9
31.5	2.902	1.890	19.1	28.821	5.512	2.059	34.9
33.6	3.014	2.002	19.1	28.821	5.517	2.116	34.9
35.7	3.126	2.114	19.1	28.821	5.522	2.172	34.9
37.8	3.238	2.226	19.1	28.821	5.526	2.228	34.9
39.9	3.350	2.338	19.1	28.821	5.530	2.284	34.9
42.0	3.462	2.450	19.1	28.821	5.534	2.340	34.9
44.1	3.574	2.562	19.1	28.821	5.537	2.396	34.9
46.2	3.686	2.673	19.1	28.821	5.540	2.452	34.9
48.3	3.798	2.785	19.1	28.821	5.542	2.508	34.9
50.4	3.910	2.897	19.1	28.821	5.545	2.564	34.9
52.5	4.021	3.009	19.1	28.821	5.547	2.620	34.9
54.6	4.133	3.121	19.1	28.821	5.549	2.676	34.9
56.7	4.245	3.233	19.1	28.821	5.551	2.732	34.9
58.8	4.357	3.345	19.1	28.821	5.553	2.788	34.9
60.9	4.469	3.457	19.1	28.821	5.554	2.844	34.9
63.0	4.581	3.569	19.1	28.821	5.556	2.901	34.9
65.1	4.693	3.681	19.1	28.821	5.557	2.957	34.9
67.2	4.805	3.793	19.1	28.821	5.559	3.013	34.9
69.3	4.917	3.905	19.1	28.821	5.560	3.069	34.9
71.4	5.029	4.016	19.1	28.821	5.561	3.125	34.9
73.5	5.141	4.128	19.1	28.821	5.562	3.181	34.9
75.6	5.253	4.240	19.1	28.821	5.563	3.237	34.9
77.7	5.364	4.352	19.1	28.821	5.564	3.293	34.9
79.8	5.476	4.464	19.1	28.821	5.565	3.349	34.9
81.9	5.588	4.576	19.1	28.821	5.566	3.405	34.9
84.0	5.700	4.688	19.1	28.821	5.567	3.461	34.9
86.1	5.812	4.800	19.1	28.821	5.568	3.517	34.9
88.2	5.924	4.912	19.1	28.821	5.568	3.573	34.9
90.3	6.036	5.024	19.1	28.821	5.569	3.628	34.9
92.4	6.148	5.136	19.1	28.821	5.570	3.684	34.9
94.5	6.260	5.248	19.1	28.821	5.571	3.740	34.9
96.6	6.372	5.359	19.1	28.821	5.571	3.796	34.9
98.7	6.484	5.471	19.1	28.821	5.572	3.852	34.9
100.8	6.595	5.583	19.1	28.821	5.572	3.908	34.9
102.9	6.707	5.695	19.1	28.821	5.573	3.964	34.9
105.0	6.819	5.807	19.1	28.821	5.574	4.020	34.9
107.1	6.931	5.919	19.1	28.821	5.574	4.076	34.9
109.2	7.043	6.031	19.1	28.821	5.575	4.132	34.9
111.3	7.155	6.143	19.1	28.821	5.575	4.188	34.9
113.4	7.267	6.255	19.1	28.821	5.575	4.244	34.9
115.5	7.379	6.367	19.1	28.821	5.576	4.300	34.9
117.6	7.491	6.479	19.1	28.821	5.576	4.356	34.9
119.7	7.603	6.590	19.1	28.821	5.577	4.412	34.9
121.8	7.715	6.702	19.1	28.821	5.577	4.468	34.9
123.9	7.827	6.814	19.1	28.821	5.578	4.524	34.9
126.0	7.938	6.926	19.1	28.821	5.578	4.580	34.9
127.2	8.000	6.988	19.1	28.821	5.578	4.611	34.9

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TANK NO. 051 : LIQUID CARGO TANK PS : NET VOLUME= 126.17 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.012	0.000	0.0	0.000	0.000	0.000	0.0
2.1	1.328	0.316	13.8	24.679	-4.439	1.229	14.0
4.2	1.467	0.455	16.3	24.713	-4.814	1.314	23.6
6.3	1.595	0.583	17.2	24.717	-4.990	1.387	26.9
8.4	1.716	0.704	18.0	24.710	-5.101	1.454	29.8
10.5	1.833	0.821	18.6	24.699	-5.182	1.518	32.2
12.6	1.947	0.935	19.0	24.688	-5.245	1.580	34.1
14.7	2.059	1.047	19.1	24.678	-5.295	1.641	34.7
16.8	2.171	1.159	19.1	24.671	-5.333	1.700	34.9
18.9	2.283	1.271	19.1	24.666	-5.362	1.759	34.9
21.0	2.395	1.383	19.1	24.661	-5.386	1.817	34.9
23.1	2.507	1.495	19.1	24.658	-5.406	1.874	34.9
25.2	2.619	1.607	19.1	24.655	-5.422	1.932	34.9
27.3	2.731	1.719	19.1	24.652	-5.436	1.989	34.9
29.4	2.843	1.830	19.1	24.650	-5.447	2.046	34.9
31.5	2.955	1.942	19.1	24.648	-5.457	2.103	34.9
33.6	3.067	2.054	19.1	24.646	-5.466	2.159	34.9
35.7	3.178	2.166	19.1	24.645	-5.474	2.216	34.9
37.8	3.290	2.278	19.1	24.643	-5.481	2.273	34.9
39.9	3.402	2.390	19.1	24.642	-5.487	2.329	34.9
42.0	3.514	2.502	19.1	24.641	-5.493	2.386	34.9
44.1	3.626	2.614	19.1	24.640	-5.498	2.442	34.9
46.2	3.738	2.726	19.1	24.639	-5.503	2.498	34.9
48.3	3.850	2.838	19.1	24.639	-5.507	2.555	34.9
50.4	3.962	2.950	19.1	24.638	-5.511	2.611	34.9
52.5	4.074	3.062	19.1	24.637	-5.514	2.667	34.9
54.6	4.186	3.173	19.1	24.637	-5.518	2.724	34.9
56.7	4.298	3.285	19.1	24.636	-5.521	2.780	34.9
58.8	4.410	3.397	19.1	24.636	-5.524	2.836	34.9
60.9	4.521	3.509	19.1	24.635	-5.526	2.892	34.9
63.0	4.633	3.621	19.1	24.635	-5.529	2.948	34.9
65.1	4.745	3.733	19.1	24.634	-5.531	3.004	34.9
67.2	4.857	3.845	19.1	24.634	-5.533	3.061	34.9
69.3	4.969	3.957	19.1	24.633	-5.535	3.117	34.9
71.4	5.081	4.069	19.1	24.633	-5.537	3.173	34.9
73.5	5.193	4.181	19.1	24.633	-5.539	3.229	34.9
75.6	5.305	4.293	19.1	24.632	-5.541	3.285	34.9
77.7	5.417	4.405	19.1	24.632	-5.542	3.341	34.9
79.8	5.529	4.516	19.1	24.632	-5.544	3.397	34.9
81.9	5.641	4.628	19.1	24.632	-5.545	3.453	34.9
84.0	5.752	4.740	19.1	24.631	-5.547	3.509	34.9
86.1	5.864	4.852	19.1	24.631	-5.548	3.566	34.9
88.2	5.976	4.964	19.1	24.631	-5.549	3.622	34.9
90.3	6.088	5.076	19.1	24.631	-5.550	3.678	34.9
92.4	6.200	5.188	19.1	24.630	-5.551	3.734	34.9
94.5	6.312	5.300	19.1	24.630	-5.552	3.790	34.9
96.6	6.424	5.412	19.1	24.630	-5.554	3.846	34.9
98.7	6.536	5.524	19.1	24.630	-5.555	3.902	34.9
100.8	6.648	5.636	19.1	24.630	-5.555	3.958	34.9
102.9	6.760	5.747	19.1	24.629	-5.556	4.014	34.9
105.0	6.872	5.859	19.1	24.629	-5.557	4.070	34.9
107.1	6.984	5.971	19.1	24.629	-5.558	4.126	34.9
109.2	7.095	6.083	19.1	24.629	-5.559	4.182	34.9
111.3	7.207	6.195	19.1	24.629	-5.560	4.238	34.9
113.4	7.319	6.307	19.1	24.629	-5.560	4.294	34.9
115.5	7.431	6.419	19.1	24.629	-5.561	4.350	34.9
117.6	7.543	6.531	19.1	24.628	-5.562	4.406	34.9
119.7	7.655	6.643	19.1	24.628	-5.562	4.462	34.9
121.8	7.767	6.755	19.1	24.628	-5.563	4.518	34.9
123.9	7.879	6.867	19.1	24.628	-5.564	4.574	34.9
126.2	8.000	6.988	19.1	24.628	-5.564	4.635	34.9

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TANK NO. 052 : LIQUID CARGO TANK SB : NET VOLUME= 126.17 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.012	0.000	0.0	0.000	0.000	0.000	0.0
2.1	1.328	0.316	13.8	24.679	4.439	1.229	14.0
4.2	1.467	0.455	16.3	24.713	4.814	1.314	23.6
6.3	1.595	0.583	17.2	24.717	4.990	1.387	26.9
8.4	1.716	0.704	18.0	24.710	5.101	1.454	29.8
10.5	1.833	0.821	18.6	24.699	5.182	1.518	32.2
12.6	1.947	0.935	19.0	24.688	5.245	1.580	34.1
14.7	2.059	1.047	19.1	24.678	5.295	1.641	34.7
16.8	2.171	1.159	19.1	24.671	5.333	1.700	34.9
18.9	2.283	1.271	19.1	24.666	5.362	1.759	34.9
21.0	2.395	1.383	19.1	24.661	5.386	1.817	34.9
23.1	2.507	1.495	19.1	24.658	5.406	1.874	34.9
25.2	2.619	1.607	19.1	24.655	5.422	1.932	34.9
27.3	2.731	1.719	19.1	24.652	5.436	1.989	34.9
29.4	2.843	1.830	19.1	24.650	5.447	2.046	34.9
31.5	2.955	1.942	19.1	24.648	5.457	2.103	34.9
33.6	3.067	2.054	19.1	24.646	5.466	2.159	34.9
35.7	3.178	2.166	19.1	24.645	5.474	2.216	34.9
37.8	3.290	2.278	19.1	24.643	5.481	2.273	34.9
39.9	3.402	2.390	19.1	24.642	5.487	2.329	34.9
42.0	3.514	2.502	19.1	24.641	5.493	2.386	34.9
44.1	3.626	2.614	19.1	24.640	5.498	2.442	34.9
46.2	3.738	2.726	19.1	24.639	5.503	2.498	34.9
48.3	3.850	2.838	19.1	24.639	5.507	2.555	34.9
50.4	3.962	2.950	19.1	24.638	5.511	2.611	34.9
52.5	4.074	3.062	19.1	24.637	5.514	2.667	34.9
54.6	4.186	3.173	19.1	24.637	5.518	2.724	34.9
56.7	4.298	3.285	19.1	24.636	5.521	2.780	34.9
58.8	4.410	3.397	19.1	24.636	5.524	2.836	34.9
60.9	4.521	3.509	19.1	24.635	5.526	2.892	34.9
63.0	4.633	3.621	19.1	24.635	5.529	2.948	34.9
65.1	4.745	3.733	19.1	24.634	5.531	3.004	34.9
67.2	4.857	3.845	19.1	24.634	5.533	3.061	34.9
69.3	4.969	3.957	19.1	24.633	5.535	3.117	34.9
71.4	5.081	4.069	19.1	24.633	5.537	3.173	34.9
73.5	5.193	4.181	19.1	24.633	5.539	3.229	34.9
75.6	5.305	4.293	19.1	24.632	5.541	3.285	34.9
77.7	5.417	4.405	19.1	24.632	5.542	3.341	34.9
79.8	5.529	4.516	19.1	24.632	5.544	3.397	34.9
81.9	5.641	4.628	19.1	24.632	5.545	3.453	34.9
84.0	5.752	4.740	19.1	24.631	5.547	3.509	34.9
86.1	5.864	4.852	19.1	24.631	5.548	3.566	34.9
88.2	5.976	4.964	19.1	24.631	5.549	3.622	34.9
90.3	6.088	5.076	19.1	24.631	5.550	3.678	34.9
92.4	6.200	5.188	19.1	24.630	5.551	3.734	34.9
94.5	6.312	5.300	19.1	24.630	5.552	3.790	34.9
96.6	6.424	5.412	19.1	24.630	5.554	3.846	34.9
98.7	6.536	5.524	19.1	24.630	5.555	3.902	34.9
100.8	6.648	5.636	19.1	24.630	5.555	3.958	34.9
102.9	6.760	5.747	19.1	24.629	5.556	4.014	34.9
105.0	6.872	5.859	19.1	24.629	5.557	4.070	34.9
107.1	6.984	5.971	19.1	24.629	5.558	4.126	34.9
109.2	7.095	6.083	19.1	24.629	5.559	4.182	34.9
111.3	7.207	6.195	19.1	24.629	5.560	4.238	34.9
113.4	7.319	6.307	19.1	24.629	5.560	4.294	34.9
115.5	7.431	6.419	19.1	24.629	5.561	4.350	34.9
117.6	7.543	6.531	19.1	24.628	5.562	4.406	34.9
119.7	7.655	6.643	19.1	24.628	5.562	4.462	34.9
121.8	7.767	6.755	19.1	24.628	5.563	4.518	34.9
123.9	7.879	6.867	19.1	24.628	5.564	4.574	34.9
126.2	8.000	6.988	19.1	24.628	5.564	4.635	34.9

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TANK NO. 053 : STORAGE TANK PS : NET VOLUME= 128.57 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.110	0.000	0.1	21.483	-4.208	1.111	0.0
2.1	1.764	0.654	6.1	21.053	-4.842	1.526	2.3
4.2	2.038	0.929	9.8	20.827	-5.019	1.719	6.6
6.3	2.231	1.121	12.7	20.654	-5.145	1.859	10.6
8.4	2.385	1.275	15.1	20.508	-5.234	1.973	14.2
10.5	2.519	1.409	17.0	20.385	-5.309	2.069	17.1
12.6	2.639	1.530	18.6	20.281	-5.372	2.154	19.3
14.7	2.750	1.641	20.0	20.190	-5.426	2.231	21.4
16.8	2.855	1.745	21.1	20.110	-5.474	2.303	23.0
18.9	2.954	1.844	22.1	20.038	-5.517	2.370	24.5
21.0	3.048	1.939	23.0	19.973	-5.556	2.433	25.7
23.1	3.140	2.030	23.8	19.913	-5.590	2.493	26.9
25.2	3.229	2.119	24.4	19.858	-5.622	2.551	27.8
27.3	3.316	2.206	25.0	19.808	-5.651	2.606	28.7
29.4	3.400	2.291	25.5	19.760	-5.678	2.660	29.7
31.5	3.483	2.374	26.1	19.716	-5.704	2.712	31.0
33.6	3.565	2.455	26.6	19.674	-5.728	2.763	32.5
35.7	3.645	2.535	26.8	19.635	-5.750	2.812	33.0
37.8	3.725	2.615	26.9	19.600	-5.771	2.861	33.2
39.9	3.804	2.694	27.0	19.568	-5.790	2.908	33.4
42.0	3.883	2.773	27.1	19.538	-5.807	2.955	33.6
44.1	3.962	2.852	27.2	19.510	-5.823	3.001	33.9
46.2	4.041	2.931	27.3	19.485	-5.837	3.047	34.1
48.3	4.119	3.010	27.3	19.461	-5.850	3.092	34.1
50.4	4.198	3.088	27.3	19.440	-5.863	3.136	34.1
52.5	4.276	3.167	27.3	19.420	-5.874	3.180	34.1
54.6	4.355	3.245	27.3	19.402	-5.884	3.224	34.1
56.7	4.434	3.324	27.3	19.385	-5.894	3.267	34.1
58.8	4.512	3.402	27.3	19.369	-5.903	3.310	34.1
60.9	4.591	3.481	27.3	19.354	-5.911	3.353	34.1
63.0	4.669	3.560	27.3	19.341	-5.919	3.396	34.1
65.1	4.748	3.638	27.3	19.328	-5.926	3.438	34.1
67.2	4.826	3.717	27.3	19.316	-5.933	3.480	34.1
69.3	4.905	3.795	27.3	19.305	-5.939	3.522	34.1
71.4	4.984	3.874	27.3	19.294	-5.945	3.564	34.1
73.5	5.079	3.969	21.6	19.267	-5.951	3.606	26.6
75.6	5.178	4.068	21.6	19.238	-5.955	3.648	26.6
77.7	5.277	4.168	21.6	19.210	-5.960	3.691	26.6
79.8	5.377	4.267	21.6	19.183	-5.964	3.734	26.6
81.9	5.476	4.366	21.6	19.158	-5.969	3.777	26.6
84.0	5.576	4.466	21.6	19.134	-5.972	3.821	26.6
86.1	5.675	4.565	21.6	19.111	-5.976	3.865	26.6
88.2	5.774	4.665	21.6	19.089	-5.980	3.909	26.6
90.3	5.874	4.764	21.6	19.069	-5.983	3.954	26.6
92.4	5.973	4.863	21.6	19.049	-5.986	3.999	26.6
94.5	6.073	4.963	21.6	19.030	-5.989	4.043	26.6
96.6	6.172	5.062	21.6	19.012	-5.992	4.089	26.6
98.7	6.271	5.162	21.6	18.995	-5.995	4.134	26.6
100.8	6.371	5.261	21.6	18.978	-5.998	4.180	26.6
102.9	6.470	5.360	21.6	18.962	-6.000	4.225	26.6
105.0	6.574	5.465	18.5	18.945	-6.003	4.271	22.8
107.1	6.690	5.580	18.5	18.925	-6.005	4.318	22.8
109.2	6.806	5.696	18.5	18.906	-6.007	4.364	22.8
111.3	6.922	5.812	18.5	18.887	-6.009	4.411	22.8
113.4	7.043	5.933	16.2	18.867	-6.011	4.459	20.0
115.5	7.175	6.066	16.2	18.842	-6.013	4.507	20.0
117.6	7.308	6.198	16.2	18.818	-6.015	4.556	20.0
119.7	7.440	6.331	16.2	18.795	-6.017	4.605	20.0
121.8	7.573	6.463	16.2	18.772	-6.019	4.655	20.0
123.9	7.705	6.596	16.2	18.751	-6.021	4.706	20.0
126.0	7.838	6.728	16.2	18.730	-6.023	4.757	20.0
128.1	7.970	6.861	16.2	18.710	-6.024	4.809	20.0
128.6	8.000	6.890	16.2	18.705	-6.025	4.820	20.0

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TANK NO. 054 : STORAGE TANK SB : NET VOLUME= 128.35 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.110	0.000	0.1	21.672	4.245	1.111	0.0
2.1	1.768	0.659	6.2	21.088	4.845	1.529	2.3
4.2	2.040	0.930	9.8	20.834	5.015	1.721	6.5
6.3	2.231	1.121	12.7	20.656	5.138	1.861	10.5
8.4	2.386	1.276	15.1	20.508	5.231	1.973	14.2
10.5	2.519	1.409	17.0	20.386	5.307	2.069	17.1
12.6	2.639	1.530	18.6	20.282	5.371	2.154	19.4
14.7	2.750	1.641	20.0	20.191	5.426	2.232	21.3
16.8	2.855	1.745	21.1	20.111	5.474	2.303	23.0
18.9	2.954	1.844	22.1	20.039	5.517	2.370	24.4
21.0	3.048	1.939	23.0	19.974	5.555	2.433	25.8
23.1	3.140	2.030	23.8	19.914	5.590	2.493	27.0
25.2	3.229	2.119	24.5	19.859	5.622	2.551	28.1
27.3	3.315	2.205	25.1	19.808	5.652	2.606	29.1
29.4	3.400	2.290	25.6	19.761	5.679	2.660	30.1
31.5	3.483	2.373	26.0	19.717	5.705	2.712	31.0
33.6	3.564	2.455	26.4	19.676	5.728	2.763	31.7
35.7	3.645	2.535	26.7	19.638	5.750	2.812	32.5
37.8	3.725	2.615	26.9	19.602	5.770	2.861	33.1
39.9	3.804	2.695	27.1	19.569	5.789	2.908	33.5
42.0	3.883	2.774	27.2	19.539	5.806	2.955	33.9
44.1	3.962	2.852	27.3	19.511	5.822	3.001	34.1
46.2	4.041	2.931	27.3	19.485	5.837	3.047	34.1
48.3	4.119	3.009	27.3	19.461	5.850	3.092	34.1
50.4	4.198	3.088	27.3	19.440	5.863	3.136	34.1
52.5	4.280	3.170	24.7	19.416	5.874	3.180	31.2
54.6	4.367	3.257	24.7	19.388	5.885	3.224	31.1
56.7	4.454	3.344	24.6	19.362	5.895	3.268	30.8
58.8	4.541	3.431	24.6	19.338	5.904	3.312	30.8
60.9	4.628	3.518	24.6	19.315	5.912	3.356	30.8
63.0	4.715	3.606	24.6	19.294	5.920	3.400	30.8
65.1	4.803	3.693	24.6	19.274	5.927	3.444	30.8
67.2	4.885	3.775	26.9	19.261	5.933	3.487	33.1
69.3	4.964	3.854	26.9	19.250	5.939	3.531	33.1
71.4	5.055	3.945	21.6	19.229	5.945	3.574	26.6
73.5	5.154	4.044	21.6	19.199	5.950	3.618	26.6
75.6	5.253	4.144	21.6	19.171	5.955	3.662	26.6
77.7	5.353	4.243	21.6	19.145	5.959	3.706	26.6
79.8	5.452	4.342	21.6	19.120	5.964	3.751	26.6
81.9	5.552	4.442	21.6	19.097	5.968	3.796	26.6
84.0	5.651	4.541	21.6	19.074	5.972	3.841	26.6
86.1	5.750	4.641	21.6	19.053	5.975	3.886	26.6
88.2	5.850	4.740	21.6	19.033	5.979	3.932	26.6
90.3	5.949	4.839	21.6	19.013	5.982	3.978	26.6
92.4	6.049	4.939	21.6	18.995	5.986	4.024	26.6
94.5	6.148	5.038	21.6	18.977	5.989	4.070	26.6
96.6	6.247	5.138	21.6	18.960	5.992	4.116	26.6
98.7	6.347	5.237	21.6	18.944	5.995	4.162	26.6
100.8	6.446	5.336	21.6	18.929	5.997	4.209	26.6
102.9	6.545	5.436	21.6	18.914	6.000	4.256	26.6
105.0	6.645	5.535	21.6	18.899	6.002	4.302	26.6
107.1	6.744	5.635	21.6	18.886	6.005	4.349	26.6
109.2	6.844	5.734	21.6	18.873	6.007	4.396	26.6
111.3	6.943	5.833	21.6	18.860	6.009	4.443	26.6
113.4	7.057	5.947	16.2	18.842	6.011	4.491	20.0
115.5	7.189	6.079	16.2	18.818	6.014	4.539	20.0
117.6	7.322	6.212	16.2	18.794	6.015	4.587	20.0
119.7	7.454	6.344	16.2	18.771	6.017	4.636	20.0
121.8	7.587	6.477	16.2	18.750	6.019	4.686	20.0
123.9	7.719	6.609	16.2	18.728	6.021	4.736	20.0
126.0	7.852	6.742	16.2	18.708	6.023	4.787	20.0
128.1	7.984	6.874	16.2	18.688	6.024	4.838	20.0
128.3	8.000	6.890	16.2	18.686	6.025	4.845	20.0

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TANK NO. 055 : LFL TANK PS : NET VOLUME= 77.65 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	5.000	0.000	36.0	12.250	-4.375	5.001	162.1
1.3	5.037	0.037	36.0	12.250	-4.375	5.018	162.1
2.6	5.074	0.074	36.0	12.250	-4.375	5.037	162.1
3.9	5.110	0.110	36.0	12.250	-4.375	5.055	162.1
5.2	5.147	0.147	36.0	12.250	-4.375	5.074	162.1
6.5	5.184	0.184	36.0	12.250	-4.375	5.092	162.1
7.8	5.221	0.221	36.0	12.250	-4.375	5.110	162.1
9.1	5.258	0.258	36.0	12.250	-4.375	5.129	162.1
10.4	5.295	0.295	36.0	12.250	-4.375	5.147	162.1
11.7	5.331	0.331	36.0	12.250	-4.375	5.166	162.1
13.0	5.368	0.368	36.0	12.250	-4.375	5.184	162.1
14.3	5.405	0.405	36.0	12.250	-4.375	5.203	162.1
15.6	5.442	0.442	36.0	12.250	-4.375	5.221	162.1
16.9	5.479	0.479	36.0	12.250	-4.375	5.239	162.1
18.2	5.516	0.516	36.0	12.250	-4.375	5.258	162.1
19.5	5.552	0.552	36.0	12.250	-4.375	5.276	162.1
20.8	5.589	0.589	36.0	12.250	-4.375	5.295	162.1
22.1	5.626	0.626	36.0	12.250	-4.375	5.313	162.1
23.4	5.663	0.663	36.0	12.250	-4.375	5.331	162.1
24.7	5.700	0.700	36.0	12.250	-4.375	5.350	162.1
26.0	5.737	0.737	36.0	12.250	-4.375	5.368	162.1
27.3	5.773	0.773	36.0	12.250	-4.375	5.387	162.1
28.6	5.810	0.810	36.0	12.250	-4.375	5.405	162.1
29.9	5.847	0.847	36.0	12.250	-4.375	5.424	162.1
31.2	5.884	0.884	36.0	12.250	-4.375	5.442	162.1
32.5	5.921	0.921	36.0	12.250	-4.375	5.460	162.1
33.8	5.958	0.958	36.0	12.250	-4.375	5.479	162.1
35.1	5.994	0.994	36.0	12.250	-4.375	5.497	162.1
36.4	6.031	1.031	36.0	12.250	-4.375	5.516	162.1
37.7	6.068	1.068	36.0	12.250	-4.375	5.534	162.1
39.0	6.105	1.105	36.0	12.250	-4.375	5.552	162.1
40.3	6.142	1.142	36.0	12.250	-4.375	5.571	162.1
41.6	6.179	1.179	36.0	12.250	-4.375	5.589	162.1
42.9	6.215	1.215	36.0	12.250	-4.375	5.608	162.1
44.2	6.252	1.252	36.0	12.250	-4.375	5.626	162.1
45.5	6.289	1.289	36.0	12.250	-4.375	5.645	162.1
46.8	6.326	1.326	36.0	12.250	-4.375	5.663	162.1
48.1	6.363	1.363	36.0	12.250	-4.375	5.681	162.1
49.4	6.400	1.400	36.0	12.250	-4.375	5.700	162.1
50.7	6.436	1.436	36.0	12.250	-4.375	5.718	162.1
52.0	6.473	1.473	36.0	12.250	-4.375	5.737	162.1
53.3	6.510	1.510	36.0	12.250	-4.375	5.755	162.1
54.6	6.547	1.547	36.0	12.250	-4.375	5.773	162.1
55.9	6.584	1.584	36.0	12.250	-4.375	5.792	162.1
57.2	6.621	1.621	36.0	12.250	-4.375	5.810	162.1
58.5	6.657	1.657	36.0	12.250	-4.375	5.829	162.1
59.8	6.694	1.694	36.0	12.250	-4.375	5.847	162.1
61.1	6.731	1.731	36.0	12.250	-4.375	5.866	162.1
62.4	6.768	1.768	36.0	12.250	-4.375	5.884	162.1
63.7	6.805	1.805	36.0	12.250	-4.375	5.902	162.1
65.0	6.842	1.842	36.0	12.250	-4.375	5.921	162.1
66.3	6.878	1.878	36.0	12.250	-4.375	5.939	162.1
67.6	6.915	1.915	36.0	12.250	-4.375	5.958	162.1
68.9	6.952	1.952	36.0	12.250	-4.375	5.976	162.1
70.2	6.989	1.989	36.0	12.250	-4.375	5.994	162.1
71.5	7.026	2.026	36.0	12.250	-4.375	6.013	162.1
72.8	7.063	2.063	36.0	12.250	-4.375	6.031	162.1
74.1	7.099	2.099	36.0	12.250	-4.375	6.050	162.1
75.4	7.136	2.136	36.0	12.250	-4.375	6.068	162.1
76.7	7.173	2.173	36.0	12.250	-4.375	6.087	162.1
77.6	7.200	2.200	36.0	12.250	-4.375	6.100	162.1

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TANK NO. 056 : LFL TANK SB : NET VOLUME= 77.65 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	5.000	0.000	36.0	12.250	4.375	5.001	162.1
1.3	5.037	0.037	36.0	12.250	4.375	5.018	162.1
2.6	5.074	0.074	36.0	12.250	4.375	5.037	162.1
3.9	5.110	0.110	36.0	12.250	4.375	5.055	162.1
5.2	5.147	0.147	36.0	12.250	4.375	5.074	162.1
6.5	5.184	0.184	36.0	12.250	4.375	5.092	162.1
7.8	5.221	0.221	36.0	12.250	4.375	5.110	162.1
9.1	5.258	0.258	36.0	12.250	4.375	5.129	162.1
10.4	5.295	0.295	36.0	12.250	4.375	5.147	162.1
11.7	5.331	0.331	36.0	12.250	4.375	5.166	162.1
13.0	5.368	0.368	36.0	12.250	4.375	5.184	162.1
14.3	5.405	0.405	36.0	12.250	4.375	5.203	162.1
15.6	5.442	0.442	36.0	12.250	4.375	5.221	162.1
16.9	5.479	0.479	36.0	12.250	4.375	5.239	162.1
18.2	5.516	0.516	36.0	12.250	4.375	5.258	162.1
19.5	5.552	0.552	36.0	12.250	4.375	5.276	162.1
20.8	5.589	0.589	36.0	12.250	4.375	5.295	162.1
22.1	5.626	0.626	36.0	12.250	4.375	5.313	162.1
23.4	5.663	0.663	36.0	12.250	4.375	5.331	162.1
24.7	5.700	0.700	36.0	12.250	4.375	5.350	162.1
26.0	5.737	0.737	36.0	12.250	4.375	5.368	162.1
27.3	5.773	0.773	36.0	12.250	4.375	5.387	162.1
28.6	5.810	0.810	36.0	12.250	4.375	5.405	162.1
29.9	5.847	0.847	36.0	12.250	4.375	5.424	162.1
31.2	5.884	0.884	36.0	12.250	4.375	5.442	162.1
32.5	5.921	0.921	36.0	12.250	4.375	5.460	162.1
33.8	5.958	0.958	36.0	12.250	4.375	5.479	162.1
35.1	5.994	0.994	36.0	12.250	4.375	5.497	162.1
36.4	6.031	1.031	36.0	12.250	4.375	5.516	162.1
37.7	6.068	1.068	36.0	12.250	4.375	5.534	162.1
39.0	6.105	1.105	36.0	12.250	4.375	5.552	162.1
40.3	6.142	1.142	36.0	12.250	4.375	5.571	162.1
41.6	6.179	1.179	36.0	12.250	4.375	5.589	162.1
42.9	6.215	1.215	36.0	12.250	4.375	5.608	162.1
44.2	6.252	1.252	36.0	12.250	4.375	5.626	162.1
45.5	6.289	1.289	36.0	12.250	4.375	5.645	162.1
46.8	6.326	1.326	36.0	12.250	4.375	5.663	162.1
48.1	6.363	1.363	36.0	12.250	4.375	5.681	162.1
49.4	6.400	1.400	36.0	12.250	4.375	5.700	162.1
50.7	6.436	1.436	36.0	12.250	4.375	5.718	162.1
52.0	6.473	1.473	36.0	12.250	4.375	5.737	162.1
53.3	6.510	1.510	36.0	12.250	4.375	5.755	162.1
54.6	6.547	1.547	36.0	12.250	4.375	5.773	162.1
55.9	6.584	1.584	36.0	12.250	4.375	5.792	162.1
57.2	6.621	1.621	36.0	12.250	4.375	5.810	162.1
58.5	6.657	1.657	36.0	12.250	4.375	5.829	162.1
59.8	6.694	1.694	36.0	12.250	4.375	5.847	162.1
61.1	6.731	1.731	36.0	12.250	4.375	5.866	162.1
62.4	6.768	1.768	36.0	12.250	4.375	5.884	162.1
63.7	6.805	1.805	36.0	12.250	4.375	5.902	162.1
65.0	6.842	1.842	36.0	12.250	4.375	5.921	162.1
66.3	6.878	1.878	36.0	12.250	4.375	5.939	162.1
67.6	6.915	1.915	36.0	12.250	4.375	5.958	162.1
68.9	6.952	1.952	36.0	12.250	4.375	5.976	162.1
70.2	6.989	1.989	36.0	12.250	4.375	5.994	162.1
71.5	7.026	2.026	36.0	12.250	4.375	6.013	162.1
72.8	7.063	2.063	36.0	12.250	4.375	6.031	162.1
74.1	7.099	2.099	36.0	12.250	4.375	6.050	162.1
75.4	7.136	2.136	36.0	12.250	4.375	6.068	162.1
76.7	7.173	2.173	36.0	12.250	4.375	6.087	162.1
77.6	7.200	2.200	36.0	12.250	4.375	6.100	162.1

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TANK NO. 061 : ROLL RED. TANK : NET VOLUME= 125.23 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	10.900	0.000	44.5	51.659	0.000	10.901	1325.3
2.1	10.949	0.049	44.5	51.659	0.000	10.924	1325.3
4.2	10.997	0.097	44.5	51.659	0.000	10.949	1325.3
6.3	11.046	0.146	44.5	51.659	0.000	10.973	1325.3
8.4	11.095	0.195	44.5	51.659	0.000	10.997	1325.3
10.5	11.143	0.243	44.5	51.659	0.000	11.022	1325.3
12.6	11.192	0.292	44.5	51.659	0.000	11.046	1325.3
14.7	11.240	0.340	44.5	51.659	0.000	11.070	1325.3
16.8	11.289	0.389	44.5	51.659	0.000	11.095	1325.3
18.9	11.338	0.438	44.5	51.659	0.000	11.119	1325.3
21.0	11.386	0.486	44.5	51.659	0.000	11.143	1325.3
23.1	11.435	0.535	44.5	51.659	0.000	11.167	1325.3
25.2	11.484	0.584	44.5	51.659	0.000	11.192	1325.3
27.3	11.532	0.632	44.5	51.659	0.000	11.216	1325.3
29.4	11.581	0.681	44.5	51.659	0.000	11.240	1325.3
31.5	11.629	0.729	44.5	51.659	0.000	11.265	1325.3
33.6	11.678	0.778	44.5	51.659	0.000	11.289	1325.3
35.7	11.727	0.827	44.5	51.659	0.000	11.313	1325.3
37.8	11.775	0.875	44.5	51.659	0.000	11.338	1325.3
39.9	11.824	0.924	44.5	51.659	0.000	11.362	1325.3
42.0	11.873	0.973	44.5	51.659	0.000	11.386	1325.3
44.1	11.921	1.021	44.5	51.659	0.000	11.411	1325.3
46.2	11.970	1.070	44.5	51.659	0.000	11.435	1325.3
48.3	12.018	1.118	44.5	51.659	0.000	11.459	1325.3
50.4	12.067	1.167	44.5	51.659	0.000	11.484	1325.3
52.5	12.116	1.216	44.5	51.659	0.000	11.508	1325.3
54.6	12.164	1.264	44.5	51.659	0.000	11.532	1325.3
56.7	12.213	1.313	44.5	51.659	0.000	11.556	1325.3
58.8	12.262	1.362	44.5	51.659	0.000	11.581	1325.3
60.9	12.310	1.410	44.5	51.659	0.000	11.605	1325.3
63.0	12.359	1.459	44.5	51.659	0.000	11.629	1325.3
65.1	12.407	1.507	44.5	51.659	0.000	11.654	1325.3
67.2	12.456	1.556	44.5	51.659	0.000	11.678	1325.3
69.3	12.505	1.605	44.5	51.659	0.000	11.702	1325.3
71.4	12.553	1.653	44.5	51.659	0.000	11.727	1325.3
73.5	12.602	1.702	44.5	51.659	0.000	11.751	1325.3
75.6	12.651	1.751	44.5	51.659	0.000	11.775	1325.3
77.7	12.699	1.799	44.5	51.659	0.000	11.800	1325.3
79.8	12.748	1.848	44.5	51.659	0.000	11.824	1325.3
81.9	12.797	1.897	44.5	51.659	0.000	11.848	1325.3
84.0	12.845	1.945	44.5	51.659	0.000	11.873	1325.3
86.1	12.894	1.994	44.5	51.659	0.000	11.897	1325.3
88.2	12.942	2.042	44.5	51.659	0.000	11.921	1325.3
90.3	12.991	2.091	44.5	51.659	0.000	11.946	1325.3
92.4	13.040	2.140	44.5	51.659	0.000	11.970	1325.3
94.5	13.088	2.188	44.5	51.659	0.000	11.994	1325.3
96.6	13.137	2.237	44.5	51.659	0.000	12.018	1325.3
98.7	13.186	2.286	44.5	51.659	0.000	12.043	1325.3
100.8	13.234	2.334	44.5	51.659	0.000	12.067	1325.3
102.9	13.283	2.383	44.5	51.659	0.000	12.091	1325.3
105.0	13.331	2.431	44.5	51.659	0.000	12.116	1325.3
107.1	13.380	2.480	44.5	51.659	0.000	12.140	1325.3
109.2	13.429	2.529	44.5	51.659	0.000	12.164	1325.3
111.3	13.477	2.577	44.5	51.659	0.000	12.189	1325.3
113.4	13.526	2.626	44.5	51.659	0.000	12.213	1325.3
115.5	13.575	2.675	44.5	51.659	0.000	12.237	1325.3
117.6	13.623	2.723	44.5	51.659	0.000	12.262	1325.3
119.7	13.672	2.772	44.5	51.659	0.000	12.286	1325.3
121.8	13.720	2.820	44.5	51.659	0.000	12.310	1325.3
123.9	13.769	2.869	44.5	51.659	0.000	12.335	1325.3
125.2	13.800	2.900	44.5	51.659	0.000	12.350	1325.3

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TANK NO. 062 : ROLL RED. TANK : NET VOLUME= 205.42 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	3.619	0.000	0.0	0.000	0.000	0.000	0.0
3.4	4.423	0.804	50.8	7.582	0.000	4.263	1082.0
6.8	4.492	0.872	51.5	7.533	0.000	4.360	1126.4
10.2	4.559	0.940	52.2	7.515	0.000	4.415	1169.5
13.6	4.626	1.007	52.9	7.506	0.000	4.460	1211.3
17.0	4.692	1.073	53.5	7.499	0.000	4.499	1251.1
20.4	4.757	1.138	54.1	7.494	0.000	4.537	1289.8
23.8	4.821	1.202	54.6	7.490	0.000	4.573	1326.6
27.2	4.885	1.266	55.1	7.487	0.000	4.608	1361.7
30.6	4.949	1.329	55.6	7.484	0.000	4.642	1395.4
34.0	5.011	1.392	56.1	7.481	0.000	4.676	1426.4
37.4	5.074	1.455	56.5	7.479	0.000	4.709	1455.8
40.8	5.136	1.516	56.8	7.477	0.000	4.742	1483.2
44.2	5.197	1.578	57.2	7.475	0.000	4.775	1508.6
47.6	5.258	1.639	57.5	7.473	0.000	4.807	1532.2
51.0	5.319	1.700	57.8	7.472	0.000	4.839	1553.9
54.4	5.380	1.760	58.0	7.470	0.000	4.871	1573.6
57.8	5.440	1.821	58.3	7.469	0.000	4.903	1591.8
61.2	5.500	1.881	58.5	7.467	0.000	4.934	1608.1
64.6	5.560	1.941	58.7	7.466	0.000	4.966	1623.1
68.0	5.619	2.000	58.8	7.465	0.000	4.997	1636.2
71.4	5.679	2.060	59.0	7.463	0.000	5.028	1647.8
74.8	5.738	2.119	59.1	7.462	0.000	5.059	1658.2
78.2	5.797	2.178	59.2	7.461	0.000	5.090	1666.6
81.6	5.857	2.237	59.3	7.460	0.000	5.120	1674.2
85.0	5.916	2.296	59.4	7.459	0.000	5.151	1679.9
88.4	5.975	2.355	59.5	7.458	0.000	5.182	1684.8
91.8	6.033	2.414	59.5	7.457	0.000	5.212	1687.9
95.2	6.092	2.473	59.5	7.456	0.000	5.242	1690.3
98.6	6.151	2.532	59.5	7.456	0.000	5.273	1692.0
102.0	6.210	2.591	59.6	7.455	0.000	5.303	1693.1
105.4	6.269	2.650	59.6	7.454	0.000	5.333	1693.7
108.8	6.328	2.709	59.6	7.453	0.000	5.363	1693.9
112.2	6.387	2.768	59.6	7.453	0.000	5.394	1694.0
115.6	6.446	2.826	59.6	7.452	0.000	5.424	1694.0
119.0	6.504	2.885	59.6	7.451	0.000	5.454	1694.0
122.4	6.563	2.944	59.6	7.451	0.000	5.484	1694.0
125.8	6.622	3.003	59.6	7.450	0.000	5.514	1694.0
129.2	6.681	3.062	59.6	7.450	0.000	5.544	1694.0
132.6	6.740	3.121	59.6	7.449	0.000	5.573	1694.0
136.0	6.799	3.179	59.6	7.449	0.000	5.603	1694.0
139.4	6.857	3.238	59.6	7.449	0.000	5.633	1694.0
142.8	6.916	3.297	59.6	7.448	0.000	5.663	1694.0
146.2	6.975	3.356	59.6	7.448	0.000	5.693	1694.0
149.6	7.034	3.415	59.6	7.447	0.000	5.723	1694.0
153.0	7.093	3.474	59.6	7.447	0.000	5.753	1694.0
156.4	7.152	3.533	59.6	7.447	0.000	5.782	1694.0
159.8	7.210	3.591	59.6	7.446	0.000	5.812	1694.0
163.2	7.269	3.650	59.6	7.446	0.000	5.842	1694.0
166.6	7.328	3.709	59.6	7.446	0.000	5.872	1694.0
170.0	7.387	3.768	59.6	7.445	0.000	5.901	1694.0
173.4	7.446	3.827	59.6	7.445	0.000	5.931	1694.0
176.8	7.505	3.886	59.6	7.445	0.000	5.961	1694.0
180.2	7.564	3.944	59.6	7.445	0.000	5.990	1694.0
183.6	7.622	4.003	59.6	7.444	0.000	6.020	1694.0
187.0	7.681	4.062	59.6	7.444	0.000	6.050	1694.0
190.4	7.740	4.121	59.6	7.444	0.000	6.079	1694.0
193.8	7.799	4.180	59.6	7.444	0.000	6.109	1694.0
197.2	7.858	4.239	59.6	7.444	0.000	6.139	1694.0
200.6	7.917	4.298	59.6	7.443	0.000	6.168	1694.0
204.0	7.975	4.356	59.6	7.443	0.000	6.198	1694.0
205.4	8.000	4.381	59.6	7.443	0.000	6.210	1694.0

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TANK NO. 201 : DRY BULK TANK : NET VOLUME= 64.97 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.346	0.000	0.1	36.750	0.000	1.348	0.0
1.1	1.850	0.504	3.0	36.750	0.000	1.655	0.8
2.2	2.144	0.798	4.3	36.750	0.000	1.820	1.5
3.3	2.373	1.027	5.6	36.750	0.000	1.968	2.5
4.4	2.557	1.210	6.7	36.750	0.000	2.093	3.6
5.5	2.712	1.366	7.7	36.750	0.000	2.202	4.8
6.6	2.849	1.502	8.7	36.750	0.000	2.298	6.1
7.7	2.971	1.625	9.6	36.750	0.000	2.386	7.4
8.8	3.082	1.736	10.5	36.750	0.000	2.466	8.8
9.9	3.185	1.839	11.3	36.750	0.000	2.540	10.2
11.0	3.281	1.935	12.1	36.750	0.000	2.610	11.8
12.1	3.370	2.024	12.9	36.750	0.000	2.675	13.3
13.2	3.455	2.108	13.7	36.750	0.000	2.736	14.9
14.3	3.534	2.188	14.4	36.750	0.000	2.795	16.6
15.4	3.610	2.264	15.1	36.750	0.000	2.850	18.1
16.5	3.684	2.337	15.5	36.750	0.000	2.903	19.1
17.6	3.756	2.409	15.7	36.750	0.000	2.954	19.5
18.7	3.827	2.481	15.7	36.750	0.000	3.004	19.6
19.8	3.899	2.552	15.7	36.750	0.000	3.051	19.6
20.9	3.970	2.624	15.7	36.750	0.000	3.098	19.6
22.0	4.042	2.696	15.7	36.750	0.000	3.143	19.6
23.1	4.113	2.767	15.7	36.750	0.000	3.188	19.6
24.2	4.185	2.839	15.7	36.750	0.000	3.231	19.6
25.3	4.256	2.910	15.7	36.750	0.000	3.274	19.6
26.4	4.328	2.982	15.7	36.750	0.000	3.317	19.6
27.5	4.400	3.053	15.7	36.750	0.000	3.359	19.6
28.6	4.471	3.125	15.7	36.750	0.000	3.400	19.6
29.7	4.543	3.196	15.7	36.750	0.000	3.441	19.6
30.8	4.614	3.268	15.7	36.750	0.000	3.482	19.6
31.9	4.686	3.339	15.7	36.750	0.000	3.522	19.6
33.0	4.757	3.411	15.7	36.750	0.000	3.562	19.6
34.1	4.829	3.483	15.7	36.750	0.000	3.602	19.6
35.2	4.900	3.554	15.7	36.750	0.000	3.641	19.6
36.3	4.972	3.626	15.7	36.750	0.000	3.680	19.6
37.4	5.043	3.697	15.7	36.750	0.000	3.719	19.6
38.5	5.115	3.769	15.7	36.750	0.000	3.758	19.6
39.6	5.187	3.840	15.7	36.750	0.000	3.797	19.6
40.7	5.258	3.912	15.7	36.750	0.000	3.836	19.6
41.8	5.330	3.983	15.7	36.750	0.000	3.874	19.6
42.9	5.401	4.055	15.7	36.750	0.000	3.912	19.6
44.0	5.473	4.126	15.7	36.750	0.000	3.950	19.6
45.1	5.544	4.198	15.7	36.750	0.000	3.988	19.6
46.2	5.616	4.269	15.7	36.750	0.000	4.026	19.6
47.3	5.687	4.341	15.7	36.750	0.000	4.064	19.6
48.4	5.759	4.413	15.7	36.750	0.000	4.102	19.6
49.5	5.830	4.484	15.7	36.750	0.000	4.139	19.6
50.6	5.902	4.556	15.7	36.750	0.000	4.177	19.6
51.7	5.973	4.627	15.7	36.750	0.000	4.214	19.6
52.8	6.045	4.699	15.7	36.750	0.000	4.252	19.6
53.9	6.117	4.770	15.7	36.750	0.000	4.289	19.6
55.0	6.188	4.842	15.6	36.750	0.000	4.326	19.3
56.1	6.261	4.915	15.4	36.750	0.000	4.364	18.8
57.2	6.335	4.988	15.0	36.750	0.000	4.401	18.0
58.3	6.410	5.064	14.6	36.750	0.000	4.438	16.9
59.4	6.489	5.143	13.9	36.750	0.000	4.475	15.5
60.5	6.572	5.226	13.0	36.750	0.000	4.512	13.6
61.6	6.662	5.316	11.8	36.750	0.000	4.550	11.0
62.7	6.766	5.420	9.8	36.750	0.000	4.588	7.6
63.8	6.900	5.553	7.0	36.750	0.000	4.627	4.0
65.0	7.231	5.885	0.0	36.750	0.000	4.670	0.0

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TANK NO. 202 : DRY BULK TANK : NET VOLUME= 64.97 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.346	0.000	0.1	31.850	0.000	1.348	0.0
1.1	1.850	0.504	3.0	31.850	0.000	1.655	0.8
2.2	2.144	0.798	4.3	31.850	0.000	1.820	1.5
3.3	2.373	1.027	5.6	31.850	0.000	1.968	2.5
4.4	2.557	1.210	6.7	31.850	0.000	2.093	3.6
5.5	2.712	1.366	7.7	31.850	0.000	2.202	4.8
6.6	2.849	1.502	8.7	31.850	0.000	2.298	6.1
7.7	2.971	1.625	9.6	31.850	0.000	2.386	7.4
8.8	3.082	1.736	10.5	31.850	0.000	2.466	8.8
9.9	3.185	1.839	11.3	31.850	0.000	2.540	10.2
11.0	3.281	1.935	12.1	31.850	0.000	2.610	11.8
12.1	3.370	2.024	12.9	31.850	0.000	2.675	13.3
13.2	3.455	2.108	13.7	31.850	0.000	2.736	14.9
14.3	3.534	2.188	14.4	31.850	0.000	2.795	16.6
15.4	3.610	2.264	15.1	31.850	0.000	2.850	18.1
16.5	3.684	2.337	15.5	31.850	0.000	2.903	19.1
17.6	3.756	2.409	15.7	31.850	0.000	2.954	19.5
18.7	3.827	2.481	15.7	31.850	0.000	3.004	19.6
19.8	3.899	2.552	15.7	31.850	0.000	3.051	19.6
20.9	3.970	2.624	15.7	31.850	0.000	3.098	19.6
22.0	4.042	2.696	15.7	31.850	0.000	3.143	19.6
23.1	4.113	2.767	15.7	31.850	0.000	3.188	19.6
24.2	4.185	2.839	15.7	31.850	0.000	3.231	19.6
25.3	4.256	2.910	15.7	31.850	0.000	3.274	19.6
26.4	4.328	2.982	15.7	31.850	0.000	3.317	19.6
27.5	4.400	3.053	15.7	31.850	0.000	3.359	19.6
28.6	4.471	3.125	15.7	31.850	0.000	3.400	19.6
29.7	4.543	3.196	15.7	31.850	0.000	3.441	19.6
30.8	4.614	3.268	15.7	31.850	0.000	3.482	19.6
31.9	4.686	3.339	15.7	31.850	0.000	3.522	19.6
33.0	4.757	3.411	15.7	31.850	0.000	3.562	19.6
34.1	4.829	3.483	15.7	31.850	0.000	3.602	19.6
35.2	4.900	3.554	15.7	31.850	0.000	3.641	19.6
36.3	4.972	3.626	15.7	31.850	0.000	3.680	19.6
37.4	5.043	3.697	15.7	31.850	0.000	3.719	19.6
38.5	5.115	3.769	15.7	31.850	0.000	3.758	19.6
39.6	5.187	3.840	15.7	31.850	0.000	3.797	19.6
40.7	5.258	3.912	15.7	31.850	0.000	3.836	19.6
41.8	5.330	3.983	15.7	31.850	0.000	3.874	19.6
42.9	5.401	4.055	15.7	31.850	0.000	3.912	19.6
44.0	5.473	4.126	15.7	31.850	0.000	3.950	19.6
45.1	5.544	4.198	15.7	31.850	0.000	3.988	19.6
46.2	5.616	4.269	15.7	31.850	0.000	4.026	19.6
47.3	5.687	4.341	15.7	31.850	0.000	4.064	19.6
48.4	5.759	4.413	15.7	31.850	0.000	4.102	19.6
49.5	5.830	4.484	15.7	31.850	0.000	4.139	19.6
50.6	5.902	4.556	15.7	31.850	0.000	4.177	19.6
51.7	5.973	4.627	15.7	31.850	0.000	4.214	19.6
52.8	6.045	4.699	15.7	31.850	0.000	4.252	19.6
53.9	6.117	4.770	15.7	31.850	0.000	4.289	19.6
55.0	6.188	4.842	15.6	31.850	0.000	4.326	19.3
56.1	6.261	4.915	15.4	31.850	0.000	4.364	18.8
57.2	6.335	4.988	15.0	31.850	0.000	4.401	18.0
58.3	6.410	5.064	14.6	31.850	0.000	4.438	16.9
59.4	6.489	5.143	13.9	31.850	0.000	4.475	15.5
60.5	6.572	5.226	13.0	31.850	0.000	4.512	13.6
61.6	6.662	5.316	11.8	31.850	0.000	4.550	11.0
62.7	6.766	5.420	9.8	31.850	0.000	4.588	7.6
63.8	6.900	5.553	7.0	31.850	0.000	4.627	4.0
65.0	7.231	5.885	0.0	31.850	0.000	4.670	0.0

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TANK NO. 203 : DRY BULK TANK : NET VOLUME= 64.97 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.346	0.000	0.1	26.950	0.000	1.348	0.0
1.1	1.850	0.504	3.0	26.950	0.000	1.655	0.8
2.2	2.144	0.798	4.3	26.950	0.000	1.820	1.5
3.3	2.373	1.027	5.6	26.950	0.000	1.968	2.5
4.4	2.557	1.210	6.7	26.950	0.000	2.093	3.6
5.5	2.712	1.366	7.7	26.950	0.000	2.202	4.8
6.6	2.849	1.502	8.7	26.950	0.000	2.298	6.1
7.7	2.971	1.625	9.6	26.950	0.000	2.386	7.4
8.8	3.082	1.736	10.5	26.950	0.000	2.466	8.8
9.9	3.185	1.839	11.3	26.950	0.000	2.540	10.2
11.0	3.281	1.935	12.1	26.950	0.000	2.610	11.8
12.1	3.370	2.024	12.9	26.950	0.000	2.675	13.3
13.2	3.455	2.108	13.7	26.950	0.000	2.736	14.9
14.3	3.534	2.188	14.4	26.950	0.000	2.795	16.6
15.4	3.610	2.264	15.1	26.950	0.000	2.850	18.1
16.5	3.684	2.337	15.5	26.950	0.000	2.903	19.1
17.6	3.756	2.409	15.7	26.950	0.000	2.954	19.5
18.7	3.827	2.481	15.7	26.950	0.000	3.004	19.6
19.8	3.899	2.552	15.7	26.950	0.000	3.051	19.6
20.9	3.970	2.624	15.7	26.950	0.000	3.098	19.6
22.0	4.042	2.696	15.7	26.950	0.000	3.143	19.6
23.1	4.113	2.767	15.7	26.950	0.000	3.188	19.6
24.2	4.185	2.839	15.7	26.950	0.000	3.231	19.6
25.3	4.256	2.910	15.7	26.950	0.000	3.274	19.6
26.4	4.328	2.982	15.7	26.950	0.000	3.317	19.6
27.5	4.400	3.053	15.7	26.950	0.000	3.359	19.6
28.6	4.471	3.125	15.7	26.950	0.000	3.400	19.6
29.7	4.543	3.196	15.7	26.950	0.000	3.441	19.6
30.8	4.614	3.268	15.7	26.950	0.000	3.482	19.6
31.9	4.686	3.339	15.7	26.950	0.000	3.522	19.6
33.0	4.757	3.411	15.7	26.950	0.000	3.562	19.6
34.1	4.829	3.483	15.7	26.950	0.000	3.602	19.6
35.2	4.900	3.554	15.7	26.950	0.000	3.641	19.6
36.3	4.972	3.626	15.7	26.950	0.000	3.680	19.6
37.4	5.043	3.697	15.7	26.950	0.000	3.719	19.6
38.5	5.115	3.769	15.7	26.950	0.000	3.758	19.6
39.6	5.187	3.840	15.7	26.950	0.000	3.797	19.6
40.7	5.258	3.912	15.7	26.950	0.000	3.836	19.6
41.8	5.330	3.983	15.7	26.950	0.000	3.874	19.6
42.9	5.401	4.055	15.7	26.950	0.000	3.912	19.6
44.0	5.473	4.126	15.7	26.950	0.000	3.950	19.6
45.1	5.544	4.198	15.7	26.950	0.000	3.988	19.6
46.2	5.616	4.269	15.7	26.950	0.000	4.026	19.6
47.3	5.687	4.341	15.7	26.950	0.000	4.064	19.6
48.4	5.759	4.413	15.7	26.950	0.000	4.102	19.6
49.5	5.830	4.484	15.7	26.950	0.000	4.139	19.6
50.6	5.902	4.556	15.7	26.950	0.000	4.177	19.6
51.7	5.973	4.627	15.7	26.950	0.000	4.214	19.6
52.8	6.045	4.699	15.7	26.950	0.000	4.252	19.6
53.9	6.117	4.770	15.7	26.950	0.000	4.289	19.6
55.0	6.188	4.842	15.6	26.950	0.000	4.326	19.3
56.1	6.261	4.915	15.4	26.950	0.000	4.364	18.8
57.2	6.335	4.988	15.0	26.950	0.000	4.401	18.0
58.3	6.410	5.064	14.6	26.950	0.000	4.438	16.9
59.4	6.489	5.143	13.9	26.950	0.000	4.475	15.5
60.5	6.572	5.226	13.0	26.950	0.000	4.512	13.6
61.6	6.662	5.316	11.8	26.950	0.000	4.550	11.0
62.7	6.766	5.420	9.8	26.950	0.000	4.588	7.6
63.8	6.900	5.553	7.0	26.950	0.000	4.627	4.0
65.0	7.231	5.885	0.0	26.950	0.000	4.670	0.0

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TANK NO. 204 : DRY BULK TANK : NET VOLUME= 64.97 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.346	0.000	0.1	22.050	0.000	1.348	0.0
1.1	1.850	0.504	3.0	22.050	0.000	1.655	0.8
2.2	2.144	0.798	4.3	22.050	0.000	1.820	1.5
3.3	2.373	1.027	5.6	22.050	0.000	1.968	2.5
4.4	2.557	1.210	6.7	22.050	0.000	2.093	3.6
5.5	2.712	1.366	7.7	22.050	0.000	2.202	4.8
6.6	2.849	1.502	8.7	22.050	0.000	2.298	6.1
7.7	2.971	1.625	9.6	22.050	0.000	2.386	7.4
8.8	3.082	1.736	10.5	22.050	0.000	2.466	8.8
9.9	3.185	1.839	11.3	22.050	0.000	2.540	10.2
11.0	3.281	1.935	12.1	22.050	0.000	2.610	11.8
12.1	3.370	2.024	12.9	22.050	0.000	2.675	13.3
13.2	3.455	2.108	13.7	22.050	0.000	2.736	14.9
14.3	3.534	2.188	14.4	22.050	0.000	2.795	16.6
15.4	3.610	2.264	15.1	22.050	0.000	2.850	18.1
16.5	3.684	2.337	15.5	22.050	0.000	2.903	19.1
17.6	3.756	2.409	15.7	22.050	0.000	2.954	19.5
18.7	3.827	2.481	15.7	22.050	0.000	3.004	19.6
19.8	3.899	2.552	15.7	22.050	0.000	3.051	19.6
20.9	3.970	2.624	15.7	22.050	0.000	3.098	19.6
22.0	4.042	2.696	15.7	22.050	0.000	3.143	19.6
23.1	4.113	2.767	15.7	22.050	0.000	3.188	19.6
24.2	4.185	2.839	15.7	22.050	0.000	3.231	19.6
25.3	4.256	2.910	15.7	22.050	0.000	3.274	19.6
26.4	4.328	2.982	15.7	22.050	0.000	3.317	19.6
27.5	4.400	3.053	15.7	22.050	0.000	3.359	19.6
28.6	4.471	3.125	15.7	22.050	0.000	3.400	19.6
29.7	4.543	3.196	15.7	22.050	0.000	3.441	19.6
30.8	4.614	3.268	15.7	22.050	0.000	3.482	19.6
31.9	4.686	3.339	15.7	22.050	0.000	3.522	19.6
33.0	4.757	3.411	15.7	22.050	0.000	3.562	19.6
34.1	4.829	3.483	15.7	22.050	0.000	3.602	19.6
35.2	4.900	3.554	15.7	22.050	0.000	3.641	19.6
36.3	4.972	3.626	15.7	22.050	0.000	3.680	19.6
37.4	5.043	3.697	15.7	22.050	0.000	3.719	19.6
38.5	5.115	3.769	15.7	22.050	0.000	3.758	19.6
39.6	5.187	3.840	15.7	22.050	0.000	3.797	19.6
40.7	5.258	3.912	15.7	22.050	0.000	3.836	19.6
41.8	5.330	3.983	15.7	22.050	0.000	3.874	19.6
42.9	5.401	4.055	15.7	22.050	0.000	3.912	19.6
44.0	5.473	4.126	15.7	22.050	0.000	3.950	19.6
45.1	5.544	4.198	15.7	22.050	0.000	3.988	19.6
46.2	5.616	4.269	15.7	22.050	0.000	4.026	19.6
47.3	5.687	4.341	15.7	22.050	0.000	4.064	19.6
48.4	5.759	4.413	15.7	22.050	0.000	4.102	19.6
49.5	5.830	4.484	15.7	22.050	0.000	4.139	19.6
50.6	5.902	4.556	15.7	22.050	0.000	4.177	19.6
51.7	5.973	4.627	15.7	22.050	0.000	4.214	19.6
52.8	6.045	4.699	15.7	22.050	0.000	4.252	19.6
53.9	6.117	4.770	15.7	22.050	0.000	4.289	19.6
55.0	6.188	4.842	15.6	22.050	0.000	4.326	19.3
56.1	6.261	4.915	15.4	22.050	0.000	4.364	18.8
57.2	6.335	4.988	15.0	22.050	0.000	4.401	18.0
58.3	6.410	5.064	14.6	22.050	0.000	4.438	16.9
59.4	6.489	5.143	13.9	22.050	0.000	4.475	15.5
60.5	6.572	5.226	13.0	22.050	0.000	4.512	13.6
61.6	6.662	5.316	11.8	22.050	0.000	4.550	11.0
62.7	6.766	5.420	9.8	22.050	0.000	4.588	7.6
63.8	6.900	5.553	7.0	22.050	0.000	4.627	4.0
65.0	7.231	5.885	0.0	22.050	0.000	4.670	0.0

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TANK NO. 252 : BLACK WATER TANK : NET VOLUME= 10.66 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	5.300	0.000	3.7	53.889	-6.915	5.301	2.1
0.2	5.356	0.056	3.7	53.889	-6.918	5.328	2.2
0.4	5.411	0.111	3.7	53.889	-6.922	5.356	2.2
0.6	5.467	0.167	3.7	53.889	-6.926	5.384	2.2
0.8	5.522	0.222	3.8	53.889	-6.929	5.411	2.3
1.0	5.577	0.277	3.8	53.889	-6.932	5.439	2.3
1.2	5.631	0.331	3.8	53.889	-6.936	5.466	2.3
1.4	5.685	0.385	3.8	53.889	-6.939	5.494	2.4
1.6	5.739	0.439	3.8	53.889	-6.942	5.521	2.4
1.8	5.793	0.493	3.8	53.889	-6.946	5.548	2.4
2.0	5.846	0.546	3.9	53.889	-6.949	5.575	2.5
2.2	5.900	0.600	3.9	53.890	-6.952	5.603	2.5
2.4	5.953	0.653	3.9	53.890	-6.955	5.629	2.5
2.6	6.006	0.706	3.9	53.890	-6.958	5.656	2.6
2.8	6.058	0.758	3.9	53.890	-6.961	5.683	2.6
3.0	6.111	0.811	3.9	53.890	-6.964	5.710	2.6
3.2	6.163	0.863	4.0	53.890	-6.967	5.737	2.6
3.4	6.215	0.915	4.0	53.890	-6.970	5.763	2.7
3.6	6.266	0.966	4.0	53.890	-6.973	5.790	2.7
3.8	6.318	1.018	4.0	53.890	-6.976	5.816	2.7
4.0	6.370	1.070	4.0	53.890	-6.979	5.843	2.8
4.2	6.421	1.121	4.0	53.890	-6.982	5.869	2.8
4.4	6.472	1.172	4.0	53.890	-6.984	5.895	2.8
4.6	6.523	1.223	4.1	53.890	-6.987	5.921	2.8
4.8	6.573	1.273	4.1	53.890	-6.990	5.947	2.9
5.0	6.624	1.324	4.1	53.891	-6.992	5.973	2.9
5.2	6.674	1.374	4.1	53.891	-6.995	5.999	2.9
5.4	6.725	1.425	4.1	53.891	-6.998	6.025	3.0
5.6	6.775	1.475	4.1	53.891	-7.000	6.051	3.0
5.8	6.825	1.525	4.1	53.891	-7.003	6.077	3.0
6.0	6.875	1.575	4.1	53.891	-7.005	6.103	3.0
6.2	6.924	1.624	4.2	53.891	-7.008	6.129	3.1
6.4	6.974	1.674	4.2	53.891	-7.010	6.154	3.1
6.6	7.023	1.723	4.2	53.891	-7.013	6.180	3.1
6.8	7.073	1.773	4.2	53.891	-7.015	6.205	3.1
7.0	7.122	1.822	4.2	53.891	-7.018	6.231	3.2
7.2	7.171	1.871	4.2	53.891	-7.020	6.256	3.2
7.4	7.220	1.920	4.2	53.891	-7.022	6.282	3.2
7.6	7.268	1.968	4.2	53.891	-7.025	6.307	3.2
7.8	7.317	2.017	4.2	53.891	-7.027	6.332	3.3
8.0	7.365	2.065	4.3	53.891	-7.029	6.357	3.3
8.2	7.414	2.114	4.3	53.892	-7.032	6.383	3.3
8.4	7.462	2.162	4.3	53.892	-7.034	6.408	3.3
8.6	7.510	2.210	4.3	53.892	-7.036	6.433	3.4
8.8	7.558	2.258	4.3	53.892	-7.038	6.458	3.4
9.0	7.606	2.306	4.3	53.892	-7.040	6.483	3.4
9.2	7.654	2.354	4.3	53.892	-7.043	6.508	3.4
9.4	7.702	2.402	4.3	53.892	-7.045	6.533	3.4
9.6	7.749	2.449	4.3	53.892	-7.047	6.557	3.5
9.8	7.797	2.497	4.3	53.892	-7.049	6.582	3.5
10.0	7.844	2.544	4.4	53.892	-7.051	6.607	3.5
10.2	7.892	2.592	4.4	53.892	-7.053	6.632	3.5
10.4	7.939	2.639	4.4	53.892	-7.055	6.656	3.6
10.6	7.986	2.686	4.4	53.892	-7.057	6.681	3.6
10.7	8.000	2.700	4.4	53.892	-7.058	6.688	3.6

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TANK NO. 254 : GREY WATER TANK : NET VOLUME= 32.26 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.400	0.000	2.4	54.630	-5.561	1.401	0.3
0.6	1.631	0.231	3.0	54.626	-5.620	1.520	0.5
1.2	1.824	0.424	3.5	54.625	-5.669	1.625	0.6
1.8	1.993	0.593	3.8	54.623	-5.711	1.720	0.8
2.4	2.147	0.747	4.2	54.623	-5.748	1.808	1.0
3.0	2.290	0.890	4.5	54.622	-5.782	1.890	1.2
3.6	2.425	1.025	4.7	54.622	-5.813	1.968	1.4
4.2	2.552	1.152	5.0	54.622	-5.841	2.043	1.6
4.8	2.674	1.274	5.2	54.621	-5.867	2.114	1.8
5.4	2.791	1.391	5.4	54.621	-5.891	2.183	2.0
6.0	2.904	1.504	5.6	54.621	-5.915	2.249	2.1
6.6	3.013	1.613	5.7	54.621	-5.936	2.314	2.3
7.2	3.119	1.719	5.9	54.621	-5.957	2.376	2.5
7.8	3.223	1.823	6.1	54.621	-5.976	2.437	2.7
8.4	3.323	1.923	6.2	54.622	-5.995	2.497	2.9
9.0	3.422	2.022	6.3	54.622	-6.013	2.556	3.0
9.6	3.519	2.119	6.5	54.622	-6.030	2.613	3.2
10.2	3.614	2.214	6.6	54.622	-6.046	2.669	3.4
10.8	3.707	2.307	6.7	54.622	-6.062	2.724	3.5
11.4	3.798	2.398	6.8	54.622	-6.077	2.778	3.7
12.0	3.889	2.489	6.9	54.622	-6.091	2.831	3.9
12.6	3.977	2.577	7.0	54.622	-6.105	2.884	4.0
13.2	4.065	2.665	7.1	54.622	-6.119	2.936	4.2
13.8	4.152	2.752	7.2	54.622	-6.132	2.987	4.3
14.4	4.237	2.837	7.3	54.622	-6.145	3.037	4.5
15.0	4.321	2.921	7.4	54.622	-6.157	3.087	4.6
15.6	4.405	3.005	7.5	54.622	-6.169	3.136	4.8
16.2	4.487	3.087	7.5	54.623	-6.180	3.184	4.9
16.8	4.569	3.169	7.6	54.623	-6.192	3.232	5.1
17.4	4.650	3.250	7.7	54.623	-6.203	3.280	5.2
18.0	4.730	3.330	7.8	54.623	-6.213	3.327	5.3
18.6	4.809	3.409	7.8	54.623	-6.223	3.373	5.5
19.2	4.888	3.488	7.9	54.623	-6.234	3.419	5.6
19.8	4.966	3.566	8.0	54.623	-6.243	3.465	5.7
20.4	5.044	3.644	8.0	54.623	-6.253	3.510	5.9
21.0	5.120	3.720	8.1	54.623	-6.262	3.555	6.0
21.6	5.197	3.797	8.1	54.623	-6.271	3.600	6.1
22.2	5.272	3.872	8.2	54.624	-6.280	3.644	6.3
22.8	5.418	4.018	3.3	54.635	-6.292	3.688	1.6
23.4	5.602	4.202	3.4	54.651	-6.305	3.735	1.7
24.0	5.781	4.381	3.5	54.667	-6.318	3.784	1.8
24.6	5.958	4.558	3.5	54.682	-6.331	3.835	1.9
25.2	6.131	4.731	3.6	54.697	-6.344	3.888	2.0
25.8	6.301	4.901	3.7	54.710	-6.357	3.942	2.1
26.4	6.469	5.069	3.7	54.724	-6.370	3.997	2.2
27.0	6.634	5.234	3.8	54.736	-6.383	4.054	2.3
27.6	6.797	5.397	3.8	54.748	-6.395	4.112	2.4
28.2	6.958	5.558	3.9	54.760	-6.407	4.171	2.5
28.8	7.116	5.716	3.9	54.771	-6.420	4.230	2.6
29.4	7.274	5.874	4.0	54.781	-6.432	4.291	2.6
30.0	7.429	6.029	4.0	54.792	-6.443	4.352	2.7
30.6	7.583	6.183	4.0	54.801	-6.455	4.414	2.8
31.2	7.735	6.335	4.1	54.811	-6.466	4.476	2.9
31.8	7.886	6.486	4.1	54.820	-6.478	4.539	3.0
32.3	8.000	6.600	4.1	54.827	-6.486	4.588	3.0

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TANK NO. 260 : WATER MIST TANK : NET VOLUME= 2.38 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	5.300	0.000	1.2	60.552	-4.296	5.301	0.1
0.1	5.384	0.084	1.2	60.552	-4.296	5.342	0.1
0.2	5.469	0.169	1.2	60.552	-4.296	5.384	0.1
0.3	5.553	0.253	1.2	60.552	-4.296	5.427	0.1
0.4	5.637	0.337	1.2	60.552	-4.296	5.469	0.1
0.5	5.722	0.422	1.2	60.552	-4.296	5.511	0.1
0.6	5.806	0.506	1.2	60.552	-4.296	5.553	0.1
0.7	5.890	0.590	1.2	60.552	-4.296	5.595	0.1
0.8	5.975	0.675	1.2	60.552	-4.296	5.637	0.1
0.9	6.059	0.759	1.2	60.552	-4.296	5.680	0.1
1.0	6.144	0.844	1.2	60.552	-4.296	5.722	0.1
1.1	6.228	0.928	1.2	60.552	-4.296	5.764	0.1
1.2	6.312	1.012	1.2	60.552	-4.296	5.806	0.1
1.3	6.397	1.097	1.2	60.552	-4.296	5.848	0.1
1.4	6.481	1.181	1.2	60.552	-4.296	5.890	0.1
1.5	6.565	1.265	1.2	60.552	-4.296	5.933	0.1
1.6	6.650	1.350	1.2	60.552	-4.296	5.975	0.1
1.7	6.734	1.434	1.2	60.552	-4.296	6.017	0.1
1.8	6.818	1.518	1.2	60.552	-4.296	6.059	0.1
1.9	6.903	1.603	1.2	60.552	-4.296	6.101	0.1
2.0	6.987	1.687	1.2	60.552	-4.296	6.144	0.1
2.1	7.071	1.771	1.2	60.552	-4.296	6.186	0.1
2.2	7.156	1.856	1.2	60.552	-4.296	6.228	0.1
2.3	7.240	1.940	1.2	60.552	-4.296	6.270	0.1
2.4	7.308	2.008	1.2	60.552	-4.296	6.304	0.1

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TANK NO. 263 : HYDRAULIC OIL STORE TANK : NET VOLUME= 5.12 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.090	0.000	0.0	0.000	0.000	0.000	0.0
0.1	1.362	0.272	1.3	43.950	-4.037	1.301	0.4
0.2	1.424	0.334	1.8	44.110	-4.182	1.348	0.7
0.3	1.481	0.391	1.8	44.177	-4.238	1.383	0.7
0.4	1.537	0.447	1.8	44.211	-4.265	1.414	0.7
0.5	1.594	0.503	1.8	44.231	-4.282	1.445	0.7
0.6	1.650	0.560	1.8	44.244	-4.293	1.474	0.7
0.7	1.706	0.616	1.8	44.254	-4.301	1.503	0.7
0.8	1.763	0.673	1.8	44.261	-4.307	1.532	0.7
0.9	1.819	0.729	1.8	44.267	-4.311	1.561	0.7
1.0	1.876	0.785	1.8	44.271	-4.315	1.590	0.7
1.1	1.932	0.842	1.8	44.275	-4.318	1.618	0.7
1.2	1.988	0.898	1.8	44.278	-4.320	1.647	0.7
1.3	2.045	0.955	1.8	44.280	-4.323	1.675	0.7
1.4	2.101	1.011	1.8	44.283	-4.324	1.704	0.7
1.5	2.158	1.067	1.8	44.285	-4.326	1.732	0.7
1.6	2.214	1.124	1.8	44.286	-4.327	1.760	0.7
1.7	2.270	1.180	1.8	44.288	-4.329	1.789	0.7
1.8	2.327	1.237	1.8	44.289	-4.330	1.817	0.7
1.9	2.383	1.293	1.8	44.290	-4.331	1.845	0.7
2.0	2.440	1.349	1.8	44.291	-4.331	1.874	0.7
2.1	2.496	1.406	1.8	44.292	-4.332	1.902	0.7
2.2	2.552	1.462	1.8	44.293	-4.333	1.930	0.7
2.3	2.609	1.519	1.8	44.294	-4.334	1.958	0.7
2.4	2.665	1.575	1.8	44.295	-4.334	1.987	0.7
2.5	2.722	1.631	1.8	44.295	-4.335	2.015	0.7
2.6	2.778	1.688	1.8	44.296	-4.335	2.043	0.7
2.7	2.834	1.744	1.8	44.297	-4.336	2.071	0.7
2.8	2.891	1.801	1.8	44.297	-4.336	2.100	0.7
2.9	2.947	1.857	1.8	44.298	-4.337	2.128	0.7
3.0	3.003	1.913	1.8	44.298	-4.337	2.156	0.7
3.1	3.060	1.970	1.8	44.298	-4.337	2.184	0.7
3.2	3.116	2.026	1.8	44.299	-4.338	2.213	0.7
3.3	3.173	2.082	1.8	44.299	-4.338	2.241	0.7
3.4	3.229	2.139	1.8	44.300	-4.338	2.269	0.7
3.5	3.285	2.195	1.8	44.300	-4.339	2.297	0.7
3.6	3.342	2.252	1.8	44.300	-4.339	2.326	0.7
3.7	3.398	2.308	1.8	44.301	-4.339	2.354	0.7
3.8	3.455	2.364	1.8	44.301	-4.339	2.382	0.7
3.9	3.511	2.421	1.8	44.301	-4.340	2.410	0.7
4.0	3.567	2.477	1.8	44.301	-4.340	2.439	0.7
4.1	3.624	2.534	1.8	44.302	-4.340	2.467	0.7
4.2	3.680	2.590	1.8	44.302	-4.340	2.495	0.7
4.3	3.737	2.646	1.8	44.302	-4.340	2.523	0.7
4.4	3.793	2.703	1.8	44.302	-4.340	2.551	0.7
4.5	3.849	2.759	1.8	44.302	-4.341	2.580	0.7
4.6	3.906	2.816	1.8	44.303	-4.341	2.608	0.7
4.7	3.962	2.872	1.8	44.303	-4.341	2.636	0.7
4.8	4.019	2.928	1.8	44.303	-4.341	2.664	0.7
4.9	4.075	2.985	1.8	44.303	-4.341	2.693	0.7
5.0	4.131	3.041	1.8	44.303	-4.341	2.721	0.7
5.1	4.188	3.098	1.8	44.304	-4.342	2.749	0.7
5.1	4.200	3.110	1.8	44.304	-4.342	2.755	0.7

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TANK NO. 271 : FO SETTLING TANK PS : NET VOLUME= 14.20 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	2.500	0.000	1.3	50.361	-6.072	2.501	0.1
0.3	2.726	0.226	1.4	50.362	-6.093	2.615	0.1
0.6	2.934	0.434	1.5	50.364	-6.114	2.723	0.2
0.9	3.129	0.629	1.6	50.365	-6.133	2.826	0.2
1.2	3.311	0.811	1.7	50.366	-6.152	2.925	0.2
1.5	3.484	0.984	1.8	50.367	-6.170	3.020	0.3
1.8	3.649	1.149	1.9	50.368	-6.187	3.111	0.3
2.1	3.806	1.306	2.0	50.369	-6.204	3.199	0.4
2.4	3.958	1.458	2.1	50.370	-6.220	3.285	0.4
2.7	4.103	1.603	2.2	50.370	-6.236	3.367	0.4
3.0	4.243	1.743	2.2	50.371	-6.251	3.448	0.5
3.3	4.379	1.879	2.3	50.372	-6.266	3.527	0.5
3.6	4.511	2.011	2.4	50.372	-6.281	3.603	0.6
3.9	4.639	2.139	2.5	50.373	-6.295	3.678	0.6
4.2	4.763	2.263	2.5	50.373	-6.309	3.751	0.7
4.5	4.885	2.385	2.6	50.374	-6.323	3.823	0.7
4.8	5.003	2.503	2.6	50.374	-6.336	3.893	0.8
5.1	5.119	2.619	2.7	50.374	-6.349	3.961	0.9
5.4	5.232	2.732	2.8	50.375	-6.362	4.029	0.9
5.7	5.342	2.842	2.8	50.375	-6.374	4.095	1.0
6.0	5.450	2.950	2.9	50.375	-6.387	4.160	1.0
6.3	5.557	3.057	2.9	50.376	-6.399	4.224	1.1
6.6	5.661	3.161	3.0	50.376	-6.411	4.287	1.2
6.9	5.763	3.263	3.1	50.376	-6.423	4.349	1.2
7.2	5.863	3.363	3.1	50.377	-6.434	4.410	1.3
7.5	5.962	3.462	3.2	50.377	-6.446	4.470	1.4
7.8	6.059	3.559	3.2	50.377	-6.457	4.529	1.4
8.1	6.155	3.655	3.3	50.378	-6.468	4.588	1.5
8.4	6.249	3.749	3.3	50.378	-6.479	4.645	1.5
8.7	6.343	3.843	3.3	50.378	-6.490	4.702	1.6
9.0	6.435	3.935	3.4	50.379	-6.500	4.759	1.6
9.3	6.526	4.026	3.4	50.379	-6.510	4.814	1.7
9.6	6.617	4.117	3.4	50.380	-6.520	4.869	1.7
9.9	6.708	4.208	3.4	50.380	-6.529	4.923	1.7
10.2	6.798	4.298	3.4	50.381	-6.537	4.977	1.7
10.5	6.888	4.388	3.4	50.381	-6.545	5.031	1.7
10.8	6.979	4.479	3.4	50.382	-6.553	5.083	1.7
11.1	7.069	4.569	3.4	50.382	-6.560	5.136	1.7
11.4	7.159	4.659	3.4	50.383	-6.567	5.188	1.7
11.7	7.249	4.749	3.4	50.383	-6.574	5.240	1.7
12.0	7.339	4.839	3.4	50.384	-6.580	5.291	1.7
12.3	7.429	4.929	3.4	50.384	-6.586	5.342	1.7
12.6	7.520	5.020	3.4	50.384	-6.592	5.393	1.7
12.9	7.610	5.110	3.4	50.385	-6.597	5.443	1.7
13.2	7.700	5.200	3.4	50.385	-6.603	5.494	1.7
13.5	7.790	5.290	3.4	50.385	-6.607	5.544	1.7
13.8	7.880	5.380	3.4	50.386	-6.612	5.593	1.7
14.1	7.970	5.470	3.4	50.386	-6.617	5.643	1.7
14.2	8.000	5.500	3.4	50.386	-6.618	5.659	1.7

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TANK NO. 272 : FO SETTLING TANK SB : NET VOLUME= 14.20 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	2.500	0.000	1.3	50.361	6.072	2.501	0.1
0.3	2.726	0.226	1.4	50.362	6.093	2.615	0.1
0.6	2.934	0.434	1.5	50.364	6.114	2.723	0.2
0.9	3.129	0.629	1.6	50.365	6.133	2.826	0.2
1.2	3.311	0.811	1.7	50.366	6.152	2.925	0.2
1.5	3.484	0.984	1.8	50.367	6.170	3.020	0.3
1.8	3.649	1.149	1.9	50.368	6.187	3.111	0.3
2.1	3.806	1.306	2.0	50.369	6.204	3.199	0.4
2.4	3.958	1.458	2.1	50.370	6.220	3.285	0.4
2.7	4.103	1.603	2.2	50.370	6.236	3.367	0.4
3.0	4.243	1.743	2.2	50.371	6.251	3.448	0.5
3.3	4.379	1.879	2.3	50.372	6.266	3.527	0.5
3.6	4.511	2.011	2.4	50.372	6.281	3.603	0.6
3.9	4.639	2.139	2.5	50.373	6.295	3.678	0.6
4.2	4.763	2.263	2.5	50.373	6.309	3.751	0.7
4.5	4.885	2.385	2.6	50.374	6.323	3.823	0.7
4.8	5.003	2.503	2.6	50.374	6.336	3.893	0.8
5.1	5.119	2.619	2.7	50.374	6.349	3.961	0.9
5.4	5.232	2.732	2.8	50.375	6.362	4.029	0.9
5.7	5.342	2.842	2.8	50.375	6.374	4.095	1.0
6.0	5.450	2.950	2.9	50.375	6.387	4.160	1.0
6.3	5.557	3.057	2.9	50.376	6.399	4.224	1.1
6.6	5.661	3.161	3.0	50.376	6.411	4.287	1.2
6.9	5.763	3.263	3.1	50.376	6.423	4.349	1.2
7.2	5.863	3.363	3.1	50.377	6.434	4.410	1.3
7.5	5.962	3.462	3.2	50.377	6.446	4.470	1.4
7.8	6.059	3.559	3.2	50.377	6.457	4.529	1.4
8.1	6.155	3.655	3.3	50.378	6.468	4.588	1.5
8.4	6.249	3.749	3.3	50.378	6.479	4.645	1.5
8.7	6.343	3.843	3.3	50.378	6.490	4.702	1.6
9.0	6.435	3.935	3.4	50.379	6.500	4.759	1.6
9.3	6.526	4.026	3.4	50.379	6.510	4.814	1.7
9.6	6.617	4.117	3.4	50.380	6.520	4.869	1.7
9.9	6.708	4.208	3.4	50.380	6.529	4.923	1.7
10.2	6.798	4.298	3.4	50.381	6.537	4.977	1.7
10.5	6.888	4.388	3.4	50.381	6.545	5.031	1.7
10.8	6.979	4.479	3.4	50.382	6.553	5.083	1.7
11.1	7.069	4.569	3.4	50.382	6.560	5.136	1.7
11.4	7.159	4.659	3.4	50.383	6.567	5.188	1.7
11.7	7.249	4.749	3.4	50.383	6.574	5.240	1.7
12.0	7.339	4.839	3.4	50.384	6.580	5.291	1.7
12.3	7.429	4.929	3.4	50.384	6.586	5.342	1.7
12.6	7.520	5.020	3.4	50.384	6.592	5.393	1.7
12.9	7.610	5.110	3.4	50.385	6.597	5.443	1.7
13.2	7.700	5.200	3.4	50.385	6.603	5.494	1.7
13.5	7.790	5.290	3.4	50.385	6.607	5.544	1.7
13.8	7.880	5.380	3.4	50.386	6.612	5.593	1.7
14.1	7.970	5.470	3.4	50.386	6.617	5.643	1.7
14.2	8.000	5.500	3.4	50.386	6.618	5.659	1.7

See ABS Singapore letter Reg 1499174 Dated 04-APR-2016

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TANK NO. 273 : FO SERVICE TANK PS : NET VOLUME= 17.65 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	2.500	0.000	1.1	51.996	-5.887	2.501	0.0
0.3	2.751	0.251	1.3	52.008	-5.910	2.629	0.1
0.6	2.969	0.469	1.5	52.018	-5.931	2.746	0.1
0.9	3.165	0.665	1.7	52.025	-5.951	2.853	0.1
1.2	3.343	0.843	1.8	52.031	-5.969	2.954	0.1
1.5	3.508	1.008	1.9	52.036	-5.987	3.048	0.2
1.8	3.663	1.163	2.1	52.041	-6.003	3.138	0.2
2.1	3.809	1.309	2.2	52.044	-6.019	3.224	0.2
2.4	3.947	1.447	2.3	52.048	-6.034	3.305	0.2
2.7	4.079	1.579	2.4	52.051	-6.049	3.384	0.3
3.0	4.206	1.706	2.5	52.054	-6.063	3.460	0.3
3.3	4.327	1.827	2.6	52.056	-6.077	3.533	0.4
3.6	4.444	1.944	2.7	52.059	-6.090	3.604	0.4
3.9	4.557	2.057	2.8	52.061	-6.103	3.673	0.4
4.2	4.667	2.167	2.9	52.063	-6.116	3.740	0.5
4.5	4.773	2.273	3.0	52.064	-6.128	3.806	0.5
4.8	4.876	2.376	3.0	52.066	-6.140	3.869	0.6
5.1	4.977	2.477	3.1	52.068	-6.152	3.932	0.6
5.4	5.075	2.575	3.2	52.069	-6.163	3.992	0.6
5.7	5.171	2.671	3.3	52.071	-6.174	4.052	0.7
6.0	5.264	2.764	3.3	52.072	-6.185	4.110	0.7
6.3	5.356	2.856	3.4	52.073	-6.196	4.167	0.8
6.6	5.445	2.945	3.5	52.074	-6.207	4.223	0.8
6.9	5.533	3.033	3.6	52.076	-6.217	4.279	0.9
7.2	5.619	3.119	3.6	52.077	-6.227	4.333	0.9
7.5	5.703	3.203	3.7	52.078	-6.237	4.386	1.0
7.8	5.786	3.286	3.8	52.079	-6.247	4.438	1.0
8.1	5.868	3.368	3.8	52.080	-6.257	4.489	1.1
8.4	5.948	3.448	3.9	52.081	-6.266	4.540	1.2
8.7	6.027	3.527	4.0	52.081	-6.276	4.590	1.2
9.0	6.104	3.604	4.0	52.082	-6.285	4.639	1.3
9.3	6.180	3.680	4.1	52.083	-6.294	4.688	1.3
9.6	6.256	3.756	4.1	52.084	-6.303	4.735	1.4
9.9	6.330	3.830	4.2	52.085	-6.312	4.783	1.4
10.2	6.403	3.903	4.3	52.085	-6.321	4.829	1.5
10.5	6.475	3.975	4.3	52.086	-6.330	4.875	1.6
10.8	6.546	4.046	4.4	52.087	-6.338	4.921	1.6
11.1	6.617	4.117	4.4	52.087	-6.347	4.966	1.7
11.4	6.686	4.186	4.5	52.088	-6.355	5.010	1.7
11.7	6.754	4.254	4.5	52.088	-6.363	5.054	1.8
12.0	6.822	4.322	4.6	52.089	-6.372	5.097	1.9
12.3	6.889	4.389	4.6	52.090	-6.380	5.140	1.9
12.6	6.955	4.455	4.7	52.090	-6.388	5.182	2.0
12.9	7.021	4.521	4.7	52.091	-6.396	5.224	2.0
13.2	7.086	4.586	4.8	52.091	-6.403	5.266	2.1
13.5	7.150	4.650	4.8	52.092	-6.411	5.307	2.1
13.8	7.214	4.714	4.9	52.093	-6.418	5.348	2.2
14.1	7.277	4.777	4.9	52.093	-6.426	5.388	2.3
14.4	7.340	4.840	4.9	52.094	-6.433	5.428	2.3
14.7	7.402	4.902	5.0	52.095	-6.440	5.468	2.4
15.0	7.464	4.964	5.0	52.096	-6.447	5.507	2.4
15.3	7.525	5.025	5.1	52.096	-6.454	5.546	2.5
15.6	7.586	5.086	5.1	52.097	-6.461	5.585	2.5
15.9	7.647	5.147	5.1	52.098	-6.468	5.623	2.5
16.2	7.708	5.208	5.1	52.098	-6.474	5.661	2.5
16.5	7.769	5.269	5.1	52.099	-6.480	5.699	2.5
16.8	7.829	5.329	5.1	52.100	-6.486	5.737	2.5
17.1	7.890	5.390	5.1	52.101	-6.492	5.774	2.5
17.4	7.950	5.450	5.1	52.101	-6.498	5.811	2.5
17.6	8.000	5.500	5.1	52.102	-6.502	5.841	2.5

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TANK NO. 274 : FO SERVICE TANK SB : NET VOLUME= 17.65 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	2.500	0.000	1.1	51.996	5.887	2.501	0.0
0.3	2.751	0.251	1.3	52.008	5.910	2.629	0.1
0.6	2.969	0.469	1.5	52.018	5.931	2.746	0.1
0.9	3.165	0.665	1.7	52.025	5.951	2.853	0.1
1.2	3.343	0.843	1.8	52.031	5.969	2.954	0.1
1.5	3.508	1.008	1.9	52.036	5.987	3.048	0.2
1.8	3.663	1.163	2.1	52.041	6.003	3.138	0.2
2.1	3.809	1.309	2.2	52.044	6.019	3.224	0.2
2.4	3.947	1.447	2.3	52.048	6.034	3.305	0.2
2.7	4.079	1.579	2.4	52.051	6.049	3.384	0.3
3.0	4.206	1.706	2.5	52.054	6.063	3.460	0.3
3.3	4.327	1.827	2.6	52.056	6.077	3.533	0.4
3.6	4.444	1.944	2.7	52.059	6.090	3.604	0.4
3.9	4.557	2.057	2.8	52.061	6.103	3.673	0.4
4.2	4.667	2.167	2.9	52.063	6.116	3.740	0.5
4.5	4.773	2.273	3.0	52.064	6.128	3.806	0.5
4.8	4.876	2.376	3.0	52.066	6.140	3.869	0.6
5.1	4.977	2.477	3.1	52.068	6.152	3.932	0.6
5.4	5.075	2.575	3.2	52.069	6.163	3.992	0.6
5.7	5.171	2.671	3.3	52.071	6.174	4.052	0.7
6.0	5.264	2.764	3.3	52.072	6.185	4.110	0.7
6.3	5.356	2.856	3.4	52.073	6.196	4.167	0.8
6.6	5.445	2.945	3.5	52.074	6.207	4.223	0.8
6.9	5.533	3.033	3.6	52.076	6.217	4.279	0.9
7.2	5.619	3.119	3.6	52.077	6.227	4.333	0.9
7.5	5.703	3.203	3.7	52.078	6.237	4.386	1.0
7.8	5.786	3.286	3.8	52.079	6.247	4.438	1.0
8.1	5.868	3.368	3.8	52.080	6.257	4.489	1.1
8.4	5.948	3.448	3.9	52.081	6.266	4.540	1.2
8.7	6.027	3.527	4.0	52.081	6.276	4.590	1.2
9.0	6.104	3.604	4.0	52.082	6.285	4.639	1.3
9.3	6.180	3.680	4.1	52.083	6.294	4.688	1.3
9.6	6.256	3.756	4.1	52.084	6.303	4.735	1.4
9.9	6.330	3.830	4.2	52.085	6.312	4.783	1.4
10.2	6.403	3.903	4.3	52.085	6.321	4.829	1.5
10.5	6.475	3.975	4.3	52.086	6.330	4.875	1.6
10.8	6.546	4.046	4.4	52.087	6.338	4.921	1.6
11.1	6.617	4.117	4.4	52.087	6.347	4.966	1.7
11.4	6.686	4.186	4.5	52.088	6.355	5.010	1.7
11.7	6.754	4.254	4.5	52.088	6.363	5.054	1.8
12.0	6.822	4.322	4.6	52.089	6.372	5.097	1.9
12.3	6.889	4.389	4.6	52.090	6.380	5.140	1.9
12.6	6.955	4.455	4.7	52.090	6.388	5.182	2.0
12.9	7.021	4.521	4.7	52.091	6.396	5.224	2.0
13.2	7.086	4.586	4.8	52.091	6.403	5.266	2.1
13.5	7.150	4.650	4.8	52.092	6.411	5.307	2.1
13.8	7.214	4.714	4.9	52.093	6.418	5.348	2.2
14.1	7.277	4.777	4.9	52.093	6.426	5.388	2.3
14.4	7.340	4.840	4.9	52.094	6.433	5.428	2.3
14.7	7.402	4.902	5.0	52.095	6.440	5.468	2.4
15.0	7.464	4.964	5.0	52.096	6.447	5.507	2.4
15.3	7.525	5.025	5.1	52.096	6.454	5.546	2.5
15.6	7.586	5.086	5.1	52.097	6.461	5.585	2.5
15.9	7.647	5.147	5.1	52.098	6.468	5.623	2.5
16.2	7.708	5.208	5.1	52.098	6.474	5.661	2.5
16.5	7.769	5.269	5.1	52.099	6.480	5.699	2.5
16.8	7.829	5.329	5.1	52.100	6.486	5.737	2.5
17.1	7.890	5.390	5.1	52.101	6.492	5.774	2.5
17.4	7.950	5.450	5.1	52.101	6.498	5.811	2.5
17.6	8.000	5.500	5.1	52.102	6.502	5.841	2.5

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TANK NO. 275 : FO DRAIN TANK : NET VOLUME= 9.34 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	0.000	0.000	6.9	45.854	-1.637	0.001	6.1
0.2	0.030	0.030	6.9	45.854	-1.637	0.015	6.1
0.4	0.060	0.060	6.9	45.854	-1.637	0.030	6.1
0.6	0.090	0.090	6.9	45.854	-1.637	0.045	6.1
0.8	0.120	0.120	6.9	45.854	-1.637	0.060	6.1
1.0	0.150	0.150	6.9	45.854	-1.637	0.075	6.1
1.2	0.180	0.180	6.9	45.854	-1.637	0.090	6.1
1.4	0.210	0.210	6.9	45.854	-1.637	0.105	6.1
1.6	0.240	0.240	6.9	45.854	-1.637	0.120	6.1
1.8	0.270	0.270	6.9	45.854	-1.637	0.135	6.1
2.0	0.300	0.300	6.9	45.854	-1.637	0.150	6.1
2.2	0.330	0.330	6.9	45.854	-1.637	0.165	6.1
2.4	0.360	0.360	6.9	45.854	-1.637	0.180	6.1
2.6	0.390	0.390	6.9	45.854	-1.637	0.195	6.1
2.8	0.420	0.420	6.9	45.854	-1.637	0.210	6.1
3.0	0.450	0.450	6.9	45.854	-1.637	0.225	6.1
3.2	0.480	0.480	6.9	45.854	-1.637	0.240	6.1
3.4	0.510	0.510	6.9	45.854	-1.637	0.255	6.1
3.6	0.540	0.540	6.9	45.854	-1.637	0.270	6.1
3.8	0.570	0.570	6.9	45.854	-1.637	0.285	6.1
4.0	0.600	0.600	6.9	45.854	-1.637	0.300	6.1
4.2	0.630	0.630	6.9	45.854	-1.637	0.315	6.1
4.4	0.660	0.660	6.9	45.854	-1.637	0.330	6.1
4.6	0.690	0.690	6.9	45.854	-1.637	0.345	6.1
4.8	0.720	0.720	6.9	45.854	-1.637	0.360	6.1
5.0	0.750	0.750	6.9	45.854	-1.637	0.375	6.1
5.2	0.780	0.780	6.9	45.854	-1.637	0.390	6.1
5.4	0.810	0.810	6.9	45.854	-1.637	0.405	6.1
5.6	0.840	0.840	6.9	45.854	-1.637	0.420	6.1
5.8	0.870	0.870	6.9	45.854	-1.637	0.435	6.1
6.0	0.900	0.900	6.9	45.854	-1.637	0.450	6.1
6.2	0.930	0.930	6.9	45.854	-1.637	0.465	6.1
6.4	0.960	0.960	6.9	45.854	-1.637	0.480	6.1
6.6	0.990	0.990	6.9	45.854	-1.637	0.495	6.1
6.8	1.020	1.020	6.9	45.854	-1.637	0.510	6.1
7.0	1.050	1.050	6.9	45.854	-1.637	0.525	6.1
7.2	1.080	1.080	6.9	45.854	-1.637	0.540	6.1
7.4	1.109	1.109	6.9	45.854	-1.637	0.555	6.1
7.6	1.139	1.139	6.9	45.854	-1.637	0.570	6.1
7.8	1.169	1.169	6.9	45.854	-1.637	0.585	6.1
8.0	1.199	1.199	6.9	45.854	-1.637	0.600	6.1
8.2	1.229	1.229	6.9	45.854	-1.637	0.615	6.1
8.4	1.259	1.259	6.9	45.854	-1.637	0.630	6.1
8.6	1.289	1.289	6.9	45.854	-1.637	0.645	6.1
8.8	1.319	1.319	6.9	45.854	-1.637	0.660	6.1
9.0	1.349	1.349	6.9	45.854	-1.637	0.675	6.1
9.2	1.379	1.379	6.9	45.854	-1.637	0.690	6.1
9.3	1.400	1.400	6.9	45.854	-1.637	0.700	6.1

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TANK NO. 277 : FO OVERFLOW TANK : NET VOLUME= 41.39 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.000	0.000	3.6	41.554	1.956	1.001	1.0
0.7	1.140	0.140	6.0	41.890	2.083	1.074	2.4
1.4	1.246	0.246	7.6	42.071	2.122	1.135	3.8
2.1	1.331	0.331	9.1	42.243	2.177	1.187	6.7
2.8	1.402	0.402	11.1	42.409	2.247	1.232	11.9
3.5	1.467	0.467	11.1	42.536	2.303	1.272	11.9
4.2	1.531	0.531	11.1	42.622	2.340	1.310	11.9
4.9	1.596	0.596	11.1	42.682	2.367	1.346	11.9
5.6	1.661	0.661	11.1	42.728	2.387	1.381	11.9
6.3	1.725	0.725	11.1	42.764	2.402	1.416	11.9
7.0	1.790	0.790	11.1	42.792	2.415	1.450	11.9
7.7	1.855	0.855	11.1	42.815	2.425	1.484	11.9
8.4	1.920	0.920	11.1	42.835	2.434	1.518	11.9
9.1	1.984	0.984	11.1	42.851	2.441	1.551	11.9
9.8	2.049	1.049	11.1	42.865	2.447	1.584	11.9
10.5	2.114	1.114	11.1	42.877	2.452	1.617	11.9
11.2	2.178	1.178	11.1	42.888	2.457	1.651	11.9
11.9	2.243	1.243	11.1	42.897	2.461	1.683	11.9
12.6	2.308	1.308	11.1	42.906	2.465	1.716	11.9
13.3	2.373	1.373	11.1	42.913	2.468	1.749	11.9
14.0	2.437	1.437	11.1	42.920	2.471	1.782	11.9
14.7	2.502	1.502	11.1	42.926	2.474	1.815	11.9
15.4	2.567	1.567	11.1	42.931	2.476	1.847	11.9
16.1	2.632	1.632	11.1	42.936	2.478	1.880	11.9
16.8	2.696	1.696	11.1	42.941	2.480	1.913	11.9
17.5	2.761	1.761	11.1	42.945	2.482	1.945	11.9
18.2	2.826	1.826	11.1	42.949	2.484	1.978	11.9
18.9	2.890	1.890	11.1	42.953	2.485	2.011	11.9
19.6	2.955	1.955	11.1	42.956	2.487	2.043	11.9
20.3	3.020	2.020	11.1	42.959	2.488	2.076	11.9
21.0	3.085	2.085	11.1	42.962	2.490	2.108	11.9
21.7	3.149	2.149	11.1	42.965	2.491	2.141	11.9
22.4	3.214	2.214	11.1	42.968	2.492	2.173	11.9
23.1	3.279	2.279	11.1	42.970	2.493	2.206	11.9
23.8	3.343	2.343	11.1	42.972	2.494	2.238	11.9
24.5	3.408	2.408	11.1	42.974	2.495	2.271	11.9
25.2	3.473	2.473	11.1	42.976	2.496	2.303	11.9
25.9	3.538	2.538	11.1	42.978	2.497	2.336	11.9
26.6	3.602	2.602	11.1	42.980	2.497	2.368	11.9
27.3	3.667	2.667	11.1	42.982	2.498	2.401	11.9
28.0	3.732	2.732	11.1	42.984	2.499	2.433	11.9
28.7	3.797	2.797	11.1	42.985	2.500	2.466	11.9
29.4	3.861	2.861	11.1	42.987	2.500	2.498	11.9
30.1	3.926	2.926	11.1	42.988	2.501	2.531	11.9
30.8	3.991	2.991	11.1	42.989	2.502	2.563	11.9
31.5	4.055	3.055	11.1	42.991	2.502	2.596	11.9
32.2	4.120	3.120	11.1	42.992	2.503	2.628	11.9
32.9	4.185	3.185	11.1	42.993	2.503	2.660	11.9
33.6	4.250	3.250	11.1	42.994	2.504	2.693	11.9
34.3	4.314	3.314	11.1	42.995	2.504	2.725	11.9
35.0	4.381	3.381	10.6	42.995	2.505	2.758	11.8
35.7	4.449	3.449	10.6	42.995	2.506	2.790	11.8
36.4	4.517	3.517	10.6	42.995	2.506	2.823	11.8
37.1	4.584	3.584	10.6	42.994	2.507	2.855	11.8
37.8	4.652	3.652	10.6	42.994	2.508	2.888	11.8
38.5	4.720	3.720	10.6	42.993	2.509	2.921	11.8
39.2	4.788	3.788	10.6	42.993	2.509	2.953	11.8
39.9	4.856	3.856	10.6	42.993	2.510	2.986	11.8
40.6	4.924	3.924	10.6	42.992	2.511	3.019	11.8
41.3	4.992	3.992	10.6	42.992	2.511	3.052	11.8
41.4	5.000	4.000	10.6	42.992	2.511	3.056	11.8

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TANK NO. 279 : EMERG. GEN. FO TANK : NET VOLUME= 2.20 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	23.350	0.000	0.9	61.600	-1.375	23.351	0.0
0.1	23.463	0.113	0.9	61.600	-1.375	23.407	0.0
0.2	23.577	0.227	0.9	61.600	-1.375	23.463	0.0
0.3	23.690	0.340	0.9	61.600	-1.375	23.520	0.0
0.4	23.803	0.453	0.9	61.600	-1.375	23.577	0.0
0.5	23.916	0.566	0.9	61.600	-1.375	23.633	0.0
0.6	24.030	0.680	0.9	61.600	-1.375	23.690	0.0
0.7	24.143	0.793	0.9	61.600	-1.375	23.747	0.0
0.8	24.256	0.906	0.9	61.600	-1.375	23.803	0.0
0.9	24.370	1.020	0.9	61.600	-1.375	23.860	0.0
1.0	24.483	1.133	0.9	61.600	-1.375	23.916	0.0
1.1	24.596	1.246	0.9	61.600	-1.375	23.973	0.0
1.2	24.709	1.359	0.9	61.600	-1.375	24.030	0.0
1.3	24.823	1.473	0.9	61.600	-1.375	24.086	0.0
1.4	24.936	1.586	0.9	61.600	-1.375	24.143	0.0
1.5	25.049	1.699	0.9	61.600	-1.375	24.200	0.0
1.6	25.163	1.813	0.9	61.600	-1.375	24.256	0.0
1.7	25.276	1.926	0.9	61.600	-1.375	24.313	0.0
1.8	25.389	2.039	0.9	61.600	-1.375	24.370	0.0
1.9	25.502	2.152	0.9	61.600	-1.375	24.426	0.0
2.0	25.616	2.266	0.9	61.600	-1.375	24.483	0.0
2.1	25.729	2.379	0.9	61.600	-1.375	24.540	0.0
2.2	25.926	2.576	0.0	61.591	-1.375	24.597	0.0

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TANK NO. 280 : BILGE WATER HOLDING TANK : NET VOLUME= 8.86 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.400	0.000	0.5	48.639	5.982	1.401	0.0
0.2	1.724	0.324	0.7	48.641	6.060	1.570	0.1
0.4	1.995	0.595	0.8	48.641	6.102	1.716	0.1
0.6	2.243	0.843	0.9	48.642	6.133	1.851	0.1
0.8	2.474	1.074	0.9	48.642	6.159	1.978	0.1
1.0	2.690	1.290	1.0	48.642	6.184	2.099	0.2
1.2	2.895	1.495	1.0	48.643	6.207	2.215	0.2
1.4	3.089	1.689	1.1	48.643	6.229	2.326	0.2
1.6	3.274	1.874	1.1	48.643	6.250	2.433	0.3
1.8	3.451	2.051	1.2	48.643	6.270	2.536	0.3
2.0	3.622	2.222	1.2	48.643	6.289	2.636	0.3
2.2	3.786	2.386	1.3	48.644	6.308	2.733	0.4
2.4	3.945	2.545	1.3	48.644	6.326	2.828	0.4
2.6	4.099	2.699	1.4	48.644	6.344	2.920	0.4
2.8	4.249	2.849	1.4	48.644	6.361	3.009	0.5
3.0	4.394	2.994	1.4	48.644	6.378	3.097	0.5
3.2	4.536	3.136	1.5	48.644	6.395	3.182	0.5
3.4	4.674	3.274	1.5	48.644	6.411	3.266	0.6
3.6	4.809	3.409	1.5	48.644	6.426	3.348	0.6
3.8	4.940	3.540	1.6	48.644	6.442	3.428	0.7
4.0	5.069	3.669	1.6	48.645	6.457	3.507	0.7
4.2	5.195	3.795	1.7	48.645	6.472	3.585	0.8
4.4	5.319	3.919	1.7	48.645	6.486	3.661	0.8
4.6	5.440	4.040	1.7	48.645	6.500	3.735	0.8
4.8	5.561	4.161	1.7	48.645	6.514	3.809	0.9
5.0	5.681	4.281	1.7	48.645	6.526	3.881	0.9
5.2	5.802	4.402	1.7	48.645	6.538	3.953	0.9
5.4	5.922	4.522	1.7	48.646	6.548	4.024	0.9
5.6	6.042	4.642	1.7	48.646	6.558	4.094	0.9
5.8	6.162	4.762	1.7	48.646	6.567	4.163	0.9
6.0	6.283	4.883	1.7	48.646	6.576	4.232	0.9
6.2	6.403	5.003	1.7	48.646	6.584	4.300	0.9
6.4	6.523	5.123	1.7	48.646	6.591	4.367	0.9
6.6	6.643	5.243	1.7	48.646	6.599	4.434	0.9
6.8	6.763	5.363	1.7	48.647	6.605	4.501	0.9
7.0	6.884	5.484	1.7	48.647	6.611	4.567	0.9
7.2	7.004	5.604	1.7	48.647	6.617	4.634	0.9
7.4	7.124	5.724	1.7	48.647	6.623	4.699	0.9
7.6	7.244	5.844	1.7	48.647	6.628	4.765	0.9
7.8	7.365	5.965	1.7	48.647	6.633	4.830	0.9
8.0	7.485	6.085	1.7	48.647	6.638	4.895	0.9
8.2	7.605	6.205	1.7	48.647	6.643	4.959	0.9
8.4	7.725	6.325	1.7	48.647	6.647	5.024	0.9
8.6	7.846	6.446	1.7	48.647	6.651	5.088	0.9
8.8	7.966	6.566	1.7	48.647	6.655	5.152	0.9
8.9	8.000	6.600	1.7	48.647	6.656	5.170	0.9

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TANK NO. 281 : BILGE WATER SETTLING TANK : NET VOLUME= 3.95 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.400	0.000	0.4	49.336	5.909	1.401	0.0
0.1	1.607	0.207	0.6	49.338	5.959	1.508	0.0
0.2	1.778	0.378	0.6	49.339	5.996	1.601	0.0
0.3	1.937	0.537	0.7	49.339	6.019	1.687	0.1
0.4	2.086	0.686	0.7	49.340	6.037	1.768	0.1
0.5	2.228	0.828	0.7	49.340	6.054	1.846	0.1
0.6	2.363	0.963	0.8	49.340	6.070	1.921	0.1
0.7	2.491	1.091	0.8	49.341	6.085	1.993	0.1
0.8	2.615	1.215	0.9	49.341	6.099	2.063	0.1
0.9	2.734	1.334	0.9	49.341	6.112	2.131	0.1
1.0	2.848	1.448	0.9	49.341	6.125	2.197	0.1
1.1	2.959	1.559	0.9	49.341	6.138	2.262	0.1
1.2	3.067	1.667	1.0	49.342	6.150	2.324	0.2
1.3	3.172	1.772	1.0	49.342	6.162	2.385	0.2
1.4	3.273	1.873	1.0	49.342	6.174	2.445	0.2
1.5	3.372	1.972	1.1	49.342	6.185	2.504	0.2
1.6	3.469	2.069	1.1	49.342	6.196	2.561	0.2
1.7	3.564	2.164	1.1	49.342	6.207	2.617	0.2
1.8	3.656	2.256	1.1	49.342	6.218	2.672	0.2
1.9	3.746	2.346	1.2	49.343	6.228	2.727	0.3
2.0	3.835	2.435	1.2	49.343	6.238	2.780	0.3
2.1	3.921	2.521	1.2	49.343	6.248	2.832	0.3
2.2	4.006	2.606	1.2	49.343	6.258	2.884	0.3
2.3	4.090	2.690	1.2	49.343	6.268	2.934	0.3
2.4	4.172	2.772	1.3	49.343	6.278	2.984	0.3
2.5	4.253	2.853	1.3	49.343	6.287	3.033	0.4
2.6	4.332	2.932	1.3	49.343	6.296	3.082	0.4
2.7	4.410	3.010	1.3	49.343	6.306	3.129	0.4
2.8	4.487	3.087	1.4	49.343	6.315	3.176	0.4
2.9	4.563	3.163	1.4	49.343	6.323	3.223	0.4
3.0	4.637	3.237	1.4	49.344	6.332	3.269	0.5
3.1	4.711	3.311	1.4	49.344	6.341	3.314	0.5
3.2	4.783	3.383	1.4	49.344	6.350	3.359	0.5
3.3	4.855	3.455	1.4	49.344	6.358	3.403	0.5
3.4	4.926	3.526	1.5	49.344	6.366	3.447	0.5
3.5	4.995	3.595	1.5	49.344	6.375	3.490	0.6
3.6	5.064	3.664	1.5	49.344	6.383	3.533	0.6
3.7	5.132	3.732	1.5	49.344	6.391	3.575	0.6
3.8	5.199	3.799	1.5	49.344	6.399	3.617	0.6
3.9	5.266	3.866	1.6	49.344	6.407	3.659	0.6
4.0	5.300	3.900	1.6	49.344	6.411	3.680	0.7

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TANK NO. 283 : SLUDGE DRAIN TANK : NET VOLUME= 9.38 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	0.000	0.000	6.9	47.950	1.645	0.001	6.2
0.2	0.030	0.030	6.9	47.950	1.645	0.015	6.2
0.4	0.060	0.060	6.9	47.950	1.645	0.030	6.2
0.6	0.090	0.090	6.9	47.950	1.645	0.045	6.2
0.8	0.119	0.119	6.9	47.950	1.645	0.060	6.2
1.0	0.149	0.149	6.9	47.950	1.645	0.075	6.2
1.2	0.179	0.179	6.9	47.950	1.645	0.090	6.2
1.4	0.209	0.209	6.9	47.950	1.645	0.104	6.2
1.6	0.239	0.239	6.9	47.950	1.645	0.119	6.2
1.8	0.269	0.269	6.9	47.950	1.645	0.134	6.2
2.0	0.298	0.298	6.9	47.950	1.645	0.149	6.2
2.2	0.328	0.328	6.9	47.950	1.645	0.164	6.2
2.4	0.358	0.358	6.9	47.950	1.645	0.179	6.2
2.6	0.388	0.388	6.9	47.950	1.645	0.194	6.2
2.8	0.418	0.418	6.9	47.950	1.645	0.209	6.2
3.0	0.448	0.448	6.9	47.950	1.645	0.224	6.2
3.2	0.477	0.477	6.9	47.950	1.645	0.239	6.2
3.4	0.507	0.507	6.9	47.950	1.645	0.254	6.2
3.6	0.537	0.537	6.9	47.950	1.645	0.269	6.2
3.8	0.567	0.567	6.9	47.950	1.645	0.284	6.2
4.0	0.597	0.597	6.9	47.950	1.645	0.298	6.2
4.2	0.627	0.627	6.9	47.950	1.645	0.313	6.2
4.4	0.657	0.657	6.9	47.950	1.645	0.328	6.2
4.6	0.686	0.686	6.9	47.950	1.645	0.343	6.2
4.8	0.716	0.716	6.9	47.950	1.645	0.358	6.2
5.0	0.746	0.746	6.9	47.950	1.645	0.373	6.2
5.2	0.776	0.776	6.9	47.950	1.645	0.388	6.2
5.4	0.806	0.806	6.9	47.950	1.645	0.403	6.2
5.6	0.836	0.836	6.9	47.950	1.645	0.418	6.2
5.8	0.865	0.865	6.9	47.950	1.645	0.433	6.2
6.0	0.895	0.895	6.9	47.950	1.645	0.448	6.2
6.2	0.925	0.925	6.9	47.950	1.645	0.463	6.2
6.4	0.955	0.955	6.9	47.950	1.645	0.477	6.2
6.6	0.985	0.985	6.9	47.950	1.645	0.492	6.2
6.8	1.015	1.015	6.9	47.950	1.645	0.507	6.2
7.0	1.045	1.045	6.9	47.950	1.645	0.522	6.2
7.2	1.074	1.074	6.9	47.950	1.645	0.537	6.2
7.4	1.104	1.104	6.9	47.950	1.645	0.552	6.2
7.6	1.134	1.134	6.9	47.950	1.645	0.567	6.2
7.8	1.164	1.164	6.9	47.950	1.645	0.582	6.2
8.0	1.194	1.194	6.9	47.950	1.645	0.597	6.2
8.2	1.224	1.224	6.9	47.950	1.645	0.612	6.2
8.4	1.253	1.253	6.9	47.950	1.645	0.627	6.2
8.6	1.283	1.283	6.9	47.950	1.645	0.642	6.2
8.8	1.313	1.313	6.9	47.950	1.645	0.657	6.2
9.0	1.343	1.343	6.9	47.950	1.645	0.671	6.2
9.2	1.373	1.373	6.9	47.950	1.645	0.686	6.2
9.4	1.400	1.400	6.9	47.950	1.645	0.700	6.2

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TANK NO. 287 : BILGE WATER DRAIN TANK : NET VOLUME= 9.34 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	0.000	0.000	6.9	45.854	1.637	0.001	6.1
0.2	0.030	0.030	6.9	45.854	1.637	0.015	6.1
0.4	0.060	0.060	6.9	45.854	1.637	0.030	6.1
0.6	0.090	0.090	6.9	45.854	1.637	0.045	6.1
0.8	0.120	0.120	6.9	45.854	1.637	0.060	6.1
1.0	0.150	0.150	6.9	45.854	1.637	0.075	6.1
1.2	0.180	0.180	6.9	45.854	1.637	0.090	6.1
1.4	0.210	0.210	6.9	45.854	1.637	0.105	6.1
1.6	0.240	0.240	6.9	45.854	1.637	0.120	6.1
1.8	0.270	0.270	6.9	45.854	1.637	0.135	6.1
2.0	0.300	0.300	6.9	45.854	1.637	0.150	6.1
2.2	0.330	0.330	6.9	45.854	1.637	0.165	6.1
2.4	0.360	0.360	6.9	45.854	1.637	0.180	6.1
2.6	0.390	0.390	6.9	45.854	1.637	0.195	6.1
2.8	0.420	0.420	6.9	45.854	1.637	0.210	6.1
3.0	0.450	0.450	6.9	45.854	1.637	0.225	6.1
3.2	0.480	0.480	6.9	45.854	1.637	0.240	6.1
3.4	0.510	0.510	6.9	45.854	1.637	0.255	6.1
3.6	0.540	0.540	6.9	45.854	1.637	0.270	6.1
3.8	0.570	0.570	6.9	45.854	1.637	0.285	6.1
4.0	0.600	0.600	6.9	45.854	1.637	0.300	6.1
4.2	0.630	0.630	6.9	45.854	1.637	0.315	6.1
4.4	0.660	0.660	6.9	45.854	1.637	0.330	6.1
4.6	0.690	0.690	6.9	45.854	1.637	0.345	6.1
4.8	0.720	0.720	6.9	45.854	1.637	0.360	6.1
5.0	0.750	0.750	6.9	45.854	1.637	0.375	6.1
5.2	0.780	0.780	6.9	45.854	1.637	0.390	6.1
5.4	0.810	0.810	6.9	45.854	1.637	0.405	6.1
5.6	0.840	0.840	6.9	45.854	1.637	0.420	6.1
5.8	0.870	0.870	6.9	45.854	1.637	0.435	6.1
6.0	0.900	0.900	6.9	45.854	1.637	0.450	6.1
6.2	0.930	0.930	6.9	45.854	1.637	0.465	6.1
6.4	0.960	0.960	6.9	45.854	1.637	0.480	6.1
6.6	0.990	0.990	6.9	45.854	1.637	0.495	6.1
6.8	1.020	1.020	6.9	45.854	1.637	0.510	6.1
7.0	1.050	1.050	6.9	45.854	1.637	0.525	6.1
7.2	1.080	1.080	6.9	45.854	1.637	0.540	6.1
7.4	1.109	1.109	6.9	45.854	1.637	0.555	6.1
7.6	1.139	1.139	6.9	45.854	1.637	0.570	6.1
7.8	1.169	1.169	6.9	45.854	1.637	0.585	6.1
8.0	1.199	1.199	6.9	45.854	1.637	0.600	6.1
8.2	1.229	1.229	6.9	45.854	1.637	0.615	6.1
8.4	1.259	1.259	6.9	45.854	1.637	0.630	6.1
8.6	1.289	1.289	6.9	45.854	1.637	0.645	6.1
8.8	1.319	1.319	6.9	45.854	1.637	0.660	6.1
9.0	1.349	1.349	6.9	45.854	1.637	0.675	6.1
9.2	1.379	1.379	6.9	45.854	1.637	0.690	6.1
9.3	1.400	1.400	6.9	45.854	1.637	0.700	6.1

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TANK NO. 288 : SLUDGE SETTLING TANK : NET VOLUME= 4.45 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	5.300	0.000	1.6	49.346	6.721	5.301	0.7
0.1	5.365	0.065	1.6	49.346	6.727	5.333	0.7
0.2	5.430	0.130	1.6	49.346	6.734	5.365	0.7
0.3	5.494	0.194	1.6	49.346	6.740	5.397	0.7
0.4	5.557	0.257	1.6	49.346	6.746	5.430	0.7
0.5	5.620	0.320	1.7	49.346	6.752	5.461	0.8
0.6	5.682	0.382	1.7	49.346	6.758	5.493	0.8
0.7	5.743	0.443	1.7	49.346	6.763	5.524	0.8
0.8	5.804	0.504	1.7	49.347	6.769	5.556	0.8
0.9	5.865	0.565	1.7	49.347	6.774	5.587	0.8
1.0	5.925	0.625	1.7	49.347	6.779	5.617	0.8
1.1	5.986	0.686	1.7	49.347	6.783	5.648	0.9
1.2	6.046	0.746	1.7	49.347	6.786	5.679	0.9
1.3	6.106	0.806	1.7	49.348	6.789	5.709	0.9
1.4	6.166	0.866	1.7	49.348	6.792	5.740	0.9
1.5	6.226	0.926	1.7	49.348	6.794	5.770	0.9
1.6	6.286	0.986	1.7	49.348	6.796	5.801	0.9
1.7	6.346	1.046	1.7	49.348	6.798	5.831	0.9
1.8	6.406	1.106	1.7	49.348	6.799	5.861	0.9
1.9	6.467	1.167	1.7	49.348	6.800	5.891	0.9
2.0	6.527	1.227	1.7	49.349	6.802	5.922	0.9
2.1	6.587	1.287	1.7	49.349	6.803	5.952	0.9
2.2	6.647	1.347	1.7	49.349	6.804	5.982	0.9
2.3	6.707	1.407	1.7	49.349	6.805	6.012	0.9
2.4	6.767	1.467	1.7	49.349	6.806	6.043	0.9
2.5	6.827	1.527	1.7	49.349	6.806	6.073	0.9
2.6	6.887	1.587	1.7	49.349	6.807	6.103	0.9
2.7	6.947	1.647	1.7	49.349	6.808	6.133	0.9
2.8	7.008	1.708	1.7	49.349	6.808	6.163	0.9
2.9	7.068	1.768	1.7	49.349	6.809	6.193	0.9
3.0	7.128	1.828	1.7	49.349	6.809	6.224	0.9
3.1	7.188	1.888	1.7	49.349	6.810	6.254	0.9
3.2	7.248	1.948	1.7	49.349	6.810	6.284	0.9
3.3	7.308	2.008	1.7	49.349	6.811	6.314	0.9
3.4	7.368	2.068	1.7	49.349	6.811	6.344	0.9
3.5	7.428	2.128	1.7	49.349	6.812	6.374	0.9
3.6	7.488	2.188	1.7	49.349	6.812	6.404	0.9
3.7	7.549	2.249	1.7	49.349	6.812	6.434	0.9
3.8	7.609	2.309	1.7	49.349	6.813	6.465	0.9
3.9	7.669	2.369	1.7	49.349	6.813	6.495	0.9
4.0	7.729	2.429	1.7	49.349	6.813	6.525	0.9
4.1	7.789	2.489	1.7	49.349	6.814	6.555	0.9
4.2	7.849	2.549	1.7	49.349	6.814	6.585	0.9
4.3	7.909	2.609	1.7	49.349	6.814	6.615	0.9
4.4	7.969	2.669	1.7	49.349	6.814	6.645	0.9
4.5	8.000	2.700	1.7	49.349	6.815	6.661	0.9

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TANK NO. 290 : LO STORE TANK MAIN ENG. : NET VOLUME= 9.10 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	2.500	0.000	0.7	55.331	5.166	2.501	0.0
0.2	2.767	0.267	0.8	55.328	5.192	2.638	0.0
0.4	2.995	0.495	1.0	55.326	5.217	2.761	0.1
0.6	3.197	0.697	1.1	55.325	5.238	2.873	0.1
0.8	3.381	0.881	1.2	55.323	5.259	2.977	0.1
1.0	3.550	1.050	1.3	55.322	5.278	3.075	0.1
1.2	3.708	1.208	1.3	55.321	5.296	3.168	0.1
1.4	3.857	1.357	1.4	55.321	5.313	3.256	0.1
1.6	3.998	1.498	1.5	55.320	5.329	3.340	0.2
1.8	4.132	1.632	1.6	55.319	5.344	3.420	0.2
2.0	4.260	1.760	1.6	55.319	5.359	3.498	0.2
2.2	4.383	1.883	1.7	55.318	5.374	3.573	0.2
2.4	4.502	2.002	1.8	55.318	5.388	3.645	0.3
2.6	4.616	2.116	1.8	55.317	5.401	3.716	0.3
2.8	4.727	2.227	1.9	55.317	5.414	3.784	0.3
3.0	4.834	2.334	1.9	55.316	5.427	3.850	0.3
3.2	4.939	2.439	2.0	55.316	5.440	3.915	0.4
3.4	5.040	2.540	2.1	55.316	5.452	3.978	0.4
3.6	5.139	2.639	2.1	55.316	5.464	4.040	0.4
3.8	5.236	2.736	2.2	55.315	5.475	4.101	0.5
4.0	5.352	2.852	1.3	55.314	5.492	4.160	0.1
4.2	5.508	3.008	1.4	55.312	5.520	4.220	0.1
4.4	5.655	3.155	1.4	55.309	5.547	4.282	0.1
4.6	5.795	3.295	1.5	55.308	5.572	4.345	0.2
4.8	5.928	3.428	1.6	55.306	5.596	4.408	0.2
5.0	6.055	3.555	1.7	55.304	5.620	4.472	0.2
5.2	6.177	3.677	1.7	55.303	5.642	4.535	0.2
5.4	6.295	3.795	1.8	55.302	5.664	4.598	0.2
5.6	6.409	3.909	1.8	55.301	5.685	4.660	0.3
5.8	6.519	4.019	1.9	55.300	5.705	4.723	0.3
6.0	6.625	4.125	2.0	55.299	5.725	4.784	0.3
6.2	6.729	4.229	2.0	55.298	5.744	4.845	0.4
6.4	6.830	4.330	2.1	55.297	5.762	4.906	0.4
6.6	6.929	4.429	2.1	55.297	5.780	4.966	0.4
6.8	7.025	4.525	2.2	55.296	5.797	5.025	0.4
7.0	7.118	4.618	2.2	55.295	5.814	5.083	0.5
7.2	7.210	4.710	2.3	55.295	5.831	5.141	0.5
7.4	7.300	4.800	2.3	55.294	5.847	5.198	0.5
7.6	7.388	4.888	2.4	55.294	5.862	5.255	0.6
7.8	7.474	4.974	2.4	55.294	5.878	5.311	0.6
8.0	7.559	5.059	2.5	55.293	5.892	5.366	0.6
8.2	7.642	5.142	2.5	55.293	5.907	5.420	0.7
8.4	7.724	5.224	2.5	55.293	5.921	5.474	0.7
8.6	7.804	5.304	2.6	55.292	5.935	5.527	0.7
8.8	7.883	5.383	2.6	55.292	5.949	5.580	0.8
9.0	7.961	5.461	2.7	55.292	5.962	5.632	0.8
9.1	8.000	5.500	2.7	55.292	5.969	5.658	0.8

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TANK NO. 291 : LO STORE TANK MAIN ENG. : NET VOLUME= 4.80 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	2.500	0.000	0.3	54.259	5.460	2.501	0.0
0.1	2.819	0.319	0.4	54.258	5.491	2.667	0.0
0.2	3.077	0.577	0.4	54.257	5.519	2.809	0.0
0.3	3.299	0.799	0.5	54.257	5.543	2.936	0.0
0.4	3.497	0.997	0.5	54.256	5.566	3.052	0.0
0.5	3.678	1.178	0.6	54.256	5.586	3.160	0.0
0.6	3.845	1.345	0.6	54.256	5.605	3.260	0.0
0.7	4.001	1.501	0.7	54.255	5.624	3.355	0.1
0.8	4.148	1.648	0.7	54.255	5.641	3.445	0.1
0.9	4.287	1.787	0.8	54.255	5.657	3.531	0.1
1.0	4.420	1.920	0.8	54.255	5.673	3.613	0.1
1.1	4.548	2.048	0.8	54.255	5.688	3.692	0.1
1.2	4.670	2.170	0.9	54.254	5.703	3.769	0.1
1.3	4.787	2.287	0.9	54.254	5.717	3.843	0.1
1.4	4.901	2.401	0.9	54.254	5.731	3.914	0.1
1.5	5.011	2.511	1.0	54.254	5.744	3.984	0.1
1.6	5.118	2.618	1.0	54.254	5.757	4.051	0.2
1.7	5.222	2.722	1.0	54.254	5.770	4.117	0.2
1.8	5.329	2.829	0.8	54.254	5.784	4.181	0.1
1.9	5.456	2.956	0.8	54.253	5.805	4.245	0.1
2.0	5.577	3.077	0.9	54.253	5.825	4.309	0.1
2.1	5.694	3.194	0.9	54.252	5.844	4.372	0.1
2.2	5.807	3.307	0.9	54.252	5.863	4.434	0.1
2.3	5.916	3.416	1.0	54.251	5.881	4.497	0.1
2.4	6.022	3.522	1.0	54.251	5.898	4.558	0.2
2.5	6.126	3.626	1.0	54.251	5.915	4.619	0.2
2.6	6.226	3.726	1.0	54.250	5.931	4.678	0.2
2.7	6.324	3.824	1.1	54.250	5.946	4.738	0.2
2.8	6.420	3.920	1.1	54.250	5.962	4.796	0.2
2.9	6.513	4.013	1.1	54.250	5.976	4.854	0.2
3.0	6.605	4.105	1.1	54.249	5.991	4.910	0.3
3.1	6.694	4.194	1.2	54.249	6.005	4.967	0.3
3.2	6.782	4.282	1.2	54.249	6.018	5.022	0.3
3.3	6.868	4.368	1.2	54.249	6.032	5.077	0.3
3.4	6.952	4.452	1.2	54.249	6.045	5.130	0.3
3.5	7.035	4.535	1.3	54.249	6.057	5.184	0.3
3.6	7.116	4.616	1.3	54.249	6.070	5.236	0.4
3.7	7.196	4.696	1.3	54.248	6.082	5.288	0.4
3.8	7.275	4.775	1.3	54.248	6.094	5.339	0.4
3.9	7.352	4.852	1.3	54.248	6.106	5.390	0.4
4.0	7.429	4.929	1.4	54.248	6.117	5.440	0.4
4.1	7.504	5.004	1.4	54.248	6.128	5.489	0.4
4.2	7.578	5.078	1.4	54.248	6.139	5.538	0.5
4.3	7.651	5.151	1.4	54.248	6.150	5.587	0.5
4.4	7.723	5.223	1.4	54.248	6.161	5.634	0.5
4.5	7.794	5.294	1.5	54.248	6.172	5.682	0.5
4.6	7.865	5.365	1.5	54.248	6.182	5.728	0.5
4.7	7.934	5.434	1.5	54.248	6.192	5.774	0.6
4.8	8.000	5.500	1.5	54.248	6.202	5.818	0.6

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TANK NO. 293 : LO STORE TANK THRUSTER : NET VOLUME= 4.96 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	2.500	0.000	0.2	53.562	5.651	2.501	0.0
0.1	2.865	0.365	0.3	53.560	5.688	2.693	0.0
0.2	3.145	0.645	0.4	53.559	5.719	2.852	0.0
0.3	3.381	0.881	0.5	53.558	5.745	2.990	0.0
0.4	3.588	1.088	0.5	53.557	5.769	3.114	0.0
0.5	3.776	1.276	0.6	53.557	5.791	3.228	0.0
0.6	3.949	1.449	0.6	53.556	5.811	3.334	0.0
0.7	4.109	1.609	0.7	53.556	5.830	3.433	0.1
0.8	4.260	1.760	0.7	53.556	5.849	3.527	0.1
0.9	4.403	1.903	0.7	53.556	5.866	3.617	0.1
1.0	4.538	2.038	0.8	53.555	5.882	3.702	0.1
1.1	4.668	2.168	0.8	53.555	5.898	3.784	0.1
1.2	4.792	2.292	0.8	53.555	5.913	3.863	0.1
1.3	4.912	2.412	0.9	53.555	5.927	3.939	0.1
1.4	5.027	2.527	0.9	53.555	5.941	4.013	0.1
1.5	5.139	2.639	0.9	53.555	5.955	4.084	0.1
1.6	5.247	2.747	1.0	53.555	5.968	4.153	0.2
1.7	5.356	2.856	0.9	53.554	5.983	4.221	0.1
1.8	5.467	2.967	0.9	53.554	5.999	4.287	0.1
1.9	5.574	3.074	1.0	53.553	6.014	4.352	0.2
2.0	5.678	3.178	1.0	53.553	6.029	4.416	0.2
2.1	5.780	3.280	1.0	53.552	6.043	4.478	0.2
2.2	5.878	3.378	1.1	53.552	6.057	4.540	0.2
2.3	5.975	3.475	1.1	53.551	6.070	4.600	0.2
2.4	6.069	3.569	1.1	53.551	6.083	4.659	0.2
2.5	6.161	3.661	1.1	53.551	6.096	4.717	0.2
2.6	6.251	3.751	1.2	53.551	6.108	4.775	0.3
2.7	6.339	3.839	1.2	53.550	6.120	4.831	0.3
2.8	6.425	3.925	1.2	53.550	6.132	4.886	0.3
2.9	6.510	4.010	1.2	53.550	6.144	4.941	0.3
3.0	6.594	4.094	1.2	53.550	6.155	4.995	0.3
3.1	6.675	4.175	1.3	53.550	6.166	5.047	0.4
3.2	6.756	4.256	1.3	53.549	6.177	5.100	0.4
3.3	6.835	4.335	1.3	53.549	6.188	5.151	0.4
3.4	6.913	4.413	1.3	53.549	6.199	5.202	0.4
3.5	6.990	4.490	1.4	53.549	6.209	5.252	0.4
3.6	7.065	4.565	1.4	53.549	6.219	5.301	0.4
3.7	7.140	4.640	1.4	53.549	6.229	5.350	0.5
3.8	7.213	4.713	1.4	53.549	6.239	5.398	0.5
3.9	7.285	4.785	1.4	53.549	6.249	5.445	0.5
4.0	7.357	4.857	1.5	53.549	6.258	5.492	0.5
4.1	7.427	4.927	1.5	53.549	6.268	5.538	0.5
4.2	7.497	4.997	1.5	53.548	6.277	5.584	0.6
4.3	7.566	5.066	1.5	53.548	6.286	5.630	0.6
4.4	7.634	5.134	1.5	53.548	6.295	5.674	0.6
4.5	7.701	5.201	1.5	53.548	6.304	5.719	0.6
4.6	7.767	5.267	1.6	53.548	6.313	5.762	0.7
4.7	7.833	5.333	1.6	53.548	6.322	5.806	0.7
4.8	7.898	5.398	1.6	53.548	6.330	5.849	0.7
4.9	7.962	5.462	1.6	53.548	6.339	5.891	0.7
5.0	8.000	5.500	1.6	53.548	6.344	5.916	0.7

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TANK NO. 295 : LO STORE TANK AZIMUTH THR. : NET VOLUME= 6.13 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	2.200	0.000	2.9	14.350	-2.100	2.201	0.5
0.1	2.235	0.035	2.9	14.350	-2.100	2.218	0.5
0.2	2.270	0.070	2.9	14.350	-2.100	2.235	0.5
0.3	2.305	0.105	2.9	14.350	-2.100	2.253	0.5
0.4	2.340	0.140	2.9	14.350	-2.100	2.270	0.5
0.5	2.375	0.175	2.9	14.350	-2.100	2.288	0.5
0.6	2.410	0.210	2.9	14.350	-2.100	2.305	0.5
0.7	2.445	0.245	2.9	14.350	-2.100	2.323	0.5
0.8	2.481	0.281	2.9	14.350	-2.100	2.340	0.5
0.9	2.516	0.316	2.9	14.350	-2.100	2.358	0.5
1.0	2.551	0.351	2.9	14.350	-2.100	2.375	0.5
1.1	2.586	0.386	2.9	14.350	-2.100	2.393	0.5
1.2	2.621	0.421	2.9	14.350	-2.100	2.410	0.5
1.3	2.656	0.456	2.9	14.350	-2.100	2.428	0.5
1.4	2.691	0.491	2.9	14.350	-2.100	2.445	0.5
1.5	2.726	0.526	2.9	14.350	-2.100	2.463	0.5
1.6	2.761	0.561	2.9	14.350	-2.100	2.481	0.5
1.7	2.796	0.596	2.9	14.350	-2.100	2.498	0.5
1.8	2.831	0.631	2.9	14.350	-2.100	2.516	0.5
1.9	2.866	0.666	2.9	14.350	-2.100	2.533	0.5
2.0	2.901	0.701	2.9	14.350	-2.100	2.551	0.5
2.1	2.936	0.736	2.9	14.350	-2.100	2.568	0.5
2.2	2.971	0.771	2.9	14.350	-2.100	2.586	0.5
2.3	3.007	0.807	2.9	14.350	-2.100	2.603	0.5
2.4	3.042	0.842	2.9	14.350	-2.100	2.621	0.5
2.5	3.077	0.877	2.9	14.350	-2.100	2.638	0.5
2.6	3.112	0.912	2.9	14.350	-2.100	2.656	0.5
2.7	3.147	0.947	2.9	14.350	-2.100	2.673	0.5
2.8	3.182	0.982	2.9	14.350	-2.100	2.691	0.5
2.9	3.217	1.017	2.9	14.350	-2.100	2.708	0.5
3.0	3.252	1.052	2.9	14.350	-2.100	2.726	0.5
3.1	3.287	1.087	2.9	14.350	-2.100	2.744	0.5
3.2	3.322	1.122	2.9	14.350	-2.100	2.761	0.5
3.3	3.357	1.157	2.9	14.350	-2.100	2.779	0.5
3.4	3.392	1.192	2.9	14.350	-2.100	2.796	0.5
3.5	3.427	1.227	2.9	14.350	-2.100	2.814	0.5
3.6	3.462	1.262	2.9	14.350	-2.100	2.831	0.5
3.7	3.497	1.297	2.9	14.350	-2.100	2.849	0.5
3.8	3.532	1.332	2.9	14.350	-2.100	2.866	0.5
3.9	3.568	1.368	2.9	14.350	-2.100	2.884	0.5
4.0	3.603	1.403	2.9	14.350	-2.100	2.901	0.5
4.1	3.638	1.438	2.9	14.350	-2.100	2.919	0.5
4.2	3.674	1.474	2.7	14.349	-2.101	2.936	0.4
4.3	3.712	1.512	2.7	14.348	-2.101	2.954	0.4
4.4	3.750	1.550	2.7	14.346	-2.102	2.972	0.4
4.5	3.787	1.587	2.7	14.345	-2.103	2.989	0.4
4.6	3.825	1.625	2.7	14.344	-2.104	3.007	0.4
4.7	3.862	1.662	2.7	14.343	-2.104	3.025	0.4
4.8	3.900	1.700	2.7	14.342	-2.105	3.043	0.4
4.9	3.938	1.738	2.7	14.341	-2.106	3.061	0.4
5.0	3.975	1.775	2.7	14.339	-2.106	3.079	0.4
5.1	4.013	1.813	2.7	14.339	-2.107	3.096	0.4
5.2	4.050	1.850	2.7	14.338	-2.107	3.114	0.4
5.3	4.088	1.888	2.7	14.337	-2.108	3.132	0.4
5.4	4.126	1.926	2.7	14.336	-2.108	3.151	0.4
5.5	4.163	1.963	2.7	14.335	-2.109	3.169	0.4
5.6	4.201	2.001	2.7	14.334	-2.109	3.187	0.4
5.7	4.238	2.038	2.7	14.333	-2.110	3.205	0.4
5.8	4.276	2.076	2.7	14.333	-2.110	3.223	0.4
5.9	4.314	2.114	2.7	14.332	-2.111	3.241	0.4
6.0	4.351	2.151	2.7	14.331	-2.111	3.259	0.4
6.1	4.389	2.189	2.7	14.330	-2.111	3.278	0.4
6.1	4.400	2.200	2.7	14.330	-2.112	3.283	0.4

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TANK NO. 297 : LO DRAIN TANK : NET VOLUME= 9.38 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	0.000	0.000	6.9	47.950	-1.645	0.001	6.2
0.2	0.030	0.030	6.9	47.950	-1.645	0.015	6.2
0.4	0.060	0.060	6.9	47.950	-1.645	0.030	6.2
0.6	0.090	0.090	6.9	47.950	-1.645	0.045	6.2
0.8	0.119	0.119	6.9	47.950	-1.645	0.060	6.2
1.0	0.149	0.149	6.9	47.950	-1.645	0.075	6.2
1.2	0.179	0.179	6.9	47.950	-1.645	0.090	6.2
1.4	0.209	0.209	6.9	47.950	-1.645	0.104	6.2
1.6	0.239	0.239	6.9	47.950	-1.645	0.119	6.2
1.8	0.269	0.269	6.9	47.950	-1.645	0.134	6.2
2.0	0.298	0.298	6.9	47.950	-1.645	0.149	6.2
2.2	0.328	0.328	6.9	47.950	-1.645	0.164	6.2
2.4	0.358	0.358	6.9	47.950	-1.645	0.179	6.2
2.6	0.388	0.388	6.9	47.950	-1.645	0.194	6.2
2.8	0.418	0.418	6.9	47.950	-1.645	0.209	6.2
3.0	0.448	0.448	6.9	47.950	-1.645	0.224	6.2
3.2	0.477	0.477	6.9	47.950	-1.645	0.239	6.2
3.4	0.507	0.507	6.9	47.950	-1.645	0.254	6.2
3.6	0.537	0.537	6.9	47.950	-1.645	0.269	6.2
3.8	0.567	0.567	6.9	47.950	-1.645	0.284	6.2
4.0	0.597	0.597	6.9	47.950	-1.645	0.298	6.2
4.2	0.627	0.627	6.9	47.950	-1.645	0.313	6.2
4.4	0.657	0.657	6.9	47.950	-1.645	0.328	6.2
4.6	0.686	0.686	6.9	47.950	-1.645	0.343	6.2
4.8	0.716	0.716	6.9	47.950	-1.645	0.358	6.2
5.0	0.746	0.746	6.9	47.950	-1.645	0.373	6.2
5.2	0.776	0.776	6.9	47.950	-1.645	0.388	6.2
5.4	0.806	0.806	6.9	47.950	-1.645	0.403	6.2
5.6	0.836	0.836	6.9	47.950	-1.645	0.418	6.2
5.8	0.865	0.865	6.9	47.950	-1.645	0.433	6.2
6.0	0.895	0.895	6.9	47.950	-1.645	0.448	6.2
6.2	0.925	0.925	6.9	47.950	-1.645	0.463	6.2
6.4	0.955	0.955	6.9	47.950	-1.645	0.477	6.2
6.6	0.985	0.985	6.9	47.950	-1.645	0.492	6.2
6.8	1.015	1.015	6.9	47.950	-1.645	0.507	6.2
7.0	1.045	1.045	6.9	47.950	-1.645	0.522	6.2
7.2	1.074	1.074	6.9	47.950	-1.645	0.537	6.2
7.4	1.104	1.104	6.9	47.950	-1.645	0.552	6.2
7.6	1.134	1.134	6.9	47.950	-1.645	0.567	6.2
7.8	1.164	1.164	6.9	47.950	-1.645	0.582	6.2
8.0	1.194	1.194	6.9	47.950	-1.645	0.597	6.2
8.2	1.224	1.224	6.9	47.950	-1.645	0.612	6.2
8.4	1.253	1.253	6.9	47.950	-1.645	0.627	6.2
8.6	1.283	1.283	6.9	47.950	-1.645	0.642	6.2
8.8	1.313	1.313	6.9	47.950	-1.645	0.657	6.2
9.0	1.343	1.343	6.9	47.950	-1.645	0.671	6.2
9.2	1.373	1.373	6.9	47.950	-1.645	0.686	6.2
9.4	1.400	1.400	6.9	47.950	-1.645	0.700	6.2

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TANK NO. 298 : DIRTY LO TANK : NET VOLUME= 6.13 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	2.200	0.000	2.9	14.350	2.100	2.201	0.5
0.1	2.235	0.035	2.9	14.350	2.100	2.218	0.5
0.2	2.270	0.070	2.9	14.350	2.100	2.235	0.5
0.3	2.305	0.105	2.9	14.350	2.100	2.253	0.5
0.4	2.340	0.140	2.9	14.350	2.100	2.270	0.5
0.5	2.375	0.175	2.9	14.350	2.100	2.288	0.5
0.6	2.410	0.210	2.9	14.350	2.100	2.305	0.5
0.7	2.445	0.245	2.9	14.350	2.100	2.323	0.5
0.8	2.481	0.281	2.9	14.350	2.100	2.340	0.5
0.9	2.516	0.316	2.9	14.350	2.100	2.358	0.5
1.0	2.551	0.351	2.9	14.350	2.100	2.375	0.5
1.1	2.586	0.386	2.9	14.350	2.100	2.393	0.5
1.2	2.621	0.421	2.9	14.350	2.100	2.410	0.5
1.3	2.656	0.456	2.9	14.350	2.100	2.428	0.5
1.4	2.691	0.491	2.9	14.350	2.100	2.445	0.5
1.5	2.726	0.526	2.9	14.350	2.100	2.463	0.5
1.6	2.761	0.561	2.9	14.350	2.100	2.481	0.5
1.7	2.796	0.596	2.9	14.350	2.100	2.498	0.5
1.8	2.831	0.631	2.9	14.350	2.100	2.516	0.5
1.9	2.866	0.666	2.9	14.350	2.100	2.533	0.5
2.0	2.901	0.701	2.9	14.350	2.100	2.551	0.5
2.1	2.936	0.736	2.9	14.350	2.100	2.568	0.5
2.2	2.971	0.771	2.9	14.350	2.100	2.586	0.5
2.3	3.007	0.807	2.9	14.350	2.100	2.603	0.5
2.4	3.042	0.842	2.9	14.350	2.100	2.621	0.5
2.5	3.077	0.877	2.9	14.350	2.100	2.638	0.5
2.6	3.112	0.912	2.9	14.350	2.100	2.656	0.5
2.7	3.147	0.947	2.9	14.350	2.100	2.673	0.5
2.8	3.182	0.982	2.9	14.350	2.100	2.691	0.5
2.9	3.217	1.017	2.9	14.350	2.100	2.708	0.5
3.0	3.252	1.052	2.9	14.350	2.100	2.726	0.5
3.1	3.287	1.087	2.9	14.350	2.100	2.744	0.5
3.2	3.322	1.122	2.9	14.350	2.100	2.761	0.5
3.3	3.357	1.157	2.9	14.350	2.100	2.779	0.5
3.4	3.392	1.192	2.9	14.350	2.100	2.796	0.5
3.5	3.427	1.227	2.9	14.350	2.100	2.814	0.5
3.6	3.462	1.262	2.9	14.350	2.100	2.831	0.5
3.7	3.497	1.297	2.9	14.350	2.100	2.849	0.5
3.8	3.532	1.332	2.9	14.350	2.100	2.866	0.5
3.9	3.568	1.368	2.9	14.350	2.100	2.884	0.5
4.0	3.603	1.403	2.9	14.350	2.100	2.901	0.5
4.1	3.638	1.438	2.9	14.350	2.100	2.919	0.5
4.2	3.674	1.474	2.7	14.349	2.101	2.936	0.4
4.3	3.712	1.512	2.7	14.348	2.101	2.954	0.4
4.4	3.750	1.550	2.7	14.346	2.102	2.972	0.4
4.5	3.787	1.587	2.7	14.345	2.103	2.989	0.4
4.6	3.825	1.625	2.7	14.344	2.104	3.007	0.4
4.7	3.862	1.662	2.7	14.343	2.104	3.025	0.4
4.8	3.900	1.700	2.7	14.342	2.105	3.043	0.4
4.9	3.938	1.738	2.7	14.341	2.106	3.061	0.4
5.0	3.975	1.775	2.7	14.339	2.106	3.079	0.4
5.1	4.013	1.813	2.7	14.339	2.107	3.096	0.4
5.2	4.050	1.850	2.7	14.338	2.107	3.114	0.4
5.3	4.088	1.888	2.7	14.337	2.108	3.132	0.4
5.4	4.126	1.926	2.7	14.336	2.108	3.151	0.4
5.5	4.163	1.963	2.7	14.335	2.109	3.169	0.4
5.6	4.201	2.001	2.7	14.334	2.109	3.187	0.4
5.7	4.238	2.038	2.7	14.333	2.110	3.205	0.4
5.8	4.276	2.076	2.7	14.333	2.110	3.223	0.4
5.9	4.314	2.114	2.7	14.332	2.111	3.241	0.4
6.0	4.351	2.151	2.7	14.331	2.111	3.259	0.4
6.1	4.389	2.189	2.7	14.330	2.111	3.278	0.4
6.1	4.400	2.200	2.7	14.330	2.112	3.283	0.4

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TANK NO. 310 : CHAIN LOCKER PS : NET VOLUME= 15.97 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	5.300	0.000	2.9	70.350	-0.700	5.301	0.5
0.3	5.405	0.105	2.9	70.350	-0.700	5.353	0.5
0.6	5.510	0.210	2.9	70.350	-0.700	5.405	0.5
0.9	5.616	0.316	2.9	70.350	-0.700	5.458	0.5
1.2	5.721	0.421	2.9	70.350	-0.700	5.510	0.5
1.5	5.826	0.526	2.9	70.350	-0.700	5.563	0.5
1.8	5.931	0.631	2.9	70.350	-0.700	5.616	0.5
2.1	6.036	0.736	2.9	70.350	-0.700	5.668	0.5
2.4	6.142	0.842	2.9	70.350	-0.700	5.721	0.5
2.7	6.247	0.947	2.9	70.350	-0.700	5.773	0.5
3.0	6.352	1.052	2.9	70.350	-0.700	5.826	0.5
3.3	6.457	1.157	2.9	70.350	-0.700	5.879	0.5
3.6	6.562	1.262	2.9	70.350	-0.700	5.931	0.5
3.9	6.668	1.368	2.9	70.350	-0.700	5.984	0.5
4.2	6.773	1.473	2.9	70.350	-0.700	6.036	0.5
4.5	6.878	1.578	2.9	70.350	-0.700	6.089	0.5
4.8	6.983	1.683	2.9	70.350	-0.700	6.142	0.5
5.1	7.088	1.788	2.9	70.350	-0.700	6.194	0.5
5.4	7.194	1.894	2.9	70.350	-0.700	6.247	0.5
5.7	7.299	1.999	2.9	70.350	-0.700	6.299	0.5
6.0	7.404	2.104	2.9	70.350	-0.700	6.352	0.5
6.3	7.509	2.209	2.9	70.350	-0.700	6.405	0.5
6.6	7.614	2.314	2.9	70.350	-0.700	6.457	0.5
6.9	7.720	2.420	2.9	70.350	-0.700	6.510	0.5
7.2	7.825	2.525	2.9	70.350	-0.700	6.562	0.5
7.5	7.930	2.630	2.9	70.350	-0.700	6.615	0.5
7.8	8.035	2.735	2.9	70.350	-0.700	6.668	0.5
8.1	8.140	2.840	2.9	70.350	-0.700	6.720	0.5
8.4	8.245	2.945	2.9	70.350	-0.700	6.773	0.5
8.7	8.351	3.051	2.9	70.350	-0.700	6.825	0.5
9.0	8.456	3.156	2.9	70.350	-0.700	6.878	0.5
9.3	8.561	3.261	2.9	70.350	-0.700	6.931	0.5
9.6	8.666	3.366	2.9	70.350	-0.700	6.983	0.5
9.9	8.771	3.471	2.9	70.350	-0.700	7.036	0.5
10.2	8.877	3.577	2.9	70.350	-0.700	7.088	0.5
10.5	8.982	3.682	2.9	70.350	-0.700	7.141	0.5
10.8	9.087	3.787	2.9	70.350	-0.700	7.194	0.5
11.1	9.192	3.892	2.9	70.350	-0.700	7.246	0.5
11.4	9.297	3.997	2.9	70.350	-0.700	7.299	0.5
11.7	9.403	4.103	2.9	70.350	-0.700	7.351	0.5
12.0	9.508	4.208	2.9	70.350	-0.700	7.404	0.5
12.3	9.613	4.313	2.9	70.350	-0.700	7.457	0.5
12.6	9.718	4.418	2.9	70.350	-0.700	7.509	0.5
12.9	9.823	4.523	2.9	70.350	-0.700	7.562	0.5
13.2	9.929	4.629	2.9	70.350	-0.700	7.614	0.5
13.5	10.034	4.734	2.9	70.350	-0.700	7.667	0.5
13.8	10.139	4.839	2.9	70.350	-0.700	7.720	0.5
14.1	10.244	4.944	2.9	70.350	-0.700	7.772	0.5
14.4	10.349	5.049	2.9	70.350	-0.700	7.825	0.5
14.7	10.455	5.155	2.9	70.350	-0.700	7.877	0.5
15.0	10.560	5.260	2.9	70.350	-0.700	7.930	0.5
15.3	10.665	5.365	2.9	70.350	-0.700	7.983	0.5
15.6	10.770	5.470	2.9	70.350	-0.700	8.035	0.5
15.9	10.875	5.575	2.9	70.350	-0.700	8.088	0.5
16.0	10.900	5.600	2.9	70.350	-0.700	8.100	0.5

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TANK NO. 311 : CHAIN LOCKER SB : NET VOLUME= 15.97 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	5.300	0.000	2.9	70.350	0.700	5.301	0.5
0.3	5.405	0.105	2.9	70.350	0.700	5.353	0.5
0.6	5.510	0.210	2.9	70.350	0.700	5.405	0.5
0.9	5.616	0.316	2.9	70.350	0.700	5.458	0.5
1.2	5.721	0.421	2.9	70.350	0.700	5.510	0.5
1.5	5.826	0.526	2.9	70.350	0.700	5.563	0.5
1.8	5.931	0.631	2.9	70.350	0.700	5.616	0.5
2.1	6.036	0.736	2.9	70.350	0.700	5.668	0.5
2.4	6.142	0.842	2.9	70.350	0.700	5.721	0.5
2.7	6.247	0.947	2.9	70.350	0.700	5.773	0.5
3.0	6.352	1.052	2.9	70.350	0.700	5.826	0.5
3.3	6.457	1.157	2.9	70.350	0.700	5.879	0.5
3.6	6.562	1.262	2.9	70.350	0.700	5.931	0.5
3.9	6.668	1.368	2.9	70.350	0.700	5.984	0.5
4.2	6.773	1.473	2.9	70.350	0.700	6.036	0.5
4.5	6.878	1.578	2.9	70.350	0.700	6.089	0.5
4.8	6.983	1.683	2.9	70.350	0.700	6.142	0.5
5.1	7.088	1.788	2.9	70.350	0.700	6.194	0.5
5.4	7.194	1.894	2.9	70.350	0.700	6.247	0.5
5.7	7.299	1.999	2.9	70.350	0.700	6.299	0.5
6.0	7.404	2.104	2.9	70.350	0.700	6.352	0.5
6.3	7.509	2.209	2.9	70.350	0.700	6.405	0.5
6.6	7.614	2.314	2.9	70.350	0.700	6.457	0.5
6.9	7.720	2.420	2.9	70.350	0.700	6.510	0.5
7.2	7.825	2.525	2.9	70.350	0.700	6.562	0.5
7.5	7.930	2.630	2.9	70.350	0.700	6.615	0.5
7.8	8.035	2.735	2.9	70.350	0.700	6.668	0.5
8.1	8.140	2.840	2.9	70.350	0.700	6.720	0.5
8.4	8.245	2.945	2.9	70.350	0.700	6.773	0.5
8.7	8.351	3.051	2.9	70.350	0.700	6.825	0.5
9.0	8.456	3.156	2.9	70.350	0.700	6.878	0.5
9.3	8.561	3.261	2.9	70.350	0.700	6.931	0.5
9.6	8.666	3.366	2.9	70.350	0.700	6.983	0.5
9.9	8.771	3.471	2.9	70.350	0.700	7.036	0.5
10.2	8.877	3.577	2.9	70.350	0.700	7.088	0.5
10.5	8.982	3.682	2.9	70.350	0.700	7.141	0.5
10.8	9.087	3.787	2.9	70.350	0.700	7.194	0.5
11.1	9.192	3.892	2.9	70.350	0.700	7.246	0.5
11.4	9.297	3.997	2.9	70.350	0.700	7.299	0.5
11.7	9.403	4.103	2.9	70.350	0.700	7.351	0.5
12.0	9.508	4.208	2.9	70.350	0.700	7.404	0.5
12.3	9.613	4.313	2.9	70.350	0.700	7.457	0.5
12.6	9.718	4.418	2.9	70.350	0.700	7.509	0.5
12.9	9.823	4.523	2.9	70.350	0.700	7.562	0.5
13.2	9.929	4.629	2.9	70.350	0.700	7.614	0.5
13.5	10.034	4.734	2.9	70.350	0.700	7.667	0.5
13.8	10.139	4.839	2.9	70.350	0.700	7.720	0.5
14.1	10.244	4.944	2.9	70.350	0.700	7.772	0.5
14.4	10.349	5.049	2.9	70.350	0.700	7.825	0.5
14.7	10.455	5.155	2.9	70.350	0.700	7.877	0.5
15.0	10.560	5.260	2.9	70.350	0.700	7.930	0.5
15.3	10.665	5.365	2.9	70.350	0.700	7.983	0.5
15.6	10.770	5.470	2.9	70.350	0.700	8.035	0.5
15.9	10.875	5.575	2.9	70.350	0.700	8.088	0.5
16.0	10.900	5.600	2.9	70.350	0.700	8.100	0.5

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TANK NO. 312 : CD VOID OUTSIDE LFL TANKS PS : NET VOLUME= 20.25 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	4.400	0.000	4.1	12.882	-8.418	4.401	0.2
0.4	4.496	0.096	4.5	12.816	-8.426	4.449	0.3
0.8	4.586	0.186	4.7	12.761	-8.433	4.495	0.3
1.2	4.671	0.271	5.0	12.714	-8.440	4.540	0.3
1.6	4.752	0.352	5.2	12.673	-8.446	4.583	0.3
2.0	4.830	0.430	5.3	12.638	-8.452	4.624	0.3
2.4	4.906	0.506	5.5	12.607	-8.458	4.665	0.4
2.8	4.981	0.581	5.6	12.578	-8.463	4.705	0.4
3.2	5.054	0.654	5.7	12.553	-8.468	4.744	0.4
3.6	5.126	0.726	5.8	12.530	-8.472	4.782	0.4
4.0	5.197	0.797	5.8	12.510	-8.476	4.820	0.4
4.4	5.268	0.868	5.9	12.491	-8.479	4.858	0.4
4.8	5.338	0.938	5.9	12.474	-8.482	4.895	0.4
5.2	5.407	1.007	5.9	12.459	-8.485	4.932	0.4
5.6	5.477	1.077	6.0	12.446	-8.488	4.968	0.4
6.0	5.546	1.146	6.0	12.433	-8.490	5.004	0.4
6.4	5.615	1.215	6.0	12.422	-8.492	5.040	0.4
6.8	5.684	1.284	6.0	12.412	-8.494	5.076	0.4
7.2	5.753	1.353	6.0	12.403	-8.496	5.112	0.4
7.6	5.822	1.422	6.0	12.395	-8.498	5.147	0.5
8.0	5.891	1.491	6.0	12.388	-8.499	5.183	0.5
8.4	5.960	1.560	6.0	12.381	-8.500	5.218	0.5
8.8	6.029	1.629	6.0	12.375	-8.501	5.253	0.5
9.2	6.097	1.697	6.0	12.370	-8.502	5.289	0.5
9.6	6.166	1.766	6.0	12.365	-8.503	5.324	0.5
10.0	6.235	1.835	6.0	12.360	-8.504	5.359	0.5
10.4	6.304	1.904	6.0	12.356	-8.505	5.394	0.5
10.8	6.373	1.973	6.0	12.352	-8.506	5.429	0.5
11.2	6.442	2.042	6.0	12.349	-8.506	5.464	0.5
11.6	6.511	2.111	6.0	12.345	-8.507	5.499	0.5
12.0	6.580	2.180	6.0	12.342	-8.508	5.534	0.5
12.4	6.649	2.249	6.0	12.339	-8.508	5.568	0.5
12.8	6.718	2.318	6.0	12.336	-8.509	5.603	0.5
13.2	6.786	2.386	6.0	12.334	-8.509	5.638	0.5
13.6	6.855	2.455	6.0	12.331	-8.510	5.673	0.5
14.0	6.924	2.524	6.0	12.329	-8.510	5.708	0.5
14.4	6.993	2.593	6.0	12.327	-8.511	5.742	0.5
14.8	7.062	2.662	6.0	12.325	-8.511	5.777	0.5
15.2	7.131	2.731	6.0	12.323	-8.511	5.812	0.5
15.6	7.200	2.800	6.0	12.321	-8.512	5.847	0.5
16.0	7.269	2.869	6.0	12.319	-8.512	5.881	0.5
16.4	7.338	2.938	6.0	12.317	-8.512	5.916	0.5
16.8	7.407	3.007	6.0	12.316	-8.513	5.951	0.5
17.2	7.475	3.075	6.0	12.314	-8.513	5.985	0.5
17.6	7.544	3.144	6.0	12.313	-8.513	6.020	0.5
18.0	7.613	3.213	6.0	12.311	-8.513	6.055	0.5
18.4	7.682	3.282	6.0	12.310	-8.514	6.089	0.5
18.8	7.751	3.351	6.0	12.309	-8.514	6.124	0.5
19.2	7.820	3.420	6.0	12.308	-8.514	6.158	0.5
19.6	7.889	3.489	6.0	12.306	-8.514	6.193	0.5
20.0	7.958	3.558	6.0	12.305	-8.515	6.228	0.5
20.2	8.000	3.600	6.0	12.305	-8.515	6.249	0.5

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TANK NO. 313 : CD VOID OUTSIDE LFL TANKS SB : NET VOLUME= 20.25 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	4.400	0.000	4.1	12.882	8.418	4.401	0.2
0.4	4.496	0.096	4.5	12.816	8.426	4.449	0.3
0.8	4.586	0.186	4.7	12.761	8.433	4.495	0.3
1.2	4.671	0.271	5.0	12.714	8.440	4.540	0.3
1.6	4.752	0.352	5.2	12.673	8.446	4.583	0.3
2.0	4.830	0.430	5.3	12.638	8.452	4.624	0.3
2.4	4.906	0.506	5.5	12.607	8.458	4.665	0.4
2.8	4.981	0.581	5.6	12.578	8.463	4.705	0.4
3.2	5.054	0.654	5.7	12.553	8.468	4.744	0.4
3.6	5.126	0.726	5.8	12.530	8.472	4.782	0.4
4.0	5.197	0.797	5.8	12.510	8.476	4.820	0.4
4.4	5.268	0.868	5.9	12.491	8.479	4.858	0.4
4.8	5.338	0.938	5.9	12.474	8.482	4.895	0.4
5.2	5.407	1.007	5.9	12.459	8.485	4.932	0.4
5.6	5.477	1.077	6.0	12.446	8.488	4.968	0.4
6.0	5.546	1.146	6.0	12.433	8.490	5.004	0.4
6.4	5.615	1.215	6.0	12.422	8.492	5.040	0.4
6.8	5.684	1.284	6.0	12.412	8.494	5.076	0.4
7.2	5.753	1.353	6.0	12.403	8.496	5.112	0.4
7.6	5.822	1.422	6.0	12.395	8.498	5.147	0.5
8.0	5.891	1.491	6.0	12.388	8.499	5.183	0.5
8.4	5.960	1.560	6.0	12.381	8.500	5.218	0.5
8.8	6.029	1.629	6.0	12.375	8.501	5.253	0.5
9.2	6.097	1.697	6.0	12.370	8.502	5.289	0.5
9.6	6.166	1.766	6.0	12.365	8.503	5.324	0.5
10.0	6.235	1.835	6.0	12.360	8.504	5.359	0.5
10.4	6.304	1.904	6.0	12.356	8.505	5.394	0.5
10.8	6.373	1.973	6.0	12.352	8.506	5.429	0.5
11.2	6.442	2.042	6.0	12.349	8.506	5.464	0.5
11.6	6.511	2.111	6.0	12.345	8.507	5.499	0.5
12.0	6.580	2.180	6.0	12.342	8.508	5.534	0.5
12.4	6.649	2.249	6.0	12.339	8.508	5.568	0.5
12.8	6.718	2.318	6.0	12.336	8.509	5.603	0.5
13.2	6.786	2.386	6.0	12.334	8.509	5.638	0.5
13.6	6.855	2.455	6.0	12.331	8.510	5.673	0.5
14.0	6.924	2.524	6.0	12.329	8.510	5.708	0.5
14.4	6.993	2.593	6.0	12.327	8.511	5.742	0.5
14.8	7.062	2.662	6.0	12.325	8.511	5.777	0.5
15.2	7.131	2.731	6.0	12.323	8.511	5.812	0.5
15.6	7.200	2.800	6.0	12.321	8.512	5.847	0.5
16.0	7.269	2.869	6.0	12.319	8.512	5.881	0.5
16.4	7.338	2.938	6.0	12.317	8.512	5.916	0.5
16.8	7.407	3.007	6.0	12.316	8.513	5.951	0.5
17.2	7.475	3.075	6.0	12.314	8.513	5.985	0.5
17.6	7.544	3.144	6.0	12.313	8.513	6.020	0.5
18.0	7.613	3.213	6.0	12.311	8.513	6.055	0.5
18.4	7.682	3.282	6.0	12.310	8.514	6.089	0.5
18.8	7.751	3.351	6.0	12.309	8.514	6.124	0.5
19.2	7.820	3.420	6.0	12.308	8.514	6.158	0.5
19.6	7.889	3.489	6.0	12.306	8.514	6.193	0.5
20.0	7.958	3.558	6.0	12.305	8.515	6.228	0.5
20.2	8.000	3.600	6.0	12.305	8.515	6.249	0.5

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TANK NO. 314 : CD VOID AROUND LFL TANKS CE : NET VOLUME= 200.50 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	4.400	0.000	101.4	12.250	0.000	4.401	2191.0
3.3	4.434	0.034	101.4	12.250	0.000	4.417	2191.0
6.6	4.467	0.067	101.4	12.250	0.000	4.434	2191.0
9.9	4.501	0.101	101.4	12.250	0.000	4.450	2191.0
13.2	4.534	0.134	101.4	12.250	0.000	4.467	2191.0
16.5	4.568	0.168	101.4	12.250	0.000	4.484	2191.0
19.8	4.601	0.201	101.4	12.250	0.000	4.501	2191.0
23.1	4.635	0.235	101.4	12.250	0.000	4.517	2191.0
26.4	4.668	0.268	101.4	12.250	0.000	4.534	2191.0
29.7	4.702	0.302	101.4	12.250	0.000	4.551	2191.0
33.0	4.735	0.335	101.4	12.250	0.000	4.568	2191.0
36.3	4.769	0.369	101.4	12.250	0.000	4.584	2191.0
39.6	4.802	0.402	101.4	12.250	0.000	4.601	2191.0
42.9	4.836	0.436	101.4	12.250	0.000	4.618	2191.0
46.2	4.870	0.470	101.4	12.250	0.000	4.635	2191.0
49.5	4.903	0.503	101.4	12.250	0.000	4.652	2191.0
52.8	4.937	0.537	101.4	12.250	0.000	4.668	2191.0
56.1	4.970	0.570	101.4	12.250	0.000	4.685	2191.0
59.4	5.013	0.613	29.5	12.250	0.000	4.702	496.7
62.7	5.128	0.728	29.5	12.250	0.000	4.721	491.6
66.0	5.244	0.844	29.4	12.250	0.000	4.745	488.0
69.3	5.360	0.960	29.4	12.250	0.000	4.771	488.0
72.6	5.475	1.075	29.4	12.250	0.000	4.800	488.0
75.9	5.591	1.191	29.4	12.250	0.000	4.832	488.0
79.2	5.707	1.307	29.4	12.250	0.000	4.866	488.0
82.5	5.822	1.422	29.4	12.250	0.000	4.902	488.0
85.8	5.938	1.538	29.4	12.250	0.000	4.940	488.0
89.1	6.054	1.654	29.4	12.250	0.000	4.979	488.0
92.4	6.170	1.770	29.4	12.250	0.000	5.019	488.0
95.7	6.285	1.885	29.4	12.250	0.000	5.061	488.0
99.0	6.401	2.001	29.4	12.250	0.000	5.104	488.0
102.3	6.517	2.117	29.4	12.250	0.000	5.148	488.0
105.6	6.632	2.232	29.4	12.250	0.000	5.192	488.0
108.9	6.748	2.348	29.4	12.250	0.000	5.238	488.0
112.2	6.864	2.464	29.4	12.250	0.000	5.284	488.0
115.5	6.980	2.580	29.4	12.250	0.000	5.330	488.0
118.8	7.095	2.695	29.4	12.250	0.000	5.378	488.0
122.1	7.203	2.803	101.4	12.250	0.000	5.426	2191.0
125.4	7.237	2.837	101.4	12.250	0.000	5.473	2191.0
128.7	7.270	2.870	101.4	12.250	0.000	5.519	2191.0
132.0	7.304	2.904	101.4	12.250	0.000	5.563	2191.0
135.3	7.337	2.937	101.4	12.250	0.000	5.606	2191.0
138.6	7.371	2.971	101.4	12.250	0.000	5.647	2191.0
141.9	7.404	3.004	101.4	12.250	0.000	5.688	2191.0
145.2	7.438	3.038	101.4	12.250	0.000	5.727	2191.0
148.5	7.472	3.072	101.4	12.250	0.000	5.766	2191.0
151.8	7.505	3.105	101.4	12.250	0.000	5.803	2191.0
155.1	7.539	3.139	101.4	12.250	0.000	5.840	2191.0
158.4	7.572	3.172	101.4	12.250	0.000	5.875	2191.0
161.7	7.606	3.206	101.4	12.250	0.000	5.910	2191.0
165.0	7.639	3.239	101.4	12.250	0.000	5.945	2191.0
168.3	7.673	3.273	101.4	12.250	0.000	5.978	2191.0
171.6	7.706	3.306	101.4	12.250	0.000	6.011	2191.0
174.9	7.740	3.340	101.4	12.250	0.000	6.043	2191.0
178.2	7.773	3.373	101.4	12.250	0.000	6.075	2191.0
181.5	7.807	3.407	101.4	12.250	0.000	6.106	2191.0
184.8	7.840	3.440	101.4	12.250	0.000	6.137	2191.0
188.1	7.874	3.474	101.4	12.250	0.000	6.167	2191.0
191.4	7.908	3.508	101.4	12.250	0.000	6.197	2191.0
194.7	7.941	3.541	101.4	12.250	0.000	6.226	2191.0
198.0	7.975	3.575	101.4	12.250	0.000	6.255	2191.0
200.5	8.000	3.600	101.4	12.250	0.000	6.277	2191.0

See ABS Shipbuilder Letter Ref 1489174 Dated 04-APR-2016

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TANK NO. 317 : CD VOID CE : NET VOLUME= 33.12 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	0.000	0.000	17.8	10.029	0.000	0.001	5.0
0.6	0.035	0.035	17.9	10.019	0.000	0.017	5.0
1.2	0.069	0.069	17.9	10.011	0.000	0.035	5.1
1.8	0.104	0.104	17.8	10.003	0.000	0.052	5.1
2.4	0.138	0.138	17.8	9.994	0.000	0.069	5.1
3.0	0.173	0.173	17.9	9.988	0.000	0.086	5.1
3.6	0.208	0.208	17.9	9.981	0.000	0.104	5.1
4.2	0.242	0.242	18.0	9.975	0.000	0.121	5.1
4.8	0.276	0.276	18.0	9.969	0.000	0.138	5.1
5.4	0.311	0.311	18.1	9.963	0.000	0.155	5.1
6.0	0.345	0.345	18.1	9.957	0.000	0.173	5.2
6.6	0.379	0.379	18.1	9.951	0.000	0.190	5.2
7.2	0.413	0.413	18.2	9.945	0.000	0.207	5.2
7.8	0.447	0.447	18.2	9.940	0.000	0.224	5.2
8.4	0.481	0.481	18.3	9.934	0.000	0.241	5.2
9.0	0.515	0.515	18.4	9.929	0.000	0.258	5.4
9.6	0.548	0.548	18.6	9.927	0.000	0.275	5.6
10.2	0.581	0.581	18.8	9.926	0.000	0.293	6.0
10.8	0.614	0.614	19.4	9.931	0.000	0.309	7.1
11.4	0.645	0.645	20.0	9.942	0.000	0.326	8.4
12.0	0.676	0.676	20.5	9.957	0.000	0.343	10.0
12.6	0.705	0.705	21.2	9.976	0.000	0.360	12.0
13.2	0.734	0.734	21.8	9.999	0.000	0.376	14.4
13.8	0.762	0.762	22.4	10.024	0.000	0.392	16.9
14.4	0.789	0.789	23.1	10.050	0.000	0.408	19.8
15.0	0.815	0.815	23.9	10.079	0.000	0.424	23.1
15.6	0.841	0.841	24.7	10.110	0.000	0.439	26.6
16.2	0.865	0.865	25.5	10.141	0.000	0.455	30.3
16.8	0.889	0.889	26.1	10.172	0.000	0.470	33.5
17.4	0.913	0.913	26.6	10.203	0.000	0.485	34.9
18.0	0.936	0.936	27.0	10.233	0.000	0.499	35.7
18.6	0.959	0.959	27.1	10.261	0.000	0.514	35.9
19.2	0.982	0.982	27.1	10.288	0.000	0.528	35.9
19.8	1.004	1.004	27.1	10.312	0.000	0.542	35.9
20.4	1.027	1.027	27.3	10.335	0.000	0.556	36.0
21.0	1.050	1.050	27.4	10.356	0.000	0.570	36.2
21.6	1.072	1.072	27.5	10.377	0.000	0.583	36.3
22.2	1.095	1.095	27.5	10.396	0.000	0.597	36.4
22.8	1.117	1.117	27.6	10.414	0.000	0.610	36.4
23.4	1.140	1.140	27.6	10.431	0.000	0.624	36.4
24.0	1.162	1.162	27.6	10.447	0.000	0.637	36.4
24.6	1.185	1.185	27.7	10.462	0.000	0.650	36.4
25.2	1.207	1.207	27.7	10.476	0.000	0.663	36.4
25.8	1.229	1.229	27.7	10.489	0.000	0.676	36.5
26.4	1.252	1.252	27.7	10.501	0.000	0.689	36.5
27.0	1.274	1.274	27.8	10.513	0.000	0.701	36.5
27.6	1.296	1.296	27.8	10.524	0.000	0.714	36.5
28.2	1.318	1.318	27.8	10.535	0.000	0.727	36.5
28.8	1.341	1.341	27.8	10.544	0.000	0.739	36.5
29.4	1.363	1.363	27.9	10.554	0.000	0.752	36.5
30.0	1.385	1.385	27.9	10.563	0.000	0.764	36.5
30.6	1.407	1.407	27.9	10.571	0.000	0.777	36.5
31.2	1.429	1.429	27.9	10.579	0.000	0.789	36.5
31.8	1.451	1.451	28.0	10.586	0.000	0.801	36.5
32.4	1.474	1.474	28.0	10.593	0.000	0.814	36.5
33.0	1.496	1.496	28.0	10.600	0.000	0.826	36.6
33.1	1.500	1.500	28.0	10.602	0.000	0.828	36.6

See ABS Singapore Letter Rpt 1469174 Dated 04-APR-2016

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TANK NO. 318 : CD VOID PS : NET VOLUME= 10.45 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.400	0.000	0.6	56.351	-5.035	1.401	0.0
0.2	1.684	0.284	0.8	56.351	-5.106	1.548	0.1
0.4	1.916	0.516	1.0	56.351	-5.164	1.676	0.2
0.6	2.119	0.719	1.1	56.351	-5.213	1.790	0.2
0.8	2.303	0.903	1.2	56.351	-5.257	1.896	0.3
1.0	2.472	1.072	1.3	56.351	-5.296	1.994	0.3
1.2	2.632	1.232	1.3	56.351	-5.332	2.087	0.4
1.4	2.782	1.382	1.4	56.351	-5.365	2.176	0.5
1.6	2.926	1.526	1.5	56.351	-5.395	2.261	0.5
1.8	3.064	1.664	1.5	56.351	-5.424	2.342	0.6
2.0	3.198	1.798	1.6	56.351	-5.450	2.421	0.7
2.2	3.327	1.927	1.6	56.351	-5.476	2.498	0.7
2.4	3.452	2.052	1.7	56.351	-5.500	2.572	0.8
2.6	3.574	2.174	1.7	56.351	-5.522	2.645	0.9
2.8	3.693	2.293	1.8	56.351	-5.544	2.715	0.9
3.0	3.809	2.409	1.8	56.351	-5.564	2.784	1.0
3.2	3.923	2.523	1.8	56.351	-5.584	2.852	1.0
3.4	4.035	2.635	1.9	56.351	-5.603	2.918	1.1
3.6	4.144	2.744	1.9	56.351	-5.621	2.983	1.2
3.8	4.252	2.852	1.9	56.351	-5.638	3.047	1.2
4.0	4.358	2.958	2.0	56.351	-5.655	3.110	1.3
4.2	4.463	3.063	2.0	56.351	-5.671	3.172	1.3
4.4	4.566	3.166	2.0	56.351	-5.686	3.233	1.4
4.6	4.668	3.268	2.0	56.351	-5.701	3.293	1.4
4.8	4.768	3.368	2.1	56.351	-5.716	3.353	1.5
5.0	4.868	3.468	2.1	56.351	-5.730	3.411	1.6
5.2	4.966	3.566	2.1	56.351	-5.744	3.469	1.6
5.4	5.063	3.663	2.1	56.351	-5.757	3.526	1.7
5.6	5.159	3.759	2.2	56.351	-5.770	3.583	1.7
5.8	5.254	3.854	2.2	56.351	-5.782	3.639	1.8
6.0	5.372	3.972	1.5	56.351	-5.803	3.695	0.6
6.2	5.509	4.109	1.5	56.351	-5.831	3.751	0.6
6.4	5.643	4.243	1.6	56.351	-5.859	3.808	0.6
6.6	5.775	4.375	1.6	56.351	-5.885	3.866	0.7
6.8	5.904	4.504	1.6	56.350	-5.910	3.924	0.7
7.0	6.032	4.632	1.6	56.350	-5.934	3.982	0.7
7.2	6.157	4.757	1.7	56.350	-5.957	4.041	0.8
7.4	6.281	4.881	1.7	56.350	-5.980	4.100	0.8
7.6	6.403	5.003	1.7	56.350	-6.001	4.159	0.8
7.8	6.523	5.123	1.7	56.350	-6.023	4.218	0.9
8.0	6.642	5.242	1.7	56.350	-6.043	4.277	0.9
8.2	6.760	5.360	1.8	56.350	-6.063	4.336	0.9
8.4	6.876	5.476	1.8	56.350	-6.082	4.395	1.0
8.6	6.991	5.591	1.8	56.350	-6.101	4.454	1.0
8.8	7.104	5.704	1.8	56.350	-6.119	4.513	1.0
9.0	7.217	5.817	1.8	56.350	-6.136	4.572	1.1
9.2	7.328	5.928	1.9	56.349	-6.153	4.631	1.1
9.4	7.438	6.038	1.9	56.349	-6.170	4.689	1.1
9.6	7.548	6.148	1.9	56.349	-6.186	4.748	1.2
9.8	7.656	6.256	1.9	56.349	-6.202	4.806	1.2
10.0	7.764	6.364	1.9	56.349	-6.217	4.864	1.2
10.2	7.870	6.470	1.9	56.349	-6.232	4.922	1.2
10.4	7.976	6.576	2.0	56.349	-6.247	4.980	1.3
10.4	8.000	6.600	2.0	56.349	-6.250	4.993	1.3

See ABS Singapore Letter Ref 1499174 Dated 04-APR-2016

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TANK NO. 319 : CD VOID SB : NET VOLUME= 32.45 M3

VOLUME NET m3	LEVEL FROM BL m	BOTTOM m	W.PLANE AREA m2	CENTRE OF GRAVITY LONG m	TRANSV m	VERT m	INERTIA MOMENT m4
0.0	1.400	0.000	1.7	54.268	5.671	1.401	0.2
0.6	1.702	0.302	2.3	54.265	5.748	1.559	0.3
1.2	1.944	0.544	2.8	54.264	5.808	1.693	0.5
1.8	2.154	0.754	3.1	54.264	5.859	1.812	0.7
2.4	2.344	0.944	3.4	54.264	5.903	1.922	0.9
3.0	2.514	1.114	4.9	54.281	5.940	2.024	1.6
3.6	2.637	1.237	5.1	54.408	5.964	2.116	1.7
4.2	2.757	1.357	5.2	54.499	5.988	2.199	1.9
4.8	2.874	1.474	5.4	54.569	6.013	2.276	2.0
5.4	2.988	1.588	5.5	54.624	6.037	2.349	2.1
6.0	3.101	1.701	5.6	54.669	6.061	2.418	2.3
6.6	3.211	1.811	5.6	54.706	6.084	2.485	2.4
7.2	3.320	1.920	5.7	54.738	6.107	2.550	2.5
7.8	3.427	2.027	5.8	54.765	6.130	2.614	2.7
8.4	3.533	2.133	5.9	54.789	6.152	2.676	2.8
9.0	3.639	2.239	5.9	54.810	6.173	2.736	2.9
9.6	3.743	2.343	6.0	54.829	6.194	2.796	3.0
10.2	3.846	2.446	6.0	54.846	6.214	2.855	3.1
10.8	3.948	2.548	6.1	54.862	6.234	2.913	3.3
11.4	4.050	2.650	6.1	54.876	6.254	2.970	3.4
12.0	4.152	2.752	6.1	54.889	6.273	3.026	3.5
12.6	4.253	2.853	6.1	54.901	6.291	3.082	3.6
13.2	4.353	2.953	6.2	54.913	6.309	3.138	3.7
13.8	4.453	3.053	6.2	54.924	6.327	3.193	3.8
14.4	4.553	3.153	6.2	54.934	6.345	3.248	3.9
15.0	4.653	3.253	6.2	54.943	6.362	3.302	4.0
15.6	4.753	3.353	6.2	54.953	6.378	3.356	4.1
16.2	4.852	3.452	6.2	54.961	6.395	3.409	4.2
16.8	4.952	3.552	6.2	54.970	6.411	3.463	4.3
17.4	5.051	3.651	6.2	54.978	6.427	3.516	4.4
18.0	5.151	3.751	6.2	54.985	6.442	3.568	4.5
18.6	5.250	3.850	6.2	54.993	6.458	3.621	4.7
19.2	5.356	3.956	5.5	54.998	6.476	3.674	1.9
19.8	5.469	4.069	5.5	55.001	6.498	3.726	2.0
20.4	5.582	4.182	5.5	55.003	6.520	3.779	2.0
21.0	5.696	4.296	5.5	55.006	6.541	3.832	2.1
21.6	5.809	4.409	5.4	55.009	6.562	3.886	2.1
22.2	5.923	4.523	5.4	55.013	6.582	3.939	2.2
22.8	6.038	4.638	5.4	55.016	6.602	3.993	2.2
23.4	6.153	4.753	5.4	55.019	6.621	4.047	2.2
24.0	6.268	4.868	5.3	55.023	6.640	4.101	2.3
24.6	6.385	4.985	5.3	55.027	6.659	4.155	2.3
25.2	6.502	5.102	5.3	55.030	6.677	4.210	2.4
25.8	6.619	5.219	5.2	55.034	6.695	4.264	2.4
26.4	6.738	5.338	5.2	55.038	6.713	4.319	2.5
27.0	6.858	5.458	5.2	55.042	6.730	4.374	2.5
27.6	6.978	5.578	5.1	55.047	6.748	4.430	2.6
28.2	7.100	5.700	5.1	55.051	6.764	4.485	2.7
28.8	7.223	5.823	5.0	55.055	6.781	4.541	2.7
29.4	7.347	5.947	5.0	55.060	6.797	4.597	2.8
30.0	7.472	6.072	4.9	55.065	6.813	4.653	2.8
30.6	7.599	6.199	4.9	55.070	6.829	4.710	2.9
31.2	7.727	6.327	4.8	55.075	6.844	4.766	2.9
31.8	7.857	6.457	4.7	55.080	6.859	4.823	3.0
32.5	8.000	6.600	4.7	55.086	6.875	4.886	3.0

See ABS Singapore Letter Ref 14699174 Dated 04-APR-2016

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12 LOADING CONDITIONS

This chapter contains detailed documentation of the loading conditions. The below briefly listed loading conditions are documented on the next pages for both intact and damage stability. The calculation of the stability curve and the metacentric height, GM, take the actual floating position into account. Therefore, small deviation to manual calculations can NOT be prevented.

The real free surface moment is calculated for the tanks at their actual filling. The correction for different heeling angles is got by calculating the real new center of gravity for the load in the heeled tank. The correction for the initial GM is got by calculating the real moment of inertia for the free surface in the tank.

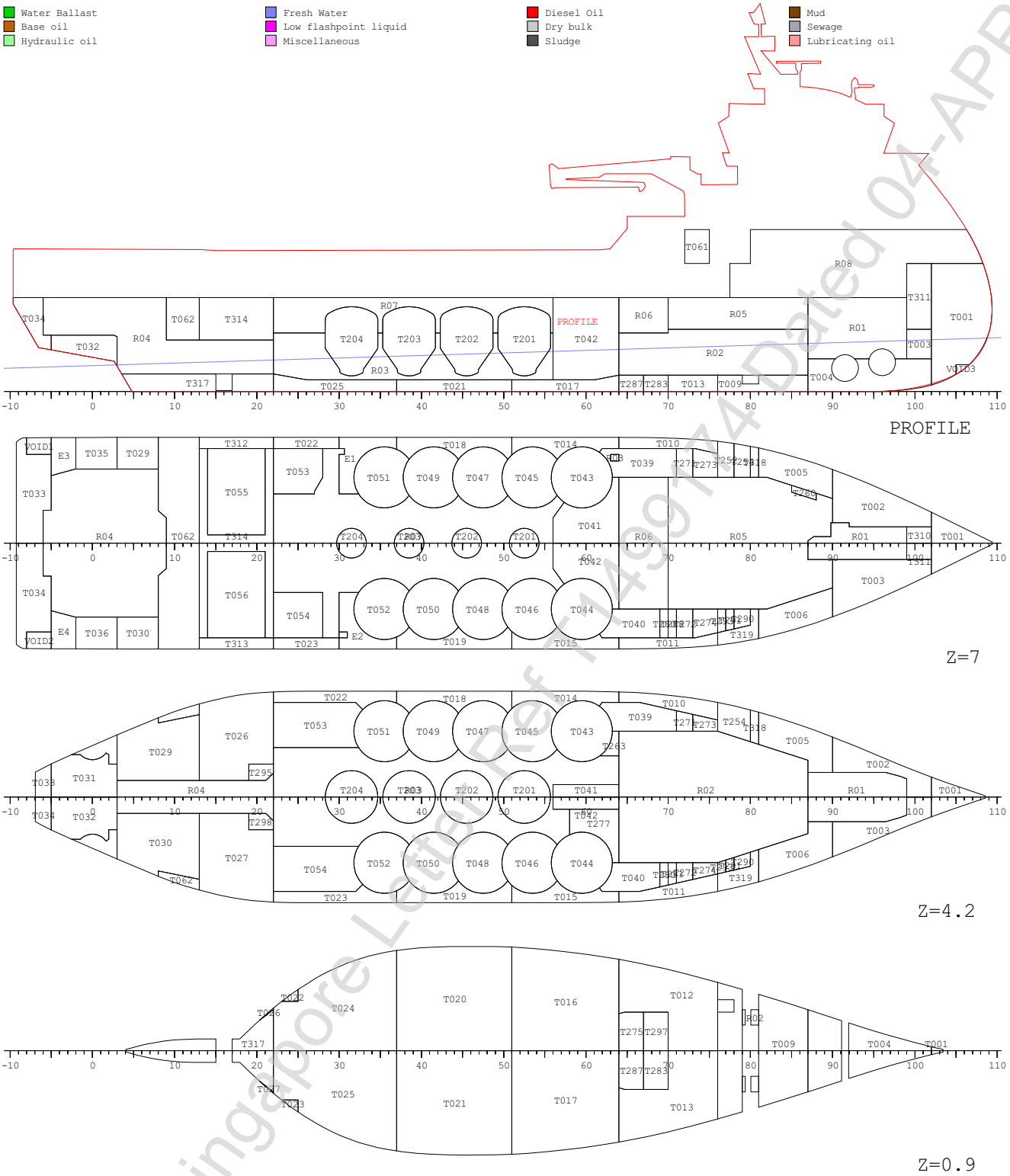
Please, refer to the chapter INSTRUCTIONS FOR MASTER for further details of the calculation methods.

NAME	DESCRIPTION	DISP t	T m	TA m	TF m	TR m	HEEL degree	GM m	GMCORR m
L01	Lightship - NOT SAILING CONDITION!	2704.9	3.346	2.122	4.571	2.449	-0.3	2.763	0.000
L02	Ballast - Departure	4826.2	5.170	5.038	5.302	0.264	0.0	2.465	-0.730
L03	Ballast - Arrival	4293.1	4.762	4.387	5.137	0.751	0.0	2.379	-0.816
L04	Fully loaded SWL - Departure	6814.3	6.700	6.567	6.832	0.266	0.0	2.288	-0.644
L05	Fully loaded SWL - Arrival	6188.3	6.208	6.207	6.209	0.003	0.0	2.301	-0.719
L06	Max deckload - Departure	6389.8	6.360	6.358	6.363	0.005	0.0	1.745	-0.046
L07	Max deckload - Arrival	6086.7	6.077	6.407	5.748	-0.659	0.0	1.722	-0.038
L08	Max deckload - roll red tk - Departure	6234.4	6.257	6.164	6.351	0.187	0.0	1.389	-0.544
L09	Max deckload - roll red tk - Arrival	5931.3	5.974	6.213	5.735	-0.478	0.0	1.352	-0.560
L10	Pipe deckload - Departure	6114.1	6.180	5.999	6.361	0.362	0.0	1.536	-0.048
L11	Pipe deckload - Arrival	5811.0	5.896	6.050	5.742	-0.308	0.0	1.510	-0.039
L12	CEM - Departure	4302.3	4.719	4.780	4.659	-0.121	0.0	1.883	-0.794
L13	CEM - Arrival	3953.8	4.438	4.419	4.457	0.039	0.0	1.821	-0.862
L14	MUD - Departure	5601.9	5.762	5.749	5.775	0.027	0.0	2.518	-0.499
L15	MUD - Arrival	5255.3	5.498	5.450	5.546	0.096	0.0	2.492	-0.531
L16	LFL-BRI-MUD - DEPARTURE	6795.2	6.680	6.582	6.777	0.195	0.0	2.532	-0.490
L17	LFL-BRI-MUD - ARRIVAL	6575.2	6.539	6.289	6.789	0.500	0.0	2.539	-0.507
L18	Transition,Fi-Fi - Departure	4250.3	4.680	4.717	4.643	-0.074	0.0	1.665	-0.796
L19	Transition,Fi-Fi - Arrival	3901.5	4.397	4.348	4.445	0.097	0.0	1.586	-0.865

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12.1 L01 -> Lightship - NOT SAILING CONDITION!



See ABS Singapore Letter Ref 1149174 Dated 04-APR-2016

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LOAD SUMMARY TABLE

Lightweight	2704.9	39.15	-0.01	7.08
Deadweight	0.0	0.00	0.00	0.00
Total weight	2704.9	39.15	-0.01	7.08

F L O A T I N G P O S I T I O N

Draught moulded	3.346	m	KM	9.84 m
Trim	2.449	m	KG	7.08 m
Heel, SB=+	-0.3	deg		
TA	2.122	m	GM0	2.76 m
TF	4.571	m	GMCORR	0.00 m
Trimming moment	8467	tonm	GM	2.76 m

STABILITY COMPONENTS

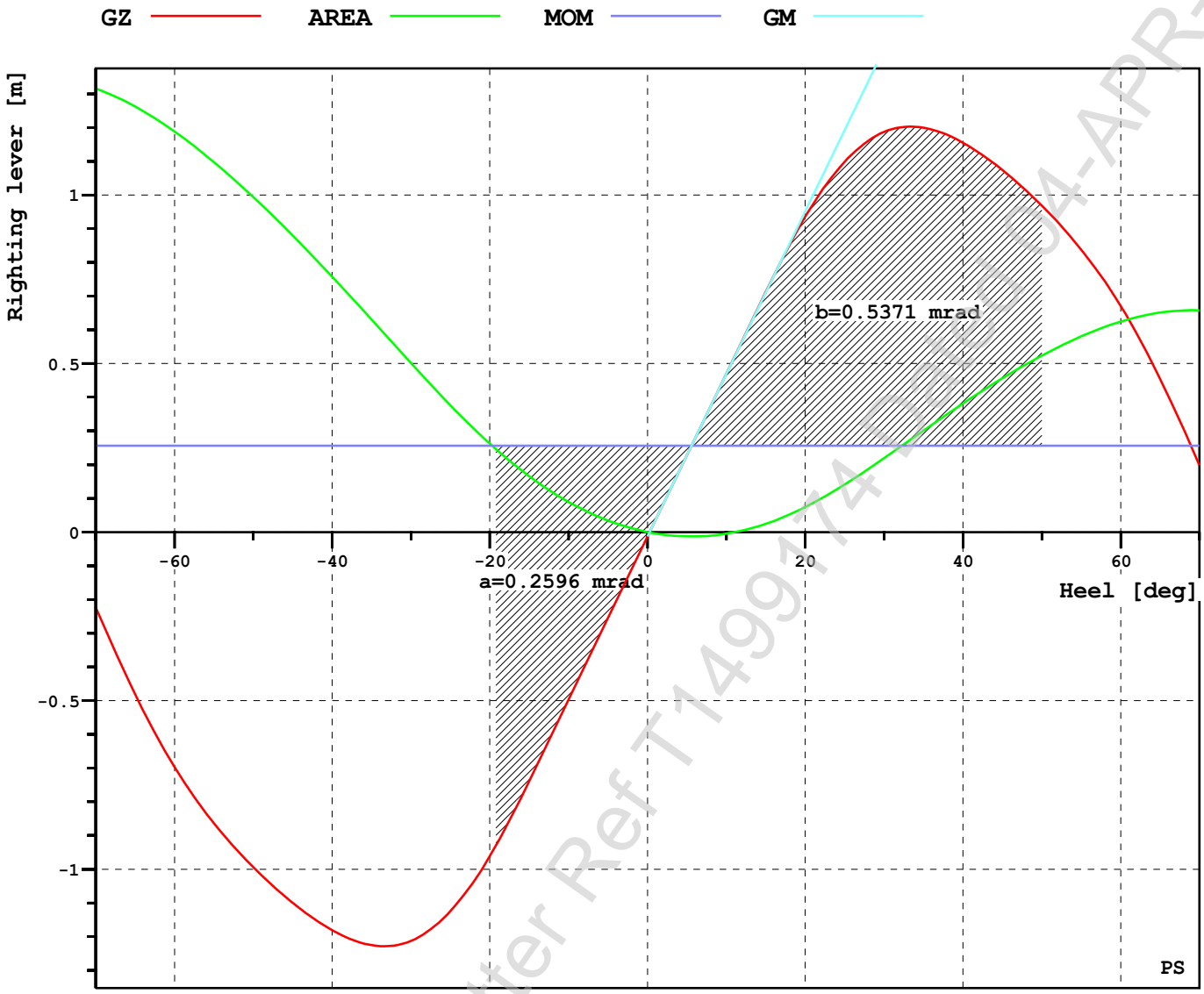
LOADING CONDITION L01, Lightship - NOT SAILING CONDITION!

HEEL degree	MS m	HPHI m	EPHI mrad	FSMOM tm	DGZ m	GMSIN m	KN m
0.0	-0.013	-0.01	0.000	0.0	0.000	0.000	0.002
1.0	-0.013	0.04	0.000	0.0	0.000	0.048	0.174
3.0	-0.013	0.13	0.003	0.0	0.000	0.145	0.517
5.0	-0.013	0.23	0.009	0.0	0.000	0.241	0.860
10.0	-0.009	0.47	0.040	0.0	0.000	0.480	1.715
15.0	-0.003	0.71	0.092	0.0	0.000	0.715	2.560
20.0	-0.010	0.94	0.164	0.0	0.000	0.945	3.371
30.0	-0.194	1.19	0.353	0.0	0.000	1.382	4.741
40.0	-0.621	1.16	0.561	0.0	0.000	1.776	5.718
50.0	-1.148	0.97	0.748	0.0	0.000	2.117	6.402
60.0	-1.722	0.67	0.893	0.0	0.000	2.393	6.811
70.0	-2.399	0.20	0.971	0.0	0.000	2.597	6.857

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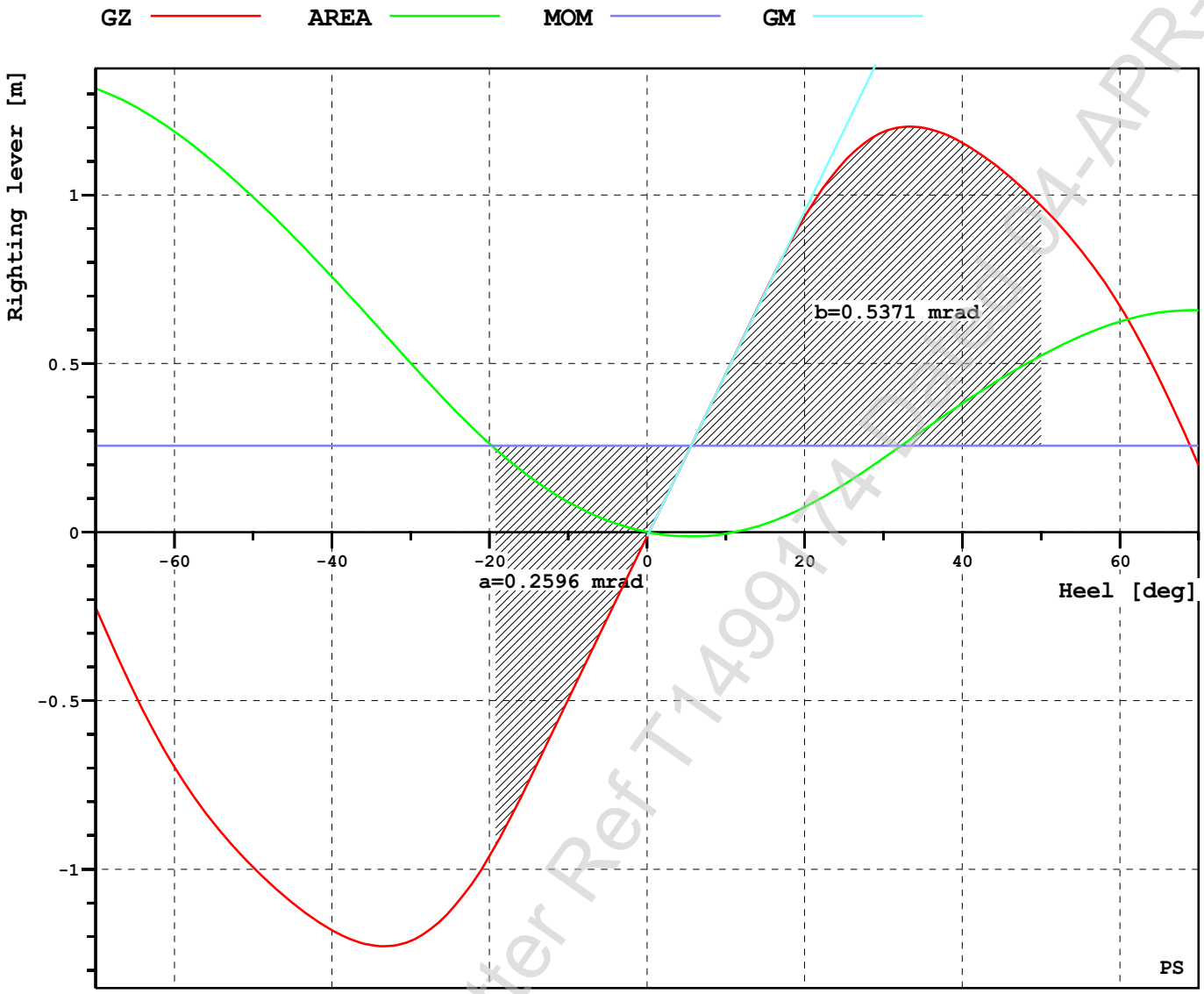
STABILITY CHECK PLOT



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IMOWEATHER CHECK PLOT



INTACT STABILITY CRITERIA

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM
V.AREA15-30	Area depending on the position of the Max. .	0.055	0.353	mrad	OK	0.827
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.208	mrad	OK	0.986
V.GZ0.2	Min. GZ > 0.2	0.200	1.203	m	OK	0.789
V.POSMAX15	Max. GZ at an angle > 15 deg.	15.000	33.373	deg	OK	-0.040
V.GM0.15	GM > 0.15 m	0.150	2.763	m	OK	0.150
IMOWEATHER	IMO weather criterion	1.000	2.069		OK	1.633
2008IS-A2.3.1.2	Heeling angle due to steady wind <16 or <=8.	16.000	3.816	deg	OK	0.627
ABSFIFI1	Angle of heel < 1/2 freeboard immersion	13.212	0.916	deg	OK	0.159
ABSFIFI2	Area of residual stability >0.09 mrad	0.090	0.539	mrad	OK	0.824
TOTAL						1.633

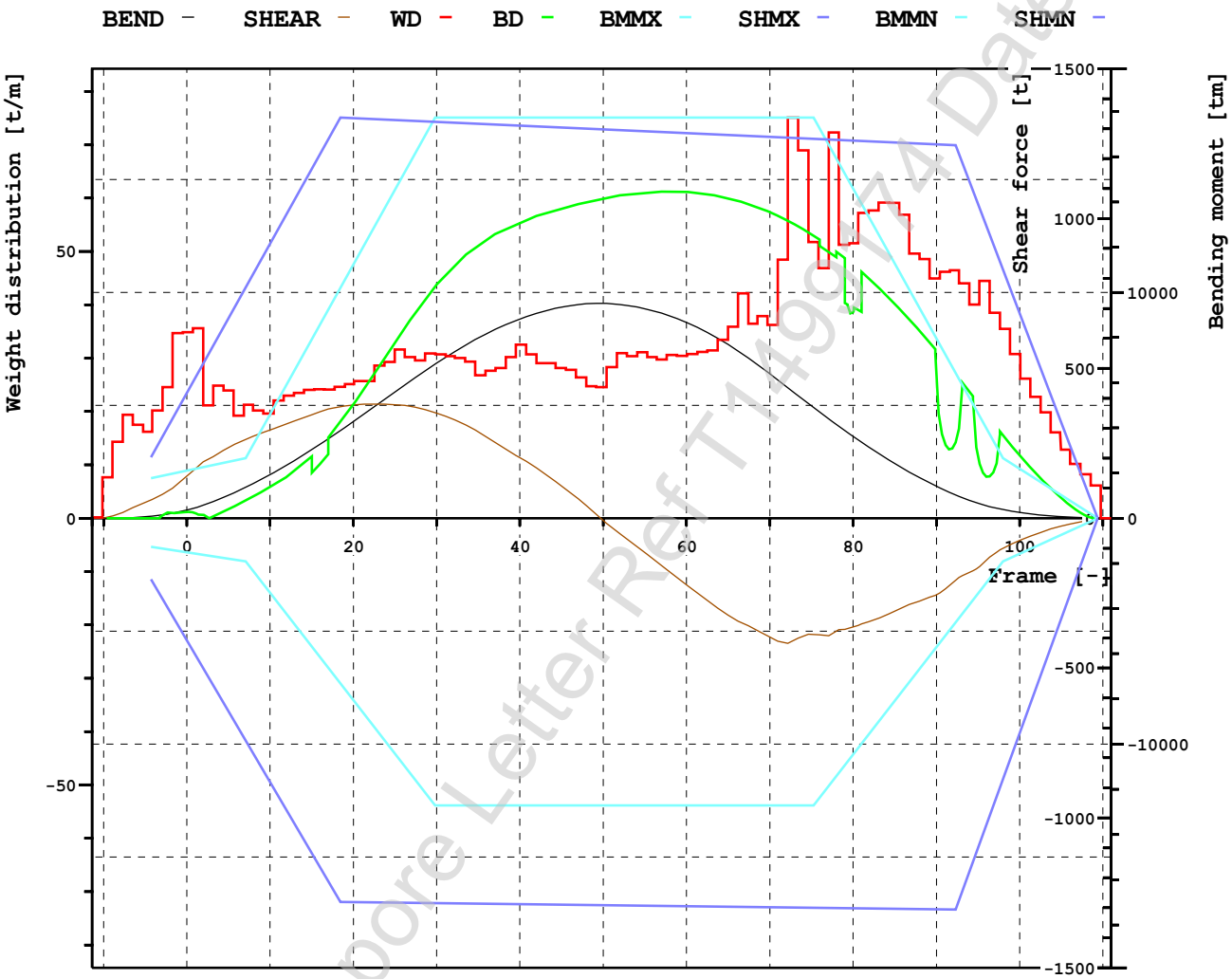
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MOST CRITICAL OPENINGS

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
V-421G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	55.890	2.093	14.288	-	10.358
V-424G	SUPPLY AIR ENGINE ROOM PS	UNPROTECTED	54.243	-3.413	14.288	-	10.383
V-425G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	54.243	3.767	14.288	-	10.416
D-651	ACCOMMODATION/C-DECK	UNPROTECTED	56.000	3.771	16.906	-	12.979
V-421E	EXHAUST AIR ENGINE ROOM PS	UNPROTECTED	55.993	-2.538	17.088	-	13.132
V-423E	EXHAUST AIR ENGINE ROOM SB	UNPROTECTED	55.993	3.423	17.088	-	13.159

STRENGTH DIAGRAM



STRENGTH SUMMARY

LOADING CONDITION L01, Lightship - NOT SAILING CONDITION!

			X	Frame
Shear force (min)	-417.2 t	(32.1%)	50.526 m	72.18
Shear force (max)	380.6 t	(28.6%)	16.461 m	23.52
Sagging moment	-6.3 tm	(0.0%)	-6.248 m	-8.93
Hogging moment	9512.3 tm	(53.6%)	34.427 m	49.18

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STRENGTH DISTRIBUTION

LOADING CONDITION L01, Lightship - NOT SAILING CONDITION!

X	FR	BEND	BMMN	BMMX	BMREL	SHEAR	SHMN	SHMX	SHREL	WD	BD
m	#	tm	tm	tm	%	t	t	t	%	t/m	t/m
-3.500	-5.00	59	0	0	0.0	53	0	0	0.0	16.23	0.00
0.000	0.00	358	-1510	2108	17.0	141	-407	418	33.7	34.85	1.15
3.500	5.00	1016	-1790	2498	40.6	236	-644	667	35.3	23.96	1.71
7.000	10.00	1930	-3292	4594	42.1	295	-881	916	32.2	19.61	5.86
10.500	15.00	3032	-5668	7911	38.3	345	-1117	1166	29.6	24.04	11.60
14.000	20.00	4290	-8045	11228	38.2	378	-1281	1335	28.3	25.16	21.37
17.500	25.00	5601	-10422	14545	38.5	379	-1283	1329	28.5	31.62	32.90
21.000	30.00	6870	-12709	17736	38.7	351	-1284	1323	26.5	30.72	43.80
24.500	35.00	7978	-12709	17736	45.0	288	-1286	1317	21.9	26.82	51.08
28.000	40.00	8816	-12709	17736	49.7	202	-1288	1310	15.4	32.53	55.32
31.500	45.00	9345	-12709	17736	52.7	108	-1290	1304	8.3	28.17	57.99
35.000	50.00	9508	-12709	17736	53.6	-7	-1292	1298	0.5	24.56	59.86
38.500	55.00	9272	-12709	17736	52.3	-115	-1293	1292	8.9	31.10	60.94
42.000	60.00	8664	-12709	17736	48.8	-223	-1295	1285	17.2	30.42	61.15
45.500	65.00	7683	-12709	17736	43.3	-324	-1297	1279	25.0	35.91	59.94
49.000	70.00	6400	-12709	17736	36.1	-397	-1299	1273	30.5	36.21	57.36
52.500	75.00	4969	-12709	17736	28.0	-388	-1300	1267	29.8	51.75	53.22
56.000	80.00	3617	-10463	14601	24.8	-363	-1302	1260	27.9	51.53	38.62
59.500	85.00	2415	-8086	11284	21.4	-310	-1304	1254	23.8	59.12	40.18
63.000	90.00	1410	-5709	7968	17.7	-256	-1306	1248	19.6	44.92	27.30
66.500	95.00	666	-3332	4651	14.3	-165	-1102	1050	14.9	40.04	11.58
70.000	100.00	252	-1571	2193	11.5	-74	-718	685	10.3	30.80	11.79
73.500	105.00	63	-733	1023	6.1	-26	-335	319	7.9	12.84	3.78

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DAMAGE STABILITY OF LOADING CONDITION L01

CRITICAL DAMAGE STABILITY CRITERIA FOR EACH DAMAGE CASE

RCR	DAM	STAGE	PHASE	REQ	ATTV	UNIT	STAT	MINGM
V.MINGZ.A469	D01	1	EQ	0.100	0.948	m	OK	0.315
V.MINGZ.A469	D02	1	EQ	0.100	0.930	m	OK	0.340
V.MINGZ.A469	D03	1	EQ	0.100	0.924	m	OK	0.418
V.MINGZ.A469	D04	1	EQ	0.100	0.851	m	OK	0.728
V.MINGZ.A469	D05	1	EQ	0.100	0.824	m	OK	0.874
V.MINGZ.A469	D06	1	EQ	0.100	0.832	m	OK	0.810
V.MINGZ.A469	D07	1	EQ	0.100	0.805	m	OK	0.951
V.RANGE.A469	D08	1	EQ	20.000	66.187	deg	OK	0.738
V.RANGE.A469	D09	1	EQ	20.000	65.651	deg	OK	0.799
V.RANGE.A469	D10	1	EQ	20.000	65.529	deg	OK	0.778
V.RANGE.A469	D11	1	EQ	20.000	64.999	deg	OK	0.833
V.RANGE.A469	D12	1	EQ	20.000	66.664	deg	OK	0.638
V.RANGE.A469	D13	1	EQ	20.000	66.127	deg	OK	0.707
V.MINGZ.A469	D14	1	EQ	0.100	0.902	m	OK	0.463
V.MINGZ.A469	D15	1	EQ	0.100	0.898	m	OK	0.536
V.MINGZ.A469	D16	1	EQ	0.100	0.938	m	OK	0.338
V.MINGZ.A469	D17	1	EQ	0.100	0.928	m	OK	0.372
V.MINGZ.A469	D18	1	EQ	0.100	0.946	m	OK	0.321
V.MINGZ.A469	D19	1	EQ	0.100	0.946	m	OK	0.321
V.MINGZ.A469	D20	1	EQ	0.100	0.946	m	OK	0.321
V.MINGZ.A469	D21	1	EQ	0.100	0.946	m	OK	0.321
TOTAL								0.951

DETERMINATION OF CRITICAL DAMAGE CASE

CRITICAL DAMAGE CASE : D07
STAGE OF CRITICAL DAMAGE CASE : 1
PHASE OF CRITICAL DAMAGE CASE : EQ

DAMAGE STABILITY CRITERIA - CRITICAL CASE

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM m
V.PROGR.A469	No progr. flooding	30.416	2.886	deg	OK	0.682
V.MAXHEEL.A469	Maximum heel	17.000	2.886	deg	OK	0.727
V.RANGE.A469	Range of pos. stab	20.000	67.114	deg	OK	0.928
V.MINGZ.A469	Min. righting lever	0.100	0.805	m	OK	0.951
TOTAL						0.951

DAMAGE CASE RESULTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	T m	TR m	HEEL degree	RESFLD m	OPEN	CROSSTM s
D07	1	EQ	PS	3.389	2.671	-2.9	4.27	D-351	-

DAMAGED COMPARTMENTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	NAME	DES	PERM	VOL m3	W t	XCG m	YCG m	ZCG m
D07	INTACT	EQ	PS				0.0	0.000			
D07	1	EQ	PS	T010	WING TANK PS	0.95	28.2	28.874	49.47	-7.39	2.79
D07	1	EQ	PS	T273	FO SERVICE TANK PS	0.95	3.0	3.049	52.06	-6.07	3.47
D07	1	EQ	PS	T252	BLACK WATER TANK	0.95	0.0	0.000	-	-	-
D07	1	EQ	PS	T254	GREY WATER TANK	0.95	14.7	15.045	54.63	-6.17	3.09
D07	1	EQ	PS	T318	CD VOID PS	0.95	3.9	4.006	56.35	-5.67	3.11

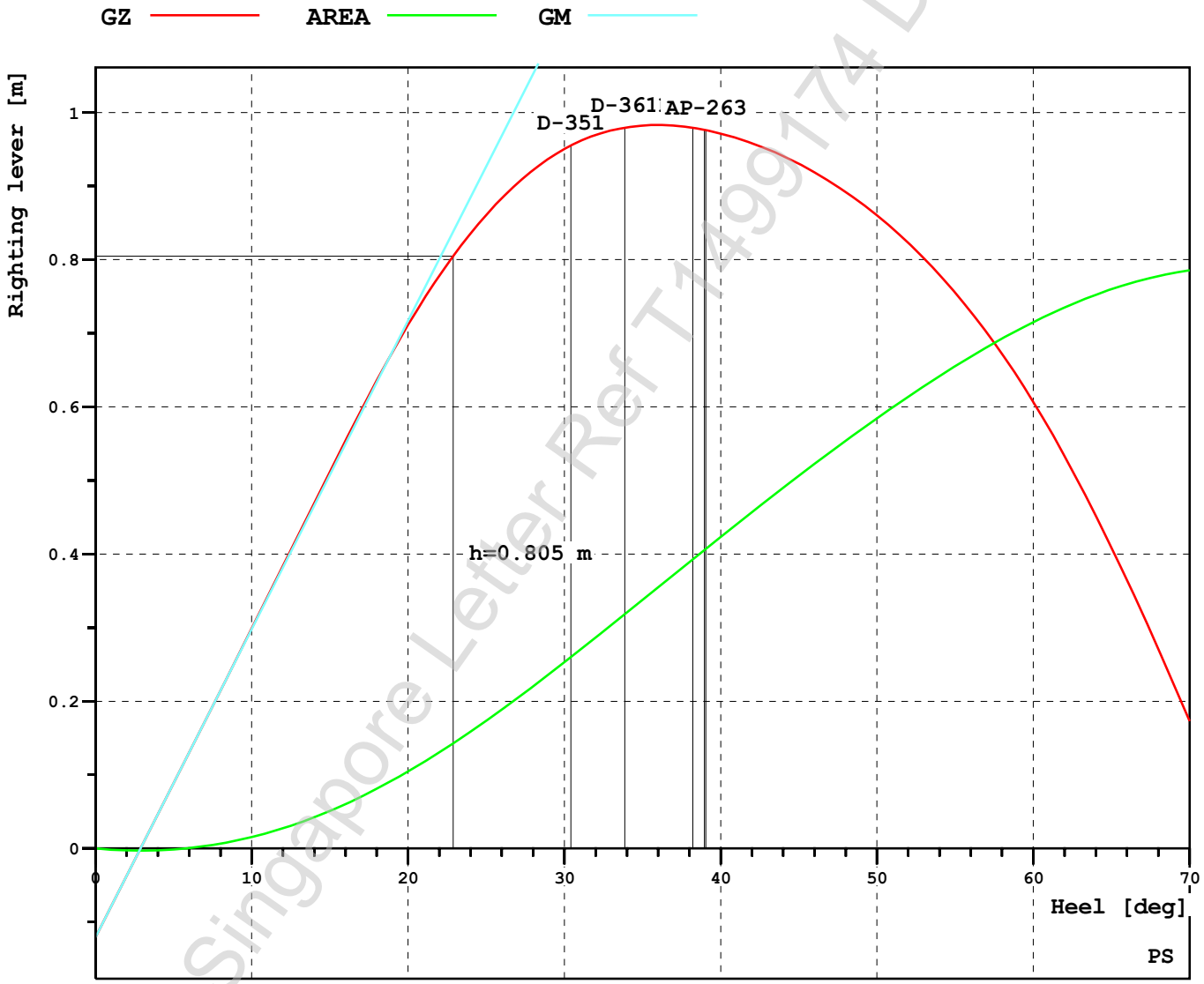
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CRITICAL OPENINGS - CRITICAL CASE

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
D-351	CO2 ROOM/M-DECK	WEATHERTIGHT	51.792	-8.360	8.600	30.4	4.272
D-361	PS CARGORAIL #57/M-DECK	WEATHERTIGHT	39.885	-8.560	8.600	33.9	4.661
D-353	DECK WORKSHOP/M-DECK	WEATHERTIGHT	53.192	-6.550	8.600	38.2	4.316
AP-012	DB TANK PS	WEATHERTIGHT	50.000	-7.250	9.047	39.0	4.834
AP-263	HYDRAULIC OIL STORE TANK	WEATHERTIGHT	47.690	-7.260	8.912	39.1	4.777
D-363	PS CARGORAIL #37/M-DECK	WEATHERTIGHT	25.900	-8.550	8.600	39.9	5.131
AP-014	WING TANK PS	WEATHERTIGHT	43.078	-7.815	9.372	41.2	5.362
AP-016	DB TANK PS	WEATHERTIGHT	43.905	-7.745	9.372	41.2	5.338

DAMAGE STABILITY CHECK PLOT - CRITICAL CASE



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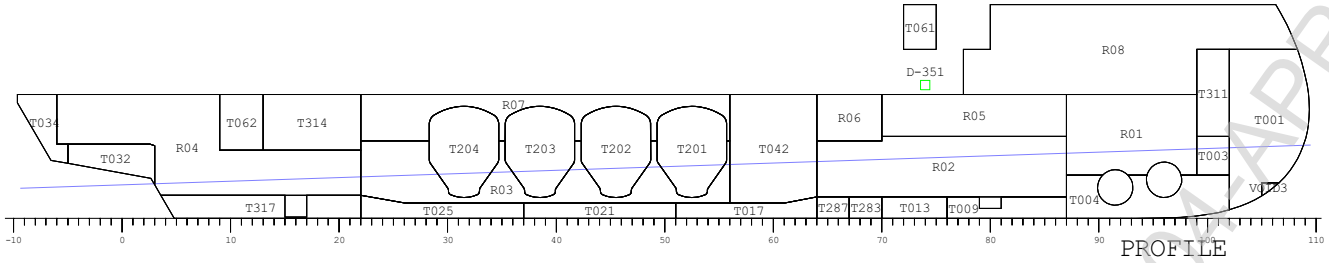
GZ-CURVE - CRITICAL CASE

HEEL	DISP	T	TR	GZ	MS	AREA
degree	t	m	m	m	m	mrad
0.0	2704.9	3.389	2.632	0.121	0.121	0.000
-1.0	2704.9	3.391	2.644	0.079	0.127	-0.002
-3.0	2704.9	3.389	2.673	-0.005	0.140	-0.003
-5.0	2704.9	3.379	2.708	-0.089	0.152	-0.001
-10.0	2704.9	3.322	2.832	-0.299	0.181	0.015
-15.0	2704.9	3.215	3.000	-0.512	0.203	0.051
-20.0	2704.9	3.050	3.204	-0.711	0.234	0.104
-30.0	2704.9	2.505	3.727	-0.950	0.431	0.253
-40.0	2704.9	1.681	4.254	-0.971	0.805	0.423
-50.0	2704.9	0.652	4.634	-0.860	1.257	0.585
-60.0	2704.9	-0.465	4.661	-0.607	1.786	0.715
-70.0	2704.9	-1.563	4.484	-0.173	2.424	0.785

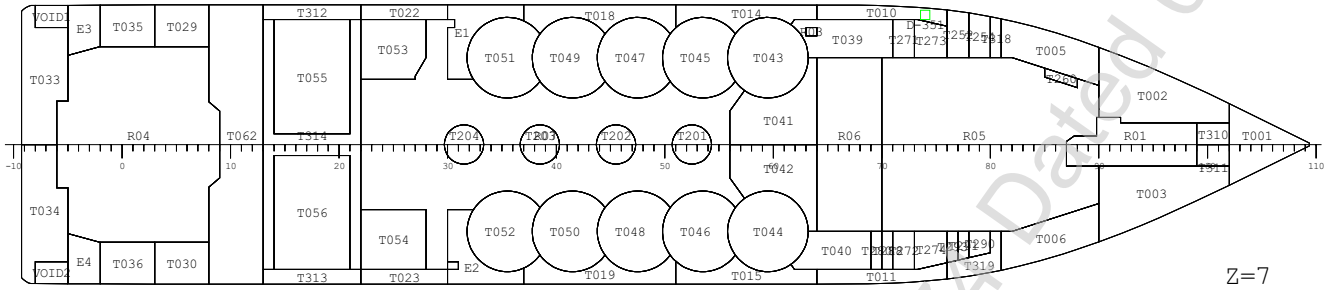
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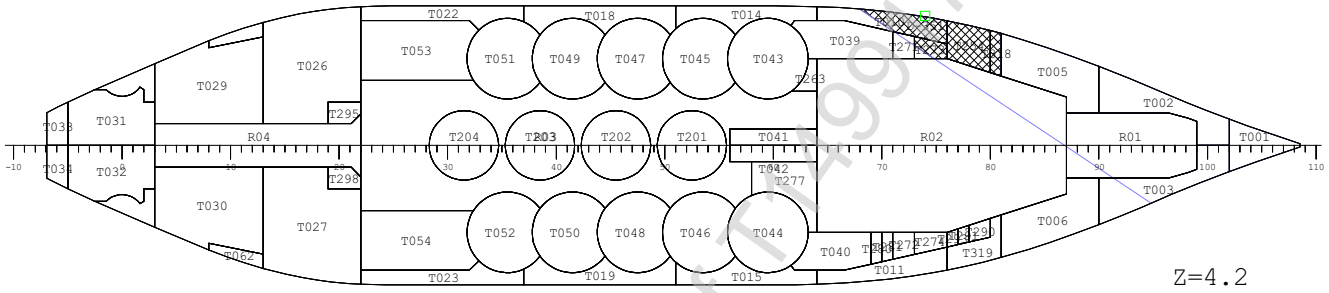
DAMAGE ARRANGEMENT - CRITICAL CASE



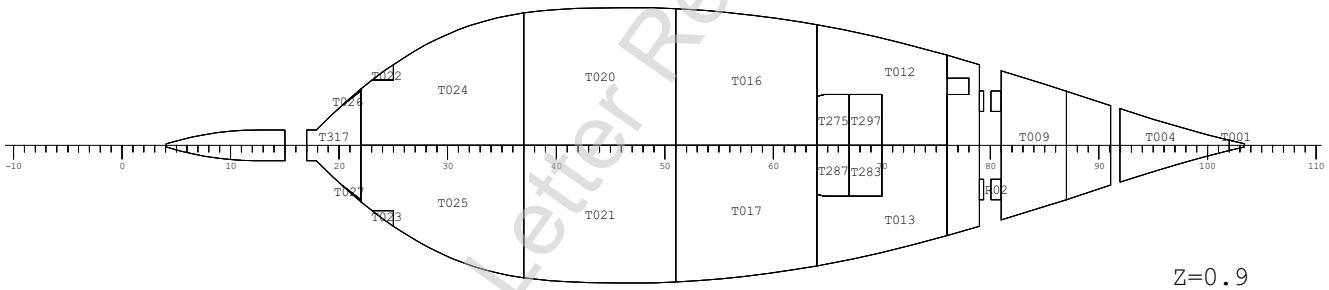
PROFILE



Z=7



Z=4.2

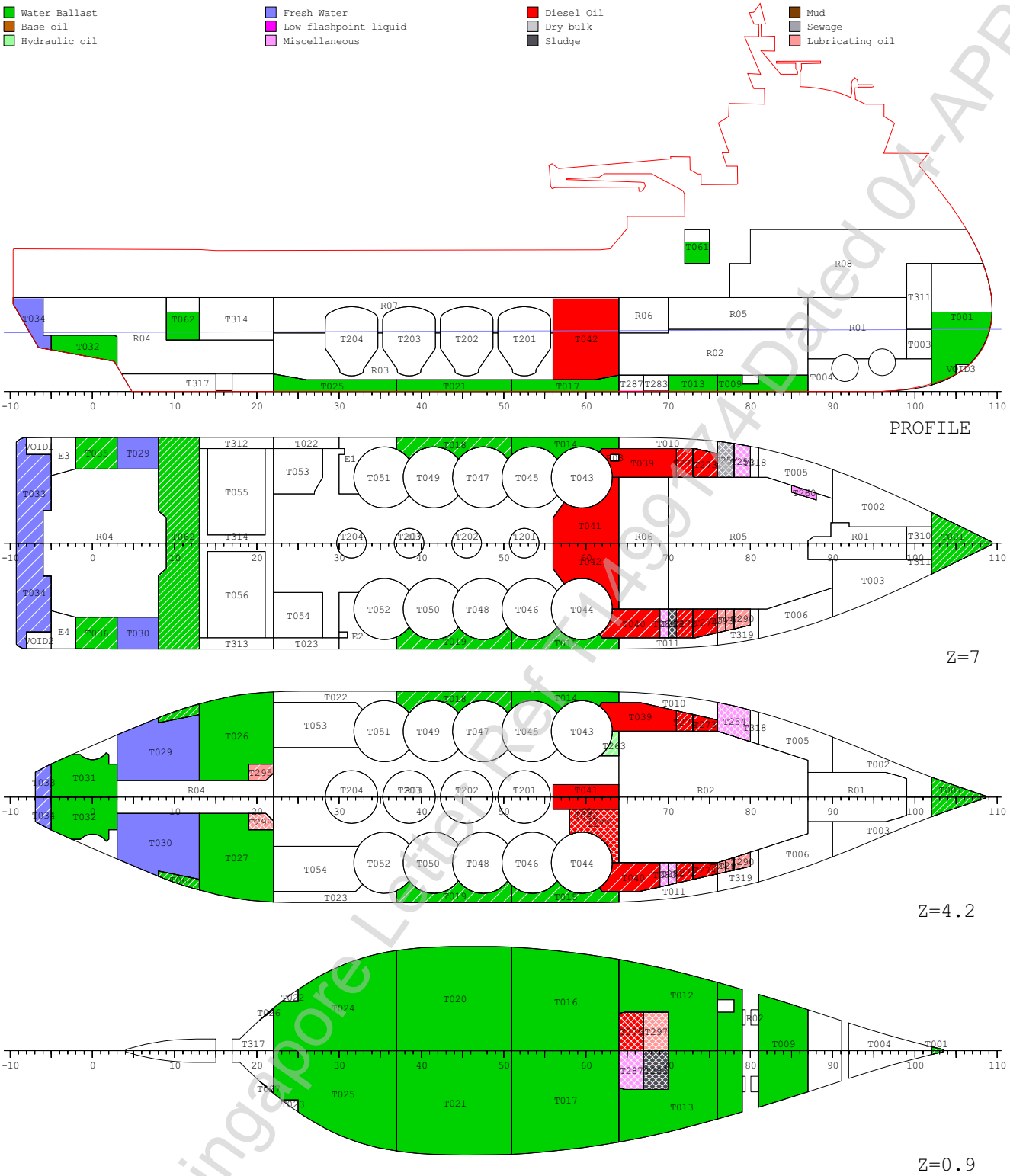


Z=0.9

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12.2 L02 -> Ballast - Departure



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LIST OF LOADS

NAME	DES	LOAD	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm	FRS	HM m
CONTENTS=Diesel Oil (RHO=0.86)										
T039	STORAGE TANK PS	FO	66.9	98.0	46.68	-6.76	4.91	6	REAL	7.88
T040	STORAGE TANK SB	FO	53.8	98.0	46.00	6.80	4.87	5	REAL	7.88
T041	STORAGE TANK PS	FO	68.5	98.0	42.16	-1.51	5.48	27	MAX	7.92
T042	STORAGE TANK SB	FO	68.5	98.0	42.16	1.51	5.48	27	MAX	7.92
T271	FO SETTLING TANK PS	FO	8.5	70.0	50.38	-6.53	4.93	1	REAL	6.72
T272	FO SETTLING TANK SB	FO	8.5	70.0	50.38	6.53	4.93	1	REAL	6.72
T273	FO SERVICE TANK PS	FO	10.6	70.0	52.09	-6.38	5.15	2	REAL	6.90
T274	FO SERVICE TANK SB	FO	10.6	70.0	52.09	6.38	5.15	2	REAL	6.90
T275	FO DRAIN TANK	FO	0.8	10.0	45.85	-1.64	0.07	5	REAL	0.14
T277	FO OVERFLOW TANK	FO	3.6	10.0	42.62	2.34	1.31	10	REAL	1.53
T279	EMERG. GEN. FO TANK	FO	1.9	98.0	61.60	-1.38	24.57	0	REAL	25.80
SUBTOTAL		FO	302.3		45.14	-0.27	5.25	86		
CONTENTS=Fresh Water (RHO=1)										
T029	WING TANK PS	FW	74.6	98.0	5.29	-4.87	4.61	6	REAL	7.83
T030	WING TANK SB	FW	74.6	98.0	5.29	4.87	4.61	6	REAL	7.83
T033	WING TANK PS	FW	57.9	98.0	-4.96	-3.91	6.49	126	MAX	7.95
T034	WING TANK SB	FW	57.9	98.0	-4.96	3.91	6.49	126	MAX	7.95
SUBTOTAL		FW	265.1		0.82	0.00	5.43	264		
CONTENTS=Hydraulic oil (RHO=0.924)										
T263	HYDRAULIC OIL STORE TANK	HO	3.3	70.0	44.30	-4.34	2.32	1	REAL	3.33
CONTENTS=Lubricating oil (RHO=0.924)										
T290	LO STORE TANK MAIN ENG.	LO	5.9	70.0	55.30	5.76	4.90	0	REAL	6.82
T291	LO STORE TANK MAIN ENG.	LO	3.1	70.0	54.25	6.04	5.11	0	REAL	6.92
T293	LO STORE TANK THRUSTER	LO	3.2	70.0	53.55	6.21	5.24	0	REAL	6.97
T295	LO STORE TANK AZIMUTH THR.	LO	4.0	70.0	14.35	-2.10	2.95	0	REAL	3.71
T297	LO DRAIN TANK	LO	0.9	10.0	47.95	-1.64	0.07	6	REAL	0.14
T298	DIRTY LO TANK	LO	0.6	10.0	14.35	2.10	2.31	0	REAL	2.41
SUBTOTAL		LO	17.6		43.89	3.64	4.24	8		
CONTENTS=Mass load (RHO=1)										
M1	CREW	MASS	2.6	0.0	57.40	0.00	15.50	0		0.00
M2	PROVISION	MASS	7.4	0.0	56.35	-6.50	12.40	0		0.00
M3	STORES	MASS	7.4	0.0	72.10	0.00	12.40	0		0.00
SUBTOTAL		MASS	17.5		63.21	-2.77	12.86	0		
CONTENTS=Miscellaneous (RHO=1)										
T280	BILGE WATER HOLDING TANK	MIS	0.9	10.0	48.64	6.17	2.03	0	REAL	2.57
T281	BILGE WATER SETTLING TANK	MIS	0.4	10.0	49.34	6.04	1.76	0	REAL	2.08
T287	BILGE WATER DRAIN TANK	MIS	0.9	10.0	45.85	1.64	0.07	6	REAL	0.14
T254	GREY WATER TANK	MIS	3.2	10.0	54.62	-5.79	1.92	1	REAL	2.34
T260	WATER MIST TANK	MIS	2.3	98.0	60.55	-4.30	6.28	0	REAL	7.27
SUBTOTAL		MIS	7.8		54.40	-2.49	3.01	8		
CONTENTS=Sewage (RHO=1)										
T252	BLACK WATER TANK	SEW	1.1	10.0	53.89	-6.93	5.45	2	REAL	5.59
CONTENTS=Sludge (RHO=1)										
T283	SLUDGE DRAIN TANK	SLU	0.9	10.0	47.95	1.64	0.07	6	REAL	0.14
T288	SLUDGE SETTLING TANK	SLU	0.4	10.0	49.35	6.75	5.44	1	REAL	5.59
SUBTOTAL		SLU	1.4		48.40	3.29	1.80	7		
CONTENTS=Water Ballast (RHO=1.025)										
T001	FOREPEAK TANK	WB	42.0	40.0	73.00	0.00	4.79	34	MAX	6.79
T009	DB TANK CE	WB	77.3	100.0	56.90	0.09	0.73	0	(REAL)	1.40
T012	DB TANK PS	WB	56.2	100.0	49.41	-3.85	0.76	0	(REAL)	1.40
T013	DB TANK SB	WB	56.2	100.0	49.41	3.85	0.76	0	(REAL)	1.40
T014	WING TANK PS	WB	60.3	98.0	40.15	-8.42	4.67	1	REAL	7.87
T015	WING TANK SB	WB	60.2	98.0	40.15	8.42	4.66	1	REAL	7.87
T016	DB TANK PS	WB	85.6	100.0	40.20	-4.33	0.62	0	REAL	1.50
T017	DB TANK SB	WB	85.6	100.0	40.20	4.33	0.62	0	REAL	1.50
T018	WING TANK PS	WB	57.9	78.0	30.80	-8.37	4.01	2	REAL	6.56
T019	WING TANK SB	WB	72.8	98.0	30.79	8.37	4.67	2	REAL	7.87
T020	DB TANK PS	WB	96.7	100.0	30.84	-4.63	0.61	0	(REAL)	1.92
T021	DB TANK SB	WB	96.7	100.0	30.84	4.63	0.61	0	(REAL)	1.92

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T024	DB TANK PS	WB	75.9	100.0	21.46	-3.85	0.81	0	(REAL)	2.75
T025	DB TANK SB	WB	75.9	100.0	21.46	3.85	0.81	0	(REAL)	2.75
T026	WING TANK PS	WB	92.4	100.0	12.42	-4.45	3.13	0	(REAL)	4.40
T027	WING TANK SB	WB	92.4	100.0	12.42	4.45	3.13	0	(REAL)	4.40
T031	WING TANK PS	WB	27.2	100.0	-0.65	-1.76	4.18	0	(REAL)	5.39
T032	WING TANK SB	WB	27.2	100.0	-0.65	1.76	4.18	0	(REAL)	5.39
T061	ROLL RED. TANK	WB	83.4	65.0	51.66	0.00	11.84	1358	REAL	12.78
T062	ROLL RED. TANK	WB	136.9	65.0	7.45	0.00	5.58	1736	REAL	6.76
T035	WING TANK PS	WB	23.2	98.0	0.39	-7.50	6.66	6	REAL	7.95
T036	WING TANK SB	WB	23.2	98.0	0.39	7.50	6.66	6	REAL	7.95
SUBTOTAL		WB	1505.3		30.29	0.09	3.12	3147		
TOTAL			2121.3		29.24	0.01	3.80	3522		

LOAD SUMMARY TABLE

ITEM	MASS t	LCG m	TCG m	VCG m	FRSM tm
Diesel Oil	302.3	45.14	-0.27	5.25	85.89
Fresh Water	265.1	0.82	0.00	5.43	264.28
Hydraulic oil	3.3	44.30	-4.34	2.32	0.61
Lubricating oil	17.6	43.89	3.64	4.24	7.62
Mass load	17.5	63.21	-2.77	12.86	0.00
Miscellaneous	7.8	54.40	-2.49	3.01	7.72
Sewage	1.1	53.89	-6.93	5.45	2.31
Sludge	1.4	48.40	3.29	1.80	6.99
Water Ballast	1505.3	30.29	0.09	3.12	3147.04
DEADWEIGHT	2121.3	29.24	0.01	3.80	3522.46
Lightweight	2704.9	39.15	-0.01	7.08	
Deadweight	2121.3	29.24	0.01	3.80	
Total weight	4826.2	34.80	0.00	5.64	

F L O A T I N G P O S I T I O N

Draught moulded	5.170	m	KM	8.83 m
Trim	0.264	m	KG	5.64 m
Heel, SB=+	0.0	deg		
TA	5.038	m	GM0	3.20 m
TF	5.302	m	GMCORR	-0.73 m
Trimming moment	1671	tonm	GM	2.47 m

STABILITY COMPONENTS

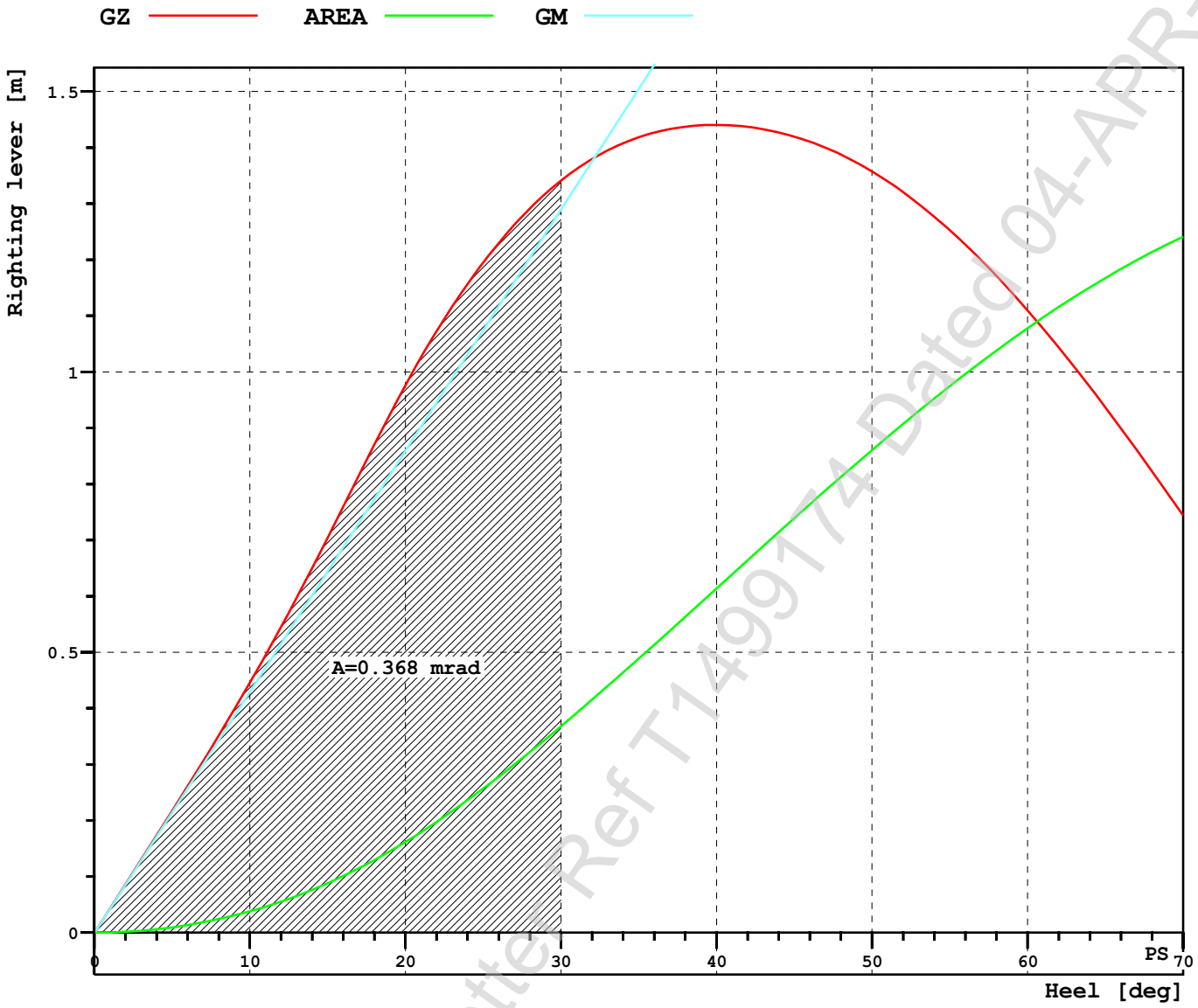
LOADING CONDITION L02, Ballast - Departure

HEEL degree	MS m	HPHI m	EPHI mrad	FSMOM tm	DGZ m	GMSIN m	KN m
0.0	-0.001	0.00	0.000	0.0	0.000	0.000	0.001
1.0	-0.001	0.04	0.000	59.6	0.012	0.056	0.155
3.0	-0.001	0.13	0.003	179.0	0.037	0.167	0.464
5.0	0.000	0.22	0.009	298.4	0.062	0.278	0.772
10.0	0.007	0.45	0.038	558.8	0.116	0.555	1.543
15.0	0.021	0.70	0.088	694.0	0.144	0.827	2.310
20.0	0.042	0.98	0.162	766.0	0.159	1.093	3.066
30.0	-0.084	1.34	0.368	833.7	0.173	1.598	4.335
40.0	-0.436	1.44	0.614	855.4	0.177	2.054	5.243
50.0	-0.917	1.36	0.860	835.6	0.173	2.448	5.851
60.0	-1.494	1.11	1.078	789.9	0.164	2.767	6.157
70.0	-2.109	0.74	1.241	723.8	0.150	3.003	6.192

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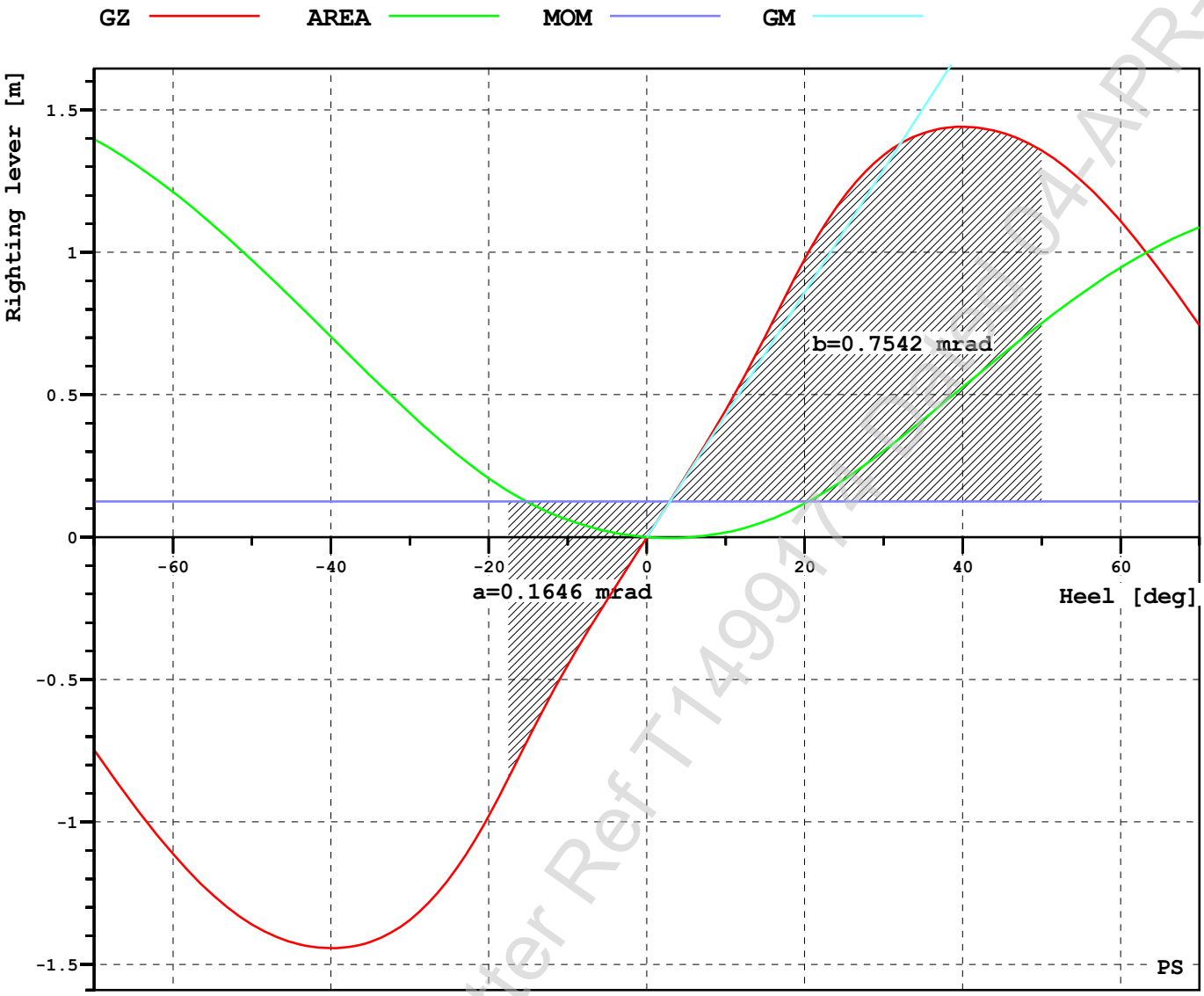
STABILITY CHECK PLOT



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IMOWEATHER CHECK PLOT



INTACT STABILITY CRITERIA

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM
V.AREA15-30	Area depending on the position of the Max. .	0.055	0.368	mrad	OK	0.308
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.246	mrad	OK	0.306
V.GZ0.2	Min. GZ > 0.2	0.200	1.440	m	OK	0.183
V.POSMAX15	Max. GZ at an angle > 15 deg.	15.000	39.904	deg	OK	-0.398
V.GM0.15	GM > 0.15 m	0.150	2.465	m	OK	0.150
IMOWEATHER	IMO weather criterion	1.000	4.583		OK	0.215
2008IS-A2.3.1.2	Heeling angle due to steady wind <16 or <=8.	13.927	1.954	deg	OK	0.129
ABSFIFI1	Angle of heel < 1/2 freeboard immersion	8.704	0.434	deg	OK	0.042
ABSFIFI2	Area of residual stability >0.09 mrad	0.090	0.602	mrad	OK	0.276
TOTAL						0.308

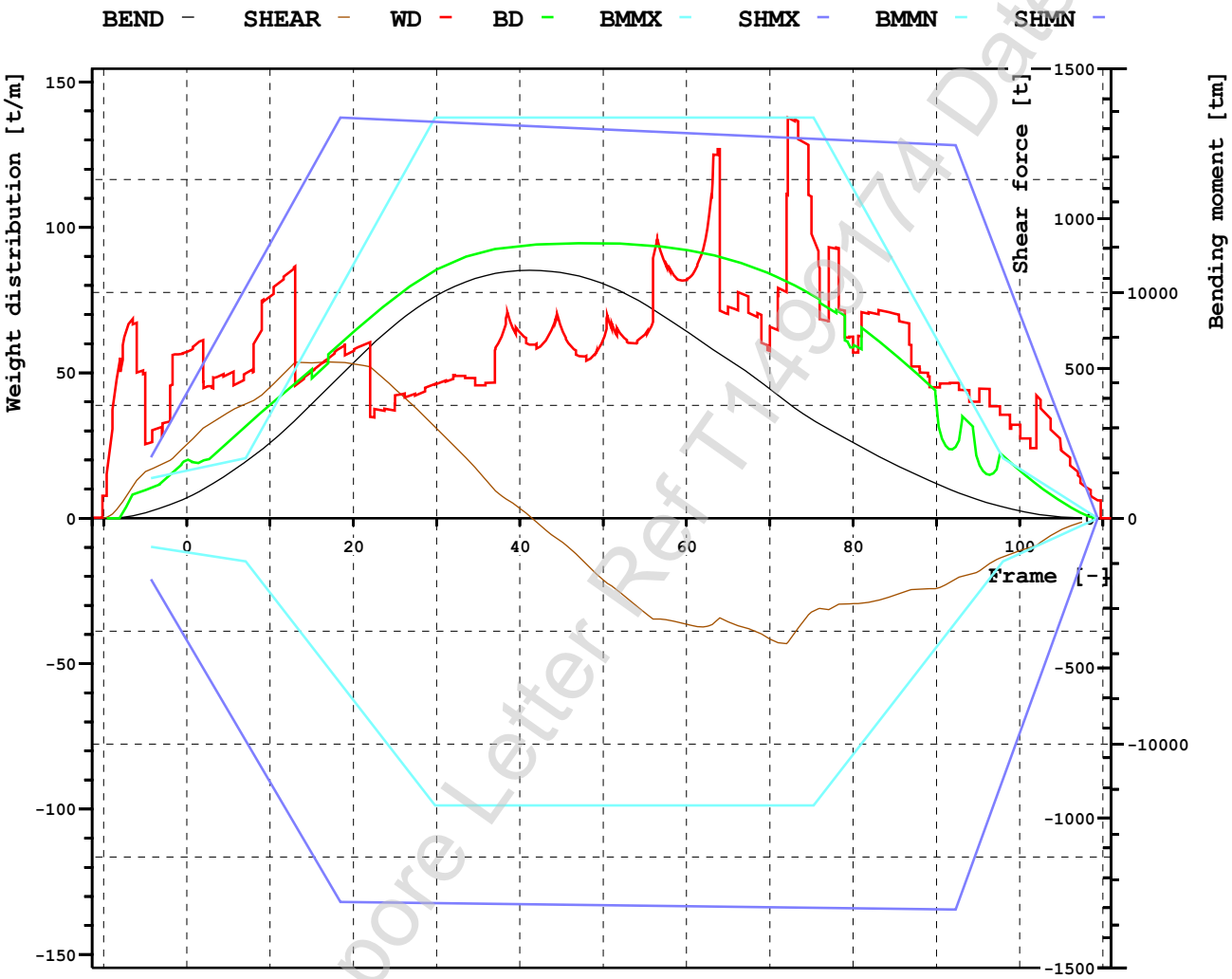
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MOST CRITICAL OPENINGS

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
V-421G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	55.890	2.093	14.288	-	9.055
V-424G	SUPPLY AIR ENGINE ROOM PS	UNPROTECTED	54.243	-3.413	14.288	-	9.059
V-425G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	54.243	3.767	14.288	-	9.061
D-651	ACCOMMODATION/C-DECK	UNPROTECTED	56.000	3.771	16.906	-	11.674
V-421E	EXHAUST AIR ENGINE ROOM PS	UNPROTECTED	55.993	-2.538	17.088	-	11.853
V-423E	EXHAUST AIR ENGINE ROOM SB	UNPROTECTED	55.993	3.423	17.088	-	11.855

STRENGTH DIAGRAM



STRENGTH SUMMARY

LOADING CONDITION L02, Ballast - Departure

			X	Frame
Shear force (min)	-418.4 t	(32.2%)	50.400 m	72.00
Shear force (max)	521.8 t	(41.2%)	11.900 m	17.00
Sagging moment	-0.7 tm	(0.0%)	-7.104 m	-10.15
Hogging moment	10974.3 tm	(61.9%)	28.796 m	41.14

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STRENGTH DISTRIBUTION

LOADING CONDITION L02, Ballast - Departure

X	FR	BEND	BMMN	BMMX	BMREL	SHEAR	SHMN	SHMX	SHREL	WD	BD
m	#	tm	tm	tm	%	t	t	t	%	t/m	t/m
-3.500	-5.00	237	0	0	0.0	155	0	0	0.0	49.87	9.63
0.000	0.00	905	-1510	2108	43.0	246	-407	418	58.9	57.16	19.96
3.500	5.00	1973	-1790	2498	79.0	354	-644	667	53.0	49.49	26.28
7.000	10.00	3336	-3292	4594	72.8	438	-881	916	47.8	76.24	39.03
10.500	15.00	5062	-5668	7911	64.0	519	-1117	1166	44.5	50.37	51.24
14.000	20.00	6879	-8045	11228	61.3	516	-1281	1335	38.6	57.50	64.06
17.500	25.00	8576	-10422	14545	59.0	431	-1283	1329	32.4	42.27	75.81
21.000	30.00	9856	-12709	17736	55.6	299	-1284	1323	22.6	46.22	85.50
24.500	35.00	10653	-12709	17736	60.1	157	-1286	1317	11.9	45.68	91.03
28.000	40.00	10958	-12709	17736	61.8	33	-1288	1310	2.5	63.79	93.50
31.500	45.00	10870	-12709	17736	61.3	-78	-1290	1304	6.1	65.20	94.40
35.000	50.00	10374	-12709	17736	58.5	-206	-1292	1298	16.0	61.55	94.49
38.500	55.00	9462	-12709	17736	53.4	-316	-1293	1292	24.4	63.15	93.80
42.000	60.00	8273	-12709	17736	46.6	-354	-1295	1285	27.4	81.87	92.21
45.500	65.00	7035	-12709	17736	39.7	-346	-1297	1279	26.7	72.67	89.00
49.000	70.00	5732	-12709	17736	32.3	-404	-1299	1273	31.1	65.88	84.01
52.500	75.00	4385	-12709	17736	24.7	-314	-1300	1267	24.1	110.55	76.87
56.000	80.00	3345	-10463	14601	22.9	-284	-1302	1260	21.8	62.37	58.74
59.500	85.00	2379	-8086	11284	21.1	-257	-1304	1254	19.7	70.36	56.37
63.000	90.00	1534	-5709	7968	19.3	-234	-1306	1248	17.9	45.02	39.42
66.500	95.00	817	-3332	4651	17.6	-178	-1102	1050	16.2	40.05	19.73
70.000	100.00	330	-1571	2193	15.0	-109	-718	685	15.1	32.12	16.44
73.500	105.00	55	-733	1023	5.4	-39	-335	319	11.5	23.73	5.67

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DAMAGE STABILITY OF LOADING CONDITION L02

CRITICAL DAMAGE STABILITY CRITERIA FOR EACH DAMAGE CASE

RCR	DAM	STAGE	PHASE	REQ	ATTV	UNIT	STAT	MINGM
V.MINGZ.A469	D01	1	EQ	0.100	1.001	m	OK	-0.203
V.MINGZ.A469	D02	1	EQ	0.100	0.974	m	OK	-0.069
V.MINGZ.A469	D03	1	EQ	0.100	0.977	m	OK	-0.075
V.MINGZ.A469	D04	1	EQ	0.100	0.954	m	OK	0.145
V.MINGZ.A469	D05	1	EQ	0.100	0.946	m	OK	0.179
V.RANGE.A469	D06	1	EQ	20.000	68.064	deg	OK	0.157
V.RANGE.A469	D07	1	EQ	20.000	67.833	deg	OK	0.188
V.RANGE.A469	D08	1	EQ	20.000	68.814	deg	OK	0.102
V.RANGE.A469	D09	1	EQ	20.000	68.874	deg	OK	0.103
V.RANGE.A469	D10	1	EQ	20.000	68.548	deg	OK	0.178
V.RANGE.A469	D11	1	EQ	20.000	69.253	deg	OK	0.106
V.RANGE.A469	D12	1	EQ	20.000	68.260	deg	OK	0.116
V.RANGE.A469	D13	1	EQ	20.000	68.192	deg	OK	0.115
V.MINGZ.A469	D14	1	EQ	0.100	0.973	m	OK	-0.116
V.MINGZ.A469	D15	1	EQ	0.100	0.975	m	OK	-0.118
V.MINGZ.A469	D16	1	EQ	0.100	0.956	m	OK	0.013
V.MINGZ.A469	D17	1	EQ	0.100	0.953	m	OK	0.020
V.RANGE.A469	D18	1	EQ	20.000	69.134	deg	OK	-0.086
V.RANGE.A469	D19	1	EQ	20.000	69.196	deg	OK	-0.084
V.RANGE.A469	D20	1	EQ	20.000	68.901	deg	OK	-0.048
V.RANGE.A469	D21	1	EQ	20.000	68.962	deg	OK	-0.045
TOTAL								0.188

DETERMINATION OF CRITICAL DAMAGE CASE

CRITICAL DAMAGE CASE : D07
STAGE OF CRITICAL DAMAGE CASE : 1
PHASE OF CRITICAL DAMAGE CASE : EQ

DAMAGE STABILITY CRITERIA - CRITICAL CASE

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM m
V.PROGR.A469	No progr. flooding	21.014	2.167	deg	OK	-0.017
V.MAXHEEL.A469	Maximum heel	17.000	2.167	deg	OK	0.130
V.RANGE.A469	Range of pos. stab	20.000	67.833	deg	OK	0.188
V.MINGZ.A469	Min. righting lever	0.100	0.944	m	OK	0.179
TOTAL						0.188

DAMAGE CASE RESULTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	T m	TR m	HEEL degree	RESFLD m	OPEN	CROSSTM s
D07	1	EQ	PS	5.230	0.478	-2.2	2.96	D-351	-

DAMAGED COMPARTMENTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	NAME	DES	PERM	VOL m3	W t	XCG m	YCG m	ZCG m
D07	INTACT	EQ	PS				0.0	0.000			
D07	1	EQ	PS	T010	WING TANK PS	0.95	40.2	41.232	49.42	-7.62	3.42
D07	1	EQ	PS	T273	FO SERVICE TANK PS	0.95	6.9	7.057	52.08	-6.23	4.30
D07	1	EQ	PS	T252	BLACK WATER TANK	0.95	1.1	1.088	53.89	-7.01	5.45
D07	1	EQ	PS	T254	GREY WATER TANK	0.95	22.9	23.494	54.65	-6.31	3.74
D07	1	EQ	PS	T318	CD VOID PS	0.95	6.2	6.367	56.35	-5.85	3.79

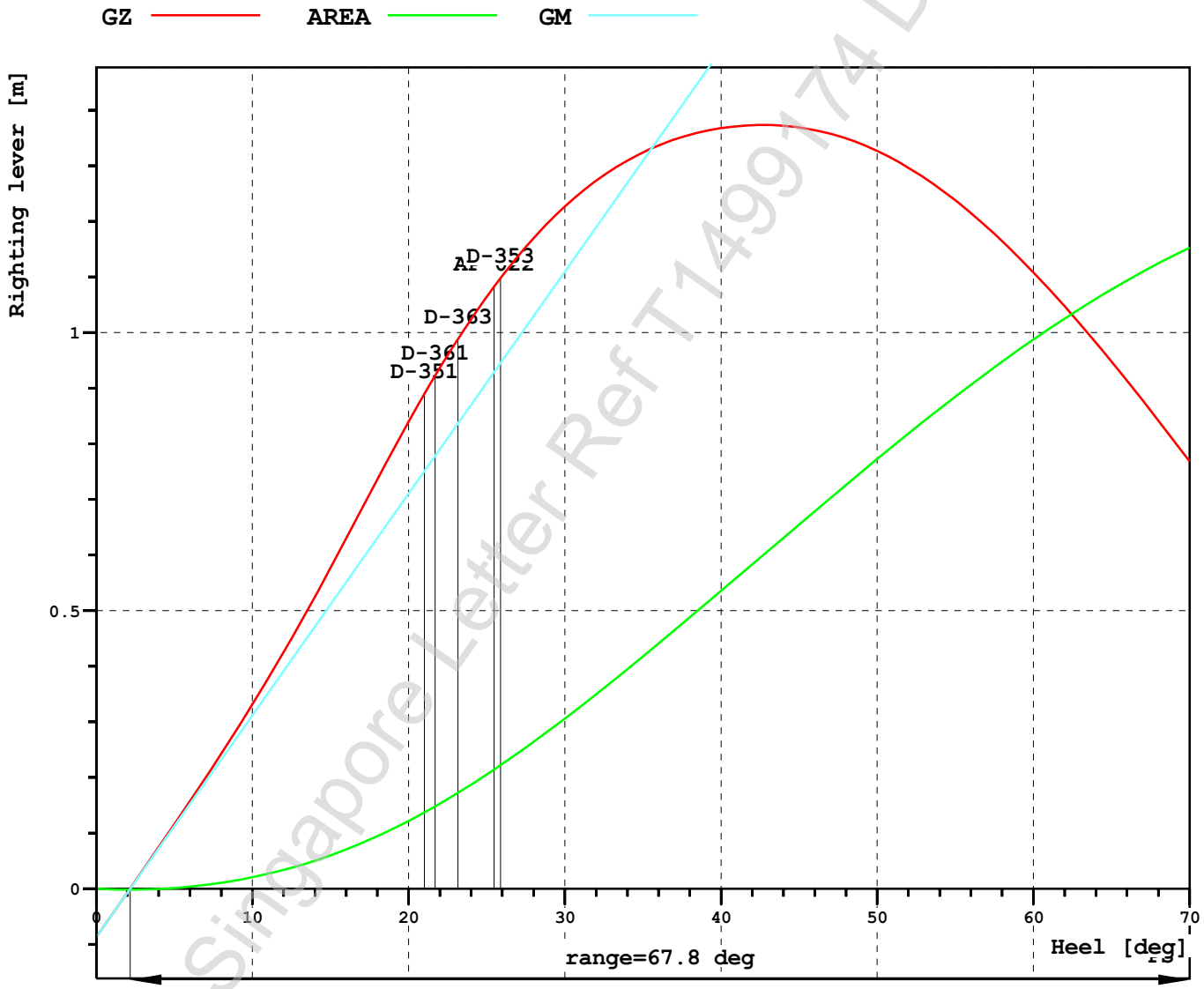
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CRITICAL OPENINGS - CRITICAL CASE

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
D-351	CO2 ROOM/M-DECK	WEATHERTIGHT	51.792	-8.360	8.600	21.0	2.958
D-361	PS CARGORAIL #57/M-DECK	WEATHERTIGHT	39.885	-8.560	8.600	21.7	3.022
D-363	PS CARGORAIL #37/M-DECK	WEATHERTIGHT	25.900	-8.550	8.600	23.2	3.106
AP-022	WING TANK PS	WEATHERTIGHT	19.200	-8.702	8.936	25.5	3.476
D-353	DECK WORKSHOP/M-DECK	WEATHERTIGHT	53.192	-6.550	8.600	25.9	3.018
AP-263	HYDRAULIC OIL STORE TANK	WEATHERTIGHT	47.690	-7.260	8.912	26.3	3.336
AP-012	DB TANK PS	WEATHERTIGHT	50.000	-7.250	9.047	26.9	3.457
AP-014	WING TANK PS	WEATHERTIGHT	43.078	-7.815	9.372	27.9	3.802

DAMAGE STABILITY CHECK PLOT - CRITICAL CASE



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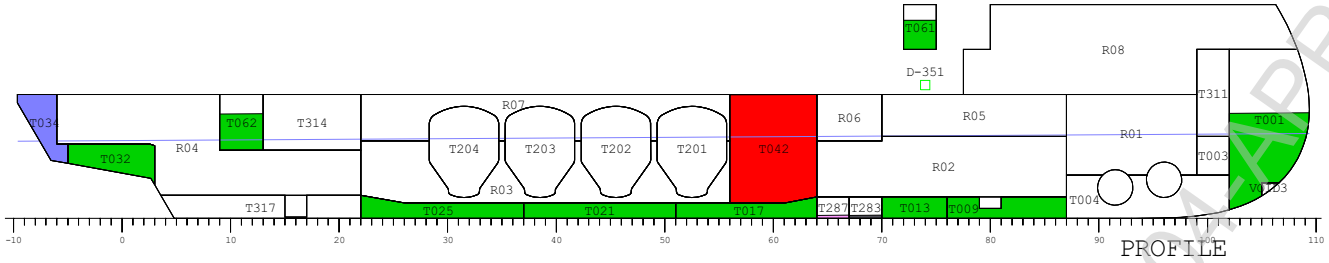
GZ-CURVE - CRITICAL CASE

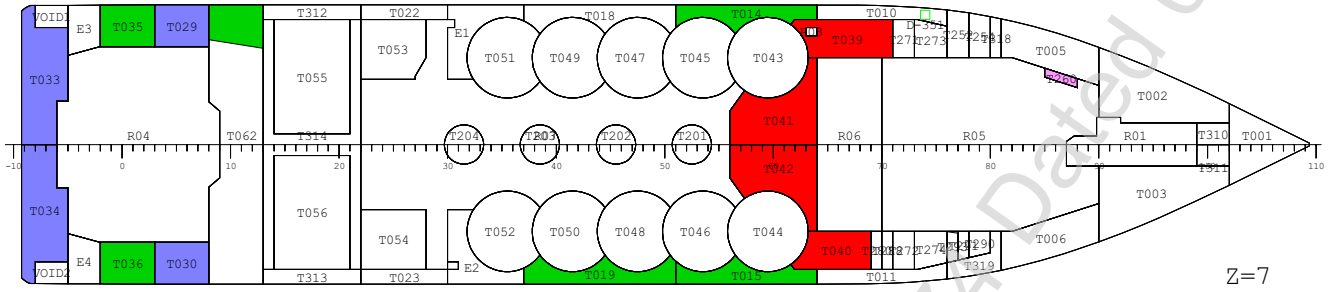
HEEL	DISP	T	TR	GZ	MS	AREA
degree	t	m	m	m	m	mrad
0.0	4826.2	5.229	0.450	0.087	0.087	0.000
-1.0	4826.2	5.231	0.461	0.047	0.103	-0.001
-3.0	4826.2	5.227	0.493	-0.034	0.133	-0.001
-5.0	4826.2	5.214	0.539	-0.116	0.163	0.001
-10.0	4826.2	5.145	0.704	-0.331	0.223	0.021
-15.0	4826.2	5.021	0.915	-0.577	0.250	0.060
-20.0	4826.2	4.845	1.150	-0.840	0.253	0.122
-30.0	4826.2	4.412	1.395	-1.227	0.371	0.306
-40.0	4826.2	3.861	1.180	-1.368	0.686	0.535
-50.0	4826.2	3.152	0.537	-1.327	1.121	0.773
-60.0	4826.2	2.346	-0.201	-1.108	1.659	0.988
-70.0	4826.2	1.482	-0.912	-0.768	2.234	1.153

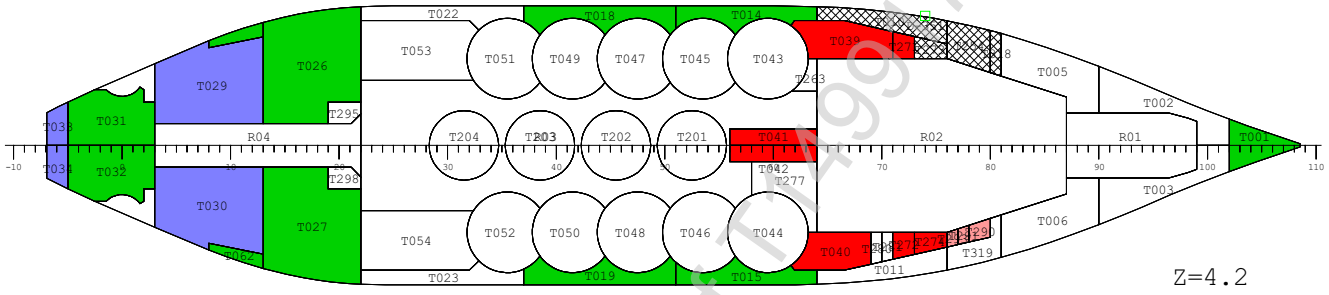
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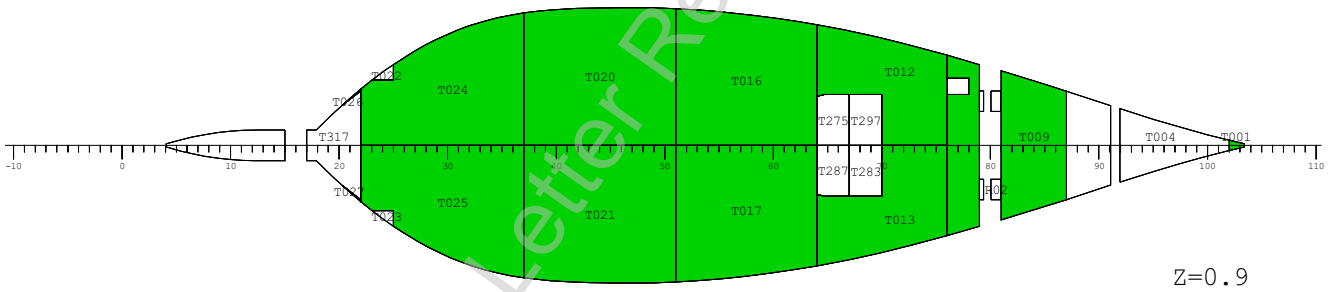
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DAMAGE ARRANGEMENT - CRITICAL CASE





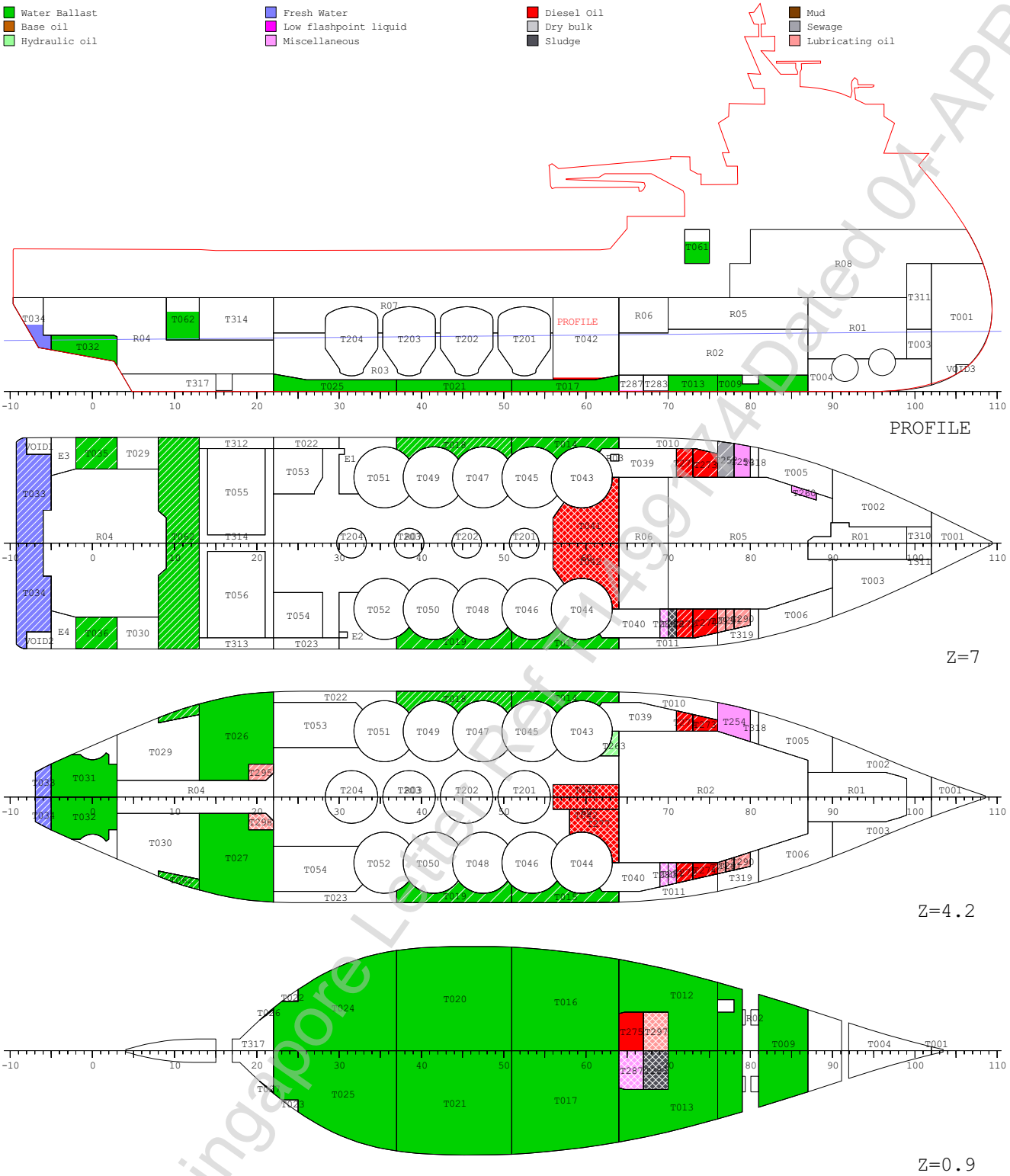




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12.3 L03 -> Ballast - Arrival



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LIST OF LOADS

NAME	DES	LOAD	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm	FRS	HM m
CONTENTS=Diesel Oil (RHO=0.86)										
T041	STORAGE TANK PS	FO	0.7	1.0	41.24	-0.52	1.10	27	MAX	1.20
T042	STORAGE TANK SB	FO	0.7	1.0	41.24	0.52	1.10	27	MAX	1.20
T271	FO SETTLING TANK PS	FO	8.5	70.0	50.38	-6.53	4.93	1	REAL	6.72
T272	FO SETTLING TANK SB	FO	8.5	70.0	50.38	6.53	4.93	1	REAL	6.72
T273	FO SERVICE TANK PS	FO	10.6	70.0	52.09	-6.38	5.15	2	REAL	6.90
T274	FO SERVICE TANK SB	FO	10.6	70.0	52.09	6.38	5.15	2	REAL	6.90
T275	FO DRAIN TANK	FO	7.9	98.0	45.85	-1.64	0.69	5	REAL	1.37
T277	FO OVERFLOW TANK	FO	3.6	10.0	42.62	2.34	1.31	10	REAL	1.53
T279	EMERG. GEN. FO TANK	FO	1.9	98.0	61.60	-1.38	24.57	0	REAL	25.80
SUBTOTAL		FO	53.0		50.02	-0.13	4.73	75		
CONTENTS=Fresh Water (RHO=1)										
T033	WING TANK PS	FW	13.2	22.4	-4.48	-2.98	5.03	126	MAX	5.69
T034	WING TANK SB	FW	13.2	22.4	-4.48	2.98	5.03	126	MAX	5.69
SUBTOTAL		FW	26.5		-4.48	0.00	5.03	253		
CONTENTS=Hydraulic oil (RHO=0.924)										
T263	HYDRAULIC OIL STORE TANK	HO	1.9	40.0	44.29	-4.33	1.89	1	REAL	2.47
CONTENTS=Lubricating oil (RHO=0.924)										
T290	LO STORE TANK MAIN ENG.	LO	3.4	40.0	55.32	5.47	4.05	0	REAL	5.16
T291	LO STORE TANK MAIN ENG.	LO	1.8	40.0	54.25	5.81	4.26	0	REAL	5.48
T293	LO STORE TANK THRUSTER	LO	1.8	40.0	53.55	6.03	4.41	0	REAL	5.66
T295	LO STORE TANK AZIMUTH THR.	LO	2.3	40.0	14.35	-2.10	2.63	0	REAL	3.06
T297	LO DRAIN TANK	LO	0.9	10.0	47.95	-1.64	0.07	6	REAL	0.14
T298	DIRTY LO TANK	LO	0.6	10.0	14.35	2.10	2.31	0	REAL	2.41
SUBTOTAL		LO	10.7		43.36	3.26	3.43	7		
CONTENTS=Mass load (RHO=1)										
M1	CREW	MASS	2.6	0.0	57.40	0.00	15.50	0		0.00
M2	PROVISION	MASS	0.8	0.0	56.35	-6.50	12.40	0		0.00
M3	STORES	MASS	0.8	0.0	72.10	0.00	12.40	0		0.00
SUBTOTAL		MASS	4.2		60.00	-1.24	14.32	0		
CONTENTS=Miscellaneous (RHO=1)										
T280	BILGE WATER HOLDING TANK	MIS	0.9	10.0	48.64	6.17	2.03	0	REAL	2.57
T281	BILGE WATER SETTLING TANK	MIS	0.4	10.0	49.34	6.04	1.76	0	REAL	2.08
T287	BILGE WATER DRAIN TANK	MIS	0.9	10.0	45.85	1.64	0.07	6	REAL	0.14
T254	GREY WATER TANK	MIS	31.6	98.0	54.82	-6.47	4.52	3	REAL	7.84
T260	WATER MIST TANK	MIS	2.3	98.0	60.55	-4.30	6.28	0	REAL	7.27
SUBTOTAL		MIS	36.2		54.74	-5.68	4.43	9		
CONTENTS=Sewage (RHO=1)										
T252	BLACK WATER TANK	SEW	10.4	97.9	53.89	-7.06	6.66	4	REAL	7.95
CONTENTS=Sludge (RHO=1)										
T283	SLUDGE DRAIN TANK	SLU	0.9	10.0	47.95	1.64	0.07	6	REAL	0.14
T288	SLUDGE SETTLING TANK	SLU	0.4	10.0	49.35	6.75	5.44	1	REAL	5.59
SUBTOTAL		SLU	1.4		48.40	3.29	1.80	7		
CONTENTS=Water Ballast (RHO=1.025)										
T001	FOREPEAK TANK	WB	0.0	0.0	73.11	0.00	7.21	34	MAX	0.00
T009	DB TANK CE	WB	77.3	100.0	56.90	0.09	0.73	0	(REAL)	1.40
T012	DB TANK PS	WB	56.2	100.0	49.41	-3.85	0.76	0	(REAL)	1.40
T013	DB TANK SB	WB	56.2	100.0	49.41	3.85	0.76	0	(REAL)	1.40
T014	WING TANK PS	WB	60.3	98.0	40.15	-8.42	4.67	1	REAL	7.87
T015	WING TANK SB	WB	60.2	98.0	40.15	8.42	4.66	1	REAL	7.87
T016	DB TANK PS	WB	85.6	100.0	40.20	-4.33	0.62	0	REAL	1.50
T017	DB TANK SB	WB	85.6	100.0	40.20	4.33	0.62	0	REAL	1.50
T018	WING TANK PS	WB	38.6	52.0	30.80	-8.37	3.16	2	REAL	4.86
T019	WING TANK SB	WB	72.8	98.0	30.79	8.37	4.67	2	REAL	7.87
T020	DB TANK PS	WB	96.7	100.0	30.84	-4.63	0.61	0	(REAL)	1.92
T021	DB TANK SB	WB	96.7	100.0	30.84	4.63	0.61	0	(REAL)	1.92
T024	DB TANK PS	WB	75.9	100.0	21.46	-3.85	0.81	0	(REAL)	2.75
T025	DB TANK SB	WB	75.9	100.0	21.46	3.85	0.81	0	(REAL)	2.75
T026	WING TANK PS	WB	92.4	100.0	12.42	-4.45	3.13	0	(REAL)	4.40
T027	WING TANK SB	WB	92.4	100.0	12.42	4.45	3.13	0	(REAL)	4.40

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T031	WING TANK PS	WB	27.2	100.0	-0.65	-1.76	4.18	0 (REAL)	5.39
T032	WING TANK SB	WB	27.2	100.0	-0.65	1.76	4.18	0 (REAL)	5.39
T061	ROLL RED. TANK	WB	83.4	65.0	51.66	0.00	11.84	1358 REAL	12.78
T062	ROLL RED. TANK	WB	136.9	65.0	7.45	0.00	5.58	1736 REAL	6.76
T035	WING TANK PS	WB	23.2	98.0	0.39	-7.50	6.66	6 REAL	7.95
T036	WING TANK SB	WB	23.2	98.0	0.39	7.50	6.66	6 REAL	7.95
SUBTOTAL		WB	1444.0		29.04	0.20	3.03	3147	
TOTAL			1588.2		30.15	0.02	3.21	3503	

LOAD SUMMARY TABLE

ITEM	MASS t	LCG m	TCG m	VCG m	FRSM tm
Diesel Oil	53.0	50.02	-0.13	4.73	74.98
Fresh Water	26.5	-4.48	0.00	5.03	252.79
Hydraulic oil	1.9	44.29	-4.33	1.89	0.61
Lubricating oil	10.7	43.36	3.26	3.43	7.30
Mass load	4.2	60.00	-1.24	14.32	0.00
Miscellaneous	36.2	54.74	-5.68	4.43	9.39
Sewage	10.4	53.89	-7.06	6.66	3.55
Sludge	1.4	48.40	3.29	1.80	6.99
Water Ballast	1444.0	29.04	0.20	3.03	3147.04
DEADWEIGHT	1588.2	30.15	0.02	3.21	3502.66
Lightweight	2704.9	39.15	-0.01	7.08	
Deadweight	1588.2	30.15	0.02	3.21	
Total weight	4293.1	35.82	0.00	5.65	

F L O A T I N G P O S I T I O N

Draught moulded	4.762	m	KM	8.84 m
Trim	0.751	m	KG	5.65 m
Heel, SB=+	0.0	deg		
TA	4.387	m	GM0	3.19 m
TF	5.137	m	GMCORR	-0.82 m
Trimming moment	4234	tonm	GM	2.38 m

STABILITY COMPONENTS

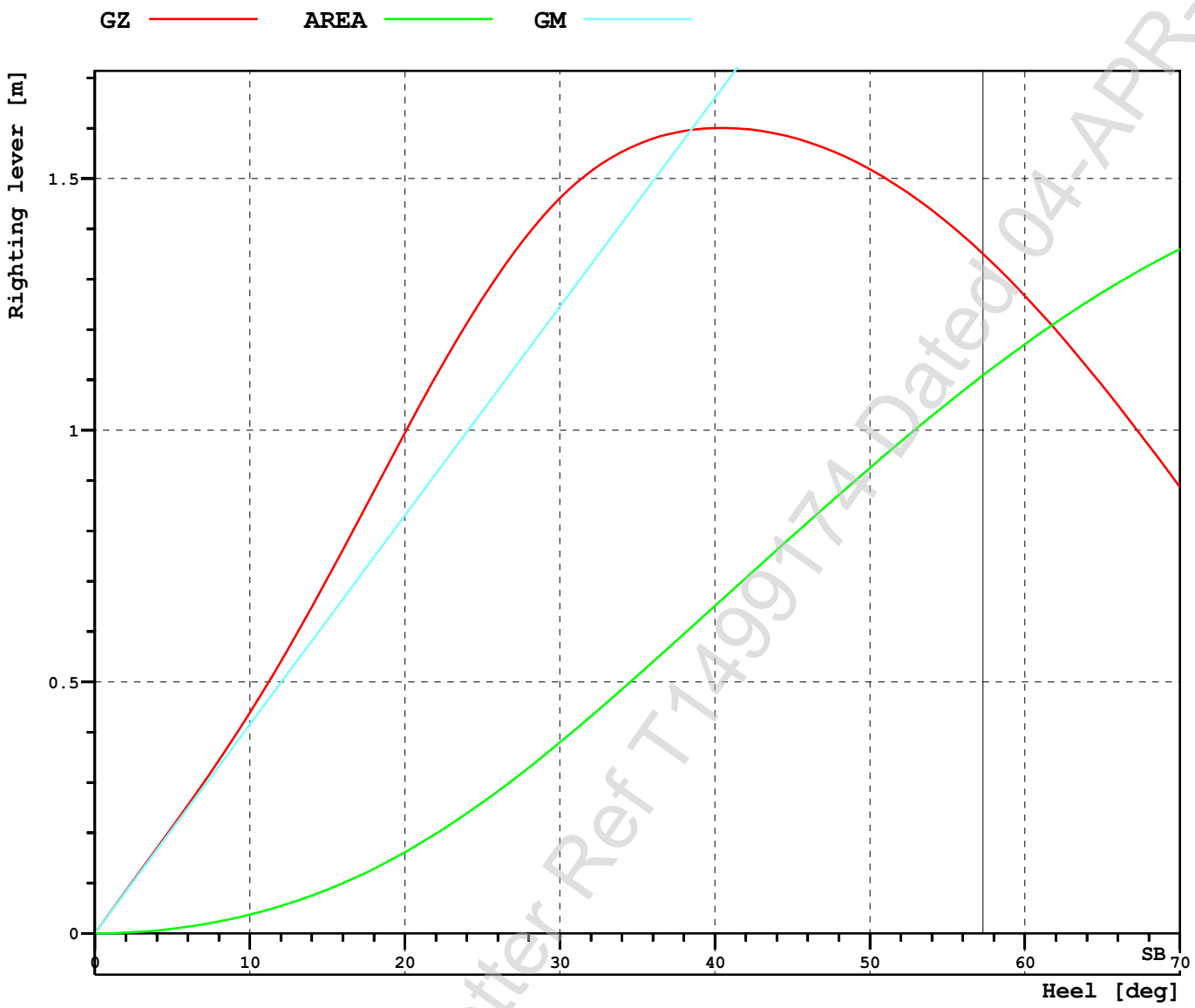
LOADING CONDITION L03, Ballast - Arrival

HEEL degree	MS m	HPHI m	EPHI mrad	FSMOM tm	DGZ m	GMSIN m	KN m
0.0	0.000	0.00	0.000	0.0	0.000	0.000	-0.002
1.0	0.001	0.04	0.000	59.3	0.014	0.056	0.153
3.0	0.001	0.13	0.003	177.9	0.041	0.167	0.462
5.0	0.002	0.21	0.009	296.4	0.069	0.278	0.771
10.0	0.013	0.44	0.037	555.5	0.129	0.555	1.547
15.0	0.039	0.71	0.087	690.2	0.161	0.827	2.326
20.0	0.080	0.99	0.161	762.2	0.178	1.093	3.103
30.0	0.057	1.46	0.379	829.9	0.193	1.597	4.477
40.0	-0.257	1.60	0.651	842.5	0.196	2.053	5.426
50.0	-0.737	1.52	0.926	823.3	0.192	2.447	6.036
60.0	-1.318	1.27	1.171	779.3	0.182	2.767	6.339
70.0	-1.948	0.89	1.360	718.0	0.167	3.002	6.361

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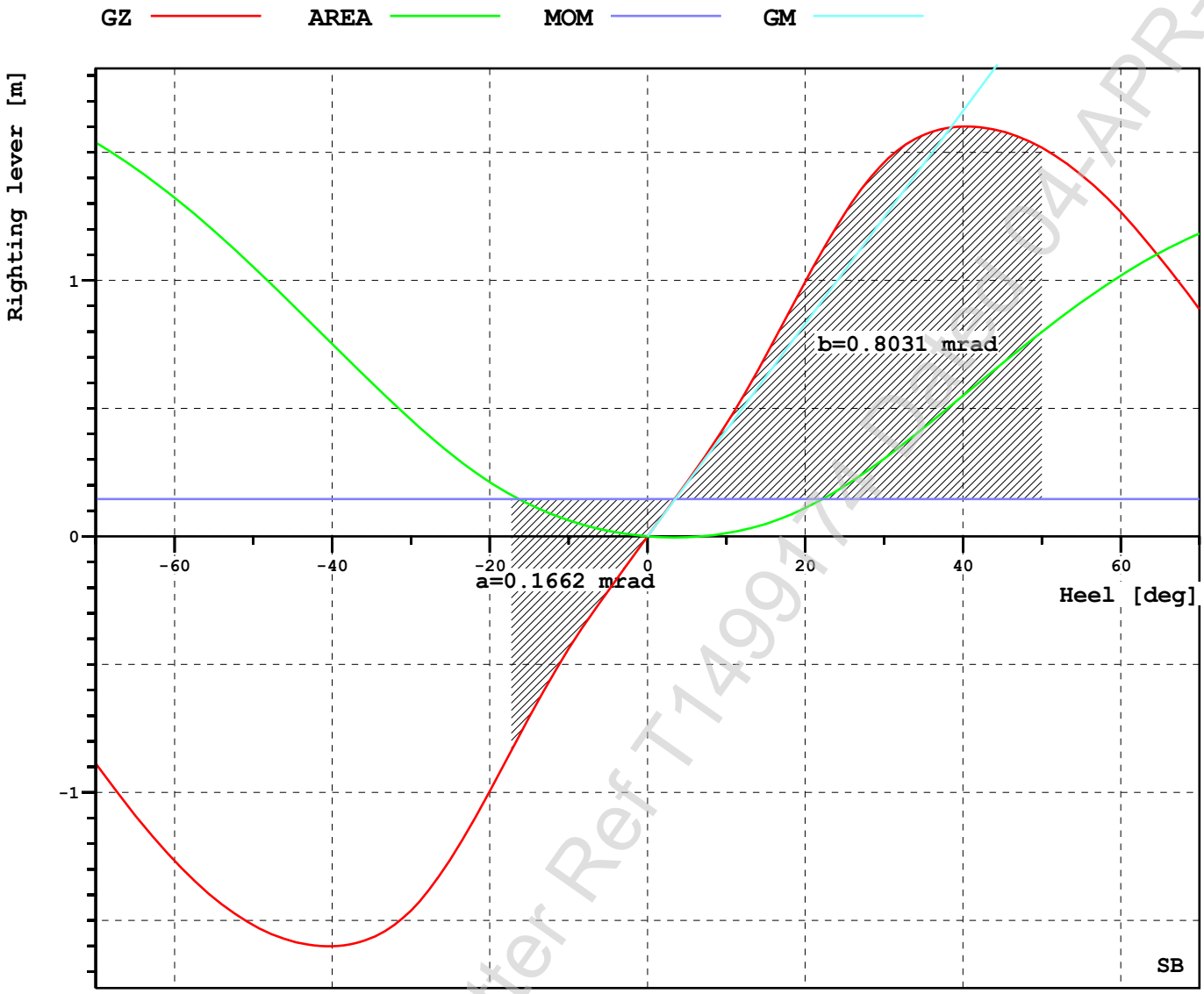
STABILITY CHECK PLOT



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IMOWEATHER CHECK PLOT



INTACT STABILITY CRITERIA

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM
V.AREA15-30	Area depending on the position of the Max. .	0.055	0.379	mrad	OK	0.017
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.271	mrad	OK	-0.037
V.GZ0.2	Min. GZ > 0.2	0.200	1.601	m	OK	-0.144
V.POSMAX15	Max. GZ at an angle > 15 deg.	15.000	40.393	deg	OK	-0.602
V.GM0.15	GM > 0.15 m	0.150	2.379	m	OK	0.150
IMOWEATHER	IMO weather criterion	1.000	4.833		OK	0.029
2008IS-A2.3.1.2	Heeling angle due to steady wind <16 or <=8.	15.479	2.296	deg	OK	-0.003
ABSFIFI1	Angle of heel < 1/2 freeboard immersion	9.674	0.464	deg	OK	-0.021
ABSFIFI2	Area of residual stability >0.09 mrad	0.090	0.637	mrad	OK	0.032
TOTAL						0.150

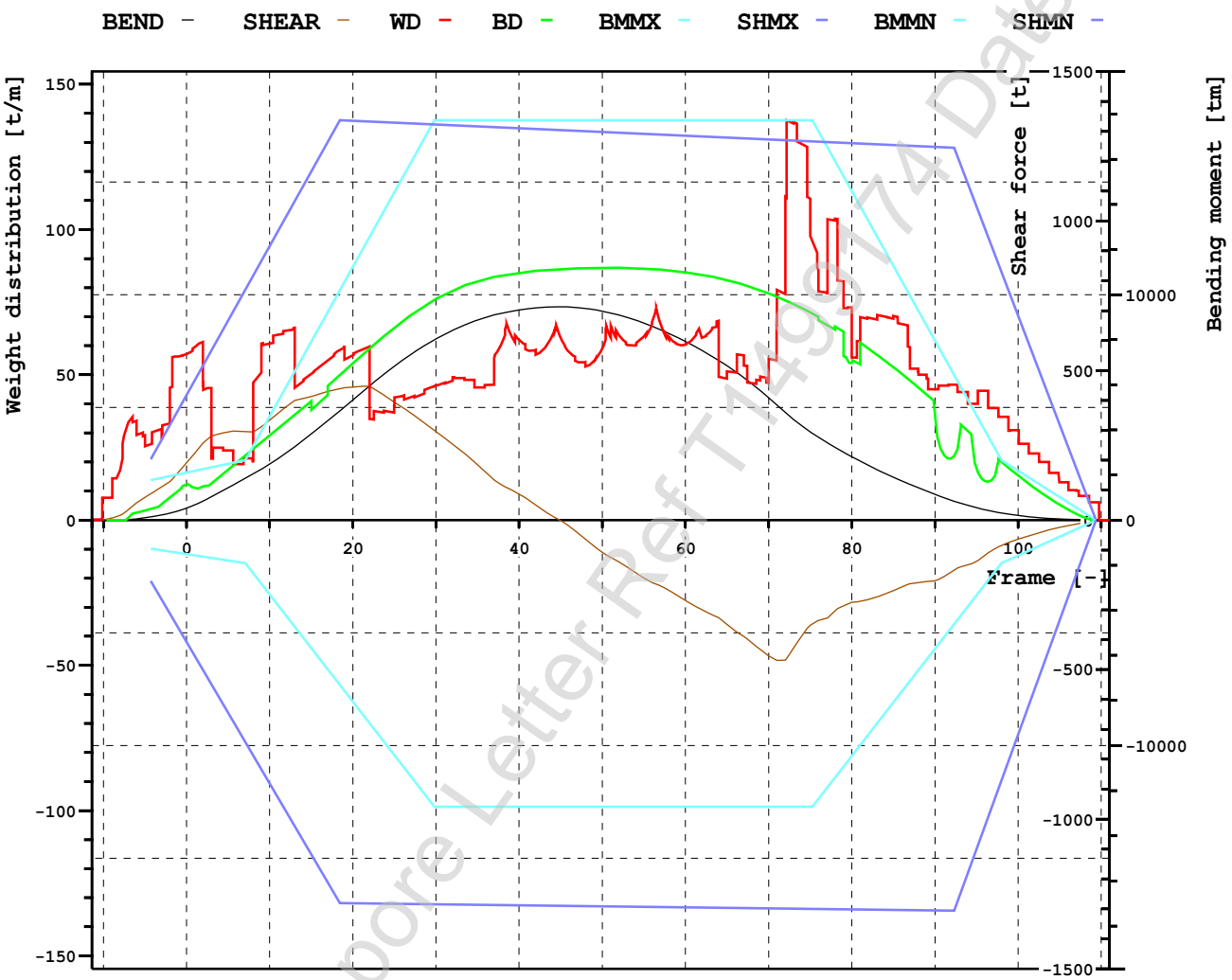
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MOST CRITICAL OPENINGS

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
V-421G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	55.890	2.093	14.288	-	9.345
V-424G	SUPPLY AIR ENGINE ROOM PS	UNPROTECTED	54.243	-3.413	14.288	-	9.361
V-425G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	54.243	3.767	14.288	-	9.361
D-651	ACCOMMODATION/C-DECK	UNPROTECTED	56.000	3.771	16.906	-	11.962
V-421E	EXHAUST AIR ENGINE ROOM PS	UNPROTECTED	55.993	-2.538	17.088	-	12.144
V-423E	EXHAUST AIR ENGINE ROOM SB	UNPROTECTED	55.993	3.423	17.088	-	12.144

STRENGTH DIAGRAM



STRENGTH SUMMARY

LOADING CONDITION L03, Ballast - Arrival

			X	Frame
Shear force (min)	-469.4 t	(36.1%)	49.700 m	71.00
Shear force (max)	447.9 t	(33.6%)	15.400 m	22.00
Sagging moment	-1.8 tm	(0.0%)	-6.750 m	-9.64
Hogging moment	9459.2 tm	(53.3%)	31.375 m	44.82

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STRENGTH DISTRIBUTION

LOADING CONDITION L03, Ballast - Arrival

X m	FR #	BEND tm	BMMN tm	BMMX tm	BMREL %	SHEAR t	SHMN t	SHMX t	SHREL %	WD t/m	BD t/m
-3.500	-5.00	99	0	0	0.0	76	0	0	0.0	29.14	3.25
0.000	0.00	530	-1510	2108	25.1	192	-407	418	45.9	57.15	12.23
3.500	5.00	1431	-1790	2498	57.2	295	-644	667	44.2	23.96	17.30
7.000	10.00	2487	-3292	4594	54.3	334	-881	916	36.4	60.73	29.15
10.500	15.00	3819	-5668	7911	48.3	413	-1117	1166	35.5	50.37	41.00
14.000	20.00	5325	-8045	11228	47.4	445	-1281	1335	33.3	56.68	53.96
17.500	25.00	6829	-10422	14545	47.0	394	-1283	1329	29.6	42.27	66.10
21.000	30.00	8035	-12709	17736	45.3	296	-1284	1323	22.4	46.22	76.17
24.500	35.00	8872	-12709	17736	50.0	185	-1286	1317	14.1	45.68	82.10
28.000	40.00	9321	-12709	17736	52.6	87	-1288	1310	6.6	62.01	84.97
31.500	45.00	9459	-12709	17736	53.3	-2	-1290	1304	0.1	62.71	86.27
35.000	50.00	9265	-12709	17736	52.2	-108	-1292	1298	8.4	59.12	86.75
38.500	55.00	8734	-12709	17736	49.2	-193	-1293	1292	14.9	63.13	86.45
42.000	60.00	7925	-12709	17736	44.7	-269	-1295	1285	20.8	60.37	85.26
45.500	65.00	6843	-12709	17736	38.6	-350	-1297	1279	27.0	51.19	82.47
49.000	70.00	5431	-12709	17736	30.6	-454	-1299	1273	35.0	55.50	77.98
52.500	75.00	3917	-12709	17736	22.1	-351	-1300	1267	27.0	110.65	71.48
56.000	80.00	2813	-10463	14601	19.3	-275	-1302	1260	21.1	73.15	54.13
59.500	85.00	1897	-8086	11284	16.8	-237	-1304	1254	18.1	70.28	52.64
63.000	90.00	1140	-5709	7968	14.3	-202	-1306	1248	15.5	45.02	36.60
66.500	95.00	546	-3332	4651	11.7	-138	-1102	1050	12.5	40.05	17.82
70.000	100.00	205	-1571	2193	9.3	-64	-718	685	8.9	30.94	15.34
73.500	105.00	46	-733	1023	4.5	-24	-335	319	7.3	12.99	5.22

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DAMAGE STABILITY OF LOADING CONDITION L03

CRITICAL DAMAGE STABILITY CRITERIA FOR EACH DAMAGE CASE

RCR	DAM	STAGE	PHASE	REQ	ATTV	UNIT	STAT	MINGM
V.RANGE.A469	D01	1	EQ	20.000	69.988	deg	OK	-0.408
V.RANGE.A469	D02	1	EQ	20.000	69.582	deg	OK	-0.292
V.RANGE.A469	D03	1	EQ	20.000	69.557	deg	OK	-0.301
V.MAXHEEL.A469	D04	1	EQ	17.000	2.137	deg	OK	-0.010
V.MAXHEEL.A469	D05	1	EQ	17.000	0.752	deg	OK	-0.151
V.MAXHEEL.A469	D06	1	EQ	17.000	2.022	deg	OK	0.004
V.MAXHEEL.A469	D07	1	EQ	17.000	0.620	deg	OK	-0.138
V.RANGE.A469	D08	1	EQ	20.000	68.474	deg	OK	-0.052
V.RANGE.A469	D09	1	EQ	20.000	68.497	deg	OK	-0.045
V.RANGE.A469	D10	1	EQ	20.000	68.091	deg	OK	0.046
V.RANGE.A469	D11	1	EQ	20.000	69.831	deg	OK	-0.119
V.RANGE.A469	D12	1	EQ	20.000	68.266	deg	OK	-0.129
V.MAXHEEL.A469	D13	1	EQ	17.000	1.760	deg	OK	-0.124
V.RANGE.A469	D14	1	EQ	20.000	69.967	deg	OK	-0.145
V.MINGZ.A469	D15	1	EQ	0.100	0.909	m	OK	0.014
V.RANGE.A469	D16	1	EQ	20.000	69.318	deg	OK	-0.318
V.RANGE.A469	D17	1	EQ	20.000	69.294	deg	OK	-0.325
V.RANGE.A469	D18	1	EQ	20.000	69.030	deg	OK	-0.294
V.RANGE.A469	D19	1	EQ	20.000	69.055	deg	OK	-0.288
V.RANGE.A469	D20	1	EQ	20.000	69.780	deg	OK	-0.379
V.RANGE.A469	D21	1	EQ	20.000	69.804	deg	OK	-0.371
TOTAL								0.046

DETERMINATION OF CRITICAL DAMAGE CASE

CRITICAL DAMAGE CASE : D10
STAGE OF CRITICAL DAMAGE CASE : 1
PHASE OF CRITICAL DAMAGE CASE : EQ

DAMAGE STABILITY CRITERIA - CRITICAL CASE

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM m
V.PROGR.A469	No progr. flooding	24.569	1.909	deg	OK	-0.219
V.MAXHEEL.A469	Maximum heel	17.000	1.909	deg	OK	0.011
V.RANGE.A469	Range of pos. stab	20.000	68.091	deg	OK	0.046
V.MINGZ.A469	Min. righting lever	0.100	0.954	m	OK	0.001
TOTAL						0.046

DAMAGE CASE RESULTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	T m	TR m	HEEL degree	RESFLD m	OPEN	CROSSTM s
D10	1	EQ	PS	4.723	0.773	-1.9	3.45	D-351	-

DAMAGED COMPARTMENTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	NAME	DES	PERM	VOL m3	W t	XCG m	YCG m	ZCG m
D10	INTACT	EQ	-				0.0	0.000			
D10	1	EQ	PS	T019	WING TANK SB	0.95	31.8	32.555	30.84	8.37	2.93
D10	1	EQ	PS	T021	DB TANK SB	0.95	92.4	94.737	30.84	4.63	0.61

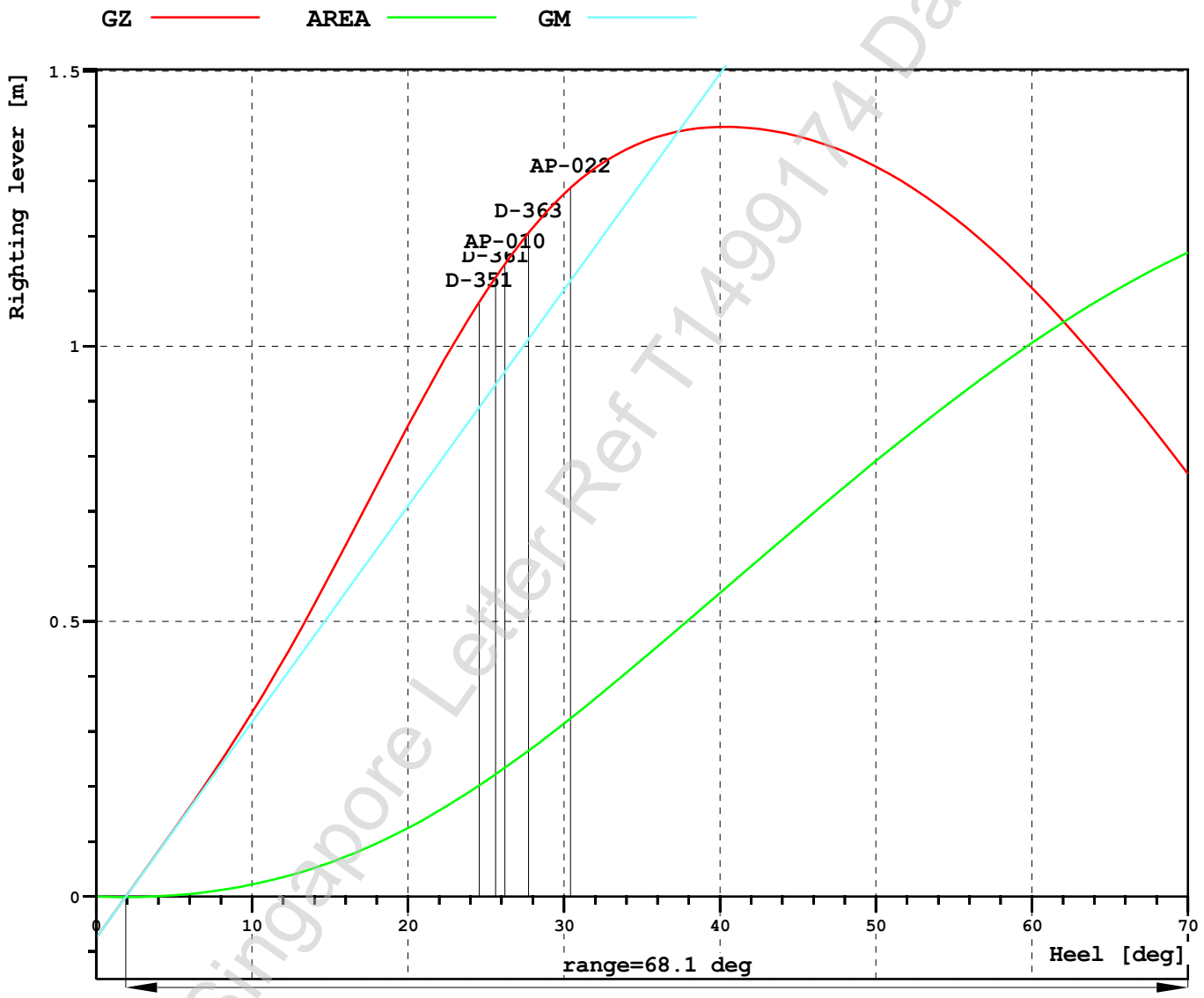
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CRITICAL OPENINGS - CRITICAL CASE

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
D-351	CO2 ROOM/M-DECK	WEATHERTIGHT	51.792	-8.360	8.600	24.6	3.448
D-361	PS CARGORAIL #57/M-DECK	WEATHERTIGHT	39.885	-8.560	8.600	25.6	3.556
AP-010	WING TANK PS	WEATHERTIGHT	47.211	-8.731	8.953	26.2	3.832
D-363	PS CARGORAIL #37/M-DECK	WEATHERTIGHT	25.900	-8.550	8.600	27.7	3.693
AP-022	WING TANK PS	WEATHERTIGHT	19.200	-8.702	8.936	30.4	4.088
D-353	DECK WORKSHOP/M-DECK	WEATHERTIGHT	53.192	-6.550	8.600	30.5	3.494
AP-263	HYDRAULIC OIL STORE TANK	WEATHERTIGHT	47.690	-7.260	8.912	30.8	3.836
AP-012	DB TANK PS	WEATHERTIGHT	50.000	-7.250	9.047	31.3	3.949

DAMAGE STABILITY CHECK PLOT - CRITICAL CASE



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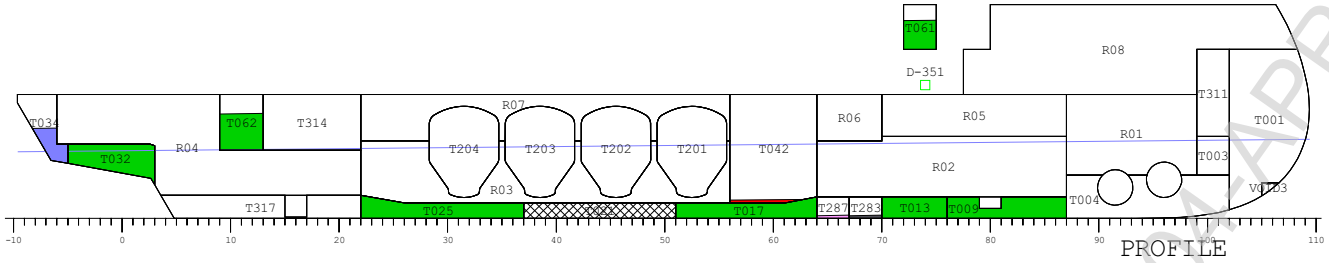
GZ-CURVE - CRITICAL CASE

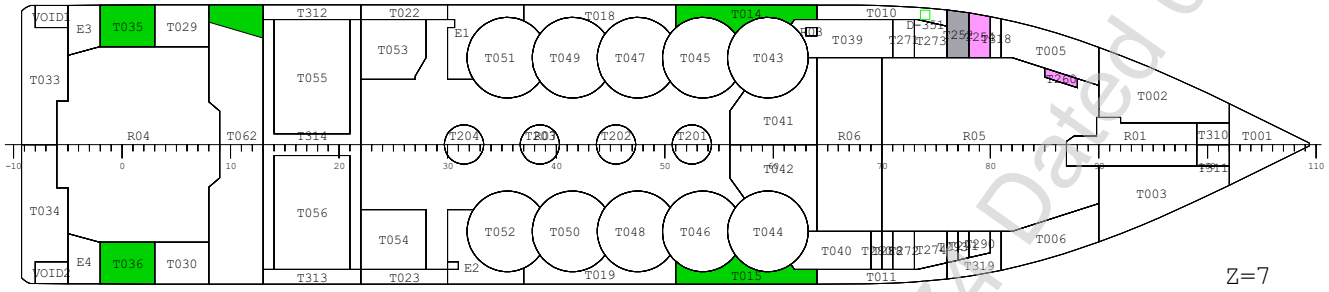
HEEL	DISP	T	TR	GZ	MS	AREA
degree	t	m	m	m	m	mrad
0.0	4293.1	4.730	0.764	0.075	0.075	0.000
-1.0	4293.1	4.727	0.767	0.036	0.092	-0.001
-3.0	4293.1	4.716	0.780	-0.043	0.124	-0.001
-5.0	4293.1	4.696	0.806	-0.122	0.156	0.002
-10.0	4293.1	4.608	0.925	-0.334	0.220	0.022
-15.0	4293.1	4.465	1.117	-0.584	0.243	0.062
-20.0	4293.1	4.270	1.343	-0.855	0.237	0.124
-30.0	4293.1	3.736	1.843	-1.277	0.320	0.314
-40.0	4293.1	3.044	2.043	-1.398	0.655	0.551
-50.0	4293.1	2.209	1.768	-1.326	1.121	0.791
-60.0	4293.1	1.292	1.249	-1.106	1.661	1.006
-70.0	4293.1	0.351	0.646	-0.768	2.234	1.171

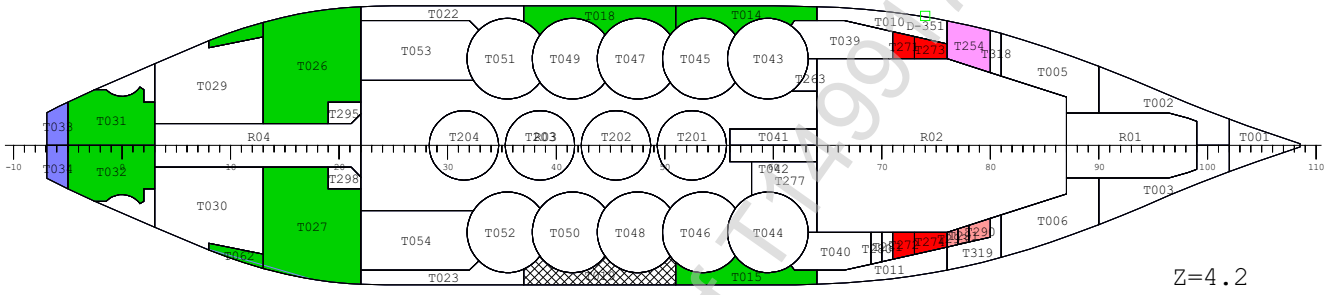
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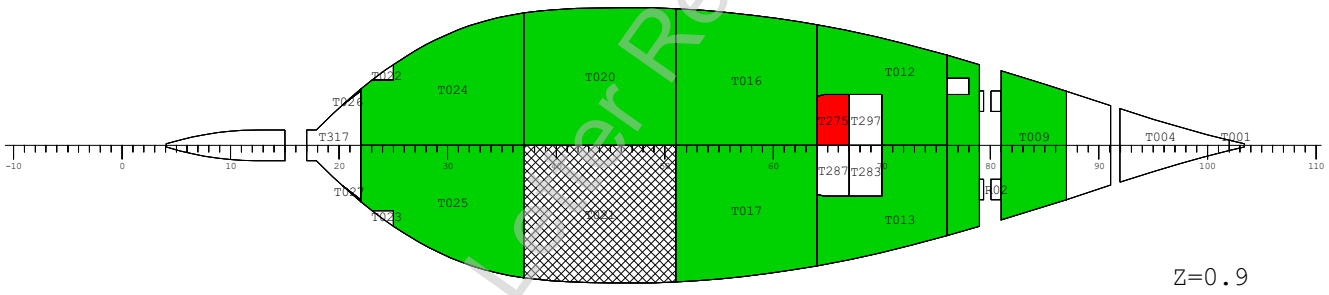
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DAMAGE ARRANGEMENT - CRITICAL CASE





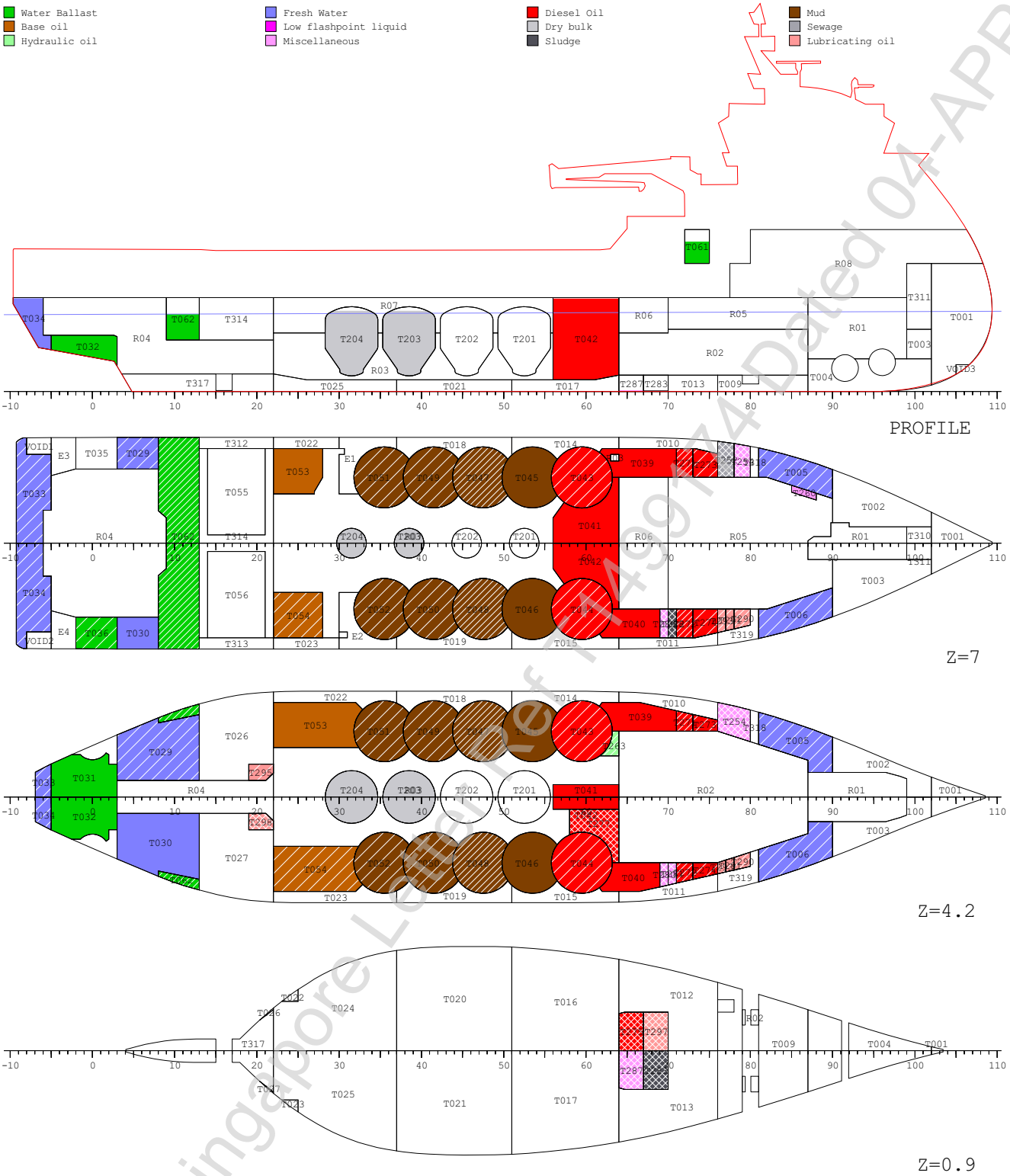




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12.4 L04 -> Fully loaded SWL - Departure



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LIST OF LOADS

NAME	DES	LOAD	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm	FRS	HM m
CONTENTS=Base oil (RHO=0.83)										
T053	STORAGE TANK PS	BSO	104.6	98.0	18.73	-6.02	4.76	17	REAL	7.84
T054	STORAGE TANK SB	BSO	104.4	98.0	18.71	6.02	4.78	17	REAL	7.84
SUBTOTAL		BSO	209.0		18.72	-0.01	4.77	33		
CONTENTS=Dry bulk (RHO=2.4)										
T203	DRY BULK TANK	CEM	152.8	98.0	26.95	0.00	4.62	11	REAL	6.88
T204	DRY BULK TANK	CEM	152.8	98.0	22.05	0.00	4.62	11	REAL	6.88
SUBTOTAL		CEM	305.6		24.50	0.00	4.62	21		
CONTENTS=Diesel Oil (RHO=0.86)										
T039	STORAGE TANK PS	FO	66.9	98.0	46.68	-6.76	4.91	6	REAL	7.88
T040	STORAGE TANK SB	FO	53.8	98.0	46.00	6.80	4.87	5	REAL	7.88
T041	STORAGE TANK PS	FO	68.5	98.0	42.16	-1.51	5.48	27	MAX	7.92
T042	STORAGE TANK SB	FO	68.5	98.0	42.16	1.51	5.48	27	MAX	7.92
T043	LIQUID CARGO TANK PS	FO	118.1	98.0	41.65	-5.56	4.56	31	REAL	7.87
T044	LIQUID CARGO TANK SB	FO	118.1	98.0	41.65	5.56	4.56	31	REAL	7.87
T271	FO SETTLING TANK PS	FO	8.5	70.0	50.38	-6.53	4.93	1	REAL	6.72
T272	FO SETTLING TANK SB	FO	8.5	70.0	50.38	6.53	4.93	1	REAL	6.72
T273	FO SERVICE TANK PS	FO	10.6	70.0	52.09	-6.38	5.15	2	REAL	6.90
T274	FO SERVICE TANK SB	FO	10.6	70.0	52.09	6.38	5.15	2	REAL	6.90
T275	FO DRAIN TANK	FO	0.8	10.0	45.85	-1.64	0.07	5	REAL	0.14
T277	FO OVERFLOW TANK	FO	3.6	10.0	42.62	2.34	1.31	10	REAL	1.53
T279	EMERG. GEN. FO TANK	FO	1.9	98.0	61.60	-1.38	24.57	0	REAL	25.80
SUBTOTAL		FO	538.5		43.61	-0.15	4.95	148		
CONTENTS=Fresh Water (RHO=1)										
T005	WING TANK PS	FW	96.2	98.0	59.97	-5.04	4.80	16	REAL	7.88
T006	WING TANK SB	FW	96.2	98.0	59.97	5.04	4.80	16	REAL	7.88
T029	WING TANK PS	FW	74.6	98.0	5.29	-4.87	4.61	6	REAL	7.83
T030	WING TANK SB	FW	74.6	98.0	5.29	4.87	4.61	6	REAL	7.83
T033	WING TANK PS	FW	57.9	98.0	-4.96	-3.91	6.49	126	MAX	7.95
T034	WING TANK SB	FW	57.9	98.0	-4.96	3.91	6.49	126	MAX	7.95
SUBTOTAL		FW	457.5		25.69	0.00	5.16	296		
CONTENTS=Hydraulic oil (RHO=0.924)										
T263	HYDRAULIC OIL STORE TANK	HO	3.3	70.0	44.30	-4.34	2.32	1	REAL	3.33
CONTENTS=Lubricating oil (RHO=0.924)										
T290	LO STORE TANK MAIN ENG.	LO	5.9	70.0	55.30	5.76	4.90	0	REAL	6.82
T291	LO STORE TANK MAIN ENG.	LO	3.1	70.0	54.25	6.04	5.11	0	REAL	6.92
T293	LO STORE TANK THRUSTER	LO	3.2	70.0	53.55	6.21	5.24	0	REAL	6.97
T295	LO STORE TANK AZIMUTH THR.	LO	4.0	70.0	14.35	-2.10	2.95	0	REAL	3.71
T297	LO DRAIN TANK	LO	0.9	10.0	47.95	-1.64	0.07	6	REAL	0.14
T298	DIRTY LO TANK	LO	0.6	10.0	14.35	2.10	2.31	0	REAL	2.41
SUBTOTAL		LO	17.6		43.89	3.64	4.24	8		
CONTENTS=Mass load (RHO=1)										
M1	CREW	MASS	2.6	0.0	57.40	0.00	15.50	0		0.00
M2	PROVISION	MASS	7.4	0.0	56.35	-6.50	12.40	0		0.00
M3	STORES	MASS	7.4	0.0	72.10	0.00	12.40	0		0.00
SUBTOTAL		MASS	17.5		63.21	-2.77	12.86	0		
CONTENTS=Miscellaneous (RHO=1)										
T280	BILGE WATER HOLDING TANK	MIS	0.9	10.0	48.64	6.17	2.03	0	REAL	2.57
T281	BILGE WATER SETTLING TANK	MIS	0.4	10.0	49.34	6.04	1.76	0	REAL	2.08
T287	BILGE WATER DRAIN TANK	MIS	0.9	10.0	45.85	1.64	0.07	6	REAL	0.14
T254	GREY WATER TANK	MIS	3.2	10.0	54.62	-5.79	1.92	1	REAL	2.34
T260	WATER MIST TANK	MIS	2.3	98.0	60.55	-4.30	6.28	0	REAL	7.27
SUBTOTAL		MIS	7.8		54.40	-2.49	3.01	8		
CONTENTS=Mud (RHO=2.8)										
T045	LIQUID CARGO TANK PS	MUD	348.9	98.0	37.22	-5.58	4.54	98	REAL	7.86
T046	LIQUID CARGO TANK SB	MUD	341.1	98.0	37.19	5.58	4.47	82	REAL	7.84
T047	LIQUID CARGO TANK PS	MUD	92.6	26.0	33.02	-5.52	2.10	98	REAL	2.99
T048	LIQUID CARGO TANK SB	MUD	92.6	26.0	33.02	5.52	2.10	98	REAL	2.99
T049	LIQUID CARGO TANK PS	MUD	348.9	98.0	28.82	-5.58	4.54	98	REAL	7.86

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T050	LIQUID CARGO TANK SB	MUD	348.9	98.0	28.82	5.58	4.54	98	REAL	7.86
T051	LIQUID CARGO TANK PS	MUD	346.2	98.0	24.63	-5.56	4.57	98	REAL	7.87
T052	LIQUID CARGO TANK SB	MUD	346.2	98.0	24.63	5.56	4.57	98	REAL	7.87
SUBTOTAL		MUD	2265.5		30.44	-0.02	4.34	765		
CONTENTS=Sewage (RHO=1)										
T252	BLACK WATER TANK	SEW	1.1	10.0	53.89	-6.93	5.45	2	REAL	5.59
CONTENTS=Sludge (RHO=1)										
T283	SLUDGE DRAIN TANK	SLU	0.9	10.0	47.95	1.64	0.07	6	REAL	0.14
T288	SLUDGE SETTLING TANK	SLU	0.4	10.0	49.35	6.75	5.44	1	REAL	5.59
SUBTOTAL		SLU	1.4		48.40	3.29	1.80	7		
CONTENTS=Water Ballast (RHO=1.025)										
T019	WING TANK SB	WB	0.0	0.0	30.79	8.37	4.73	2	MAX	0.00
T031	WING TANK PS	WB	27.2	100.0	-0.65	-1.76	4.18	0	(REAL)	5.39
T032	WING TANK SB	WB	27.2	100.0	-0.65	1.76	4.18	0	(REAL)	5.39
T061	ROLL RED. TANK	WB	80.9	63.0	51.66	0.00	11.81	1358	REAL	12.73
T062	ROLL RED. TANK	WB	126.3	60.0	7.45	0.00	5.49	1736	REAL	6.58
T036	WING TANK SB	WB	23.2	98.0	0.39	7.50	6.66	6	REAL	7.95
SUBTOTAL		WB	284.8		17.88	0.61	7.13	3103		
TOTAL			4109.4		29.99	0.01	4.78	4391		

LOAD SUMMARY TABLE

ITEM	MASS t	LCG m	TCG m	VCG m	FRSM tm
Base oil	209.0	18.72	-0.01	4.77	33.16
Dry bulk	305.6	24.50	0.00	4.62	21.37
Diesel Oil	538.5	43.61	-0.15	4.95	147.62
Fresh Water	457.5	25.69	0.00	5.16	295.74
Hydraulic oil	3.3	44.30	-4.34	2.32	0.61
Lubricating oil	17.6	43.89	3.64	4.24	7.62
Mass load	17.5	63.21	-2.77	12.86	0.00
Miscellaneous	7.8	54.40	-2.49	3.01	7.72
Mud	2265.5	30.44	-0.02	4.34	765.42
Sewage	1.1	53.89	-6.93	5.45	2.31
Sludge	1.4	48.40	3.29	1.80	6.99
Water Ballast	284.8	17.88	0.61	7.13	3102.62
DEADWEIGHT	4109.4	29.99	0.01	4.78	4391.19
Lightweight	2704.9	39.15	-0.01	7.08	
Deadweight	4109.4	29.99	0.01	4.78	
Total weight	6814.3	33.63	0.00	5.69	

F L O A T I N G P O S I T I O N

Draught moulded	6.700	m	KM	8.62 m
Trim	0.266	m	KG	5.69 m
Heel, SB=+	0.0	deg		
TA	6.567	m	GM0	2.93 m
TF	6.832	m	GMCORR	-0.64 m
Trimming moment	2012	tonm	GM	2.29 m

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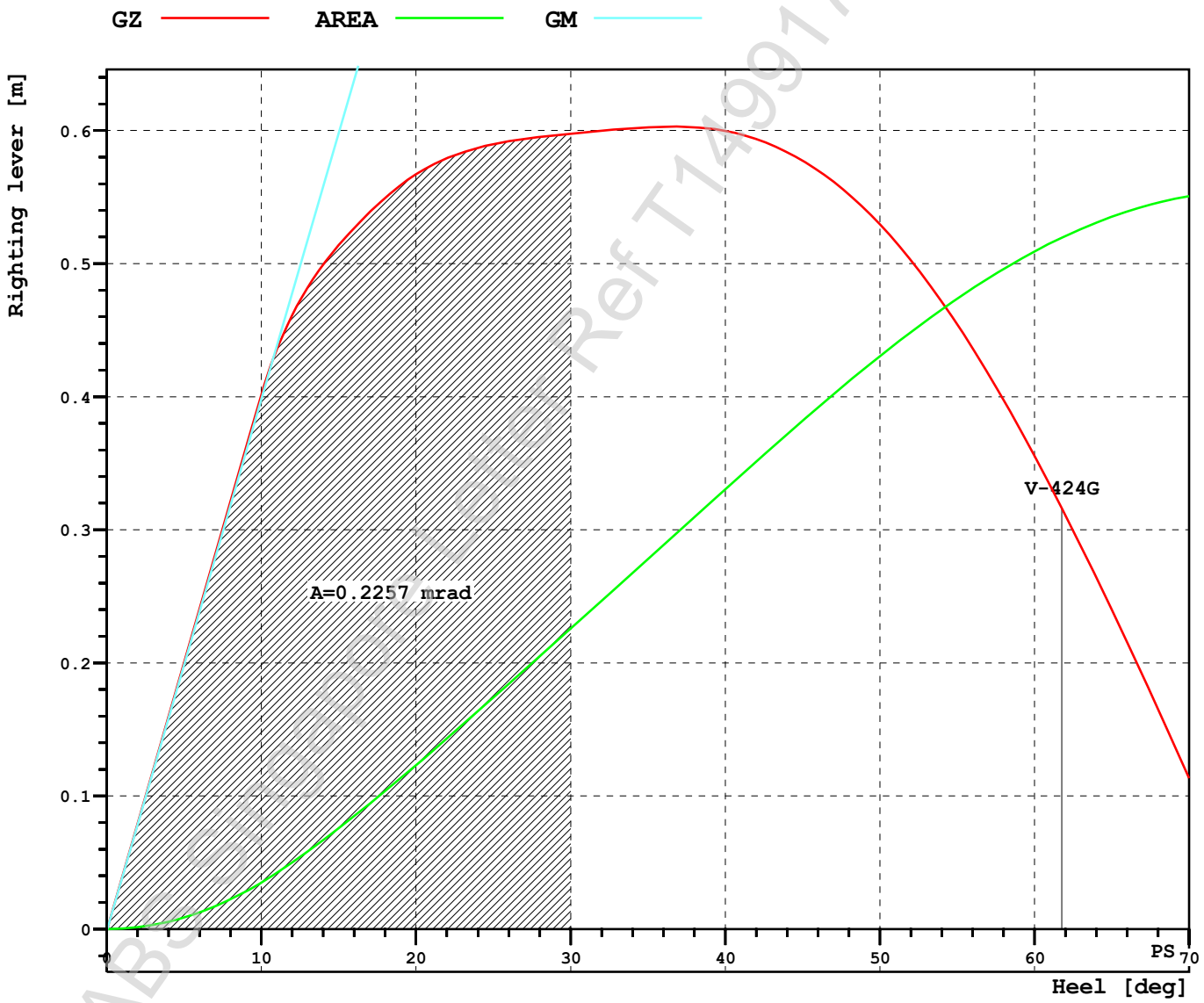
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STABILITY COMPONENTS

LOADING CONDITION L04, Fully loaded SWL - Departure

HEEL degree	MS m	HPHI m	EPHI mrad	FSMOM tm	DGZ m	GMSIN m	KN m
0.0	-0.001	0.00	0.000	0.0	0.000	0.000	0.001
1.0	-0.001	0.04	0.000	74.5	0.011	0.051	0.151
3.0	-0.001	0.12	0.003	223.6	0.033	0.153	0.452
5.0	-0.001	0.20	0.009	367.8	0.054	0.256	0.752
10.0	-0.008	0.40	0.035	676.1	0.099	0.509	1.491
15.0	-0.120	0.51	0.076	853.8	0.125	0.759	2.114
20.0	-0.296	0.57	0.123	955.2	0.140	1.003	2.656
30.0	-0.713	0.60	0.226	1062.4	0.156	1.466	3.602
40.0	-1.120	0.60	0.331	1126.1	0.165	1.885	4.426
50.0	-1.549	0.53	0.431	1147.0	0.168	2.246	5.060
60.0	-2.017	0.36	0.509	1140.9	0.167	2.539	5.454
70.0	-2.477	0.11	0.551	1128.1	0.166	2.756	5.629

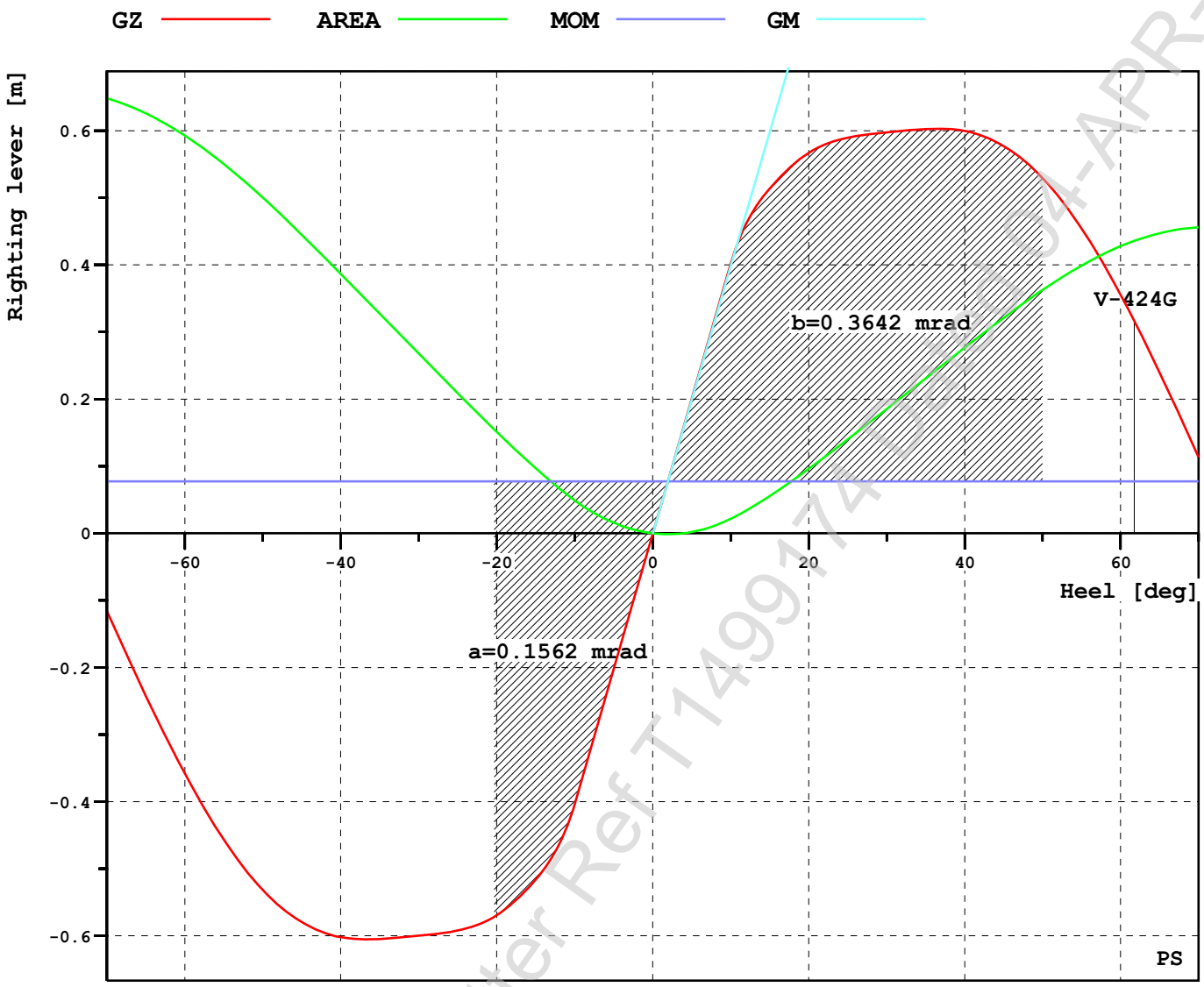
STABILITY CHECK PLOT



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IMOWEATHER CHECK PLOT



INTACT STABILITY CRITERIA

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM
V.AREA15-30	Area depending on the position of the Max. .	0.055	0.226	mrad	OK	1.632
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.105	mrad	OK	1.539
V.GZ0.2	Min. GZ > 0.2	0.200	0.603	m	OK	1.493
V.POSMAX15	Max. GZ at an angle > 15 deg.	15.000	36.540	deg	OK	1.490
V.GM0.15	GM > 0.15 m	0.150	2.288	m	OK	0.150
IMOWEATHER	IMO weather criterion	1.000	2.331		OK	1.512
2008IS-A2.3.1.2	Heeling angle due to steady wind <16 or <=8.	6.416	1.318	deg	OK	0.446
ABSFIFI1	Angle of heel < 1/2 freeboard immersion	4.010	0.330	deg	OK	0.175
ABSFIFI2	Area of residual stability >0.09 mrad	0.090	0.322	mrad	OK	1.289
TOTAL						1.632

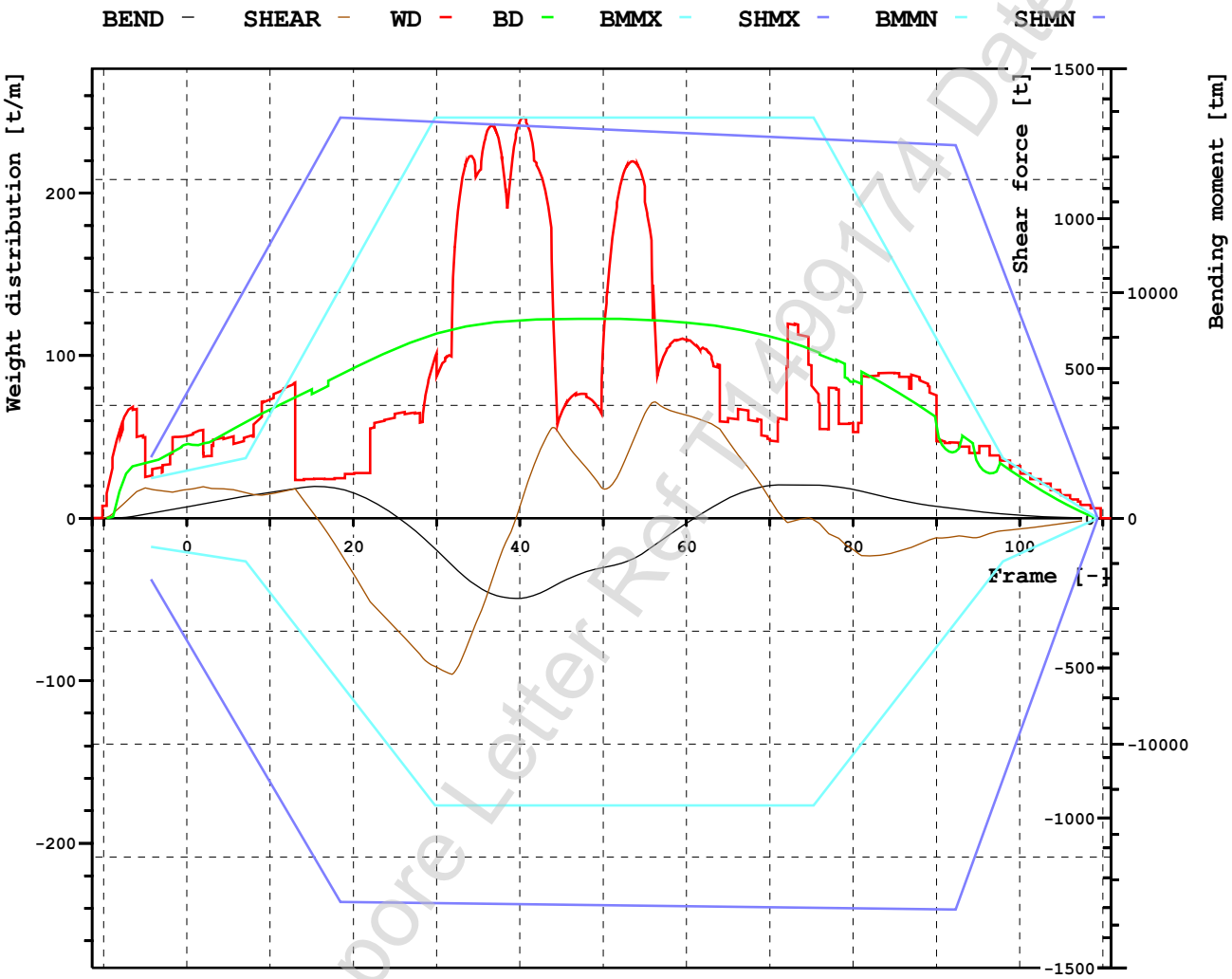
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MOST CRITICAL OPENINGS

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
V-421G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	55.890	2.093	14.288	-	7.525
V-424G	SUPPLY AIR ENGINE ROOM PS	UNPROTECTED	54.243	-3.413	14.288	-61.8	7.529
V-425G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	54.243	3.767	14.288	-	7.532
D-651	ACCOMMODATION/C-DECK	UNPROTECTED	56.000	3.771	16.906	-	10.144
V-421E	EXHAUST AIR ENGINE ROOM PS	UNPROTECTED	55.993	-2.538	17.088	-	10.323
V-423E	EXHAUST AIR ENGINE ROOM SB	UNPROTECTED	55.993	3.423	17.088	-	10.326

STRENGTH DIAGRAM



STRENGTH SUMMARY

LOADING CONDITION L04, Fully loaded SWL - Departure

			X	Frame
Shear force (min)	-520.7 t	(40.5%)	22.269 m	31.81
Shear force (max)	387.4 t	(30.0%)	39.296 m	56.14
Sagging moment	-3556.0 tm	(28.0%)	27.662 m	39.52
Hogging moment	1478.8 tm	(8.3%)	50.400 m	72.00

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STRENGTH DISTRIBUTION

LOADING CONDITION L04, Fully loaded SWL - Departure

X	FR	BEND	BMMN	BMMX	BMREL	SHEAR	SHMN	SHMX	SHREL	WD	BD
m	#	tm	tm	tm	%	t	t	t	%	t/m	t/m
-3.500	-5.00	173	0	0	0.0	101	0	0	0.0	49.87	33.82
0.000	0.00	493	-1510	2108	23.4	95	-407	418	22.9	50.63	45.54
3.500	5.00	839	-1790	2498	33.6	97	-644	667	14.5	49.49	53.23
7.000	10.00	1131	-3292	4594	24.7	81	-881	916	8.8	73.06	66.86
10.500	15.00	1400	-5668	7911	17.7	24	-1117	1166	2.0	24.04	79.41
14.000	20.00	1127	-8045	11228	10.0	-185	-1281	1335	14.5	27.32	92.25
17.500	25.00	124	-10422	14545	0.8	-365	-1283	1329	28.5	64.20	103.99
21.000	30.00	-1412	-12709	17736	11.1	-497	-1284	1323	38.7	101.49	113.66
24.500	35.00	-3041	-12709	17736	23.9	-332	-1286	1317	25.8	212.90	119.19
28.000	40.00	-3550	-12709	17736	27.9	38	-1288	1310	2.9	244.28	121.66
31.500	45.00	-2819	-12709	17736	22.2	269	-1290	1304	20.7	65.21	122.56
35.000	50.00	-2182	-12709	17736	17.2	98	-1292	1298	7.6	107.96	122.65
38.500	55.00	-1494	-12709	17736	11.8	349	-1293	1292	27.1	204.26	121.97
42.000	60.00	-222	-12709	17736	1.7	344	-1295	1285	26.8	109.62	120.38
45.500	65.00	877	-12709	17736	4.9	257	-1297	1279	20.1	61.58	117.13
49.000	70.00	1447	-12709	17736	8.2	66	-1299	1273	5.2	48.63	111.86
52.500	75.00	1465	-12709	17736	8.3	-1	-1300	1267	0.1	94.04	103.77
56.000	80.00	1273	-10463	14601	8.7	-105	-1302	1260	8.0	58.35	83.78
59.500	85.00	846	-8086	11284	7.5	-114	-1304	1254	8.7	89.40	78.55
63.000	90.00	525	-5709	7968	6.6	-65	-1306	1248	5.0	75.56	57.90
66.500	95.00	301	-3332	4651	6.5	-64	-1102	1050	5.8	40.05	33.81
70.000	100.00	136	-1571	2193	6.2	-36	-718	685	5.0	32.12	25.69
73.500	105.00	35	-733	1023	3.4	-18	-335	319	5.5	14.17	10.08

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DAMAGE STABILITY OF LOADING CONDITION L04

CRITICAL DAMAGE STABILITY CRITERIA FOR EACH DAMAGE CASE

RCR	DAM	STAGE	PHASE	REQ	ATTV	UNIT	STAT	MINGM
V.RANGE.A469	D01	1	EQ	20.000	61.147	deg	OK	0.591
V.RANGE.A469	D02	1	EQ	20.000	57.218	deg	OK	0.826
V.RANGE.A469	D03	1	EQ	20.000	59.277	deg	OK	0.811
V.RANGE.A469	D04	1	EQ	20.000	58.950	deg	OK	0.752
V.RANGE.A469	D05	1	EQ	20.000	60.551	deg	OK	0.818
V.RANGE.A469	D06	1	EQ	20.000	56.317	deg	OK	1.105
V.RANGE.A469	D07	1	EQ	20.000	57.876	deg	OK	1.136
V.RANGE.A469	D08	1	EQ	20.000	55.678	deg	OK	1.152
V.RANGE.A469	D09	1	EQ	20.000	57.722	deg	OK	1.135
V.RANGE.A469	D10	1	EQ	20.000	55.333	deg	OK	1.260
V.RANGE.A469	D11	1	EQ	20.000	57.378	deg	OK	1.243
V.RANGE.A469	D12	1	EQ	20.000	56.076	deg	OK	1.276
V.RANGE.A469	D13	1	EQ	20.000	58.103	deg	OK	1.261
V.RANGE.A469	D14	1	EQ	20.000	57.682	deg	OK	1.094
V.RANGE.A469	D15	1	EQ	20.000	59.738	deg	OK	1.078
V.RANGE.A469	D16	1	EQ	20.000	61.370	deg	OK	0.703
V.RANGE.A469	D17	1	EQ	20.000	59.308	deg	OK	0.707
V.RANGE.A469	D18	1	EQ	20.000	61.444	deg	OK	0.688
V.RANGE.A469	D19	1	EQ	20.000	61.350	deg	OK	0.829
V.RANGE.A469	D20	1	EQ	20.000	61.185	deg	OK	0.703
V.RANGE.A469	D21	1	EQ	20.000	59.118	deg	OK	0.709
TOTAL								1.276

DETERMINATION OF CRITICAL DAMAGE CASE

CRITICAL DAMAGE CASE : D12
STAGE OF CRITICAL DAMAGE CASE : 1
PHASE OF CRITICAL DAMAGE CASE : EQ

DAMAGE STABILITY CRITERIA - CRITICAL CASE

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM m
V.PROGR.A469	No progr. flooding	11.726	3.009	deg	OK	0.684
V.MAXHEEL.A469	Maximum heel	17.000	3.009	deg	OK	0.684
V.RANGE.A469	Range of pos. stab	20.000	56.076	deg	OK	1.276
V.MINGZ.A469	Min. righting lever	0.100	0.426	m	OK	1.198
TOTAL						1.276

DAMAGE CASE RESULTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	T m	TR m	HEEL degree	RESFLD m	OPEN	CROSSTM s
D12	1	EQ	SB	6.787	0.075	3.0	1.35	D-352	-

DAMAGED COMPARTMENTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	NAME	DES	PERM	VOL m3	W t	XCG m	YCG m	ZCG m
D12	INTACT	EQ	PS				0.0	0.000			
D12	1	EQ	SB	T023	WING TANK SB	0.95	54.8	56.202	19.35	7.84	3.64
D12	1	EQ	SB	T025	DB TANK SB	0.95	72.5	74.304	21.46	3.85	0.81
D12	1	EQ	SB	E2	Emrg. exit 1 SB	0.95	21.1	21.670	22.57	7.30	6.08

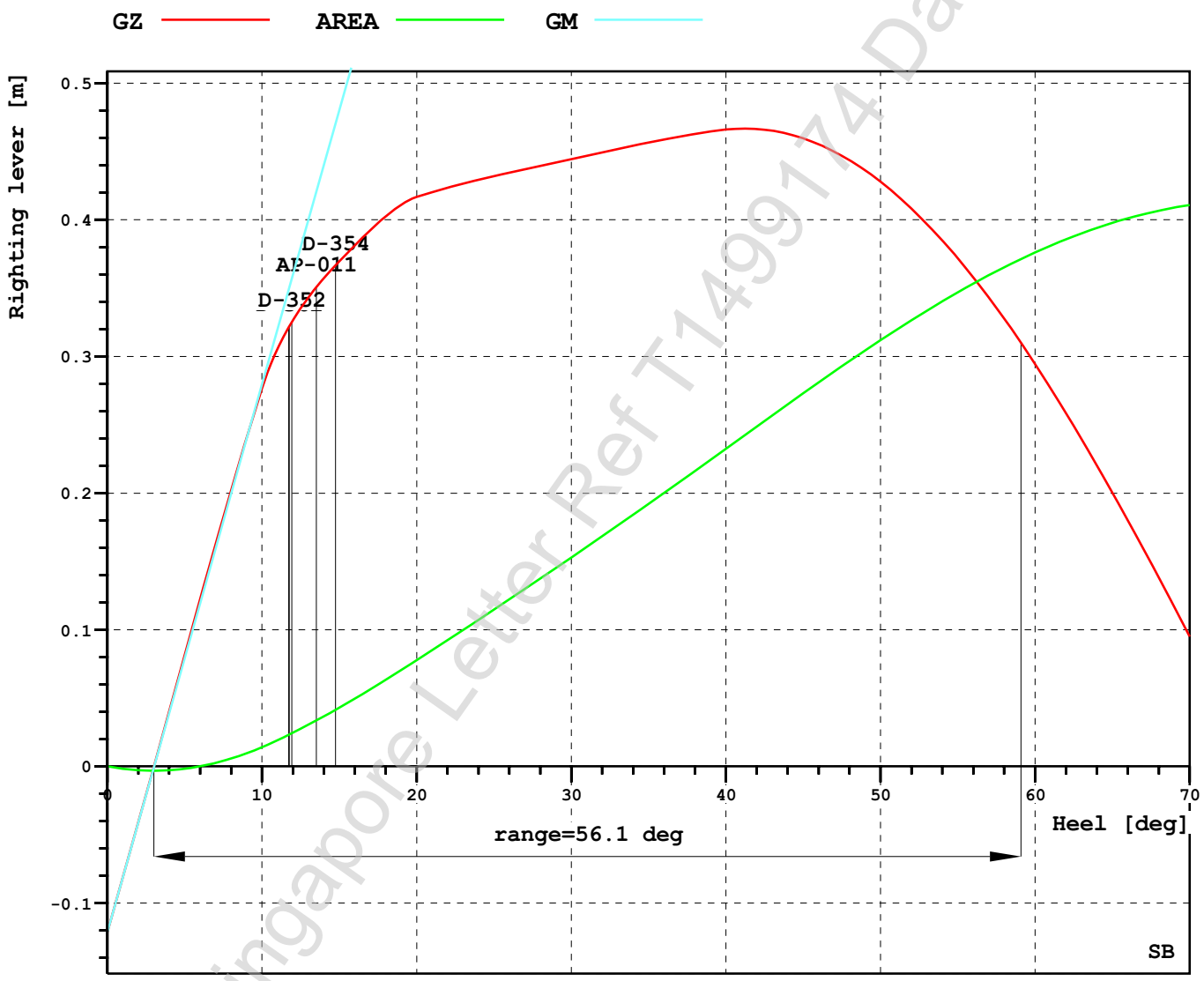
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CRITICAL OPENINGS - CRITICAL CASE

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
D-360	SB CARGORAIL #57/M-DECK	WEATHERTIGHT	39.885	8.560	8.600	11.7	1.349
D-362	SB CARGORAIL #37/M-DECK	WEATHERTIGHT	25.900	8.550	8.600	11.8	1.362
D-352	CHEMICAL/M-DECK	WEATHERTIGHT	53.192	8.375	8.600	11.9	1.346
AP-011	WING TANK SB	WEATHERTIGHT	47.211	8.754	8.953	13.5	1.684
D-354	PAINT/M-DECK	WEATHERTIGHT	53.192	6.600	8.600	14.8	1.439
AP-290	LO STORE TANK MAIN ENG	WEATHERTIGHT	52.985	7.410	8.927	15.5	1.723
AP-030	WING TANK SB	WEATHERTIGHT	5.250	7.801	9.168	15.8	1.988
AP-280	BILGE WATER HOLDING TANK	WEATHERTIGHT	47.738	7.275	8.951	15.9	1.759

DAMAGE STABILITY CHECK PLOT - CRITICAL CASE



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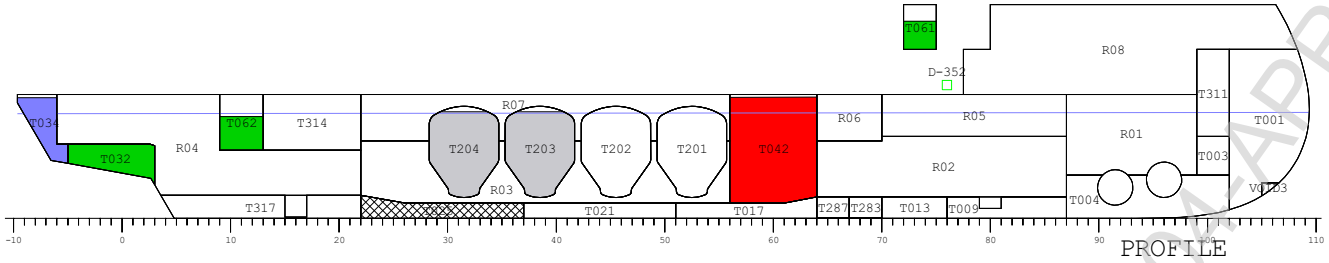
GZ-CURVE - CRITICAL CASE

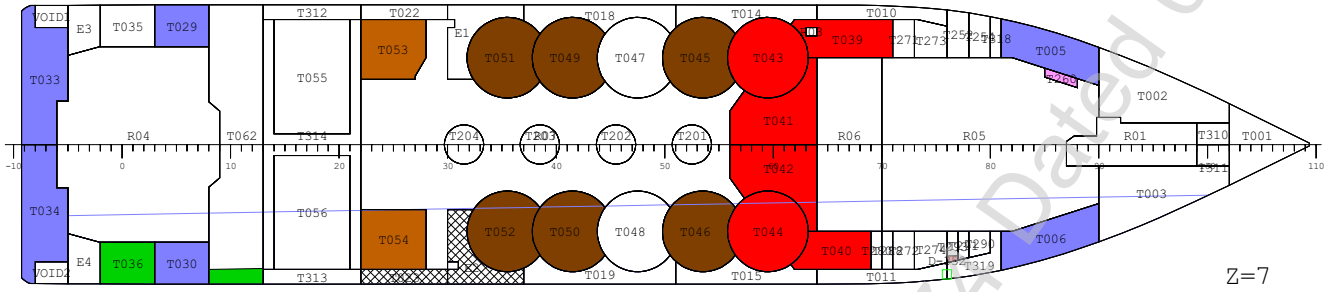
HEEL	DISP	T	TR	GZ	MS	AREA
degree	t	m	m	m	m	mrad
0.0	6814.3	6.794	0.082	-0.122	-0.122	0.000
1.0	6814.3	6.794	0.079	-0.081	-0.133	-0.002
3.0	6814.3	6.787	0.075	0.000	-0.154	-0.003
5.0	6814.3	6.771	0.074	0.082	-0.174	-0.002
10.0	6814.3	6.700	0.064	0.276	-0.233	0.014
15.0	6814.3	6.649	-0.089	0.370	-0.389	0.043
20.0	6814.3	6.599	-0.379	0.417	-0.586	0.078
30.0	6814.3	6.449	-1.326	0.444	-1.022	0.153
40.0	6814.3	6.173	-2.740	0.466	-1.419	0.232
50.0	6814.3	5.787	-4.272	0.428	-1.818	0.312
60.0	6814.3	5.278	-5.686	0.294	-2.245	0.376
70.0	6814.3	4.637	-6.886	0.095	-2.660	0.411

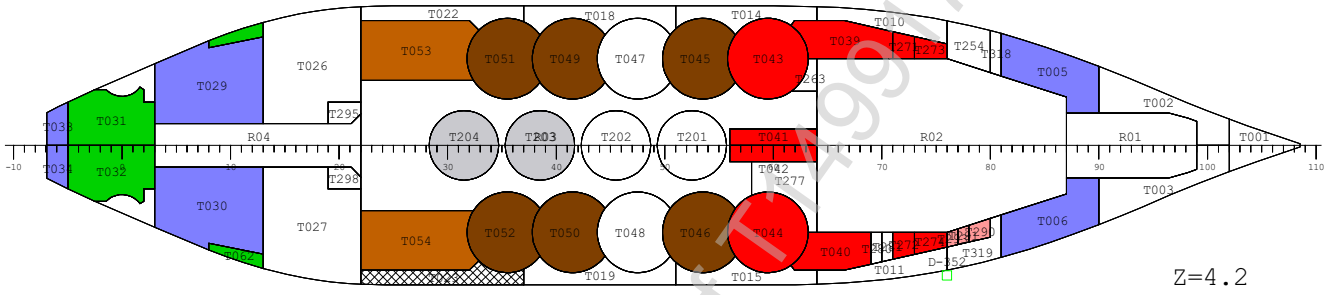
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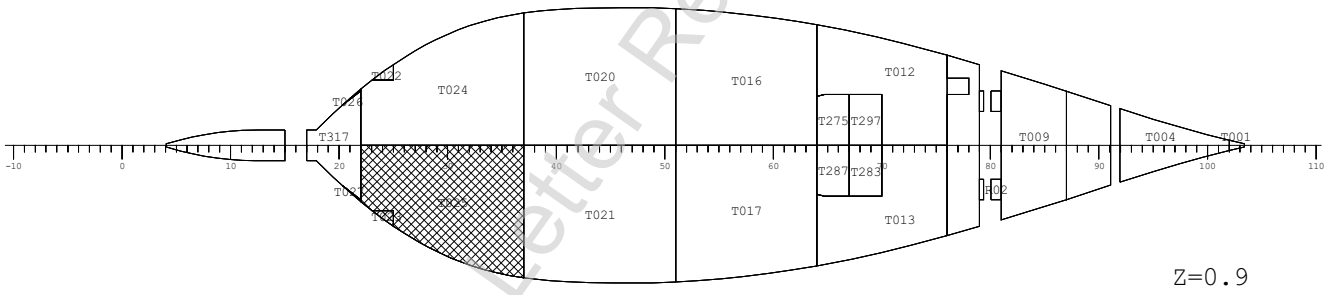
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DAMAGE ARRANGEMENT - CRITICAL CASE





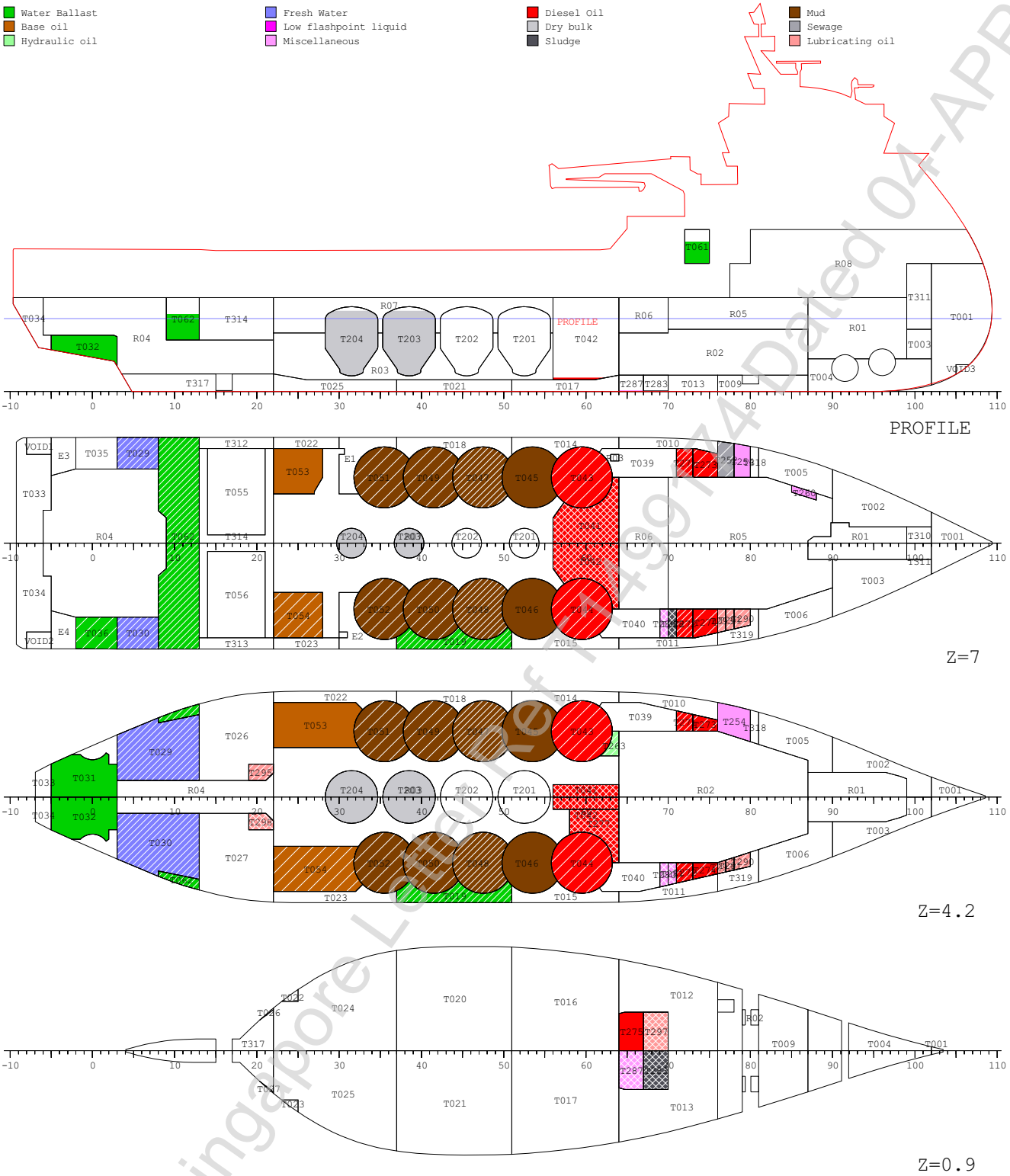




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12.5 L05 -> Fully loaded SWL - Arrival



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LIST OF LOADS

NAME	DES	LOAD	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm	FRS	HM m
CONTENTS=Base oil (RHO=0.83)										
T053	STORAGE TANK PS	BSO	104.6	98.0	18.73	-6.02	4.76	17	REAL	7.84
T054	STORAGE TANK SB	BSO	104.4	98.0	18.71	6.02	4.78	17	REAL	7.84
SUBTOTAL		BSO	209.0		18.72	-0.01	4.77	33		
CONTENTS=Dry bulk (RHO=2.4)										
T203	DRY BULK TANK	CEM	152.8	98.0	26.95	0.00	4.62	11	REAL	6.88
T204	DRY BULK TANK	CEM	152.8	98.0	22.05	0.00	4.62	11	REAL	6.88
SUBTOTAL		CEM	305.6		24.50	0.00	4.62	21		
CONTENTS=Diesel Oil (RHO=0.86)										
T041	STORAGE TANK PS	FO	0.7	1.0	41.24	-0.52	1.10	27	MAX	1.20
T042	STORAGE TANK SB	FO	0.7	1.0	41.24	0.52	1.10	27	MAX	1.20
T043	LIQUID CARGO TANK PS	FO	118.1	98.0	41.65	-5.56	4.56	31	REAL	7.87
T044	LIQUID CARGO TANK SB	FO	118.1	98.0	41.65	5.56	4.56	31	REAL	7.87
T271	FO SETTLING TANK PS	FO	8.5	70.0	50.38	-6.53	4.93	1	REAL	6.72
T272	FO SETTLING TANK SB	FO	8.5	70.0	50.38	6.53	4.93	1	REAL	6.72
T273	FO SERVICE TANK PS	FO	10.6	70.0	52.09	-6.38	5.15	2	REAL	6.90
T274	FO SERVICE TANK SB	FO	10.6	70.0	52.09	6.38	5.15	2	REAL	6.90
T275	FO DRAIN TANK	FO	7.9	98.0	45.85	-1.64	0.69	5	REAL	1.37
T277	FO OVERFLOW TANK	FO	3.6	10.0	42.62	2.34	1.31	10	REAL	1.53
T279	EMERG. GEN. FO TANK	FO	1.9	98.0	61.60	-1.38	24.57	0	REAL	25.80
SUBTOTAL		FO	289.2		43.18	-0.02	4.59	137		
CONTENTS=Fresh Water (RHO=1)										
T029	WING TANK PS	FW	22.9	30.0	6.60	-3.00	3.15	54	REAL	3.71
T030	WING TANK SB	FW	22.9	30.0	6.60	3.00	3.15	54	REAL	3.71
T033	WING TANK PS	FW	0.0	0.0	-4.96	-3.92	6.52	126	MAX	0.00
T034	WING TANK SB	FW	0.0	0.0	-4.96	3.92	6.52	126	MAX	0.00
SUBTOTAL		FW	45.7		6.60	0.00	3.15	362		
CONTENTS=Hydraulic oil (RHO=0.924)										
T263	HYDRAULIC OIL STORE TANK	HO	1.9	40.0	44.29	-4.33	1.89	1	REAL	2.47
CONTENTS=Lubricating oil (RHO=0.924)										
T290	LO STORE TANK MAIN ENG.	LO	3.4	40.0	55.32	5.47	4.05	0	REAL	5.16
T291	LO STORE TANK MAIN ENG.	LO	1.8	40.0	54.25	5.81	4.26	0	REAL	5.48
T293	LO STORE TANK THRUSTER	LO	1.8	40.0	53.55	6.03	4.41	0	REAL	5.66
T295	LO STORE TANK AZIMUTH THR.	LO	2.3	40.0	14.35	-2.10	2.63	0	REAL	3.06
T297	LO DRAIN TANK	LO	0.9	10.0	47.95	-1.64	0.07	6	REAL	0.14
T298	DIRTY LO TANK	LO	0.6	10.0	14.35	2.10	2.31	0	REAL	2.41
SUBTOTAL		LO	10.7		43.36	3.26	3.43	7		
CONTENTS=Mass load (RHO=1)										
M1	CREW	MASS	2.6	0.0	57.40	0.00	15.50	0		0.00
M2	PROVISION	MASS	0.8	0.0	56.35	-6.50	12.40	0		0.00
M3	STORES	MASS	0.8	0.0	72.10	0.00	12.40	0		0.00
SUBTOTAL		MASS	4.2		60.00	-1.24	14.32	0		
CONTENTS=Miscellaneous (RHO=1)										
T280	BILGE WATER HOLDING TANK	MIS	0.9	10.0	48.64	6.17	2.03	0	REAL	2.57
T281	BILGE WATER SETTLING TANK	MIS	0.4	10.0	49.34	6.04	1.76	0	REAL	2.08
T287	BILGE WATER DRAIN TANK	MIS	0.9	10.0	45.85	1.64	0.07	6	REAL	0.14
T254	GREY WATER TANK	MIS	31.6	98.0	54.82	-6.47	4.52	3	REAL	7.84
T260	WATER MIST TANK	MIS	2.3	98.0	60.55	-4.30	6.28	0	REAL	7.27
SUBTOTAL		MIS	36.2		54.74	-5.68	4.43	9		
CONTENTS=Mud (RHO=2.8)										
T045	LIQUID CARGO TANK PS	MUD	348.9	98.0	37.22	-5.58	4.54	98	REAL	7.86
T046	LIQUID CARGO TANK SB	MUD	341.1	98.0	37.19	5.58	4.47	82	REAL	7.84
T047	LIQUID CARGO TANK PS	MUD	92.6	26.0	33.02	-5.52	2.10	98	REAL	2.99
T048	LIQUID CARGO TANK SB	MUD	92.6	26.0	33.02	5.52	2.10	98	REAL	2.99
T049	LIQUID CARGO TANK PS	MUD	348.9	98.0	28.82	-5.58	4.54	98	REAL	7.86
T050	LIQUID CARGO TANK SB	MUD	348.9	98.0	28.82	5.58	4.54	98	REAL	7.86
T051	LIQUID CARGO TANK PS	MUD	346.2	98.0	24.63	-5.56	4.57	98	REAL	7.87
T052	LIQUID CARGO TANK SB	MUD	346.2	98.0	24.63	5.56	4.57	98	REAL	7.87

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SUBTOTAL	MUD	2265.4		30.44	-0.02	4.34	765		
CONTENTS=Sewage (RHO=1)									
T252 BLACK WATER TANK	SEW	10.4	97.9	53.89	-7.06	6.66	4	REAL	7.95
CONTENTS=Sludge (RHO=1)									
T283 SLUDGE DRAIN TANK	SLU	0.9	10.0	47.95	1.64	0.07	6	REAL	0.14
T288 SLUDGE SETTLING TANK	SLU	0.4	10.0	49.35	6.75	5.44	1	REAL	5.59
SUBTOTAL	SLU	1.4		48.40	3.29	1.80	7		
CONTENTS=Water Ballast (RHO=1.025)									
T019 WING TANK SB	WB	19.3	26.0	30.82	8.36	2.31	2	MAX	3.16
T031 WING TANK PS	WB	27.0	99.4	-0.64	-1.74	4.17	1	REAL	5.23
T032 WING TANK SB	WB	27.0	99.4	-0.64	1.74	4.17	1	REAL	5.23
T061 ROLL RED. TANK	WB	80.9	63.0	51.66	0.00	11.81	1358	REAL	12.73
T062 ROLL RED. TANK	WB	126.3	60.0	7.45	0.00	5.49	1736	REAL	6.58
T036 WING TANK SB	WB	23.2	98.0	0.39	7.50	6.66	6	REAL	7.95
SUBTOTAL	WB	303.8		18.73	1.11	6.83	3104		
TOTAL		3483.4		29.35	0.01	4.63	4450		

LOAD SUMMARY TABLE

ITEM	MASS t	LCG m	TCG m	VCG m	FRSM tm
Base oil	209.0	18.72	-0.01	4.77	33.16
Dry bulk	305.6	24.50	0.00	4.62	21.37
Diesel Oil	289.2	43.18	-0.02	4.59	136.72
Fresh Water	45.7	6.60	0.00	3.15	361.72
Hydraulic oil	1.9	44.29	-4.33	1.89	0.61
Lubricating oil	10.7	43.36	3.26	3.43	7.30
Mass load	4.2	60.00	-1.24	14.32	0.00
Miscellaneous	36.2	54.74	-5.68	4.43	9.39
Mud	2265.4	30.44	-0.02	4.34	765.42
Sewage	10.4	53.89	-7.06	6.66	3.55
Sludge	1.4	48.40	3.29	1.80	6.99
Water Ballast	303.8	18.73	1.11	6.83	3103.80
DEADWEIGHT	3483.4	29.35	0.01	4.63	4450.02
Lightweight	2704.9	39.15	-0.01	7.08	
Deadweight	3483.4	29.35	0.01	4.63	
Total weight	6188.3	33.64	0.00	5.70	

F L O A T I N G P O S I T I O N

Draught moulded	6.208	m	KM	8.72 m
Trim	0.003	m	KG	5.70 m
Heel, SB=+	0.0	deg		
TA	6.207	m	GM0	3.02 m
TF	6.209	m	GMCORR	-0.72 m
Trimming moment	19	tonm	GM	2.30 m

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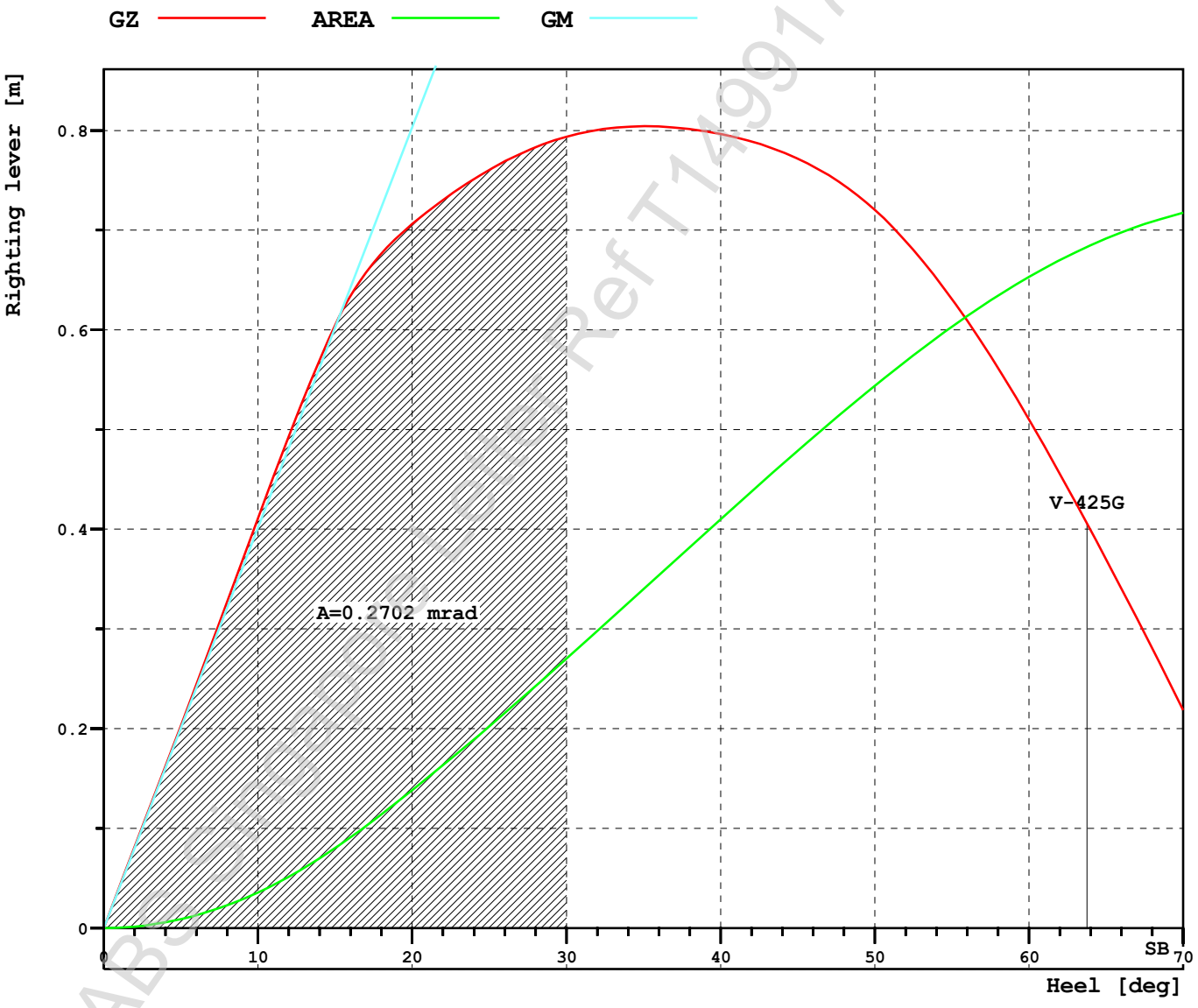
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STABILITY COMPONENTS

LOADING CONDITION L05, Fully loaded SWL - Arrival

HEEL degree	MS m	HPHI m	EPHI mrad	FSMOM tm	DGZ m	GMSIN m	KN m
0.0	0.000	0.00	0.000	0.0	0.000	0.000	-0.002
1.0	0.000	0.04	0.000	75.5	0.012	0.053	0.151
3.0	0.000	0.12	0.003	226.5	0.037	0.158	0.455
5.0	0.000	0.20	0.009	372.6	0.060	0.263	0.758
10.0	-0.002	0.41	0.036	687.9	0.111	0.524	1.510
15.0	-0.036	0.60	0.080	873.8	0.141	0.782	2.220
20.0	-0.167	0.71	0.138	986.3	0.159	1.033	2.814
30.0	-0.534	0.79	0.270	1125.8	0.182	1.510	3.824
40.0	-0.950	0.80	0.410	1208.5	0.195	1.941	4.654
50.0	-1.393	0.72	0.544	1243.2	0.201	2.314	5.287
60.0	-1.906	0.51	0.653	1238.3	0.200	2.616	5.646
70.0	-2.422	0.22	0.717	1222.5	0.198	2.838	5.772

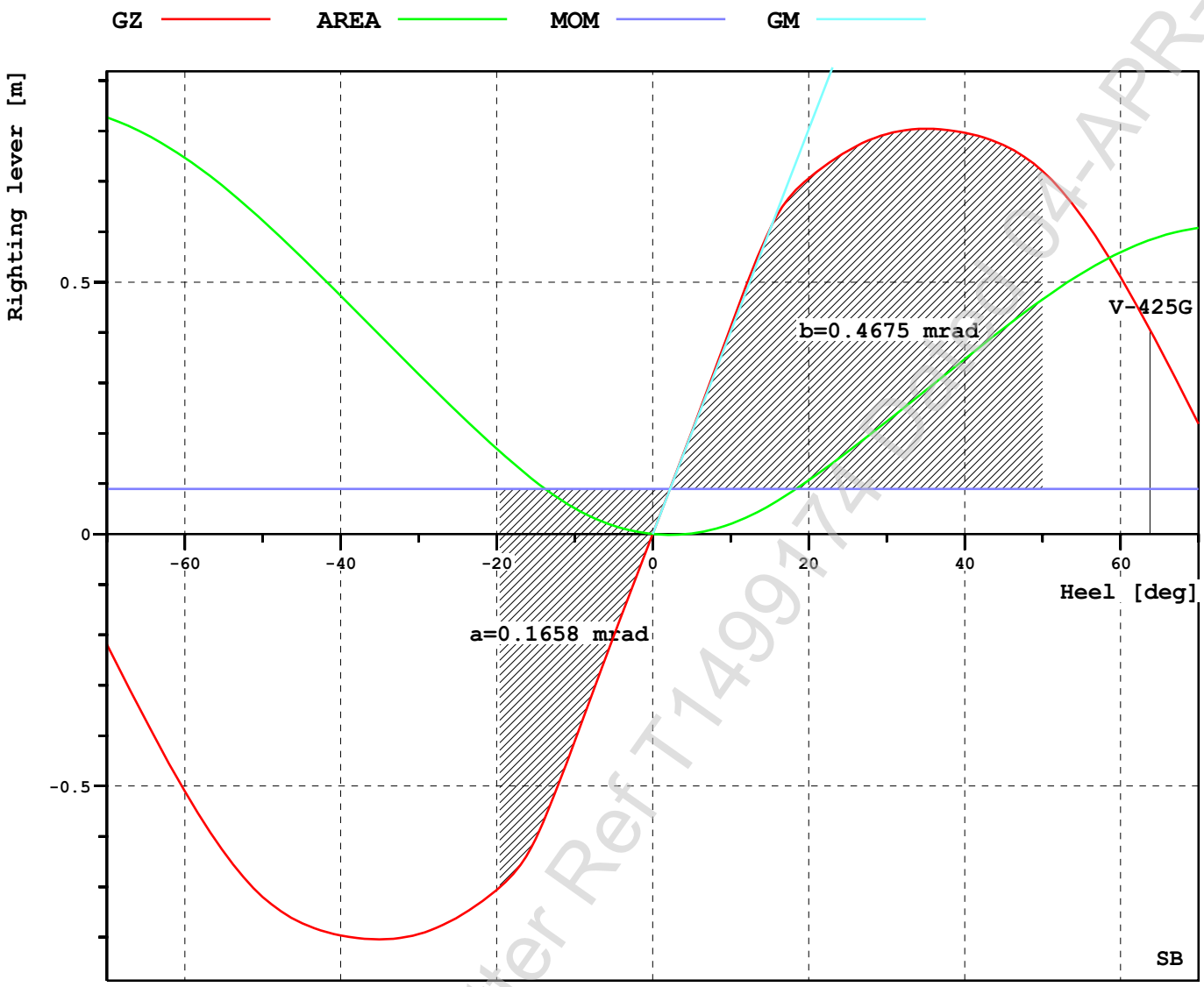
STABILITY CHECK PLOT



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IMOWEATHER CHECK PLOT



INTACT STABILITY CRITERIA

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM
V.AREA15-30	Area depending on the position of the Max. .	0.055	0.270	mrad	OK	1.301
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.140	mrad	OK	1.203
V.GZ0.2	Min. GZ > 0.2	0.200	0.804	m	OK	1.114
V.POSMAX15	Max. GZ at an angle > 15 deg.	15.000	35.115	deg	OK	0.357
V.GM0.15	GM > 0.15 m	0.150	2.301	m	OK	0.150
IMOWEATHER	IMO weather criterion	1.000	2.819		OK	1.077
2008IS-A2.3.1.2	Heeling angle due to steady wind <16 or <=8.	9.070	1.463	deg	OK	0.321
ABSFIFI1	Angle of heel < 1/2 freeboard immersion	5.669	0.334	deg	OK	0.109
ABSFIFI2	Area of residual stability >0.09 mrad	0.090	0.401	mrad	OK	0.963
TOTAL						1.301

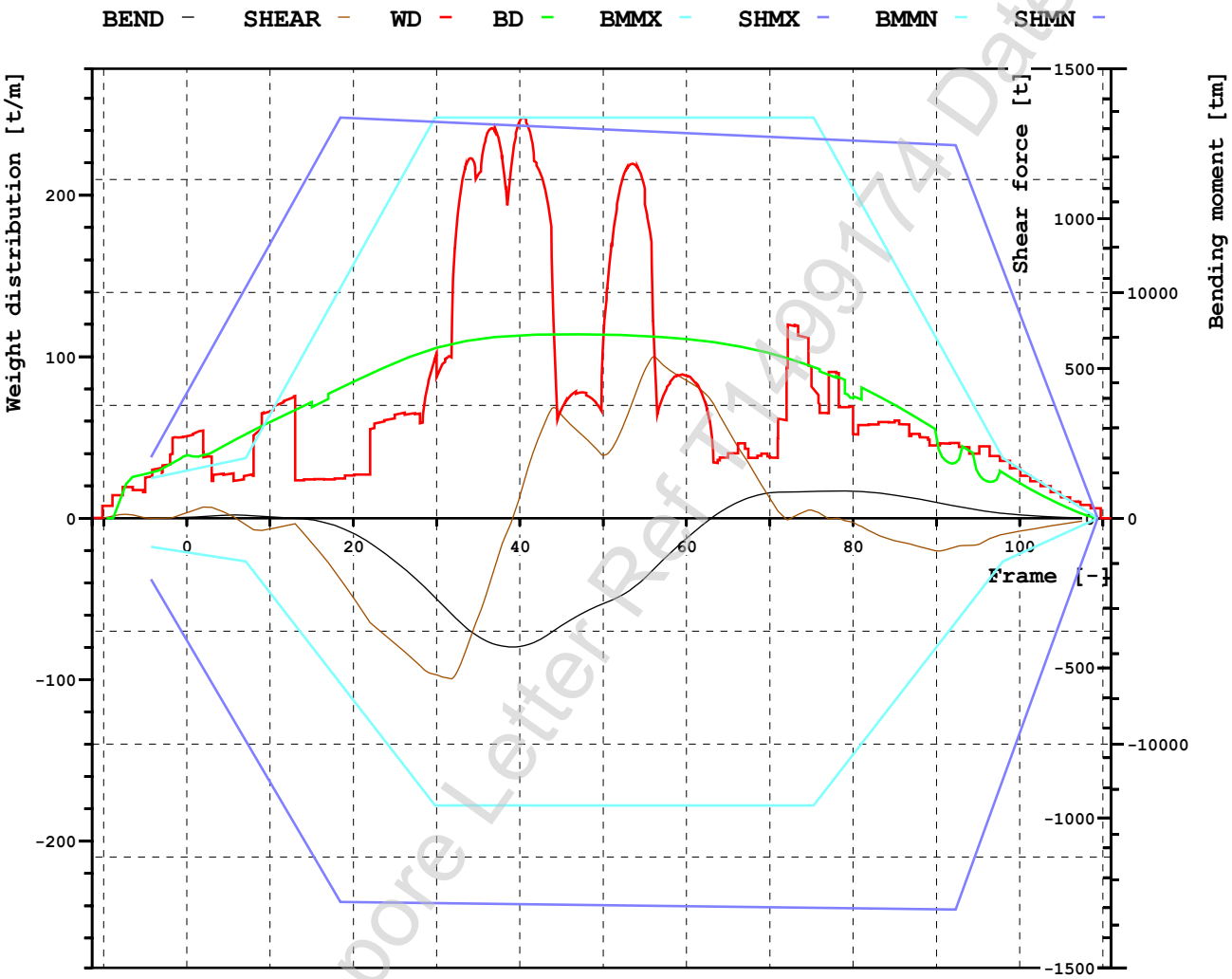
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MOST CRITICAL OPENINGS

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
V-421G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	55.890	2.093	14.288	-	8.079
V-424G	SUPPLY AIR ENGINE ROOM PS	UNPROTECTED	54.243	-3.413	14.288	-	8.080
V-425G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	54.243	3.767	14.288	63.8	8.080
D-651	ACCOMMODATION/C-DECK	UNPROTECTED	56.000	3.771	16.906	-	10.697
V-421E	EXHAUST AIR ENGINE ROOM PS	UNPROTECTED	55.993	-2.538	17.088	-	10.879
V-423E	EXHAUST AIR ENGINE ROOM SB	UNPROTECTED	55.993	3.423	17.088	-	10.879

STRENGTH DIAGRAM



STRENGTH SUMMARY

LOADING CONDITION L05, Fully loaded SWL - Arrival

			X	Frame
Shear force (min)	-535.9 t	(41.7%)	22.250 m	31.79
Shear force (max)	538.3 t	(41.7%)	39.294 m	56.13
Sagging moment	-5700.9 tm	(44.9%)	27.406 m	39.15
Hogging moment	1205.4 tm	(7.3%)	53.900 m	77.00

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STRENGTH DISTRIBUTION

LOADING CONDITION L05, Fully loaded SWL - Arrival

X	FR	BEND	BMMN	BMMX	BMREL	SHEAR	SHMN	SHMX	SHREL	WD	BD
m	#	tm	tm	tm	%	t	t	t	%	t/m	t/m
-3.500	-5.00	27	0	0	0.0	1	0	0	0.0	25.14	27.43
0.000	0.00	39	-1510	2108	1.9	18	-407	418	4.4	50.64	38.83
3.500	5.00	137	-1790	2498	5.5	13	-644	667	2.0	27.19	46.21
7.000	10.00	58	-3292	4594	1.3	-36	-881	916	4.1	66.19	59.59
10.500	15.00	-71	-5668	7911	1.3	-82	-1117	1166	7.4	24.04	71.92
14.000	20.00	-666	-8045	11228	8.3	-265	-1281	1335	20.7	26.51	84.54
17.500	25.00	-1902	-10422	14545	18.3	-419	-1283	1329	32.7	64.20	96.07
21.000	30.00	-3574	-12709	17736	28.1	-522	-1284	1323	40.7	101.51	105.53
24.500	35.00	-5240	-12709	17736	41.2	-329	-1286	1317	25.6	212.91	110.85
28.000	40.00	-5678	-12709	17736	44.7	76	-1288	1310	5.8	246.08	113.10
31.500	45.00	-4750	-12709	17736	37.4	344	-1290	1304	26.4	67.85	113.79
35.000	50.00	-3785	-12709	17736	29.8	210	-1292	1298	16.2	110.49	113.66
38.500	55.00	-2641	-12709	17736	20.8	495	-1293	1292	38.3	204.13	112.76
42.000	60.00	-889	-12709	17736	7.0	460	-1295	1285	35.8	88.09	110.96
45.500	65.00	494	-12709	17736	2.8	293	-1297	1279	22.9	40.12	107.50
49.000	70.00	1121	-12709	17736	6.3	67	-1299	1273	5.3	38.24	102.09
52.500	75.00	1185	-12709	17736	6.7	27	-1300	1267	2.2	93.99	94.08
56.000	80.00	1194	-10463	14601	8.2	-13	-1302	1260	1.0	69.10	74.51
59.500	85.00	1026	-8086	11284	9.1	-75	-1304	1254	5.8	60.41	70.12
63.000	90.00	702	-5709	7968	8.8	-109	-1306	1248	8.4	45.02	50.67
66.500	95.00	361	-3332	4651	7.8	-88	-1102	1050	7.9	40.05	28.13
70.000	100.00	147	-1571	2193	6.7	-43	-718	685	6.0	30.94	21.83
73.500	105.00	35	-733	1023	3.4	-20	-335	319	5.9	12.99	8.17

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DAMAGE STABILITY OF LOADING CONDITION L05

CRITICAL DAMAGE STABILITY CRITERIA FOR EACH DAMAGE CASE

RCR	DAM	STAGE	PHASE	REQ	ATTV	UNIT	STAT	MINGM
V.MINGZ.A469	D01	1	EQ	0.100	0.728	m	OK	0.300
V.RANGE.A469	D02	1	EQ	20.000	61.454	deg	OK	0.460
V.MINGZ.A469	D03	1	EQ	0.100	0.694	m	OK	0.453
V.RANGE.A469	D04	1	EQ	20.000	59.945	deg	OK	0.750
V.MINGZ.A469	D05	1	EQ	0.100	0.660	m	OK	0.632
V.RANGE.A469	D06	1	EQ	20.000	60.513	deg	OK	0.769
V.RANGE.A469	D07	1	EQ	20.000	63.791	deg	OK	0.649
V.RANGE.A469	D08	1	EQ	20.000	59.679	deg	OK	0.807
V.RANGE.A469	D09	1	EQ	20.000	61.845	deg	OK	0.790
V.RANGE.A469	D10	1	EQ	20.000	60.031	deg	OK	0.828
V.RANGE.A469	D11	1	EQ	20.000	61.470	deg	OK	0.883
V.RANGE.A469	D12	1	EQ	20.000	60.119	deg	OK	0.902
V.RANGE.A469	D13	1	EQ	20.000	62.261	deg	OK	0.888
V.RANGE.A469	D14	1	EQ	20.000	61.722	deg	OK	0.721
V.RANGE.A469	D15	1	EQ	20.000	63.886	deg	OK	0.707
V.RANGE.A469	D16	1	EQ	20.000	63.069	deg	OK	0.498
V.RANGE.A469	D17	1	EQ	20.000	65.236	deg	OK	0.490
V.MINGZ.A469	D18	1	EQ	0.100	0.712	m	OK	0.399
V.MINGZ.A469	D19	1	EQ	0.100	0.678	m	OK	0.482
V.MINGZ.A469	D20	1	EQ	0.100	0.670	m	OK	0.464
V.MINGZ.A469	D21	1	EQ	0.100	0.674	m	OK	0.464
TOTAL								0.902

DETERMINATION OF CRITICAL DAMAGE CASE

CRITICAL DAMAGE CASE : D12
STAGE OF CRITICAL DAMAGE CASE : 1
PHASE OF CRITICAL DAMAGE CASE : EQ

DAMAGE STABILITY CRITERIA - CRITICAL CASE

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM m
V.PROGR.A469	No progr. flooding	14.896	3.074	deg	OK	0.458
V.MAXHEEL.A469	Maximum heel	17.000	3.074	deg	OK	0.458
V.RANGE.A469	Range of pos. stab	20.000	60.119	deg	OK	0.902
V.MINGZ.A469	Min. righting lever	0.100	0.606	m	OK	0.847
TOTAL						0.902

DAMAGE CASE RESULTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	T m	TR m	HEEL degree	RESFLD m	OPEN	CROSSTM s
D12	1	EQ	SB	6.292	-0.181	3.1	1.81	D-362	-

DAMAGED COMPARTMENTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	NAME	DES	PERM	VOL m3	W t	XCG m	YCG m	ZCG m
D12	INTACT	EQ	-				0.0	0.000			
D12	1	EQ	SB	T023	WING TANK SB	0.95	52.7	53.994	19.40	7.82	3.50
D12	1	EQ	SB	T025	DB TANK SB	0.95	72.5	74.304	21.46	3.85	0.81
D12	1	EQ	SB	E2	Emrg. exit 1 SB	0.95	16.8	17.248	22.57	7.31	5.86

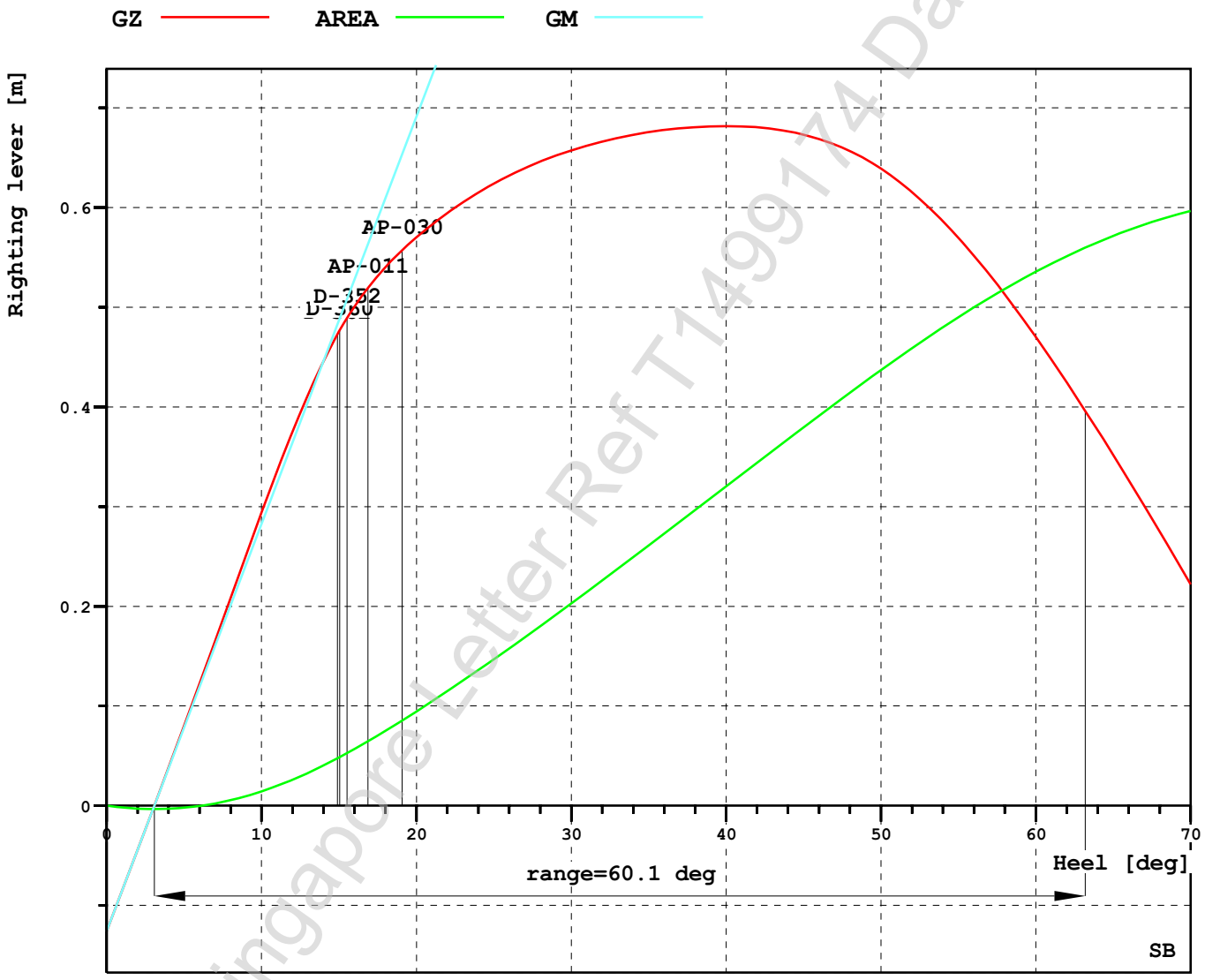
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CRITICAL OPENINGS - CRITICAL CASE

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
D-362	SB CARGORAIL #37/M-DECK	WEATHERTIGHT	25.900	8.550	8.600	14.9	1.812
D-360	SB CARGORAIL #57/M-DECK	WEATHERTIGHT	39.885	8.560	8.600	15.0	1.844
D-352	CHEMICAL/M-DECK	WEATHERTIGHT	53.192	8.375	8.600	15.5	1.884
AP-011	WING TANK SB	WEATHERTIGHT	47.211	8.754	8.953	16.9	2.202
AP-030	WING TANK SB	WEATHERTIGHT	5.250	7.801	9.168	19.1	2.373
D-354	PAINT/M-DECK	WEATHERTIGHT	53.192	6.600	8.600	19.2	1.979
AP-290	LO STORE TANK MAIN ENG	WEATHERTIGHT	52.985	7.410	8.927	19.5	2.262
V-353A	EXHAUST FAN SWITCHBOARD ROOM	WEATHERTIGHT	47.146	7.297	8.959	19.8	2.286

DAMAGE STABILITY CHECK PLOT - CRITICAL CASE



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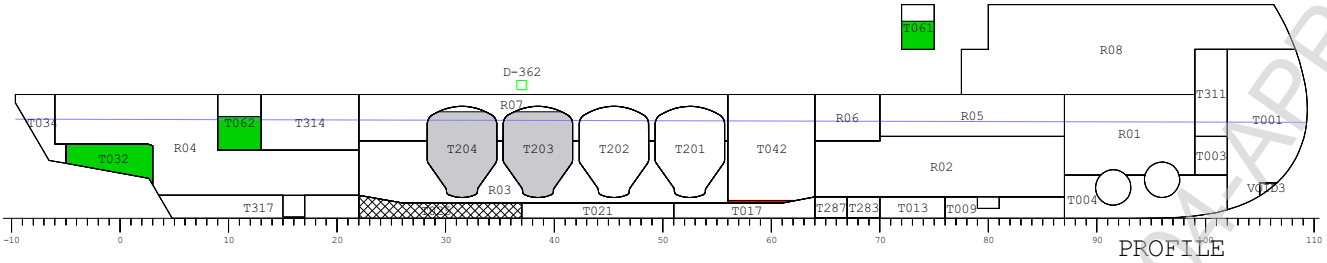
GZ-CURVE - CRITICAL CASE

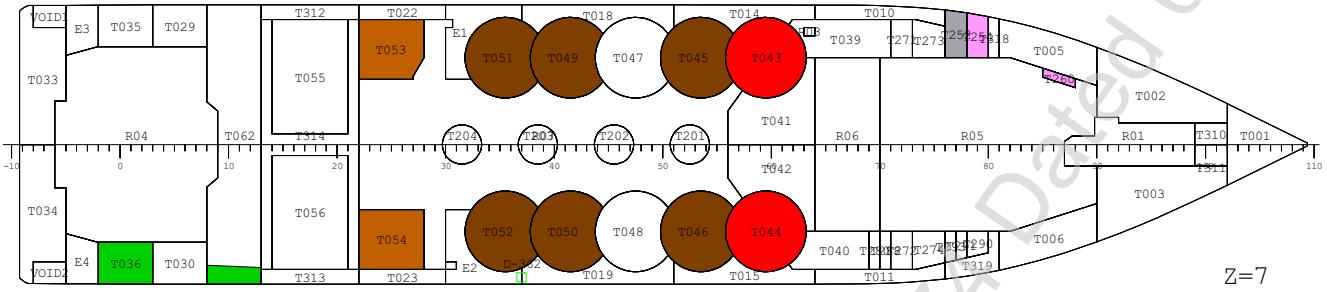
HEEL	DISP	T	TR	GZ	MS	AREA
degree	t	m	m	m	m	mrad
0.0	6188.3	6.299	-0.179	-0.126	-0.126	0.000
1.0	6188.3	6.299	-0.179	-0.085	-0.138	-0.002
3.0	6188.3	6.292	-0.181	-0.003	-0.161	-0.003
5.0	6188.3	6.277	-0.177	0.080	-0.183	-0.002
10.0	6188.3	6.198	-0.146	0.294	-0.231	0.014
15.0	6188.3	6.090	-0.157	0.476	-0.306	0.048
20.0	6188.3	5.992	-0.292	0.570	-0.463	0.094
30.0	6188.3	5.763	-0.889	0.657	-0.853	0.203
40.0	6188.3	5.436	-1.968	0.682	-1.260	0.320
50.0	6188.3	4.965	-3.284	0.639	-1.675	0.437
60.0	6188.3	4.369	-4.533	0.470	-2.146	0.536
70.0	6188.3	3.665	-5.593	0.222	-2.616	0.597

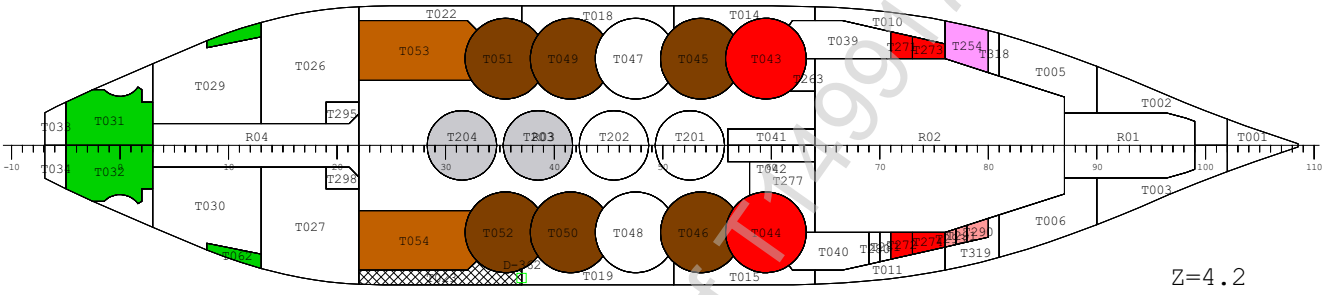
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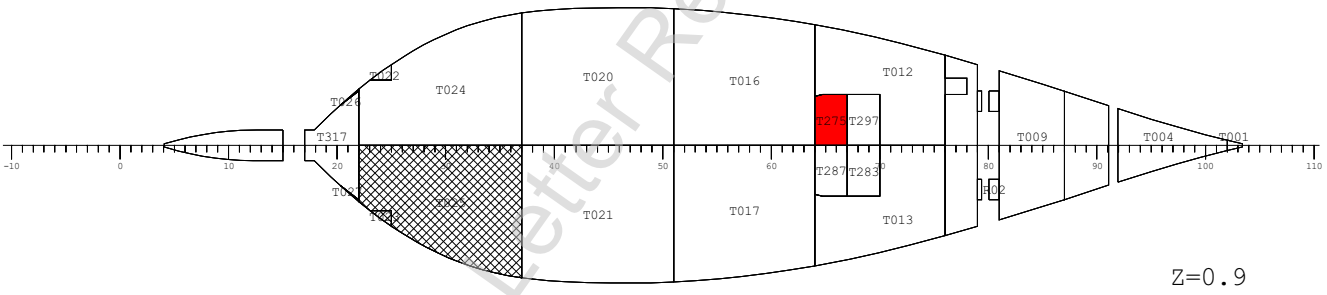
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DAMAGE ARRANGEMENT - CRITICAL CASE





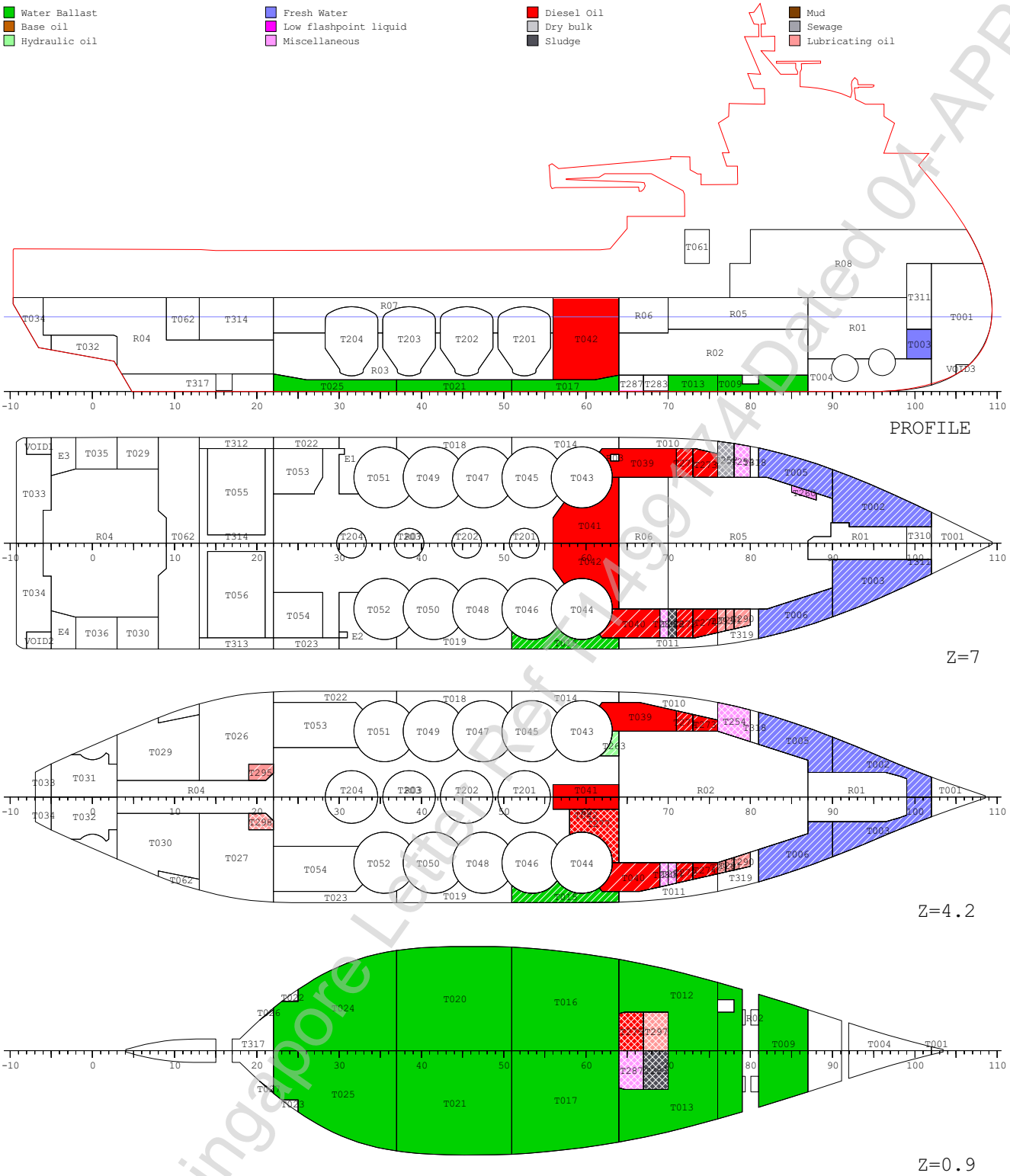




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12.6 L06 -> Max deckload - Departure



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LIST OF LOADS

NAME	DES	LOAD	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm	FRS	HM m
CONTENTS=Diesel Oil (RHO=0.86)										
T039	STORAGE TANK PS	FO	66.9	98.0	46.68	-6.76	4.91	6	REAL	7.88
T040	STORAGE TANK SB	FO	53.8	98.0	46.00	6.80	4.87	5	REAL	7.88
T041	STORAGE TANK PS	FO	68.5	98.0	42.16	-1.51	5.48	27	MAX	7.92
T042	STORAGE TANK SB	FO	68.5	98.0	42.16	1.51	5.48	27	MAX	7.92
T271	FO SETTLING TANK PS	FO	8.5	70.0	50.38	-6.53	4.93	1	REAL	6.72
T272	FO SETTLING TANK SB	FO	8.5	70.0	50.38	6.53	4.93	1	REAL	6.72
T273	FO SERVICE TANK PS	FO	10.6	70.0	52.09	-6.38	5.15	2	REAL	6.90
T274	FO SERVICE TANK SB	FO	10.6	70.0	52.09	6.38	5.15	2	REAL	6.90
T275	FO DRAIN TANK	FO	0.8	10.0	45.85	-1.64	0.07	5	REAL	0.14
T277	FO OVERFLOW TANK	FO	3.6	10.0	42.62	2.34	1.31	10	REAL	1.53
T279	EMERG. GEN. FO TANK	FO	1.9	98.0	61.60	-1.38	24.57	0	REAL	25.80
SUBTOTAL		FO	302.3		45.14	-0.27	5.25	86		
CONTENTS=Fresh Water (RHO=1)										
T002	WING TANK PS	FW	56.8	50.0	66.71	-2.66	4.87	45	MAX	6.35
T003	WING TANK SB	FW	57.5	50.0	66.69	2.65	4.88	46	MAX	6.35
T005	WING TANK PS	FW	49.1	50.0	60.05	-4.39	3.34	29	REAL	4.79
T006	WING TANK SB	FW	49.1	50.0	60.05	4.39	3.34	29	REAL	4.79
SUBTOTAL		FW	212.5		63.63	0.01	4.17	149		
CONTENTS=Hydraulic oil (RHO=0.924)										
T263	HYDRAULIC OIL STORE TANK	HO	3.3	70.0	44.30	-4.34	2.32	1	REAL	3.33
CONTENTS=Lubricating oil (RHO=0.924)										
T290	LO STORE TANK MAIN ENG.	LO	5.9	70.0	55.30	5.76	4.90	0	REAL	6.82
T291	LO STORE TANK MAIN ENG.	LO	3.1	70.0	54.25	6.04	5.11	0	REAL	6.92
T293	LO STORE TANK THRUSTER	LO	3.2	70.0	53.55	6.21	5.24	0	REAL	6.97
T295	LO STORE TANK AZIMUTH THR.	LO	4.0	70.0	14.35	-2.10	2.95	0	REAL	3.71
T297	LO DRAIN TANK	LO	0.9	10.0	47.95	-1.64	0.07	6	REAL	0.14
T298	DIRTY LO TANK	LO	0.6	10.0	14.35	2.10	2.31	0	REAL	2.41
SUBTOTAL		LO	17.6		43.89	3.64	4.24	8		
CONTENTS=Mass load (RHO=1)										
M1	CREW	MASS	2.6	0.0	57.40	0.00	15.50	0		0.00
M2	PROVISION	MASS	7.4	0.0	56.35	-6.50	12.40	0		0.00
M3	STORES	MASS	7.4	0.0	72.10	0.00	12.40	0		0.00
M4	DECK LOAD 10T/M2 AFT OF #35	MASS	1330.0	0.0	9.10	0.00	~~9.00~~ 8.90	0	GIVEN	0.00
M5	DECK LOAD 5T/M2 FWD OF #35	MASS	1070.0	0.0	37.26	0.00	~~9.00~~ 8.90	0		0.00
SUBTOTAL		MASS	2417.5		21.95	-0.02	9.03	AMENDED BY ABS		
CONTENTS=Miscellaneous (RHO=1)										
T280	BILGE WATER HOLDING TANK	MIS	0.9	10.0	48.64	6.17	2.03	0	REAL	2.57
T281	BILGE WATER SETTLING TANK	MIS	0.4	10.0	49.34	6.04	1.76	0	REAL	2.08
T287	BILGE WATER DRAIN TANK	MIS	0.9	10.0	45.85	1.64	0.07	6	REAL	0.14
T254	GREY WATER TANK	MIS	3.2	10.0	54.62	-5.79	1.92	1	REAL	2.34
T260	WATER MIST TANK	MIS	2.3	98.0	60.55	-4.30	6.28	0	REAL	7.27
SUBTOTAL		MIS	7.8		54.40	-2.49	3.01	8		
CONTENTS=Sewage (RHO=1)										
T252	BLACK WATER TANK	SEW	1.1	10.0	53.89	-6.93	5.45	2	REAL	5.59
CONTENTS=Sludge (RHO=1)										
T283	SLUDGE DRAIN TANK	SLU	0.9	10.0	47.95	1.64	0.07	6	REAL	0.14
T288	SLUDGE SETTLING TANK	SLU	0.4	10.0	49.35	6.75	5.44	1	REAL	5.59
SUBTOTAL		SLU	1.4		48.40	3.29	1.80	7		
CONTENTS=Water Ballast (RHO=1.025)										
T001	FOREPEAK TANK	WB	0.0	0.0	73.11	0.00	7.21	34	MAX	0.00
T009	DB TANK CE	WB	77.3	100.0	56.90	0.09	0.73	0	(REAL)	1.40
T012	DB TANK PS	WB	56.2	100.0	49.41	-3.85	0.76	0	(REAL)	1.40
T013	DB TANK SB	WB	56.2	100.0	49.41	3.85	0.76	0	(REAL)	1.40
T015	WING TANK SB	WB	15.4	25.0	40.17	8.33	2.27	1	REAL	3.09
T016	DB TANK PS	WB	85.6	100.0	40.20	-4.33	0.62	0	REAL	1.50
T017	DB TANK SB	WB	85.6	100.0	40.20	4.33	0.62	0	REAL	1.50
T020	DB TANK PS	WB	96.7	100.0	30.84	-4.63	0.61	0	(REAL)	1.92
T021	DB TANK SB	WB	96.7	100.0	30.84	4.63	0.61	0	(REAL)	1.92
T024	DB TANK PS	WB	75.9	100.0	21.46	-3.85	0.81	0	(REAL)	2.75

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T025	DB TANK SB	WB	75.9	100.0	21.46	3.85	0.81	0 (REAL)	2.75
SUBTOTAL		WB	721.5		36.97	0.19	0.73	35	
TOTAL			3684.9		29.41	0.01	6.77	296	

LOAD SUMMARY TABLE

ITEM	MASS t	LCG m	TCG m	VCG m	FRSM tm
Diesel Oil	302.3	45.14	-0.27	5.25	85.89
Fresh Water	212.5	63.63	0.01	4.17	149.15
Hydraulic oil	3.3	44.30	-4.34	2.32	0.61
Lubricating oil	17.6	43.89	3.64	4.24	7.62
Mass load	2417.5	21.95	-0.02	9.03	0.00
Miscellaneous	7.8	54.40	-2.49	3.01	7.72
Sewage	1.1	53.89	-6.93	5.45	2.31
Sludge	1.4	48.40	3.29	1.80	6.99
Water Ballast	721.5	36.97	0.19	0.73	35.31
DEADWEIGHT	3684.9	29.41	0.01	6.77	295.60
Lightweight	2704.9	39.15	-0.01	7.08	
Deadweight	3684.9	29.41	0.01	6.77	
Total weight	6389.8	33.54	0.00	6.90	

F L O A T I N G P O S I T I O N

Draught moulded	6.360	m	KM	8.69 m
Trim	0.005	m	KG	6.90 m
Heel, SB=+	0.0	deg		
TA	6.358	m	GM0	1.79 m
TF	6.363	m	GMCORR	-0.05 m
Trimming moment	38	tonm	GM	1.75 m

STABILITY COMPONENTS

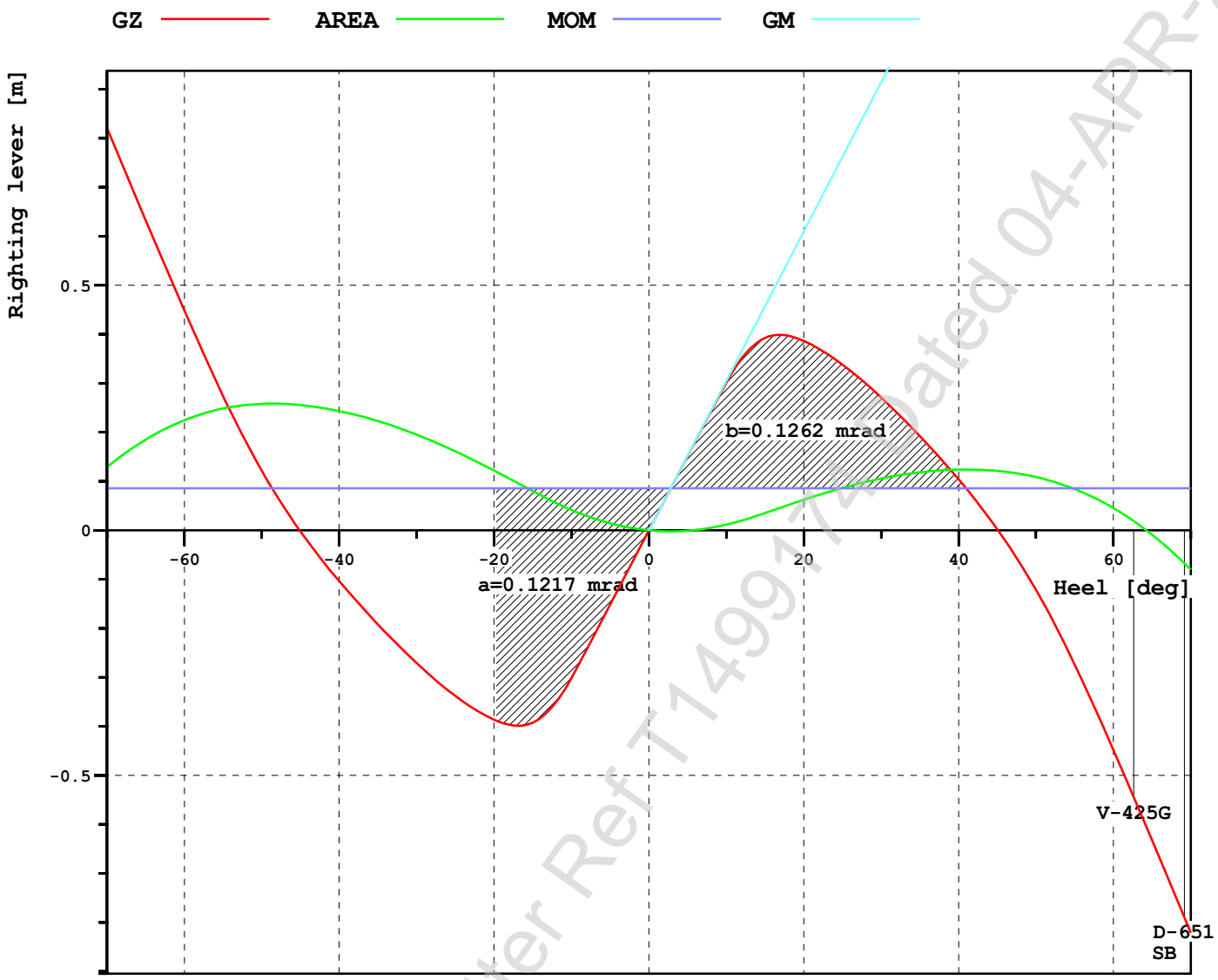
LOADING CONDITION L06, Max deckload - Departure

HEEL degree	MS m	HPHI m	EPHI mrad	FSMOM tm	DGZ m	GMSIN m	KN m
0.0	0.000	0.00	0.000	0.0	0.000	0.000	-0.001
1.0	0.000	0.03	0.000	5.0	0.001	0.031	0.151
3.0	0.000	0.09	0.002	15.0	0.002	0.094	0.453
5.0	-0.001	0.15	0.007	25.0	0.004	0.156	0.756
10.0	-0.002	0.30	0.026	48.9	0.008	0.311	1.506
15.0	-0.060	0.39	0.058	71.4	0.011	0.464	2.188
20.0	-0.212	0.39	0.092	92.9	0.015	0.613	2.760
30.0	-0.604	0.27	0.150	134.2	0.021	0.896	3.741
40.0	-1.020	0.10	0.184	177.0	0.028	1.152	4.566
50.0	-1.458	-0.12	0.183	226.5	0.035	1.372	5.199
60.0	-1.957	-0.45	0.135	276.6	0.043	1.552	5.570
70.0	-2.454	-0.82	0.024	324.6	0.051	1.684	5.713

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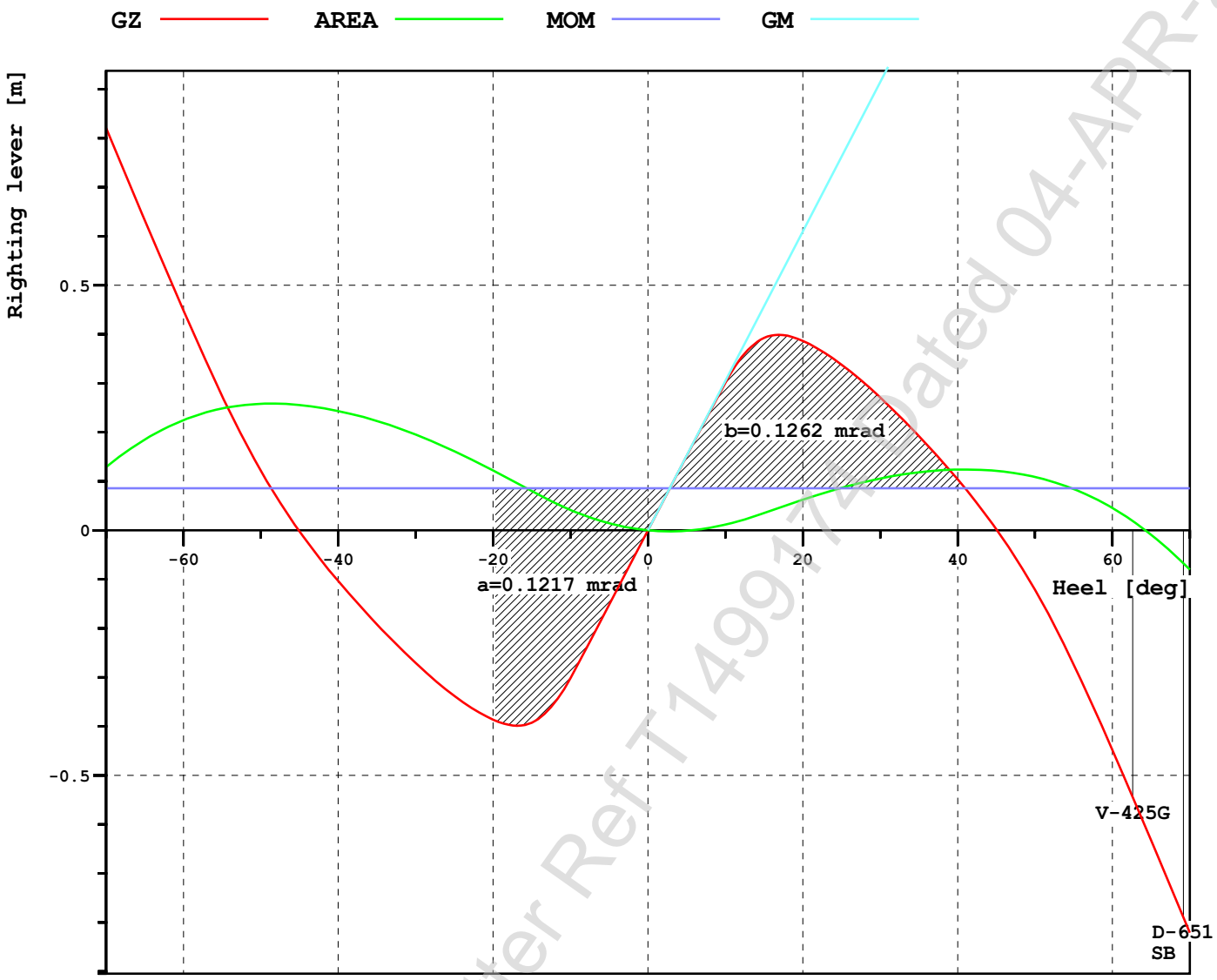
STABILITY CHECK PLOT



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IMOWEATHER CHECK PLOT



INTACT STABILITY CRITERIA

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM
V.AREA15-30	Area depending on the position of the Max. .	0.068	0.071	mrad	OK	1.715
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.033	mrad	OK	1.713
V.GZ0.2	Min. GZ > 0.2	0.200	0.271	m	OK	1.604
V.POSMAX15	Max. GZ at an angle > 15 deg.	15.000	16.909	deg	OK	1.330
V.GM0.15	GM > 0.15 m	0.150	1.745	m	OK	0.150
IMOWEATHER	IMO weather criterion	1.000	1.037		OK	1.723
2008IS-A2.3.1.2	Heeling angle due to steady wind <16 or <=8.	8.309	1.879	deg	OK	0.400
ABSFIFI1	Angle of heel < 1/2 freeboard immersion	5.193	0.437	deg	OK	0.152
ABSFIFI2	Area of residual stability >0.09 mrad	0.090	0.174	mrad	OK	1.337
TOTAL						1.723

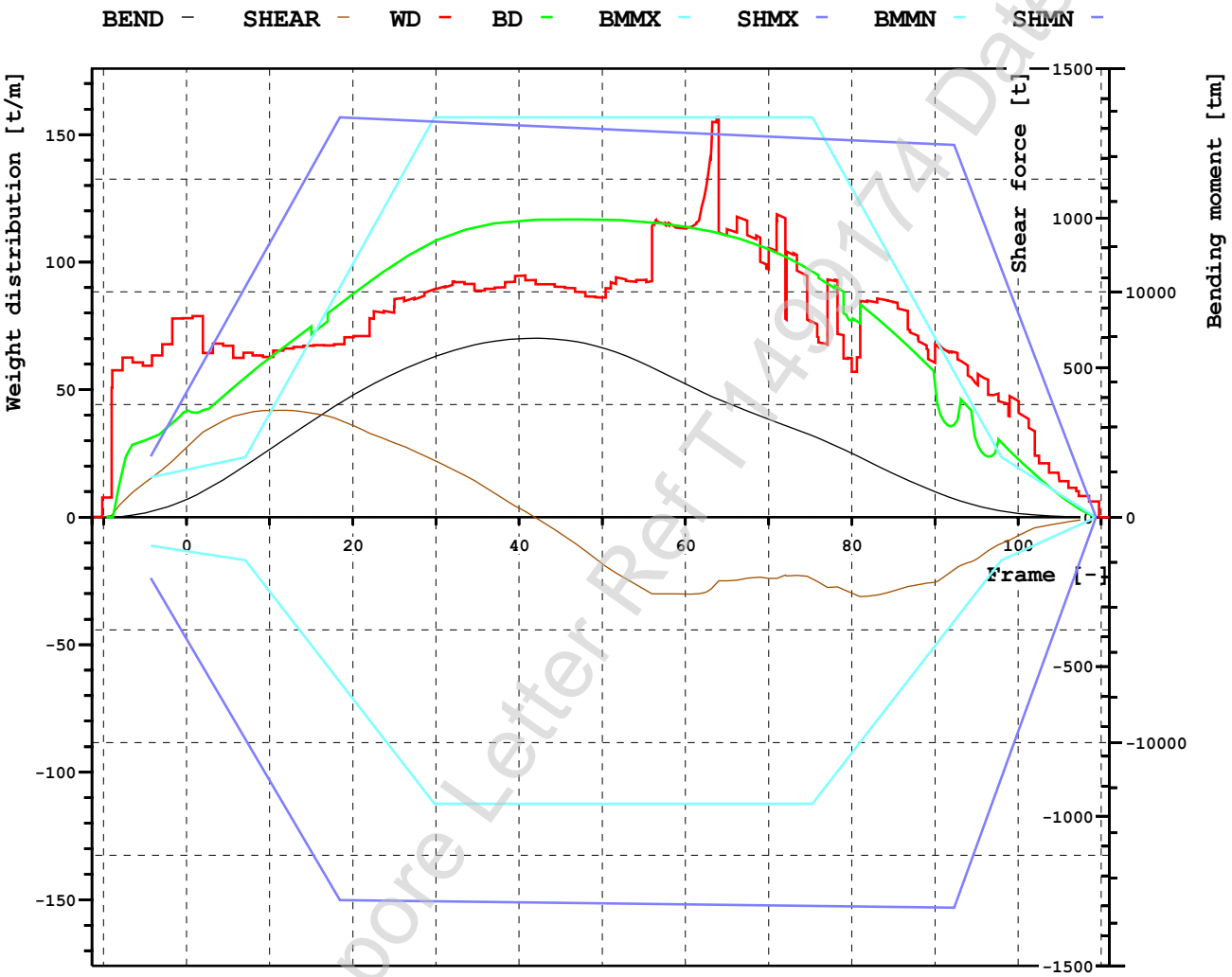
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MOST CRITICAL OPENINGS

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
V-421G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	55.890	2.093	14.288	-	7.926
V-424G	SUPPLY AIR ENGINE ROOM PS	UNPROTECTED	54.243	-3.413	14.288	-	7.927
V-425G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	54.243	3.767	14.288	62.6	7.927
D-651	ACCOMMODATION/C-DECK	UNPROTECTED	56.000	3.771	16.906	69.1	10.544
V-421E	EXHAUST AIR ENGINE ROOM PS	UNPROTECTED	55.993	-2.538	17.088	-	10.726
V-423E	EXHAUST AIR ENGINE ROOM SB	UNPROTECTED	55.993	3.423	17.088	-	10.726

STRENGTH DIAGRAM



STRENGTH SUMMARY

LOADING CONDITION L06, Max deckload - Departure

			X	Frame
Shear force (min)	-265.7 t	(20.4%)	56.700 m	81.00
Shear force (max)	357.7 t	(35.9%)	8.138 m	11.63
Sagging moment	-0.4 tm	(0.0%)	-7.104 m	-10.15
Hogging moment	7939.6 tm	(44.8%)	29.375 m	41.96

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STRENGTH DISTRIBUTION

LOADING CONDITION L06, Max deckload - Departure

X	FR	BEND	BMMN	BMMX	BMREL	SHEAR	SHMN	SHMX	SHREL	WD	BD
m	#	tm	tm	tm	%	t	t	t	%	t/m	t/m
-3.500	-5.00	187	0	0	0.0	115	0	0	0.0	59.41	30.12
0.000	0.00	784	-1510	2108	37.2	233	-407	418	55.8	78.04	41.59
3.500	5.00	1792	-1790	2498	71.7	329	-644	667	49.4	67.14	49.02
7.000	10.00	3007	-3292	4594	65.7	357	-881	916	39.0	62.79	62.43
10.500	15.00	4248	-5668	7911	53.7	348	-1117	1166	29.9	67.22	74.78
14.000	20.00	5406	-8045	11228	48.2	307	-1281	1335	23.0	70.51	87.41
17.500	25.00	6373	-10422	14545	43.8	248	-1283	1329	18.7	85.46	98.95
21.000	30.00	7137	-12709	17736	40.2	187	-1284	1323	14.1	89.40	108.42
24.500	35.00	7670	-12709	17736	43.2	116	-1286	1317	8.8	88.86	113.75
28.000	40.00	7919	-12709	17736	44.6	30	-1288	1310	2.3	94.64	116.01
31.500	45.00	7877	-12709	17736	44.4	-55	-1290	1304	4.3	90.30	116.69
35.000	50.00	7511	-12709	17736	42.3	-156	-1292	1298	12.0	86.12	116.57
38.500	55.00	6812	-12709	17736	38.4	-241	-1293	1292	18.6	93.00	115.67
42.000	60.00	5917	-12709	17736	33.4	-257	-1295	1285	19.9	113.19	113.88
45.500	65.00	5072	-12709	17736	28.6	-213	-1297	1279	16.4	112.87	110.41
49.000	70.00	4349	-12709	17736	24.5	-205	-1299	1273	15.8	105.56	104.98
52.500	75.00	3651	-12709	17736	20.6	-203	-1300	1267	15.6	77.12	96.87
56.000	80.00	2841	-10463	14601	19.5	-253	-1302	1260	19.4	62.36	77.11
59.500	85.00	1932	-8086	11284	17.1	-247	-1304	1254	18.9	84.43	72.43
63.000	90.00	1127	-5709	7968	14.1	-218	-1306	1248	16.7	67.75	52.60
66.500	95.00	510	-3332	4651	11.0	-140	-1102	1050	12.7	52.22	29.60
70.000	100.00	166	-1571	2193	7.5	-62	-718	685	8.6	45.57	22.80
73.500	105.00	36	-733	1023	3.5	-21	-335	319	6.2	14.17	8.64

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DAMAGE STABILITY OF LOADING CONDITION L06

CRITICAL DAMAGE STABILITY CRITERIA FOR EACH DAMAGE CASE

RCR	DAM	STAGE	PHASE	REQ	ATTV	UNIT	STAT	MINGM
V.RANGE.A469	D01	1	EQ	20.000	46.065	deg	OK	0.595
V.RANGE.A469	D02	1	EQ	20.000	45.149	deg	OK	0.652
V.RANGE.A469	D03	1	EQ	20.000	44.215	deg	OK	0.627
V.RANGE.A469	D04	1	EQ	20.000	41.842	deg	OK	1.004
V.RANGE.A469	D05	1	EQ	20.000	41.601	deg	OK	1.046
V.RANGE.A469	D06	1	EQ	20.000	38.947	deg	OK	1.165
V.RANGE.A469	D07	1	EQ	20.000	38.690	deg	OK	1.198
V.RANGE.A469	D08	1	EQ	20.000	41.832	deg	OK	0.939
V.RANGE.A469	D09	1	EQ	20.000	41.168	deg	OK	1.017
V.RANGE.A469	D10	1	EQ	20.000	38.931	deg	OK	1.107
V.RANGE.A469	D11	1	EQ	20.000	39.589	deg	OK	1.089
V.RANGE.A469	D12	1	EQ	20.000	37.051	deg	OK	1.171
V.RANGE.A469	D13	1	EQ	20.000	37.684	deg	OK	1.156
V.RANGE.A469	D14	1	EQ	20.000	37.465	deg	OK	1.127
V.RANGE.A469	D15	1	EQ	20.000	38.152	deg	OK	1.109
V.RANGE.A469	D16	1	EQ	20.000	31.288	deg	OK	1.268
V.RANGE.A469	D17	1	EQ	20.000	31.990	deg	OK	1.248
V.RANGE.A469	D18	1	EQ	20.000	41.845	deg	OK	0.838
V.RANGE.A469	D19	1	EQ	20.000	42.428	deg	OK	0.824
V.RANGE.A469	D20	1	EQ	20.000	40.414	deg	OK	0.858
V.RANGE.A469	D21	1	EQ	20.000	41.033	deg	OK	0.844
TOTAL								1.268

DETERMINATION OF CRITICAL DAMAGE CASE

CRITICAL DAMAGE CASE : D16
STAGE OF CRITICAL DAMAGE CASE : 1
PHASE OF CRITICAL DAMAGE CASE : EQ

DAMAGE STABILITY CRITERIA - CRITICAL CASE

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM m
V.PROGR.A469	No progr. flooding	13.354	1.680	deg	OK	0.507
V.MAXHEEL.A469	Maximum heel	17.000	1.680	deg	OK	0.507
V.RANGE.A469	Range of pos. stab	20.000	31.288	deg	OK	1.268
V.MINGZ.A469	Min. righting lever	0.100	0.255	m	OK	1.038
TOTAL						1.268

DAMAGE CASE RESULTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	T m	TR m	HEEL degree	RESFLD m	OPEN	CROSSTM s
D16	1	EQ	SB	6.454	-0.622	1.7	1.81	D-362	-

DAMAGED COMPARTMENTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	NAME	DES	PERM	VOL m3	W t	XCG m	YCG m	ZCG m
D16	INTACT	EQ	-				0.0	0.000			
D16	1	EQ	SB	T062	ROLL RED. TANK	0.95	126.8	129.975	7.45	0.37	5.55
D16	1	EQ	SB	T030	WING TANK SB	0.95	65.1	66.686	5.47	4.53	4.26

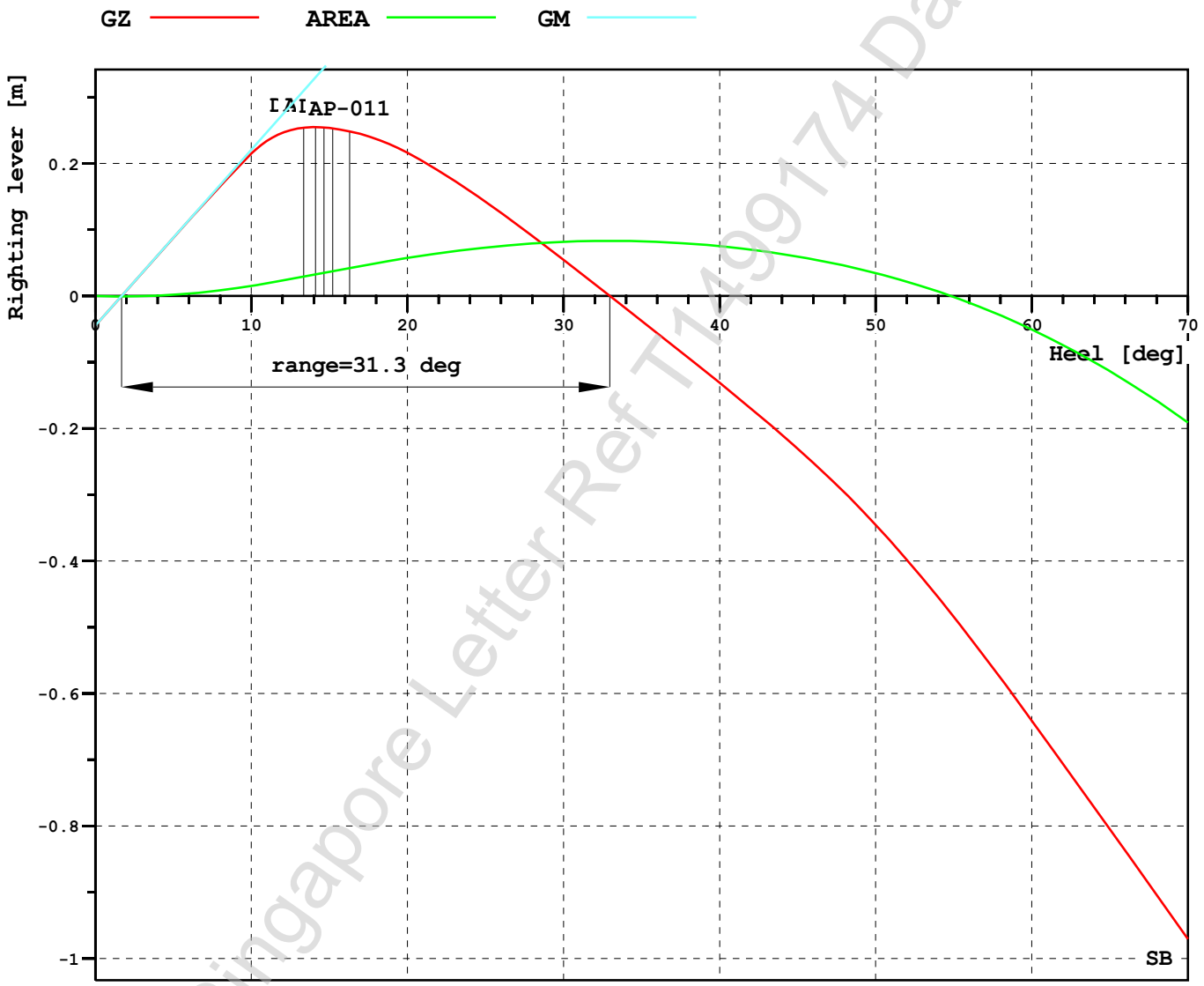
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CRITICAL OPENINGS - CRITICAL CASE

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
D-362	SB CARGORAIL #37/M-DECK	WEATHERTIGHT	25.900	8.550	8.600	13.4	1.807
D-360	SB CARGORAIL #57/M-DECK	WEATHERTIGHT	39.885	8.560	8.600	14.1	1.916
AP-023	WING TANK SB	WEATHERTIGHT	19.200	8.702	8.936	14.7	2.086
D-352	CHEMICAL/M-DECK	WEATHERTIGHT	53.192	8.375	8.600	15.2	2.025
AP-011	WING TANK SB	WEATHERTIGHT	47.211	8.754	8.953	16.3	2.320
AP-034	WING TANK SB	WEATHERTIGHT	-5.873	7.880	9.372	16.7	2.350
AP-036	WING TANK SB	WEATHERTIGHT	1.163	7.810	9.372	17.2	2.407
AP-027	WING TANK SB	WEATHERTIGHT	14.986	7.800	9.372	18.1	2.515

DAMAGE STABILITY CHECK PLOT - CRITICAL CASE



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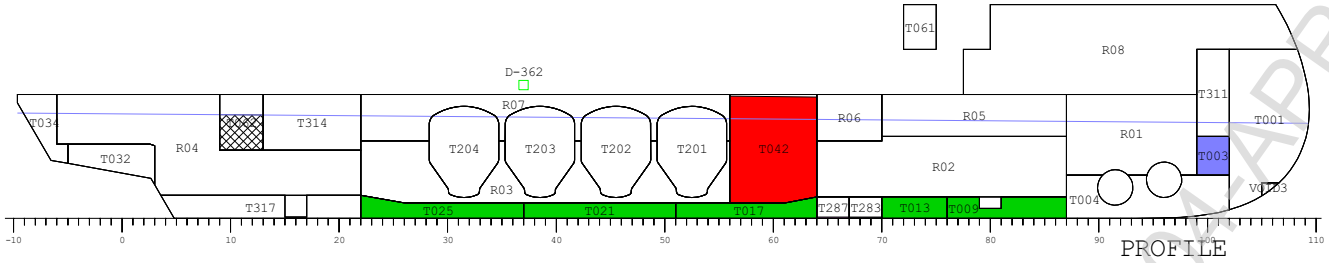
GZ-CURVE - CRITICAL CASE

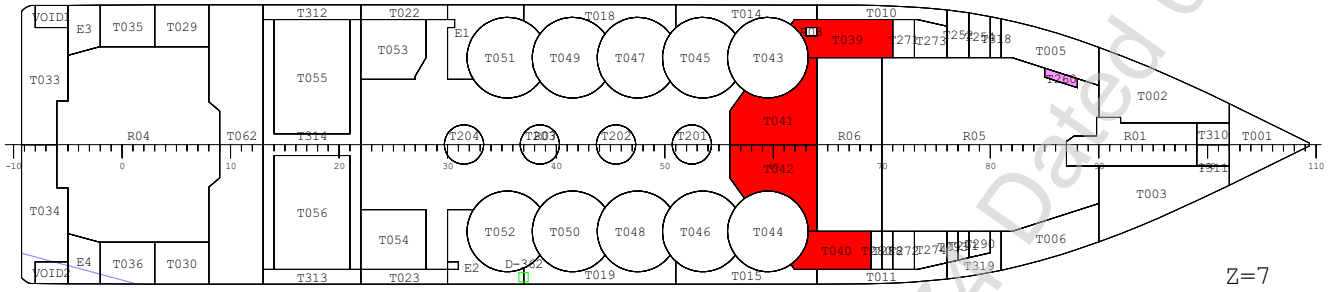
HEEL	DISP	T	TR	GZ	MS	AREA
degree	t	m	m	m	m	mrad
0.0	6389.8	6.456	-0.614	-0.045	-0.045	0.000
1.0	6389.8	6.455	-0.619	-0.018	-0.049	-0.001
3.0	6389.8	6.448	-0.627	0.035	-0.059	0.000
5.0	6389.8	6.431	-0.630	0.088	-0.068	0.002
10.0	6389.8	6.351	-0.641	0.215	-0.096	0.015
15.0	6389.8	6.269	-0.787	0.254	-0.210	0.037
20.0	6389.8	6.199	-1.049	0.216	-0.397	0.057
30.0	6389.8	6.030	-1.908	0.054	-0.841	0.082
40.0	6389.8	5.761	-3.258	-0.131	-1.283	0.075
50.0	6389.8	5.358	-4.838	-0.345	-1.718	0.035
60.0	6389.8	4.826	-6.308	-0.641	-2.192	-0.050
70.0	6389.8	4.173	-7.518	-0.970	-2.654	-0.191

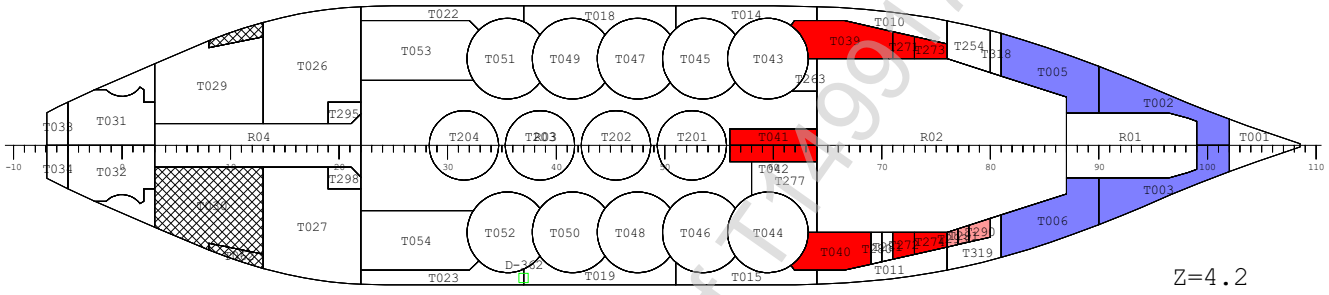
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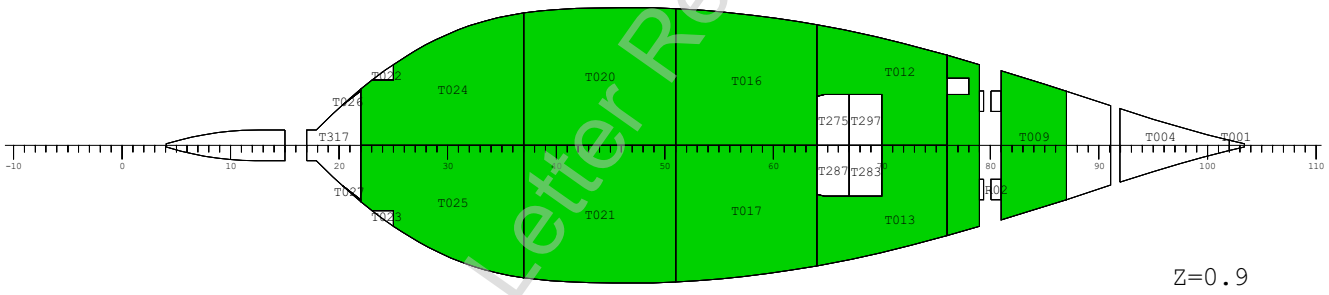
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DAMAGE ARRANGEMENT - CRITICAL CASE





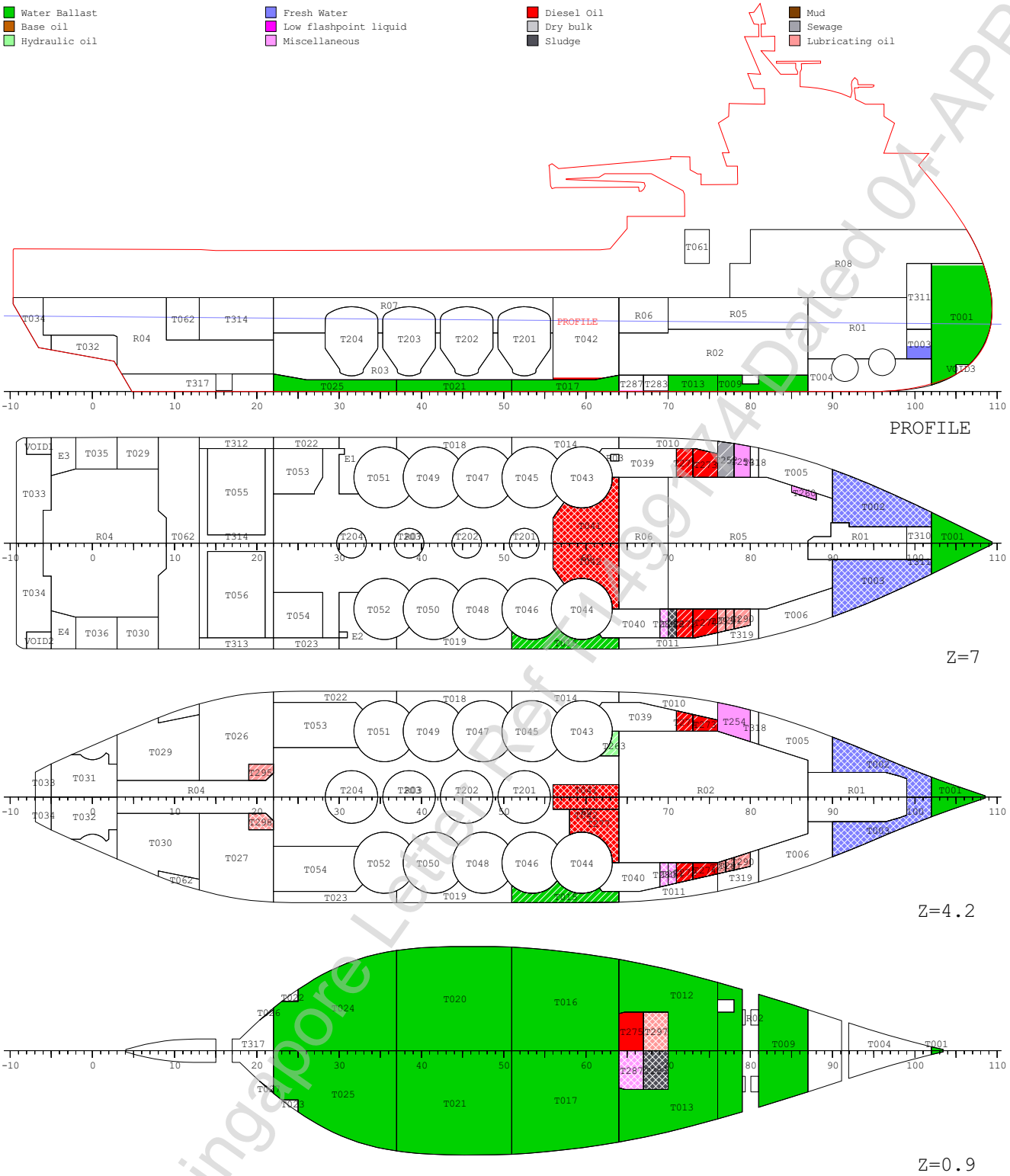




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12.7 L07 -> Max deckload - Arrival



See ABS Singapore Letter Ref 11499X4 Dated 04-APR-2016

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LIST OF LOADS

NAME	DES	LOAD	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm	FRS	HM m
CONTENTS=Diesel Oil (RHO=0.86)										
T041	STORAGE TANK PS	FO	0.7	1.0	41.24	-0.52	1.10	27	MAX	1.20
T042	STORAGE TANK SB	FO	0.7	1.0	41.24	0.52	1.10	27	MAX	1.20
T271	FO SETTLING TANK PS	FO	8.5	70.0	50.38	-6.53	4.93	1	REAL	6.72
T272	FO SETTLING TANK SB	FO	8.5	70.0	50.38	6.53	4.93	1	REAL	6.72
T273	FO SERVICE TANK PS	FO	10.6	70.0	52.09	-6.38	5.15	2	REAL	6.90
T274	FO SERVICE TANK SB	FO	10.6	70.0	52.09	6.38	5.15	2	REAL	6.90
T275	FO DRAIN TANK	FO	7.9	98.0	45.85	-1.64	0.69	5	REAL	1.37
T277	FO OVERFLOW TANK	FO	3.6	10.0	42.62	2.34	1.31	10	REAL	1.53
T279	EMERG. GEN. FO TANK	FO	1.9	98.0	61.60	-1.38	24.57	0	REAL	25.80
SUBTOTAL		FO	53.0		50.02	-0.13	4.73	75		
CONTENTS=Fresh Water (RHO=1)										
T002	WING TANK PS	FW	10.6	9.3	67.25	-2.14	3.41	45	MAX	3.84
T003	WING TANK SB	FW	10.6	9.2	67.25	2.14	3.41	46	MAX	3.84
SUBTOTAL		FW	21.2		67.25	0.00	3.41	91		
CONTENTS=Hydraulic oil (RHO=0.924)										
T263	HYDRAULIC OIL STORE TANK	HO	1.9	40.0	44.29	-4.33	1.89	1	REAL	2.47
CONTENTS=Lubricating oil (RHO=0.924)										
T290	LO STORE TANK MAIN ENG.	LO	3.4	40.0	55.32	5.47	4.05	0	REAL	5.16
T291	LO STORE TANK MAIN ENG.	LO	1.8	40.0	54.25	5.81	4.26	0	REAL	5.48
T293	LO STORE TANK THRUSTER	LO	1.8	40.0	53.55	6.03	4.41	0	REAL	5.66
T295	LO STORE TANK AZIMUTH THR.	LO	2.3	40.0	14.35	-2.10	2.63	0	REAL	3.06
T297	LO DRAIN TANK	LO	0.9	10.0	47.95	-1.64	0.07	6	REAL	0.14
T298	DIRTY LO TANK	LO	0.6	10.0	14.35	2.10	2.31	0	REAL	2.41
SUBTOTAL		LO	10.7		43.36	3.26	3.43	7		
CONTENTS=Mass load (RHO=1)										
M1	CREW	MASS	2.6	0.0	57.40	0.00	15.50	0		0.00
M2	PROVISION	MASS	0.8	0.0	56.35	-6.50	12.40	0		0.00
M3	STORES	MASS	0.8	0.0	72.10	0.00	12.40	0		0.00
M4	DECK LOAD 10T/M2 AFT OF #35	MASS	1330.0	0.0	9.10	0.00	9.00	0	GIVEN	0.00
M5	DECK LOAD 5T/M2 FWD OF #35	MASS	1070.0	0.0	37.26	0.00	9.00	0		0.00
SUBTOTAL		MASS	2404.2		21.72	0.00	9.01	0		
CONTENTS=Miscellaneous (RHO=1)										
T280	BILGE WATER HOLDING TANK	MIS	0.9	10.0	48.64	6.17	2.03	0	REAL	2.57
T281	BILGE WATER SETTLING TANK	MIS	0.4	10.0	49.34	6.04	1.76	0	REAL	2.08
T287	BILGE WATER DRAIN TANK	MIS	0.9	10.0	45.85	1.64	0.07	6	REAL	0.14
T254	GREY WATER TANK	MIS	31.6	98.0	54.82	-6.47	4.52	3	REAL	7.84
T260	WATER MIST TANK	MIS	2.3	98.0	60.55	-4.30	6.28	0	REAL	7.27
SUBTOTAL		MIS	36.2		54.74	-5.68	4.43	9		
CONTENTS=Sewage (RHO=1)										
T252	BLACK WATER TANK	SEW	10.4	97.9	53.89	-7.06	6.66	4	REAL	7.95
CONTENTS=Sludge (RHO=1)										
T283	SLUDGE DRAIN TANK	SLU	0.9	10.0	47.95	1.64	0.07	6	REAL	0.14
T288	SLUDGE SETTLING TANK	SLU	0.4	10.0	49.35	6.75	5.44	1	REAL	5.59
SUBTOTAL		SLU	1.4		48.40	3.29	1.80	7		
CONTENTS=Water Ballast (RHO=1.025)										
T001	FOREPEAK TANK	WB	102.9	98.0	73.11	0.00	7.14	34	MAX	10.74
T009	DB TANK CE	WB	77.3	100.0	56.90	0.09	0.73	0	(REAL)	1.40
T012	DB TANK PS	WB	56.2	100.0	49.41	-3.85	0.76	0	(REAL)	1.40
T013	DB TANK SB	WB	56.2	100.0	49.41	3.85	0.76	0	(REAL)	1.40
T015	WING TANK SB	WB	33.8	55.0	40.14	8.39	3.26	1	REAL	5.06
T016	DB TANK PS	WB	85.6	100.0	40.20	-4.33	0.62	0	REAL	1.50
T017	DB TANK SB	WB	85.6	100.0	40.20	4.33	0.62	0	REAL	1.50
T020	DB TANK PS	WB	96.7	100.0	30.84	-4.63	0.61	0	(REAL)	1.92
T021	DB TANK SB	WB	96.7	100.0	30.84	4.63	0.61	0	(REAL)	1.92
T024	DB TANK PS	WB	75.9	100.0	21.46	-3.85	0.81	0	(REAL)	2.75
T025	DB TANK SB	WB	75.9	100.0	21.46	3.85	0.81	0	(REAL)	2.75
SUBTOTAL		WB	842.8		41.45	0.35	1.58	35		

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TOTAL	3381.8	27.91	0.01	6.98	229

LOAD SUMMARY TABLE

ITEM	MASS t	LCG m	TCG m	VCG m	FRSM tm
Diesel Oil	53.0	50.02	-0.13	4.73	74.98
Fresh Water	21.2	67.25	0.00	3.41	91.23
Hydraulic oil	1.9	44.29	-4.33	1.89	0.61
Lubricating oil	10.7	43.36	3.26	3.43	7.30
Mass load	2404.2	21.72	0.00	9.01	0.00
Miscellaneous	36.2	54.74	-5.68	4.43	9.39
Sewage	10.4	53.89	-7.06	6.66	3.55
Sludge	1.4	48.40	3.29	1.80	6.99
Water Ballast	842.8	41.45	0.35	1.58	35.38
DEADWEIGHT	3381.8	27.91	0.01	6.98	229.44
Lightweight	2704.9	39.15	-0.01	7.08	
Deadweight	3381.8	27.91	0.01	6.98	
Total weight	6086.7	32.91	0.00	7.02	

F L O A T I N G P O S I T I O N

Draught moulded	6.077	m	KM	8.78 m
Trim	-0.659	m	KG	7.02 m
Heel, SB=+	0.0	deg		
TA	6.407	m	GM0	1.76 m
TF	5.748	m	GMCORR	-0.04 m
Trimming moment	-4743	tonm	GM	1.72 m

STABILITY COMPONENTS

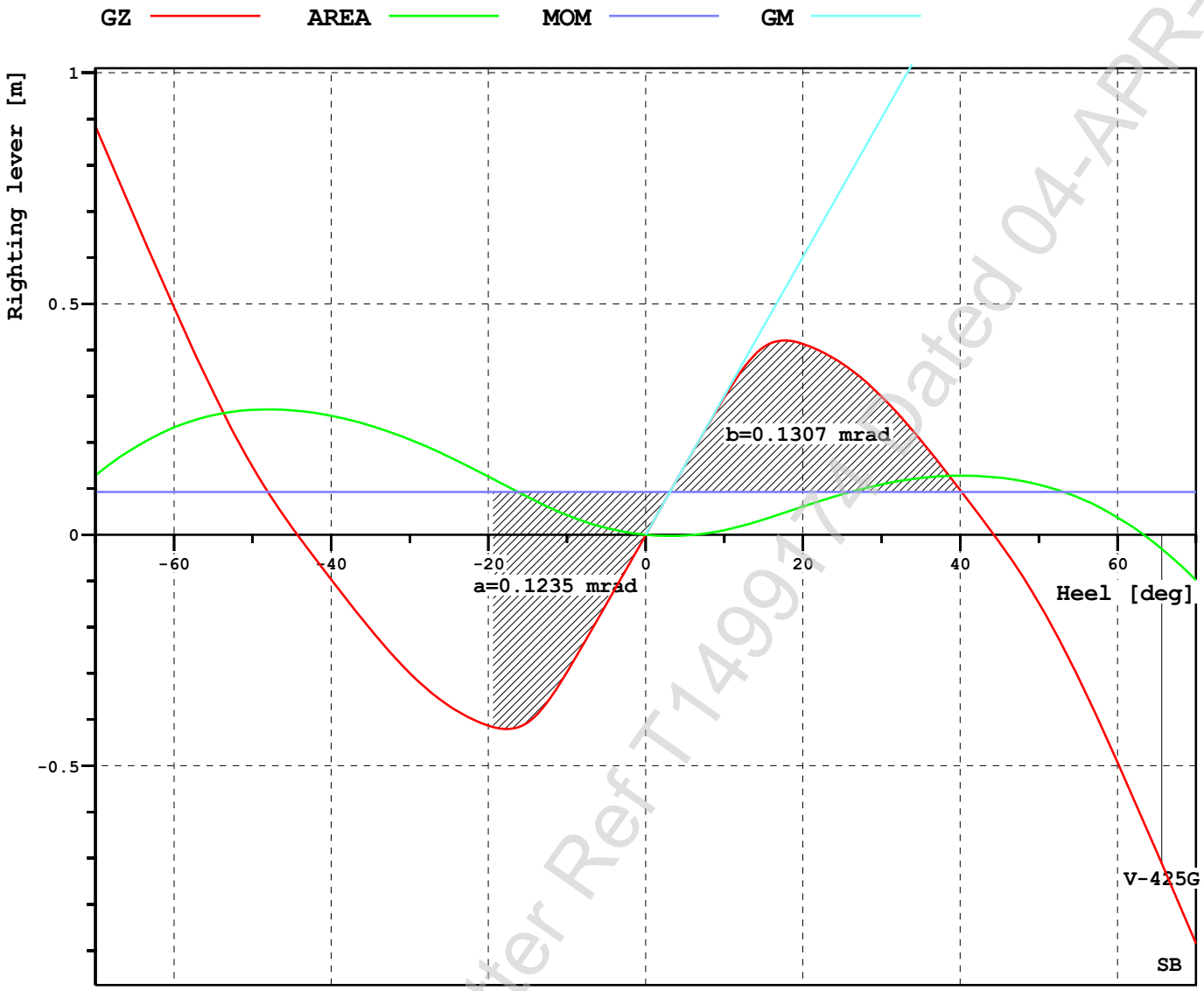
LOADING CONDITION L07, Max deckload - Arrival

HEEL degree	MS m	HPHI m	EPHI mrad	FSMOM tm	DGZ m	GMSIN m	KN m
0.0	0.000	0.00	0.000	0.0	0.000	0.000	-0.002
1.0	0.000	0.03	0.000	3.9	0.001	0.031	0.152
3.0	0.000	0.09	0.002	11.5	0.002	0.092	0.458
5.0	0.000	0.15	0.007	19.0	0.003	0.153	0.764
10.0	-0.002	0.30	0.026	36.9	0.006	0.306	1.521
15.0	-0.040	0.41	0.057	54.0	0.009	0.456	2.231
20.0	-0.177	0.41	0.094	70.5	0.012	0.602	2.825
30.0	-0.564	0.30	0.158	101.5	0.017	0.880	3.826
40.0	-1.013	0.10	0.193	130.0	0.021	1.131	4.631
50.0	-1.469	-0.15	0.190	156.2	0.026	1.348	5.257
60.0	-1.987	-0.49	0.135	180.6	0.030	1.524	5.618
70.0	-2.505	-0.88	0.015	206.1	0.034	1.654	5.748

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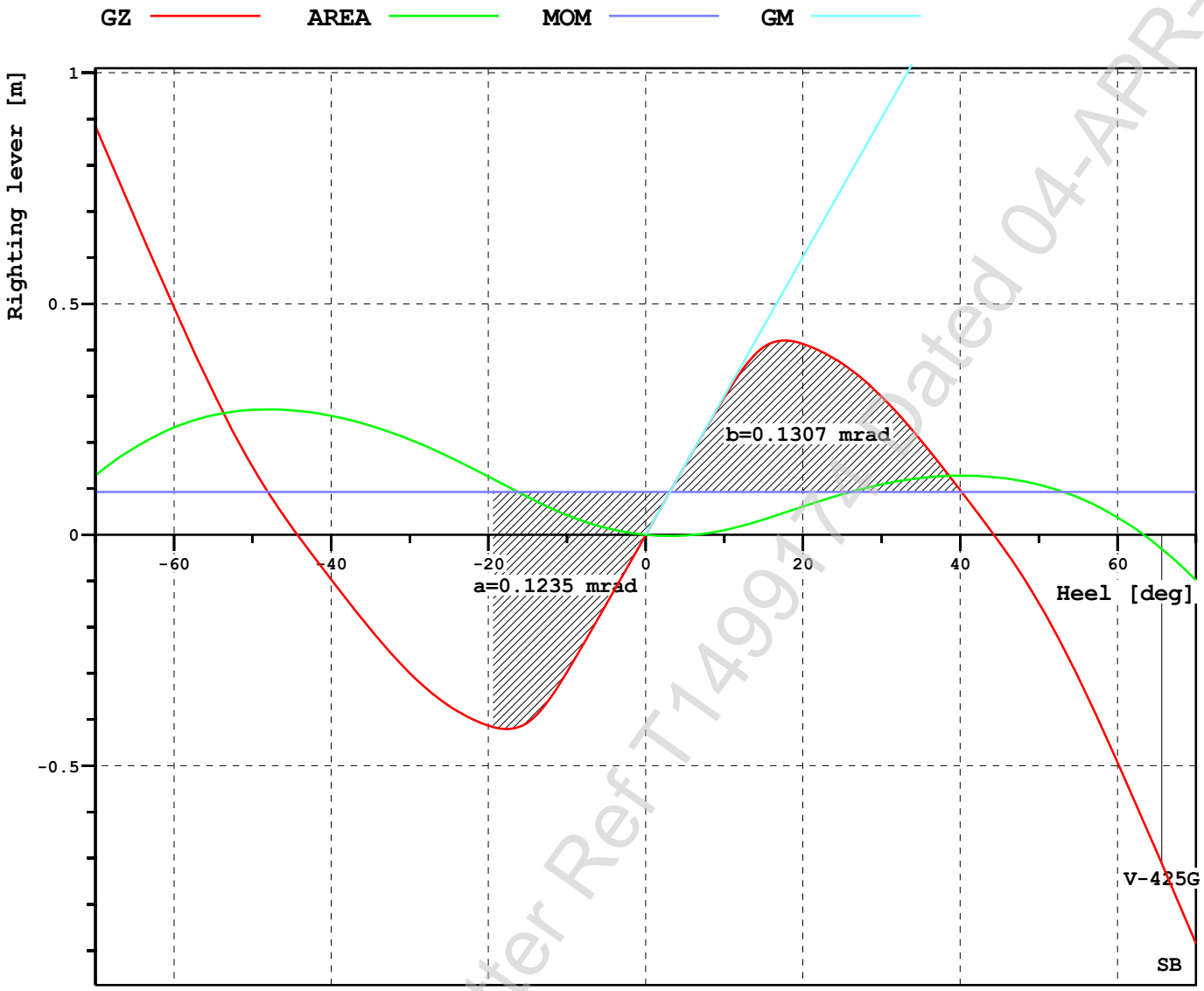
STABILITY CHECK PLOT



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IMOWEATHER CHECK PLOT



INTACT STABILITY CRITERIA

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM
V.AREA15-30	Area depending on the position of the Max. .	0.067	0.077	mrad	OK	1.609
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.035	mrad	OK	1.670
V.GZ0.2	Min. GZ > 0.2	0.200	0.300	m	OK	1.523
V.POSMAX15	Max. GZ at an angle > 15 deg.	15.000	17.700	deg	OK	1.033
V.GM0.15	GM > 0.15 m	0.150	1.722	m	OK	0.150
IMOWEATHER	IMO weather criterion	1.000	1.058		OK	1.673
2008IS-A2.3.1.2	Heeling angle due to steady wind <16 or <=8.	8.186	2.040	deg	OK	0.438
ABSFIFI1	Angle of heel < 1/2 freeboard immersion	5.116	0.456	deg	OK	0.156
ABSFIFI2	Area of residual stability >0.09 mrad	0.090	0.183	mrad	OK	1.263
TOTAL						1.673

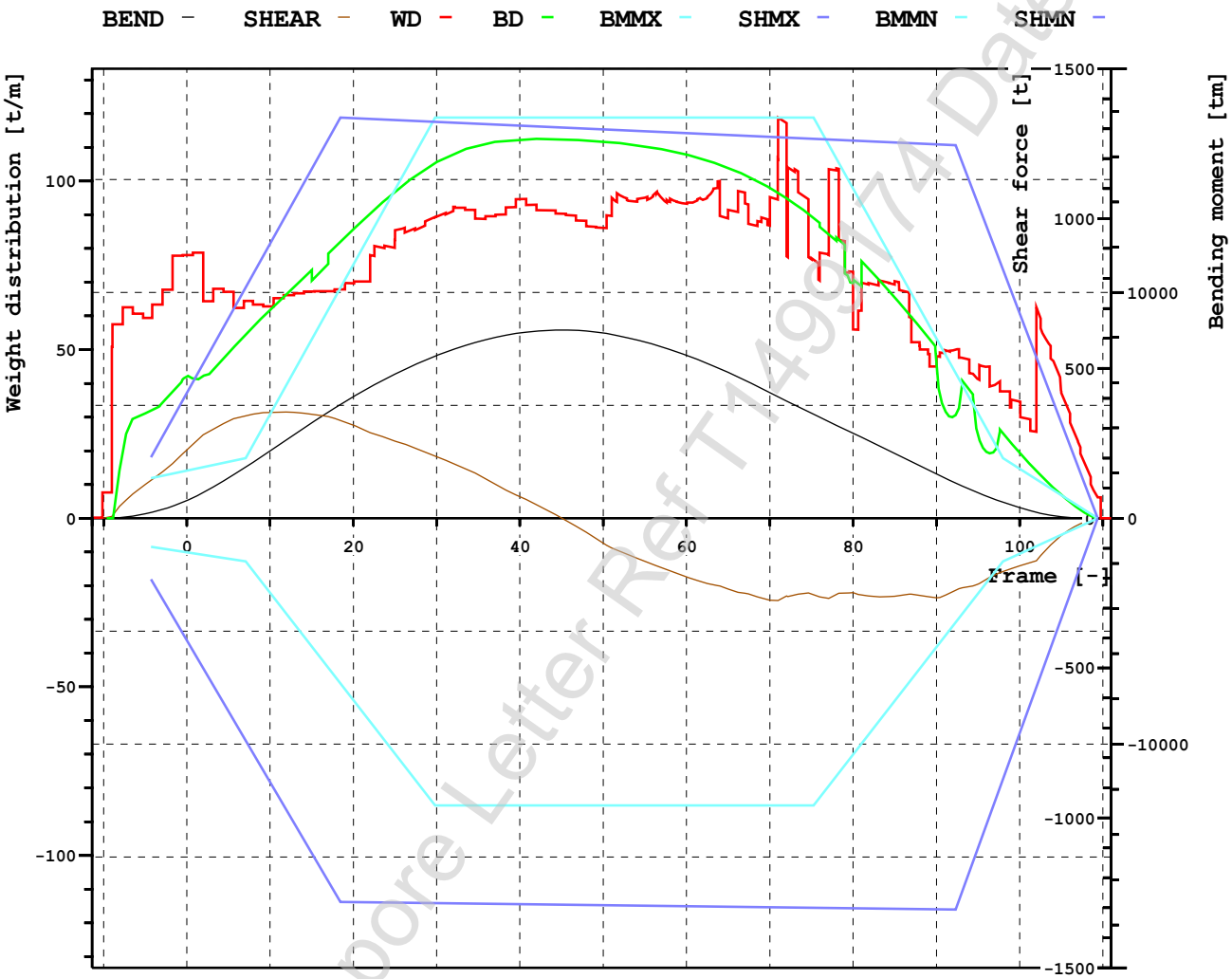
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MOST CRITICAL OPENINGS

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
V-424G	SUPPLY AIR ENGINE ROOM PS	UNPROTECTED	54.243	-3.413	14.288	-	8.355
V-425G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	54.243	3.767	14.288	65.6	8.355
V-421G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	55.890	2.093	14.288	-	8.369
D-651	ACCOMMODATION/C-DECK	UNPROTECTED	56.000	3.771	16.906	-	10.988
V-421E	EXHAUST AIR ENGINE ROOM PS	UNPROTECTED	55.993	-2.538	17.088	-	11.170
V-423E	EXHAUST AIR ENGINE ROOM SB	UNPROTECTED	55.993	3.423	17.088	-	11.170

STRENGTH DIAGRAM



STRENGTH SUMMARY

LOADING CONDITION L07, Max deckload - Arrival

			X	Frame
Shear force (min)	-275.6 t	(21.2%)	49.669 m	70.96
Shear force (max)	354.2 t	(35.0%)	8.348 m	11.93
Sagging moment	-1.3 tm	(0.0%)	76.567 m	109.38
Hogging moment	8334.8 tm	(47.0%)	31.858 m	45.51

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STRENGTH DISTRIBUTION

LOADING CONDITION L07, Max deckload - Arrival

X	FR	BEND	BMMN	BMMX	BMREL	SHEAR	SHMN	SHMX	SHREL	WD	BD
m	#	tm	tm	tm	%	t	t	t	%	t/m	t/m
-3.500	-5.00	193	0	0	0.0	113	0	0	0.0	59.41	31.06
0.000	0.00	782	-1510	2108	37.1	228	-407	418	54.6	78.04	42.01
3.500	5.00	1777	-1790	2498	71.1	324	-644	667	48.6	67.14	48.92
7.000	10.00	2981	-3292	4594	65.1	353	-881	916	38.5	62.79	61.79
10.500	15.00	4218	-5668	7911	53.3	347	-1117	1166	29.8	67.22	73.60
14.000	20.00	5387	-8045	11228	48.0	310	-1281	1335	23.2	69.70	85.69
17.500	25.00	6381	-10422	14545	43.9	257	-1283	1329	19.3	85.46	96.69
21.000	30.00	7200	-12709	17736	40.6	205	-1284	1323	15.5	89.40	105.62
24.500	35.00	7821	-12709	17736	44.1	145	-1286	1317	11.0	88.86	110.40
28.000	40.00	8199	-12709	17736	46.2	72	-1288	1310	5.5	94.64	112.13
31.500	45.00	8333	-12709	17736	47.0	1	-1290	1304	0.0	90.30	112.27
35.000	50.00	8199	-12709	17736	46.2	-83	-1292	1298	6.4	86.12	111.61
38.500	55.00	7799	-12709	17736	44.0	-145	-1293	1292	11.2	95.02	110.17
42.000	60.00	7209	-12709	17736	40.6	-196	-1295	1285	15.1	93.26	107.84
45.500	65.00	6455	-12709	17736	36.4	-237	-1297	1279	18.3	91.41	103.84
49.000	70.00	5567	-12709	17736	31.4	-274	-1299	1273	21.1	95.22	97.93
52.500	75.00	4657	-12709	17736	26.3	-253	-1300	1267	19.5	77.11	89.53
56.000	80.00	3763	-10463	14601	25.8	-248	-1302	1260	19.1	73.10	69.79
59.500	85.00	2861	-8086	11284	25.4	-260	-1304	1254	19.9	70.29	65.52
63.000	90.00	1960	-5709	7968	24.6	-266	-1306	1248	20.4	48.03	46.48
66.500	95.00	1121	-3332	4651	24.1	-220	-1102	1050	20.0	41.23	24.67
70.000	100.00	475	-1571	2193	21.7	-159	-718	685	22.1	34.63	19.39
73.500	105.00	65	-733	1023	6.3	-57	-335	319	17.1	37.65	6.94

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DAMAGE STABILITY OF LOADING CONDITION L07

CRITICAL DAMAGE STABILITY CRITERIA FOR EACH DAMAGE CASE

RCR	DAM	STAGE	PHASE	REQ	ATTV	UNIT	STAT	MINGM
V.RANGE.A469	D01	1	EQ	20.000	44.921	deg	OK	0.473
V.RANGE.A469	D02	1	EQ	20.000	43.956	deg	OK	0.686
V.RANGE.A469	D03	1	EQ	20.000	44.412	deg	OK	0.680
V.RANGE.A469	D04	1	EQ	20.000	39.662	deg	OK	1.057
V.RANGE.A469	D05	1	EQ	20.000	42.087	deg	OK	0.910
V.RANGE.A469	D06	1	EQ	20.000	38.628	deg	OK	1.078
V.RANGE.A469	D07	1	EQ	20.000	41.109	deg	OK	0.933
V.RANGE.A469	D08	1	EQ	20.000	43.149	deg	OK	0.671
V.RANGE.A469	D09	1	EQ	20.000	40.605	deg	OK	0.940
V.RANGE.A469	D10	1	EQ	20.000	38.562	deg	OK	1.024
V.RANGE.A469	D11	1	EQ	20.000	39.053	deg	OK	1.014
V.RANGE.A469	D12	1	EQ	20.000	36.849	deg	OK	1.091
V.RANGE.A469	D13	1	EQ	20.000	37.285	deg	OK	1.084
V.RANGE.A469	D14	1	EQ	20.000	37.260	deg	OK	1.046
V.RANGE.A469	D15	1	EQ	20.000	37.754	deg	OK	1.036
V.RANGE.A469	D16	1	EQ	20.000	32.256	deg	OK	1.183
V.RANGE.A469	D17	1	EQ	20.000	32.726	deg	OK	1.176
V.RANGE.A469	D18	1	EQ	20.000	41.423	deg	OK	0.751
V.RANGE.A469	D19	1	EQ	20.000	41.872	deg	OK	0.748
V.RANGE.A469	D20	1	EQ	20.000	40.164	deg	OK	0.772
V.RANGE.A469	D21	1	EQ	20.000	40.612	deg	OK	0.770
TOTAL								1.183

DETERMINATION OF CRITICAL DAMAGE CASE

CRITICAL DAMAGE CASE : D16
STAGE OF CRITICAL DAMAGE CASE : 1
PHASE OF CRITICAL DAMAGE CASE : EQ

DAMAGE STABILITY CRITERIA - CRITICAL CASE

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM m
V.PROGR.A469	No progr. flooding	14.566	1.774	deg	OK	0.523
V.MAXHEEL.A469	Maximum heel	17.000	1.774	deg	OK	0.523
V.RANGE.A469	Range of pos. stab	20.000	32.256	deg	OK	1.183
V.MINGZ.A469	Min. righting lever	0.100	0.269	m	OK	1.024
TOTAL						1.183

DAMAGE CASE RESULTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	T m	TR m	HEEL degree	RESFLD m	OPEN	CROSSTM s
D16	1	EQ	SB	6.168	-1.281	1.8	1.99	D-362	-

DAMAGED COMPARTMENTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	NAME	DES	PERM	VOL m3	W t	XCG m	YCG m	ZCG m
D16	INTACT	EQ	-				0.0	0.000			
D16	1	EQ	SB	T062	ROLL RED. TANK	0.95	124.4	127.478	7.44	0.40	5.53
D16	1	EQ	SB	T030	WING TANK SB	0.95	65.1	66.678	5.47	4.53	4.26

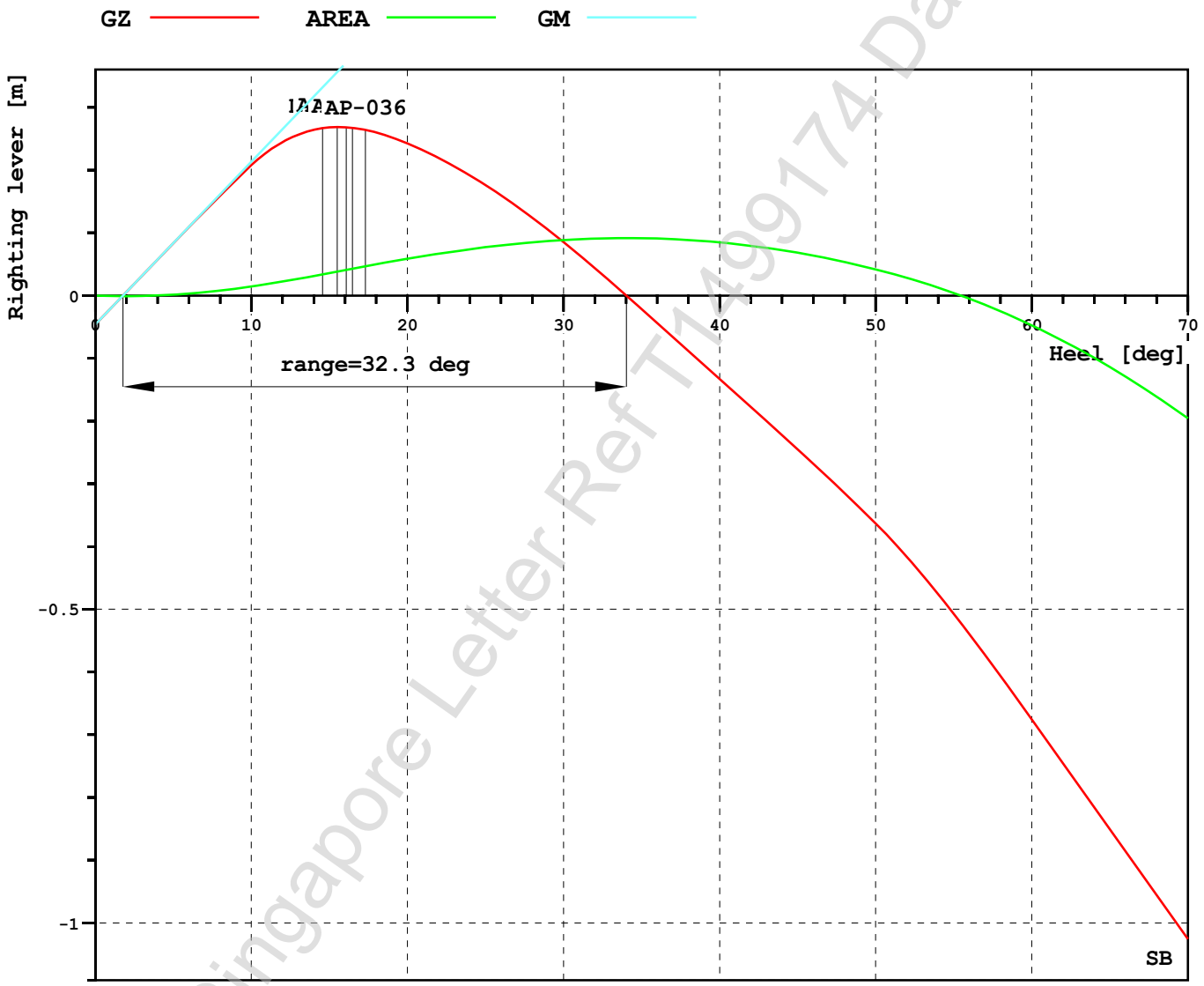
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CRITICAL OPENINGS - CRITICAL CASE

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
D-362	SB CARGORAIL #37/M-DECK	WEATHERTIGHT	25.900	8.550	8.600	14.6	1.989
AP-023	WING TANK SB	WEATHERTIGHT	19.200	8.702	8.936	15.5	2.212
D-360	SB CARGORAIL #57/M-DECK	WEATHERTIGHT	39.885	8.560	8.600	16.1	2.213
AP-034	WING TANK SB	WEATHERTIGHT	-5.873	7.880	9.372	16.5	2.270
AP-036	WING TANK SB	WEATHERTIGHT	1.163	7.810	9.372	17.3	2.385
D-352	CHEMICAL/M-DECK	WEATHERTIGHT	53.192	8.375	8.600	18.0	2.433
AP-011	WING TANK SB	WEATHERTIGHT	47.211	8.754	8.953	18.6	2.678
AP-027	WING TANK SB	WEATHERTIGHT	14.986	7.800	9.372	18.8	2.608

DAMAGE STABILITY CHECK PLOT - CRITICAL CASE



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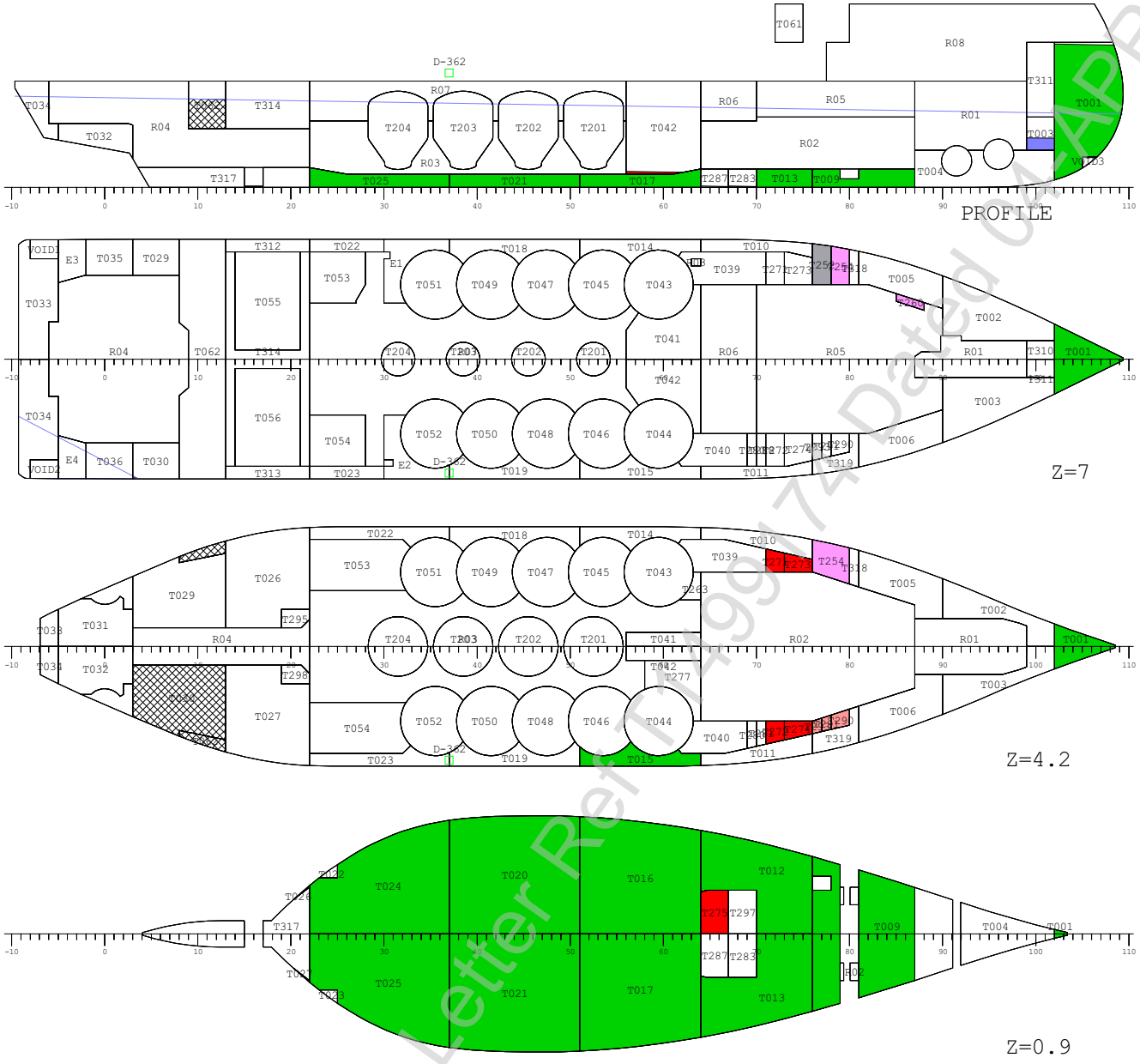
GZ-CURVE - CRITICAL CASE

HEEL	DISP	T	TR	GZ	MS	AREA
degree	t	m	m	m	m	mrad
0.0	6086.7	6.170	-1.273	-0.046	-0.046	0.000
1.0	6086.7	6.170	-1.278	-0.020	-0.051	-0.001
3.0	6086.7	6.162	-1.285	0.032	-0.060	0.000
5.0	6086.7	6.146	-1.288	0.084	-0.070	0.002
10.0	6086.7	6.069	-1.295	0.208	-0.098	0.014
15.0	6086.7	5.971	-1.442	0.268	-0.187	0.036
20.0	6086.7	5.885	-1.695	0.243	-0.359	0.059
30.0	6086.7	5.696	-2.478	0.085	-0.795	0.089
40.0	6086.7	5.425	-3.707	-0.133	-1.264	0.085
50.0	6086.7	4.994	-5.225	-0.363	-1.712	0.042
60.0	6086.7	4.430	-6.655	-0.675	-2.200	-0.047
70.0	6086.7	3.751	-7.846	-1.025	-2.679	-0.196

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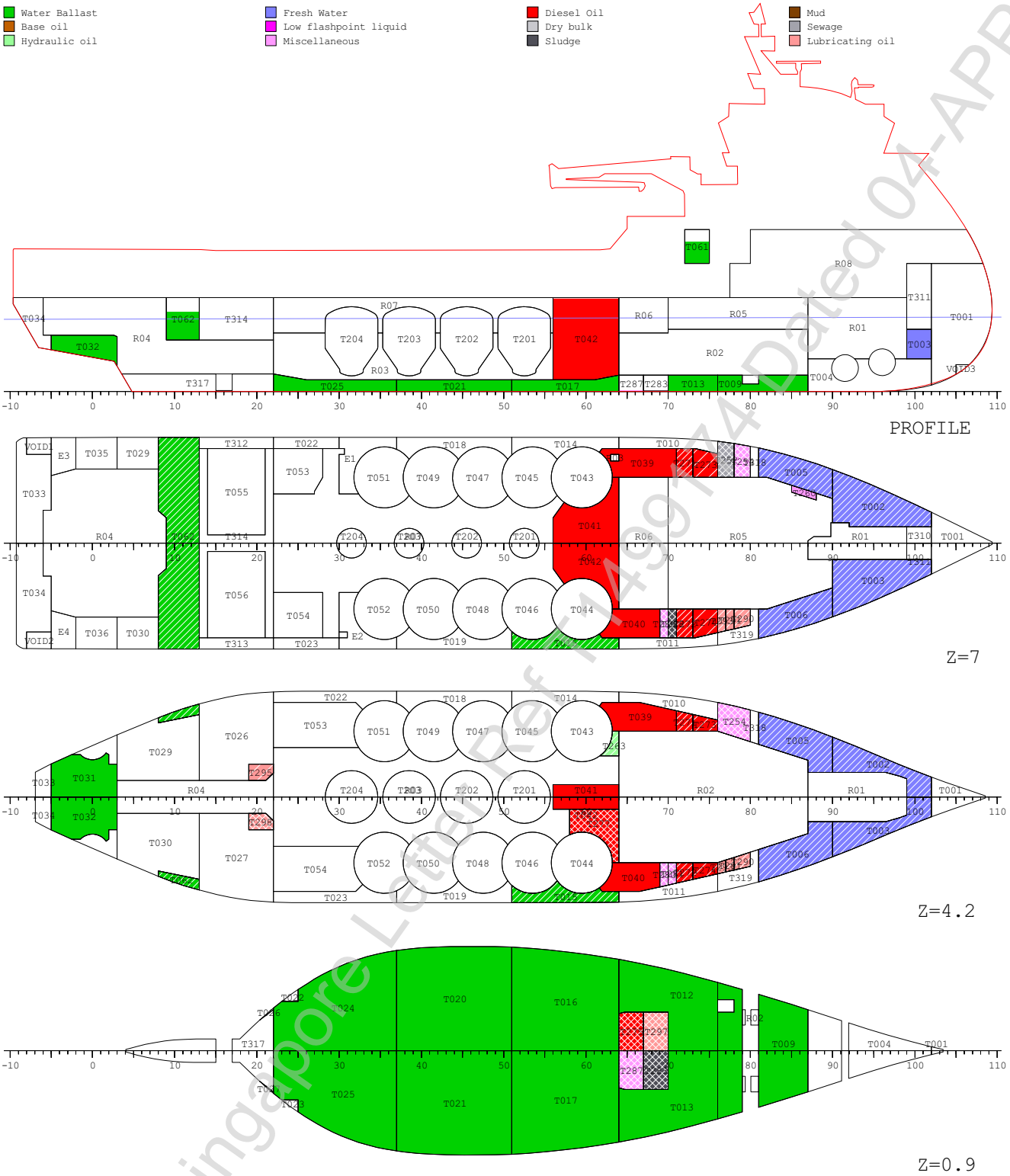
DAMAGE ARRANGEMENT - CRITICAL CASE



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12.8 L08 -> Max deckload - roll red tk - Departure



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LIST OF LOADS

NAME	DES	LOAD	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm	FRS	HM m
CONTENTS=Diesel Oil (RHO=0.86)										
T039	STORAGE TANK PS	FO	66.9	98.0	46.68	-6.76	4.91	6	REAL	7.88
T040	STORAGE TANK SB	FO	53.8	98.0	46.00	6.80	4.87	5	REAL	7.88
T041	STORAGE TANK PS	FO	68.5	98.0	42.16	-1.51	5.48	27	MAX	7.92
T042	STORAGE TANK SB	FO	68.5	98.0	42.16	1.51	5.48	27	MAX	7.92
T271	FO SETTLING TANK PS	FO	8.5	70.0	50.38	-6.53	4.93	1	REAL	6.72
T272	FO SETTLING TANK SB	FO	8.5	70.0	50.38	6.53	4.93	1	REAL	6.72
T273	FO SERVICE TANK PS	FO	10.6	70.0	52.09	-6.38	5.15	2	REAL	6.90
T274	FO SERVICE TANK SB	FO	10.6	70.0	52.09	6.38	5.15	2	REAL	6.90
T275	FO DRAIN TANK	FO	0.8	10.0	45.85	-1.64	0.07	5	REAL	0.14
T277	FO OVERFLOW TANK	FO	3.6	10.0	42.62	2.34	1.31	10	REAL	1.53
T279	EMERG. GEN. FO TANK	FO	1.9	98.0	61.60	-1.38	24.57	0	REAL	25.80
SUBTOTAL		FO	302.3		45.14	-0.27	5.25	86		
CONTENTS=Fresh Water (RHO=1)										
T002	WING TANK PS	FW	56.8	50.0	66.71	-2.66	4.87	45	MAX	6.35
T003	WING TANK SB	FW	57.5	50.0	66.69	2.65	4.88	46	MAX	6.35
T005	WING TANK PS	FW	49.1	50.0	60.05	-4.39	3.34	29	REAL	4.79
T006	WING TANK SB	FW	49.1	50.0	60.05	4.39	3.34	29	REAL	4.79
SUBTOTAL		FW	212.5		63.63	0.01	4.17	149		
CONTENTS=Hydraulic oil (RHO=0.924)										
T263	HYDRAULIC OIL STORE TANK	HO	3.3	70.0	44.30	-4.34	2.32	1	REAL	3.33
CONTENTS=Lubricating oil (RHO=0.924)										
T290	LO STORE TANK MAIN ENG.	LO	5.9	70.0	55.30	5.76	4.90	0	REAL	6.82
T291	LO STORE TANK MAIN ENG.	LO	3.1	70.0	54.25	6.04	5.11	0	REAL	6.92
T293	LO STORE TANK THRUSTER	LO	3.2	70.0	53.55	6.21	5.24	0	REAL	6.97
T295	LO STORE TANK AZIMUTH THR.	LO	4.0	70.0	14.35	-2.10	2.95	0	REAL	3.71
T297	LO DRAIN TANK	LO	0.9	10.0	47.95	-1.64	0.07	6	REAL	0.14
T298	DIRTY LO TANK	LO	0.6	10.0	14.35	2.10	2.31	0	REAL	2.41
SUBTOTAL		LO	17.6		43.89	3.64	4.24	8		
CONTENTS=Mass load (RHO=1)										
M1	CREW	MASS	2.6	0.0	57.40	0.00	15.50	0		0.00
M2	PROVISION	MASS	7.4	0.0	56.35	-6.50	12.40	0		0.00
M3	STORES	MASS	7.4	0.0	72.10	0.00	12.40	0		0.00
M4	MAX DECK LOAD 10T/M2 AFT OF #35	MASS	1070.0	0.0	9.10	0.00	9.00	0	GIVEN	0.00
M5	MAX DECK LOAD 5T/M2 FWD OF #35	MASS	900.0	0.0	37.26	0.00	9.00	0		0.00
SUBTOTAL		MASS	1987.5		22.33	-0.02	9.03	0		
CONTENTS=Miscellaneous (RHO=1)										
T280	BILGE WATER HOLDING TANK	MIS	0.9	10.0	48.64	6.17	2.03	0	REAL	2.57
T281	BILGE WATER SETTLING TANK	MIS	0.4	10.0	49.34	6.04	1.76	0	REAL	2.08
T287	BILGE WATER DRAIN TANK	MIS	0.9	10.0	45.85	1.64	0.07	6	REAL	0.14
T254	GREY WATER TANK	MIS	3.2	10.0	54.62	-5.79	1.92	1	REAL	2.34
T260	WATER MIST TANK	MIS	2.3	98.0	60.55	-4.30	6.28	0	REAL	7.27
SUBTOTAL		MIS	7.8		54.40	-2.49	3.01	8		
CONTENTS=Sewage (RHO=1)										
T252	BLACK WATER TANK	SEW	1.1	10.0	53.89	-6.93	5.45	2	REAL	5.59
CONTENTS=Sludge (RHO=1)										
T283	SLUDGE DRAIN TANK	SLU	0.9	10.0	47.95	1.64	0.07	6	REAL	0.14
T288	SLUDGE SETTLING TANK	SLU	0.4	10.0	49.35	6.75	5.44	1	REAL	5.59
SUBTOTAL		SLU	1.4		48.40	3.29	1.80	7		
CONTENTS=Water Ballast (RHO=1.025)										
T001	FOREPEAK TANK	WB	0.0	0.0	73.11	0.00	7.21	34	MAX	0.00
T009	DB TANK CE	WB	77.3	100.0	56.90	0.09	0.73	0	(REAL)	1.40
T012	DB TANK PS	WB	56.2	100.0	49.41	-3.85	0.76	0	(REAL)	1.40
T013	DB TANK SB	WB	56.2	100.0	49.41	3.85	0.76	0	(REAL)	1.40
T015	WING TANK SB	WB	15.4	25.0	40.17	8.33	2.27	1	REAL	3.09
T016	DB TANK PS	WB	85.6	100.0	40.20	-4.33	0.62	0	REAL	1.50
T017	DB TANK SB	WB	85.6	100.0	40.20	4.33	0.62	0	REAL	1.50
T020	DB TANK PS	WB	96.7	100.0	30.84	-4.63	0.61	0	(REAL)	1.92
T021	DB TANK SB	WB	96.7	100.0	30.84	4.63	0.61	0	(REAL)	1.92
T024	DB TANK PS	WB	75.9	100.0	21.46	-3.85	0.81	0	(REAL)	2.75

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T025	DB TANK SB	WB	75.9	100.0	21.46	3.85	0.81	0	(REAL)	2.75
T031	WING TANK PS	WB	27.2	100.0	-0.65	-1.76	4.18	0	(REAL)	5.39
T032	WING TANK SB	WB	27.2	100.0	-0.65	1.76	4.18	0	(REAL)	5.39
T061	ROLL RED. TANK	WB	83.4	65.0	51.66	0.00	11.84	1358	REAL	12.78
T062	ROLL RED. TANK	WB	136.9	65.0	7.45	0.00	5.58	1736	REAL	6.76
SUBTOTAL		WB	996.1		32.09	0.14	2.51	3130		
TOTAL			3529.5		29.74	0.01	6.53	3390		

LOAD SUMMARY TABLE

ITEM	MASS t	LCG m	TCG m	VCG m	FRSM tm
Diesel Oil	302.3	45.14	-0.27	5.25	85.89
Fresh Water	212.5	63.63	0.01	4.17	149.15
Hydraulic oil	3.3	44.30	-4.34	2.32	0.61
Lubricating oil	17.6	43.89	3.64	4.24	7.62
Mass load	1987.5	22.33	-0.02	9.03	0.00
Miscellaneous	7.8	54.40	-2.49	3.01	7.72
Sewage	1.1	53.89	-6.93	5.45	2.31
Sludge	1.4	48.40	3.29	1.80	6.99
Water Ballast	996.1	32.09	0.14	2.51	3130.08
DEADWEIGHT	3529.5	29.74	0.01	6.53	3390.37
Lightweight	2704.9	39.15	-0.01	7.08	
Deadweight	3529.5	29.74	0.01	6.53	
Total weight	6234.4	33.83	0.00	6.77	

F L O A T I N G P O S I T I O N

Draught moulded	6.257	m	KM	8.70 m
Trim	0.187	m	KG	6.77 m
Heel, SB=+	0.0	deg		
TA	6.164	m	GM0	1.93 m
TF	6.351	m	GMCORR	-0.54 m
Trimming moment	1356	tonm	GM	1.39 m

STABILITY COMPONENTS

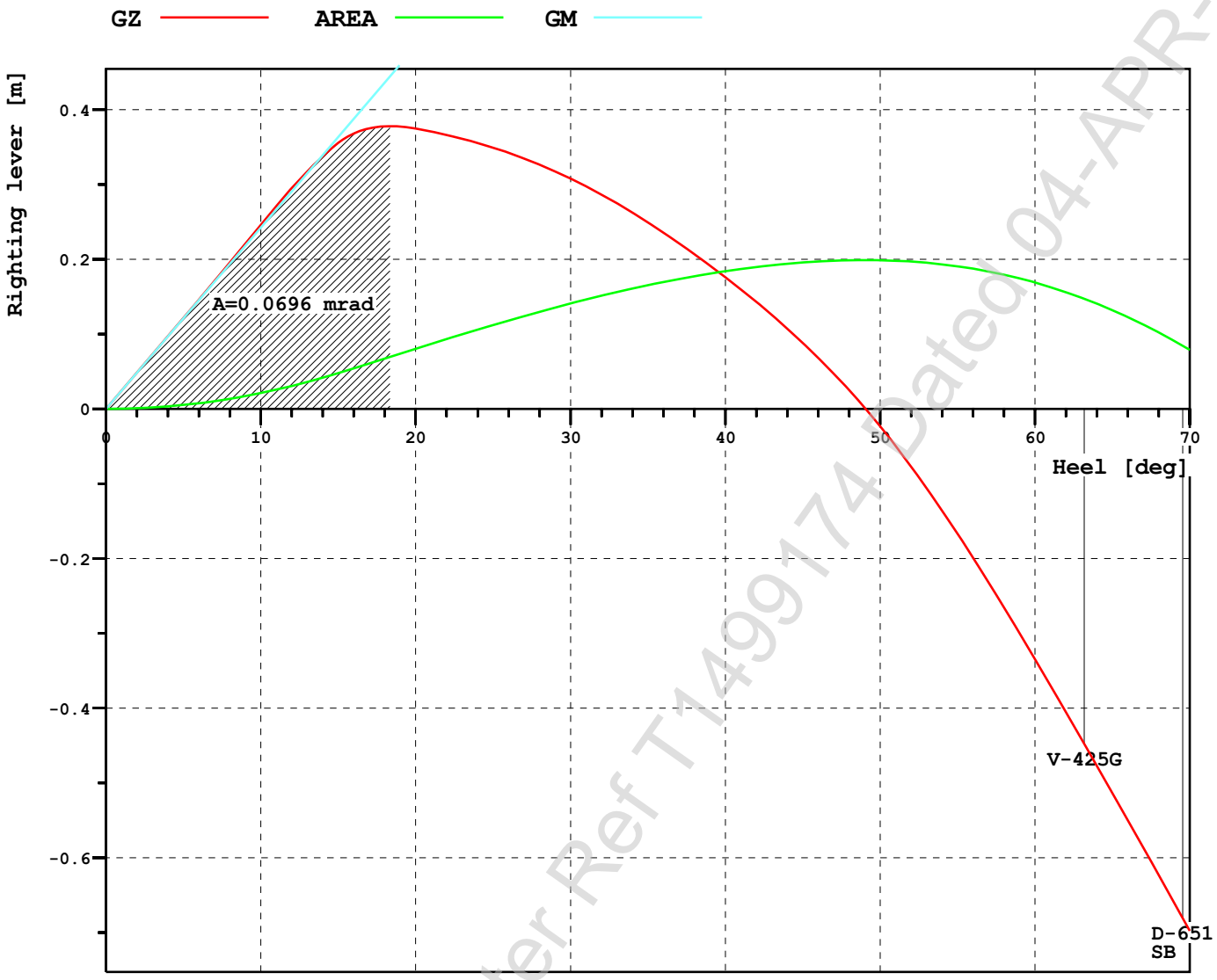
LOADING CONDITION L08, Max deckload - roll red tk - Departure

HEEL degree	MS m	HPHI m	EPHI mrad	FSMOM tm	DGZ m	GMSIN m	KN m
0.0	0.000	0.00	0.000	0.0	0.000	0.000	-0.001
1.0	0.000	0.02	0.000	57.4	0.009	0.034	0.151
3.0	0.000	0.07	0.002	172.3	0.028	0.101	0.454
5.0	-0.001	0.12	0.005	287.6	0.046	0.168	0.757
10.0	-0.003	0.25	0.021	538.5	0.086	0.336	1.507
15.0	-0.038	0.36	0.048	664.9	0.107	0.500	2.213
20.0	-0.169	0.37	0.081	728.4	0.117	0.661	2.806
30.0	-0.533	0.31	0.141	781.2	0.125	0.966	3.817
40.0	-0.940	0.18	0.184	785.4	0.126	1.242	4.652
50.0	-1.380	-0.02	0.199	769.8	0.123	1.481	5.284
60.0	-1.891	-0.33	0.169	735.2	0.118	1.674	5.644
70.0	-2.404	-0.70	0.079	683.3	0.110	1.816	5.772

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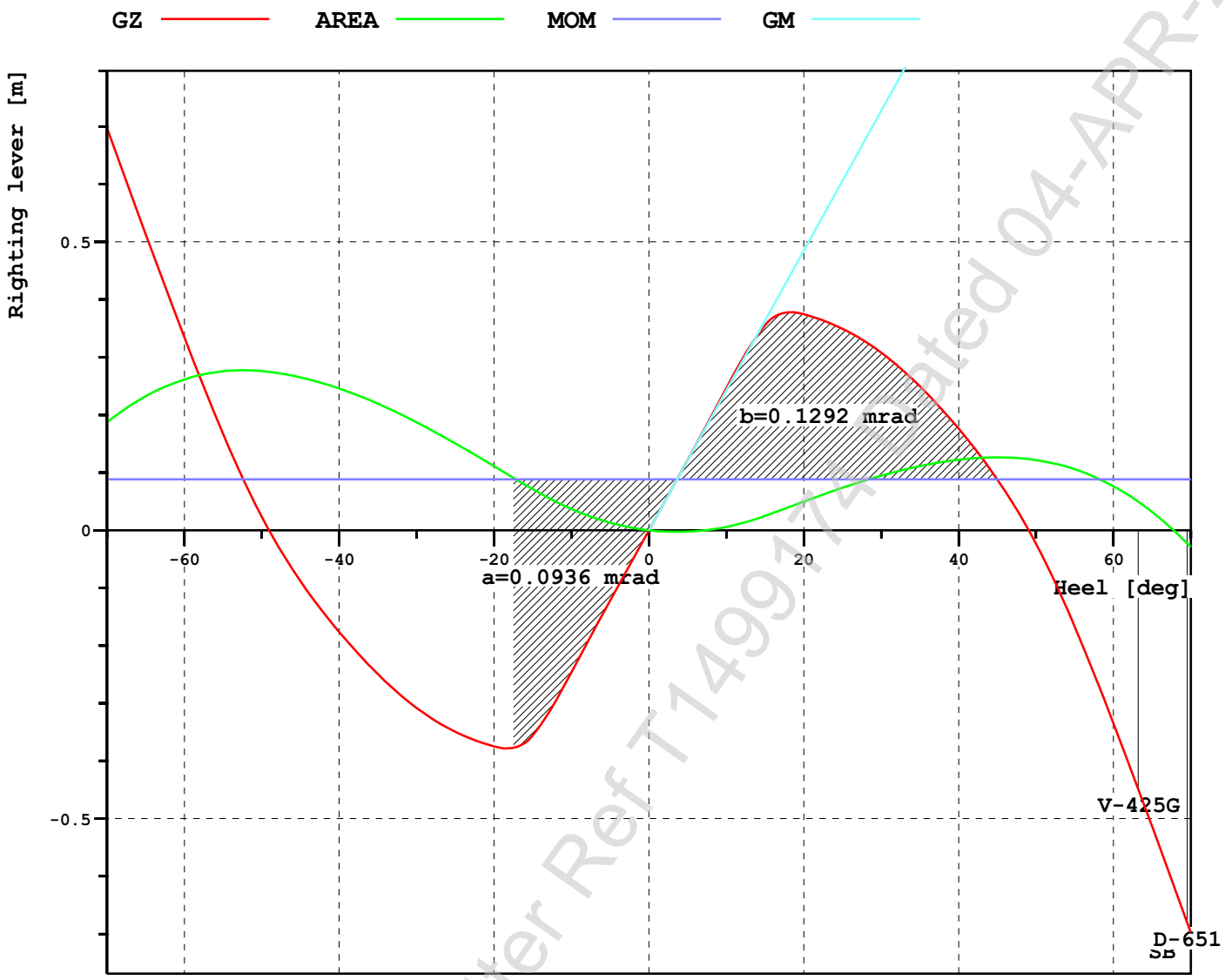
STABILITY CHECK PLOT



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IMOWEATHER CHECK PLOT



INTACT STABILITY CRITERIA

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM
V.AREA15-30	Area depending on the position of the Max. .	0.067	0.070	mrad	OK	1.358
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.043	mrad	OK	1.258
V.GZ0.2	Min. GZ > 0.2	0.200	0.308	m	OK	1.173
V.POSMAX15	Max. GZ at an angle > 15 deg.	15.000	18.341	deg	OK	0.454
V.GM0.15	GM > 0.15 m	0.150	1.389	m	OK	0.150
IMOWEATHER	IMO weather criterion	1.000	1.379		OK	1.168
2008IS-A2.3.1.2	Heeling angle due to steady wind <16 or <=8.	8.684	2.413	deg	OK	0.366
ABSFIFI1	Angle of heel < 1/2 freeboard immersion	5.427	0.554	deg	OK	0.136
ABSFIFI2	Area of residual stability >0.09 mrad	0.090	0.175	mrad	OK	1.016
TOTAL						1.358

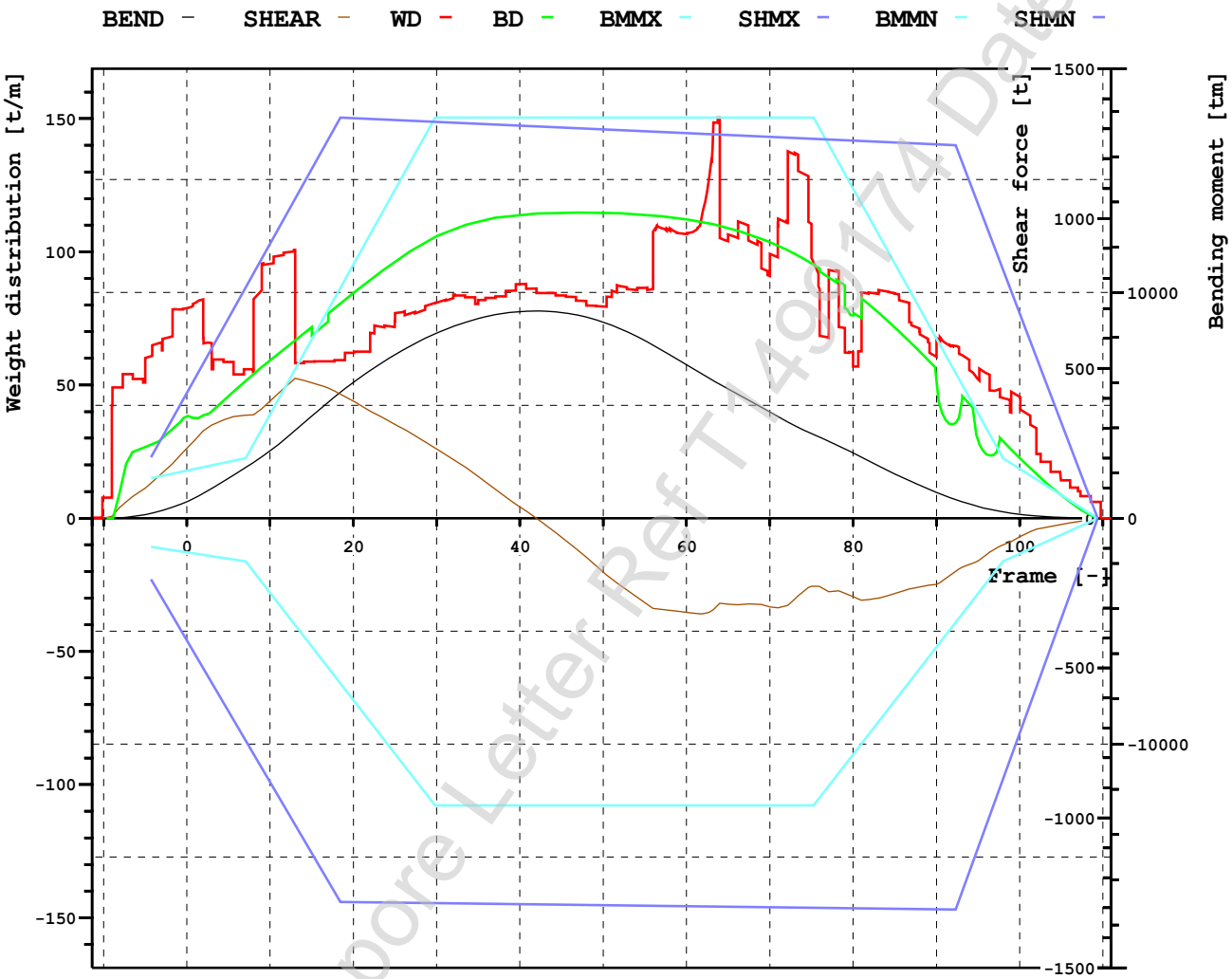
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MOST CRITICAL OPENINGS

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
V-421G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	55.890	2.093	14.288	-	7.986
V-424G	SUPPLY AIR ENGINE ROOM PS	UNPROTECTED	54.243	-3.413	14.288	-	7.989
V-425G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	54.243	3.767	14.288	63.2	7.989
D-651	ACCOMMODATION/C-DECK	UNPROTECTED	56.000	3.771	16.906	69.5	10.603
V-421E	EXHAUST AIR ENGINE ROOM PS	UNPROTECTED	55.993	-2.538	17.088	-	10.785
V-423E	EXHAUST AIR ENGINE ROOM SB	UNPROTECTED	55.993	3.423	17.088	-	10.785

STRENGTH DIAGRAM



STRENGTH SUMMARY

LOADING CONDITION L08, Max deckload - roll red tk - Departure

			X	Frame
Shear force (min)	-319.0 t	(24.6%)	43.200 m	61.71
Shear force (max)	466.8 t	(43.8%)	9.100 m	13.00
Sagging moment	-0.7 tm	(0.0%)	-7.104 m	-10.15
Hogging moment	9171.4 tm	(51.7%)	29.375 m	41.96

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STRENGTH DISTRIBUTION

LOADING CONDITION L08, Max deckload - roll red tk - Departure

X	FR	BEND	BMMN	BMMX	BMREL	SHEAR	SHMN	SHMX	SHREL	WD	BD
m	#	tm	tm	tm	%	t	t	t	%	t/m	t/m
-3.500	-5.00	163	0	0	0.0	100	0	0	0.0	60.17	26.69
0.000	0.00	733	-1510	2108	34.8	233	-407	418	55.8	78.85	38.22
3.500	5.00	1760	-1790	2498	70.4	333	-644	667	50.0	58.70	45.73
7.000	10.00	2989	-3292	4594	65.2	391	-881	916	42.7	95.53	59.27
10.500	15.00	4529	-5668	7911	57.2	452	-1117	1166	38.7	58.78	71.77
14.000	20.00	6014	-8045	11228	53.6	391	-1281	1335	29.3	62.06	84.54
17.500	25.00	7240	-10422	14545	49.8	312	-1283	1329	23.5	77.01	96.23
21.000	30.00	8193	-12709	17736	46.2	231	-1284	1323	17.5	80.96	105.85
24.500	35.00	8843	-12709	17736	49.9	139	-1286	1317	10.6	81.96	111.32
28.000	40.00	9144	-12709	17736	51.6	38	-1288	1310	2.9	87.86	113.73
31.500	45.00	9100	-12709	17736	51.3	-63	-1290	1304	4.9	83.60	114.56
35.000	50.00	8677	-12709	17736	48.9	-180	-1292	1298	13.9	79.50	114.59
38.500	55.00	7864	-12709	17736	44.3	-281	-1293	1292	21.8	86.45	113.84
42.000	60.00	6796	-12709	17736	38.3	-315	-1295	1285	24.3	106.73	112.19
45.500	65.00	5720	-12709	17736	32.3	-287	-1297	1279	22.1	106.47	108.87
49.000	70.00	4707	-12709	17736	26.5	-296	-1299	1273	22.8	99.26	103.60
52.500	75.00	3748	-12709	17736	21.1	-227	-1300	1267	17.5	110.54	95.68
56.000	80.00	2897	-10463	14601	19.8	-261	-1302	1260	20.1	62.37	76.13
59.500	85.00	1964	-8086	11284	17.4	-252	-1304	1254	19.3	84.43	71.68
63.000	90.00	1143	-5709	7968	14.3	-221	-1306	1248	16.9	67.69	52.06
66.500	95.00	518	-3332	4651	11.1	-142	-1102	1050	12.9	52.22	29.27
70.000	100.00	169	-1571	2193	7.7	-62	-718	685	8.7	45.60	22.63
73.500	105.00	37	-733	1023	3.7	-21	-335	319	6.2	14.17	8.58

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DAMAGE STABILITY OF LOADING CONDITION L08

CRITICAL DAMAGE STABILITY CRITERIA FOR EACH DAMAGE CASE

RCR	DAM	STAGE	PHASE	REQ	ATTV	UNIT	STAT	MINGM
V.MINGZ.A469	D01	1	EQ	0.100	0.383	m	OK	0.386
V.MINGZ.A469	D02	1	EQ	0.100	0.365	m	OK	0.438
V.MINGZ.A469	D03	1	EQ	0.100	0.375	m	OK	0.413
V.RANGE.A469	D04	1	EQ	20.000	45.702	deg	OK	0.707
V.RANGE.A469	D05	1	EQ	20.000	45.356	deg	OK	0.748
V.RANGE.A469	D06	1	EQ	20.000	42.603	deg	OK	0.855
V.RANGE.A469	D07	1	EQ	20.000	42.211	deg	OK	0.885
V.RANGE.A469	D08	1	EQ	20.000	45.743	deg	OK	0.642
V.RANGE.A469	D09	1	EQ	20.000	44.928	deg	OK	0.716
V.RANGE.A469	D10	1	EQ	20.000	42.782	deg	OK	0.797
V.RANGE.A469	D11	1	EQ	20.000	43.374	deg	OK	0.779
V.RANGE.A469	D12	1	EQ	20.000	40.972	deg	OK	0.852
V.RANGE.A469	D13	1	EQ	20.000	41.541	deg	OK	0.839
V.RANGE.A469	D14	1	EQ	20.000	41.538	deg	OK	0.803
V.RANGE.A469	D15	1	EQ	20.000	42.176	deg	OK	0.785
V.RANGE.A469	D16	1	EQ	20.000	44.241	deg	OK	0.655
V.RANGE.A469	D17	1	EQ	20.000	44.805	deg	OK	0.639
V.MINGZ.A469	D18	1	EQ	0.100	0.332	m	OK	0.561
V.MINGZ.A469	D19	1	EQ	0.100	0.335	m	OK	0.556
V.MINGZ.A469	D20	1	EQ	0.100	0.332	m	OK	0.538
V.MINGZ.A469	D21	1	EQ	0.100	0.335	m	OK	0.534
TOTAL								0.885

DETERMINATION OF CRITICAL DAMAGE CASE

CRITICAL DAMAGE CASE : D07
STAGE OF CRITICAL DAMAGE CASE : 1
PHASE OF CRITICAL DAMAGE CASE : EQ

DAMAGE STABILITY CRITERIA - CRITICAL CASE

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM m
V.PROGR.A469	No progr. flooding	14.250	4.293	deg	OK	0.501
V.MAXHEEL.A469	Maximum heel	17.000	4.293	deg	OK	0.492
V.RANGE.A469	Range of pos. stab	20.000	42.211	deg	OK	0.885
V.MINGZ.A469	Min. righting lever	0.100	0.259	m	OK	0.859
TOTAL						0.885

DAMAGE CASE RESULTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	T m	TR m	HEEL degree	RESFLD m	OPEN	CROSSTM s
D07	1	EQ	PS	6.328	0.478	-4.3	1.53	D-351	-

DAMAGED COMPARTMENTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	NAME	DES	PERM	VOL m3	W t	XCG m	YCG m	ZCG m
D07	INTACT	EQ	-				0.0	0.000			
D07	1	EQ	PS	T010	WING TANK PS	0.95	51.1	52.387	49.36	-7.78	4.04
D07	1	EQ	PS	T273	FO SERVICE TANK PS	0.95	12.3	12.591	52.09	-6.40	5.17
D07	1	EQ	PS	T252	BLACK WATER TANK	0.95	6.3	6.454	53.89	-7.05	6.16
D07	1	EQ	PS	T254	GREY WATER TANK	0.95	27.7	28.400	54.76	-6.42	4.18
D07	1	EQ	PS	T318	CD VOID PS	0.95	8.4	8.616	56.35	-6.11	4.45

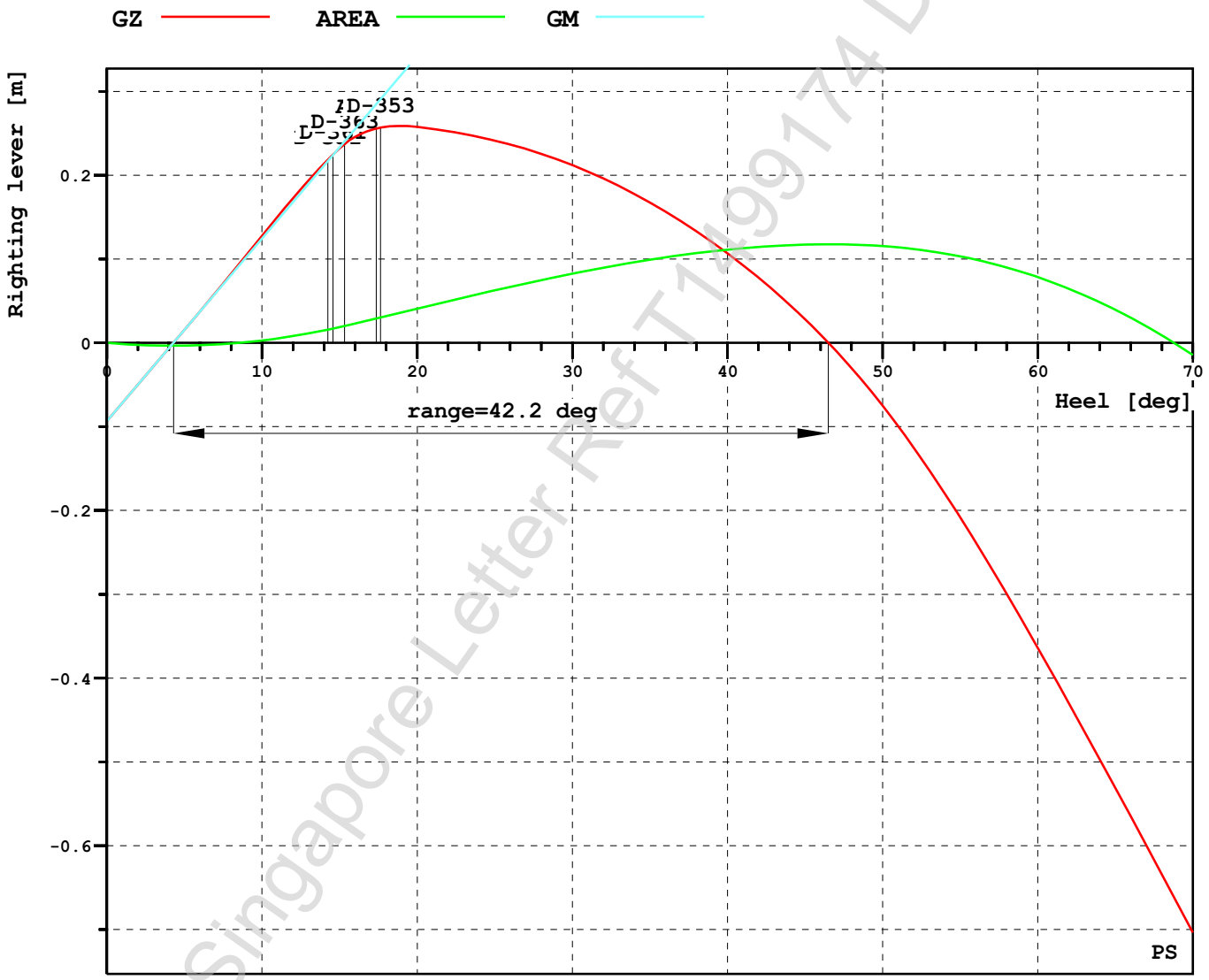
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CRITICAL OPENINGS - CRITICAL CASE

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
D-351	CO2 ROOM/M-DECK	WEATHERTIGHT	51.792	-8.360	8.600	14.2	1.532
D-361	PS CARGORAIL #57/M-DECK	WEATHERTIGHT	39.885	-8.560	8.600	14.6	1.589
D-363	PS CARGORAIL #37/M-DECK	WEATHERTIGHT	25.900	-8.550	8.600	15.3	1.673
AP-022	WING TANK PS	WEATHERTIGHT	19.200	-8.702	8.936	17.4	2.037
D-353	DECK WORKSHOP/M-DECK	WEATHERTIGHT	53.192	-6.550	8.600	17.6	1.659
AP-263	HYDRAULIC OIL STORE TANK	WEATHERTIGHT	47.690	-7.260	8.912	18.5	1.950
AP-012	DB TANK PS	WEATHERTIGHT	50.000	-7.250	9.047	19.3	2.072
V-612E	EXHAUST PROPULSION ROOM	WEATHERTIGHT	6.600	-7.796	8.936	19.7	2.181

DAMAGE STABILITY CHECK PLOT - CRITICAL CASE



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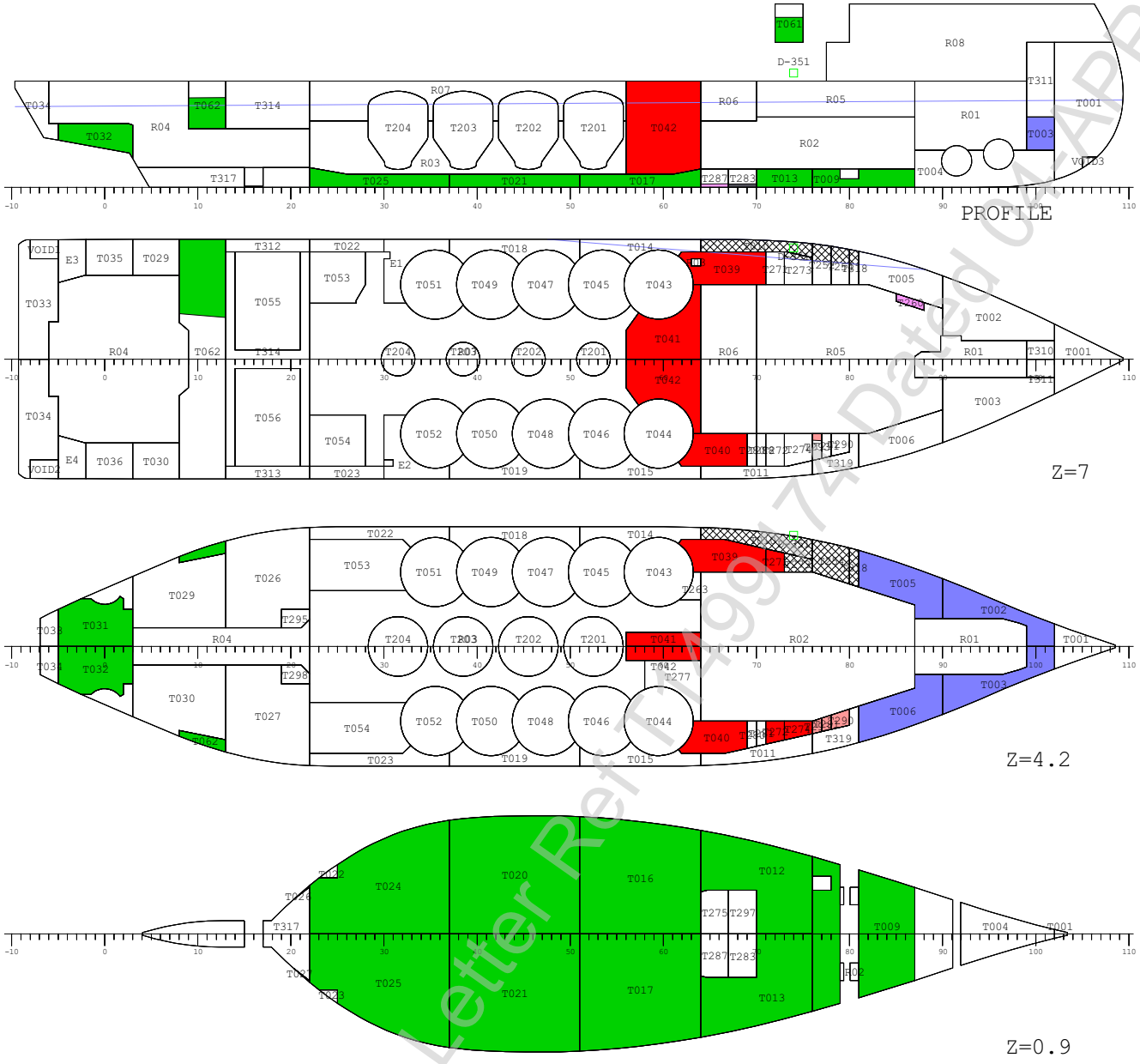
GZ-CURVE - CRITICAL CASE

HEEL	DISP	T	TR	GZ	MS	AREA
degree	t	m	m	m	m	mrad
0.0	6234.4	6.337	0.425	0.093	0.093	0.000
-1.0	6234.4	6.339	0.436	0.072	0.105	-0.001
-3.0	6234.4	6.335	0.460	0.028	0.129	-0.003
-5.0	6234.4	6.322	0.487	-0.015	0.153	-0.003
-10.0	6234.4	6.249	0.586	-0.128	0.208	0.003
-15.0	6234.4	6.133	0.675	-0.232	0.268	0.019
-20.0	6234.4	6.020	0.620	-0.257	0.404	0.041
-30.0	6234.4	5.754	0.120	-0.212	0.754	0.082
-40.0	6234.4	5.390	-0.904	-0.107	1.136	0.111
-50.0	6234.4	4.896	-2.144	0.075	1.556	0.115
-60.0	6234.4	4.285	-3.318	0.364	2.038	0.078
-70.0	6234.4	3.566	-4.338	0.704	2.520	-0.014

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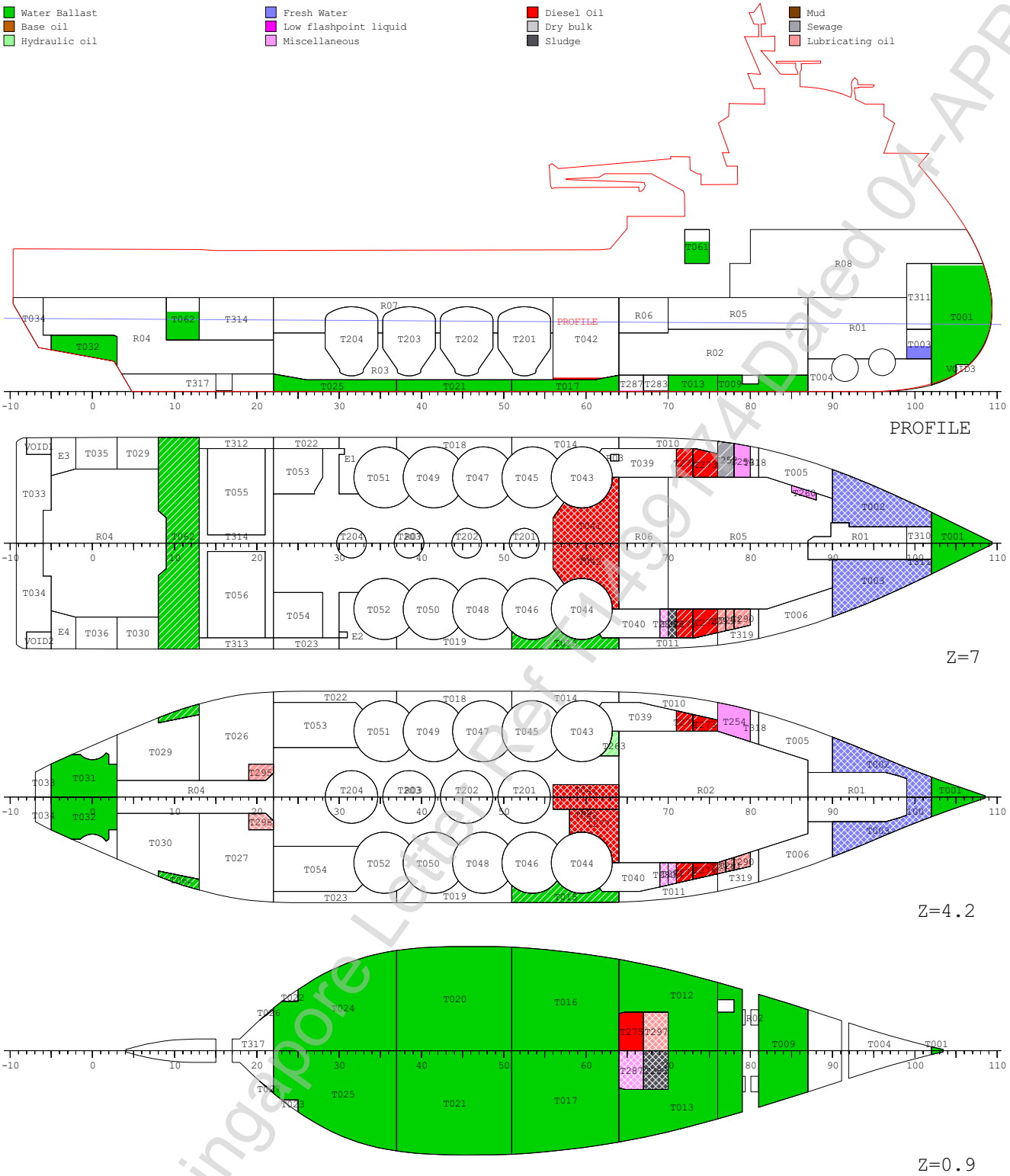
DAMAGE ARRANGEMENT - CRITICAL CASE



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12.9 L09 -> Max deckload - roll red tk - Arrival



See ABS Singapore Letter Ref 11499 X4 Dated 04-APR-2016

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LIST OF LOADS

NAME	DES	LOAD	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm	FRS	HM m
CONTENTS=Diesel Oil (RHO=0.86)										
T041	STORAGE TANK PS	FO	0.7	1.0	41.24	-0.52	1.10	27	MAX	1.20
T042	STORAGE TANK SB	FO	0.7	1.0	41.24	0.52	1.10	27	MAX	1.20
T271	FO SETTLING TANK PS	FO	8.5	70.0	50.38	-6.53	4.93	1	REAL	6.72
T272	FO SETTLING TANK SB	FO	8.5	70.0	50.38	6.53	4.93	1	REAL	6.72
T273	FO SERVICE TANK PS	FO	10.6	70.0	52.09	-6.38	5.15	2	REAL	6.90
T274	FO SERVICE TANK SB	FO	10.6	70.0	52.09	6.38	5.15	2	REAL	6.90
T275	FO DRAIN TANK	FO	7.9	98.0	45.85	-1.64	0.69	5	REAL	1.37
T277	FO OVERFLOW TANK	FO	3.6	10.0	42.62	2.34	1.31	10	REAL	1.53
T279	EMERG. GEN. FO TANK	FO	1.9	98.0	61.60	-1.38	24.57	0	REAL	25.80
SUBTOTAL		FO	53.0		50.02	-0.13	4.73	75		
CONTENTS=Fresh Water (RHO=1)										
T002	WING TANK PS	FW	10.6	9.3	67.25	-2.14	3.41	45	MAX	3.84
T003	WING TANK SB	FW	10.6	9.2	67.25	2.14	3.41	46	MAX	3.84
SUBTOTAL		FW	21.2		67.25	0.00	3.41	91		
CONTENTS=Hydraulic oil (RHO=0.924)										
T263	HYDRAULIC OIL STORE TANK	HO	1.9	40.0	44.29	-4.33	1.89	1	REAL	2.47
CONTENTS=Lubricating oil (RHO=0.924)										
T290	LO STORE TANK MAIN ENG.	LO	3.4	40.0	55.32	5.47	4.05	0	REAL	5.16
T291	LO STORE TANK MAIN ENG.	LO	1.8	40.0	54.25	5.81	4.26	0	REAL	5.48
T293	LO STORE TANK THRUSTER	LO	1.8	40.0	53.55	6.03	4.41	0	REAL	5.66
T295	LO STORE TANK AZIMUTH THR.	LO	2.3	40.0	14.35	-2.10	2.63	0	REAL	3.06
T297	LO DRAIN TANK	LO	0.9	10.0	47.95	-1.64	0.07	6	REAL	0.14
T298	DIRTY LO TANK	LO	0.6	10.0	14.35	2.10	2.31	0	REAL	2.41
SUBTOTAL		LO	10.7		43.36	3.26	3.43	7		
CONTENTS=Mass load (RHO=1)										
M1	CREW	MASS	2.6	0.0	57.40	0.00	15.50	0		0.00
M2	PROVISION	MASS	0.8	0.0	56.35	-6.50	12.40	0		0.00
M3	STORES	MASS	0.8	0.0	72.10	0.00	12.40	0		0.00
M4	MAX DECK LOAD 10T/M2 AFT OF #35	MASS	1070.0	0.0	9.10	0.00	9.00	0	GIVEN	0.00
M5	MAX DECK LOAD 5T/M2 FWD OF #35	MASS	900.0	0.0	37.26	0.00	9.00	0		0.00
SUBTOTAL		MASS	1974.2		22.05	0.00	9.01	0		
CONTENTS=Miscellaneous (RHO=1)										
T280	BILGE WATER HOLDING TANK	MIS	0.9	10.0	48.64	6.17	2.03	0	REAL	2.57
T281	BILGE WATER SETTLING TANK	MIS	0.4	10.0	49.34	6.04	1.76	0	REAL	2.08
T287	BILGE WATER DRAIN TANK	MIS	0.9	10.0	45.85	1.64	0.07	6	REAL	0.14
T254	GREY WATER TANK	MIS	31.6	98.0	54.82	-6.47	4.52	3	REAL	7.84
T260	WATER MIST TANK	MIS	2.3	98.0	60.55	-4.30	6.28	0	REAL	7.27
SUBTOTAL		MIS	36.2		54.74	-5.68	4.43	9		
CONTENTS=Sewage (RHO=1)										
T252	BLACK WATER TANK	SEW	10.4	97.9	53.89	-7.06	6.66	4	REAL	7.95
CONTENTS=Sludge (RHO=1)										
T283	SLUDGE DRAIN TANK	SLU	0.9	10.0	47.95	1.64	0.07	6	REAL	0.14
T288	SLUDGE SETTLING TANK	SLU	0.4	10.0	49.35	6.75	5.44	1	REAL	5.59
SUBTOTAL		SLU	1.4		48.40	3.29	1.80	7		
CONTENTS=Water Ballast (RHO=1.025)										
T001	FOREPEAK TANK	WB	102.9	98.0	73.11	0.00	7.14	34	MAX	10.74
T009	DB TANK CE	WB	77.3	100.0	56.90	0.09	0.73	0	(REAL)	1.40
T012	DB TANK PS	WB	56.2	100.0	49.41	-3.85	0.76	0	(REAL)	1.40
T013	DB TANK SB	WB	56.2	100.0	49.41	3.85	0.76	0	(REAL)	1.40
T015	WING TANK SB	WB	33.8	55.0	40.14	8.39	3.26	1	REAL	5.06
T016	DB TANK PS	WB	85.6	100.0	40.20	-4.33	0.62	0	REAL	1.50
T017	DB TANK SB	WB	85.6	100.0	40.20	4.33	0.62	0	REAL	1.50
T020	DB TANK PS	WB	96.7	100.0	30.84	-4.63	0.61	0	(REAL)	1.92
T021	DB TANK SB	WB	96.7	100.0	30.84	4.63	0.61	0	(REAL)	1.92
T024	DB TANK PS	WB	75.9	100.0	21.46	-3.85	0.81	0	(REAL)	2.75
T025	DB TANK SB	WB	75.9	100.0	21.46	3.85	0.81	0	(REAL)	2.75
T031	WING TANK PS	WB	27.2	100.0	-0.65	-1.76	4.18	0	(REAL)	5.39
T032	WING TANK SB	WB	27.2	100.0	-0.65	1.76	4.18	0	(REAL)	5.39
T061	ROLL RED. TANK	WB	83.4	65.0	51.66	0.00	11.84	1358	REAL	12.78

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T062	ROLL RED. TANK	WB	136.9	65.0	7.45	0.00	5.58	1736 REAL	6.76
SUBTOTAL		WB	1117.5		36.00	0.26	2.97	3130	
TOTAL			3226.4		28.20	0.01	6.73	3324	

LOAD SUMMARY TABLE

ITEM	MASS t	LCG m	TCG m	VCG m	FRSM tm
Diesel Oil	53.0	50.02	-0.13	4.73	74.98
Fresh Water	21.2	67.25	0.00	3.41	91.23
Hydraulic oil	1.9	44.29	-4.33	1.89	0.61
Lubricating oil	10.7	43.36	3.26	3.43	7.30
Mass load	1974.2	22.05	0.00	9.01	0.00
Miscellaneous	36.2	54.74	-5.68	4.43	9.39
Sewage	10.4	53.89	-7.06	6.66	3.55
Sludge	1.4	48.40	3.29	1.80	6.99
Water Ballast	1117.5	36.00	0.26	2.97	3130.16
DEADWEIGHT	3226.4	28.20	0.01	6.73	3324.22
Lightweight	2704.9	39.15	-0.01	7.08	
Deadweight	3226.4	28.20	0.01	6.73	
Total weight	5931.3	33.20	0.00	6.89	

F L O A T I N G P O S I T I O N

Draught moulded	5.974	m	KM	8.80 m
Trim	-0.478	m	KG	6.89 m
Heel, SB=+	0.0	deg		
TA	6.213	m	GM0	1.91 m
TF	5.735	m	GMCORR	-0.56 m
Trimming moment	-3409	tonm	GM	1.35 m

STABILITY COMPONENTS

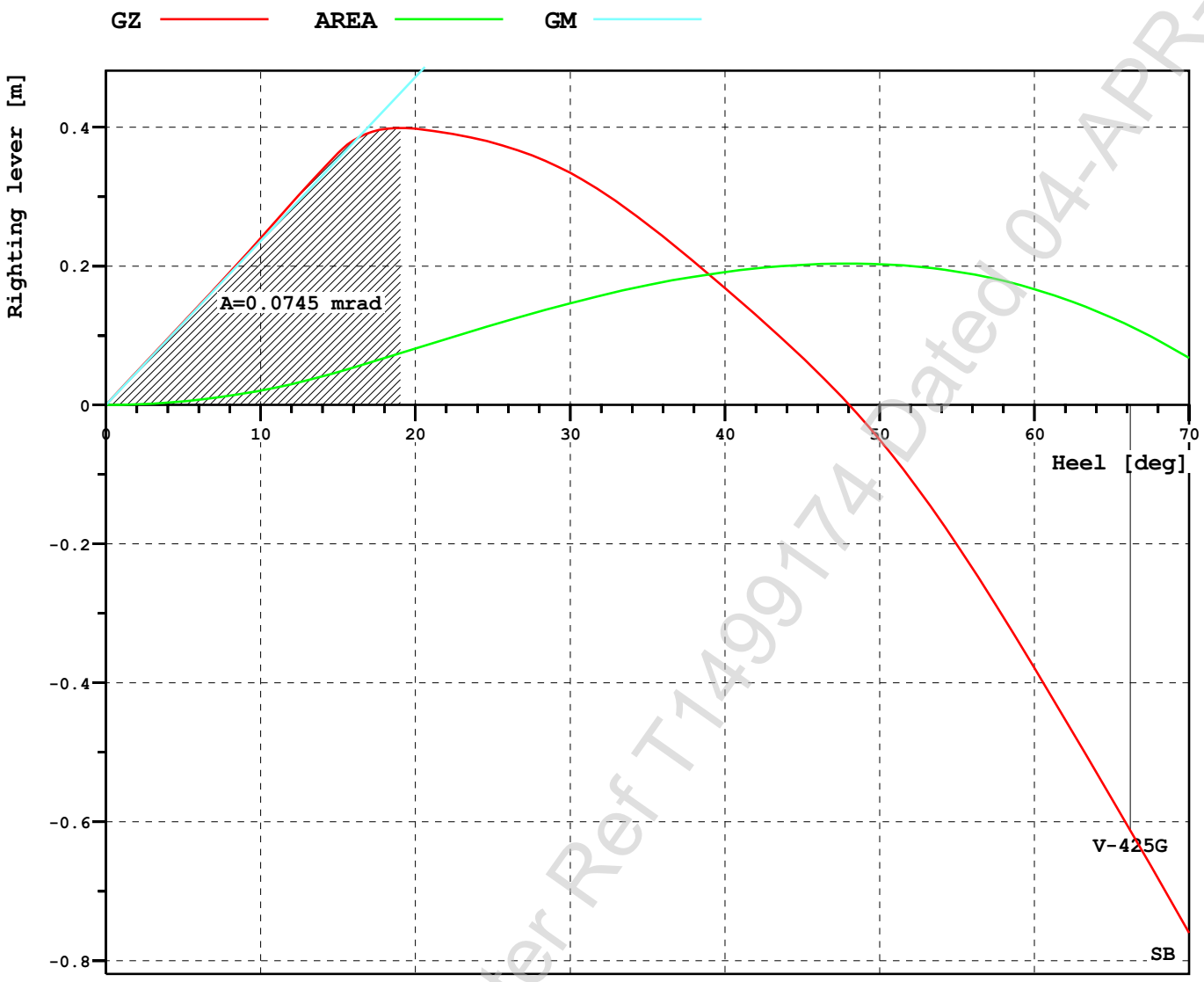
LOADING CONDITION L09, Max deckload - roll red tk - Arrival

HEEL degree	MS m	HPHI m	EPHI mrad	FSMOM tm	DGZ m	GMSIN m	KN m
0.0	0.000	0.00	0.000	0.0	0.000	0.000	-0.002
1.0	0.000	0.02	0.000	56.3	0.009	0.033	0.152
3.0	0.000	0.07	0.002	168.8	0.028	0.100	0.459
5.0	0.000	0.12	0.005	281.6	0.047	0.167	0.765
10.0	-0.003	0.24	0.021	526.5	0.089	0.332	1.523
15.0	-0.023	0.36	0.047	647.4	0.109	0.495	2.253
20.0	-0.137	0.40	0.081	706.1	0.119	0.654	2.871
30.0	-0.496	0.33	0.147	748.5	0.126	0.956	3.903
40.0	-0.937	0.17	0.191	738.4	0.124	1.230	4.719
50.0	-1.397	-0.05	0.203	699.5	0.118	1.465	5.343
60.0	-1.927	-0.38	0.167	639.3	0.108	1.657	5.693
70.0	-2.462	-0.76	0.068	564.8	0.095	1.797	5.807

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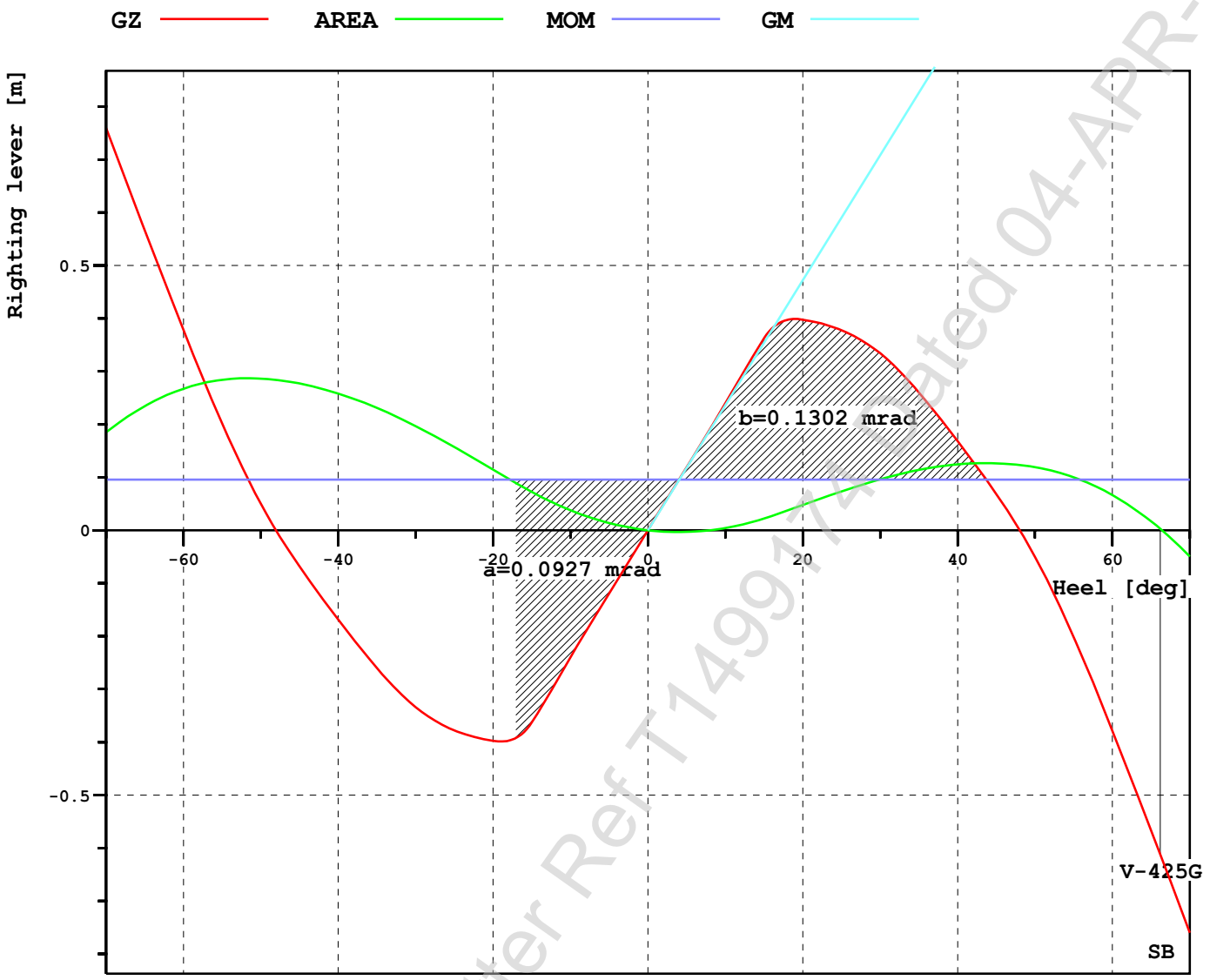
STABILITY CHECK PLOT



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IMOWEATHER CHECK PLOT



INTACT STABILITY CRITERIA

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM
V.AREA15-30	Area depending on the position of the Max. .	0.066	0.075	mrad	OK	1.263
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.045	mrad	OK	1.203
V.GZ0.2	Min. GZ > 0.2	0.200	0.334	m	OK	1.084
V.POSMAX15	Max. GZ at an angle > 15 deg.	15.000	19.030	deg	OK	0.055
V.GM0.15	GM > 0.15 m	0.150	1.352	m	OK	0.150
IMOWEATHER	IMO weather criterion	1.000	1.404		OK	1.078
2008IS-A2.3.1.2	Heeling angle due to steady wind <16 or <=8.	9.327	2.665	deg	OK	0.369
ABSFIFI1	Angle of heel < 1/2 freeboard immersion	5.829	0.587	deg	OK	0.129
ABSFIFI2	Area of residual stability >0.09 mrad	0.090	0.181	mrad	OK	0.943
TOTAL						1.263

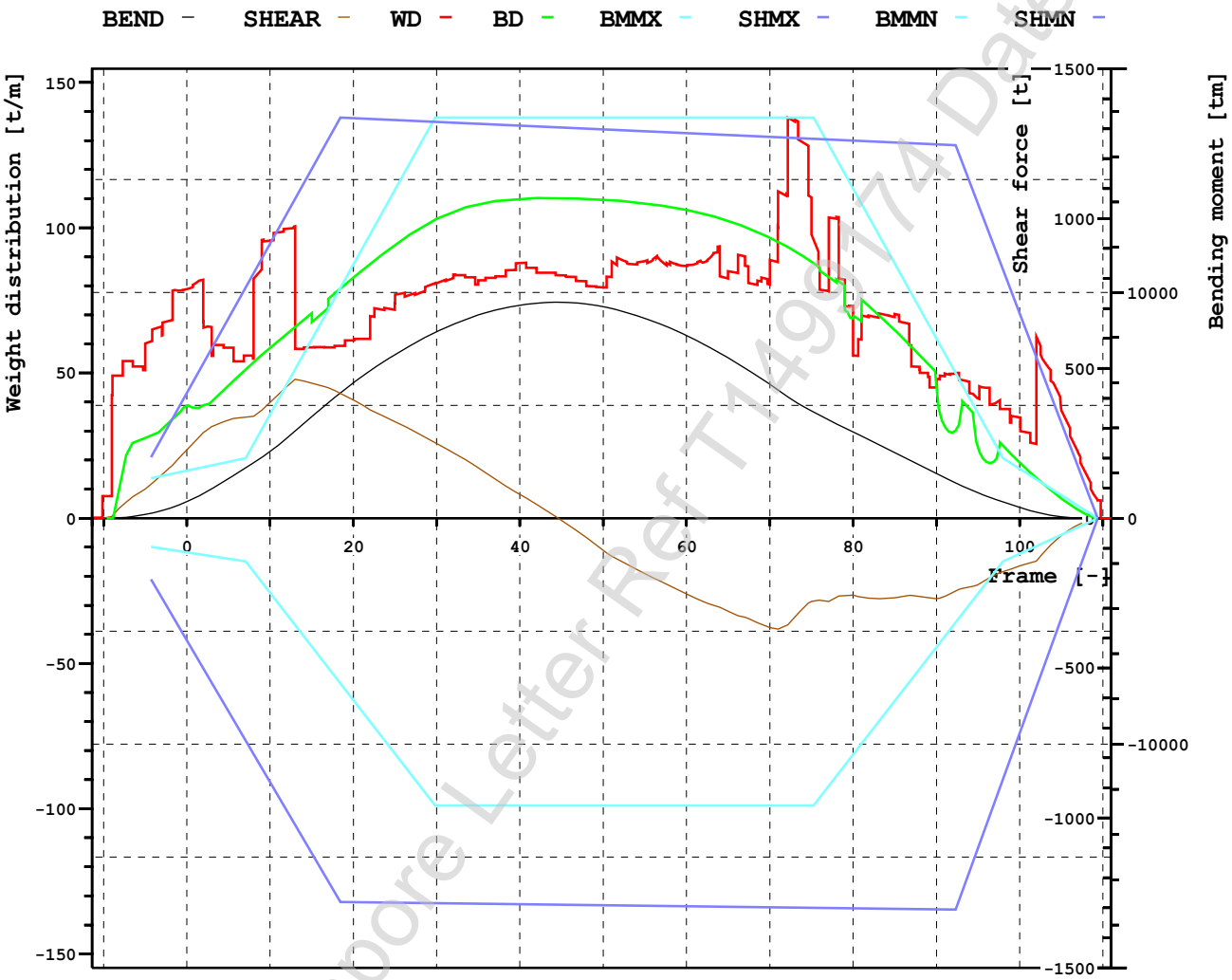
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MOST CRITICAL OPENINGS

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
V-424G	SUPPLY AIR ENGINE ROOM PS	UNPROTECTED	54.243	-3.413	14.288	-	8.419
V-425G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	54.243	3.767	14.288	66.2	8.419
V-421G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	55.890	2.093	14.288	-	8.429
D-651	ACCOMMODATION/C-DECK	UNPROTECTED	56.000	3.771	16.906	-	11.047
V-421E	EXHAUST AIR ENGINE ROOM PS	UNPROTECTED	55.993	-2.538	17.088	-	11.229
V-423E	EXHAUST AIR ENGINE ROOM SB	UNPROTECTED	55.993	3.423	17.088	-	11.229

STRENGTH DIAGRAM



STRENGTH SUMMARY

LOADING CONDITION L09, Max deckload - roll red tk - Arrival

			X	Frame
Shear force (min)	-369.9 t	(28.5%)	49.669 m	70.96
Shear force (max)	464.1 t	(43.5%)	9.100 m	13.00
Sagging moment	-0.9 tm	(0.0%)	76.567 m	109.38
Hogging moment	9558.2 tm	(53.9%)	31.002 m	44.29

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STRENGTH DISTRIBUTION

LOADING CONDITION L09, Max deckload - roll red tk - Arrival

X	FR	BEND	BMMN	BMMX	BMREL	SHEAR	SHMN	SHMX	SHREL	WD	BD
m	#	tm	tm	tm	%	t	t	t	%	t/m	t/m
-3.500	-5.00	168	0	0	0.0	98	0	0	0.0	60.17	27.62
0.000	0.00	730	-1510	2108	34.6	228	-407	418	54.5	78.85	38.65
3.500	5.00	1745	-1790	2498	69.8	327	-644	667	49.1	58.70	45.64
7.000	10.00	2963	-3292	4594	64.7	387	-881	916	42.3	95.59	58.63
10.500	15.00	4499	-5668	7911	56.9	450	-1117	1166	38.6	58.78	70.59
14.000	20.00	5995	-8045	11228	53.4	394	-1281	1335	29.5	61.26	82.82
17.500	25.00	7248	-10422	14545	49.8	322	-1283	1329	24.2	77.01	93.97
21.000	30.00	8255	-12709	17736	46.5	249	-1284	1323	18.8	80.96	103.05
24.500	35.00	8992	-12709	17736	50.7	168	-1286	1317	12.7	81.96	107.98
28.000	40.00	9423	-12709	17736	53.1	79	-1288	1310	6.1	87.86	109.84
31.500	45.00	9556	-12709	17736	53.9	-8	-1290	1304	0.6	83.60	110.14
35.000	50.00	9363	-12709	17736	52.8	-107	-1292	1298	8.3	79.50	109.62
38.500	55.00	8849	-12709	17736	49.9	-185	-1293	1292	14.3	88.47	108.33
42.000	60.00	8087	-12709	17736	45.6	-253	-1295	1285	19.5	86.80	106.14
45.500	65.00	7102	-12709	17736	40.0	-311	-1297	1279	24.0	85.02	102.29
49.000	70.00	5924	-12709	17736	33.4	-365	-1299	1273	28.1	88.91	96.54
52.500	75.00	4753	-12709	17736	26.8	-277	-1300	1267	21.3	110.40	88.34
56.000	80.00	3819	-10463	14601	26.2	-256	-1302	1260	19.7	73.11	68.81
59.500	85.00	2892	-8086	11284	25.6	-265	-1304	1254	20.3	70.29	64.77
63.000	90.00	1977	-5709	7968	24.8	-269	-1306	1248	20.6	47.98	45.95
66.500	95.00	1129	-3332	4651	24.3	-221	-1102	1050	20.1	41.23	24.35
70.000	100.00	479	-1571	2193	21.8	-159	-718	685	22.2	34.66	19.23
73.500	105.00	67	-733	1023	6.5	-58	-335	319	17.2	37.65	6.88

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DAMAGE STABILITY OF LOADING CONDITION L09

CRITICAL DAMAGE STABILITY CRITERIA FOR EACH DAMAGE CASE

RCR	DAM	STAGE	PHASE	REQ	ATTV	UNIT	STAT	MINGM
V.MINGZ.A469	D01	1	EQ	0.100	0.416	m	OK	0.290
V.MINGZ.A469	D02	1	EQ	0.100	0.373	m	OK	0.414
V.MINGZ.A469	D03	1	EQ	0.100	0.375	m	OK	0.415
V.RANGE.A469	D04	1	EQ	20.000	42.980	deg	OK	0.755
V.MINGZ.A469	D05	1	EQ	0.100	0.320	m	OK	0.623
V.RANGE.A469	D06	1	EQ	20.000	41.910	deg	OK	0.774
V.MINGZ.A469	D07	1	EQ	0.100	0.315	m	OK	0.644
V.MINGZ.A469	D08	1	EQ	0.100	0.368	m	OK	0.448
V.RANGE.A469	D09	1	EQ	20.000	44.054	deg	OK	0.640
V.RANGE.A469	D10	1	EQ	20.000	41.917	deg	OK	0.719
V.RANGE.A469	D11	1	EQ	20.000	42.408	deg	OK	0.708
V.RANGE.A469	D12	1	EQ	20.000	40.077	deg	OK	0.776
V.RANGE.A469	D13	1	EQ	20.000	40.520	deg	OK	0.768
V.RANGE.A469	D14	1	EQ	20.000	40.657	deg	OK	0.726
V.RANGE.A469	D15	1	EQ	20.000	41.182	deg	OK	0.714
V.RANGE.A469	D16	1	EQ	20.000	43.471	deg	OK	0.576
V.RANGE.A469	D17	1	EQ	20.000	43.933	deg	OK	0.569
V.MINGZ.A469	D18	1	EQ	0.100	0.353	m	OK	0.488
V.MINGZ.A469	D19	1	EQ	0.100	0.354	m	OK	0.492
V.MINGZ.A469	D20	1	EQ	0.100	0.353	m	OK	0.474
V.MINGZ.A469	D21	1	EQ	0.100	0.353	m	OK	0.477
TOTAL								0.776

DETERMINATION OF CRITICAL DAMAGE CASE

CRITICAL DAMAGE CASE : D12
STAGE OF CRITICAL DAMAGE CASE : 1
PHASE OF CRITICAL DAMAGE CASE : EQ

DAMAGE STABILITY CRITERIA - CRITICAL CASE

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM m
V.PROGR.A469	No progr. flooding	16.443	3.522	deg	OK	0.342
V.MAXHEEL.A469	Maximum heel	17.000	3.522	deg	OK	0.342
V.RANGE.A469	Range of pos. stab	20.000	40.077	deg	OK	0.776
V.MINGZ.A469	Min. righting lever	0.100	0.291	m	OK	0.711
TOTAL						0.776

DAMAGE CASE RESULTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	T m	TR m	HEEL degree	RESFLD m	OPEN	CROSSTM s
D12	1	EQ	SB	6.004	-0.563	3.5	1.98	D-362	-

DAMAGED COMPARTMENTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	NAME	DES	PERM	VOL m3	W t	XCG m	YCG m	ZCG m
D12	INTACT	EQ	-				0.0	0.000			
D12	1	EQ	SB	T023	WING TANK SB	0.95	52.0	53.320	19.41	7.81	3.46
D12	1	EQ	SB	T025	DB TANK SB	0.95	72.5	74.304	21.46	3.85	0.81
D12	1	EQ	SB	E2	Emrg. exit 1 SB	0.95	15.3	15.637	22.58	7.32	5.78

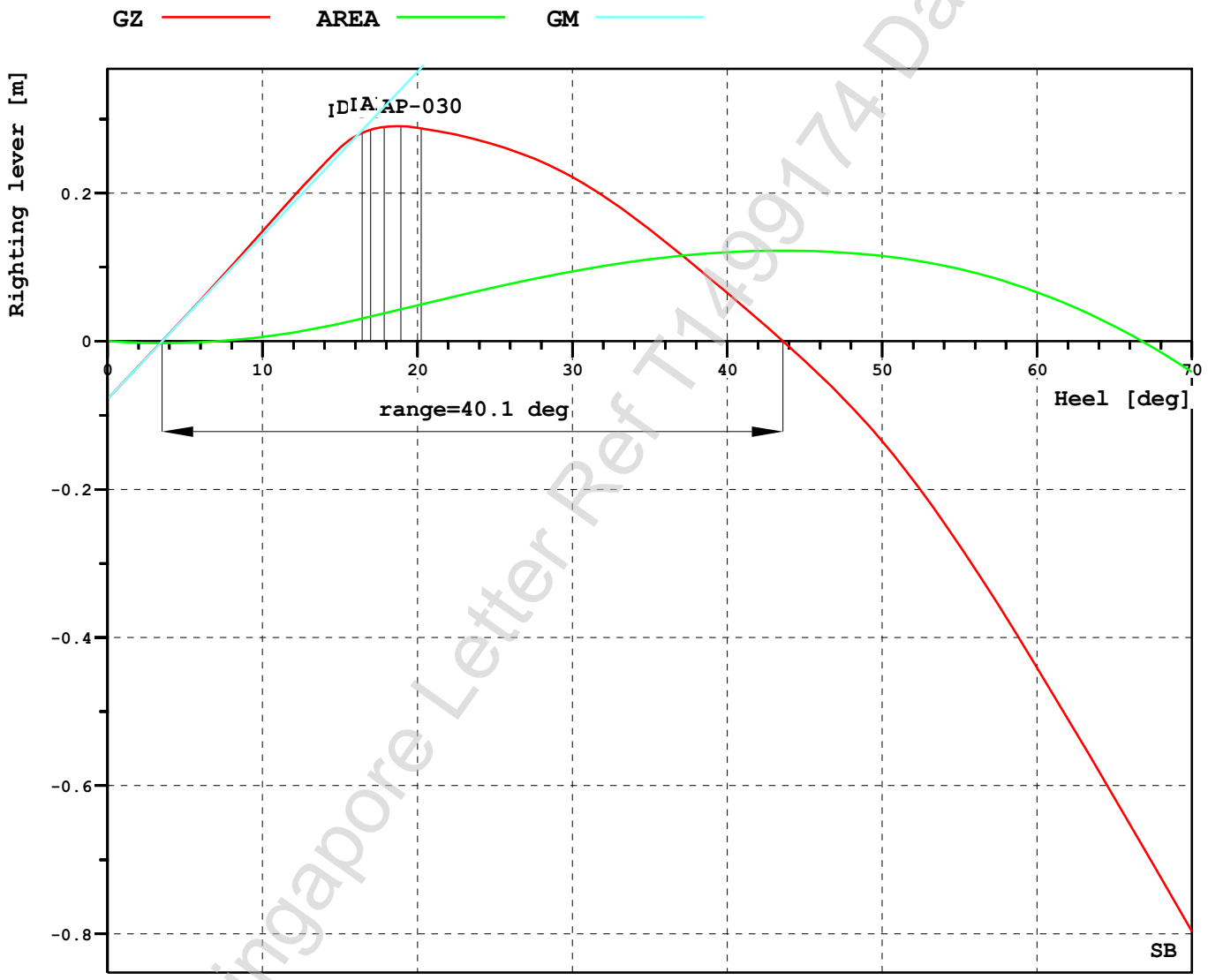
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CRITICAL OPENINGS - CRITICAL CASE

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
D-362	SB CARGORAIL #37/M-DECK	WEATHERTIGHT	25.900	8.550	8.600	16.4	1.978
D-360	SB CARGORAIL #57/M-DECK	WEATHERTIGHT	39.885	8.560	8.600	17.0	2.076
D-352	CHEMICAL/M-DECK	WEATHERTIGHT	53.192	8.375	8.600	17.9	2.182
AP-011	WING TANK SB	WEATHERTIGHT	47.211	8.754	8.953	18.9	2.469
AP-030	WING TANK SB	WEATHERTIGHT	5.250	7.801	9.168	20.2	2.445
AP-034	WING TANK SB	WEATHERTIGHT	-5.873	7.880	9.372	20.7	2.565
AP-036	WING TANK SB	WEATHERTIGHT	1.163	7.810	9.372	21.1	2.619
AP-027	WING TANK SB	WEATHERTIGHT	14.986	7.800	9.372	21.8	2.717

DAMAGE STABILITY CHECK PLOT - CRITICAL CASE



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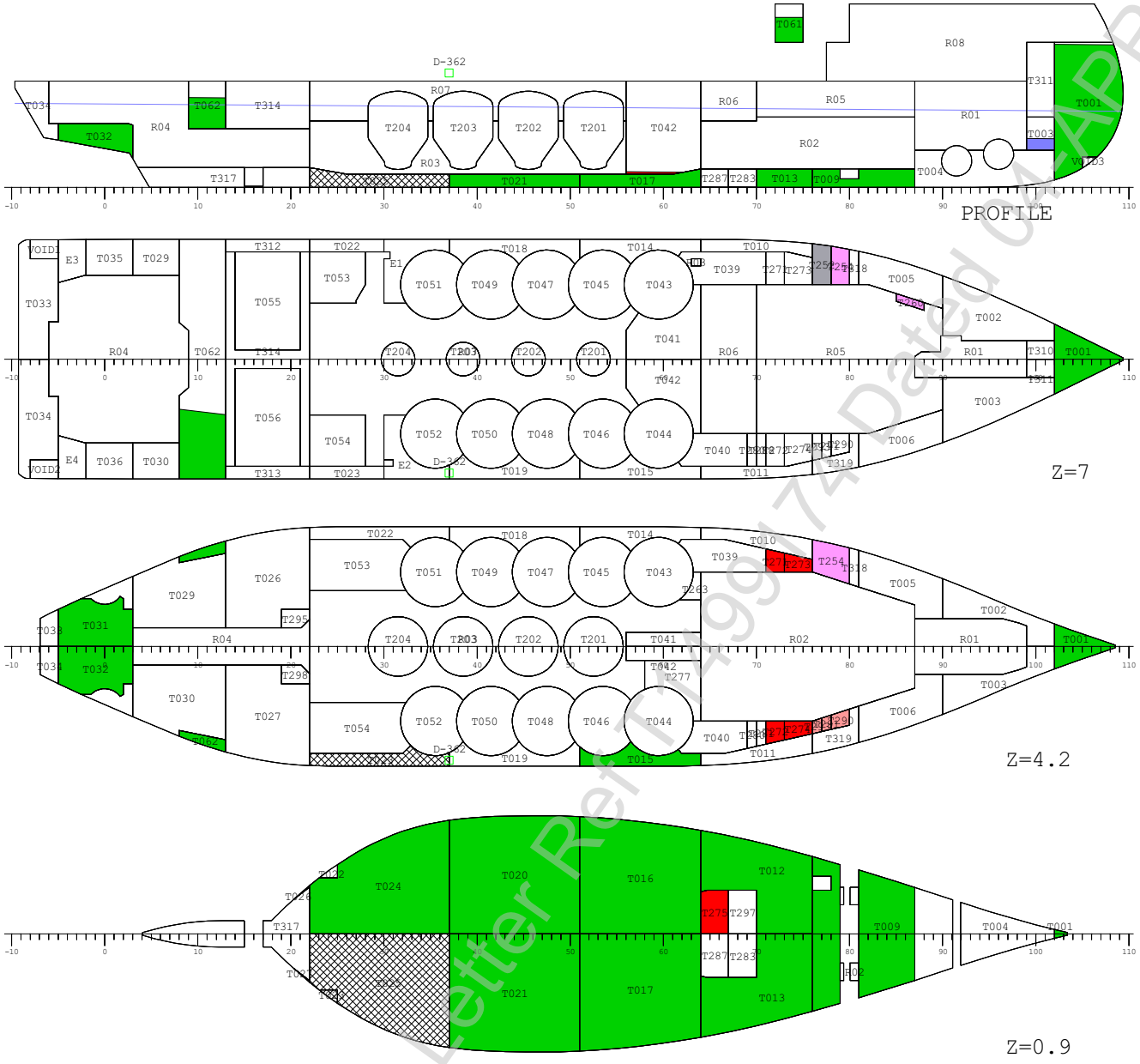
GZ-CURVE - CRITICAL CASE

HEEL	DISP	T	TR	GZ	MS	AREA
degree	t	m	m	m	m	mrad
0.0	5931.3	6.013	-0.565	-0.078	-0.078	0.000
1.0	5931.3	6.013	-0.564	-0.056	-0.089	-0.001
3.0	5931.3	6.007	-0.564	-0.012	-0.112	-0.002
5.0	5931.3	5.992	-0.559	0.033	-0.134	-0.002
10.0	5931.3	5.916	-0.515	0.148	-0.184	0.006
15.0	5931.3	5.794	-0.491	0.261	-0.234	0.024
20.0	5931.3	5.676	-0.582	0.288	-0.366	0.049
30.0	5931.3	5.415	-1.054	0.222	-0.735	0.094
40.0	5931.3	5.069	-1.985	0.065	-1.164	0.120
50.0	5931.3	4.558	-3.219	-0.135	-1.600	0.115
60.0	5931.3	3.923	-4.408	-0.441	-2.097	0.066
70.0	5931.3	3.190	-5.413	-0.797	-2.594	-0.041

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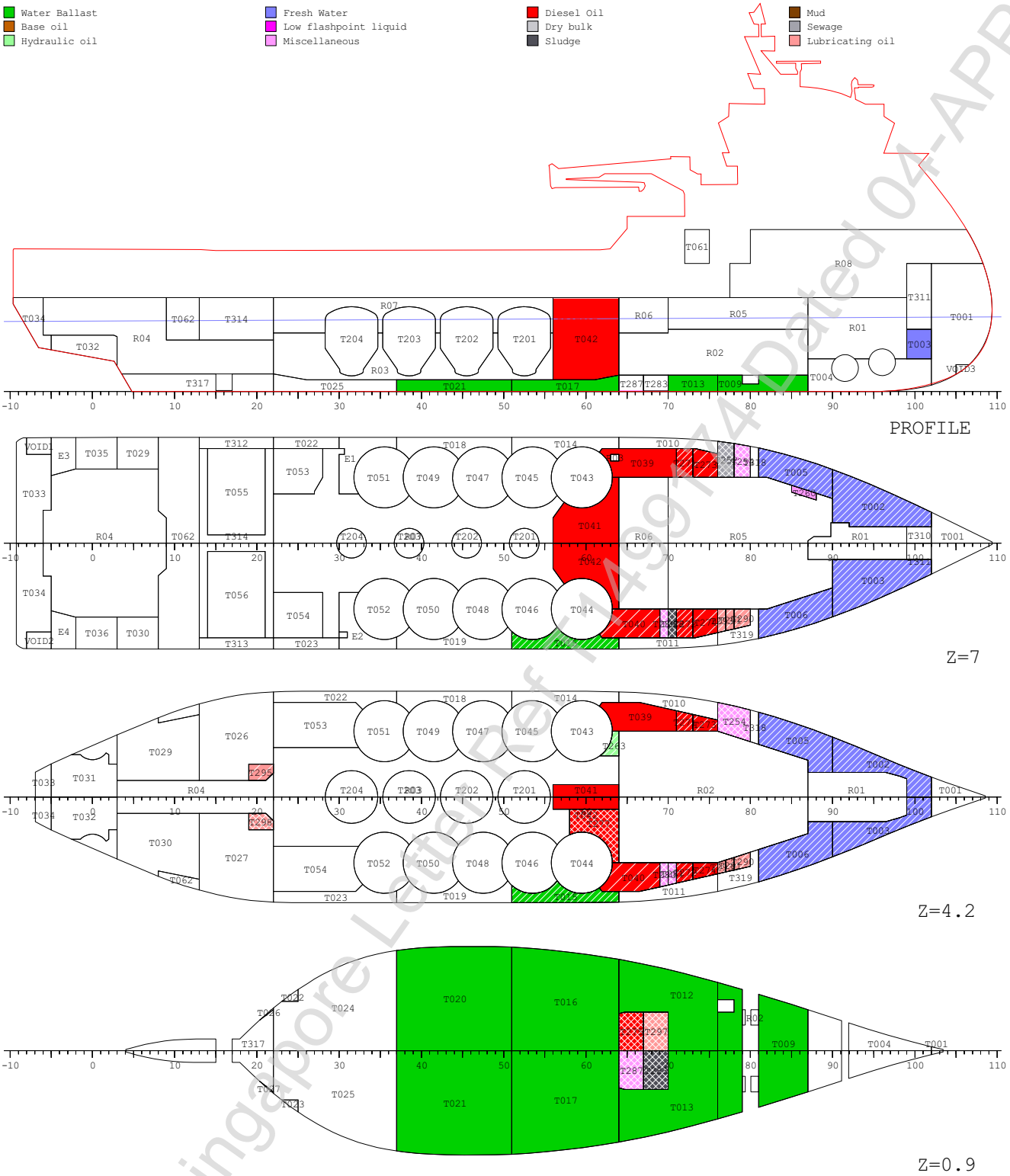
DAMAGE ARRANGEMENT - CRITICAL CASE



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12.10 L10 -> Pipe deckload - Departure



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LIST OF LOADS

NAME	DES	LOAD	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm	FRS	HM m
CONTENTS=Diesel Oil (RHO=0.86)										
T039	STORAGE TANK PS	FO	66.9	98.0	46.68	-6.76	4.91	6	REAL	7.88
T040	STORAGE TANK SB	FO	53.8	98.0	46.00	6.80	4.87	5	REAL	7.88
T041	STORAGE TANK PS	FO	68.5	98.0	42.16	-1.51	5.48	27	MAX	7.92
T042	STORAGE TANK SB	FO	68.5	98.0	42.16	1.51	5.48	27	MAX	7.92
T271	FO SETTLING TANK PS	FO	8.5	70.0	50.38	-6.53	4.93	1	REAL	6.72
T272	FO SETTLING TANK SB	FO	8.5	70.0	50.38	6.53	4.93	1	REAL	6.72
T273	FO SERVICE TANK PS	FO	10.6	70.0	52.09	-6.38	5.15	2	REAL	6.90
T274	FO SERVICE TANK SB	FO	10.6	70.0	52.09	6.38	5.15	2	REAL	6.90
T275	FO DRAIN TANK	FO	0.8	10.0	45.85	-1.64	0.07	5	REAL	0.14
T277	FO OVERFLOW TANK	FO	3.6	10.0	42.62	2.34	1.31	10	REAL	1.53
T279	EMERG. GEN. FO TANK	FO	1.9	98.0	61.60	-1.38	24.57	0	REAL	25.80
SUBTOTAL		FO	302.3		45.14	-0.27	5.25	86		
CONTENTS=Fresh Water (RHO=1)										
T002	WING TANK PS	FW	56.8	50.0	66.71	-2.66	4.87	45	MAX	6.35
T003	WING TANK SB	FW	57.5	50.0	66.69	2.65	4.88	46	MAX	6.35
T005	WING TANK PS	FW	49.1	50.0	60.05	-4.39	3.34	29	REAL	4.79
T006	WING TANK SB	FW	49.1	50.0	60.05	4.39	3.34	29	REAL	4.79
SUBTOTAL		FW	212.5		63.63	0.01	4.17	149		
CONTENTS=Hydraulic oil (RHO=0.924)										
T263	HYDRAULIC OIL STORE TANK	HO	3.3	70.0	44.30	-4.34	2.32	1	REAL	3.33
CONTENTS=Lubricating oil (RHO=0.924)										
T290	LO STORE TANK MAIN ENG.	LO	5.9	70.0	55.30	5.76	4.90	0	REAL	6.82
T291	LO STORE TANK MAIN ENG.	LO	3.1	70.0	54.25	6.04	5.11	0	REAL	6.92
T293	LO STORE TANK THRUSTER	LO	3.2	70.0	53.55	6.21	5.24	0	REAL	6.97
T295	LO STORE TANK AZIMUTH THR.	LO	4.0	70.0	14.35	-2.10	2.95	0	REAL	3.71
T297	LO DRAIN TANK	LO	0.9	10.0	47.95	-1.64	0.07	6	REAL	0.14
T298	DIRTY LO TANK	LO	0.6	10.0	14.35	2.10	2.31	0	REAL	2.41
SUBTOTAL		LO	17.6		43.89	3.64	4.24	8		
CONTENTS=Mass load (RHO=1)										
M1	CREW	MASS	2.6	0.0	57.40	0.00	15.50	0		0.00
M2	PROVISION	MASS	7.4	0.0	56.35	-6.50	12.40	0		0.00
M3	STORES	MASS	7.4	0.0	72.10	0.00	12.40	0		0.00
M4	PIPE LOAD	MASS	2276.1	0.0	21.70	0.00	9.29	0		0.00
SUBTOTAL		MASS	2293.5		22.02	-0.02	9.32	0		
CONTENTS=Miscellaneous (RHO=1)										
T280	BILGE WATER HOLDING TANK	MIS	0.9	10.0	48.64	6.17	2.03	0	REAL	2.57
T281	BILGE WATER SETTLING TANK	MIS	0.4	10.0	49.34	6.04	1.76	0	REAL	2.08
T287	BILGE WATER DRAIN TANK	MIS	0.9	10.0	45.85	1.64	0.07	6	REAL	0.14
T254	GREY WATER TANK	MIS	3.2	10.0	54.62	-5.79	1.92	1	REAL	2.34
T260	WATER MIST TANK	MIS	2.3	98.0	60.55	-4.30	6.28	0	REAL	7.27
SUBTOTAL		MIS	7.8		54.40	-2.49	3.01	8		
CONTENTS=Sewage (RHO=1)										
T252	BLACK WATER TANK	SEW	1.1	10.0	53.89	-6.93	5.45	2	REAL	5.59
CONTENTS=Sludge (RHO=1)										
T283	SLUDGE DRAIN TANK	SLU	0.9	10.0	47.95	1.64	0.07	6	REAL	0.14
T288	SLUDGE SETTLING TANK	SLU	0.4	10.0	49.35	6.75	5.44	1	REAL	5.59
SUBTOTAL		SLU	1.4		48.40	3.29	1.80	7		
CONTENTS=Water Ballast (RHO=1.025)										
T001	FOREPEAK TANK	WB	0.0	0.0	73.11	0.00	7.21	34	MAX	0.00
T009	DB TANK CE	WB	77.3	100.0	56.90	0.09	0.73	0	(REAL)	1.40
T012	DB TANK PS	WB	56.2	100.0	49.41	-3.85	0.76	0	(REAL)	1.40
T013	DB TANK SB	WB	56.2	100.0	49.41	3.85	0.76	0	(REAL)	1.40
T015	WING TANK SB	WB	15.4	25.0	40.17	8.33	2.27	1	REAL	3.09
T016	DB TANK PS	WB	85.6	100.0	40.20	-4.33	0.62	0	REAL	1.50
T017	DB TANK SB	WB	85.6	100.0	40.20	4.33	0.62	0	REAL	1.50
T020	DB TANK PS	WB	96.7	100.0	30.84	-4.63	0.61	0	(REAL)	1.92
T021	DB TANK SB	WB	96.7	100.0	30.84	4.63	0.61	0	(REAL)	1.92
SUBTOTAL		WB	569.7		41.10	0.24	0.70	35		

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TOTAL	3409.2	30.08	0.01	7.14	296

LOAD SUMMARY TABLE

ITEM	MASS t	LCG m	TCG m	VCG m	FRSM tm
Diesel Oil	302.3	45.14	-0.27	5.25	85.89
Fresh Water	212.5	63.63	0.01	4.17	149.15
Hydraulic oil	3.3	44.30	-4.34	2.32	0.61
Lubricating oil	17.6	43.89	3.64	4.24	7.62
Mass load	2293.5	22.02	-0.02	9.32	0.00
Miscellaneous	7.8	54.40	-2.49	3.01	7.72
Sewage	1.1	53.89	-6.93	5.45	2.31
Sludge	1.4	48.40	3.29	1.80	6.99
Water Ballast	569.7	41.10	0.24	0.70	35.31
DEADWEIGHT	3409.2	30.08	0.01	7.14	295.60
Lightweight	2704.9	39.15	-0.01	7.08	
Deadweight	3409.2	30.08	0.01	7.14	
Total weight	6114.1	34.09	0.00	7.12	

F L O A T I N G P O S I T I O N

Draught moulded	6.180	m	KM	8.70 m
Trim	0.362	m	KG	7.12 m
Heel, SB=+	0.0	deg		
TA	5.999	m	GM0	1.58 m
TF	6.361	m	GMCORR	-0.05 m
Trimming moment	2583	tonm	GM	1.54 m

STABILITY COMPONENTS

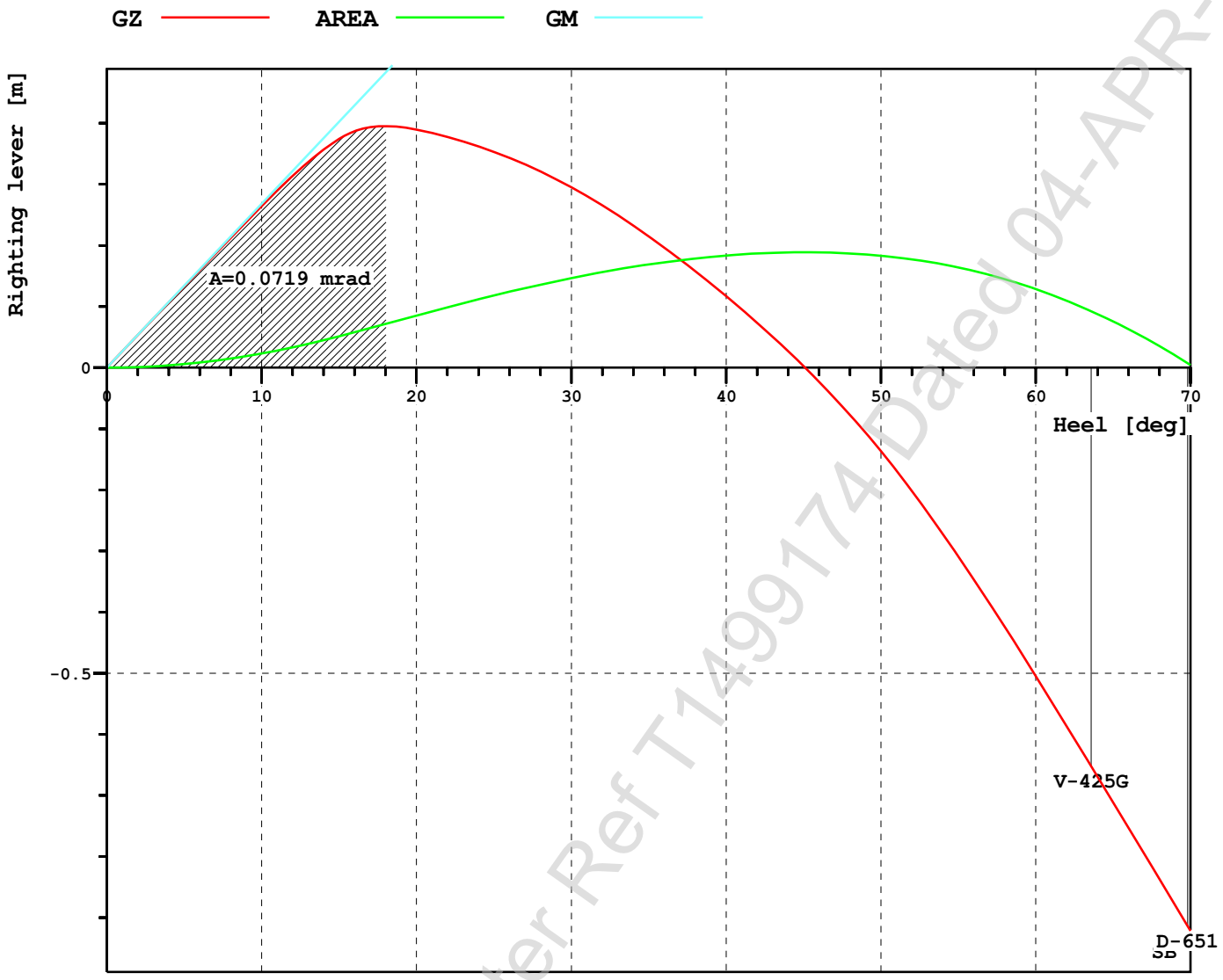
LOADING CONDITION L10, Pipe deckload - Departure

HEEL degree	MS m	HPHI m	EPHI mrad	FSMOM tm	DGZ m	GMSIN m	KN m
0.0	0.000	0.00	0.000	0.0	0.000	0.000	-0.001
1.0	0.000	0.03	0.000	5.0	0.001	0.028	0.151
3.0	0.000	0.08	0.002	15.0	0.002	0.083	0.454
5.0	-0.001	0.13	0.006	25.0	0.004	0.138	0.757
10.0	-0.003	0.26	0.023	48.9	0.008	0.275	1.507
15.0	-0.024	0.37	0.051	71.4	0.012	0.410	2.227
20.0	-0.137	0.39	0.085	92.9	0.015	0.542	2.838
30.0	-0.475	0.30	0.146	134.2	0.022	0.792	3.874
40.0	-0.873	0.12	0.183	177.0	0.029	1.018	4.720
50.0	-1.313	-0.14	0.183	226.5	0.037	1.214	5.351
60.0	-1.832	-0.50	0.129	276.6	0.045	1.372	5.703
70.0	-2.357	-0.92	0.004	324.6	0.053	1.489	5.819

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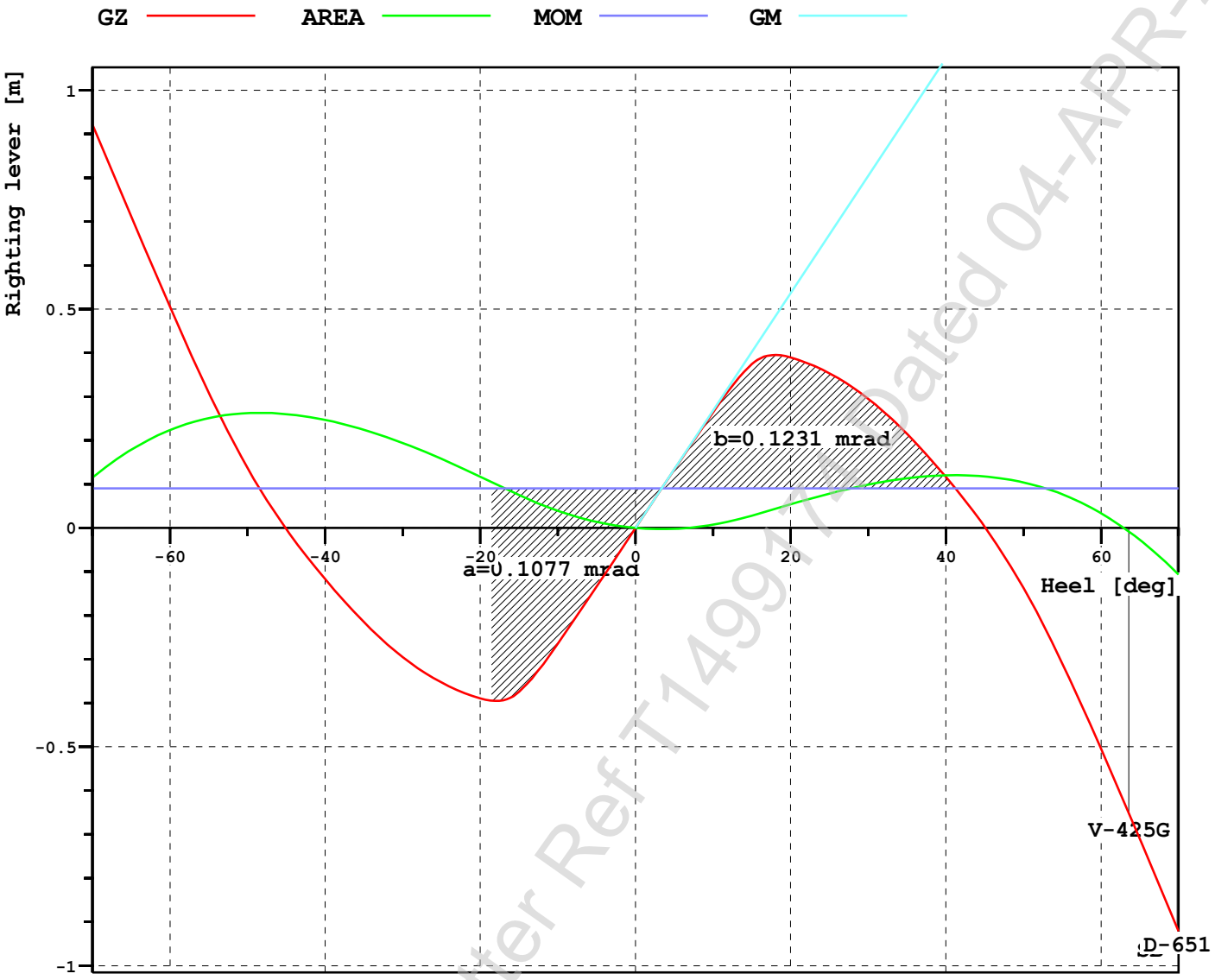
STABILITY CHECK PLOT



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IMOWEATHER CHECK PLOT



INTACT STABILITY CRITERIA

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM
V.AREA15-30	Area depending on the position of the Max. .	0.067	0.072	mrad	OK	1.479
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.037	mrad	OK	1.466
V.GZ0.2	Min. GZ > 0.2	0.200	0.295	m	OK	1.346
V.POSMAX15	Max. GZ at an angle > 15 deg.	15.000	18.024	deg	OK	0.615
V.GM0.15	GM > 0.15 m	0.150	1.536	m	OK	0.150
IMOWEATHER	IMO weather criterion	1.000	1.144		OK	1.431
2008IS-A2.3.1.2	Heeling angle due to steady wind <16 or <=8.	8.925	2.254	deg	OK	0.401
ABSFIFI1	Angle of heel < 1/2 freeboard immersion	5.578	0.515	deg	OK	0.150
ABSFIFI2	Area of residual stability >0.09 mrad	0.090	0.174	mrad	OK	1.145
TOTAL						1.479

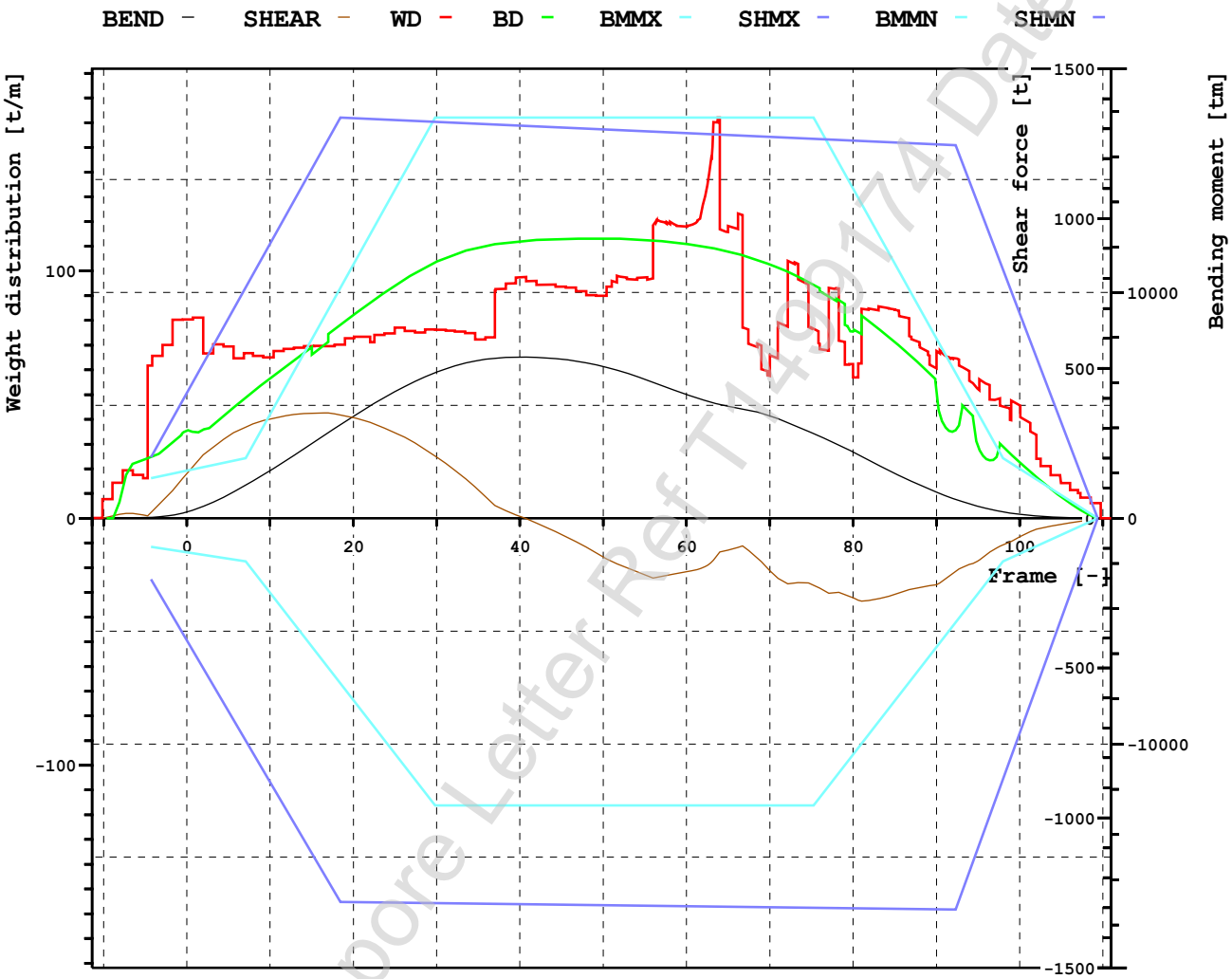
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MOST CRITICAL OPENINGS

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
V-421G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	55.890	2.093	14.288	-	8.021
V-424G	SUPPLY AIR ENGINE ROOM PS	UNPROTECTED	54.243	-3.413	14.288	-	8.029
V-425G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	54.243	3.767	14.288	63.6	8.029
D-651	ACCOMMODATION/C-DECK	UNPROTECTED	56.000	3.771	16.906	69.8	10.639
V-421E	EXHAUST AIR ENGINE ROOM PS	UNPROTECTED	55.993	-2.538	17.088	-	10.821
V-423E	EXHAUST AIR ENGINE ROOM SB	UNPROTECTED	55.993	3.423	17.088	-	10.821

STRENGTH DIAGRAM



STRENGTH SUMMARY

LOADING CONDITION L10, Pipe deckload - Departure

			X	Frame
Shear force (min)	-276.6 t	(21.2%)	56.700 m	81.00
Shear force (max)	351.7 t	(28.3%)	11.563 m	16.52
Sagging moment	-1.3 tm	(0.0%)	-6.750 m	-9.64
Hogging moment	7128.4 tm	(40.2%)	28.350 m	40.50

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STRENGTH DISTRIBUTION

LOADING CONDITION L10, Pipe deckload - Departure

X	FR	BEND	BMMN	BMMX	BMREL	SHEAR	SHMN	SHMX	SHREL	WD	BD
m	#	tm	tm	tm	%	t	t	t	%	t/m	t/m
-3.500	-5.00	33	0	0	0.0	10	0	0	0.0	16.23	23.84
0.000	0.00	278	-1510	2108	13.2	148	-407	418	35.5	80.38	35.42
3.500	5.00	1039	-1790	2498	41.5	274	-644	667	41.1	69.48	42.99
7.000	10.00	2108	-3292	4594	46.0	331	-881	916	36.1	65.13	56.63
10.500	15.00	3303	-5668	7911	41.8	350	-1117	1166	30.0	69.56	69.27
14.000	20.00	4512	-8045	11228	40.2	336	-1281	1335	25.1	72.84	82.19
17.500	25.00	5602	-10422	14545	38.5	285	-1283	1329	21.4	77.15	94.02
21.000	30.00	6464	-12709	17736	36.4	204	-1284	1323	15.4	76.24	103.77
24.500	35.00	6986	-12709	17736	39.4	94	-1286	1317	7.1	72.34	109.39
28.000	40.00	7127	-12709	17736	40.2	6	-1288	1310	0.5	97.52	111.94
31.500	45.00	7040	-12709	17736	39.7	-55	-1290	1304	4.2	93.67	112.92
35.000	50.00	6718	-12709	17736	37.9	-130	-1292	1298	10.0	89.98	113.08
38.500	55.00	6151	-12709	17736	34.7	-189	-1293	1292	14.6	97.35	112.47
42.000	60.00	5482	-12709	17736	30.9	-179	-1295	1285	13.8	118.03	110.97
45.500	65.00	4957	-12709	17736	27.9	-106	-1297	1279	8.2	118.18	107.79
49.000	70.00	4532	-12709	17736	25.5	-175	-1299	1273	13.5	65.88	102.67
52.500	75.00	3793	-12709	17736	21.4	-221	-1300	1267	17.0	77.12	94.91
56.000	80.00	2928	-10463	14601	20.1	-265	-1302	1260	20.4	62.37	75.54
59.500	85.00	1983	-8086	11284	17.6	-254	-1304	1254	19.5	84.42	71.27
63.000	90.00	1155	-5709	7968	14.5	-222	-1306	1248	17.0	67.62	51.82
66.500	95.00	525	-3332	4651	11.3	-142	-1102	1050	12.9	52.23	29.15
70.000	100.00	174	-1571	2193	7.9	-62	-718	685	8.7	45.63	22.60
73.500	105.00	40	-733	1023	3.9	-21	-335	319	6.2	14.17	8.59

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DAMAGE STABILITY OF LOADING CONDITION L10

CRITICAL DAMAGE STABILITY CRITERIA FOR EACH DAMAGE CASE

RCR	DAM	STAGE	PHASE	REQ	ATTV	UNIT	STAT	MINGM
V.MINGZ.A469	D01	1	EQ	0.100	0.399	m	OK	0.461
V.MINGZ.A469	D02	1	EQ	0.100	0.395	m	OK	0.483
V.MINGZ.A469	D03	1	EQ	0.100	0.392	m	OK	0.486
V.RANGE.A469	D04	1	EQ	20.000	41.737	deg	OK	0.817
V.RANGE.A469	D05	1	EQ	20.000	41.420	deg	OK	0.862
V.RANGE.A469	D06	1	EQ	20.000	38.810	deg	OK	0.979
V.RANGE.A469	D07	1	EQ	20.000	38.455	deg	OK	1.011
V.RANGE.A469	D08	1	EQ	20.000	41.903	deg	OK	0.745
V.RANGE.A469	D09	1	EQ	20.000	41.112	deg	OK	0.826
V.RANGE.A469	D10	1	EQ	20.000	39.102	deg	OK	0.911
V.RANGE.A469	D11	1	EQ	20.000	39.675	deg	OK	0.893
V.RANGE.A469	D12	1	EQ	20.000	34.410	deg	OK	1.108
V.RANGE.A469	D13	1	EQ	20.000	35.032	deg	OK	1.091
V.RANGE.A469	D14	1	EQ	20.000	38.054	deg	OK	0.912
V.RANGE.A469	D15	1	EQ	20.000	38.642	deg	OK	0.894
V.RANGE.A469	D16	1	EQ	20.000	33.303	deg	OK	1.016
V.RANGE.A469	D17	1	EQ	20.000	33.978	deg	OK	1.001
V.MINGZ.A469	D18	1	EQ	0.100	0.352	m	OK	0.622
V.MINGZ.A469	D19	1	EQ	0.100	0.355	m	OK	0.618
V.RANGE.A469	D20	1	EQ	20.000	41.253	deg	OK	0.603
V.RANGE.A469	D21	1	EQ	20.000	41.776	deg	OK	0.590
TOTAL								1.108

DETERMINATION OF CRITICAL DAMAGE CASE

CRITICAL DAMAGE CASE : D12
STAGE OF CRITICAL DAMAGE CASE : 1
PHASE OF CRITICAL DAMAGE CASE : EQ

DAMAGE STABILITY CRITERIA - CRITICAL CASE

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM m
V.PROGR.A469	No progr. flooding	15.189	4.693	deg	OK	0.584
V.MAXHEEL.A469	Maximum heel	17.000	4.693	deg	OK	0.584
V.RANGE.A469	Range of pos. stab	20.000	34.410	deg	OK	1.108
V.MINGZ.A469	Min. righting lever	0.100	0.255	m	OK	0.982
TOTAL						1.108

DAMAGE CASE RESULTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	T m	TR m	HEEL degree	RESFLD m	OPEN	CROSSTM s
D12	1	EQ	SB	6.250	0.180	4.7	1.60	D-352	-

DAMAGED COMPARTMENTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	NAME	DES	PERM	VOL m3	W t	XCG m	YCG m	ZCG m
D12	INTACT	EQ	-				0.0	0.000			
D12	1	EQ	SB	T023	WING TANK SB	0.95	53.3	54.657	19.38	7.82	3.54
D12	1	EQ	SB	T025	DB TANK SB	0.95	72.5	74.305	21.46	3.85	0.81
D12	1	EQ	SB	E2	Emrg. exit 1 SB	0.95	17.9	18.365	22.59	7.34	5.92

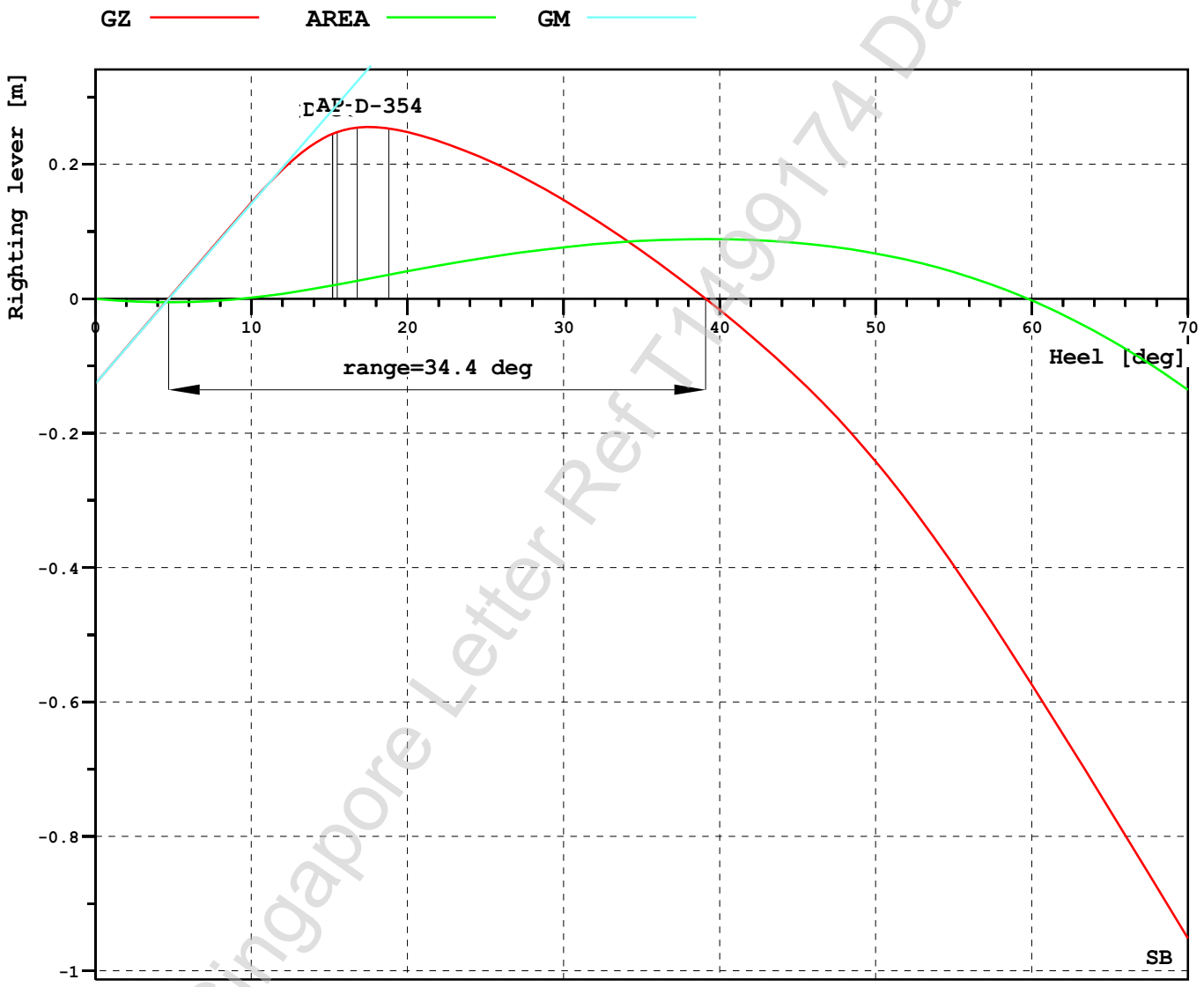
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CRITICAL OPENINGS - CRITICAL CASE

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
D-360	SB CARGORAIL #57/M-DECK	WEATHERTIGHT	39.885	8.560	8.600	15.2	1.613
D-352	CHEMICAL/M-DECK	WEATHERTIGHT	53.192	8.375	8.600	15.2	1.598
D-362	SB CARGORAIL #37/M-DECK	WEATHERTIGHT	25.900	8.550	8.600	15.5	1.646
AP-011	WING TANK SB	WEATHERTIGHT	47.211	8.754	8.953	16.8	1.933
D-354	PAINT/M-DECK	WEATHERTIGHT	53.192	6.600	8.600	18.8	1.744
AP-290	LO STORE TANK MAIN ENG	WEATHERTIGHT	52.985	7.410	8.927	19.1	2.004
AP-291	LO STORE TANK MAIN ENG	WEATHERTIGHT	52.985	7.170	8.927	19.7	2.023
AP-280	BILGE WATER HOLDING TANK	WEATHERTIGHT	47.738	7.275	8.951	19.7	2.051

DAMAGE STABILITY CHECK PLOT - CRITICAL CASE



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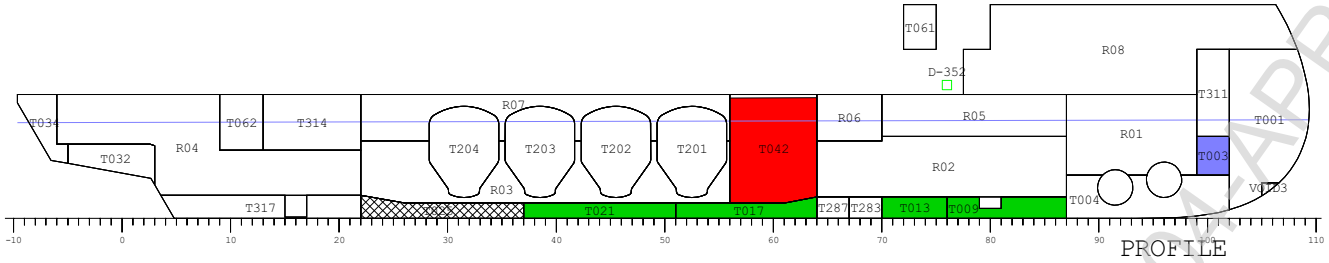
GZ-CURVE - CRITICAL CASE

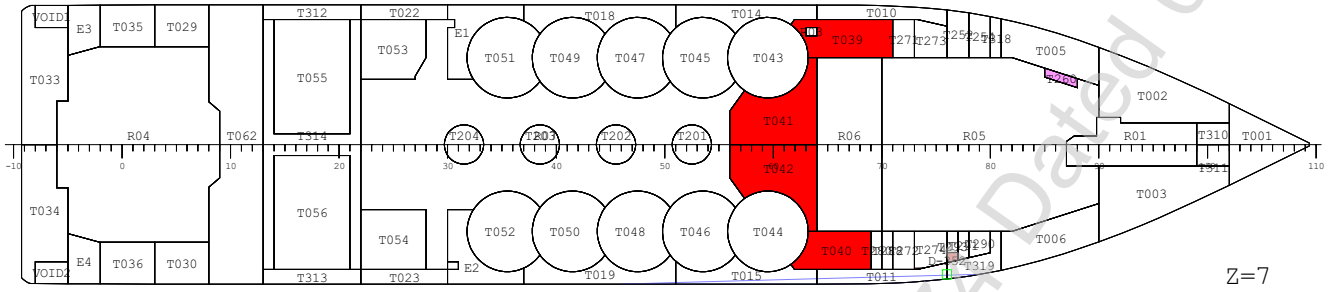
HEEL	DISP	T	TR	GZ	MS	AREA
degree	t	m	m	m	m	mrad
0.0	6114.1	6.270	0.175	-0.126	-0.126	0.000
1.0	6114.1	6.270	0.175	-0.100	-0.127	-0.002
3.0	6114.1	6.263	0.177	-0.046	-0.129	-0.005
5.0	6114.1	6.248	0.180	0.008	-0.130	-0.005
10.0	6114.1	6.168	0.221	0.143	-0.132	0.001
15.0	6114.1	6.051	0.267	0.243	-0.167	0.019
20.0	6114.1	5.942	0.194	0.248	-0.294	0.041
30.0	6114.1	5.688	-0.310	0.147	-0.645	0.076
40.0	6114.1	5.337	-1.328	-0.016	-1.035	0.089
50.0	6114.1	4.846	-2.589	-0.242	-1.456	0.067
60.0	6114.1	4.233	-3.788	-0.574	-1.946	-0.003
70.0	6114.1	3.516	-4.810	-0.951	-2.440	-0.136

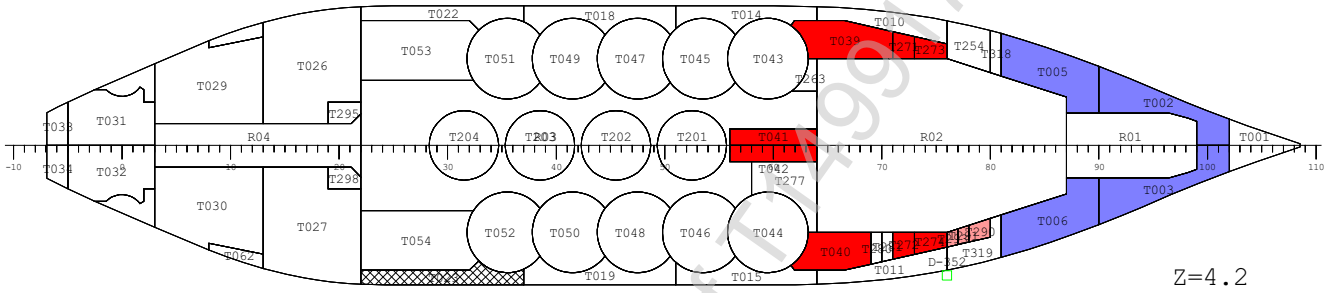
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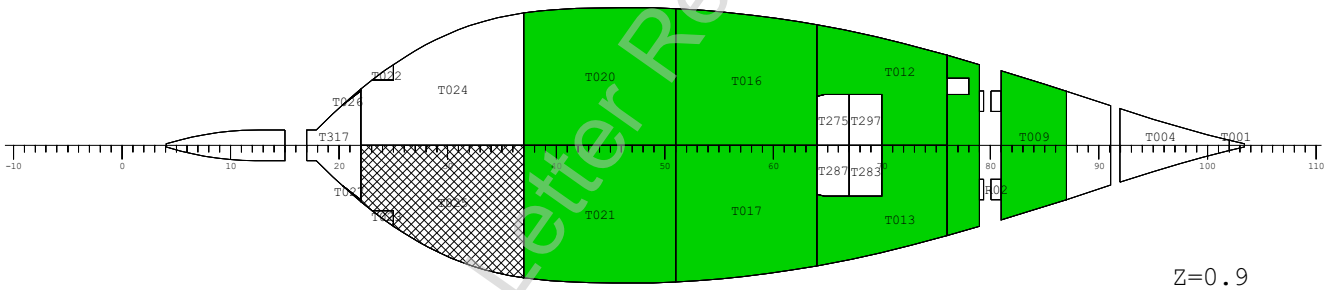
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DAMAGE ARRANGEMENT - CRITICAL CASE









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CALCULATION OF PIPE LOAD WITH ENTRAPPED WATER

INPUT DATA:

Loading condition name	L10	
Length of the vessel acc. ICLL 1966	79.614	[m]
Total weight of pipe cargo	1969.8	[t]
Total number of pipes	98	[-]
No. of pipe lenghts	4	[-]
Weight of each pipe	20.1	[t]
Length of each pipe	12.5	[m]
Outer diameter of pipe	1.235	[m]
Inner diameter of pipe	0.966	[m]
Breadth between cargo rails	15.076	[m]
Freeboard	1.3	[m]
Depth to cargo deck	8	[m]
Thickeness of deck covering	75	[mm]

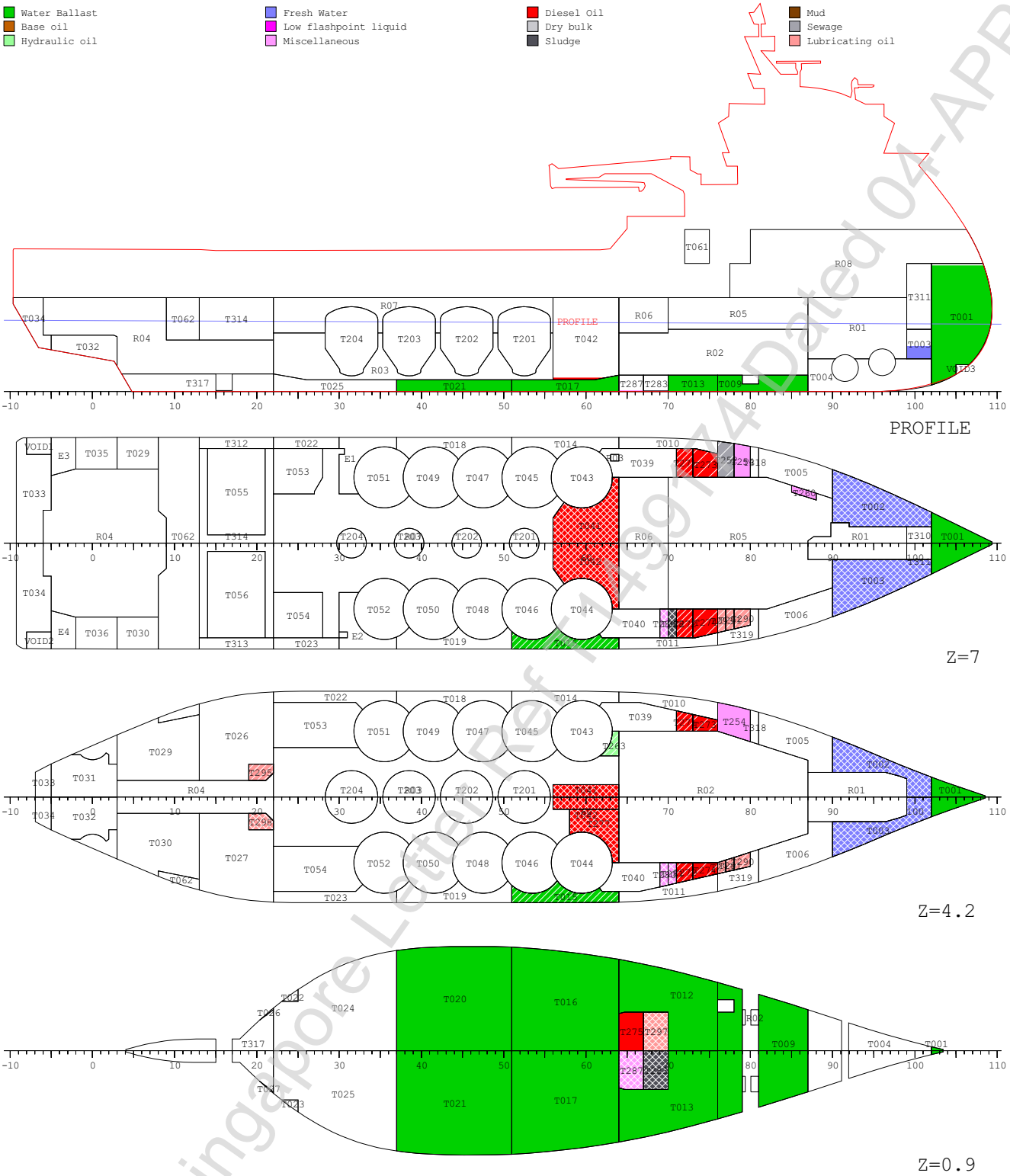
OUTPUT DATA:

Layers [-]	No. of pipes [-]	Weight [t]	VCG [m]	Moment [tm]
Weight of pipes :				
1. Layer	48	964.8	8.6925	8386.52
2. Layer	44	884.4	9.76204	8633.55
3. Layer	6	120.6	10.8316	1306.29
Total	98	1969.8	9.30367	18326.4
Weight of water :				
Inside pipes	98	920.246	9.30367	8561.66
Between pipes	92	72.4782	9.30367	674.313
Between pipes and deck	44	92.2622	8.21293	757.743
Total		1084.99	9.21092	9993.72
Percent coefficient of entrapped water 28.2 %				
Entrapped water (assumed percentage)		306.273	9.21092	2821.06
Total moment (pipes+water)		2276.07	9.29119	21147.4

See ABS Singapore Letter Ref T 499174 Dated 04-APR-2016

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12.11 L11 -> Pipe deckload - Arrival



See ABS Singapore Letter Ref 11499 X4 Dated 04-APR-2016

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LIST OF LOADS

NAME	DES	LOAD	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm	FRS	HM m
CONTENTS=Diesel Oil (RHO=0.86)										
T041	STORAGE TANK PS	FO	0.7	1.0	41.24	-0.52	1.10	27	MAX	1.20
T042	STORAGE TANK SB	FO	0.7	1.0	41.24	0.52	1.10	27	MAX	1.20
T271	FO SETTLING TANK PS	FO	8.5	70.0	50.38	-6.53	4.93	1	REAL	6.72
T272	FO SETTLING TANK SB	FO	8.5	70.0	50.38	6.53	4.93	1	REAL	6.72
T273	FO SERVICE TANK PS	FO	10.6	70.0	52.09	-6.38	5.15	2	REAL	6.90
T274	FO SERVICE TANK SB	FO	10.6	70.0	52.09	6.38	5.15	2	REAL	6.90
T275	FO DRAIN TANK	FO	7.9	98.0	45.85	-1.64	0.69	5	REAL	1.37
T277	FO OVERFLOW TANK	FO	3.6	10.0	42.62	2.34	1.31	10	REAL	1.53
T279	EMERG. GEN. FO TANK	FO	1.9	98.0	61.60	-1.38	24.57	0	REAL	25.80
SUBTOTAL		FO	53.0		50.02	-0.13	4.73	75		
CONTENTS=Fresh Water (RHO=1)										
T002	WING TANK PS	FW	10.6	9.3	67.25	-2.14	3.41	45	MAX	3.84
T003	WING TANK SB	FW	10.6	9.2	67.25	2.14	3.41	46	MAX	3.84
SUBTOTAL		FW	21.2		67.25	0.00	3.41	91		
CONTENTS=Hydraulic oil (RHO=0.924)										
T263	HYDRAULIC OIL STORE TANK	HO	1.9	40.0	44.29	-4.33	1.89	1	REAL	2.47
CONTENTS=Lubricating oil (RHO=0.924)										
T290	LO STORE TANK MAIN ENG.	LO	3.4	40.0	55.32	5.47	4.05	0	REAL	5.16
T291	LO STORE TANK MAIN ENG.	LO	1.8	40.0	54.25	5.81	4.26	0	REAL	5.48
T293	LO STORE TANK THRUSTER	LO	1.8	40.0	53.55	6.03	4.41	0	REAL	5.66
T295	LO STORE TANK AZIMUTH THR.	LO	2.3	40.0	14.35	-2.10	2.63	0	REAL	3.06
T297	LO DRAIN TANK	LO	0.9	10.0	47.95	-1.64	0.07	6	REAL	0.14
T298	DIRTY LO TANK	LO	0.6	10.0	14.35	2.10	2.31	0	REAL	2.41
SUBTOTAL		LO	10.7		43.36	3.26	3.43	7		
CONTENTS=Mass load (RHO=1)										
M1	CREW	MASS	2.6	0.0	57.40	0.00	15.50	0		0.00
M2	PROVISION	MASS	0.8	0.0	56.35	-6.50	12.40	0		0.00
M3	STORES	MASS	0.8	0.0	72.10	0.00	12.40	0		0.00
M4	PIPE LOAD	MASS	2276.1	0.0	21.70	0.00	9.29	0		0.00
SUBTOTAL		MASS	2280.3		21.77	0.00	9.30	0		
CONTENTS=Miscellaneous (RHO=1)										
T280	BILGE WATER HOLDING TANK	MIS	0.9	10.0	48.64	6.17	2.03	0	REAL	2.57
T281	BILGE WATER SETTLING TANK	MIS	0.4	10.0	49.34	6.04	1.76	0	REAL	2.08
T287	BILGE WATER DRAIN TANK	MIS	0.9	10.0	45.85	1.64	0.07	6	REAL	0.14
T254	GREY WATER TANK	MIS	31.6	98.0	54.82	-6.47	4.52	3	REAL	7.84
T260	WATER MIST TANK	MIS	2.3	98.0	60.55	-4.30	6.28	0	REAL	7.27
SUBTOTAL		MIS	36.2		54.74	-5.68	4.43	9		
CONTENTS=Sewage (RHO=1)										
T252	BLACK WATER TANK	SEW	10.4	97.9	53.89	-7.06	6.66	4	REAL	7.95
CONTENTS=Sludge (RHO=1)										
T283	SLUDGE DRAIN TANK	SLU	0.9	10.0	47.95	1.64	0.07	6	REAL	0.14
T288	SLUDGE SETTLING TANK	SLU	0.4	10.0	49.35	6.75	5.44	1	REAL	5.59
SUBTOTAL		SLU	1.4		48.40	3.29	1.80	7		
CONTENTS=Water Ballast (RHO=1.025)										
T001	FOREPEAK TANK	WB	102.9	98.0	73.11	0.00	7.14	34	MAX	10.74
T009	DB TANK CE	WB	77.3	100.0	56.90	0.09	0.73	0	(REAL)	1.40
T012	DB TANK PS	WB	56.2	100.0	49.41	-3.85	0.76	0	(REAL)	1.40
T013	DB TANK SB	WB	56.2	100.0	49.41	3.85	0.76	0	(REAL)	1.40
T015	WING TANK SB	WB	33.8	55.0	40.14	8.39	3.26	1	REAL	5.06
T016	DB TANK PS	WB	85.6	100.0	40.20	-4.33	0.62	0	REAL	1.50
T017	DB TANK SB	WB	85.6	100.0	40.20	4.33	0.62	0	REAL	1.50
T020	DB TANK PS	WB	96.7	100.0	30.84	-4.63	0.61	0	(REAL)	1.92
T021	DB TANK SB	WB	96.7	100.0	30.84	4.63	0.61	0	(REAL)	1.92
SUBTOTAL		WB	691.1		45.84	0.42	1.75	35		
TOTAL			3106.1		28.51	0.01	7.41	229		

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LOAD SUMMARY TABLE

ITEM	MASS t	LCG m	TCG m	VCG m	FRSM tm
Diesel Oil	53.0	50.02	-0.13	4.73	74.98
Fresh Water	21.2	67.25	0.00	3.41	91.23
Hydraulic oil	1.9	44.29	-4.33	1.89	0.61
Lubricating oil	10.7	43.36	3.26	3.43	7.30
Mass load	2280.3	21.77	0.00	9.30	0.00
Miscellaneous	36.2	54.74	-5.68	4.43	9.39
Sewage	10.4	53.89	-7.06	6.66	3.55
Sludge	1.4	48.40	3.29	1.80	6.99
Water Ballast	691.1	45.84	0.42	1.75	35.38
DEADWEIGHT	3106.1	28.51	0.01	7.41	229.44
Lightweight	2704.9	39.15	-0.01	7.08	
Deadweight	3106.1	28.51	0.01	7.41	
Total weight	5811.0	33.46	0.00	7.26	

F L O A T I N G P O S I T I O N

Draught moulded	5.896	m	KM	8.81 m
Trim	-0.308	m	KG	7.26 m
Heel, SB=+	0.0	deg		
TA	6.050	m	GM0	1.55 m
TF	5.742	m	GMCORR	-0.04 m
Trimming moment	-2162	tonm	GM	1.51 m

STABILITY COMPONENTS

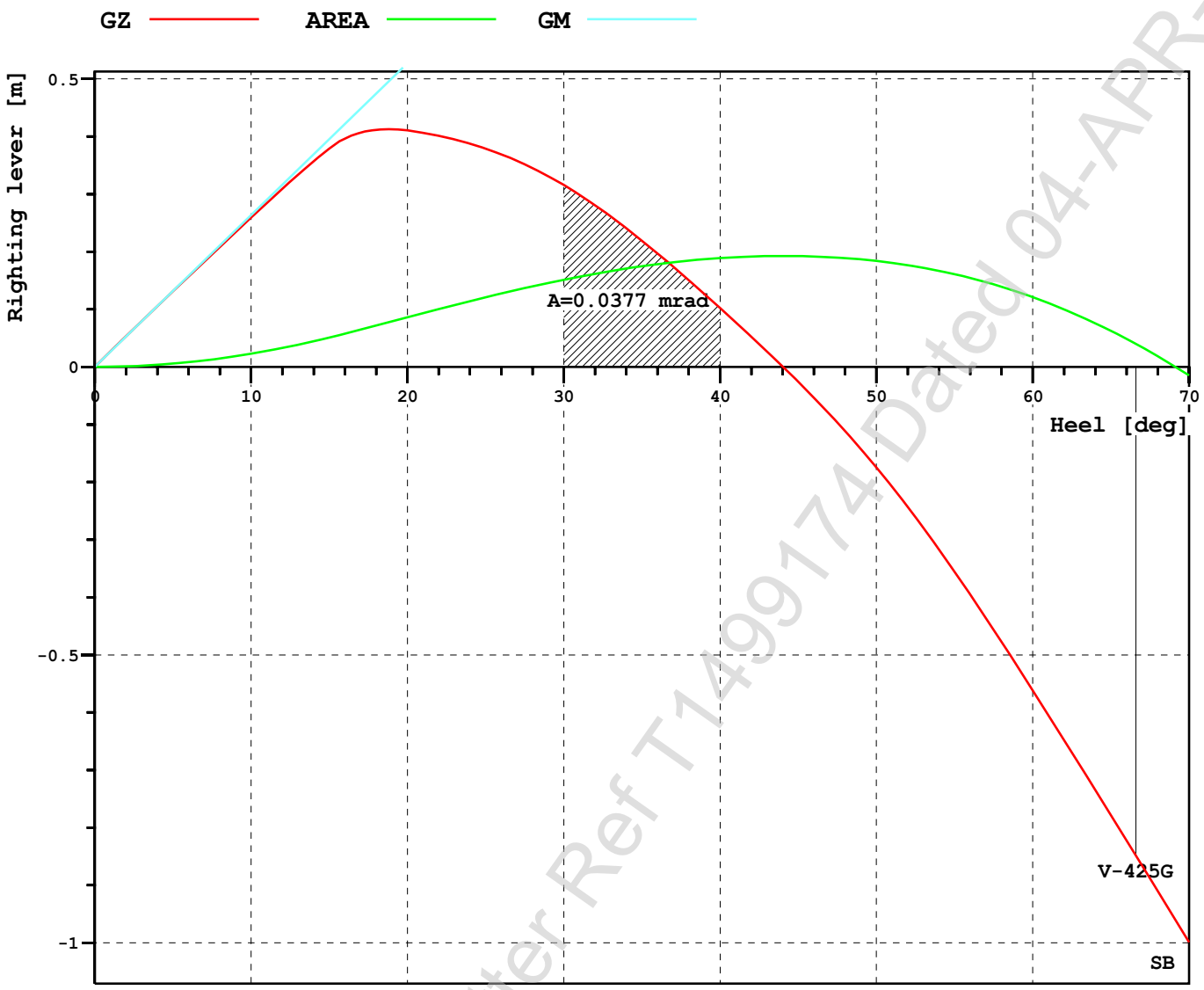
LOADING CONDITION L11, Pipe deckload - Arrival

HEEL degree	MS m	HPHI m	EPHI mrad	FSMOM tm	DGZ m	GMSIN m	KN m
0.0	0.000	0.00	0.000	0.0	0.000	0.000	-0.002
1.0	0.001	0.03	0.000	3.9	0.001	0.027	0.153
3.0	0.000	0.08	0.002	11.5	0.002	0.081	0.460
5.0	0.000	0.13	0.006	19.0	0.003	0.135	0.765
10.0	-0.004	0.26	0.023	36.9	0.006	0.269	1.524
15.0	-0.013	0.38	0.051	54.0	0.009	0.401	2.265
20.0	-0.107	0.41	0.086	70.5	0.012	0.530	2.903
30.0	-0.442	0.32	0.151	101.5	0.017	0.775	3.960
40.0	-0.872	0.10	0.189	130.0	0.022	0.996	4.788
50.0	-1.335	-0.17	0.184	156.2	0.027	1.187	5.410
60.0	-1.873	-0.56	0.121	180.6	0.031	1.342	5.753
70.0	-2.420	-1.00	-0.015	206.1	0.035	1.456	5.855

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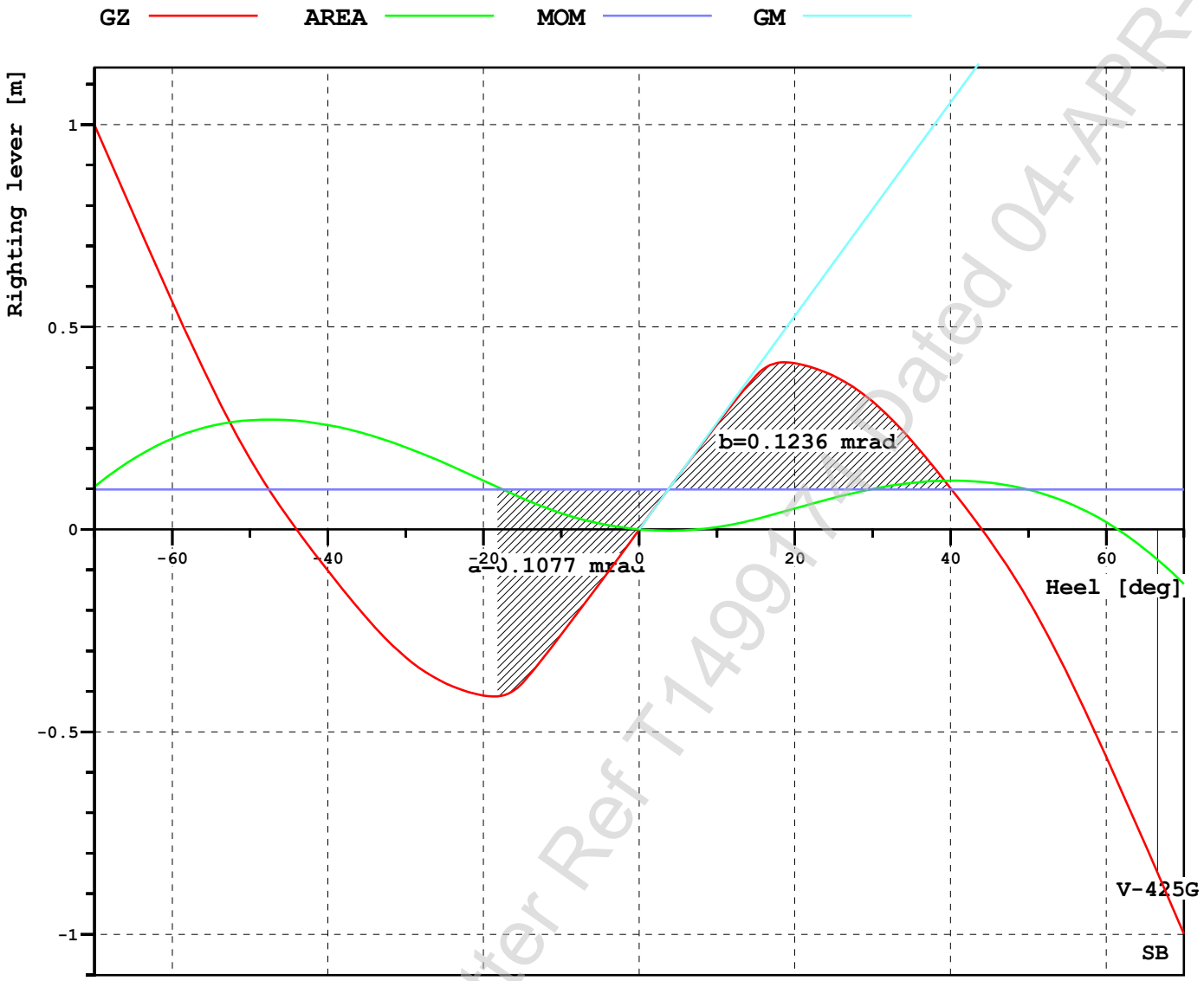
STABILITY CHECK PLOT



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IMOWEATHER CHECK PLOT



INTACT STABILITY CRITERIA

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM
V.AREA15-30	Area depending on the position of the Max. .	0.066	0.077	mrad	OK	1.387
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.038	mrad	OK	1.434
V.GZ0.2	Min. GZ > 0.2	0.200	0.316	m	OK	1.279
V.POSMAX15	Max. GZ at an angle > 15 deg.	15.000	18.760	deg	OK	0.281
V.GM0.15	GM > 0.15 m	0.150	1.510	m	OK	0.150
IMOWEATHER	IMO weather criterion	1.000	1.147		OK	1.393
2008IS-A2.3.1.2	Heeling angle due to steady wind <16 or <=8.	10.327	2.466	deg	OK	0.384
ABSFIFI1	Angle of heel < 1/2 freeboard immersion	6.455	0.541	deg	OK	0.138
ABSFIFI2	Area of residual stability >0.09 mrad	0.090	0.179	mrad	OK	1.087
TOTAL						1.434

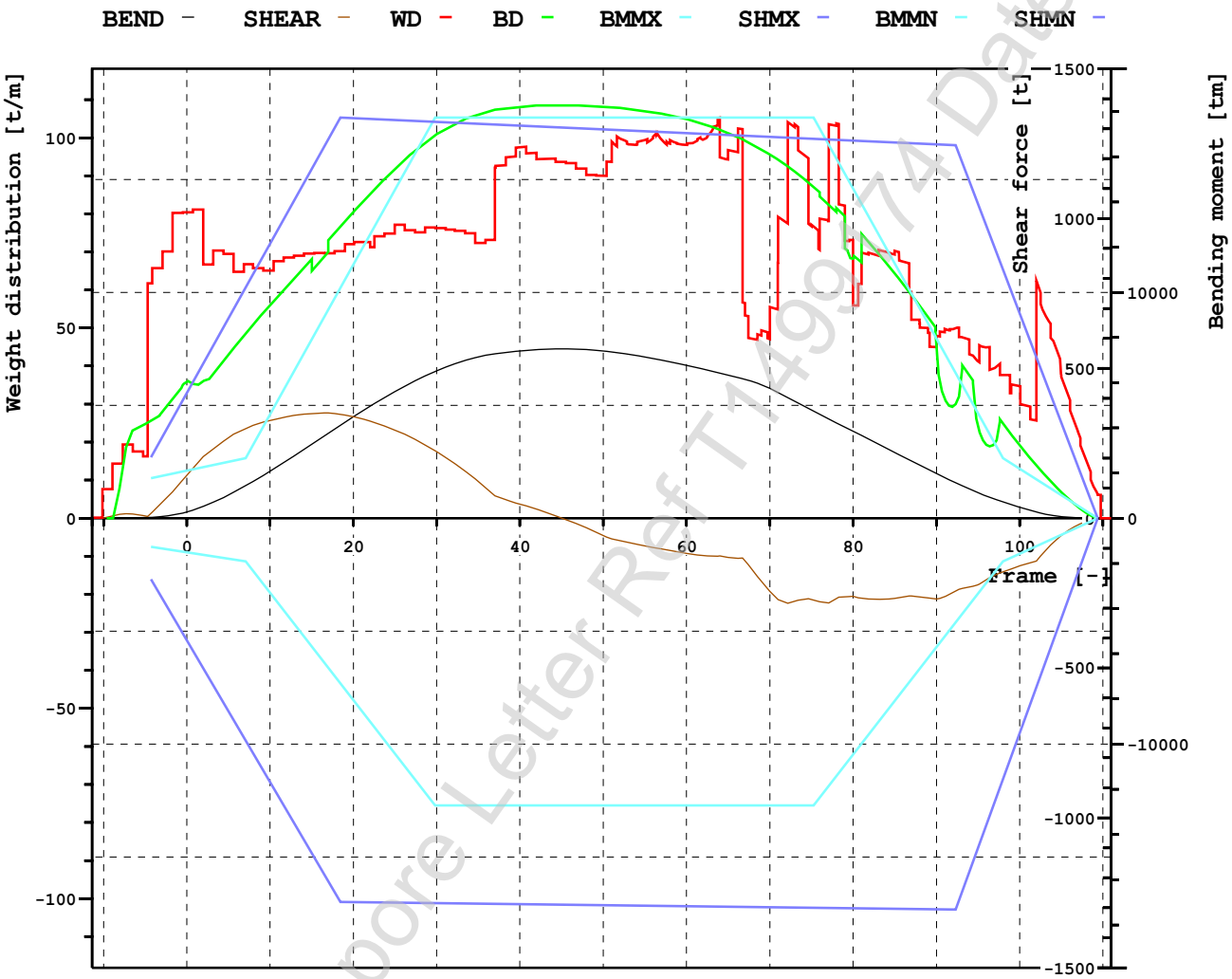
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MOST CRITICAL OPENINGS

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
V-424G	SUPPLY AIR ENGINE ROOM PS	UNPROTECTED	54.243	-3.413	14.288	-	8.459
V-425G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	54.243	3.767	14.288	66.6	8.459
V-421G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	55.890	2.093	14.288	-	8.466
D-651	ACCOMMODATION/C-DECK	UNPROTECTED	56.000	3.771	16.906	-	11.084
V-421E	EXHAUST AIR ENGINE ROOM PS	UNPROTECTED	55.993	-2.538	17.088	-	11.266
V-423E	EXHAUST AIR ENGINE ROOM SB	UNPROTECTED	55.993	3.423	17.088	-	11.266

STRENGTH DIAGRAM



STRENGTH SUMMARY

LOADING CONDITION L11, Pipe deckload - Arrival

			X	Frame
Shear force (min)	-284.1 t	(21.9%)	50.526 m	72.18
Shear force (max)	351.6 t	(27.8%)	11.900 m	17.00
Sagging moment	-0.5 tm	(0.0%)	76.814 m	109.73
Hogging moment	7495.7 tm	(42.3%)	31.858 m	45.51

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STRENGTH DISTRIBUTION

LOADING CONDITION L11, Pipe deckload - Arrival

X	FR	BEND	BMMN	BMMX	BMREL	SHEAR	SHMN	SHMX	SHREL	WD	BD
m	#	tm	tm	tm	%	t	t	t	%	t/m	t/m
-3.500	-5.00	39	0	0	0.0	7	0	0	0.0	16.23	24.78
0.000	0.00	278	-1510	2108	13.2	143	-407	418	34.3	80.38	35.87
3.500	5.00	1027	-1790	2498	41.0	269	-644	667	40.3	69.48	42.92
7.000	10.00	2085	-3292	4594	45.5	326	-881	916	35.6	65.13	56.02
10.500	15.00	3276	-5668	7911	41.4	349	-1117	1166	29.9	69.56	68.11
14.000	20.00	4495	-8045	11228	40.0	339	-1281	1335	25.4	72.04	80.48
17.500	25.00	5612	-10422	14545	38.6	294	-1283	1329	22.1	77.15	91.77
21.000	30.00	6526	-12709	17736	36.8	222	-1284	1323	16.8	76.24	100.98
24.500	35.00	7136	-12709	17736	40.2	122	-1286	1317	9.3	72.34	106.05
28.000	40.00	7405	-12709	17736	41.8	47	-1288	1310	3.6	97.52	108.05
31.500	45.00	7495	-12709	17736	42.3	1	-1290	1304	0.1	93.67	108.49
35.000	50.00	7402	-12709	17736	41.7	-58	-1292	1298	4.5	89.98	108.11
38.500	55.00	7134	-12709	17736	40.2	-93	-1293	1292	7.2	99.36	106.96
42.000	60.00	6769	-12709	17736	38.2	-117	-1295	1285	9.0	98.11	104.91
45.500	65.00	6335	-12709	17736	35.7	-131	-1297	1279	10.1	96.74	101.20
49.000	70.00	5744	-12709	17736	32.4	-244	-1299	1273	18.8	55.52	95.60
52.500	75.00	4794	-12709	17736	27.0	-271	-1300	1267	20.9	77.11	87.56
56.000	80.00	3846	-10463	14601	26.3	-260	-1302	1260	20.0	73.11	68.21
59.500	85.00	2907	-8086	11284	25.8	-267	-1304	1254	20.5	70.29	64.35
63.000	90.00	1986	-5709	7968	24.9	-270	-1306	1248	20.7	47.94	45.69
66.500	95.00	1134	-3332	4651	24.4	-222	-1102	1050	20.1	41.23	24.22
70.000	100.00	482	-1571	2193	22.0	-159	-718	685	22.2	34.69	19.19
73.500	105.00	68	-733	1023	6.6	-58	-335	319	17.2	37.65	6.88

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DAMAGE STABILITY OF LOADING CONDITION L11

CRITICAL DAMAGE STABILITY CRITERIA FOR EACH DAMAGE CASE

RCR	DAM	STAGE	PHASE	REQ	ATTV	UNIT	STAT	MINGM
V.MINGZ.A469	D01	1	EQ	0.100	0.431	m	OK	0.379
V.MINGZ.A469	D02	1	EQ	0.100	0.386	m	OK	0.520
V.MINGZ.A469	D03	1	EQ	0.100	0.387	m	OK	0.521
V.RANGE.A469	D04	1	EQ	20.000	39.009	deg	OK	0.891
V.RANGE.A469	D05	1	EQ	20.000	41.585	deg	OK	0.739
V.RANGE.A469	D06	1	EQ	20.000	38.118	deg	OK	0.908
V.RANGE.A469	D07	1	EQ	20.000	40.761	deg	OK	0.756
V.MINGZ.A469	D08	1	EQ	0.100	0.381	m	OK	0.547
V.RANGE.A469	D09	1	EQ	20.000	40.199	deg	OK	0.763
V.RANGE.A469	D10	1	EQ	20.000	38.336	deg	OK	0.842
V.RANGE.A469	D11	1	EQ	20.000	38.764	deg	OK	0.833
V.RANGE.A469	D12	1	EQ	20.000	33.851	deg	OK	1.049
V.RANGE.A469	D13	1	EQ	20.000	34.264	deg	OK	1.038
V.RANGE.A469	D14	1	EQ	20.000	37.370	deg	OK	0.847
V.RANGE.A469	D15	1	EQ	20.000	37.790	deg	OK	0.837
V.RANGE.A469	D16	1	EQ	20.000	33.570	deg	OK	0.946
V.RANGE.A469	D17	1	EQ	20.000	33.989	deg	OK	0.942
V.MINGZ.A469	D18	1	EQ	0.100	0.369	m	OK	0.578
V.MINGZ.A469	D19	1	EQ	0.100	0.370	m	OK	0.582
V.MINGZ.A469	D20	1	EQ	0.100	0.373	m	OK	0.549
V.MINGZ.A469	D21	1	EQ	0.100	0.373	m	OK	0.553
TOTAL								1.049

DETERMINATION OF CRITICAL DAMAGE CASE

CRITICAL DAMAGE CASE : D12
STAGE OF CRITICAL DAMAGE CASE : 1
PHASE OF CRITICAL DAMAGE CASE : EQ

DAMAGE STABILITY CRITERIA - CRITICAL CASE

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM m
V.PROGR.A469	No progr. flooding	16.724	4.883	deg	OK	0.552
V.MAXHEEL.A469	Maximum heel	17.000	4.883	deg	OK	0.552
V.RANGE.A469	Range of pos. stab	20.000	33.851	deg	OK	1.049
V.MINGZ.A469	Min. righting lever	0.100	0.272	m	OK	0.924
TOTAL						1.049

DAMAGE CASE RESULTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	T m	TR m	HEEL degree	RESFLD m	OPEN	CROSSTM s
D12	1	EQ	SB	5.965	-0.483	4.9	1.81	D-362	-

DAMAGED COMPARTMENTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	NAME	DES	PERM	VOL m3	W t	XCG m	YCG m	ZCG m
D12	INTACT	EQ	-				0.0	0.000			
D12	1	EQ	SB	T023	WING TANK SB	0.95	52.8	54.131	19.39	7.82	3.51
D12	1	EQ	SB	T025	DB TANK SB	0.95	72.5	74.304	21.46	3.85	0.81
D12	1	EQ	SB	E2	Emrg. exit 1 SB	0.95	16.5	16.942	22.59	7.35	5.85

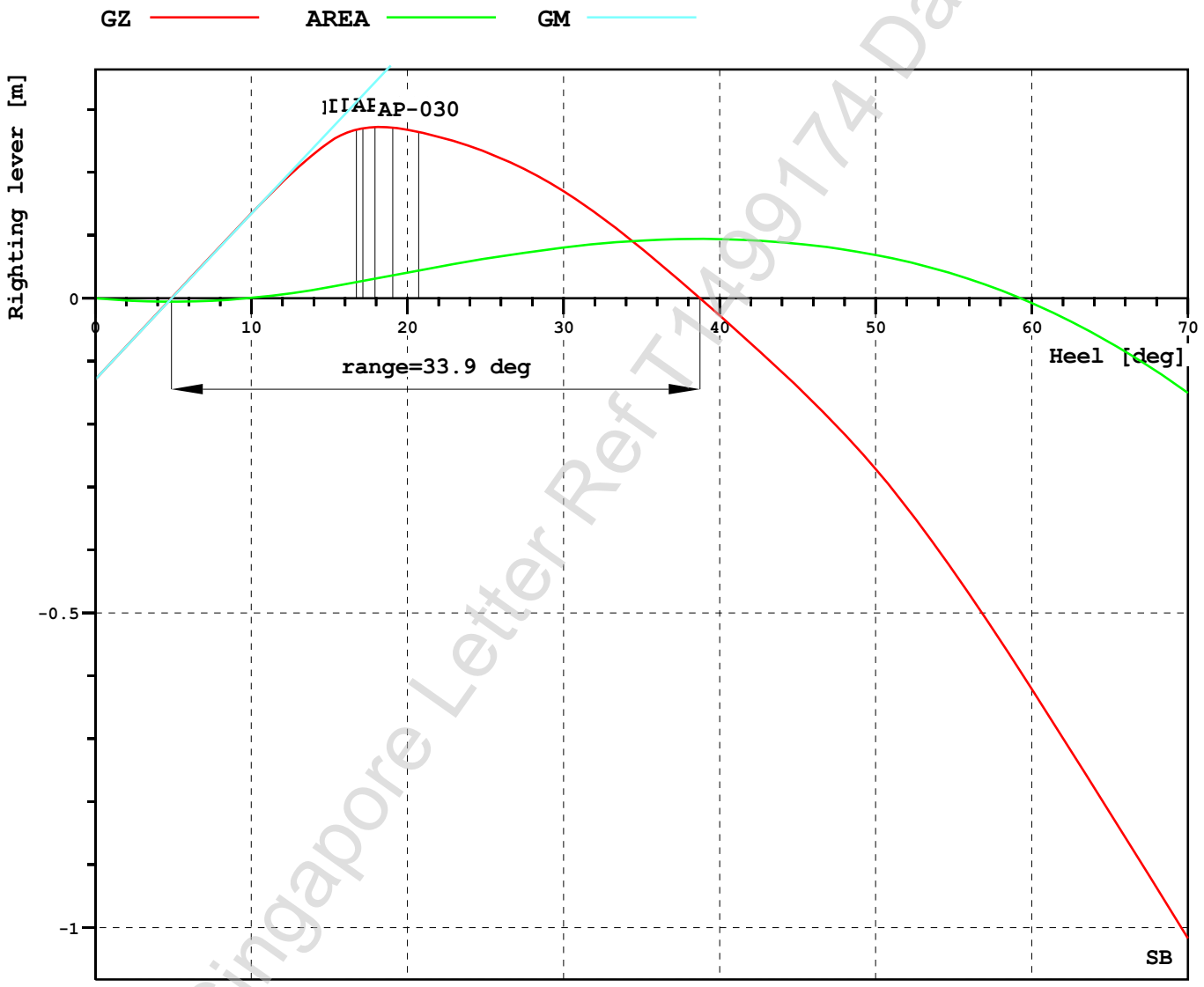
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CRITICAL OPENINGS - CRITICAL CASE

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
D-362	SB CARGORAIL #37/M-DECK	WEATHERTIGHT	25.900	8.550	8.600	16.7	1.810
D-360	SB CARGORAIL #57/M-DECK	WEATHERTIGHT	39.885	8.560	8.600	17.1	1.894
D-352	CHEMICAL/M-DECK	WEATHERTIGHT	53.192	8.375	8.600	17.9	1.991
AP-011	WING TANK SB	WEATHERTIGHT	47.211	8.754	8.953	19.0	2.274
AP-030	WING TANK SB	WEATHERTIGHT	5.250	7.801	9.168	20.7	2.315
AP-034	WING TANK SB	WEATHERTIGHT	-5.873	7.880	9.372	21.3	2.444
AP-036	WING TANK SB	WEATHERTIGHT	1.163	7.810	9.372	21.7	2.493
AP-290	LO STORE TANK MAIN ENG	WEATHERTIGHT	52.985	7.410	8.927	22.1	2.397

DAMAGE STABILITY CHECK PLOT - CRITICAL CASE



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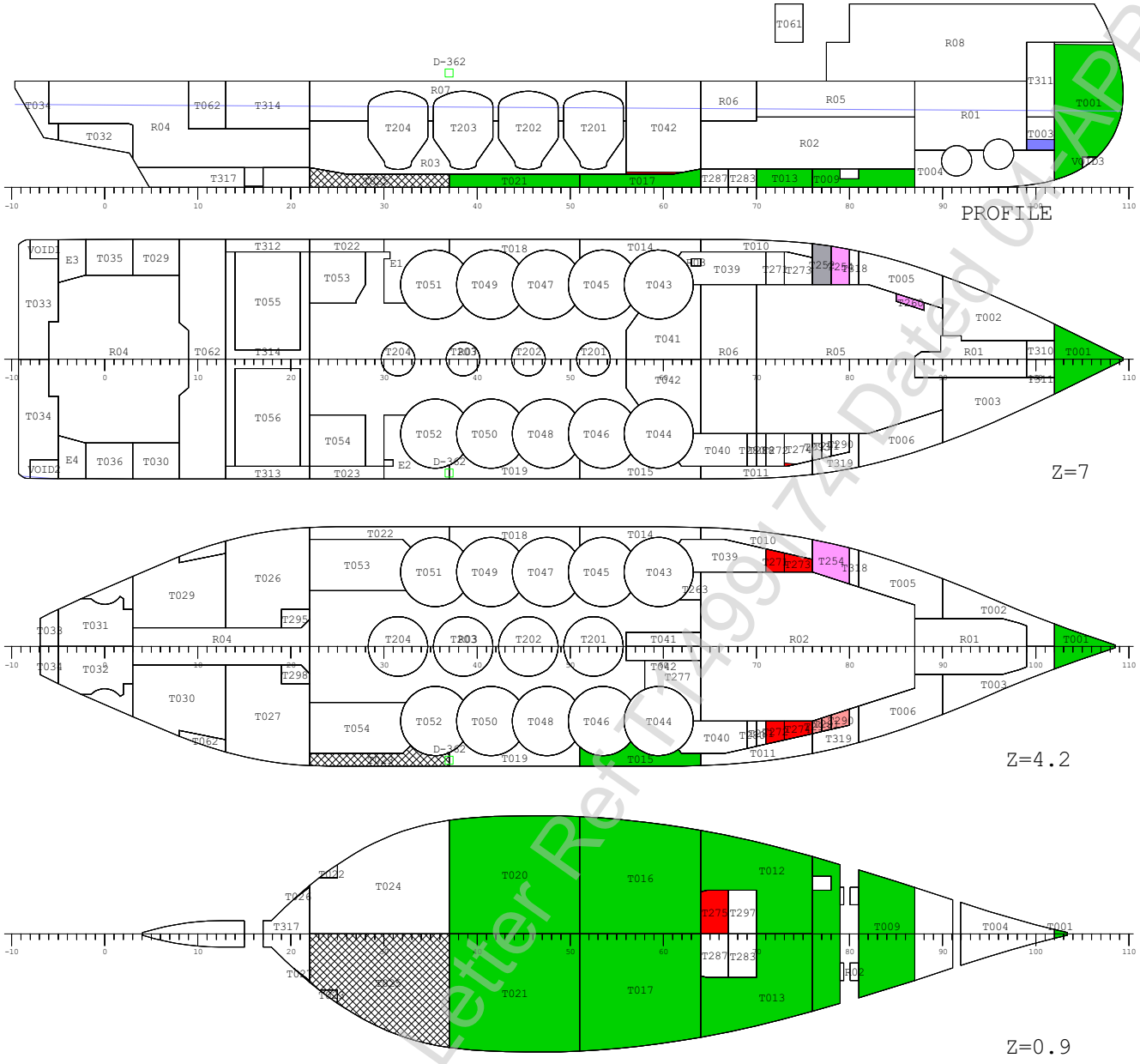
GZ-CURVE - CRITICAL CASE

HEEL	DISP	T	TR	GZ	MS	AREA
degree	t	m	m	m	m	mrad
0.0	5811.0	5.985	-0.490	-0.129	-0.129	0.000
1.0	5811.0	5.985	-0.489	-0.103	-0.130	-0.002
3.0	5811.0	5.979	-0.488	-0.050	-0.131	-0.005
5.0	5811.0	5.964	-0.482	0.003	-0.132	-0.006
10.0	5811.0	5.887	-0.435	0.135	-0.135	0.001
15.0	5811.0	5.763	-0.394	0.248	-0.153	0.018
20.0	5811.0	5.640	-0.463	0.268	-0.263	0.041
30.0	5811.0	5.371	-0.900	0.170	-0.605	0.081
40.0	5811.0	5.016	-1.801	-0.027	-1.024	0.094
50.0	5811.0	4.497	-3.011	-0.272	-1.459	0.069
60.0	5811.0	3.854	-4.181	-0.621	-1.963	-0.008
70.0	5811.0	3.115	-5.172	-1.017	-2.473	-0.150

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DAMAGE ARRANGEMENT - CRITICAL CASE



See ABS Singapore Letter Ref 17499174 Dated 04-APR-2016

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CALCULATION OF PIPE LOAD WITH ENTRAPPED WATER

INPUT DATA:

Loading condition name	L11	
Length of the vessel acc. ICLL 1966	79.614	[m]
Total weight of pipe cargo	1969.8	[t]
Total number of pipes	98	[-]
No. of pipe lenghts	4	[-]
Weight of each pipe	20.1	[t]
Length of each pipe	12.5	[m]
Outer diameter of pipe	1.235	[m]
Inner diameter of pipe	0.966	[m]
Breadth between cargo rails	15.076	[m]
Freeboard	1.3	[m]
Depth to cargo deck	8	[m]
Thickeness of deck covering	75	[mm]

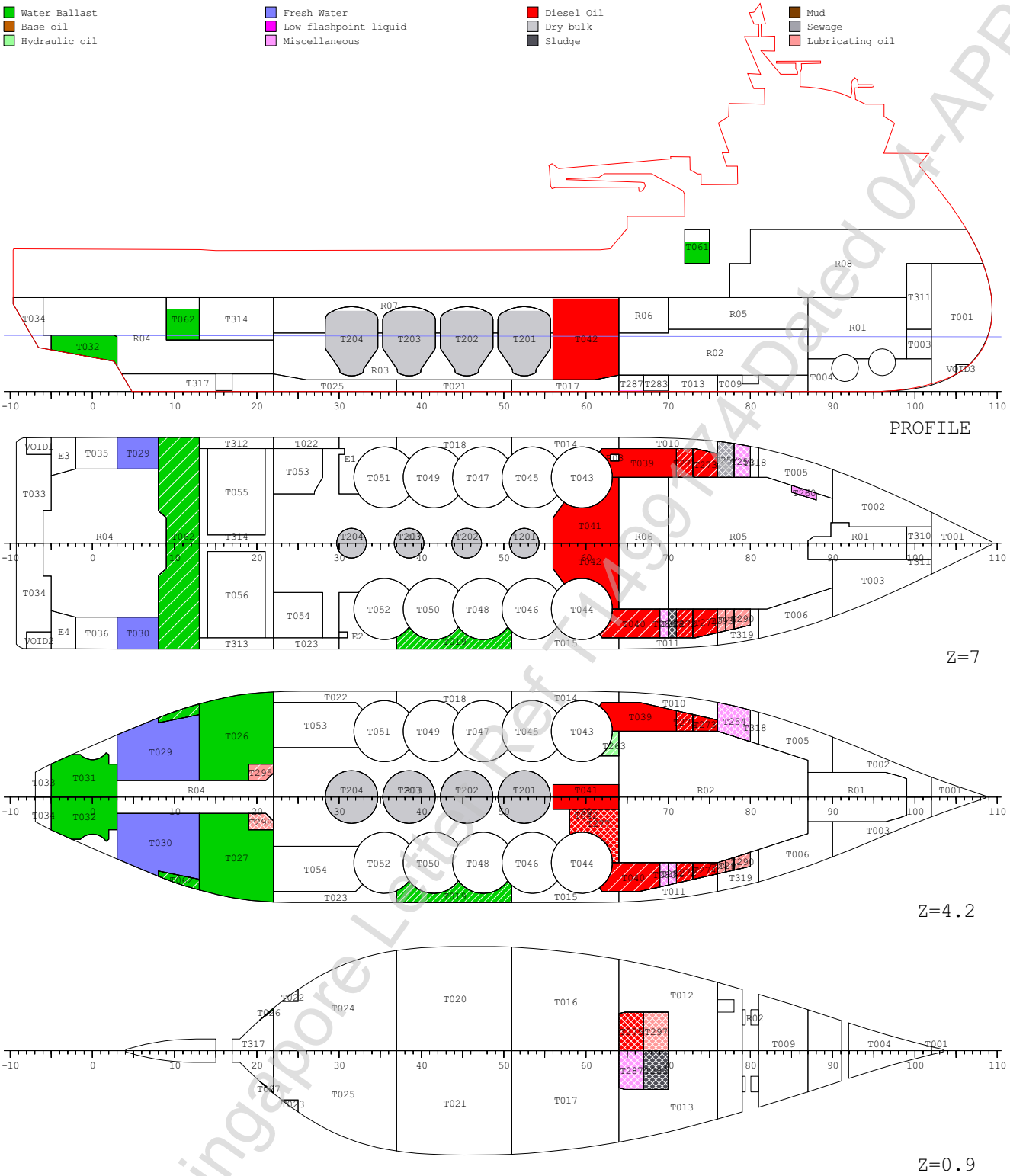
OUTPUT DATA:

Layers [-]	No. of pipes [-]	Weight [t]	VCG [m]	Moment [tm]
Weight of pipes :				
1. Layer	48	964.8	8.6925	8386.52
2. Layer	44	884.4	9.76204	8633.55
3. Layer	6	120.6	10.8316	1306.29
Total	98	1969.8	9.30367	18326.4
Weight of water :				
Inside pipes	98	920.246	9.30367	8561.66
Between pipes	92	72.4782	9.30367	674.313
Between pipes and deck	44	92.2622	8.21293	757.743
Total		1084.99	9.21092	9993.72
Percent coefficient of entrapped water 28.2 %				
Entrapped water (assumed percentage)		306.273	9.21092	2821.06
Total moment (pipes+water)		2276.07	9.29119	21147.4

See ABS Singapore Letter Ref TA 499174 Dated 04-APR-2016

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12.12 L12 -> CEM - Departure



See ABS Singapore Letter Ref 14991X4 Dated 04-APR-2016

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LIST OF LOADS

NAME	DES	LOAD	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm	FRS	HM m
CONTENTS=Dry bulk (RHO=2.4)										
T201	DRY BULK TANK	CEM	152.8	98.0	36.75	0.00	4.62	11	REAL	6.88
T202	DRY BULK TANK	CEM	152.8	98.0	31.85	0.00	4.62	11	REAL	6.88
T203	DRY BULK TANK	CEM	152.8	98.0	26.95	0.00	4.62	11	REAL	6.88
T204	DRY BULK TANK	CEM	152.8	98.0	22.05	0.00	4.62	11	REAL	6.88
SUBTOTAL		CEM	611.2		29.40	0.00	4.62	43		
CONTENTS=Diesel Oil (RHO=0.86)										
T039	STORAGE TANK PS	FO	66.9	98.0	46.68	-6.76	4.91	6	REAL	7.88
T040	STORAGE TANK SB	FO	53.8	98.0	46.00	6.80	4.87	5	REAL	7.88
T041	STORAGE TANK PS	FO	68.5	98.0	42.16	-1.51	5.48	27	MAX	7.92
T042	STORAGE TANK SB	FO	68.5	98.0	42.16	1.51	5.48	27	MAX	7.92
T271	FO SETTLING TANK PS	FO	8.5	70.0	50.38	-6.53	4.93	1	REAL	6.72
T272	FO SETTLING TANK SB	FO	8.5	70.0	50.38	6.53	4.93	1	REAL	6.72
T273	FO SERVICE TANK PS	FO	10.6	70.0	52.09	-6.38	5.15	2	REAL	6.90
T274	FO SERVICE TANK SB	FO	10.6	70.0	52.09	6.38	5.15	2	REAL	6.90
T275	FO DRAIN TANK	FO	0.8	10.0	45.85	-1.64	0.07	5	REAL	0.14
T277	FO OVERFLOW TANK	FO	3.6	10.0	42.62	2.34	1.31	10	REAL	1.53
T279	EMERG. GEN. FO TANK	FO	1.9	98.0	61.60	-1.38	24.57	0	REAL	25.80
SUBTOTAL		FO	302.3		45.14	-0.27	5.25	86		
CONTENTS=Fresh Water (RHO=1)										
T029	WING TANK PS	FW	74.6	98.0	5.29	-4.87	4.61	83	MAX	7.83
T030	WING TANK SB	FW	74.6	98.0	5.29	4.87	4.61	83	MAX	7.83
SUBTOTAL		FW	149.3		5.29	0.00	4.61	165		
CONTENTS=Hydraulic oil (RHO=0.924)										
T263	HYDRAULIC OIL STORE TANK	HO	3.3	70.0	44.30	-4.34	2.32	1	REAL	3.33
CONTENTS=Lubricating oil (RHO=0.924)										
T290	LO STORE TANK MAIN ENG.	LO	5.9	70.0	55.30	5.76	4.90	0	REAL	6.82
T291	LO STORE TANK MAIN ENG.	LO	3.1	70.0	54.25	6.04	5.11	0	REAL	6.92
T293	LO STORE TANK THRUSTER	LO	3.2	70.0	53.55	6.21	5.24	0	REAL	6.97
T295	LO STORE TANK AZIMUTH THR.	LO	4.0	70.0	14.35	-2.10	2.95	0	REAL	3.71
T297	LO DRAIN TANK	LO	0.9	10.0	47.95	-1.64	0.07	6	REAL	0.14
T298	DIRTY LO TANK	LO	0.6	10.0	14.35	2.10	2.31	0	REAL	2.41
SUBTOTAL		LO	17.6		43.89	3.64	4.24	8		
CONTENTS=Mass load (RHO=1)										
M1	CREW	MASS	2.6	0.0	57.40	0.00	15.50	0		0.00
M2	PROVISION	MASS	7.4	0.0	56.35	-6.50	12.40	0		0.00
M3	STORES	MASS	7.4	0.0	72.10	0.00	12.40	0		0.00
SUBTOTAL		MASS	17.5		63.21	-2.77	12.86	0		
CONTENTS=Miscellaneous (RHO=1)										
T280	BILGE WATER HOLDING TANK	MIS	0.9	10.0	48.64	6.17	2.03	0	REAL	2.57
T281	BILGE WATER SETTLING TANK	MIS	0.4	10.0	49.34	6.04	1.76	0	REAL	2.08
T287	BILGE WATER DRAIN TANK	MIS	0.9	10.0	45.85	1.64	0.07	6	REAL	0.14
T254	GREY WATER TANK	MIS	3.2	10.0	54.62	-5.79	1.92	1	REAL	2.34
T260	WATER MIST TANK	MIS	2.3	98.0	60.55	-4.30	6.28	0	REAL	7.27
SUBTOTAL		MIS	7.8		54.40	-2.49	3.01	8		
CONTENTS=Sewage (RHO=1)										
T252	BLACK WATER TANK	SEW	1.1	10.0	53.89	-6.93	5.45	2	REAL	5.59
CONTENTS=Sludge (RHO=1)										
T283	SLUDGE DRAIN TANK	SLU	0.9	10.0	47.95	1.64	0.07	6	REAL	0.14
T288	SLUDGE SETTLING TANK	SLU	0.4	10.0	49.35	6.75	5.44	1	REAL	5.59
SUBTOTAL		SLU	1.4		48.40	3.29	1.80	7		
CONTENTS=Water Ballast (RHO=1.025)										
T019	WING TANK SB	WB	16.0	21.5	30.83	8.36	2.17	2	MAX	2.87
T026	WING TANK PS	WB	92.4	100.0	12.42	-4.45	3.13	0	(REAL)	4.40
T027	WING TANK SB	WB	92.4	100.0	12.42	4.45	3.13	0	(REAL)	4.40
T031	WING TANK PS	WB	27.2	100.0	-0.65	-1.76	4.18	0	(REAL)	5.39
T032	WING TANK SB	WB	27.2	100.0	-0.65	1.76	4.18	0	(REAL)	5.39
T061	ROLL RED. TANK	WB	83.4	65.0	51.66	0.00	11.84	1358	REAL	12.79

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T062	ROLL RED. TANK	WB	147.4	70.0	7.45	0.00	5.67	1736 REAL	6.93
SUBTOTAL		WB	486.0		16.79	0.28	5.48	3097	
TOTAL			1597.4		27.01	0.02	5.07	3416	

LOAD SUMMARY TABLE

ITEM	MASS t	LCG m	TCG m	VCG m	FRSM tm
Dry bulk	611.2	29.40	0.00	4.62	42.74
Diesel Oil	302.3	45.14	-0.27	5.25	85.89
Fresh Water	149.3	5.29	0.00	4.61	165.03
Hydraulic oil	3.3	44.30	-4.34	2.32	0.61
Lubricating oil	17.6	43.89	3.64	4.24	7.62
Mass load	17.5	63.21	-2.77	12.86	0.00
Miscellaneous	7.8	54.40	-2.49	3.01	7.72
Sewage	1.1	53.89	-6.93	5.45	2.31
Sludge	1.4	48.40	3.29	1.80	6.99
Water Ballast	486.0	16.79	0.28	5.48	3096.74
DEADWEIGHT	1597.4	27.01	0.02	5.07	3415.65
Lightweight	2704.9	39.15	-0.01	7.08	
Deadweight	1597.4	27.01	0.02	5.07	
Total weight	4302.3	34.64	0.00	6.33	

F L O A T I N G P O S I T I O N

Draught moulded	4.719	m	KM	9.01 m
Trim	-0.121	m	KG	6.33 m
Heel, SB=+	0.0	deg		
TA	4.780	m	GM0	2.68 m
TF	4.659	m	GMCORR	-0.79 m
Trimming moment	-714	tonm	GM	1.88 m

STABILITY COMPONENTS

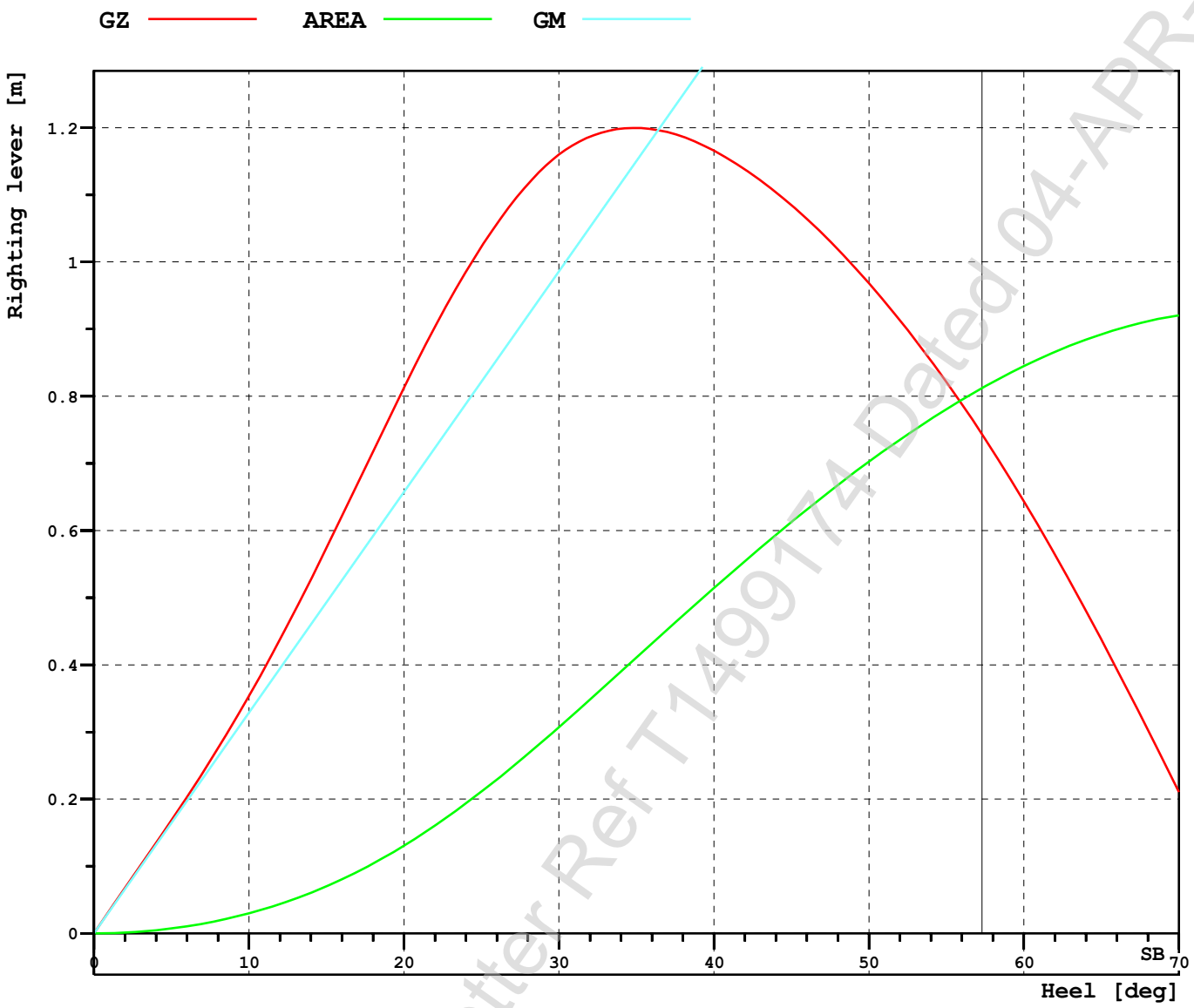
LOADING CONDITION L12, CEM - Departure

HEEL degree	MS m	HPHI m	EPHI mrad	FSMOM tm	DGZ m	GMSIN m	KN m
0.0	0.000	0.00	0.000	0.0	0.000	0.000	-0.002
1.0	0.001	0.03	0.000	57.8	0.013	0.047	0.156
3.0	0.001	0.10	0.003	173.6	0.040	0.140	0.471
5.0	0.002	0.17	0.007	289.8	0.067	0.233	0.786
10.0	0.012	0.35	0.030	529.0	0.123	0.465	1.575
15.0	0.031	0.57	0.070	642.6	0.149	0.693	2.361
20.0	0.060	0.81	0.131	700.4	0.163	0.916	3.140
30.0	-0.004	1.16	0.307	750.2	0.174	1.339	4.500
40.0	-0.380	1.17	0.514	752.8	0.175	1.721	5.411
50.0	-0.914	0.97	0.703	723.2	0.168	2.051	5.988
60.0	-1.519	0.64	0.845	670.1	0.156	2.319	6.284
70.0	-2.166	0.21	0.921	598.7	0.139	2.516	6.302

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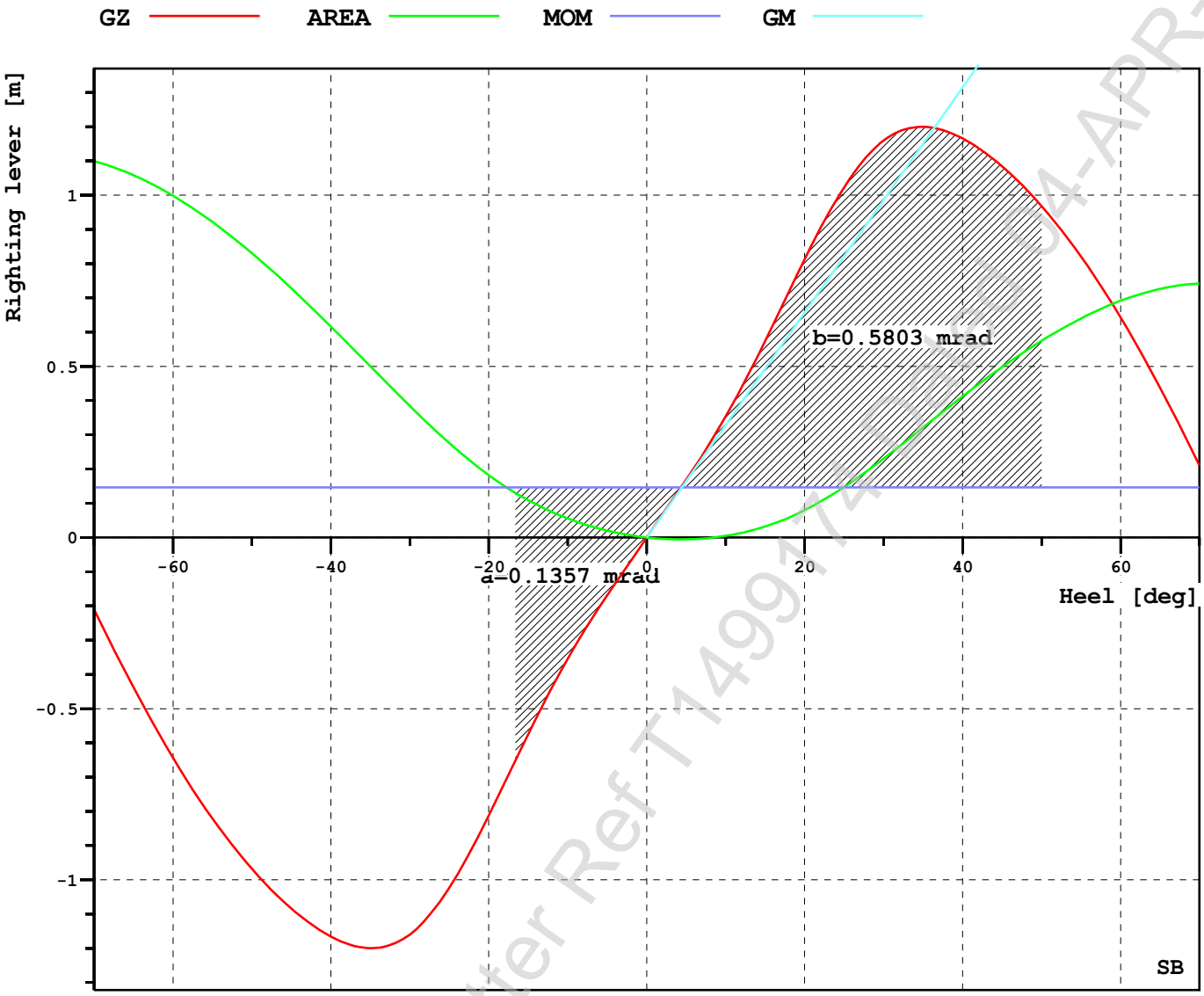
STABILITY CHECK PLOT



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IMOWEATHER CHECK PLOT



INTACT STABILITY CRITERIA

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM
V.AREA15-30	Area depending on the position of the Max. .	0.055	0.307	mrad	OK	0.107
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.207	mrad	OK	0.111
V.GZ0.2	Min. GZ > 0.2	0.200	1.200	m	OK	-0.037
V.POSMAX15	Max. GZ at an angle > 15 deg.	15.000	34.870	deg	OK	-0.536
V.GM0.15	GM > 0.15 m	0.150	1.883	m	OK	0.150
IMOWEATHER	IMO weather criterion	1.000	4.278		OK	0.128
2008IS-A2.3.1.2	Heeling angle due to steady wind <16 or <=8.	16.000	2.900	deg	OK	-0.018
ABSFIFI1	Angle of heel < 1/2 freeboard immersion	10.234	0.577	deg	OK	-0.052
ABSFIFI2	Area of residual stability >0.09 mrad	0.090	0.501	mrad	OK	0.124
TOTAL						0.150

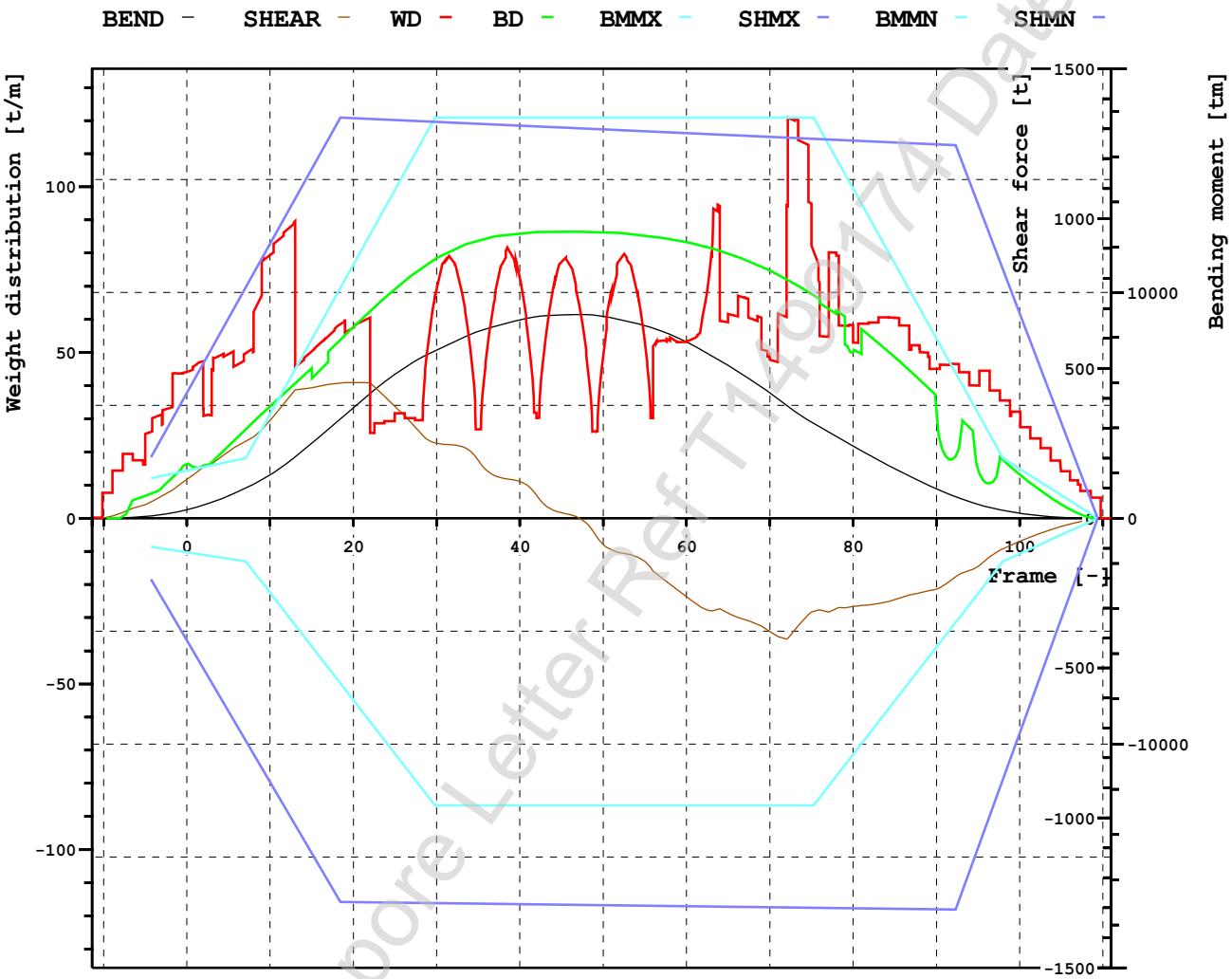
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MOST CRITICAL OPENINGS

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
V-424G	SUPPLY AIR ENGINE ROOM PS	UNPROTECTED	54.243	-3.413	14.288	-	9.595
V-425G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	54.243	3.767	14.288	-	9.595
V-421G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	55.890	2.093	14.288	-	9.598
D-651	ACCOMMODATION/C-DECK	UNPROTECTED	56.000	3.771	16.906	-	12.216
V-421E	EXHAUST AIR ENGINE ROOM PS	UNPROTECTED	55.993	-2.538	17.088	-	12.398
V-423E	EXHAUST AIR ENGINE ROOM SB	UNPROTECTED	55.993	3.423	17.088	-	12.398

STRENGTH DIAGRAM



STRENGTH SUMMARY

LOADING CONDITION L12, CEM - Departure

			X	Frame
Shear force (min)	-403.8 t	(31.1%)	50.400 m	72.00
Shear force (max)	453.1 t	(33.9%)	14.181 m	20.26
Sagging moment	-			
Hogging moment	9016.9 tm	(50.8%)	32.996 m	47.14

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STRENGTH DISTRIBUTION

LOADING CONDITION L12, CEM - Departure

X	FR	BEND	BMMN	BMMX	BMREL	SHEAR	SHMN	SHMX	SHREL	WD	BD
m	#	tm	tm	tm	%	t	t	t	%	t/m	t/m
-3.500	-5.00	77	0	0	0.0	45	0	0	0.0	25.43	6.76
0.000	0.00	369	-1510	2108	17.5	129	-407	418	30.9	44.10	16.33
3.500	5.00	985	-1790	2498	39.4	222	-644	667	33.3	49.50	21.79
7.000	10.00	1924	-3292	4594	41.9	327	-881	916	35.7	79.46	33.74
10.500	15.00	3316	-5668	7911	41.9	435	-1117	1166	37.3	50.37	45.31
14.000	20.00	4879	-8045	11228	43.5	453	-1281	1335	33.9	57.51	57.70
17.500	25.00	6380	-10422	14545	43.9	373	-1283	1329	28.1	31.62	69.12
21.000	30.00	7445	-12709	17736	42.0	250	-1284	1323	18.9	69.25	78.48
24.500	35.00	8269	-12709	17736	46.6	193	-1286	1317	14.6	26.82	83.70
28.000	40.00	8773	-12709	17736	49.5	123	-1288	1310	9.4	72.48	85.85
31.500	45.00	9001	-12709	17736	50.7	18	-1290	1304	1.4	77.36	86.44
35.000	50.00	8939	-12709	17736	50.4	-88	-1292	1298	6.8	49.08	86.21
38.500	55.00	8528	-12709	17736	48.1	-145	-1293	1292	11.2	53.51	85.21
42.000	60.00	7802	-12709	17736	44.0	-262	-1295	1285	20.2	53.18	83.31
45.500	65.00	6761	-12709	17736	38.1	-317	-1297	1279	24.4	61.61	79.83
49.000	70.00	5555	-12709	17736	31.3	-377	-1299	1273	29.1	48.58	74.71
52.500	75.00	4255	-12709	17736	24.0	-314	-1300	1267	24.1	95.00	67.71
56.000	80.00	3190	-10463	14601	21.8	-294	-1302	1260	22.6	58.33	50.09
59.500	85.00	2190	-8086	11284	19.4	-272	-1304	1254	20.8	60.48	48.60
63.000	90.00	1304	-5709	7968	16.4	-237	-1306	1248	18.2	45.02	32.90
66.500	95.00	617	-3332	4651	13.3	-161	-1102	1050	14.6	40.05	14.79
70.000	100.00	220	-1571	2193	10.0	-77	-718	685	10.7	32.12	13.24
73.500	105.00	41	-733	1023	4.0	-28	-335	319	8.3	14.17	4.18

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DAMAGE STABILITY OF LOADING CONDITION L12

CRITICAL DAMAGE STABILITY CRITERIA FOR EACH DAMAGE CASE

RCR	DAM	STAGE	PHASE	REQ	ATTV	UNIT	STAT	MINGM
V.RANGE.A469	D01	1	EQ	20.000	69.976	deg	OK	-0.327
V.RANGE.A469	D02	1	EQ	20.000	69.611	deg	OK	-0.228
V.RANGE.A469	D03	1	EQ	20.000	69.562	deg	OK	-0.231
V.RANGE.A469	D04	1	EQ	20.000	67.562	deg	OK	0.017
V.RANGE.A469	D05	1	EQ	20.000	67.323	deg	OK	0.051
V.RANGE.A469	D06	1	EQ	20.000	67.610	deg	OK	0.034
V.RANGE.A469	D07	1	EQ	20.000	67.386	deg	OK	0.067
V.MAXHEEL.A469	D08	1	EQ	17.000	4.227	deg	OK	0.025
V.MAXHEEL.A469	D09	1	EQ	17.000	4.274	deg	OK	0.030
V.RANGE.A469	D10	1	EQ	20.000	65.844	deg	OK	0.013
V.MAXHEEL.A469	D11	1	EQ	17.000	5.035	deg	OK	0.093
V.RANGE.A469	D12	1	EQ	20.000	65.888	deg	OK	0.032
V.RANGE.A469	D13	1	EQ	20.000	65.835	deg	OK	0.025
V.RANGE.A469	D14	1	EQ	20.000	70.000	deg	OK	-0.252
V.RANGE.A469	D15	1	EQ	20.000	69.937	deg	OK	-0.252
V.RANGE.A469	D16	1	EQ	20.000	68.397	deg	OK	-0.054
V.RANGE.A469	D17	1	EQ	20.000	68.447	deg	OK	-0.050
V.RANGE.A469	D18	1	EQ	20.000	69.977	deg	OK	-0.325
V.RANGE.A469	D19	1	EQ	20.000	69.975	deg	OK	-0.237
V.RANGE.A469	D20	1	EQ	20.000	69.977	deg	OK	-0.272
V.RANGE.A469	D21	1	EQ	20.000	69.929	deg	OK	-0.271
TOTAL								0.093

DETERMINATION OF CRITICAL DAMAGE CASE

CRITICAL DAMAGE CASE : D11
STAGE OF CRITICAL DAMAGE CASE : 1
PHASE OF CRITICAL DAMAGE CASE : EQ

DAMAGE STABILITY CRITERIA - CRITICAL CASE

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM m
V.PROGR.A469	No progr. flooding	24.562	5.035	deg	OK	-0.230
V.MAXHEEL.A469	Maximum heel	17.000	5.035	deg	OK	0.093
V.RANGE.A469	Range of pos. stab	20.000	64.965	deg	OK	0.055
V.MINGZ.A469	Min. righting lever	0.100	0.897	m	OK	-0.029
TOTAL						0.093

DAMAGE CASE RESULTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	T m	TR m	HEEL degree	RESFLD m	OPEN	CROSSTM s
D11	1	EQ	PS	4.806	-0.089	-5.0	3.00	D-363	-

DAMAGED COMPARTMENTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	NAME	DES	PERM	VOL m3	W t	XCG m	YCG m	ZCG m
D11	INTACT	EQ	-				0.0	0.000			
D11	1	EQ	PS	T018	WING TANK PS	0.95	44.6	45.689	30.80	-8.37	3.52
D11	1	EQ	PS	T020	DB TANK PS	0.95	92.4	94.737	30.84	-4.63	0.61

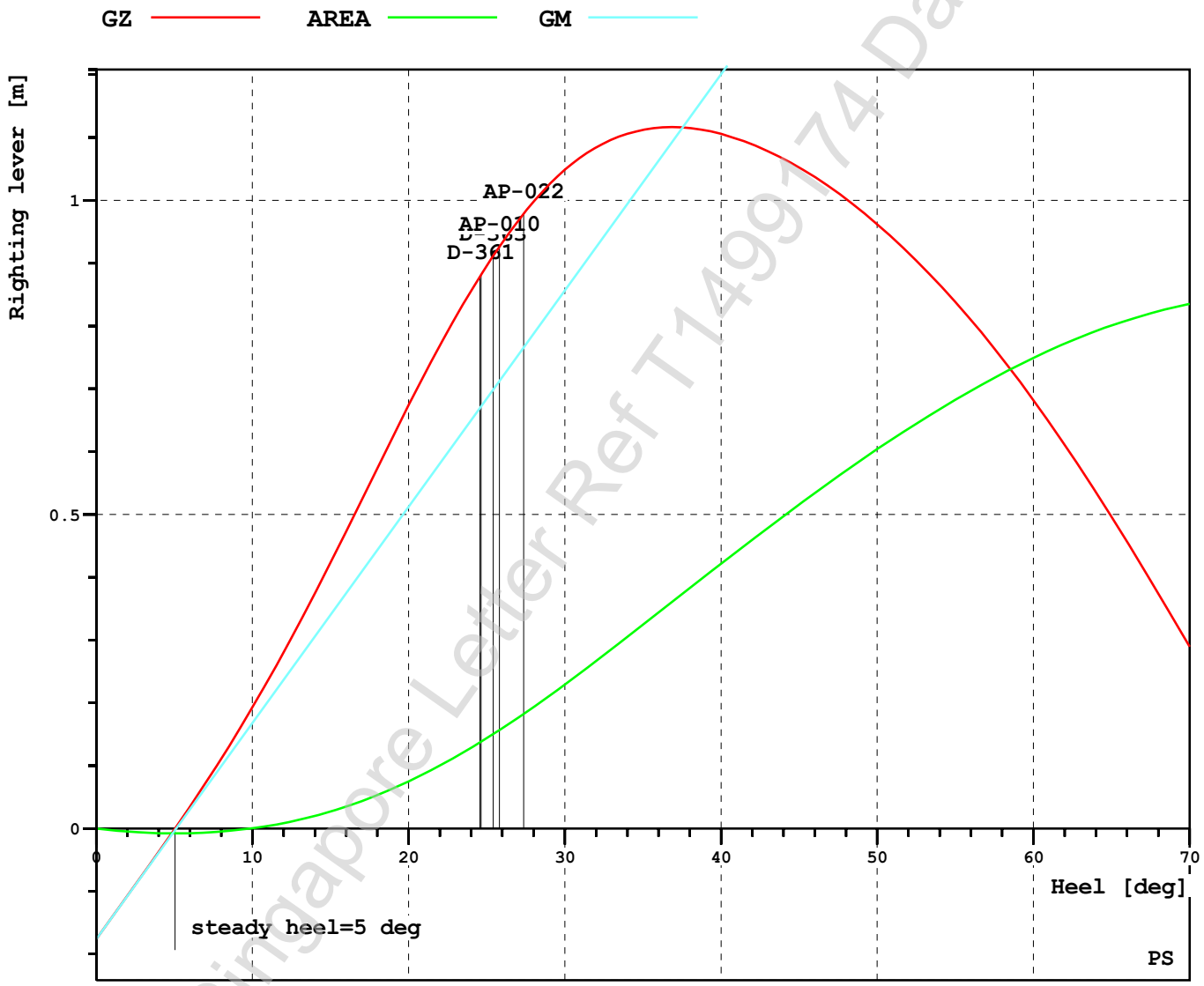
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CRITICAL OPENINGS - CRITICAL CASE

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
D-351	CO2 ROOM/M-DECK	WEATHERTIGHT	51.792	-8.360	8.600	24.6	3.044
D-361	PS CARGORAIL #57/M-DECK	WEATHERTIGHT	39.885	-8.560	8.600	24.6	3.013
D-363	PS CARGORAIL #37/M-DECK	WEATHERTIGHT	25.900	-8.550	8.600	25.4	2.999
AP-010	WING TANK PS	WEATHERTIGHT	47.211	-8.731	8.953	25.8	3.358
AP-022	WING TANK PS	WEATHERTIGHT	19.200	-8.702	8.936	27.4	3.313
AP-263	HYDRAULIC OIL STORE TANK	WEATHERTIGHT	47.690	-7.260	8.912	30.2	3.447
D-353	DECK WORKSHOP/M-DECK	WEATHERTIGHT	53.192	-6.550	8.600	30.4	3.205
AP-012	DB TANK PS	WEATHERTIGHT	50.000	-7.250	9.047	30.9	3.585

DAMAGE STABILITY CHECK PLOT - CRITICAL CASE



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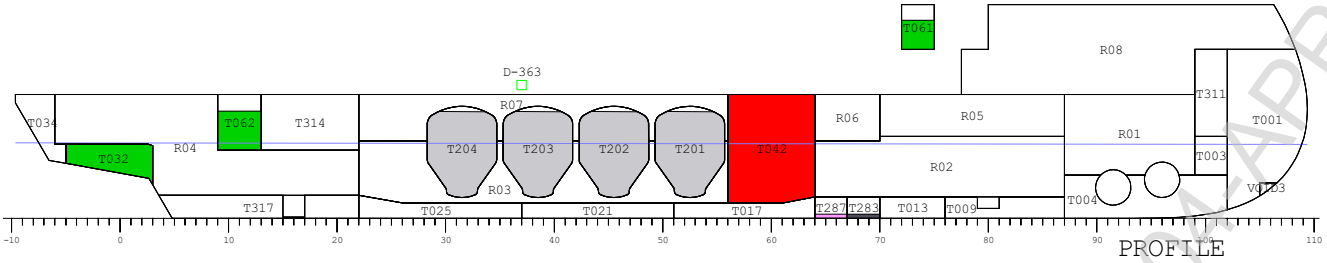
GZ-CURVE - CRITICAL CASE

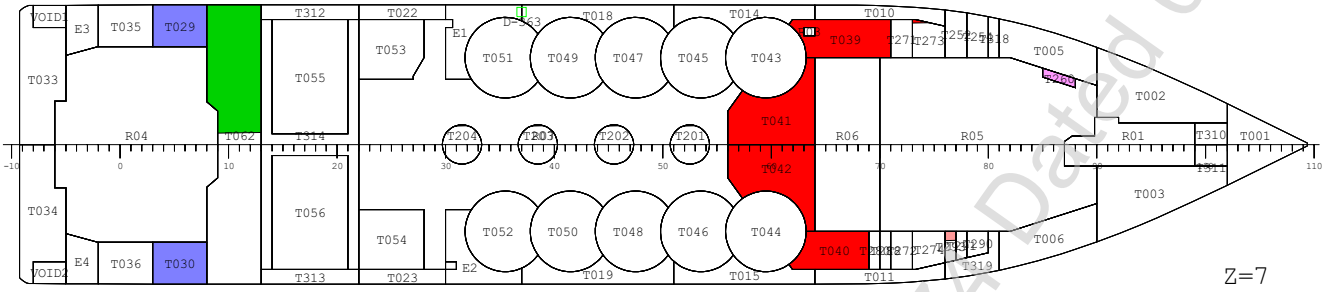
HEEL	DISP	T	TR	GZ	MS	AREA
degree	t	m	m	m	m	mrad
0.0	4302.3	4.827	-0.138	0.176	0.176	0.000
-1.0	4302.3	4.827	-0.136	0.141	0.188	-0.003
-3.0	4302.3	4.821	-0.121	0.072	0.212	-0.006
-5.0	4302.3	4.806	-0.090	0.001	0.235	-0.008
-10.0	4302.3	4.731	0.044	-0.193	0.272	0.000
-15.0	4302.3	4.604	0.230	-0.423	0.269	0.027
-20.0	4302.3	4.425	0.446	-0.674	0.242	0.075
-30.0	4302.3	3.960	0.776	-1.049	0.290	0.229
-40.0	4302.3	3.362	0.729	-1.106	0.615	0.421
-50.0	4302.3	2.614	0.236	-0.962	1.089	0.604
-60.0	4302.3	1.761	-0.431	-0.682	1.637	0.749
-70.0	4302.3	0.858	-1.105	-0.289	2.226	0.835

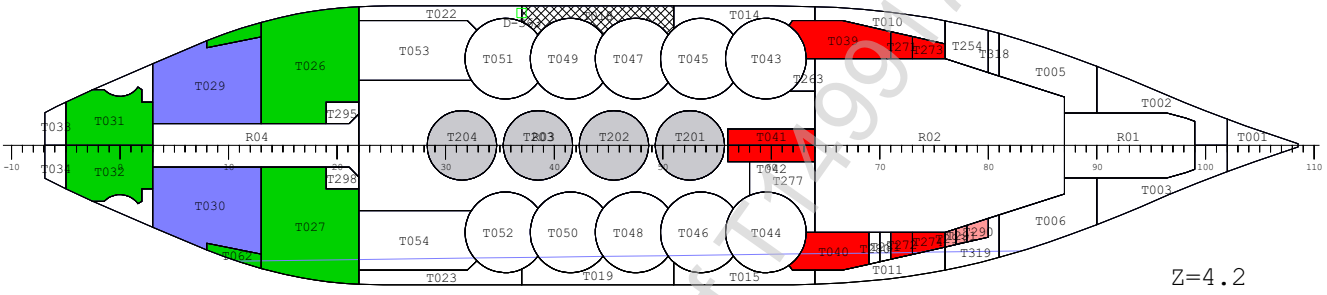
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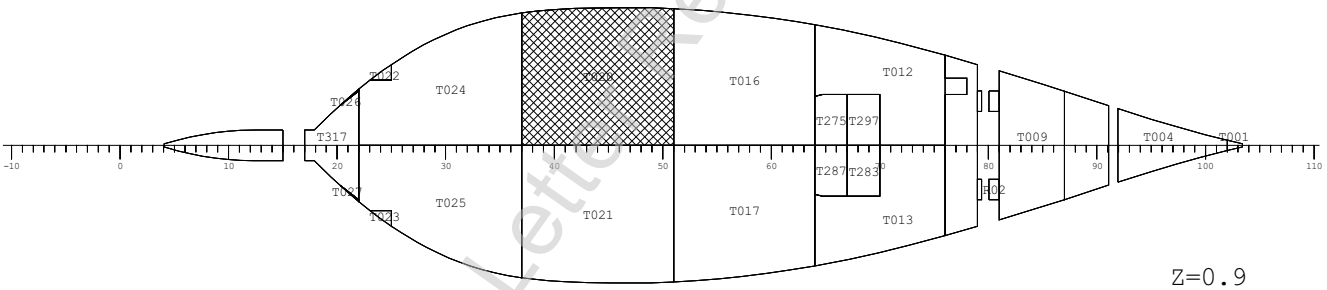
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DAMAGE ARRANGEMENT - CRITICAL CASE





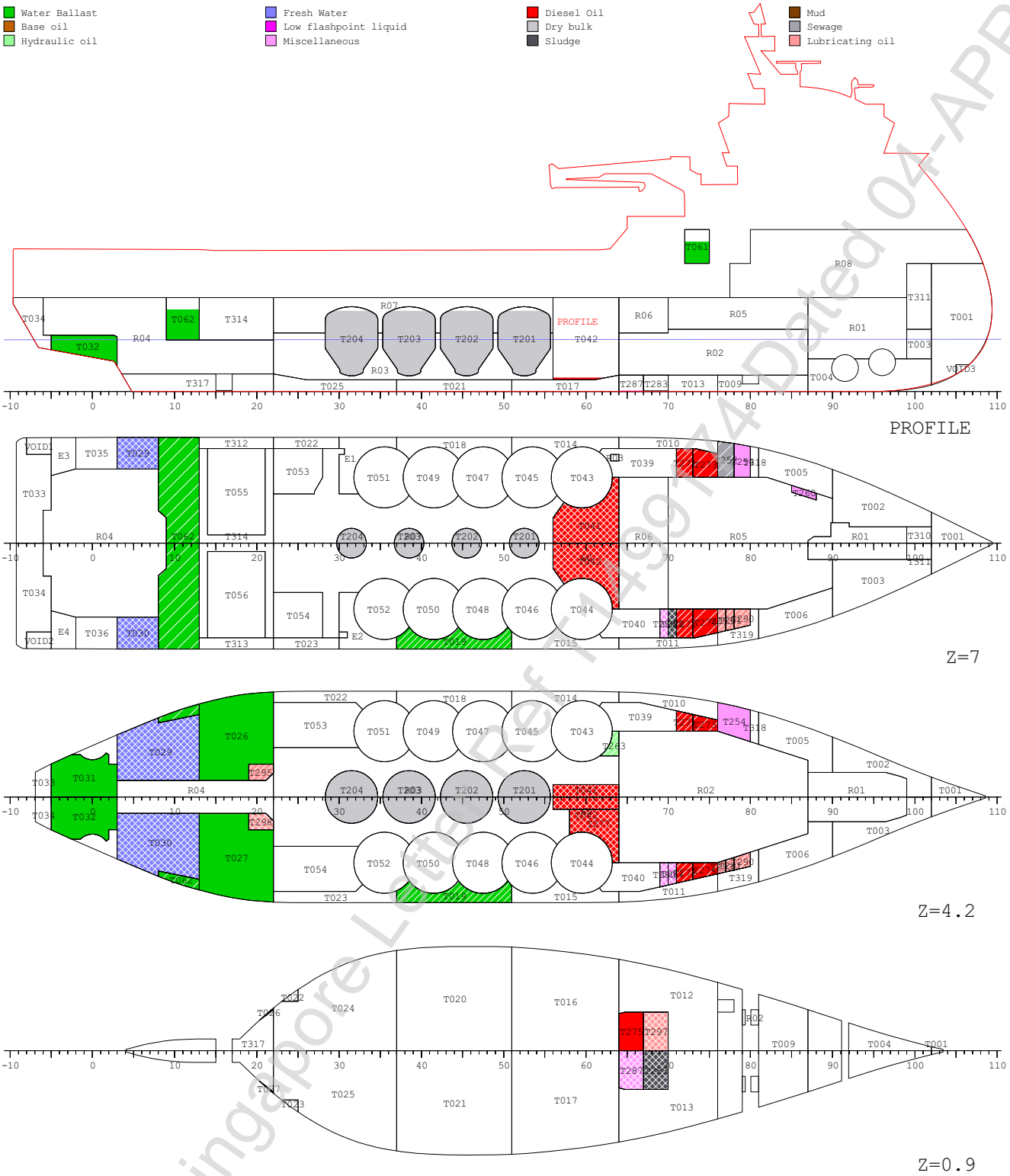




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12.13 L13 -> CEM - Arrival



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LIST OF LOADS

NAME	DES	LOAD	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm	FRS	HM m
CONTENTS=Dry bulk (RHO=2.4)										
T201	DRY BULK TANK	CEM	152.8	98.0	36.75	0.00	4.62	11	REAL	6.88
T202	DRY BULK TANK	CEM	152.8	98.0	31.85	0.00	4.62	11	REAL	6.88
T203	DRY BULK TANK	CEM	152.8	98.0	26.95	0.00	4.62	11	REAL	6.88
T204	DRY BULK TANK	CEM	152.8	98.0	22.05	0.00	4.62	11	REAL	6.88
SUBTOTAL		CEM	611.2		29.40	0.00	4.62	43		
CONTENTS=Diesel Oil (RHO=0.86)										
T041	STORAGE TANK PS	FO	0.7	1.0	41.24	-0.52	1.10	27	MAX	1.20
T042	STORAGE TANK SB	FO	0.7	1.0	41.24	0.52	1.10	27	MAX	1.20
T271	FO SETTLING TANK PS	FO	8.5	70.0	50.38	-6.53	4.93	1	REAL	6.72
T272	FO SETTLING TANK SB	FO	8.5	70.0	50.38	6.53	4.93	1	REAL	6.72
T273	FO SERVICE TANK PS	FO	10.6	70.0	52.09	-6.38	5.15	2	REAL	6.90
T274	FO SERVICE TANK SB	FO	10.6	70.0	52.09	6.38	5.15	2	REAL	6.90
T275	FO DRAIN TANK	FO	7.9	98.0	45.85	-1.64	0.69	5	REAL	1.37
T277	FO OVERFLOW TANK	FO	3.6	10.0	42.62	2.34	1.31	10	REAL	1.53
T279	EMERG. GEN. FO TANK	FO	1.9	98.0	61.60	-1.38	24.57	0	REAL	25.80
SUBTOTAL		FO	53.0		50.02	-0.13	4.73	75		
CONTENTS=Fresh Water (RHO=1)										
T029	WING TANK PS	FW	7.5	9.8	7.28	-2.50	2.65	83	MAX	3.01
T030	WING TANK SB	FW	7.5	9.8	7.28	2.50	2.65	83	MAX	3.01
SUBTOTAL		FW	14.9		7.28	0.00	2.65	165		
CONTENTS=Hydraulic oil (RHO=0.924)										
T263	HYDRAULIC OIL STORE TANK	HO	1.9	40.0	44.29	-4.33	1.89	1	REAL	2.47
CONTENTS=Lubricating oil (RHO=0.924)										
T290	LO STORE TANK MAIN ENG.	LO	3.4	40.0	55.32	5.47	4.05	0	REAL	5.16
T291	LO STORE TANK MAIN ENG.	LO	1.8	40.0	54.25	5.81	4.26	0	REAL	5.48
T293	LO STORE TANK THRUSTER	LO	1.8	40.0	53.55	6.03	4.41	0	REAL	5.66
T295	LO STORE TANK AZIMUTH THR.	LO	2.3	40.0	14.35	-2.10	2.63	0	REAL	3.06
T297	LO DRAIN TANK	LO	0.9	10.0	47.95	-1.64	0.07	6	REAL	0.14
T298	DIRTY LO TANK	LO	0.6	10.0	14.35	2.10	2.31	0	REAL	2.41
SUBTOTAL		LO	10.7		43.36	3.26	3.43	7		
CONTENTS=Mass load (RHO=1)										
M1	CREW	MASS	2.6	0.0	57.40	0.00	15.50	0		0.00
M2	PROVISION	MASS	0.8	0.0	56.35	-6.50	12.40	0		0.00
M3	STORES	MASS	0.8	0.0	72.10	0.00	12.40	0		0.00
SUBTOTAL		MASS	4.2		60.00	-1.24	14.32	0		
CONTENTS=Miscellaneous (RHO=1)										
T280	BILGE WATER HOLDING TANK	MIS	0.9	10.0	48.64	6.17	2.03	0	REAL	2.57
T281	BILGE WATER SETTLING TANK	MIS	0.4	10.0	49.34	6.04	1.76	0	REAL	2.08
T287	BILGE WATER DRAIN TANK	MIS	0.9	10.0	45.85	1.64	0.07	6	REAL	0.14
T254	GREY WATER TANK	MIS	31.6	98.0	54.82	-6.47	4.52	3	REAL	7.84
T260	WATER MIST TANK	MIS	2.3	98.0	60.55	-4.30	6.28	0	REAL	7.27
SUBTOTAL		MIS	36.2		54.74	-5.68	4.43	9		
CONTENTS=Sewage (RHO=1)										
T252	BLACK WATER TANK	SEW	10.4	97.9	53.89	-7.06	6.66	4	REAL	7.95
CONTENTS=Sludge (RHO=1)										
T283	SLUDGE DRAIN TANK	SLU	0.9	10.0	47.95	1.64	0.07	6	REAL	0.14
T288	SLUDGE SETTLING TANK	SLU	0.4	10.0	49.35	6.75	5.44	1	REAL	5.59
SUBTOTAL		SLU	1.4		48.40	3.29	1.80	7		
CONTENTS=Water Ballast (RHO=1.025)										
T019	WING TANK SB	WB	35.0	47.1	30.81	8.37	3.00	2	MAX	4.54
T026	WING TANK PS	WB	92.4	100.0	12.42	-4.45	3.13	0	(REAL)	4.40
T027	WING TANK SB	WB	92.4	100.0	12.42	4.45	3.13	0	(REAL)	4.40
T031	WING TANK PS	WB	27.2	100.0	-0.65	-1.76	4.18	0	(REAL)	5.39
T032	WING TANK SB	WB	27.2	100.0	-0.65	1.76	4.18	0	(REAL)	5.39
T061	ROLL RED. TANK	WB	83.4	65.0	51.66	0.00	11.84	1358	REAL	12.79
T062	ROLL RED. TANK	WB	147.4	70.0	7.45	0.00	5.67	1736	REAL	6.93

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SUBTOTAL	WB	505.0	17.32	0.58	5.41	3097
TOTAL		1248.9	26.33	0.03	4.95	3407

LOAD SUMMARY TABLE

ITEM	MASS t	LCG m	TCG m	VCG m	FRSM tm
Dry bulk	611.2	29.40	0.00	4.62	42.74
Diesel Oil	53.0	50.02	-0.13	4.73	74.98
Fresh Water	14.9	7.28	0.00	2.65	165.03
Hydraulic oil	1.9	44.29	-4.33	1.89	0.61
Lubricating oil	10.7	43.36	3.26	3.43	7.30
Mass load	4.2	60.00	-1.24	14.32	0.00
Miscellaneous	36.2	54.74	-5.68	4.43	9.39
Sewage	10.4	53.89	-7.06	6.66	3.55
Sludge	1.4	48.40	3.29	1.80	6.99
Water Ballast	505.0	17.32	0.58	5.41	3096.74
DEADWEIGHT	1248.9	26.33	0.03	4.95	3407.33
Lightweight	2704.9	39.15	-0.01	7.08	
Deadweight	1248.9	26.33	0.03	4.95	
Total weight	3953.8	35.10	0.00	6.41	

F L O A T I N G P O S I T I O N

Draught moulded	4.438	m	KM	9.09 m
Trim	0.039	m	KG	6.41 m
Heel, SB=+	0.0	deg		
TA	4.419	m	GM0	2.68 m
TF	4.457	m	GMCORR	-0.86 m
Trimming moment	212	tonm	GM	1.82 m

STABILITY COMPONENTS

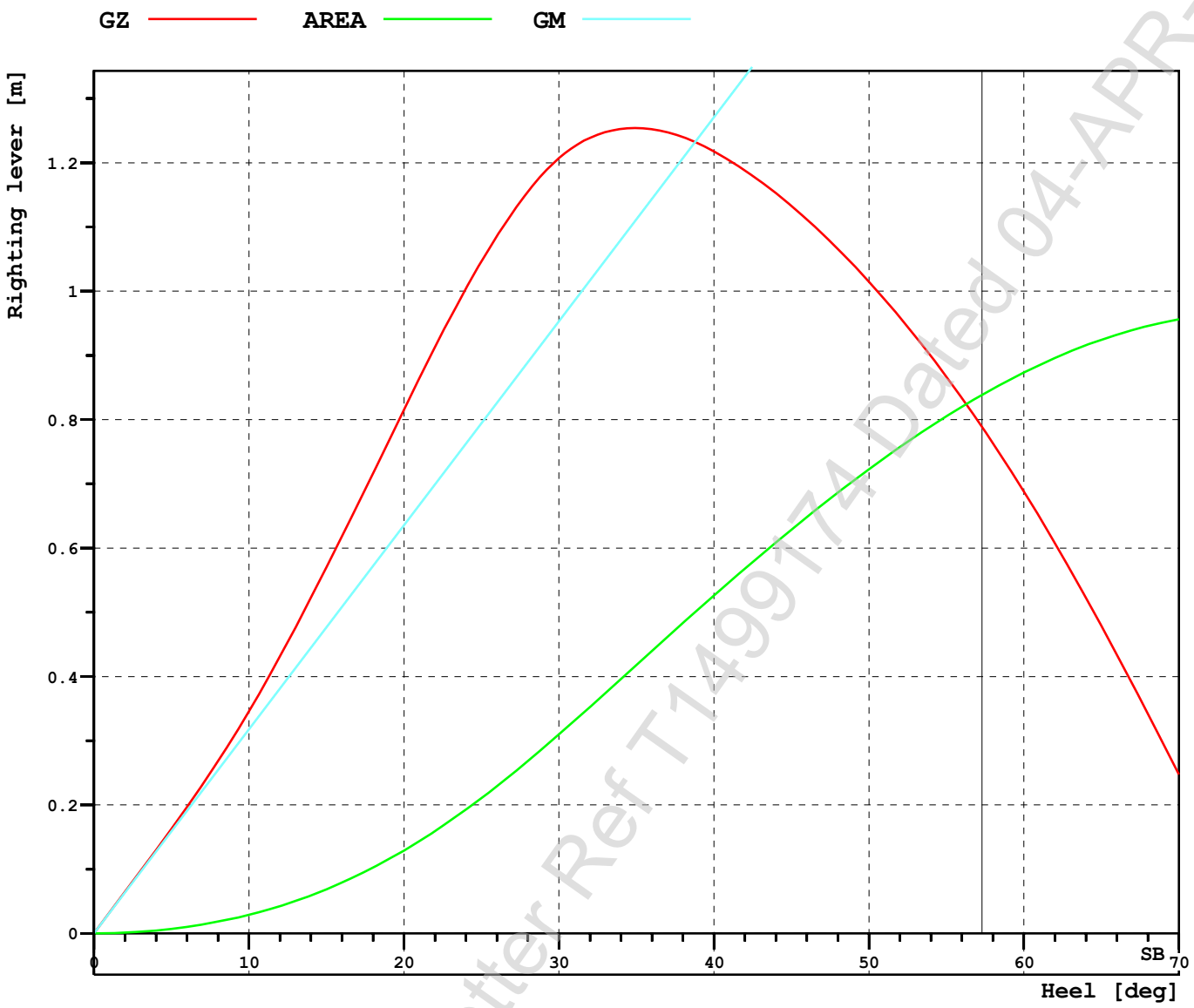
LOADING CONDITION L13, CEM - Arrival

HEEL degree	MS m	HPHI m	EPHI mrad	FSMOM tm	DGZ m	GMSIN m	KN m
0.0	0.000	0.00	0.000	0.0	0.000	0.000	-0.002
1.0	0.001	0.03	0.000	57.7	0.015	0.047	0.157
3.0	0.001	0.10	0.003	173.1	0.044	0.140	0.475
5.0	0.002	0.16	0.007	288.7	0.073	0.234	0.792
10.0	0.014	0.35	0.029	527.0	0.133	0.466	1.590
15.0	0.038	0.57	0.069	640.0	0.162	0.694	2.389
20.0	0.075	0.82	0.129	697.4	0.176	0.917	3.182
30.0	0.055	1.21	0.310	746.8	0.189	1.341	4.598
40.0	-0.317	1.22	0.526	749.0	0.189	1.724	5.524
50.0	-0.858	1.01	0.723	719.1	0.182	2.055	6.104
60.0	-1.466	0.69	0.873	666.6	0.169	2.323	6.405
70.0	-2.122	0.25	0.956	597.6	0.151	2.521	6.419

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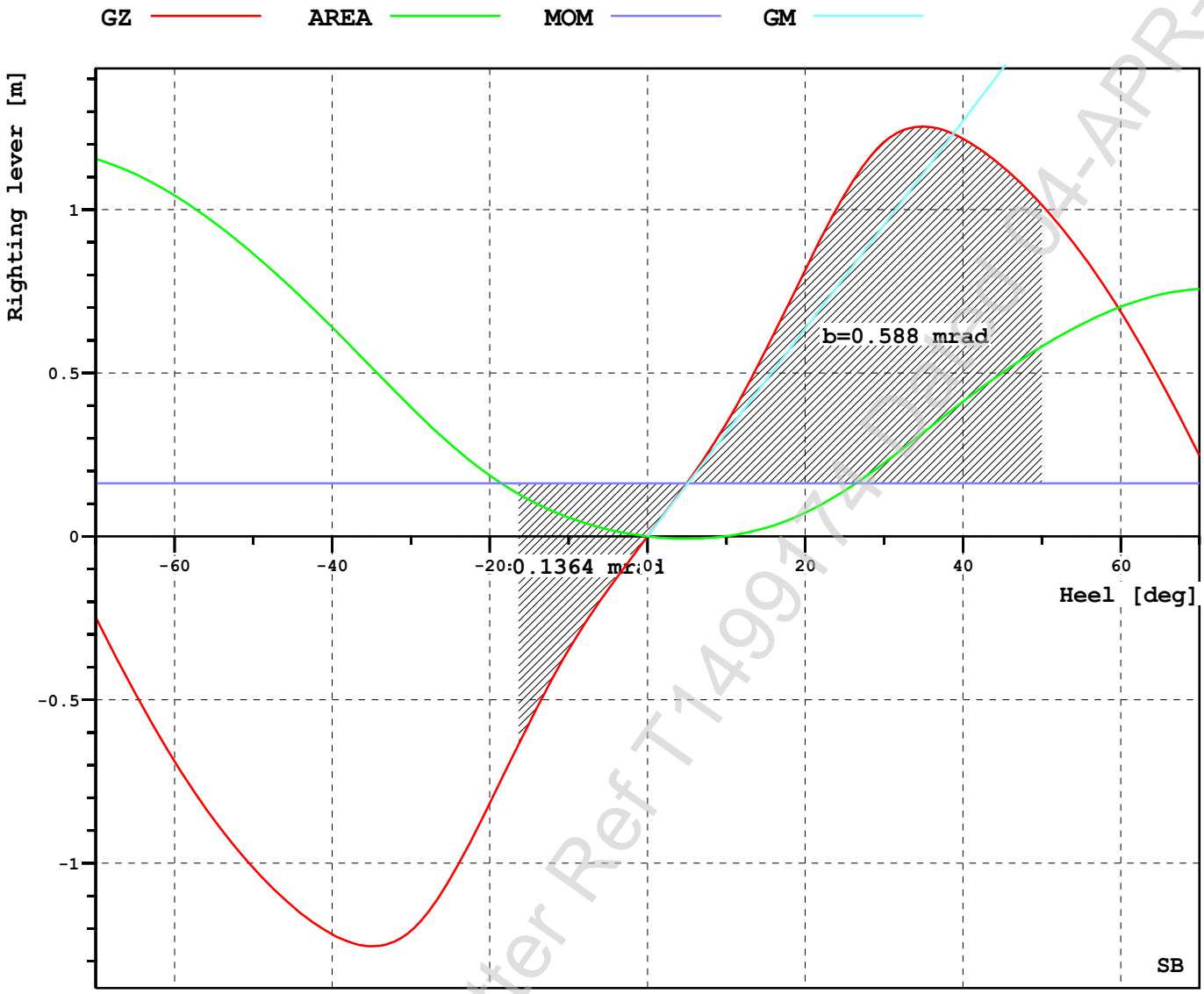
STABILITY CHECK PLOT



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IMOWEATHER CHECK PLOT



INTACT STABILITY CRITERIA

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM
V.AREA15-30	Area depending on the position of the Max. .	0.055	0.310	mrad	OK	-0.016
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.216	mrad	OK	-0.045
V.GZ0.2	Min. GZ > 0.2	0.200	1.254	m	OK	-0.194
V.POSMAX15	Max. GZ at an angle > 15 deg.	15.000	34.922	deg	OK	-0.657
V.GM0.15	GM > 0.15 m	0.150	1.821	m	OK	0.150
IMOWEATHER	IMO weather criterion	1.000	4.312		OK	0.065
2008IS-A2.3.1.2	Heeling angle due to steady wind <16 or <=8.	16.000	3.344	deg	OK	-0.030
ABSFIFI1	Angle of heel < 1/2 freeboard immersion	11.001	0.654	deg	OK	-0.100
ABSFIFI2	Area of residual stability >0.09 mrad	0.090	0.511	mrad	OK	0.010
TOTAL						0.150

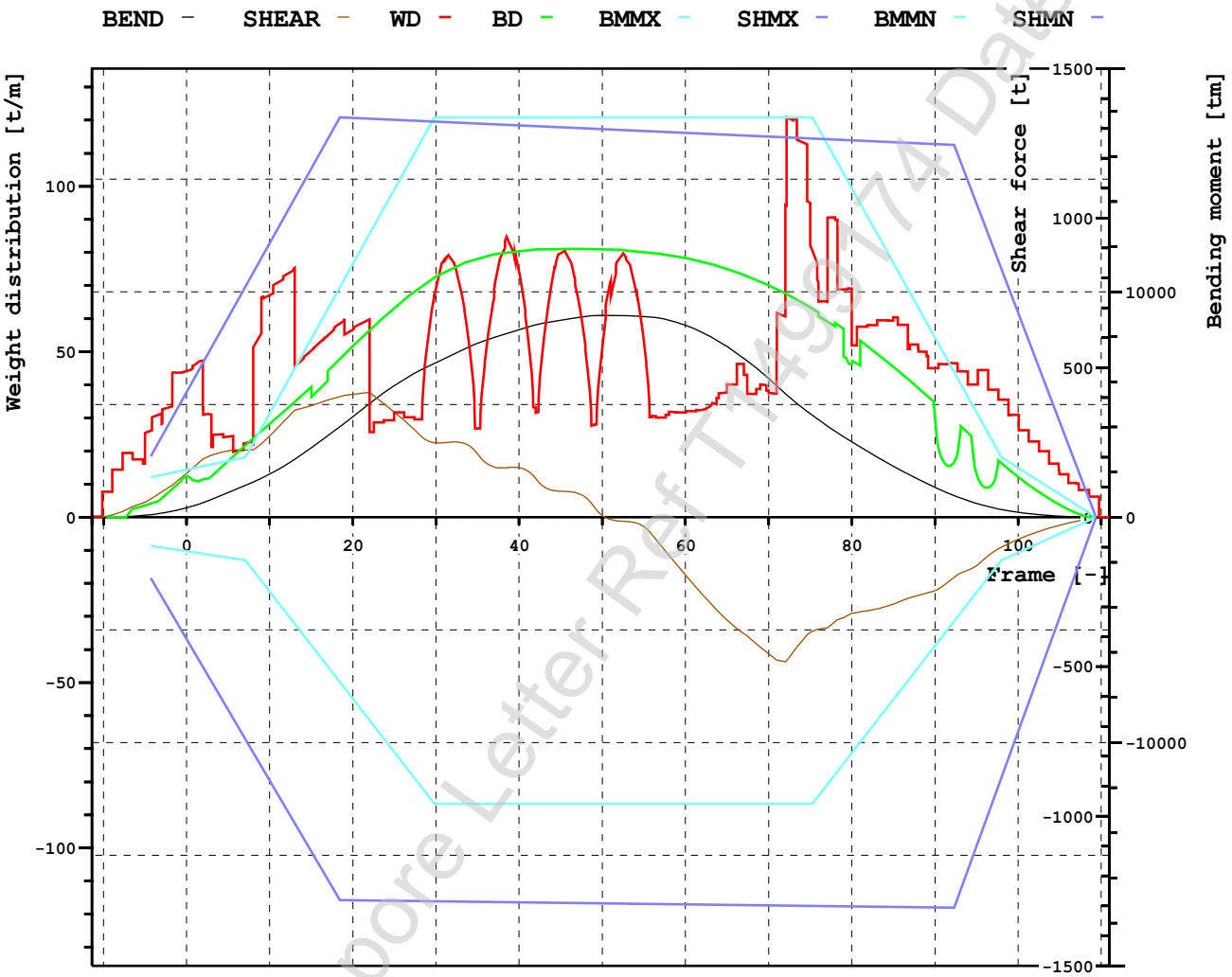
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MOST CRITICAL OPENINGS

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
V-421G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	55.890	2.093	14.288	-	9.841
V-424G	SUPPLY AIR ENGINE ROOM PS	UNPROTECTED	54.243	-3.413	14.288	-	9.842
V-425G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	54.243	3.767	14.288	-	9.842
D-651	ACCOMMODATION/C-DECK	UNPROTECTED	56.000	3.771	16.906	-	12.459
V-421E	EXHAUST AIR ENGINE ROOM PS	UNPROTECTED	55.993	-2.538	17.088	-	12.641
V-423E	EXHAUST AIR ENGINE ROOM SB	UNPROTECTED	55.993	3.423	17.088	-	12.641

STRENGTH DIAGRAM



STRENGTH SUMMARY

LOADING CONDITION L13, CEM - Arrival

			X	Frame
Shear force (min)	-483.9 t	(37.2%)	50.400 m	72.00
Shear force (max)	417.2 t	(31.3%)	15.400 m	22.00
Sagging moment	-0.3 tm	(0.0%)	-7.104 m	-10.15
Hogging moment	8953.2 tm	(50.5%)	35.210 m	50.30

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STRENGTH DISTRIBUTION

LOADING CONDITION L13, CEM - Arrival

X	FR	BEND	BMMN	BMMX	BMREL	SHEAR	SHMN	SHMX	SHREL	WD	BD
m	#	tm	tm	tm	%	t	t	t	%	t/m	t/m
-3.500	-5.00	79	0	0	0.0	50	0	0	0.0	25.43	3.51
0.000	0.00	408	-1510	2108	19.4	147	-407	418	35.1	44.10	12.26
3.500	5.00	1087	-1790	2498	43.5	222	-644	667	33.2	24.31	16.91
7.000	10.00	1906	-3292	4594	41.6	273	-881	916	29.7	67.13	28.19
10.500	15.00	3073	-5668	7911	38.8	372	-1117	1166	32.0	50.37	39.37
14.000	20.00	4452	-8045	11228	39.7	411	-1281	1335	30.8	56.69	51.68
17.500	25.00	5838	-10422	14545	40.1	351	-1283	1329	26.4	31.62	63.20
21.000	30.00	6859	-12709	17736	38.7	248	-1284	1323	18.8	69.23	72.69
24.500	35.00	7711	-12709	17736	43.5	211	-1286	1317	16.0	26.82	78.03
28.000	40.00	8316	-12709	17736	46.9	165	-1288	1310	12.6	74.20	80.32
31.500	45.00	8735	-12709	17736	49.2	87	-1290	1304	6.6	79.83	81.03
35.000	50.00	8953	-12709	17736	50.5	5	-1292	1298	0.4	51.56	80.94
38.500	55.00	8904	-12709	17736	50.2	-32	-1293	1292	2.4	53.51	80.07
42.000	60.00	8513	-12709	17736	48.0	-194	-1295	1285	15.0	31.65	78.30
45.500	65.00	7560	-12709	17736	42.6	-344	-1297	1279	26.5	40.12	74.98
49.000	70.00	6156	-12709	17736	34.7	-457	-1299	1273	35.2	38.25	70.09
52.500	75.00	4581	-12709	17736	25.8	-385	-1300	1267	29.6	95.03	63.46
56.000	80.00	3333	-10463	14601	22.8	-322	-1302	1260	24.7	69.10	46.32
59.500	85.00	2251	-8086	11284	19.9	-291	-1304	1254	22.3	60.41	45.43
63.000	90.00	1316	-5709	7968	16.5	-246	-1306	1248	18.8	45.02	30.42
66.500	95.00	610	-3332	4651	13.1	-162	-1102	1050	14.7	40.05	13.04
70.000	100.00	216	-1571	2193	9.8	-74	-718	685	10.4	30.94	12.19
73.500	105.00	42	-733	1023	4.1	-27	-335	319	8.0	12.99	3.74

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DAMAGE STABILITY OF LOADING CONDITION L13

CRITICAL DAMAGE STABILITY CRITERIA FOR EACH DAMAGE CASE

RCR	DAM	STAGE	PHASE	REQ	ATTV	UNIT	STAT	MINGM
V.RANGE.A469	D01	1	EQ	20.000	69.984	deg	OK	-0.419
V.RANGE.A469	D02	1	EQ	20.000	69.634	deg	OK	-0.325
V.RANGE.A469	D03	1	EQ	20.000	69.602	deg	OK	-0.331
V.MAXHEEL.A469	D04	1	EQ	17.000	2.463	deg	OK	-0.052
V.RANGE.A469	D05	1	EQ	20.000	69.659	deg	OK	-0.189
V.MAXHEEL.A469	D06	1	EQ	17.000	2.424	deg	OK	-0.031
V.RANGE.A469	D07	1	EQ	20.000	69.734	deg	OK	-0.176
V.MAXHEEL.A469	D08	1	EQ	17.000	4.584	deg	OK	-0.002
V.MAXHEEL.A469	D09	1	EQ	17.000	4.614	deg	OK	0.001
V.MAXHEEL.A469	D10	1	EQ	17.000	3.302	deg	OK	-0.143
V.MAXHEEL.A469	D11	1	EQ	17.000	5.418	deg	OK	0.063
V.MAXHEEL.A469	D12	1	EQ	17.000	4.369	deg	OK	-0.048
V.MAXHEEL.A469	D13	1	EQ	17.000	4.402	deg	OK	-0.042
V.RANGE.A469	D14	1	EQ	20.000	69.911	deg	OK	-0.110
V.RANGE.A469	D15	1	EQ	20.000	69.949	deg	OK	-0.105
V.RANGE.A469	D16	1	EQ	20.000	69.288	deg	OK	-0.321
V.RANGE.A469	D17	1	EQ	20.000	69.259	deg	OK	-0.323
V.RANGE.A469	D18	1	EQ	20.000	69.984	deg	OK	-0.420
V.RANGE.A469	D19	1	EQ	20.000	69.984	deg	OK	-0.350
V.RANGE.A469	D20	1	EQ	20.000	70.000	deg	OK	-0.381
V.RANGE.A469	D21	1	EQ	20.000	69.968	deg	OK	-0.383
TOTAL								0.063

DETERMINATION OF CRITICAL DAMAGE CASE

CRITICAL DAMAGE CASE : D11
STAGE OF CRITICAL DAMAGE CASE : 1
PHASE OF CRITICAL DAMAGE CASE : EQ

DAMAGE STABILITY CRITERIA - CRITICAL CASE

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM m
V.PROGR.A469	No progr. flooding	26.151	5.418	deg	OK	-0.366
V.MAXHEEL.A469	Maximum heel	17.000	5.418	deg	OK	0.063
V.RANGE.A469	Range of pos. stab	20.000	64.582	deg	OK	-0.054
V.MINGZ.A469	Min. righting lever	0.100	0.933	m	OK	-0.175
TOTAL						0.063

DAMAGE CASE RESULTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	T m	TR m	HEEL degree	RESFLD m	OPEN	CROSSTM s
D11	1	EQ	PS	4.521	0.058	-5.4	3.23	D-361	-

DAMAGED COMPARTMENTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	NAME	DES	PERM	VOL m3	W t	XCG m	YCG m	ZCG m
D11	INTACT	EQ	-				0.0	0.000			
D11	1	EQ	PS	T018	WING TANK PS	0.95	42.0	43.044	30.80	-8.37	3.40
D11	1	EQ	PS	T020	DB TANK PS	0.95	92.4	94.737	30.84	-4.63	0.61

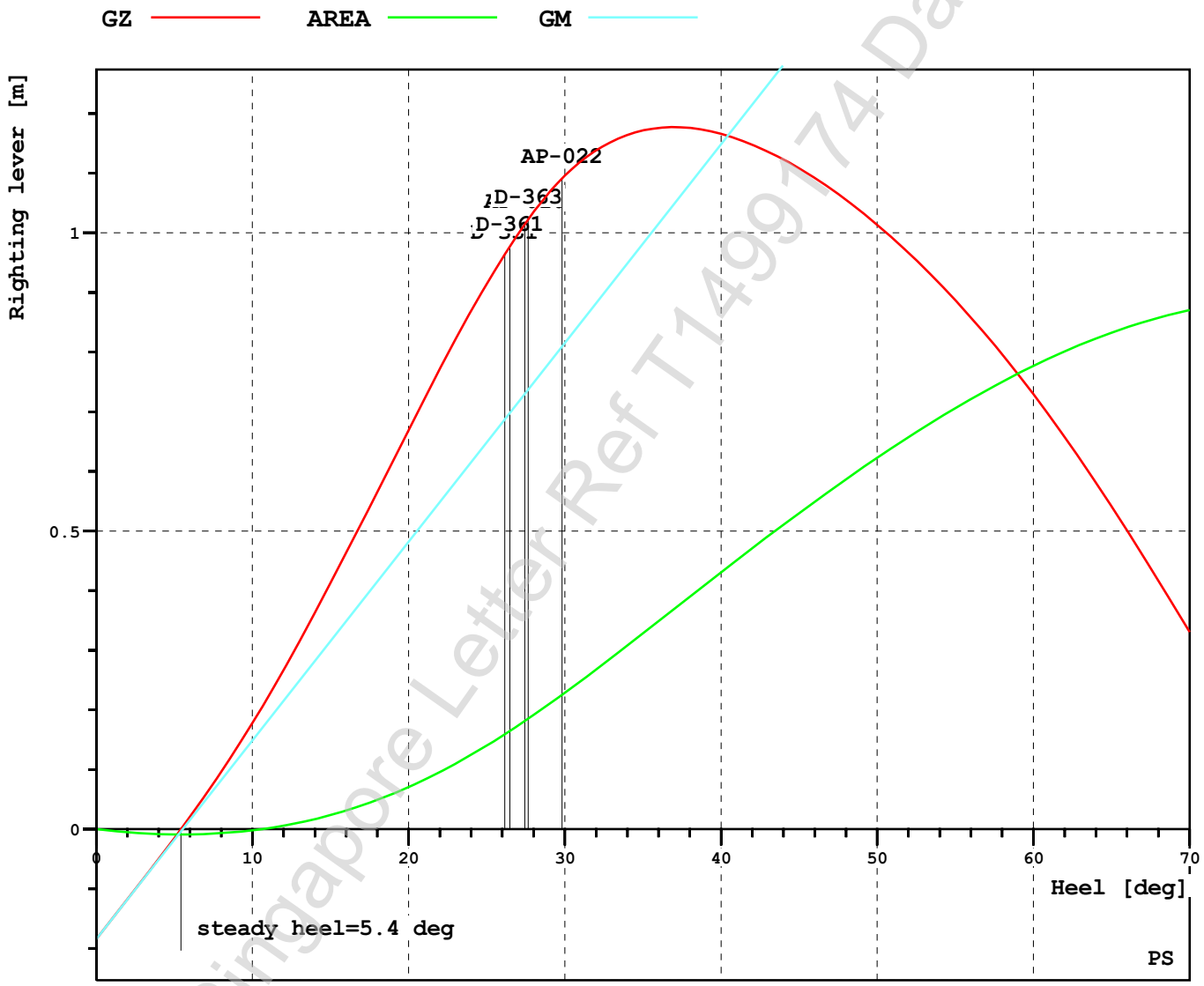
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CRITICAL OPENINGS - CRITICAL CASE

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
D-351	CO2 ROOM/M-DECK	WEATHERTIGHT	51.792	-8.360	8.600	26.2	3.240
D-361	PS CARGORAIL #57/M-DECK	WEATHERTIGHT	39.885	-8.560	8.600	26.5	3.230
AP-010	WING TANK PS	WEATHERTIGHT	47.211	-8.731	8.953	27.4	3.560
D-363	PS CARGORAIL #37/M-DECK	WEATHERTIGHT	25.900	-8.550	8.600	27.7	3.241
AP-022	WING TANK PS	WEATHERTIGHT	19.200	-8.702	8.936	29.8	3.566
AP-263	HYDRAULIC OIL STORE TANK	WEATHERTIGHT	47.690	-7.260	8.912	32.2	3.658
D-353	DECK WORKSHOP/M-DECK	WEATHERTIGHT	53.192	-6.550	8.600	32.5	3.410
AP-012	DB TANK PS	WEATHERTIGHT	50.000	-7.250	9.047	32.8	3.792

DAMAGE STABILITY CHECK PLOT - CRITICAL CASE



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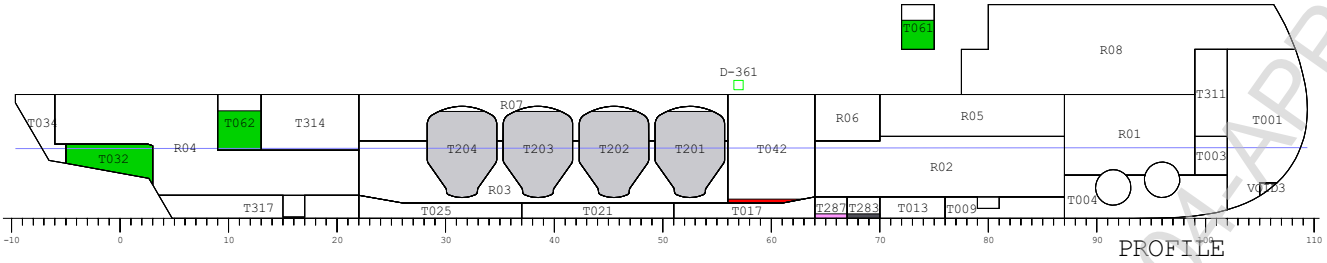
GZ-CURVE - CRITICAL CASE

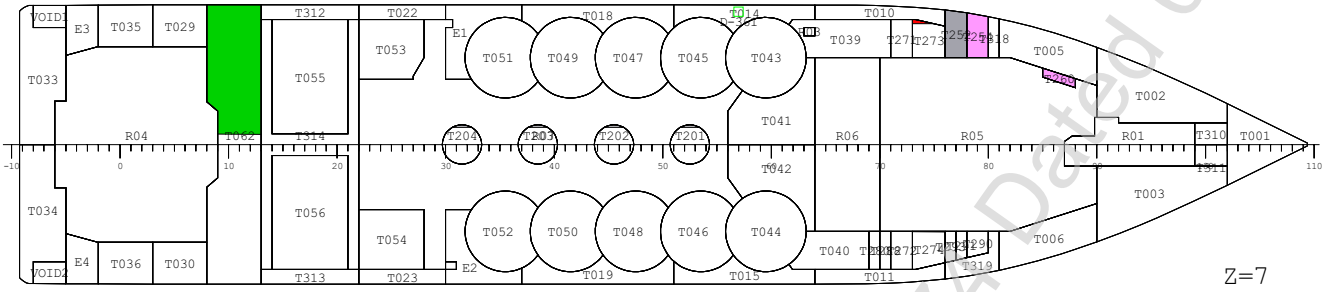
HEEL	DISP	T	TR	GZ	MS	AREA
degree	t	m	m	m	m	mrad
0.0	3953.8	4.545	0.006	0.184	0.184	0.000
-1.0	3953.8	4.545	0.007	0.151	0.198	-0.003
-3.0	3953.8	4.539	0.021	0.083	0.224	-0.007
-5.0	3953.8	4.525	0.050	0.015	0.249	-0.009
-10.0	3953.8	4.452	0.182	-0.178	0.288	-0.002
-15.0	3953.8	4.326	0.385	-0.413	0.282	0.024
-20.0	3953.8	4.148	0.619	-0.669	0.248	0.071
-30.0	3953.8	3.651	1.071	-1.096	0.245	0.229
-40.0	3953.8	2.997	1.199	-1.166	0.558	0.431
-50.0	3953.8	2.199	0.878	-1.013	1.041	0.623
-60.0	3953.8	1.301	0.308	-0.729	1.594	0.777
-70.0	3953.8	0.375	-0.350	-0.330	2.190	0.871

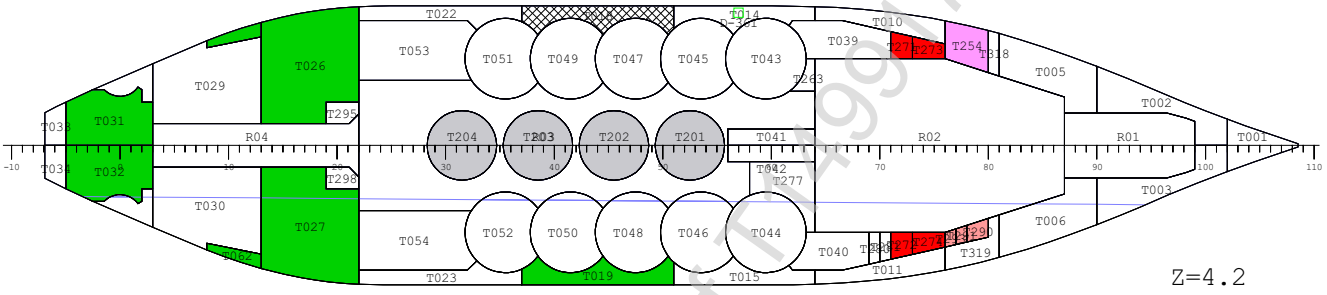
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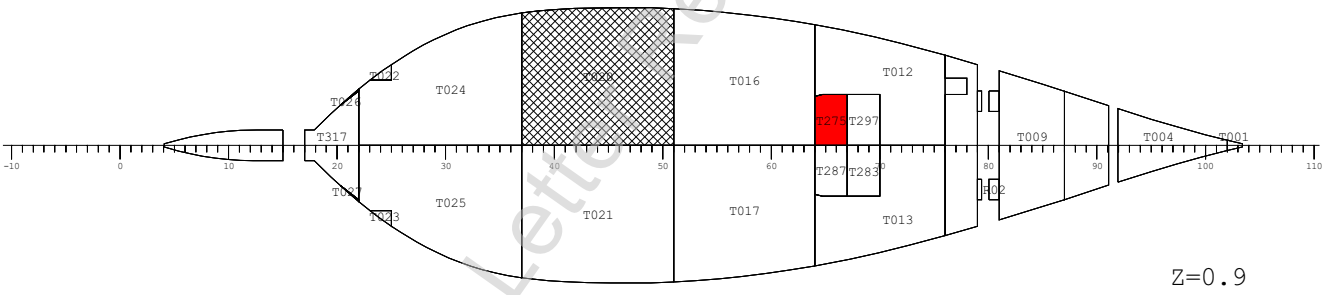
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DAMAGE ARRANGEMENT - CRITICAL CASE





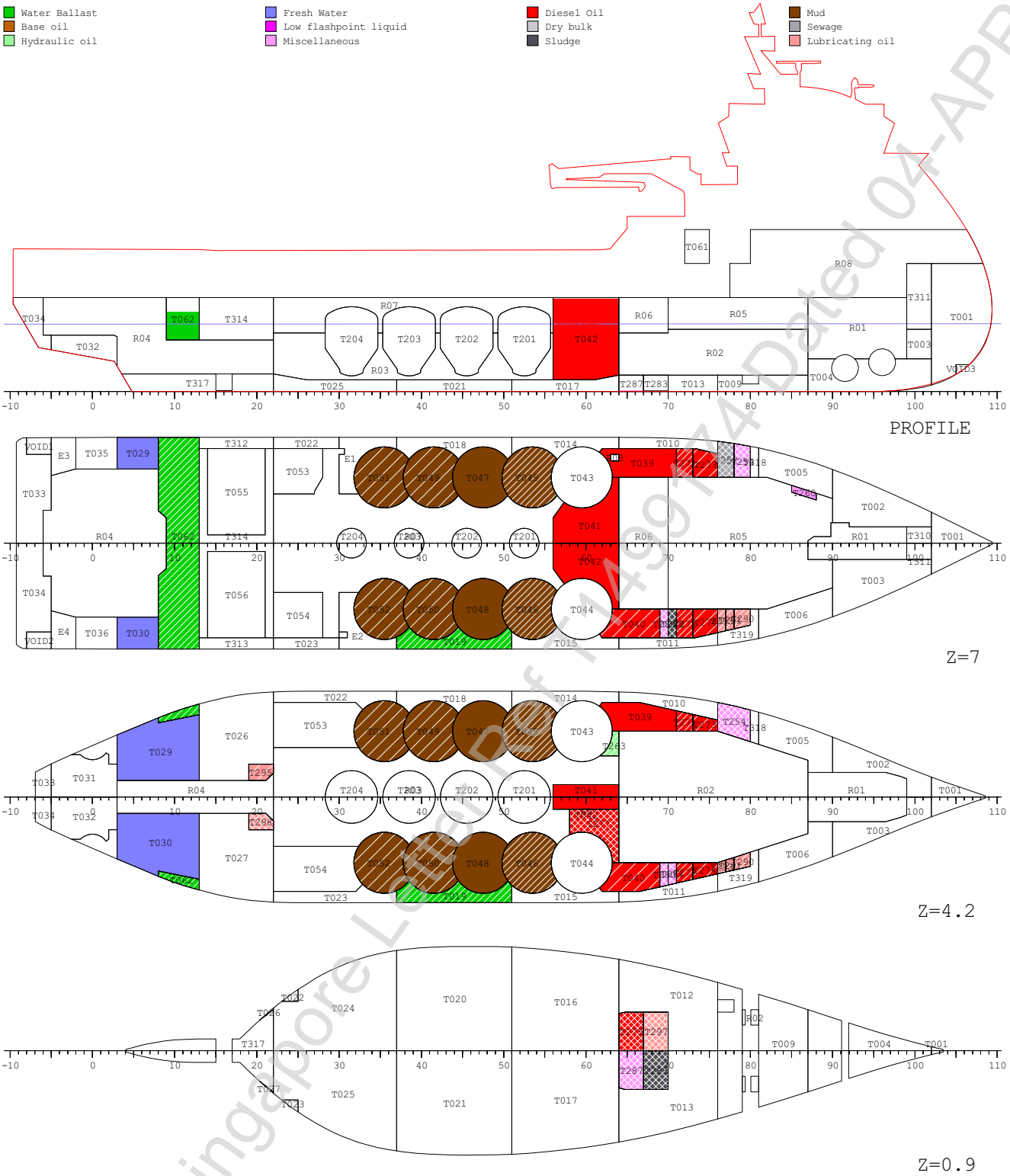




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12.14 L14 -> MUD - Departure



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LIST OF LOADS

NAME	DES	LOAD	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm	FRS	HM m
CONTENTS=Diesel Oil (RHO=0.86)										
T039	STORAGE TANK PS	FO	66.9	98.0	46.68	-6.76	4.91	6	REAL	7.88
T040	STORAGE TANK SB	FO	53.8	98.0	46.00	6.80	4.87	5	REAL	7.88
T041	STORAGE TANK PS	FO	68.5	98.0	42.16	-1.51	5.48	27	MAX	7.92
T042	STORAGE TANK SB	FO	68.5	98.0	42.16	1.51	5.48	27	MAX	7.92
T271	FO SETTLING TANK PS	FO	8.5	70.0	50.38	-6.53	4.93	1	REAL	6.72
T272	FO SETTLING TANK SB	FO	8.5	70.0	50.38	6.53	4.93	1	REAL	6.72
T273	FO SERVICE TANK PS	FO	10.6	70.0	52.09	-6.38	5.15	2	REAL	6.90
T274	FO SERVICE TANK SB	FO	10.6	70.0	52.09	6.38	5.15	2	REAL	6.90
T275	FO DRAIN TANK	FO	0.8	10.0	45.85	-1.64	0.07	5	REAL	0.14
T277	FO OVERFLOW TANK	FO	3.6	10.0	42.62	2.34	1.31	10	REAL	1.53
T279	EMERG. GEN. FO TANK	FO	1.9	98.0	61.60	-1.38	24.57	0	REAL	25.80
SUBTOTAL		FO	302.3		45.14	-0.27	5.25	86		
CONTENTS=Fresh Water (RHO=1)										
T029	WING TANK PS	FW	74.6	98.0	5.29	-4.87	4.61	83	MAX	7.83
T030	WING TANK SB	FW	74.6	98.0	5.29	4.87	4.61	83	MAX	7.83
SUBTOTAL		FW	149.3		5.29	0.00	4.61	165		
CONTENTS=Hydraulic oil (RHO=0.924)										
T263	HYDRAULIC OIL STORE TANK	HO	3.3	70.0	44.30	-4.34	2.32	1	REAL	3.33
CONTENTS=Lubricating oil (RHO=0.924)										
T290	LO STORE TANK MAIN ENG.	LO	5.9	70.0	55.30	5.76	4.90	0	REAL	6.82
T291	LO STORE TANK MAIN ENG.	LO	3.1	70.0	54.25	6.04	5.11	0	REAL	6.92
T293	LO STORE TANK THRUSTER	LO	3.2	70.0	53.55	6.21	5.24	0	REAL	6.97
T295	LO STORE TANK AZIMUTH THR.	LO	4.0	70.0	14.35	-2.10	2.95	0	REAL	3.71
T297	LO DRAIN TANK	LO	0.9	10.0	47.95	-1.64	0.07	6	REAL	0.14
T298	DIRTY LO TANK	LO	0.6	10.0	14.35	2.10	2.31	0	REAL	2.41
SUBTOTAL		LO	17.6		43.89	3.64	4.24	8		
CONTENTS=Mass load (RHO=1)										
M1	CREW	MASS	2.6	0.0	57.40	0.00	15.50	0		0.00
M2	PROVISION	MASS	7.4	0.0	56.35	-6.50	12.40	0		0.00
M3	STORES	MASS	7.4	0.0	72.10	0.00	12.40	0		0.00
SUBTOTAL		MASS	17.5		63.21	-2.77	12.86	0		
CONTENTS=Miscellaneous (RHO=1)										
T280	BILGE WATER HOLDING TANK	MIS	0.9	10.0	48.64	6.17	2.03	0	REAL	2.57
T281	BILGE WATER SETTLING TANK	MIS	0.4	10.0	49.34	6.04	1.76	0	REAL	2.08
T287	BILGE WATER DRAIN TANK	MIS	0.9	10.0	45.85	1.64	0.07	6	REAL	0.14
T254	GREY WATER TANK	MIS	3.2	10.0	54.62	-5.79	1.92	1	REAL	2.34
T260	WATER MIST TANK	MIS	2.3	98.0	60.55	-4.30	6.28	0	REAL	7.27
SUBTOTAL		MIS	7.8		54.40	-2.49	3.01	8		
CONTENTS=Mud (RHO=2.8)										
T045	LIQUID CARGO TANK PS	MUD	79.8	22.4	37.22	-5.50	1.98	98	REAL	2.74
T046	LIQUID CARGO TANK SB	MUD	72.1	20.7	37.22	5.49	1.90	98	REAL	2.59
T047	LIQUID CARGO TANK PS	MUD	348.9	98.0	33.02	-5.58	4.54	98	REAL	7.86
T048	LIQUID CARGO TANK SB	MUD	348.9	98.0	33.02	5.58	4.54	98	REAL	7.86
T049	LIQUID CARGO TANK PS	MUD	348.9	98.0	28.82	-5.58	4.54	98	REAL	7.86
T050	LIQUID CARGO TANK SB	MUD	348.9	98.0	28.82	5.58	4.54	98	REAL	7.86
T051	LIQUID CARGO TANK PS	MUD	346.2	98.0	24.63	-5.56	4.57	98	REAL	7.87
T052	LIQUID CARGO TANK SB	MUD	346.2	98.0	24.63	5.56	4.57	98	REAL	7.87
SUBTOTAL		MUD	2240.0		29.40	-0.02	4.37	781		
CONTENTS=Sewage (RHO=1)										
T252	BLACK WATER TANK	SEW	1.1	10.0	53.89	-6.93	5.45	2	REAL	5.59
CONTENTS=Sludge (RHO=1)										
T283	SLUDGE DRAIN TANK	SLU	0.9	10.0	47.95	1.64	0.07	6	REAL	0.14
T288	SLUDGE SETTLING TANK	SLU	0.4	10.0	49.35	6.75	5.44	1	REAL	5.59
SUBTOTAL		SLU	1.4		48.40	3.29	1.80	7		
CONTENTS=Water Ballast (RHO=1.025)										
T019	WING TANK SB	WB	20.0	26.9	30.82	8.36	2.34	2	MAX	3.22
T062	ROLL RED. TANK	WB	136.9	65.0	7.45	0.00	5.58	1736	REAL	6.76

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SUBTOTAL	WB	156.9	10.43	1.07	5.17	1738
TOTAL		2897.0	29.17	0.01	4.56	2796

LOAD SUMMARY TABLE

ITEM	MASS t	LCG m	TCG m	VCG m	FRSM tm
Diesel Oil	302.3	45.14	-0.27	5.25	85.89
Fresh Water	149.3	5.29	0.00	4.61	165.03
Hydraulic oil	3.3	44.30	-4.34	2.32	0.61
Lubricating oil	17.6	43.89	3.64	4.24	7.62
Mass load	17.5	63.21	-2.77	12.86	0.00
Miscellaneous	7.8	54.40	-2.49	3.01	7.72
Mud	2240.0	29.40	-0.02	4.37	781.14
Sewage	1.1	53.89	-6.93	5.45	2.31
Sludge	1.4	48.40	3.29	1.80	6.99
Water Ballast	156.9	10.43	1.07	5.17	1738.33
DEADWEIGHT	2897.0	29.17	0.01	4.56	2795.65
Lightweight	2704.9	39.15	-0.01	7.08	
Deadweight	2897.0	29.17	0.01	4.56	
Total weight	5601.9	33.99	0.00	5.78	

F L O A T I N G P O S I T I O N

Draught moulded	5.762	m	KM	8.80 m
Trim	0.027	m	KG	5.78 m
Heel, SB=+	0.0	deg		
TA	5.748	m	GM0	3.02 m
TF	5.775	m	GMCORR	-0.50 m
Trimming moment	188	tonm	GM	2.52 m

STABILITY COMPONENTS

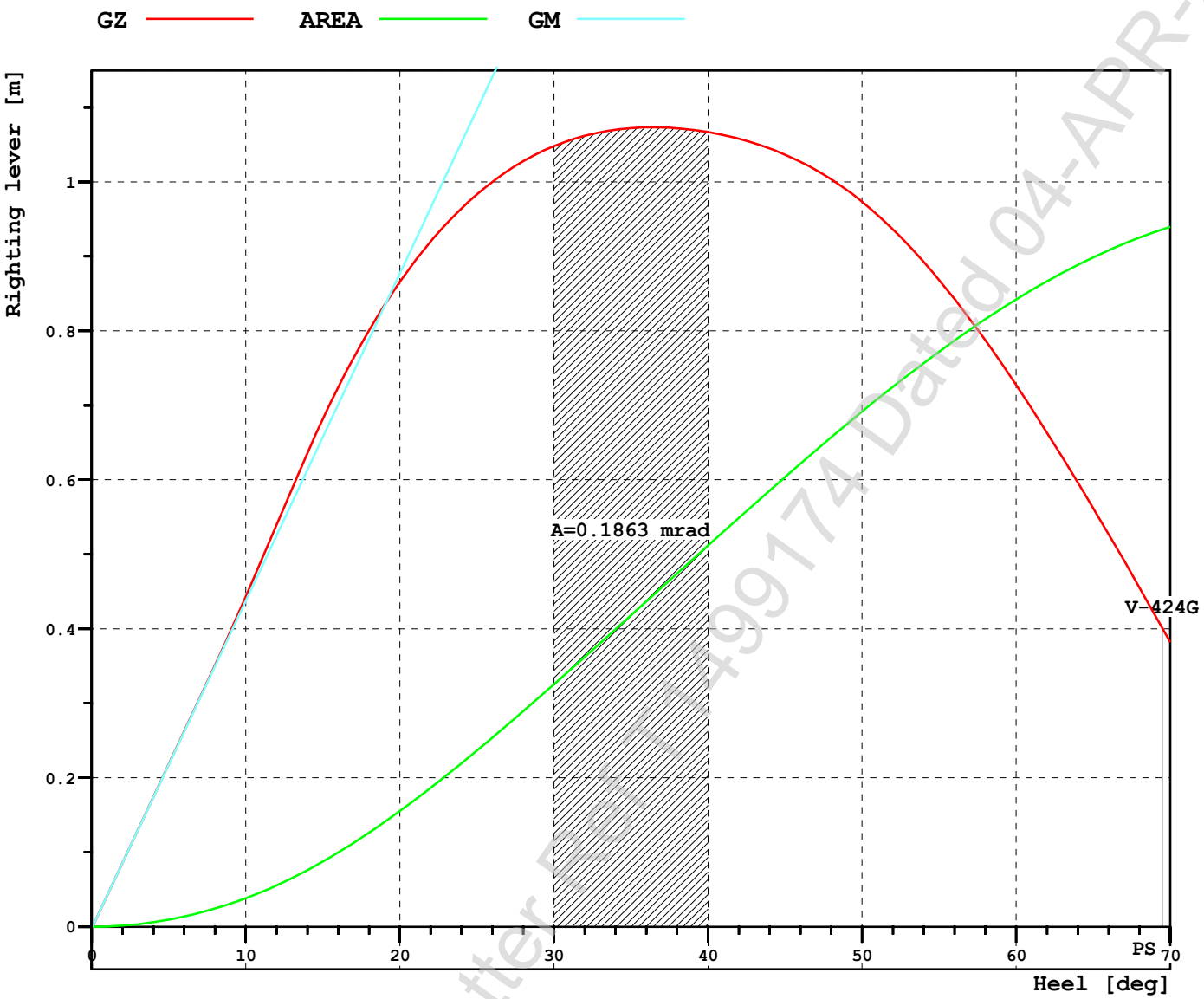
LOADING CONDITION L14, MUD - Departure

HEEL degree	MS m	HPHI m	EPHI mrad	FSMOM tm	DGZ m	GMSIN m	KN m
0.0	-0.002	0.00	0.000	0.0	0.000	0.000	0.001
1.0	-0.002	0.04	0.000	47.5	0.008	0.053	0.155
3.0	-0.003	0.13	0.003	142.5	0.025	0.158	0.461
5.0	-0.003	0.22	0.009	232.8	0.042	0.263	0.767
10.0	-0.005	0.44	0.038	423.6	0.076	0.524	1.526
15.0	-0.004	0.68	0.087	535.9	0.096	0.781	2.276
20.0	-0.058	0.87	0.155	607.7	0.108	1.032	2.954
30.0	-0.335	1.05	0.325	701.6	0.125	1.509	4.066
40.0	-0.738	1.07	0.512	754.8	0.135	1.940	4.919
50.0	-1.199	0.97	0.692	780.0	0.139	2.311	5.542
60.0	-1.745	0.73	0.842	786.7	0.140	2.613	5.874
70.0	-2.312	0.38	0.940	794.1	0.142	2.835	5.956

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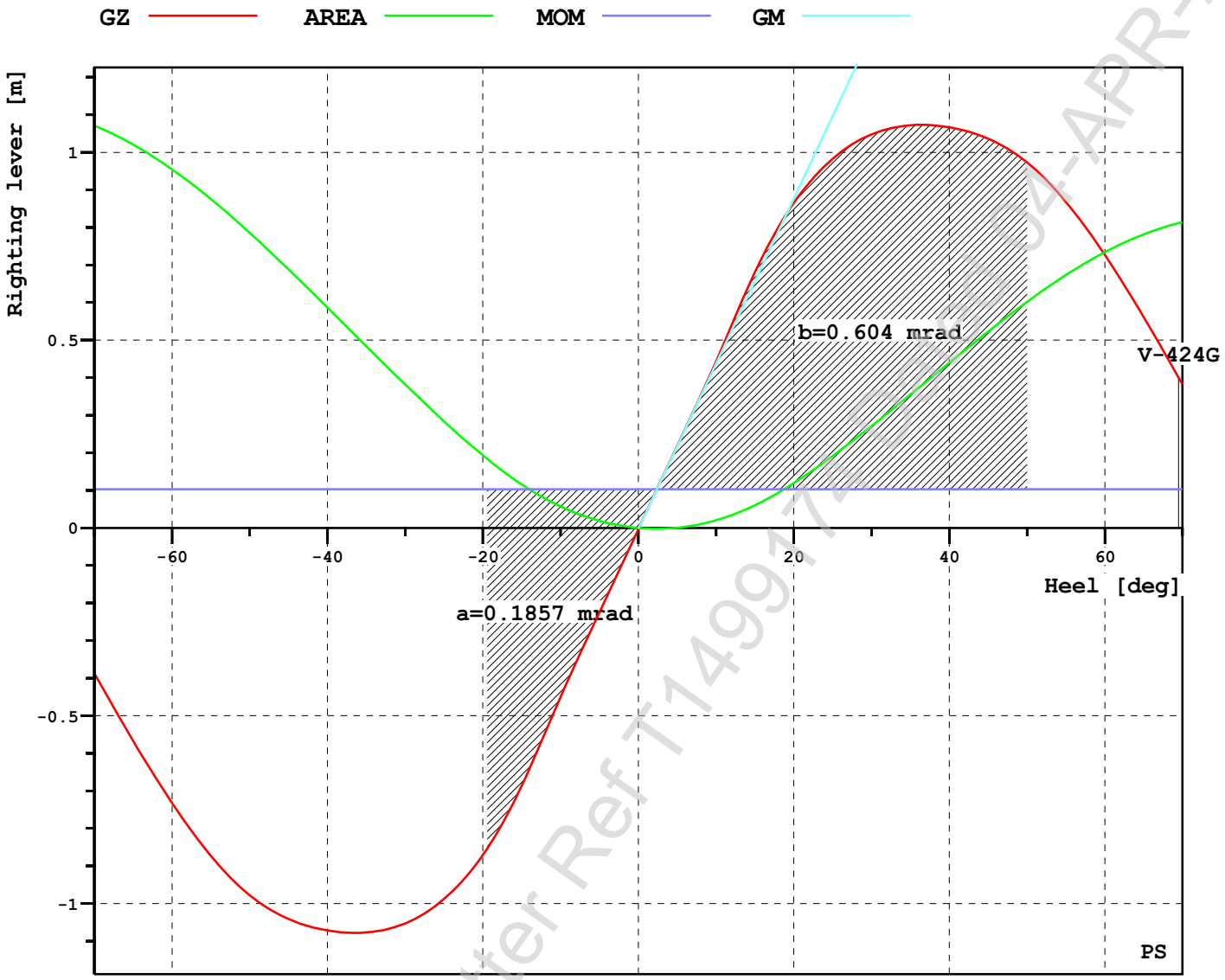
STABILITY CHECK PLOT



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IMOWEATHER CHECK PLOT



INTACT STABILITY CRITERIA

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM
V.AREA15-30	Area depending on the position of the Max. .	0.055	0.325	mrad	OK	0.891
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.186	mrad	OK	0.955
V.GZ0.2	Min. GZ > 0.2	0.200	1.073	m	OK	0.822
V.POSMAX15	Max. GZ at an angle > 15 deg.	15.000	36.412	deg	OK	-0.109
V.GM0.15	GM > 0.15 m	0.150	2.518	m	OK	0.150
IMOWEATHER	IMO weather criterion	1.000	3.253		OK	0.736
2008IS-A2.3.1.2	Heeling angle due to steady wind <16 or <=8.	11.253	1.615	deg	OK	0.290
ABSFIFI1	Angle of heel < 1/2 freeboard immersion	7.033	0.388	deg	OK	0.127
ABSFIFI2	Area of residual stability >0.09 mrad	0.090	0.501	mrad	OK	0.750
TOTAL						0.955

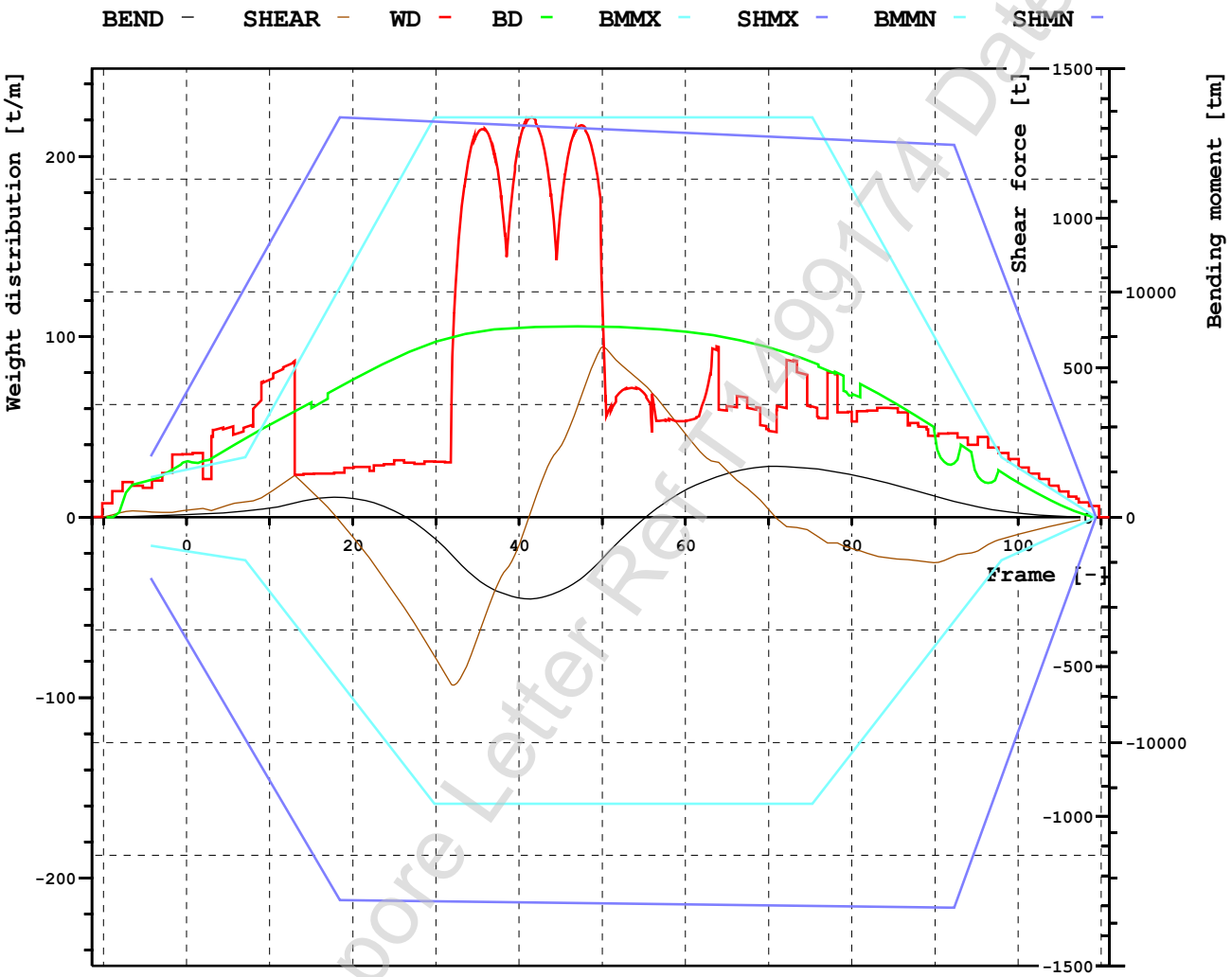
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MOST CRITICAL OPENINGS

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
V-424G	SUPPLY AIR ENGINE ROOM PS	UNPROTECTED	54.243	-3.413	14.288	-69.5	8.517
V-421G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	55.890	2.093	14.288	-	8.521
V-425G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	54.243	3.767	14.288	-	8.523
D-651	ACCOMMODATION/C-DECK	UNPROTECTED	56.000	3.771	16.906	-	11.140
V-421E	EXHAUST AIR ENGINE ROOM PS	UNPROTECTED	55.993	-2.538	17.088	-	11.317
V-423E	EXHAUST AIR ENGINE ROOM SB	UNPROTECTED	55.993	3.423	17.088	-	11.322

STRENGTH DIAGRAM



STRENGTH SUMMARY

LOADING CONDITION L14, MUD - Departure

			X	Frame
Shear force (min)	-561.6 t	(43.7%)	22.447 m	32.07
Shear force (max)	569.5 t	(43.9%)	35.047 m	50.07
Sagging moment	-3625.2 tm	(28.5%)	28.812 m	41.16
Hogging moment	2259.7 tm	(12.7%)	49.598 m	70.85

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STRENGTH DISTRIBUTION

LOADING CONDITION L14, MUD - Departure

X m	FR #	BEND tm	BMMN tm	BMMX tm	BMREL %	SHEAR t	SHMN t	SHMX t	SHREL %	WD t/m	BD t/m
-3.500	-5.00	48	0	0	0.0	20	0	0	0.0	16.23	19.75
0.000	0.00	108	-1510	2108	5.1	23	-407	418	5.5	34.85	30.88
3.500	5.00	205	-1790	2498	8.2	41	-644	667	6.1	49.50	38.00
7.000	10.00	399	-3292	4594	8.7	83	-881	916	9.0	76.26	51.24
10.500	15.00	786	-5668	7911	9.9	87	-1117	1166	7.5	24.04	63.58
14.000	20.00	834	-8045	11228	7.4	-66	-1281	1335	5.2	27.33	76.23
17.500	25.00	279	-10422	14545	1.9	-255	-1283	1329	19.9	31.62	87.78
21.000	30.00	-986	-12709	17736	7.8	-473	-1284	1323	36.8	30.72	97.27
24.500	35.00	-2767	-12709	17736	21.8	-403	-1286	1317	31.3	212.91	102.61
28.000	40.00	-3586	-12709	17736	28.2	-92	-1288	1310	7.1	207.62	104.89
31.500	45.00	-3261	-12709	17736	25.7	237	-1290	1304	18.2	170.77	105.59
35.000	50.00	-1840	-12709	17736	14.5	569	-1292	1298	43.9	117.81	105.49
38.500	55.00	-70	-12709	17736	0.5	445	-1293	1292	34.4	68.84	104.61
42.000	60.00	1202	-12709	17736	6.8	278	-1295	1285	21.6	53.17	102.83
45.500	65.00	1930	-12709	17736	10.9	155	-1297	1279	12.1	61.60	99.40
49.000	70.00	2253	-12709	17736	12.7	26	-1299	1273	2.0	48.60	94.08
52.500	75.00	2170	-12709	17736	12.2	-48	-1300	1267	3.7	61.64	86.36
56.000	80.00	1881	-10463	14601	12.9	-104	-1302	1260	8.0	58.34	67.34
59.500	85.00	1443	-8086	11284	12.8	-139	-1304	1254	10.6	60.48	63.79
63.000	90.00	933	-5709	7968	11.7	-152	-1306	1248	11.7	45.02	45.41
66.500	95.00	469	-3332	4651	10.1	-114	-1102	1050	10.4	40.05	24.13
70.000	100.00	182	-1571	2193	8.3	-58	-718	685	8.0	32.12	19.22
73.500	105.00	39	-733	1023	3.8	-23	-335	319	7.0	14.17	6.94

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DAMAGE STABILITY OF LOADING CONDITION L14

CRITICAL DAMAGE STABILITY CRITERIA FOR EACH DAMAGE CASE

RCR	DAM	STAGE	PHASE	REQ	ATTV	UNIT	STAT	MINGM
V.MINGZ.A469	D01	1	EQ	0.100	0.879	m	OK	0.215
V.MINGZ.A469	D02	1	EQ	0.100	0.847	m	OK	0.338
V.MINGZ.A469	D03	1	EQ	0.100	0.850	m	OK	0.347
V.MINGZ.A469	D04	1	EQ	0.100	0.809	m	OK	0.594
V.MINGZ.A469	D05	1	EQ	0.100	0.801	m	OK	0.650
V.MINGZ.A469	D06	1	EQ	0.100	0.808	m	OK	0.598
V.MINGZ.A469	D07	1	EQ	0.100	0.799	m	OK	0.654
V.RANGE.A469	D08	1	EQ	20.000	63.202	deg	OK	0.617
V.RANGE.A469	D09	1	EQ	20.000	65.347	deg	OK	0.613
V.RANGE.A469	D10	1	EQ	20.000	63.619	deg	OK	0.624
V.RANGE.A469	D11	1	EQ	20.000	65.001	deg	OK	0.691
V.RANGE.A469	D12	1	EQ	20.000	63.729	deg	OK	0.686
V.RANGE.A469	D13	1	EQ	20.000	65.870	deg	OK	0.683
V.RANGE.A469	D14	1	EQ	20.000	65.232	deg	OK	0.508
V.RANGE.A469	D15	1	EQ	20.000	67.408	deg	OK	0.510
V.MINGZ.A469	D16	1	EQ	0.100	0.849	m	OK	0.390
V.MINGZ.A469	D17	1	EQ	0.100	0.846	m	OK	0.383
V.MINGZ.A469	D18	1	EQ	0.100	0.832	m	OK	0.346
V.MINGZ.A469	D19	1	EQ	0.100	0.833	m	OK	0.359
V.MINGZ.A469	D20	1	EQ	0.100	0.841	m	OK	0.309
V.MINGZ.A469	D21	1	EQ	0.100	0.842	m	OK	0.323
TOTAL								0.691

DETERMINATION OF CRITICAL DAMAGE CASE

CRITICAL DAMAGE CASE : D11
STAGE OF CRITICAL DAMAGE CASE : 1
PHASE OF CRITICAL DAMAGE CASE : EQ

DAMAGE STABILITY CRITERIA - CRITICAL CASE

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM m
V.PROGR.A469	No progr. flooding	17.826	3.411	deg	OK	0.363
V.MAXHEEL.A469	Maximum heel	17.000	3.411	deg	OK	0.364
V.RANGE.A469	Range of pos. stab	20.000	65.001	deg	OK	0.691
V.MINGZ.A469	Min. righting lever	0.100	0.818	m	OK	0.677
TOTAL						0.691

DAMAGE CASE RESULTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	T m	TR m	HEEL degree	RESFLD m	OPEN	CROSSTM s
D11	1	EQ	PS	5.864	0.047	-3.4	2.21	D-361	-

DAMAGED COMPARTMENTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	NAME	DES	PERM	VOL m3	W t	XCG m	YCG m	ZCG m
D11	INTACT	EQ	PS				0.0	0.000			
D11	1	EQ	PS	T018	WING TANK PS	0.95	53.3	54.604	30.80	-8.37	3.92
D11	1	EQ	PS	T020	DB TANK PS	0.95	92.4	94.737	30.84	-4.63	0.61

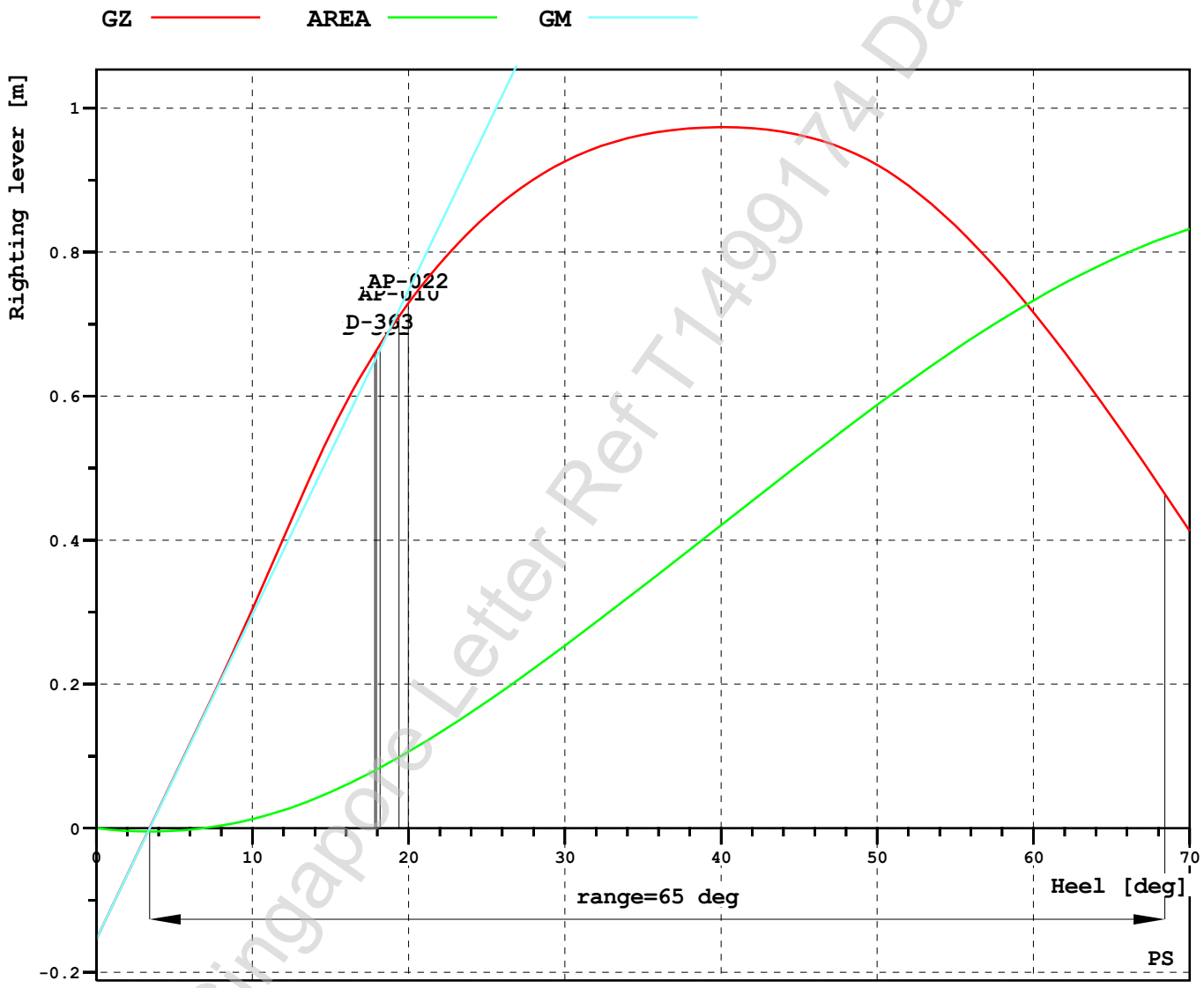
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CRITICAL OPENINGS - CRITICAL CASE

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
D-361	PS CARGORAIL #57/M-DECK	WEATHERTIGHT	39.885	-8.560	8.600	17.8	2.210
D-351	CO2 ROOM/M-DECK	WEATHERTIGHT	51.792	-8.360	8.600	17.9	2.215
D-363	PS CARGORAIL #37/M-DECK	WEATHERTIGHT	25.900	-8.550	8.600	18.2	2.219
AP-010	WING TANK PS	WEATHERTIGHT	47.211	-8.731	8.953	19.4	2.548
AP-022	WING TANK PS	WEATHERTIGHT	19.200	-8.702	8.936	20.0	2.549
D-353	DECK WORKSHOP/M-DECK	WEATHERTIGHT	53.192	-6.550	8.600	22.3	2.322
V-612E	EXHAUST PROPULSION ROOM	WEATHERTIGHT	6.600	-7.796	8.936	22.3	2.611
AP-263	HYDRAULIC OIL STORE TANK	WEATHERTIGHT	47.690	-7.260	8.912	22.5	2.594

DAMAGE STABILITY CHECK PLOT - CRITICAL CASE



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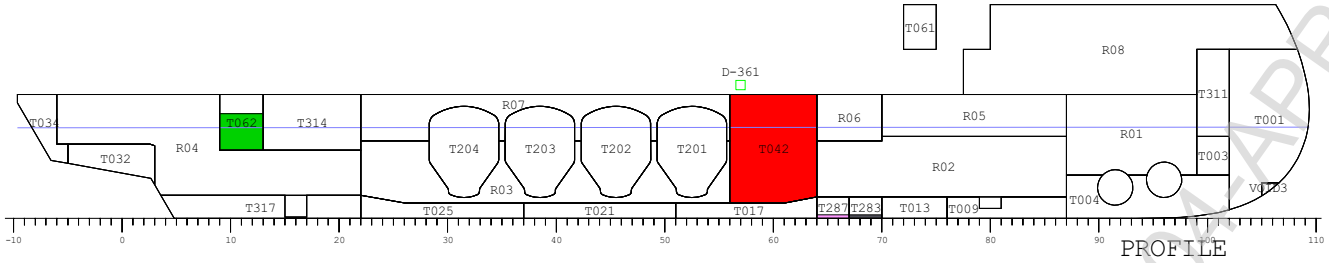
GZ-CURVE - CRITICAL CASE

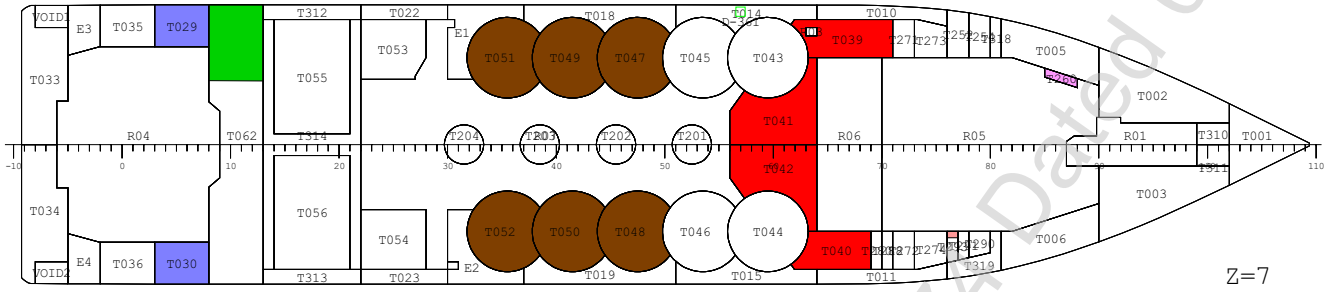
HEEL	DISP	T	TR	GZ	MS	AREA
degree	t	m	m	m	m	mrad
0.0	5601.9	5.873	0.031	0.154	0.154	0.000
-1.0	5601.9	5.873	0.033	0.109	0.162	-0.002
-3.0	5601.9	5.866	0.044	0.019	0.177	-0.005
-5.0	5601.9	5.851	0.062	-0.072	0.191	-0.004
-10.0	5601.9	5.772	0.138	-0.304	0.220	0.013
-15.0	5601.9	5.639	0.250	-0.548	0.232	0.050
-20.0	5601.9	5.490	0.295	-0.730	0.302	0.106
-30.0	5601.9	5.168	0.048	-0.926	0.583	0.253
-40.0	5601.9	4.758	-0.677	-0.974	0.966	0.421
-50.0	5601.9	4.188	-1.729	-0.921	1.390	0.588
-60.0	5601.9	3.505	-2.778	-0.716	1.897	0.733
-70.0	5601.9	2.736	-3.702	-0.413	2.423	0.832

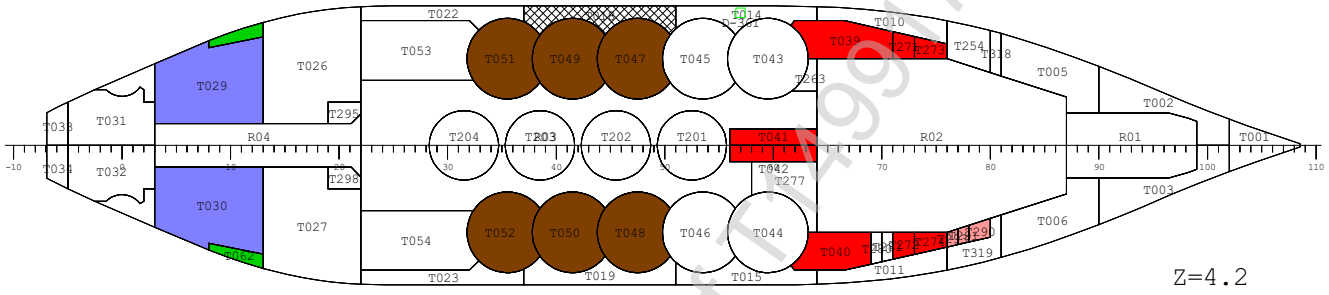
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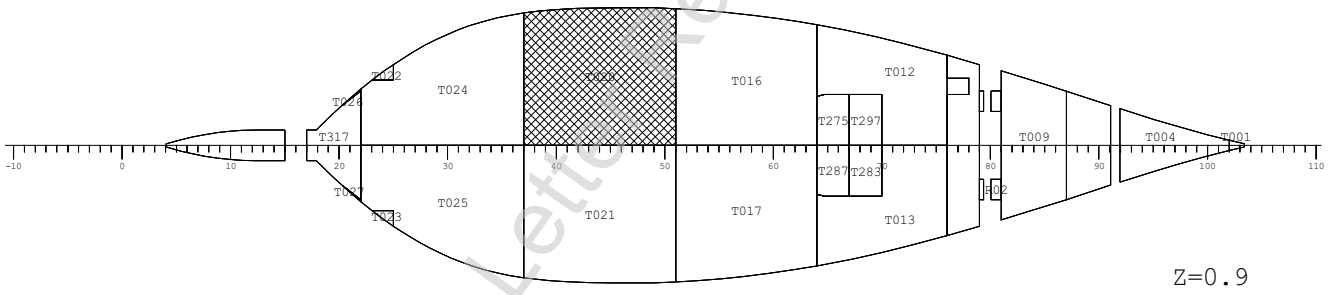
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DAMAGE ARRANGEMENT - CRITICAL CASE





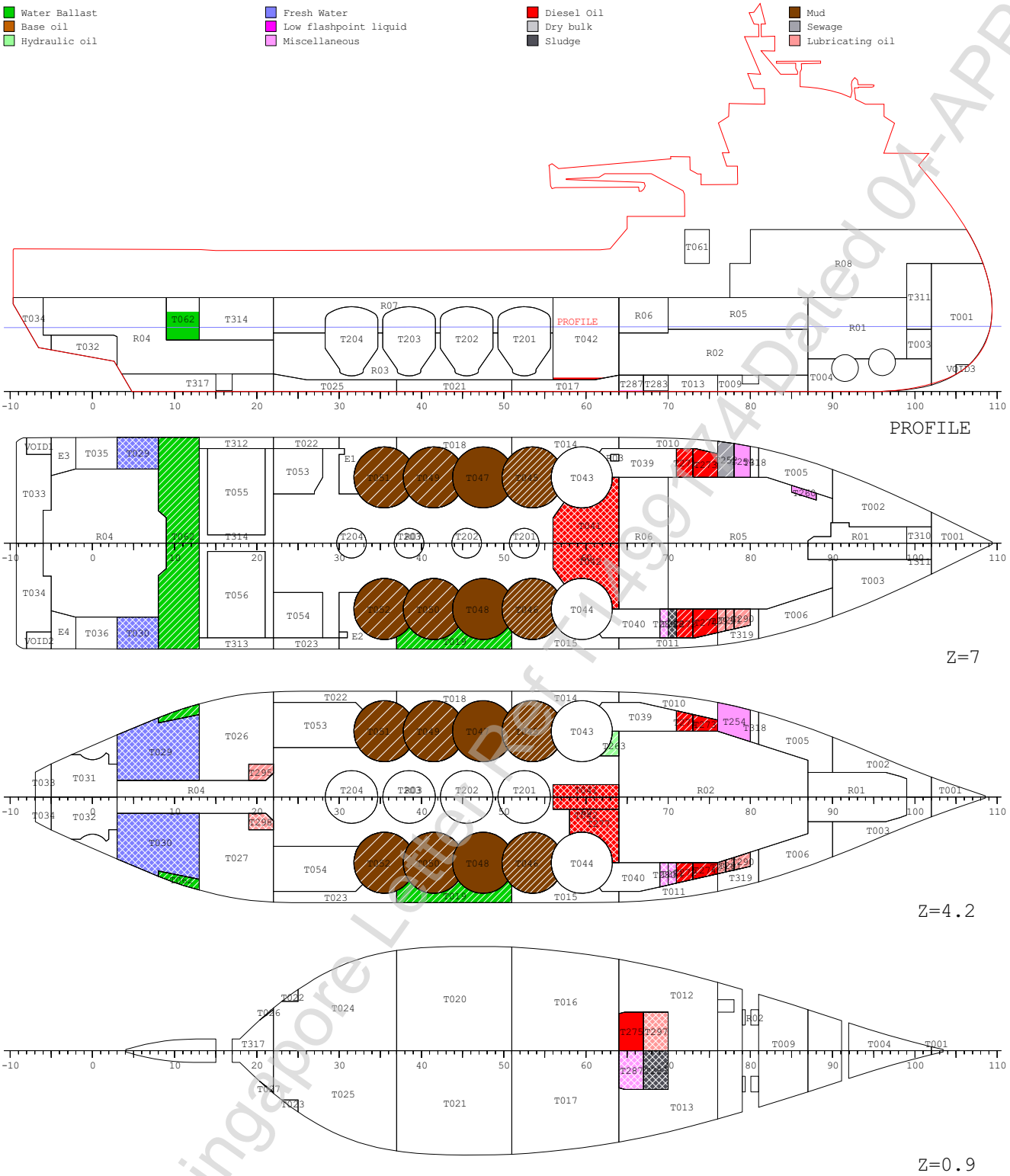




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12.15 L15 -> MUD - Arrival



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LIST OF LOADS

NAME	DES	LOAD	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm	FRS	HM m
CONTENTS=Diesel Oil (RHO=0.86)										
T041	STORAGE TANK PS	FO	0.7	1.0	41.24	-0.52	1.10	27	MAX	1.20
T042	STORAGE TANK SB	FO	0.7	1.0	41.24	0.52	1.10	27	MAX	1.20
T271	FO SETTLING TANK PS	FO	8.5	70.0	50.38	-6.53	4.93	1	REAL	6.72
T272	FO SETTLING TANK SB	FO	8.5	70.0	50.38	6.53	4.93	1	REAL	6.72
T273	FO SERVICE TANK PS	FO	10.6	70.0	52.09	-6.38	5.15	2	REAL	6.90
T274	FO SERVICE TANK SB	FO	10.6	70.0	52.09	6.38	5.15	2	REAL	6.90
T275	FO DRAIN TANK	FO	7.9	98.0	45.85	-1.64	0.69	5	REAL	1.37
T277	FO OVERFLOW TANK	FO	3.6	10.0	42.62	2.34	1.31	10	REAL	1.53
T279	EMERG. GEN. FO TANK	FO	1.9	98.0	61.60	-1.38	24.57	0	REAL	25.80
SUBTOTAL		FO	53.0		50.02	-0.13	4.73	75		
CONTENTS=Fresh Water (RHO=1)										
T029	WING TANK PS	FW	7.5	9.8	7.28	-2.50	2.65	83	MAX	3.01
T030	WING TANK SB	FW	7.5	9.8	7.28	2.50	2.65	83	MAX	3.01
SUBTOTAL		FW	14.9		7.28	0.00	2.65	165		
CONTENTS=Hydraulic oil (RHO=0.924)										
T263	HYDRAULIC OIL STORE TANK	HO	1.9	40.0	44.29	-4.33	1.89	1	REAL	2.47
CONTENTS=Lubricating oil (RHO=0.924)										
T290	LO STORE TANK MAIN ENG.	LO	3.4	40.0	55.32	5.47	4.05	0	REAL	5.16
T291	LO STORE TANK MAIN ENG.	LO	1.8	40.0	54.25	5.81	4.26	0	REAL	5.48
T293	LO STORE TANK THRUSTER	LO	1.8	40.0	53.55	6.03	4.41	0	REAL	5.66
T295	LO STORE TANK AZIMUTH THR.	LO	2.3	40.0	14.35	-2.10	2.63	0	REAL	3.06
T297	LO DRAIN TANK	LO	0.9	10.0	47.95	-1.64	0.07	6	REAL	0.14
T298	DIRTY LO TANK	LO	0.6	10.0	14.35	2.10	2.31	0	REAL	2.41
SUBTOTAL		LO	10.7		43.36	3.26	3.43	7		
CONTENTS=Mass load (RHO=1)										
M1	CREW	MASS	2.6	0.0	57.40	0.00	15.50	0		0.00
M2	PROVISION	MASS	0.8	0.0	56.35	-6.50	12.40	0		0.00
M3	STORES	MASS	0.8	0.0	72.10	0.00	12.40	0		0.00
SUBTOTAL		MASS	4.2		60.00	-1.24	14.32	0		
CONTENTS=Miscellaneous (RHO=1)										
T280	BILGE WATER HOLDING TANK	MIS	0.9	10.0	48.64	6.17	2.03	0	REAL	2.57
T281	BILGE WATER SETTLING TANK	MIS	0.4	10.0	49.34	6.04	1.76	0	REAL	2.08
T287	BILGE WATER DRAIN TANK	MIS	0.9	10.0	45.85	1.64	0.07	6	REAL	0.14
T254	GREY WATER TANK	MIS	31.6	98.0	54.82	-6.47	4.52	3	REAL	7.84
T260	WATER MIST TANK	MIS	2.3	98.0	60.55	-4.30	6.28	0	REAL	7.27
SUBTOTAL		MIS	36.2		54.74	-5.68	4.43	9		
CONTENTS=Mud (RHO=2.8)										
T045	LIQUID CARGO TANK PS	MUD	79.8	22.4	37.22	-5.50	1.98	98	REAL	2.74
T046	LIQUID CARGO TANK SB	MUD	72.1	20.7	37.22	5.49	1.90	98	REAL	2.59
T047	LIQUID CARGO TANK PS	MUD	348.9	98.0	33.02	-5.58	4.54	98	REAL	7.86
T048	LIQUID CARGO TANK SB	MUD	348.9	98.0	33.02	5.58	4.54	98	REAL	7.86
T049	LIQUID CARGO TANK PS	MUD	348.9	98.0	28.82	-5.58	4.54	98	REAL	7.86
T050	LIQUID CARGO TANK SB	MUD	348.9	98.0	28.82	5.58	4.54	98	REAL	7.86
T051	LIQUID CARGO TANK PS	MUD	346.2	98.0	24.63	-5.56	4.57	98	REAL	7.87
T052	LIQUID CARGO TANK SB	MUD	346.2	98.0	24.63	5.56	4.57	98	REAL	7.87
SUBTOTAL		MUD	2240.0		29.40	-0.02	4.37	781		
CONTENTS=Sewage (RHO=1)										
T252	BLACK WATER TANK	SEW	10.4	97.9	53.89	-7.06	6.66	4	REAL	7.95
CONTENTS=Sludge (RHO=1)										
T283	SLUDGE DRAIN TANK	SLU	0.9	10.0	47.95	1.64	0.07	6	REAL	0.14
T288	SLUDGE SETTLING TANK	SLU	0.4	10.0	49.35	6.75	5.44	1	REAL	5.59
SUBTOTAL		SLU	1.4		48.40	3.29	1.80	7		
CONTENTS=Water Ballast (RHO=1.025)										
T018	WING TANK PS	WB	0.0	0.0	30.79	-8.37	4.73	2	MAX	0.00
T019	WING TANK SB	WB	40.9	55.0	30.80	8.37	3.26	2	MAX	5.06
T062	ROLL RED. TANK	WB	136.9	65.0	7.45	0.00	5.58	1736	REAL	6.76

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SUBTOTAL	WB	177.7	12.82	1.92	5.05	1740
TOTAL		2550.4	29.14	0.01	4.44	2789

LOAD SUMMARY TABLE

ITEM	MASS t	LCG m	TCG m	VCG m	FRSM tm
Diesel Oil	53.0	50.02	-0.13	4.73	74.98
Fresh Water	14.9	7.28	0.00	2.65	165.03
Hydraulic oil	1.9	44.29	-4.33	1.89	0.61
Lubricating oil	10.7	43.36	3.26	3.43	7.30
Mass load	4.2	60.00	-1.24	14.32	0.00
Miscellaneous	36.2	54.74	-5.68	4.43	9.39
Mud	2240.0	29.40	-0.02	4.37	781.14
Sewage	10.4	53.89	-7.06	6.66	3.55
Sludge	1.4	48.40	3.29	1.80	6.99
Water Ballast	177.7	12.82	1.92	5.05	1740.29
DEADWEIGHT	2550.4	29.14	0.01	4.44	2789.29
Lightweight	2704.9	39.15	-0.01	7.08	
Deadweight	2550.4	29.14	0.01	4.44	
Total weight	5255.3	34.29	0.00	5.80	

F L O A T I N G P O S I T I O N

Draught moulded	5.498	m	KM	8.82 m
Trim	0.096	m	KG	5.80 m
Heel, SB=+	0.0	deg		
TA	5.450	m	GM0	3.02 m
TF	5.546	m	GMCORR	-0.53 m
Trimming moment	646	tonm	GM	2.49 m

STABILITY COMPONENTS

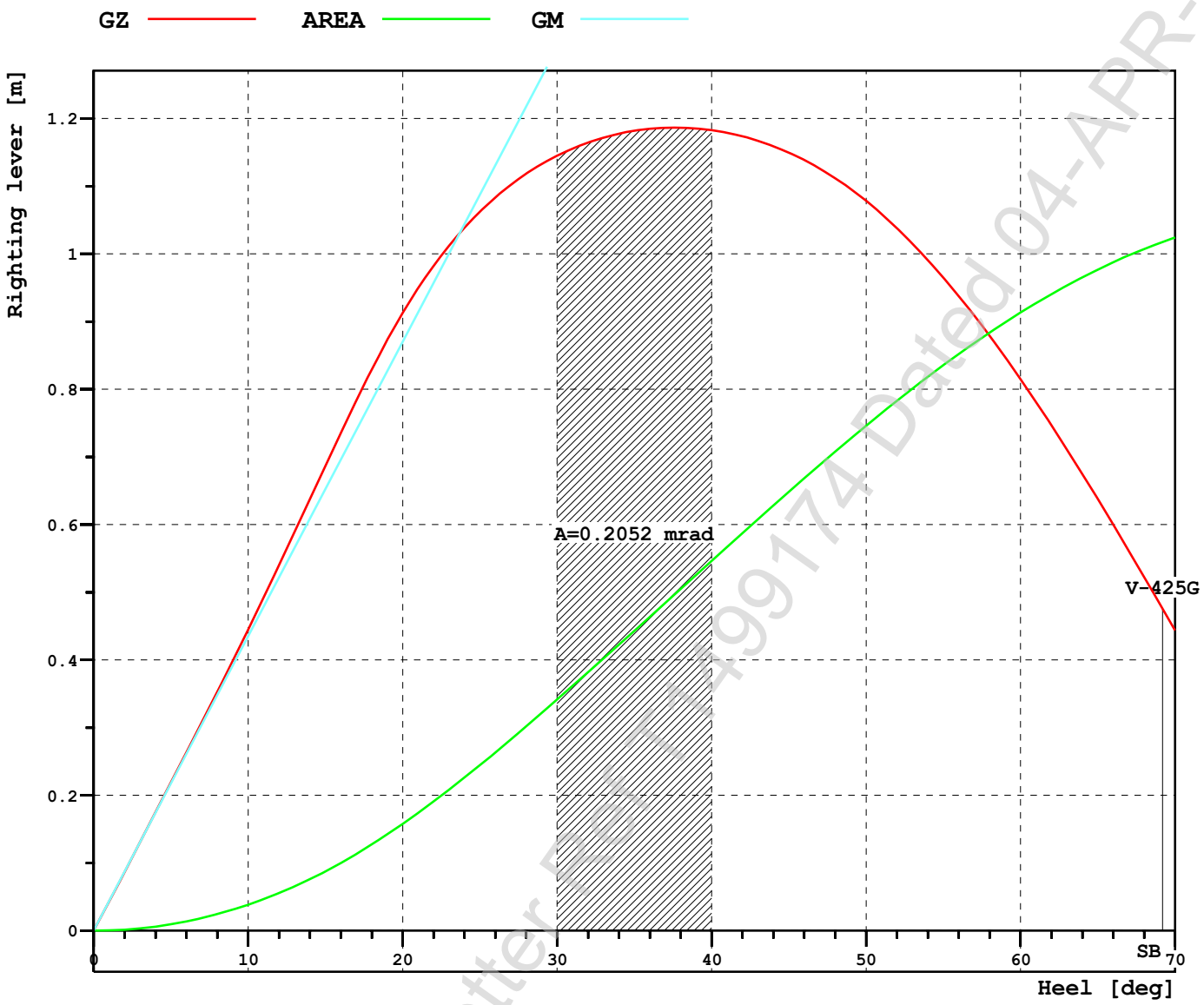
LOADING CONDITION L15, MUD - Arrival

HEEL degree	MS m	HPHI m	EPHI mrad	FSMOM tm	DGZ m	GMSIN m	KN m
0.0	0.000	0.00	0.000	0.0	0.000	0.000	0.000
1.0	-0.001	0.04	0.000	47.4	0.009	0.053	0.153
3.0	-0.001	0.13	0.003	142.1	0.027	0.158	0.460
5.0	-0.001	0.22	0.009	232.0	0.044	0.263	0.768
10.0	0.000	0.44	0.038	422.4	0.080	0.525	1.532
15.0	0.006	0.69	0.088	534.8	0.102	0.782	2.289
20.0	-0.006	0.91	0.158	607.0	0.115	1.034	3.011
30.0	-0.233	1.15	0.341	702.2	0.134	1.512	4.177
40.0	-0.617	1.18	0.547	756.5	0.144	1.943	5.053
50.0	-1.089	1.08	0.746	781.2	0.149	2.316	5.668
60.0	-1.653	0.81	0.913	788.3	0.150	2.618	5.986
70.0	-2.245	0.44	1.024	798.7	0.152	2.841	6.044

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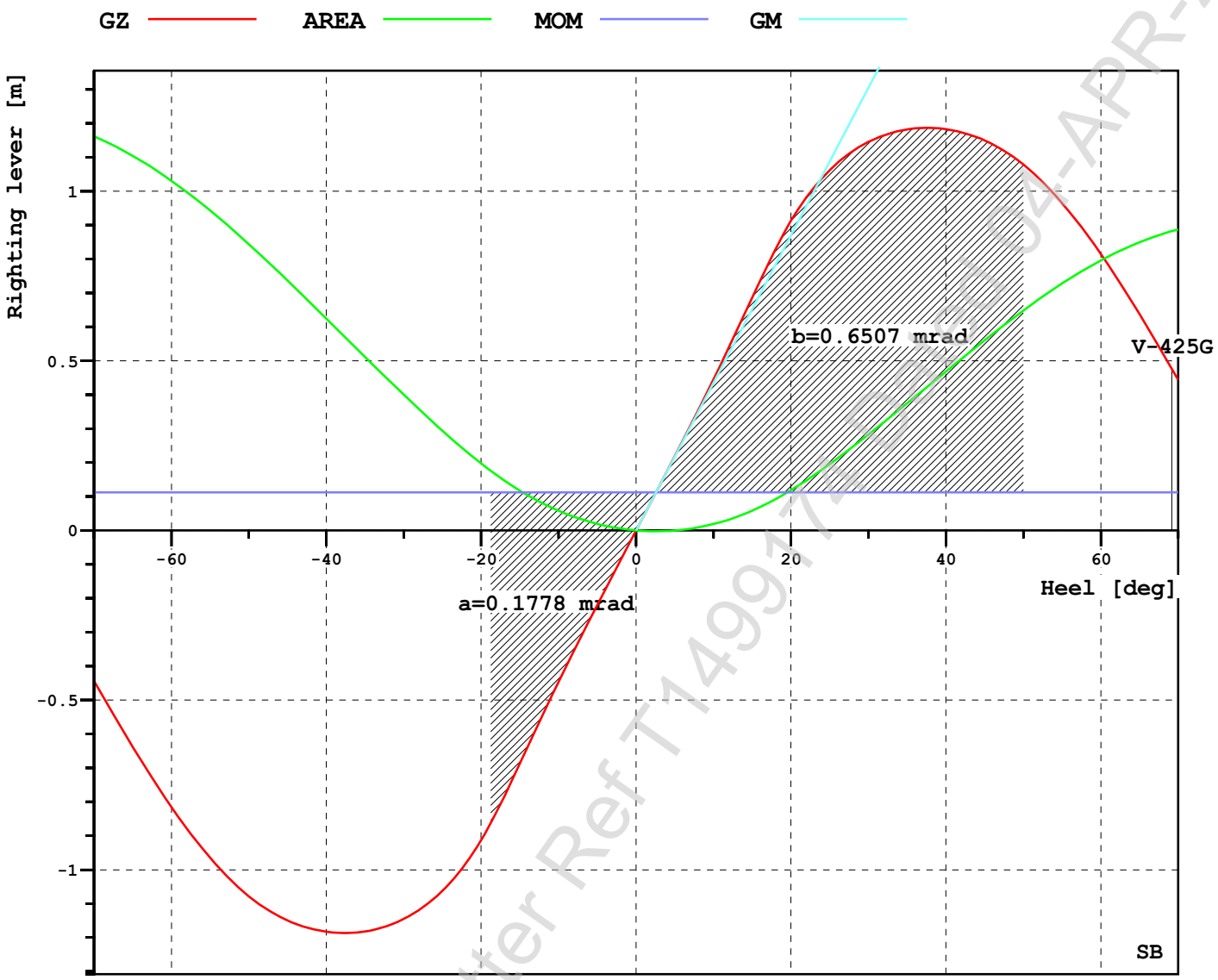
STABILITY CHECK PLOT



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IMOWEATHER CHECK PLOT



INTACT STABILITY CRITERIA

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM
V.AREA15-30	Area depending on the position of the Max. .	0.055	0.341	mrad	OK	0.681
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.205	mrad	OK	0.740
V.GZ0.2	Min. GZ > 0.2	0.200	1.187	m	OK	0.602
V.POSMAX15	Max. GZ at an angle > 15 deg.	15.000	37.577	deg	OK	-0.191
V.GM0.15	GM > 0.15 m	0.150	2.492	m	OK	0.150
IMOWEATHER	IMO weather criterion	1.000	3.659		OK	0.519
2008IS-A2.3.1.2	Heeling angle due to steady wind <16 or <=8.	12.478	1.731	deg	OK	0.231
ABSFIFI1	Angle of heel < 1/2 freeboard immersion	7.799	0.375	deg	OK	0.078
ABSFIFI2	Area of residual stability >0.09 mrad	0.090	0.535	mrad	OK	0.583
TOTAL						0.740

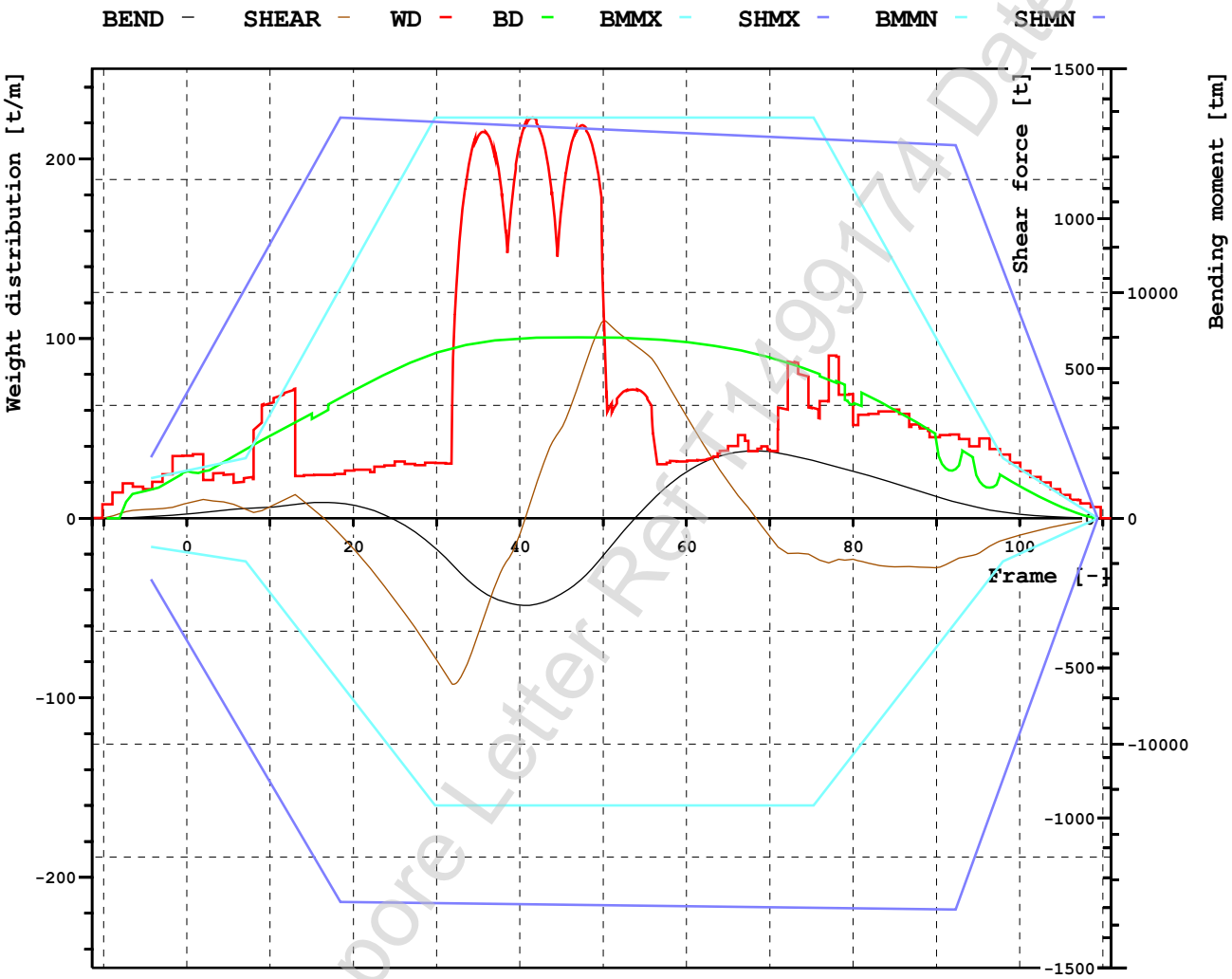
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MOST CRITICAL OPENINGS

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
V-421G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	55.890	2.093	14.288	-	8.767
V-424G	SUPPLY AIR ENGINE ROOM PS	UNPROTECTED	54.243	-3.413	14.288	-	8.769
V-425G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	54.243	3.767	14.288	69.2	8.769
D-651	ACCOMMODATION/C-DECK	UNPROTECTED	56.000	3.771	16.906	-	11.385
V-421E	EXHAUST AIR ENGINE ROOM PS	UNPROTECTED	55.993	-2.538	17.088	-	11.567
V-423E	EXHAUST AIR ENGINE ROOM SB	UNPROTECTED	55.993	3.423	17.088	-	11.567

STRENGTH DIAGRAM



STRENGTH SUMMARY

LOADING CONDITION L15, MUD - Arrival

			X	Frame
Shear force (min)	-554.7 t	(43.2%)	22.381 m	31.97
Shear force (max)	658.8 t	(50.8%)	35.097 m	50.14
Sagging moment	-3859.8 tm	(30.4%)	28.419 m	40.60
Hogging moment	2984.4 tm	(16.8%)	47.950 m	68.50

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STRENGTH DISTRIBUTION

LOADING CONDITION L15, MUD - Arrival

X	FR	BEND	BMMN	BMMX	BMREL	SHEAR	SHMN	SHMX	SHREL	WD	BD
m	#	tm	tm	tm	%	t	t	t	%	t/m	t/m
-3.500	-5.00	58	0	0	0.0	30	0	0	0.0	16.23	15.16
0.000	0.00	181	-1510	2108	8.6	49	-407	418	11.8	34.85	26.02
3.500	5.00	376	-1790	2498	15.1	51	-644	667	7.6	24.30	32.88
7.000	10.00	489	-3292	4594	10.7	38	-881	916	4.2	63.95	45.97
10.500	15.00	686	-5668	7911	8.7	34	-1117	1166	2.9	24.04	58.28
14.000	20.00	578	-8045	11228	5.1	-102	-1281	1335	7.9	26.51	70.98
17.500	25.00	-74	-10422	14545	0.7	-273	-1283	1329	21.3	31.62	82.60
21.000	30.00	-1373	-12709	17736	10.8	-473	-1284	1323	36.8	30.72	92.14
24.500	35.00	-3125	-12709	17736	24.6	-385	-1286	1317	30.0	212.90	97.53
28.000	40.00	-3848	-12709	17736	30.3	-52	-1288	1310	4.0	209.48	99.87
31.500	45.00	-3341	-12709	17736	26.3	302	-1290	1304	23.2	173.45	100.63
35.000	50.00	-1652	-12709	17736	13.0	658	-1292	1298	50.7	120.52	100.58
38.500	55.00	464	-12709	17736	2.6	553	-1293	1292	42.8	68.87	99.76
42.000	60.00	2053	-12709	17736	11.6	340	-1295	1285	26.4	31.65	98.03
45.500	65.00	2846	-12709	17736	16.0	121	-1297	1279	9.4	40.12	94.67
49.000	70.00	2948	-12709	17736	16.6	-60	-1299	1273	4.6	38.25	89.46
52.500	75.00	2568	-12709	17736	14.5	-125	-1300	1267	9.6	61.65	81.97
56.000	80.00	2074	-10463	14601	14.2	-137	-1302	1260	10.5	69.10	63.33
59.500	85.00	1536	-8086	11284	13.6	-162	-1304	1254	12.4	60.41	60.30
63.000	90.00	963	-5709	7968	12.1	-165	-1306	1248	12.6	45.02	42.56
66.500	95.00	470	-3332	4651	10.1	-118	-1102	1050	10.7	40.05	22.02
70.000	100.00	180	-1571	2193	8.2	-56	-718	685	7.8	30.94	17.87
73.500	105.00	40	-733	1023	3.9	-23	-335	319	6.8	12.99	6.31

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DAMAGE STABILITY OF LOADING CONDITION L15

CRITICAL DAMAGE STABILITY CRITERIA FOR EACH DAMAGE CASE

RCR	DAM	STAGE	PHASE	REQ	ATTV	UNIT	STAT	MINGM
V.MINGZ.A469	D01	1	EQ	0.100	0.923	m	OK	0.086
V.MINGZ.A469	D02	1	EQ	0.100	0.897	m	OK	0.212
V.MINGZ.A469	D03	1	EQ	0.100	0.901	m	OK	0.199
V.MINGZ.A469	D04	1	EQ	0.100	0.864	m	OK	0.450
V.MINGZ.A469	D05	1	EQ	0.100	0.861	m	OK	0.362
V.RANGE.A469	D06	1	EQ	20.000	66.071	deg	OK	0.454
V.MINGZ.A469	D07	1	EQ	0.100	0.860	m	OK	0.362
V.RANGE.A469	D08	1	EQ	20.000	64.996	deg	OK	0.462
V.RANGE.A469	D09	1	EQ	20.000	67.084	deg	OK	0.444
V.RANGE.A469	D10	1	EQ	20.000	66.316	deg	OK	0.373
V.RANGE.A469	D11	1	EQ	20.000	66.531	deg	OK	0.515
V.RANGE.A469	D12	1	EQ	20.000	65.590	deg	OK	0.513
V.RANGE.A469	D13	1	EQ	20.000	67.021	deg	OK	0.498
V.RANGE.A469	D14	1	EQ	20.000	67.100	deg	OK	0.351
V.RANGE.A469	D15	1	EQ	20.000	68.099	deg	OK	0.335
V.MINGZ.A469	D16	1	EQ	0.100	0.901	m	OK	0.238
V.MINGZ.A469	D17	1	EQ	0.100	0.903	m	OK	0.228
V.MINGZ.A469	D18	1	EQ	0.100	0.882	m	OK	0.215
V.MINGZ.A469	D19	1	EQ	0.100	0.884	m	OK	0.206
V.MINGZ.A469	D20	1	EQ	0.100	0.897	m	OK	0.176
V.MINGZ.A469	D21	1	EQ	0.100	0.899	m	OK	0.168
TOTAL								0.515

DETERMINATION OF CRITICAL DAMAGE CASE

CRITICAL DAMAGE CASE : D11
STAGE OF CRITICAL DAMAGE CASE : 1
PHASE OF CRITICAL DAMAGE CASE : EQ

DAMAGE STABILITY CRITERIA - CRITICAL CASE

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM m
V.PROGR.A469	No progr. flooding	19.516	3.469	deg	OK	0.213
V.MAXHEEL.A469	Maximum heel	17.000	3.469	deg	OK	0.268
V.RANGE.A469	Range of pos. stab	20.000	66.531	deg	OK	0.515
V.MINGZ.A469	Min. righting lever	0.100	0.898	m	OK	0.483
TOTAL						0.515

DAMAGE CASE RESULTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	T m	TR m	HEEL degree	RESFLD m	OPEN	CROSSTM s
D11	1	EQ	PS	5.598	0.116	-3.5	2.46	D-351	-

DAMAGED COMPARTMENTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	NAME	DES	PERM	VOL m3	W t	XCG m	YCG m	ZCG m
D11	INTACT	EQ	-				0.0	0.000			
D11	1	EQ	PS	T018	WING TANK PS	0.95	50.4	51.685	30.80	-8.37	3.79
D11	1	EQ	PS	T020	DB TANK PS	0.95	92.4	94.737	30.84	-4.63	0.61

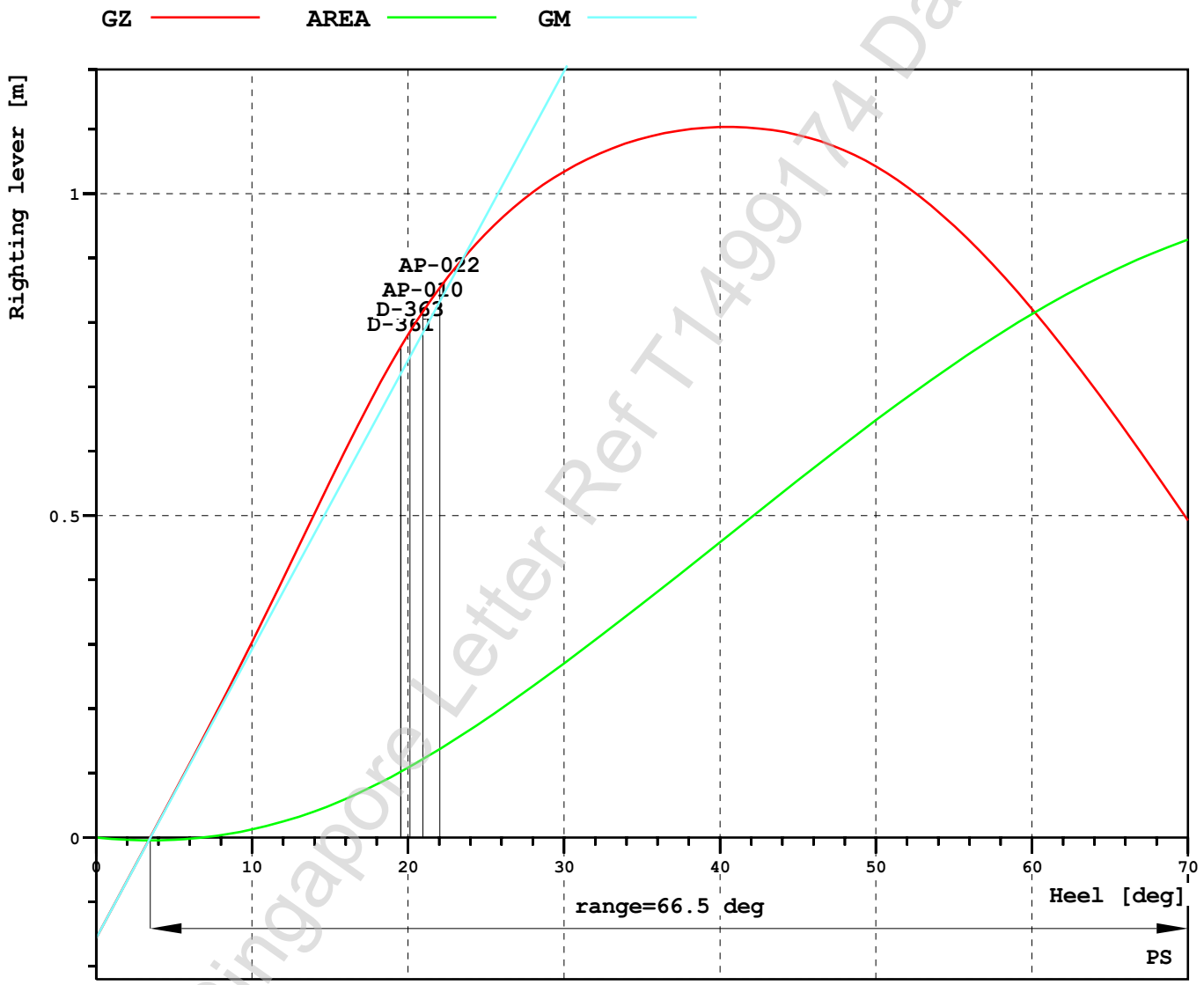
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CRITICAL OPENINGS - CRITICAL CASE

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
D-351	CO2 ROOM/M-DECK	WEATHERTIGHT	51.792	-8.360	8.600	19.5	2.458
D-361	PS CARGORAIL #57/M-DECK	WEATHERTIGHT	39.885	-8.560	8.600	19.6	2.464
D-363	PS CARGORAIL #37/M-DECK	WEATHERTIGHT	25.900	-8.550	8.600	20.1	2.485
AP-010	WING TANK PS	WEATHERTIGHT	47.211	-8.731	8.953	21.0	2.795
AP-022	WING TANK PS	WEATHERTIGHT	19.200	-8.702	8.936	22.0	2.820
D-353	DECK WORKSHOP/M-DECK	WEATHERTIGHT	53.192	-6.550	8.600	24.2	2.566
AP-263	HYDRAULIC OIL STORE TANK	WEATHERTIGHT	47.690	-7.260	8.912	24.4	2.842
V-612E	EXHAUST PROPULSION ROOM	WEATHERTIGHT	6.600	-7.796	8.936	24.8	2.894

DAMAGE STABILITY CHECK PLOT - CRITICAL CASE



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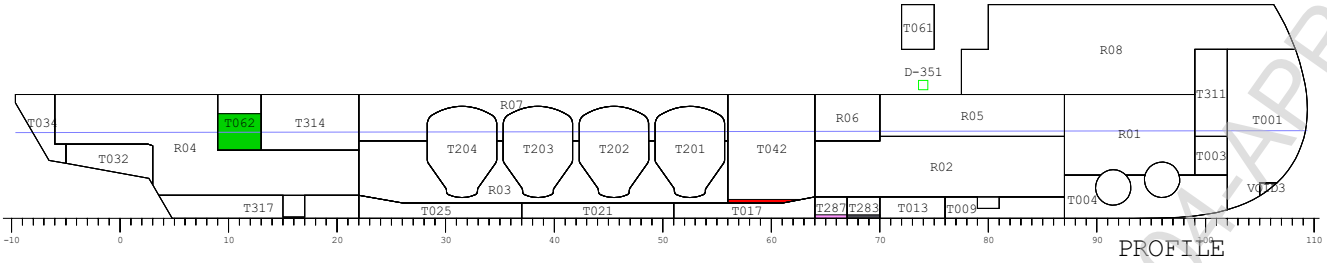
GZ-CURVE - CRITICAL CASE

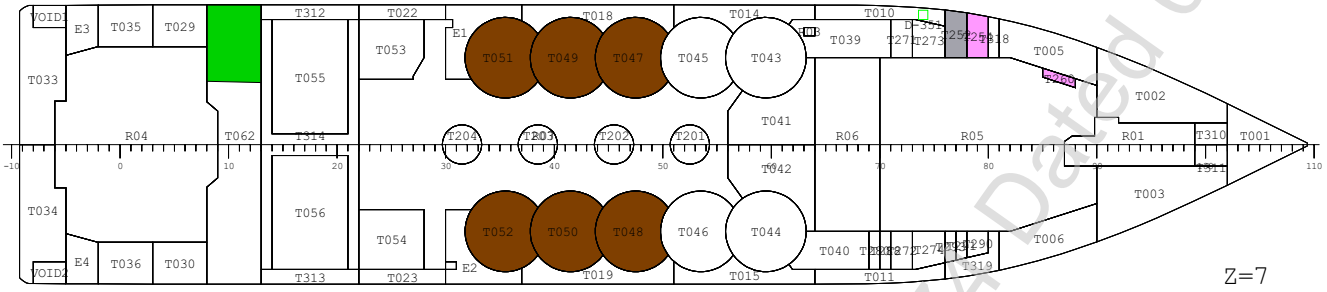
HEEL	DISP	T	TR	GZ	MS	AREA
degree	t	m	m	m	m	mrad
0.0	5255.3	5.608	0.095	0.156	0.156	0.000
-1.0	5255.3	5.608	0.097	0.111	0.164	-0.002
-3.0	5255.3	5.601	0.110	0.021	0.180	-0.005
-5.0	5255.3	5.586	0.135	-0.070	0.194	-0.004
-10.0	5255.3	5.508	0.229	-0.303	0.222	0.012
-15.0	5255.3	5.374	0.363	-0.553	0.229	0.050
-20.0	5255.3	5.205	0.492	-0.781	0.253	0.108
-30.0	5255.3	4.839	0.433	-1.034	0.477	0.270
-40.0	5255.3	4.380	-0.100	-1.104	0.840	0.459
-50.0	5255.3	3.757	-1.005	-1.042	1.273	0.648
-60.0	5255.3	3.024	-1.933	-0.821	1.797	0.813
-70.0	5255.3	2.216	-2.771	-0.493	2.348	0.928

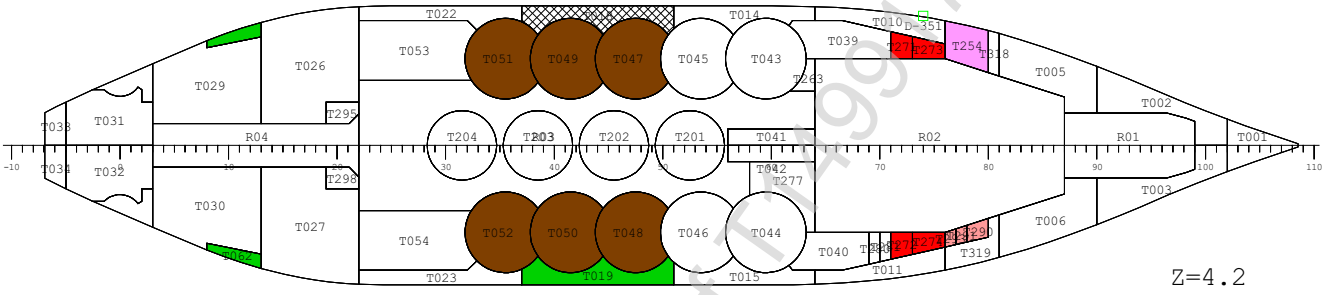
See ABS Singapore Letter Ref T1499174 Dated 04-APR-2016

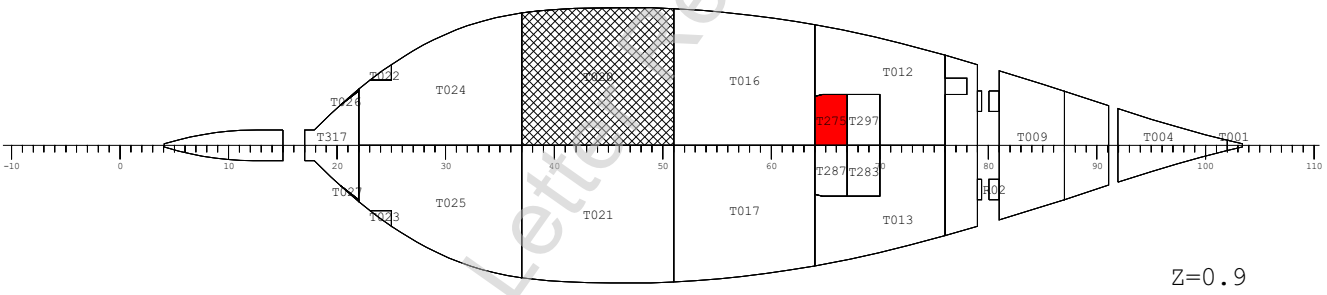
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DAMAGE ARRANGEMENT - CRITICAL CASE





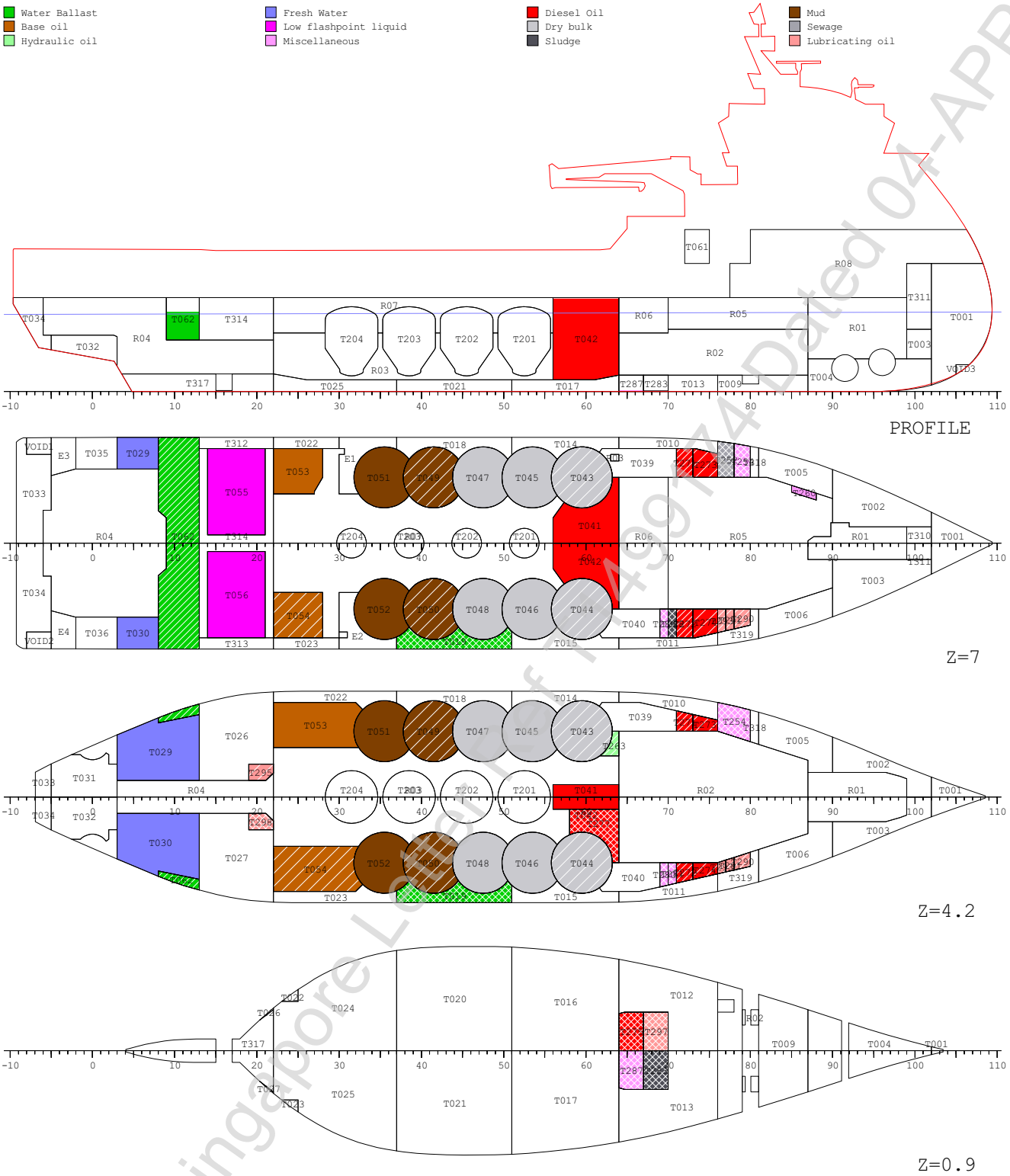




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12.16 L16 -> LFL-BRI-MUD - DEPARTURE



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LIST OF LOADS

NAME	DES	LOAD	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm	FRS	HM m
CONTENTS=Brine (RHO=2.5)										
T043	LIQUID CARGO TANK PS	BRINE	270.4	77.2	41.65	-5.56	3.86	90	REAL	6.47
T044	LIQUID CARGO TANK SB	BRINE	270.4	77.2	41.65	5.56	3.86	90	REAL	6.47
T045	LIQUID CARGO TANK PS	BRINE	311.5	98.0	37.22	-5.58	4.54	87	REAL	7.86
T046	LIQUID CARGO TANK SB	BRINE	304.6	98.0	37.19	5.58	4.47	73	REAL	7.84
T047	LIQUID CARGO TANK PS	BRINE	311.5	98.0	33.02	-5.58	4.54	87	REAL	7.86
T048	LIQUID CARGO TANK SB	BRINE	311.5	98.0	33.02	5.58	4.54	87	REAL	7.86
SUBTOTAL		BRINE	1779.9		37.09	-0.02	4.32	514		
CONTENTS=Base oil (RHO=0.83)										
T053	STORAGE TANK PS	BSO	104.6	98.0	18.73	-6.02	4.76	17	REAL	7.84
T054	STORAGE TANK SB	BSO	104.4	98.0	18.71	6.02	4.78	17	REAL	7.84
SUBTOTAL		BSO	209.0		18.72	-0.01	4.77	33		
CONTENTS=Diesel Oil (RHO=0.86)										
T041	STORAGE TANK PS	FO	68.5	98.0	42.16	-1.51	5.48	27	MAX	7.92
T042	STORAGE TANK SB	FO	68.5	98.0	42.16	1.51	5.48	27	MAX	7.92
T271	FO SETTLING TANK PS	FO	8.5	70.0	50.38	-6.53	4.93	1	REAL	6.72
T272	FO SETTLING TANK SB	FO	8.5	70.0	50.38	6.53	4.93	1	REAL	6.72
T273	FO SERVICE TANK PS	FO	10.6	70.0	52.09	-6.38	5.15	2	REAL	6.90
T274	FO SERVICE TANK SB	FO	10.6	70.0	52.09	6.38	5.15	2	REAL	6.90
T275	FO DRAIN TANK	FO	0.8	10.0	45.85	-1.64	0.07	5	REAL	0.14
T277	FO OVERFLOW TANK	FO	3.6	10.0	42.62	2.34	1.31	10	REAL	1.53
T279	EMERG. GEN. FO TANK	FO	1.9	98.0	61.60	-1.38	24.57	0	REAL	25.80
SUBTOTAL		FO	181.6		44.32	0.02	5.48	75		
CONTENTS=Fresh Water (RHO=1)										
T029	WING TANK PS	FW	74.6	98.0	5.29	-4.87	4.61	83	MAX	7.83
T030	WING TANK SB	FW	74.6	98.0	5.29	4.87	4.61	83	MAX	7.83
SUBTOTAL		FW	149.3		5.29	0.00	4.61	165		
CONTENTS=Hydraulic oil (RHO=0.924)										
T263	HYDRAULIC OIL STORE TANK	HO	3.3	70.0	44.30	-4.34	2.32	1	REAL	3.33
CONTENTS=Low flashpoint liquid (RHO=1.2)										
T055	LFL TANK PS	LFL	91.3	98.0	12.25	-4.38	6.08	195	REAL	7.16
T056	LFL TANK SB	LFL	91.3	98.0	12.25	4.38	6.08	195	REAL	7.16
SUBTOTAL		LFL	182.6		12.25	0.00	6.08	389		
CONTENTS=Lubricating oil (RHO=0.924)										
T290	LO STORE TANK MAIN ENG.	LO	5.9	70.0	55.30	5.76	4.90	0	REAL	6.82
T291	LO STORE TANK MAIN ENG.	LO	3.1	70.0	54.25	6.04	5.11	0	REAL	6.92
T293	LO STORE TANK THRUSTER	LO	3.2	70.0	53.55	6.21	5.24	0	REAL	6.97
T295	LO STORE TANK AZIMUTH THR.	LO	4.0	70.0	14.35	-2.10	2.95	0	REAL	3.71
T297	LO DRAIN TANK	LO	0.9	10.0	47.95	-1.64	0.07	6	REAL	0.14
T298	DIRTY LO TANK	LO	0.6	10.0	14.35	2.10	2.31	0	REAL	2.41
SUBTOTAL		LO	17.6		43.89	3.64	4.24	8		
CONTENTS=Mass load (RHO=1)										
M1	CREW	MASS	2.6	0.0	57.40	0.00	15.50	0		0.00
M2	PROVISION	MASS	7.4	0.0	56.35	-6.50	12.40	0		0.00
M3	STORES	MASS	7.4	0.0	72.10	0.00	12.40	0		0.00
SUBTOTAL		MASS	17.5		63.21	-2.77	12.86	0		
CONTENTS=Miscellaneous (RHO=1)										
T280	BILGE WATER HOLDING TANK	MIS	0.9	10.0	48.64	6.17	2.03	0	REAL	2.57
T281	BILGE WATER SETTLING TANK	MIS	0.4	10.0	49.34	6.04	1.76	0	REAL	2.08
T287	BILGE WATER DRAIN TANK	MIS	0.9	10.0	45.85	1.64	0.07	6	REAL	0.14
T254	GREY WATER TANK	MIS	3.2	10.0	54.62	-5.79	1.92	1	REAL	2.34
T260	WATER MIST TANK	MIS	2.3	98.0	60.55	-4.30	6.28	0	REAL	7.27
SUBTOTAL		MIS	7.8		54.40	-2.49	3.01	8		
CONTENTS=Mud (RHO=2.8)										
T049	LIQUID CARGO TANK PS	MUD	348.9	98.0	28.82	-5.58	4.54	98	REAL	7.86
T050	LIQUID CARGO TANK SB	MUD	348.9	98.0	28.82	5.58	4.54	98	REAL	7.86
T051	LIQUID CARGO TANK PS	MUD	346.2	98.0	24.63	-5.56	4.57	98	REAL	7.87

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T052	LIQUID CARGO TANK SB	MUD	346.2	98.0	24.63	5.56	4.57	98	REAL	7.87
SUBTOTAL		MUD	1390.3		26.73	0.00	4.56	391		
CONTENTS=Sewage (RHO=1)										
T252	BLACK WATER TANK	SEW	1.1	10.0	53.89	-6.93	5.45	2	REAL	5.59
CONTENTS=Sludge (RHO=1)										
T283	SLUDGE DRAIN TANK	SLU	0.9	10.0	47.95	1.64	0.07	6	REAL	0.14
T288	SLUDGE SETTLING TANK	SLU	0.4	10.0	49.35	6.75	5.44	1	REAL	5.59
SUBTOTAL		SLU	1.4		48.40	3.29	1.80	7		
CONTENTS=Water Ballast (RHO=1.025)										
T019	WING TANK SB	WB	12.3	16.6	30.84	8.36	2.00	2	MAX	2.55
T062	ROLL RED. TANK	WB	136.9	65.0	7.45	0.00	5.58	1736	REAL	6.76
SUBTOTAL		WB	149.2		9.38	0.69	5.29	1738		
TOTAL			4090.3		29.86	0.01	4.63	3331		

LOAD SUMMARY TABLE

ITEM	MASS t	LCG m	TCG m	VCG m	FRSM tm
Brine	1779.9	37.09	-0.02	4.32	514.14
Base oil	209.0	18.72	-0.01	4.77	33.16
Diesel Oil	181.6	44.32	0.02	5.48	74.98
Fresh Water	149.3	5.29	0.00	4.61	165.03
Hydraulic oil	3.3	44.30	-4.34	2.32	0.61
Low flashpoint liquid	182.6	12.25	0.00	6.08	389.12
Lubricating oil	17.6	43.89	3.64	4.24	7.62
Mass load	17.5	63.21	-2.77	12.86	0.00
Miscellaneous	7.8	54.40	-2.49	3.01	7.72
Mud	1390.3	26.73	0.00	4.56	390.57
Sewage	1.1	53.89	-6.93	5.45	2.31
Sludge	1.4	48.40	3.29	1.80	6.99
Water Ballast	149.2	9.38	0.69	5.29	1738.33
DEADWEIGHT	4090.3	29.86	0.01	4.63	3330.59
Lightweight	2704.9	39.15	-0.01	7.08	
Deadweight	4090.3	29.86	0.01	4.63	
Total weight	6795.2	33.56	0.00	5.61	

F L O A T I N G P O S I T I O N

Draught moulded	6.680	m	KM	8.63 m
Trim	0.195	m	KG	5.61 m
Heel, SB=+	0.0	deg		
TA	6.582	m	GM0	3.02 m
TF	6.777	m	GMCORR	-0.49 m
Trimming moment	1475	tonm	GM	2.53 m

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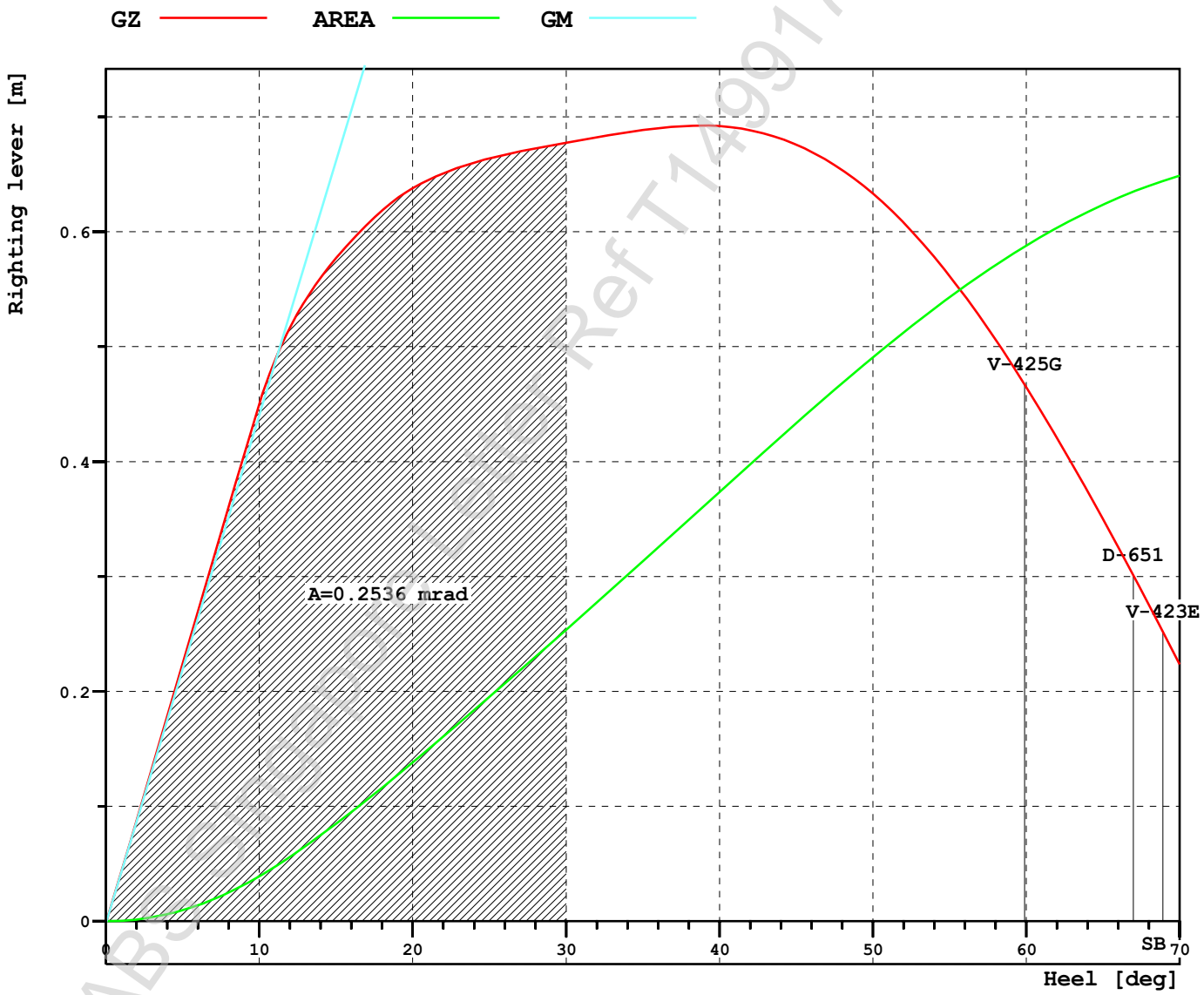
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STABILITY COMPONENTS

LOADING CONDITION L16, LFL-BRI-MUD - DEPARTURE

HEEL degree	MS m	HPHI m	EPHI mrad	FSMOM tm	DGZ m	GMSIN m	KN m
0.0	-0.002	0.00	0.000	0.0	0.000	0.000	-0.001
1.0	-0.002	0.04	0.000	56.1	0.008	0.053	0.150
3.0	-0.002	0.13	0.003	159.3	0.023	0.158	0.451
5.0	-0.002	0.22	0.010	254.8	0.038	0.263	0.751
10.0	-0.009	0.45	0.039	450.9	0.066	0.525	1.491
15.0	-0.122	0.58	0.085	564.9	0.083	0.782	2.112
20.0	-0.302	0.64	0.138	637.0	0.094	1.034	2.650
30.0	-0.726	0.68	0.254	729.9	0.107	1.511	3.589
40.0	-1.136	0.69	0.374	782.8	0.115	1.943	4.412
50.0	-1.563	0.63	0.491	808.2	0.119	2.315	5.048
60.0	-2.033	0.46	0.588	817.5	0.120	2.618	5.441
70.0	-2.495	0.22	0.649	827.6	0.122	2.840	5.614

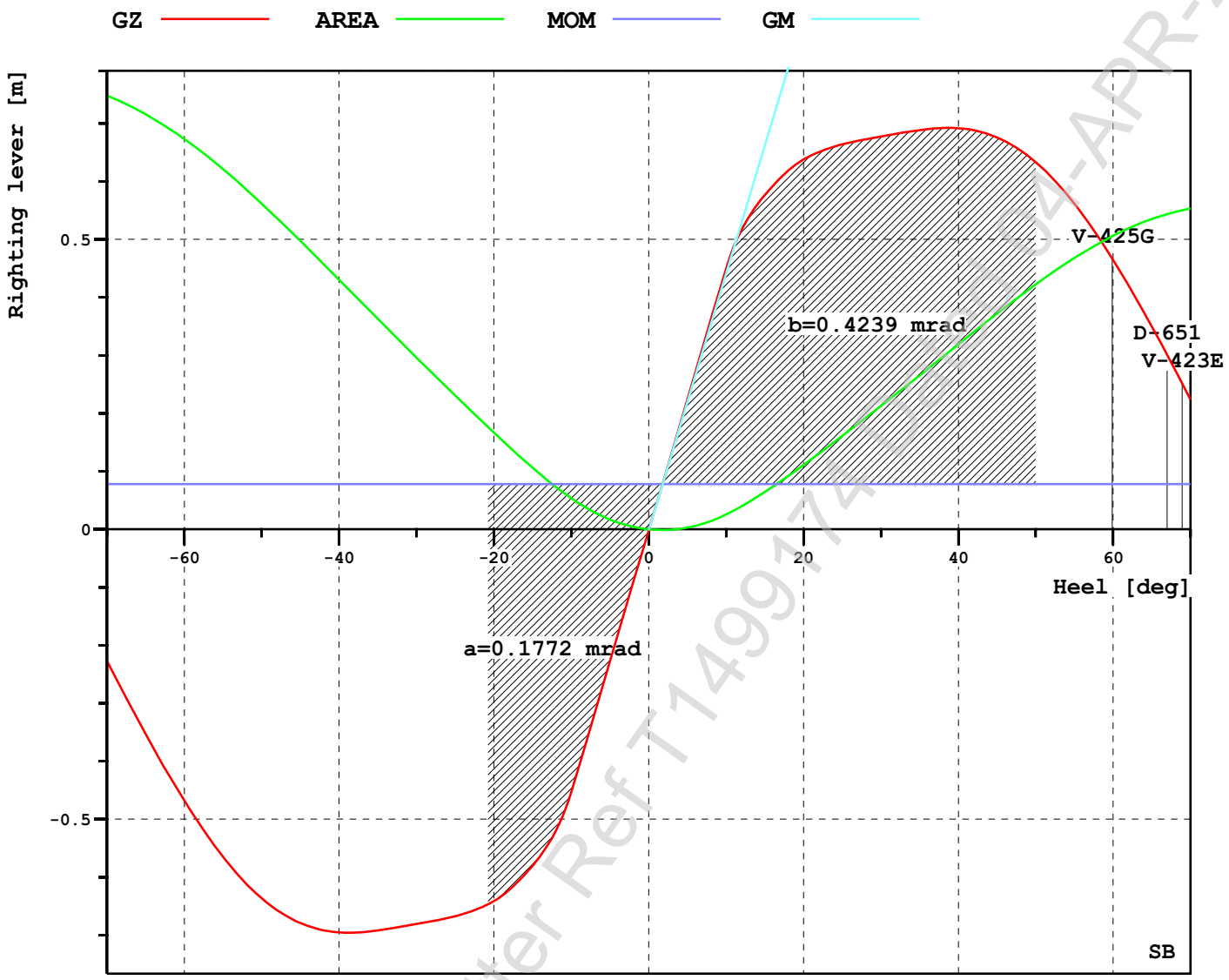
STABILITY CHECK PLOT



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IMOWEATHER CHECK PLOT



INTACT STABILITY CRITERIA

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM
V.AREA15-30	Area depending on the position of the Max. .	0.055	0.254	mrad	OK	1.720
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.120	mrad	OK	1.633
V.GZ0.2	Min. GZ > 0.2	0.200	0.692	m	OK	1.577
V.POSMAX15	Max. GZ at an angle > 15 deg.	15.000	38.830	deg	OK	1.594
V.GM0.15	GM > 0.15 m	0.150	2.532	m	OK	0.150
IMOWEATHER	IMO weather criterion	1.000	2.392		OK	1.620
2008IS-A2.3.1.2	Heeling angle due to steady wind <16 or <=8.	6.575	1.208	deg	OK	0.403
ABSFIFI1	Angle of heel < 1/2 freeboard immersion	4.110	0.307	deg	OK	0.146
ABSFIFI2	Area of residual stability >0.09 mrad	0.090	0.365	mrad	OK	1.339
TOTAL						1.720

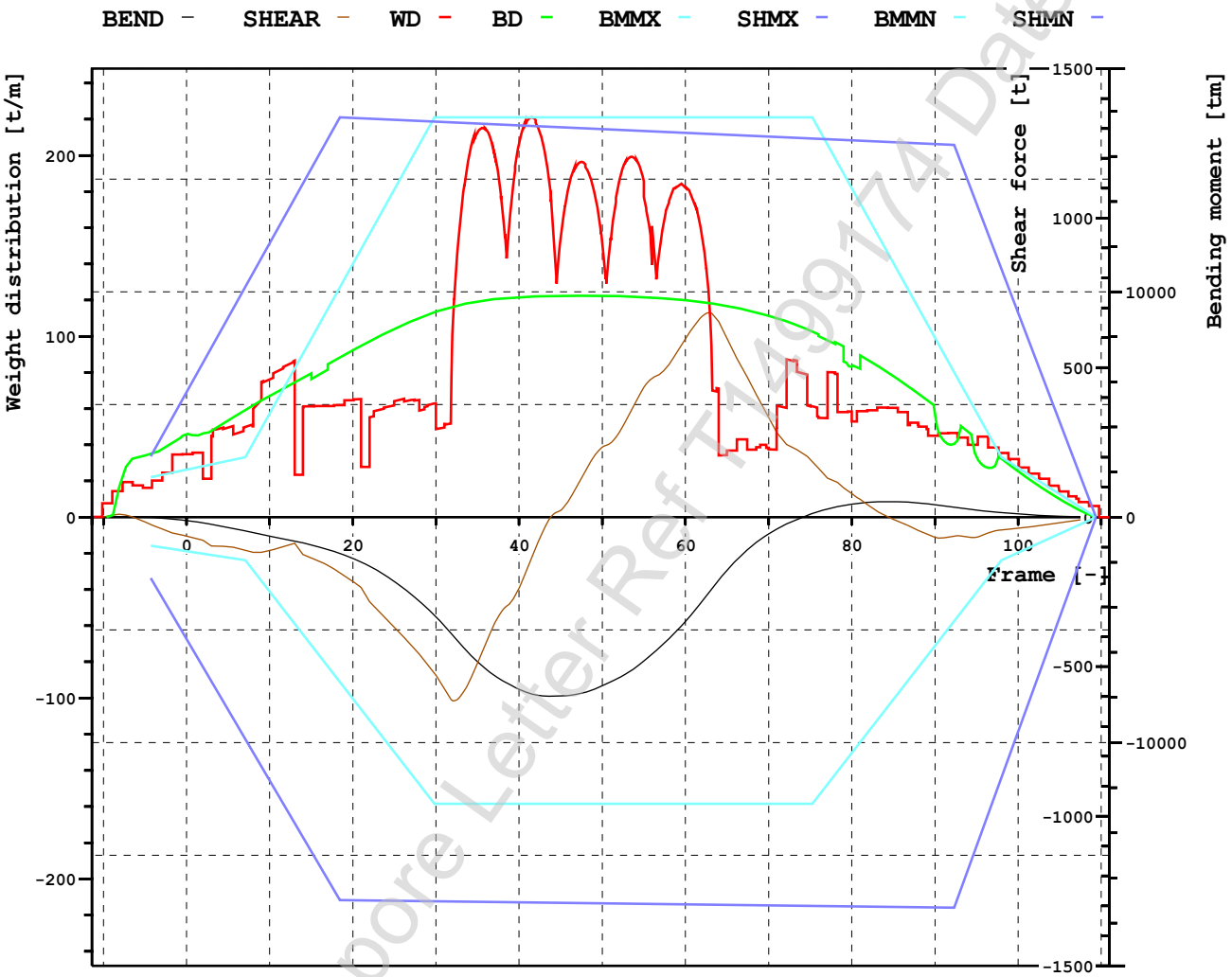
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MOST CRITICAL OPENINGS

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
V-421G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	55.890	2.093	14.288	-	7.560
V-425G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	54.243	3.767	14.288	59.9	7.563
V-424G	SUPPLY AIR ENGINE ROOM PS	UNPROTECTED	54.243	-3.413	14.288	-	7.567
D-651	ACCOMMODATION/C-DECK	UNPROTECTED	56.000	3.771	16.906	67.0	10.177
V-423E	EXHAUST AIR ENGINE ROOM SB	UNPROTECTED	55.993	3.423	17.088	68.9	10.359
V-421E	EXHAUST AIR ENGINE ROOM PS	UNPROTECTED	55.993	-2.538	17.088	-	10.363

STRENGTH DIAGRAM



STRENGTH SUMMARY

LOADING CONDITION L16, LFL-BRI-MUD - DEPARTURE

			X	Frame
Shear force (min)	-614.4 t	(47.8%)	22.477 m	32.11
Shear force (max)	684.0 t	(53.4%)	44.006 m	62.87
Sagging moment	-7943.4 tm	(62.5%)	30.683 m	43.83
Hogging moment	688.3 tm	(5.9%)	59.003 m	84.29

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STRENGTH DISTRIBUTION

LOADING CONDITION L16, LFL-BRI-MUD - DEPARTURE

X	FR	BEND	BMMN	BMMX	BMREL	SHEAR	SHMN	SHMX	SHREL	WD	BD
m	#	tm	tm	tm	%	t	t	t	%	t/m	t/m
-3.500	-5.00	1	0	0	0.0	-16	0	0	0.0	16.23	34.11
0.000	0.00	-153	-1510	2108	10.1	-64	-407	418	15.6	34.85	45.79
3.500	5.00	-454	-1790	2498	25.4	-99	-644	667	15.4	49.50	53.42
7.000	10.00	-846	-3292	4594	25.8	-112	-881	916	12.7	76.25	66.99
10.500	15.00	-1227	-5668	7911	21.7	-136	-1117	1166	12.2	61.23	79.49
14.000	20.00	-1834	-8045	11228	22.8	-215	-1281	1335	16.8	64.67	92.27
17.500	25.00	-2861	-10422	14545	27.4	-369	-1283	1329	28.8	64.20	103.95
21.000	30.00	-4422	-12709	17736	34.8	-529	-1284	1323	41.2	62.99	113.57
24.500	35.00	-6423	-12709	17736	50.5	-487	-1286	1317	37.9	212.90	119.05
28.000	40.00	-7642	-12709	17736	60.1	-236	-1288	1310	18.3	206.88	121.46
31.500	45.00	-7933	-12709	17736	62.4	23	-1290	1304	1.8	154.71	122.30
35.000	50.00	-7483	-12709	17736	58.9	234	-1292	1298	18.0	151.09	122.33
38.500	55.00	-6383	-12709	17736	50.2	435	-1293	1292	33.7	185.68	121.59
42.000	60.00	-4621	-12709	17736	36.4	597	-1295	1285	46.4	182.20	119.94
45.500	65.00	-2353	-12709	17736	18.5	596	-1297	1279	46.6	36.73	116.63
49.000	70.00	-736	-12709	17736	5.8	334	-1299	1273	26.2	38.26	111.31
52.500	75.00	128	-12709	17736	0.7	193	-1300	1267	15.2	61.65	103.18
56.000	80.00	578	-10463	14601	4.0	79	-1302	1260	6.3	58.34	83.17
59.500	85.00	686	-8086	11284	6.1	-8	-1304	1254	0.6	60.48	77.96
63.000	90.00	549	-5709	7968	6.9	-67	-1306	1248	5.2	45.02	57.36
66.500	95.00	313	-3332	4651	6.7	-67	-1102	1050	6.1	40.05	33.36
70.000	100.00	140	-1571	2193	6.4	-37	-718	685	5.1	32.12	25.36
73.500	105.00	37	-733	1023	3.6	-19	-335	319	5.6	14.17	9.91

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DAMAGE STABILITY OF LOADING CONDITION L16

CRITICAL DAMAGE STABILITY CRITERIA FOR EACH DAMAGE CASE

RCR	DAM	STAGE	PHASE	REQ	ATTV	UNIT	STAT	MINGM
V.RANGE.A469	D01	1	EQ	20.000	59.267	deg	OK	0.643
V.RANGE.A469	D02	1	EQ	20.000	57.488	deg	OK	0.868
V.RANGE.A469	D03	1	EQ	20.000	59.689	deg	OK	0.824
V.RANGE.A469	D04	1	EQ	20.000	56.178	deg	OK	1.130
V.RANGE.A469	D05	1	EQ	20.000	57.905	deg	OK	1.140
V.RANGE.A469	D06	1	EQ	20.000	56.747	deg	OK	1.153
V.RANGE.A469	D07	1	EQ	20.000	58.489	deg	OK	1.162
V.RANGE.A469	D08	1	EQ	20.000	56.146	deg	OK	1.203
V.RANGE.A469	D09	1	EQ	20.000	58.328	deg	OK	1.167
V.RANGE.A469	D10	1	EQ	20.000	56.245	deg	OK	1.265
V.RANGE.A469	D11	1	EQ	20.000	58.037	deg	OK	1.277
V.RANGE.A469	D12	1	EQ	20.000	56.569	deg	OK	1.328
V.RANGE.A469	D13	1	EQ	20.000	58.736	deg	OK	1.295
V.RANGE.A469	D14	1	EQ	20.000	58.051	deg	OK	1.141
V.RANGE.A469	D15	1	EQ	20.000	60.243	deg	OK	1.103
V.RANGE.A469	D16	1	EQ	20.000	61.725	deg	OK	0.711
V.RANGE.A469	D17	1	EQ	20.000	59.518	deg	OK	0.754
V.RANGE.A469	D18	1	EQ	20.000	59.517	deg	OK	0.885
V.RANGE.A469	D19	1	EQ	20.000	61.720	deg	OK	0.842
V.RANGE.A469	D20	1	EQ	20.000	59.656	deg	OK	0.919
V.RANGE.A469	D21	1	EQ	20.000	61.859	deg	OK	0.877
TOTAL								1.328

DETERMINATION OF CRITICAL DAMAGE CASE

CRITICAL DAMAGE CASE : D12
STAGE OF CRITICAL DAMAGE CASE : 1
PHASE OF CRITICAL DAMAGE CASE : EQ

DAMAGE STABILITY CRITERIA - CRITICAL CASE

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM m
V.PROGR.A469	No progr. flooding	11.833	2.765	deg	OK	0.671
V.MAXHEEL.A469	Maximum heel	17.000	2.765	deg	OK	0.671
V.RANGE.A469	Range of pos. stab	20.000	56.569	deg	OK	1.328
V.MINGZ.A469	Min. righting lever	0.100	0.502	m	OK	1.189
TOTAL						1.328

DAMAGE CASE RESULTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	T m	TR m	HEEL degree	RESFLD m	OPEN	CROSSTM s
D12	1	EQ	SB	6.769	0.005	2.8	1.41	D-360	-

DAMAGED COMPARTMENTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	NAME	DES	PERM	VOL m3	W t	XCG m	YCG m	ZCG m
D12	INTACT	EQ	SB				0.0	0.000			
D12	1	EQ	SB	T023	WING TANK SB	0.95	54.6	55.993	19.35	7.84	3.63
D12	1	EQ	SB	T025	DB TANK SB	0.95	72.5	74.304	21.46	3.85	0.81
D12	1	EQ	SB	E2	Emrg. exit 1 SB	0.95	20.8	21.284	22.57	7.29	6.06

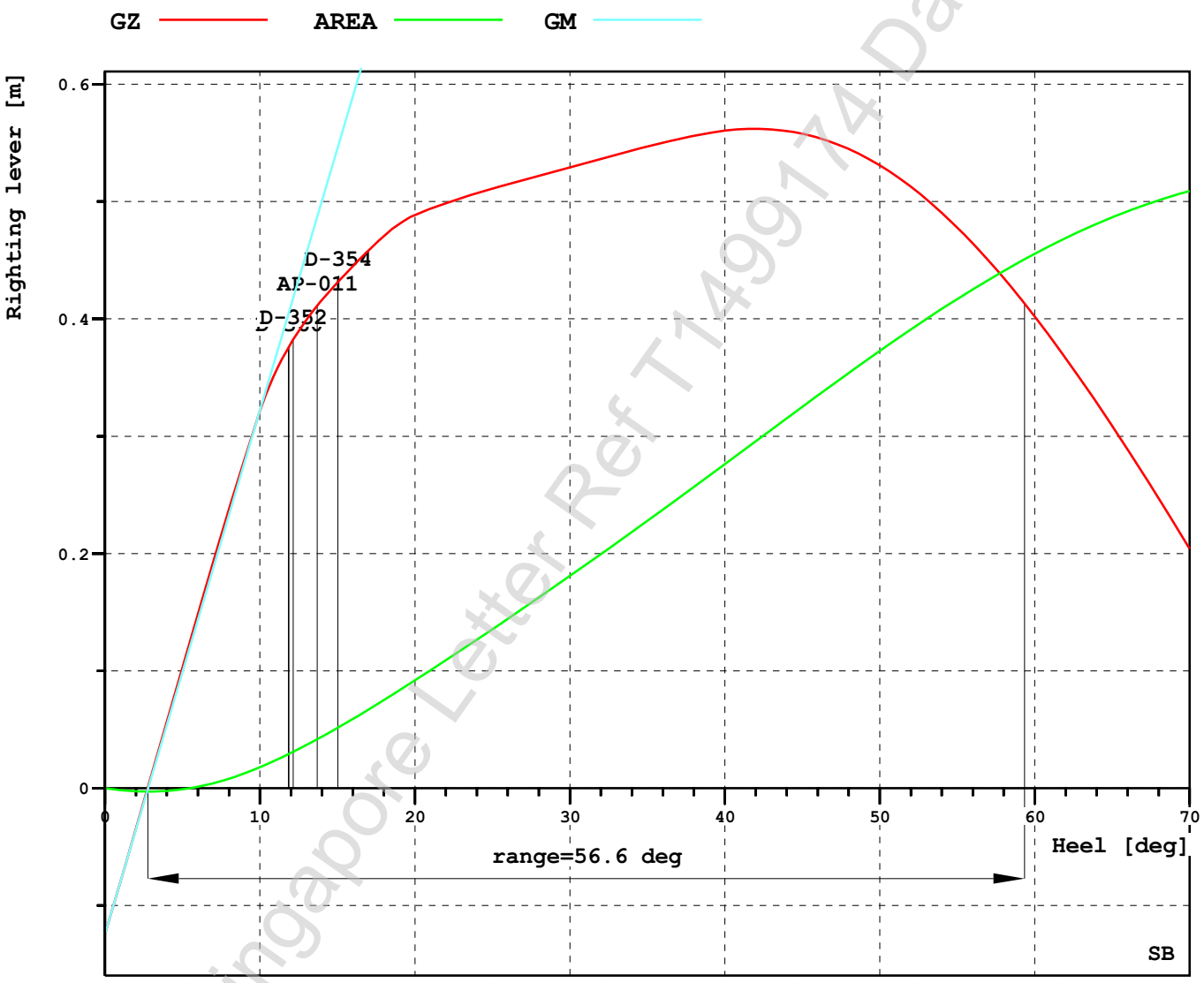
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CRITICAL OPENINGS - CRITICAL CASE

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
D-362	SB CARGORAIL #37/M-DECK	WEATHERTIGHT	25.900	8.550	8.600	11.8	1.409
D-360	SB CARGORAIL #57/M-DECK	WEATHERTIGHT	39.885	8.560	8.600	11.9	1.408
D-352	CHEMICAL/M-DECK	WEATHERTIGHT	53.192	8.375	8.600	12.2	1.416
AP-011	WING TANK SB	WEATHERTIGHT	47.211	8.754	8.953	13.7	1.751
D-354	PAINT/M-DECK	WEATHERTIGHT	53.192	6.600	8.600	15.0	1.502
AP-290	LO STORE TANK MAIN ENG	WEATHERTIGHT	52.985	7.410	8.927	15.8	1.789
AP-030	WING TANK SB	WEATHERTIGHT	5.250	7.801	9.168	15.8	2.014
V-353A	EXHAUST FAN SWITCHBOARD ROOM	WEATHERTIGHT	47.146	7.297	8.959	16.1	1.827

DAMAGE STABILITY CHECK PLOT - CRITICAL CASE



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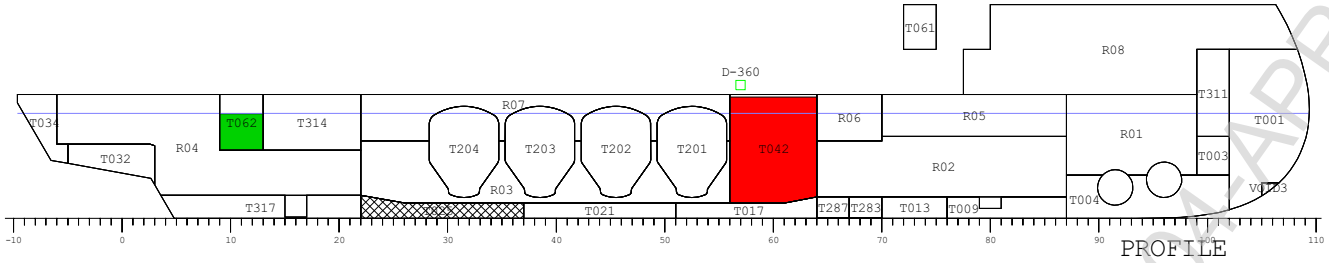
GZ-CURVE - CRITICAL CASE

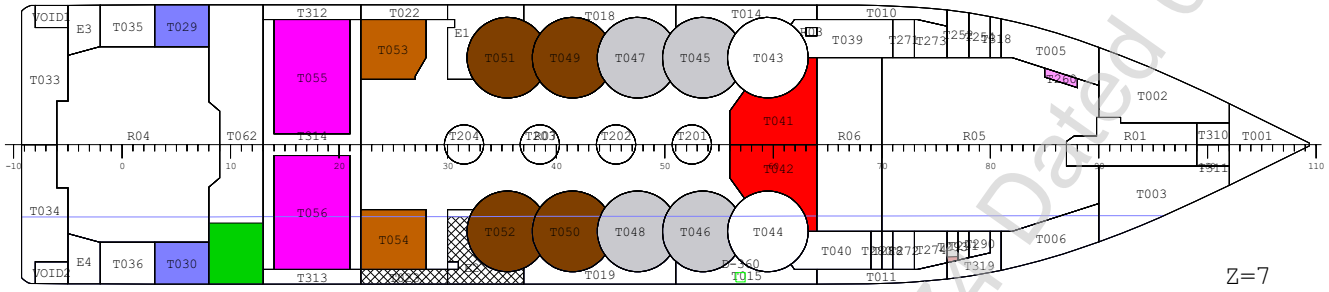
HEEL	DISP	T	TR	GZ	MS	AREA
degree	t	m	m	m	m	mrad
0.0	6795.2	6.774	0.012	-0.125	-0.125	0.000
1.0	6795.2	6.774	0.009	-0.080	-0.133	-0.002
3.0	6795.2	6.768	0.004	0.011	-0.147	-0.003
5.0	6795.2	6.751	0.003	0.103	-0.161	-0.001
10.0	6795.2	6.679	-0.009	0.321	-0.204	0.018
15.0	6795.2	6.628	-0.168	0.431	-0.351	0.051
20.0	6795.2	6.578	-0.460	0.489	-0.545	0.092
30.0	6795.2	6.428	-1.406	0.529	-0.982	0.181
40.0	6795.2	6.154	-2.817	0.560	-1.382	0.276
50.0	6795.2	5.767	-4.354	0.531	-1.784	0.373
60.0	6795.2	5.258	-5.775	0.402	-2.216	0.455
70.0	6795.2	4.616	-6.982	0.204	-2.636	0.509

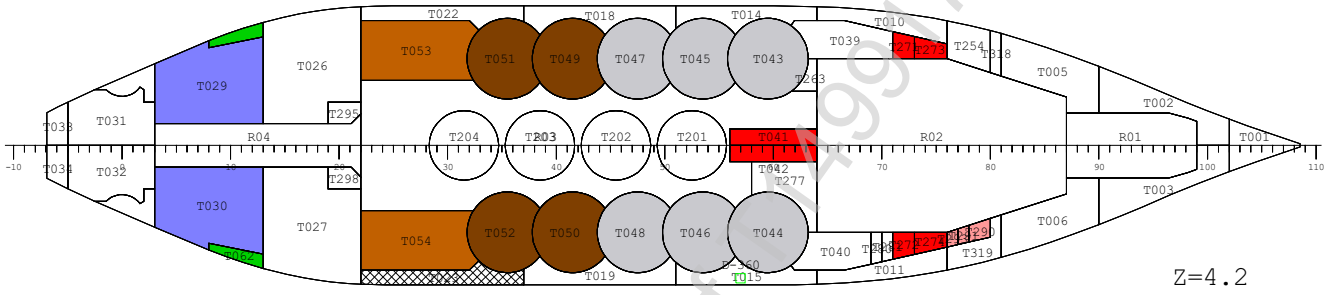
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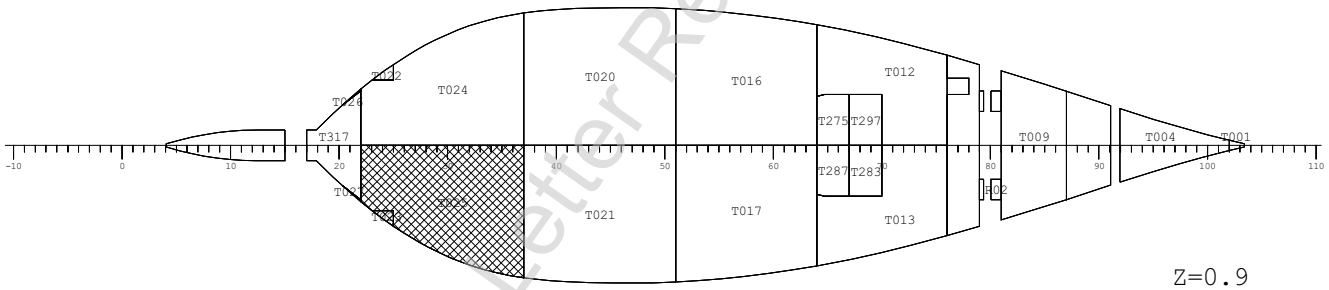
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DAMAGE ARRANGEMENT - CRITICAL CASE





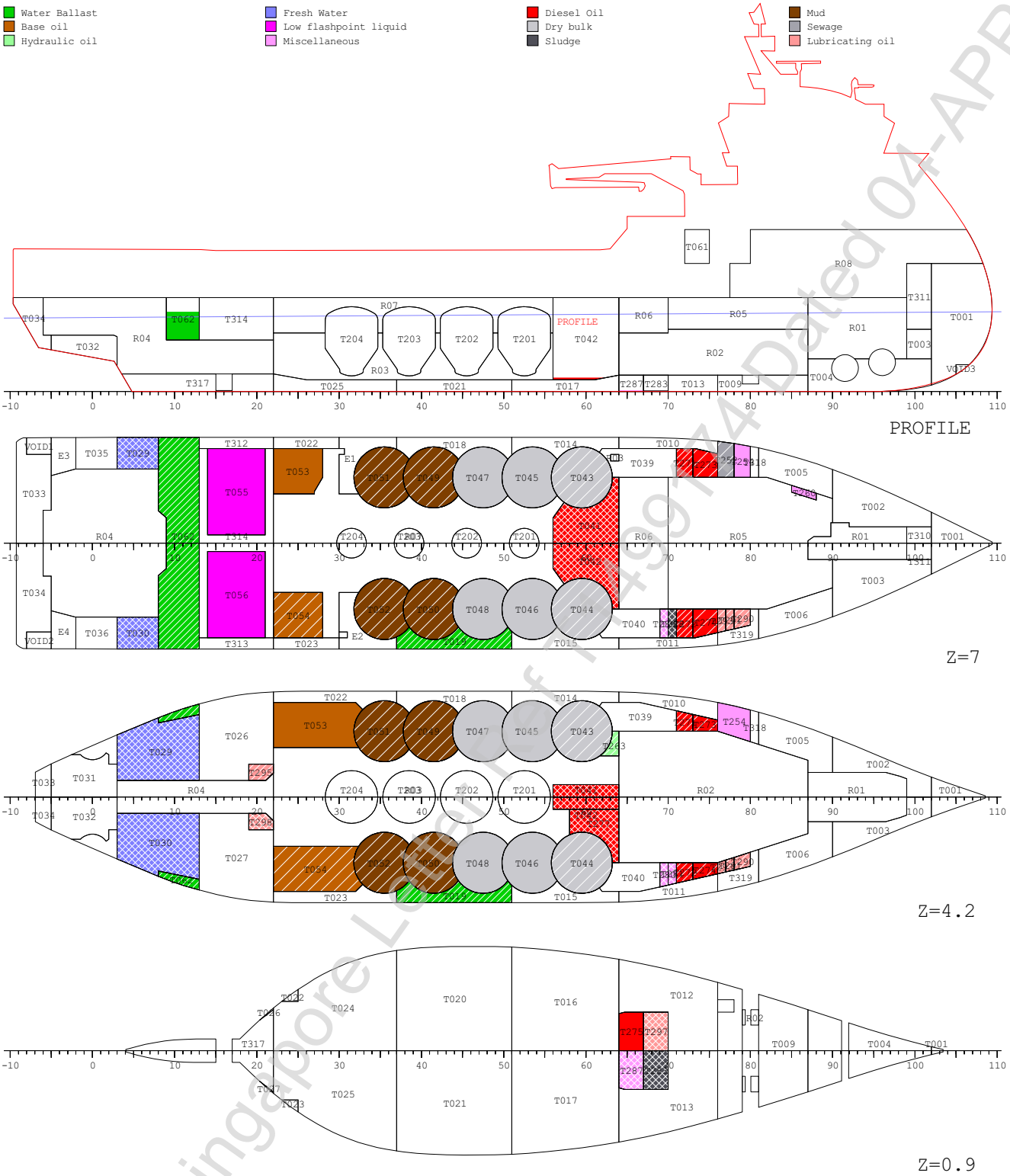




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12.17 L17 -> LFL-BRI-MUD - ARRIVAL



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LIST OF LOADS

NAME	DES	LOAD	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm	FRS	HM m
CONTENTS=Brine (RHO=2.5)										
T043	LIQUID CARGO TANK PS	BRINE	270.4	77.2	41.65	-5.56	3.86	90	REAL	6.47
T044	LIQUID CARGO TANK SB	BRINE	270.4	77.2	41.65	5.56	3.86	90	REAL	6.47
T045	LIQUID CARGO TANK PS	BRINE	311.5	98.0	37.22	-5.58	4.54	87	REAL	7.86
T046	LIQUID CARGO TANK SB	BRINE	304.6	98.0	37.19	5.58	4.47	73	REAL	7.84
T047	LIQUID CARGO TANK PS	BRINE	311.5	98.0	33.02	-5.58	4.54	87	REAL	7.86
T048	LIQUID CARGO TANK SB	BRINE	311.5	98.0	33.02	5.58	4.54	87	REAL	7.86
SUBTOTAL		BRINE	1779.9		37.09	-0.02	4.32	514		
CONTENTS=Base oil (RHO=0.83)										
T053	STORAGE TANK PS	BSO	104.6	98.0	18.73	-6.02	4.76	17	REAL	7.84
T054	STORAGE TANK SB	BSO	104.4	98.0	18.71	6.02	4.78	17	REAL	7.84
SUBTOTAL		BSO	209.0		18.72	-0.01	4.77	33		
CONTENTS=Diesel Oil (RHO=0.86)										
T041	STORAGE TANK PS	FO	0.7	1.0	41.24	-0.52	1.10	27	MAX	1.20
T042	STORAGE TANK SB	FO	0.7	1.0	41.24	0.52	1.10	27	MAX	1.20
T271	FO SETTLING TANK PS	FO	8.5	70.0	50.38	-6.53	4.93	1	REAL	6.72
T272	FO SETTLING TANK SB	FO	8.5	70.0	50.38	6.53	4.93	1	REAL	6.72
T273	FO SERVICE TANK PS	FO	10.6	70.0	52.09	-6.38	5.15	2	REAL	6.90
T274	FO SERVICE TANK SB	FO	10.6	70.0	52.09	6.38	5.15	2	REAL	6.90
T275	FO DRAIN TANK	FO	7.9	98.0	45.85	-1.64	0.69	5	REAL	1.37
T277	FO OVERFLOW TANK	FO	3.6	10.0	42.62	2.34	1.31	10	REAL	1.53
T279	EMERG. GEN. FO TANK	FO	1.9	98.0	61.60	-1.38	24.57	0	REAL	25.80
SUBTOTAL		FO	53.0		50.02	-0.13	4.73	75		
CONTENTS=Fresh Water (RHO=1)										
T029	WING TANK PS	FW	7.5	9.8	7.28	-2.50	2.65	83	MAX	3.01
T030	WING TANK SB	FW	7.5	9.8	7.28	2.50	2.65	83	MAX	3.01
SUBTOTAL		FW	14.9		7.28	0.00	2.65	165		
CONTENTS=Hydraulic oil (RHO=0.924)										
T263	HYDRAULIC OIL STORE TANK	HO	1.9	40.0	44.29	-4.33	1.89	1	REAL	2.47
CONTENTS=Low flashpoint liquid (RHO=1.2)										
T055	LFL TANK PS	LFL	91.3	98.0	12.25	-4.38	6.08	195	REAL	7.16
T056	LFL TANK SB	LFL	91.3	98.0	12.25	4.38	6.08	195	REAL	7.16
SUBTOTAL		LFL	182.6		12.25	0.00	6.08	389		
CONTENTS=Lubricating oil (RHO=0.924)										
T290	LO STORE TANK MAIN ENG.	LO	3.4	40.0	55.32	5.47	4.05	0	REAL	5.16
T291	LO STORE TANK MAIN ENG.	LO	1.8	40.0	54.25	5.81	4.26	0	REAL	5.48
T293	LO STORE TANK THRUSTER	LO	1.8	40.0	53.55	6.03	4.41	0	REAL	5.66
T295	LO STORE TANK AZIMUTH THR.	LO	2.3	40.0	14.35	-2.10	2.63	0	REAL	3.06
T297	LO DRAIN TANK	LO	0.9	10.0	47.95	-1.64	0.07	6	REAL	0.14
T298	DIRTY LO TANK	LO	0.6	10.0	14.35	2.10	2.31	0	REAL	2.41
SUBTOTAL		LO	10.7		43.36	3.26	3.43	7		
CONTENTS=Mass load (RHO=1)										
M1	CREW	MASS	2.6	0.0	57.40	0.00	15.50	0		0.00
M2	PROVISION	MASS	0.8	0.0	56.35	-6.50	12.40	0		0.00
M3	STORES	MASS	0.8	0.0	72.10	0.00	12.40	0		0.00
SUBTOTAL		MASS	4.2		60.00	-1.24	14.32	0		
CONTENTS=Miscellaneous (RHO=1)										
T280	BILGE WATER HOLDING TANK	MIS	0.9	10.0	48.64	6.17	2.03	0	REAL	2.57
T281	BILGE WATER SETTLING TANK	MIS	0.4	10.0	49.34	6.04	1.76	0	REAL	2.08
T287	BILGE WATER DRAIN TANK	MIS	0.9	10.0	45.85	1.64	0.07	6	REAL	0.14
T254	GREY WATER TANK	MIS	31.6	98.0	54.82	-6.47	4.52	3	REAL	7.84
T260	WATER MIST TANK	MIS	2.3	98.0	60.55	-4.30	6.28	0	REAL	7.27
SUBTOTAL		MIS	36.2		54.74	-5.68	4.43	9		
CONTENTS=Mud (RHO=2.8)										
T049	LIQUID CARGO TANK PS	MUD	348.9	98.0	28.82	-5.58	4.54	98	REAL	7.86
T050	LIQUID CARGO TANK SB	MUD	348.9	98.0	28.82	5.58	4.54	98	REAL	7.86
T051	LIQUID CARGO TANK PS	MUD	346.2	98.0	24.63	-5.56	4.57	98	REAL	7.87

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T052	LIQUID CARGO TANK SB	MUD	346.2	98.0	24.63	5.56	4.57	98	REAL	7.87
SUBTOTAL		MUD	1390.3		26.73	0.00	4.56	391		
CONTENTS=Sewage (RHO=1)										
T252	BLACK WATER TANK	SEW	10.4	97.9	53.89	-7.06	6.66	4	REAL	7.95
CONTENTS=Sludge (RHO=1)										
T283	SLUDGE DRAIN TANK	SLU	0.9	10.0	47.95	1.64	0.07	6	REAL	0.14
T288	SLUDGE SETTLING TANK	SLU	0.4	10.0	49.35	6.75	5.44	1	REAL	5.59
SUBTOTAL		SLU	1.4		48.40	3.29	1.80	7		
CONTENTS=Water Ballast (RHO=1.025)										
T019	WING TANK SB	WB	39.0	52.5	30.80	8.37	3.18	2	MAX	4.90
T062	ROLL RED. TANK	WB	136.9	65.0	7.45	0.00	5.58	1736	REAL	6.76
SUBTOTAL		WB	175.9		12.63	1.86	5.05	1738		
TOTAL			3870.3		30.42	0.01	4.56	3333		

LOAD SUMMARY TABLE

ITEM	MASS t	LCG m	TCG m	VCG m	FRSM tm
Brine	1779.9	37.09	-0.02	4.32	514.14
Base oil	209.0	18.72	-0.01	4.77	33.16
Diesel Oil	53.0	50.02	-0.13	4.73	74.98
Fresh Water	14.9	7.28	0.00	2.65	165.03
Hydraulic oil	1.9	44.29	-4.33	1.89	0.61
Low flashpoint liquid	182.6	12.25	0.00	6.08	389.12
Lubricating oil	10.7	43.36	3.26	3.43	7.30
Mass load	4.2	60.00	-1.24	14.32	0.00
Miscellaneous	36.2	54.74	-5.68	4.43	9.39
Mud	1390.3	26.73	0.00	4.56	390.57
Sewage	10.4	53.89	-7.06	6.66	3.55
Sludge	1.4	48.40	3.29	1.80	6.99
Water Ballast	175.9	12.63	1.86	5.05	1738.33
DEADWEIGHT	3870.3	30.42	0.01	4.56	3333.18
Lightweight	2704.9	39.15	-0.01	7.08	
Deadweight	3870.3	30.42	0.01	4.56	
Total weight	6575.2	34.01	0.00	5.60	

F L O A T I N G P O S I T I O N

Draught moulded	6.539	m	KM	8.64 m
Trim	0.500	m	KG	5.60 m
Heel, SB=+	0.0	deg		
TA	6.289	m	GM0	3.05 m
TF	6.789	m	GMCORR	-0.51 m
Trimming moment	3736	tonm	GM	2.54 m

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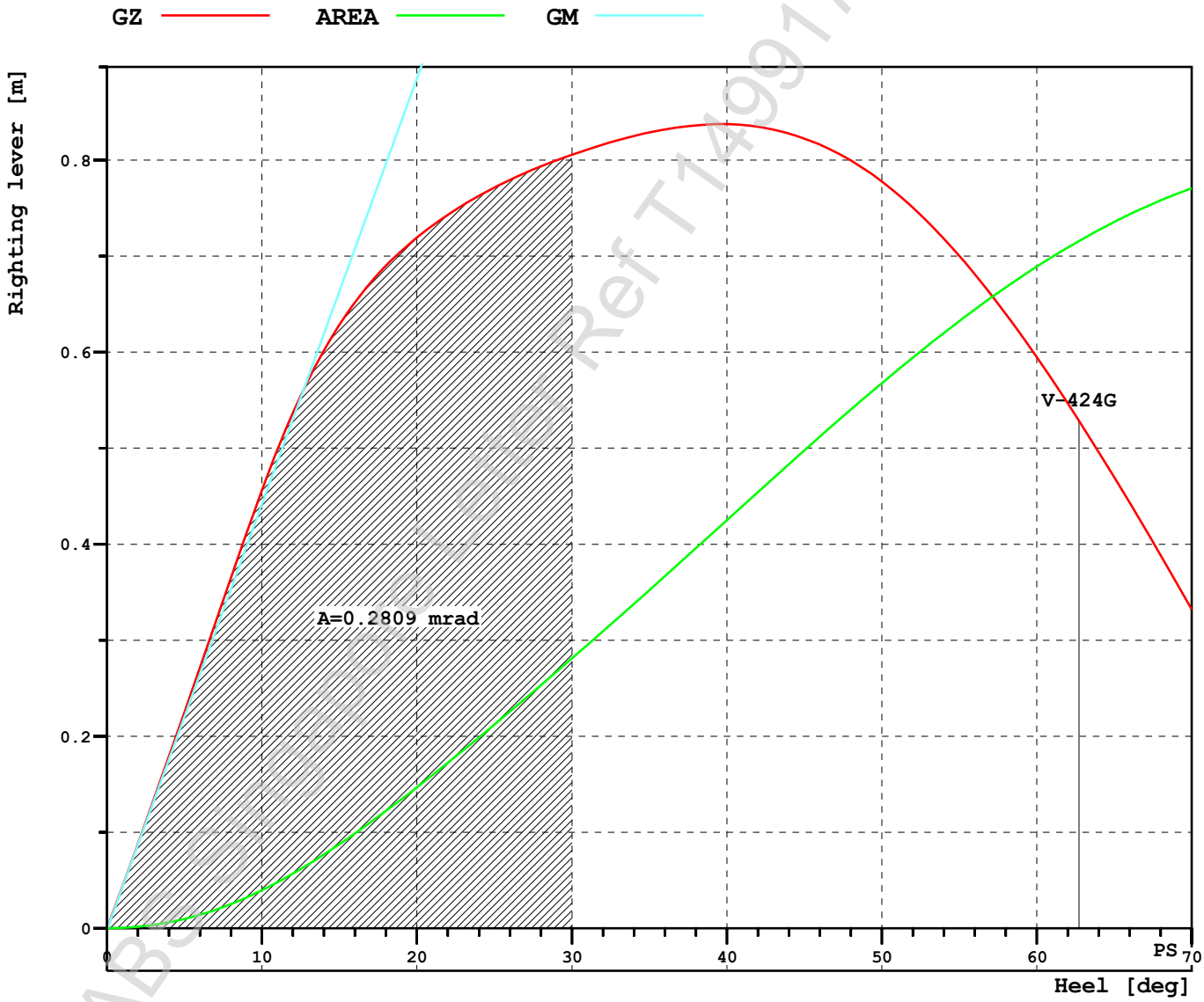
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STABILITY COMPONENTS

LOADING CONDITION L17, LFL-BRI-MUD - ARRIVAL

HEEL degree	MS m	HPHI m	EPHI mrad	FSMOM tm	DGZ m	GMSIN m	KN m
0.0	-0.001	0.00	0.000	0.0	0.000	0.000	0.001
1.0	-0.002	0.04	0.000	56.1	0.009	0.053	0.152
3.0	-0.002	0.13	0.003	159.3	0.024	0.159	0.453
5.0	-0.002	0.22	0.010	254.6	0.039	0.265	0.753
10.0	-0.004	0.46	0.039	450.0	0.068	0.529	1.499
15.0	-0.074	0.63	0.087	563.0	0.086	0.788	2.165
20.0	-0.226	0.72	0.147	634.3	0.096	1.042	2.732
30.0	-0.607	0.81	0.281	725.7	0.110	1.523	3.716
40.0	-1.002	0.84	0.425	777.2	0.118	1.958	4.555
50.0	-1.434	0.78	0.568	800.7	0.122	2.333	5.188
60.0	-1.921	0.59	0.689	807.3	0.123	2.638	5.566
70.0	-2.407	0.33	0.771	813.7	0.124	2.862	5.716

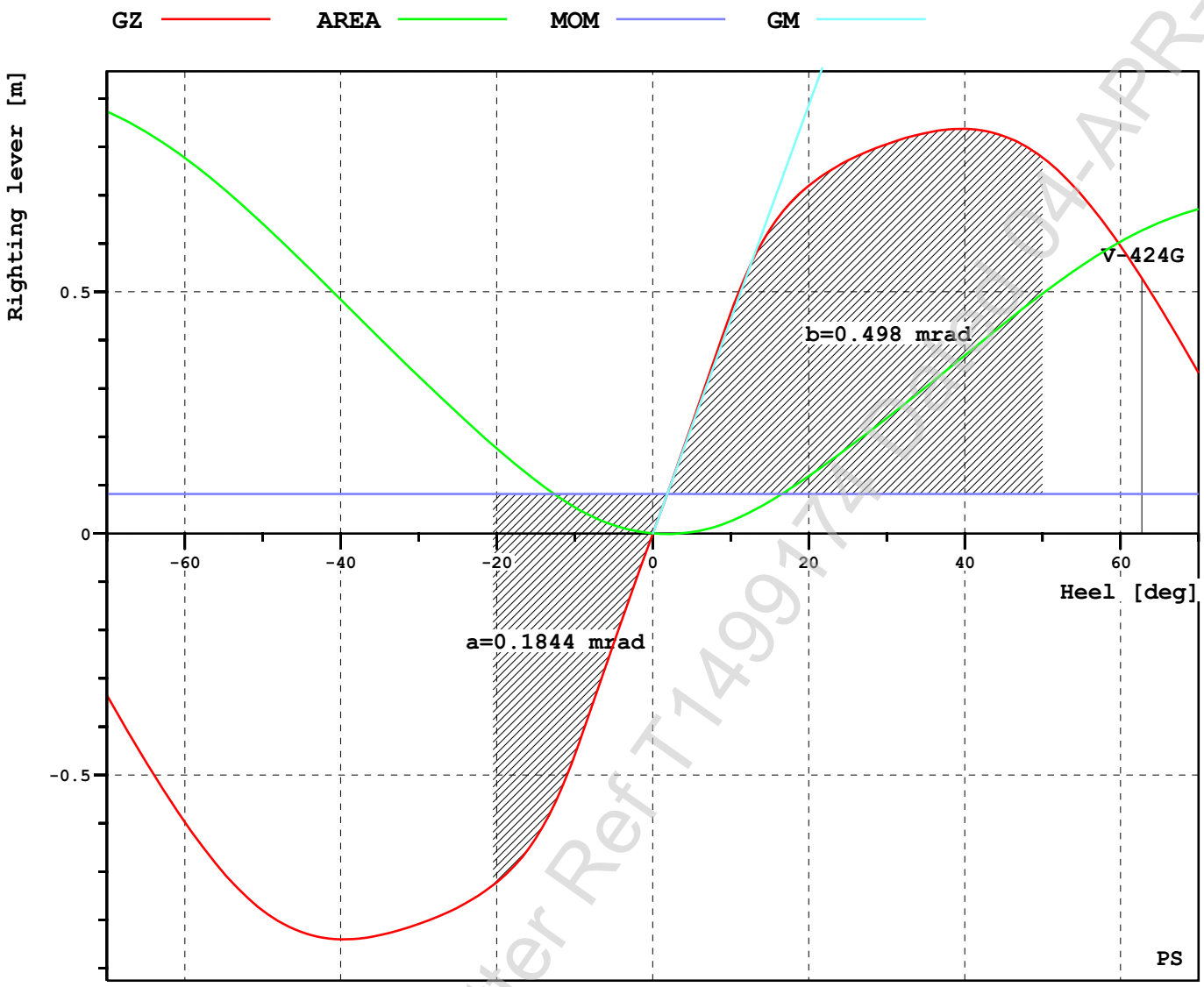
STABILITY CHECK PLOT



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IMOWEATHER CHECK PLOT



INTACT STABILITY CRITERIA

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM
V.AREA15-30	Area depending on the position of the Max. .	0.055	0.281	mrad	OK	1.481
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.144	mrad	OK	1.397
V.GZ0.2	Min. GZ > 0.2	0.200	0.837	m	OK	1.328
V.POSMAX15	Max. GZ at an angle > 15 deg.	15.000	39.653	deg	OK	1.066
V.GM0.15	GM > 0.15 m	0.150	2.539	m	OK	0.150
IMOWEATHER	IMO weather criterion	1.000	2.701		OK	1.342
2008IS-A2.3.1.2	Heeling angle due to steady wind <16 or <=8.	7.015	1.255	deg	OK	0.374
ABSFIFI1	Angle of heel < 1/2 freeboard immersion	4.384	0.308	deg	OK	0.137
ABSFIFI2	Area of residual stability >0.09 mrad	0.090	0.416	mrad	OK	1.132
TOTAL						1.481

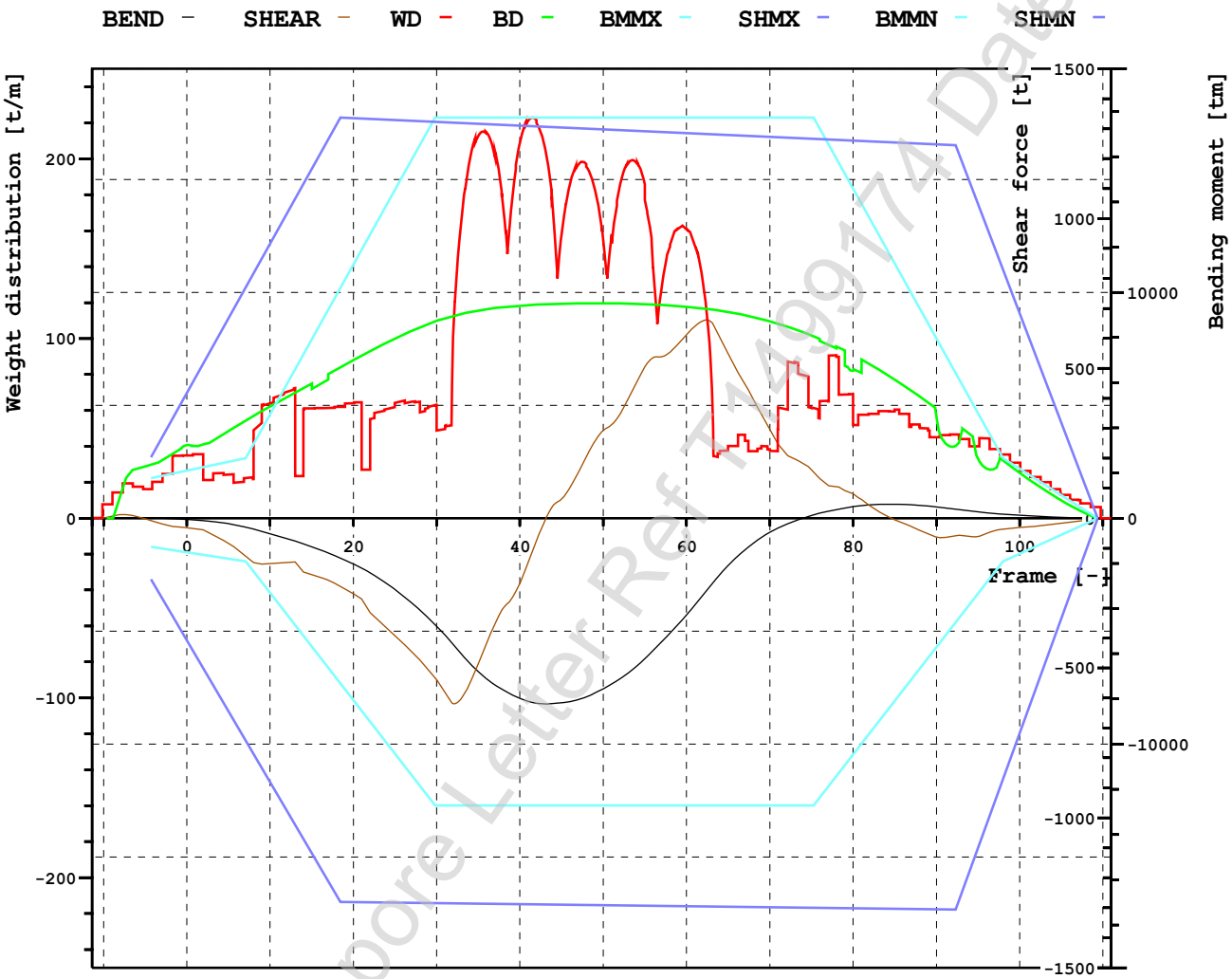
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MOST CRITICAL OPENINGS

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
V-421G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	55.890	2.093	14.288	-	7.630
V-424G	SUPPLY AIR ENGINE ROOM PS	UNPROTECTED	54.243	-3.413	14.288	-62.7	7.638
V-425G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	54.243	3.767	14.288	-	7.641
D-651	ACCOMMODATION/C-DECK	UNPROTECTED	56.000	3.771	16.906	-	10.248
V-421E	EXHAUST AIR ENGINE ROOM PS	UNPROTECTED	55.993	-2.538	17.088	-	10.427
V-423E	EXHAUST AIR ENGINE ROOM SB	UNPROTECTED	55.993	3.423	17.088	-	10.430

STRENGTH DIAGRAM



STRENGTH SUMMARY

LOADING CONDITION L17, LFL-BRI-MUD - ARRIVAL

			X	Frame
Shear force (min)	-619.4 t	(48.2%)	22.447 m	32.07
Shear force (max)	661.0 t	(51.5%)	43.600 m	62.29
Sagging moment	-8219.8 tm	(64.7%)	30.187 m	43.12
Hogging moment	627.0 tm	(5.3%)	59.003 m	84.29

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STRENGTH DISTRIBUTION

LOADING CONDITION L17, LFL-BRI-MUD - ARRIVAL

X	FR	BEND	BMMN	BMMX	BMREL	SHEAR	SHMN	SHMX	SHREL	WD	BD
m	#	tm	tm	tm	%	t	t	t	%	t/m	t/m
-3.500	-5.00	18	0	0	0.0	-2	0	0	0.0	16.23	28.83
0.000	0.00	-58	-1510	2108	3.9	-32	-407	418	7.8	34.85	40.66
3.500	5.00	-242	-1790	2498	13.5	-83	-644	667	12.9	24.27	48.47
7.000	10.00	-699	-3292	4594	21.2	-152	-881	916	17.2	63.92	62.27
10.500	15.00	-1260	-5668	7911	22.2	-188	-1117	1166	16.9	61.12	75.02
14.000	20.00	-2023	-8045	11228	25.1	-252	-1281	1335	19.7	63.97	88.04
17.500	25.00	-3159	-10422	14545	30.3	-393	-1283	1329	30.6	64.20	99.97
21.000	30.00	-4782	-12709	17736	37.6	-539	-1284	1323	42.0	62.99	109.84
24.500	35.00	-6798	-12709	17736	53.5	-485	-1286	1317	37.7	212.89	115.56
28.000	40.00	-7983	-12709	17736	62.8	-216	-1288	1310	16.7	209.19	118.22
31.500	45.00	-8170	-12709	17736	64.3	64	-1290	1304	4.9	158.07	119.31
35.000	50.00	-7547	-12709	17736	59.4	293	-1292	1298	22.6	154.60	119.59
38.500	55.00	-6218	-12709	17736	48.9	506	-1293	1292	39.1	185.81	119.09
42.000	60.00	-4283	-12709	17736	33.7	613	-1295	1285	47.7	160.71	117.70
45.500	65.00	-2091	-12709	17736	16.5	548	-1297	1279	42.8	40.10	114.63
49.000	70.00	-620	-12709	17736	4.9	296	-1299	1273	23.3	38.29	109.57
52.500	75.00	123	-12709	17736	0.7	161	-1300	1267	12.7	61.65	101.73
56.000	80.00	511	-10463	14601	3.5	82	-1302	1260	6.5	69.12	82.03
59.500	85.00	626	-8086	11284	5.5	-6	-1304	1254	0.4	60.40	77.14
63.000	90.00	500	-5709	7968	6.3	-63	-1306	1248	4.8	45.02	56.84
66.500	95.00	281	-3332	4651	6.0	-61	-1102	1050	5.5	40.05	33.09
70.000	100.00	129	-1571	2193	5.9	-31	-718	685	4.3	30.94	25.27
73.500	105.00	37	-733	1023	3.6	-17	-335	319	5.1	12.99	9.91

See ABS Singapore Letter Ref T1499174 Dated 04-MAR-2016

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DAMAGE STABILITY OF LOADING CONDITION L17

CRITICAL DAMAGE STABILITY CRITERIA FOR EACH DAMAGE CASE

RCR	DAM	STAGE	PHASE	REQ	ATTV	UNIT	STAT	MINGM
V.RANGE.A469	D01	1	EQ	20.000	62.069	deg	OK	0.415
V.RANGE.A469	D02	1	EQ	20.000	58.211	deg	OK	0.677
V.RANGE.A469	D03	1	EQ	20.000	60.301	deg	OK	0.667
V.RANGE.A469	D04	1	EQ	20.000	56.888	deg	OK	0.946
V.RANGE.A469	D05	1	EQ	20.000	59.874	deg	OK	0.839
V.RANGE.A469	D06	1	EQ	20.000	57.474	deg	OK	0.966
V.RANGE.A469	D07	1	EQ	20.000	60.478	deg	OK	0.858
V.RANGE.A469	D08	1	EQ	20.000	56.847	deg	OK	1.005
V.RANGE.A469	D09	1	EQ	20.000	58.919	deg	OK	0.990
V.RANGE.A469	D10	1	EQ	20.000	57.850	deg	OK	0.943
V.RANGE.A469	D11	1	EQ	20.000	58.633	deg	OK	1.090
V.RANGE.A469	D12	1	EQ	20.000	57.317	deg	OK	1.111
V.RANGE.A469	D13	1	EQ	20.000	59.371	deg	OK	1.099
V.RANGE.A469	D14	1	EQ	20.000	58.786	deg	OK	0.925
V.RANGE.A469	D15	1	EQ	20.000	60.870	deg	OK	0.914
V.RANGE.A469	D16	1	EQ	20.000	59.799	deg	OK	0.756
V.RANGE.A469	D17	1	EQ	20.000	61.886	deg	OK	0.749
V.RANGE.A469	D18	1	EQ	20.000	60.368	deg	OK	0.649
V.RANGE.A469	D19	1	EQ	20.000	62.458	deg	OK	0.648
V.RANGE.A469	D20	1	EQ	20.000	60.500	deg	OK	0.666
V.RANGE.A469	D21	1	EQ	20.000	62.590	deg	OK	0.665
TOTAL								1.111

DETERMINATION OF CRITICAL DAMAGE CASE

CRITICAL DAMAGE CASE : D12
STAGE OF CRITICAL DAMAGE CASE : 1
PHASE OF CRITICAL DAMAGE CASE : EQ

DAMAGE STABILITY CRITERIA - CRITICAL CASE

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM m
V.PROGR.A469	No progr. flooding	12.672	2.700	deg	OK	0.544
V.MAXHEEL.A469	Maximum heel	17.000	2.700	deg	OK	0.544
V.RANGE.A469	Range of pos. stab	20.000	57.317	deg	OK	1.111
V.MINGZ.A469	Min. righting lever	0.100	0.599	m	OK	1.012
TOTAL						1.111

DAMAGE CASE RESULTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	T m	TR m	HEEL degree	RESFLD m	OPEN	CROSSTM s
D12	1	EQ	SB	6.626	0.311	2.7	1.51	D-352	-

DAMAGED COMPARTMENTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	NAME	DES	PERM	VOL m3	W t	XCG m	YCG m	ZCG m
D12	INTACT	EQ	PS				0.0	0.000			
D12	1	EQ	SB	T023	WING TANK SB	0.95	53.5	54.833	19.38	7.83	3.56
D12	1	EQ	SB	T025	DB TANK SB	0.95	72.5	74.304	21.46	3.85	0.81
D12	1	EQ	SB	E2	Emrg. exit 1 SB	0.95	18.8	19.223	22.57	7.30	5.96

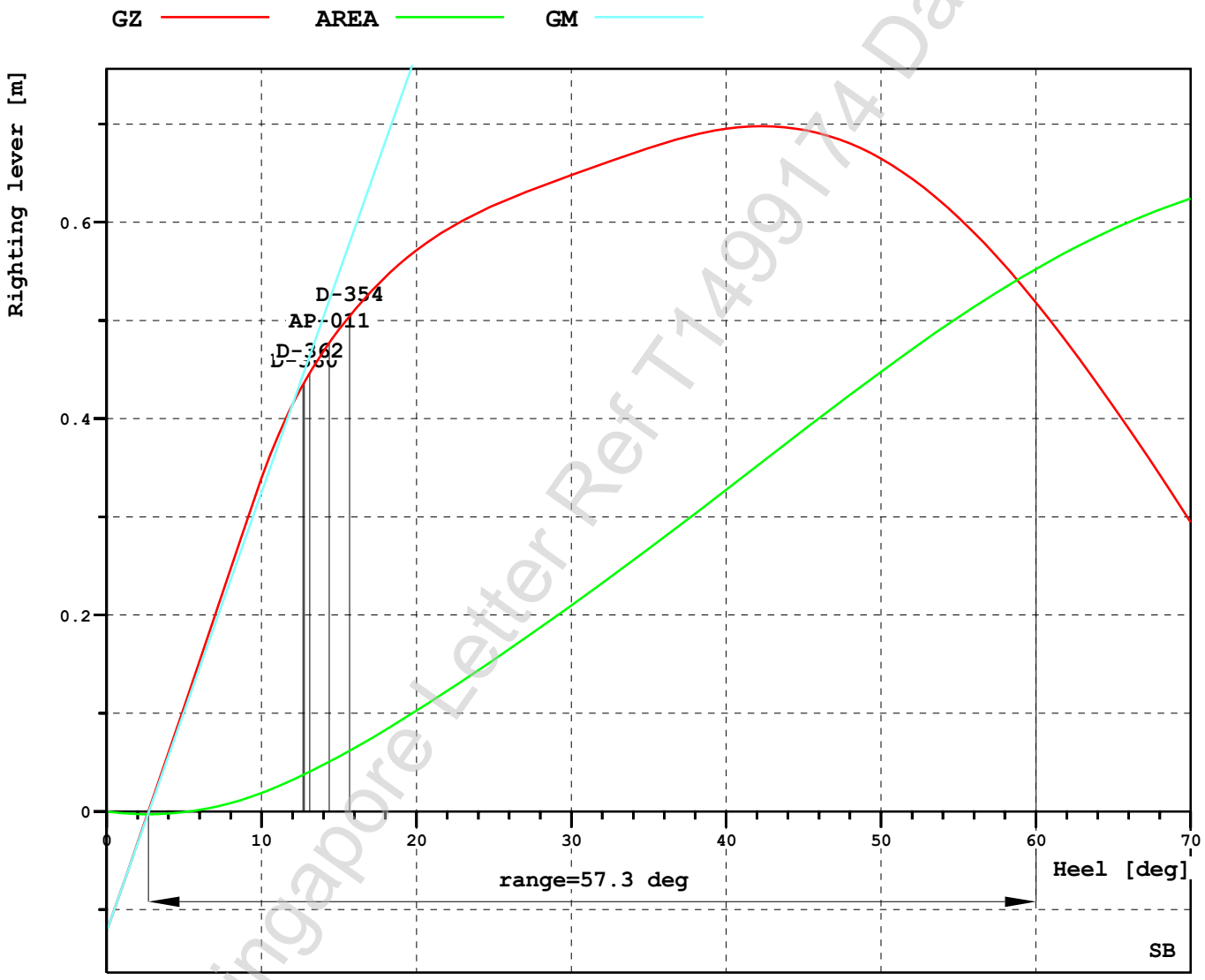
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CRITICAL OPENINGS - CRITICAL CASE

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
D-352	CHEMICAL/M-DECK	WEATHERTIGHT	53.192	8.375	8.600	12.7	1.505
D-360	SB CARGORAIL #57/M-DECK	WEATHERTIGHT	39.885	8.560	8.600	12.8	1.549
D-362	SB CARGORAIL #37/M-DECK	WEATHERTIGHT	25.900	8.550	8.600	13.1	1.604
AP-011	WING TANK SB	WEATHERTIGHT	47.211	8.754	8.953	14.4	1.863
D-354	PAINT/M-DECK	WEATHERTIGHT	53.192	6.600	8.600	15.7	1.589
AP-290	LO STORE TANK MAIN ENG	WEATHERTIGHT	52.985	7.410	8.927	16.3	1.878
AP-291	LO STORE TANK MAIN ENG	WEATHERTIGHT	52.985	7.170	8.927	16.8	1.889
AP-280	BILGE WATER HOLDING TANK	WEATHERTIGHT	47.738	7.275	8.951	16.9	1.929

DAMAGE STABILITY CHECK PLOT - CRITICAL CASE



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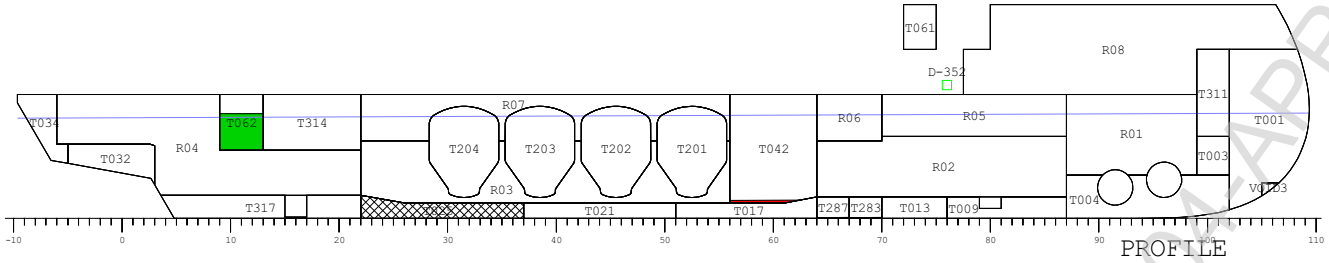
GZ-CURVE - CRITICAL CASE

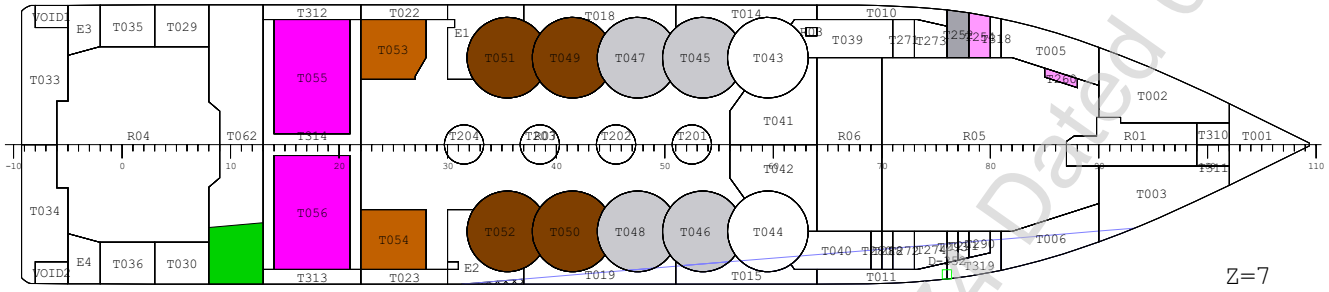
HEEL	DISP	T	TR	GZ	MS	AREA
degree	t	m	m	m	m	mrad
0.0	6575.2	6.631	0.315	-0.122	-0.122	0.000
1.0	6575.2	6.632	0.314	-0.077	-0.131	-0.002
3.0	6575.2	6.625	0.311	0.014	-0.146	-0.003
5.0	6575.2	6.609	0.312	0.106	-0.159	-0.001
10.0	6575.2	6.529	0.336	0.339	-0.190	0.019
15.0	6575.2	6.449	0.292	0.491	-0.298	0.056
20.0	6575.2	6.374	0.106	0.572	-0.470	0.102
30.0	6575.2	6.178	-0.645	0.648	-0.875	0.210
40.0	6575.2	5.867	-1.895	0.695	-1.263	0.327
50.0	6575.2	5.440	-3.299	0.665	-1.669	0.448
60.0	6575.2	4.892	-4.612	0.518	-2.120	0.552
70.0	6575.2	4.220	-5.745	0.295	-2.568	0.624

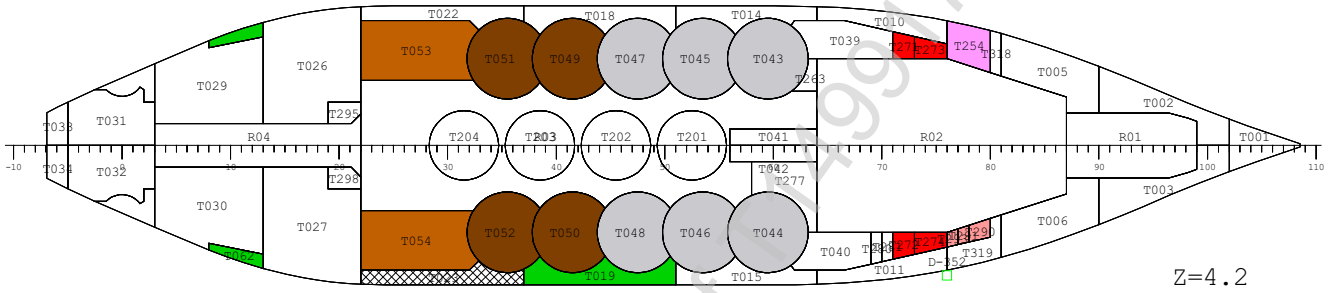
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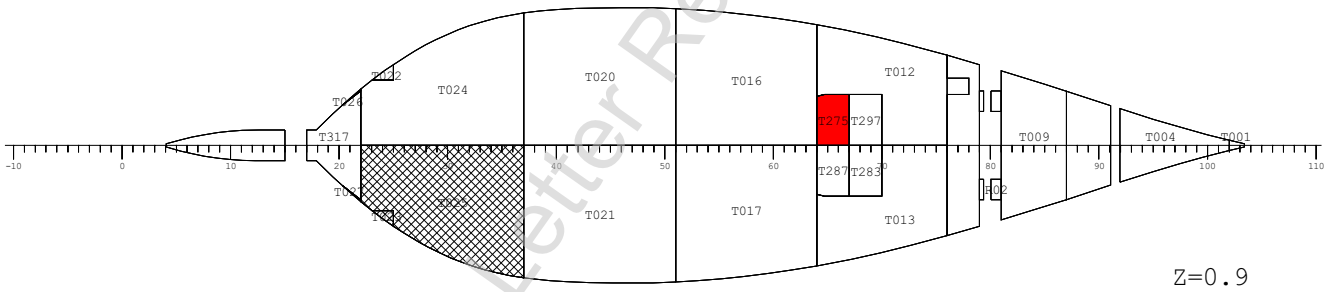
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DAMAGE ARRANGEMENT - CRITICAL CASE





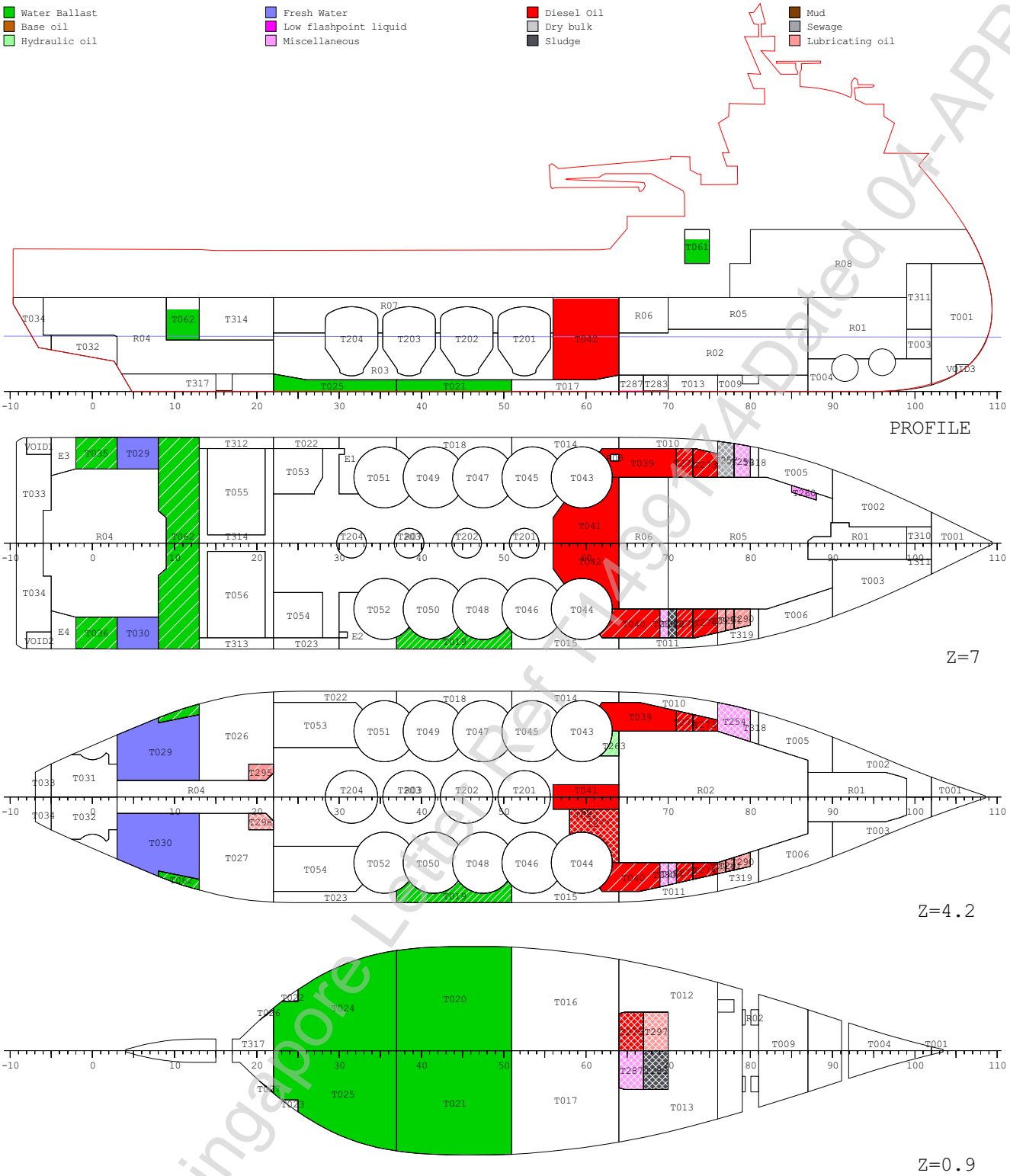




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12.18 L18 -> Transition,Fi-Fi - Departure



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LIST OF LOADS

NAME	DES	LOAD	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm	FRS	HM m
CONTENTS=Diesel Oil (RHO=0.86)										
T039	STORAGE TANK PS	FO	66.9	98.0	46.68	-6.76	4.91	6	REAL	7.88
T040	STORAGE TANK SB	FO	53.8	98.0	46.00	6.80	4.87	5	REAL	7.88
T041	STORAGE TANK PS	FO	68.5	98.0	42.16	-1.51	5.48	27	MAX	7.92
T042	STORAGE TANK SB	FO	68.5	98.0	42.16	1.51	5.48	27	MAX	7.92
T271	FO SETTLING TANK PS	FO	8.5	70.0	50.38	-6.53	4.93	1	REAL	6.72
T272	FO SETTLING TANK SB	FO	8.5	70.0	50.38	6.53	4.93	1	REAL	6.72
T273	FO SERVICE TANK PS	FO	10.6	70.0	52.09	-6.38	5.15	2	REAL	6.90
T274	FO SERVICE TANK SB	FO	10.6	70.0	52.09	6.38	5.15	2	REAL	6.90
T275	FO DRAIN TANK	FO	0.8	10.0	45.85	-1.64	0.07	5	REAL	0.14
T277	FO OVERFLOW TANK	FO	3.6	10.0	42.62	2.34	1.31	10	REAL	1.53
T279	EMERG. GEN. FO TANK	FO	1.9	98.0	61.60	-1.38	24.57	0	REAL	25.80
SUBTOTAL		FO	302.3		45.14	-0.27	5.25	86		
CONTENTS=Fresh Water (RHO=1)										
T029	WING TANK PS	FW	74.6	98.0	5.29	-4.87	4.61	83	MAX	7.83
T030	WING TANK SB	FW	74.6	98.0	5.29	4.87	4.61	83	MAX	7.83
SUBTOTAL		FW	149.3		5.29	0.00	4.61	165		
CONTENTS=Hydraulic oil (RHO=0.924)										
T263	HYDRAULIC OIL STORE TANK	HO	3.3	70.0	44.30	-4.34	2.32	1	REAL	3.33
CONTENTS=Lubricating oil (RHO=0.924)										
T290	LO STORE TANK MAIN ENG.	LO	5.9	70.0	55.30	5.76	4.90	0	REAL	6.82
T291	LO STORE TANK MAIN ENG.	LO	3.1	70.0	54.25	6.04	5.11	0	REAL	6.92
T293	LO STORE TANK THRUSTER	LO	3.2	70.0	53.55	6.21	5.24	0	REAL	6.97
T295	LO STORE TANK AZIMUTH THR.	LO	4.0	70.0	14.35	-2.10	2.95	0	REAL	3.71
T297	LO DRAIN TANK	LO	0.9	10.0	47.95	-1.64	0.07	6	REAL	0.14
T298	DIRTY LO TANK	LO	0.6	10.0	14.35	2.10	2.31	0	REAL	2.41
SUBTOTAL		LO	17.6		43.89	3.64	4.24	8		
CONTENTS=Mass load (RHO=1)										
M1	CREW	MASS	2.6	0.0	57.40	0.00	15.50	0		0.00
M2	PROVISION	MASS	7.4	0.0	56.35	-6.50	12.40	0		0.00
M3	STORES	MASS	7.4	0.0	72.10	0.00	12.40	0		0.00
M4	DECK LOAD 10T/M2 AFT OF #35	MASS	200.0	0.0	9.10	0.00	9.00	0		0.00
M5	DECK LOAD 5T/M2 FWD OF #35	MASS	200.0	0.0	37.26	0.00	9.00	0		0.00
SUBTOTAL		MASS	417.5		24.85	-0.12	9.16	0		
CONTENTS=Miscellaneous (RHO=1)										
T280	BILGE WATER HOLDING TANK	MIS	0.9	10.0	48.64	6.17	2.03	0	REAL	2.57
T281	BILGE WATER SETTLING TANK	MIS	0.4	10.0	49.34	6.04	1.76	0	REAL	2.08
T287	BILGE WATER DRAIN TANK	MIS	0.9	10.0	45.85	1.64	0.07	6	REAL	0.14
T254	GREY WATER TANK	MIS	3.2	10.0	54.62	-5.79	1.92	1	REAL	2.34
T260	WATER MIST TANK	MIS	2.3	98.0	60.55	-4.30	6.28	0	REAL	7.27
SUBTOTAL		MIS	7.8		54.40	-2.49	3.01	8		
CONTENTS=Sewage (RHO=1)										
T252	BLACK WATER TANK	SEW	1.1	10.0	53.89	-6.93	5.45	2	REAL	5.59
CONTENTS=Sludge (RHO=1)										
T283	SLUDGE DRAIN TANK	SLU	0.9	10.0	47.95	1.64	0.07	6	REAL	0.14
T288	SLUDGE SETTLING TANK	SLU	0.4	10.0	49.35	6.75	5.44	1	REAL	5.59
SUBTOTAL		SLU	1.4		48.40	3.29	1.80	7		
CONTENTS=Water Ballast (RHO=1.025)										
T019	WING TANK SB	WB	16.3	22.0	30.83	8.36	2.18	2	MAX	2.90
T020	DB TANK PS	WB	96.7	100.0	30.84	-4.63	0.61	0	(REAL)	1.92
T021	DB TANK SB	WB	96.7	100.0	30.84	4.63	0.61	0	(REAL)	1.92
T024	DB TANK PS	WB	75.9	100.0	21.46	-3.85	0.81	0	(REAL)	2.75
T025	DB TANK SB	WB	75.9	100.0	21.46	3.85	0.81	0	(REAL)	2.75
T061	ROLL RED. TANK	WB	89.9	70.0	51.66	0.00	11.91	1358	REAL	12.93
T062	ROLL RED. TANK	WB	147.4	70.0	7.45	0.00	5.67	1736	REAL	6.93
T035	WING TANK PS	WB	23.2	98.0	0.39	-7.50	6.66	6	REAL	7.95
T036	WING TANK SB	WB	23.2	98.0	0.39	7.50	6.66	6	REAL	7.95
SUBTOTAL		WB	645.2		24.00	0.21	3.86	3109		

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TOTAL	1545.4	27.02	0.02	5.63	3385

LOAD SUMMARY TABLE

ITEM	MASS t	LCG m	TCG m	VCG m	FRSM tm
Diesel Oil	302.3	45.14	-0.27	5.25	85.89
Fresh Water	149.3	5.29	0.00	4.61	165.03
Hydraulic oil	3.3	44.30	-4.34	2.32	0.61
Lubricating oil	17.6	43.89	3.64	4.24	7.62
Mass load	417.5	24.85	-0.12	9.16	0.00
Miscellaneous	7.8	54.40	-2.49	3.01	7.72
Sewage	1.1	53.89	-6.93	5.45	2.31
Sludge	1.4	48.40	3.29	1.80	6.99
Water Ballast	645.2	24.00	0.21	3.86	3108.51
DEADWEIGHT	1545.4	27.02	0.02	5.63	3384.68
Lightweight	2704.9	39.15	-0.01	7.08	
Deadweight	1545.4	27.02	0.02	5.63	
Total weight	4250.3	34.74	0.00	6.55	

F L O A T I N G P O S I T I O N

Draught moulded	4.680	m	KM	9.02 m
Trim	-0.074	m	KG	6.55 m
Heel, SB=+	0.0	deg		
TA	4.717	m	GM0	2.46 m
TF	4.643	m	GMCORR	-0.80 m
Trimming moment	-431	tonm	GM	1.66 m

STABILITY COMPONENTS

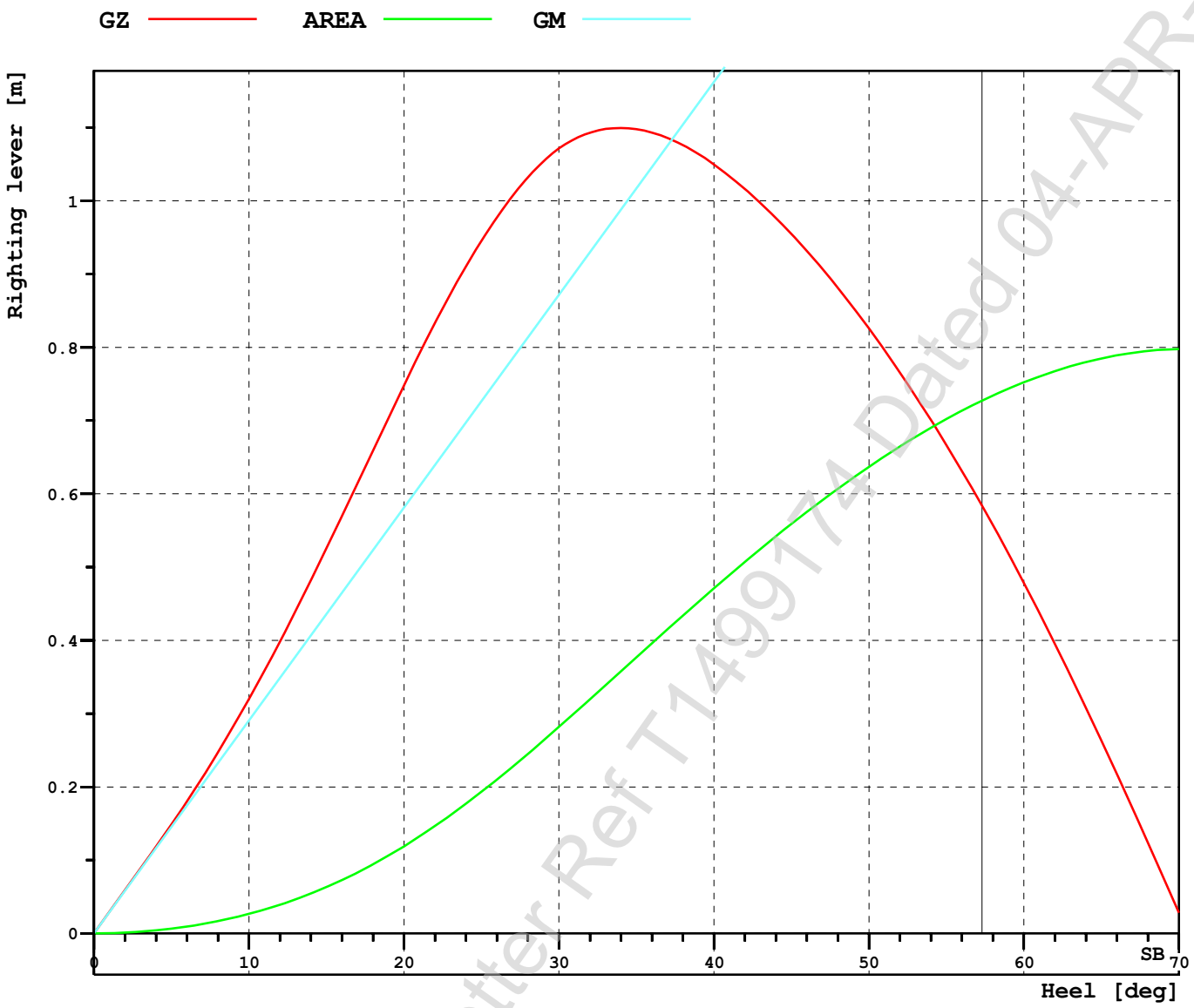
LOADING CONDITION L18, Transition,Fi-Fi - Departure

HEEL degree	MS m	HPHI m	EPHI mrad	FSMOM tm	DGZ m	GMSIN m	KN m
0.0	0.000	0.00	0.000	0.0	0.000	0.000	-0.002
1.0	0.000	0.03	0.000	57.3	0.013	0.043	0.156
3.0	0.001	0.09	0.002	172.0	0.040	0.129	0.471
5.0	0.002	0.15	0.006	286.9	0.067	0.214	0.786
10.0	0.012	0.32	0.027	506.6	0.119	0.427	1.576
15.0	0.032	0.53	0.063	608.8	0.143	0.637	2.364
20.0	0.063	0.75	0.119	664.2	0.156	0.842	3.145
30.0	0.008	1.07	0.282	711.6	0.167	1.231	4.515
40.0	-0.364	1.05	0.471	714.2	0.168	1.582	5.430
50.0	-0.898	0.83	0.637	686.5	0.162	1.885	6.007
60.0	-1.503	0.48	0.752	636.6	0.150	2.131	6.304
70.0	-2.150	0.03	0.798	568.8	0.134	2.313	6.321

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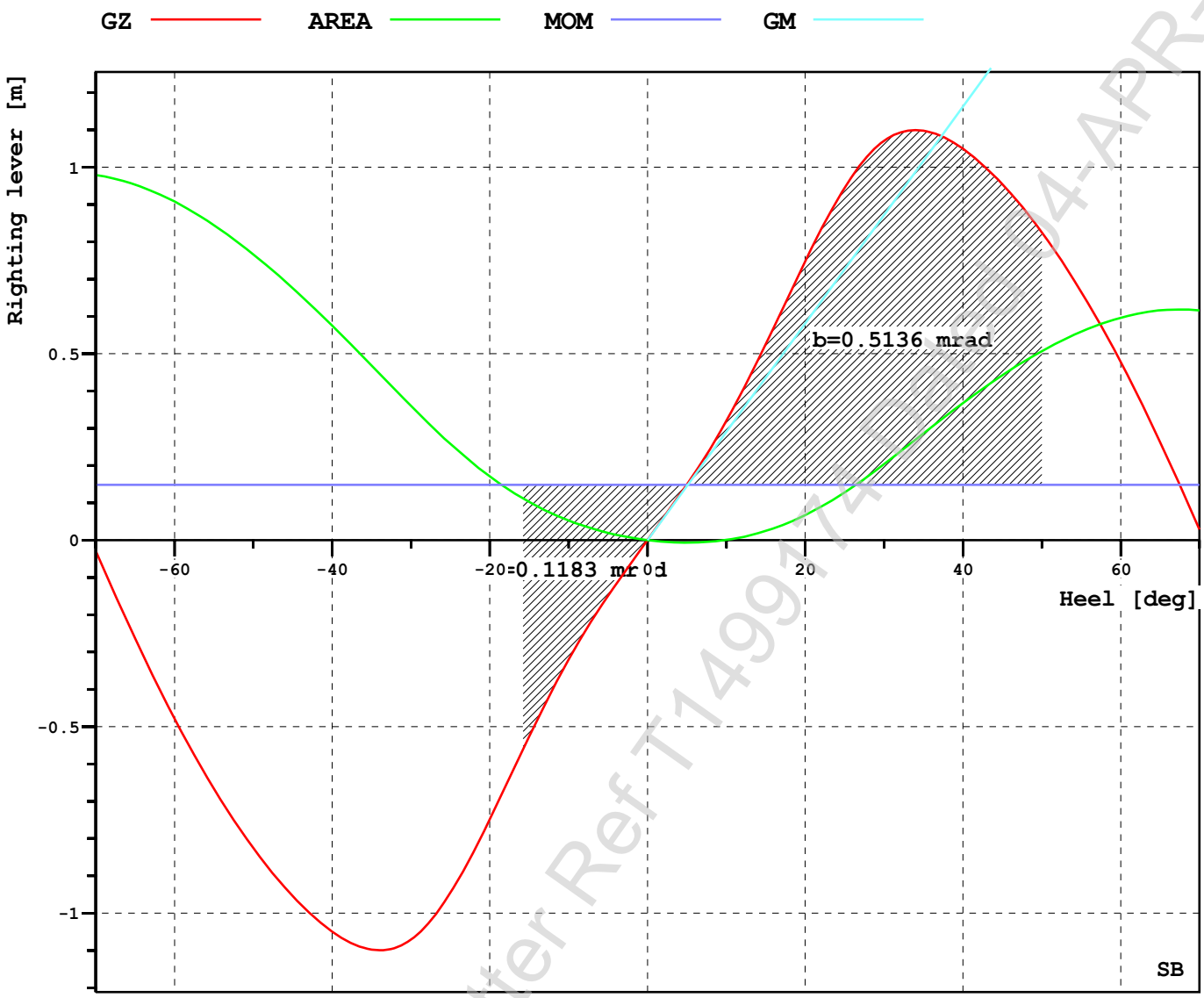
STABILITY CHECK PLOT



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IMOWEATHER CHECK PLOT



INTACT STABILITY CRITERIA

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM
V.AREA15-30	Area depending on the position of the Max. .	0.055	0.282	mrad	OK	0.070
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.189	mrad	OK	0.070
V.GZ0.2	Min. GZ > 0.2	0.200	1.099	m	OK	-0.078
V.POSMAX15	Max. GZ at an angle > 15 deg.	15.000	33.953	deg	OK	-0.574
V.GM0.15	GM > 0.15 m	0.150	1.665	m	OK	0.150
IMOWEATHER	IMO weather criterion	1.000	4.341		OK	0.102
2008IS-A2.3.1.2	Heeling angle due to steady wind <16 or <=8.	16.000	3.348	deg	OK	-0.043
ABSFIFI1	Angle of heel < 1/2 freeboard immersion	10.330	0.672	deg	OK	-0.080
ABSFIFI2	Area of residual stability >0.09 mrad	0.090	0.457	mrad	OK	0.088
TOTAL						0.150

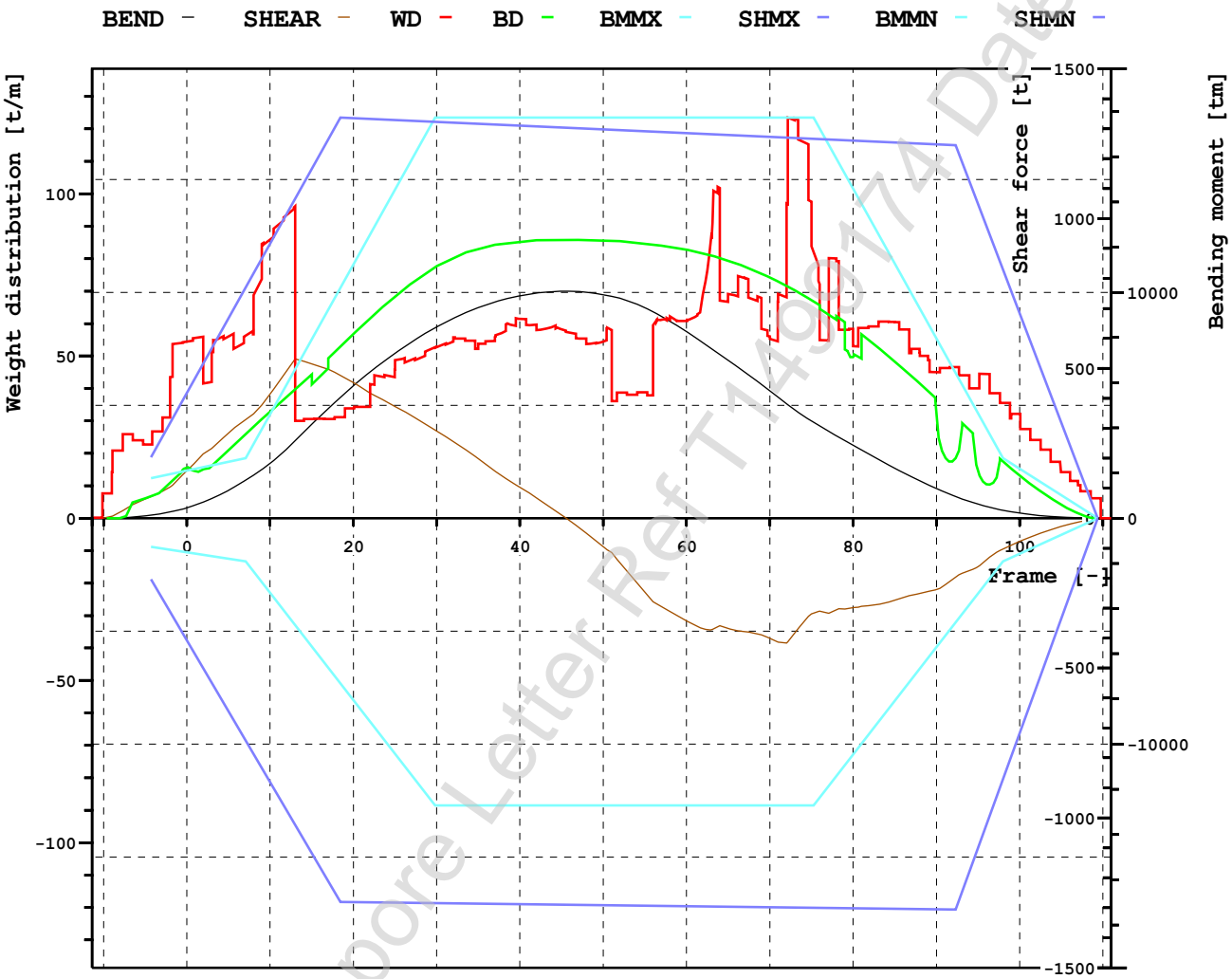
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MOST CRITICAL OPENINGS

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
V-424G	SUPPLY AIR ENGINE ROOM PS	UNPROTECTED	54.243	-3.413	14.288	-	9.625
V-425G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	54.243	3.767	14.288	-	9.625
V-421G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	55.890	2.093	14.288	-	9.626
D-651	ACCOMMODATION/C-DECK	UNPROTECTED	56.000	3.771	16.906	-	12.244
V-421E	EXHAUST AIR ENGINE ROOM PS	UNPROTECTED	55.993	-2.538	17.088	-	12.426
V-423E	EXHAUST AIR ENGINE ROOM SB	UNPROTECTED	55.993	3.423	17.088	-	12.426

STRENGTH DIAGRAM



STRENGTH SUMMARY

LOADING CONDITION L18, Transition,Fi-Fi - Departure

			X	Frame
Shear force (min)	-416.2 t	(32.0%)	50.400 m	72.00
Shear force (max)	531.6 t	(49.9%)	9.100 m	13.00
Sagging moment	-0.3 tm	(0.0%)	-7.104 m	-10.15
Hogging moment	10046.6 tm	(56.6%)	31.858 m	45.51

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STRENGTH DISTRIBUTION

LOADING CONDITION L18, Transition,Fi-Fi - Departure

X	FR	BEND	BMMN	BMMX	BMREL	SHEAR	SHMN	SHMX	SHREL	WD	BD
m	#	tm	tm	tm	%	t	t	t	%	t/m	t/m
-3.500	-5.00	102	0	0	0.0	64	0	0	0.0	22.72	6.12
0.000	0.00	458	-1510	2108	21.7	159	-407	418	38.0	54.41	15.59
3.500	5.00	1237	-1790	2498	49.5	284	-644	667	42.6	56.00	20.94
7.000	10.00	2437	-3292	4594	53.1	415	-881	916	45.3	85.95	32.81
10.500	15.00	4157	-5668	7911	52.6	515	-1117	1166	44.2	30.53	44.36
14.000	20.00	5862	-8045	11228	52.2	452	-1281	1335	33.9	33.82	56.78
17.500	25.00	7301	-10422	14545	50.2	372	-1283	1329	28.0	48.77	68.25
21.000	30.00	8465	-12709	17736	47.7	291	-1284	1323	22.0	52.71	77.65
24.500	35.00	9324	-12709	17736	52.6	199	-1286	1317	15.1	53.74	82.91
28.000	40.00	9843	-12709	17736	55.5	103	-1288	1310	7.9	61.45	85.11
31.500	45.00	10044	-12709	17736	56.6	11	-1290	1304	0.8	58.18	85.73
35.000	50.00	9900	-12709	17736	55.8	-95	-1292	1298	7.4	54.40	85.55
38.500	55.00	9314	-12709	17736	52.5	-247	-1293	1292	19.1	38.79	84.58
42.000	60.00	8260	-12709	17736	46.6	-342	-1295	1285	26.4	60.78	82.72
45.500	65.00	6989	-12709	17736	39.4	-368	-1297	1279	28.4	69.11	79.27
49.000	70.00	5650	-12709	17736	31.9	-401	-1299	1273	30.9	56.00	74.20
52.500	75.00	4304	-12709	17736	24.3	-320	-1300	1267	24.6	97.61	67.26
56.000	80.00	3222	-10463	14601	22.1	-298	-1302	1260	22.9	58.33	49.70
59.500	85.00	2210	-8086	11284	19.6	-274	-1304	1254	21.0	60.48	48.29
63.000	90.00	1316	-5709	7968	16.5	-238	-1306	1248	18.3	45.02	32.67
66.500	95.00	624	-3332	4651	13.4	-161	-1102	1050	14.6	40.05	14.63
70.000	100.00	224	-1571	2193	10.2	-77	-718	685	10.8	32.12	13.15
73.500	105.00	43	-733	1023	4.2	-28	-335	319	8.3	14.17	4.15

See ABS Singapore Letter Ref T1499174 Dated 04-MAR-2016

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DAMAGE STABILITY OF LOADING CONDITION L18

CRITICAL DAMAGE STABILITY CRITERIA FOR EACH DAMAGE CASE

RCR	DAM	STAGE	PHASE	REQ	ATTV	UNIT	STAT	MINGM
V.RANGE.A469	D01	1	EQ	20.000	70.000	deg	OK	-0.361
V.RANGE.A469	D02	1	EQ	20.000	69.540	deg	OK	-0.262
V.RANGE.A469	D03	1	EQ	20.000	69.530	deg	OK	-0.270
V.RANGE.A469	D04	1	EQ	20.000	67.187	deg	OK	-0.016
V.RANGE.A469	D05	1	EQ	20.000	66.960	deg	OK	0.015
V.RANGE.A469	D06	1	EQ	20.000	67.240	deg	OK	0.001
V.MAXHEEL.A469	D07	1	EQ	17.000	2.970	deg	OK	0.034
V.MAXHEEL.A469	D08	1	EQ	17.000	4.805	deg	OK	-0.001
V.MAXHEEL.A469	D09	1	EQ	17.000	4.814	deg	OK	-0.001
V.RANGE.A469	D10	1	EQ	20.000	68.663	deg	OK	-0.132
V.RANGE.A469	D11	1	EQ	20.000	67.508	deg	OK	-0.078
V.RANGE.A469	D12	1	EQ	20.000	67.398	deg	OK	-0.055
V.RANGE.A469	D13	1	EQ	20.000	67.385	deg	OK	-0.061
V.RANGE.A469	D14	1	EQ	20.000	66.886	deg	OK	-0.137
V.RANGE.A469	D15	1	EQ	20.000	66.876	deg	OK	-0.145
V.RANGE.A469	D16	1	EQ	20.000	68.151	deg	OK	-0.086
V.RANGE.A469	D17	1	EQ	20.000	68.160	deg	OK	-0.079
V.RANGE.A469	D18	1	EQ	20.000	68.611	deg	OK	-0.238
V.RANGE.A469	D19	1	EQ	20.000	68.621	deg	OK	-0.231
V.RANGE.A469	D20	1	EQ	20.000	69.960	deg	OK	-0.310
V.RANGE.A469	D21	1	EQ	20.000	69.950	deg	OK	-0.315
TOTAL								0.034

DETERMINATION OF CRITICAL DAMAGE CASE

CRITICAL DAMAGE CASE : D07
STAGE OF CRITICAL DAMAGE CASE : 1
PHASE OF CRITICAL DAMAGE CASE : EQ

DAMAGE STABILITY CRITERIA - CRITICAL CASE

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM m
V.PROGR.A469	No progr. flooding	24.401	2.970	deg	OK	-0.213
V.MAXHEEL.A469	Maximum heel	17.000	2.970	deg	OK	0.034
V.RANGE.A469	Range of pos. stab	20.000	67.030	deg	OK	0.031
V.MINGZ.A469	Min. righting lever	0.100	0.722	m	OK	-0.021
TOTAL						0.034

DAMAGE CASE RESULTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	T m	TR m	HEEL degree	RESFLD m	OPEN	CROSSTM s
D07	1	EQ	PS	4.726	0.126	-3.0	3.41	D-351	-

DAMAGED COMPARTMENTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	NAME	DES	PERM	VOL m3	W t	XCG m	YCG m	ZCG m
D07	INTACT	EQ	-				0.0	0.000			
D07	1	EQ	PS	T010	WING TANK PS	0.95	36.6	37.491	49.42	-7.56	3.22
D07	1	EQ	PS	T273	FO SERVICE TANK PS	0.95	5.3	5.460	52.07	-6.17	4.00
D07	1	EQ	PS	T252	BLACK WATER TANK	0.95	0.0	0.000	-	-	-
D07	1	EQ	PS	T254	GREY WATER TANK	0.95	20.4	20.942	54.62	-6.27	3.55
D07	1	EQ	PS	T318	CD VOID PS	0.95	5.3	5.458	56.35	-5.77	3.54

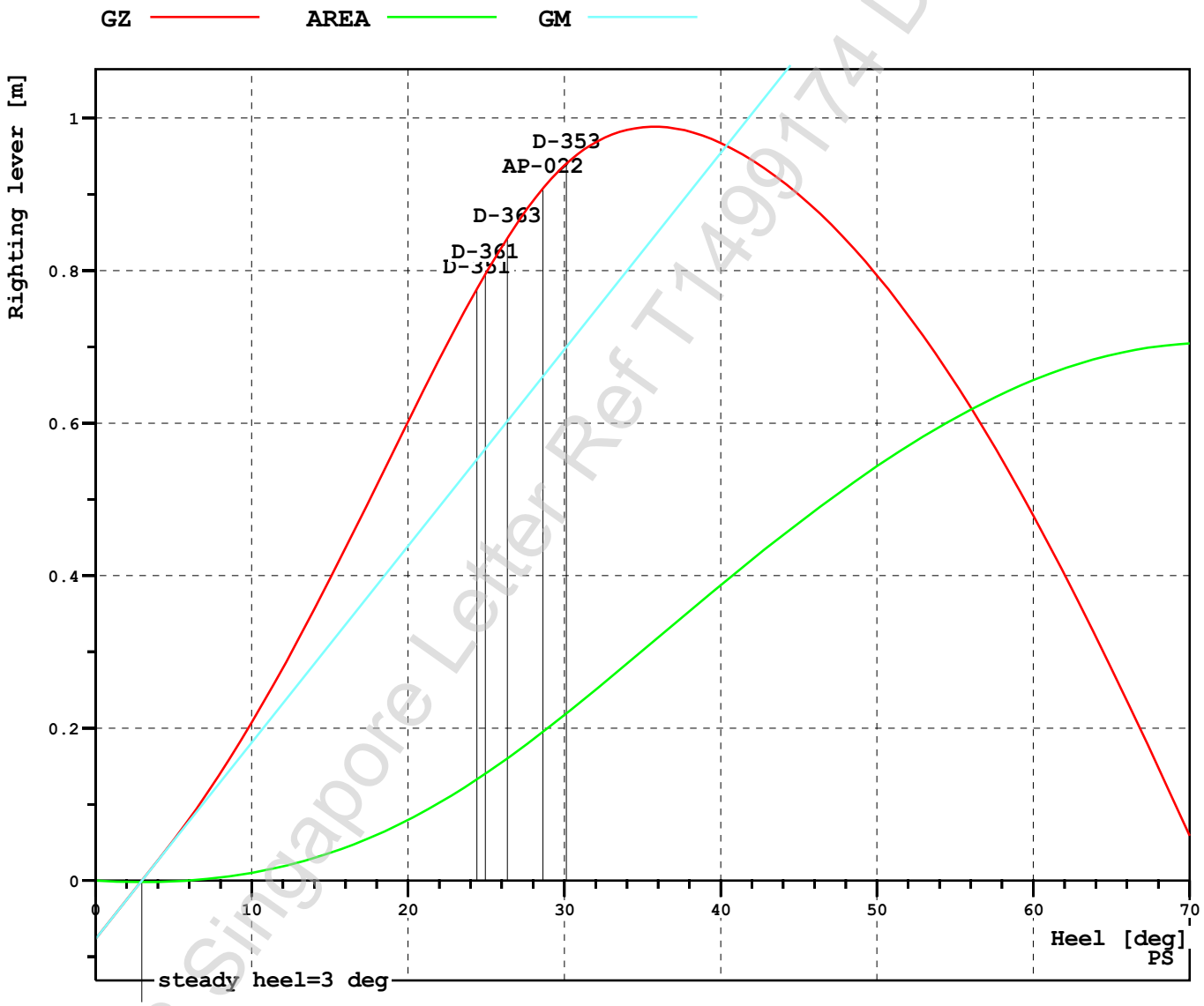
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CRITICAL OPENINGS - CRITICAL CASE

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
D-351	CO2 ROOM/M-DECK	WEATHERTIGHT	51.792	-8.360	8.600	24.4	3.405
D-361	PS CARGORAIL #57/M-DECK	WEATHERTIGHT	39.885	-8.560	8.600	24.9	3.414
D-363	PS CARGORAIL #37/M-DECK	WEATHERTIGHT	25.900	-8.550	8.600	26.3	3.436
AP-022	WING TANK PS	WEATHERTIGHT	19.200	-8.702	8.936	28.6	3.775
D-353	DECK WORKSHOP/M-DECK	WEATHERTIGHT	53.192	-6.550	8.600	30.1	3.497
AP-263	HYDRAULIC OIL STORE TANK	WEATHERTIGHT	47.690	-7.260	8.912	30.2	3.780
AP-012	DB TANK PS	WEATHERTIGHT	50.000	-7.250	9.047	30.8	3.912
AP-014	WING TANK PS	WEATHERTIGHT	43.078	-7.815	9.372	31.5	4.218

DAMAGE STABILITY CHECK PLOT - CRITICAL CASE



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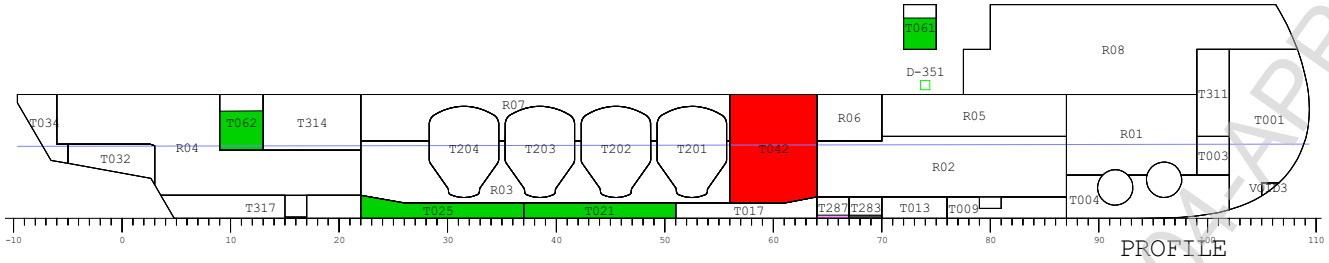
GZ-CURVE - CRITICAL CASE

HEEL	DISP	T	TR	GZ	MS	AREA
degree	t	m	m	m	m	mrad
0.0	4250.3	4.727	0.081	0.077	0.077	0.000
-1.0	4250.3	4.729	0.092	0.051	0.094	-0.001
-3.0	4250.3	4.726	0.127	-0.001	0.128	-0.002
-5.0	4250.3	4.715	0.176	-0.053	0.161	-0.001
-10.0	4250.3	4.648	0.357	-0.207	0.220	0.010
-15.0	4250.3	4.529	0.601	-0.396	0.241	0.036
-20.0	4250.3	4.359	0.877	-0.602	0.240	0.080
-30.0	4250.3	3.882	1.327	-0.938	0.293	0.217
-40.0	4250.3	3.255	1.374	-0.967	0.615	0.388
-50.0	4250.3	2.479	0.955	-0.794	1.091	0.543
-60.0	4250.3	1.605	0.346	-0.479	1.652	0.656
-70.0	4250.3	0.695	-0.309	-0.059	2.254	0.704

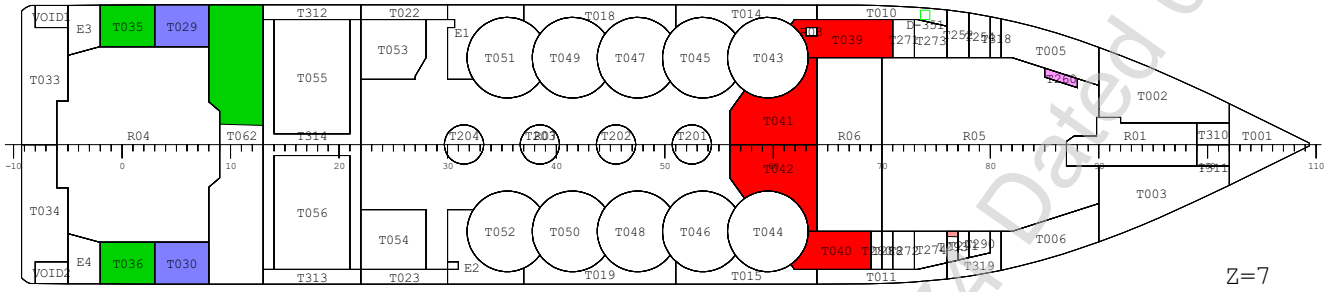
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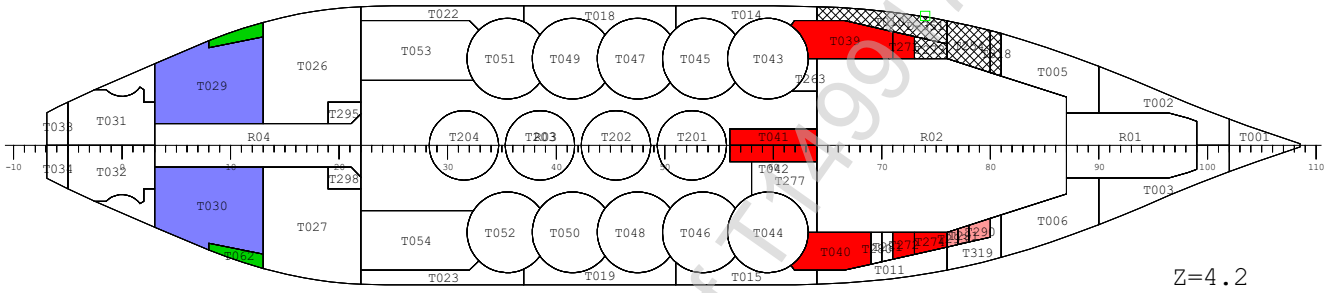
DAMAGE ARRANGEMENT - CRITICAL CASE



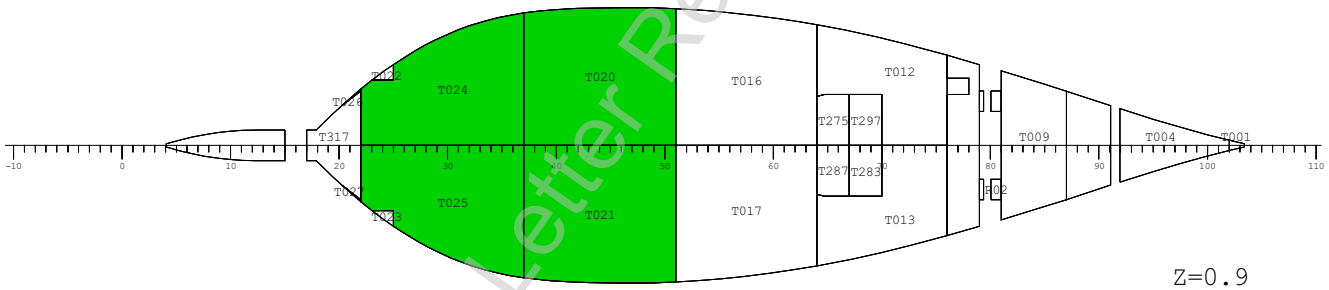
PROFILE



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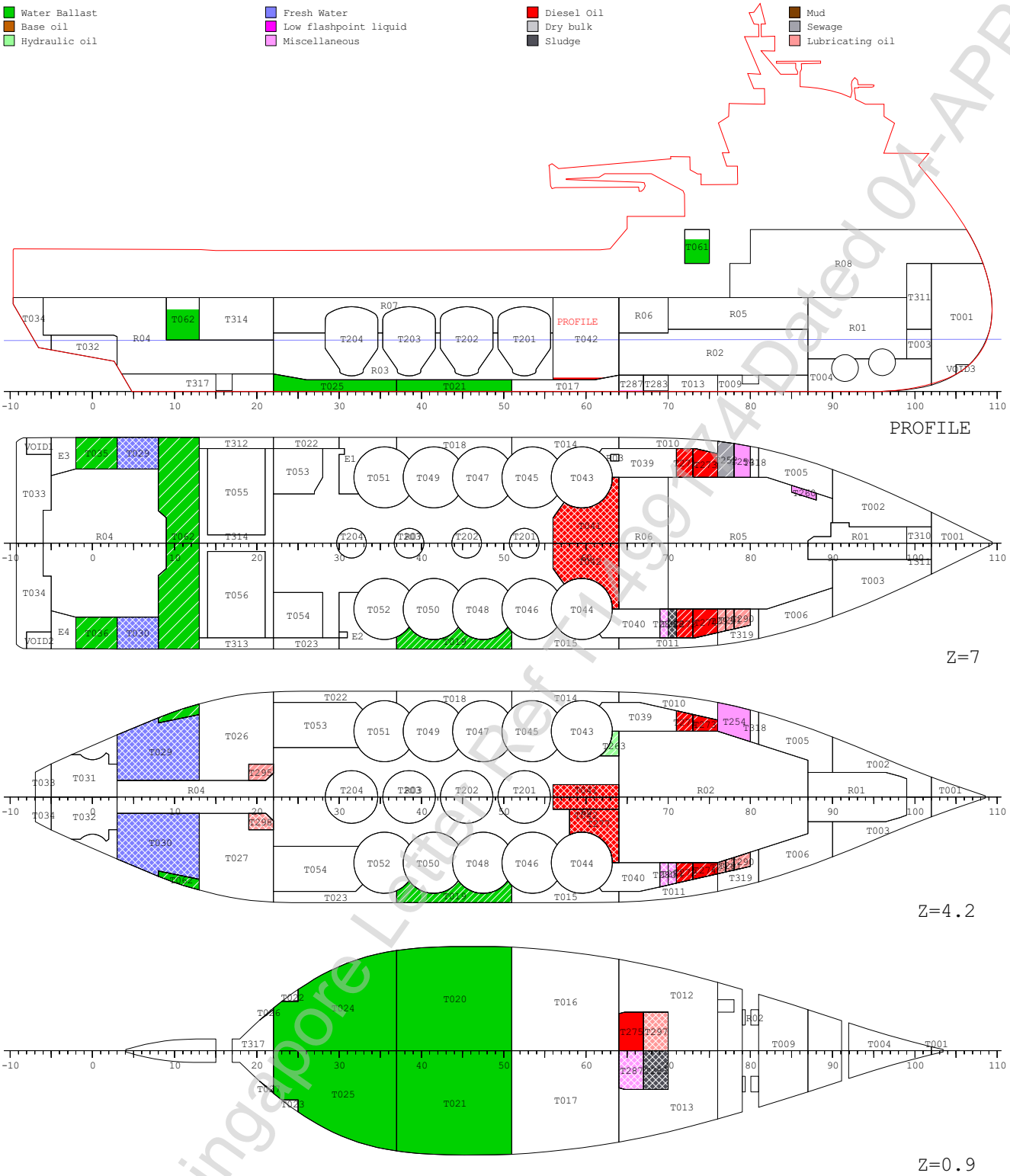


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12.19 L19 -> Transition,Fi-Fi - Arrival



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LIST OF LOADS

NAME	DES	LOAD	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm	FRS	HM m
CONTENTS=Diesel Oil (RHO=0.86)										
T041	STORAGE TANK PS	FO	0.7	1.0	41.24	-0.52	1.10	27	MAX	1.20
T042	STORAGE TANK SB	FO	0.7	1.0	41.24	0.52	1.10	27	MAX	1.20
T271	FO SETTLING TANK PS	FO	8.5	70.0	50.38	-6.53	4.93	1	REAL	6.72
T272	FO SETTLING TANK SB	FO	8.5	70.0	50.38	6.53	4.93	1	REAL	6.72
T273	FO SERVICE TANK PS	FO	10.6	70.0	52.09	-6.38	5.15	2	REAL	6.90
T274	FO SERVICE TANK SB	FO	10.6	70.0	52.09	6.38	5.15	2	REAL	6.90
T275	FO DRAIN TANK	FO	7.9	98.0	45.85	-1.64	0.69	5	REAL	1.37
T277	FO OVERFLOW TANK	FO	3.6	10.0	42.62	2.34	1.31	10	REAL	1.53
T279	EMERG. GEN. FO TANK	FO	1.9	98.0	61.60	-1.38	24.57	0	REAL	25.80
SUBTOTAL		FO	53.0		50.02	-0.13	4.73	75		
CONTENTS=Fresh Water (RHO=1)										
T029	WING TANK PS	FW	7.5	9.8	7.28	-2.50	2.65	83	MAX	3.01
T030	WING TANK SB	FW	7.5	9.8	7.28	2.50	2.65	83	MAX	3.01
SUBTOTAL		FW	14.9		7.28	0.00	2.65	165		
CONTENTS=Hydraulic oil (RHO=0.924)										
T263	HYDRAULIC OIL STORE TANK	HO	1.9	40.0	44.29	-4.33	1.89	1	REAL	2.47
CONTENTS=Lubricating oil (RHO=0.924)										
T290	LO STORE TANK MAIN ENG.	LO	3.4	40.0	55.32	5.47	4.05	0	REAL	5.16
T291	LO STORE TANK MAIN ENG.	LO	1.8	40.0	54.25	5.81	4.26	0	REAL	5.48
T293	LO STORE TANK THRUSTER	LO	1.8	40.0	53.55	6.03	4.41	0	REAL	5.66
T295	LO STORE TANK AZIMUTH THR.	LO	2.3	40.0	14.35	-2.10	2.63	0	REAL	3.06
T297	LO DRAIN TANK	LO	0.9	10.0	47.95	-1.64	0.07	6	REAL	0.14
T298	DIRTY LO TANK	LO	0.6	10.0	14.35	2.10	2.31	0	REAL	2.41
SUBTOTAL		LO	10.7		43.36	3.26	3.43	7		
CONTENTS=Mass load (RHO=1)										
M1	CREW	MASS	2.6	0.0	57.40	0.00	15.50	0		0.00
M2	PROVISION	MASS	0.8	0.0	56.35	-6.50	12.40	0		0.00
M3	STORES	MASS	0.8	0.0	72.10	0.00	12.40	0		0.00
M4	DECK LOAD 10T/M2 AFT OF #35	MASS	200.0	0.0	9.10	0.00	9.00	0		0.00
M5	DECK LOAD 5T/M2 FWD OF #35	MASS	200.0	0.0	37.26	0.00	9.00	0		0.00
SUBTOTAL		MASS	404.2		23.56	-0.01	9.06	0		
CONTENTS=Miscellaneous (RHO=1)										
T280	BILGE WATER HOLDING TANK	MIS	0.9	10.0	48.64	6.17	2.03	0	REAL	2.57
T281	BILGE WATER SETTLING TANK	MIS	0.4	10.0	49.34	6.04	1.76	0	REAL	2.08
T287	BILGE WATER DRAIN TANK	MIS	0.9	10.0	45.85	1.64	0.07	6	REAL	0.14
T254	GREY WATER TANK	MIS	31.6	98.0	54.82	-6.47	4.52	3	REAL	7.84
T260	WATER MIST TANK	MIS	2.3	98.0	60.55	-4.30	6.28	0	REAL	7.27
SUBTOTAL		MIS	36.2		54.74	-5.68	4.43	9		
CONTENTS=Sewage (RHO=1)										
T252	BLACK WATER TANK	SEW	10.4	97.9	53.89	-7.06	6.66	4	REAL	7.95
CONTENTS=Sludge (RHO=1)										
T283	SLUDGE DRAIN TANK	SLU	0.9	10.0	47.95	1.64	0.07	6	REAL	0.14
T288	SLUDGE SETTLING TANK	SLU	0.4	10.0	49.35	6.75	5.44	1	REAL	5.59
SUBTOTAL		SLU	1.4		48.40	3.29	1.80	7		
CONTENTS=Water Ballast (RHO=1.025)										
T019	WING TANK SB	WB	35.0	47.1	30.81	8.37	3.00	2	MAX	4.54
T020	DB TANK PS	WB	96.7	100.0	30.84	-4.63	0.61	0	(REAL)	1.92
T021	DB TANK SB	WB	96.7	100.0	30.84	4.63	0.61	0	(REAL)	1.92
T024	DB TANK PS	WB	75.9	100.0	21.46	-3.85	0.81	0	(REAL)	2.75
T025	DB TANK SB	WB	75.9	100.0	21.46	3.85	0.81	0	(REAL)	2.75
T061	ROLL RED. TANK	WB	89.9	70.0	51.66	0.00	11.91	1358	REAL	12.93
T062	ROLL RED. TANK	WB	147.4	70.0	7.45	0.00	5.67	1736	REAL	6.93
T035	WING TANK PS	WB	23.2	98.0	0.39	-7.50	6.66	6	REAL	7.95
T036	WING TANK SB	WB	23.2	98.0	0.39	7.50	6.66	6	REAL	7.95
SUBTOTAL		WB	663.9		24.19	0.44	3.86	3109		
TOTAL			1196.6		26.32	0.03	5.67	3376		

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LOAD SUMMARY TABLE

ITEM	MASS t	LCG m	TCG m	VCG m	FRSM tm
Diesel Oil	53.0	50.02	-0.13	4.73	74.98
Fresh Water	14.9	7.28	0.00	2.65	165.03
Hydraulic oil	1.9	44.29	-4.33	1.89	0.61
Lubricating oil	10.7	43.36	3.26	3.43	7.30
Mass load	404.2	23.56	-0.01	9.06	0.00
Miscellaneous	36.2	54.74	-5.68	4.43	9.39
Sewage	10.4	53.89	-7.06	6.66	3.55
Sludge	1.4	48.40	3.29	1.80	6.99
Water Ballast	663.9	24.19	0.44	3.86	3108.51
DEADWEIGHT	1196.6	26.32	0.03	5.67	3376.37
Lightweight	2704.9	39.15	-0.01	7.08	
Deadweight	1196.6	26.32	0.03	5.67	
Total weight	3901.5	35.22	0.00	6.65	

F L O A T I N G P O S I T I O N

Draught moulded	4.397	m	KM	9.10 m
Trim	0.097	m	KG	6.65 m
Heel, SB=+	0.0	deg		
TA	4.348	m	GM0	2.45 m
TF	4.445	m	GMCORR	-0.87 m
Trimming moment	520	tonm	GM	1.59 m

STABILITY COMPONENTS

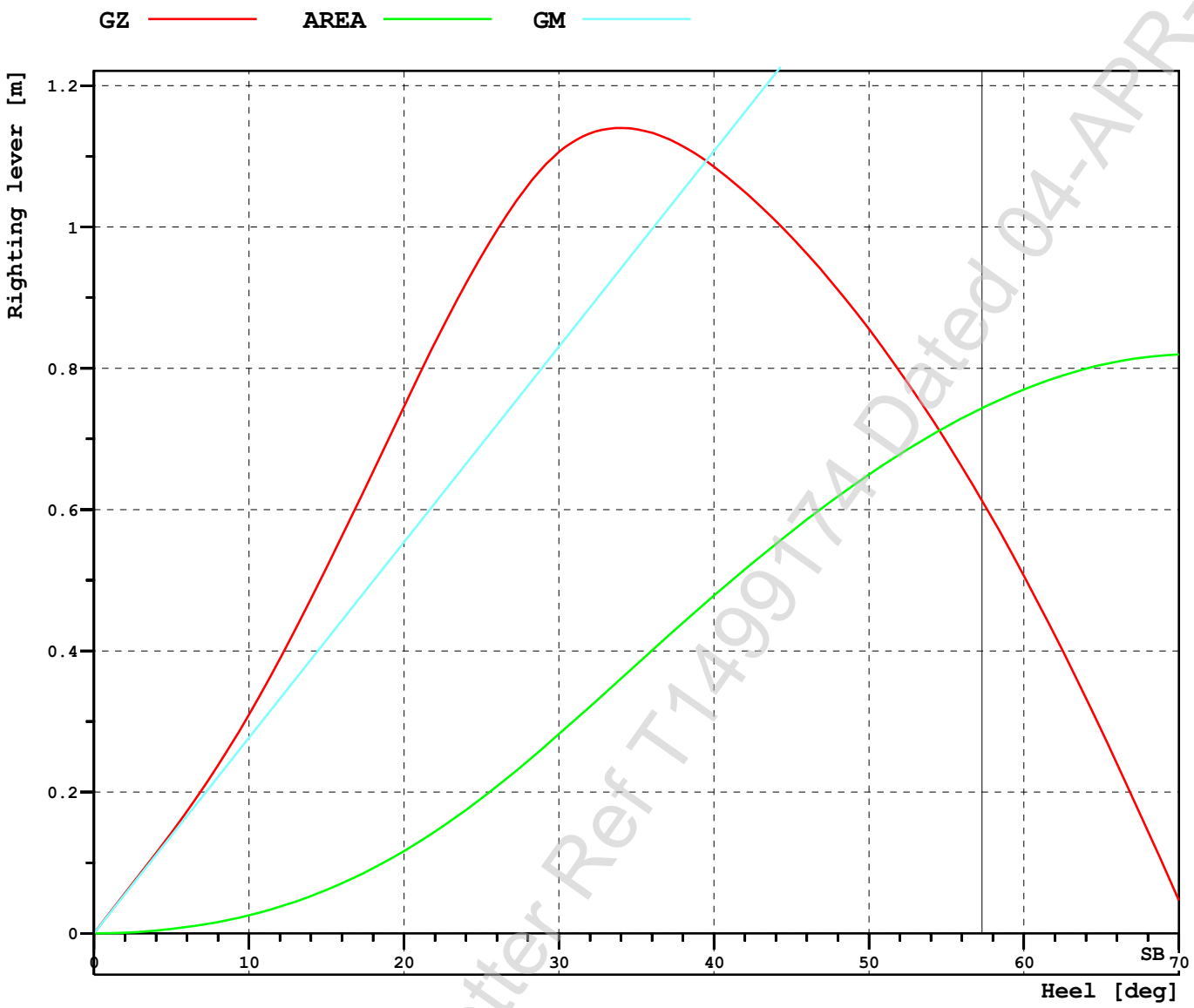
LOADING CONDITION L19, Transition,Fi-Fi - Arrival

HEEL degree	MS m	HPHI m	EPHI mrad	FSMOM tm	DGZ m	GMSIN m	KN m
0.0	0.000	0.00	0.000	0.0	0.000	0.000	-0.002
1.0	0.001	0.03	0.000	57.2	0.015	0.043	0.157
3.0	0.001	0.09	0.002	171.5	0.044	0.128	0.475
5.0	0.002	0.14	0.006	285.8	0.073	0.214	0.793
10.0	0.013	0.31	0.026	504.5	0.129	0.426	1.592
15.0	0.039	0.52	0.062	606.2	0.155	0.635	2.392
20.0	0.077	0.75	0.117	661.2	0.169	0.839	3.187
30.0	0.062	1.11	0.282	708.1	0.181	1.226	4.610
40.0	-0.308	1.09	0.478	710.4	0.182	1.576	5.539
50.0	-0.848	0.86	0.650	682.4	0.175	1.878	6.122
60.0	-1.455	0.51	0.770	633.1	0.162	2.123	6.425
70.0	-2.112	0.05	0.820	567.8	0.146	2.304	6.439

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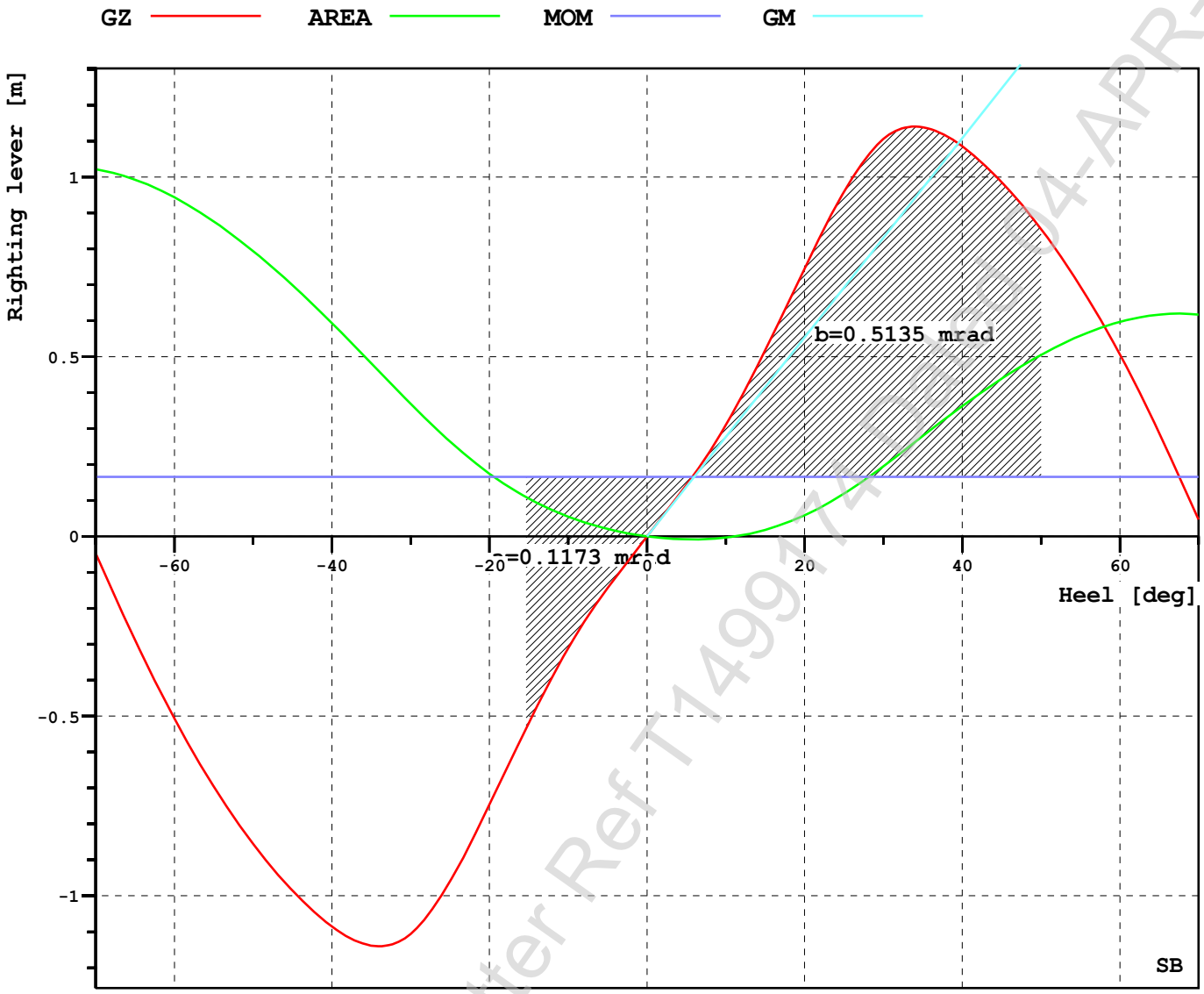
STABILITY CHECK PLOT



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IMOWEATHER CHECK PLOT



INTACT STABILITY CRITERIA

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM
V.AREA15-30	Area depending on the position of the Max. .	0.055	0.282	mrad	OK	-0.049
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.196	mrad	OK	-0.076
V.GZ0.2	Min. GZ > 0.2	0.200	1.140	m	OK	-0.226
V.POSMAX15	Max. GZ at an angle > 15 deg.	15.000	33.987	deg	OK	-0.684
V.GM0.15	GM > 0.15 m	0.150	1.586	m	OK	0.150
IMOWEATHER	IMO weather criterion	1.000	4.376		OK	0.046
2008IS-A2.3.1.2	Heeling angle due to steady wind <16 or <=8.	16.000	3.885	deg	OK	-0.054
ABSFIFI1	Angle of heel < 1/2 freeboard immersion	11.098	0.757	deg	OK	-0.132
ABSFIFI2	Area of residual stability >0.09 mrad	0.090	0.463	mrad	OK	-0.020
TOTAL						0.150

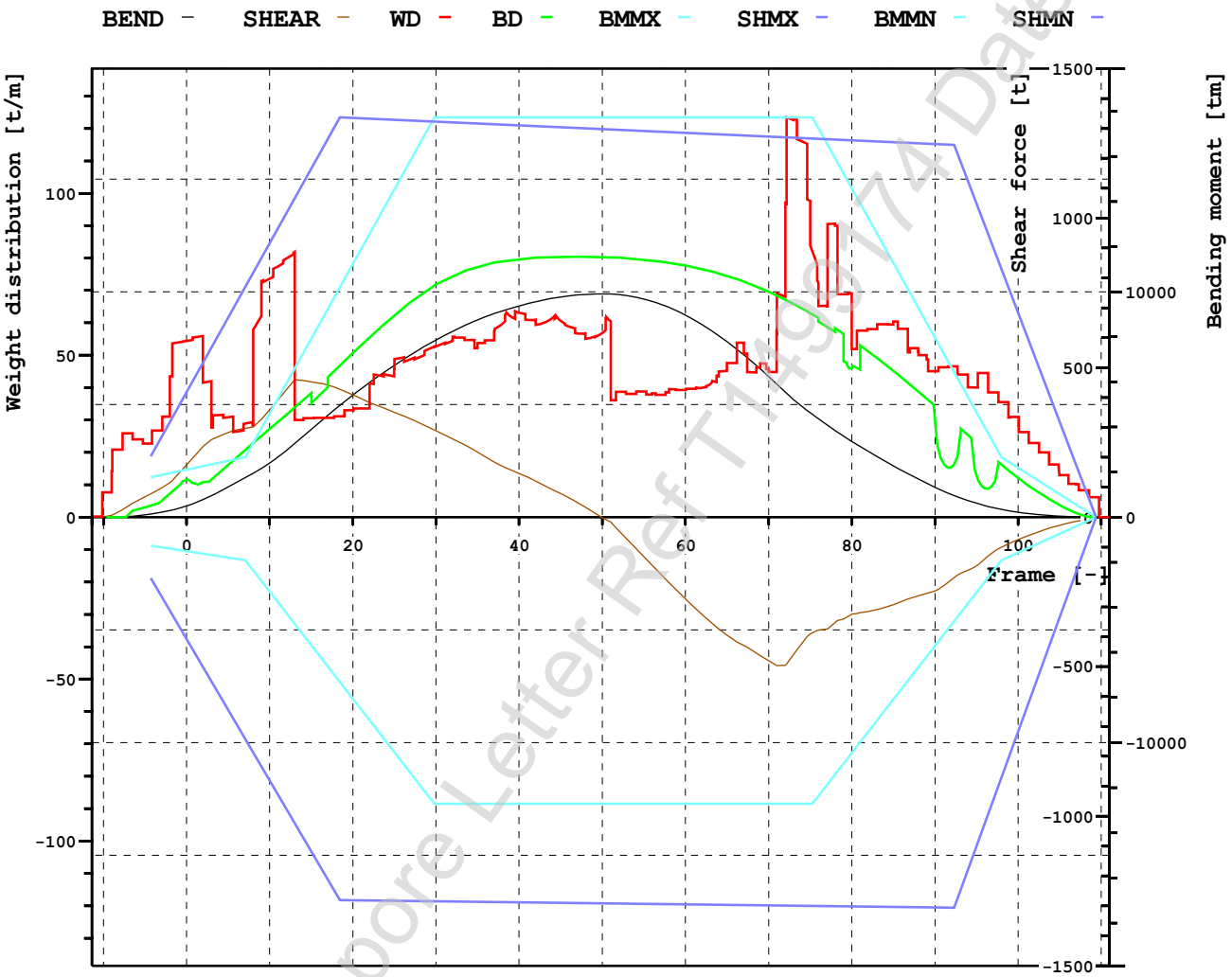
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MOST CRITICAL OPENINGS

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
V-421G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	55.890	2.093	14.288	-	9.868
V-424G	SUPPLY AIR ENGINE ROOM PS	UNPROTECTED	54.243	-3.413	14.288	-	9.870
V-425G	SUPPLY AIR ENGINE ROOM SB	UNPROTECTED	54.243	3.767	14.288	-	9.870
D-651	ACCOMMODATION/C-DECK	UNPROTECTED	56.000	3.771	16.906	-	12.486
V-421E	EXHAUST AIR ENGINE ROOM PS	UNPROTECTED	55.993	-2.538	17.088	-	12.668
V-423E	EXHAUST AIR ENGINE ROOM SB	UNPROTECTED	55.993	3.423	17.088	-	12.668

STRENGTH DIAGRAM



STRENGTH SUMMARY

LOADING CONDITION L19, Transition,Fi-Fi - Arrival

			X	Frame
Shear force (min)	-496.3 t	(38.2%)	49.700 m	71.00
Shear force (max)	460.0 t	(43.2%)	9.100 m	13.00
Sagging moment	-0.7 tm	(0.0%)	-7.104 m	-10.15
Hogging moment	9901.7 tm	(55.8%)	34.850 m	49.79

See ABS Singapore Letter Ref T1499174 Dated 04-APR-2016

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STRENGTH DISTRIBUTION

LOADING CONDITION L19, Transition,Fi-Fi - Arrival

X	FR	BEND	BMMN	BMMX	BMREL	SHEAR	SHMN	SHMX	SHREL	WD	BD
m	#	tm	tm	tm	%	t	t	t	%	t/m	t/m
-3.500	-5.00	103	0	0	0.0	68	0	0	0.0	22.72	3.00
0.000	0.00	494	-1510	2108	23.4	175	-407	418	42.0	54.41	11.58
3.500	5.00	1332	-1790	2498	53.3	283	-644	667	42.4	30.80	16.08
7.000	10.00	2409	-3292	4594	52.6	360	-881	916	39.2	73.62	27.24
10.500	15.00	3901	-5668	7911	49.3	452	-1117	1166	38.8	30.53	38.36
14.000	20.00	5420	-8045	11228	48.3	410	-1281	1335	30.7	33.00	50.68
17.500	25.00	6742	-10422	14545	46.4	350	-1283	1329	26.3	48.77	62.25
21.000	30.00	7862	-12709	17736	44.3	289	-1284	1323	21.8	52.71	71.80
24.500	35.00	8749	-12709	17736	49.3	217	-1286	1317	16.5	53.74	77.19
28.000	40.00	9371	-12709	17736	52.8	146	-1288	1310	11.1	63.13	79.53
31.500	45.00	9764	-12709	17736	55.1	79	-1290	1304	6.1	60.61	80.29
35.000	50.00	9901	-12709	17736	55.8	-2	-1292	1298	0.1	56.83	80.24
38.500	55.00	9678	-12709	17736	54.6	-133	-1293	1292	10.3	38.79	79.42
42.000	60.00	8962	-12709	17736	50.5	-274	-1295	1285	21.1	39.25	77.69
45.500	65.00	7779	-12709	17736	43.9	-395	-1297	1279	30.5	47.63	74.42
49.000	70.00	6244	-12709	17736	35.2	-480	-1299	1273	36.9	45.67	69.59
52.500	75.00	4625	-12709	17736	26.1	-390	-1300	1267	30.0	97.64	63.02
56.000	80.00	3361	-10463	14601	23.0	-325	-1302	1260	24.9	69.10	45.95
59.500	85.00	2270	-8086	11284	20.1	-293	-1304	1254	22.4	60.41	45.16
63.000	90.00	1327	-5709	7968	16.7	-247	-1306	1248	18.9	45.02	30.22
66.500	95.00	617	-3332	4651	13.3	-163	-1102	1050	14.8	40.05	12.90
70.000	100.00	220	-1571	2193	10.0	-75	-718	685	10.4	30.94	12.12
73.500	105.00	44	-733	1023	4.3	-27	-335	319	8.0	12.99	3.71

See ABS Singapore Letter Ref T1499174 Dated 04-APR-2016

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DAMAGE STABILITY OF LOADING CONDITION L19

CRITICAL DAMAGE STABILITY CRITERIA FOR EACH DAMAGE CASE

RCR	DAM	STAGE	PHASE	REQ	ATTV	UNIT	STAT	MINGM
V.RANGE.A469	D01	1	EQ	20.000	69.982	deg	OK	-0.451
V.RANGE.A469	D02	1	EQ	20.000	69.584	deg	OK	-0.356
V.RANGE.A469	D03	1	EQ	20.000	69.547	deg	OK	-0.361
V.MAXHEEL.A469	D04	1	EQ	17.000	2.858	deg	OK	-0.079
V.RANGE.A469	D05	1	EQ	20.000	69.640	deg	OK	-0.219
V.MAXHEEL.A469	D06	1	EQ	17.000	2.817	deg	OK	-0.059
V.RANGE.A469	D07	1	EQ	20.000	69.735	deg	OK	-0.207
V.MAXHEEL.A469	D08	1	EQ	17.000	5.242	deg	OK	-0.029
V.MAXHEEL.A469	D09	1	EQ	17.000	5.275	deg	OK	-0.026
V.RANGE.A469	D10	1	EQ	20.000	67.520	deg	OK	-0.134
V.RANGE.A469	D11	1	EQ	20.000	67.388	deg	OK	-0.169
V.RANGE.A469	D12	1	EQ	20.000	67.293	deg	OK	-0.157
V.RANGE.A469	D13	1	EQ	20.000	67.255	deg	OK	-0.160
V.RANGE.A469	D14	1	EQ	20.000	66.603	deg	OK	-0.247
V.RANGE.A469	D15	1	EQ	20.000	66.568	deg	OK	-0.251
V.RANGE.A469	D16	1	EQ	20.000	69.273	deg	OK	-0.358
V.RANGE.A469	D17	1	EQ	20.000	69.239	deg	OK	-0.361
V.RANGE.A469	D18	1	EQ	20.000	68.418	deg	OK	-0.315
V.RANGE.A469	D19	1	EQ	20.000	68.454	deg	OK	-0.310
V.RANGE.A469	D20	1	EQ	20.000	70.000	deg	OK	-0.417
V.RANGE.A469	D21	1	EQ	20.000	69.968	deg	OK	-0.419
TOTAL								-0.026

DETERMINATION OF CRITICAL DAMAGE CASE

CRITICAL DAMAGE CASE : D09
STAGE OF CRITICAL DAMAGE CASE : 1
PHASE OF CRITICAL DAMAGE CASE : EQ

DAMAGE STABILITY CRITERIA - CRITICAL CASE

RCR	TEXT	REQ	ATTV	UNIT	STAT	MINGM m
V.PROGR.A469	No progr. flooding	26.105	5.275	deg	OK	-0.431
V.MAXHEEL.A469	Maximum heel	17.000	5.275	deg	OK	-0.026
V.RANGE.A469	Range of pos. stab	20.000	64.725	deg	OK	-0.127
V.MINGZ.A469	Min. righting lever	0.100	0.855	m	OK	-0.243
TOTAL						-0.026

DAMAGE CASE RESULTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	T m	TR m	HEEL degree	RESFLD m	OPEN	CROSSTM s
D09	1	EQ	PS	4.477	0.313	-5.3	3.26	D-351	-

DAMAGED COMPARTMENTS - CRITICAL CASE

DAM	STAGE	PHASE	SIDE	NAME	DES	PERM	VOL m3	W t	XCG m	YCG m	ZCG m
D09	INTACT	EQ	-				0.0	0.000			
D09	1	EQ	PS	T016	DB TANK PS	0.95	81.8	83.828	40.20	-4.33	0.62
D09	1	EQ	PS	T014	WING TANK PS	0.95	34.3	35.205	40.15	-8.40	3.37

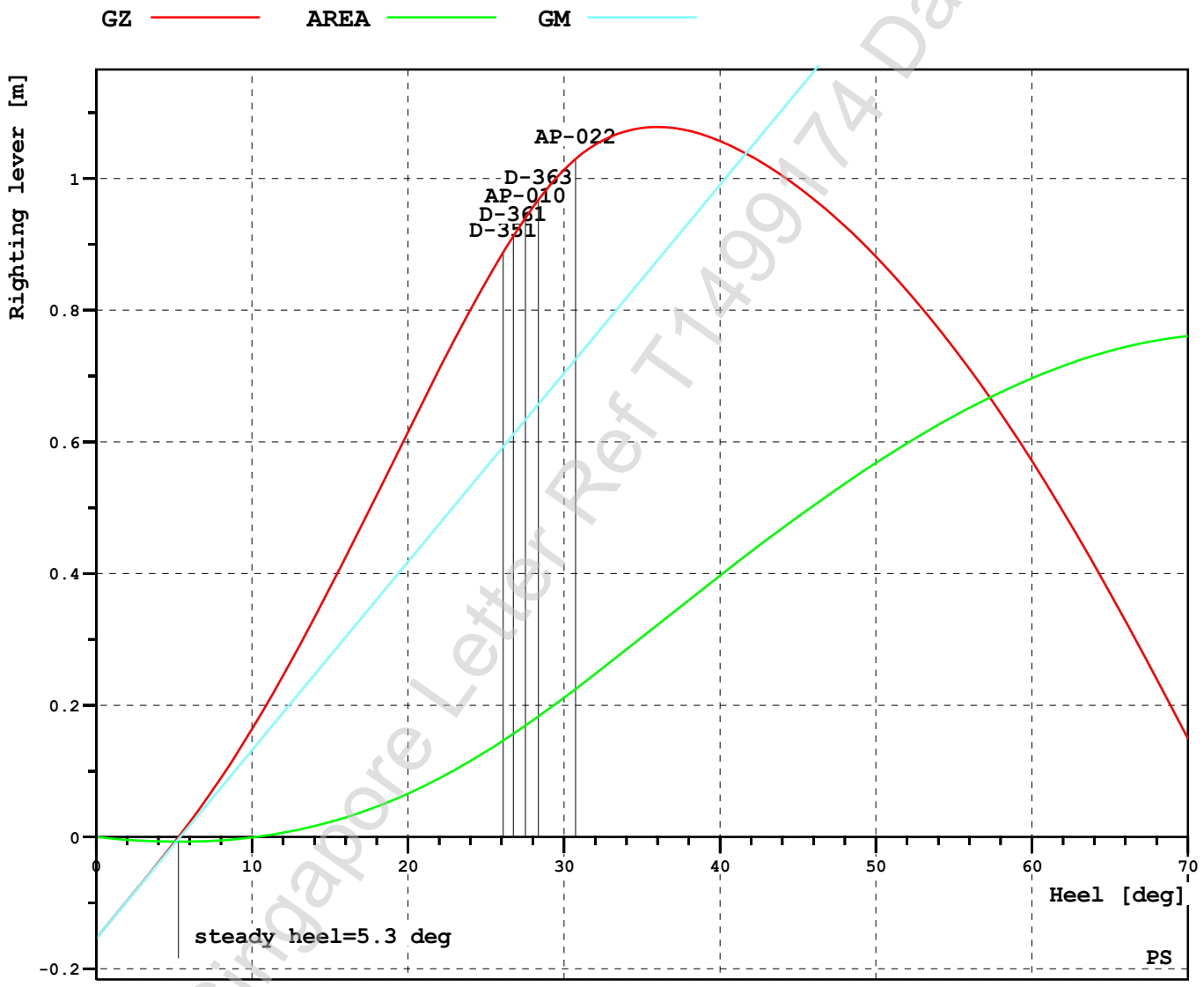
See ABS Singapore Letter Ref T14991 Dated 04-APR-2016

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CRITICAL OPENINGS - CRITICAL CASE

NAME	TEXT	WT	X m	Y m	Z m	IMMA degree	IMMR m
D-351	CO2 ROOM/M-DECK	WEATHERTIGHT	51.792	-8.360	8.600	26.1	3.259
D-361	PS CARGORAIL #57/M-DECK	WEATHERTIGHT	39.885	-8.560	8.600	26.7	3.287
AP-010	WING TANK PS	WEATHERTIGHT	47.211	-8.731	8.953	27.5	3.594
D-363	PS CARGORAIL #37/M-DECK	WEATHERTIGHT	25.900	-8.550	8.600	28.3	3.343
AP-022	WING TANK PS	WEATHERTIGHT	19.200	-8.702	8.936	30.7	3.690
AP-263	HYDRAULIC OIL STORE TANK	WEATHERTIGHT	47.690	-7.260	8.912	32.3	3.686
D-353	DECK WORKSHOP/M-DECK	WEATHERTIGHT	53.192	-6.550	8.600	32.4	3.419
AP-012	DB TANK PS	WEATHERTIGHT	50.000	-7.250	9.047	32.9	3.813

DAMAGE STABILITY CHECK PLOT - CRITICAL CASE



See ABS Singapore Letter Ref T1499174 Dated 04-APR-2016

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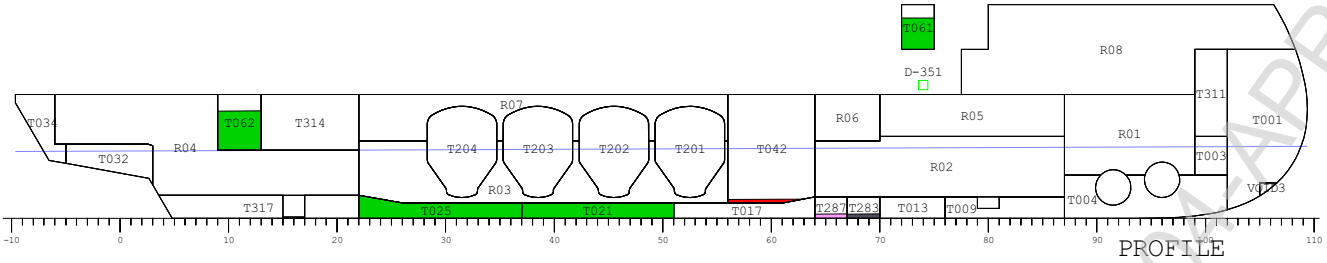
GZ-CURVE - CRITICAL CASE

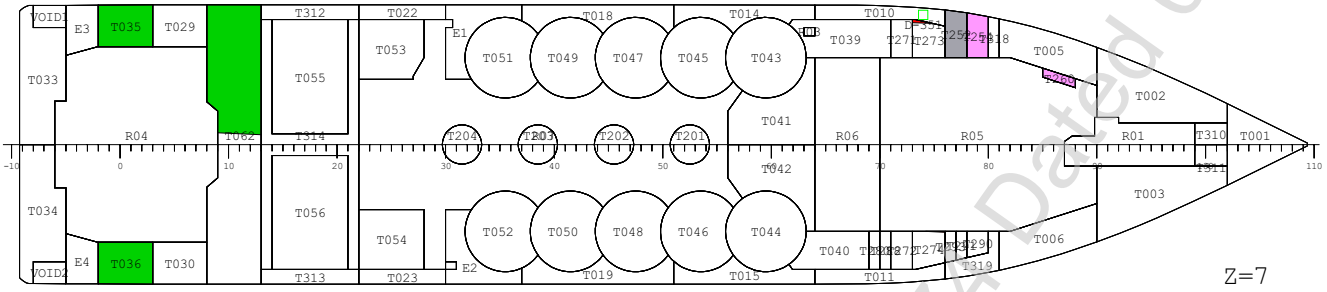
HEEL	DISP	T	TR	GZ	MS	AREA
degree	t	m	m	m	m	mrad
0.0	3901.5	4.500	0.255	0.154	0.154	0.000
-1.0	3901.5	4.500	0.259	0.125	0.168	-0.002
-3.0	3901.5	4.494	0.276	0.067	0.196	-0.006
-5.0	3901.5	4.480	0.307	0.008	0.222	-0.007
-10.0	3901.5	4.407	0.443	-0.164	0.262	-0.001
-15.0	3901.5	4.281	0.656	-0.378	0.256	0.023
-20.0	3901.5	4.102	0.902	-0.615	0.223	0.066
-30.0	3901.5	3.594	1.402	-1.013	0.213	0.212
-40.0	3901.5	2.921	1.594	-1.057	0.519	0.397
-50.0	3901.5	2.104	1.322	-0.881	0.997	0.568
-60.0	3901.5	1.195	0.776	-0.571	1.552	0.697
-70.0	3901.5	0.263	0.137	-0.150	2.154	0.761

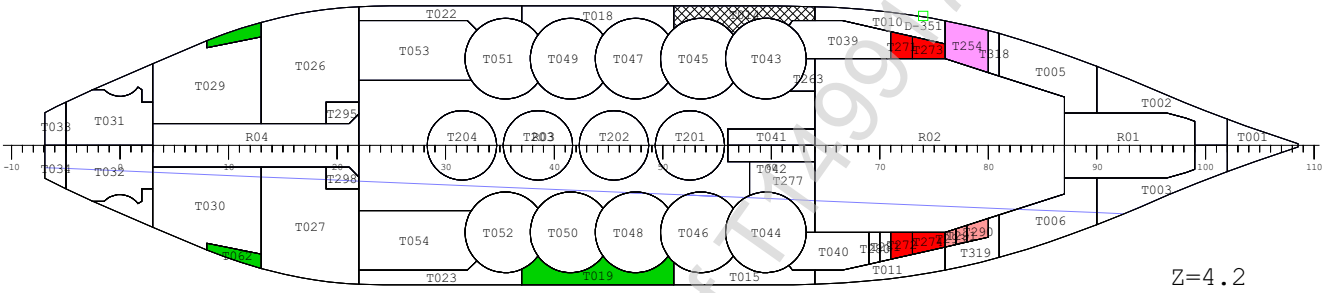
See ABS Singapore Letter Ref T1499174 Dated 04-APR-2016

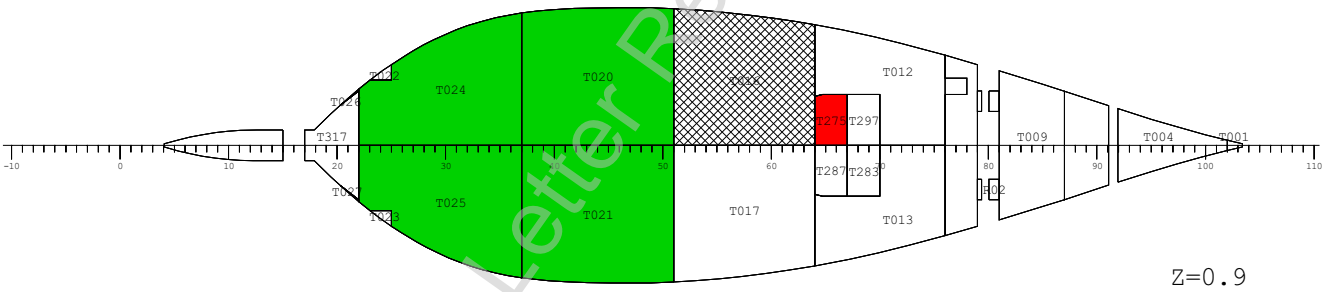
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DAMAGE ARRANGEMENT - CRITICAL CASE









See ABS Singapore Letter Ref No 19974 Dated 04-APR-2016

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13 ATTACHMENTS

13.1 LIGHTSHIP SURVEY REPORT

'

See ABS Singapore Letter Ref T1499174 Dated 04-APR-2016

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Electronically published by ABS Singapore.
Reference T1486145, dated 18-FEB-2016.

FOUNDED 1862

COSCO (GUANGDONG) SHIPYARD CO., LTD
Dashen Mashong, Dongguan,
Gunagdong, China

Attn.: Mr. Zhijian Ruan

Via ABS Eagle Construct Client Interface
Date: 18 February 2016
File Ref.: T-8-2
Refer to: ZL/T1486145
Oracle ID: 3201405
Class No.: YY254739

Re: **PX121 Platform Supply Vessel (Flag: Netherlands)**
83.4m (Loa) x 18.0m x 8.0m
Cosco (Guangdong) Shipyard Co., Ltd, Hull Nos.: N571

Gentlemen,

We have received your electronic transmittal CR1493581 dated 05 Februdary 2016 submitting the following document:

Item	*Drawing No.*	*Rev*	*Drawing Title*
1.	U10747.4_152-128-01	A	Lightship Survey Report

We have reviewed the report and are pleased to advise that the following lightship characteristics, as calculated by you, are satisfactory:

Lightship Weight	2704.878	Metric Tonnes
Lightship VCG	7.081	Meters above Baseline
Lightship LCG	39.154	Meters fwd of Fr. 0

An electronic copy of the above document, stamped to indicate our review, is available through the ABS Eagle Construct Engineering Manager (O2E), Web Portal.

Please be advised that the final stability booklets prepared using the above lightship particulars are to be submitted for our review.

Very truly yours,

K. M. Wong
Senior Vice President of Engineering
ABS Pacific Division

By: _____
L. W. Cheung
Senior Principal Engineer
Ship Engineering Department

cc: ABS Surveyor (Via ABS Eagle Construct)

See ABS Singapore Letter Ref T1499174 Dated 04-APR-2016

PACIFIC DIVISION
438B ALEXANDRA ROAD, #06-05, BLK B ALEXANDRA TECHNOPARK, SINGAPORE 119968
TEL: 65-6276-8700 FAX: 65-6272 2195 (SED) 65-6276 8130 (ESD) EMAIL: ABSSingapore@eagle.org WEB: www.eagle.org
CO. UEN: S58FC1284D

Ulstein Design & Solutions AS
Postboks 278
N-6067 ULSTEINVIK
NORWAY

RESTRICTED

ULSTEIN®

LIGHTWEIGHT SURVEY REPORT

for

ULSTEIN PX121

YARD NO. N571

"VOS PASSION"

REVIEWED
on behalf of the government of the vessel's registry as indicated in ABS letter

ABS FOUNDED 1862

Calculation :	**Approved :**
Adrian Fotache	

Ulsteinvik, 4.2.2016

Dwg. no. U10747.4_152-128-01_A

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See ABS Singapore Letter Ref 1480145 Dated 08-FEB-2016

1 GENERAL PARTICULARS

SHIP IDENTIFICATION

Ship name	VOS PASSION
Ship type	ULSTEIN PX121
Ship purpose	OFFSHORE SERVICE VESSEL
Signal letters	PBHU
Official number (IMO number)	9709130
Port of registry	BRESKENS
Flag	NL
Classification society	ABS
Classification society identification number	YY254739

OWNERS IDENTIFICATION

Name of the owner	VROON B.V.
Postal address	HAVEN WESTZIDE 21, 4511 AR BRESKENS
Country	THE NETHERLANDS

BUILDERS IDENTIFICATION

Name of the builder	COSCO (GUANGDONG) SHIPYARD CO., LTD
Postal address	DASHENG INDUSTRIAL ZONE, MACHONG TOWN
Country	DONGGUAN, GUANGDONG PROVINCE, CHINA

Yard number	N571
Date keel laid	10.11.2014
Name of the sister vessel (if any)	VOS PACE

MAIN PARTICULARS, MOULDED VALUES

Length over all	83.314 m
Length between perpendiculars	79.612 m
Length of the vessel acc. ICLL 1966	79.614 m
Length of the vessel according to class	79.554 m
Breadth moulded	18 m
Maximum breadth	18 m
Moulded depth to maindeck	8 m
Summer load draught	6.7 m
Keel thickness	12 mm
Mean thickness of the shell	12 mm

PARAMETERS AT SUMMER LOAD DRAUGHT

Block coefficient	0.693
Volume, moulded	6651.1 cbm
Volume, total	6676.4 cbm

2 RESULTS SUMMARY

LIGHTSHIP PARTICULARS

Lightship weight	2704.878 t
Lightship Longhitudinal Centre of Gravity	39.154 m
Lightship Transverse Centre of Gravity	-0.015 m
Lightship Vertical Centre of Gravity	7.081 m

LIGHTSHIP FLOATING POSITION

Lightship metacentric height	2.763 m
Lightship draught	3.346 m
Lightship trim	2.448 m
Lightship heel	-0.27 deg

*NOTE: As a basis of calculation VCG=7.081 m of a sister vessel has been used.

See ABS Singapore Letter Ref T1498145 Dated 08-FEB-2016

3 GENERAL INFORMATION

ENVIRONMENTAL CONDITIONS

Weather conditions in general	GOOD, CLOUDY
Sea state	CALM
Wind velocity	1.5 M/S
Current	WEAK
Mooring arrangement	SLACK MOORING ROPES
Density measured?	YES
Density of the water	0.9975 t/m3

OTHER INFORMATION

Test carried out at	COSCO (GUANGDONG) SHIPYARD CO. LTD.
Address:	DASHENG INDUSTRIAL ZONE, MACHONG TOWN DONGGUAN, GUANGDONG PROVINCE, 523146 CHINA
Date of Survey/test dd.mm.yyyy	03.02.2016
Survey/test commenced from hh:mm	14:30
Survey/test commenced to hh:mm	16:30
Report date dd.mm.yyyy	04.02.2016
Survey/test conducted by (name and company)	WANG FENG, COSCO (GUANGDONG) SHIPYARD CO., LTD.
Survey/test report prepared by (name and company)	ADRIAN FOTACHE, ULSTEIN DESIGN & SOLUTIONS AS
Attending surveyor	ERMAN KAMASOGLU
Classification society	ABS
If existing ship, state reason of rebuilding	
Year of last hull modification	
Amount of permanent ballast during the test	0 t
Amount of permanent ballast taken onboard after the test	0 t

SIGN CONVENTION

Zero references are as follows:
In longitudinal direction frame 0 (#0), positive direction is forward, negative aft
In transverse direction center line (CL), positive is SB, negative is PS
In vertical direction base line (BL), positive upwards, negative downwards

Trim by bow is assumed as positive (+)
Trim by stern is assumed as negative (-)
Heel towards starboard side (SB) is assumed as positive (+)
Heel towards port side (PS) is assumed as negative (-)

NOTES ON USED COMPUTATION METHODS

Hogging/sagging correction is considered directly on draughts by using mean of mean draught method. This method is determining the final draught as (draught aft + draught fwd + 6*draught amidship)/8

See ABS Singapore Letter Ref T1489715 Dated 08-FEB-2016

GENERAL REQUIREMENTS FOR THE SURVEY

The following items shall be checked and found in order by the person in charge of the survey before this report is submitted to the surveyor for approval:

1. Wind velocity during the test should not exceed 10 knots (5.144 m/s). There should be no significant swell or current. There should be now ice or snow which may influence the test. The vessel must be free floating; i.e. mooring lines must be slack during draught readings. Gangways, cranes etc. are to be in stowed position. As few cables, hoses etc. as possible should be connected to the shore. Those that are connected should be slack. The loading condition during the test is to be specified.
2. If no reliable drawings are available, position of draught marks should be checked prior to the test by measuring against a known datum point(s). Draught measurements should be performed on both sides aft, forward and midship. The depth of water under the hull should be sufficient to ensure that the hull will be entirely free from bottom.
3. Hydrometer for water sampling at approximately 25 % and 75 % of the draught are required to determine the specific gravity of seawater during test. The hydrometer or salinometer should be checked for accuracy prior to test. Seawater temperature is to be recorded if the density is not measured immediately.
4. Tanks should in principle be either empty or full; i.e. as few tanks as possible shall be partly filled. Full tanks should be topped up and checked through air pipes or by sounding. Ballast water tanks and fresh water tanks to be full should be filled to overflow where practically possible. The specific gravity of liquids in tanks must be recorded before test is commenced. All empty tanks and void spaces should be visually inspected before the tests. When the soundings have been taken, no movement of liquid onboard is allowed until the test is completed. Valves should be closed to eliminate any possibility of cross flow between the tanks. All system pipes and tanks are to be filled to normal operation condition.
5. Minimum number of personnel is to be onboard during the test. Persons onboard are to remain in the same position during the test. Deadweight is to be reduced to a minimum. A detailed list of remaining items of stores etc. should be prepared; samples of multiple items should be weighed as a check, as should also large items which are difficult to estimate. Loose weights (which may roll or slide) must be secured. List of foreign weights (weights that are not part of the lightweight) including position of their centre of gravity is to be prepared.
   Foreign weights include:
   - Liquids in tanks
   - Stores, provisions, personnel etc.
   - Temporary equipment and weights
   - Inclining weights
   - Any other variable weights
6. Degree of lightweight completion at the test; minimum about 98 % of the final lightweight. Solid foreign weights; maximum about 2 % of final lightweight. The values given here are intended as guidance. Deviations may be discussed with Class or National Authorities.

See ABS Singapore Letter Ref T14867AS Dated 08-FEB-2016

4 DRAUGHT READINGS

DRAUGHT MARKS LOCATION FROM #0:

Draught mark forward	72.1 m
Draught mark amidhsip	38.5 m
Draught mark aft	4.2 m

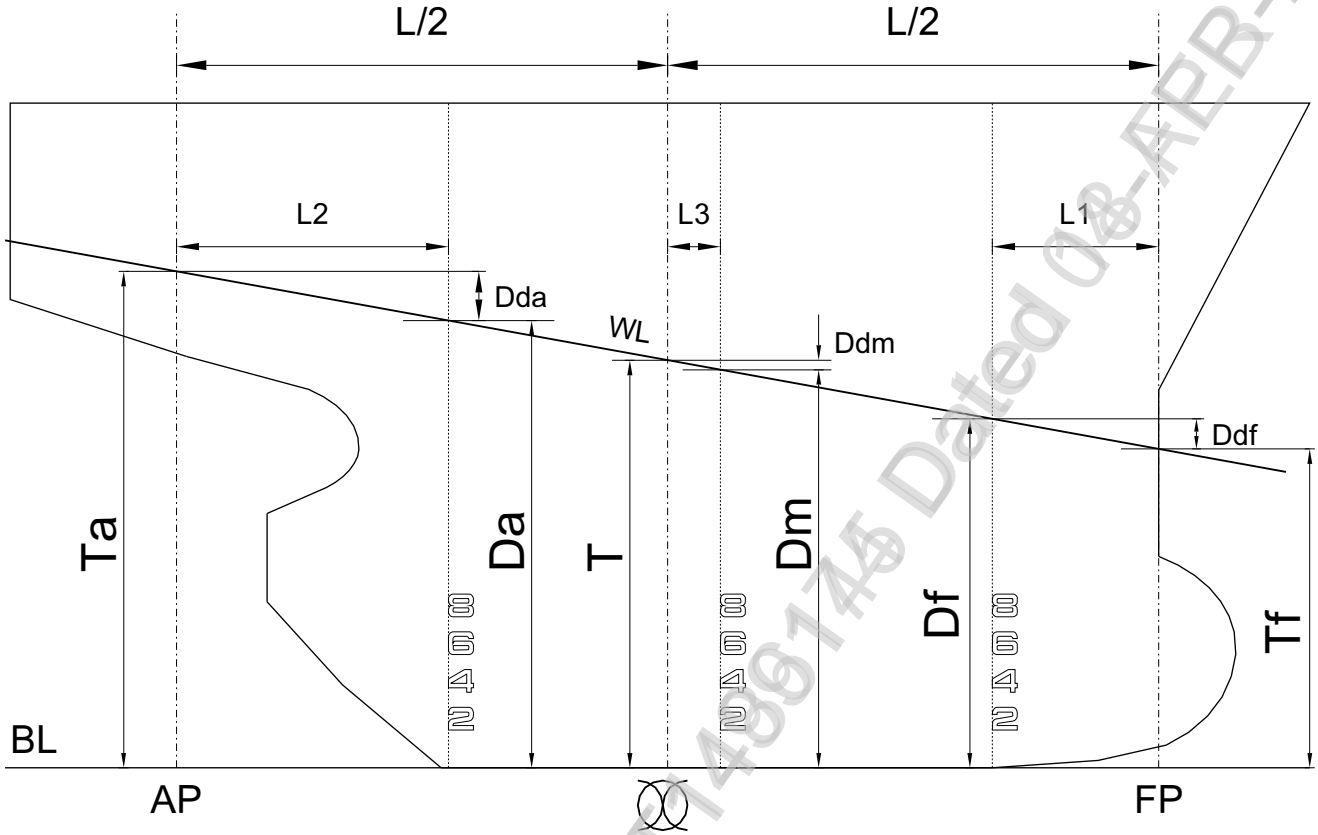
DRAUGHT READINGS:

	PS	SB
Draught reading foreship	3.87 m	3.87 m
Draught reading midship	3.59 m	3.61 m
Draught reading aftship	3.32 m	3.32 m

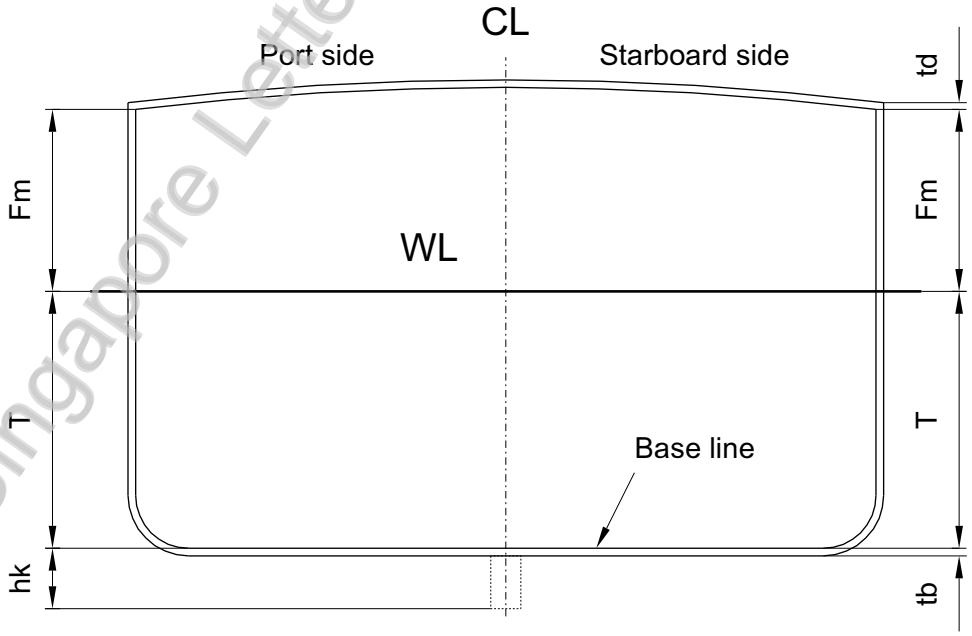
Correction for long. location of draught marks foreship	0.036 m
Correction for long. location of draught marks midship	-0.014 m
Correction for long. location of draught marks aftship	-0.059 m
Keel thickness correction	0.012 m
Moulded mean corrected draught at FP	3.894 m
Moulded mean corrected draught at L/2	3.574 m
Moulded mean corrected draught at AP	3.249 m
Hogging/Sagging deflection from draught marks	0.002 m
Moulded draught at L/2 including hogg./sagg. correction	3.573 m
Trim	0.645 m
Heeling angle	0.064 deg

See ABS Singapore Letter Ref T1486145 Dated 08-FEB-2016

EXPLANATORY PROFILE SKETCH:



EXPLANATORY FRAME SKETCH AMIDSHIP (L/2) WITH PLATE THICKNESS:



See ABS Singapore Letter Ref TH436145 Dated 08-FEB-2016

5 HYDROSTATICS AS SURVEYED

MAIN CHARACTERISTICS OF THE VESSEL:

Length between perpendiculars	79.61	m
Breadth (moulded)	18.00	m
Design draught (moulded)	6.00	m
X-coordinate of aft perpendicular	-3.05	m
X-coordinate of reference point (XREF)	36.75	m
X-coordinate of midship section (XMID)	36.75	m
Thickness of keelplate	0.012	m
Mean thickness of shell plating	0.012	m
Seawater density	0.998	ton/m3

Calculations are based on STABHULL date 2015-10-21 time 09:58

Shell thickness used in the calculation	12.0	mm
X-coord. of aft end of DWL	-5.90	m
X-coord. of fore end of DWL	76.50	m
Calc. sections	143	
Plate thickness	12.0	mm

EXPLANATION OF SYMBOLS:

T	draught, moulded	m
DISP	total displacement	t
VOLT	total volume	m3
LCB	longitudinal centre of buoyancy	m
TCB	transverse center of buoy.	m
VCB	vertical center of buoyancy	m
KMT	transv. metac. height	m
KML	long. metac. height	m
CB	block coefficient	
MCT	moment to change trim	tm/cm
TPC	change of displacement/change of draught	t/cm

Calculated for trim=0.64 heel=0.06

T	DISP	VOLT	LCB	TCB	VCB	KMT	KML	CB	MCT	TPC
m	t	m3	m	m	m	m	m		tm/cm	t/cm
3.573	2877.0	2884.2	36.657	0.007	1.955	9.700	116.146	0.5599	41.3	10.3

See ABS Singapore Letter Ref T1498145 Dated 08-FEB-2016

6 LIGHTSHIP SURVEY RESULTS

VERTICAL CENTRE OF GRAVITY as surveyed (VCG from sister vessel corrected for missign/foreign weights): 7.051 m

LONGITUDINAL CENTRE OF GRAVITY as surveyed (trim=0.645 m) (LCG=LCB-(VCG-VCB)*trim/L): 36.616 m

TRANSVERSE CENTRE OF GRAVITY as surveyed (heel=0.064 deg) (TCG=TCB-(VCG-VCB)*tan(heel)): 0.001 m

RESULT TABLE FOR FINAL LIGHTSHIP PARTICULARS

DESCRIPTION	WEIGHT t	XCG m	YCG m	ZCG m
Vessel as surveyed	2877.030	36.615	0.001	7.051
Missing weights to be added	19.160	46.332	-0.192	11.933
Foreign weights to be deducted	-56.110	0.817	-1.085	9.137
Tanks content to be deducted	-135.202	2.226	0.760	6.356
Shifted weights to be relocated from	-5.740	55.068	1.835	11.770
Shifted weights to be relocated to	5.740	58.780	2.061	13.384
Final lightship	2704.878	39.154	-0.015	7.081

Deviation from lightship weight of sister vessel 2723.22 t: 0.674 %
Deviation from sister vessel LCG=39.087 m: 0.08 %

Subdivision length Ls used for LCG ratio: 83.317 m

See ABS Singapore Letter Ref T1486145 Dated 08-FEB-2016

7 TANKS CONTENT DETAILS

TANK	Tank description	METHOD	VALUE	RHO t/m3	WEIGHT t	XCG m	YCG m	ZCG m	FSM tm
T033	WING TANK PS	FULL	0.000	0.998	58.961	-4.962	-3.920	6.516	0.00
T034	WING TANK SB	FULL	0.000	0.998	58.961	-4.962	3.920	6.516	0.00
T271	FO SETTLING TANK PS	SOUNDING	0.350	0.860	0.408	50.367	-6.105	2.678	0.13
T272	FO SETTLING TANK SB	SOUNDING	4.720	0.860	9.983	50.383	6.572	5.224	1.48
T273	FO SERVICE TANK PS	SOUNDING	0.030	0.860	0.027	52.098	-5.876	2.515	0.03
T274	FO SERVICE TANK SB	SOUNDING	3.210	0.860	6.473	52.079	6.238	4.390	0.86
T279	EMERG. GEN. FO TANK	SOUNDING	0.520	0.860	0.390	61.603	-1.375	23.607	0.03
TOTAL					135.202	2.226	0.760	6.356	2.52

See ABS Singapore Letter Ref T1486174 Dated 08-APR-2016

8 WEIGHTS TO BE ADDED

Item description	WEIGHT t	XCG m	YCG m	ZCG DECK m
Chairs in cabins	0.780	58.100	-0.100	10.700
Pilot chair in wheelhouse	0.200	61.030	0.000	21.000
Mattress	0.910	59.500	0.100	13.850
Safety equipment	1.200	53.900	0.000	8.000
Gymnasium equipment	0.500	58.800	0.000	7.300
Equipment in galley and mess room	0.120	60.200	-6.100	11.900
BOOKS AND DOCUMENTS	0.280	63.600	-3.400	11.340
Wooden plate on shelf	0.130	57.400	-1.500	9.350
Steel lockers in WARDROBE	1.020	57.370	6.340	9.100
Steel lockers in GYM	0.050	66.880	-1.170	9.100
Steel lockers in CORRIDOR B BECK	0.170	59.400	-4.640	14.900
Steel lockers in DECK WORKSHOP	0.090	54.600	-6.750	9.100
TV set & DVD	0.070	66.410	0.000	11.500
MOORING ROPE	0.270	66.700	0.000	15.500
Steel wire on TUGGER WHICH	0.300	51.000	0.000	8.600
Shelf and tools in deck work shop	0.130	54.600	-5.650	8.900
PVC floor,epoxy floor	1.200	63.700	0.000	20.000
Steel plate of wall in way of deck workshop	0.050	54.500	-6.300	9.500
Steel plate of wall in way of ER.workshop	0.040	61.100	2.950	6.900
Deck cover in water mist room	0.130	61.100	-6.250	6.200
Manhole covers on main deck	1.520	28.150	0.030	8.150
Storm cover of windows in front of wheelhouse	0.600	68.800	0.000	20.100
GRP grating on the radar mast platform	0.150	57.100	0.000	29.100
GRP grating on the platform in front of wheelhouse	0.180	64.390	0.000	21.180
GRP grating on Main deck in way of cargo rail	1.900	22.100	0.000	8.080
GRP grating on Main deck in front of FR72	0.810	51.800	0.000	8.080
GRP grating on A deck in way of cargo rail	2.000	22.200	0.000	19.800
Steel mesh on FR72	0.250	50.500	0.000	8.850
Oxygen bottle in HOSPITAL	0.100	60.200	5.500	8.600
BED IN HOSPITAL	0.050	60.200	5.400	8.400
Gangway (short)	0.100	50.600	0.000	14.200
Gangway (long)	0.200	38.720	-8.730	11.600
Waste COMPACTOR	0.300	58.400	-6.670	8.480
High lashing	0.410	-0.300	0.000	8.500
Low lashing	0.250	71.400	0.000	11.260
OX/AC EQUIPMENT	0.450	53.598	-4.394	9.273
Crane in engine room	0.300	50.015	-4.953	4.400
Wood in E/R workshop	0.100	61.958	1.786	6.777
Washing table in workshop	0.050	62.520	2.800	6.220
Net for louver in accommodation	0.100	56.000	-0.325	1.651
Air pipes	0.020	20.486	2.434	11.824
NET CAB.	0.200	60.225	2.290	17.357
VSAT CAB.	0.150	60.180	2.240	18.370
ANTENNA	0.050	57.160	0.190	25.850
ELECTRIC EQUIPMENT OF BRIDGE DECK	0.180	59.460	-0.250	21.320
TELEPHONE	0.010	61.290	-0.530	16.520
DECK LIGHT	0.012	61.286	-0.525	16.518
Valves for sounding pipes	0.010	29.993	0.000	2.210
Tools in workshop	0.100	61.958	1.786	6.777
Pain on tank top	0.248	51.400	0.000	1.750
Paint one coat for wall and ceiling in engine room	0.078	51.400	0.000	4.700
Remain one coat for tween deck in engine room	0.011	51.400	0.000	5.300
Remain one coat below tween deck in cargo room	0.045	30.000	0.000	3.150
Remain two coats for Bulwark on walkway top (PS)	0.113	25.200	-8.275	11.300
Remain two coats for Bulwark on walkway top (SB)	0.113	25.200	8.275	11.300
Remain one coat in walkway (PS)	0.084	26.600	-8.275	8.000
Remain one coat in walkway (SB)	0.084	26.600	8.275	8.000
Paint on open decks on accomodation	0.192	51.800	0.000	13.800
TOTAL	19.160	46.332	-0.192	11.933

See ABS Singapore Letter Ref T149514 Dated 08-AEB-2016

9 WEIGHTS TO BE REMOVED

Item description	WEIGHT t	XCG m	YCG m	ZCG DECK m
Scaffolding arround bridge deck	1.835	60.900	0.000	21.275
Accommodation protection	1.200	57.450	0.000	13.180
Protect under maindeck	0.200	52.500	0.000	3.300
Weight Block #72	13.090	-3.230	-6.170	8.600
Weight Block #73	13.290	-3.250	0.500	8.600
Weight Block #78	12.870	-3.250	-0.700	8.600
Weight Block #94	13.100	-3.350	1.700	8.700
5 pers	0.375	49.000	0.000	11.800
Vent hose	0.150	49.700	0.000	8.000
TOTAL	56.110	0.817	-1.085	9.137

See ABS Singapore Letter Ref T1498174 Dated 08-FEB-2016

10 WEIGHTS TO BE RELOCATED

Item description	WEIGHT t	XCG0 m	YCG0 m	ZCG0 m	XCG m	YCG m	ZCG m	DECK
Liferafts	0.910	26.950	0.000	8.400	49.300	0.000	12.660	
Pannel in Wheelhouse	1.100	65.380	0.300	21.510	62.800	0.000	21.450	
PANNEL IN LAUNDRY	0.350	60.500	4.770	8.900	63.750	3.690	8.900	
Handrail in accomodation	0.800	57.110	5.680	8.300	60.400	0.150	13.370	
Shelves in wooden box in.	0.250	54.100	0.400	14.270	61.650	1.200	16.230	
Galley cabinets	1.120	66.860	-0.200	11.340	63.470	3.790	11.340	
Furniture in HOSPITAL & .	1.080	60.200	4.750	8.500	62.000	4.500	9.340	
Rope aft MDK	0.130	-4.900	-7.800	8.100	-4.900	7.800	8.100	
TOTAL	5.740	55.068	1.835	11.770	58.780	2.061	13.384	

See ABS Singapore Letter Ref T1498174 Dated 08-APR-2016

11 PARTICULAR WEIGHTS INCLUDED IN THE LIGHT SHIP

The light ship as calculated shall exclude cargo, fuel, lubricating oil, water ballast or other fluid or pumpable ballast, potable water, feed water, sewage, provisions and other consumables as well as crew, passengers and their effects. However, the minimum amount of mooring equipment, spares etc. required by national authority or the classification society as well as the minimum amount of water in boilers, system oil in piping etc. is to be included in the light ship.

The light ship data calculated in this report includes the following items related to special operations and shall not be included in the ship's loading conditions:

Included in lightship weight:

MOB-boat
Liferafts
Mattresses in beds
Hydr./Lub. oil in systems
Cooling water in systems
Gas bottles in stores (Ac./Ox. and CO2)
Two anchors and chain
Galley/ mess equipm.
Crane and wire on crane

Not included in lightship weight:

Fuel oil in day and settling tank
Hydr./Lub. oil in stores tanks
Personal equipment carried onboard by crew
Stores
Spare parts
Various tools etc.

Note:In this table items such as fishing gear, towing gear, fixed hotel appliaces, removable modules and solid movable ballast which may be taken ashore in case of change of operation shall be specified.

Comments on the survey/test (if any):

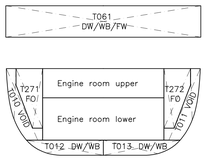
See ABS Singapore Letter Ref T1498174 Dated 04-APR-2016

13.2 TANK PLAN DRAWING

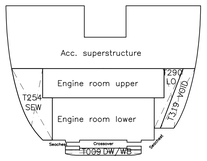
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See ABS Singapore Letter Ref T1499174 Dated 04-APR-2016

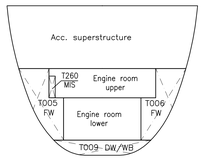
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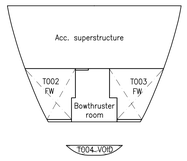
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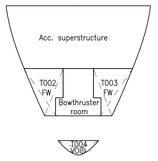
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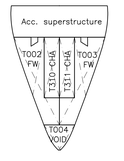
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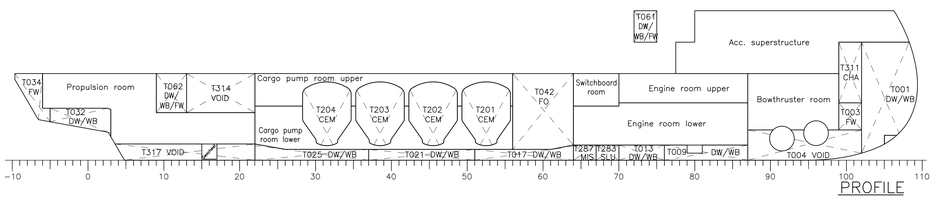
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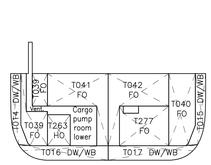
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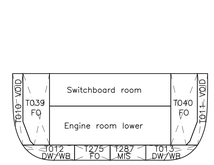
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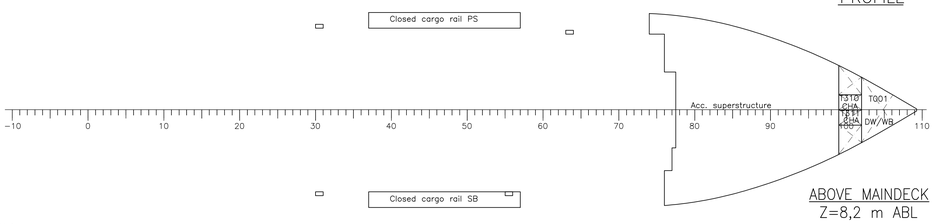
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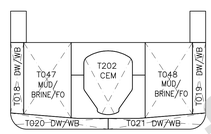
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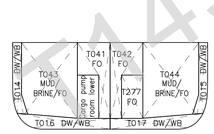
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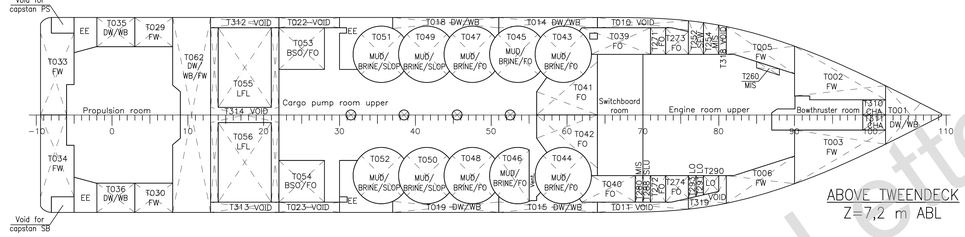
ABOVE MAINDECK
Z=8,2 m ABL



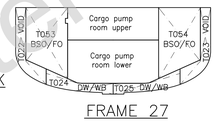
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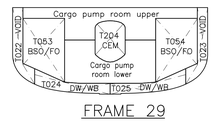
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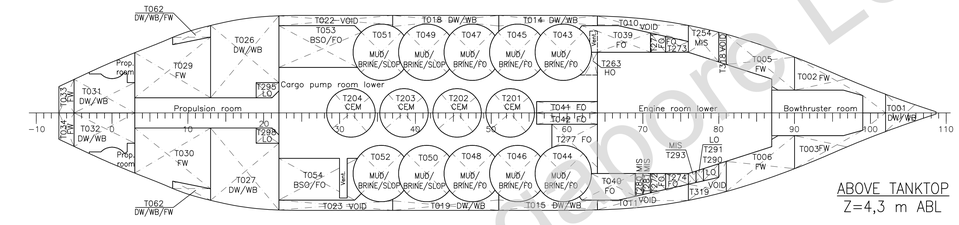
ABOVE TWEENDECK
Z=7,2 m ABL



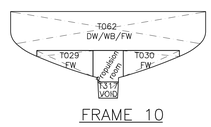
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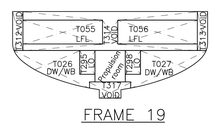
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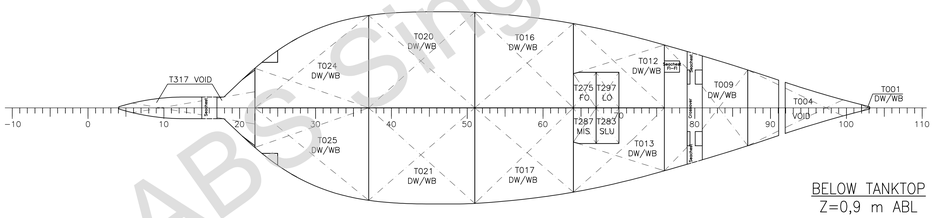
ABOVE TANKTOP
Z=4,3 m ABL



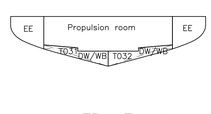
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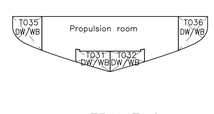
FRAME 19



BELOW TANKTOP
Z=0,9 m ABL



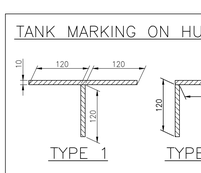
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FRAME 0

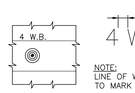
TANK NO.	MAX. AIR PIPE HEIGHT ABL. (m)	TANK ID.	FRAME MIN.#	FRAME MAX.#	CONTENTS	DENSITY (t/m3)	MAX DENSITY (t/m3)	VOLUME (m3)	WEIGHT (t) BRINE	BSO	CEM	DW	FO	FW	HO	LFL	LO	MIS	MUD	SEW	SLOP	SLU	WB	CENTER OF GRAVITY (m) LCG(AP)	TCG(C1)	VCG(BL)	IT (m4)	TANK NO.
001	14,56	FOREPEAK TANK	102	109	DW/WB	1,000	1,025	102,3				102,31												104,87	73,11	0,00	7,22	33,1	001
002	14,56	WING TANK PS	90	102	FW	1,000	1,000	113,6						113,64											67,06	-2,88	6,32	44,9	002
003	14,56	WING TANK SB	90	102	FW	1,000	1,000	114,9						114,92											67,02	2,86	6,33	46,5	003
004	14,56	DB TANK CE	87	102	VOID	1,025	1,025	73,1																	65,08	0,00	1,39	92,6	004
005	14,56	WING TANK PS	81	91	FW	1,000	1,000	98,1						98,06											59,97	-5,07	4,86	33,6	005
006	14,56	WING TANK SB	81	91	FW	1,000	1,000	98,1						98,06											59,97	5,07	4,86	33,6	006
009	14,56	DB TANK CE	76	87	DW/WB	1,000	1,025	75,4				75,37												77,25	56,90	0,09	0,73	592,6	009
010	11,66	WING TANK PS	64	76	VOID	1,025	1,025	59,1																	49,30	-7,86	4,45	7,6	010
011	11,66	WING TANK SB	64	76	VOID	1,025	1,025	59,1																	49,30	7,86	4,45	7,6	011
012	11,66	DB TANK PS	64	76	DW/WB	1,000	1,025	54,9				54,85												56,22	49,41	-3,85	0,76	200,4	012
013	11,66	DB TANK SB	64	76	DW/WB	1,000	1,025	54,9				54,85												56,22	49,41	3,85	0,76	200,4	013
014	11,66	WING TANK PS	51	64	DW/WB	1,000	1,025	60,0				60,02												61,52	40,15	-8,42	4,73	1,3	014
015	11,66	WING TANK SB	51	64	DW/WB	1,000	1,025	59,9				59,94												61,44	40,15	8,42	4,73	1,3	015
016	11,66	DB TANK PS	51	64	DW/WB	1,000	1,025	83,5				83,50												85,59	40,20	-4,33	0,62	462,2	016
017	11,66	DB TANK SB	51	64	DW/WB	1,000	1,025	83,5				83,50												85,59	40,20	4,33	0,62	462,2	017
018	11,66	WING TANK PS	37	51	DW/WB	1,000	1,025	72,5				72,47												74,28	30,79	-8,37	4,73	1,9	018
019	11,66	WING TANK SB	37	51	DW/WB	1,000	1,025	72,5				72,47												74,28	30,79	8,37	4,73	1,9	019
020	11,66	DB TANK PS	37	51	DW/WB	1,000	1,025	94,4				94,39												96,75	30,84	-4,63	0,61	567,8	020
021	11,66	DB TANK SB	37	51	DW/WB	1,000	1,025	94,4				94,39												96,75	30,84	4,63	0,61	567,8	021
022	11,66	WING TANK PS	22	37	VOID	1,025	1,025	60,3																	19,27	-7,88	3,89	21,0	022
023	11,66	WING TANK SB	22	37	VOID	1,025	1,025	60,3																	19,27	7,88	3,89	21,0	023
024	11,66	DB TANK PS	22	37	DW/WB	1,000	1,025	74,3				74,34												76,19	21,46	-3,86	0,81	345,3	024
025	11,66	DB TANK SB	22	37	DW/WB	1,000	1,025	74,3				74,34												76,19	21,46	3,86	0,81	345,3	025
026	11,66	WING TANK PS	13	22	DW/WB	1,000	1,025	89,9				89,95												92,20	12,42	-4,45	3,13	177,7	026
027	11,66	WING TANK SB	13	22	DW/WB	1,000	1,025	89,9				89,95												92,20	12,42	4,45	3,13	177,7	027
029	11,66	WING TANK PS	3	13	FW	1,000	1,000	76,1						76,14											5,26	-4,92	4,67	82,5	029
030	11,66	WING TANK SB	3	13	FW	1,000	1,000	76,1						76,14											5,26	4,92	4,67	82,5	030
031	11,66	WING TANK PS	-5	3	DW/WB	1,000	1,025	26,4				26,37												27,03	-0,64	-1,75	4,18	33,8	031
032	11,66	WING TANK SB	-5	3	DW/WB	1,000	1,025	26,4				26,37												27,03	-0,64	1,75	4,18	33,8	032
033	11,66	WING TANK PS	-10	-5	FW	1,000	1,000	59,0						58,96											-4,96	-3,92	6,52	125,8	033
034	11,66	WING TANK SB	-10	-5	FW	1,000	1,000	59,0						58,96											-4,96	3,92	6,52	125,8	034
035	11,66	WING TANK PS	-2	3	DW/WB	1,000	1,025	23,1				23,11												23,69	0,39	-7,51	6,69	5,7	035
036	11,66	WING TANK SB	-2	3	DW/WB	1,000	1,025	23,1				23,11												23,69	0,39	7,51	6,69	5,7	036
039	11,66	STORAGE TANK PS	62	71	FO	0,860	0,860	79,4					68,28												46,68	-6,76	4,97	7,2	039
040	11,66	STORAGE TANK SB	62	69	FO	0,860	0,860	63,9					54,95												46,00	6,80	4,93	5,5	040
041	11,66	STORAGE TANK PS	56	64	FO	0,860	0,860	81,3					69,90												42,17	-1,52	5,53	30,9	041
042	11,66	STORAGE TANK SB	56	64	FO	0,860	0,860	81,3					69,90												42,17	1,52	5,53	30,9	042
043	11,66	LIQUID CARGO TANK PS	56	63	MUD/BRINE/FO	0,860	2,800	140,1	350,21				120,47							392,24					41,65	-5,57	4,63	35,9	043
044	11,66	LIQUID CARGO TANK SB	56	63	MUD/BRINE/FO	0,860	2,800	140,1	350,21				120,47							392,24					41,65	5,57	4,63	35,9	044
045	11,66	LIQUID CARGO TANK PS	50	56	MUD/BRINE/FO	0,860	2,800	127,2	317,89				109,35							356,03					37,22	-5,58	4,61	34,9	045
046	11,66	LIQUID CARGO TANK SB	50	56	MUD/BRINE/FO	0,860	2,800	124,3	310,78				106,91							348,07					37,18	5,58	4,54	34,9	046
047	11,66	LIQUID CARGO TANK PS	44	50	MUD/BRINE/FO	0,860	2,800	127,2	317,89				109,35							356,03					33,02	-5,58	4,61	34,9	047
048	11,66	LIQUID CARGO TANK SB	44	50	MUD/BRINE/FO	0,860	2,800	127,2	317,89				109,35							356,03					33,02	5,58	4,61	34,9	048
049	11,66	LIQUID CARGO TANK PS	38	44	MUD/BRINE/SLOP	1,000	2,800	127,2	317,89											356,03		127,15			28,82	-5,58	4,61	34,9	049
050	11,66	LIQUID CARGO TANK SB	38	44	MUD/BRINE/SLOP	1,000	2,800	127,2	317,89											356,03		127,15			28,82	5,58	4,61	34,9	050
051	11,66	LIQUID CARGO TANK PS	32	38	MUD/BRINE/SLOP	1,000	2,800	126,2	315,43											353,28		126,17			24,63	-5,56	4,63	34,9	051
052	11,66	LIQUID CARGO TANK SB	32	38	MUD/BRINE/SLOP	1,000	2,800	126,2	315,43											353,28		126,17			24,63	5,56	4,63	34,9	052
053	11,66	STORAGE TANK PS	22	33	BSO/FO	0,830	0,860	128,6		106,71			110,57												18,71	-6,02	4,82	34,1	053
054	11,66	STORAGE TANK SB	22	33	BSO/FO	0,830	0,860	128,3		106,53			110,38												18,69	6,02	4,84	34,1	054
055	11,66	LFL TANK PS	14	21	LFL	1,200	1,200	77,6								93,18									12,25	-4,38	6,10	162,1	055
056	11,66	LFL TANK SB	14	21	LFL	1,200	1,200	77,6								93,18									12,25	4,38	6,10	162,1	056
061	11,66	ROLL RED. TANK	72	76	DW/WB/FW	1,000	1,025	125,2				125,23		125,23										128,37	51,66	0,00	11,25	325,3	061
062	11,66	ROLL RED. TANK	8	13	DW/WB/FW	1,000	1,025	205,6				205,63		205,63										210,77	7,44	0,00	6,21	1694,0	062
201	11,66	DRY BULK TANK	49	56	CEM	2,400	2,400	65,0			155,92														36,75	0,00	4,67	19,6	201
202	11,66	DRY BULK TANK	42	49	CEM	2,400	2,400	65,0			155,92														31,85	0,00	4,67	19,6	202
203	11,66	DRY BULK TANK	35	42	CEM	2,400	2,400	65,0			155,92														26,95	0,00	4,67	19,6	203
204	11,66	DRY BULK TANK	28	35	CEM	2,400	2,400	65,0			155,92														22,05	0,00	4,67	19,6	204
252	11,66	BLACK WATER TANK	76	78	SEW	1,000	1,000	10,7													10,65				53,89	-7,06	6,69	3,6	252
254	11,66	GREY WATER TANK	76	80	MIS	1,000	1,000	32,3											32,26						54,83	-6,49	4,59	6,3	254
260	-	WATER MIST TANK	85	88	MIS	1,000	1,000	2,4											2,38						60,55	-4,30	6,30	0,1	260
263	11,66	HYDRAULIC OIL STORE TANK	62	64	HO	0,924	0,924	5,1								4,73									44,30	-4,34	2,76	0,7	263
271	11,66	FO SETTLING TANK PS	71	73	FO	0,860	0,860	14,2					12,21												50,39	-6,62	5,66	1,7	271
272	11,66	FO SETTLING TANK SB	71	73	FO	0,860	0,860	14,2					12,21												50,39	6,62	5,66	1,7	272
273	11,66	FO SERVICE TANK PS	73	76	FO	0,860	0,860	17,6					15,18												52,10	-6,50	5,84	2,5	273
274	11,66	FO SERVICE TANK SB	73	76	FO	0,860	0,860	17,6					15,18												52,10	6,50	5,84	2,5	274
275	11,66	FO DRAIN TANK	64	67	FO	0,860	0,860	9,3					8,03												45,85	-1,64	0,70	6,1	275
277	11,66	FO OVERFLOW TANK	58	64	FO	0,860	0,860	41,4					35,59												42,99	2,51	3,06	11,9	277
279	-	EMERG. GEN. FO TANK	87	89	FO	0,860	0,860	2,2					1,89												61,59	-1,37	24,60	0,0	279
280	11,66	BILGE WATER HOLDING TANK	69	70	MIS	1,000	1,000	8,9											8,86						48,65	6,66	5,17	0,9	280
281	11,66	BILGE WATER SETTLING TANK	70	71	MIS	1,000	1,000	4,0											3,95						49,34	6,41	3,68	0,7	281
283	11,66	SLUDGE DRAIN TANK	67	70	SLU	1,000	1,000	9,4															9,38		47,95	1,64	0,70	6,2	283
287	11,66	BILGE WATER DRAIN TANK	64	67	MIS	1,000	1,000	9,3											9,34						45,85	1,64	0,70	6,1	287
288	11,66	SLUDGE SETTLING TANK	70	71	SLU	1,000	1,000	4,5														4,45			49,35	6,81	6,66	0,9	288
290	11,66	LO STORE TANK MAIN ENG.	78	80	LO	0,924	0,924	9,1										8,41							55,29	5,97	5,66	0,8	290
291	11,66	LO STORE TANK MAIN ENG.	77	78	LO	0,924	0,924	4,8										4,43							54,25	6,20	5,82	0,6	291
293	11,66	LO STORE TANK THRUSTER	76	77	LO	0,924	0,924	5,0										4,58							53,55	6,34	5,92	0,7	293
295	11,66	LO STORE TANK AZIMUTH THR.	19	22	LO	0,924	0,924	6,1										5,66							14,33	-2,11	3,28	0,5	295
297	11,66	LO DRAIN TANK	67	70	LO	0,924	0,924	9,4										8,67							47,95	-1,64	0,70	6,2	297
298	11,66	DIRTY LO TANK	19	22	LO	0,924	0,924	6,1										5,66							14,33	2,11	3,28	0,5	298
310	-	CHAIN LOCKER PS	99	102	CHA	1,025	1,025	16,0																	70,35	-0,70	8,10	0,5	310
311	-	CHAIN LOCKER SB	99	102	CHA	1,025	1,025	16,0																	70,35	0,70	8,10	0,5	311
312	11,66	CD VOID OUTSIDE LFL TANKS PS	13	22	N2	0,001	0,001	20,2																	12,30	-8,51	6,25	0,5	312
313	11,66	CD VOID OUTSIDE LFL TANKS SB	13	22	N2	0,001	0,001	20,2																	12,30	8,51	6,25	0,5	313
314	11,66	CD VOID AROUND LFL TANKS CE	13	22	N2	0,001	0,001	200,5																	12,25	0,00	6,28	2191,0	314
317	11,66	CD VOID CE	4	22	VOID	1,025	1,025	33,1																	10,60	0,00	0,83	36,6	317
318	11,66	CD VOID PS	80	81	VOID	1,025	1,025	10,4																	56,35	-6,25	4,99	1,8	318
319	11,66	CD VOID SB	76	81	VOID	1,025	1,025	32,5																	55,09	6,87	4,89	4,7	319
		TOTAL WEIGHTS							**3231,51**	**213,24**	**623,69**	**1666,45**	**1260,18**	**1025,73**	**4,73**	**186,36**	**37,42**	**56,79**	**3619,29**	**10,65**	**506,65**	**13,83**	**1708,11**					
		TOTAL VOLUMES							**1292,60**	**256,92**	**259,87**	**1666,45**	**1465,33**	**1025,73**	**5,12**	**155,30**	**40,50**	**56,79**	**1292,60**	**10,65**	**506,65**	**13,83**	**1666,45**					
									BRINE	BSO	CEM	DW	FO	FW	HO	LFL	LO	MIS	MUD	SEW	SLOP	SLU	WB					

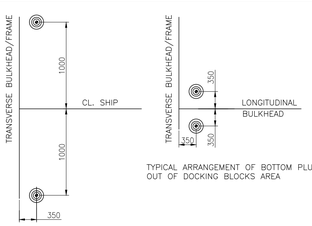
TANK MARKING ON HULL



EXAMPLE TANK MARKING BY EACH BOTTOM PLUG

(SEE MANHOLE PLAN, DWG NO. U10747_304−310−01)





TYPICAL ARRANGEMENT OF BOTTOM PLUGS OUT OF DOCKING BLOCKS AREA

BOTTOM PLUGS M42x2 (STAINLESS STEEL) ACC. TO NS 2573 (ISO 5483).
PLUGS TO BE PREVENTED FROM UNLOCKING
WELDING SEAMS WITH TANK NUMBERS BY EACH BOTTOM PLUG
WELDING SEAM ON EACH CORNER OF TANK.

NOTE:
PLUGS FOR DW/WB AND FW TANKS TO BE OF HEXAGONAL TYPE, OTHER TANKS TO BE OF QUADRATIC TYPE.

PRINCIPAL DIMENSIONS:

L.O.A. .....APPROX.. :	83,4 m
L.P.P. (Design).....:	76,5 m
LF. (DNV Pt.3 Ch.1 Sec.1)..:	79,7 m
BREADTH MOULDED ..... :	18,0 m
DEPTH TO MAIN DECK..... :	8,0 m
MAX. DRAUGHT (SLW).... :	6,7 m
DESIGN DRAUGHT ..... :	6,0 m

REV.	DATE	DESCRIPTION	DRAWN
E	23.11.14	Changes according to as built	RAG
D	08.07.14	Voids T320,T321 removed and T279 moved, capstan modified	RAG, KHS
C	18.06.14	Updated position T061	KHS
B	14.10.13	Changed position fore bow thruster and T062	KHS

DRAWING TITLE: TANK PLAN PLATFORM SUPPLY VESSEL

DESIGN: ULSTEIN PX121

CLIENT INFO: VROON

DRAWN: TVA | DATE: 25.10.12 | NUMBER OF SHEET: 1

SCALE: 1:200 | FORMAT: A0 | REVISION: E

DRAWING NO: U10747_101 - 105 - 01

PROJECT NO. | SFI CODE (DOC. CODE) | SHEET NO.

APPROVED BY:

ULSTEIN
Ulstein Design & Solutions AS. Postbox 278, 6067 Ulsteinvik, Norway
www.ulsteingroup.com

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See ABS Singapore Letter Ref T1499 dated 04-APR-2016

13.3 DRAUGHT MARKS DRAWING

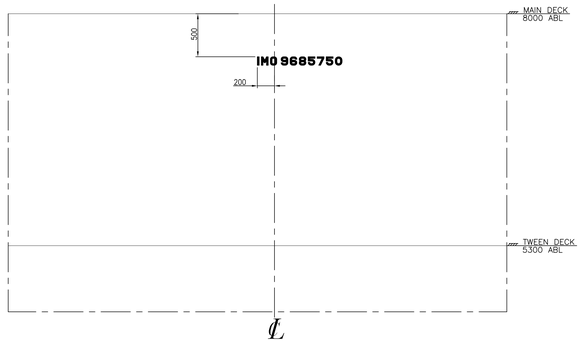
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See ABS Singapore Letter Ref T1499174 Dated 04-APR-2016

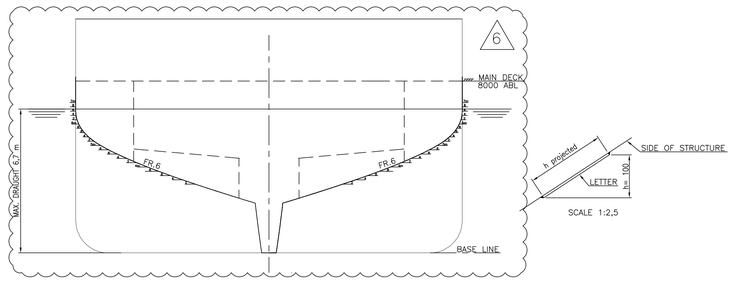
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N568-261-050

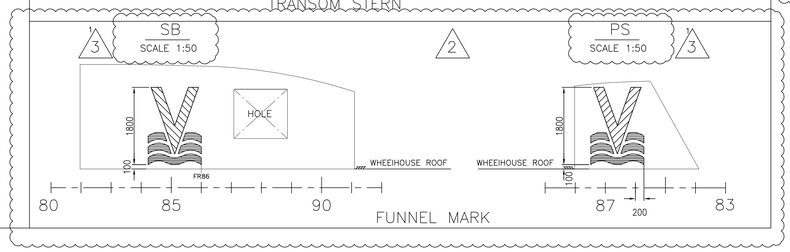
FR70
SCALE 1:25



IMO NUMBER IN ENGINE ROOM

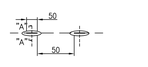


TRANSOM STERN



FUNNEL MARK

FUNNEL MARK & BOW MARK WELDED BEADS DETAIL



NOTE :
WELD BEAD – TO BE PROVIDED AT ALL CORNERS AND INTERMITTENTLY ON THE STRAIGHT LINES.
BEADS SHALL BE FREE FROM SCATTERS AND UNDERCUTS.

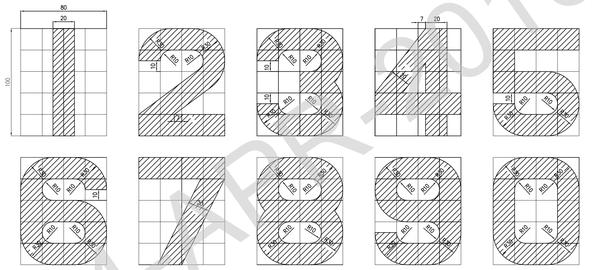
V MARK WELDED BEADS DETAIL

1



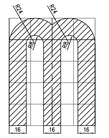
NOTE :
WELD BEAD – TO BE PROVIDED AT ALL CORNERS AND INTERMITTENTLY ON THE STRAIGHT LINES.
BEADS SHALL BE FREE FROM SCATTERS AND UNDERCUTS.

LETTER OF DRAFT MARK



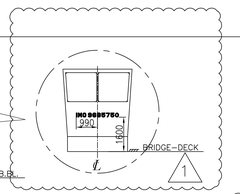
DETAIL OF DRAFT MARK

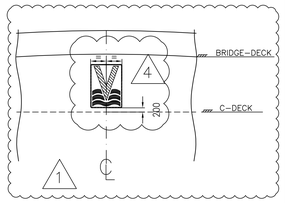




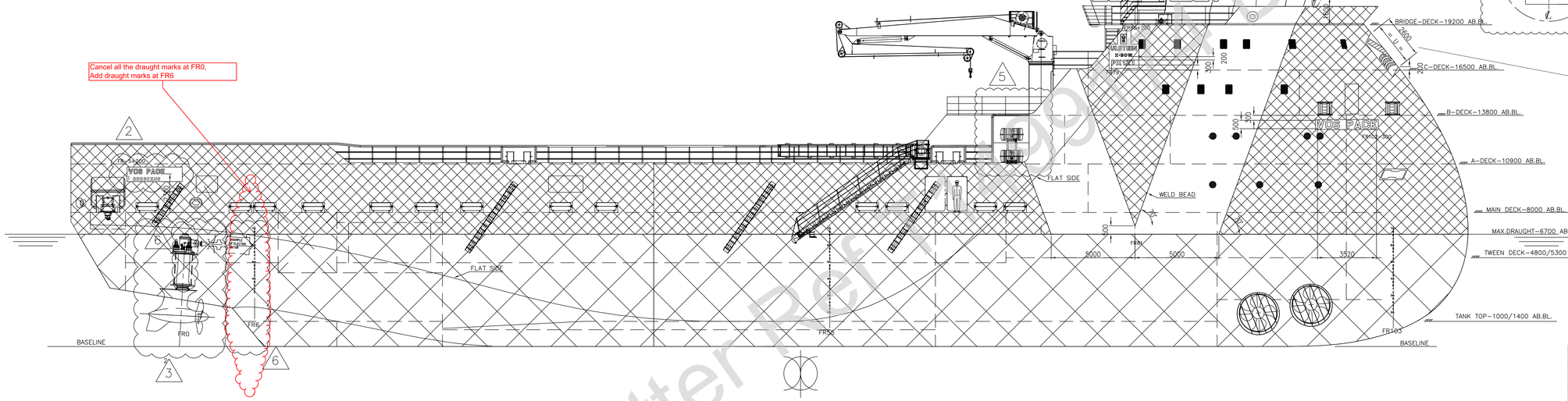
2
6m = DRAUGHT MARKS
8

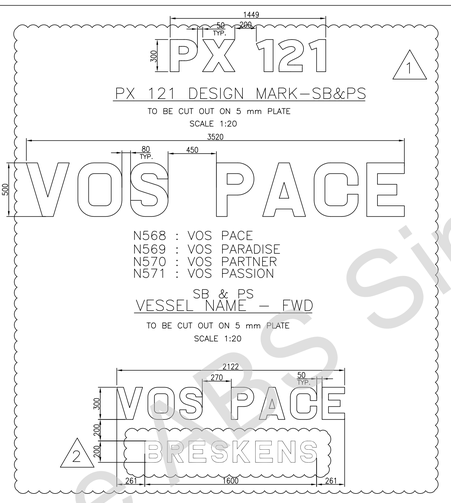
= 2 PCS.





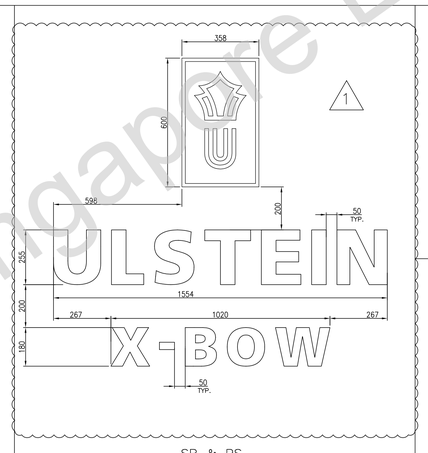
HOLD!!!
ITEMS:
- Load line cert



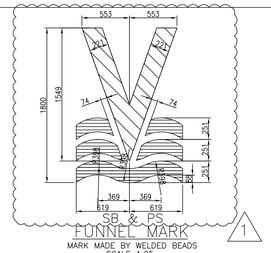


SB & PS
VESSEL NAME – FWD
TO BE CUT OUT ON 5 mm PLATE
SCALE 1:20

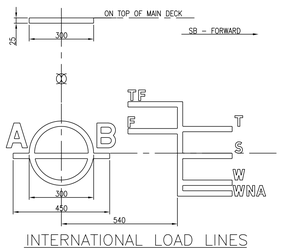
SB & PS
VESSEL NAME & PORT OF REGISTRY – AFT
TO BE CUT OUT ON 5 mm PLATE
SCALE 1:20



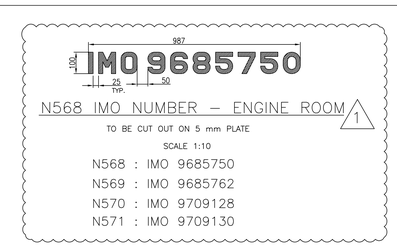
SB & PS
UISTEIN DESING MARK &
DESING LETTERS & X–BOW LETTERS
TO BE CUT OUT ON 5 mm PLATE
SCALE: 1:10



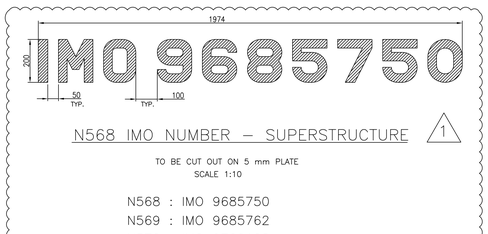
SB & PS
FUNNEL MARK
MARK MADE BY WELDED BEADS
SCALE 1:25



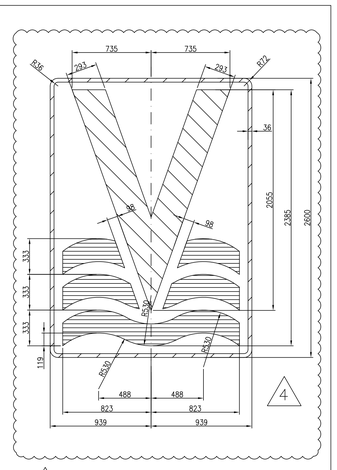
INTERNATIONAL LOAD LINES
STARBOARD SIDE
PORT SIDE MIRROR
TO BE CUT OUT ON 5 mm PLATE
SCALE 1:10



N568 IMO NUMBER – ENGINE ROOM
TO BE CUT OUT ON 5 mm PLATE
SCALE 1:10
N568 : IMO 9685750
N569 : IMO 9685762
N570 : IMO 9709128
N571 : IMO 9709130



N568 IMO NUMBER – SUPERSTRUCTURE
TO BE CUT OUT ON 5 mm PLATE
SCALE 1:10
N568 : IMO 9685750
N569 : IMO 9685762
N570 : IMO 9709128
N571 : IMO 9709130



BOW MARK
MARK MADE BY WELDED BEADS
SCALE 1:20

Class Comment Tracking List			
Approval Letter Reference No.			
Comment No.	Comments	Check	Remark
	NO COMMENT		

Owner Comment Tracking List			
Approval Letter Reference No.			N568-261-050-PCC-VROONCGS-0201
Comment No.	Comments	Check	Remark
1	Please find CAD drawing of V mark enclosed. Please update the drawing accordingly.	CLOSE	Please see N568-DT-CGSVROON-0152 ,31th,Dec 2013, the rev.2 had been sert to you for comment.
2	Funnel mark PS and SB is switched, please correct.	CLOSE	Agree, Please see rev.3 drawing.

NO	DESCRIPTIONS & NOTES	BY	DATE
6	BECAUSE PROPULSION AFFECT CANCEL DRAUGHT MARKS AT FR0, AND ADD DRAUGHT MARKS AT FR6.	H.M.Z	2014.6.27
5	UPDATE HULL BACKGROUND	H.M.Z	2014.6.12
4	BOW MARK UPDATE TO OWNER'S COMMENT	H.M.Z	2014.4.23
3	PROPULSION UPDATE	H.M.Z	2014.4.21
2	REVISED ACCORDING TO OWNER'S COMMENT	H.M.Z	2014.4.21
1	ADD HONE PORT MARK	H.M.Z	2013.12.28
1	REVISED ACCORDING TO OWNER'S COMMENT	H.M.Z	2013.11.22

LIST OF ALTERATIONS

1	N568-101-100	GENERAL ARRANGEMENT
ITEM	DWG NO	DRAWING NAME OR DOCUMENTS TITLE

LIST OF REFERENCED DRAWINGS AND DOCUMENTS

DATE	REV.	DESCRIPTION	DWN	CHKD	APPD
2014.6.27	6	ISSUED FOR COUSTRUCTION	H.M.Z	Wei.Cai	M.L.P
2014.6.12	5	ISSUED FOR COUSTRUCTION	H.M.Z	Wei.Cai	M.L.P
2014.4.23	4	ISSUED FOR COUSTRUCTION	H.M.Z	Wei.Cai	M.L.P
2014.4.21	3	ISSUED FOR COUSTRUCTION	H.M.Z	Wei.Cai	M.L.P
2013.12.28	2	ISSUED FOR COUSTRUCTION	H.M.Z	Wei.Cai	M.L.P
2013.11.22	1	ISSUED FOR APPROVAL	H.M.Z	Wei.Cai	M.L.P
2013.07.17	0	ISSUED FOR APPROVAL	H.M.Z	Wei.Cai	M.L.P

BUILDER	COSCO(GUANGDONG) SHIPYARD CO. LTD	HULL NO.	N568-N569
		CLASS	ABS
OWNER	VROON	VOS PASSION	
PROJECT	PX121 PLATFORM SUPPLY VESSEL		

DWG TITLE:	DWG No.	N571-261-050	
NAME&MARKING INCL IMO NUMBER &DRAUGHT MARKS	SFI No.	N/A	
	SCALE	SHEET	STATUS
	1:100	1/1	IFC

COSCO (GUANGDONG) SHIPYARD CO.,LTD
Dashang Industrial Park, Machang, Dongguan, Guangdong, China, 523146
Tel: +86 769 8828 6988 Fax: +86 769 8823 1981 Website: www.cosco-shipyard.com

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OWNER	–
CLASS	–
HULL	–
MACH	–
PIPING	–
OUTF'G	–
HVAC	–
E&I	–
SHOP	–
PJM	–
QA/QC	–
HSE	–
ORDC	–
Total	

CLASS APP. DATE:

OWNER APP. DATE:

A0

See RS Singapore Letter Ref No. dated 04 APR 2016

Table of content is located at the beginning of the booklet.

See ABS Singapore Letter Ref T1499174 Dated 04-APR-2016

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